CLINICAL SPORTS NUTRITION

5TH EDITION

LOUISE BURKE • VICKI DEAKIN

20.

FIFTH EDITION

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Preface

Clinical Sports Nutrition, now in its fifth edition, continues to update the science and practice of sports nutrition. The text has been thoroughly revised by internationally recognised experts, incorporating the latest research and scientific principles relevant to the elite athlete. At the end of each chapter, experienced sports dietitians provide practical strategies for translating this theory into practice.

Topics include the physiology of sports; nutritional assessment of athletes; measuring physique; weight loss and weight making; post-exercise recovery; nutritional strategies before and during competition; iron depletion; micronutrient needs; eating disorders in athletes; gastrointestinal issues; supplements and sports foods; requirements for specific athletic populations (children, vegetarians and masters athletes); and considerations in different environmental conditions (travel, cold, heat and altitude).

New chapters have been added on periodisation of nutrition; gastrointestinal issues; food allergies and intolerances; immunity; infective illness and injury; and a revised approach to the female athlete triad. New commentaries include discussions on the molecular link with exercise; measuring energy availability; the physiological effects of exercise on gut function; antioxidant supplements; and updated fluid requirements.

This book is targeted at students interested in a career in sports nutrition. It is also for sports nutrition professionals who need to translate science into their practice with athletes and coaches. We wish all of you excellence in your endeavours and hope that *Clinical Sports Nutrition* can assist you on this pathway.

Louise Burke
Vicki Deakin

Acknowledgments

Thank you to our long-standing sports nutrition experts and to the emerging researchers in sports nutrition who have contributed their knowledge and experience in updating this edition. This edition has been an endurance event for all of us. Once again, we have had to shut ourselves away from the daily needs of our families, friends and workplaces to make this project happen. We thank many people for their understanding—in particular, the men in our lives—Lachlan Deakin, and John and Jack Hawley.

Thank you to Sarah Long, Claire Linsdell and Jane Roy at McGraw-Hill Education and to freelance copyeditor Rosemary Moore, for their understanding and support in helping us to bring this edition together. We appreciate your patience.

Thank you also to our work colleagues. The team in the Department of Sports Nutrition at the Australian Institute of Sport undertakes a large variety of inspiring activities each day. To that we have added writing book chapters. All things are possible with the best team in the world.

Finally we thank all the coaches, athletes and other sport service providers with whom we have worked, and who continue to challenge us and keep us pursuing new opportunities to improve the performance, health and well-being of ourselves and our athletes through nutrition intervention.

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DIGITAL RESOURCES

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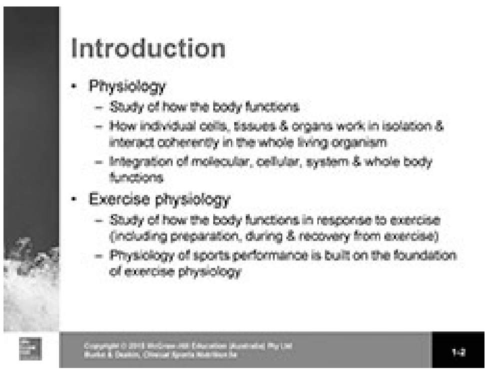


CHAPTER ONE

Physiology of sports

Ronald J Maughan and Susan M Shirreffs

1.1 Introduction

Physiology is the study of how the body functions: how individual cells, tissues and organs work in isolation and how they interact in a coherent way in the whole living organism. An understanding of physiology is of central importance in medicine and all of the related health sciences, including nutrition. The scope of physiology ranges from understanding events at the molecular level (e.g. how muscles contract or how cells sense nutrients) to the integrative physiology of organs and systems (e.g. the brain and the cardiovascular and respiratory systems) and how they are regulated and adjust to stress or change (e.g. in response to exercise or to environmental extremes such as the microgravity of space flight). The emphasis of physiology is on the integration of molecular, cellular, system and whole-body function. Understanding normal function allows prediction of how the organism will respond to any stimulus that is applied.

Exercise physiology is the study of how the body functions in response to the challenge of exercise. Again, this includes the study of how cells, tissues and organs work, but specifically in preparation for exercise, during exercise and in the recovery period after exercise. Research in the field of exercise physiology has led to understanding in a wide range of topics related to sport and exercise nutrition. The physiology of sports performance is built on the foundation of exercise physiology and includes the study of elite athletes in an effort to understand the physiological characteristics that contribute to successful performance in different sports and of the various factors, including training and nutrition, that can modify these characteristics and so determine performance in various types of sports activities.

1.2 Homeostasis

Homeostasis refers to the maintenance of a constant internal environment of the body. It is the central feature of the body's physiological processes. These processes aim to reverse any imposed change and so maintain a constant internal environment. The existence of these mechanisms allows the human body to cope with conditions far beyond those that are normally encountered; typically, we use only a small part of our functional capabilities in normal day-to-day activities. A bout of hard exercise is one of the biggest threats to this homeostasis that the healthy individual is likely to encounter. Exercise, if it is sufficiently severe and prolonged, will result in changes to the partial pressure of oxygen in the tissues, blood volume and blood pressure, temperature, acid-base balance and many other parameters. While some of these changes may be beneficial to exercise performance, all will cause system failure if they are too great. Homeostatic mechanisms act to limit these changes in order to allow exercise to continue.

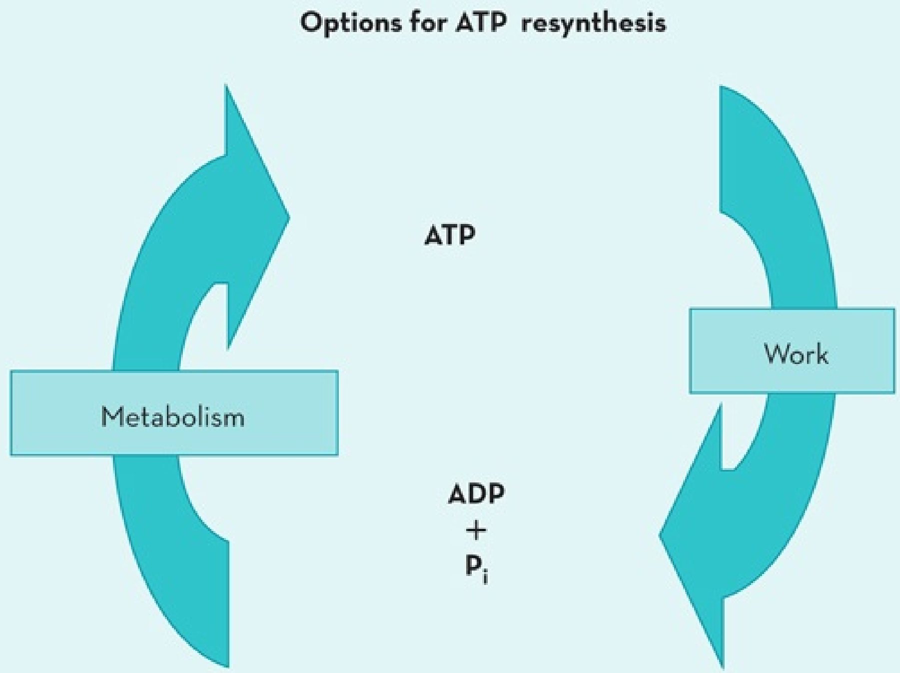
All exercise involves an increase in the rate of energy use by the active muscles: if the muscles cannot meet the energy demand then the exercise cannot be sustained. In both high-intensity exercise and long-duration exercise, fatigue will result if the supply of energy cannot keep pace with the demand. Meeting the demands of exercise requires an integrated response by all of the body's major organs. The limiting factor will depend on the nature of the activity and of the physiological characteristics of the individual, but exercise cannot continue past the point at which the body can keep its internal environment within the rather narrow limits of the physiological norm. The only option in this situation is to reduce the exercise intensity or to stop altogether.

1.3 Acute responses to exercise

1.3.1 Metabolic

At rest in a comfortable environment, the average human body consumes about 200–300 mL of oxygen each minute. This oxygen is used in the chemical reactions that provide the energy necessary to maintain physiological function—the processes that keep us alive. When at complete rest, we do no external work but chemical work is done in maintaining chemical and electrical gradients across cell membranes, and various biosynthetic and catabolic reactions occur: all of the energy used in these processes appears as heat, allowing us to maintain body temperature at a higher level than that of our surroundings. During exercise, the muscles require additional energy to generate force or to do work, the heart has to work harder to increase blood supply, the respiratory muscles face an increased demand for moving air in and out of the lungs, and so the metabolic rate must increase accordingly, with a corresponding increase in the rate of heat production. In sustained exercise, an increased rate of energy turnover—and therefore of heat production—must be maintained for the duration of the exercise and for some time afterwards: depending on the task and on the fitness level of the individual, this may be from 5–20 times the resting metabolic rate. In very high intensity activity, the demand for energy may be more than 100 times the resting level, though such intense efforts can be sustained for only very short periods of time. In spite of these large changes in metabolic activity, the internal environment of the body changes little: effective buffering systems work to limit any change.

All the body's cells require a constant input of energy to maintain homeostasis. The ionic and chemical gradients that exist across cell membranes and between intracellular compartments must be maintained, biosynthetic chemical reactions proceed, and other energy-demanding processes must be supported. The ultimate energy source for all of these reactions is the chemical energy contained in the foods that we have eaten. This energy is made available by the oxidation of the fat, carbohydrate (CHO), protein and alcohol present in those foods, in the process of which they are converted to degradation products, primarily carbon dioxide and water. The immediate source of energy used in the body's cells is the hydrolysis of the terminal phosphate bond in the high-energy phosphate compound adenosine triphosphate (ATP), releasing a phosphate group, adenosine diphosphate (ADP) and energy (Figure 1.1). ATP is a large molecule, and it is stored in cells in only very small amounts, so the challenge is to regenerate the cellular ATP content as fast as it is being used. In some cells, ATP is used at a more or less constant rate, but other tissues, especially skeletal muscle, have a relatively low energy demand while at rest and an extremely high rate of ATP demand when active. It is the sum of all these ATP-consuming reactions that determines the total energy turnover, or metabolic rate.



Capacity (the amount of work that can be done) and power (the rate at which work can be done) of the three energy-supplying metabolic processes in human skeletal muscle

	Capacity (J/kg)	Power(W/kg)
ATP/CP hydrolysis	400	800
Lactate formation	1000	325
Oxidative metabolism		200

Figure 1.1 Options for ATP resynthesis

Note: These values are expressed per kg of muscle. They are approximations only and will be greatly influenced by training status and other factors.
ATP = adenosine triphosphate, CP = creatine phosphate

The regeneration of ATP from ADP requires an input of energy, and there are three main ways in which this energy can be made available (Figure 1.1). Each offers some advantages and disadvantages to the cell, but together they allow the muscle to produce very high rates of power for short periods or to sustain lower power outputs for prolonged periods. Muscle cells contain large amounts of creatine, and indeed about 95% of the total amount of creatine in the body is found in the skeletal muscles, which explains why meat is a good source of creatine in the diet. Degradation of the stored creatine to creatinine is an irreversible reaction that occurs at a rate of about 1.6% per day (2 g/d for an average individual, but in proportion to the total muscle mass), and the creatinine that is produced is excreted in the urine. This reaction is so constant that the urinary creatinine excretion can be used as a marker for muscle mass. The animal muscle present in a typical non-vegetarian diet provides about 1 g of creatine each day, and the remainder of the requirement can be synthesised from amino acids (methionine, arginine and glycine) obtained from the diet.

At rest, about two-thirds of this creatine is in the form of creatine phosphate (CP), and one phosphate group (Pi) can be transferred from CP to ADP to form free creatine (C) and ATP as shown here in a reaction catalysed by the enzyme creatine kinase:

$$ATP \Rightarrow ADP + Pi$$

$$CP + ADP \Rightarrow ATP + C$$

This single-enzyme reaction can occur extremely rapidly in the presence of a high concentration of CP (Figure 1.1) but the CP store is limited and the rate at which this reaction can sustain activity begins to fall within a few seconds. Even the 100-m sprinter generally slows down towards the end of the race. In low-intensity exercise, little CP is used. Most of the energy comes from aerobic metabolism. In very high intensity exercise, the muscle CP concentration falls to very low levels within 30–60 seconds, but this allows the ATP concentration to be maintained.

This is not a full description of the reaction taking place, and it ignores the important role of this reaction in intracellular buffering during high-intensity exercise. A proton is absorbed during this reaction, and this can help buffer the protons released by the formation of lactate when high rates of anaerobic glycolysis occur:

$$CP^{2+} + ADP^{3-} + H^{+} \Rightarrow ATP^{4-} + C$$

It is important also to recognise that the majority of the energy used during exercise is generated by oxidative phosphorylation in mitochondria, but ATP utilisation during muscle contraction occurs in the cytoplasm. CP shuttles phosphate groups across the mitochondrial membrane, thus serving as a spatial buffer to distribute energy through the cell. The muscle creatine (and therefore CP) content can be increased by supplementing the diet with creatine for a few days (at doses of 10–20 g/d) or a few weeks (at 3–5 g/d), leading to increases in performance of high-intensity efforts.

Two key elements in producing energy are the power (rate of work) that can be achieved and the capacity (amount of work) of the system. CP hydrolysis can support high power outputs as the resynthesis of ATP by this mechanism is very fast, but it has a low capacity, so fatigue soon intervenes (Figure 1.1). The second source of energy is the breakdown of CHO (primarily glycogen stored in the muscle cells) to pyruvate by glycolysis and further conversion of this pyruvate to lactate (often referred to as lactic acid even though it is dissociated at physiological pH). Glycolysis converts one six-carbon glucose molecule to two three-carbon molecules, and some of the energy liberated in this sequence of reactions is conserved in the form of the high-energy phosphate bond in ATP. The chemical energy that is lost as heat at each stage of the series of reactions is important in ensuring the reactions proceed in one direction only—when the reactions need to be reversed, energy in the form of ATP must be fed back into the system. The breakdown of glycogen to pyruvate is accompanied by conversion of Nicotinamide adenine dinucleotide (NAD), an essential co-factor within the pathway, to NADH (the reduced form of NAD). Conversion of pyruvate to lactate allows regeneration of NAD, thus maintaining the cellular concentration of NAD and allowing glycolysis to continue. For each glucose residue converted to lactate, three ATP molecules are formed if glycogen is the starting point. Two are formed if glucose is the substrate. Muscle pH falls as lactate accumulates, and this has a variety of effects on the muscle: it can inhibit some of the key enzyme-catalysed reactions (though their rate is kept high by other chemical changes taking place within the cells), it can interfere directly with the contractile process by affecting calcium binding, and it produces the characteristic sensation of discomfort that accompanies very intense muscular effort. In spite of the negative effects of a falling pH, the energy made available by anaerobic glycolysis allows a higher intensity of exercise than would otherwise be possible. These pathways are anaerobic—no molecular oxygen is involved in the process of regenerating ATP.

Alternatively, instead of being metabolised to lactate, pyruvate is oxidised to CO_2 and H_2O. As shown in Figure 1.1, this is a much slower process, but it generates more energy and has a virtually unlimited capacity. Endogenous fuel stores for oxidation (glycogen in muscle and liver, plus triglyceride in muscle and adipose tissues, as well as the carbon skeletons of some amino acids) are large and can be replenished by ingestion of foods containing these substrates during exercise. Complete oxidation of one molecule of glucose to carbon dioxide and water leads to the formation of 38 molecules of ATP. Oxidation of one molecule of palmitic acid, as an example of a typical fatty acid, results in the generation of 127 molecules of ATP. The contribution of protein to oxidative energy supply over the day will normally equal the fraction that dietary protein contributes to the total daily energy intake: this must be so if a steady state is to be maintained. If the availability of CHO is limited, however, the contribution of protein to energy metabolism will be increases and there will be a net loss of protein unless this is balanced by an increased intake. In exercise lasting more than a few minutes, aerobic metabolism predominates. In prolonged exercise such as a marathon race, anaerobic metabolism makes a significant contribution to the total energy requirement only at periods of high energy demand when the energy demand is transiently increased (e.g. going uphill, or in an intermediate sprint or a finishing sprint). Team games consist of multiple short sprints, when anaerobic energy supply is important, but aerobic recovery must follow each sprint: during the recovery period, aerobic metabolism allows correction of any fall in the cellular ATP concentration, recovery of the muscle CP stores, and oxidation of the lactate produced. Until these processes are essentially complete, subsequent sprint performance will be impaired. The capacity of oxidative metabolism is essentially unlimited, as the system can be continually refuelled even during exercise.

The power that can be generated by aerobic metabolism varies greatly between individuals and is usually characterised by the maximum rate of oxygen consumption (VO_{2max}) that can be achieved. The VO_{2max} can be expressed either in absolute terms (litres of oxygen per minute) or relative to body mass (mL of oxygen per kg of body mass per minute). The former is more relevant when absolute power output must be considered, but the latter becomes more relevant where body mass must be moved against gravity, such as when running or cycling uphill. The highest rate of oxygen consumption that can be achieved will vary greatly between individuals and is influenced by genetic endowment, training and health status, gender, age, body composition and other factors (Table 1.1).

TABLE 1.1 Typical VO_{2max} values for different subject groups

Subject group	VO_{2max} value
Functionally impaired	15–25 mL/kg/min
Typical sedentary	30–40 mL/kg/min
Recreationally active	40–60 mL/kg/min
Elite endurance athlete	65–85 mL/kg/min

Note: Values for men are typically somewhat larger (perhaps by 5–15%) than those for women

Endurance activities require a high rate of aerobic metabolism and this is achieved by having a high maximal aerobic capacity and by working at a high fraction of that capacity. The fraction of aerobic capacity that can be sustained declines progressively as the duration of exercise increases. It is important to recognise that this is perhaps related more to distance than to time. The elite marathon runner who takes less than about 2 hours 10 minutes to complete the race will use an average of about 80–85% of his VO_{2max} over the whole distance, but the slowest participants (5 hours or even more), whether male or female, probably use only about 60–65% of their (much lower) VO_{2max} (Maughan & Leiper 1983). In a race over a distance of 100 km, however, where the world record is a little outside 6 hours, the elite performer in these events probably uses about 60–65% of VO_{2max} over the whole distance. In such prolonged exercise, a high VO_{2max} is not absolutely necessary for success as the ability to work at a high fraction of VO_{2max} can compensate. Not surprisingly, therefore, athletes in events that last only a few minutes (middle distance running, rowing, pursuit cycling) often record higher values for VO_{2max} than those achieved by endurance athletes.

If the oxygen supply is limited, it is important to make effective use of the available oxygen. In this regard, CHO is a better fuel than fat: per litre of oxygen, 21.1 kJ are available when CHO is the fuel, but oxidation of fat generates only 19.5 kJ. This difference may appear small and it has been largely ignored, but it is important when competing at the limits of what is possible. It provides a compelling reason for ensuring an adequate supply of CHO. For almost 50 years, there has been a strong emphasis in sports nutrition on sparing CHO by increasing the contribution of fat to energy metabolism. It might be more appropriate to ensure that as much CHO as possible is available and is used to maximise the efficiency of exercise performance.

The various options open to the muscle for providing energy do not operate independently: they are fully integrated to ensure that, as far as possible, the energy demand is met with the smallest threat to the cell's homeostasis. Even in a 100-m sprint some (perhaps about 10%) of the energy is provided by oxidative metabolism, using primarily oxygen stored in the tissues.

The marathon runner who accelerates in mid-race will almost certainly generate some ATP from anaerobic metabolism. Table 1.2 shows the relative contributions of anaerobic and aerobic energy supply in races over different distances: these values are only approximations, but indicate how the balance of energy supply shifts as the duration of exercise increases.

TABLE 1.2 Approximate contributions of anaerobic and aerobic energy supply to total energy demand in races over different distances

Distance	Duration (min:s)	% Anaerobic	% Aerobic
100 m	0:9.58	90	10
400 m	0:43.18	70	30
800 m	1:40.91	40	60
1500 m	3:26.00	20	80
5000 m	12:37.35	5	95
10 000 m	26:17.53	3	97
42.2 km	122:57	1	99

Note: The times given for each distance are the current (November 2014) men's world records for these events

The metabolic response to exercise is dictated largely by the biochemical characteristics of the muscle fibres and their recruitment pattern. In low-intensity work, only a few motor units are activated and these will involve type 1 fibres. These fibres have a high oxidative capacity, a relatively low glycolytic capacity and a good supply of oxygen. This has some important implications for the choice of substrate used. Most of the energy required by these fibres is derived from the oxidation of fatty acids that can come either from the relatively large amounts of fat stored as triglyceride in adipose tissue via the plasma or from the much smaller amounts of triglyceride present in the form of intracellular fat stores. CHO breakdown makes only a small contribution to the energy needs of these fibres. As progressively more motor units are recruited, those with a lower capacity for fat oxidation and a greater reliance on CHO as a fuel begin to be activated. Eventually, a point is reached where, even though the oxidative type 1 fibres are still contributing, some of the fibres being recruited are breaking glycogen down to pyruvate faster than it can be oxidised in the mitochondria. Some of this excess pyruvate is converted to lactate, to regenerate the coenzyme NAD within the cytoplasm of the cells and thus allow glycolysis to continue. Some of this lactate diffuses out of the muscle cells and a progressive rise in the blood lactate concentration is observed.

The pattern of substrate use is therefore dictated primarily by the exercise intensity. It is not fixed, however, and will change over time as well as being modulated by a number of factors, including prior diet and exercise, fitness level and environmental conditions. Increasing the muscle glycogen content by feeding a high-CHO diet for a few days will lead to an increased rate of glycolysis at rest and during exercise: blood lactate will be elevated and CHO oxidation also increased. Likewise, feeding a high-fat, low-CHO diet will shift metabolism in favour of fat oxidation. Increasing aerobic fitness levels as a result of endurance training has a number of cardiovascular and metabolic effects, but one of the key adaptations is to increase that oxidative capacity of the muscle, and in particular to increase the ability to oxidise fatty acids. This results in a marked shift in the pattern of substrate use in favour of fat oxidation, but, as highlighted above, this means that there will be an increased oxygen requirement at any given power output.

1.3.2 Muscle

Skeletal muscle makes up about 40% of total body mass in a typical lean male and about 35% in a typical lean female, but these proportions will vary greatly depending on the sport and on the individual athlete. The successful bodybuilder is characterised by an exceptionally high muscle mass combined with an extremely low body fat content. The sumo wrestler and the rugby front-row forward also have a high muscle mass, but have also a relatively high body fat content. Muscle is a highly plastic tissue and responds rapidly to use or disuse. Both muscle mass and muscle composition (and therefore functional properties) respond to training and detraining. These changes are generally specific to the training stimulus and proportional to the training load.

Each muscle contains hundreds or even many thousands of individual muscle cells: these specialised cells are long and thin, and are usually referred to as fibres rather than as cells. Two proteins, actin and myosin, make up much of the muscle mass, and the interaction of these two proteins allows muscle cells to generate force. These proteins are arranged to form long overlapping microfilaments within the skeletal muscle fibres and these filaments can slide past each other, allowing the muscle to shorten. A number of other proteins are involved in the regulation of the interaction of these filaments, allowing precise control of activation and relaxation. Part of the myosin molecule functions as an ATPase, breaking down ATP and so making energy available to power muscle activity.

The unit of contraction is the sarcomere. The maximum force that a muscle can generate is closely related to its physiological cross-sectional area—to the number of sarcomeres in parallel; peak velocity of shortening is proportional to muscle fibre length—to the number of sarcomeres in series (Maughan et al. 1983 and Figure 1.2). The strength of a joint, however, is determined by a number of biomechanical parameters, including the distance between muscle insertions and pivot points and muscle size.

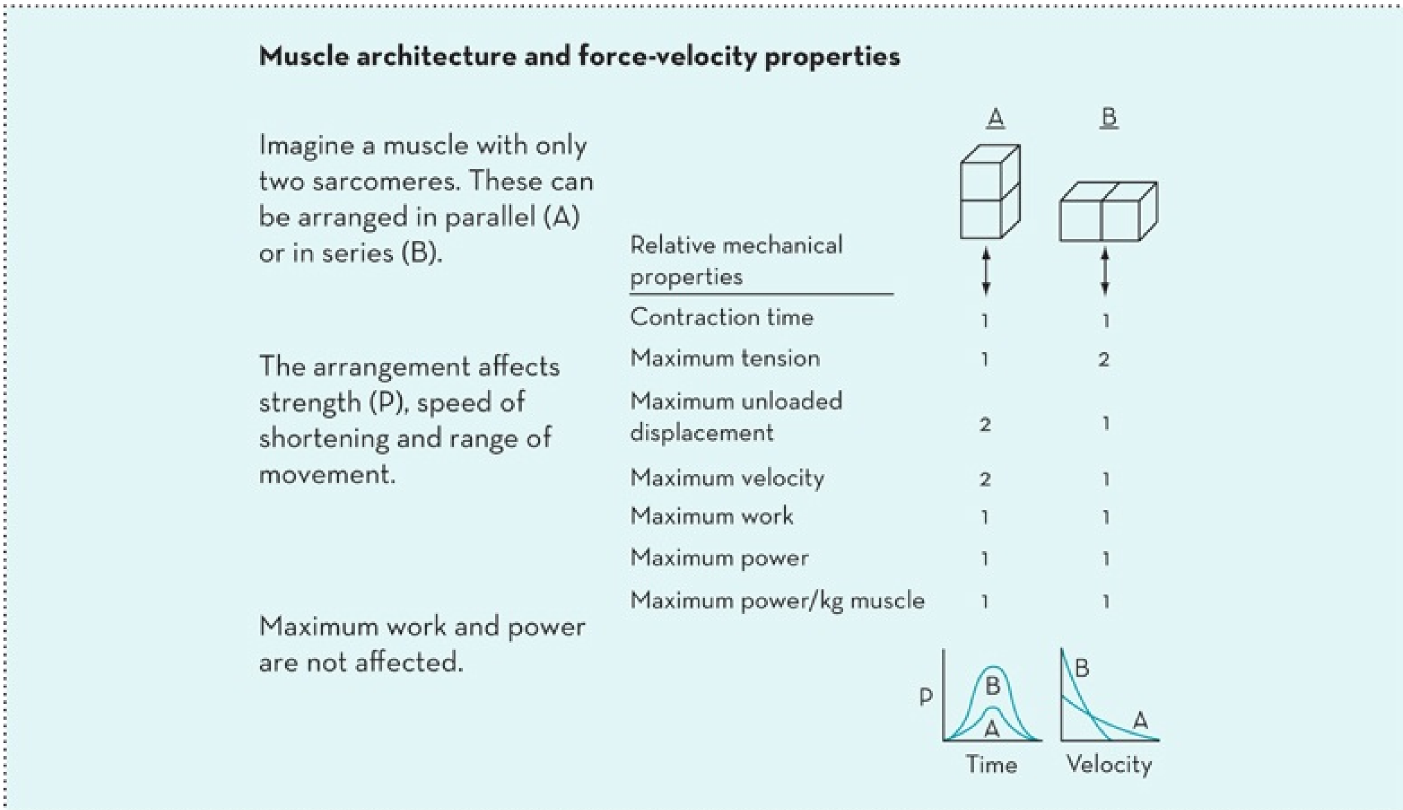


Figure 1.2 Muscle architecture and force–velocity synthesis

All exercise requires activation of several muscle groups. The extent of activation is determined by the demand placed on the muscle in terms of the force to be generated or the speed of movement. Not all of the muscle fibres are used in most tasks—only enough to generate the force necessary. The higher the force required, the greater the number of individual muscle fibres that must be activated. The fibres with the lowest activation threshold (i.e. the first to be recruited) are those with a low speed of contraction and a high fatigue resistance: this makes sense as these fibres will be used most often in daily tasks. As the weight to be moved is increased or the power output increases (i.e. an increased speed in running or cycling), progressively more motor units are recruited. At very high forces, all the fibres are likely to be active. In prolonged exercise, some of the fibres that were recruited in the early stages may become fatigued and will cease to contribute to work performance and be replaced by others.

The maximum force that a muscle can generate is closely related to its physiological cross-sectional area: that is, to the number of sarcomeres (units of contraction) in parallel. Peak velocity of shortening is proportional to muscle fibre length: that is, to the number of sarcomeres in series. Muscles are typically arranged in opposing groups so that as one group of muscles contracts, another group acting across the same joint relaxes so as not to oppose lengthening as it is stretched. The organisation of the nerve impulse transmission to the muscles means that it is impossible to stimulate the contraction of two antagonistic muscles at any one time.

Receptors in the muscle and in the tendons can sense the length of the muscle and also the force being generated, allowing the brain to know how many fibres need to be recruited and how to coordinate the recruitment of the different muscles. In well-trained and highly motivated individuals, it can be demonstrated that the muscle is working maximally—adding an electrical stimulus to the nerve does not generate any additional force.

With appropriate training, the muscle will adapt, but the nature of the adaptation is determined by the training stimulus applied. Although an individual's muscle fibre distribution is to some degree an inherited characteristic, training can change both the size of the individual fibres and their contractile properties.

Muscle fibre types

Human muscle fibres can be classified in a number of ways, depending on their maximum speed of contraction, their biochemical characteristics and their resistance to fatigue. Contraction occurs by interaction of actin and myosin filaments within the fibres, and the speed of contraction is determined largely by the ATPase activity of the myosin: the faster ATP can be hydrolysed to release energy, the faster contraction can occur. Three main fibre types are generally recognised in skeletal muscle, but it is important to recognise that substantial changes can be induced by training. The type II muscle fibres of an elite marathon runner may have a higher oxidative capacity than the type I fibres of the sprinter, but they are still distinct from the type I fibres present in the same muscle. The primary muscle fibre types are as follows.

- *Type I* slow oxidative (also called slow twitch or fatigue-resistant fibres). These muscle fibres are dark red because of their high myoglobin content and high density of blood capillaries, contain many mitochondria and so have a high oxidative capacity, have a slow peak contraction velocity and are relatively fatigue resistant. They occur in higher numbers in postural muscle. Elite endurance athletes have higher than normal numbers of these fibres.
- *Type IIa* fast oxidative (also called fast twitch A or fatigue resistant fibres). These muscle fibres also have a high myoglobin content and high density of blood capillaries, contain many mitochondria and so have a high oxidative capacity, but they can hydrolyse ATP at a high rate and so have a fast peak contraction velocity. They are resistant to fatigue, but less so than the type 1 fibres.
- *Type IIX* fast glycolytic (also called fast twitch B or fatiguable fibres). These muscle fibres have a low myoglobin content, low capillary density and few mitochondria. They can hydrolyse ATP at a high rate and so have a fast peak contraction velocity. They have a high activity of glycolytic enzymes and contain large amount of glycogen. They are useful when high power outputs are needed but fatigue rapidly. The muscles of elite sprinters have high proportions of these fibres.

1.3.3 Respiratory

A key element of performance in most forms of exercise is the individual's maximum oxygen uptake (VO_{2max}). This represents the highest rate of aerobic energy production that can be achieved and the energy required for any power output in excess of this must come entirely from anaerobic metabolism. The importance of VO_{2max} for endurance athletes such as marathon runners lies in the fact that endurance capacity is largely a function of the rate of aerobic energy metabolism that can be sustained for prolonged periods: the higher the fraction of aerobic capacity that must be used, the shorter the time for which a given pace can be sustained. Improving performance requires either an increase in VO_{2max}, an increase in the fraction of VO_{2max} that can be sustained for the duration of the race, or a decrease in the energy cost of running. In practice, all of these can be achieved with suitable training. It is important to note that the athlete engaged in very prolonged exercise can compensate to some degree for a low VO_{2max} by working at a high fraction of that aerobic capacity. For the middle distance runner and other athletes in sports lasting a few minutes (e.g. rowing, pursuit in cycling) a high VO_{2max} is an absolute requirement for successful performance.

Limiting factors to VO_{2max} have been discussed and debated over the years. Part of the reason for the debate is that the limiting factor may vary in different types of exercise, in different environments and in different individuals. Typically, the lungs are not considered to limit performance at sea level in the absence of lung disease, so attention has focused primarily on whether the limitation lies in the delivery of oxygen by the cardiovascular system or in the ability of the working muscles themselves to utilise oxygen. However, at altitude, the picture is clearer: the oxygen content of the inspired air falls as the elevation above sea level increases, leading to a progressive fall in arterial oxygen saturation, decreased oxygen transport and a fall in VO_{2max}.This accounts for the reduction in performance in events lasting more than a few minutes that is generally seen at altitudes above about 1500 metres. Some highly trained runners, however, show arterial desaturation—clear evidence of a limitation at the level of the lungs—in maximal exercise even at sea level and this is reversed, and VO_{2max} increased, by breathing air with an increased O_2 content. This effect is not normally seen in trained but non-elite runners, suggesting there may be a pulmonary limitation in elite endurance-trained runners, perhaps because all the other steps in the oxygen transport chain have adapted (Powers et al. 1989).

Research investigating the responses to training of the inspiratory muscles also provides some support for the idea that there may be a pulmonary limitation. For example, when the effects of 4 weeks of inspiratory muscle training for 30 min/d on cycle ergometer endurance time at 77% of VO_{2max} were measured, untrained subjects increased their endurance time at the same power output from 26.8 minutes before training to 40.2 minutes after training. In trained subjects, who worked at a higher absolute power output, endurance time was increased from 22.8 minutes to 31.5 minutes (Boutellier et al. 1992). However, it is important to note that not all research has reported the same findings so the picture is not entirely clear at present. As in other cases where divergent results are found, this may relate to the exercise task or to the specific subject population. Both of these factors are likely to affect the limitation to both VO_{2max} and performance.

1.3.4 Cardiovascular

The cardiovascular system's functions, among others, are to deliver oxygen and nutrients to all tissues of the body, to remove the products of metabolism from those tissues, to control heat flux within the body, and to circulate hormones from the sites of their production to the sites of their action. It remains debatable whether VO_{2max} is limited by the delivery of oxygen to the active muscles or by the ability of those muscles to use oxygen. As usual when there is debate and uncertainty, it is likely that there may be different answers in different situations. It will depend, for example, on whether the exercise involves large muscle groups, such as in cross-country skiing, or whether only a small muscle mass is involved. In the latter case, the heart is clearly capable of delivering more oxygenated blood than the muscle can accommodate, but in the former example, it is easier to believe that oxygen supply might be limiting. Laboratory studies involving arm, leg and arm plus leg exercise models show that the VO_{2max} increases as the muscle mass increases. When the arms only are exercising, the heart is clearly capable of providing a higher rate of oxygen delivery than the muscles can use.

There is strong experimental evidence to support the idea that the limitations to oxygen delivery in most whole-body exercise situations are imposed by the cardiovascular system and the limitation may lie at any one or more of several stages. The key element seems to be the maximum cardiac output that can be achieved, as this is related to both VO_{2max} and endurance performance. The size of the heart, and more specifically the size of the left ventricle, is also important as this determines the stroke volume—the amount of blood ejected with each beat of the heart. Because maximum heart rate varies little, the cardiac output, which is the product of heart rate and stroke volume, is determined primarily by the stroke volume. Cross-sectional studies of athletes of varying levels of ability from different sports and comparisons with sedentary individuals show that the maximum cardiac output that can be achieved is closely correlated with both VO_{2max} and endurance performance. In elite endurance athletes, the cardiac output can exceed 40 L/min, compared with the maximum of about 20 L/min that the sedentary individual can achieve (Table 1.3). As maximum heart rate does not change much, this difference is accounted for almost entirely by the greater stroke volume of the endurance athlete. The high stroke volume of the endurance athlete is also responsible for the characteristically low heart rate displayed by these individuals at rest. Resting cardiac output is the same for the elite athlete as for the sedentary individual, and can be achieved at a much lower heart rate if the stroke volume is high. A high blood volume will also benefit the endurance athlete by helping to maintain the central venous pressure, thus maintaining stroke volume (Coyle et al. 1990).

TABLE 1.3 Cardiovascular characteristics at rest and during maximal exercise at varying levels of endurance fitness

	Heart rate (b/min)	Stroke volume (mL)	Cardiac output (L/min)
Sedentary			
Rest	70	70	5
Max	200	100	20
Moderately trained			
Rest	40–60	100	5
Max	200	150	30
Elite endurance			
Rest	30–40	150	5
Max	200	200	40

The oxygen-carrying capacity of the blood is also important, and this is influenced by the haemoglobin concentration and the total blood volume. Almost all of the oxygen in the blood is transported bound to haemoglobin and each gram of haemoglobin can bind a fixed amount (1.34 mL) of oxygen. The average male has a higher haemoglobin concentration (about 140 g/L) than the average female (about 120 g/L) and so has about 15% more oxygen in the blood when it leaves the lungs. This difference accounts in part for the generally higher aerobic capacity of males and it also explains the various strategies used by athletes to increase the haemoglobin content of the blood: these strategies include altitude training, the use of agents such as erythropoietin that stimulate the formation of new red blood cells, and the use of blood transfusions prior to competition. Even though some of these strategies are prohibited by the World Anti-Doping Agency (WADA), they have been used—and almost certainly still are being used—by some athletes. In the older literature, a low haematocrit, which was ascribed to a disproportionate expansion of the plasma volume, was described as one of the characteristics of the endurance athlete. More recently, however, a high haematocrit has typically been seen in successful endurance athletes, at least in cycling, where the haematocrit is carefully monitored and where a value of over 50 will cause a rider to be prevented from starting a race. The delivery of oxygen to the muscles is also influenced by the density of the capillary network within the muscles and by the size of the individual muscle fibres. An increase in the number of capillaries or smaller muscle fibres means less distance for oxygen (and substrate) to diffuse from the capillary to the mitochondria within the muscle where it is used. Muscles that rely on aerobic metabolism are generally characterised by smaller fibres and by a large number of capillaries: in response to endurance training, the capillary density can increase substantially.

1.3.5 Thermoregulatory

The efficiency of the conversion of metabolic energy to external work in most forms of exercise is only a little over 20% so almost 80% of the energy available from the breakdown of substrates appears as heat. In a cold environment, this is useful for the maintenance of body temperature and the preservation of muscle function but it can present a challenge in situations such as prolonged hard exercise in hot environments, where heat is generated at high rates and heat loss to the environment is more difficult. Laboratory studies have shown that increases in both ambient temperature and humidity can result in a restriction of the capacity to perform endurance exercise. This is associated with an increased skin and core temperature and greater cardiovascular strain. One important observation that is often ignored is that there is also a greater sensation of effort, even in the early stages of exercise, in hot and humid conditions, and the same subjective response is seen if exercise begins in a hypohydrated state or if hypohydration is allowed to develop. Several extensive analyses of the effects of weather conditions on marathon running performance have confirmed these findings, with suggestions that the optimum temperature for elite marathon runners is less than 10°C, with higher temperatures being better for the slower participants.

Heat stress during exercise poses a major challenge to the cardiovascular system as there is a need for a continued supply of blood to the working muscles, the brain and other tissues, while at the same time there is a greatly increased demand for blood flow to the skin. If effective mechanisms to promote heat loss were not in place, the body temperature would rise rapidly. The elite marathon runner, for example, produces so much heat that body temperature would increase by about 1°C every 10 minutes if there was no heat lost from the body. Even in warm environments, body temperature does not rise by more than 2–3°C over the course of the whole race, and even this increase may be regulated to ensure effective heat loss can take place. A key issue seems to be the maintenance of skin temperature to ensure an adequate vapour pressure to allow effective evaporation of sweat from the skin surface. This requires an increased cardiac output and also means that a significant part of the blood volume is distributed to the skin so the central blood volume is decreased. Reductions in blood flow to non-essential regions will help, but these effects are relatively minor when the demands on cardiac output are high: reduction of blood flow to the splanchnic region can contribute at most about 1 litre per minute of cardiac output. When the demand for cardiac output is 30 or even 40 L/min, that extra litre is useful, but is a small fraction of the total requirement. Peripheral pooling of the blood in turn may reduce the return of blood to the heart and result in a fall in stroke volume. If the heart rate cannot increase to compensate, cardiac output will fall. If this happens there must be either a reduced blood flow to the muscles, and hence a reduced supply of oxygen and substrate, or a reduced blood flow to the skin, which will reduce heat loss and accelerate the development of hyperthermia. It seems likely that the temperature of the brain is the most relevant parameter, but there seems to be no set temperature at which exercise must be terminated and fatigue occurs across a wide range of core temperatures. Allowing some increase in core body temperature may be important for promoting heat loss by increasing the temperature gradient from core to skin and from skin to environment. The importance of maintaining body temperature during exercise in warm environments is well demonstrated by the effects of manipulating body temperature prior to exercise. Increasing body temperature by exposure to a hot environment or by immersion in warm water will reduce exercise capacity. Conversely, per-exercise cooling by exposure to cold air or cold water or by ingestion of cold or iced drinks can improve endurance performance in conditions of heat stress.

Sweat evaporation from the surface of the skin promotes heat loss and limits the rise in core temperature. As this occurs, there is a loss of body water and electrolytes, particularly sodium, though sweat is invariably hypotonic with respect to body fluids. The loss of body water results in a reduction in the water content of all body water compartments (intracellular, interstitial and plasma), and it is the loss of plasma volume that perhaps has the most immediate impact. Blood volume can increase substantially, resulting in a fall in cardiac filling, and heart rate must increase to maintain cardiac output. One immediate effect is an increase in the subjective perception of effort when hypohydrated—the exercise simply feels harder, even in the early stages of a prolonged exercise challenge. Hypohydration and hyperthermia will both, if sufficiently severe, impair physical and cognitive function, but low levels of hypohydration are probably of little significance in most exercise tasks (Judelson et al. 2007). Some of the adverse effects of sweat loss can be offset by ingesting sufficient fluid during exercise to limit the development of hypohydration to less than about 2–3% of body mass (Sawka et al. 2007). In most sporting contexts, the salt losses in sweat are small and can be replaced from food and drinks eaten after exercise. Some individuals, though, can lose large amounts of salt and there is some evidence for a link between high salt losses and muscle cramp, at least in some exercise situations. These issues are explored further in Chapter 13.

1.3.6 Fatigue

Whatever the exercise challenge and no matter what the environment or the training status of the individual, exercise inevitably leads to fatigue if the exercise is sufficiently intense or prolonged. The nature of the fatigue process is not well understood and it is unlikely that any single factor is directly responsible for fatigue. There are, however, some interventions that can enhance performance and, by implication, affect specific aspects of the fatigue process, and these observations may give some clues as to where the limitation lies. In very high intensity exercise that leads to fatigue within 1–2 minutes, there is a rapid decline in the intracellular concentration of CP as high energy phosphate groups are transferred to ADP to maintain the muscle ATP concentration. ATP concentration will fall, but only slightly, and increasing concentrations of ADP may impair muscle contractility. Increasing the pre-exercise muscle CP content by feeding creatine supplements for a few days can lead to higher power outputs and a delay in fatigue, suggesting that the fall in the contribution of CP to energy supply is a factor in fatigue. In exercise that causes fatigue within about 1–10 minutes, the very high rates of anaerobic glycolysis that occur lead to a marked acidosis within the muscle cells as the high rate of hydrogen ion formation overwhelm the buffer capacity of the muscle. Increasing the body's capacity to buffer the acidity (by ingestion of bicarbonate or citrate) can allow greater amounts of lactate to be formed before the pH within the cell becomes limiting. In contrast to the old idea that lactate production was 'a bad thing' that caused fatigue, it is now recognised that without producing the lactate, there will not be sufficient power to allow high power outputs to be achieved.

In prolonged exercise, it is more difficult to identify a single factor that might be responsible for fatigue. We know that performance in cycling tests lasting about 1–3 hours can be improved by increasing the muscle glycogen store and is impaired if exercise begins in a glycogen-depleted state (Bergstrom et al. 1967). Feeding CHO during this type of exercise can also delay fatigue, and these findings suggest that there is a metabolic component to fatigue. We also know that performance of this type of exercise is progressively impaired as the ambient temperature increases above about 10°C (Galloway & Maughan 1997). When the temperature is high, there seems still to be an adequate amount of glycogen remaining in the muscle, suggesting that glycogen depletion is unlikely to be the cause of fatigue in prolonged exercise in the heat, even though this may be the case in cool environments. Nevertheless, feeding a high-CHO diet in the days prior to exercise can improve endurance performance in the heat even when glycogen availability should not be limiting, so there may well be other factors involved (Pitsiladis & Maughan 1999). Pre-exercise cooling, either by immersion in cold water or by ingestion of cold drinks, can improve endurance performance in warm environments, apparently by delaying the time until a critical elevation of core temperature occurs. Acclimatisation improves performance in the heat by a number of mechanisms, including an increased sweating sensitivity, an increased plasma volume and also a lowering of the basal pre-exercise core temperature (Nielsen et al. 1993).

From studies of muscle fatigue carried out in the late nineteenth century, it was generally concluded that fatigue was in part a local phenomenon occurring within the active muscle but that there was also a primary role for the brain in terminating exercise, or at least reducing the intensity, before irreversible damage was caused. Fundamental to this conclusion was the observation that direct electrical stimulation of the muscle or its motor nerve could still produce a strong contraction even when voluntary activation of the muscle was impossible. Technical developments that allowed the collection and analysis of samples from muscle are perhaps responsible for the focus on muscle fatigue that developed in the twentieth century: it continues to be much harder to study events occurring within the brain. Results of muscle biopsy analysis, for example, showed a clear link between the depletion of the muscle glycogen store and the onset of fatigue, at least in prolonged cycling exercise. More recently, however, there has been a renewed recognition of the role of the brain in fatigue, even though the mechanisms remain uncertain. This has been described as the action of a 'central governor' that acts to regulate pace and effort to optimise performance (Swart et al. 2009). This is reminiscent of the work of Lagrange, who, in 1889, referred to fatigue as a 'regulator, warning us that we are exceeding the limits of useful exercise, and that work will soon become dangerous'. The danger referred to here is that of irreversible damage to the muscles or other tissues, with the attainment of excessively high body temperatures being a real concern.

A number of pharmacological interventions have been shown to affect exercise performance without any obvious cardiovascular or metabolic effects that could explain this. A range of stimulant drugs, including amphetamines, for example, can enhance performance: their actions on neurones in the brain that use dopamine as a neurotransmitter seem likely to be the explanation for this (Roelands & Meeusen 2010). Paroxetine, a drug that acts on neurones that use serotonin as a neurotransmitter, can reduce performance (Wilson & Maughan 1992) and other drugs with opposing actions will likely enhance performance. One of the dangers associated with the use of drugs that can override the sensation of fatigue is that the outcome that Lagrange referred to may occur. This seems to be the case with amphetamines, which may result in fatal hyperthermia during hard exercise in the heat, as happened to Tom Simpson in the 1967 Tour de France.

1.4 Adaptations to exercise training

The aim of training is, or at least should be, to increase functional capacity and to bring about event-specific improvements in performance. Training affects every organ and tissue of the body but the adaptation is specific to the training stimulus and to the muscles being trained. A well-designed strength training program will have little effect on endurance and vice versa; one leg can be specifically trained for strength and the other for endurance with relatively little cross-over. Training should therefore be designed to address the event-specific limitations to performance, and this will differ between individuals as well as between events. It is important to note, however, that training is not entirely specific, as the effects on the cardiovascular system will be similar whether the athlete is running, cycling or performing any other activity that engages a large muscle mass. The performance improvement is generally proportional to the training load (i.e. the intensity, duration and frequency of the training sessions). Generally, the harder an athlete trains, the greater the improvements in performance that result. There is a limit, however, beyond which further increases in training will result in poorer performances: this is usually referred to as an overtraining syndrome, but few athletes ever reach this level of training. Where it does happen, overtraining is associated with impaired performance, chronic fatigue leading to an inability to sustain the training program, and an increased risk of infectious illness.

It used to be thought that a primary role of nutrition in the athlete's diet was to support consistent, intensive training by promoting recovery between training sessions. While it is undoubtedly true that recovery is an important element, there is a growing recognition that nutrition has a key role in promoting the adaptations that take place in muscle and other tissues in response to each training session. Training provides the stimulus to turn on the genes responsible for the expression of functional proteins: strength training leads to synthesis of more actin and myosin, making muscles bigger and stronger, while endurance training leads to synthesis of more oxidative enzymes and of all the other components necessary for endurance performance. A selective stimulation of protein synthesis and degradation must be taking place. The response is modulated by the nutrient, metabolic and hormonal environment, and this can be modified by food intake before, during and after training. There is good evidence that feeding a small amount of protein or essential amino acids after a training session can stimulate protein synthesis for up to 24 hours after training. There is a need, though, for more studies with functional outcomes rather than simply measuring protein turnover rates. Without the training stimulus, though, it is clear that adaptation will not take place in the muscle.

Training should aim to address the factors that limit exercise performance, shifting the barriers to allow better performance. In the case of strength training, a significant part of the adaptation that takes place, especially in the early stages, is within the nervous system: strength improves after only a few training sessions, before any measurable changes in muscle structure have taken place. In the case of endurance training, a large number of adaptations have been identified, both in the central circulation and in the muscles themselves. The pumping capacity of the heart is increased, primarily by an increase in stroke volume as a result of an increase in left ventricular volume. Blood volume and red blood cell mass both increase, thus increasing the total oxygen carrying capacity. New capillaries grow in the endurance-trained muscle, shortening the diffusion distance for oxygen and nutrients between the circulation and the muscle fibres. Mitochondrial mass increases, and with it the activity of the enzymes involved in the oxidation of CHO and fat. There is, in particular, an increase in the capacity of the trained muscle to oxidise fat, thus decreasing the reliance on CHO during exercise, though this adaptation can be reversed to some extent by feeding a high-CHO diet. As mentioned above, this may be beneficial when the availability of CHO is limited but is of questionable value in other situations as it will lead to an increased energy cost of exercise. Tissues respond to disuse with a reversal of the central and local adaptations caused by training.

Regular training also has important effects on the brain, though these are less well understood than many of the peripheral adaptations. One important learned response is the ability to judge pace, so that effort can be distributed evenly across the whole duration of an event: this is not an innate ability but is learned by repeated experience. A common mistake of novice athletes is to set off too fast and then to fade badly in the later stages or to finish with too much still in reserve. The limitation to performance in this situation may be very different from that affecting the experienced athlete.

Summary

Exercise physiology, while a scientific discipline in its own right, is intrinsically linked to many aspects of sports nutrition. An understanding of the body's physiological processes at rest, during exercise and in the recovery period after exercise are key to understanding many of the nutrition requirements of an exercising person and important in identifying nutrients that may help the exercising individual to achieve their aims, be they training adaptations, exercise performance or recovery after exercise.

Practice tips

ADAM ZEMSKI

- A comprehensive understanding of the physiology of a sport should underpin any plan to develop nutritional strategies to assist an athlete to maximise favourable training adaptations and to optimise competition performance.
- The tailoring of advice for a specific athlete requires an appreciation of their individual role within a team or their unique characteristics in a group environment.
- The sports dietitian should be aware that an athlete's nutrition needs and goals are not static, but can change from day to day, within the various components of a macrocycle, over the season, and over their career. Periodisation of training is a key element in the preparation of the modern athlete, and should be reflected in a periodised approach to nutrition.
- It is useful for the sports dietitian to have a list of key pieces of information that can be used to construct an overview of an individual athlete's nutrition demands and challenges. Probes that can be used to identify this information are summarised in Boxes 1.1 (training) and 1.2 (competition).
- It can be difficult for the sports dietitian to build up an intimate knowledge of a sport/event to allow them to find answers to the questions in Boxes 1.1 and 1.2. Table 1.4 summarises how best to access information on the physiological demands and dynamics of a sport. Gathering information from a variety of sources will help to build a more comprehensive picture.

Box 1.1 Identifying nutrition issues and challenges for optimising the effectiveness of an athlete's training program and everyday health goals

Underpinning ideas

- *Is there evidence in this sport or for specific training sessions undertaken in this sport that that specific fuelling, recovery, hydration or supplement strategies before, during, or after training may assist in promoting training performance or training adaptations?*
- *Is there evidence that nutrient requirements are altered by the athlete's commitment to exercise or that their everyday eating practices might predispose them to nutrition-related problems?*

Training overview

- What types of training sessions does the athlete undertake and which energy systems do they use in these sessions?
- Do athletes undertake special training programs, such as altitude training, heat acclimatisation?
- What are the frequency, duration and intensity of these sessions?
- How are training sessions periodised over the week, month, season and year?
- Is there opportunity to consume drinks/food during training. How is this made available?

Energy and carbohydrate needs

- Based on the considerations of the training program, what are the total energy requirements for training and do these differ throughout the week, macrocycle and season?
- Are there other issues (e.g. growth, weight loss, active job, lifestyle) that will require manipulation of energy needs?
- Based on the considerations of the training program, what are the general carbohydrate requirements for

training? Which key workouts should be targeted for special fuelling practices? How should the athlete consume extra CHO before, during or after training sessions to meet extra CHO needs for key sessions? How will this change throughout the macrocycle and season?
- What are the needs and opportunities to consume CHO during key workouts? Is there a need or opportunity to practise competition fuelling practices during training?

Hydration considerations

- What is the training environment: indoor/outdoor, time of day, summer/winter?
- What are the typical sweat losses during different types of training?
- What opportunities are available to consume fluid during training?

Body composition considerations

- How important is body composition to performance in this sport?
- What are the typical physique characteristics of elite performers in the sport: body mass, lean mass, fat mass, regional distribution of lean and fat mass?
- What is the current physique of the athlete and what is their history of physique change?
- What is the athlete's desirable body composition based on past history and other input?
- Does the athlete need to maintain this throughout a season or for a short time at particular competitions?

Cultural approach to nutrition within the sport

- What is the typical level of nutrition awareness and knowledge within the sport?
- Where do athletes from the sport usually seek nutritional advice?
- Is there a culture of particular nutritional strategies, supplement or ergogenic aid use within the sport?

Practicality and food availability considerations

- What is the athlete's domestic situation: who shops and cooks?
- Does the athlete have other commitments: work, study, travel to/from training?
- What time of the day does the athlete train and does this create difficulties in managing the daily food plan?
- Do other factors influence nutrition choices: financial constraints, religious/social customs, vegetarianism?

Health considerations

- What is the general risk that athletes in this sport develop nutrition-related problems: nutrient deficiencies (particularly iron and vitamin D), menstrual dysfunction, compromised bone status, disordered eating?
- Does the athlete have a specific risk for the development of nutrient deficiencies?
- Does the athlete have any reasons for restricting the variety of foods in their diet (e.g. fussiness, food intolerances, allergies)?
- Do gastrointestinal considerations or appetite issues limit food intake at particular times?

Box 1.2 Identifying nutritional strategies that will help to optimise competition performance

Underpinning ideas

- *What are the limiting factors for performance in this event/sport: what are the physiological factors that* *explain the occurrence of fatigue or a loss of performance throughout the event?*

- *Is there evidence that specific fuelling, recovery, hydration or supplement strategies before, during and after events may improve competition performance?*

Event characteristics

- What is the frequency of competition: how often does the athlete need to be at their best and how often do they get to practise their nutrition tactics?
- What are the typical environmental conditions in which competition is undertaken: temperature, indoors/outdoors, time of day?
- Is competition undertaken as a single event, a series of events (heats and finals), a weekly fixture or other format?
- Is the athlete competing in multiple events on the same day?
- What factors interfere with post-exercise recovery eating (e.g. media commitments, recovery strategies)?

Fuelling considerations

- What are the energy requirements of competition?
- What are the duration and intensity of the movement patterns in competition?
- Does this differ depending on playing position, style, strategy or any other variables?
- Is there evidence that low CHO availability might limit performance or that strategies to increase CHO availability can enhance performance?
- Does the athlete have an opportunity to consume drinks or foods containing CHO during the event to provide additional fuel?

Hydration considerations

- What are the typical sweat losses and fluid deficits experienced during competition?
- Is there evidence that dehydration can limit performance in this event?
- Does the athlete have the opportunity to consume fluid during competition and are they currently making optimal use of this?

Body composition considerations

- Does the sport involve weight classes which require the athlete to weigh-in prior to competing?
- If so, when is the weigh-in, how often do they weigh-in and how long is the interval between weigh-in and competition?

Practicality and food availability considerations

- Is the athlete in familiar surroundings or do they have to travel to their competition?
- Will food and fluid be provided during the event, or will the athlete need to supply it?
- Has the athlete practised a competition fluid/fuel plan that integrates their likely needs and the available opportunities to eat/drink?

Gut comfort

- Does the athlete experience gastrointestinal problems or gut discomfort that might limit performance or their ability to follow a nutrition plan during the event?

Ergogenic aids

- Is their evidence or a strong hypothesis that ergogenic aids (e.g. caffeine, bicarbonate, beetroot juice/nitrate) can enhance performance in this event?
- Has the athlete ever trialled the use of such products?

TABLE 1.4 Ways to gather information on the physiology of sport

Source	Tips
Internet searches	• Many common resources such as Wikipedia can provide a wealth of information on the history and characteristics of a sport/event. • Not all sources of information on the internet are reliable, and a check on the credibility of the information should be undertaken. • In the case of 'fringe' sites, there is still often some value in reading commentary on chat rooms and other posts. Even if the information is not fact based, it may reveal the kinds of information being provided to athletes that may need to be addressed in a more critical way.
Journal search engines (e.g. PubMed, Google Scholar)	• Various comprehensive databases of references and abstracts are available. • It is useful to know the strengths and weaknesses of various databases: for example, PubMed offers a more extensive search strategy while Google Scholar offers access to 'grey literature' such as preprint archives, conference proceedings, and institutional repositories. • Tutorials on how to maximise the effectiveness of your search are available, for example, via PubMed or with help from library resources.
Peer-reviewed journal articles and reviews	• Articles can be located using the journal search engines discussed above. To find out if an article is peer reviewed, you may need to read the editorial guidelines of the journal in which it was published. • Review articles are useful in summarising the published literature on a sport with some insights from the author(s). They may provide a comprehensive reference list of articles on the same topic that is useful as a resource or to cross-check your own search strategies.
Textbooks and sports encyclopaedia	• Textbooks and sports encyclopaedia are useful in gaining a history or summary of the rules and conditions of a sport/event. This is often a good starting point for developing an understanding of the sport. • Although textbooks quickly become outdated, many provide a long-lasting reference on aspects of a sport and its scientific basis.
National and international federations and governing bodies	• Most sports host an official website, containing information about their sport, and also links and contact details to aid additional research. • In Australia, a directory of national sporting organisations and contact details can be obtained from the Australian Sports Commission. This may allow contact with personnel who can provide important insights.
Direct contact with coaches, athletes and sports scientists	• It can be extremely useful to have direct communication with qualified sports science/medicine professionals or other high performance managers and coaches. They may provide insights and important new knowledge that is yet to be available in current literature. • Many athletes are also insightful about their sport and may provide good knowledge about the bigger context as well as their own issues. • It is important to recognise the limitations of information supplied by a single coach, athlete or other sports-involved person. Many may be 'old school' or have unfounded beliefs about aspects of their sport. Although anecdotes and testimonials provide some interesting information, the limitations of such material should be recognised.

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Commentary 1

A molecular view of exercise

During the early part of the twentieth century, the emerging methods of biochemistry were applied to questions in exercise physiology. August Krogh and Johannes Lindhard in Copenhagen assessed the relative contributions of carbohydrate and fat to exercise metabolism and were engaged in debate with scientists from Cambridge over fuel sources during exercise. The latter group included AV Hill, who became famous for his work on the biophysics of muscle contraction, lactic acid generation from glycogen during contraction and the ‘oxygen debt’ hypothesis.

In 1947, the UK Medical Research Council established the Unit for Research on the Molecular Structure of Biological Systems at Cambridge, with an initial focus on the structure of proteins, notably insulin and haemoglobin. Research activities diversified to include the structure of DNA, the molecular mechanism of muscle contraction and virus structure. It became one of the birthplaces of modern molecular biology and in 1962 a new building opened as the Laboratory of Molecular Biology. Over the years, it pioneered many of the fundamental techniques used within molecular biology, including sequencing methods that were crucial for the Human Genome Project, and work undertaken there resulted in the awarding of 10 Nobel Prizes!

In 1967, Professor John Holloszy published a seminal paper describing the effects of endurance exercise training on muscle mitochondria and oxidative capacity (Holloszy 1967). Over nearly 5 decades, he and his postdoctoral fellows and collaborators have made an enormous contribution to our understanding of the cellular and molecular bases of the effects of exercise on muscle, with implications for both health and exercise performance. Mitochondria, metabolism, glucose transport and GLUT4 expression, and insulin action, were just some of the topics they investigated. A postdoctoral fellow from this laboratory, Professor Frank Booth, pioneered the application of the ever-expanding range of molecular biology techniques to research questions on muscle adaptation to exercise and disuse (see Booth 1988).

In the late 1960s, Jonas Bergstrom and Eric Hultman used the percutaneous needle biopsy technique to obtain muscle samples from human subjects before, during and after exercise for the later measurement of muscle metabolites. Use of this technique soon increased within exercise science and numerous researchers, notably David Costill, Phil Gollnick and Bengt Saltin, examined the effects of acute and chronic exercise on muscle fibre type composition and morphology, metabolic enzyme activity and substrate utilisation. An important sports nutrition practice to emerge from such studies was carbohydrate or glycogen loading. As the number of molecular biology techniques increased, and they were adapted for use on human muscle samples, more studies examined effects of acute and chronic exercise on parameters such as gene and protein expression, activation of key signalling kinases and pathways, intracellular expression and localisation of key transcription factors and coactivators, gene transcription, protein synthesis and metabolite kinetics (see Egan & Zierath 2013 for review).

The use of tracer technology has allowed sports scientists to examine turnover of substrates and metabolites in the blood, as well as muscle oxidation of substrates. One example in a sports nutrition context has been the enhanced understanding of the metabolic, and potential ergogenic, effects of ingesting multiple transportable carbohydrates in terms of promoting carbohydrate (and fluid) bioavailability and endurance performance (Currell & Jeukendrup 2008). The findings have had implications for both athlete practices as well as commercial product development.

The developments in the exercise field are apparent when I reflect on my own scientific journey. As a graduate student working with Professor David Costill in the early 1980s, we studied the metabolic and performance effects of carbohydrate ingestion during exercise by measuring substrate oxidation using indirect calorimetry and muscle glycogen levels in muscle biopsy samples (Hargreaves et al. 1984). As a junior faculty member back in Melbourne, we measured expression of the glucose transporter GLUT4, the carrier protein that facilitates sarcolemmal glucose transport in response to muscle contraction and insulin stimulation, in muscle samples obtained from untrained, trained and detrained subjects (McCoy et al. 1994). We then described the effects of a single exercise bout on GLUT4 gene expression in muscle (Kraniou et al. 2000) and the molecular regulation of exercise-induced GLUT4 transcription (McGee & Hargreaves 2004; McGee et al. 2008). It is now generally accepted that these transient changes in gene expression in response to repeated exercise bouts are crucial for the increased expression of key proteins in skeletal muscle following exercise training (Egan et al. 2013; Perry et al. 2010). We now have a much greater understanding of the molecular and cellular mechanisms underpinning the responses and adaptations to acute and chronic exercise (see Baar 2009; Coffey & Hawley 2007; Egan & Zierath 2013).

In undertaking modern studies of the interaction of training and nutrition, it has become common for sports scientists to measure changes in the transcription of genes, translation of proteins, and/or changes in the activity of proteins within skeletal muscle. These measurements can often help to explain the mechanisms for changes in functional outcomes such as muscle morphology, endurance and strength, or exercise performance; however, it is important to recognise that many studies report changes in muscle characteristics from the molecular view, without a concomitant change in performance. This just emphasises again the complex, multifactorial nature of sports performance and the need for caution in extrapolating too far from such muscle measurements to sports performance.

Increased understanding of the molecular bases of exercise adaptations and advances in biotechnology and bioengineering has seen the development of genetically modified mice with enhanced exercise performance and improved health outcomes. For example, muscle-specific overexpression of the transcriptional co-activator PGC-1α results in increased muscle oxidative capacity and running endurance (Calvo et al. 2008). The ability to systemically deliver adenoviruses harbouring genes specifically targeted to skeletal muscle raises the possibility of novel therapeutic strategies to enhance health (Williams & Kraus 2005), but also the spectre of ‘gene doping’ to enhance sporting performance. Indeed, the current World Anti-Doping Agency prohibited list excludes ‘the transfer of polymers of nucleic acids or nucleic acid analogues’ and ‘the use of normal or genetically modified cells’. Another development building on enhanced understanding of the molecular bases of exercise responses and adaptations is interest in a so-called ‘exercise pill’ that might pharmacologically recapitulate some, or many, of the beneficial effects of exercise, primarily in a health context. While there are promising candidate molecules, it is unlikely that a single one of them can fully replicate the effects of exercise (Goodyear 2008).

Finally, there has long been interest in the relative importance on ‘nature’ versus ‘nurture’ in the sporting context (see Tucker & Collins 2012 for review), examination of the heritability and potential genetic bases of physiological and performance traits and the possibility that genetic testing may be applied to sport (Roth 2012). Many research groups have taken a candidate gene approach and one important gene linked with sports performance, at least from an Australian perspective, is α-actinin 3, which has been associated with elite sprinting performance (MacArthur & North 2005). Another interesting development has been the greater recognition of individual responses to training programs and their biological bases (Bouchard & Rankinen 2001) and the potential to predict such responses (Timmons et al. 2010) so as to better inform ‘personalised training’ (Buford et al. 2013).

The rapid advances in gene sequencing technology, partly catalysed by the Human Genome Project, have led to rapid and cheap sequencing of whole genomes and exomes. Increased numbers of potential ‘sports genes’ and genome-wide association studies have emerged, and the technology really has outpaced the biology. The challenge will be to link and integrate the large amount of genomic and molecular data with traditional phenotypic measures of physiological and sporting performance.

It is important to recognise that environmental influences, such as exercise training and diet, can modify gene expression and ultimately phenotype via mechanisms that don’t involve changes in the fundamental gene sequence. These include DNA methylation (Barres et al. 2012), histone modifications (McGee & Hargreaves 2011) and interactions with microRNAs (Zacharewicz et al. 2013). The application of sophisticated mathematical and statistical techniques to problems in exercise biology (i.e. computational and systems biology) will facilitate understanding of the complex gene–gene–environment–environment interactions that underpin exercise and sports performance (see Ghosh et al. 2013). As stated eloquently recently, ‘Advances in DNA sequencing approaches alone are not enough, of course: humans are more than just a product of their genomes. True predictive medicine will require integrating risk factors from both genetic and environmental sources … Fortunately, there is a machine that accurately integrates both genetic and environmental risk: the human body itself’ (MacArthur & Lek 2012). Coaches, sports scientists and sports dietitians have long observed and measured the performance of their athletes and this is the ultimate integration of these multiple inputs. The developments in molecular biology provide exciting new opportunities to better understand the biological bases of such performance and, at least the potential, to enhance it.

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CHAPTER TWO

Dietary assessment of athletes: clinical and research perspectives

Vicki Deakin, Deborah Kerr and Carol Boushey

2.1 Introduction

A nutritional assessment of an individual athlete, as well as a medical check-up, a musculo-skeletal assessment and a psychological assessment, is now routine in many sporting organisations. For dietitians, it is the first of four steps in the Nutrition Care Process. The Nutrition Care Process is a structural framework for dietitians to provide nutrition care to patients, clients and groups or communities and is easily adapted for use in athletes (Academy of Nutrition and Dietetics 2013). A nutrition assessment is the basis for a nutrition diagnosis or evaluation, nutrition intervention (if warranted) and nutrition monitoring and evaluation—the four stages that complete the Nutrition Care Process.

Dietary assessment is a key component of the nutritional assessment of an individual athlete. An assessment of dietary intake is not simply an evaluation of what a person eats and drinks. The process may include the collection of social, medical and psychological influences on food choice. This information may be further supplemented with energy expenditure analysis (in training and competition), anthropometric assessment, biochemical data and health history. The outcomes of nutrition assessment are to identify nutrition-related problems and their probable causes. For individual athletes, this forms the foundation for specific strategies for nutrition intervention that enhance performance and training capacity.

In a healthy athlete, symptoms of lethargy, fatigue, poor performance capacity, poor concentration and slow recovery from a hard training session can be nutrition-related. Increased incidence of injury or infection or large losses in body mass (BM) may also be linked to suboptimal nutrient intake or energy imbalance. Often inconsistency in performance and during training is a signal for a coach to refer an individual athlete to a dietitian for a nutrition assessment. In addition to dietary assessment, biochemical tests, anthropometric measurements and nutrition-focused physical examination findings all form part of the overall nutrition assessment. A sports dietitian is trained to diagnose nutrition-related problems taking into account the physiological, medical, social and psychological issues that affect performance and health, and to plan the nutritional intervention, monitoring and evaluation of the athlete accordingly.

The main goals of nutrition assessment, the first step in the nutritional care process, are to:

- identify athletes who require nutritional support to restore or maintain nutritional status (e.g. at-risk athletes who may have an underlying condition/disorder)
- compare assessment data with reference standards where feasible
- provide appropriate nutrition intervention (e.g. individual dietary therapy including behaviour modification or group education)
- monitor the progress and efficacy of dietary intervention and effects on performance.

2.1.1 Nutrition-focused examination

The purpose of a nutrition-focused examination is to uncover any medical condition or physiological factors that interfere with food intake, digestion and metabolism. Recent or chronic illness, anxiety, depression and some drugs interfere with absorption of nutrients and thus affect nutritional status. Diarrhoea, loss of appetite, gastrointestinal disturbances and BM loss could be associated with an underlying illness. Psychosocial stress also affects appetite and eating behaviour. However, loss of appetite is a normal physiological response for some athletes and can persist for up to 1–2 hours after hard workouts. Some athletes also experience discomfort or nausea when food is consumed before strenuous exercise, and may avoid eating anything for 2 or more hours before training. This behaviour can adversely affect an athlete's recovery and ability to meet daily nutrient and energy requirements.

2.1.2 Biochemical tests

Dietary biomarkers are increasingly used as complementary or substitute measures of dietary intake. Although these tests are objective and used as external criteria for validating dietary intake methods, they are not always diagnostic of nutrient depletion or deficiency (for review, see Lee & Nieman 2007).

Low blood levels of some micronutrients may reflect low dietary intake, defective absorption, increased utilisation or excretion. Biomarkers for many nutrients have large diurnal variations (e.g. serum iron) or are under such strict homeostatic control (e.g. calcium) that interpretation is misleading. The interpretation of biomarkers relevant to athletes is discussed in detail in other chapters: vitamins and minerals (see Chapter 11), iron (see Chapter 10), protein (see Chapter 4) and calcium and vitamin D (see Chapter 9).

Population reference ranges for biomarkers are still used in clinical practice and research studies of athletes. These ranges may be inapplicable for athletes involved in strenuous training with high turnover or losses of some nutrients, and need some adjustment.

Several diagnostic criteria should be investigated to establish the micronutrient status of an individual. A one-off test that indicates a low nutrient biomarker may not necessarily be diagnostic of a clinical or even subclinical condition. Use of several tests and several biomarkers, if available, gives more meaningful information about individual status and may reveal trends. Nutrient depletion or deficiency from biochemical tests is confirmed in individuals when large deviations from their usual level or from the population reference level are seen.

2.1.3 Anthropometric measurements

Anthropometry involves the application of physical measurements to appraise human size, shape, proportion, body composition, maturation and gross function. Physical measurements, including height, BM, skinfolds, mid-arm muscle circumference, girth and frame size, are well-recognised measures used for athletes. These are used indirectly to estimate body composition and predict estimates of energy requirements. Comparison of height and BM parameters with ‘ideal’ or reference standards such as the body mass index (BMI) (height (m)/weight (kg)^2) is inappropriate for many athletes. Those athletes with large muscle bulk are categorised by these standards as overweight or, in some cases, obese, despite having low body fat. BMI and growth charts for children and adolescents can be useful for showing very lean athletes that they are not overweight.

The sum of skinfolds is a practical, inexpensive and reliable method to estimate body fat or to monitor changes in body composition of athletes over time, thus, skinfold measures are a particularly useful skill for sports dietitians. Other methods for measuring body composition are described in Chapter 3. Using absolute values for skinfolds, such as the sum of six to eight skinfold sites, is preferable to using the numerous equations that calculate body fat percentage used extensively in the past. Because these equations are derived from cadavers and assume that the fat mass (FM) has the same density as the fat-free mass (FFM)—an invalid assumption—they are no longer favoured. For more on skinfolds and BM, see Chapter 3.

This chapter focuses on the applications, strengths and limitations of the main methods used for conducting a nutrition assessment and in particular measuring dietary intakes in clinical practice and in research, and their relevance to athletes.

2.2 Dietary assessment

Prior to undertaking dietary assessment, a dietitian should be aware of the eating habits, food quantities consumed, and the food attitudes and beliefs of different groups of athletes relating to the culture of the sport in either a research or clinical environment. This knowledge is crucial for deciding on the method used for data collection, improving accuracy of data collection (e.g. serve sizes) and conducting high-quality research on dietary intakes, dietary attitudes and beliefs in any population group. For example, under-reporting of food intake or under-eating is more likely to occur in female athletes, particularly in those who are overweight or who have body image issues (Edwards et al. 1993; Hill & Davies 2002); Fudge et al. 2006; Singh et al. 2009). Weighed food records are particularly prone to under-reporting; this may be partly attributable to the high respondent burden of keeping a food record, especially for high energy consumers such as endurance-trained athletes. The challenge for sports dietitians is to understand that in the pursuit of an accurate food record, greater respondent burden may lead to reactivity, causing the individual to change their behaviour (Thomson & Subar 2013).

Dietary assessment involves collecting data on food and beverage intakes and then evaluating nutrient, energy or food group intakes against population or athlete reference measures that are age-, gender- and country-specific for the general population. Nutrient intake in athlete groups is usually compared with age-, gender- and country-specific nutrient intake recommendations developed for the general population. Collecting dietary intake data and comparing to population references is not a simple process (as it may appear to the untrained observer), but one that requires care, precision and a high degree of skill and knowledge. Although athletes may have similar eating habits to non-athletes, their requirements for some nutrients and the volume of food consumed is often higher. Dietary data collection methods and nutrient reference standards for evaluating food or nutrient intakes may need to be modified when applying to an athlete population if not already validated in that population.

2.2.1 Why measure dietary intake?

Table 2.1 defines the broad uses and applications of measuring dietary intakes in athletes. The main objectives for measuring dietary intakes in individuals or groups of athletes are to assess the probability of inadequate nutrient intakes or inappropriate food choices, for determining intervention of risk-related dietary behaviours or to evaluate dietary interventions. Dietary assessment can also be useful for increasing an athlete's awareness of their eating habits and forming the basis for nutrition education.

TABLE 2.1 Purpose and application of measuring dietary intakes

Reason for use	Practical application
Determining nutritional status	Calculating average nutrient intakes in groups of athletes
	Comparing probabilities of inadequate nutrient intake of individuals and groups with population nutrient standards
	Combining dietary intake assessment with other parameters (e.g. biochemical, anthropometric and medical) to assess nutritional status of individuals and groups
Assessing the links with performance, diet and health status	Comparing and contrasting indices of nutritional status with the incidence and prevalence of health problems or performance measures in groups
Evaluating nutrition education and intervention	Providing feedback on the efficacy of dietary intervention programs
Assessing the effect of different dietary	Determining potential ergogenic effects of diets, components of diets or

regimens on performance measures or metabolic responses	supplements
Assessing the effect of different training periods or intensities on dietary intake	Determining the turnover of nutrient requirements at different training intensities and duration in combination with other parameters

Research applications

In any dietary assessment for research into groups of athletes or for individual assessment, the reasons for assessing dietary intakes need to be clearly defined and matched with the most valid data collection method. To date, the main method used to assess current dietary intakes of athletes in research has been food records using household measures. Because of the analysis time involved, paper-based food records are best suited to small samples (Burke et al. 2001, 2003). In the future, as more digital and image-based food records become available, it is likely that these types of food records will make it possible to undertake larger data collections.

Food frequency questionnaires (FFQs), although more suitable than food records due to their low burden and for surveying larger samples, have been infrequently used in dietary surveys of athletes. Where FFQs have been used, researchers have mostly adapted questionnaires that have been validated in other population groups, to match the eating habits of the athletes surveyed (for more detail, see section 2.4.2). There is a real need to validate both conventional paper and interview techniques specifically for use in athletes in both clinical practice and research. Rather than developing and validating new methods, most researchers use modifications of previously validated dietary survey instruments or technologies but adapt them for another population group. From a research perspective, any previously validated dietary assessment tool that has subsequently been modified for use in another population, such as athletes, should be re-validated in that population, particularly if changes to the original food list are substantial. However, the difficulties inherent in collecting accurate information on food intake in human subjects and the logistics and cost associated with the design, sampling and implementation of any validity study probably preclude the development and validation of any new assessment tool in athletes.

Innovative technologies for measuring diet such as mobile devices (Wang et al. 2006; Six et al. 2010; Weiss et al. 2010; Schap et al. 2011; Boushey et al. 2012) and other web-based software (Subar et al. 2012), although still developing, are likely to improve dietary assessment for research applications (Thompson et al. 2010; Illner et al. 2012). These methods are yet to be evaluated in athletes. An important step is to conduct usability testing to determine whether athletes will be fully cooperative with the method.

2.2.2 Methods for measuring dietary intakes

Methods for measuring dietary intake are categorised into two main types: current dietary intakes and past dietary intakes (retrospective short- or long-term recall of foods consumed) and have been reviewed elsewhere (see Bingham et al. 1994; Thompson & Subar 2013). Over the last few years, many of the conventional paper and interview-based methods listed in Table 2.2 have been modified by innovative technology into self-reported formats using either interactive computer-based technologies (for example, the ASA24) or mobile telephones or devices. With the widespread use of mobile telephones and wireless transmission, several free dietary assessment apps are now available. (For recent reviews of communication technologies for dietary assessment, see Ngo et al. 2009; Illner et al. 2012; Stumbo 2013.) There is a real need to validate both conventional paper and interview techniques as well as the newer technological tools specifically for use in athletes in both clinical practice and research. Sources of error associated with these new methods are likely to be similar to the conventional methods (see section 2.5).

TABLE 2.2 Comparison of dietary assessment methods for measuring dietary intakes in groups and individuals

	Diet history	FFQ	24-hour recall	Paper food record	Digital food record	Image-based food record
Overview	Respondent recalls the frequency of consumption of core food groups over 7 days with 3, 6 or 12 m cross-check plus a 24-hr recall	Respondent recalls, over a set time period, the frequency of consumption of foods and beverages from a set list. Some FFQ incorporate portion sizes for foods/beverages	Respondent recalls all food and beverages consumed in the previous 24 hours Conducted in-person or by telephone with an interviewer or web-based; can be done for more than 1 day	Respondent records foods and beverages consumed after either weighing or estimating with household measures; usually done for 3–7 days	Respondent keeps a digital log of all foods and beverages consumed with an app that provides household measures or weights for portion sizes; usually done for 3–7 days	Respondent takes before or before/after images of all foods and beverages consumed with an app using an embedded camera or a camera; usually for 3–7 days
Application/purpose	Provides assessment of usual intakes of individuals in dietetic clinical practice Details of meal preparation and meal patterns can be obtained	Provides an estimate of total intake over a specific time period Used to rank individuals by their consumption of nutrients, foods, food groups or dietary patterns	Used mainly as a population survey method A single 24-hour recall can be used to provide population average of nutrients, foods, food groups or dietary patterns Multiple days required for individual estimates	Assesses food and beverages consumed at the time of eating Can be used to estimate nutrients, foods, food groups, or dietary patterns for populations or individuals	Assesses food and beverages consumed in real-time Can be used to estimate nutrients, foods, food groups or dietary patterns for populations or individuals	Assesses food and beverages consumed in real time Can be used to estimate nutrients, foods, food groups or dietary patterns for populations or individuals Images can provide additional contextual information
Training required	Interviewer training required	Can be self-administered without training or can be interviewer administered	Interviewer training required for the structured interview with	Respondent needs to be instructed on how to record foods and	Respondent needs to be instructed on how to use the app to record foods and amounts and the	Respondent needs to be instructed on how to use the app and

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			specific probes (e.g. multi-pass method) Respondent training recommended for web-based administration	amounts and the level of detail required	level of detail required	capture a usable image and place a fiducial marker (if used)
Respondent burden	*Low* (20–40 minutes to complete) Requires an interview by a dietitian	*Low* (15–30 minutes to complete depending on length of questionnaire)	*Low to moderate* (30–60 minutes to complete) depending on type (e.g. interview versus web)	*High* for hand recording all foods and beverages in every eating occasion, additional burden for weighing foods and moderate for estimating amounts	*High*—requires selection of individual food and beverages in every eating occasion and their corresponding amounts	*Moderate*—requires image capture of food and beverages in every eating occasion
Data collection burden	*High*—trained staff needed for collection and data entry	*Low*—particularly if instrument is scannable or administered via the web	*High*—trained staff needed for collection and data entry *Low* for web-based	*High*—trained staff needed for resolutions and data entry	*Low*—if raw data are available from app and linked to food composition database *High*—if data are unavailable, need to be reconstructed and linked to database	*Low*—if automatic analyses and linked to food composition database. *Moderate*—if trained analysts are used to link to food composition database
Strengths and limitations for the clinician or researcher						
Strengths	Literacy is not required	Suitable for large populations due to the low respondent burden Relatively inexpensive to administer and process	When conducted as an interview literacy is not required Low burden on the respondent	Does not rely on memory if recorded at time of eating Potential to provide detailed accurate information on food and beverages consumed	Provides less burden on researcher/clinician for analysis if linked to food composition database Provides metadata on the time of eating if completed correctly	Literacy is not required Provides real-time dietary assessment that includes metadata on the time of eating and additional contextual information Has potential for more accurate automated estimation of food volume
Limitations	Methodology needs to be standardised for suitability for research	Potential for measurement error is high Subject to recall bias Estimation of portion size difficult for respondents to	Subject to recall bias Previous day may not be representative Multiple days needed to capture ‘usual’ intake	Potential for reactivity high High burden, especially for respondents recording high energy intakes Verification of time of	Potential for reactivity yet to be fully tested	Potential for reactivity yet to be fully tested Automation of food and volume recognition not fully developed

		quantify Episodically consumed foods may not be captured if not on food list	Episodically consumed foods may not be captured	recording not possible (e.g. if recorded after eating)		

2.3 Dietary intakes in the present (diet record methods)

Where current diet is of interest, written diet records (either weighed measures or estimated measures, using household measures) are mainly used. More recently, widespread use of computers, hand-held digital devices, the internet and mobile telephones with or without cameras has allowed the development of new technology for real-time recording and direct analysis of dietary records.

2.3.1 In research

The main conventional method used in research to assess current dietary intakes of athletes has been diet records using household measures, which are best suited to small samples (Burke et al. 2001, 2003). In a review of dietary surveys of athletes by Burke and colleagues (2001), 3- or 4-day diet records using household measures were the method of choice. Periods longer than 3 to 4 days of food records reduce compliance and accuracy, and have a high drop-out rate (Krall & Dwyer 1987). Diet records, however, are not representative of usual diet unless repeated several times, usually 2 to 3 months apart, using non-consecutive random days (including weekends) and conducted over different seasons (Block 1989; Buzzard 1998).

The main disadvantages of traditional paper-based diet records as a research tool are that they take a long time to complete, and require a literate and cooperative respondent and a trained interviewer. Acceptability of paper-based food records has been shown to decrease with the number of recording days, with higher dropout rates after 3 days of recording (Gersovitz et al. 1978). For a 4-day food record, the time required for collecting data equates to a 1-hour respondent training session, 30 min/d to complete and an additional 20-minute interview for review (Kristal et al. 1990). Weighed food records are more accurate for measuring food intake than household measures, providing respondents are trained and motivated. Even in motivated subjects, poor compliance and a distortion of food choice limits their usefulness in providing a valid measure (Dennis & Shifflet 1985). The effect of the assessment method on behaviour has been termed reactivity (Thompson & Subar 2013). Although records using household measures are less accurate, they have acceptable validity for use in research and are more representative of what people actually eat than weighed records (Lee & Nieman 2007). Unfortunately, respondent inaccuracy in describing food portions is high and can deviate between 20% and 50% below the true weight (Livingstone & Black 2003) so an under-estimation of amounts consumed is an inherent problem.

2.3.2 Use of innovative technologies as an alternative to diet records

In research

Innovative technology (e.g. digital image-based food recording apps running on mobile devices) provide opportunities for real-time recording, which reduces respondent burden of written recording. The increased computing power of mobile devices has made it possible to run complex apps.

Two types of apps have emerged. These are the digital food record and the image-based food record. The digital food record, which links to a food composition database, requires the respondent to select from a drop-down menu the food or beverage and the amount consumed. Some apps also have the capacity to scan a barcode. This method requires the respondent to be knowledgeable of food and assumes they will accurately select the correct portion size they have consumed. The image-based food record under development requires the respondent to take images of their food and beverage intake. There are several mobile apps using digitised images under development, funded by the US National Institutes of Health (Six et al. 2010; Weiss et al. 2010). The TADA app, which operates on an iPhone or iPod touch (see www.tadaproject.org), requires the user to capture before and after images of food and beverages consumed. These images are automatically uploaded to a server using a Wi-Fi or 3G/4G connection for direct analysis. Inclusion of a fiducial marker, which acts as a reference of known dimension and markings (e.g. colour), assists in the identification of the food and the volume of food present in the image. Together the information from image analysis and volume estimation is linked to a nutrient database to estimate energy and nutrients consumed. The TADA app is currently being trialled in community-dwelling populations in Australia and the USA but the full app with automatic image analysis is not yet available for clinical application.

Early usability and reliability studies of image-based food recording apps running on mobile devices have shown high acceptability, especially in children and adolescents (Boushey et al. 2009; Six et al. 2010; Weiss et al. 2010; Schap et al. 2011; Boushey et al. 2012). These technology apps, still in development, have potential to improve accuracy and reduce burden on the respondent and researcher (Boushey et al. 2009). To date, their use in athletes has not been published.

In clinical practice

Diet records are useful to:

- raise awareness of eating habits
- provide a benchmark for evaluating diet and follow-up counselling
- provide a self-monitoring tool
- avoid the problems of memory bias inherent in recall methods
- assess diet where an individual has an erratic or inconsistent food intake and the usual recall methods used in practice are inapplicable (i.e. targeted diet history).

Adequate training in data collection should be used to collect quality food intake data in clinical practice, similar to the protocols used in research.

2.3.3 Use of innovative technologies as an alternative to diet records

In clinical practice

Nutrition monitoring and calorie tracker apps are one of the most rapidly expanding areas of app development and are becoming more accessible and increasing in use in clinical practice. Most are simply digital or electronic methods for food recording and require individuals to select each food item from a drop-down menu of foods (Mobihealth News 2010). Telephone apps are usually linked to a food composition database of the country of origin of the application. Some apps may have additional capacity to scan barcodes on food labels, take images of food and allow automatic uploading to a server or transfer via email to the dietitian, who can upload into compatible software.

For self-monitoring of diet or adherence to dietary intervention, mobile apps have demonstrated an improved compliance compared with conventional or paper-based methods (Lieffers & Hanning 2012). Other advantages include lower respondent burden than conventional methods, suitability for low-literacy clients and ease of use. In our experience, most athletes are compliant with digital recording of dietary intake using a mobile telephone with reasonable accuracy over 7 days in combination with a training diary. However, athletes still need to be trained to record dietary intake accurately using any device and a follow-up appointment is required to cross-check data entry, portion sizes and substitute foods. For digital apps, errors are high, especially with regard to choosing appropriate food substitutes (see Practice tips).

2.4 Dietary intakes in the past (recall methods)

These methods include the 24-hour recall, the food frequency questionnaire (FFQ) and the diet history. The first two methods have been adapted to computer-based approaches, which are predominantly used in epidemiological research rather than clinical practice. The diet history is the most common tool used by the dietitian to assess dietary intake in clinical practice.

2.4.1 24-hour recall

For a 24-hour recall, respondents are asked to remember and describe quantities of food and beverages consumed in the previous 24 hours (i.e. yesterday) and may include information on the timing of meals and snacks, eating environment and food preparation. Usually a 24-hour recall is administered face-to-face, by telephone interview or more recently using interactive computer or web-based software.

In research

A 24-hour recall method is used mainly in epidemiological research or national population surveys in combination with FFQs to estimate group rather than individual nutrient or food intakes. When 24-hour recalls are repeated a number of times at random, they can be used to assess usual nutrient intake. Based on predictive modelling of dietary intakes from a US sample of adults ($n = 965$), Carroll et al. (2012) found that four to six administrations of a 24-hour recall is optimal for measurement of most nutrients and food groups. Repeated 24-hour recalls are valuable aids in classifying dietary intake in groups of people and can also be used to provide an accurate assessment of an individual's food intake, if sufficient numbers of recalls are obtained.

In 2005, the National Cancer Institute in the USA commenced the adaptation of the 24-hour recall method to a web-based, automated, self-administered tool. This tool, called ASA24 (Automated Self-Administered 24-hour dietary recall), which is based on using the automated multi-pass method now used by the US National Health and Nutrition Examination Surveys, has reduced the burden and cost of using trained professionals for data collection of large population groups. The ASA24 was modified for use for the Australian Health Survey 2012–2013. Another similar web-based program for collecting 24-hour recall data, specifically designed from food intakes of children in US children's surveys, called the Food Intake Recording Software System or FIRSSt4, is completed and available for use at the same website that hosts the ASA24 (Subar et al. 2012). This new tool is called the ASA24-Kids. These online tools could be adapted for athletes and the local food supply, and may be invaluable for capturing variation of intake at different training phases and during competition for groups of athletes (Heaney et al. 2010). To date there is no published research using 24-hour recall measures in athletes.

In clinical practice

Where usual dietary intake is erratic or inconsistent, which is not uncommon in athletes, a 24-hour recall is useful and forms the basis for pursuing a broader or more detailed assessment of usual intake using a diet history.

2.4.2 Food frequency questionnaires (FFQs)

FFQs contain a predetermined food list, with or without portion sizes, plus a frequency response option for respondents to report how often each food was eaten. The questionnaire may also provide an option for respondents to report on foods not on the list and about food preparation, supplement use and other food-related behaviours.

In research

FFQs have been designed and validated in different population groups to retrospectively assess intakes of nutrients, specific foods or selected food groups consumed in the recent or distant past (24 hours to 20 years). The most common use is to measure usual diet of groups over the past 6 to 12 months. FFQs have been adapted and validated for new and different uses in research other than to gain a picture of usual intake (e.g. as screening tools to detect or rank intake of one or several nutrients, foods or selected food groups, compare or document nutrient intake during an intervention trial). FFQs are infrequently used as a survey method in athletes despite their utility and widespread use in epidemiological research, although condensed FFQs are an acceptable method of assessing specific micronutrient intake (Willett 1998). FFQS can be easily adapted for online use and analysis, are self-administered and do not required expensive respondent training or trained personnel to administer.

The limited number of FFQs used in dietary surveys or intervention trials of athletes have used modifications of previously validated questionnaires (e.g. Fogelholm & Lahti-Koski 1991; Telford et al. 1993; Hinton et al. 2004; Ward et al. 2004; Heaney et al. 2010). There has been one exception, which was developed and validated specifically to measure antioxidant intake in 113 male and female rowers, aged 17−36 years (Braakhuis et al. 2011). Ideally, use of any previously validated FFQ in a group of athletes should be modified to suit the eating habits of the target group and revalidated, although validation studies are not always feasible. Cade and colleagues (2004) and others (Molag et al. 2007) have provided guidelines to assist researchers to improve the design, application and validation of new FFQs or to modify existing FFQs for use in different population groups.

In clinical practice

FFQs are not useful for assessing usual dietary intake of individuals and are not accurate for estimating energy intake (Willett 1998). They are useful in clinical practice for other purposes:

- as a rapid screening tool or rapid assessment method for selected foods or nutrients
- to evaluate the impact of dietary intervention counselling or intervention programs
- as an education resource, for self-help intervention
- to monitor compliance with diet intervention
- for a variety of clinical situations (such as recent large loss/gain of BM) where there is a need to gather dietary information over a short period of time
- to generate inexpensive and rapid behavioural feedback to an individual on dietary intake.

2.4.3 Diet history

A diet history involves a combination of a 24-hour recall and an FFQ to determine usual eating patterns.

In research

Although a diet history provides a comprehensive assessment of usual diet and can include seasonal influences, it is time consuming, dependent on a skilled interviewer and relies on memory and the cooperation of the respondent. These factors limit its use in research.

In clinical practice

A diet history, which takes about 20 minutes to complete, is the technique of choice used by most dietitians in clinical practice to assess usual dietary intake of individuals in the past. Apart from assessing dietary intake during training or competition, a diet history provides an opportunity to explore the social, behavioural and medical influences on food choice and investigate an athlete's knowledge, beliefs and attitudes.

2.5 Sources of error in dietary measurement

Error or bias is inherent in all dietary survey methods. Recognising and reducing error when collecting and analysing dietary intake data is crucial in both research and clinical practice. In research, the magnitude of errors should be addressed in dietary intake studies, although all too frequently this is overlooked (for reviews, see Livingston & Black 2003; Kerr et al. 2013; Thompson & Subar 2013).

2.6 Sources of error common to all methods of dietary intake data collection

The limitations of the methods of collecting and interpreting dietary intake data are not always fully appreciated or described in either clinical practice or journals. The largest source of error in collecting dietary intake data is respondent inaccuracy in either recalling or reporting actual food intake (Block 1989). The ability of any respondent, including athletes, to provide accurate dietary intake data depends on their motivation, literacy, memory, communication skills and awareness of food intake. The perception of foods eaten, both in type and quantity, is crucial to the success of data collection. Athletes may give biased responses because they may not want to reveal inappropriate food choice to the coach or dietitian, or because they want to impress. Biased reporting towards socially desirable foods (such as fresh fruits and vegetables) rather than sweet or fatty foods is common in surveys of the general population (Worsley et al. 1984) and is likely to be similar or even more pronounced in athletes. The process of collecting dietary intake data also substantially distorts usual eating behaviour. The personality of the dietitian, the way questions are asked, possibly poor communication skills, and failure to develop rapport or gain the confidence of the athlete introduce additional response bias.

Most dietary survey methods underestimate energy and nutrient intakes (see Table 2.2). Techniques used to check the magnitude of under-reporting that can be applied to a research or clinical situation are discussed in section 2.7.

2.6.1 Errors using food records

Few respondents are experts at recording food intake and providing accurate food records. Motivated and educated volunteers provide the most accurate and reliable food intake data (Black et al. 1991). Athletes are unlikely to be motivated or see the relevance of recording food intakes if someone else (such as a concerned coach or parent) has sent them to see a dietitian. Self-recording food intake changes eating behaviour. Completing a food record discourages snacking and inhibits spontaneous food selection and consumption of mixed meals, and consequently distorts true food intake. In one study, over 50% of respondents completing weighed food records freely admitted altering their intakes. Inconvenience, being self-conscious or being ashamed were the main reasons (Macdiarmid & Blundell 1997). Athletes are usually far too busy to complete a weighed food record, especially if they perceive nutrition as a low priority and have not sought dietary assistance of their own accord.

The household measure method of self-recording dietary intake or using a mobile telephone app is more appropriate than weighed diet records for the busy athlete. Instructing athletes to quantify serve sizes and then cross-checking entry prior to analysis or interpretation is essential. In the future, image-based food recording methods running on mobile devices may reduce the need to quantify serve sizes. These tools will one day accurately estimate the volume of food, thus eliminating this burdensome task for those completing food records.

Under-reporting, either intentionally or unintentionally, is highest using food records compared to other methods in surveys of the general population (Buzzard 1998). Similar under-reporting also occurs in athletes, although few studies have applied measures to check this. Based on the Goldberg equation (see section 2.7.2), underestimates of energy (and hence nutrient) intake using food records have been reported in female endurance athletes (Haggarty et al. 1988; Schoeller 1995), female gymnasts (Jonnalagadda et al. 2000), cyclists in the Tour de France (Westerterp et al. 1986), Australian rugby players (Lundy et al. 2006) and lightweight female rowers (Hill & Davies 2002).

In elite athletes preoccupied with weight or in those who need to maintain a lean BM, distortion of energy intake is likely to be high, although there is little research to confirm this assumption. In one study, 61% of gymnasts were classified as under-reporters (Jonnalagadda et al. 2000), which is not surprising for this sport.

2.6.2 Errors in recall methods

Recall methods tend to overestimate the intakes of those with low energy intakes and underestimate intakes of those with high energy intakes compared with the food records (Gersovitz et al. 1978). However, validity studies of recall methods used in athletes to confirm or refute this claim have not been undertaken.

FFQs that measure nutrients in the usual diet have been criticised for their lower accuracy compared with diet records and 24-hour recalls. Recall bias and varying abilities of respondents to remember past intake and quantify foods accurately are the main reasons for this. Respondents consistently have difficulty quantifying foods, particularly meat and breakfast cereals that are not presented in packaged serves. Athletes are reported to be no different from the general population in this respect (Fogelholm & Lahti-Koski 1991). We have also observed large discrepancies in athletes' perceptions of weights of foods and volumes of beverages compared to actual weights and volumes, irrespective of age and gender.

2.6.3 Errors and limitations in converting foods into nutrients using food composition databases and dietary analysis apps

The conversion of food into nutrients is a major source of error in dietary surveys and is a reflection of the skills and knowledge of the user, the method of data collection and the food composition database. A sound knowledge of food composition and skill in data collection and analysis is essential to undertake this task. Compounding the error are lack of specificity

in the description of food or quantities consumed, together with insufficient knowledge about common preparation methods, edible portions, weight for volume and how to select foods that are not instantly matched by the food composition database.

Food composition databases do not contain the large number of foods that are consumed in real life, so inappropriate food substitutes, omission of foods and guesswork substantially distort accuracy. Defining a food substitute protocol to address these issues is crucial for minimising error before data analysis when undertaking research. Although researchers usually report the food composition database used in a study, few acknowledge other factors that might influence the accuracy of the nutrient analysis in their publications. When interpreting nutrient data derived from food composition data and entering food intake data for nutrient analysis, the following limitations should be recognised.

(a) Food composition data are only estimates of nutrient composition

Nutrient intakes calculated from food composition data are estimates only. Nutrient composition of any given food is not constant in a single raw food grown in the same environment. This variation is highest for micronutrients, especially β-carotene, vitamin C and selenium, and also varies considerably for foods produced within the same country of origin (Cashel 1990). Natural, biological, geographical and agricultural factors affect the nutrient composition of raw foods, and different cooking and processing technologies introduce wider differences. Additives and brand names may not be included, although databases include some commercial foods.

(b) Food composition data are specific to the country of origin

Nutrient values on food composition databases are not appropriate for use in other countries because of differences in biological variants, agricultural practices and food regulations.

Hence using food composition data from one country to analyse dietary intakes of athletes living in another country is unacceptable.

(c) Information on nutrients and foods is incomplete

Food composition data rapidly evolve because of changes to plant and animal breeding and to food regulation and processing techniques, and so do not contain all foods (or nutrients in food) available for consumption in the food supply. The cost of updating a rapidly changing food supply precludes comprehensive testing. Some mobile apps can scan bar codes which reflect only those nutrients included on the nutrition information panel. Adding nutrient information from food labels to food composition data which occurs when scanning food labels using mobile applications is unacceptable in research.

(d) Substitute foods may be used

Where a specific food or ingredient is unavailable in the food composition tables, substitutes have to be made. When several substitutes are made, inaccuracy is compounded. No food should go unentered because the food's contribution to the diet would be nothing (or zero); thus, a best match substitute is mandatory and is a better approximation of the truth than ignoring the food entirely.

(e) Serve sizes are difficult to standardise

Athletes (and most people) do not necessarily eat the serve sizes described on food labels and in dietary analysis software. Use of the default serve size is inappropriate and requires manual changes to food weights/volumes in the program to avoid systematic error.

(f) There may be errors in selecting the best food substitute

Inexperienced and untrained people such as athletes can introduce many errors when entering foods for nutrient analysis. As noted above, not all foods are available in a food database, thus there is opportunity to select foods that are not a best match with regard to energy and nutrient profiles. Large differences in entering foods are also a problem for trained people. Substantial variability in nutrient analysis was reported between experienced sports dietitians entering dietary intake data from self-recorded dietary intake records collected from the 1997 Australian Olympic team (Braakhuis et al. 2003). The sports dietitians were given no formal instructions or protocols when entering data, although strict protocols were given to respondents when recording food intake. The errors in dietary data entry were attributed to different interpretations of foods and food substitutes. The outcome of this study highlights the importance of developing substitution protocols and cross-checking for translation errors.

(g) Using diet apps

The widespread public access of diet analysis apps on mobile telephone and the internet raises further concerns about misuse and misinterpretation of food intake data. The app may be linked to a food composition database in another country where the food supply, food composition and food laws differ. These apps are made for popular use, thus protocols for consistent data entry are not included.

2.7 Methods used to measure under-reporting of dietary intakes

Misreporting is inherent in almost every dietary survey method. Most methods demonstrate a high frequency of under-reporting energy intakes (and subsequent nutrient intakes) when validated against more accurate measures. Several external measures or criteria are used to determine the extent of under-reporting of dietary intakes. These include the doubly labelled water method, biochemical indices (e.g 24-hour urinary nitrogen test) and EI:BMR ratio. For a comprehensive review of these and other external validation methods, see Livingston & Black (2003).

2.7.1 The doubly labelled water method for validating total energy intakes

The doubly labelled water (DLW) ($^{2}H_{2}{}^{18}O$) method is considered the gold standard for measuring community dwelling energy expenditure and validating other methods for assessing energy intake. Study participants drink water containing a load dose of stable isotopes (deuterium $^{2}H_{2}$ and oxygen ^{18}O) and provide periodic urine samples to measure the rate of elimination of these isotopes. This method is expensive and requires specialised equipment and investigator training.

2.7.2 The ratio of energy intake to basal metabolic rate (EI:BMR)

A less expensive method to validate energy intakes than DLW and urinary nitrogen is reported energy intake (EI) divided by basal metabolic rate (BMR), originally described by Goldberg and colleagues in 1991 and often referred to as the Goldberg method. A recent evaluation of the Goldberg technique, using previously collected data from an FFQ, two 24-hour recalls and DLW, found in the absence of objective measures of thermic effect of exercise (TEE) the Goldberg method was able to characterise under-reporting (Tooze et al. 2012). The final EI:BMR ratio determines whether reported energy intakes using a food record method are consistent with energy intakes required for a person to live a normal (not bed-bound) lifestyle (McLennan & Podger 1998). This ratio is represented by minimum cut-off values derived from whole-body calorimetry and DLW measurements for determining energy expenditure (Goldberg et al. 1991). The cut-off limits for the EI:BMR ratio are adjusted for sample size and duration of measurement of dietary intakes (Goldberg et al. 1991). These researchers defined two categories for cut-off values: those based on habitual or usual intake (cut-off 1) and those based on actual intake over a specified measurement period as used in food records (cut-off 2). For example, a sample size for one individual using a 3-day diet record can be assessed using cut-off 2. BMR is estimated using predictive equations. For athletes, the Cunningham equation is the best predictor where lean body mass (LBM) or fat-free mass (FFM) is available. Where only height and BM are available, the Harris–Benedict equation is recommended.

In summary, although the EI:BMR ratio is crude and dependent on estimates of BMR, which are themselves inaccurate, this method is well accepted in research and clinical practice for checking under-reporting of dietary intakes in individuals and groups.

2.8 Criteria for interpreting dietary intakes

Several criteria are used to interpret dietary intake data. Dietary guidelines and food guides provide a qualitative means of evaluating diet as well as educating both athletes and the general population about food choice. Population nutrient references can be applied to interpreting dietary intakes of most athletes, although for some nutrients (e.g. iron, carbohydrate, protein) there are specific and absolute benchmarks, which are discussed in detail in other chapters.

2.8.1 Nutrient targets/goals for athletes

Absolute quantitative amounts (in grams: g) of CHO and protein intakes are now accepted as a benchmark for recommending and assessing nutrient intakes of athletes. These values can be adjusted for BM and type, intensity and duration of physical activity (see Chapters 4 and 13).

In dietary surveys of athletes, CHO (and other macronutrient intakes) have been reported in numerous ways: from absolute amounts usually g/d, percent of energy from macronutrient to total energy in the diet, and more recently in g CHO/kg of body weight or in nutrient density terms (g CHO/1000 kJ). Adjusting for energy and BM standardises these values and makes comparison between athletes and controls more meaningful.

2.8.2 Dietary guidelines

Dietary guidelines are qualitative evidence-based policy statements that describe the major areas of dietary (and lifestyle) practices needed to help healthy people achieve recommended food choices and lifestyles (e.g. eat plenty of vegetables, legumes and fruits; be physically active; and eat according to your energy needs). These recommendations are translated into practical advice to consumers about food choice and are published by governments of many countries (see www.fao.org/ag/humannutrition/nutritioneducation/fbdg/en). The intended outcome of the dietary guidelines is to improve nutritional health of the population by helping to reduce the risk of chronic nutrition-related diseases.

Dietary guidelines provide a baseline for consistent nutrition education messages to the consumer. These same messages are appropriate for nutrition education of athletes and are used by dietitians in combination with food selection guides when assessing the nutrient density of the diet and advising athletes about food choice.

2.8.3 Food selection guides

Food selection guides provide recommendations for a suggested number of serves from each food group and are based on the estimated nutrient requirements for each age and gender group of different heights and activity levels (NHMRC & DHA 2013). They can be used to plan and select a nutrient-dense diet consistent with the dietary guidelines and form the basis of a healthy training diet for athletes. Most countries have devised food selection guides specific to the food supply, nutrient recommendations and cultural needs of their populations. Examples of food guides include *The Australian guide to healthy eating* (NHMRC & DHA 2013), *MyPyramid and MyPlate* (USDA & USDHHS 2010).

2.9 Population nutrient standards: are they relevant to athletes?

Population nutrient standards termed Nutrient Reference Values (NRV) in Australia and New Zealand (Commonwealth Department of Health and Ageing et al. 2006) and Dietary Reference Intakes (DRI) in the USA and Canada (Institute of Medicine 2000a) for macro- and micronutrients are published and frequently reviewed. Modifications to vitamin D and calcium reference standards have recently been updated (Ross et al. 2011). Prior to these publications, one nutrient standard was used to represent the amount of a nutrient needed to prevent deficiency and meet the nutrient requirements of most (97.5%) of the population (Institute of Medicine 2000b). This standard was termed Recommended Dietary Intake (RDI) in Australia and New Zealand, Recommended Dietary Allowance (RDA) in the USA and Recommended Nutrient Intake (RNI) in Canada. The nutrient reference values (i.e. DRI/NRV) are now broader in scope and application than RDA/RDI/RNI and include multiple levels of nutrient standards based on four nutrient-based reference values: Estimated Average Requirements (EAR); Adequate Intake (AI); and Upper Level of Intake (UL) in Australia and New Zealand or Upper Intake Level (UIL) in the US and Canada.

Irrespective of these multiple levels of nutrient standards and differences in terminology between countries, the applications are similar (see Table 2.3). For an individual and group, the EAR is considered the best estimate of a nutrient requirement (Institute of Medicine 2000b; Murphy & Poos 2002). Although EARs may not be applicable for athletes without some adjustment, they can serve as a benchmark to *assess* the probability that usual nutrient intake is 'inadequate' or 'adequate' for an individual athlete or for groups of athletes (Institute of Medicine 2000b). Where an EAR is unavailable for a nutrient, AI is used for individual assessment. The RDI or AI is used to indicate the intake at which, or above which, there is a low probability of inadequacy (Commonwealth Department of Health and Ageing et al. 2006). The AI is of limited value in assessing nutrient adequacy and should not be used to assess the prevalence of inadequacy (Murphy & Poos 2002).

TABLE 2.3 Recommended uses of DRI/NRV for assessing individuals and groups

For an individual	For a group
EAR: Examines the probability that usual intake of a nutrient is inadequate	EAR: Estimates the prevalence of inadequate nutrient intake within a group
RDA/RDI: Usual intake of a nutrient at or above this level has a low probability of inadequacy	RDA/RDI: Do not use to assess nutrient intake of groups
AI: Usual intake of a nutrient at or above this level has a low probability of nutrient inadequacy	AI: Mean usual intake at or above this level implies a low prevalence of inadequate nutrient intakes
UL/UIL: Usual nutrient intake above this level may place an individual at risk of adverse effects from excessive nutrient intake	UL/UIL: Estimates the percentage of the population at potential risk of adverse effects from excessive nutrient intake

AI = Adequate Intake, DRI = Dietary Reference Intakes (USA and Canada), EAR = Estimated Average Requirements, NRV = Nutrient Reference Values (Australia and New Zealand), RDA = Recommended Dietary Allowance, RDI = Recommended Dietary Intake, UIL = Tolerable Upper Intake Level (US/Canada), UL = Upper Level of Intake (Australia)

Source: Adapted from Institute of Medicine 2000b

For detailed information about the derivation, use, limitations and interpretation of the DRI for use in assessing nutrient adequacy in research or clinical practice, see the Institute of Medicine report (2000a). For a summary of these applications for use in dietary assessment, see Murphy & Poos (2002) and Murphy et al. (2006).

Using the DRI/NRV for *planning* diets for individuals and groups is more complex than assessing adequacy of nutrient intakes for individual and groups of athletes and requires a different approach and is beyond the scope of this chapter. For guidelines, see Murphy and Barr (2005) and Institute of Medicine (2003).

There are further limitations to applying population nutrient reference values for *assessing* apparent nutrient inadequacy in an athlete population, if the physical characteristics and energy demands of the athlete group deviate substantially from the general population. Athletes involved in strenuous endurance training or with a large BM, for example, are likely to need higher intakes of micronutrients than suggested by population EAR values, although definitive recommendations have not yet been established. Evidence supports a slight increase in some micronutrient requirements for athletes involved in strenuous endurance training compared to non-athletes, to compensate for high nutrient turnover and increases in free radical formation induced by exercise (see Commentary 6). However, this evidence is based mainly on biochemical and physiological indices of micronutrients, which are highly variable between individual athletes and often difficult to interpret. Of the *micronutrients* investigated, nutrient requirements (not recommendations) for antioxidant vitamins (C, E and β-carotene), B group vitamins, magnesium, zinc and iron may need to be slightly higher in athletes than non-athletes, but are unlikely to exceed RDAs/RDIs. More research is needed on large groups of athletes participating in different sports to allow micronutrient recommendations to be better quantified.

Evidence for slightly higher *macronutrient* requirements (protein and CHO) for endurance athletes compared to non-athletes is well documented (see Chapters 4 and 13). Definitive quantitative targets are specified for these nutrients for those athletes involved in endurance and power sports and are recommended for use in preference to population nutrient standards or population goals and targets.

In summary, the DRI/NRV for micronutrients is appropriate for most athletes, because of the wide safety margins for nutrient recommendations (American College of Sports Medicine 2010). Adjustment may need to be made for some nutrients in athletes with very high energy expenditures or diets that have poor nutrient bioavailability (such as vegetarian diets). For example, the EAR for iron is 1.3–1.7 times higher for athletes and 1.8 times higher for vegetarians (non-athletes) to account for low bioavailability (Institute of Medicine 2000a). The SDT (Suggested Dietary Targets) for antioxidant vitamins for those athletes involved in high-intensity training programs may be warranted (see Commentary 6).

Summary

The collection of nutritional status measures in sportspeople is critical to our understanding of the association between nutrition, health and sports performance. Accurate data collection of these measures requires highly trained people who are familiar with the protocols for data collection and limitations in data collection and interpretation.

In dietary assessment of groups and individuals, only estimates of food or nutrient intakes are possible. Therefore techniques chosen for evaluating food or nutrient intakes in athletes are related to the intended purpose of the dietary assessment and the data collection method used.

To improve accuracy when collecting food intake data, researchers and dietitians need to be familiar with the dietary habits of athletes so that appropriate serves and quantities of food likely to be consumed can be better quantified. Despite the use of food models and other techniques for improving accuracy of data collection (such as training and standardised household measures) under-reporting and misreporting is a major problem in all dietary survey methods. Future image-based food recording apps may improve the accuracy, however evaluation against biomarkers is still forthcoming.

Population standards for nutrients and biochemical indices can be applied to athletes, with caution, and with few exceptions. Because of the extensive research on CHO and protein intakes in athletes, specific recommended amounts for average daily intakes of these nutrients are available. More research is needed in varying groups of athletes before such recommendations can be made for micronutrient intakes, except perhaps iron. The recommended nutrient values for CHO, protein and iron represent the upper limits as they are derived from laboratory or field studies of mostly elite or semi-elite athletes involved in regular and often intensive training programs. Athletes involved in less rigorous training programs, or those involved in intermittent training, such as in team sports, are unlikely to need these upper limits.

Data on physique and skinfold measures in elite international- and national-level athletes involved in different sports are available elsewhere for comparative purposes (see Chapter 3). Again, comparison of skinfolds of individual athletes with group data of other elite athletes needs to be interpreted with caution because of the large standard deviations observed. A large standard deviation between and within sports indicates a high variability in individual differences. Individual athletes may not fit within these values or have optimum physiques for their chosen sport but can still become world champions. Therefore, such values should not be the basis for prescriptive targets for BM and skinfolds, impossible for many individuals to attain.

In conclusion, assessment of nutrient adequacy or recommendations for nutrient requirements for any individual (or group) is important in insuring health and performance. However, the methods used can be imprecise and need to be interpreted using professional judgment and in combination with biomedical and medical information.

USEFUL WEBSITES

- **http://andevidencelibrary.com/category.cfm?cid=2&cat=0**
  Academy of Nutrition and Dietetics. Nutrition care process for the international community
- **www.health.gov/dietaryguidelines/2010.asp**
  US Dietary Guidelines, 2010
- **www.eatforhealth.gov.au**
  Australian dietary guidelines and Australian guide to healthy eating
- **www.fao.org/ag/humannutrition/nutritioneducation/fbdg/en**
  Dietary guidelines of different countries. FAO website

Practice tips

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COLLECTING INFORMATION

- Collecting a range of information from a variety of sources (such as family, biochemical and medical) about an individual's eating behaviours and about factors influencing eating behaviours may contribute to a more accurate assessment. Such comprehensive nutritional assessments are not always appropriate, necessary or feasible. Nutritional assessment can involve minimal to comprehensive screening, and can be tailor-made for an individual athlete or team.

DIETARY ASSESSMENT OF AN INDIVIDUAL ATHLETE

- Identify the athlete's reasons for dietary consultation early in the interview.
- Find out the athlete's attitude and beliefs about nutrition (e.g. has the athlete been sent by the coach? Does the athlete consider nutrition an important part of the training program?). Often beliefs are based on testimony from other athletes or convictions of their coaches.
- Make a preliminary assessment of expected outcomes of the dietary assessment—knowing what the athlete expects to gain by seeing a dietitian is always helpful.

- Enquire about the type, intensity and duration of the athlete's training program and determine the influence of training on eating habits, timing of eating occasions and food preparation (e.g. early-morning training sessions may mean that the athlete skips breakfast; late afternoon or early evening sessions are associated with a reliance on take-away foods). Allocating a scheduled time for shopping and cooking to fit in with the training program is a worthwhile strategy.
- Assessment of daily fluid intake is a critical factor and requires investigation even in cold climates. Cramps, for instance, are associated with low fluid intakes or high sweat rates. School-aged athletes often do not drink enough fluid at school and before training, and then tend to drink large volumes during and after training to compensate. Hydration kits are useful for checking urine colour and hydration status and are routinely used at elite training centres.
- The use of vitamin and mineral supplements and other sports food supplements needs to be investigated during the interview. Often these are unnecessary and used inappropriately.
- Specific nutritional assessment forms (e.g. diet record sheets, FFQs and self-assessment checklists for specific nutrients) facilitate the interview process. In clinical practice, dietary histories are the most frequently used method for assessing diet. A 3- to 7-day diet record accompanied by a training diary is often necessary to gain a full understanding of training commitments, timing of eating occasions and dietary practices. Comprehensive instruction and a data collection protocol are needed to make the diet record worthwhile, especially if dietary analysis is expected. Compliance in reporting intake accurately for as long as 7 days needs to be encouraged. Although dietary intake can be distorted or under-reported (see section 2.6), it is useful for providing a window into eating habits, especially in an athlete who eats erratically.
- The sports dietitian needs to be aware of the accessibility of nutrition monitoring apps to athletes and enquire whether they are being used to monitor diet.
  - The decision to use a device (mobile app) that allows direct recording of food intake (using digital food record or image-based food record) that shows the nutrient and energy content of the diet to the athlete should be made with caution, especially with at-risk athletes who are preoccupied with food or energy intake.
  - If the nutrient content of the foods come from barcode scanning using a mobile app, use caution because the nutrient content from the food label may be incomplete.
- Athletes need specific training and instructions on how to handle a range of issues when recording food intakes, for example weighing or estimating serve size of individual foods, quantifying components of mixed dishes and recipes, reporting wastage and foods eaten, and how to report foods eaten away from home. The necessity for precision in all aspects of recording needs to be explained clearly. Including instructions about cooked or uncooked foods, type of cooking (e.g. grilled or baked), level of fat trimming, use of added sugar, brand name, food descriptor (e.g. reduced or modified fat), type and quantity of oil/fat in cooking and beverages is important for improving accuracy. Determining the day of the week and the number of days of data collection should also be clearly specified.
- Serve sizes on commercial foods or dietary analysis programs are generally not applicable to athletes. A standard bowl of breakfast cereal could be a pudding basin for an athlete! To enhance accuracy and decrease respondent error, when using a manual food record, it is essential to provide standard measuring cups and spoons and/or calibrated scales, or use grids, photographs of food or food models, rather than relying on respondents' perceptions.
- Concurrent interaction with computer software for dietary analysis during an interview raises an athlete's awareness of food composition and demonstrates the effects of modifications to usual eating habits. This is especially useful for follow-up consultations.
- An assortment of sports supplements and sports foods is useful as an education tool. However, displaying these in a consulting room can be counterproductive and detract from the interview. They can inappropriately influence athletes looking for that 'extra edge' or quick fix.
- The factorial method can be used to estimate average energy requirements (see Chapter 5). This involves estimating BMR and assigning an activity factor or PAL (physical activity level) value represented as a ratio to BMR. (See Commonwealth Department of Health and Ageing et al. 2006 or Institute of Medicine 2002 for PAL ratios.) This calculation is a crude estimate that can be used only as a guideline. Other wearable devices are available for measuring energy expenditure (see Practice tips, Chapter 5).
- Nutrient reference standards (DRI/NRV) are used with caution in some groups of athletes especially when used to assess the probability of nutrient inadequacy. The EAR/RDA cut-offs may not be applicable to an athlete with a large BM and very high energy expenditure. Protein, calcium and iron cut-offs may need some

adjustment when planning diets or assessing intakes (see Chapters 4, 9 and 10).

CLINICAL OBSERVATION AND MEDICAL HISTORY

- Despite the availability of medical records or medical referrals in some dietetic practices, confirmation of clinical symptoms is warranted. Athletes often do not mention their low-grade symptoms to their doctors. Athletes may reveal chronic gastrointestinal symptoms, post-training loss of appetite, recurring mild infections, nausea, headache, fatigue and bowel or menstrual problems to a dietitian without seeking medical attention.

BIOCHEMICAL MEASUREMENTS

- Interpretation of nutrient status from a single blood test can be misleading, especially if the athlete is dehydrated when tested and the nutrient tested is under homeostatic control. Dehydration is associated with haemoconcentration, resulting in falsely high readings.

ANTHROPOMETRIC MEASURES

- Techniques and standards for assessing body composition are found in Chapter 3. In clinical practice, height, BM and sometimes body frame and skinfolds are measured. Measuring skinfolds in young or adolescent athletes, who are rapidly growing, especially prior to menstruation, is not recommended. A sudden increase in skinfold measurements can have devastating psychological effects on an adolescent female. She may think she is suddenly getting very fat, whereas the increase in skinfold measurements is merely a reflection of normal growth and development. When used appropriately and conducted by a trained kinanthropometrist, skinfolds are more accurate than weight measurement for setting realistic weight targets, which vary between sports. Average values or ranges for skinfolds or weights for elite national and international athletes are not necessarily applicable or realistic for some individuals (see Chapter 3) and are better suited as targets for teams.
- Many elite athletes are preoccupied with BM and body composition. Having a BMI chart prominently displayed can be unnecessarily stressful for large-framed muscular athletes, who should be reassured that many athletes do not fit into the usual population reference standards. A BMI chart can be useful for showing very lean athletes of slight build, who perceive themselves as overweight, that they are not overweight.
- One technique that is useful for discouraging a preoccupation with daily weighing is to ask athletes to record their BM before and after each training session for 1 or 2 weeks. Fluctuations of one or more kilograms before and after training sessions and large fluctuations between continuous training days and rest days are common. Athletes soon realise that large diurnal and weekly BM fluctuations occur and that interpretation of changes in BM is difficult and may not be due to food ingestion alone.

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CHAPTER THREE

Physique assessment of the athlete

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3.1 Introduction

The physical attributes of humans have been of interest to artists and scientists for centuries. Although the human form has been measured and described extensively, interest in manipulating its components through diet and exercise has increased within the last hundred years. There has been significant interest in describing the optimal morphology and/or composition of athletes for peak performance in specific sports, while accurately measuring changes in physique characteristics influenced by genetics, growth, training and nutrition.

A relationship between sports performance and specific physique traits has been demonstrated in swimming (Carter & Ackland 1994; Anderson et al. 2008), track and field (Kyriazis et al. 2010; Watts et al. 2012), rowing (Slater et al. 2005; Kerr et al. 2007), sprint canoe and kayak (Ackland et al. 2003), track cycling (McLean & Parker 1989) and aesthetically judged and skill-based sports (Douda et al. 2008). Understanding the impact body shape and composition has on performance is essential for sports science, biomechanics and nutrition experts working with athletes.

For a substantial part of the twentieth century, body fat was a major focus of body composition research and dietary interventions in athletes to enhance sports performance. This was attributed to the technology available (underwater weighing, skinfold callipers) and the nutrition focus at the time (Drinkwater & Ross 1980). In the early part of the twentieth century, anthropologist Jindřich Matiegka described a methodology for fractionating the body into three compartments (Matiegka 1921), which was added to many years later (Drinkwater & Ross 1980). Newer technologies are now available that allow reliable measurement of individual compartments of body composition which take the place of the *gold standard* of composition measurement, cadaver analysis (see section 3.5). It is now less common for sports science and sports nutrition professionals to solely focus on fat mass (FM) when measuring and discussing body composition and sports performance. The more widespread availability of new techniques and the enhanced understanding of how various components of body composition can be influenced by nutrition interventions (see Chapters 4 and 9) means the need to monitor physique, and assess the impact of interventions over time, has become an essential skill for sports dietitians.

This chapter outlines the tools currently available to measure physique attributes of athletes.

It focuses on techniques for body composition measurement, their validity, reliability and practicality for use in everyday and research settings. Although this chapter focuses on the measurement of physique and its contribution to sports performance, it should be noted that in some sports where physique may not be a key performance metric, such assessments may form part of routine health monitoring.

3.2 Why assess physique?

Body composition and physique traits are two factors known to influence success in sports (Novak et al. 1968; Sprynarova & Parizkova 1971; Raven et al. 1976; Fleck 1983; Farmosi et al. 1984; Siders et al. 1993; Cosgrove et al. 1999; Olds 2001; Douda et al. 2008; Legaz-Arrese et al. 2005) and their assessment or manipulation is one of the primary reasons for an athlete to seek the services of a sports nutrition professional. In fact, data from a recent survey among international practitioners confirmed that the majority of sports professionals measure body composition in athletes, and assessments were mostly undertaken by the sport dietitian/nutritionist and physiologist/scientist (Meyer et al. 2013).

Physique characteristics known to influence performance include height, body mass, absolute and relative measures of muscle, fat and bone, as well as other characteristics such as limb lengths, circumferences, bone breadths and composition of various regions (Kumagai et al. 2000; Legaz & Eston 2005). Ultimately, according to Kerr and Ackland (2010), the assessment of physique in sports science has four fundamental applications:

1. to identify and understand physique characteristics critical to performance
2. to assess and monitor growth, especially in athletes on talent trajectories
3. to monitor the effectiveness of training programs and nutrition interventions
4. to determine safe and achievable body compositions for weight-category sports.

3.3 Physique attributes important for sports performance

Athletes who reach Olympic or world-class standard represent the optimum combination of ethnicity, heredity and environment to produce peak performance (Carter 1984).

Physique differences have been described globally to compare between, or even within, sports such as through somatotyping (Carter & Honeyman-Heath 1990) and proportionality (Ross & Wilson 1974). Somatotyping is a method of describing morphology and body composition concurrently via a three-number rating system, in the following order: endomorphy (relative fatness), mesomorphy (relative muscularity) and ectomorphy (relative linearity). For example, tall, muscular swimmers (ecto-mesomorphs) are more successful than their peers (Carter & Ackland 1994). Proportionality utilises a unisex phantom with which a variety of anthropometrical measures can be scaled and compared within and across sports. This provides a useful tool to describe the variance of physique traits between successful and less successful athletes, over time, or compared to a sample of the general population (Norton et al. 1996; Sedeaud et al. 2014).

With the majority of sports observing a morphological optimisation in their population over time (Norton et al. 1996), somatotypes and proportional differences within sports and events have become less apparent, requiring more detailed measures of physique components. Not surprisingly, components such as fat or lean mass have been associated with athletic success in a number of studies. For example, Australian lightweight rowers who had a lower percentage body fat and a greater total muscle mass were shown to be more successful over 2000 m compared to their peers (Slater et al. 2005). In some sports, the relationship between performance and body composition is even correlated to the success of specific technical skills. For example, shot putters using a power position were shown to benefit more from an increase in fat-free mass compared to shot putters who utilised a rotational technique (Kyriazis et al. 2010). More recently, the advent of new techniques in combination with readily available technologies have led to a greater understanding of how specific body segments can impact performance (Lee et al. 2009). For example, leg muscle volume has been shown to be significantly correlated with maximal power across a large age range in cycling (Martin et al. 2000).

Certain physique attributes that have been correlated with success in different sports, such as arm span and its correlation with successful swimming performance (Carter & Ackland 1994), are unlikely to be influenced by training or nutrition interventions. In contrast, total mass, muscle mass, fat mass and bone mass are all factors that the sports dietitians, in combination with a coach and/or exercise scientist, can influence. Clearly, the ability to accurately measure changes in body compartments, and hence physique attributes, is an essential tool for measuring the effectiveness of dietary and training interventions. Practitioners are encouraged to identify a range of techniques that may quantify changes in body composition, including morphology and segmental compositions.

3.4 Physique assessment for talent identification

Following the increased understanding around specific physique attributes associated with performance, many sporting organisations have attempted to use physique assessment as a way of identifying potential athletes in formalised talent identification programs. Numerous anthropometric measures have been suggested as talent markers. For example, peak height velocity has been used as a way of estimating developmental age (Mirwald et al. 2002). A number of morphological characteristics are genetically determined and are unlikely to be altered by growth and development. Examples include *brachial index* (the ratio of forearm length to upper arm length), *arm span* relative to height and *sitting height* relative to stretch stature. A higher brachial index is advantageous in canoe/kayak and sprint swimming (Ridge et al. 2007). A longer arm span is advantageous in basketball, rowing and boxing. A higher proportion of sitting height to total height is advantageous for swimming and a lower proportion is advantageous to distance running (Norton et al. 1996). Hence these measures can be used as markers of potential suitability for a sport.

However, it is important for organisations using physique measures for talent identification purposes to understand the composition changes that might occur post-puberty, especially in females. Females are likely to increase adiposity post-puberty compared to males who have a larger increase in lean tissue mass and a potential reduction in fat mass during and after puberty. Coaches, and female athletes on a talent trajectory, need to be aware of expected maturation changes in body composition, especially in sports where pre-pubertal physique traits are desirable. Similarly, predictions of potential suitability for various sports may not be possible pre-puberty because of the physical changes that occur post-puberty (e.g. ideal weight category for a weight-category sport). Therefore physique assessment tools for talent identification purposes need to be used with caution and not prior to the completion of pubertal development.

3.5 Methods for assessing body composition

Although numerous physique attributes are associated with sports performance, the one of interest to sports nutrition professionals is the ability to measure and manipulate the components of body composition. Several technologies and techniques are used in athletes that are able to measure and track changes in body composition. A detailed understanding of not only the validity of a technique, but more importantly its reliability, is essential to ensure their appropriate use in research and everyday practice.

Before choosing an assessment tool, the following questions assist selection of the most suitable measurement technique:

- What is the purpose of assessment and which body composition variables need to be quantified? Is it:
  – a one-off assessment to capture cross-sectional data at a single point in time?
  – intended to track changes using repeated assessments?
  – intended to evaluate the effectiveness of an intervention?
- Is the assessment tool valid and reliable? Validity refers to the agreement between the value of a measurement and its true value (Hopkins 2000) or, more broadly, how well a measure is representative of the most accurate method or *gold standard*. *Validity* is important for the precision of a single measurement. A consequence of poor measurement validity is the reduction in ability to characterise relationships between variables in descriptive studies (Hopkins 2000). *Reliability* refers to the reproducibility of the observed value when the measurement is repeated (Hopkins 2000). Reliability results from high precision of a single measurement and also facilitates the researcher's or clinician's ability to detect changes between serial measurements on the same athlete. This is an important consideration if an assessment tool is to be used to assess body composition throughout an athlete's maturation, training history or as a result of a specific intervention.

- Does the measurement require a skilled technician and hence extensive training? For example, the International Society for the Advancement of Kinanthropometry (ISAK) has developed international standards for anthropometric assessment. In the case of dual-energy X-ray absorptiometry (DXA) scans, technicians are required to complete a densitometry training course, a qualification that does not train them specifically in techniques for body composition analysis.
- What is required of the subject undertaking the test, including their level of comfort? For example, subjects are required to completely exhale the air in their lungs under water for underwater weighing, therefore, this method is not suitable for young children, the elderly or people uncomfortable in aquatic environments. Surface anthropometry (skinfolds, girths, breadths and lengths) requires the subject to wear minimum clothing; hence sensitivity about body image or religious/cultural beliefs may exclude this option. Furthermore, in some populations, males cannot measure females, and vice versa.
- What is the accessibility and ease of operation of the measurement device? Does the measurement (e.g. DXA, bioelectrical impedance, air displacement plethysmography) require a strict standardisation protocol?
- What costs are associated with the technique? Costs include initial capital cost of the machine/tool, running costs (maintenance, calibration or electricity) or consumable costs.
- How long does the measurement take? All aspects of the measurement must be considered. For example, time taken to calibrate the machine, carry out the measurement, analyse/interpret results and maintain it (e.g. cleaning, recalibration). Some measurements (e.g. three-dimensional photon scanning or DXA) require extensive analysis which can range from 5–30 minutes per person. Time to complete and analyse is important when considering the logistics of undertaking an assessment on a squad or group of athletes at the same time.

The various methods of body composition assessment, their methodological assumptions and considerations are presented in Table 3.1.

TABLE 3.1 Overview of commonly available body composition assessment techniques

Technique	Measurements	Level	Strengths	Weaknesses	Checklist		
					Practical	Reliable	Valid
Research/laboratory-based							
Cadaver dissection	Anatomical (skin, muscle, adipose tissue, bone and organs) or chemical (water, fat, protein and mineral elements) approaches	Direct	Considered the *gold standard*	Based on small sample size (dissection study (*n* = 51) and chemical study (*n* = 8) Large inter-subject variations (e.g. age) and sex differences (Clarys et al. 1999) Tests are expensive and	No	No	Yes

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				complex Ethical issues Results are not applicable to individuals			
Multi-compartment models	Fat mass, total body water, bone mineral and residual (protein, some non-bone mineral, some glycogen and essential lipid)	Direct	Currently accepted as the reference method	Initial outlay for equipment is expensive Tests are complex and time consuming	No	Yes	Yes
Hydrodensitometry (underwater weighing: UWW)	Body density (can be used to estimate percent body fat via regression equations)	Direct (body density) Indirect (percent body fat)	Initially accepted method	Initial outlay for equipment is expensive Tests are time consuming High subject burden (exhale during submersion) Assumes density of fat mass and fat-free mass (FFM) to be constant, which was based on very small sample size (Brozek et al. 1963) There is error associated with estimation of residual lung volume Regression equations are not suitable for calculation of body composition from body density measurements, especially in athletic populations	No	Yes	Yes
Air-displacement plethysmography (ADP via BodPod)	Total body volume (can be used to estimate percent body fat via regression equations)	Direct (body density) Indirect (percent body fat)	Tests are fast with less subject burden than UWW (more suitable for children and elderly)	Initial outlay of equipment is expensive Careful consideration in the methodologies is required (clothing choice, regulated temperature and humidity of room) Assumes density of fat mass and FFM to be constant There is error associated with estimation of residual lung	Yes	Yes (WS)	Yes

				volume Regression equations are not suitable for calculation of body composition from body density measurements, especially in athletic populations Some studies found calculation of body fat to be different to UWW			
Dual-energy X-ray absorptiometry (DXA)	Total and regional bone mineral content (bone tissue), fat mass (total adipose tissue) and lean mass (fat-free soft tissue mass)	Direct	Tests are fast and provide regional body composition estimates (i.e. left and right sides of arms, legs and trunk)	Initial outlay of equipment is expensive Requires a trained technician Technique exposes subjects to small amount of radiation Careful consideration in the methodologies is required (subject presentation, subject positioning on the scanner, analysis technique) Scanner is relatively small in size and may not be practical for tall and/or broad athletes	Yes	Yes (WS)	Yes
Field							
Surface anthropometry (skinfolds, lengths, girths, breadths)	Skinfolds: thickness of double layer of skin and compressed subcutaneous tissue Can use fractionation technique in conjunction with lengths, girths and breadths measurements to estimate total body fat and lean mass	Indirect (skinfolds) Doubly indirect (percent body fat)	Initial outlay for equipment is inexpensive Portable and suitable in the field	Indirect measure of absolute fat and muscle mass Measurement uncertainty from adipose tissue compressibility Duration of test depends on the number of variables being measured, and requires a trained technician Quantification of changes in regional muscle mass is crude	Yes	Yes (WS)	Yes
Bio-electrical impedance (BIA) Single frequency BIA (50 kHz) – does not measure TBW, but rather a	Water fluid compartments (TBW, ECW and ICW) and FFM	Indirect	Initial outlay of equipment is inexpensive Tests are fast and portable in the field	Results are greatly influenced by hydration status and applications to athletic populations are	Yes	Yes (WS)	Unsure

weighted sum of ECW water and ICW resistivities Multiple frequency BIA (0, 1, 5, 50, 100, 200 to 500 kHz) Bioelectrical spectroscopy (BIS): a more complex model with wide range of frequencies				unknown Careful consideration in the methodologies is required Mixed results when comparing BIA with other methods such as DXA			
Ultrasound (US)	Tissue thickness measurement (dermis, subcutaneous adipose tissue, fascia, muscle)	Direct	Initial outlay for equipment is inexpensive Tests are fast and portable in the field Measurement of uncompressed subcutaneous adipose tissue	Required trained technician to correctly apply speed of sound and detect tissue layer boundaries Not all skinfold sites are suitable	Yes	Yes (WS)	Yes

BIA = bioelectrical impedance, BIS = bioelectrical spectroscopy, ECW = extra-cellular water, FFM = fat-free mass, ICW = intracellular water, TBW = total body water, WS = with standardisation

3.5.1 Level of body composition assessment

Body composition assessment techniques generally fall into three types of method based on their primary purpose/functionality: *reference*, *laboratory* and *field*. Within each method, techniques can be further categorised into three levels of analysis: *direct, indirect (a surrogate parameter is measured to estimate tissue or molecular component)* or *doubly indirect* (one indirect measure is used to predict another indirect measure, e.g. via regression equations). Each technique measures a number of compartments (i.e. two-, three-, four- or five-compartment models), which are based on either a *chemical/molecular* or *anatomical* approach (see Figure 3.1).

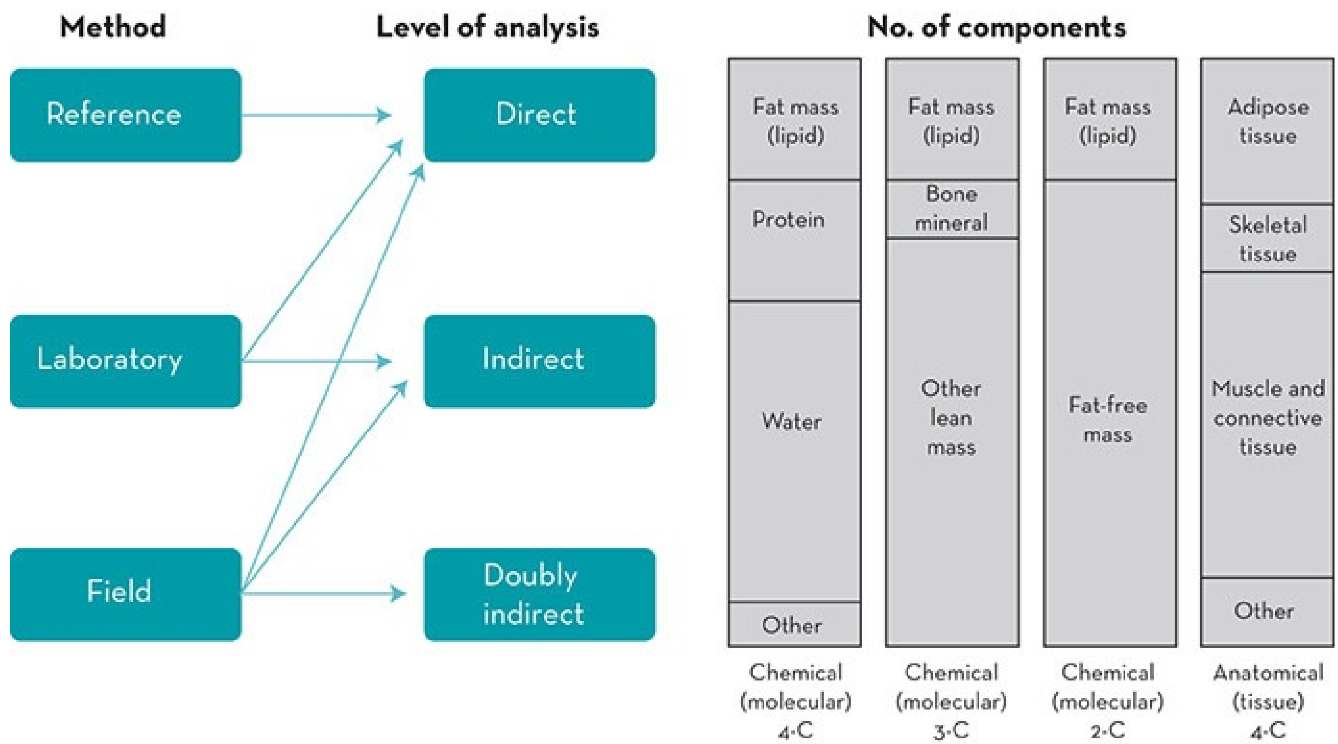

Figure 3.1 Possible levels of categorisation of body composition analysis techniques
Source: Adapted from Ackland et al. (2012)

3.5.2 Reference methods

The three assessment techniques that are regarded as reference methods for composition analysis (often referred to as the *gold standard*) are cadaver dissection/analysis, multi-technique multi-compartment models and medical imaging. All of these methods directly measure body compartments; however, none is easily accessible or practical for everyday sports practitioners.

Although considered the ultimate *gold standard*, data from cadaver dissection are not only sparse, but also difficult to access. Pooled data from the Brussels Cadaver Analysis Study and others, which was based on a small number of subjects (n = 51, age range 16–94 yrs), were used to derive many subsequent key body composition assumptions for the basis of other indirect body composition assessment tools (Clarys et al. 1999; Clarys et al. 2005). Furthermore, this analysis technique is very complex, time consuming and expensive.

The availability of newer technologies has enabled the development of multi-compartment models (four-compartment is the current standard) to assess body composition. Four-compartment models, whereby fat and components of fat-free mass (FFM) can be estimated by measuring body volume from underwater weighing (UWW) or air displacement plethysmography, total body water from deuterium dilution and total bone mineral content (BMC) from DXA (Lohman & Going 1993) is currently regarded as the criterion method for body composition assessment (Withers et al. 1999). However, this method is time consuming (e.g. deuterium dilution requires ~6 hours to equilibrate in the body) and expensive, therefore

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its use is mostly restricted to research purposes.

Medical imaging such as computed tomography (CT) or magnetic resonance imaging (MRI) are alternative reference methods. The cost of each scan is high and accessibility for non-clinical use is limited. CT scans also expose subjects to high levels of radiation and may not be suitable for athletes who might already receive high levels of radiation exposure from other sources (e.g. from travelling, body composition scan by DXA and medical imaging from injury) (Cross et al. 2003; Orchard et al. 2005).

3.5.3 Laboratory methods

Most laboratory tools are usually accessible through sports institutes, universities or commercial testing laboratories. Hydrodensitometry (better known as UWW) was traditionally used to measure body density for the estimation of percentage body fat, based on a two-component model. However, the technique requires the subjects to be completely submerged under water while fully exhaling their breath, therefore it is not practical for young children, the elderly and people uncomfortable in aquatic environments.

A more popular alternative to UWW, and based on the similar principles, is air displacement plethysmography (ADP), commonly measured using a BodPod. Subject clothing (Shafer et al. 2008; Peeters & Claessens 2011) and movement within the device (Tegenkamp et al. 2011) can add significant error to body composition measurements. Both UWW and ADP methods allow measurement of body density; however, inherent assumptions around residual lung volume and the assumption of a similar density of fat and FFM introduce limitations when translating body density into percentage body fat (Withers et al. 1999; Ellis 2000).

A novel technology available to measure body volume is three-dimensional (3D) photonic scanning. 3D scanners use either laser or white light in combination with a series of cameras to capture reflected light, developing a cloud map of individual points each with an XYZ coordinate (Schranz et al. 2010; Stewart 2010). This three-dimensional model can be analysed using computer software to extract anthropometric measures (Schranz et al. 2010) and has even been validated for percent body fat predictions (Wang et al. 2006). Although the same assumptions and errors associated with densitometry measures are also concerns with this technique, it offers a quick and low-subject-burden alternative. Interesting is the emerging availability of this technology in the living room through high-definition motion-capture cameras from gaming stations to produce a 3D image.

Another recently popular technique is DXA due to its rapid measurement (~5 minutes) and its ability to generate detailed estimates of whole and regional fat and lean mass. One of the limitations of the DXA machine is, however, the size of the active scanning area. This is clearly problematic for some subgroups in the athletic populations, especially those who are tall and/or broad. Nevertheless, several studies have investigated practical strategies to accommodate athletes with these physiques (Evans et al. 2005; Rothney et al. 2009; Nana et al. 2012b).

Although these techniques have traditionally been considered laboratory methods due to the limited access to the devices, the increased public health interest in body composition has meant devices such as BodPod, DXA and 3D scanners are readily available to athletes and practitioners in real-world settings. Of concern with this increased availability of these techniques is that some users do not recognise their limitations and consider that they have less error than other doubly indirect field-based methods. Poor control around standardisation can lead to significant issues with error of measurement reducing the ability to interpret changes effectively (Nana et al. 2014). Hence, caution should be taken around standardisation of subjects when using commercial devices to measure change in body composition over time.

3.5.4 Field methods

One of the most important aspects of a tool to assess body composition is its practicality, which includes its portability in the field. The most common tool currently used by sports practitioners is surface anthropometry (Meyer et al. 2013), which includes measurements of skinfolds, girths, lengths and breadths. Although basic, measurement of skinfold thickness has been considered a valuable proxy for whole-body adiposity and a reliable tool to track fat patterning over time (Clarys et al. 2005; Ackland et al. 2012). However, skinfold measures do not provide accurate information on the absolute amount of fat mass.

Skinfolds correlate poorly with FFM estimates (Roche 1996) and offer little insight into quantitative changes in total or regional muscle mass (Cisar et al. 1989; Stewart & Hannan 2000). One benefit of this technique is the standardisation of measurement by an international body (Marfell-Jones & Stewart 2012). ISAK is an international organisation formed to educate practitioners on standardised techniques, ensuring protocols for landmarking (the location and marking of skinfold, length, girth and breadth sites) and assessment are maintained between practitioners.

Some practitioners have attempted to validate indices that are useful in the field setting to interpret proportional changes in lean mass over time. The lean mass index (LMI) is an empirical measure that allows practitioners to track within-subject proportional changes in body mass adjusted for changes in skinfold thickness (Slater et al. 2006). This index allows practitioners to track relative changes in lean mass from nothing more than a measure of body mass and sum of seven skinfolds. However, limitations associated with body mass variability and hydration status and the fact that LMI has only been validated in a male population mean it should be used with caution.

Bioelectrical impedance analysis (BIA) is widely available, has a relatively low capital cost and the additional benefit of portability in some models. However, BIA results vary with hydration status, hence measurement protocols are required to control for this (Kyle et al. 2004a, 2004b). Furthermore, there are different methods using this technique. For example, single-frequency BIA, usually at 50 kHz, is passed between surface electrodes placed on hand and foot, or in the case of commonly available portable devices, from foot to foot or hand to hand (Kyle et al. 2004b). In contrast, multiple-frequency BIA uses impedance at multiple frequencies, utilising linear regression models to estimate water and tissue compartments; while bioelectrical spectroscopy (BIS) uses mathematical modelling and a mixture of equations (Kyle et al. 2004a).

Recently, ultrasound devices have been used to measure subcutaneous fat thickness, similar to skinfolds (Muller et al. 2013a; 2013b). Ultrasound accurately measures uncompressed subcutaneous adipose tissue, however the method requires a well-trained technician to apply the correct speed of sound and accurately detect tissue layer boundaries. Furthermore, it was noted that not all skinfold sites are suitable and regression equations developed from skinfold data (which involves a double layer of fat and skin, as opposed to the single layer that ultrasound measures) are currently used to estimate body fat (Muller & Maughan 2013). This makes ultrasound devices designed for body composition analysis potentially highly accurate at measuring subcutaneous fat at specific skinfold sites but invalid in projecting these to estimates of whole body fat.

3.6 Using body composition assessments to monitor training programs

Monitoring the change in overall lean and/or fat mass is a necessary technique to determine the success of an intervention, whether that be training and/or nutrition. However, poor standardisation and control with the assessment tool can result in poor precision, or an over-exaggeration of small changes in body composition. For example, errors in the measurement of lean mass using BIA occur if an individual is measured in a state of slight dehydration. By failing to account for changes in hydration status in this situation an apparent loss of lean mass may be reported by this technology. This may result in unnecessary stress being placed on an athlete whose training and nutrition goal has been bulking up combined with a loss of faith in the practitioner prescribing the dietary intervention. There is not a single laboratory or field technique for measuring body characteristics that that does not require some form of subject or equipment standardisation.

Perhaps surprisingly, a sum of skinfolds is relatively robust in the face of standardisation concerns and can be considered a valid proxy for adiposity, especially when maintained as a sum of skinfolds and not converted to a percent body fat (Kerr & Stewart 2009). Tables 3.2 and 3.3 present sum of skinfold, body mass and height data (where available) for elite level athletes. While the sample size of some data sets presented may not allow for this data to be considered *normative*, they are representative of the elite competitors within the sports and provide indications for the variance between and within sports. For other normative data, readers are directed to other reviews (Norton et al. 1996; Santos et al. 2014).

TABLE 3.2 Representative surface anthropometry data for female athletes

Sport	Level	Position/event	Timing of data	Height (cm)	Body mass	Sum of 7 skinfolds	Key features	Source

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			collection	(mean; range)	(kg) (mean; range)	(mm) (mean; range)		
Athletics	National	Sprint hurdle (100–400 m, *n* > 61)	2001–2010	172.5; 162.1–181.1	62.4; 56.1–73.4	52.9; 36.8–81.9		Tanner & Gore 201
		Distance (800 m–marathon, *n* > 25)		167.8; 157.8–184.2	55.6; 46.2–67.5	56.4; 28.0–96.0		Tanner & Gore 201
		Walk (10–50 km; *n* > 31)		167.9; 159.2–174.0	55.5; 47.7–62.1	77.3; 43.0–111.7		Tanner & Gore 201
Basketball	International senior	Guard (*n* = 64)	1994 World Championships	171.9 ± 6.1	66.1 ± 6.2	76.6; 36.4–143.5		Ackland al. 1997
		Forward (*n* = 57)		181.3 ± 5.9	73.3 ± 5.1	76.0; 49–131.7		
		Centre (*n* = 47)		189.8 ± 6.4	82.6 ± 8.2	88.0; 45.7–146.8		
	National junior (U20, *n* = 500)		Residential program 2003–2010	182.8; 158.8–204.0	75.0; 47.0–118.0	92.0; 47.0–165.0		Tanner & Gore 201
Bobsled	International (*n* = 5)	Driver + push combined		175.3 ± 2.1	77.8 ± 2.8	70.5; 53.3–93.0		US unpublis data
	National (*n* = 8)	Driver + push combined		171.0 ± 4.7	76.5 ± 4.8	89.8; 58.0–118.2		US unpublis data
Boxing	National	48 kg (*n* = 3)	2012–2013	159.9; 154.6–163.0	51.5; 49.1–55.3	72.2; 57.8–81.5		AIS unpublis data
		57 kg (*n* = 5)	2012–2013	166.1; 164.2–167.2	57.2; 55.9–58.3	84.0; 75.3–94.0		AIS unpublis data
		60 kg (*n* = 3)	2012–2013	165.9; 163.1–171.0	60.2; 59.4–61.5	83.0; 75.9–88.4		AIS unpublis data
		64 kg (*n* = 4)	2012–2013	165.9; 163.4–170.9	64.0; 61.2–66.2	84.3; 76.2–94.1		AIS unpublis data
		69 kg (*n* = 3)	2012–2013	174.0; 170.9–178.6	70.6; 69.9–71.8	107.8; 66.9–130.0		AIS unpublis data
		75 kg (*n* = 3)	2012–2013	174.2; 167.7–178.0	72.4; 70.2–73.9	87.4; 80.6–97.3		AIS unpublis data
Cricket	International (*n* > 46)		2007–2010	168.3; 153.6–177.5	65.4; 55.2–86.0	105.2; 49.6–193.2		Tanner & Gore 201

	National (*n* > 89)		2007–2010	167.6; 153.6–176.0	65.2; 48.5–97.0	109.1; 49.5–213.5		Tanner & Gore 20
Cycling	International senior	Track sprint (*n* = 7)	Close to competition, 2002–2012		69.6; 62.6–73.5	81.5; 72.8–97.3		AIS unpublis data
		Road (*n* = 9)	Competitive season		58.2; 50.9–64.9	55.2; 30.8–73.6		Tanner & Gore 20
		MTB (*n* = 6)	Close to competition		57.4; 53.4–62.8	68.0; 41.6–92.2		Tanner & Gore 20
		BMX (*n* = 13)			65.1; 56.4–74.3	91.3; 59.4–130.0		AIS and unpublis data
	National	Track sprint (*n* = 7)	Close to competition 2002–2012		64.4; 58.3–76.1	81.8; 52.1–120.0		AIS unpublis data
	Junior	MTB (*n* = 5)	Close to competition		58.2; 49.9–65.2	85.0; 69.5–94.9		Tanner & Gore 20
Diving	National senior (*n* = 5)			157; 148–164	52.0; 47.9–57.4	58.4; 47.6–74.7		AIS unpublis data
Football (Soccer)	International senior (*n* = 38)		2006–2009	166.8; 156.2–178.3	61.3; 52.5–72.7	75.2; 53.7–103.1		Tanner & Gore 20
	National U17 (*n* = 42)		2005–2009	163.1; 154.5–170.1	52.9; 44.6–68.4	98.6; 63.6–151.7		Tanner & Gore 20
Gymnastics	International senior (*n* = 5)			158.2; 150.8–162.5	50.4; 47.3–55.1	47.4; 36.4–54.7		AIS unpublis data
Hockey	National	Striker (*n* = 4)			65.2; 58.9–70.1	65.5; 56.4–85.7		Tanner & Gore 20
		Midfield (*n* = 9)			60.4; 57.0–62.3	76.6; 62.4–104.3		
		Defender (*n* = 6)			62.0; 57.7–67.3	68.4; 49.5–82.5		
		Goalkeeper (*n* = 2)			65.0; 59.8–70.2	66.7; 54.5–78.9		
	National U21 (*n* = 23)				61.2; 54.9–71.7	81.3; 60.1–112.6		Tanner & Gore 20
	National U17 (*n* = 17)				61.9; 50.5–79.8	87.8; 52.6–128.9		Tanner & Gore 20

Kayak	Olympic	Flatwater (*n* = 20)	15-day period before Olympics	170.4; 159.2–184.2	67.7; 59.1–80.7	80.0; 52.9–103.7a	Proportionally large upper body	Ackland al. 2001
		Slalom (*n* = 12)	15-day period before Olympics	168; 158–176	59.0; 53.3–68.6	68.9; 46.0–99.0a	High brachial index	Ridge et 2007
	International senior (*n* = 14)	Flatwater	2006–2013	172.6; 162.0–185.6	68.5; 56.7–77.0	64.9; 42.4–80.7		AIS unpublis data
	National U23 (*n* = 6)	Flatwater	2006–2013	172.1; 162.9–182.2	70.1; 61.3–81.1	97.6; 72.1–128.9		AIS unpublis data
Luge	Junior (*n* = 4)			161.6 ± 5.0	60.9 ± 5.3	87.1; 60.4–117.0		US unpublis data
Netball	National senior (*n* = 30)		2010	182.4; 170.0–196.0	72.6; 58.6–92.4	82.9; 46.4–136.6		Tanner & Gore 20
	National U21 (*n* = 46)		2010	181.5; 170.5–186.6	73.7; 64.8–91.9	99.8; 62.6–169.9		Tanner & Gore 20
Rowing	International (Olympic)	Lightweight (*n* = 14)	15-day period before Olympics	169.0 ± 0.05	58.5 ± 1.5	59.7; 40.1–77.9a	Proportionally large chest, waist and thigh but smaller hip girth	Ackland al. 2001; Kerr et a 2007
		Open (69)	15-day period before Olympics	181 ± 0.05	76.6 ± 5.2	89.0; 56.5–135.3a	Proportionally large chest, waist and thigh but smaller hip girth	Ackland al. 2001; Kerr et a 2007
	National senior	Lightweight (*n* = 17)	2003 National championships	170.3 ± 3.5	57.9 ± 1.1	68.5 ± 17.1a	Short sitting height relative to stature and longer limb lengths	Slater et 2005
		Open (*n* = 50)			75.5; 65.4–87.7	82.4; 49.0–139.4		Tanner & Gore 20
	National U23	Lightweight (*n* = 28)	2003 National championships	170.0 ± 5.3	57.4 ± 1.6	73.6 ± 15.5a		Slater et 2005
Rugby League	National	Forwards (*n* = 16)	2 weeks after 2005 National championships	169.0 ± 6.6	75.5 ± 12.5	141.2 ± 37.2		Gabbett al. 2009
		Backs (*n* = 16)		166.1 ± 5.4	64.7 ± 7.6	114.8 ± 20.2		
Sailing	National/international	470 – Helm (*n* = 6)	2006–2010	165.1; 159.9–173.1	55.1; 48.2–60.2	86.3; 55.8–116.4		Tanner & Gore 20

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		470 – Crew (*n* = 5)	2006–2010	175.5; 170.5–179.5	67.2; 63.8–73.4	76.9; 60.5–94.0		
		Laser Radial (*n* = 4)	2006–2010	172.0; 169.7–174.6	66.6; 63.3–71.6	89.1; 76.7–110.5		
		RSX (*n* = 3)	2006–2010	170.1; 168.1–172.1	62.7; 55.3–70.6	87.3; 69.5–110.7		
		Match Racing (*n* = 9)	2006–2010	167.5; 161.8–175.1	66.4; 61.8–70.5	108.2; 90.3–138.7		
Skeleton	International (*n* = 3)			164.2 ± 3.9	65.1 ± 3.1	79.4; 65.1–99.6		US unpublis data
	National (*n* = 8)			167.7 ± 7.4	64.9 ± 4.0	80.4; 52.2–128.0		US unpublis data
Speed skating	National/international (*n* = 8)	Short track		165.0; 151.5–175.0	58.9; 48.0–74.5	77.7; 43.3–131.0		AIS unpublis data
Swimming	National/international (*n* = 31)				64.9; 51.4–78.1	67.6; 49.9–100.5		Pyne et 2006
Tennis	National academy	16+ y (*n* > 11)	2008–2010	170.6; 161.4–183.0	62.7; 47.1–77.0	94.1; 53.3–122.9		Tanner & Gore 20
		15–16 y (*n* > 8)	2008–2010	167.5; 160.0–179.0	57.4; 46.7–73.0	105.5; 73.1–151.8		Tanner & Gore 20
Triathlon	National/international	U23/Elite (*n* = 10)	Race season	165.7; 159.2–170.5	55.0; 49.0-64.0	52.0; 40.0–70.0		Tanner & Gore 20
	National	Junior (*n* = 12)		167.3 ± 5.4	52.8 ± 6.4	75.8 ± 23.4[a]		Landers al. 2013
Volleyball	National junior & senior (*n* = 11)	Indoor		183.0; 173.7–194.2	73.0; 65.0–84.0	101.0; 81.0–136.0		Tanner & Gore 20
	National junior & senior (*n* = 16)	Beach		182.4; 172.7–193.7	71.4; 62.2–77.9	79.4; 47.4–113.2		AIS unpublis data
Water Polo	National/international	Centre (*n* = 7)	2007–2008	177; 170–183	80.5; 71.7–91.7	102.0; 65.6–126.6		Tanner & Gore 20
		Perimeter (*n* = 12)	2007–2008	174; 167–182	70.2; 61.7–83.6	90.7; 65.9–142.7		
		Goalkeeper (*n* = 4)	2007–2008	174; 169–183	76.8; 69.5–82.1	98.9; 75.7–114.0		

[a]Sum of 8 skinfolds (includes iliac crest)

TABLE 3.3 Representative surface anthropometry data for male athletes

Sport	Level	Position/event	Timing of data collection	Height (cm) (mean; range)	Body mass (kg) (mean; range)	Sum of 7 skinfolds (mm) (mean; range)	Key features	Source
Athletics	National/international	Sprints/hurdles (100–400 m) (*n* > 154)	2001–2010	182.4; 170.7–192.0	77.5; 64.0–96.2	39.4; 25.9–90.6		Tanner Gore 2
		Distance (800 m, marathon) (*n* > 80)	2001–2010	178.9; 169.0–192.4	66.2; 54.0–82.7	37.4; 23.5–54.8		
		Walks (10–50 km) (*n* > 53)	2001–2010	180.0; 166.2–191.7	65.4; 55.0–81.2	38.2; 30.3–55.5		
Australian Rules Football	~ 18 y (*n* = 284)		Draft program 2006–2009	186; 167–202	81; 63–107	56; 31–113		Tanner Gore 2
Basketball	National junior (U20)		Residential program 2003–2010	197.5; 174.1–221.8	91.0; 59.0–141.0	57.0; 34.0–220.0		Tanner Gore 2
Bobsled	International	Driver (*n* = 4)		184.1 ± 6.0	99.3 ± 4.3	82.6; 67.6–104.4		US unpubli data
		Push (*n* = 9)		187.5 ± 5.1	101.8 ± 4.2	52.4; 38.1–89.8		US unpubli data
	National	Driver (*n* = 3)		181.2 ± 8.9	89.5 ± 2.1	69.7; 54.1–85.7		US unpubli data
		Push (*n* = 3)		186.2 ± 5.3	96.7 ± 4.0	50.1; 37.8–63.8		US unpubli data
Boxing	National	54 kg (*n* = 5)	2003–2013	167.9; 165.5–170.6	57.0; 55.0–59.5	38.2; 31.4–44.6		AIS unpubli data
		56 kg (*n* = 13)	2003–2013	171.5; 161.2–179.6	59.2; 56.9–62.6	42.0; 30.3–54.5		AIS unpubli data
		60 kg (*n* = 13)	2003–2013	171.7; 168.4–176.6	63.2; 60.7–65.7	43.8; 39.6–49.1		AIS unpubli data
		64 kg (*n* = 11)	2003–2013	175.0; 169.6–181.3	66.5; 64.3–68.7	43.6; 27.4–59.9		AIS unpubli data
		69 kg (*n* = 13)	2003–2013	179.0; 173.1–189.9	71.7; 68.9–78.1	47.1; 33.9–64.5		AIS unpubli data

		75 kg (*n* = 18)	2003–2013	182.9; 177.6–190.2	77.0; 74.1–80.5	48.2; 31.7–71.5		
		81 kg (*n* = 14)		184.7; 180.6–190.9	82.6; 76.5–84.9	53.0; 39.2–82.8		AIS unpubli data
		91+ kg (*n* = 18)		190.2; 179.2–200.5	99.5; 90.5–114.8	84.2; 42.9–155.0		AIS unpubli data
Cricket	International (*n* > 151)			182.4; 170.0–197.0	85.8; 72.7–102.0	71.7; 40.8–125.9		Tanner Gore 2
	National (*n* > 445)			183.8; 169.6–203.0	86.0; 63.5–110.5	70.8; 30.2–152.8		
	U19 (*n* > 132)			181.9; 170.0–196.8	84.0; 61.5–103.5	69.3; 38.0–122.0		
Cycling	International senior	Track endurance (*n* = 6)	Close to competition		72.6; 69.0–80.2	38.1; 30.5–53.4		Tanner Gore 2
		Track sprint (*n* = 13)	Close to competition 2002–2012		90.8; 85.2–100.6	47.6; 36.5–63.3		AIS unpubli data
		BMX (*n* = 22)			80.9; 68.0–93.0	46.9; 31.8–83.0		AIS an USA unpubli data
	National	Track sprint (*n* = 9)	Close to competition 2002–2012		80.7; 67.8–91.9	50.4; 35.0–68.0		AIS unpubli data
	U23	Road (*n* = 7)	Competitive season		73.4; 58.2–77.4	45.1; 32.5–56.2		Tanner Gore 2
		MTB (*n* = 11)	Close to competition		72.5; 63.4–88.5	42.9; 32.5–53.9		Tanner Gore 2
	Junior	MTB (*n* = 9)	Close to competition		67.0; 56.7–80.5	43.0; 32.5–53.9		Tanner Gore 2
Diving	National (*n* = 3)			173	69.0; 63.0–74.1	39.7; 31.5–51.2		AIS unpubli data
Football (Soccer)	International senior (*n* = 44)		2007–2010	182.3; 166.1–197.8	78.3; 61.2–96.8	47.5; 32.3–73.6		Tanner Gore 2
	Olympic (U23; *n* = 54)		2003–2008	181.8; 165.0–192.2	77.5; 60.0–96.4	57.8; 37.2–78.1		Tanner Gore 2
	National U17 (*n* = 74)		2003–2009	176.7; 163.3–	69.6; 58.0–	56.4; 43.2–		Tanner Gore 2

				178.3	82.4	79.6		
Gymnastics	International (*n* = 5)			160.3; 150.0–168.1	59.6; 46.6–68.4	34.0; 31.6–37.6		AIS unpubli data
Hockey	National	Striker (*n* = 8)	2009–2010		80.3; 74.9–89.3	56.7; 44.6–74.7		Tanner Gore 2
		Midfield (*n* = 12)	2009–2010		76.0; 67.0–85.4	50.2; 32.2–70.3		
		Defender (*n* = 9)	2009–2010		80.3; 62.2–90.9	60.0; 42.9–79.3		
		Goalkeeper (*n* = 4)	2009–2010		83.5; 74.5–95.1	77.0; 55.6–110.6		
	National U21 (*n* = 20)		2009–2010		74.9; 58.9–86.5	55.3; 32.6–85.7		Tanner Gore 2
Kayak/Canoe	Olympic	Flatwater (*n* = 50)	15-day period before Olympics	184.3; 169.7–195.8	85.2; 73.6–99.8	55.4; 30.9–116.1a	Proportionally large upper body and narrow hips	Acklan al. 200
		Slalom kayak (*n* = 12)	15-day period before Olympics	177; 172–190	71.7; 62.7–79.0	45.8; 32.4–63.5a	High brachial index	Ridge e 2007
		Slalom canoe (*n* = 19)	15-day period before Olympics	177; 159–194	73.1; 59.6–84.3	57.1; 38.8–73.7a	High brachial index	Ridge e 2007
	International (*n* = 14)	Flatwater	2006–2013	185.5; 178.5–191.0	86.8; 81.5–92.0	40.1; 29.4–53.3	Proportionally narrow hips	AIS unpubli data
	National U23 (*n* = 5)	Flatwater	2006–2013	184.3; 181.5–189.0	79.6; 78.0–84.5	43.9; 37.1–57.2		AIS unpubli data
Luge	International (*n* = 5)	Singles & doubles		182.4 ± 4.6	87.7 ± 6.2	71.0; 60.8–86.9		US unpubli data
	National (*n* = 4)	Singles & doubles		177.8	78.1 ± 5.7	78.5; 43.5–113.1		US unpubli data
	Junior (*n* = 8)	Singles & doubles			77.5 ± 7.7	61.3; 43.4–89.5		US unpubli data
Rowing	Olympic	Lightweight (*n* = 50)	15-day period before Olympics	182 ± 0.04	72.5 ± 1.8	44.7; 31.2–62.3a	Proportionally large chest, waist and thigh but smaller hip girth	Acklan al. 200 Kerr et 2007
		Open (*n* = 140)	15-day period	194 ±	94.3 ±	65.3;	Proportionally	Acklan

			before Olympics	0.05	5.9	42.6–97.4a	large chest, waist and thigh but smaller hip girth	al. 200 Kerr et 2007
	National senior	Lightweight (n = 27)	2003 National Championships	180.7 ± 3.9	71.2 ± 1.1	42.7 ± 4.2a	Short sitting height relative to stature and longer limb lengths	Slater e 2005
		Open (n = 76)			92.2; 77.4–108.7	54.0; 36.5–90.4		Tanner Gore 2
	National U23	Lightweight (n = 35)	2003 National Championships	181.6 ± 5.2	70.6 ± 1.9	44.5 ± 7.1a		Slater e 2005
Rugby League	National	Senior (n = 68)		184.1; 173.0–197.0	96.0; 82.0–121.0	53.0; 37.0–101.0		Tanner Gore 2
		Junior (n = 28)		178.0; 167.0–191.5	78.0; 64.0–104.0	67.0; 43.0–107.0		Tanner Gore 2
Rugby Union	International 7s (n = 18)			183 ± 0.06	89.7 ± 7.6	52.2 ± 11.5		Higham al. 201
	International 15s	Forwards (n = 20)	2012–2013	191 ± 7.5	111.7 ± 8.0	73.1 ± 12.2		Zemski al. (in review)
		Backs (n = 17)	2012–2013	182.6 ± 5.6	91.7 ± 5.6	49.0 ± 7.4		Zemski al. (in review)
Sailing	National/international	470 – Helm (n = 9)	2006–2010	173.0; 167.1–182.9	61.8; 53.0–68.5	50.7; 37.9–89.1		Tanner Gore 2
		470 – Crew (n = 7)	2006–2010	186.5; 179.1–194.4	72.1; 66.5–81.7	47.7; 31.4–73.8		
		49er – Helm (n = 6)	2006–2010	178.8; 173.0–185.7	73.9; 69.1–78.9	54.5; 39.8–60.4		
		49er – Crew (n = 6)	2006–2010	184.5; 179.8–187.6	80.1; 75.9–82.7	53.9; 43.0–66.9		
		Laser (n = 8)	2006–2010	182.4; 176.9–187.6	80.5; 76.3–86.0	60.5; 37.7–87.0		
		Finn (n = 3)	2006–2010	186.3; 184.6–189.3	96.2; 93.3–101.2	80.6; 71.9–94.4		
Skeleton	International (n = 4)			179.1 ± 5.4	81.3 ± 6.8	64.0; 44.2–88.0		US unpubli data
Speed	National/international	Short track	2006–2013	175.0;	71.7;	49.8;		AIS

Skating	(*n* = 20)			164.3–188.8	60.0–83.4	33.1–85.4		unpubli data
Swimming	National/international (*n* = 46)				82.1; 66.1–99.6	49.2; 32.8–86.0		Pyne e 2006
Tennis	National academy	16+ y (*n* > 14)		183.6; 174.6–199.2	76.5; 62.8–97.4	46.5; 35.9–67.5		Tanner Gore 2
		15–16 y (*n* > 5)		180.4; 174.0–189.8	71.0; 61.4–82.0	59.5; 52.1–74.0		Tanner Gore 2
Triathlon	National/international	U23/Elite (*n* = 18)	Race season	178.0; 169.3–193.8	66.0; 53.0–78.0	38.0; 27.0–50.0		Tanner Gore 2
	National	Junior (*n* = 28)		178.4 ± 5.5	65.8 ± 6.4	51.1 ± 13.3[a]		Landers al. 201
Volleyball	National junior & senior (*n* = 22)	Indoor		200.4; 187.3–207.5	95.0; 82.0–109.0	57.0; 40.0–72.0		Tanner Gore 2
Volleyball	National junior & senior (*n* = 13)	Beach		196.9; 186.6–214.6	91.6; 84.3–103.2	46.0; 36.7–56.6		AIS unpubli data

Note: Unless otherwise specified, data provided with permission from Australian and/or US team databases between 2000 and 2013, with thanks to the United States Olympic Committee, Australian Institute of Sport and National Sports governing bodies. Only one data point per athlete is used. Unless specified, data was taken at various times of the season.

[a]Sum of 8 skinfolds (includes iliac crest)

3.7 Determining optimal body composition for weight-category sports

Some weight-category sports, where athletes try to gain a competitive advantage by minimising unnecessary body mass loss through reduction in body fat and body water, have implemented policies for minimal body compositions. Body composition measurement is central to estimating a realistic and safe weight-category classification for individual athletes. For example, the National Collegiate Athletic Association (NCAA) has established a process of determining a minimal weight (MW) for college wrestlers in an attempt to stop unhealthy weight management practices. By using surface anthropometry and body density regression equations, a body mass is estimated for a wrestler in a euhydrated state (USG <1.020 specific gravity units), at a minimum of 5% or 12% body fat (males and females respectively). Although this process seems intuitive and well formulated, its flaws have been addressed in the literature (Loenneke et al. 2011).

Body composition analysis allows practitioners, coaches and athletes to determine whether a weight category is truly possible for an athlete. For example, a male rower who has <7%

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body fat based on a DXA scan and weighs 75 kg may not be able to reach a lightweight cut off of 72 kg through body fat loss alone and may require the reduction of lean mass over time to achieve this goal. Practitioners working with athletes in weight-category sports are encouraged to use body composition assessment to identify safe and realistic weights for competition in consultation with the coach, with recognition of the limitations and potential errors of the various techniques available.

3.8 Potential impact of anthropometric measures

Athletes are now routinely subjected to physique assessment at regular intervals throughout their development. Body composition manipulation is a key tool regularly used by sports dietitians. It is not uncommon for athletes to seek nutrition advice to increase lean mass and reduce fat mass: hence body composition needs to be measured and monitored. Recently, the IOC Medical Commission Ad Hoc Research Working Group on Body Composition, Health and Performance had the following comments when assessing body composition as part of a holistic screening process with athletes suspected of eating disorders or disordered eating (Sundgot-Borgen et al. 2013). These comments could be considered useful for all uses of anthropometric analysis in athletes, particularly those in weight-sensitive sports.

- Use valid and reliable methods of body composition analysis (e.g. DXA, skinfold assessment using the ISAK standards; four-compartment models assessing fat, fat-free and lean tissue mass and total body water; estimation of hydration status using urine specific gravity recommended for all anthropometric assessments).
- Careful reflection is required to ascertain whether assessment of body mass and composition may trigger anxiety or disordered eating behaviours and body image issues.
- Body mass index (BMI) is an index of *relative body mass* (ponderosity) and is not a measure of body composition.

Coaches, sport dietitians and sport science practitioners should be aware that measuring body composition or physique characteristics of an athlete, irrespective of the method chosen, can be stressful and provoke anxiety for some individuals. Informing athletes of the rationale for the measurements, the expectations of the measurement process, how the results will be used and how the measurement variables will assist in targeting intervention helps to deflect the potential stress and anxiety experienced by many athletes. Conducting any body composition measurement should be adequately justified and the reporting and discussion of the results undertaken with the sensitivity and care expected for any individual client. In an elite sporting environment, it is not unusual for coaches and other sport science professionals to have access to athlete’s physique measurements; hence, consent is required. Where possible, physique assessment should not be undertaken in public or in front of a group of peers. The results of tests of individual athletes should be shared with coaches or other sports professionals at a time suitable to discuss the implications of the assessment appropriately, not in the presence of other athletes.

The use and reporting of images produced by techniques such as DXA or 3D also needs to be done with caution. Some athletes may be sensitive to the perceived exaggeration of their body shape by the image produced (e.g. exaggeration of the hip when laying flat on a DXA scanning bed, or a confronting 3D avatar that exposes all aspects of the human form). An extra caution and explanation of various aspects of the assessment may be required when carrying out these assessments for the first time, especially in less experienced athletes.

Summary

The assessment of an athlete's physique and body composition is an important tool for the sports science practitioner or sports dietitian. There are many methods to choose from, and it is important for practitioners to understand the limitations of the measures and recognise these when reporting changes over time to athletes and coaches. Consequently, the practitioner is encouraged to employ rigour in their standardisation of measurement protocols, decide upon a suitable frequency of measurement, and manage the expectations of coaches, sports scientists and athletes accordingly.

Practice tips

ALISA NANA, GREG SHAW AND ELIZABETH BROAD

VALIDITY VERSUS RELIABILITY VERSUS PRACTICALITY OF PHYSIQUE MEASURES

- When choosing a method to measure physique characteristics and/or body composition of an athlete and interpreting the measurements, practitioners need to understand the validity, reliability (repeatability) and practicality of the measurements as well as the type of errors involved with its use. Simply stated:
  - *Validity*—does the method measure what it says it's measuring?
  - *Reliability*—does the method measure the same thing over time?
  - *Practicality*—is the method easy to use and accessible?
- An ideal measurement instrument or technique should have good reliability and validity but this is not always possible with physique and body composition measurements, except for body mass and stature (using a high-quality and regularly calibrated scale and stadiometer). Reliability of the measure is more important than validity when measuring and monitoring other body composition changes. Therefore, strategies to minimise measurement errors are important.
- DXA is a valid and reliable measure of body composition when strict protocols are adhered to. The protocols require that the subject is fasted, preferably hydrated and has not trained (i.e. prior to morning training). From a practical perspective, adhering to these protocols may be difficult if the device is not readily accessible to the athlete. Also, the scanning beds on most DXA machines are likely to be too small for very tall and/or broad athletes so the measure of body composition of the whole body will not be valid without applying a modified protocol. For athletes who fall outside the area of the scanning bed, summation of partial scans is required. This takes longer than a normal scan and requires a meticulous scanning protocol using a technician experienced in body composition scans. Partial scanning also increases the duration of each assessment

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and hence risk of radiation exposure (Nana et al. 2012b).

- Skinfolds are not a valid measure of absolute fat mass but have good reliability and are highly practical as a measure of changes in fat mass when taken by an ISAK-accredited anthropometrist. Estimating total fat mass and percentage body fat from skinfold measures using predictive equations do not provide accurate information on the absolute amount of fat mass.

UNDERSTANDING AND MINIMISING MEASUREMENT ERROR WHEN MEASURING PHYSIQUE

- Variability or error in physique and body composition measurements come from two main sources: technical (machine and technician) and biological (day-to-day/random) variations. An example of technical variation is the inherent *noise* of the measurement instrument. When undertaking skinfold measures, for example, even a minor difference in the placement site for the calliper between measures is an example of technician error or variation. A damaged or old calliper may be faulty and represents a machine error. Biological factors that contribute to error in anthropometric measures include hydration status, stage of menstrual cycle, diurnal variation, physical activity, growth, food intake, and supplements that alter intracellular content such as creatine. Some of these variables can be controlled or minimised; others are residual (e.g. growth, diurnal variation) and affect the validity of the measure.
- The magnitude of the technical error (or variability of the measure) of any measurement tool—termed *the technical error of the measurement (TEM)*—can be calculated by undertaking a short-term reliability study to determine the variation in repeated measurements on one or several subjects (e.g. skinfolds). Ideally the instrument/machine used for body composition measurement needs to be standardised when used by a new technician (see Table 3.4). In summary:
  - The TEM (of the repeated measures from instrument/machine or technician) is converted to the coefficient of variation (CV), which expresses the standard deviation (SD) of the repeated measures as a percentage of the mean measure or score (i.e. SD/mean × 100). The TEM can then be used to determine the confidence limits around the measure, which is essential for interpreting whether any measured change in skinfold, for example, is a real change or attributed to technical error.
  - Hence, the TEM and CV is a measure of reliability of within-subject variation of repeated measures.
  - An ISAK-accredited anthropometrist calculates his/her TEM when undertaking an ISAK course and at re-accreditation (see www.isakonline.com/accreditation_scheme). Inter-tester TEM targets for each ISAK level must be met to meet accreditation standards.
  - When using DXA to measure body composition, the TEM and CV for each machine/technician can be calculated by scanning a group of targeted subjects at least twice back-to-back, with repositioning of subjects in between (i.e. subjects stand up then lie back down for the second scan).
  - Further information and methods used to calculate TEM, CV and confidence limits are found at www.sportsci.org/resource/stats/index.html and Norton and Olds (1996). The URL provides a template that computes these statistics from raw data input using an Excel spreadsheet (see reliability spreadsheet on the site).
  - The TEM should always be reported on body composition measures of athletes and in scientific journals.

TABLE 3.4 Summary of standardisation protocols of popular body composition assessment techniques

Surface anthropometry	DXA	BIA
• Equipment is calibrated. • Subjects wear appropriate clothing, which usually consists of underwear and, for females, preferably a racer back crop top. • Measurements can be taken any time of day. Measures should be taken at the same time of day (if tracking changes); preferably before exercise (sweaty skin), any heavy weights session (may alter girth measurements), and physical therapy session (oily skin).	• DXA machine is calibrated according to the manufacturer's guidelines. • Subjects need to be fasted, rested and euhydrated. • Subjects are positioned in a standardised manner on the scanning bed with regional composition being analysed by one experienced technician.	• BIA is calibrated according to the manufacturer's guidelines. • Subjects are fasted/no alcohol/no exercise for >8 hours. • For longitudinal tracking, measurements should

• Technique needs to be standardised using ISAK guidelines; the measurer needs to report his/her TEM with the results (Marfell-Jones & Stewart 2012). (Small variations, by as little as 1 cm away for the actual ISAK skinfold site, add significant error to the measurement (Hume & Marfell-Jones 2008).)	• For longitudinal tracking, measurements can only be compared when taken on the same machine and ideally assessed and analysed by the same technician. • DXA should be measured in conjunction with surface anthropometry. This enhances the understanding of the relationship between the two assessments and may help to explain incongruent results. • For further details on standardisation protocols, refer to Nana et al. (2014).	be undertaken at the same time of day. • Subjects supine for 5–10 minutes prior to measurement. • More detailed research with relation to standardisation protocols for the newer multi-frequency devices and stand and hold devices is required. • For further details, refer to Kyle et al. (2004b).

BIA = bioelectrical impedance analysis, DXA = dual-energy x-ray absorptiometry

Note: Surface anthropometry = body mass, girths, skinfolds (ISAK sites), bone breadths and lengths, including height

WORTHWHILE CHANGE IN ANTHROPOMETRIC MEASURES

- The interpretation of differences or changes in estimates of body composition needs to consider the smallest *worthwhile change* that has clinical or practical relevance to an athlete (note that this change may not be statistically significant, just as a statistically significant change may not be worthwhile). This change is defined as the smallest worthwhile change that can influence an outcome of interest, such as performance, strength, metabolic function or adherence to a weight category.
- Coaches (and athletes) often request specific body mass, lean mass or skinfold targets when referring individual athletes for dietary advice or anthropometric assessment. In some instances, these targets are unrealistic for that individual and may have no *worthwhile* or even measurable effect on improving performance (and may even reduce performance).
- There are some measures in which a large change or difference relative to the TEM is required before a measurable outcome is detectable. For example, a relatively large change/difference in leg lean mass might be required before changes/differences in strength can be detected or a large change in trunk fat mass may be needed before a difference in the power needed to cycle uphill is evident.
- Alternatively, some body composition measures may exist where a small change/difference relative to the TEM may have an effect (e.g. a very small difference in body mass may result in failure to meet a competition weigh-in for an athlete in a weight division sport).

STANDARDISATION PROTOCOLS

To minimise measurement error, each assessment tool should follow a standardisation protocol as described above. Examples of standardisation protocols of popular anthropometric methods are shown in Table 3.4.

INTERPRETATION OF CHANGES IN BODY COMPOSITION

Surface anthropometry

- Measurement of skinfolds, girths, bone lengths and breadths is robust and not influenced by hydration status (Norton et al. 2000). In contrast, body mass measurement can vary (e.g. time of day, stage of menstrual cycle, dehydration/overhydration), which can confound interpretations of change. Where possible, these factors need to be controlled according to the protocols outlined in Table 3.4. Ideally, body mass measurements should be taken at the same time of day in conjunction with surface anthropometry to address and improve reliability of the measure. The direction of changes in body mass and skinfolds and their possible interpretation is summarised in Table 3.5.

- Comparison of sum of skinfold measures of an individual athlete with data from other elite athletes (see Tables 3.2 and 3.3) should be done with caution. Data from groups of elite athletes should not be used to set definitive targets for an individual, particularly a developing and adolescent athlete. Use these data as a guideline only.
- For those athletes who exceed the skinfold range or are in the upper level for the sum of skinfolds (compared with Tables 3.2 and 3.3), providing assurance that body composition is only one factor contributing to performance helps diminish concern about body image. Most athletes are sensitive about body composition. Hence, sports dietitians should instigate dietary interventions that encourage a slow reduction in fat mass (or skinfolds) and monitor the changes in conjunction with other performance and health parameters.
- The rate of weight loss (and fat mass loss) in relation to skinfold sum shows large variation in individual athletes. As a general guideline, at higher skinfolds (>80 mm for the sum of seven sites), a 1 kg weight loss equates to around 10 mm reduction in skinfold sum. At a lower skinfold sum (<80 mm for sum of seven sites), a 1 kg weight loss is much less than the 10 mm change observed at the higher level (Kerr & Ackland 2010). However, some athletes have a high proportion of intra-abdominal to subcutaneous body fat, where waist girth and body mass changes may provide a better estimate of body fat loss than skinfolds alone.
- While the above benchmark can be used as a crude guide for determining whether a specific weight target is possible, DXA is a better predictor of total fat mass and is the preferred method for assessing the suitability of an athlete for a specific weight category in a weight-category sport.

TABLE 3.5 Interpretation of changes in skinfolds and body mass

Changes			Interpretation		
Body mass		**Skinfolds**	**Muscle mass**		**Body fat**
Increase	and	Stable	Gain	and	No change
Decrease	and	Stable	Loss	and	No change
Stable	and	Increase	Loss	and	Gain
Stable	and	Decrease	Gain	and	Loss
Increase	and	Increase	Potential gain	and	Gain
Increase	and	Decrease	Gain	and	Loss
Decrease	and	Increase	Loss	and	Gain
Decrease	and	Decrease	Potential loss	and	Loss

Source: Tanner & Gore 2013

DXA

- When interpreting changes in body composition estimates from DXA, the TEM of the particular DXA machine used is required to determine whether the apparent change is valid or confounded by error (see measurement error above). For example, the lowest CV is reported for bone mineral content (~0.7–1.2%) and lean mass (~0.8–1.7%); the highest for total fat mass and estimates of regional body composition (~1.5–5%) (Nana et al. 2012a).
- Errors from biological variation specifically affect the accuracy of lean mass values from DXA analysis. Although standardisation protocols minimise this type of error, the impact of muscle glycogen content, menstrual cycle and hydration status is still unclear. Controlling for these factors may aid in interpreting changes in lean tissue and improve accuracy.

Frequency of anthropometric measurement

- The frequency of taking anthropometric measures depends on the magnitude of any real change measured in an individual athlete. Changes in body composition are perceived as *real* when they exceed the *noise* or CV of the measure. For example, as a rule of thumb, a change of 500 g in lean mass measured by DXA under standardised scanning protocol is regarded as a real change, because the CV of total lean mass calculated in a specific DXA machine is ~250 g (Nana et al. 2012a).
- Therefore, an athlete undertaking a dietary intervention with the goal of lean mass gain of ~250 g/wk would not undertake a second DXA scan within 2 weeks. If a worthwhile change was considered to be a total gain of >1 kg lean mass, a second DXA scan would not be undertaken in anything under 4 weeks.
- Frequency of skinfold assessment should be based on whether body composition change is desired or the purpose is monitoring of progression with growth or a training phase. The practitioner should decide what a reasonable frequency of measurement is based on current skinfold sum, the desired rate of body composition change and the practitioner's TEM. In most circumstances, it is unlikely that retesting within a 2-week period will be worthwhile.
- Generally, body composition assessment is recommended at regular intervals throughout a training year (e.g. before and after a high-intensity strength and conditioning or endurance phase of training, with sudden growth spurt, with substantial weight loss or gain, before and after competition and to coincide with performance testing).
- For devices that present a potential safety risk (e.g. radiation exposure) the frequency of measurement should be guided by minimising exposure of athletes over their athletic life. The increased overall radiation exposure per year (particularly in longitudinal monitoring over time) could be substantial for athletes who are already exposed to ionising radiation from other diagnostic imaging techniques (Cross et al. 2003; Orchard et al. 2005), frequent air travel (cosmic radiation) or those who are too tall and/or broad for the DXA scanner (multiple partial scanning).
- An important role of the sports dietitian is to educate athletes, coaches and other sports service staff about the benefits and potential drawbacks of measuring physique characteristic of athletes and to consider the reasons for measurement when deciding upon the method of measurement and its frequency.

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CHAPTER FOUR

Protein

Daniel Moore, Stuart Phillips and Gary Slater

4.1 Introduction

Proteins are composed of 20 different amino acids that are joined together via peptide bonds between their carboxyl and amino terminals to form polypeptide strands of varying amino acid sequences and lengths. The specific sequence of the amino acids is determined by messenger RNA templates and ultimately dictates the final shape and function of the protein. Proteins have a wide variety of structural and regulatory functions that are crucial to normal cellular metabolism and function. In the case of skeletal muscle, this includes, for example:

- contractile myofibrillar proteins (e.g. actin and myosin), which convert chemical energy to mechanical work to allow for muscular contractions
- structural proteins (e.g. collagen), which provide the physical scaffolding for the muscle cell to transit the forces to the bone levers
- energetic mitochondrial proteins, which collectively generate the ATP energy that is vital for essentially all cellular function
- regulatory enzymes, which catalyse the metabolism of a plethora of chemical reactions essential for cellular function (e.g. ATP hydrolysis and synthesis).

While it is essential for all proteins within the body (e.g. immune system, cardiovascular system) to be present in the proper proportion and be of high quality and function for an individual to excel at their chosen sport, the proteins within skeletal muscle are of primary importance to the athlete, given that the composition of skeletal muscle typically determines athletic success (e.g. strength and/or more fatigue resistance).

4.2 Protein turnover

Proteins within the body are incredibly dynamic as they are continually being broken down into their constituent amino acids and resynthesised from amino acids within the free amino acid pool. Proteins that are damaged (e.g. through oxidation, nitrosylation and/or mechanical forces) are targeted for degradation by the ubiquitin-proteosome system. These amino acids liberated from the breakdown of skeletal proteins can be:

- reutilised as substrates for new muscle protein synthesis (and this happens to an underappreciated extent)
- exported from the cell to sustain other vital body functions (e.g. synthesis of circulating proteins or for gluconeogenesis)
- transaminated and utilised as a source of fuel (e.g. the branched chain amino acids leucine, isoleucine, valine) or to replete the tricarboxylic acid cycle intermediates within the mitochondria of the muscle.

Alternatively, amino acids within the free pool can be combined with charged tRNA molecules and incorporated into new polypeptide chains through the translation of mRNA templates to yield new proteins, a process that is regulated primarily by the mammalian target of rapamycin (mTORC1) signalling pathway (Walker et al. 2011).

Beyond childhood growth, sustaining muscle is a matter of balancing the rates of protein synthesis and breakdown. If an athlete has the goal of gaining muscle, however, a shift towards a net positive muscle protein balance needs to occur (Breen et al. 2012; Phillips 2012). Ultimately, the algebraic difference between protein synthesis and protein breakdown determines whether a muscle (or specific pool of muscle proteins) is in a *positive* (i.e. net new proteins made), *neutral* (i.e. no net change in the protein pool) or *negative* (i.e. net proteins lost) balance. Much of what is discussed in this chapter relates to the net gain of new muscle proteins to achieve a gain in mitochondrial mass (primarily as a result of endurance training) or myofibrillar proteins (primarily as a result of resistance training). It should be appreciated that these synthetic processes are not mutually exclusive and can occur to a certain extent along a continuum of training modalities (Wilkinson et al. 2008).

Muscle protein turnover has been measured in humans through the use of stable isotope-labelled amino acids, which are amino acids that are 'heavier' than normal due to an increased number of neutrons than their naturally occurring parent isotope (e.g. ^{13}C versus ^{12}C, ^{15}N versus ^{14}N, or ^{2}H versus ^{1}H). The labelled amino acids are typically infused intravenously or orally ingested and measured using sensitive mass spectrometers in different biofluids (e.g. blood, muscle or other proteins) to model their metabolism. The most common approaches involve measuring the concentration and enrichment in blood and muscle of an exercised limb (e.g. leg) to indirectly assess changes in muscle protein synthesis and, less commonly, muscle protein breakdown or to measure the presence of the isotope in the muscle protein fraction of interest (e.g. myofibrillar, mitochondrial, mixed muscle) to directly determine the relative rate of muscle protein synthesis. Direct measurement of muscle protein synthesis has the distinct advantage over whole-body estimates of protein balance (e.g. nitrogen balance) as it allows for the assessment of an intervention on the target tissue of interest, in most cases skeletal muscle. Therefore, this chapter will evaluate how exercise and nutrition alter skeletal muscle protein turnover to facilitate skeletal muscle remodelling and adaptation.

4.3 Protein turnover with exercise

Resistance training unequivocally enhances muscle strength and hypertrophy and increases lean body mass, processes that require the remodelling of skeletal muscle. As such, resistance exercise enhances the rates of skeletal muscle protein turnover (synthesis and degradation) in the hours to days after a training session (Biolo et al. 1995b; Burd et al. 2010; Phillips et al. 1997). This increase in protein turnover ultimately functions to break down older or potentially damaged proteins and to synthesise new functional skeletal muscle proteins to replace the older proteins. This protein remodelling occurs for variety of proteins, including, for example: the collagen structural support proteins (Moore et al. 2005), sarcoplasmic proteins (including intracellular enzymes and sarcoplasmic and endoplasmic reticuli) (Burd et al. 2010), mitochondrial proteins (Wilkinson et al. 2008), and especially the force-generating myofibrillar proteins (Kim et al. 2005; Wilkinson et al. 2008; Burd et al. 2010), which constitute ~60% of all muscle proteins. Importantly, these enhanced rates of protein synthesis after exercise are primarily mediated via local mechanisms *within the muscle*, since circulating physiological concentrations of 'anabolic' hormones such as growth hormone, insulin-like growth factor-1, and testosterone have a negligible effect on muscle remodelling acutely (West et al. 2009) or muscle growth chronically (West et al. 2010; West et al. 2012), which is in contrast with the anabolic effects of pharmacological administration of some of these hormones (Bhasin et al. 1996).

Increased rates of post-exercise protein turnover, especially of the energy producing mitochondrial proteins (Wilkinson et al. 2008), are also a feature of the recovery from endurance training (Harber et al. 2010); however, similar to resistance exercise, there is little change in whole body, as opposed to muscle, protein turnover after exercise (Tarnopolsky et al. 1991; Bowtell et al. 1998), which highlights the need to study the effect of an exercise stimulus on the target tissue of interest, specifically skeletal muscle. While muscle protein synthesis is elevated after both resistance and endurance exercise, in the absence of protein ingestion there is a concomitant stimulation of muscle protein breakdown presumably to provide the amino acid substrates for the synthetic pathways (Carraro et al. 1990; Phillips et al. 1997). Thus, without protein consumption, muscle net protein balance, while improved relative to a rested condition, remains negative with no net new proteins synthesised. Therefore, one of the most important aspects of recovery from a muscle remodelling perspective is the consumption of protein to result in aminoacidaemia and enhance protein turnover and net protein balance.

4.4 Amino acid metabolism

Carbohydrates (both endogenous stores and exogenously consumed) represent the primary metabolic fuel during aerobic exercise with a significant contribution also coming from lipid with lower exercise intensities or increased exercise duration. As such, these macronutrients often take centre stage for the recovery from exercise (and subsequent performance) for the endurance athlete. Nonetheless, amino acid oxidation can provide ~2–6% of total energy during aerobic exercise (Tarnopolsky 2004). This enhanced amino acid oxidation arises to a significant extent from the suppression of muscle protein synthesis and small elevation of muscle protein breakdown during exercise liberating amino acids (Howarth et al. 2010) that can subsequently be utilised as substrates for gluconeogenesis, oxidised within the muscle mitochondria as a direct source of fuel (this is the main fate of the branched chain amino acids and especially leucine), and/or replenish the tricarboxylic cycle intermediates via anapleurosis (Gibala 2001). The oxidation of endogenous amino acids can be enhanced by a variety of factors such as greater exercise intensity and/or duration (Lamont et al. 2001), low muscle glycogen availability (Howarth et al. 2010), a high protein (i.e. ≥1.8 g/kg/d) diet (Bowtell et al. 1998) and male sex, with oxidation being higher in males (Phillips et al. 1993). In contrast, leucine oxidation can be reduced with training (McKenzie et al. 2000). Additionally, the common practice of consuming exogenous carbohydrate during exercise may attenuate amino acid oxidation somewhat but rates still remain significantly elevated above rest (Bowtell et al. 2000). In fact, leucine, an amino acid with key regulatory roles in protein metabolism (see below), can be oxidised at a rate of ~8 mg/kg/h in athletes performing just moderate (~60%) intensity exercise, resulting in total body oxidative losses of ~1.2 g over 2 hours (Bowtell et al. 1998). Assuming muscle protein is ~9% leucine, this would be the equivalent of ~13 g of protein that would either not be synthesised or require direct catabolism with only moderate-intensity exercise, a factor that would likely be increased further with exercise of longer duration and/or greater intensity. Therefore, aerobic exercise can result in a significant use of body amino acids as a source of fuel, which is commonly cited as the underlying basis for increased protein requirements with endurance training (Tarnopolsky 2004). Given that these oxidised amino acids (and especially leucine) are essentially lost from the body and unable to participate in normal protein metabolism (such as increased muscle protein synthesis in recovery), they must ostensibly be replaced through dietary sources.

4.5 Nutritional regulation of protein turnover with exercise

Intravenous infusion of insulin enhances post-exercise rates of muscle protein synthesis and breakdown (Biolo et al. 1995a), which led to the belief that large amounts of carbohydrates (resulting in hyperinsulinaemia) were required to maximise post-exercise muscle anabolism. This contention was supported by earlier work suggesting that carbohydrate, either in isolation (Borsheim et al. 2004) or with relatively low amounts of exogenous amino acids (Borsheim et al. 2002; Miller et al. 2003), enhanced net protein balance in the post-exercise recovery period primarily through an insulin-induced suppression of protein breakdown and not increased protein synthesis. However, recent work has revealed that when protein or essential amino acid intake is high, additional carbohydrate has little effect on the post-exercise stimulation of muscle protein synthesis or enhancement of net protein balance (Koopman et al. 2007; Glynn et al. 2010; Staples et al. 2011). Therefore, while ingestion of additional energy (most likely in the form of carbohydrates) would be essential for complete muscle recovery after exercise (e.g. maximising glycogen resynthesis), the enhancement of muscle protein synthesis is primarily determined by the essential amino acid content of the ingested protein source.

It was first observed by Biolo and colleagues (1997) that the provision of exogenous amino acids (in the form of an intravenous infusion) markedly enhanced muscle net protein balance after resistance exercise through the stimulation of protein synthesis with little effect on protein breakdown. Subsequent seminal studies from Robert Wolfe's research group demonstrated that oral crystalline amino acids, and especially the essential amino acids, had a similar effect on enhancing muscle protein turnover and net protein balance (Tipton et al. 1999; Rasmussen et al. 2000). For the most part, this enhancement in net protein balance occurs primarily through the stimulation of muscle protein synthesis, which varies 3–4-fold across the course of a day, rather than the suppression of muscle protein breakdown (Rasmussen & Phillips 2003). This observation highlights that muscle protein synthesis is the more regulated and 'flux-determining' variable in the remodelling of skeletal muscle with exercise. Nevertheless, the increase in net protein balance over 3 hours after exercise with essential amino acid ingestion has been shown to be additive to the 24-hour response (Tipton et al. 2003) demonstrating this acute (i.e. 3–4 h) augmentation of net protein balance is sustained over an entire day, a condition that would be a prerequisite for chronic gains in muscle mass (i.e. hypertrophy) with resistance training. Additional support for the importance of enhanced nutrient-induced muscle protein remodelling comes from the finding that the nutritionally mediated post-exercise increase in muscle protein synthesis has been shown to qualitatively predict chronic increases in muscle mass and fibre growth (Hartman et al. 2007; Wilkinson et al. 2007). Therefore, the following sections outline the key dietary factors, as they pertain to protein and amino acid ingestion, which augment rates of muscle protein turnover with exercise as a means to facilitate the remodelling, and subsequent recovery and adaptation, of skeletal muscle.

4.6 Protein dose

Providing the amino acid building blocks through the ingestion of dietary protein and/or crystalline amino acids is well established to increase rates of muscle protein synthesis to induce a net positive protein balance (Burd et al. 2009). This anabolic effect of dietary protein can be observed with the equivalent of ~10 g of protein, both after resistance (Borsheim et al. 2002; Tang et al. 2007) and aerobic-based exercise (Levenhagen et al. 2002), suggesting even small, snack-like meals would help facilitate skeletal muscle remodelling. Nevertheless, the goal of any athlete aiming to enhance the recovery from exercise should be to maximally stimulate muscle protein synthesis. To this end, recent dose–response studies have revealed that 20 g of high-quality protein (i.e. egg or whey) are sufficient to maximally stimulate muscle protein synthesis after resistance exercise in average weight (~80 kg) young men (Moore et al. 2009; Witard et al. 2014). Moreover, doubling the protein intake to 40 g has no further effect on muscle protein synthesis but rather results in dietary amino acids being directed towards oxidative catabolism (Moore et al. 2009) and ureagenesis (Witard et al. 2014), which is the normal response to the consumption of quantities of protein over and above that which can be used for protein synthesis. Therefore, the 'optimal' protein intake, defined as one that maximally stimulates muscle protein synthesis yet minimally increases amino acid oxidation, is currently at ~20–25 g in average weight athletes (or the equivalent of ~0.25 g/kg), a level that is not markedly different from what is required in rested skeletal muscle (Cuthbertson et al. 2005).

While no study has currently evaluated the ingested protein dose–response relationship with muscle protein synthesis after aerobic exercise, it has been shown that ~20 g of protein (or the equivalent of ~0.25 g/kg) can elicit a robust stimulation of muscle protein synthesis after aerobic exercise in trained (Breen et al. 2011) and recreationally active individuals (Lunn et al. 2012). Underscoring this requirement for protein after endurance exercise, lower protein intakes (~10–16 g) can increase protein synthesis and net protein balance at the muscle level despite whole-body protein balance remaining negative (Lunn et al. 2012) or in neutral balance (Levenhagen et al. 2002). This could suggest that dietary amino acids provided post-exercise are preferentially utilised by the exercised muscle to facilitate the repair and remodelling, perhaps at the relative expense of whole-body (i.e. non-muscle) protein turnover. Therefore, based on currently available data, acute protein requirements to enhance muscle remodelling and recovery appear to be similar after both aerobic and resistance exercise.

4.7 Protein timing

The timing of ingested food to support an athlete's training and recovery can generally be viewed to occur in three eating occasions relative to exercise: before, during and/or after. While post-exercise is arguably that best time at which to consume food, rates of protein turnover (especially within the muscle) may be enhanced for up to 24–48 hours after a single bout of exercise (Phillips et al. 1997; Kim et al. 2005; Miller et al. 2005). This would mean that for the athlete training repeatedly and on consecutive days, nutrition consumed outside of the training bouts would arguably fall into the 'after' exercise category, although the line between 'after' and 'before' becomes blurred somewhat with multiple training sessions per day. Nevertheless, this section will discuss the current understanding of how best to consume dietary protein during these three feeding opportunities.

4.7.1 Before exercise

It was first observed that muscle net protein balance was greater over 3 hours when crystalline amino acids were consumed before as compared to immediately after a bout of resistance exercise, an effect that was presumably mediated via a greater delivery of amino acids to the working muscle as a result of the exercise-induced hyperaemia (Tipton et al. 2001). However, subsequent studies have failed to replicate these initial findings: no clear advantage for enhancing net muscle protein balance or protein synthesis has been observed with pre- as compared to post-exercise protein or essential amino acid ingestion (Tipton et al. 2007; Fujita et al. 2009). If the exercise bout is of relatively short duration (e.g. ≤60 min) then it is possible that pre-exercise feeding may provide a source of amino acids to facilitate immediate post-exercise muscle protein remodelling due to the time required to digest the protein and for its constituent amino acids to appear in the circulation; this could be likened to 'priming' the amino acid pool so that there are substrates to support the exercise-induced stimulation of muscle protein synthesis, as has been shown in previous studies (Tipton et al. 2001; Tipton et al. 2007; Coffey et al. 2011; Burke et al. 2012a). Nevertheless, most laboratory-based studies often employ strict dietary control in the hours before investigation and therefore study individuals in an overnight fasted state, one that is unlikely to reflect how athletes generally train.

4.7.2 During exercise

As highlighted previously, aerobic exercise is associated with an enhanced oxidative disposal of amino acids (especially the branched-chain amino acids), the source of which is primarily a suppression of protein synthesis and possibly a small increase in the breakdown of muscle tissue as suggested by the net efflux of amino acids from the working muscle (Howarth et al. 2010). As such, aerobic exercise generates a net catabolic environment within the muscle which is generally reflected at the whole-body level as a negative protein balance (Howarth et al. 2010). For athletes who train for relatively long bouts of exercise (e.g. ≥1.5 h) and/or perform multiple training bouts per day, the inclusion of protein in their workout nutrition may help limit the endogenous use of amino acids as a source of fuel to improve whole-body protein balance during exercise (Koopman et al. 2004; Beelen et al. 2011). However, whether protein intake may also improve muscle protein balance and/or muscle protein remodelling during exercise is currently a point of contention, with separate studies reporting either an enhancement (Beelen et al. 2011) or no change (Hulston et al. 2011) in protein synthesis during the exercise bout itself. Nevertheless, the endogenous protein sparing effect of protein ingestion during aerobic exercise may potentially improve an athlete's nutritional profile so that any post-exercise protein ingestion can more easily be utilised to facilitate muscle protein remodelling.

In contrast to the high constant ATP utilisation of working muscle during aerobic exercise, an environment that is generally not conducive to supporting the energetically expensive process of protein turnover (Rose et al. 2009), resistance exercise is typically punctuated by rest periods of varying duration during individual training sets. These brief 'rest' (i.e. recovery) periods may represent an opportunity to initiate skeletal muscle remodelling under the right nutritional conditions, as suggested by studies reporting elevated rates of muscle protein synthesis during a relatively prolonged (i.e. 2-h) resistance training session with protein ingestion throughout the exercise bout (Beelen et al. 2008). However, for the athlete who trains for a shorter duration and/or perhaps at a higher metabolic intensity (e.g. circuit training) with shorter inter-set rest intervals, protein ingestion during exercise is of uncertain physiological value given that muscle protein remodelling predominantly occurs in the hours to days after an exercise bout rather than the minutes between exercise sets during the training bout.

4.7.3 After exercise

It is without question that protein ingestion after exercise enhances muscle protein synthesis and net protein balance after all forms of exercise and should be a viewed as a core and critical component of 'recovery' nutrition for all athletes (Burd et al. 2009; Phillips & van Loon 2011). While it would be prudent for athletes to consume a source of protein as soon after exercise as possible to rapidly initiate the muscle remodelling process, protein consumed even up to 24 hours after an exercise bout would contribute to the enhanced remodelling of skeletal muscle (Burd et al. 2011) as the 'sensitising' effect of exercise is quite persistent. In fact, a recent meta-analysis suggests that for athletes aiming to enhance muscle growth and strength with resistance training there is a relatively minimal benefit to consuming protein around an exercise bout (e.g. within ~2 h before and/or after) in the ability to augment training adaptations (Schoenfeld et al. 2013). Therefore, aside from the immediate pre-/post-exercise feeding period, protein ingestion outside of this potential 'anabolic window of opportunity' may have a greater bearing on the ability to augment rates of skeletal muscle remodelling over a prolonged 12–24-hour recovery period, as will be discussed below.

It has recently been demonstrated that the pattern of protein intake (and not merely the absolute amount) can influence how efficiently muscle protein remodelling is supported during an extended (i.e. up to 12-h) recovery period. For example, the repeated ingestion of 20 g of protein every 3 hours over a 12-hour post-exercise recovery period supported greater rates of myofibrillar protein synthesis and induced a more positive whole-body protein balance after a bout of resistance exercise than the identical amount of protein ingested as 10 g feedings every 1.5 hours or 40 g feedings every 6 hours (Areta et al. 2013; Moore et al. 2012). While similar studies have not been performed after aerobic exercise, it is likely that a similar feeding pattern (both of protein amount and frequency) would also support the greatest rates of muscle protein synthesis after this type of training given that 20–25 g of protein (0.25–0.30 g protein/kg/meal) saturates the protein synthetic capacity of the muscle in multiple studies at rest and after exercise (Cuthbertson et al. 2005; Moore et al. 2009; Witard et al. 2014). Nevertheless, western diets are typically characterised by an unbalanced distribution of protein during the day with lowest amounts being consumed in the morning and greatest amounts in the evening (de Castro et al. 1997). Such a 'skewed' distribution of protein may preclude the ability to support maximal rates of muscle protein synthesis over a 24-hour period (Mamerow et al. 2014) and underscores the importance of athletes paying attention to how they consume their daily protein intake on a meal-to-meal basis in order to maximise muscle remodelling and recovery. In fact, many elite athletes often periodise their meals throughout the day in ~3–4-hour intervals as a means to ensure sufficient energy intake and therefore may only have to pay closer attention to the protein content of each meal (Hawley et al. 1997). Finally, athletes aiming to identify alternative ‘feeding opportunities’ or those who train in the evening (due to scheduling and/or preference) may also consider pre-bedtime protein consumption as a means to support enhanced rates of muscle protein synthesis during overnight recovery (Res et al. 2012). Therefore, given the similar enhancements of muscle protein synthesis and net protein balance with protein ingestion after endurance and resistance exercise, athletes aiming to support the greatest rates of muscle remodelling would benefit from targeting protein consumption immediately after exercise and every 3–4 hours thereafter.

4.8 Dietary protein sources

Dietary proteins can differ in their amino acid composition and rate of digestion. For example, isolated soy protein and animal-based proteins such as beef, egg, and dairy (e.g. soluble whey proteins, insoluble casein) have relatively high essential amino acid compositions compared to plant-based proteins such as rice and legumes. Alternatively, the digestion rate (i.e. the time required to digest a protein and for its constituent amino acids to appear in the blood) is most rapid with acid-soluble proteins such as whey and soy and is the slowest with micellar casein (which clots in the acidic pH of the stomach). In general, proteins that are enriched in essential amino acids (and especially leucine) and are rapidly digested are able to stimulate a more rapid rise in muscle protein synthesis after exercise (Tang et al. 2009; West et al. 2011). This has led to the suggestion that a certain ‘leucine threshold’ must be reached within the blood or muscle intracellular pool in order to maximally activate muscle protein synthesis (Phillips 2011), although this remains a theoretical concept at the present time with no firm understanding of the absolute concentration required.

Nevertheless, mixed protein sources and whole foods often have markedly different amino acid profiles compared to isolated protein fractions (Burke et al. 2012b) but are still likely to support enhanced rates of muscle protein synthesis and positive net protein balance. As evidence of this, milk (and its constituent whey and casein protein fractions) or even protein blends containing predominantly dairy can enhance post-exercise rates of muscle protein synthesis and net protein balance (Elliot et al. 2006; Wilkinson et al. 2007; Lunn et al. 2012; Reidy et al. 2013) and improve gains in lean mass with resistance training (Hartman et al. 2007; Josse et al. 2010). Therefore, whole foods containing an adequate amount of protein would enhance muscle protein remodelling after exercise (and certainly throughout the remainder of the day), although athletes aiming to initiate a rapid post-exercise increase in muscle protein synthesis would consume a rapidly digested, leucine-rich protein. An important point to make is that many lab-based findings on the effectiveness of how individual isolated protein sources can stimulate muscle protein synthesis (Elliot et al. 2006; Wilkinson et al. 2007; Lunn et al. 2012; Reidy et al. 2013) are not easily transferable to foods, particularly if consumed in meals with other macronutrients, since the patterns of aminoacidaemia (i.e. digestion) are markedly different with different foods and with complete meals (Burke et al. 2012b).

4.9 Effect of training

Exercise training has the effect of refining the remodelling effects of an exercise bout by attenuating the post-exercise increase in mixed muscle protein turnover (Phillips et al. 1999; Kim et al. 2005) but enhancing the synthesis of specific muscle protein fractions, such as a sustained stimulation of myofibrillar protein synthesis with resistance exercise (Kim et al. 2005; Wilkinson et al. 2008) and a preferential remodelling of mitochondrial proteins with aerobic exercise (Wilkinson et al. 2008). However, the ability of dietary protein ingestion to enhance post-exercise muscle protein synthesis has been reported to be heightened in the trained state in the immediate (i.e. 4-h) but not prolonged (i.e. ~28-h) recovery period (Tang et al. 2008); this could suggest that trained, more so than untrained, athletes should prioritise protein ingestion immediately after a training session to capitalise on this enhanced anabolic sensitivity, the duration of which appears to be abbreviated relative to their untrained peers. Nevertheless, this apparently greater post-exercise sensitivity to dietary protein would not affect the dose of protein (i.e. ~20 g) required to maximise muscle protein synthesis as previous dose–response studies have utilised trained populations (Moore et al. 2009; Witard et al. 2014).

4.10 General protein requirements

A critical question is: How much protein might athletes require to function optimally? This is a different question than that posed by dietary guideline committees setting protein intake recommendations, who are looking to provide the lowest level of protein to offset deficiency. As has been pointed out, there is a sharp difference for athletes who are looking to optimise adaptation rather than offset deficiency (Phillips 2012). Thus, the fundamental question of the protein intake required to promote optimal athletic performance, which would encompass a potentially far greater scope than for a non-athletic person, is discussed here.

Requirements for protein for the general population are defined by various agencies but generally appear in the range 0.8–0.9 g/kg/d. In Canada and the USA, the Recommended Dietary Allowance (RDA) is 'the average daily intake level that is sufficient to meet the nutrient requirement of nearly all [98 %] healthy individuals' (Food and Nutrition Board, Institute of Medicine 2002). The Institute of Medicine panel also states that 'no additional dietary protein is suggested for healthy adults undertaking resistance or endurance exercise' (2002). It may be true that the basal 'requirement' for protein, even for the most intensely training athlete, is satisfied by the protein RDA. That is, 0.8 g/kg/d of protein can satisfy the needs for all amino acid-requiring processes and most athletes could likely even achieve nitrogen balance when consuming this intake. However, it is pertinent to ask whether such a state would result, due to an adaptive process, in down-regulation of amino acid-requiring processes and whether this adaptive change would compromise some goal in an athlete. This is not a question that is easy to answer, however, as a number of reviews (Phillips 2004, 2012;

Tarnopolsky 2004) and position stands (Rodriguez et al. 2009) have concluded higher protein intake is required by athletes. Thus, protein 'requirements' is a misnomer when referring to an athlete and a more precise term may be to define an optimal protein intake for an athletic population versus a protein intake to achieve nitrogen balance (i.e. requirement). This sentiment may be particularly true during an energy deficit, when the choice of which macronutrients to consume may be even more critical, at least from an athlete's perspective.

Data from Moore and colleagues (2009) in ~87 kg males indicated that that a dose of egg protein that saturated muscle protein synthesis was 20 g. More recently, Witard and colleagues (2014) confirmed, using whey protein, that the same dose of protein was sufficient to maximally stimulate post-exercise muscle protein synthesis. While many, correctly, point out that humans have the capacity to be able to digest more protein, there is obviously an upper limit to the rate (and capacity) at which amino acids can be used in muscle. In fact, Atherton et al. (2010) have described a 'muscle full' phenomenon following protein ingestion, which shows that muscle (and likely other tissues) simply cannot assimilate amino acids beyond a certain dose. The importantly question, however, is what the protein dose per meal might be on a body-weight basis to allow adjustment for smaller or larger athletes. Estimations based on the data we have at present (Cuthbertson et al. 2005; Moore et al. 2009; Witard et al. 2014) are that a per-meal 'dose' of protein of ~0.25–0.30 g protein/kg/d would optimally stimulate protein synthesis. With this per meal 'dose' in mind one can begin to formulate a protein consumption strategy based around periodic stimulation of protein synthesis, which is in fact what was tested in young men post-exercise (Moore et al. 2012; Areta et al. 2013). In this investigation, a group of young men during recovery from resistance exercise had the largest stimulation of muscle protein synthesis (MPS) and increase in whole-body net protein balance with protein ingestion of 20 g (~0.25 g/kg) every 4 hours versus 10 g (~0.12 g/kg) every 2 hours or 40 g (~0.48 g/kg) every 8 hours (Moore et al. 2012; Areta et al. 2013). While not definitive, these data provide some proof of concept that ~0.25 g protein/kg/meal seems to be optimally effective, at least in stimulating MPS, over a relatively prolonged recovery period.

One important consideration in interpreting the results from acute feeding trials, such as the one described above (Moore et al. 2012; Areta et al. 2013), is that they represent an acute response to feeding an isolated source of protein and the influence of other nutrients and energy balance are unknown. In addition, the quality of protein consumed in most omnivorous diets may also need to be considered given that ~34% may be lower quality plant-based protein sources (Lin et al. 2010), which may subsequently require a marginally greater meal-protein intake to maximally stimulate muscle protein synthesis (Tang et al. 2009; Reidy et al. 2013). Finally, the long-term translation of acute findings to chronic phenotypic changes requires further study and subsequently caution in interpretation. Nonetheless, if we accept that a dose of 0.25 g protein/kg/meal is a reasonable estimate and means of defining an optimal protein intake, this could allow the calculation of daily recommendation for an athlete looking for optimal protein intake. Using this approach and including four discrete eating occasions per day featuring high-quality protein sources as well as one pre-sleep meal that is twice as large (i.e. 0.5 g/kg/meal protein) to offset catabolic losses during sleeping (Res et al. 2012), a 90-kg athlete would be consuming four meals of ~22.5 g of protein (i.e. 0.25 g/kg) plus one meal of 45 g of protein (i.e. 0.5 g/kg) for ~135 g of high biological value protein daily or ~1.5 g/kg/d, which could be ~25–30% greater if both low and high biological value proteins are included in the athlete's diet. One could argue that more eating occasions could be required and be more optimal, but it appears that such a feeding pattern would result in a relatively sustained daily hyperaminoacidaemia, which has been shown to result in a refractory response of muscle protein synthesis (Bohe et al. 2001) at rest and a suboptimal protein synthesis after exercise (Areta et al. 2013).

4.11 Negative energy balance

Muscle protein turnover is energetically 'expensive' and, as such, is generally depressed during periods of energy restriction (Pasiakos et al. 2010; Areta et al. 2014); this can be a concern for athletes who wish to maintain muscle mass and performance during voluntary weight loss. While resistance exercise helps maintain muscle mass during negative energy balance, it has recently been demonstrated that post-exercise protein ingestion is essential to enhance muscle protein synthesis during the recovery from an acute resistance training bout (Areta et al. 2014). Currently, it is unclear if the dose of protein required to maximally stimulate muscle protein synthesis after exercise while in a negative energy balance is greater than what is required during energy balance (i.e. ~0.25g/kg) (Moore et al. 2009; Witard et al. 2014). However, it has been demonstrated that 30 g of protein (~0.36–0.42 g/kg) elicits a greater dose-dependent stimulation of muscle protein synthesis compared to 15 g of protein (~0.18–0.21g/kg) (Areta et al. 2014), which suggests that the acute protein intake while in negative energy balance to enhance muscle protein synthesis is slightly greater. Although these preliminary acute dose–response findings require further confirmation, they are not without precedence as protein intakes ≥1.6 g/kg/d day support a greater retention of lean body mass during diet and exercise-induced negative energy balance (Mettler et al. 2010; Pasiakos et al. 2013). A recent systematic review (albeit based on a low number of studies) suggested that a protein target of ≥2.3 g/kg lean body mass/d (or the equivalent of ~1.9 g/kg/d in an average athlete with 15% body fat) should be achieved to help retain muscle mass during weight loss in athletes (Helms et al. 2013), although this relatively simplified recommendation does not take into consideration the optimal feeding pattern to elicit maximal rates of muscle protein synthesis, as outlined above. Therefore, athletes aiming to maintain muscle mass during energy restriction should focus on a balanced daily protein distribution with a slightly greater meal-protein intake (e.g. 0.3–0.4 g/kg) to enhance muscle protein synthesis and increase daily protein intake. An additional benefit of this greater meal protein intake may be its favourable effect on meal satiety, which would be a desirable outcome for athletes attempting to facilitate a negative energy balance (Davidenko et al. 2013).

4.12 Higher protein diets: are they a health risk?

It is hard to define what constitutes a higher protein diet for an athlete. The Acceptable Macronutrient Distribution Range (AMDR) defines that protein should constitute 10-35% of total energy intake, which, for a 90-kg individual consuming energy to meet their basic needs, would represent ~0.6–2.1 g/kg/d. However, athletes tend to have a large distribution of energy intakes based on their training and competition schedules. As such, a diet containing twice the RDA (i.e. 1.6 g/kg/d) in a 90-kg athlete may represent ~14% of total energy during intense training and/or positive energy balance, ~20% during a weight-stable, energy neutral period, or ~28% during a period of negative energy balance for weight loss. Therefore, a relative protein intake of approximately twice the RDA has previously been suggested to represent a reasonable lower boundary for the classification of a 'higher' protein diet (Martin et al. 2005).

There is a common (and pervasive) conception that high-protein diets are detrimental to renal function and pose a significant health risk to those who chronically consume protein in excess of the RDA. It is presumed this belief stems from clinical populations with compromised renal function who unquestionably benefit from lower protein diets. However, in healthy individuals with normal renal function (presumably most if not all athletes) there is no substantiation for a link between protein intake and renal damage (Martin et al. 2005); this is reflected in multiple international expert consensus statements from the Food and Nutrition Board, Institute of Medicine (2002) to the World Health Organization (2007). Aside from using the AMDR as an upper limit for protein intake, it has recently been suggested that the upper limit for leucine intake may be as high as ~550 mg/kg/d (Pencharz et al. 2012). Assuming a liberal leucine content of ~10% for dietary protein (which would give a conservative upper limit for total protein) would be equivalent to ~5.5 g protein/kg/d, this level of intake is far in excess of the recommendations herein and what most athletes have habitually been reported to consume (i.e. ~1.4–1.9 g protein/kg/d) (Burke et al. 2003), even those in more extreme physique sports such as bodybuilding (i.e. ≥3 g protein/kg/d) (Tarnopolsky et al. 1988).

Nevertheless, more recent data on current protein intakes of athletes may suggest diet trends that prioritise this macronutrient in the diet through its allocation of up to half the daily energy intake (i.e. ~3.8 g/kg/d) (Rossow et al. 2013). Subsequently, more research is needed in humans to understand the potential health consequences (if any) of chronically (e.g. >1 month) very high (e.g. 4–5 g/kg/d) protein diets, which anecdotally can be observed in sub-populations of athletes who typically ascribe to the 'more is better' mentality despite any scientific substantiation for an ergogenic effect of such a dietary practice.

Summary

The protein needs of athletes continues to be an evolving and high-profile topic in sports nutrition, with new information targeting eating strategies that promote muscle protein synthesis in response to a training session rather than the previously valued metric of nitrogen balance. Current guidelines promote the intake of

0.3 g/kg of protein of high biological value soon after key exercise sessions, with the continuation of such doses over the day and prior to bed. Future research may continue to refine these guidelines, but for the moment they represent practice that can be shown to produce acute enhancement of muscle protein synthetic rates leading to chronic outputs such as enhanced muscle mass and strength.

Practice tips

GARY SLATER

- There are few other nutrients that have captivated the attention of the general population, fitness enthusiasts and athletes alike, than dietary protein and specific amino acids. The evolution of a multi-billion dollar protein supplement industry, combined with recent dietary trends emphasising the proposed virtues of a higher protein intake, have further enhanced the mystique of dietary protein. While this has been true for strength power athletes for many years, greater emphasis is now also being placed on protein by endurance athletes in an attempt to manage recovery and body composition. Given this, it is critical the sports dietitian has both an excellent understanding of the scientific evidence relating to protein intake, but also the practical implementation of this in developing meal plans giving consideration to optimising protein intake.
- Meeting the increased protein needs of both strength and endurance athletes in hard training is essential if adaptation and recovery are to be optimised. Fortunately, the higher food intake of most athletes ensures a generous protein intake, usually well above requirements typically prescribed for athletes (i.e. 1.2–1.7 g/kg/d).
- It is too simplistic to assess the adequacy of an athlete’s dietary protein intake via total daily intake alone. For any given protein intake, the metabolic response is dependent on other factors, including protein distribution throughout the day, individual dose at each meal/snack and amino acid composition of ingested proteins, plus digestion rate and total energy intake. The type of training undertaken, desired outcomes of training and training status of the athlete should also be considered when assessing protein needs. If changes in body composition are desired, this will also influence dietary protein intake guidance, both when attempting to facilitate skeletal muscle hypertrophy and when attempting to promote body fat loss.
- The timing of protein intake may be just as important as total protein intake over the day. As consumption of large amounts of protein at any one time merely stimulates protein oxidation, the inclusion of small amounts of food rich in high biological value protein at most meals and snacks throughout the day may result in enhanced adaptations. The meal plans in Table 4.1 illustrate both a poorly distributed protein intake emphasising the majority of daily protein intake at the evening meal, common among many athletes, and a more evenly distributed daily protein intake.
- Intra-session amino acid supplementation has become increasingly popular among athletes, a trend driven primarily by supplement industry hype rather than strong scientific rationale. Higher priority should be given to more immediate, proven needs, such as carbohydrate ingestion during prolonged and/or intense exercise. Furthermore, high protein intakes during sustained exercise have been associated with increased incidence of gastrointestinal distress.
- While adding carbohydrate to a post-training meal or snack does not influence the protein synthetic response, co-ingestion of carbohydrate with a post-exercise protein snack is generally advocated, especially when recovery time is brief, as it facilitates restoration of muscle glycogen stores. Furthermore, the co-ingestion of both protein (0.3 g/kg) and carbohydrate post-exercise moderates the amount of carbohydrate required to facilitate maximal rates of muscle glycogen restoration by approximately one-third, which may be particularly attractive for athletes with smaller energy needs. Inclusion of protein in the post-training meal/snack may also assist in moderating post-exercise appetite, a response attractive to weight-conscious athletes.
- As training influences protein metabolism for upwards of 48 hours afterwards, athletes should be encouraged to follow a meal plan with an even protein distribution throughout the week, not just on training days.

- While high biological value proteins like meat, poultry, seafood, eggs and dairy products provide approximately two-thirds of protein intake in a typical western diet, other sources—such as cereal products like bread, pasta, rice, legumes and breakfast cereals—also contribute approximately one-third of daily protein (albeit low biological value) as well as other important nutrients; see Table 4.2. Ideally, the individual dose of protein at a meal or snack advocated within the range of 0.25–0.3 g/kg body mass should come from high biological value choices. If low biological value protein sources predominate in the meal plan, protein intake goals at a meal or snack should be adjusted towards the upper range of this guidance. Similarly, athletes following reduced energy meal plans may require a proportionally higher protein intake, especially when attempting to minimise loss of skeletal muscle mass. Aiming for the upper range of guidelines may be advisable but will depend on total available energy budget and consideration of other nutrients essential for fuelling and recovery.
- While most athletes readily achieve daily protein targets often advocated for athletes within the range of 1.2–1.7 g/kg/d, some consume more than 4 g/kg, believing this will further facilitate adaptations. Current research suggests such extremes in protein intake are neither necessary nor beneficial, as excess dietary protein will simply be oxidised for energy production. However, routine consumption of a high-protein diet does not appear to negatively impact renal function in healthy individuals, nor have a significant impact on hydration status or bone health. A likely disadvantage for the athlete is that a high protein intake may displace other important nutrients necessary to support their training/competition demands and add to their weekly shopping bill, and create a greater acidic load on the body, which may have adverse metabolic implications. The later can be somewhat overridden by emphasising the intake of foods rich in potassium salts, specifically fruit and vegetables.
- Emotive labelling on products promoted in many gyms, sporting magazines and health food stores ensures amino acids and protein supplements are very popular among athletes. While there is compelling evidence of benefit associated with the ingestion of rapidly digested proteins rich in leucine, such as whey protein, in the post-exercise period, focus should remain on the selection of whole–food sources of protein, facilitating favourable metabolic adaptations while also promoting the achievement of other nutrient needs. Athletes should be guided to choose meal combinations that simultaneously match protein requirements with other nutrient needs. The fortification of meal plans emphasising whole foods with isolated protein sources may be necessary during periods of intentional weight loss where energy intake is moderated but protein intake increased in an attempt to both facilitate retention of muscle mass and function, while also promoting greater satiety. For example, the addition of 15 g whey protein to a 200 g tub of natural yoghurt doubles the protein content of this nutrition snack for an additional energy cost of just over 200 kJ or ~50 kcal. The protein dose facilitating maximal satiety may not align with the intake proven to optimise protein synthesis; thus the individual dose should be adjusted based on the primary outcome desired for the individual athlete at that particular meal or snack.
- Athletes who are attempting to promote skeletal muscle hypertrophy can become overly focused on dietary protein intake. While protein dose, distribution and source are important variables to consider in developing a meal plan conducive to promoting gains in muscle mass, an evaluation of the athlete's energy intake is essential. For any given protein intake, increasing the energy content of the meal plan will enhance nitrogen balance. This can be a challenge for some athletes. Frequent and/or prolonged training sessions can limit opportunities for meals and snacks while intense training may suppress an athlete's appetite. Novel strategies like an increased reliance on energy-dense snacks and drinks may be required to overcome such obstacles. The following tips may help to increase energy density of a meal plan for an athlete attempting to promote muscle hypertrophy:
  - Increase meal or snack frequency. Intestinal tolerance is generally higher when the frequency of meals is increased rather than the size of existing meals and snacks. Eating frequently should become a priority, even during busy days. Meal plans with five or more meals and snacks (including pre- and post-training snacks) throughout the day should be encouraged, with the inclusion of ~20–25 g high biological value (HBV) protein in the majority of these eating occasions advocated.
  - Make use of energy-dense drinks (such as smoothies, milk shakes, powdered liquid meal supplements, UHT packs of flavoured milk) and other energy-dense foods (e.g. low fat dairy snacks like yoghurt, cereal or sports bars, fruit bread, dried fruit and nuts or trail mix). Commercial liquid meal supplements are a convenient option for athletes on the run; products fortified with vitamins and minerals plus a combination of both carbohydrate and protein are most suitable. Alternatively, homemade milk drinks can be fortified with skim milk powder to add an extra protein and energy boost. These drinks can be particularly useful for athletes unable to tolerate solid food before and/or after training or those with small appetites due to the

lower satiety of drinks.

- The sports dietitian should ensure that short-term and long-term weight gain goals are realistic. An increase in body mass of 0.25–0.5 kg/wk may be possible but will depend on genetics and the resistance-training history of the athlete; significant prior gains inevitably ensure only smaller future gains are possible. Significantly greater rates of gain are likely to include increments in body fat stores that may have to be reduced at a later stage. It is also important to ensure overall body composition goals are realistic. Far too many athletes want to increase muscle mass and decrease body fat simultaneously. This is difficult for many athletes to achieve as the two goals are mutually exclusive: one demands an increase in energy availability while the other requires a reduction in energy availability. Priorities must be set and dietary intervention applied accordingly.
- While the energy surplus required to support a 1 kg gain in muscle mass is likely to vary between individuals (depending on e.g. current training and diet, genetic profile), a conservative 1500–2000 kJ (360–480 kcal) increase in daily energy intake for an athlete with a stable body mass (and thus presumably in energy balance) is a sensible starting point. However, more aggressive increments in daily energy intake of 4000 kJ (960 kcal) or more may be warranted for some athletes, especially if the energy cost of resistance training has to be accounted for (e.g. if the athlete had not previously been undertaking regular resistance training). Regular body mass and composition monitoring will provide invaluable feedback on the requirement for further adjustments in energy intake to support muscle hypertrophy. See Chapter 3 for a closer examination of physique assessment tools.
- Many athletes will require advice about lifestyle and time management to allow them to achieve adequate time for eating, sleeping, training and their other daily commitments. Planning the day's intake—what and when—may assist some athletes in ensuring suitable foods and drinks are on hand when required. A ready supply of non-perishable snacks in a locker or training bag can be a great idea, for example, tetra packs of UHT-flavoured milk/fruit juice, cereal bars, powdered liquid meal supplements and sports drinks. The sports dietitian should assess the potential of the athlete's environment for purchasing suitable foods, or for storing snacks brought from home.

TABLE 4.1 Two meal plans with similar total nutritional value (~14 000 kJ, 190 g protein, 110 g fat, 420 g carbohydrate) but differing in their distribution of protein throughout the day for a 90-kg athlete

Adequate protein: Poor distribution		Adequate protein: Optimal distribution	
Meal	**Protein**	**Meal**	**Protein**
Breakfast 2 cups cereal, low fat milk 2 slices toast, jam and margarine	28 (10)	Breakfast 3 poached eggs 4 slices toast, 2 with jam and margarine	37 (21)
Morning tea 1 cereal bar	1 (0)	Morning tea 200 g low-fat Greek yoghurt + 1 fruit	21 (20)
Lunch 2 ham (90 g), cheese (30 g) and salad rolls Orange juice (300 mL)	37 (24)	Lunch 2 ham (60 g), cheese (40 g) and salad rolls Orange juice (300 mL)	34 (21)
Afternoon tea 2 slices fruit loaf	5 (0)	Afternoon tea 2 slices fruit loaf	5 (0)
Training Water, sports drink (600 mL)	0	Training Water, sports drink (600 mL)	0
Post-training	0	Post-training Whey protein isolate (25 g) + 2 fruit	23 (23)
Dinner 400 g lean steak (cooked) 2 cups steamed rice, 1 cup vegetables	116 (108)	Dinner 100 g lean steak (cooked) 2 cups steamed rice, 1 cup vegetables	36 (27)
Supper	0	Supper	38 (38)

		200 g low-fat Greek yoghurt 20 g whey protein isolate	

Note: The value in brackets indicates the amount of high biological value protein (g) at an eating occasion.

TABLE 4.2 Common food sources of protein, including both low and high biological value sources

Food	Amount	Protein (g)	Energy (kJ)	Cost (A$)
Milk (skim)	600 mL	22	900	0.75
Soy beverage	900 mL	33	1600	1.80
Milk powder (skim)	60 g	22	880	0.39
Cheese (red. fat cheddar)	70 g	22	770	1.10
Cheese (cottage)	140 g	25	530	0.90
Yoghurt (skim, Greek)	200 g	20	500	1.42
Yoghurt (skim, flavoured)	400 g	21	1290	1.72
Whey protein isolate	17 g	16	290	0.88
Egg (whole)	3 eggs	19	890	0.73
Egg white	175 g	20	350	1.17
Beef, poultry, seafood (raw)[a]	120 g	25	640	1.80
Almonds	130 g	26	3200	2.60
Tofu	400 g	48	1900	2.56
Kidney beans (drained)	350 g	23	1300	1.39
Lentils	380 g	18	820	1.39
Bread	9 slices	28	3000	0.84
Rice (white, cooked)	6 cups	26	6000	1.42

Note: Each of the foods in the amounts specified provides ~3 g of leucine, which equates to ~20–25 g of protein from high biological value sources. The associated energy and financial cost of these options is also provided as these are likely important considerations when making protein source selections.

[a]When cooked via conventional methods like grilling or frying, red meat, poultry and other animal flesh will decrease in weight by approximately 25%.

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CHAPTER FIVE

Energy requirements of the athlete: assessment and evidence of energy efficiency

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5.1 Introduction

Most individuals, including athletes, maintain a stable body mass over long periods of time, while paying little attention to the amount of energy consumed or expended each day. However, energy balance is of primary concern to the athlete who wants to alter body mass and/or composition to improve their exercise performance or meet a designated weight requirement for their sport. When energy consumption is insufficient to match that expended, much of the effort of training can be lost, since both muscle and fat will be used for energy. In addition, if energy intake is limited or restricted, the ability to obtain other essential nutrients, such as carbohydrate, protein, fat, vitamins and minerals—which are necessary for optimal sport performance and good health—will also be compromised.

Many athletes, especially female athletes, feel pressured by their coaches, parents, peers and themselves to reduce body mass. To maintain a low body mass, these athletes restrict energy intake even though their energy expenditure is high. Athletes of any age must consume enough energy to cover the energy costs of daily living, the energy cost of their sport and the energy costs associated with building and repairing muscle tissue. Females of reproductive age must also cover the costs of menstruation, whereas younger athletes must cover the additional costs of growth. What are the consequences of restricting energy intake when energy expenditure is high? It has been hypothesised that one consequence of this behaviour is an increase in energy efficiency, thus decreasing the amount of energy actually required to maintain body mass.

This chapter will briefly review the dynamic nature of energy balance and the many factors, such as macronutrient balance, that contribute to energy balance in an athlete or active individual. Manipulating the energy balance equation for either gain or loss of body mass for an individual athlete is covered in other sections of this book (see Chapter 4 for body mass gain and Chapters 6 and 7 for body mass loss). We will then briefly discuss the concept of energy efficiency and the research evidence for and against this phenomenon in the athlete.

5.2 Energy balance

At first, the concept of energy balance appears straightforward and simplistic. For body mass to be maintained, energy in (total kilojoules or kilocalories consumed and those drawn from body stores) must equal the energy expended. Under these conditions, an individual is considered to be in energy balance. This concept can be stated using the equation below, where 'energy in' means 'metabolisable energy' (energy intake minus energy lost in faeces and urine). For most individuals, metabolisable energy is about 90–95% of energy intake (Jequier & Tappy 1999).

$$\text{Energy balance occurs when } E_{in} = E_{out}$$

where E_{in} = energy in (kJ/d or kcal/d) and E_{out} = energy expended (kJ/d or kcal/d).

However, the ability of the body to regulate body weight within a narrow range and maintain energy balance is more complicated than it initially appears. Energy balance is a dynamic process whereby altering one component of the energy balance equation (e.g. energy intake or composition of the diet) can affect the physiological and biological components of the other (e.g. energy expenditure) in unpredictable and unintended ways (Galgani & Ravussin 2008; Hall et al. 2011). Thus, overall energy balance is influenced by a number of internal (e.g. genetic, epigenetic, metabolic, hormonal, neural) and external (e.g. environmental, social, behavioural) factors that can vary between individuals (Galgani & Ravussin 2008; Stensel 2010). This concept is shown in Figure 5.1. For example, a number of dietary factors can influence total energy intake, such a diet composition, timing of nutrient intake, timing and intensity of exercise, and the types of foods consumed (e.g. high- or low-energy-dense foods). The same is true for energy expenditure. A number of factors can influence energy expenditure, including composition of the diet, intensity of exercise and factors influencing resting metabolic rate. Thus, determining the exact impact of each of these factors on energy intake and expenditure is impossible for any one individual, since these factors are variable, change daily and have underlying metabolic controls that work to regulate energy balance to maintain weight stability.

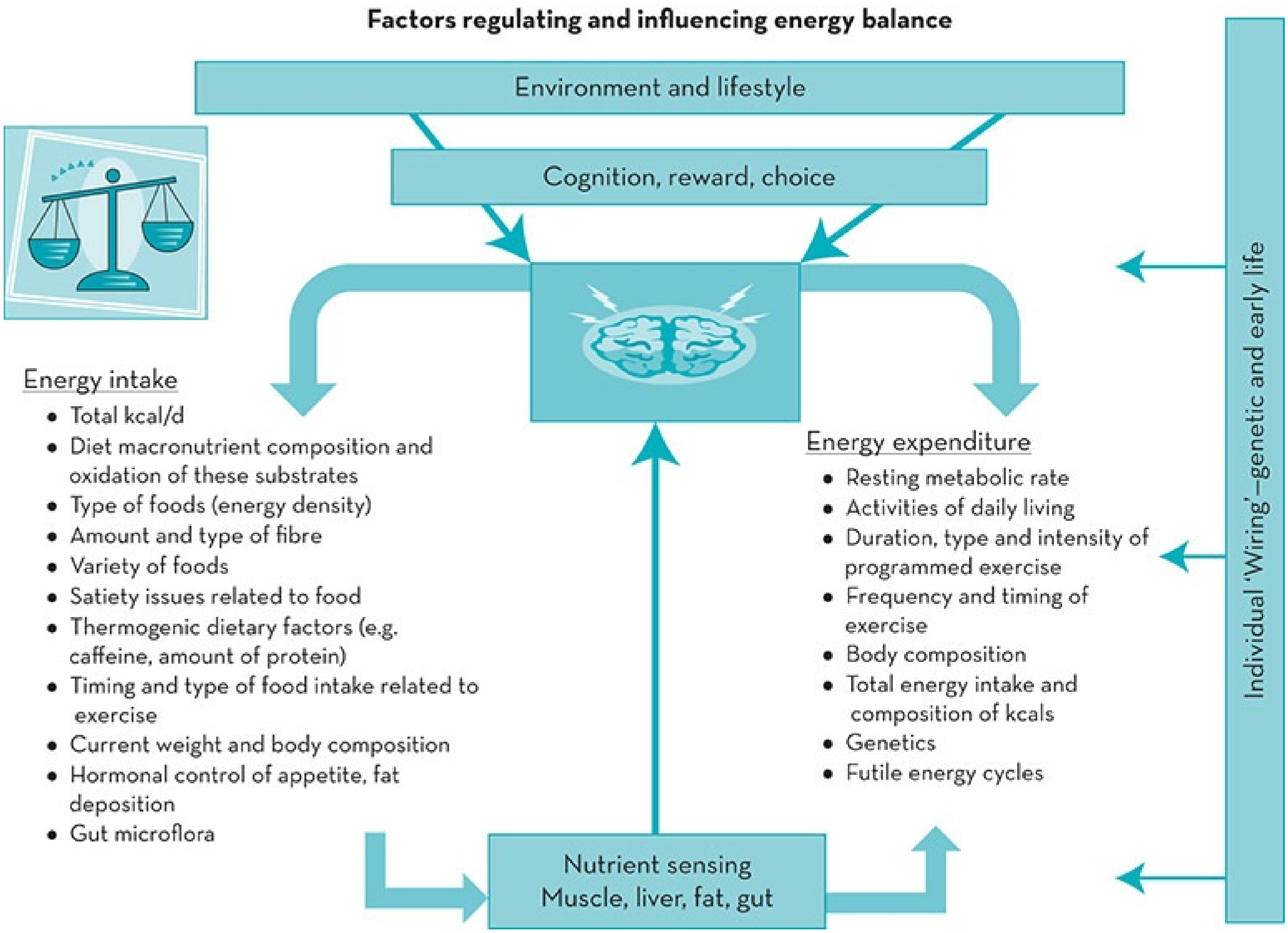

Figure 5.1 Factors that regulate and influence energy balance: changes in the environment that can influence subsequent generations (e.g. genetic and epigenetics) and current habitual lifestyle factors that influence diet and physical activity

If an athlete is trying to increase body mass, then E_{in} must exceed E_{out}. Conversely, if an athlete is trying to reduce body mass, E_{out} must exceed E_{in}. Thus, inducing an imbalance between E_{in} and E_{out} will result in the most dramatic changes in body mass. However, as indicated above, there are numerous factors that influence E_{in} and E_{out} besides the designated energy intake and expenditure that is programmed into an exercise training routine and diet. These unquantifiable factors make it difficult to determine exactly how an individual will respond to a designated diet and/or training routine, or how these things will interact and influence each other if they are implemented simultaneously.

As mentioned above, the energy balance equation is both dynamic and time-dependent, and allows for the effect of changing energy stores on energy expenditure over time. The following example, given by Swinburn and Ravussin (1993), illustrates this point. What would happen if an individual decided to consume an extra 413 kJ (100 kcal) a day for 40 years? The amount of extra energy consumed in this time would equal about 6 million kJ or 1.5 million kcal. If one assumes that there are 31,786 kJ/kg (7700 kcal/kg) of body fat, the theoretical gain in body mass would equal 190 kg over this 40-year period, yet the actual gain would be about 2.7 kg.

After a period of positive energy balance, extra energy intake would cause a gain in body mass (both fat and lean tissue). The larger body size would cause an increase in energy expenditure that would eventually balance the extra energy consumed. Of course, the actual gain in body mass will depend on the amount of extra energy consumed, and to a lesser extent on the composition of this energy (the amount of fat, carbohydrate, protein or alcohol) and overall energy expenditure. Therefore, a gain in body mass is the consequence of an initial positive energy balance, but can also be a mechanism whereby energy balance is eventually restored at a higher body mass and energy requirement.

5.3 Macronutrient balance

Macronutrient composition of the diet also influences long-term changes in body mass and composition. We now know that the maintenance of body mass and body composition over time requires that $E_{in} = E_{out}$ and that intakes of protein, carbohydrate, fat and alcohol equal their oxidation rates (Jequier & Tappy 1999; Astrup 2011). Therefore, macronutrient balance occurs when:

$$\begin{aligned} \text{protein}_{\text{in}} &= \text{protein}_{\text{oxidation}} \\ \text{carbohydrate}_{\text{in}} &= \text{carbohydrate}_{\text{oxidation}} \\ \text{fat}_{\text{in}} &= \text{fat}_{\text{oxidation}} \\ \text{alcohol}_{\text{in}} &= \text{alcohol}_{\text{oxidation}} \end{aligned}$$

where $\text{protein}_{\text{in}}$ is the amount of protein intake (g/d) and $\text{protein}_{\text{oxidation}}$ is the amount of protein oxidised (g/d). These same notations apply to carbohydrate, fat and alcohol. Alterations in either energy intake or expenditure are the primary determinants of body mass. Changes in the type and amount of macronutrients consumed (protein, fat, carbohydrate and alcohol) and the oxidation of these macronutrients within the body must be considered when examining long-term weight maintenance (Flatt 2001; Astrup 2011). Under normal physiological conditions, carbohydrate, protein and alcohol are not easily converted to body fat (Swinburn & Ravussin 1993; Flatt 2001). Thus, increases in the intake of non-fat nutrients stimulate their oxidation rates proportionally. Conversely, an increase in dietary fat intake does not immediately stimulate fat oxidation, thus increasing the probability that excess dietary fat will be stored as adipose tissue (Astrup 2011; Saris & Tarnopolsky 2003; Schutz 2004a; Westerterp 1993). In this way, the type of food eaten can play a role in the amount of energy consumed and expended each day (Acheson et al. 1984; Swinburn & Ravussin 1993; Schutz 2004a; Galgani et al. 2010).

5.3.1 Carbohydrate balance

Under energy balance conditions, carbohydrate balance is precisely regulated such that carbohydrate intake matches oxidation (Acheson et al. 1984; Flatt 2001; Jebb et al. 1996; Galgani et al. 2010). The ingestion of carbohydrate stimulates both glycogen storage and glucose oxidation, and inhibits fat oxidation. Glucose not stored as glycogen is oxidised directly in almost equal balance to that consumed (Flatt et al. 1985). The conversion of excess dietary carbohydrate and protein to triglycerides (de novo lipogenesis or DNL) is limited in normal-weight humans except under non-physiological situations (Acheson et al. 1987; Schutz 2004b; Hellerstein 1999). However, if large amounts of carbohydrate are consumed over several consecutive days, and E_{in} exceeds E_{out}, fat in the form of triglyceride is synthesised (Acheson et al. 1988). Schutz (2004b) and Strable and Ntambi (2010) have reviewed DNL in detail and discussed how carbohydrate intake influences fat balance and DNL.

5.3.2 Protein balance

As with carbohydrate, the body adjusts to a wide range of protein intakes by altering the rate of oxidation of dietary protein. After body protein needs are met, the carbon skeletons of any excess amino acids are diverted into the energy substrate pool and used for energy. The adequacy of total energy intake, and carbohydrate intake in particular, appear to affect this process dramatically. Inadequate intakes of either energy or carbohydrate result in negative protein balance (Krempf et al. 1993). Conversely, excess intake of either energy or carbohydrate will spare protein. This protein is then available to support brief periods of protein accumulation until the protein pool is expanded to a new balance point. At this point, the degradation of endogenous protein matches the available exogenous protein. The excess protein consumed or the protein made available through protein sparing may contribute indirectly to fat storage by diverting dietary fat for storage. Thus, protein will be used for energy preferentially over fat, leaving the excess dietary fat to be stored as adipose tissue.

5.3.3 Fat balance

Fat balance is not as precisely regulated as carbohydrate and protein balance (Schutz 2004a; Galgani et al. 2010). As dietary fat intake increases, the short-term oxidation of fat does not always increase proportionately (Schrauwen et al. 1997; Galgani et al. 2010; Astrup 2011). In addition, when energy expenditure is fixed and protein held constant, there is a precise inverse relationship between carbohydrate and fat oxidation. Under these conditions, the greater the proportion of carbohydrate oxidised, the less fat is oxidised (Schutz 2004a, 2000b; Galgani et al. 2010). Over the long term, a positive fat balance, due to excess energy intake from a palatable high-fat diet, will lead to a progressive increase in total body-fat stores as the body attempts to achieve energy balance (Schutz 2004a). Strong data support the hypothesis that increases in body-fat stores are due to low rates of fat oxidation relative to fat intake (Flatt 2001; Astrup 2011). As fat stores expand, they increase the free fatty acid concentrations in the blood as a consequence of the constant flux of triglycerides occurring in these stores. This increase in circulation of free fatty acids may increase fat oxidation slightly; thus, the larger adipose tissue mass promotes increased fat oxidation. When the new rate of fat oxidation equals the rate of fat intake, the individual will achieve fat balance, and hence energy balance, but at a significantly higher body weight (Schutz 2004a).

5.3.4 Alcohol balance

Alcohol consumption causes a rapid rise in alcohol oxidation until all the alcohol is cleared from the body. Thus alcohol is used preferentially as an energy source over other substrates and can suppress the oxidation of fat and, to a lesser degree, that of protein and carbohydrate (Shelmet et al. 1988; Suter 2005). Alcohol is not converted to triglycerides and stored as adipose tissue, nor can it contribute to the formation of muscle or liver glycogen. It may, however, indirectly divert dietary fat to storage by providing an alternative and preferred energy source for the body (Suter 2005). Alcohol has an energy density of about 29 kJ/g (7 kcal/g) and thus can contribute significantly to total daily energy intake. A review of the impact of alcohol on energy intake by Yeomans (2004) found that all the research to date fails to show a reduction in food intake in response to alcohol ingestion either before or with a meal. Therefore, individuals who consume alcohol must reduce their consumption of energy from other dietary components in order to maintain energy balance. The review by Suter (2005) provides a more complete description of the effect of alcohol on weight gain and obesity.

5.4 Energy expenditure

Determining energy balance requires the direct measurement or estimate of energy in (energy intake from the diet and from stored energy) and energy expended. This section reviews the various components of energy expenditure and how the components are measured, and discusses how physical activity may influence these components. We will also cover methods for predicting energy expenditure based on age, gender and body size.

5.4.1 Components of energy expenditure

The components of total daily energy expenditure are generally divided into three main categories (see Figure 5.2):

1. basal energy expenditure or basal metabolic rate (BMR)
2. thermic effect of food (TEF)

3. energy expended in planned physical activity and non-exercise activity thermogenesis (NEAT), which when combined covers the thermic effect of activity (TEA). The energy expended in fidgeting, which is also called spontaneous physical activity (SPA), would also be included in the total TEA.

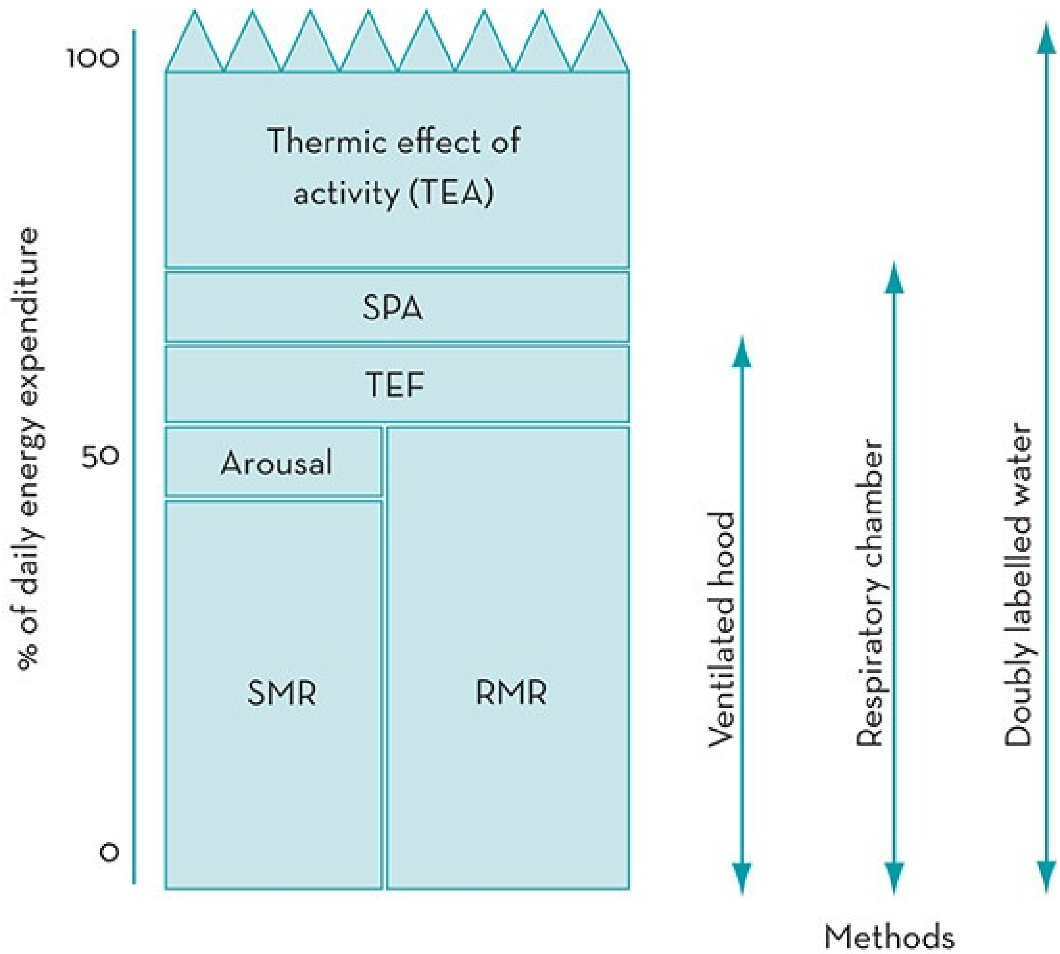

Figure 5.2 Components of daily energy expenditure in humans
SPA = spontaneous physical activity; TEF = thermic effect of food;
SMR = sleep metabolic rate; RMR = resting metabolic rate

Source: Adapted from Ravussin & Swinburn 1993

BMR is the energy required to maintain the systems of the body and to regulate body temperature at rest. BMR is measured in the morning after an overnight fast, while the individual is resting in a bed. The individual must be comfortable and free from stress, medications or any other stimulation that would increase metabolic activity. In addition, the room where BMR is measured needs to be quiet, temperature-controlled and free of distractions. Because assessment of BMR requires the individual to stay overnight in the laboratory, many researchers measure basal metabolism under conditions that are given the terminology resting metabolic rate (RMR) instead. Assessment of RMR usually means that the individual sleeps at home and drives or is driven to the research laboratory, where they rest for a period of time before metabolic rate is assessed. Like BMR, subjects need to have fasted overnight, refrained from strenuous exercise the day before assessment, and be measured in a quiet, temperature-controlled room. In general, BMR and RMR usually differ by less than 10%. In this chapter, we will use the term RMR (except when a research study specifically reports that BMR was measured), since it is more frequently measured. It should be noted that although some people use these terms interchangeably, in fact, they are measured differently.

RMR accounts for approximately 60–80% of total daily energy expenditure in most sedentary healthy adults (Ravussin et al. 1986; Ravussin & Bogardus 1989, 2000). However, in an active individual, this percentage will vary greatly. Many elite athletes easily expend 4100–8300 kJ/d (1000–2000 kcal/d) in sport-related activities. For example, Thompson and colleagues (1993) reported that RMR represented only 38–47% of total daily energy expenditure in 24 elite male endurance athletes. In female endurance athletes, primarily runners, Guebels and colleagues (2013) and Beidleman and colleagues (1995) found that RMR represented only 54% and 42% of total energy expenditure, respectively. During days of repetitive, heavy competition, such as ultramarathons, RMR may represent less than 20% of total energy expenditure (Rontoyannis et al. 1989).

The TEF (sometimes called diet-induced thermogenesis (DIT)) is the increase in energy expenditure above RMR that results from the consumption of food throughout the day. The TEF includes the energy cost of food digestion, absorption, transport, metabolism and storage within the body. TEF usually accounts for ~6–10% of total daily energy expenditure. However, the TEF for an individual will vary, depending on the energy content of the meal or the food eaten over the day, the types of foods eaten, macronutrient composition of the diet and the degree of obesity (Westerterp et al. 1999). Although TEF is frequently used interchangeably with the thermic effect of a meal (TEM), the terms are not synonymous. TEM represents the increase in metabolic rate above RMR after eating a meal. Most researchers measure TEM instead of TEF because of the difficulties in trying to assess the cumulative energy cost of all foods consumed within a day. Thus, most of the research literature examining the energy costs of active individuals reports TEM, unless a metabolic chamber is used to collect data.

TEA is the most variable component of energy expenditure in humans. It includes the energy cost of daily activities above RMR and TEF, such as planned exercise events (e.g. running, swimming weightlifting) and activities such as walking or bike riding. TEA also includes purposeful activities of daily living (e.g. dressing, shopping, cooking, standing). These types of daily life activities are also called non-exercise activity thermogenesis or NEAT. Research now suggests that for some individuals, the energy expended in NEAT may play a significant role in helping to maintain energy balance (Levine et al. 2005; Levine 2007). Finally, TEA includes the energy cost of involuntary muscular activity such as shivering and fidgeting (SPA). TEA may be only 10–15% of total daily energy expenditure in sedentary individuals, but may be as high as 50% in active individuals. Levine (2004a, 2004b) provides an in-depth review of NEAT and the environmental and biological factors that influence it. Donahoo and colleagues (2004) have reviewed the variability in total energy expenditure and its components.

A number of factors can increase energy expenditure above normal baseline levels, such as cold (via shivering), heat, fear, stress and various medications or drugs (e.g. caffeine, alcohol, smoking) (Manore et al. 2009). The thermic effect of these factors is frequently referred to as adaptive thermogenesis (AT). AT represents a temporary increase in thermogenesis that may last for hours or even days, depending on the duration and magnitude of the stimulus. In athletes, a serious physical injury, the stress associated with an upcoming event, going to a higher altitude, performance or training in extreme environmental temperatures, or the use of certain medications may all increase RMR above normal levels. For detailed reviews on the impact of weight loss on AT and how alterations in AT due to dieting may contribute to weight regain after dieting, see Dulloo and colleagues (2012), Müller and Bosy-Westphal (2013) and Westerterp (2013).

Factors that influence RMR

A variety of factors can influence RMR for a given individual on any given day; however, some factors appear to have more of an influence than others. It is well documented that RMR is influenced by age, sex and body size, including the size of an individual's fat-free mass (FFM) and fat mass. In fact, these factors are usually included in prediction equations for RMR. Three of these variables (age, sex and FFM) generally explain about 80% of the variability in RMR (Bogardus et al. 1986). Since FFM, especially organ tissue, is very metabolically active, any change in FFM can dramatically influence RMR (Henry 2000). In general, males have larger RMRs than females because of an increased size and greater FFM; however, there may be other contributing factors to the differences in RMR besides gender (Blanc et al. 2004). Ferraro and colleagues (1992) report that females have a lower BMR than males (413 kJ or 100 kcal/d less), even after controlling for differences in FFM, fat mass and age. Conversely, Blanc and colleagues (2004) found no difference in RMR when comparing elderly men and women (70–79 yr) after controlling for FFM. Age is known to influence BMR, with an estimated decline in BMR of about 1–2% per decade from the second through to the seventh decade of life (Keys et al. 1987). This reduction in RMR is attributed to decreases in organ mass and losses in FFM that occurs with ageing, especially if an individual leads a more sedentary lifestyle (Henry 2000; Manini 2010).

RMR also has a genetic component, which means that individuals within families may have similar RMRs. For example, Bogardus and colleagues (1986) found that family membership explained 11% of the variability in RMR ($p < 0.0001$) when they examined 130 non-diabetic adult south-western Native Americans from 54 families. Bouchard and colleagues (1989) also found that heritability explained approximately 40% of the variability in RMR in Canadian twins and parent–child pairs after adjusting for age, gender and FFM.

Phases of the menstrual cycle may also influence RMR and total energy balance. Although the current research is equivocal, some studies report that RMR values are lowest during the follicular phase of the cycle (beginning of the cycle) and highest during the luteal phase (end of the cycle) (Solomon et al. 1982; Bisdee et al. 1989). The difference in RMR between these two phases is estimated to be approximately 413–1238 kJ/d (100–300 kcal/d); however, adaptations in energy intake appear to mimic the changes in RMR. A study by Barr and colleagues (1995) reports that females consume approximately 1238 kJ/d (300 kcal/d) more during the luteal phase of the menstrual cycle compared with the follicular phase. Thus, the increased energy expenditure, due to a higher RMR during the luteal phase, is compensated for by an increase in energy intake during this period. Additional evidence supporting the impact of menstrual cycle on RMR comes from studies examining the impact of menstrual dysfunction on energy expenditure. Lebenstedt and colleagues (1999) found that RMR was significantly lower (about 460 kJ or 111 kcal/d) in female athletes with menstrual dysfunction (9 menstrual periods/y) compared to active controls (12 menstrual periods/y). Reed and colleagues (2011) report a similar finding. Exercising women with exercise-induced menstrual disturbances had significantly lower RMR (1191 kcal/d) than ovulatory controls (1316 kcal/d). The results were the same when normalised for kcals/FFM/d. Conversely, Weststrate (1993) and Li and colleagues (1999) found no effect of menstrual cycle on RMR. These data are supported by more recent work by Guebels and colleagues (2013). They found no difference in RMR between endurance athletes with exercise-induced menstrual disturbances (1514 kcal/d) and eumenorrhoeic endurance athletes (1491 kcal/d). However, when data were expressed as kcal/FFM/d, female athletes with exercise-induced menstrual disturbances actually had higher RMR values ($p = 0.015$). Thus, until these issues are resolved, menstrual status and the phase of the menstrual cycle should be documented using some type of hormonal data and/or recorded when measuring RMR or energy intake in females, especially active females.

There are a number of ways that exercise might indirectly or directly change RMR. First, exercise may increase RMR indirectly by increasing an individual's FFM, which is a strong determinant of RMR. It is well documented in the research literature that active individuals, especially elite athletes, are leaner (lower percentage body fat) and have greater FFM than their sedentary counterparts. Thus, for a given body mass, an athlete with a lower percentage of body fat and higher percentage of FFM will have a higher RMR. Second, it has also been hypothesised that exercise training influences RMR; however, data comparing RMR in exercise-trained and sedentary controls have not shown consistent increases in RMR when subjects (athletes and controls) are matched for size and FFM (Manore et al. 2009). The discrepancies in these results may be due to a number of factors, including level of fitness, type of exercise training program, methods used to measure RMR, and level of energy flux (the amount of energy expended in exercise compared with the amount of energy consumed each day) (Bullough et al. 1995; Manore et al. 2009). Third, strenuous exercise may cause muscle tissue damage that requires building and repair after exercise is over, thus indirectly causing an increase in RMR.

An acute bout of strenuous exercise has also been hypothesised to directly influence RMR. It has been observed that RMR is increased for a period of time (minutes or hours) after strenuous exercise; this phenomenon is termed excess post-exercise oxygen consumption (EPOC). How quickly oxygen consumption returns to baseline after exercise is over may depend on a number of factors including level of training, age, environmental conditions, and intensity and duration of the exercise. It appears that to produce a significant increase in EPOC, exercise intensity must be high and/or the duration of exercise must be long. A normal exercise bout of 30–60 minutes of moderate intensity (50–65% $VO_{2\ max}$) does not appear to significantly elevate EPOC for any appreciable amount of time after the exercise is over (Manore et al. 2009). After this type of exercise, oxygen levels usually return to normal within 1 hour. However, if exercise (either aerobic or strength training) is of high intensity and/or of long duration, EPOC appears to be elevated for hours after exercise (Chad & Quigley 1991; Melby et al. 1993; Gillette et al. 1994).

Factors that influence TEF

A number of factors can influence how our bodies respond metabolically when we consume food. The TEF can last for several hours after a meal and depends on the energy content of the meal consumed and the composition of the meal (percentage of energy from protein, fat and carbohydrate). In general, the thermic effect of a mixed meal is estimated to be 6–10% of total daily energy intake; however, the total TEF will also depend on the macronutrient composition of the diet. For example, the thermogenic effect of glucose is 5–10%, fat is 3–5% and protein is 20–30% (Flatt 1992). The lower thermic response for fat is due to the lower energy requirement to store fat as triglyceride as compared to the synthesis of proteins from amino acids or glycogen from carbohydrate.

5.4.2 Measurement of total daily energy expenditure

Total daily energy expenditure or its components can be measured in the laboratory or estimated using prediction equations. The following section discusses the most commonly used laboratory techniques for measuring the components of energy expenditure. When laboratory facilities are not available, prediction equations can be used to estimate total daily energy expenditure.

Indirect calorimetry

Energy expenditure in humans is commonly assessed using indirect calorimetry, which measures the rate of oxygen consumption (L/minute) and carbon dioxide production (L/minute) either at rest or during exercise. The ratio between the volume of carbon dioxide produced (VCO_2) and the volume of oxygen consumed (VO_2) can be calculated (VCO_2/VO_2). This ratio, when considered at the cellular level, is termed the non-protein respiratory quotient (RQ) and represents the ratio between oxidation of carbohydrate and lipid. By knowing the amount of each energy substrate oxidised and the amount of oxygen consumed and carbon dioxide produced, total energy expenditure can be estimated using various published formulae. In general, the consumption of 1 litre of oxygen results in the expenditure of approximately 19.86 kJ (4.81 kcal) if the fuels oxidised represent a mixture of protein, fat and carbohydrate. Since RQ cannot be directly determined at the cellular level in humans, an indirect measurement is taken by measuring gas exchange at the mouth. The relationship of VCO_2/VO_2 measured by this means is termed the respiratory exchange ratio (RER). RER is considered an accurate reflection of RQ under steady-state conditions. Using the indirect calorimetry method, one can measure total daily energy expenditure in a metabolic chamber, or measure RMR by using a mask, hood or mouthpiece in which gases are collected and analysed for a specified period of time. Reviews by Westerterp (1993), Pinheiro Volp and colleagues (2011) and Shephard and Aoyagi (2012) provide additional information on the methods of indirect calorimetry.

RER values depend on the substrate being utilised, ranging from values of 0.7 (oxidation of fat only) to 1.0 (oxidation of pure carbohydrate). Most individuals consuming a mixed diet of protein, fat and carbohydrate will have an RER value of 0.82–0.87 at rest. However, during times of high exercise intensity, RER will increase and be closer to 1.0, while during times of fasting or low energy intake RER will decrease and be closer to 0.7. Thus, RER depends on the composition of the foods consumed, the energy demands placed on the body and whether BM is being maintained.

Doubly labelled water

Because indirect calorimetry requires that an individual be confined to a laboratory setting or a metabolic chamber, it is difficult to measure an individual's free-living energy expenditure. The development of the doubly labelled water (DLW) ($^2H_2{}^{18}O$) method for use in humans has become a valuable tool in determining free-living energy expenditure (Speakman 1998). This method was first developed for use in animals and eventually applied to humans (Schoeller et al. 1986). The DLW method is a form of indirect calorimetry based on the differential elimination of deuterium ($^2H_2$) and ^{18}oxygen (^{18}O) from body water, following a load dose of water labelled with these two stable isotopes. The deuterium is eliminated as water, while the ^{18}O is eliminated as both water and carbon dioxide. The difference between the two elimination rates is a measure of carbon dioxide production (Speakman 1998; Schoeller 2002). This method differs from traditional indirect calorimetry in that it only measures carbon dioxide production and not oxygen consumption. One advantage of this method is that it can be used to measure energy expenditure in free-living subjects for 3 days to 3 weeks, and only requires the periodic collection of urine for measurement of the isotope elimination rates. Another advantage is that it is free of bias, and subjects can engage in normal daily activities and sports without the interruption of writing down activities or wearing a heart-rate monitor. This method has become a valuable tool for the validation of other less expensive field methods of measuring energy expenditure, such as accelerometers (Ojiambo et al. 2012; Ainslie et al. 2003). The major disadvantage of this technique is expense. Another disadvantage is that there is a five times greater potential for error in estimating energy expenditure because it uses only the energy equivalent of carbon dioxide instead of the energy equivalent of oxygen (Jequier et al. 1987). Finally, the experimental variability of the DLW technique in adult humans appears to be high (5–8.5%) (Speakman 1998; Ainslie et al. 2003). This variability is high both when repeating the technique in the same individual and between individuals (Goran et al. 1994; Scagliusi et al. 2008).

Predicting total daily energy expenditure

When laboratory facilities are not available for assessing total energy expenditure, it can be estimated by applying prediction equations to estimate RMR, then multiplying RMR by an appropriate activity factor. A number of prediction equations have been developed to estimate RMR (see Table 5.1). These prediction equations have been developed for different populations that vary in age, gender, level of obesity and activity level. In general, it is best to use a prediction equation that is the most representative of the population or group of individuals with whom you are working. Table 5.1 summarises some of the commonly used RMR prediction equations and the population from which these equations were derived (Manore et al. 2009). It should be noted that most of the prediction equations have been developed using sedentary individuals. In an effort to determine which of these equations works best for active individuals and athletes, Thompson and Manore (1996) compared the actual RMR values measured in the laboratory with predicted RMR values, using equations listed in Table 5.1. They found, for both active males and active females, that the Cunningham (1980) equation best predicted RMR in this population, with the Harris–Benedict (1919) equation being the next best predictor. Figure 5.3 shows how closely these equations predicted RMR in a group of endurance-trained males and females. Because the Cunningham (1980) equation requires the measurement of lean body mass (LBM) or FFM in kilograms, the Harris–Benedict (1919) equation is easier to use in settings where FFM cannot be directly measured.

TABLE 5.1 Equations for estimating resting metabolic rate (RMR) in healthy adults

Harris–Benedict (1919)[a]
Males: RMR (kcal/d) = 66.47 + 13.75 (wt) + 5 (ht) – 6.76 (age)
Females: RMR (kcal/d) = 655.1 + 9.56 (wt) + 1.85 (ht) – 4.68 (age)
Owen et al. (1986)[b]
Active females: RMR (kcal/d) = 50.4 + 21.1 (wt)
Inactive females: RMR (kcal/d) = 795 + 7.18 (wt)
Owen et al. (1987)[c]
Males: RMR (kcal/d) = 290 + 22.3 (LBM)
Males: RMR (kcal/d)= 879 + 10.2 (wt)
Mifflin-St. Jeor et al. (1990)[d]
RMR (kcal/d) = 9.99 (wt) + 6.25 (ht) – 4.92 (age) + 166 (sex: male = 1, female = 0) – 161
Cunningham (1980)[e]
RMR (kcal/d) = 500 + 22 (LBM)
World Health Organization (1985)[f]
Sex and age (years) range equation to derive RMR in kcal/day:

Males	18–30	(15.3 × wt) + 679	Females	18–30	(14.7 × wt) + 496
	30–60	(11.6 × wt) + 879		30–60	(8.7 × wt) + 829
	> 60	(13.5 × wt) + 487		> 60	(10.5 × wt) + 596
where wt = weight (kg), ht = height (cm), age = age (years) and LBM = lean body mass (kg)					
Schofield (1985)					
Males (18–30 y): BMR (MJ/d) = 0.063 (wt) + 2.896					
Males (30–60 y): BMR (MJ/d) = 0.048 (wt) + 3.653					
Females (18–30 y): BMR (MJ/d) = 0.062 (wt) + 2.896					
Females (30–60 y): BMR (MJ/d) = 0.034 (wt) + 3.538					
where wt = weight (kg)					

[a]Harris and Benedict (1919) based on 136 men (mean age 27 ± 9 years; mean wt 64 ± 10 kg) and 103 women (mean age 31 ± 14; mean wt 56.5 ± 1.5) (n = 239 subjects). Included trained male athletes. Research indicates equation frequently over-predicts RMR by >15% (Frankenfield et al. 2005). Units of measurement expressed as basal energy expenditure (BEE), but the methods used were those of RMR.

[b]Owen et al. (1986) used 44 lean and obese women; eight women were trained athletes (ages 18–65 years; weight range 48–143 kg). No women were menstruating during the study; all were weight stable for at least 1 month.

[c]Owen et al. (1987) used 60 lean and obese men (ages: 18–82 years; weight range 60–171 kg). All were weight stable for at least 1 month. No athletes were included.

[d]Mifflin and colleagues (1990) used 498 healthy lean and obese subjects (247 females and 251 males), aged 18–78 years; weight ranged from 46–120 kg for the women and 58–143 kg for the men. Physical activity levels were not reported. This equation is more likely to estimate RMR to within 10% of measured values in both obese and non-obese individuals (Frankenfield et al. 2005).

[e]Cunningham (1980) used 223 subjects (120 males and 103 females) from the 1919 Harris and Benedict database. They eliminated 16 males who were identified as trained athletes. In this study, LBM accounted for 70% of the variability of BMR. The age variable did not add much because group age range was narrow. LBM was not calculated in the Harris–Benedict equation, so they estimated LBM based on body mass (kg) and age.

[f]World Health Organization (1985) derived these equations from BMR data.

[g]Schofield (1985) used 7393 subjects (~65% males).

Source: Adapted from Manore, Meyer & Thompson, 2009.

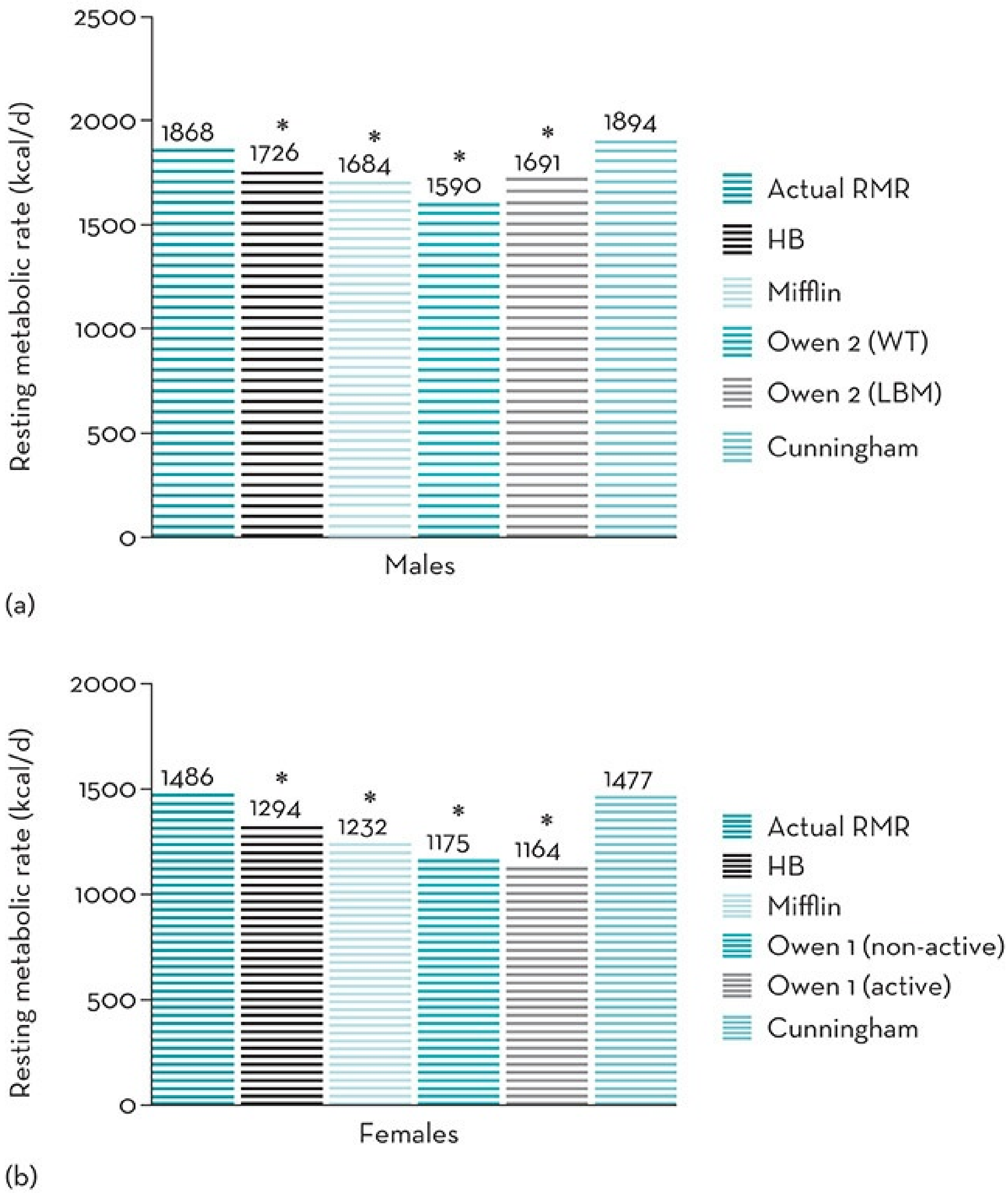

Figure 5.3 Mean group differences between actual and predicted resting metabolic rate (RMR) for (a) 24 male and (b) 13 female highly trained endurance athletes

*Indicates values were significantly different from actual measured RMR ($p < 0.05$). HB = Harris–Benedict equation (1919); Mifflin = Mifflin et al. equation (1990); Owen 1 = Owen et al. equation (1986) for active and non-active women; Owen 2 = Owen et al. equation (1997) for men using either body weight (WT) or lean body mass (LBM); Cunningham = Cunningham (1980) equation. Equations are listed in Table 5.1.

Source: Adapted from Thompson & Manore 1996

Once RMR has been estimated, total daily energy expenditure can then be estimated by a variety of different factorial methods. These methods vary in how labour intensive they are to use, and the level of subject burden. Manore and colleagues (2009) provide a detailed description of these methods. The easiest method for assessing total energy expenditure multiplies RMR by an appropriate activity factor, with the resulting value representing total daily energy expenditure. This factor may range from as low as 10–20% (0.10–0.20) of RMR for a bed-ridden individual to >100% (>1.0) for a very active individual. Although many laboratories establish their own activity factor for their particular research setting, factors of 1.3–1.6 are commonly used with sedentary individuals or individuals doing only light activity. With the activity factor methods, RMR is multiplied by a designated physical activity level or PAL. One activity factor can be applied to the whole day or a weighted activity factor can be determined. This activity factor is then multiplied by the RMR to provide 24-hour energy expenditure. For example, if an individual has a RMR of 6192 kJ/d (1500 kcal/d) and an activity factor of 1.5, then the daily energy expenditure would be 50% above RMR or 9288 kJ/d (2250 kcal/d) (6192 kJ × 1.5 = 9288 kJ/d).

The Institute of Medicine (2005) Food and Nutrition Board has also published equations of Estimated Energy Requirements (EER) to predict total daily energy expenditure. These equations are:

$$\text{adult males: EER} = 662 - (9.53 \times \text{AGE}) + \text{PA} \times (15.91 \times \text{WT} + 539.6 \times \text{HT})$$

$$\text{adult females: EER} = 354 - (6.91 \times \text{AGE}) + \text{PA} \times (9.36 \times \text{WT} + 726 \times \text{HT})$$

where AGE is age in years, PA is the physical activity quotient based on the person's PAL, WT is weight in kg, and HT is height in metres. PA is equal to 1.0 if PAL is 1.0 to 1.39, 1.11 if PAL is 1.4 to 1.59, 1.25 if PAL is 1.6 to 1.89, and 1.48 if PAL is 1.9 to 2.49.

More recently, the National Institutes of Health have developed a mathematical model for predicting energy needs based on the dynamic energy balance approach (Hall et al. 2011). This web-based tool (http://bwsimulator.niddk.nih.gov) is available to help estimate energy needs and energy expenditure in individuals. The calculator uses estimates of activity level or PAL value as outlined above. Goals can be set for weight gain or loss, in addition to input on how one may want to alter their diet or exercise based on lifestyle changes. Based on these inputs, the model will estimate and plot how weight will change over time. This tool can be useful when working with athletes, as it provides a guide for the amount of time it may take to change body weight depending on the lifestyle changes they decide to make. Further information can be found in Hall (2010) and Hall et al. (2012).

Regardless of the method used to calculate energy expenditure, it should be noted that all values are estimates. How accurate these values are depends on how accurately activity is recorded or reported, the accuracy of the database that is used to generate the energy expended per activity, and how accurately the required calculations are done.

5.5 Energy efficiency: does it exist?

The potential for energy efficiency among athletes was brought to the attention of researchers through a number of research studies in which active women reported energy intakes that appeared inadequate to meet total daily energy expenditures (Drinkwater et al. 1984; Deuster et al. 1986; Kaiserauer et al. 1989; Dahlstrom et al. 1990; Mulligan & Butterfield 1990; Myerson et al. 1991; Wilmore et al. 1992; Beidleman et al. 1995; Kopp-Woodroffe et al. 1999; Beals & Manore 1998). In these studies, active women (running 20–60 miles per week, or participating in gymnastics, swimming, triathlons or dancing) were reported to be consuming ≤147 kJ (35 kcal) per kg body mass. Despite these low energy intakes and apparent energy deficits, these individuals reported the maintenance of body mass over relatively long periods of time.

There are a number of possible explanations for an athlete's ability to maintain body mass despite the discrepancy between reported energy intake and energy expenditure. First, this discrepancy may be due to inaccuracies in reported estimates of energy expenditure or energy intakes, particularly due to athlete's under-reporting or under-consuming their usual intake during the period of monitoring (Dahlstrom et al. 1990; Wilmore et al. 1992; Schulz et al. 1992). A second explanation is that active individuals become more sedentary during non-exercising portions of the day, thus expending less energy than estimated (Westerterp 2013). Finally, these differences may be due to increased metabolic efficiency (Mulligan & Butterfield 1990; Myerson et al. 1991; Thompson et al. 1993, 1995). If metabolic efficiency is present, then the actual energy requirements of these athletes are lower than those estimated by traditional means and would partially explain their ability to maintain body weight despite a seemingly low energy intake. Thus, they may expend less energy at rest, while performing various daily tasks and during exercise than those whose energy intake appears adequate.

5.5.1 Evidence for

Thompson and colleagues (1993) reported evidence of energy efficiency in 24 male endurance athletes. In this study, the low-energy-balance athletes reported eating 6150 kJ/d (1490 kcal/d) less than the adequate-energy-balance athletes, while the estimated activity level of both groups was similar. Despite these energy intake differences, both groups had similar FFM and had been weight stable for at least 2 years. RMR was significantly lower in the low-energy-intake group compared with the adequate-energy-intake groups (about 4.9 versus 5.3 kJ/FFM/h or 1.19 versus 1.29 kcal/FFM/h, respectively).

A second study was completed on another group of male endurance athletes classified as having either low or adequate energy intakes (Thompson et al. 1995). This study aimed to determine if there were differences in 24-hour energy expenditure (EE), sleep EE, RMR and spontaneous physical activity (SPA), with these measurements being determined in a respiratory chamber. All subjects were of similar body size and composition. The low-energy-intake athletes reported a daily energy intake of 6446 ± 2163 kJ (1535 ± 524 kcal) less than estimated EE. The daily 24-hour EE, RMR, sleep EE and SPA of the low-energy-intake athletes were significantly lower than the adequate-energy-intake athletes. Thus, part of the ability of the low-energy-intake athletes to maintain body mass on a seemingly low energy intake appears due to a lower daily sedentary EE.

Myerson and colleagues (1991) found that amenorrhoeic runners had a significantly lower RMR than eumenorrhoeic runners and inactive controls, and the energy intake of these runners was similar to the inactive controls despite higher activity levels. Lebenstedt and colleagues (1999) studied eumenorrhoeic and oligomenorrhoeic active women (runners and triathletes). The menstrual function of the women was determined by assessing salivary progesterone levels, and the women were classified as having either normal menstrual function (12 periods per year) or menstrual disturbances (nine or fewer periods per year). Although the reported energy intake and activity level of these women was not different, the women with menstrual disturbances had a significantly lower RMR and reported significantly higher restrained eating scores. Work by De Souza and colleagues (2007) indicated that physically active women (exercising more than 2 h/wk) who also reported a high drive for thinness had significantly lower RMR values and were at greater risk for menstrual dysfunction than active women or sedentary women reporting a normal drive for thinness, despite all three groups reporting similar levels of energy intake and being of similar body weight. The results of these studies suggest that amenorrhoeic women and women with menstrual disturbances exhibit energy efficiency, which may be either a cause or a consequence of menstrual cycle disturbances.

Nattiv and colleagues (2007) propose that the mechanism explaining the increased energy efficiency observed in active females with menstrual dysfunction is due to reduced energy availability. Energy availability is defined as the amount of dietary energy remaining to support other body functions after exercise training. When energy availability is too low to support all body functions in addition to exercise, the amount of energy expended for maintaining cellular function, thermoregulation, growth and reproduction is reduced. This compensation in energy expenditure (or increased energy efficiency) restores energy balance and maintenance of body weight and promotes survival, but may negatively affect health. The assessment of energy availability and the detrimental effects of low energy availability are discussed in more detail in Commentaries 2 and 3.

Finally, evidence from research examining obesity and dieting for weight loss supports the presence of increased energy efficiency in formerly obese subjects (Rosenbaum & Leibel 2010; Müller & Bosy-Westphal 2013; Westerterp 2013). A meta-analysis of this research literature by Astrup and colleagues (1999) found that formerly obese subjects have a 3–5% lower relative RMR value compared to control subjects. In a review of the literature by Rosenbaum and Leibel (2010), the authors reported that 24-hour energy expenditure decreases by 20–25% in healthy men and women who have maintained a ≥10% weight loss. This decrease in energy expenditure is 10–15% below what would be predicted based on losses in fat and lean tissue. This means that a formerly obese person will need fewer calories each day to maintain a body weight that is similar to an individual who has never dieted for weight loss and has the same body weight and composition.

5.5.2 Evidence against

The results of a number of studies of female athletes do not support the existence of energy efficiency. Wilmore and colleagues (1992) and Schulz and colleagues (1992) found that female athletes reported significantly lower energy intakes than expected for their activity level, but measuring energy expenditure showed no evidence of energy efficiency, indicating they under-reported their energy intake. Beidleman and colleagues (1995) also found large differences between reported energy intake and energy expenditure in female distance runners, but could not attribute these differences to metabolic efficiency (lower RMR and EE during exercise) as compared to untrained controls. However, the data collection period was very brief (3 days), and may not have been long enough to detect true differences. Fogelholm and colleagues (1995) found that gymnasts reported a significantly lower energy balance (energy intake minus energy expenditure) than sedentary controls and soccer players, but the RMR was similar between all groups of athletes.

In most of these studies, metabolic efficiency was examined by comparing the RMR, 24-hour energy expenditure or energy expenditure during exercise of female athletes to sedentary controls. One criticism is that there was no attempt to compare the athletes who reported significant energy deficits to the athletes within the group who reported an adequate energy intake. A second criticism is that energy expenditure was not measured during the same time during the menstrual cycle in all studies (Schulz et al. 1992; Wilmore et al. 1992). RMR can change over the menstrual cycle, and is reported to be the lowest in the follicular phase and highest in the luteal phase (Bisdee et al. 1989; Solomon et al. 1982; Barr et al. 1995). Failure to compare women during the same phase of the menstrual cycle or to clearly document and hormonally assess menstrual status could mask any differences in energy expenditure that may exist. Finally, only four studies (Mulligan & Butterfield 1990; Myerson et al. 1991; Beidleman et al. 1995; Lebenstedt et al. 1999) verified ovulation in eumenorrhoeic athletes, and only Myerson and colleagues (1991) and Lebenstedt and colleagues (1999) screened for eating disorders. As demonstrated in the study by Lebenstedt and colleagues (1999), active females may report regular menstrual bleeding and still have some type of menstrual dysfunction, a condition that may decrease energy expenditure if hormone responses are blunted (Dueck et al. 1996).

Evidence from the weight-loss literature suggests that regular physical activity may help prevent the increased energy efficiency that results from inadequate energy intakes resulting in weight loss. Redman and colleagues (2009) reported a significant decrease in total daily energy expenditure in free-living healthy individuals losing an average of 10% to 14% body weight through caloric restriction over 6 months, but those in the treatment group that combined caloric restriction with structured aerobic exercise 5 times/wk experienced no decrease in total daily energy expenditure despite losing the same amount of weight as those in the caloric-restriction-only groups. Although the participants in this study were not active individuals or competitive athletes, these results suggest that becoming physically active during a period of weight loss may protect some individuals from increased energy efficiency that occurs as a result of energy restriction.

Summary

This chapter has discussed the components that determine energy balance; those on both the energy input side (dietary energy plus the contribution of energy stores within the body) and the energy expenditure side. In addition, we have discussed how the various components of energy expenditure can be measured. It appears that some athletes may have an increased energy efficiency, which can influence the energy intake needed to maintain body mass. For any one individual, the factors that influence energy balance may be numerous, including gender, age, family history, dietary choices, level of daily activity and stress level. If an individual wishes to permanently change body size, then one or more of the components of energy balance need to be altered over an extended time. Methods for doing this are discussed in Chapters 4 and 6.

Practice tips

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OVERVIEW

- Energy expenditure (EE) measurements are used to determine total energy requirements and the energy requirements for specific physical activities.
- The gold standard for measuring total daily EE and validating other EE measurement techniques is the doubly labelled water method (Speakman 1998), which is expensive and requires specialised equipment and investigator training (Schoeller & Hnilicka 1996). It is primarily used for research.
- Traditional laboratory-based methods used to estimate EE for short-term and total daily EE include direct and indirect calorimetry. These methods are used mainly in research or physiological testing of elite athletes and are restricted to treadmill, cycle or rowing ergometers. They have limitations when applied to athletes in other sports. EE values obtained are unlikely to be a true reflection of physical activity undertaken in the field, particularly in those athletes who repeatedly accelerate, decelerate and change direction.
- For the estimation of total daily EE of an individual athlete or group of athletes, EE values from population reference data (i.e. Nutrient Reference Values (NRV)/Dietary Reference Intakes (DRI)) can be used as a guideline. These are crude estimates using predictive equations (based on basal metabolic rate (BMR) and physical activity levels (PAL)) and are derived from groups of people (commonly sedentary individuals) using direct or indirect calorimetry, or doubly labelled water (Institute of Medicine 2002).
- EE values are available in the NRV/DRI tables for a range of average activity levels. These values are crude estimates.
- Innovative technology has allowed the development of several new methods that are useful for estimating EE under field or real-life conditions. Examples include portable indirect calorimetry systems, motion sensors such as pedometers, accelerometers, global positioning systems, armbands and heart rate monitors. These methods of EE assessment, which are readily available to athletes, also offer sports dietitians affordable and feasible options to accurately quantify EE for specific and varying activities including sleep and rest or for total energy expended over time.
- Energy requirements are determined from measures of energy expenditure because the measurement error

is much less than measuring energy intake from dietary assessment methods.

- Prior to selecting the most appropriate method or device to use for an individual or groups of athletes, it is worthwhile to consider the following questions and clearly define the measurement variables of interest (see Table 5.2).

TABLE 5.2 Rationale for measuring energy expenditure of an athlete

What is the purpose of measuring the athlete's energy expenditure?	• Estimate energy requirements. • Measure average daily energy expenditure (>3–4 days). • Assess variation in total daily EE between training, competition and rest/recovery. • Modify body composition. • Determine energy cost of specific activities. • Assess sleep patterns.
What method of assessment is most appropriate for the athlete and their sport?	• EE assessment should be sport-specific. • When trying to determine the EE in a team-sport athlete, estimating an athlete's EE using a metabolic cart in a laboratory is not feasible or sport-specific. • When determining the EE by a cyclist completing a time trial, indirect calorimetry such as using a metabolic cart in a laboratory is the most accurate and sport-specific method.
How feasible/practical is the assessment?	• Is there a cost associated with the assessment? • Is specialist equipment required? • Is financial support and access to the equipment and a technician available?

CALCULATING ENERGY REQUIREMENTS WITH PREDICTIVE EQUATIONS

- The Cunningham equation is the best predictor for estimating RMR and allocation of a PAL value for calculating daily EE and hence total daily energy requirements in athletes, compared with other equations (see Chapter 5).
- The second-best predictor for estimating EE using equations is the Harris–Benedict equation. This can be used with minimal information including the athlete's gender, weight, height and age.
- The last resort is the factorial method. This method forms the basis for determining the Estimated Energy Requirements (EER) in the tables of the NRVs for Australia and New Zealand. The EER are based on the following predictive equation:

$$\text{EER (kJ/d)} = \text{BEE} \times \text{PAL} \times \text{TEF}$$

where BEE = basal energy expenditure (kJ), PAL = physical activity level and TEF = thermic effect of food.

- As this method determines BEE from another predictive equation (i.e. Schofield's equation) (Schofield et al. 1985) and an estimated daily PAL (see Chapter 5), it tends to underreport EE and has many limitations (Institute of Medicine 2002).
- When determining the PAL value, the highest values (i.e. 2.0–2.2) should be reserved for those individuals undertaking multiple training sessions/d or who have full-time labouring jobs and one or more training sessions/d. It is not appropriate for estimating EE of a specific activity or training session but can be used as a benchmark for total energy values or range of values for designing individual meal plans. Adding or subtracting kilojoules from the calculated energy requirements can assist with body composition manipulation.
- To enhance accuracy of PAL values, activity diaries can be kept, which involves recording all activities continuously (usually every 10 minutes) for up to 7 days. This task is time consuming for both athlete and clinician, requires athletes training and cross-checking to enhance accuracy and detailed analysis, and therefore is not commonly used.

CALCULATING ENERGY EXPENDITURE FROM OTHER METHODS

Respiratory exchange

- *Metabolic carts* are the most commonly used form of indirect calorimetry used in a laboratory setting. Metabolic carts calculate the EE during an exercise bout through measuring the rate of O_2 consumption and CO_2 production.
- *Portable indirect calorimetry systems* were developed as a method to assess O_2 consumption and CO_2 production outside a laboratory setting. Portable systems provide a valid and reliable calculation of EE in athletes who do not run in a straight line or ride cycle ergometers. For example, EE can be calculated during resistance training, during repeated sprints or during outdoor activities such as skiing (Doyon et al. 2001; Zanetti et al. 2013; Reeve et al. 2013).

Physical activity monitors

- *Heart-rate monitors* are commonly used by athletes to estimate EE and monitor the intensity of a training session. The relationship between heart rate and oxygen consumption is normally linear for workloads above BMR and below maximum work output (Livingstone et al. 1990). The validity of this technique and hence the EE estimate is questionable in athletes who train at high intensities. Other factors that affect the validity of heart-rate monitors include emotional stress, high ambient temperature and humidity, dehydration, posture and illness, and the volume of muscle mass being utilised (Ainslie et al. 2003). These factors can cause changes to an athlete's HR without increasing the volume of O_2 consumed, and therefore lead to an overestimation of EE.
- *Accelerometers* are objective measurement devices that can estimate how much energy an individual is expending and quantify the amount of time spent in light, moderate or vigorous physical activity (Crouter et al. 2006). One of the most common accelerometers used for assessing physical activity is called the Actigraph, which detects intensity, frequency and duration of physical activity. Several regression equations have been developed using this instrument to provide predictive estimates of EE; however, no single Actigraph equation is valid for the prediction of all activities. Actigraph and other similar accelerometers have a tendency to underestimate the predicted EE during higher-intensity physical activity (Crouter et al. 2006). Although Actigraph is the most popular accelerometer device used in research, there are many commercial devices available such as the FitBit, which have yet to be rigorously evaluated.
- *Global positioning systems* (GPS) technology was originally designed to provide location information on military personnel (Aughey 2011). These devices have been modified to quantify training loads and are attached to the athlete. Location is determined when signals are received from at least four earth-orbiting satellites. Information about position and time is recorded, which allows calculation of distance and speed. GPS watches have become popular among elite and amateur athletes. These allow conversion of measurements of time, location, speed and elevation into EE estimates. Recent research has suggested GPS watches underestimate EE compared with reference methods (Hongu et al. 2013).
- *Motion sensor technology* is a new and innovative method to objectively assess the EE of free-living individuals outside a laboratory setting (Chen et al. 2005). The SenseWear Armband is one such device that is worn on the upper arm and incorporates a tri-axial accelerometer with a range of heat sensors, that when combined into patented algorithms provides estimates of EE and physical activity levels including sleep (Johannsen et al. 2010). Although the device has good validity for calculating EE at rest and during low and moderate levels of physical activity, it underestimates EE during high-intensity activity (Drenowatz & Eisenmann 2011; Koehler et al. 2011; Zanetti et al. 2013). The SenseWear armband has demonstrated good validity during resistance training that combined both upper body and lower body exercise (Reeve et al. 2013); however, its validity for EE of lower-body movement (e.g. in cyclists) in isolation is questionable (Koehler et al. 2011; Fruin et al. 2004). These studies have demonstrated that the armband significantly underestimates EE when the predominant activities are lower limb, which may have implications for cyclists and athletes completing isolated lower-body activities (e.g. a lower-body resistance training session).
- Table 5.3 provides a summary of the features, advantages and disadvantage of the methods and devices available for measuring energy expenditure.

TABLE 5.3 Advantages and disadvantages of methods and devices available to estimate energy expenditure

Method	Features	Advantages	Disadvantages
Predictive equations	Based on general population data, these equations estimate an individual's energy requirements	• Low cost and can be implemented with basic information such as the athlete's age, weight, height and level of physical activity • Can estimate an athlete's EE independent of their sport	• The most accurate equation requires lean body mass that requires DXA scanning. DXA scanning requires the use of expensive equipment and a licensed technologist • Predictive equations are not as accurate as direct or indirect calorimetry
Metabolic carts	Indirect calorimetry used in a laboratory setting to estimate an individual's energy requirements from measuring the rate of O_2 consumption and CO_2 production	• Can be used to accurately determine respiratory exchange ratio in a laboratory setting and therefore identify the type of fuel (carbohydrate and fat) being oxidised • Is most valid and reliable for the EE assessment of athletes such as cyclists and runners, as sport-specific protocols can be accurately implemented	• Limited to the laboratory, therefore unable to assess *free-living* EE • Generally limited to treadmill and cycle ergometers, therefore will not be sport-specific for some athletes • Requires expensive equipment and qualified technicians • Metabolic carts are often found in university or high-performance sport science facilities, therefore not easily accessed by sports dietitians
Portable indirect calorimetry systems	Enable the assessment of EE through the measurement of O_2 consumption and CO_2 production outside a laboratory setting	• Portable systems enable EE to be determined more accurately outside the laboratory • Portable systems are deemed valid and reliable when compared to the metabolic cart and Douglas Bag systems (Duffield et al. 2004; McLaughlin et al. 2001) • Portable systems can be used for calculating EE in non-contact activities such as resistance training, skiing, open-road running	• Portable calorimetry systems are expensive and specialist knowledge is required to complete the assessment • Athlete cannot consume any fluids while wearing the device, therefore can impact on hydration status if EE is being assessed over a long duration • Have been reported to affect balance and peripheral vision (Reeve et al. 2013)
Heart-rate monitors	Utilise heart-rate measurement to estimate EE, which is based on the assumption that heart rate and O_2 consumption are in a linear relationship above BMR and below maximum work output	• Easy to operate, low cost and reusable • Provide accurate measurements of heart rate that can be used to provide a general picture of overall physical activity	• HR is affected by factors other than exercise, which do not necessarily increase oxygen requirements and therefore EE. This can lead to the overestimation of EE • HR will be higher in activities that only involve the upper limb, for example arm cycling. Due to the smaller muscle mass utilised in isolated upper body exercise, EE can again be overestimated
Accelero meters	Estimate how much energy an individual is expending by quantifying the amount of time spent in light, moderate or vigorous physical activity	• Low cost and reusable • Excellent means to evaluate interventions aimed at increasing physical activity	• Tend to underestimate EE during high-intensity exercise • Uni-axial accelerometers are not sufficiently sensitive to quantify EE

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			• Tri-axial accelerometers, which measure acceleration in the vertical, horizontal and mediolateral planes, provide more meaningful information regarding physical activity, however have low precision
Global positioning systems	Convert measurements of time, location, speed and elevation into energy expenditure estimates	• GPS watches are easy to operate, cost effective and provide information about location, speed and elevation	• Typically underestimate EE, therefore not an effective method of accurately assessing EE
Motion sensor technology	Incorporates a tri-axial accelerometer with a range of heat sensors that, when combined into patented algorithms and analysed using computer software, allow estimates of EE, physical activity levels and sleep	• Good validity and excellent reliability for everyday physical activity and EE information in the general population • Non-invasive and can be worn during free-living conditions • A study by Reeve and colleagues (2013) has demonstrated good validity and reliability of the commercial version of the SenseWear armband during resistance training • Operating these devices does not require specialist technicians	• The most recent update has demonstrated an underestimation of EE during high-intensity physical activity and overestimation of low-intensity physical activity in well-trained subjects • The SenseWear armbands are quite expensive

Note: DXA = dual energy X-ray absorptiometry, EE = energy expenditure, GPS = global positioning systems, HR = heart rate

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Commentary 2

Female athlete triad and energy availability

ANNE LOUCKS

In the female athlete triad (the triad), energy deficiency impairs reproductive and skeletal health. The particular type of energy deficiency in the triad is low energy availability; the particular type of reproductive disorder is functional hypothalamic menstrual disorders; and the particular type of skeletal impairment is the uncoupling of bone turnover, with an increased rate of bone resorption and a reduced rate of bone formation. Each component of the triad is understood to span the range from health to disease (Figure C2.1), with the population of athletes distributed across these spectra. Individually, athletes travel along these spectra in different directions at different rates in response to their changing diet and exercise habits. Energy availability can change in a day, but effects on menstrual function may not be noticed for a month, while effects on bone mineral density may not be measurable for 6 months or more. On any given day, an athlete's menstrual function reflects her recent energy availability, but her bone mineral density is the cumulative effect of her entire history of diet and exercise behaviour.

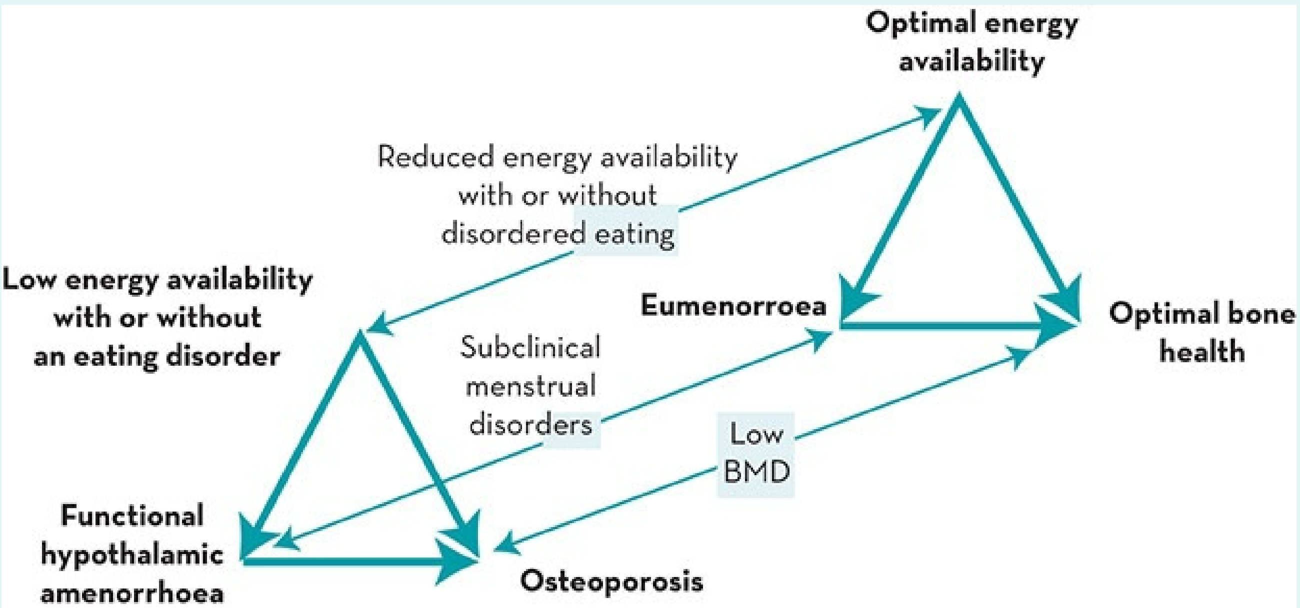

Figure C2.1 The spectra of the female athlete triad. Each component of the triad spans a range from health to disease, with low energy availability, functional hypothalamic amenorrhoea and osteoporosis at the pathological end of the spectra.
Source: Nattiv, Loucks et al. 2007

Components of the triad

The concept of energy availability derives from the recognition that mammals expend dietary energy in several basic physiological processes, including thermoregulation, cellular maintenance, immunity, growth, reproduction and locomotion. A large volume of observational and experimental evidence from humans and other mammals demonstrates that physiological processes depend on energy availability, and not on energy intake, energy expenditure, energy balance or energy stores (Wade & Schneider 1992; Wade & Jones 2004). For athletes, it is useful to define energy availability as the amount of dietary energy remaining after exercise training for all of the body's other physiological processes, and then to account for differences in body size and tissue energy expenditure by normalising this difference to fat-free mass (FFM). For healthy young adults, energy balance and resting metabolic rate occur at energy availabilities of about 45 and 30 kilocalories per kilogram of FFM per day (188 and 125 kJ/kgFFM/d).

Functional hypothalamic menstrual disorders result from the functional response of certain neurons in the hypothalamus of the brain to information about energy availability received from various neural and hormonal signals. Because menstrual disorders can also be symptoms of many medical conditions, menstrual disorders should not be assumed to be functional hypothalamic menstrual disorders. Athletes with menstrual disorders should be properly diagnosed through a series of hormone measurements and neuroendocrine stimulation tests that can exclude other conditions (ASRM 2008). Menstrual disorders may be clinical or subclinical. Because clinical menstrual disorders (i.e. amenorrhoea and oligomenorrhoea) have symptoms, they can be detected by monitoring menstrual cycles. Athletes with subclinical menstrual disorders (i.e. anovulation and luteal phase deficiency) are unaware that they have a menstrual disorder, because they menstruate regularly. Subclinical menstrual disorders are only detected by measuring hormone concentrations. Subclinical menstrual disorders were found in one or more of three consecutive menstrual cycles of almost 80% of regularly menstruating runners (De Souza et al. 1998).

Like other tissues, bone constantly turns over as it grows, repairs itself and serves as an exchangeable store of minerals. A decline in oestrogen, after menopause or for any other reason, releases its inhibition of osteoclast activity, increases the rate of bone resorption, reduces bone mass and increases the risk of fracture. In addition to suppressing oestrogen levels, however, low energy availability also suppresses anabolic hormones that promote bone formation. Such combinations of an increased rate of bone resorption and reduced rate of bone formation can cause irreversible bone loss (Compston 2001), predisposing women to stress fractures in the near term and to the premature onset of osteoporosis later in life. If this occurs during adolescence, a girl can become osteoporotic even while her bone mass is increasing, as she falls further and further behind her peers in accruing bone mass.

When physicians measure bone mass, they measure the mass of bone mineral, but bone is composed of equal parts of mineral and protein by volume, so that the skeleton comprises the largest store of protein in the body after skeletal muscle. In response to low energy availability, the body reduces energy expenditure by suppressing the synthesis of new protein and by mobilising energy stored in existing protein and fat. Protein stores are mobilised for gluconeogenesis to support the high energy needs of the brain, because fatty acids do not cross the blood–brain barrier. Recent direct measurements of the rate of bone protein synthesis in bone biopsies have found it to be 40 times greater than the previously assumed equilibrium resorption rate indicated by plasma and urine concentrations of peptides used as markers of bone resorption (Babraj et al. 2005). This finding suggests that most bone protein resorption products are further metabolised and utilised rather than excreted. Like skeletal muscle, therefore, the skeleton appears to function as an exchangeable store of protein as well as minerals.

From this perspective, the skeletal demineralisation that occurs during energy deficiency appears to be only a side effect of skeletal deproteinisation: it is the price of survival.

Reproductive and skeletal mechanisms of low energy availability

The physiological mechanisms linking the components of the triad operate in men as well as women, and also in other mammals, but the mammalian dependence of reproductive function on energy availability operates principally in females (Bronson 1985), and for the reasons described below, more women than men engage in diet and exercise behaviours that reduce energy availability. Furthermore, even though severe dietary restriction alone is sufficient to disrupt reproductive function, the more physically active a woman is, the less dietary restriction is required, and if she expends enough energy in exercise, she does not need to restrict her diet at all (Loucks et al. 1998).

Ovarian function critically depends on the frequency with which pulses of luteinising hormone (LH) are secreted into the bloodstream by the pituitary gland. In turn, LH pulse frequency depends on the frequency with which certain neurons in the hypothalamus of the brain secrete pulses of gonadotropin-releasing hormone into a system of portal vessels between the hypothalamus and the pituitary gland. LH pulsatility has been disrupted in healthy women by extreme dietary restriction alone (Loucks and Heath 1994), by extreme exercise energy expenditure alone (Loucks et al. 1998), and by the combination of moderate amounts of both (Loucks & Thuma 2003). Moreover, LH pulsatility has been preserved in women subjected to an extreme exercise regimen by increasing their dietary energy intake in compensation for their exercise energy expenditure (Loucks et al. 1998). Similarly, amenorrhoea has been induced in monkeys by increasing their exercise energy expenditure without restricting their dietary energy intake (Williams et al. 2001), and then their menstrual cycles have been restored by increasing their dietary energy intake without moderating their exercise regimen (Williams et al. 2001). These experiments demonstrate that exercise has no suppressive effect on reproductive function apart from the impact of its energy cost on energy availability.

Figure C2.2 illustrates the dose–response effects of energy availability on LH pulse frequency and plasma glucose with the relative magnitudes of effects normalised to the maximal responses (Loucks & Thuma 2003). LH pulse frequency and plasma glucose are suppressed together below a threshold of energy availability at 30 kcal/kgFFM/d (125 kJ/kgFFM/d). By comparison to this 33% reduction from energy balance, amenorrhoeic athletes have been reported to practise diet and exercise regimens that reduce energy availability by as much as 65% (Thong et al. 2000).

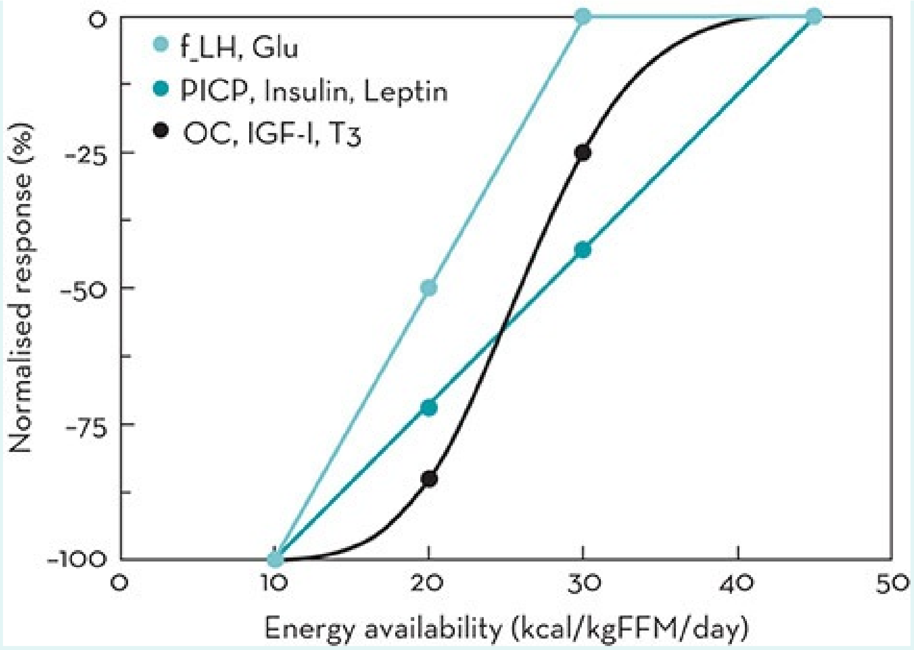

Figure C2.2 Dose – response effects of energy availability on LH pulse frequency (f_LH), plasma glucose (Glu), markers of bone protein synthesis (PICP) and mineralisation (OC), and anabolic hormones that regulate bone formation (insulin, leptin, tri-iodothyronine (T_3), and insulin-like growth factor-1 (IGF-1)). Created from data in Loucks & Thuma 2003; Ihle & Loucks 2004).

Figure C2.2 also shows the dose–response effects of energy availability on certain biochemical markers of bone formation and on anabolic hormones that regulate it. The rate of bone protein (i.e. type I collagen) synthesis by osteoblast cells is indicated by the concentration in the blood of a peptide known as type I procollagen carboxyterminal propeptide (PICP), which is cleaved off the precursor type I procollagen molecule during collagen synthesis. PICP declines linearly with energy availability in parallel with the decline in insulin, which stimulates osteoblast differentiation (Lu et al. 2003). The rate of bone protein mineralisation is indicated by the concentration in the blood of the protein osteocalcin (OC), which is secreted by osteoblasts and functions as a glue binding the mineral to the protein matrix. OC declines non-linearly with energy availability, with most of the decline occurring between 20 and 30 kcal/kgFFM/d, in parallel with the non-linear responses of insulin-like growth factor-1 (IGF-1) and tri-iodothyronine (T_3). OC secretion is mediated by IGF-1, the hepatic production of which is stimulated by growth hormone and modulated by T_3.

Peripheral signals of low energy availability—such as falling insulin, rising ghrelin, rising peptide YY (PYY) and falling leptin levels—also act centrally to activate neuropeptide-Y secreting neurons in the hypothalamus, which act via sympathetic pathways on osteoblastic Y1 receptors to repress osteoblast activity and bone formation throughout the skeleton (Shi & Baldock 2012). The rate of bone resorption also displays a diurnal rhythm that peaks during sleep and declines during the day. The reduction during the day is greater with feeding (Schlemmer & Hassager 1999), and the rate even declines acutely in response to a single meal (Scott et al. 2012). These acute responses are mediated by incretin hormones (i.e. glucose-dependent insulinotropic peptide, glucagon-like peptide 1 and glucagon-like peptide 2) secreted by K-cells and L-cells in the lining of the small intestine in response to the changing contents of the gut (Yavropoulou & Yovos 2013). Incretin hormones act on osteoblasts and osteoclasts to favour formation within minutes after a meal and resorption later.

Origins of low energy availability in athletes

In athletes, low energy availability derives from three sources that might be described as obsessive, intentional and inadvertent. The obsessive source is anorexia nervosa and other restrictive eating disorders; the intentional source is cognitive efforts to reduce weight or body fat; and the inadvertent source is the suppression of appetite by diet and exercise. Anorexia nervosa is a clinical mental illness, often accompanied by other mental illnesses (Klump et al. 2009). It requires psychiatric treatment, sometimes unwilling inpatient treatment with forced feeding (Carney et al. 2008), because it has one of the highest risks of premature death of any mental illness (Harris & Barraclough 1998) with a mortality 10 times higher than that of age- and sex-matched peers (Birmingham et al. 2005). Sixty per cent of deaths in anorexia nervosa are due to medical consequences of the disease, for which the mortality risk is increased four times (Harris & Barraclough 1998). The other 40%, due to accident, misadventure, homicide and suicide, are increased 11 times, and the specific risk of suicide is increased 32 times (Harris & Barraclough 1998).

Because the mortality of anorexia nervosa is so high, and because sports dietitians, coaches and team physicians are not competent to care for clinical mental illnesses, they should not be expected, and should not attempt, to manage them. Instead, sports organisations should establish procedures for identifying athletes who may have anorexia nervosa, for referring them for psychiatric evaluation and care, and for excluding them from participation until they receive psychiatric clearance. Energy-deficient athletes who do not cooperate with recommendations to modify diet and exercise behaviour to increase energy availability should be referred for such evaluation.

Sports dietitians should focus their efforts on helping cooperative athletes to improve their performance. Athletic performance is improved, in part, by acquiring an optimum sport-specific (and, in team sports, position-specific) body size, body composition and mix of energy stores. For many female athletes, these objectives may include a reduction in fat mass that will require an intentional reduction in energy availability that may place their reproductive and skeletal health at risk. That risk should be clearly understood by sports dietitians, fully disclosed to athletes and coaches, and carefully minimised through the cooperation of all involved.

Athletes reduce their weight or body fat by reducing their energy availability, either by reducing their dietary energy intake or by increasing their exercise energy expenditure. Many female athletes do both, but athletes in aesthetic sports tend to emphasise dietary restriction, while high energy expenditure is inherent in endurance sports. Some poorly informed athletes may also practise the same disordered eating behaviours practised by women with anorexia nervosa (e.g. skipping meals, fasting, vomiting, using laxatives) in impatient pursuit of unhealthful objectives (Rauh et al. 2010; Thein-Nissenbaum et al. 2011). Part of the job of sports dietitians is to help rational athletes to correct these errors.

Sports dietitians should keep in mind that before female athletes are athletes they are female. Worldwide, about twice as many young women as young men *at every decile of body mass index* perceive themselves to be overweight, and the numbers of women and men actively trying to lose weight are even more disproportionate (Wardle et al. 2006). The disproportion even *increases* as BMI declines so that almost nine times as many lean women as lean men are actively trying to lose weight (Wardle et al. 2006). Indeed, more young female athletes report improvement of appearance than improvement of performance as a reason for dieting (Martinsen et al. 2010). This means that social issues may need to be addressed in order to persuade energy-deficient athletes to increase their energy availability.

Even after women with eating disorders have been excluded from participation, and even when weight or fat loss is not a training objective, athletes of both sexes are susceptible to inadvertent reductions in energy availability due to the suppression of appetite by diet and exercise (Loucks et al. 2011). This hazard is not described in the current joint position stand of the American College of Sports Nutrition, the American Dietetic Association and Dietitians of Canada on sports nutrition (Rodriguez et al. 2009). Briefly, after exercise, ad libitum energy intake does not increase sufficiently to compensate for exercise energy expenditure, and athletes who expend the most energy are susceptible to the largest deficiencies. Appetite is also suppressed by diets containing a high percentage (e.g. 65%) of carbohydrates, with the counter-intuitive consequence that diets containing such high percentages of carbohydrates can reduce ad libitum carbohydrate intake and athletic performance. These effects of diet and exercise are large and additive so that in endurance sports, appetite and ad libitum energy intake can be suppressed enough to reduce energy availability below 30 kcal/kgFFM/d (125 kJ/kgFFM/d). Therefore, sports dietitians should teach endurance athletes that appetite is not a reliable indicator of energy requirements, and train them to eat according to a meal plan with specified amounts of selected foods at scheduled times.

Summary

Athletic performance is best improved, and reproductive and skeletal health best preserved, by managing energy availability in a periodised training program that aims to achieve period goals on the way to long-term objectives. Some of these goals will call for energy availability to be raised and others will call for it to be lowered, but there is a level below which energy availability should not be reduced. Sports dietitians have important roles to play in helping athletes with eating disorders to receive appropriate care, in helping other athletes to achieve weight and fat loss objectives in a healthy manner, and in helping athletes in endurance sports to avoid inadvertent energy deficiency.

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Commentary 3

Measuring energy availability

ANNA MELIN AND BRONWEN LUNDY

Introduction

As outlined in Commentary 2, energy availability (EA) is defined as the ingested energy (EI) remaining for all other metabolic processes after the energy cost of exercise (EEE) has been subtracted. The concept of EA is valuable in understanding the underlying causes of the syndrome of the female athlete triad (the triad), or the more recently named umbrella term of Relative Energy Deficiency in Sports (RED-S) in the International Olympic Committee's Consensus Statement (Mountjoy et al. 2014). EA helps to illustrate the impact of inadvertently failing to balance energy expenditure (EE) with adequate EI for performance and health. There are, however, some challenges to its measurement such as the ability to capture habitual EA, the ability to discriminate between eating disorders/disordered eating behaviour and under-reporting when performing nutritional surveys, and determining which definition and methods to use when calculating EEE. At the time of preparing this commentary, a standardised protocol for measuring EA in athletes has yet to be established. The goal of this commentary is, therefore, to discuss some of the issues involved in determining EA in field conditions including research and counselling scenarios. It will first discuss the measurement of the factors involved in calculation of EA: fat free mass (FFM), EI and EEE. In light of the difficulties undertaking these measures, and the potential errors in the calculation of EA, it will discuss other parameters that may indicate the presence of energy deficiency.

Scenarios in which EA has been measured

The concept of EA was initially established in contrived laboratory conditions where EEE and EI were controlled to induce stepwise reductions in EA with the purpose of identifying the levels of EA at which metabolic and endocrine systems were affected (see Commentary 2). In healthy eumenorrhoeic weight-stable sedentary women, the average EA has been reported to be ~188 ± 25 kJ (45 kcal)/kg FFM/d (Loucks et al. 1998; Loucks & Thuma 2003), while 5 days of EA <125 kJ (30 kcal)/kg FFM/d has been shown to reduce blood glucose levels and to suppress the pulsatility of gonadotropin-releasing hormone (GnRH), and hypothalamic-pituitary-axis hormones like triiodothyronine (T_3), luteinizing hormone (LH), and to elevate cortisol (Loucks et al. 1994), defining low EA (Table C3.1). If maintained for a long time, low EA can lead to oligomenorrhoea or functional hypothalamic amenorrhoea (FHA) (Cumming et al. 2001; Loucks et al. 2003). Physically active women are recommended to have an EA of at least 188 kJ (45 kcal)/kg FFM/d to ensure adequate energy for all physiological functions (De Souza et al. 2014). Subclinical EA has been suggested to be an EA less than 188 kJ (45 kcal)/kg FFM/d (Gibbs et al. 2013b), and it is recommended that athletes aiming to lose body weight or body fat follow a diet and exercise regimen that provides EA of 125–188 kJ (30-45 kcal)/kg FFM/d (Loucks et al. 2011).

TABLE C3.1 Defining low energy availability

Energy availability (kJ/kg FFM/d)	Zone	Comments
≥188	Optimal	Ensure adequate energy for all physiological functions (De Souza et al. 2014)
125–188	Reduced or subclinical	May be tolerated for short periods such as a well-constructed weight-loss program (Loucks et al. 2011)
<125	Low	There is a high risk of inadequate energy support for many body functions leading to impairment of health and performance (Mountjoy et al. 2014; De Souza et al. 2014)

FFM = fat-free mass

Several studies have investigated EA in female athletes using a variety of techniques to measure this characteristic (Table C3.2). Some studies have reported current EA related to reproductive function (Doyle-Lucas et al. 2010; Lagowska et al. 2014; Van Heest et al. 2014; Melin et al. 2014a). There are, however, great variations in the studies and overlap between the groups in the reported EA: 3–161 kJ/kg FFM/d in subjects with MD and 24–178 kJ/kg FFM/d in eumenorrhoeic subjects. Regardless of the major differences in the reported EA, these results indicate that the reproductive function in some women might be more robust or more sensitive than it is in others (Gibbs et al. 2013a; Ciadella-Kam et al. 2014). Furthermore, the threshold for unsafe low EA established by Loucks and Thuma (2003) in a clinical setting may be higher or not similarly manifest in athletes under free-living conditions using self-reporting and field methods to determine EA (Ciadella-Kam et al. 2014; Mountjoy et al. 2014; Gibbs et al. 2013a).

TABLE C3.2 Studies that have measured energy availability in athletes

		Methods						
Reference	**Subjects**	**Energy intake**	**Exercise energy expenditure**	**Exercise definition**	**DE**	**Reproductive function**	**BMD**	**Remarks**
Hoch et al. 2009	Mixed female athletes (*n* = 80), Sedentary controls (*n* = 80)	3 days prospective weighed dietary record	Estimated by activity logs and calculated using METs	Sport participation	EAT-26	Menstrual history and sex hormones	Yes	EEE not adjusted f FFM dete via dual-er x-ray absorption
Doyle-Lucas et al. 2010	Female ballet dancers (*n* = 15), Recreationally active non-dancer controls (*n*= 5)	4 days prospective dietary records	Estimated by activity logs and calculated using METs	All PA	TFEQEAT-26	Menstrual History	Yes	EEE not adjusted f RMR mea FFM dete via dual-er x-ray absorption
Hoch et al. 2011	Professional female ballet dancers (*n* = 22)	3 days prospective dietary record	Simultaneously estimated by accelerometers, individually calibrated at two self-selected intensities of exercise	All PA	EDE-Q	Menstrual history and sex hormones	Yes	Low/negat defined as negative v EEE not adjusted f Endotheli function (F determine not adjust FFM
Schaal et al. 2011	Runners and triathletes; MD (*n* = 6), EUM (*n* = 6)	7 days prospective weighed dietary record	Simultaneously estimated by activity logs, HR and intensity using the 10-point RPE scale EEE was estimated on the basis of RPE and HR during exercise to matched O_2 consumption and RER during laboratory testing	Exercise verified by RPE	EDE-Q	Menstrual history and diagnosis verified by physician	-	EEE not adjusted f Catechola and BP measured questionn FFM dete via DXA
Reed et al. 2013	Female soccer players (*n* = 19)	3 × 3 days prospective dietary records	Simultaneously estimated using HR during team exercise	Team exercise sessions and	EDI	Menstrual history	-	T_3 measu FFM was determine dual-energ

			sessions HR and activity logs during individual exercise METs for exercise sessions without HR monitoring	purposeful exercise sessions				absorption
Koehler et al. 2013	Athletes from mixed sports; Men (n = 160), Women (n = 185)	7 days prospective dietary record with 193 foods listed with standard portion sizes.	Simultaneously estimated using activity records, including a list of exercise-related activities with different intensity levels and SA and METs	Exercise-related activities	-	-	–	EA correc with age-s factors for Leptin, IG and insulin measured determine bioimpeda
Woodruff & Meloche 2013	Volleyball players (n = 10)	2 × 7 days prospective dietary records	Simultaneously assessed from accelerometers and METs, based on dates and times of practices, warm-ups, and games	Volleyball practice, warm-ups, and games	–	Menstrual history	–	EEE not adjusted f FFM deter via air-displacem plethysmo
Lagowska et al. 2014	Mixed athletes with MD (n = 45)	7 days prospective dietary record using a photographic album of dishes	HR monitors for 3 days. For each subject, the relationship between HR and VO_2 was established for lying in supine position, sitting quietly, standing quietly, and continuous graded exercise on a cycle ergometer	Exercise	–	Menstrual history, sex hormones and gynaecological ultrasound examination	–	EEE not adjusted f RMR mea FFM dete via bio impedanc
Van Heest et al. 2014	Female swimmers: MD (n = 5) EUM (n = 5)	7 × 3 days prospective weighed dietary records	7 days exercise logs and swimming specific EE table	Swimming practice	–	Prospective menstrual cycle record during 12 weeks, sex hormones at 0 and 2 weeks	–	EEE not adjusted f Performar assessed 12-week t program R T_3, IGF-1 measured calculated skinfold assessme
Melin et al. 2014a	Female elite endurance athletes MD (n = 24) EUM (n = 16)	7 days prospective weighed dietary record	Simultaneously estimated by activity logs and HR. EEE was estimated on the basis of HR	Exercise	EDI and EDE	Menstrual history, sex hormones and gynaecological ultrasound examination	yes	RMR, gluc insulin, lep IGF-1, cor T_3 and blo lipids mea

05:34:04.

			during exercise matched O_2 consumption and RER during laboratory testing						FFM dete[r] via DXA

[a] $p < 0.05$

BP = blood pressure, BMD = bone mineral density, DE = disordered eating behaviour, DXA = dual-energy X-ray absorptiometry, EA = energy availability, EAT-26 = eating attitudes test-26, EE = energy expenditure, EDE = eating disorder examination, EDE-Q = eating disorder examination questionnaire, EEE = exercise energy expenditure, EUM = eumenorrhoea, FFM = fat-free mass, FMD = flow mediated dilation, HR = heart rate, IGF-1 = Insulin growth factor-1, MD = menstrual dysfunction, MET = Metabolic Equivalent of Task, PA = physical activity, POMS = profile of mood states, RMR = resting metabolic rate, RER = respiratory exchange ratio, RPE = ratings of perceived exertion, SA = sedentary activities, T_3 = Triiodothyronine, TFEQ = Three-Factor Eating Questionnaire

Assessing energy intake

Studies assessing habitual EI as part of an EA assessment have used diet records, either weighing food or estimating portions for between 3 and 7 days. However, there are known errors when measuring dietary intake and these are compounded by the wide between-person and within-person variability in EI (see Chapter 2). For example, to capture seasonal variability in dietary intake, Reed and colleagues (2013) used three separate recording periods to represent different stages of the season and found significant differences in both EI and EEE at recording points. Compliance with diet records is poor after 4 days (see Chapter 2), but only some of the studies reporting EA have actually assessed dietary record validity (Schaal et al. 2011; Woodruff & Meloche 2013; Melin et al. 2014a) using the standard Goldberg or Black cut-offs (Black et al. 1991; Black 2000).

Assessing exercise energy expenditure (EEE)

One of the challenges in the calculation of EEE includes lack of access to valid data on the energy cost of exercise. Furthermore, the intention of the originator of the concept, Professor Anne Loucks, was that EEE should only represent the amount of additional energy that has been expended as a result of exercise activity (Loucks et al. 1998; Loucks 2014). Thus, the energy cost of daily living during the period of exercise should be subtracted from the estimates of EEE (Loucks et al. 1998). The failure to do this subtraction might only create a small problem in athletes who do shorter sessions of high-intensity work, but might create a major overestimation of the EEE and, thereby, underestimate EA in athletes who do many hours of low-intensity training, illustrated in two examples given in Table C3.3.

TABLE C3.3 Examples of calculations of energy availability

Athlete A: distance runner		
Feature	**Comments on measurement**	**Value**
Body mass	Measured on scales on waking	55 kg
Fat free mass	DXA derived value of total mass minus fat mass	48 kg
Energy intake	Average daily intake derived from 7-day food diary	12.5 MJ
Exercise energy expenditure (Energy cost of doing exercise = total EE during exercise minus cost of sedentary behaviour during exercise period)	Nine sessions per week running = total of 11 hours; EEE derived from HR monitor = 33.5 MJ for 11 hours (~3 MJ/h) EE during 11 hours for sedentary behaviour =~4.2 MJ (estimated from prediction equations) EEE = 33.5 MJ – 4.2 MJ = 29 MJ over 7 days = 4.2 MJ/d	4.2 MJ
Energy availability	EA = (EI – EEE)kJ/FFM (kg) = (12.500 – 4.200)/48	~173 kJ/kg FFM/d
Comments	Note that the failure to remove sedentary activities during training sessions would have changed the calculation of EA in a moderate way: 33.5 MJ per week = EEE of 4.8 MJ/d: EA = 161 kJ/kg FFM/d.	

Athlete B: Dancer		
Feature	**Comments on measurement**	**Value**
Body mass	Measured on scales on waking	55 kg
Fat free mass	DXA derived value of total mass minus fat mass	48 kg
Energy intake	Average daily intake derived from 7-day food diary	10.5 MJ
Exercise energy expenditure (Energy cost of doing exercise = total EE during exercise minus cost of sedentary behaviour during exercise period)	Nine sessions per week dancing = total of 22 hours; EEE derived from HR monitor = 33.5 MJ for 22 hours (~1.5 MJ/h) EE during 22 hours for sedentary behaviour = ~8.4 MJ (estimated from prediction equations) EEE = 33.5 – 8.4 = 25.0 MJ over 7 days = 3.6 MJ/d	3.6 MJ
Energy availability	EA = (EI – EEE)kJ/FFM (kg) = (10500 – 3600)/48	~144 kJ/kg FFM/d
Comments	Note that the failure to remove sedentary activities during training sessions would have changed the calculation of EA in a substantial way: 33.5 MJ per week = EEE of 4.8 MJ/d EA = 119 kJ/kg FFM/d.	

Note: Values have been rounded to simple numbers to allow easier understanding.
DXA = dual-energy X-ray absorptiometry, EA = energy availability, EE = energy expenditure, EEE = exercise energy expenditure, EI = ingested energy, HR = heart rate

The method used to estimate EEE in the existing literature varies. While some studies have used training records and heart-rate monitors and have estimated EEE on the basis of heart rate in relation to O_2 consumption and the respiratory exchange ratio during laboratory testing (Schaal et al. 2011; Lagowska et al. 2014; Melin et al. 2014a), others have used accelerometers monitoring bodily movements (Hoch et al. 2011; Woodruff & Meloche 2013). A frequently used method, although less precise, is prospective activity logs to calculate EEE from the corresponding Metabolic Equivalent of Task (MET)

(Ainsworth et al. 2000) or other tables of EEE (Hoch et al. 2009; Koehler et al. 2013; Van Heest et al. 2014; Doyle-Lucas et al. 2010), while Reed and colleagues (2013) used a combination of activity logs and heart rate monitors and/or accelerometers. The recommended adjustment of EEE for the EE of sedentary activity, which would have occurred instead of exercise (Loucks et al. 1998), has only been performed in a few studies (Reed et al. 2013; Koehler et al. 2013; Melin et al. 2014a).

Another methodological issue when assessing EA is how to define exercise. In the existing literature, different terminology and definitions have been used, such as all sport or exercise sessions/activities (Hoch et al. 2009; Schaal et al. 2011; Koehler et al. 2013; Van Heest et al. 2014; Melin et al. 2014a), physical activity (Hoch et al. 2011), or all physical activity except daily living activities (e.g. cleaning the house or walking the dog) (Reed et al. 2013).

All methods used to calculate EEE, including Sensewear™ (see Chapter 5, Practice tips), provide estimates of EE and EEE. If measuring EEE in athletes, the precision and accuracy of heart-rate monitors and accelerometers during specific training such as running at different speeds, rowing or dancing can be improved by simultaneously measuring EEE by indirect calorimetry (Horner et al. 2011), since accelerometers may underestimate EEE at more rigorous exercise levels (Abel et al. 2008). As athletes often cross-train in addition to their primary sport, the methods used to estimate EEE should be applicable to all types of exercise. In practice, when estimating the EEE of individual athletes, the same method and the same monitoring device should be used to check relative differences using repeated measures.

Calculation of fat-free mass

The different techniques for calculating FFM, including hydrostatic weighing, bioelectrical impedance, BodPod and dual energy X-ray absorptiometry, are found in Chapter 3. The different methods used to determine FFM in the studies reporting EA in female athletes are presented in Table C3.2. A high standard of body composition assessment is required as over or underestimation of FFM distorts the estimation of EA.

Screening for symptoms of relative energy deficiency

As an alternative or complement to measuring EA, there may be benefits to screening athletes for physiological symptoms frequently associated with symptoms of relative energy deficiency (RED-S). The following measurements may be useful.

Presence of menstrual dysfunction

The causal link between low energy and glucose availability and the endocrine perturbation related to oligomenorrhoea/FHA, especially in young women, is well documented (Loucks 2006; Mountjoy et al. 2014; De Souza et al. 2014). The presence of oligomenorrhoea/FHA has, furthermore, been reported to be a good predictor of eating disorders/disordered eating behaviour among elite athletes (Sundgot-Borgen et al. 2007) and could therefore be used in identifying female athletes at risk for RED-S.

Menstrual dysfunction is, as described in Chapter 8, common among female athletes, especially in leanness-demanding sports—for example, 60–63% in endurance athletes (Pollock et al. 2010; Melin et al. 2014a)—but is often ignored, despite the increased risk of premature osteoporosis (Nattiv et al. 2007; Mountjoy et al. 2014). Polycystic ovarian syndrome (PCOS) is, as oligomenorrhoea/FHA, also reported in female elite athletes (Hagmar et al. 2009; Melin et al. 2014b). Since the aetiology of PCOS is not related to energy deficiency, a clinical diagnosis of oligomenorrhoea/FHA requires a pertinent evaluation of sex hormones and ultrasound examination by a skilled gynaecologist in order to establish if the origin of the menstrual dysfunction is due to a suppression of the hypothalamic pituitary ovarian axis (De Souza et al. 2014).

Reduced resting metabolic rate (ratio of mRMR:pRMR)

When EI is inadequate for maintenance of all basal physiological processes, the allocation is prioritised to those processes that are essential for immediate survival (Wade & Jones 2004). Most female athletes are lean, but have a body weight and body composition within the normal range, independent of their reproductive function (Redman & Loucks 2005). Body weight therefore seems to be preserved during long-term low or reduced EA involving several metabolic mechanisms such as a reduction in resting metabolic rate (RMR) (Redman et al. 2009; Shetty 1999) and a stable body weight, therefore, might be an unreliable indicator of the level of EA in female athletes.

Some studies show that RMR changes up to 10% during the menstrual cycle (Solomon et al. 1982; Meijer et al. 1992; Henry et al. 2003), with an increase in the last part of the luteal phase (7–10 days before menstruation), followed by a sudden drop in the early follicular phase (the first days of bleeding), with a gradual return to normal levels within 7–10 days (Solomon et al. 1982). The change in RMR is suggested to be mediated by the changes in progesterone (Solomon et al. 1982); additionally, Dalvit (1981) reported that the increased EE in the luteal phase is compensated for by an increase in food intake during the 10 days following, in the follicular phase (Dalvit 1981). Henry and colleagues (2003) measured RMR in 19 premenopausal women throughout the complete menstrual cycle and demonstrated an inter-individual variation of 12% and a wide difference in the intra-individual variation (CV range 2–10%). For 10 subjects, the variability in RMR was small (CV 2–4%), while the CV in nine subjects was high (CV 5–10%), indicating a significant variation in RMR during the menstrual cycle in certain subjects (Henry et al. 2003).

Basal metabolic rate (BMR) can be predicted using standard equations such as the Harris–Benedict (Harris & Benedict 1919) or the Cunningham (including FFM) (Cunningham 1980). Since FFM is the main determinant of RMR (Speakman & Selman 2003), the Cunningham equation has been found to be the best predictive equation for RMR in female athletes (Thompson & Manore 1996). This is covered in more detail in

Chapter 5.

The ratio (RMR_{ratio}) between the measured RMR (mRMR) and the predicted RMR (pRMR) related to FFM has been reported to be low in patients with anorexia nervosa (0.60 to 0.80) (Platte et al. 2000; Marra et al. 2002) and an RMR_{ratio} <0.90 has been used in some studies with recreationally active women as an indicator of low EA instead of assessing EA (De Souza et al. 2007; De Souza et al. 2008; Schneid et al. 2009; Gibbs et al. 2011).

In the laboratory, RMR is measured, after an overnight fast, using a ventilated open hood system and calculated from the VO_2 and VCO_2 (Weir 1990), as described in Chapter 5. Direct measurement of RMR in an individual athlete will provide a specific estimate, but may be biased by:

- *A potential acute increase in RMR in response to high-intensity exercise* (Westerterp et al. 1992) However, in highly trained endurance athletes in energy balance, RMR can remain elevated for 1–2 days after training, compared to sedentary subjects matched for FFM (Sjodin et al. 1996). Therefore, RMR in elite athletes might be more or less constantly elevated compared to non-elite athletes.
- *Lack of standardisation of measurements* Since RMR in some individuals changes significantly during the different phases of the menstrual cycle (Solomon et al. 1982), measurements undertaken in eumenorrhoeic athletes need to be standardised. Some of the studies reporting RMR in eumenorrhoeic athletes and recreation active women have performed the measurements in the early follicular phase (day 0–6 of the menstrual cycle) when the RMR is as lowest (Thompson & Manore 1996; De Souza et al. 2008; O’Donnell et al. 2009; Schneid et al. 2009; Melin et al. 2014a), while others have used a longer span (day 0–10 of the menstrual cycle) (Myerson et al. 1991; Biedleman et al. 1995; Lebenstedt et al. 1999; Doyle-Lucas et al. 2010; Guebels et al. 2014).
- *Athletes with menstrual dysfunction may have metabolic adaptation resulting in lower RMR* Several studies confirm that RMR in eumenorrhoeic active women and athletes is higher than in active women and athletes with oligomenorrhoea/FHA (124–134 kJ/kg FFM/d versus 110–124 kJ/kg FFM) (Lebenstedt et al. 1999; Reed et al. 2011; Thompson et al. 1996; Doyle-Lucas et al. 2010; Melin et al. 2014a). In a group of female elite endurance athletes, we recently reported a lower RMR_{ratio} in subjects with current low or reduced EA as well as in subjects with oligomenorrhoea/FHA compared to subjects with current optimal EA and eumenorrhoeic subjects (Melin et al. 2014a). Therefore, athletes with MD (Loucks et al. 2011) as well as those with current energy deficiency might have metabolic adaptation resulting in a lower RMR than expected.

Therefore, a lowered RMR_{ratio} seems to be a useful biomarker of both current as well as long-term energy deficiency that could be used to identify athletes at risk for RED-S.

However, when assessing RMR in athletes, a standardised protocol including time limits for exercise and food intake the day prior to assessment is needed. Furthermore, when assessing RMR of female athletes, measures should be taken at the same time in the menstrual cycle (e.g. early follicular phase) to control for variability.

Drive for Thinness score

Disordered eating, thoroughly described in Chapter 8, increases the risk for restricted EI and excessive EEE and exists in male and female athletes (Filaire et al. 2007; Goltz et al. 2013; Bratland-Sanda & Sundgot-Borgen 2013; Mountjoy et al. 2014). Several studies have found elevated Drive for Thinness score (DT) measured by the self-reported questionnaire Eating Disorder Inventory-2 (EDI) or increased dietary restraint in athletes with menstrual dysfunction compared to eumenorrhoeic athletes (Lebenstedt et al. 1999; Warren et al. 1999; Cobb et al. 2003; Gibson et al. 2004; Nichols et al. 2006; Reed et al. 2011; Gibbs et al. 2011). De Souza and colleagues used an elevated DT as a proxy indicator for energy deficiency in exercising women and found no difference in the current EI between exercising women with elevated versus normal DT levels, while the prevalence of MD was higher in women with an elevated DT compared to those with a normal DT (De Souza et al. 2007). These observations indicate that athletes with oligomenorrhoea/FHA are more concerned with weight. However, the DT is often not pathologically high; therefore, to only screen athletes for disordered eating behaviour is probably not sufficiently sensitive as an indicator of athletes at risk for RED-S.

Other biomarkers

A prospective study on young female athletes reported an increased risk for injuries in those athletes with restricted eating behaviour and oligomenorrhoea/FHA (Rauh et al. 2010). The authors concluded that restricted eating behaviour and oligomenorrhoea/FHA also increase the risk for impaired bone health and stress fractures as well as muscular and joint overload. Screening for recurrent injuries could, therefore, be useful in identifying athletes at risk for RED-S.

Irregular meal patterns, dehydration, high fibre intake and inflexibility in food choice are commonly reported among females who restrict EI and with disordered eating behaviours (Nattiv et al. 2007). Low EA in combination with these types of restricted and/or irregular eating patterns is also associated with gastrointestinal symptoms including bloating, cramps and constipation (Black et al. 2003; Bonci et al. 2008; Melin et al. 2014b). Screening for gastrointestinal problems could be useful in identifying athletes at risk for RED-S.

Hypotension is a typical cardiovascular complication of anorexia nervosa (Meczekalski et al. 2013), but also associated with low/reduced EA as well as oligomenorrhoea/FHA in athletes with a BMI within the normal range (Melin et al. 2014b).

Subclinical or clinically low levels of T_3 and IGF-1, hypoglycaemia and low leptin per kg fat mass, as well as elevated levels of LDL-cholesterol and cortisol, are some of the biomarkers that can be indicative of RED-S in both male and female athletes (Mountjoy et al. 2014).

Summary

EA is a key concept in sports nutrition that provides the practitioner with a framework to assess the underlying cause and therefore management plan of athletes with conditions associated with RED-S. In the field, assessment or screening of EA is time consuming and includes a degree of error even using the best techniques available to the practitioner. Screening of athletes for symptoms included in the syndrome of RED-S may, therefore, be more cost-effective and increase the possibility of correctly identifying those at risk and if needed, allow for a more detailed assessment of EA in fewer individuals.

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Commentary 4

Treating low energy availability

ANNA MELIN AND INA GARTHE

Typical consequences of persistent energy deficiency in both female athletes and recreationally active women are functional hypothalamic oligomenorrhoea or amenorrhoea (FHA), injuries, impaired performance, impaired bone health, decreased resting metabolic rate (RMR), an increase in cardiovascular risk factors, gastrointestinal problems and deficiencies or suboptimal status of micronutrients such as iron and calcium (Eichner 1992; Beals 2004; Rickenlund et al. 2005; Nattiv et al. 2007; Rauh et al. 2010; Van Heest et al. 2014; Mountjoy et al. 2014; Melin et al. 2014). FHA is common among female athletes and recreationally active women but is often ignored and regarded as a natural result of intense training (Beals & Meyer 2007). Exercise or other stresses per se have been demonstrated not to disrupt reproductive function as long as the energy intake is not restricted (Loucks et al. 1998; Loucks & Redman 2004), while persistent limited availability of metabolic fuels has frequently been reported to be the primary event in the development of FHA (Loucks et al. 1998; Cumming & Cumming 2001; Loucks & Thuma 2003; Wade & Jones 2004). Hormonal synthesis, follicular development, endometrial proliferation and luteal phase thermogenesis are energy-consuming processes (Harber 2004), and experimental studies on healthy women have shown that energy availability (EA) of <125 kJ (<30 kcal)/kg FFM/d for 5 days is enough to disrupt the pulsatility of luteinising hormone (LH) and to reduce oestrogen and progesterone levels in healthy sedentary women (Loucks & Thuma 2003). The LH pulsatility is reported to be more sensitive to low energy availability in young women with a gynaecological age (time since menarche) <14 years, as compared to women with a gynaecological age >14 years (Loucks 2006). In recent years, there have been reports of increased prevalence of restricted eating or disordered eating behaviour and low bone mineral density (BMD) in male athletes as well (Hetland et al. 1993; Smathers et al. 2009; Dolan et al. 2012; Hagmar et al. 2013). Relative Energy Deficiency in Sports (RED-S) is an umbrella term that has been recently proposed to encompass the female athlete triad and extend the recognition of this problem among both male and female athletes (Mountjoy et al. 2014). No pharmacological agent adequately corrects the metabolic or endocrine abnormalities that impair health and performance in athletes with RED-S and securing sufficient energy availability is therefore the main treatment (Nattiv et al. 2007; Mountjoy et al. 2014; Ducher et al. 2011).

At the present time, there is limited research on the extent of the problem in male athletes and how it should be treated. Therefore, this commentary will focus on therapies available for female athletes with energy deficiency.

Treatment

A multidisciplinary approach

A multidisciplinary health care team which includes a physician, a registered dietitian, and, for athletes with eating disorders, a mental health practitioner, is necessary to estimate the magnitude and severity of individual risk factors and to secure adequate treatment (Nattiv et al. 2007; Mountjoy et al. 2014). Assessment of BMD (whole body and sport specific sites), body composition as fat mass and fat-free mass (FFM), the ratio between measured and predicted resting metabolic rate (RMR_{ratio}) and biomarkers associated to energy deficiency—such as thyroid function, glucose homeostasis, cholesterol and cortisol—might provide important information concerning the magnitude of the effects on the endocrine and metabolic system. A more complete medical and physical status might also increase compliance in a nutritional intervention program and, furthermore, provide an objective assessment of progress in the program. More details concerning biomarkers for low energy availability are presented in Commentary 3.

Body weight and body composition influence performance in many sports and there might be sport-specific demands of weight loss or recurrent weight regulation. Low energy availability might therefore be due to a deliberate restriction of food in a chronic attempt to lose or maintain a low body weight, and a realistic and healthy goal weight has to be thoroughly discussed. From a health as well as from a performance perspective, there are no general accepted values for optimal body weight or body composition in different sports. It is, therefore, of great importance that the athlete health care team is familiar with the specific sport, its cultures and demands. Furthermore, the team should be aware of how to approach athletes who seek to achieve an unrealistic body composition or body weight and how to prevent restrictive eating practices from developing into an eating disorder (Sundgot-Borgen et al. 2013).

Disordered eating or just difficult to increase energy intake in periods

with high training loads?

The content of nutritional treatment depends on the origin to energy deficiency. If there is any suspicion of eating disorders, a psychologist or psychiatrist, physician and registered dietitian specialised in eating disorders should be involved for investigation and treatment (see Chapter 8). Athletes with eating disorders should be required to meet established criteria to continue exercising, and their training and competition may need to be modified (Mountjoy et al. 2014). However, athletes don't have to have disordered eating to have low energy availability, since athletes with high training loads might have an increased risk for relative energy deficiency per se. There is some evidence that energy intake does not always track energy expenditure (EE) at high or unaccustomed levels of exercise, representing a situation of inadvertent low EA (Stubbs et al. 2004). High-intensity exercise has been shown to have an acute suppressive effect on appetite (Larson-Meyer et al. 2012), rather than the converse. Lifestyle or practical factors that can exacerbate the difficulties of increasing EI include the inhibitory effect of fatigue on the effort required to obtain and prepare food. Furthermore, it is difficult to consume large amounts of bulky fibre-rich foods, and there are often reduced opportunities for food-related activities on days in which a substantial number of hours are devoted to exercise (Burke, in press).

Female athletes and recreationally active women with FHA have been reported to have a lower energy intake compared to eumenorrhoeic subjects, mostly due to a lower fat intake (Deuster et al. 1986; Nelson et al. 1986; Kaiserauer et al. 1989; Myerson et al. 1991; Thong et al. 2000; Tomten & Høstmark 2006; Reed et al. 2011), or an increased training volume and thereby a potentially lower EA (Drinkwater et al. 1984; Harber et al. 1998; Cobb et al. 2003). Exercising women with FHA have, furthermore, been reported to eat a diet characterised by a low energy density (Reed et al. 2011). Although Laughlin and Yen (1996) found similar EI and training volumes in female athletes with FHA as in eumenorrhoeic athletes, the diet of FHA athletes was characterised by a lower fat content as well as a higher intake of dietary fibre. Low energy density, low fat and high fibre intake are dietary factors that are likely to increase the feeling of satiation and prolong the feeling of satiety, and may make it more difficult for athletes to reach EA ≥188 kJ (≥45 kcal)/kg FFM/d, especially in periods with high training loads.

Menstrual dysfunction

If there is menstrual dysfunction, a thorough gynaecological examination is needed to rule out polycystic ovarian syndrome (PCOS) or other menstrual dysfunction not associated to endocrine suppression due to persistent energy deficiency. The recommended firstline treatment strategy for athletes with FHA consists in increasing EA by increasing dietary intake, decreasing exercise EE, or both (Ducher et al. 2011; Mountjoy et al. 2014; De Souza et al. 2014). Diet and exercise interventions which have shown promising results in reversing FHA in athletes include adding 1500 kJ (360 kcal)/d) by supplementation and reducing training by one session/week for 15 weeks (Dueck et al. 1996). Regarding treatment of FHA, increasing the EI by 1250–2500 kJ/d (300–600 kcal/d) in order to restore endocrine and reproductive function is therefore recommended. FHA in female endurance athletes has been linked to a high intake of dietary fibre (Laughlin & Yen 1996; Couzinet et al. 1999) and several studies have reported a negative association between fibre intake and oestrogen levels in sedentary premenopausal women (Laughlin et al. 1998; Aubertin-Leheudre et al. 2008, Gaskins et al. 2009). The presumed mechanism involves oestrogens binding to dietary fibre in the colon, which reduces reabsorption of circulating oestrogen (Aubertin-Leheudre et al. 2008) and hence increases the excretion of oestrogens in faeces. Oestrogen excreted from the bile needs to be hydrolysed before it can be reabsorbed. A presumed indirect effect of dietary fibre, which can lower oestrogen, is via a reduced β-glucuronidase activity in the colon and thereby reduced reabsorption of oestrogens (Aubertin-Leheudre et al. 2008). These results indicate that an excessive intake of dietary fibre, independent of its effect on energy density, potentially could further increase the risk of FHA in female athletes, especially in athletes with persistent energy deficiency.

Bone health

Achievement of optimal BMD and regulation of bone maintenance depend upon a combination of mechanical, hormonal and dietary factors (Lebrun 2007). Adequate hormonal status and sufficient nutrition (calcium, protein and other bone-building materials), especially during adolescence, are therefore essential (Barrack et al. 2010) and it is vital to intervene within the first year of the onset of FHA when bone loss is most rapid (Drinkwater et al. 2006). Regarding treatment of low BMD, the strongest evidence that supports bone health improvements comes from longitudinal data obtained in girls or women with anorexia nervosa. Implementing recovery strategies including both a normalisation of body weight (BMI ≥18.5) and the resumption of menses are thought to have independent and additive effects on bone health (Drinkwater et al. 2006; Howgate et al. 2013). This is covered in more detail in Chapter 9.

Practical guidelines for increasing energy availability

Due to the physique requirements of many sports and/or eating disorders/disordered eating, many female athletes are scared of gaining weight and may therefore be reluctant to increase their energy intake or to reduce their exercise energy expenditure. To help athletes change unhealthy eating and dieting behaviour, it is important to involve the coach in the discussion and decision in order to establish consensus regarding realistic goals concerning weight, health and performance. Without the support from the coach, the compliance in a nutritional treatment program with the aim of increasing EA can easily be compromised. A small weight gain might be expected initially, before the endocrine and metabolic functions are restored. In normal-weight athletes, therefore, we recommend a stepwise increase of EA, to avoid unwarranted weight gain. Studies show that re-feeding in anorexia nervosa patients may lead to excessive abdominal fat (Mayer et al. 2005). Therefore, for underweight athletes, we recommend a slow weight gain of no more than 0.5–1 kg per week, induced by an increased energy intake of 1250–2500 kJ (300–600 kcal)/d), decreased training loads or a combination of the two. It may also be positive for both exercise energy expenditure and bone mass to replace some endurance sessions with strength training. Athletes are, in general, more accepting of an increase in body weight if some of the weight gain consists of FFM and thereby increased strength and power.

Many female athletes follow the general nutritional guidelines for a healthy diet (low in fat and added sugar and high in fibre). When EA needs to be increased to meet the goal of ≥188 kJ/kg FFM/d (≥45 kcal/kg FFM/d), nutritional needs should be calculated based on total energy expenditure, carbohydrate and protein requirements (g/kg/d) derived from individual goals and sport-specific training regimes (Rodriguez et al. 2009). Dietary fat and carbohydrate-rich foods can then be added to the diet to manipulate total energy content. To assist in increasing energy density, it is often valuable to reduce fibre intake (g/MJ or g/1000 kcal) so that it does not exceed the recommended 25–35 g/d. If the athlete experiences difficulties in adding more food to the diet, it may be easier to add liquid alternatives, such as meal replacement products (e.g. Ensure™) or smoothies between meals, and fruit juice instead of water at meals. A frequent meal pattern that provides energy-dense snacks and drinks between meals is also preferable, instead of a few large meals.

Energy availability changes during the season, mainly due to changes in training regimens but also due to the variations in daily life routines which change the availability of food (e.g. travelling) as well as the time needed to prepare food. Therefore, according to the lifestyle of the athlete, the meal plan needs to be based on foods that are simple to prepare and eat, and in many cases, portable. Information gathered from the athlete regarding their previous dietary patterns is important to consider; this includes socioeconomic situation, habitual meal frequency, food intake, food preferences as well as taboos, and feelings towards the different meals (e.g. recovery meals, eating alone, eating at training camp). Knowledge of these factors facilitates the planning of all the different elements and the cognitive development required for the nutritional intervention to be successful.

Summary

The treatment of female athletes with low energy availability requires a multidisciplinary team approach to identify the factors that underpin the eating and exercise behaviour that has caused this problem, and then to change this behaviour using appropriate support strategies. Dietary changes to increase energy intake should also address other suboptimal nutrition features and thus provide direct and indirect benefits to athletic performance.

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CHAPTER SIX

Weight loss and the athlete

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6.1 Introduction

The stereotype of the lean, toned and strong athlete paints a picture of a population that controls weight and body composition within tight limits with relative ease. This is not necessarily the case, with many studies providing evidence that athletes often experience difficulty achieving and controlling desired levels of body weight (body mass) and fat (fat mass) (see O'Connor, 2014 for a review). For most individual athletes, the perception of being overweight or over-fat is not a health risk, unless there is an underlying eating disorder. Usually, the desired level of fatness is less than that which would be considered healthy or *normal* within the context of public health guidelines.

Athletes and coaches are susceptible to just as much misinformation about weight loss and dieting as the rest of the community. Some athletes may use unsafe methods to reduce weight or attain a weight well below healthy levels, which increases the risk of negative short- and long-term health outcomes, psychological effects and performance decrements. This chapter considers a range of issues and concerns related to weight loss in athletes and evaluates popular weight loss approaches and diets and their suitability for an athlete. Specific diets and problems related to *making weight* for competition are covered in Chapter 7.

6.2 Justification for weight loss in athletes

Body mass or fat mass reduction in athletes is generally motivated by a desire either to achieve a pre-designated weight to compete in a specific weight class or category (e.g. in horse racing, lightweight rowing, boxing, weightlifting) or a specific body composition to optimise performance by improving power to weight ratio (e.g. in jumping events, distance running, triathlon, road cycling). In aesthetic sports like gymnastics, diving and figure skating, attainment of a desired body composition and physical appearance is considered important. Adding to these performance issues are current societal trends that encourage the pursuit of leanness for both men and women (Ballor & Keesey 1991). There is an unrealistic community perception and expectation of female body size, in particular, which is at odds with reality (Cash et al. 1994). Unfortunately, the expectation of leanness has become increasingly important in sports where an image of physical attractiveness is created for promotion or advertising. Athletes, or the sport itself, frequently derive substantial financial rewards for delivering an image and wearing clothing that accentuates physique (Simmers et al. 2009).

Despite the apparent preoccupation with body weight and fat levels in many athletes, there is limited empirical evidence of the effect of body composition on performance. As success in competitive sport is often decided by exceptionally small margins, testing the independent effect of specific physique characteristics relative to other factors such as talent and training is difficult. At the elite level, the range of specific physique attributes within a sport is quite narrow. Research to test the influence of body mass and/or fat mass on sports performance therefore has to be carefully designed using measures sensitive enough to assess the independent contribution of physique to competition success.

The relative importance and challenge associated with achieving a particular body or fat mass varies between sports and can be generally assessed by considering how different the desired characteristics are from the general or ‘source’ population. Body mass and fat mass may differ from the source population mean or have a narrow coefficient of variation (CV).

The body mass of winners in the Boston Marathon (the oldest annual marathon race) is tightly clustered and has remained static over many decades despite a substantial increase in the body mass of the source population. This suggests that low body mass is critical to competition success; see review by O’Connor and colleagues (2007). Competitive gradients or *best* versus the *rest* analysis (international versus state level performers) demonstrate clear trends in body mass and fat mass across athletic groups and provides evidence that is often used (sometimes inappropriately) by coaches and athletes.

Experimentally, a high level of fat mass has shown adverse effects on performance from the perspectives of heat exchange, mechanics and energy cost (O’Connor et al. 2007). Using a mathematical model, Olds and colleagues (1993) estimated that an increased fat mass of 2 kg would increase 4000 m pursuit cycling performance by about 1.5 s (20 m) and a 40 km time trial by about 15 s (180 m) The energy cost is also altered by the additional body mass, which increases the rolling resistance.

In sports where leanness is desired for aesthetic reasons, the reduction of fat mass to extremely low levels may not actually benefit performance per se. However, coaches and athletes believe that an appropriate level of leanness is assessed by the *trained eye* of the judges, which is an important characteristic that then influences the score for artistic impression. Regardless of the reason, athletes and coaches even at the recreational level frequently place too much importance on the attainment of desired body mass and fat mass levels. Yet even in athletes, as in the general population, the distribution and amount of fat mass is influenced by genetic and environmental factors as well as the training schedule, the sport and any attempts at weight control or reduction.

6.3 Factors influencing the ability to achieve optimal body weight and composition in athletes

6.3.1 Genetic factors

A significant proportion of the inter-individual variance in fat mass is attributable to genetic factors. The heritability of body fatness and body composition from epidemiological studies varies; however, studies of body mass index correlations in monozygotic, dizygotic and adoptive siblings suggest a 50–90% heritability (Barsch et al. 2000). Short- and long-term intervention studies examining energy balance in pairs of identical twins suggest that body weight and fat gains in response to overfeeding are under significant genetic control, evidenced by a threefold higher between-pair (compared to within-pair) variance in weight and fat gain (Bouchard et al. 1990). Comparable results are found in the converse situation with similar losses of adiposity in twin pairs subjected to a period of relative underfeeding and increased activity (Bouchard & Tremblay 1997).

Twin studies have also shown a greater similarity in dietary intake and food preference in monozygotic compared with dizygotic twins, suggesting genetic factors also contribute to dietary intake (Heller et al. 1988; Perusse et al. 1988). High correlations in dietary variables are observed in individuals and twins sharing the same environment, which is expected. However, differences in dietary intake between twins (and other siblings living in the same household) involved in different sports would also be expected because of the other external influences such as the culture and type of sport, nutrition or body composition expectations, which may be very different. Clearly, environment also plays a role.

A number of genes are associated and/or linked with human obesity. These *susceptibility genes* increase the risk for obesity, but not necessarily its expression. An individual with deficient or unfavourable alleles in a large number of susceptibility genes will be at a higher risk of developing greater levels of fatness than a person with a smaller number who is likely to be more resistant (Bouchard 1993). There is also a hypothesis that human obesity may be a neurobehavioural disease mediated by genes that affect hypothalamic pathways (Pritchard et al. 2002). A number of single gene mutations producing obesity have been discovered, which provide evidence for this link. These include mutations of the melanocortin-4 receptor (MCR-4), of pro-opiomelanocortin (POMC), of leptin, and of prohormone convertase. These mutations tend to be associated with gross obesity and hypogonadotropic hypogonadism. POMC is an important regulator of appetite, and is expressed in the arcuate nucleus of the hypothalamus, (and numerous other organs). POMC activates peptides that stimulate endogenous ligands of MCR-4 receptor (Pritchard et al. 2002). Mutations of genes mediating hypothalamic pathways are rare, except for the MCR-4 mutation, which has been described in 2–5% of those with morbid obesity in some studies. Nevertheless, the prevalence of this mutation in those with morbid obesity is still relatively low.

The gap between gene discovery and the application of this knowledge to obesity in humans is demonstrated by our current knowledge of the protein leptin. In 1973, mouse studies demonstrated the presence of a *circulating factor* that appeared to be a satiety factor (Coleman 1973). Some animals (e.g. ob/ob mice) appeared to be deficient in this circulating factor and became obese; others appeared insensitive to it but also became obese (db/db mice). In 1994, Zhang and colleagues isolated and described this circulating factor, which was a protein named leptin, produced by the adipocyte. Serum leptin levels in both rodents and humans correlate closely with percent body fat and body mass index (Considine et al. 1996; Rosenbaum et al. 1996). Elevated serum leptin signals adequacy of energy stores in cells and via activation of the hypothalamus is associated with decreased energy intake and increased basal metabolic activity in mice (Weigle et al. 1995). Early hopes that obesity was linked to leptin deficiency in humans as a potential cause of obesity has since been refuted, except in a very few obese individuals who have no leptin (Montague et al. 1997). Serum leptin levels are elevated in obese humans, although some are resistant to its effects on decreasing satiety and increasing metabolic rate. Leptin levels are higher in females than males (Considine et al. 1996) (which is linked to higher body fat than males) and are elevated by increasing energy intake and particularly by increasing CHO intake (Jenkins et al. 1997).

In humans, it appears that, rather than being a satiety factor, leptin acts as a signal of nutritional adequacy and protection against famine. It signals the level of fat stores and is important in the initiation of puberty and fertility. Leptin levels are low in anorexia nervosa sufferers (Grinspoon et al. 1996) and likely to be low in athletes who habitually restrict energy intakes. The long-term effect of suppressed leptin levels on reproductive potential, appetite and metabolic activity is not yet known. Leptin also affects peripheral metabolism by increasing the rate of fatty acid oxidation (Ahima & Lazar 2008). In athletes, leptin levels appear consistent with reduced body fat, although animal studies suggest there may be an independent effect of exercise on reducing serum leptin levels (Kowlalska et al. 1999). Interestingly, leptin levels are substantially reduced after a marathon, which suggests that short-term increases in energy expenditure may alter leptin levels (Leal-Cerro et al. 1998).

Another often-overlooked observation from population studies is that body weight tends to remain stable in many people for a long time, despite varying food intake and energy expenditure. This phenomena or theory is called the set point theory, which suggests that body weight is regulated and that there are a series of set points for an individual's body weight throughout life. Reductions or increases in weight away from the current baseline or set point result in metabolic alterations that resist the maintenance of a new weight and promote weight loss or gain towards the set point. This theory may explain why it is difficult for an underweight person to maintain weight gains and an overweight person to maintain weight loss. However, it does not explain the increased prevalence of obesity in both developing and developed countries.

Although most athletes have lower fat mass than the general population, genetic makeup is a key determinant of an individual's capacity to reach the desired body mass or fat mass that is expected in athletes involved in the elite sport. Genes ultimately influence an athlete's capacity to successfully and safely achieve and maintain a desired body composition.

6.3.2 Environmental factors

Energy intake and macronutrient composition

Athletes are, to some extent, protected from excess gains in fat mass because of high energy expenditure and gains in lean body mass (LBM) that result from resistance training programs. Aerobic exercise, in particular, increases oxygen delivery through improved capillarisation) (Andersen & Henriksson 1977), mitochondria density (Hoppeler et al. 1973) and elevated concentrations of enzymes required for fat metabolism (Henriksson & Reitman 1976). An increase in LBM and mitochondrial density increases metabolic rate, which increases energy metabolism and hence energy expenditure. Despite these adaptations, some athletes still gain an inappropriate amount of body mass or fat mass. In western countries, avoiding over-consumption of energy-dense fast or convenience foods, sweetened beverages and snacks requires discipline and dedication, even for athletes. Imbalance in energy intake is often experienced during tapering, illness, injury or the 'off season' when expenditure from training may be substantially reduced.

For some athletes, particularly those in aesthetic or skill sports (e.g. diving, figure skating, gymnastics), energy intake may be appropriate to maintain a *normal* body mass and fat mass (that falls within usual population reference ranges) but a lighter or leaner physique may be desired for their sport. Training programs that are skill based may also provide less opportunity to enhance fat oxidation or to substantially elevate energy expenditure. Even after large reductions in energy intake, some athletes still struggle with their body composition and never achieve the desired physique requirements of their sport. To a degree, the struggle with weight and skinfolds is partly offset at the elite level, where there is selective survival of individuals with the genetic background to be extremely light and/or lean. However, even those more suited to the physique expectations of a sport often still need to limit energy intake as the desired levels of leanness, at least in females, is below what is biologically natural.

Estimating an appropriate energy intake for an individual athlete is challenging because of the variability in energy requirement that is genetically determined and also the variable energy daily expenditure associated with periodised training programs. Regardless of the approach used, an energy deficit is still required for body weight or fat to be lost. Nevertheless, an energy-restricted diet needs to meet nutrient requirements and provide enough energy to recover for the next training session. Energy restriction, depending on the deficit, increases the risk for loss of LBM and for disruption to endocrine and immune function. Referral to a sports dietitian is recommended to ensure an appropriate level of energy restriction and a safe rate of weight loss that does not compromise metabolic functions and recovery for the next training session.

Manipulation of dietary macronutrient composition is also used to assist with body mass and fat mass loss. The optimal balance of macronutrients is currently a hotly debated area in weight management. Although early researchers strongly favoured the need to maintain tight regulation over CHO requirements to ensure energy balance (Jebb et al. 1996), there is evidence for the critical role of protein; see review by Simpson and Raubenheimer 2005. Central to this hypothesis is that protein, rather than CHO, is the most tightly regulated nutrient, despite the mechanism of glucose homeostasis. When faced with unbalanced diets, humans (and other vertebrates) prioritise protein. Over-consumption of energy intake occurs when fat- and/or CHO-rich foods are more accessible, affordable or available in highly desirable, palatable foods. This eating environment may result in passive, excess consumption of CHO and fat as an innate or unconscious mechanism to obtain the obligatory requirement for protein. As protein intake constitutes a smaller proportion of dietary energy compared to fat intake, a small decrease in the percentage of dietary protein, according to this hypothesis, drives this excess consumption until the protein requirement is satisfied. This is termed the *protein-leverage effect*. It is theoretically possible that the increased protein requirements of elite athletes (see Chapter 4) further influences this over-consumption behaviour. This effect has not been confirmed in athletes but may, at least from a physiological perspective, explain why some athletes (and the general population) often overeat.

Increased availability and intake of manufactured CHO (e.g. high-fructose corn syrup products) has been proposed as another link with the increasing prevalence of obesity, particularly low-fat, high-CHO diet products that have relatively high energy density (Van Baak & Astrup 2009). Many sports foods (e.g. gels, sports bars) are modelled on these criteria to provide fuel in a compact form suitable for consumption during and around training sessions. For some athletes, the convenience of these foods promotes regular and possibly over-consumption, which may result in excess energy intakes and a positive energy balance.

Exercise training and appetite

Although regular exercise assists in weight management for athletes and the general population (American College of Sports Medicine 2009a), some athletes eat too much food to compensate or reward themselves, particularly after a hard exercise session. Usually, the energy content of food consumed after exercise results in partial, complete and even over-compensation of the energy expended during exercise, which helps promote weight gain.

For some athletes, exercise-induced suppression of appetite is a commonly reported short-term response to exercise, which increases risk of suboptimal energy intake and weight loss. It may be related to elevated body temperature (Andersson & Larson 1961), increased levels of lactic acid (Baile et al. 1970) or even higher concentrations of tumour necrosis factor (Grunfield & Feingold 1991). Appetite control is a complex physiological process involving an interaction between hormones and neuropeptides (leptin, ghrelin, glucagon-like peptide-1, pancreatic polypeptide). Hence, isolating specific reasons for any effects of exercise on appetite is difficult; see review by Martins and colleagues (2008).

Most short-term (1–2 d) and medium-term (7–16 d) studies demonstrate that men and women can tolerate a substantial negative energy balance associated with exercise, with minimal to no measurable changes in body composition. Compensation for energy expended tends to be partial and incomplete up to around 2 weeks although there appear to be large individual differences; see review by King and colleagues (2007). This wide individual variability in compensatory responses to exercise suggests that some individuals are more susceptible to weight loss through physical activity while others are resistant. The reasons for this are not understood (see Martins et al. 2008) although there is some evidence that dietary restraint and total body mass or fat mass may contribute (King et al. 2007). Variability in the hedonic response or implicit desire for food may explain why some individuals over-compensate with food intake compared with non-compensators. These behaviours are not explained by differences in subjective feelings of hunger (Finlayson et al. 2009). Although not specifically demonstrated in studies of elite athletes, habitual exercise could improve the sensitivity of the appetite regulatory system and potentially provide better innate control of over-compensation after exercise (Long et al. 2002). In athletes, this type of study would be difficult to replicate because of the nature of periodised programs or the seasonality of sport, where habitual activity is usually highly variable.

Few studies have investigated the effect of mode of exercise on appetite. King and Blundell (1995) compared walking/running on a treadmill with cycling exercise and failed to observe a difference in appetite between groups. Although there is anecdotal evidence of an increase in appetite after swimming, there is a paucity of data comparing swimming to other modes of exercise. However, a recent study reported that 1 hour of moderate-intensity swimming in competent but not competitive swimmers resulted in a biphasic effect on appetite—that is, decreased appetite during exercise followed by an increase a few hours after the exercise (King et al. 2011). Despite these changes in appetite ratings, the swimming session did not alter the post-exercise energy or macronutrient intake, when compared with usual intakes at rest. Water temperature may also alter food choices consumed after exercise and hence energy intake. This behaviour was observed in a study of 11 adult college men (24.6 ± 5.4 yrs), using an unusual experimental protocol (i.e. cycling while submerged in water). The subjects consumed a higher energy intake from self-selection of snack foods after exercise in cold water (20°C) compared with energy intakes after the warm (33°C) water protocol (White et al. 2005). The calorie intake after the cold water trial was two times greater than the energy expended during the exercise itself. No subjective rankings of hunger were measured in this study. The authors speculated that cold water may have released hormonal factors that increased appetite. Similarly athletes as well as the general population tend to consume more food in cold weather conditions than in the heat and humidity.

There is some evidence that exercise also alters the type of food consumed and favours CHO-rich food choices. Whether this behaviour is related to appetite or an innate physiological drive for CHO to aid replenishment of limited glycogen stores or because of media or marketing is unclear. Tremblay and colleagues (1985) reported a relationship between exercise-induced changes in respiratory quotient (RQ)—an indicator of macronutrient intake—and energy intake. Individuals with the greatest reduction in RQ during exercise showed a lower increase in post-exercise energy intake than those with a high RQ. Taken a step further, eating low-fat foods after exercise resulted in a negative energy balance compared with a positive balance after consuming high-fat foods (Tremblay et al. 1985; King & Blundell 1995). This outcome may be attributed to alterations in appetite associated with the fuel mix oxidised or passive over-consumption as a result of a low-CHO diet because of its low satiety (King & Blundell 1995). The evidence relating to a direct link between appetite, substrate metabolism and exercise is limited and conflicting (see review by King et al. 2012).

Finally, gender may influence energy intake in response to exercise training. Women have demonstrated a higher compensatory increase in dietary intake than men, which may be explained by differences in baseline body fat levels and potential hormonal influences on basal metabolic rate (BMR); see review by King and colleagues (2007). Whether this increase is driven by an increase in appetite per se or other physiological, psychological or hormonal factors is unclear. These findings may help explain gender differences in the capacity for fat loss and may arise from biologically based evolutionary differences (King et al. 1997).

Physical activity, energy expenditure and substrate utilisation

The amount of energy expended during exercise depends on characteristics of the individual athlete (e.g. body mass, efficiency of performing a particular activity) and the physical activity undertaken (e.g. frequency, duration, intensity). A large body mass expends more energy to perform weight-bearing activities than a low body mass. However, a trained athlete, particularly in skill-based sports and to a lesser extent in weight-based sports, uses less energy because of improved efficiency. The frequency, intensity and duration of exercise determine the overall energy expended during activity. Exercise intensity in particular affects the magnitude of the post-exercise elevation in metabolic rate (Bahr & Sejersted 1991). Post-exercise energy expenditure may be significantly elevated in athletes who perform high-intensity, long-duration exercise, even though this component of expenditure is considered to be trivial for most non-athletes (Freedman-Akabas et al. 1985).

Total daily energy expenditure (and hence daily energy requirement) includes the energy required for metabolic functions including the thermic effect of food or digestion (TEF), for training and for the activities of daily living (sometimes referred to as non-exercise activity thermogenesis or NEAT) or incidental exercise. If elite athletes include a daytime rest and have sedentary work, study or other leisure activities outside training, then NEAT may be quite low, which can substantially decrease total daily energy requirement (see Chapters 2 and 5). Seasonal influence as well as injury and illness can also substantially decrease energy expenditure if immobile or inactive during recovery, which may result in unnecessary weight gain.

Athletes with a higher LBM have a higher energy requirement and RMR (resting metabolic rate) than athletes of the same weight with a higher fat mass. Exercise can increase the TEF although there is no consistent evidence that physical activity has a biologically important or substantial effect on TEF; see review by Melby and colleagues (1998). Also, substrate utilisation varies depending on the type (e.g. aerobic or anaerobic), intensity, duration of exercise and the adaptation induced by training. Detailed discussion of these factors is beyond the scope of this chapter.

Social and behavioural factors

Eating behaviours are important determinants in the maintenance of desired body weight (Wing

& Jeffery 1979). In many athletes who tend to have regimented lifestyles that revolve around training and competition schedules, food is often used as reward. Foods available at sports venues and when travelling to competition (e.g. takeaway, restaurants, buffet-style eating) provide opportunities for inappropriate food choices. These types of eating environments, where the food is high in fat and palatable, have a strong link with over-consumption in the general population (Stunkard & Kaplan 1977). For some athletes, pressure to perform or achieve a particular body mass or fat mass results in rebellion against a dietary regimen designed to control body composition. Alternatively, obsession with weight loss may result in disordered eating behaviours, which can lead to eating disorders (see Chapter 8). Typically, adolescent and young adults demonstrate age-related diet preferences that lean towards high-fat, fast foods and take-away foods (Truswell & Darnton-Hill 1981). At the same time, alcohol consumption—particularly in team sports, where alcohol use (and misuse) may be seen as part of a *team bonding* experience—may become an issue (Burke & Read 1987).

Growth and pubertal changes

Severe restriction of dietary energy to control body mass and fat mass may retard and possibly stunt growth in young athletes (Daly et al. 2002). Pubertal development in girls may influence their ability to achieve physique requirements in sports that require extreme leanness. The effect of energy restriction and weight loss on growth, maturation, health and performance in young athletes is covered in Chapter 17.

Tapering for competition

Tapered training to facilitate physical recovery and restoration of fuel reserves may result in a substantial reduction in energy expenditure and increase the risk for loss of LBM and an increase in fat mass (McConell et al. 1993; Margaritis et al. 2003). Anecdotally, gains in body mass during tapering are reported in different groups of athletes. This outcome is not confirmed in several studies, where weight remained relatively stable over the short term of tapering, despite some expected diurnal fluctuations and outliers (Houmard et al. 1990; Flynn et al. 1994; Dressendorfer et al. 2002). These studies did not measure other changes in body composition (e.g. skinfolds) or fat mass, so any real changes to body composition were unknown (see review by Mujika et al. 2004). Nevertheless, some athletes may be predisposed to an increase in body mass and potentially fat mass during tapering. Tapering usually occurs when an athlete is travelling to competition where dietary intake is often difficult to control. Practical strategies for managing weight status during tapering are considered in the Practice tips for this chapter.

6.4 Dietary approaches to weight and fat loss in athletes

There is a plethora of dietary approaches and fad diets marketed and recommended for weight and fat loss targeted at the general population. Most research on the efficacy of these methods has been conducted on overweight and obese people, not athletes. This section considers the suitability and efficacy of common weight loss methods/diets and their relevance to an athlete wanting to reduce body mass and fat mass.

6.4.1 Energy versus fat reduction and energy density

Ad libitum reduction in total fat intake results in a modest decrease in both energy intake and body mass and fat mass in obese individuals; see review by Astrup and colleagues (2000). Although not specifically studied in athletes, an ad libitum low-fat diet typically produces a satisfactory result in those who have long-term, modest weight or fat-reduction goals, particularly in those with energy expenditures. Clearly, this approach avoids jeopardising CHO, protein and micronutrient intake. However, ad libitum intakes of a low-fat diet will not produce the energy deficit required for those athletes desiring more extreme losses of body mass and fat mass. When planned energy restriction is required, careful dietary design is required to optimise nutrient intake and prevent low energy availability and its consequent risks (American College of Sports Medicine 2009b). The least energy restriction that will achieve the desired result is recommended and a deficit of 2100 kJ (500 kcal) from theoretical requirements may be a useful guide (Jakicic et al. 2001). However, gradual energy reduction of 10% or 20% of total estimated energy requirements is often used as an alternative approach to a sudden energy deficit. This approach helps prevent the athlete from feeling too hungry.

Diets that emphasise nutrient-rich, low-energy-density foods are also recommended. In an elegant series of studies, subjects fed diets containing 60%, 40% and 30% of energy from fat, while living in a 24-hour calorimeter, initially consumed excess energy intake, which was attributed to the energy density of food choices, not fat content of the diet per se (Prentice & Poppit 1996). This effect has been termed *passive over-consumption*—that is, eating more energy from the same food weight or volume. This over-consumption occurs in situations where athletes on energy-restricted diets favour sports foods (e.g. sports drinks, energy bars and gels) as their preferred snacks around training and recovery. Although these foods are a compact and convenient choice for busy athletes, they are energy dense, have low satiety, are often micronutrient-poor and are easy to overeat. Athletes on energy-restricted diets benefit from a meal plan with examples of suitable snacks around training to match reduced energy goals. The principles of sports nutrition still apply to athletes on energy-restricted diets.

6.4.2 Intermittent energy restriction (intermittent fasting)

A commonly used approach to reduce weight is chronic energy restriction, where a pre-planned, identical energy deficit is maintained each day until the weight target is reached. Although this approach is effective, many individuals find chronic energy restriction challenging. For elite athletes, this approach does not address the variability of training schedules and if used, should be tailored to match periodisation and rest days.

More recently, intermittent energy restriction has been promoted as an alternative approach to weight loss. Proponents claim that it enhances long-term compliance and may help prevent the fall in RMR associated with chronic energy restriction (Varady 2011). This approach involves normal eating and fasting from 1 to several days. The normal eating day is termed a *feed* day, where usual food choices can be consumed ad libitum (i.e. normal diet) for a prescribed period, followed by 1 to 4 days of fasting, where food intake is either completely or partially reduced (Varady 2011).

Based on a systematic review of intermittent energy-restricted diets in the general population, Varady (2011) concluded that there was no difference in weight or fat loss in short-term studies (3–12 wks), although only seven studies met the selection criteria for analysis. Evidence for efficacy for >12 weeks on the diet is not yet available. Four of the seven studies showed that losses of LBM were lower from intermittent energy restriction than from chronic energy restriction, which is a potential positive finding. Claimed effects of improved compliance and a decrease in RMR have not been confirmed. Further research is needed to test this claim.

The use of intermittent energy restriction as a weight-loss approach in athletes has not been studied in this context. In theoretical and practical terms, this approach is counterproductive to maintaining adequate glycogen reserves for the type of training programs undertaken by elite or recreational athletes involved in regular high-intensity training programs. Fasting or even extreme restrictive energy intake may even be contraindicated on an athlete's rest day, although evidence from Muslim athletes fasting from dawn to dusk during Ramadan for a 1-month period has shown minimal adverse physiological or performance effects, despite some complaints of fatigue (Mujuka et al. 2010). In Muslim athletes, eating more food after sunset may compensate for the energy deficit during the day.

6.4.3 Reduced-carbohydrate diets

The Zone diet

The Zone diet (*The Zone: a dietary road map*), which was popularised in the USA in the mid-1990s, is a relatively low-CHO diet based on a specific macronutrient distribution of 40% of energy from CHO, 30% from protein and 30% from fat from each meal and snack. The instigator of this diet claims that eating this ratio of macronutrients balances the insulin to glucagon ratio, which increases lipolysis and controls eicosanoid regulation. Although some claims made by the author are plausible, many are scientifically invalid (e.g. misclassification of the glycaemic index of some foods). The claim that a reduction in CHO intake to 40% of energy—based mainly around low to moderate glycaemic index foods—will significantly decrease plasma insulin levels is misleading because a reduction in CHO intake to less than 25% of energy is required before insulin levels are significantly reduced (Coulston et al.

1983).

In practice, most meals and snack foods do not comply with the 40:30:30 ratio of macronutrients designated in this diet, so food choice needs to be manipulated. This ratio is quite different to the *ideal* diet recommended for training. Several recipes in the diet plan also do not conform to this ratio. This diet as prescribed as a low-calorie diet, providing between 4200 and 8400 kJ/d (1000–2000 kcal/d) so it is not surprising that followers lose weight, irrespective of the macronutrient ratio (for review, see Cheuvront, 2003). Evidence that eating *in the Zone* results in superior exercise performance or weight loss is lacking (Jarvis et al. 2002; Bosse et al. 2004). Popularity of this diet has diminished since its inception, although snack foods with the prescribed ratio are still available online.

The Atkins diet

Surpassing the popularity of the Zone diet and most other popular diets is the Atkins diet. It originated in the 1970s with resurgence in early 2000; there have been several modifications since then with the release of the latest version in 2010 called *New Atkins for a new you*. The most rigorous version of this diet provides only 30 g CHO/d; however, it is not energy restricted. The diet permits unlimited quantities of high-protein/high-fat foods, providing they contain no or minimal CHO. This induces ketosis, which Atkins claims is critical to promote substantial weight loss and to assist with appetite control. The diet has been evaluated by a number of randomised controlled trials in obese adults (see review Astrup et al. 2004). Weight loss at 3 and 6 months is typically higher in the Atkins-style diet compared to a conventional low-fat diet. However, weight loss at 12 and at 24 months is modest and not differential except in one study (Gardner et al. 2007). Weight loss on the Atkins diet has not been greater than programs that include professional support and behavioural therapy (Wadden & Foster 2000), although weight loss was often achieved with minimal professional intervention.

A higher rate of short-term weight loss on a low-CHO diets has been related to high satiety, increased thermogenesis and hence slightly increased energy expenditure, and even the simplicity and monotony of food choices (Astrup et al. 2004). A reduction in glycaemic load, because of the restriction of foods, may also contribute to the apparent success of short-term weight loss (Ebbeling et al. 2003). Ketosis associated with restricting CHO is unlikely to be a key factor (Astrup et al. 2004). Probably the most important contributor to weight loss in this plan is dietary adherence. Although compliance is not often reported, the results of three dietary comparison studies including reduced-CHO diets, similar to the Atkins-style of diet, concluded that compliance was associated with more successful weight loss than diet composition (Dansinger et al. 2005; Alhassan et al. 2008; Sacks et al. 2009). The proscriptive regimen of the different versions of the Atkins diet may explain this high level of compliance.

One concern with restricting CHO is that endogenous protein (i.e. glucogenic amino acids) is catabolised for maintaining glucose homeostasis and glucose oxidation. Hence there is the potential for loss of LBM on these diets. Some but not all studies refute that theory and have demonstrated a preservation of LBM, which may be an adaptive effect (Layman et al. 2003). Any loss of LBM would be a negative outcome for athletes, resulting in a potential loss of power and strength. In studies of athletes who need to lose weight for competition (termed *making weight*), a high-protein diet in the presence of low CHO intakes enhances retention of LBM, when energy intake is restricted (see systematic review by Helms et al., 2014). Typically, diets recommended for making weight for competition are not as low in CHO as the Atkins diet, which may explain this effect (see Chapter 7).

Few studies have evaluated the Atkins (or a ketogenic) diet in elite athletes (Paoli et al. 2012). Firstly, the diet is extremely low in CHO. Studies evaluating the effect of low-CHO diets designed to enhance fat oxidation (via fat adaptation) as a means to improve endurance capacity usually show no performance benefits (see Chapter 15). In the short term, typical side effects of extremely low CHO diets (e.g. headaches, fatigue, nausea) are unlikely to promote high-quality training sessions (Sumithran & Proietto 2008). There is still insufficient evidence to support the safety of low-CHO diets in the long term. Recent evidence suggests an increase in risk of all-cause mortality (Noto et al. 2013). Chronic ketosis is also potentially harmful, with possible long-term consequences of hyperlipidaemia, particularly LDL, impaired neutrophil function, optic neuropathy and osteoporosis, as well as alterations in cognitive function (Denke 2001; Sumithran & Proietto 2008). Although improvement in cardiovascular risk factors has recently been reported in obese populations on reduced-CHO diets, it is likely to be attributed to weight loss rather than the macronutrient composition of the diet. High activity levels may be protective and attenuate cardiovascular disease risk (Sarna & Kaprio 1994) and override these potentially adverse effects; however, this prediction is speculative and not supported by direct evidence.

Suboptimal micronutrient intake and risk of nutrient inadequacy of the Atkins diet, particularly where several CHO-rich food groups are restricted (cereals, grains, fruits, starchy vegetables) is another concern. A review of 20 popular diets demonstrated that the Atkins induction diet phase was below the male adult RDI for fibre (13% RDI), vitamins B1 and B2 (47% and 77% RDI), vitamin C (72% RDI), calcium (89% RDI), magnesium (54% RDI) and iron (72% RDI). The Atkins ongoing weight-loss and maintenance diets fare better, but are still below the RDI for several nutrients (Williams & Williams 2003). Limiting dairy, grains, fruit and starchy vegetables also decreases phytochemicals and antioxidant intake, which have other health and performance benefits (Ralph & Provan 2000). See Chapter 11 and Commentary 5.

In summary, low-CHO, high-protein, Atkins-style diets are effective for short-term weight loss. The consequences for an athlete with extreme CHO restriction are chronic glycogen depletion and, as a consequence, fatigue, delayed recovery and possibly a reduction in LBM and impaired immune function. Nutrient intake will almost certainly be inadequate for several nutrients involved in energy metabolism. Long-term adherence may lead to serious health risks. For these reasons, low-CHO diets for weight loss in athletes are not recommended.

The Paleo diet

The Paleo diet (Cordain 2002) is another popular diet which was slightly modified and rebranded in 2005 (Cordain & Friel 2005). It is based on the type of diet consumed by our stone age ancestors. The rebranded Paleo Diet for Athletes is not an energy-restricted diet. The authors recommend avoidance of grains, dairy foods and legumes in favour of lean meats, fish, non-starchy fruits and vegetables with a macronutrient ratio of 35–45% of energy from CHO, 19–35% from protein and the remaining energy from fat, particularly saturated and monounsaturated fatty acids. Other features include generous amounts of potassium, limited sodium, balanced acid/base ratios, a high omega 3:omega 6 fatty acid ratio, and high micronutrient and phytochemical density. Modifications for the athletes' version include more carbohydrate foods before, during and after exercise, and limited consumption of bread, pasta, rice, starchy vegetables and dried fruit, which were not featured in the original version. During exercise, intake of high glycaemic index foods (including sports drinks) and electrolyte replacement is included.

Similar to other energy-restricted diets, several aspects of the Paleo diet have good scientific validity and potential health benefits, which are associated with the high vegetable intake, use of lean meats and high intake of omega-3 fatty acids (NHMRC 2013). The recommended intake of red meat in the diet exceeds recommended serve sizes specified by the World Cancer Research Fund and American Institute of Cancer Research (2009) and would not be considered ecologically and economically sustainable for widespread use (Pimentel & Pimentel 2003). Liberal use of saturated fat is also inconsistent with current population recommendations (NHMRC 2013). Similar to the Atkins diet, the high satiety value of the large meat serve of the Paleo diet helps followers lose weight by minimising hunger, which presumably translates into eating less food and calories. For athletes who do not compensate adequately with the extra helpings of CHO-rich food, weight loss may be excessive and too rapid to support the energy requirements of training and conserve LBM.

It is worth noting that the composition of the Stone Age Palaeolithic diet still remains a topic of scientific debate and is likely to be highly variable depending on the season and location of people in that era. The advantages or disadvantages of the Paleo diet and its potential health outcomes, although recognised in principle as potentially positive, have not been confirmed. For a review, see Lindeberg (2012).

6.4.4 Manipulation of the glycaemic index (GI)

Evidence supporting (McMillan-Price & Brand Miller 2006) and questioning (Sloth & Astrup 2006) the effectiveness of low-GI diets for weight management is available. A meta-analysis of the effects of the glycaemic index of foods on weight loss showed a small positive effect (Thomas et al. 2009). The effect of low-GI diets on satiety may be particularly helpful for hungry athletes or those needing to chronically restrict energy intake to maintain a low body mass for their sport (e.g. jockeys, ballet dancers, gymnasts). Satiety effects have not been specifically investigated in athletes, but have been observed in several studies of other population groups (for review, see McMillan-Price & Brand Miller, 2006). Unlike restriction of CHO, reduction of dietary GI is not associated with negative health consequences, although the impact on glycogen storage and performance in athletes has not been evaluated. More medium- to long-term studies, including those on athletes, need to be conducted.

6.4.5 Role of calcium and dairy products in weight management

Several large cross-sectional population studies support an inverse relationship between dietary calcium/dairy intake and body mass index (BMI), body fat and obesity (for review, see Major and colleagues, 2008). Two recent systematic reviews which included meta-analyses confirmed a beneficial effect of higher dairy intake for weight and fat loss when energy intake was restricted but no beneficial effect on body composition changes with an ad libitum energy intake (Abargouei et al. 2012; Chen et al. 2012).

The biological mechanisms underpinning the weight-loss benefits of higher dairy intake are uncertain but several plausible biological mechanisms have been proposed. These include the role of intracellular calcium in the regulation of adipocyte lipid metabolism and triglyceride storage. Calcium itself, with its capacity to stimulate lipolysis, inhibit fatty acid synthase and de novo lipogenesis, may also mediate beneficial effects through an increase in uncoupling protein 2 expression, which is thought to influence fat cell apoptosis, increase fat oxidation and attenuate the decline in thermogenesis that occurs with energy restriction (Shi et al. 2001). High dietary calcium has also been observed to increase faecal fat excretion (Major et al. 2008; see Major et al. 2008 for a review). Unfortunately, athletes striving to reduce body mass and fat mass are often reported to consume inadequate calcium (Barr 1987).

6.4.6 Other popular diets

In their evaluation of 20 popular diets, Williams and Williams (2003) scored diets based on nutrient adequacy, flexibility, physical activity and supplement recommendations as well as scientific support for their claims. Popular diets with point scores below 50 out of a total of 100 points included Slim Forever (32), Dr Atkins' New Diet Revolution (35), Eat Right for Your Type (36), Sugar Busters (40), The Fit for Life Diet (41), The Zone and Liver Cleansing Diet (42), The Carbohydrate Addicts Diet (43), The Complete Scarsdale Medical Diet (44) and Diet Signs—The Health Signs Diet (45). Top-scoring diets included The Volumetrics Weight Control Plan (97), Licence to Eat (96), Fat Loss for Life (87), The Diet that Works, Eat More Weigh Less (85) and The Fat Stripping Diet (80).

Adaptation to high-fat diets and *train low* strategies

Adaptation to high-fat diets has been used to delay fatigue in endurance exercise by enhancing fat oxidation and sparing CHO stores. Although not typically employed to promote weight loss, athletes may believe that fat adaptation or *train low* strategies might result in a permanent up-regulation of fat metabolism and subsequent improvement in body-fat control. This belief is speculative (see Chapter 16).

Very low energy diets (VLEDs)

Very low energy diets (VLEDs) or very low calorie diets (VLCDs) diets provide 1600–2400 kJ/d (400–600 kcal/d) and less than 100 g CHO/d in the form of a liquid meal (such as Modifast™ or Optifast™) with added vitamins and minerals to levels approximating the RDI/RDA (Brodoff & Hendler 1992). Weight loss induced by VLEDs is large and rapid (1.5–2 kg/wk), but often not sustained (see review by Saris, 2001). These diets are used as a short-term strategy for the morbidly obese. Rather than just being used as a *total* meal replacement system, these diets may include partial use where one or perhaps two meals a day are replaced by liquid meals (Ditschuneit et al. 1999). Side effects of VLEDs include nausea, halitosis (bad breath), hunger (which may decrease after the initiation of ketosis), headaches, hypotension, light-headedness and precipitation of gout (Brodoff & Hendler 1992). Other effects, including glycogen depletion, loss of LBM, dehydration, electrolyte imbalance, hypotension and risk of inadequate energy availability, make VLEDs unsuitable for athletes and explain why they have been rarely used.

Weight loss organisations

These organisations (such as Weight Watchers and Jenny Craig) are designed for the general public to provide support for weight loss and management. Many have professional input from dietitians, psychologists and medical practitioners to aid in keeping their programs safe, effective and relevant for clients. The programs are not designed specifically for athletes; however, some have the facility to provide additional energy and CHO in their meal plans or meal selections. They are suitable for athletes provided an accredited program that has professional input is chosen which has the capacity to cater for their special needs.

6.5 Exercise prescriptions for weight and fat loss

6.5.1 High-intensity versus low-intensity exercise for fat loss

Although it would seem logical for athletes to try to increase the volume of low- to moderate-intensity activity to maximise lipid oxidation, low-intensity activity may not be the most efficient prescription for fat loss in fit individuals. While the proportion of fat utilised is greater at low to moderate intensities, well-trained athletes still oxidise substantial proportions of fat at higher intensities, providing this is less than their anaerobic threshold. The effectiveness of high- versus low-intensity training on body fatness and skeletal muscle metabolism was first investigated by Tremblay and colleagues (1994). Despite a lower daily energy cost of the high compared to the lower intensity exercise program (57.9 versus 120.4 MJ/d), the former induced greater fat oxidation and produced a nine-fold higher reduction in skinfold measurements over 15–20 weeks. Two more recent studies are consistent with these findings (Yoshioka et al. 2001; Trapp et al. 2008).

Although a lower proportion of fat is utilised in the fuel mix oxidised at higher exercise intensities, a greater energy deficit occurs in less time, which results in a higher total fat utilisation (Romijn et al. 1993). The efficiency of high-intensity exercise may be attractive to athletes who have little time to incorporate extra lower intensity training that expends additional energy just for fat loss. Interestingly, a study by Knechtlet and colleagues (2008) reported a significant association between race intensity and body-fat loss (measured by skinfold thicknesses) during an Ironman triathlon. Unfortunately, energy intake during the race was not assessed.

6.5.2 Mode of exercise and fat loss

Weight-bearing exercise, in particular running, is often used to promote weight and fat loss among athletes. Apart from the practicality of running, runners tend to be leaner and lighter than other highly trained athletes in aerobically based sports such as swimming or cycling. Flynn and colleagues (1990) demonstrated that swimming utilised similar energy to running, but proposed that the higher proportion of body fat in swimmers may be attributed to differences in post-exercise energy expenditure or appetite control after exercise. It is also possible that control of thermoregulation and the recovery from the impact of running contribute to higher post-exercise energy expenditure in runners compared with swimmers. Although there is no current explanation for the superior weight-control benefits of running, it is frequently used for this purpose. The choice of other modes of exercise is often determined by avoidance of the high impact and injury risk associated with running or because running is not preferred by the athlete.

6.6 Negative aspects of weight control in athletes

Inadequate energy intakes as a means to induce loss of body mass and body fat may produce a wide range of negative physical and psychological consequences, depending on the severity and duration of energy restriction. Serious adverse effects occur when energy availability falls below 30 kcal/kg of fat-free mass/d, at least in women (Mountjoy et al. 2014). This state of low energy availability, if maintained chronically, is likely to result in a reduction in metabolic rate, lowered sex hormone levels (in both men and women), compromised immune function and bone health, decreased protein synthesis and possible electrolyte imbalance (Mountjoy et al. 2014). Depression and disordered eating are also possible outcomes (see Chapter 8). Some negative effects (e.g. reduction in metabolic rate, changes to hormones, electrolyte levels, protein synthesis, immune function) are reversible when energy intake is normalised. However, bone health and the negative psychological consequences (especially disordered eating) may not be entirely reversible or can be more difficult to normalise (see Chapters 5, 8 and 9).

Weight loss induced by modest energy restriction also results in some reduction in metabolic rate (Leibel et al. 1995). Loss of LBM can still occur although the magnitude of this loss is minimised by maintaining resistance exercise and avoiding severe energy deficits (Heyward et al. 1989). Slightly higher amounts of protein in energy-restricted diets help to minimise further losses of LBM and help adaptation to energy restriction (Walberg-Rankin et al. 1994; Helms et al. 2014). Some athletes, particularly those who need to *make weight* for competition, experience large fluctuations in body mass with repeated episodes of energy-restricted diets and often gain more weight between each successive diet. This phenomenon is called weight cycling or yo-yo dieting and has also been observed in overweight/obese people who frequently go on energy-restricted diets. The potential negative and adaptive effects of weight cycling on metabolic and general health are considered in Chapter 7.

Unsupervised energy restriction with or without substantial weight loss can disturb immune function, because of the potential for chronic malnutrition associated with an acute depletion or deficiency in specific nutrients involved in immune function (e.g. iron and zinc). See Chapters 11 and 22.

6.7 Adjunctive agents for weight and fat loss

6.7.1 Dietary supplements

A large number of supplements have been touted and are marketed for weight control or fat loss. Most claims are exaggerated or biologically implausible. Some supplements may contain natural or biologically active ingredients that are banned by sports drugs agencies. Many supplements have not been subjected to the same quality control measures as pharmacological agents or any rigorous testing to assess their efficacy or safety. For a review, see Dwyer et al. 2005.

6.7.2 Pharmacological agents

Pharmacological agents/drugs are available that regulate or alter weight for short-term, medium-term and long-term use. Most are not permitted by the World Anti-Doping Agency (WADA) and should be used only in the appropriate situation under medical supervision.

Drugs associated with short-term weight loss

Diuretics, mainly indicated for people with hypertension, cause excretion of large amounts of fluid (and sodium and potassium), which is associated with rapid short-term weight loss. Their effects can last hours to a day. They are banned by WADA because of their potential misuse as masking agents for illicit substances (Ciocca et al. 2011) and for their rapid weight loss effect, which is considered an unfair advantage in making weight for competition. Side effects include high risk of dehydration, hypokalaemia, hyponatraemia and hypotension.

Drugs indicated for medium-term weight loss

These drugs can act either locally (on the gastrointestinal tract) or centrally (on appetite control or thermogenic mechanisms).

Drugs acting on the gastrointestinal tract

Over-the-counter drugs containing bulking agents, which absorb large amounts of water in the gut, were used in the past to provide a sense of fullness and satiety as a means to inhibit the volume of food consumed. These bulking agents are no longer available because of their potential to swell and lodge in the oesophagus, requiring surgical removal.

Other types of drugs such as acarbose, a glucosidase inhibitor, prevent the breakdown of sucrose and therefore the absorption of sugar and has helped obese patients maintain lower body weight when taken in high doses (Caterson 1990). There are many other agents in this category, with no evidence of effectiveness (Dwyer et al. 2005). Orlistat is gastrointestinal lipase inhibitor that prevents fat digestion and therefore its absorption. Around 30% of ingested fat is malabsorbed using a course of orlistat. In one study, around 10 kg of weight was lost over 6 to 12 months in the orlistat treatment group compared to 6 kg weight loss in the placebo group (Sjostrom et al. 1998). Indications for treatment with orlistat (120 mg three times/d) include a low-fat eating plan. Potential side effects from malabsorbed fat include abdominal discomfort, diarrhoea, anal leakage and urgent bowel movements, which limit the duration of treatment. These can be reduced by adhering to a low-fat diet. If used for longer than 1 year, supplementation with fat-soluble vitamins should be considered. Orlistat has been used continuously by subjects in the XENDOS trial for as long as 4 years with no significant side effects and has helped subjects maintain weight loss (Torgerson et al. 2004). Other beneficial effects of orlistat include a lowering of low-density lipoprotein (LDL) cholesterol, reduction in blood pressure and waist circumference, and improved control of diabetes. These benefits are partly attributed to the elimination of fat, which results in fewer circulating fatty acids and a reduction in insulin resistance (Kelley et al. 2004). Orlistat is now marketed over the counter in many countries and is available in a lower dose formulation.

There are no data available on the use of orlistat in athletes. Although it reduces fat absorption, it does not interfere with CHO absorption or the accumulation of muscle glycogen, if used by athletes. As most athletes are not usually considered a health risk for obesity, it becomes an ethical question whether locally acting drugs like orlistat or acarbose should be used, even though they are permitted by WADA.

Drugs acting centrally or on thermogenesis

Ephedrine, ephedra (ma huang) and caffeine are used for weight reduction in obese patients. Ephedrine is a sympathomimetic drug and acts on the catecholamine receptors. In addition to its other properties, such as bronchodilation, it can suppress appetite and increase thermogenesis (Astrup & Raben 1992). Ephedra (ma huang)-containing dietary supplements, either with or without caffeine, were widely used from the mid-1990s to early 2000 for stimulating weight loss until it was banned by the US Food and Drug Administration (FDA) in 2005. A systematic review of the safety and efficacy of ephedrine and ephedra, with or without caffeine, concluded that their use promoted modest but significant weight loss of 0.9 kg/m (compared to placebo) over a period of less than 6 months. No studies have assessed their long-term effects (Shekelle et al. 2003). These substances were also associated with two to three times the incidence of nausea, vomiting, psychiatric symptoms, autonomic hyperactivity and palpitations, compared to placebo. Although useful for those with asthma, ephedrine should not be used by those with diabetes, hypertension or hyperthyroidism. These drugs have been shown to be effective in obese patients—a total weight loss of around 16 kg in 24 weeks when used with a reduced energy intake (Astrup & Raben 1992). In lean subjects, ephedrine and caffeine (together with theophylline) have been shown to increase thermogenesis and energy expenditure, but there are no data on weight loss using these agents in athletes. Both ephedrine and weight loss supplements containing ephedra are banned by WADA.

In 2006, methylhexanamine (also known as DMAA or geranium extract), an amphetamine derivative, was reintroduced as a weight loss and pre-workout dietary supplement in the USA, having previously been used as a nasal decongestant. It is now banned by the US FDA and Australian Therapeutic Goods Administration (TGA). Used in combination with caffeine, it can elevate blood pressure and induce myocardial infarction and stroke. DMAA is also banned by WADA.

Nicotine may also suppress appetite and increase resting energy expenditure. Smokers have higher resting energy expenditure at night and for a short while after smoking (Audrain et al. 1995). Smokers tend to be thinner and gain weight after cessation of smoking (Flegal et al. 1995; Rasky et al. 1996). The health consequences of smoking (particularly increased risk of vascular disease and cancer) and local effect on athletic performance make smoking an inappropriate option as a weight loss measure for athletes.

Phentermine, one of the early appetite suppressant drugs available, activates the sympathetic nervous system, resulting in decreased food intake and increased resting energy expenditure. Experience with the drug is for use in obese individuals, with little data on use in athletes because it is prohibited by WADA. It is still available although not recommended for long-term use (>12 wks). Tolerance can develop with long-term use, rendering the drug ineffective. It is contraindicated in patients with cardiovascular disease because of the risk of tachycardia and hypertension.

Another of the older appetite suppressants, phenylpropanolamine, is available in some countries in over-the-counter preparations for weight loss. It is related to both noradrenalin and amphetamine derivatives. It can be effective treatment option, but has the potential to stimulate the cardiovascular and central nervous systems, and is no longer available in Australia.

Sibutramine, which is both a selective serotonin reuptake inhibitor (SSRI) and a selective noradrenaline reuptake inhibitor (SNRI), was previously used as an effective weight loss agent, with both serotonergic (appetite suppressant) and noradrenergic (thermogenic) effects. At a dose of 10–15 mg/d, combined with an appropriate lifestyle program of diet and exercise, it produced significant weight loss in the treatment group (~12 kg compared to 6 kg in the placebo group), which was maintained for 2 years (James et al. 2000). It was, however, withdrawn from the market in 2010 following the adverse results of the Sibutramine Cardiovascular Outcomes Trial (SCOUT) (James & Caterson 2010). While sibutramine was effective for weight loss in the long term, there was an increase in non-fatal myocardial infarction and non-fatal stroke in those in the trial with pre-existing cardiovascular conditions. This was a major concern for its continued use in obese individuals, hence the withdrawal. There are no reports of its use for weight control in athletes.

In summary, several effective pharmacologic measures are available for reducing weight, but there is little if any information about the use of such agents in athletes. The use of these drugs by athletes must be considered carefully and raises an ethical question. These drugs were designed to assist those at health risk, not young, fit individuals. It is better that they are not used except where there is a risk to health of excess weight.

6.8 Guidelines for weight loss

Recommendations for a safe rate of weight loss in non-athletes are around 0.5–1 kg/wk. These limits seem reasonable for most athletes. Loss of this weight equates to an approximate energy deficit of 2100–4200 kJ/d (500–1000 kcal/d), which can be achieved by diet, training or both. Moderate energy restriction without compromising CHO or nutrient intake is optimal and best achieved with a diet low in fat (15–25% of energy), with adequate CHO intake to support daily training or competition requirements. Protein intake should be approximately 1.5–2 g/kg body mass/d with the upper level recommended if energy restriction is substantial, which may assist with satiety and maintenance of LBM. Foods high in fibre and/or of low glycaemic index may also assist with appetite control. Calcium intake at or above the RDA/RDI, ideally from dairy foods, may also assist with weight management.

Exercise complements dietary approaches to weight loss by increasing energy expenditure and inducing negative energy balance. Although lower intensity, aerobically based exercise has been recommended as *best* for loss of fat mass, this prescription may not be the most efficient for athletes. Higher intensity exercise that can be maintained for a reasonable duration (30–60 min/d), in addition to an athlete’s standard training, may be a useful and perhaps an even better approach to weight loss (see section 6.5). However, the risk of fatigue, injury and the mode of additional exercise need to be considered. Exercise prescriptions should be tailored to an athlete’s individual needs.

Summary

Athletes may attempt to lose weight to improve power to weight ratio, attain a desired body composition in a sport that has an aesthetic ideal or make a pre-designated weight to meet weight requirements for competition. Most will not need to lose weight for health reasons. Body mass and composition are determined by many factors, both biological and environmental. Genes have a significant influence on the susceptibility for weight or fat gain and indeed the potential to attain a particular physique. Genes may act through food preferences and appetite, through energy expenditure or through metabolic factors.

Currently, alteration in environmental or lifestyle factors is the only way to manipulate body mass and fat mass. The two major factors influencing energy balance are energy intake and expenditure. Interventions focusing on reduction of fat intake are sufficient to produce the desired weight or fat loss in most athletes. This approach is in line with public health guidelines and is also less likely to result in glycogen depletion. Some athletes, however, will also need a planned restriction of their total energy intake to achieve their desired goals. To maintain satiety, adequate protein intake and the use of low-glycaemic index foods may be beneficial. Adequate intake of dairy foods has also been found to support weight loss when total energy intake is restricted.

Diets advocating severe CHO restriction are currently popular in the community and with athletes. Unfortunately, rapid, short-term (1–6 months) weight loss in studies using obese participants demonstrate weight loss is typically not maintained. Long-term health risks of reduced-CHO diets have not been adequately evaluated in the general population or in athletes. Based on substrate use during exercise, athletes following these types of diets are likely to decrease exercise performance capacity and have impaired recovery and a disturbance in immune function. Nutrient adequacy may also be compromised.

Physical activity directly increases total daily energy expenditure from the exercise directly and also indirectly by potential increases in RMR and TEF. Although it is assumed that additional exercise will result in an energy deficit, it is possible that an individual may compensate, partly or even fully, for increased energy expenditure by consuming more food. This may occur through changes in appetite although the precise nature of the effect of exercise on appetite is yet to be established, particularly over the medium to long term. There appears to be an acute, post-exercise suppression of appetite and, in some studies, an increase in preference for CHO. Athletes may be susceptible to weight gain when physical activity ceases or decreases, especially during the off-season or through periods of illness or injury. Careful planning of food intake and the maintenance of some physical activity is important to help prevent weight or fat gain at these times.

Exercise prescriptions for weight loss in athletes are similar to those designed for the overweight, incorporating low-intensity (50% $VO_{2\ max}$), medium-duration (~60 min) aerobic activity. This prescription may not be the most efficient for fat loss in fit individuals. Exercise regimens using shorter duration, medium- to high-intensity exercise, possibly in combination with resistance training, may also be effective in reducing excess fat mass. This latter approach may be attractive to many athletes because of the limited time available to incorporate extra training for the specific purpose of weight or fat reduction.

Athletes and coaches are open to just as much misinformation about weight loss and dieting as the rest of the community. Unfortunately this often results in the use of unbalanced dietary regimens, dietary supplements that lack scientific support or are not permitted by sports drug agencies. Such approaches may result in decreased performance and have negative health consequences. The promotion of safe and effective weight-loss strategies is an important role of the sports medicine team.

Practice tips

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These tips are designed to assist practitioners with assessment and management of the athlete presenting for weight or fat loss.

ASSESSMENT OF THE NEED TO LOSE WEIGHT OR FAT

- The initial assessment for weight or fat loss in an athlete should include measurement of weight, stature, adiposity and muscularity using a standardised measurement protocols. Surface anthropometry is frequently used because it is portable, inexpensive and non-invasive. Anthropometric measures should be performed in duplicate by an accredited anthropometrist using an appropriate protocol (the International Society for the Advancement of Kinanthropometry protocol is recommended; see Chapter 3). Other methods of assessing body composition and their relative benefits and pitfalls are also discussed in Chapter 3.
- Error associated with body composition measurement should be explained to the athlete. This is especially important for interpreting change when serial measurements are performed.
- Body composition assessment should be complemented with a history of personal and family weight patterns including overweight/obesity through to eating disorders and other signs of energy deficiency (see Chapter 5). Previous methods of weight management used should also be collected. Recent biochemical, haematological and bone density results should also be considered. The athlete's goals need to be evaluated within the context of their history, current body composition and the optimal ranges of weight and fat in their sport.
- Goals for weight and fat loss need to be realistic, ideally not greater than 0.5 kg/wk or for body fat a maximum of 5 mm/wk reduction in total skinfold (over seven or eight sites). The goal of weight or fat loss should not compromise health or performance.

DIETARY ASSESSMENT

- Dietary assessment should focus on recent eating patterns. A dietary history and food frequency checklist is ideal for this purpose. Food diaries can complement this information.
- Dietary histories should include an assessment of eating away from home, eating habits when travelling, cooking skills, comfort or stress eating, hunger, use of sports drinks/foods and alcohol.
- Direct measurement of RMR (see Chapter 5) or estimation using prediction equations should be considered for athletes experiencing difficulty with weight loss. The ratio of energy intake to RMR would be expected to be greater than 1:3 as a ratio less than this is implausible even for sedentary individuals (Goldberg et al. 1991). Ratios greater than 1:5 would typically be expected and in very heavy training may be close to 3. If under-reporting is suspected, issues of body image or other stressors need to be explored. Discussing the implausible nature of reported food intake with the athlete may help to reveal the practical or psychological issues impeding weight or fat loss.
- Athletes experiencing great difficulty with weight loss may be better to direct their energies towards sports where the physique requirements are less extreme.

TRAINING, EXERCISE AND PHYSICAL ACTIVITY ASSESSMENT

- An athlete's daily energy expenditure can be estimated from prediction equations (see Chapter 5), tables of energy expenditure for different sporting activities, and activity history or diaries. Accelerometers, heart-rate monitors and global positioning system technology can also be used. Many athletes now own personal devices or use mobile (cellular) phones to monitor training. Energy expenditure assessed using these approaches can provide approximate estimates to guide meal planning. Alternatively, an estimate can be obtained from the athlete's current energy intake. Although under-reporting may be an issue, a history of energy intake is often the most immediate and practical method in a clinical setting.
- Ideally an approximate training program for the following 12-month period is available. Although weight- or fat-loss goals may be more immediate, it is critical to adjust the diet for different training phases, including tapering for competition and the off-season. In some sports, weight- or body-fat levels can be effectively periodised, allowing an athlete to be strategic about when they need to be at their lightest or leanest. This is helpful when physique requirements are quite rigorous or extreme yet training can still be optimised without constant physique maintenance. Large swings in body weight or fat are not recommended, but for many athletes, periodising weight or fat tapers help them to consume a more enjoyable, less restricted and balanced diet at other less-critical times.

DIETARY APPROACH

- It is preferable to start most athletes on a diet that is only moderately (10–20%) reduced in energy, compared with estimated usual requirements. This should produce a gradual weight loss (0.5–1 kg/wk at most).
- More rigorous restriction of energy intake may be required when extreme leanness is necessary or when an athlete needs to make weight for competition. Careful planning of meals is important to ensure the diet provides adequate macro and micronutrients.
- Inclusion of a protein choice at each meal and snack may assist with satiety and also the maintenance of lean mass (see Chapter 4). High-fibre, low glycaemic index foods also help to satisfy the appetite of a hungry athlete. Regular consumption of nutrient-rich, lower-energy-dense meals and snacks is recommended to prevent the build-up of hunger over the day. Organisation is important as impulse eating is often a problem, especially for the busy athlete.
- Planning meals or snacks to follow training sessions will promote recovery using nutritious and satiating food choices within the athlete's daily kilojoule budget. Energy-dense sports foods/beverages, while convenient, may not be the most nutritious or satisfying snack or recovery choice for hungry athletes on a daily basis. Use of sports foods and beverages can be planned to coincide with longer, more strenuous sessions or during competition where consumption of lower kilojoule, bulkier foods is less effective or acceptable.
- Emerging evidence supports a beneficial role for dairy foods/calcium, particularly in those with habitual low intakes. It is therefore recommended that athletes consume at least their RDI of calcium (ideally from dairy food sources) each day. Supplemental calcium should be considered if this is not possible.
- Dietary consultations should assess the athlete's (or their parent's/guardian's) shopping and cooking skills. Shopping trips and cooking classes provide a practical and enjoyable form of athlete education. A list of suitable cookbooks and resources is also useful.

TRAINING, EXERCISE AND PHYSICAL ACTIVITY

- Additional training or physical activity is a means of creating an energy deficit and should be combined with a sensible eating plan to offset any increase in appetite that could negate the energy deficit. Assessment of appetite and desire to eat using visual analogue scales is also useful.
- Additional exercise of medium to longer duration (60 min) at low to moderate intensity (50–60% $VO_{2\ max}$) is usually recommended as the best choice for weight or fat loss. Medium-duration (30–60 min), higher-intensity activity (65–70% $VO_{2\ max}$ or below anaerobic threshold) uses similar or more kilojoules and fat (even though the proportion of fat oxidised appears less). High-intensity exercise can therefore be an effective and time-efficient means of enhancing daily energy expenditure in athletes.
- Resistance training may increase LBM and therefore RMR. These benefits should be considered within the context of the exercise program devised.
- The best mode of exercise is generally determined by considering issues of access, time, injury risk and propensity for overuse of certain muscle groups. While running or weight-bearing activity appears (at least anecdotally) to produce superior weight loss results, other forms of aerobic exercise, such as swimming or cycling, may be selected as they result in less impact on the skeleton.

PSYCHOLOGICAL ASPECTS AND BEHAVIOUR MODIFICATION

- Approaches used to examine food behaviour—including food diaries and visual analogue scales to measure hunger, emotional eating and food cravings—can help athletes identify triggers to inappropriate eating. Behaviour modification techniques can then be used to manage, control or avoid problem eating behaviours.
- Referral to a clinical or sports psychologist is often integral to the success of weight-loss programs as the psychological approach is a key determinant to improving athlete confidence and subsequent compliance.

TRAVELLING AND SPECIAL ATHLETE CONSIDERATIONS

- Travelling may disrupt weight loss, as appropriate food is not always available and the daily routine is changed. Strategies that may assist include healthier (generally low-fat, low-energy) meals on airline flights, a travel meal plan and shopping list (if meals are self-catered), and a menu plan for professional caterers. A list of best choices from restaurants or take-away outlets is also useful. Avoidance of inappropriate snacks is easier if nutritious alternatives have been pre-organised. Many restaurants post menus on their website, which can help individual athletes and teams plan ahead (see Chapter 23).

- A meal plan to maintain energy balance when training is reduced (e.g. tapered) or interrupted due to injury, illness or during the off-season is beneficial to athletes prone to weight gain at these times.
- Athletes are exposed to buffet-style eating, especially in athlete villages and residential dining halls. Buffets encourage over-consumption and inclusion of foods that might not be typically consumed. Weight gain and *buffet boredom* can be offset by advising the athlete to avoid sampling every dish on offer at each meal and select a protein, CHO and several vegetable options, similar to a home-cooked meal.
- Fad or unscientific dietary regimens often appeal to athletes because they promise quick, effective weight loss. Providing advice to both athletes and coaches (and parents) on sound diet and training approaches to facilitate weight loss helps address misconceptions, mixed messages and potentially unsafe practices. The use of the sports medicine team, including a sports dietitian, is essential.
- The use of supplements and/or drugs to assist weight loss is a temptation for athletes. Despite claims, dietary supplements promoted for weight loss lack scientific support. Although a number of weight loss drugs have proven efficacy, they are either banned by WADA or their use would seem unethical or unsafe in athletes who are not at a medical risk because of excess weight.

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CHAPTER SEVEN

Making weight

Janet Walberg Rankin and Jennifer Gibson

7.1 Introduction

Most athletes who compete in weight-category sports reduce weight rapidly over 1 to several days to achieve their weight target. Loss of 5% or more of body weight is not unusual for athletes competing in sports such as wrestling, lightweight rowing, judo and boxing. These athletes are weighed in the presence of an official prior to their match or competition to compete in their designated weight category. If body mass (BM) is even slightly higher than the category allows, the athlete is disqualified from competing, which provides a powerful motivation to achieve the required weight.

Familiarity with the culture, rules and weight divisions/restrictions and the typical practices of athletes to reduce BM in weight-category sports is essential for sports dietitians and medical personnel to assist athletes to make weight safely. This chapter reviews the research about the weight-loss behaviours of athletes in making weight and the potential health and performance effects of rapid loss BM. Although BM is the scientific term, we interchange it with the commonly used terms, *weight* or *body weight*. Strategies are provided to guide advice about safe and effective means of making weight for competition and recovering afterwards.

7.2 Sports with weight divisions or restrictions

As seen in Table 7.1, a variety of sports involve weight divisions for competition. Each sport has specified procedures for prescribing and monitoring competition weights for participants. The intention of weight categories is to provide an *even playing field* for sports to match athletes—at least for weight—so that the athletes with the higher BM will not have a competitive advantage. In combative and strength sports, an athlete with a greater muscle mass and reach can generate more power than a smaller and lighter opponent. Thus, matching individuals of similar body weight should theoretically make these sports safer and fairer. The reality is that many athletes train well above their competitive weight class to maximise strength and power. To meet competition weight class, such athletes may practise extreme weight-loss methods to achieve rapid weight loss, hoping to recover between the weigh-in and the competition, and compete with an advantage over a smaller opponent. Others may be chronically dieting throughout the season to maintain a weight close to their competitive weight class.

TABLE 7.1 Summary of specific issues in making weight for weight-division sports

Sport	Competition	Weight categories		Weigh-in procedures	Personal insight
Wrestling—senior international	1 bout = 2 × 3 min rounds for international competition Competition over 1 d Athletes compete in 6 or fewer bouts each day of competition	Olympic weight classes Freestyle/Greco Roman wrestling (kg)* 57/59 65/66 74/75 85/86 97/98 125/130 Two non-Olympic categories per style will be used in all official FILA competitions (world/continental championships, world cups, golden grand prix): 61 kg/71 kg 70 kg/80 kg *adopted January 2014	Olympic weight classes female wrestling (kg)* 48 53 58 63 69 75 Two non-Olympic categories will be used in all official FILA competitions (world/continental championships, world cups, golden grand prix): 55 kg 60 kg *adopted January 2014	Weigh-in on the evening before competition High school/university weigh-in is day of competition Expected to weigh-in only once at start of competition	Typically wrestlers undertake minimal aerobic training in their daily training schedule. Therefore, it may be problematic for them to add substantial amounts of aerobic activity close to competition to make weight
Boxing—amateur	Each bout is 3 × 3-min rounds Competitors box every second day and may be expected to box 4–5 times during the tournament Men's headgear was removed and considered optional for international and Olympic competition	Male (kg) Limit 49 Light fly 52 Fly 56 Bantam 60 Light 64 Light welter 69 Welter 75 Middle 81 Light heavy 91 Heavy 91+ Super heavy	Women (kg) Limit 45 51* 54 57 60* 64 69 75* 81 81+ *Only 3 weights at Olympic Games	All competitors must attend general weigh-in on the morning of the first day of competition. On the following days only those who are drawn to box shall appear to weigh-in. Boxing does not commence earlier than 3 hours after the time appointed for the close of weigh-in on these days.	Amateur boxers are required to weigh-in on the morning of each bout. The requirement for further weigh-ins may limit their ability to rehydrate and refuel after a bout.
Boxing—professional	Australian National and the various World Association Title fights are fought over 12 × 3-min rounds Fights may vary between 4 and	17 currently recognised weight divisions/classes for professional male boxers and 13 for women, but they may vary by country Four major sanctioning bodies are: The World Boxing Association (WBA), World Boxing Council, International Boxing Federation and the World Boxing Organization		Competition weigh-in procedures vary between professional boxing associations. For WBA world title fights,	Since boxers typically compete only 3–4 times/yr, there is considerable opportunity to overindulge between fights. This pattern of 'on and off' dieting behaviour leads to large weight swings.

	12 × 3-min rounds Competitors usually fight 3–4 times per year			competitors weigh-in the day prior to the fight, between 4 and 8 pm.	
Horse racing—jockeys	Races are conducted over various distances, usually 1000–2000 m (lasting 1–2 min), but may be as long as 3200 m (the Melbourne Cup). Jockeys may have up to 6–8 rides during one racing session, which lasts roughly 5–6 h.	Minimum weight for any race in Australia as sanctioned by the Australian Rules is 43.5 kg, including saddle and accessories Minimum weight varies between states and within states (country versus city) for local races Minimum weight also varies between countries Horses are weight handicapped according to ability or age		All jockeys weigh-in no later than 45 min before racing for each race. Jockeys on horses that earn prize money (plus the next best finisher) have to weigh-in directly after the race.	As jockeys need to weigh-in *after* the race, recovery strategies cannot be employed prior to their event to address any short-term weight-loss measures used to make weight.
Olympic weightlifting	Competitors have three lifts in each discipline: clean-and-jerk and snatch Competition over 1 d	Male (kg) Limit 56 62 69 77 85 94 105+	Female (kg) Limit 48 53 58 63 69 75 75+	The weigh-in of each bodyweight category begins 2 h before competition.	With limited aerobic activity in the daily training regimen, these athletes face similar issues to those of wrestlers.
Karate	Have team and individual events 3-min bout for males and a 2-min bout for females Competitors may be expected to fight 6–8 times Competition over 1 d	Male (kg) Limit 60 67 75 84 100+	Female (kg) Limit 50 55 61 68 68+	Same day, usually 1–2 h before start of competition	Skill and reach are likely to have a greater performance advantage than body weight.
Lightweight rowing	Race is over 2000 m course Competition is over 7 d Expected to compete every second day	Male Average weight of crew not exceed 70 kg No individual >72.5 kg No single sculler >72.5 kg Female Average weight of crew not exceed 57 kg No individual >59 kg No single sculler >59 kg Coxswain Minimum 55 kg (M) and 50 kg (F/mixed crews) May carry up to 10 kg of dead weight		Weigh-in is not less than 1 h and not more than 2 h before Competitors are expected to weigh-in each day and for each event they are competing in	Rowers are required to make weight on the morning of each heat, for repechage and for the final in a regatta, requiring a decision to either stay down for the duration of the competition, or to make weight repeatedly for each race. It may be best to optimise recovery following weigh-in and then return back to weight following each race.
Judo	Each bout for senior competitors is 5 min (females	Male (kg) Limit 60 66	Female (kg) Limit 48 52	Weigh-in period for senior international competitions is	

	and males) Competition is on 1 d Can be expected to contest 4–5 bouts during the competition Minimum of 10 min between bouts	73 81 90 100 100+ Max of 4 weight classes at Olympic Games	57 63 70 78 78+	conducted the day before Other competitions may be morning of competition	
Taekwondo	Each bout involves 3 × 3-min rounds with 1 min between rounds Competition is on 1 d only Can be expected to contest 5–8 bouts during the competition	Male (kg) Limit 54 Fin 58 Fly 62 Bantam 67 Feather 72 Light 78 Welter 84 Middle 84+ Heavy Olympic weights 58 68 80 80+	Female (kg) Limit 42 Fin 51 Fly 55 Bantam 59 Feather 63 Light 67 Welter 72 Middle 72+ Heavy Olympic weights 49 57 67 67+	Weigh-in for senior international competitions is conducted the day before	The nature of the event requires aerobically based training. As a result, athletes are able to cope with increased aerobic activity in the lead-up to competition to make weight.

Many wrestlers, for example, believe that weight loss is a critical part of the culture of wrestling. In one study, 70% of the high school wrestlers interviewed from nine rural teams claimed that losing weight during the season was *very important* for winning (Marquart & Sobal 1994). Most thought that making weight was *hard*; 31% *very hard* while 47% *somewhat hard*. Hence, losing weight can exacerbate the physiological and potential psychological stress and risks of competing.

7.3 Methods used to make weight

7.3.1 Combat sports: wrestling, judo, taekwondo, boxing and martial arts

Some of the earliest studies of weight-loss methods of wrestlers were undertaken with high school wrestlers in Iowa in the late 1960s. In one study of 528 wrestlers, the main methods used to lose weight included food restriction (83%), fluid restriction (77%) and increased exercise (83%) (Tipton & Tcheng 1970). Athletes may use a combination of several methods and so have multiple responses. In another study of weight-loss methods of male wrestlers (n = 2600, age 15–18 yrs) competing at a national championship, additional aerobic exercise (running 91%, swimming 24%, cycling 33%), dehydration (saunas 55%, exercising in vapour-impermeable suits 49%) and laxatives (11%) were used (Alderman et al. 2004). In this study, the average weight gain after weigh-in at competition was 4.8% of BM, with some athletes gaining up to 16.7 kg or 13.4% of BM.

Artioli and colleagues (2009a) surveyed the prevalence and methods used for rapid weight loss in 822 Brazilian judo athletes (607 M, 215 F, 19 ± 5.3 yrs). Popular weight-loss methods reported as *always* undertaken by athletes included increased exercise (62%), heated training rooms (26%), restricting fluids (21%), skipping one or two meals (19%), gradual dieting (18%), spitting (19%), training with rubber/plastic suits (18%) and fasting (12%). Around 47% of athletes reported losing between 2% and 5% BM, 39% reported losses of 5.0–9.9% and 5% lost more than 10% of body weight.

In another survey of eating behaviours and weight-making practices in 30 international and national taekwondo athletes (23.4 ± 4.6 yrs), who competed 6–9 times/yr, 87% needed to reduce body weight for competition (Fleming & Costarelli 2009). These athletes reported similar weight-loss methods to judo athletes—that is, combining increased training with or without fluid restriction (~37%), saunas (20%) sweat suits (13%) and restricted food intake (10%). In a cross-sectional study of 580 regional, national and international Brazilian athletes from grappling (judo, jujitsu) and striking (karate and taekwondo) sports, 91% used increased activity and 68% used a low-calorie diet (45% restricted carbohydrate; 33% restricted fat) (Brito et al. 2012). Half of the athletes used saunas and plastic suits and 34% used laxatives or diuretics. The striker sports started these methods in adolescence.

In a simulated competition environment, 16 amateur boxers (23.5 ± 4.8 yrs) were given 1 week to make weight using their preferred methods (Hall & Lane 2001). Boxers lost 5.2% of body weight, with all restricting food and fluid intake and 73% using increased exercise. In a case study report of an elite professional boxer, only one meal/d was consumed and sweat suits were worn during every training session for 6–8 weeks prior to competition (Morton et al. 2010). From 24–48 hours pre-competition, he consumed no food or fluid and continued to exercise in sweat suits in the hours before weigh-in. Clearly, inappropriate weight loss methods and fairly large weight losses are endemic in combat sports, which suggests that many are training well above their competitive weights.

7.3.2 Lightweight rowers

Many lightweight rowers use similar weight-loss methods to athletes in combat sports to meet weight goals. Morris and Payne (1996) reported average weight losses of 5.9% BM in female lightweight rowers and 7.8% in men during the competitive season compared to pre-season. Additional exercise (73%), food restriction (71%) and fluid restriction (63%) were the most popular methods used to lose weight in this group. In another study of 107 lightweight Australian rowers, most athletes (76% M, 84% F in the U23 category) needed to lose weight to meet their competition weight (Slater et al. 2005). Prior to competition, the reported maximum weight loss in the preceding 4 weeks was 6 kg in men and 4.5 kg for women. Most athletes used moderate energy restriction and additional exercise to lose weight, but dehydration was also popular (83%). Female rowers were more likely to restrict CHO-rich foods, salt and fibre than males.

7.3.3 Jockeys

Jockeys are a high-risk group because of their high frequency of racing and repeated use of inappropriate weight-loss methods to meet race weight or for ongoing weight control. Typical methods include appetite suppressants and other drugs (e.g. diuretics, laxatives, smoking); extended fasting and skipping meals; chronic low-energy intakes; and restrictive diets, including avoidance of specific food groups (King & Mezey 1987). These weight-cycling practices are associated with an elevated rate of bone loss and reduced bone mass compared with controls (Dolan et al. 2012). In one survey of 21 professional jockeys, most (86%) claimed that they could lose around 2 kg between 24–48 hours before or on the designated race day to meet weight requirements (Dolan et al. 2011). In this study, rapid-weight-loss methods included a combination of saunas (86%), exercising to induce sweating (81%) and restricted energy intake (71%). Many tend to binge eat after races and engage in these extreme last-minute weight-loss measures. These methods increase the risk of dehydration and reduced aerobic capacity on race day (Dolan et al. 2013). In another study of 14 English jockeys across 48 stables, 13% were below the population weight for their height; the lightest jockey was 21% below the cut-off (King & Mezey 1987). In this study, jockeys reported that weight control was their highest priority in life during the racing season.

7.4 Beliefs about weight loss and competitive success

Some athletes in weight-division sports believe that the acute weight loss pre-competition and then the rapid regain after weigh-in gives them the winning edge. Several studies have investigated whether this perception is predictive of success in wrestling matches (Horswill et al. 1994; Scott et al. 1994; Utter & Kang 1998). The magnitude of weight gain did not affect the outcome of the match for 11 collegiate wrestlers over four matches in the season (Utter & Kang 1998) or for 668 wrestlers at a National Collegiate Athletic Association tournament in the US (Horswill et al. 1994; Scott et al. 1994). However, given the absence of a control group and the relatively small weight change reported after weigh-in reported in these studies, there is room for only a cautious interpretation of a real advantage of acute weight loss.

7.5 Effects of weight loss: research perspectives

Most weight-category sports are based on anaerobic energy utilisation with an emphasis on muscle power. The ideal goal for these athletes is to reduce total BM through loss of body fat or fluid, rather than lean tissue, and to recover any water and glycogen lost just before competition. Also, these athletes usually intend to maintain training capacity and health while making weight, or at least to recover their performance capacity to pre-weight loss levels after the weigh-in. However, some athletes do not achieve these goals and experience impairment of performance or health. This section considers the limited research available on the physiological, psychological and performance effects of acute- and long-term weight loss in athletes participating in weight-making sports.

7.5.1 Effects on plasma volume and electrolyte loss and susceptibility to heat illness

As mentioned in section 7.3, dehydration is the most practised method to rapidly reduce the last few kilograms to make weight before competition. The risk of heat injury is high with this method and was indirectly responsible for the death of three young wrestlers seeking to make their target weight in the early 1990s (American Medical Association 1998). Dehydration can produce a large decrease in plasma volume, which proportionally can be 60–80% higher than the reduction in body weight (Greiwe et al. 1998; Yankanich et al. 1998). A low plasma volume at the start of competition reduces the amount of fluid available for sweat loss and hence cooling, which contributes to an increased risk of heat injury.

The effect of dehydration on body electrolyte balance depends on the method of dehydration used. A comparison of several rapid weight-loss strategies over 24 hours demonstrated that sweating significantly increased loss of electrolytes while reduction in fluid intake did not severely impact electrolyte balance (James & Shirreffs 2013). Although there was no difference in plasma volume change or urinary loss of sodium, potassium or chloride between weight-loss treatments (10% of control fluid intake, 25% of control energy intake, or combination of both), electrolyte balance and plasma osmolality was greater for fluid restriction than the energy restriction trial. In summary, fluid restriction as an acute weight-loss strategy allows more retention of electrolytes and more rapid rehydration after weigh-in and weight regain than using energy restriction or sweating.

7.5.2 Effects on lean tissue maintenance

It is difficult to maintain lean tissue mass with rapid and severe weight loss. Several studies have verified that wrestlers (Roemmich & Sinning 1997) and rowers (McCargar et al. 1993) had lower lean body mass (LBM) during the competitive season, compared to pre-season or non-athlete classmates. Fortunately, during the off-season, muscle growth and hormonal levels rebounded to *normal* levels and were no different in matched controls.

The rate and magnitude of weight loss in the short- and long-term, the frequency of food intake and the amount of protein intake consumed while losing weight can inhibit protein synthesis and reduce LBM and potentially muscle growth. The long-term effects of repeated weight-loss episodes on lean tissue mass, changes in body composition and other health effects is less clear in weight-making athletes.

The magnitude of energy deficit and rate of weight loss on lean tissue

The magnitude of energy deficit and rate of weight loss influences the loss of LBM. In one study, 24 male and female elite Norwegian athletes (age 18–35 yr) followed an individualised energy-restricted diet and strength training program (Garthe et al. 2011a). The length of the intervention (4–12 wks) was determined by the rate of weight loss and the desired weight loss (minimum target for weight loss = 4% BM). Those who lost weight slowly (0.7% BM/wk, 19% energy deficit from usual energy intake) increased LBM and lost more fat mass than those at the faster rate of weight loss (1.4% BM/wk, 30% energy deficit), whose LBM slightly decreased (Garthe et al. 2011a). However those who lost weight slowly spent an extra 3 weeks in the strength program, which may have accounted for the increase in LBM. Although controlled studies of weight loss are scarce, this study supports a slow rate of weight loss (~0.7% BM/wk) in combination with a strength program if the intention is to gain LBM. Conversely, athletes who want to maintain or slightly decrease LBM might increase their rate of weight loss to 1.0–1.4% BM/wk. This equates to 0.5–1 kg/wk weight loss through moderate energy restriction.

Frequency of food intake

The popular concept that eating frequent small meals is a superior way to lose weight and maintain LBM has been examined in a cohort of boxers. Total weight loss over 2 weeks from a similar restricted energy intake of ~5020 kJ/d (1200 kcal/d) was no different when consumed in two or six meals/d (Iwao et al. 1996). However, more reduction in LBM occurred for the two meals/d option, providing some support for a small, frequent meal pattern. More research is needed to confirm the outcome of this study, although eating small amounts of protein and carbohydrate after training increases muscle protein synthesis and helps prevent protein degradation, which may explain this response.

Repetition of weight-making practices (weight cycling)

Based on studies of weight-making athletes reported in section 7.3, the average weight loss per period varied from 7.4% for wrestlers to 4.5% for lightweight rowers. Compared to other athletes—except perhaps jockeys—boxers reported the highest number of weight-loss episodes (26.4) over their career. The observed change in BMI for these athletes (18–50 yr) was, as expected, less than that reported in studies of those among the general population who are obese or overweight.

A meta-analysis of US weight-loss studies indicated that repeated weight cycling in overweight and obese people is associated with progressive increases in BM after each successive restrictive diet (Anderson et al. 2001). This response is often called *yo-yo* dieting. Other studies have shown similar observations (Bosy-Westphal et al. 2013) and substantial losses in LBM, especially in older men (Lee at al. 2010). The hypothesis that repeated weight cycling as a young athlete will permanently reduce LBM mass and enhance body fat gain in later life or with each successive weight loss episode was not supported in a study of 136 retired French athletes who competed in weight class sports (Marquet et al. 2013).

Nevertheless, severe fat and BM loss in normal-weight people that comprises a large loss of LBM may shift the regain of BM towards an increase in total fat and abdominal fat mass (Bosy-Westphal et al. 2013). In a controlled study of four repeated weight-loss episodes in five non-obese young women (mean age 24–26 yr, BMI 20.5kg/m^2), significant decreases from baseline in LBM, serum triiodothyronine, serum total thyroxine and resting energy expenditure were reported after the fourth episode of weight loss at 180 days (Kajioka et al. 2002). The subjects lost 4 kg with energy restriction alone (first diet period) and, in the subsequent 2 weeks, regained more than the weight they had lost by eating ad libitum. Although this study had a small sample size, it mimics the typical practices of weight-making athletes and highlights the potential side effects and need for ongoing monitoring. Clearly, longitudinal and controlled studies of athletes are needed to confirm whether repeated weight cycling has any real long-term effects on health outcomes and body composition.

Protein intake

Acute and rapid energy restriction and suboptimal protein intake reduce muscle protein synthesis (MPS) and LBM (Pasiakos et al. 2010). The amount and timing of intake of dietary protein can stimulate MPS and help prevent protein degradation associated with high-intensity exercise and moderately restrictive energy deficits (see Chapter 4). We found that resistance-training athletes who lost weight over a week while consuming a formula diet of 75 kJ/kg BM (18 kcal/kg) lost body protein when they were given the recommended dietary allowance (RDA) of 0.8 g/kg for protein, but were in nitrogen balance (no net body protein loss) when the diet contained twice the RDA for protein (Walberg et al. 1988). This finding was similar to more recent research that noted a greater loss of LBM in athletes over a week of reduced energy intake (60% of habitual) when the diet contained 1 g/kg protein compared to 2.3 g/kg (Mettler et al. 2010). A longer (21-d) energy-restriction period with a similar energy deficit caused less LBM loss when the diet had twice the RDA for protein (1.6 g/kg), compared to those subjects consuming the RDA for protein (Pasiakos et al. 2013). A higher protein intake did not further enhance this benefit.

Evaluation of older athletes suggests that there is catch-up of lean growth during the off-season that does not permanently influence body weight. In summary, protein intake should be up to twice the protein RDA during energy restriction to help minimise loss of LBM.

7.5.3 Effects on cognitive function and fatigue

Not surprisingly, most people do not feel mentally sharp and at their peak while losing weight. Mood, perceived exertion and memory can be negatively affected by energy restriction, dehydration or both. A group of 14 collegiate wrestlers tested for mood and cognitive ability before and after loss of an average of 6.2% BM reported impaired mood (more tension, depression, anger, fatigue and confusion) and poorer short-term memory after weight loss

(Choma et al. 1998). The changes were temporary and returned to baseline after 72 hours of recovery. Similarly, elite judo athletes who lost an average of 4% BM prior to competition had higher confusion, tension and decreased vigour while a comparison group who did not lose weight had no change in mood (Koral & Dosseville 2009). In another group of wrestlers who rapidly lost at least 4% BM weight using exercise, caloric restriction or fluid deprivation, an increase in confusion using the Brunel Mood Scale was evident while those who lost less than 4% BM had no impairment in mood (Marttinen et al. 2011). Using the Profile of Mood States, rapid weight loss of 5.2% BM in 16 amateur boxers was associated with significant decreased performance (in a circuit training task), increased anger, fatigue and tension and reduced vigour ($p < 0.01$) (Hall & Lane 2001). However no impairment in cognitive function was evident in jockeys who lost ~4% BM in 48 hours compared with controls (Dolan et al. 2013).

In summary, depending on experience and the rate and magnitude of acute weight loss, athletes are at risk of experiencing adverse effects on mood and motor function and decreased capacity for performing mental tasks during periods of weight loss and potentially at competition. These effects may be a consequence of energy restriction or dehydration. Training may be affected, which can translate to poor motivation and quality of workout.

7.5.4 Effects on performance

Self-determined weight loss

Athletes typically perceive that substantial weight loss over the season impairs performance, despite the belief of rapid recovery following acute weight loss and regain after weigh-in at competition (see section 7.4) Sixty-three per cent of high school wrestlers surveyed reported a perceived decrease in muscle strength, speed, agility and concentration as a result of weight loss (Marquart & Sobal 1994). Other experimental studies confirmed a reduction in muscle strength for adolescent wrestlers compared to their non-wrestling classmates over the season (Roemmich & Sinning 1997).

Observational studies of elite athletes where diet and/or fluid intake is not controlled during weight loss have high ecological validity to real-life sports practices and provide some insight into performance effects. For example, Marttinen and colleagues (2011) did not detect any effect of acute weight loss up to 8.1% on grip strength or 30-second Wingate sprint test in wrestlers who used *self-selected* methods of weight loss. Similarly, Koral and Dosseville (2009) compared performance of male and female judo athletes who lost ~4% BM using individual strategies over a month prior to national competition. Although there was no randomly assigned control group in this study, performance measures were compared to judo athletes who did not lose weight. Compared to the control group, performance of the brief, intense tests (standing jump, countermovement jump) was not affected by weight loss, although weight loss was associated with a ~6% reduction in short-burst judo moves. Although these studies have high practical value, they provide limited understanding of the advantages or disadvantages of unsupervised weight-loss strategies.

Controlled dehydration

Although scarce, experimental studies with controlled fluid restriction protocols suggest that rapid weight loss negatively impacts performance. The impact appears dependent on the magnitude of weight loss, the dehydration method used and the type of exercise performance test utilised (Folgelholm 1993; Walberg Rankin 1998). The detrimental effect of dehydration on aerobic performance is well documented, but the effect on muscle power, strength and agility is less clear. One study reported a reduction in muscle strength following weight loss involving dehydration (Viitasalo et al. 1987) while another study found no effect of dehydration on muscle isometric strength and endurance (Greiwe et al. 1998).

Energy restriction

Adding energy restriction to dehydration, or dieting alone, appears more consistent in causing impairments of muscle performance. Muscle strength was reduced in athletes who lost weight using energy restriction alone (Walberg et al. 1988; Walberg Rankin et al. 1994) and in those who energy restricted first and then dehydrated close to competition (Viitasalo et al. 1987). In addition, several studies using intermittent bouts of high-intensity exercise developed to mimic the work patterns of sporting events, including wrestling, showed that weight loss using energy restriction alone or with dehydration over several days significantly reduced the amount of work accomplished during an intermittent upper body sprint test (Horswill et al. 1990; Hickner et al. 1991; Walberg Rankin et al. 1996).

It makes theoretical sense that the magnitude of energy intake reduction would negatively influence performance. However, no difference in performance was measured in judo and wrestler athletes who lost ~6% BM at a fast rate (over 2.4 d) compared to a slow rate of weight loss (over 3 wk) both through energy restriction and dehydration (Fogelholm et al. 1993). A similar hypothesis was tested by Garthe et al. (2011a) in elite Norwegian athletes who were counselled to reduce energy intake to result in either slow (0.7% BM/wk for 8.5 weeks) or fast (1.4% BM/wk for 5.3 weeks) weight loss with a target of ~5.5% BM. At the end of the intervention, both groups of athletes lost an average of 5.6–5.7% BM. Sprint speed was not altered by either weight loss regime but muscle strength rose more for the athletes on the 19% reduction in energy intake than those following a 30% deficit. This increase in muscle strength in the slow weight-loss group may be linked to an increase in LBM or to the extra 3 weeks spent in the strength training program. A subset of these athletes provided follow-up data to assess the long-term effect of weight loss on performance 1 year after the 6% weight loss (Garthe et al. 2011b). In this cohort, no differences in performance of various muscle strength- and power-related events were observed at follow-up.

In another study, Artioli et al. (2010b) investigated the effect of rapid 5% BM loss on subsequent (4 h) judo performance in 12 experienced judoka athletes matched with controls. Athletes were given 5 days to use self-selected regimens for weight loss and replenishment after weigh-in. There was no significant impact of weight loss on judo or physiological test markers. Together, these studies suggest that the magnitude of energy restriction and rapidity of weight loss negatively impact high-intensity performance. However, athletes can apparently recover if provided sufficient time.

Dietary carbohydrate

Several studies have demonstrated the value of adequate carbohydrate in a weight-loss diet. Wrestlers maintained high-power performance on a high-CHO diet (66–70% of energy), but not when they consumed a modest CHO (41–55% of energy) diet for weight loss (Horswill et al. 1990; McMurray et al. 1991). A practical suggestion, based on these studies, is to maximise carbohydrate intake while maintaining other dietary goals in an energy-restricted diet. In summary, acute weight loss as low as 2% of body weight, especially if rapid and low in carbohydrate, may reduce performance in athletes, particularly in repeated high-intensity sprint tests. Many athletes already believe that weight loss has adverse effects on performance but assume that they can recover in time for competition. This belief has not been supported in observational studies (see section 7.4).

7.6 Recovery strategies

7.6.1 Fluids and electrolytes

Recovery of fluids lost through dehydration may take 24–48 hours, which is longer than is commonly appreciated by athletes (Costill & Sparks 1973). Athletes who intend to use dehydration for weight loss and who have less than 24 hours to recover are cautioned to limit weight loss to not more than around 2% BM (Sawka et al. 2007). Performance, tolerance to heat and ability to fully rehydrate can be impaired with greater degrees of dehydration (see Chapter 13). For example, a 2-hour rehydration period with 1.5 L of water is inadequate to restore a loss of 5% of BM, accompanied by a 12.5% reduction in plasma volume (Burge et al. 1993). The most effective strategy to restore fluid balance includes an intake of fluid equivalent to 125–150% of the fluid deficit, together with replacement of lost electrolytes, principally sodium (Shirreffs et al. 1996; Shirreffs & Maughan 2000). This is covered in more detail in Chapters 13 and 14.

Few studies have characterised the typical dietary recovery strategy of athletes who practise weight cycling before competition. Slater and colleagues (2005) questioned lightweight rowers concerning their food and fluid intake during the 1–2 hours between weigh-in and competition. Energy intake varied by gender and competitive category (e.g. U23 or Open) but ranged from 30–53 kJ/kg. Fluid and especially sodium intake (about 8–14 mL/kg fluid and 10–20 mmol/L sodium) were below recommendations for optimal rehydration.

7.6.2 Carbohydrate

Weight loss in athletes has been associated with reduced muscle glycogen storage. Wrestlers losing 5% BM with food and fluid restriction had a 54% decline in muscle glycogen (Tarnopolsky et al. 1996). A similar weight loss over just 24 hours in wrestlers was accompanied by a 30% drop in muscle glycogen (Burge et al. 1993). Suboptimal muscle glycogen levels can be overcome with time and adequate CHO intake (see Chapter 14), but recovery may not be realistically achieved between the weigh-in and the start of competition. Wrestlers who lost about 5% BM with energy restriction and were re-fed a diet with 50% of energy from CHO over 5 hours did not recover their performance to baseline levels while those fed a higher CHO diet (70% of energy intake) had a performance similar to baseline after recovery (Walberg Rankin et al. 1996). We found that most collegiate wrestlers consumed a high-CHO diet (66% of energy intake) between weigh-in and the match, but the range in reported energy intake from CHO during re-feeding was variable (28–87% of energy intake) (Pesce et al. 1996). Thus, since energy restriction to produce weight loss is likely to cause a drop in muscle and liver glycogen, which may contribute to impairment of performance if recovery time is short, athletes should be counselled to eat a high-CHO diet during recovery from energy restriction.

7.7 Role of the sports dietitian to reduce unsafe weight-loss practices

Many athletes could use assistance to develop a plan for weight loss. When a group of high school wrestlers were asked who helped them plan their weight-loss efforts, *nobody* was the second highest answer (42%), after coaches (44%) (Marquart & Sobol 1994). A minority of these wrestlers used trainers (11%), doctors (7%) or dietitians (3%). In a more recent study of elite lightweight rowers questioned on their sources of information about weight loss, *other rowers* was the most common response (Slater et al. 2005). Interestingly, about one-fifth of the female rowers but few of the male rowers ranked dietitians as *very influential*.

Sports dietitians or other dietitians/nutritionists therefore need to be pro-active in marketing their services to athletes. Most coaches are not tertiary educated in nutrition intervention and thus may not be the best choice to advise the athletes on safe weight-loss methods. Sossin and colleagues (1997) found a poor knowledge about making weight in a group of high school wrestling coaches: the percentage of correct answers on a nutrition knowledge survey was 64% for weight loss, 59% for training diets, 57% for dehydration and 52% for body composition. Although nutrition knowledge could be improved through coach education, the best approach is to use sports dietitians for counselling and intervention.

In a recent consensus statement, Sundgot-Borgen and colleagues (2013) recommended that there is a need for action by coaches, healthcare professionals and sport bodies to minimise health risks for athletes in weight-sensitive sports. The authors suggested sport-specific and gender-specific preventive programs, specific criteria for better identifying athletes with eating disorders and further modification to the regulations in at risk sports. The US National Collegiate Athletic Association (NCAA) has implemented an education program and new rules to address unsafe weight-cutting practices, which was in response to three young wrestlers dying in 1998. These new rules have helped to reduce inappropriate weight-loss practices but have not substantially changed the culture in weight-cutting sports. In one study of 741 collegiate wrestlers from 43 teams undertaken the year after implementation of the new rules, the majority used more appropriate methods including gradual dieting (79%) or increased exercise (75%) to lose weight, although some were still using unsafe practices (55% fasting, 28% saunas, 27% impermeable suits) (Oppliger et al. 2003). The authors concluded that the weight-loss amount and methods were less extreme than those observed prior to the rule changes. The need for a weight management control program in judo has also been proposed using similar guidelines as the NCAA program (Artioli et al. 2010c).

It is likely that collegiate and especially Olympic-level athletes have greater access to medical and nutrition professionals. However, there may be resistance by some coaches and athletes to use sports dietitians for fear they will discourage weight loss.

Summary

Athletes competing in weight-category sports are highly motivated to lose weight acutely prior to weigh-in. Many will use drastic energy-restricted diets or dehydration to achieve a temporary weight loss. Rapid losses of 5% of body weight, typical in many of these sports, can result in reductions in physical performance, impairments in cognitive function and increased susceptibility to heat illness. Rapid weight loss, especially when dietary protein is at the RDA or below, will likely reduce lean tissue mass. Gradual weight loss has fewer negative health consequences and should be recommended when weight loss is desirable and time allows.

Weight-loss goals and strategies should be determined with the assistance of healthcare professionals (such as team physicians, sports dietitians) to ensure appropriate and safe weight loss. Body composition assessment using skinfold measurements, validation of euhydration with urine specific gravity and diet records can be part of the assessment used by professionals to develop a goal weight and strategy for change. Regular monitoring of health and performance of the athlete will help in making decisions to adjust the weight-loss plan. Athletes should understand that substantial weight loss through heat exposure and dehydration causes reductions in plasma volume that are difficult to remedy within several hours between weigh-in and competition. Those athletes who lose weight acutely for weigh-in should attempt to recover as rapidly as possible using fluids and food adequate in sodium and carbohydrate.

Practice tips

HANNAH EVERY-HALL

Weight loss for any reason in any individual or group of people is multifaceted. Making weight for competition is no different, although most athletes in this situation are not overweight and have low body fat. Nutrition guidelines for weight-making sports vary from sport to sport and from individual to individual. These practice tips largely reflect the personal and professional experience of the author, a national and international lightweight rower for 10 years and a dietitian/sports dietitian who has worked with a variety of weight-making sports and weight-loss clients in private practice.

SUITABILITY FOR WEIGHT RESTRICTION

Typically, successful athletes are those with a higher muscle mass and lower body fat. Consequently, it is enticing for coaches to push athletes on the *fringe* of weight divisions into the lower weight category to try to maximise lean mass and improve power:weight ratios. For some individuals, this option is achievable and involves minor modification to eating habits and training programs. For others, who are already very lean, have low fat mass and increase muscle mass readily in response to strength programs, a lower weight division may never be achievable without compromising health status, performance capacity and wellbeing. Of concern are the athletes whose weight is in the mid-range of two weight classes. For these athletes, their training weight is much higher than their competitive weight but their frame size, musculature, strength and power is no match for the athletes in the higher weight category. Some sports (e.g. combat, boxing, wrestling, judo) have several weight divisions whereas in rowing there is only ‘lightweight’ or ‘heavyweight’, which limits options for changing weight classes (see Table 7.1). Too often, weight categories are determined by coaches without impartial analysis, such as body composition measures. Determining the *ideal* weight category for an individual athlete relies on a combination of body composition data, weight history, current eating behaviour and extent of dietary restriction required.

The outcomes of the nutrition, anthropometric, psychological and medical assessment/review needs to be considered and discussed with the athlete (and coach) before a decision is made to reduce BM to a lower weight class or to increase to a higher weight class. Modification to the strength training program may also be required.

Body size, shape and composition (anthropometric measures)

- Body composition measurements provide a predictive estimate about how much fat mass (FM) and/or LBM can be lost and whether the target weight class is achievable.
- A complete anthropometric profile that includes standardised protocols for measuring height, weight, skinfolds, girths, lengths and breadths should be undertaken and interpreted by an International Society for the Advancement of Kinanthropometry (ISAK)-accredited anthropometrist. Measurement includes frame size assessment and provides estimates of LBM and FM, which can be compared to international standards for elite athletes in the relevant weight-class sport (see Chapter 3, Tables 3.2 and 3.3).
- Another recently popular technique now available in many elite sports organisations is dual energy X-ray absorptiometry (DXA), which allows rapid body composition measurement (~5 min). DXA generates more accurate estimates of whole and regional fat distribution and lean mass than anthropometry. Devices such as BodPod, ultrasound and 3D scanners are readily available but may not be standardised, which increases measurement and interpretation errors.
- If a complete anthropometric profile or DXA is not feasible, skinfold measures are the next best option, despite their limitations. See Chapter 3 for the use and limitations of all anthropometric methods.
- Interpretation of body composition measures should also consider maturational timing and the potential for an increase in LBM from growth or in response to the training program.
- Generally, body composition assessment is recommended at regular intervals throughout a training year (e.g. before and after a high-intensity strength and conditioning or endurance phase of training, with sudden growth spurt, with substantial weight loss or gain, before and after competition, and to coincide with performance testing).

A DIETARY ASSESSMENT

A dietary assessment should be conducted by an experienced sports dietitian, preferably with experience in working with athletes in weight-making sports. Key areas to assess and examples of questions to ask that are specific to weight-making athletes include:

- *Current dietary energy intake and balance* Is the athlete already cutting back on volume of food or type of food choices to achieve the weight target? Is the athlete unrestricted in their food choices? Is there scope in dietary habits and current food choices for further energy restriction? Is the athlete likely to sustain a 'living weight' close to that required for competing?
- *Nutrition knowledge and attitudes* Does the athlete have an interest in or the background knowledge about how diet relates to capacity to train and recover, to mood and to health and wellbeing? They will already have many ideas (often conflicting) from what they have observed in their peers, what they have read and what they have been told.
- *Social support, organisation and life skills* Does the athlete have the organisational skills, time, money, social support and food skills to modify food choice and achieve good nutrition status while losing weight? What is their past experience in making weight? What has been problematic and how have they addressed those problems? What strategies have been effective? What other life challenges or barriers are likely to affect food choice (e.g. disruptive home environment, study)?
- *Psychological aspects* Assess attitude about body image, eating behaviours and dealing with stress. Screen for symptoms of low energy availability (see Commentary 7) and for signs of disordered eating behaviour (see Chapter 8). If aberrant eating behaviour, depression or instability is suspected or evident, refer to a clinical sports psychologist. In a susceptible individual, planned weight loss even under supervision can trigger disordered eating behaviours that can develop into an overt clinical eating disorder.
- Weight-making athletes should have regular nutrition reviews (i.e. at least four times a year).
- If the athlete has a history of chronic suboptimal nutrient intake and low energy availability, refer for medical review.

A PSYCHOLOGICAL REVIEW

Planned and unsupervised weight loss can trigger the development of disordered eating behaviour. Both males and females in weight-category sports are at high risk of disordered eating behaviour and eating disorders (see Chapter 8, section 8.5). The ongoing issue of making weight or maintaining a weight well below usual weight can also increase the risk of anxiety, sleep disturbances and depression and become a major focus and obsession. Recognising the warning signs of eating disorders and psychological issues and early referral to a psychologist is essential.

Ideally a psychological assessment should be conducted on all athletes participating in weight-restricted sports early in their career to assess risk factors, resilience and coping mechanisms under physical and mental duress, which are exacerbated during competition.

A MEDICAL REVIEW

A medical review is recommended for a weight-class athlete, particularly if there has been a history of chronic dieting or risk of suboptimal nutritional status and the athlete is intending to further restrict energy intake. This review might include an assessment of:

- iron status
- bone health, including a family history of poor bone health (e.g. osteoporosis); ideally this would also include an objective measure of bone health such as a DXA scan
- reproductive health (e.g. pubertal development, menstrual status)
- medical/physical factors (e.g. injury risk, medical conditions which may alter ability to change body weight, GIT disturbances).

DIETARY INTERVENTION FOR WEIGHT-MAKING ATHLETES: CONSIDERATIONS AND STRATEGIES

Weight-making athletes present with an array of experiences, influences and knowledge. Many have strong attitudes and behaviours regarding what they should and should not be doing to make weight, which are difficult to change.

Suggestions for effective dietary intervention relevant to these athletes are described below.

Know your athlete and the rules of their sport

- Athletes in weight-category sports are strongly influenced by the anecdotes and strategies used by other athletes more experienced in making weight. Educate the athlete and help them to discriminate fact from fiction.
- Build rapport and trust with the athlete. Explain the principles behind your advice. Providing justification and reinforcement for your advice helps to address inappropriate and misguided attitudes. Many are blinded by the end goal of losing weight.
- Athletes in weight-category sports usually seek your advice, are highly motivated to lose weight and are compliant but also often take advice to the extreme.
- For some athletes, making weight can itself become a competition.
- Allow for growth and development of adolescent athletes.
- Discuss body composition changes that are expected with growth and maturation and from the effects of training.
- The rules and regulations that govern the weigh-in procedures in sports differ (see Table 7.1). Weigh-in times range from 1–24 hours before competition. In rowing, the weigh-in may be a one off over a 7-day regatta. The timing of weigh-in and recovery time underpins dietary intervention.

Develop a periodisation and eating plan

- Long-term weight reduction and fat loss to achieve the *ideal* competitive weight target is the primary goal.
- In liaison with the athlete, develop a yearly periodisation plan for weight loss that includes long- and short-term body composition/weight targets over the whole season in conjunction with competition dates and intense training blocks. This practice allows time for modification for varied nutritional strategies and safe rapid weight loss approaches that can be tried in training.
  – Tabulate this plan to allow the athlete to actively participate and provide ongoing feedback.
  – Include practical dietary strategies and realistic targets that can be modified because weight is not usually lost in a linear pattern.
- Because of the potential for chronic restrictive eating behaviour in weight-making athletes, a periodisation plan allows the opportunity to challenge the athlete with extra foods which are not usually consumed and to monitor the outcome, such as effect on performance, fatigue (if present) and weight/body composition. Schedule these modifications during the off-season when weight is not critical.

Techniques to make weight just prior to competition

- One or several techniques can be used for last-minute weight loss just before competition although individual responses are highly variable.
- As with any competition nutrition strategy, last minute weight-making and recovery strategies should be trialled during training, ideally in simulated competition situations, in case of adverse side effects or a non-response.
- Psychological and physiological responses and stressors are enhanced at competition for many inexperienced (and often experienced) athletes, which can increase the risk of illness and induce other adverse effects such as gut discomfort.

Low residue diet

- A low- or minimal-residue diet used for a short-term (2–3 days only) before competition may help to decrease bowel contents and reduce weight by around 300–700 g. Faecal matter consists largely of dead bacteria, undigested fibre, remnants of dead cells and other gut secretions. Foods high in fibre, sugar and fat contribute to faecal mass and are limited or avoided. However, not all athletes respond.
- Side effects include constipation, small infrequent stools and an increase in intra-luminal pressure causing abdominal distension and bloating. Some athletes experience mood changes and irritability.

- Long-term restriction of fibre is associated with other bowel disorders associated with prolonged constipation and increased intra-luminal pressure (e.g. diverticulitis, piles, hiatus hernia) and may even aggravate the symptoms of irritable bowel conditions (see Chapter 20).

Sweating

- Small weight loss of (~0.1–0.2 kg) can be achieved from sweating, which is best done as close to weigh-in as possible. Advise about the risks of sweating in excessive heat situations or very hot climates. Most athletes who use this technique restrict fluid at the same time.

Fluid restriction

- Fluid losses of ~5% BM can be life threatening, although small losses over a short time can be tolerated (Slater et al. 2014). In the 24 hours before competition, the volume of fluid intake can be limited outside training times, although the 24-hour total volume should be within safe limits depending on BM and activity levels of the athlete.
- A fluid-restriction diet limits daily intake of liquids as well as foods that contain a high volume of fluid. A total fluid volume of ~1000 mL/24 h is considered extreme fluid restriction in some clinical conditions. An adequate amount of fluid is needed for removal of nitrogenous waste and kidney function. The minimum amount needed for excretion is highly variable among individuals.
- Fluid restriction as opposed to sweating may retain more electrolytes, thus allowing more rapid rehydration after weigh-in.

Salt restriction

- Reducing salt intake by limiting foods high in salt and avoiding added salt to meals may help minimise fluid retention and contribute to weight loss.

Increasing exercise in addition to the current training program

- Increasing the time and intensity of additional exercise as a weight-loss method just prior to competition is counterproductive in endurance athletes competing in hot conditions (see Chapter 13). Tapering training before competition is needed to conserve glycogen stores.

Recovery after weigh-in

- Athletes benefit from a prescribed recovery fluid and food plan (before and) after weigh-in, especially those athletes who need to weigh-in for several days over competition.
- The role of the sports dietitian is to:
  - individualise the recovery diet and fluid plan after weigh-in, in consultation with the athlete
  - prepare the athlete for the effects of unexpected environmental conditions or circumstances, including the effects of peer pressure on last minute food choice, which may override dietary advice
  - monitor the response (in the trial period and at follow-up after each competition) to help fine-tune future nutrition intervention.
- Sports drinks and liquid meal replacements, high-carbohydrate, low-fat (and low-fibre) foods should be used for recovery (see Chapter 13). Aggressive replacement of both food and fluid may be required although some athletes experience gastrointestinal distress after prolonged and even short-term restrictive eating and drinking.

Consider a range of options outside the optimal *textbook* approach to dietary management

- Effective counselling involves negotiating, building client confidence and trust, and potentially compromising conventional dietary management protocols to improve compliance and maintain ongoing supervision of athletes in weight-class sports.

ENCOURAGE A FLEXIBLE APPROACH WITH OTHER SPORTS SERVICE PROVIDERS

Coaches and sports staff need to be flexible when working with athletes in weight-category sports. An adult athlete who competes in a lower weight category may eventually need to move to the higher weight category because of substantial increases in LBM (and BM). Athletes who were in a lower weight class as adolescents or young adults often expect to compete at the same class when competing at a more mature age. An understanding of the usual response to training or time off on body composition, to changes in body composition with maturation, and to previous weight-making strategies in a specific individual athlete is needed to explain and potentially alter the ingrained philosophy and training approach of coaches and also athletes involved in weight-category sports. For some athletes, it is time to move to a higher weight class.

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CHAPTER EIGHT

Eating disorders and disordered eating in athletes

Nikki A Jeacocke and Katherine A Beals

8.1 Introduction

'Citius, Altius, Fortius' is the Olympic credo, which means 'faster, higher, stronger'. It is what all athletes strive for and many are willing to go to great lengths to achieve it. To be a successful athlete, you must be goal oriented, strong willed, disciplined and diligent, have the ability to withstand pain and be willing to do whatever it takes to win. Unfortunately, these same qualities also make athletes susceptible to disordered eating (Byrne & Mclean 2001; Beals 2004; Mazzeo & Bulik 2009).

Athletes also face considerable pressures to conform to specific aesthetic requirements and/or performance demands of their sports. In some sports, a certain physique or a low body mass is considered to be essential for optimal performance, leading to the mantra that athletes must be 'thin to win'. Such pressure placed upon a vulnerable athlete can lead to the development of disordered eating behaviours, which may eventually develop into a full-blown, clinical eating disorder (Beals 2004).

The majority of existing research indicates that the prevalence of disordered eating is higher among athletes than non-athletic controls (Bratland-Sanda & Sungot-Borgen 2013). And while current estimates indicate that 85–90% of clinical eating disorders occur in females, males are not immune to body weight and shape pressures, particularly those imposed by sport (Carlat & Carmago 1991). Research suggests that the prevalence of disordered eating among both female and male athletes has been increasing slightly but nonetheless steadily over the past two decades (Bratland-Sanda & Sungot-Borgen 2013). Thus it is important that those who work with athletes have a clear understanding of the nature and scope of disordered eating, including the aetiology, health consequences and methods for prevention, intervention and treatment.

This chapter investigates the prevalence and suggested causes for the range of disordered eating and dysfunctional body image problems in athletes, extending the discussion begun in Chapters 6 and 7 about the challenges of achieving ideal body mass and physique in sport.

8.2 Disordered eating categories/classifications

Although frequently used interchangeably, the terms 'disordered eating' and 'eating disorder' are not one and the same. ‘Disordered eating’ is a general term used to describe the spectrum of abnormal and harmful eating behaviours that are used in a misguided attempt to lose weight and/or maintain a lower than normal body weight. On the other hand, the term ‘eating disorder’ refers to one of the six clinically diagnosable conditions recognised in the most recent, fifth edition of the *Diagnostic and statistical manual of mental disorders* (DSM-5) (American Psychiatric Association 2013) including avoidant/restrictive food intake behaviour, anorexia nervosa, bulimia nervosa, binge eating disorder, other specified feeding or eating disorder and unspecified feeding or eating disorder. To be diagnosed with one of the clinical eating disorders, an individual must meet a very specific and standardised set of criteria as outlined in the DSM-5, whereas the criteria for disordered eating are much less well defined and are generally based on behaviours and symptoms associated with eating disorders but which may be less severe or less frequent. These conditions will be further described in the subsequent paragraphs.

8.3 The clinical eating disorders

According to the DSM-5, the clinical eating disorders are characterised by *severe* disturbances in eating behaviour and body image (American Psychiatric Association 2013). It must be emphasised that the clinical eating disorders are psychiatric conditions and, as such, they go beyond simple body weight/shape dissatisfaction and involve more than just abnormal eating patterns and pathogenic weight-control behaviours. Individuals with clinical eating disorders often experience co-morbid psychological conditions, such as obsessive–compulsive disorder, depression and anxiety disorder (Fairburn & Brownell 2001). In addition, they often display intense feelings of insecurity, personal ineffectiveness and worthlessness, have trouble identifying and displaying emotions, and have an underdeveloped or limited sense of identity. Common personality traits seen in people who develop eating disorders, such as perfectionism, obsessiveness, hypervigilance, discipline and achievement-orientation, are also traits that contribute to athletic success.

Although the DSM-5 recognises six clinically significant eating disorders, some have not been identified in athletes (and are unlikely to occur in this population). For example, avoidant restrictive food intake disorder replaces and extends the DSM-IV diagnosis of feeding disorder in infancy or early childhood and does not include an experience of body weight or shape dissatisfaction. Similarly, binge-eating disorder involves eating a large amount of food in a discrete period of time (larger than most people would eat in a similar period of time under similar circumstances) with no subsequent compensatory or purging behaviour. The diagnostic criteria for the clinical eating disorders that have been or are likely to be seen in athletes are presented in Tables 8.1, 8.2 and 8.3.

TABLE 8.1 Diagnostic criteria for anorexia nervosa

- Restriction of energy intake relative to requirements, leading to a significantly low body weight in the context of age, sex, developmental trajectory and physical health (significantly low weight is defined as a weight that is less than minimally normal or, for children and adolescents, less than that minimally expected)

- Intense fear of gaining weight or of becoming fat, or persistent behaviour that interferes with weight gain, even though at a significant low weight
- Disturbance in the way in which one's body weight or shape is experienced, undue influence of body weight or shape on self-evaluation, or persistent lack of recognition of the seriousness of the current low body weight

Source: Adapted from American Psychiatric Association 2013

TABLE 8.2 Diagnostic criteria for bulimia nervosa

- Recurrent episodes of binge eating. An episode of binge eating is characterised by both of the following:
  – eating in a discrete period of time (e.g. within any 2-hour period) an amount of a food that is definitely larger than what most individuals would eat in a similar period of time under similar circumstances
  – a sense of lack of control over eating during the episode (e.g. a feeling that one cannot stop eating or control what or how much one is eating)
- Recurrent inappropriate compensatory behaviours in order to prevent weight gain, such as self-induced vomiting; misuse of laxatives, diuretics or other medications; fasting; or excessive exercise
- The binge eating and inappropriate compensatory behaviours both occur, on average, at least once a week for 3 months
- Self-evaluation is unduly influenced by body shape and weight
- The disturbance does not occur exclusively during episodes of anorexia nervosa

Source: Adapted from American Psychiatric Association 2013

TABLE 8.3 Other specified feeding or eating disorder

This category is used to describe feeding and eating disorders that cause clinically significant distress or impairment in social, occupational, or other important areas of functioning but do not meet the full criteria for any of the other feeding or eating disorders. Examples of presentations that can be specified using the 'other specified feeding or eating disorder' category include:

- *Atypical anorexia nervosa* All of the criteria for anorexia nervosa are met, except that despite significant weight loss, the individual's weight is within or above the normal range
- *Bulimia nervosa of low frequency and/or limited duration* All of the criteria for bulimia nervosa are met except that the binge eating and inappropriate compensatory behaviours occur, on average, less than once per week and/or for less than 3 months
- *Purging disorder* Recurrent purging behaviour to influence weight or shape (e.g. self-induced vomiting; misuse of laxatives, diuretics, or other medications) in the absence of binge eating

Source: Adapted from American Psychiatric Association 2013

Athletes with clinical eating disorders resemble their non-athlete counterparts in many ways. They are extremely dissatisfied with their body weight/shape, are obsessed with the desire to be thin, and are willing to go to any lengths (restrictive eating/starvation or binging and purging) in an attempt to achieve their illusive body weight ideal. However, unlike non-athletes with eating disorders, who generally view thinness as the only goal, athletes with eating disorders strive for thinness *and* the improvement in performance that they believe

will accompany it. This is particularly (although not exclusively) true for female athletes, especially those participating in sports that emphasise leanness. Although starving and/or bingeing and purging in the name of improved performance may seem counterproductive to the objective eye, the athlete with anorexia and/or bulimia nervosa is not logical when it comes to body weight and often has come to embrace (and embody) the notion that thinner is better (faster, stronger, more pleasing to the judges, and so on), no matter how it is achieved (Beals 2004).

8.4 Subclinical eating disorders

The term 'subclinical eating disorder' has frequently been used by researchers and practitioners to describe individuals, both athletes and non-athletes, who present with considerable eating pathology and body weight concerns, but do not demonstrate significant psychopathology and/or fail to meet all of the diagnostic criteria for a clinical eating disorder (Bunnell et al. 1990; Beals & Manore 1994, 1999, 2000; Williamson et al. 1995). For example, one collegiate field hockey player reported routinely eating a daily energy intake of ~1200 kcal (~5 MJ) while training 2–3 hours/d. She ate similar foods every day and severely limited her fat intake (no more than 20 g/d). Occasionally she would 'binge' by eating a forbidden food (such as a piece of cake or an order of French fries) and she would feel the need to restrict her food intake more strictly the following day. Although she was openly dissatisfied with her body weight and shape, she did not display any significant psychological disturbance. The athlete definitely displays disordered eating behaviours; however, she does not meet the diagnostic criteria for either anorexia nervosa or bulimia nervosa. In fact, depending on the context of the evaluation, she might not even meet the criteria necessary for a diagnosis of feeding or eating disorders not elsewhere classified (FED-NEC). Nonetheless, her eating behaviours could not be considered 'normal'. And, more importantly, they could negatively impact her athletic performance as well as her health.

8.5 Prevalence of disordered eating among athletes

The prevalence of disordered eating among athletes has been estimated to range from 6–45% in females and 0–19% in males (Bratland-Sanda & Sundgot-Borgen 2013). This wide range of prevalence estimates is largely due to differences in the screening instruments/assessment tools used, definitions of eating disorders employed and the athlete populations studied. Most studies examining the prevalence of disordered eating among athletes have utilised one or more of the many self-report instruments currently available, such as the Eating Attitudes Test (EAT), Eating Disorder Inventory (EDI), Three-Factor-Eating Questionnaire or Bulimia Test Revised (BULIT-R). Not only do self-reported questionnaires likely under-estimate disordered eating prevalence, but most have not been validated in an athletic population. The variety of different measures used in these studies make inconsistencies in prevalence estimates more likely and comparisons between studies more difficult. Studies have also varied widely in the definitions of the term 'eating disorder'. While some studies adhered to the strict DSM-IV criteria, most used the clinical term 'eating disorder' to characterise a wide range of abnormal eating behaviours that would be more appropriately labelled 'disordered eating'. Of course, using more strict criteria will result in lower prevalence estimates, while using more liberal criteria will result in higher prevalence estimates. Finally, studies have varied greatly in the sample populations used, including the type of athlete (e.g. collegiate versus high school athlete, elite athlete versus recreational athlete versus 'physically active' individual) as well as the number of sports studied. To get an accurate, unbiased estimate of prevalence, it is important to use a sufficiently large sample size ($n \geq 100$) as well as a variety of sports in order to obtain adequate heterogeneity. Table 8.4 presents recent (post-1990) prevalence studies that meet the above-mentioned criteria.

TABLE 8.4 Summary of studies (since 1990) examining the prevalence of disordered eating in athletes participating in a variety of sports

Study/country	Sample	Instrument	Eating disorder prevalence
Fortes et al. 2014 Portugal	580 male and female high school athletes and 362 age-matched non-athletes	EAT-26	18% of female athletes and 14% of male athletes showed eating disorder symptoms based on EAT-26 scores compared with 26% and 15% of female and male non-athletes, respectively.
Chatterton and Petrie 2013 USA	732 male collegiate athletes	Q-EDD; 7 items from the BULIT-R	16% reported disordered eating symptoms; 1% were classified as eating disordered (EDNOS).
Goltz et al. 2013 Brazil	150 male athletes in aesthetic and weight-dependent sports	EAT-26, Bulimic Investigatory Test, BSQ	Disordered eating behaviours and body image dissatisfaction were found in 43 (28%) and 23 athletes (15%), respectively.
Martinsen and Sundgot-Borgen 2013 Norway	611 elite adolescent male and female athletes and 355 adolescent male and female controls	A two-stage screening process including a questionnaire developed by the authors, including subscales of the EDI-2, weight history, and self-reported history of eating disorders (stage 1) followed by a clinical interview using the EDE and EDE-Q (stage 2)	51% of the controls and 25% of the athletes were classified as 'at risk' for an eating disorder after the initial screening. In stage 2, the prevalence of ED among the total population of athletes and controls was estimated to be 7% versus 2%, with the ED prevalence being higher for female than male athletes (14% versus 3%, $p < 0.001$) and female and male controls (5% versus 0%, $p < 0.001$).
Schaal et al. 2011 France	2067 male and female elite athletes	Psychological evaluation using DSM-IV criteria for clinical eating disorders	6% of female athletes and 4% of male athletes indicated a current/ongoing eating disorder; 11% of female athletes and 5.5% of male athletes reported a history of eating disorders.
Thein-Nissenbaum et al. 2011 USA	311 female high school athletes	EDE-Q	35% of the female athletes presented with symptoms of disordered eating.
Martinsen et al. 2010 Norway	606 elite female high school athletes and 355 female controls	Questionnaire developed by the authors including dieting, use of pathogenic weight control methods and the drive for thinness (DT) and body dissatisfaction (BD) subscales from the Eating Disorders Inventory	45% of female athletes and 13% of male athletes reported symptoms of disordered eating compared with 71% and 30% of female and male controls, respectively.

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Greenleaf et al. 2009 USA	204 female collegiate athletes	Q-EDD and BULIT-R	2% of athletes were classified as having an eating disorder; 26% had symptoms consistent with disordered eating.
Rosendahl et al. 2009 Germany	576 male and female high school athletes and 291 adolescent male and female controls	EAT-26	27% of female athletes and 10% met the criteria for disordered eating based on EAT-26 scores compared with 36% of and 12% of female and male controls, respectively.
Petrie et al. 2008 USA	203 collegiate male athletes	Q-EDD and BULIT-R	None of the athletes presented with a clinical eating disorder; 19% showed symptoms of eating disorders based on the two self-report instruments.
Torstveit et al. 2008 Norway	186 adolescent and adult female elite athletes and 145 non-athlete controls	EDE-Q and clinical interview	33% of athletes and 21% of non-athletic controls were classified with an eating disorder based on the clinical interview.
Nichols et al. 2007 USA	423 female high school athletes	EDE-Q	20% of athletes met the EDE-Q criteria for disordered eating.
Nichols et al. 2006 USA	170 female high school athletes	In-depth interview developed by the author using the EDE	18% met the criteria for disordered eating.
Beals and Hill 2006 USA	112 female collegiate athletes	Questionnaire developed by the authors including the EDE-Q and Eating Disorder Symptom Checklist	3% of athletes self-reported a clinical eating disorder; 23% of athletes met the criteria for disordered eating.
Toro et al. 2005 UK	283 elite female athletes competing in 20 different sports	EAT and the Eating Disorder Assessment Questionnaire (CETCA) (based on DSM-III-R criteria)	11% of the athletes exceeded the EAT cut-off score. 2.5% and 20% of the athletes met the CETCA criteria for anorexia nervosa and bulimia nervosa, respectively.
Sundgot-Borgen and Torstveit 2004 Norway	660 elite female athletes	A two-stage screening process including a questionnaire developed by the authors, including subscales of the EDI, weight history, and self-reported history of eating disorders (stage 1) followed by a clinical interview using the EDE (stage 2)	21% (n = 121) of the female athletes were classified 'at risk' after the initial screening. Results of the clinical interview indicated that 2% met the criteria for anorexia nervosa, 6% for bulimia nervosa, 8% for eating disorders not otherwise specified (EDNOS) and 4% for anorexia athletica.
Byrne and McClean 2002 Australia	263 elite male and female athletes	Composite International Diagnostic Interview (CIDI)	22% of female athletes and 4% of male athletes were classified with an eating disorder compared to 5.5% of female controls and none of the male controls.

	and 263 non-athlete male and female controls		
Beals and Manore 2002 USA	425 female collegiate athletes	EAT-26 and EDI-BD	3% and 2% of the athletes self-reported a diagnosis of clinical anorexia and bulimia nervosa, respectively; 15% and 32% of the athletes scored above the designated cut-off scores on the EAT-26 and EDI-BD, respectively.
Johnson et al. 1999	562 female collegiate athletes	EDI-2 and questionnaire developed by the authors using DSM-IV criteria	None of the athletes met the DSM-IV criteria for anorexia nervosa, while 1% met the criteria for bulimia nervosa. 2% and 6% of the athletes believed they might have anorexia nervosa and bulimia nervosa, respectively. Subclinical anorexia and bulimia were identified in 3% and 9% of the women, respectively. 35–38% demonstrated disordered eating behaviours (e.g. binge-eating, vomiting, laxatives, diuretics, diet pills, elevated Drive for Thinness (EDI-DT) subscale score, elevated Body Dissatisfaction score).
Sundgot-Borgen 1993 Norway	522 elite female athletes	EDI and in-depth interview developed by the author based on DSM-III criteria	1%, 8% and 8% were diagnosed with anorexia nervosa, bulimia nervosa and anorexia athletica, respectively.

BSQ = Body Shape Questionnaire (Cooper et al. 1986), BULIT-R = Bulimia Test Revised (Thelen et al. 1991), DSM-IV = Diagnostic and statistical manual of psychiatric disorders IV (American Psychiatric Association 1994), EAT-26 = Eating Attitudes Test (Garner et al. 1982), EDI = Eating Disorder Inventory (Garner et al. 1983), EDI-BD = Body Dissatisfaction subscale (Garner et al. 1983), EDI-DT = Drive for Thinness subscale (Garner et al. 1983), EDI-2 = Eating Disorder Inventory-2 (Garner et al. 1991), EDE = Eating Disorder Examination (Fairburn & Cooper 1993), EDE-Q = Eating Disorder Examination Questionnaire (Fairburn & Beglin 1994), Q-EDD = Questionnaire for Eating Disorder Diagnosis (Mintz et al. 1997)

Despite myriad methodological differences and inconsistencies between studies, some general conclusions can be made regarding the prevalence of disorder eating and eating disorders among athletes. First, the prevalence of disordered eating is significantly greater than the prevalence of clinical eating disorders among athletes. Second, with few exceptions, the research suggests that the prevalence of disordered eating is greater among athletes compared to non-athlete controls. The exceptions seem to be found most often in high school athletes, which could be explained by the shorter period of exposure to sport-specific disordered eating triggers as well as the use of self-reported questionnaires that have been validated in non-athlete populations (Bratland-Sanda & Sundgot-Borgen 2013). Among athletes, the prevalence of disordered eating has consistently been shown to be higher in those that participate in ‘thin-build’ or ‘weight-dependent’ sports. These sports can be categorised into three groups: (1) ‘aesthetic sports’ such as diving, figure skating, gymnastics and synchronised swimming; (2) sports in which low body mass and body-fat levels are considered a physical or biomechanical advantage, such as distance running, road cycling and triathlon; and (3) sports that require weight categories for competition, such as lightweight rowing, weightlifting and wrestling (Beals 2004). Finally, the prevalence of eating disorders and disordered eating is higher among female compared to male athletes. Interestingly, research suggests that there is an interaction between gender and sport in terms of disordered eating prevalence. For example, Schaal et al. (2011) found in female athletes that the prevalence of disordered eating was highest in endurance sports, while in males disordered eating was most prevalent in the weight-dependent sports.

8.6 Performance and health consequences of disordered eating

The effects of disordered eating on an athlete's health and performance can be surprisingly variable, although they are generally reflective of the severity of the disorder as well as how long it has persisted. Some athletes may be able to engage in disordered eating behaviours for extended periods of time with few long-term negative effects (Thompson & Sherman 1993). For most, however, it is simply a matter of time before the food restriction, weight loss and purging practices negatively affect their physical performance and, more importantly, their physical and emotional health. The health consequences associated with disordered eating are directly related to the methods employed by the athlete within their disordered eating. The course and outcome for those with eating disorders are highly variable. Some individuals fully recover after a single bout of anorexia nervosa, others exhibit a fluctuating pattern of weight gain followed by a relapse, and others experience a chronically deteriorating course of the illness over many years. In some cases, hospitalisation is required to help restore weight and to manage fluid and electrolyte imbalances. The long-term consequences of intractable anorexia nervosa can be fatal. A meta-analysis of 36 published studies systematically compiled and analysed the mortality rates in individuals with anorexia nervosa, bulimia nervosa and eating disorder not otherwise specified (Arcelus et al. 2011). The weighted mortality rates (i.e. deaths per 1000 person years) were 5.1 for anorexia nervosa, 1.7 for bulimia nervosa and 3.3 for eating disorder not otherwise specified. One in five individuals with anorexia nervosa who died had committed suicide (Arcelus et al. 2011).

From a performance point of view, perhaps surprisingly, anecdotal evidence (reports from coaches and personal accounts by athletes with disordered eating) suggests that during the early stages of disordered eating, some athletes may actually experience an initial, albeit short-lived, improvement in performance (Beals 2004). The reasons for this transient performance enhancement are not completely understood, but are hypothesised to be related to the initial physiological and psychological effects of starvation and purging (Beals 2004). Both starvation and purging are physiological stressors and, as such, produce an up-regulation of the hypothalamic–pituitary–adrenal axis (the 'fight-or-flight response') and an increase in the adrenal hormones cortisol, epinephrine and norepinephrine. These hormones have a stimulatory effect on the central nervous system that can mask fatigue and evoke feelings of euphoria in the eating-disordered athlete (Beals 2004). In addition, the initial decrease in body weight (particularly before there is a significant decrease in muscle mass) may induce a transient increase in relative maximal oxygen uptake per kilogram of body weight ($VO_{2\ max}$) (Ingjer & Sundgot-Borgen 1991). Moreover, with weight loss, athletes may feel lighter, which may afford them a psychological boost, particularly if they believe that lighter is always better in terms of performance.

Unfortunately, the eating-disordered athlete often equates these temporary performance improvements with the disordered eating behaviours, causing the behaviours to become more entrenched and significantly more difficult to treat. Thus it must be emphasised to the athlete that any initial improvements in performance are transient. Eventually, the energy deficiencies and purging behaviours will cause the body to break down and performance will suffer. An overview of a number of the possible health and performance consequences associated with disordered eating and eating disorders are summarised in Table 8.5.

TABLE 8.5 Health and performance consequences of disordered eating

Health-related consequences
Macronutrient and micronutrient deficiencies (including iron deficiency)
Low energy availability
Irregular menstruation
Overuse injuries and stress fractures
Hypothermia
Dehydration
Electrolyte imbalances
Cardiac arrhythmias
Increased risk of illnesses and infection
Poor concentration and fatigue
Skeletal demineralisation
Performance-related consequences
Atrophy and loss of lean body mass
Decreased muscle contraction time
Decreased reaction time
Decreased strength and power
Decreased aerobic endurance
Increased recovery time
Decreased blood flow to skeletal muscle
Decreased delivery of oxygen to muscle
Fatigue and risk of injuries associated with fatigue
Increased number of training days missed through injury, illness and infections
Impaired oxidative metabolism in skeletal muscle
Decreased concentration

Sources: Adapted from Sundgot-Borgen 2002; Torstveit 2004; Thompson & Sherman 2010; Torstveit & Sundgot-Borgen 2014

8.7 The female athlete triad

The female athlete triad (triad) describes the interrelationship between energy availability, menstrual function and bone health, and represents a spectrum from health to disease within each category (Nattiv et al. 2007). The triad is an important concept that should be considered when discussing disordered eating in athletes. This issue is therefore discussed in greater detail in Commentaries 2–4.

8.8 Prevention of disordered eating among athletes

As discussed above, there is a range of health and performance consequences related to disordered eating in athletes. Some may struggle with disordered eating for many, many years and never fully recover. Even in those who have recovered from eating disorders, health problems can persist (Beals 2004). For these reasons, efforts to combat disordered eating in athletes should focus on prevention.

The prevention of disordered eating normally involves targeting the risk factors for disordered eating and then trying to eliminate or at the very least modify them. As was previously described, many of the factors that predispose an athlete to disordered eating (such as biological and personality factors, and family environment) are outside the direct control of coaches, athletic support staff or health professionals. Thus, prevention efforts should focus on those predisposing factors that can be controlled, including the overemphasis on body weight and thinness, unrealistic body weight ideals, unhealthy eating and weight-control practices, and stigmatisation of disordered eating.

8.8.1 De-emphasise weight and body composition

One of the most widely held misconceptions that continues to permeate the athletic environment is that reducing body weight invariably leads to improved performance. While no one would argue that an extreme or unhealthy excess of body fat will negatively impact performance, that does not mean that a lower body weight is always more advantageous (Wilmore 1992), particularly if that lower body weight is achieved via severe or pathogenic weight-loss methods (see Chapter 7). Moreover, the stress of the ‘evaluation process’ involved in body weight and composition measurements, when placed upon a vulnerable athlete, can be enough to trigger or exacerbate disordered eating behaviours.

There are several ways that coaches, trainers and athletic staff can help de-emphasise body weight and composition among their athletes. The most obvious is to simply eliminate such measurements altogether (Carson & Bridges 2001). In many sport settings, eliminating anthropometric assessments may not be a viable or even the most appropriate solution. In such cases, measurements should be taken by a qualified health professional not directly connected with the team (not the coach or trainer) who can thoughtfully and objectively interpret and confidentially communicate the results to each individual athlete. In addition, the athletes as well as the coaches and trainers need to be educated about the measurements being taken. This includes providing information on exactly what is being measured, how the results will be used and how they can be useful in tracking an athlete's adaptation to their training program. The limitations and potential errors in the measurements should also be clearly explained to both athletes and coaches.

8.8.2 Dispel nutrition myths and promote healthy eating behaviours

Athletes often suffer from a dearth of nutrition knowledge, particularly as it relates to athletic performance. Moreover, much of the knowledge they do possess is derived from peers and popular fitness and/or sports magazines (Chapman et al. 1997; Jacobson et al. 2001). Thus, nutrition education should be provided to all athletes, focusing on dispelling the myths and misconceptions about dieting and the impact of these factors on athletic performance. Equally important is providing accurate and appropriate nutritional information and dietary guidelines to promote optimal health and athletic performance.

To successfully promote healthy eating behaviours among athletes, nutrition education and information must be reinforced by practice. All those involved in the management of athletes must therefore practise what they preach. Coaches are probably in the best position to reinforce nutrition education messages by bringing healthy foods to practice, choosing healthy restaurants before and after competitions and, of course, eating healthily themselves.

8.8.3 Destigmatise eating disorders

Coaches, trainers, and other athletic personnel can help reduce the stigma of disordered eating by creating an atmosphere in which athletes feel comfortable discussing their concerns about body image, eating and weight control. Athletic personnel should strive to promote understanding and foster trust between themselves and their athletes. The goal is to create an atmosphere in which athletes feel comfortable confiding an eating problem. In short, coaches, trainers and athletic administrators must make it clear that they place the athletes' health and wellbeing ahead of athletic performance. Coaches may feel uncomfortable or unqualified to discuss topics such as body image, nutrition and body composition. It is therefore very important to ensure there is an appropriately trained health professional that athletes, coaches and other support staff can access to discuss any issues or concerns.

8.8.4 Prevention programs

Prevention of disordered eating normally focuses on targeting and changing environmental factors rather than delivering a specific prevention program. A recent study investigated the effect of a one-year intervention program to prevent the development of new cases of eating disorders and symptoms associated with eating disorders among adolescent male and female elite athletes (Martinsen et al. 2014a). The intervention program targeted a range of topics including motivation, self-esteem, nutrition, physiology, prevention and mental training. The school-based program was delivered as lectures, group and individual assignments, and included both theoretical and practical assignments. The intervention program also included a component for coaches. The authors concluded the program was effective at preventing new cases of eating disorders and symptoms associated with eating disorders in adolescent female elite athletes. Replication of this program in a range of elite athlete populations would be beneficial to determine the effectiveness in different populations, but in early stages offers nutrition educators a promising eating disorder prevention program.

8.9 Management

Management of disordered eating in athletes has been described as encompassing the range of intervention tactics beginning with identification, following with referral and treatment, and concluding with post-treatment follow-up (Thompson & Sherman 1993). Each of these will be described briefly below.

8.9.1 Identification

The shame and secrecy that often shrouds disordered eating makes identifying those who suffer from it often difficult at best. It is rare that an athlete will willingly admit to a disorder and agree to treatment. Thus it is up to others to recognise the signs and symptoms of disordered eating and initiate intervention. A number of different assessment tools have been developed to screen for disordered eating. Some of the most common are described in Table 8.6. It is important to recognise that none of the instruments listed in Table 8.6 were originally designed to make clinical diagnoses of eating disorders. Rather, they were developed for use in non-clinical settings as screening devices to assess attitudes and behaviours exhibited by individuals who met the clinical diagnoses of anorexia nervosa or bulimia nervosa (Leon 1991). In addition, the self-report nature of the questionnaires renders them susceptible to untruthful or biased responses. Finally, most are of questionable use with athletes as they have not been sufficiently validated in athletic populations.

TABLE 8.6 Common self-report surveys and questionnaires for identifying disordered eating

Instrument	Description
Bulimia Test-Revised (BULIT-R) (Thelen et al. 1991)	A 28-item multiple-choice questionnaire designed to assess the severity of symptoms and behaviours associated with bulimia nervosa (e.g. weight preoccupation and bingeing and purging frequency). Respondents rate each item on a five-point Likert scale in which higher scores are more indicative of bulimia nervosa.
Eating Attitudes Test, 40 items (EAT-40) (Garner & Garfinkel 1979)	A 40-item inventory designed to assess the thoughts, feelings and behaviours associated with eating disorders. Items are scored on a six-point Likert scale ranging from *never* to *always*. A score of >30 indicates risk of an eating disorder.
Eating Attitudes Test, 26 items (EAT-26) (Garner et al. 1982)	A shortened (26-item) version of the EAT-40 that also identifies thoughts, feelings and behaviours associated with eating disorders. Uses a 6-point Likert scale ranging from *rarely* to *always*. A score of >20 indicates risk of an eating disorder.
Eating Disorder Inventory (EDI) (Garner et al. 1983)	A 64-item questionnaire with eight subscales. The first three subscales (Drive for Thinness, Bulimia and Body Dissatisfaction) assess behaviours regarding body image, eating and weight-control practices. The remaining five subscales (Interpersonal Distrust, Perfectionism, Interoceptive Awareness, Maturity Fears and Ineffectiveness) assess the various psychological disturbances characteristic of those with clinical eating disorders. Items are answered using a six-point Likert scale ranging from *always* to *never*.
Eating Disorder Inventory-2 (EDI-2) (Garner 1991)	A 91-item multidimensional inventory designed to assess the symptoms of anorexia nervosa and bulimia nervosa. The EDI-2 contains the same eight subscales as the EDI and adds three additional subscales (27 more items): Asceticism, Impulse Regulation and Social Insecurity. Items are answered using a six-point Likert scale ranging from *always* to *never*.
Eating Disorder Inventory-3 (EDI-3) (Garner 2004)	A 91-item inventory representing an expansion and improvement of EDI-1 and EDI-2. The EDI-3 has been updated to be more consistent with modern theories regarding the diagnosis of eating disorders. The EDI-3 is divided into 12 primary scales: Drive for Thinness, Bulimia, Body Dissatisfaction, Low Self-Esteem, Personal Alienation, Interpersonal Insecurity, Interpersonal Alienation, Interoceptive Deficits, Emotional Dysregulation, Perfectionism, Asceticism and Maturity Fears.
Three-Factor Eating Questionnaire (TFEQ) (Stunkard 1981)	A 58-item true/false and multiple-choice questionnaire that measures the tendency towards voluntary and excessive restriction of food intake as a means of controlling body weight. The questionnaire contains three subscales: Restrained Eating (e.g. 'I often stop eating when I am not full as a conscious means of controlling my weight'), Tendency towards Disinhibition (e.g. 'When I feel lonely, I console myself by eating') and Perceived Hunger (e.g. 'I am always hungry enough to eat at any time').

Source: Adapted from Beals 2004

Because of the limitations imposed by self-report questionnaires, many researchers and practitioners have suggested that interviewing athletes may be a more accurate and effective method of identifying disordered eating behaviours (Sundgot-Borgen 1993). As was the case with screening tools, there are a number of structured interviews available for identifying disordered eating. Considered the gold standard for the diagnosis of eating disorders is the Eating Disorder Examination (EDE-16) (Fairburn et al. 2008). The EDE is a semi-structured interview for assessing the symptoms associated with anorexia and bulimia nervosa. It contains four subscales: (1) dietary restraint, (2) eating concern, (3) shape concern and (4) weight concern. The items derived from the interview are converted into 23 symptom ratings made by the interviewer. Although it has been used to identify disordered eating in athletes (Sundgot-Borgen 1993; Beals & Manore 2000) it has not been formally validated in an athlete population. Moreover, it requires a qualified professional to conduct the interview and interpret the results.

Attention should be drawn to a very recent screening tool developed specifically for use in athletes. The Brief Eating Disorder in Athletes Questionnaire (BEDA-Q) (Martinsen et al. 2014b) is a newly developed and validated screening tool for distinguishing between adolescent female elite athletes with and without eating disorders/disordered eating. Version 1 consists of seven items from the EDI-Body dissatisfaction, EDI-Drive for thinness, and questions regarding dieting. Version 2 includes an additional two items from the EDI-Perfectionism subscale. While showing promising results in its initial stages, further studies are required to determine if this tool is useful in different groups of athletes.

Sometimes simply observing the athlete's behaviour can be the most simple and effective method for identifying disordered eating behaviours. Individuals who have daily contact with athletes (e.g. coaches, trainers, team-mates, family, friends) are in the best position to recognise behaviours that are consistent with disordered eating. Table 8.7 lists some of the common warning signs and symptoms. Research supports early identification and intervention for better outcomes and shorter timeframes for recovery from an eating disorder.

TABLE 8.7 Warning signs for eating disorders

Dramatic weight loss or gain
A preoccupation with food, calories, body shape and weight
Wearing baggy or layered clothing
Relentless, excessive exercise
Mood swings
Avoiding food-related social activities
Restrictive eating
Bathroom visits after meals
Depressive moods
Strict dieting followed by eating binges
Increased criticism of one's body
Evidence of binge eating (large amounts of food disappearing)
Secretive behaviour regarding food intake
Feeling out of control in regards to food
Body image dissatisfaction and distortion
Perfectionism

8.9.2 Referral and treatment

The *preservation* of the athlete's health and mental wellbeing is the first goal of treatment (Nattiv & Lynch 1994). A multidisciplinary team involving people experienced in the management of eating disorders provides the ideal treatment approach (Johnson 1986). Each team member should have a specific role:

- A physician should monitor medical status, rule on athletic participation and often coordinate the care provided by the team.
- A registered dietitian who specialises in eating disorders should provide appropriate nutritional guidance.
- A psychologist, psychiatrist or counsellor should address issues of mental wellbeing.
- Trainers, coaches and exercise physiologists should assist with and support training program or performance monitoring as appropriate.
- In the case of young athletes (adolescents 19 years and under) who live at home, family involvement in treatment is essential.

Because eating disorders are psychological disorders, psychological counselling is considered the cornerstone of treatment. A variety of psychological approaches have been used successfully to treat eating disorders, including psychodynamic, cognitive–behavioural and behavioural methods. Additional variables to consider when selecting a treatment approach include the treatment setting (e.g. inpatient versus outpatient) and format (e.g. individual versus group, with or without family). While psychological counselling aims to uncover and correct the underlying mental and emotional issues fuelling the eating disorder, nutrition counselling focuses on changing the disordered eating behaviours (the energy restriction, bingeing and/or purging), treating any nutritional deficiencies, addressing nutrition beliefs and thoughts about food and body, and re-educating the athlete about sound nutritional practices.

In the case of adolescents with anorexia or bulimia, family-based treatment (also known as the Maudsley Approach) is considered best practice in achieving the best outcomes of recovery (Lock et al. 2006). This treatment needs to be delivered with the whole family by a therapist trained in this model, in conjunction with medical monitoring, preferably by a paediatrician. If family-based treatment is not available, family therapy should be included in the multidisciplinary approach for treatment in children and adolescents.

8.9.3 Post-treatment follow-up

Recovery from disordered eating can take months or, more typically, years. Thus, treatment can continue for at least as long. Nonetheless, active or intensive treatment, particularly if done on an inpatient basis, generally lasts for a more finite period. Managing the transition of the athlete from active treatment back to 'daily' life and to their sport requires careful planning and monitoring. The athlete will probably feel self-conscious and ashamed, convinced of the disappointment of their coach and team-mates. Understanding and reassurance from the coach and team-mates is thus essential for the athlete's successful transition and ultimate recovery.

The issues of returning to training and competing must also be addressed. The decision regarding the degree of training and competition that an athlete may undertake during recovery should be based on their physical, psychological and emotional health as well as their degree of readiness to return to competition. If the athlete is still experiencing lingering physical and/or psychological complications as a result of the eating disorder, competition should be postponed. Similarly, if the athlete refuses to follow post-treatment requirements (such as following the agreed treatment plan or counselling schedule), training and/or competition should be postponed or minimised. A contract that outlines specific terms and conditions under which the athlete may train and/or compete is sometimes helpful to ensure that the athlete returns to their sport in the best psychological and physical shape possible (Mountjoy et al. 2014).

Summary

In the world of athletics, a fraction of a second or one-tenth of a point can mean the difference between winning and losing. These high stakes can place enormous pressure on athletes. Athletes who are pressured to meet a rigid definition of ideal physique, or who lose weight because they think it will improve performance, are at risk of developing dysfunctional eating and exercise practices. Unfortunately, these weight-loss behaviours are often self-defeating. Any initial improvement in performance (as a result of weight loss) is transient. Disordered eating practices will eventually take their toll on the athlete's health and performance.

Prevention is considered the key to stemming the growing prevalence of disordered eating among athletes. Disordered eating prevention involves the development of educational programs and strategies designed to dispel the myths and misconceptions surrounding nutrition, dieting, body weight and body composition, and their impact on performance, as well as stressing the role of nutrition in promoting health and optimal physical performance.

Unfortunately, until society in general, and sport leaders in particular, eliminate the pressures that encourage these behaviours, prevention efforts will probably be largely unsuccessful. Thus, there will continue to be a need to recognise and treat disordered eating practices. Early identification and intervention is paramount in limiting the progression and shortening the duration of the disordered eating. A familiarity with the warning signs and symptoms of disorders is crucial. Treatment for disordered eating involves a combination of psychological and nutritional counselling along with appropriate medical care and family involvement for adolescents. The primary treatment goals for eating disorders in athletes are to normalise eating behaviours and body weight, and identify and correct the underlying psychological issues that initiated and perpetuate the eating disorder.

Practice tips

NIKKI JEACOCKE

- Although early identification is important for positive outcomes for the athlete's health and future performance, the often secretive aspects of disordered eating and eating disorders can challenge timely intervention. All members of the multidisciplinary team play an important role in early identification. One team member may hear or witness a behaviour that on its own is not concerning, however, when combined with a number of other behaviours, there may be a real concern. Communication between multidisciplinary team members is crucial to ensure these 'red flags' are not overlooked. It is useful to have an appropriately trained member of the support staff responsible for 'collecting' the red flags and initiating appropriate treatment as required.
- A common misconception is that weight loss is the key identifier to an eating disorder. While weight loss can certainly be a sign of a problem, there are many other warning signs that should not be ignored. Waiting for weight loss to occur, or ignoring a situation because of the absence of weight loss, can inhibit timely identification of disordered eating in an athlete.
- Body mass ('weight') is often discussed inappropriately with athletes. Before an athlete receives advice that they need to lose weight, consideration needs to be given as to whether an alternative and more effective message could be given. For example, the focus of the message could be changed to the improvement of fitness rather than altering body weight. Negative comments regarding an athlete's weight and/or body shape should be avoided.
- Although anthropometry is a useful tool in monitoring athlete body composition, when it is undertaken inappropriately it can aggravate underlying stress or problem behaviours.
  - Care should be taken when undertaking anthropometric measurements in any athlete group. However, in the case of athletes with an identified problem, there is a risk of exacerbating body image concerns and inhibiting their recovery. Therefore, anthropometry should only be undertaken if the treatment team deems it appropriate.
  - Anthropometric measurements should be undertaken by an appropriately trained clinician, using careful language to explain the process. Consideration needs to be given to how the results are interpreted and discussed, and the range of people who are given access to data.
- The importance of the menstrual cycle in optimal health and athletic performance is often poorly understood by athletes and coaches. Athletes and/or coaches may feel uncomfortable in discussing the menstrual cycle; they may ignore opportunities for education about the problems of disruption to regular menstrual cycles or feel reluctant to report it. Given its role as an indicator of low energy availability, which may or may not be a result of disordered eating, menstrual health should be discussed openly with an appropriate health professional who can evaluate an athlete's menstrual function and provide recommendations regarding irregularities or dysfunction.
- Athletes are very competitive, both on and off the field. In many sporting teams, it is common to observe comparisons of physical characteristics (weight and body composition) and eating habits between athletes: this may occur as a team-sanctioned activity or as informal interaction between athletes.
  - Ideally, systems should be in place to allow body composition results to remain private, and athletes should be encouraged to maintain these systems and avoid inappropriate comparisons with each other.
  - It is also important to encourage athletes to eat for themselves. Eating in a team setting can result in unhealthy comparisons of what is on someone else's plate, either expressed aloud or pondered internally. Athletes can fall into concerning behaviours such as competing to 'eat the least' or 'the most healthy'. Teaching athletes to focus on meeting their own requirements and to develop resilience in group settings may them to deal more effectively with challenging peer-group behaviour.
- It is important that the nutrition service provider for an athlete/team has appropriate and evidence-based knowledge and ideals on nutrition, weight, body image and sports performance. Passing on someone's own personal body hang-ups to athletes should be avoided.
- In the absence of an embedded nutrition service provider within a sporting environment, the coach needs to be educated about the early warning signs of disordered eating and have access to an appropriately trained professional to discuss concerns about an athlete.
- When raising concerns with an athlete regarding their disordered eating behaviour, it is often beneficial to bring the discussion back to performance. Negative performance consequences often resonate with an athlete to a greater extent than negative health outcomes. Although health issues are obviously of major concern, the athlete may be more motivated to consider changing their behaviour to influence performance outcomes.
- The nutrition expert needs to have a thorough understanding of the practices and culture of the sport with

which they are working, so they are able to identify and understand whether an athlete's eating practices are appropriate for the sport or indicative of an eating problem. In some sports, there are a number of eating practices in which athletes engage that may be considered 'normal', but would otherwise be considered as concerning outside this context. For example, the process of making weight may be appropriate for an athlete in a weight-division sport, but not suitable for individuals who do not compete under such rules (see Chapter 7 for more information on making weight).

- The nutrition expert should also be aware of the latest fad diets and nutrition beliefs in the general community since they may also infiltrate into athletic circles.
- Knowing what is 'normal' for an individual athlete in terms of thoughts and behaviours related to food is also important. Identifying a deviation from this 'normal' may indicate a problem, allowing timely intervention to be initiated. This is also true when discussing excessive exercise. Characteristics of training programs that are quite normal when associated with athletic preparation (e.g. long duration, early morning starts, in inclement weather or while injured) could indicate excessive exercise when taken out of this context. It is important to know what training has been prescribed to the athlete through discussion with coaches, trainers and potentially medical staff (if injured). Deviation from a prescribed training regimen may indicate excessive exercise and raise a 'red flag'.
- Rapport with the athlete is important for both identifying and treating disordered eating. Ideally, the athlete should be familiar with the nutrition expert and have developed a relationship prior to the situation of requiring nutrition counselling for disordered eating. When this is not the case, early sessions should focus on developing rapport and gaining the athlete's trust to build up the working relationship rather than aggressive nutritional counselling.
- It can be insightful to observe athletes in practical situations involving food, since this often reveals information about their thoughts and beliefs about nutrition as well as their actual eating practices. Conducting cooking classes and supermarket visits or being involved in team travel or sharing meals can provide information that is more useful than simply sitting in consultation discussing the athlete's eating practices.
- It can be useful to identify various scenarios involving stress which can trigger or exacerbate disordered eating behaviours in an athlete. Such scenarios include, but are not limited to:
  - injury (new or recurrent)
  - moving away from home (especially for the first time)
  - moving overseas
  - preparation for a major tournament or competition (both in the selection process and the period prior to the actual competition)
  - a change in the daily training environment (e.g. a change in coach)
  - selection to an elite or senior team at an early age
  - stressful life events at home or in the work, school or study environment.
- Creating an environment within a sporting team where seeking help early is encouraged rather than shunned is very important for the prevention and early identification of disordered eating.

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CHAPTER NINE

Bone, calcium, vitamin D and exercise

Deborah Kerr and Enette Larson-Meyer

9.1 Introduction

Bone is a dynamic tissue that reflects the biological principle of adaptation of structure to function and the metabolic role of mineral homeostasis. The skeleton is made up of two types of bone. The outer bone is known as cortical and the inner softer core is known as trabecular (the more metabolically active bone). The skeleton is designed to provide the strength needed to withstand the mechanical forces of daily weight bearing. Structurally, the long bones of the skeleton are often referred to as appendicular bones and the bones of the trunk as the axial skeleton.

Bone is continually being broken down and rebuilt in a process known as remodelling, under the regulation of systemic hormones and local growth factors. The remodelling cycle consists of five successive events: quiescence, activation, resorption, reversal and formation (Parfitt 1984; Raisz 1999). Following resorption of a packet of bone by the osteoclast, new bone is laid down by the osteoblast. When bone resorption exceeds formation, bone loss occurs, which, if prolonged, can lead to osteoporosis and increased risk of fracture. Further reading on bone biology and other topics related to bone health is available at the American Society for Bone and Mineral Research Bone Curriculum website (see Useful websites and resources).

9.2 The measurement of bone mineral density

Bone densitometry measures the average bone mineral within the region scanned, known as the bone mineral density (BMD). The sites measured are the hip, forearm and lumbar spine, which are the most common fracture sites. The whole body scan can estimate the total bone density and body composition. The BMD can be measured by dual energy X-ray absorptiometry (DXA) and quantitative computed tomography (CT). Both these methods involve ionising radiation, a consideration in athletes who may be undergoing other testing procedures that involve ionising radiation, such as X-rays and CT scans. Quantitative ultrasound (QUS) is useful for screening and has the advantage of no ionising radiation but is only validated at the heel site (Schousboe et al. 2013). Low bone mass is defined in terms of how far the measurement falls below the reference range for the young healthy female. A *fracture threshold* or a cut-off point is used to define osteoporosis and is based on the range of BMD measurements in the population with vertebral or hip fractures.

9.2.1 Definitions: osteoporosis and low bone mass

Osteoporosis is a condition of low bone mass associated with greater bone fragility and increased risk of fractures (WHO 1994). The risk of developing osteoporosis and subsequent fractures is largely determined by the peak bone mass achieved in adolescence and early adulthood. Between 50% and 80% of the variability in bone density has been attributed to genetic factors (Pocock et al. 1987; Zhai et al. 2009). BMD is strongly linked to bone strength and resistance to fracture but people with low mass are not always at high risk of fracture. Various fracture risk calculators, for example the WHO Fracture Risk Assessment Tool (FRAX), are now available to assist clinicians in decision making but are still being adapted for use in specific countries.

Clinically, osteoporosis is defined in terms of the BMD that is below the age-adjusted reference range. An individual is considered osteoporotic if their BMD is 2.5 standard deviations (SD) or more (T-score <-2.5) below the young adult mean for bone density. Low bone mass (or osteopaenia) is defined as BMD more than one SD below the young adult mean, but less than 2.5 SD below this value. In premenopausal women and children reporting of BMD Z-scores, which compares individuals to age- and sex-matched controls rather than T-scores, should be used. The term *sports osteopaenia* has been used in the past to refer to low bone mass. The International Society for Clinical Densitometry (ISCD) recommends the terms *low bone mass* or *low bone density* be used in preference to *osteopaenia* (Schousboe et al. 2013).

Indications for bone density testing have been formulated by the ISCD (Schousboe et al. 2013). Weight-bearing sites (spine, total hip, femoral neck) are the preferred sites for screening. Generally BMD testing is not indicated in premenopausal women unless they have had a prior fragility fracture, are taking medications known to cause bone loss (e.g. glucocorticoids) or have a disease or condition associated with bone loss. Secondary risk factors for low bone mass may be associated with low bone mass in athletes and include chronic under-nutrition, eating disorders, amenorrhoea, glucocorticoid exposure and previous fractures. Inflammatory bowel disease and coeliac disease are also causes of secondary osteoporosis in younger adults.

9.2.2 Low bone mass in athletes

Generally athletes would be expected to have higher bone densities compared to less active individuals as a response to the positive effects of weight-bearing exercise on bone. However, optimal bone health in females requires adequate menstrual function and energy balance. Oestrogen deficiency related to either athletic amenorrhoea, anorexia or the menopause results in bone loss in women. The female athlete triad, or the more recently named umbrella term–Relative Energy Deficiency in Sports (RED-S)–is a condition involving low energy availability with or without disordered eating behaviour, menstrual or endocrine dysfunction and low bone mineral density (Nattiv et al. 2007; De Souza et al. 2014; Mountjoy et al. 2014). The term RED-S encompasses the female athlete triad and includes other adverse physiological effects associated with energy deficiency and recognises that male athletes are also a risk group (Mountjoy et al. 2014 and commentaries 3 and 4). The impact of energy deficiency on endocrine function and on BMD in male athletes is not as well known as female athletes and requires more research. Although low testosterone levels are associated with low BMD in men, oestradiol is a more important hormone for bone health (Irwig 2013). Low oestradiol levels are a better predictor of stress fractures and BMD in older men (Vanderschueren et al. 2014), in male collegiate athletes and in pubertal boys (Doneray & Orbk 2010; Ackerman et al. 2012).

An elevated rate of bone loss, reduced BMD and a decrease in testosterone was reported in male jockeys on energy-restricted diets, compared with matched controls (Dolan et al. 2013). Oestradiol was not measured in this study, although disturbances in other hormones (i.e. sex hormone-binding globulin and insulin-like growth factor) suggest an association with chronic energy deficiency from repeated weight cycling practices. Nattiv and colleagues (2007) have stated that the BMD of a female athlete is a reflection of the cumulative effects of low energy availability, menstrual disturbance, genetic predisposition and exposure to other nutritional, behavioural and environmental factors. A recent publication on the female athlete triad provided recommendations for BMD testing: those athletes with more than one triad risk factor, history of eating disorders, BMI ≤ 17.5, menstrual irregularity (<6 menses over 12 m), prior stress fractures and prior BMD z-score <-2.0 (De Souza et al. 2014). The ACSM recommend further investigation of athletes with a BMD z-score of <-1.0 (Nattiv et al. 2007).

The sports dietitian needs to be aware of the risk factors associated with low bone mass in both male and female athletes and indications for medical follow-up and treatment, particularly in regards to treatment of the RED-S (see Commentary 4). Menstrual dysfunction, low serum oestradiol and a history of eating disorders are detrimental to bone health, particularly if the onset occurs around adolescence, which is a critical time for bone accrual. It is not clear whether athletes with a low BMD z-score at adolescence ever recover their BMD as adults. As peak bone mass is thought to occur late in adolescence (Baxter-Jones et al. 2011), anorexia nervosa (chronic low energy availability) and menstrual dysfunction, and potential alterations in endocrine function in males, may impact on the attainment of peak bone mass and the ongoing ability to accrue bone. A follow-up study of adolescent runners found 90% of girls with low bone mass during adolescence also had low bone mass after three years (Barrack et al. 2011). Greater increases in bone mass were associated with higher increase in lean body mass and fat mass and having more regular menses, suggesting that increasing energy intake is important for optimising bone health.

ACSM has identified specific recommendations for athletes with low BMD, including increasing energy intake, resumption of menses and specified intakes of calcium and vitamin D. As low body weight is seen as desirable for many sports, this performance goal may be at odds with optimal weight or body composition for bone health. The challenge for the sports dietitian is to educate athletes and coaches about the long-term consequences of the triad (and RED-S) and the detrimental effects on bone health and ultimately performance. The reader is referred to Commentary 2 and several key publications on the female athlete triad and the recently published RED-S (Nattiv et al. 2007; Javed et al. 2013; De Souza et al. 2014; Mountjoy et al. 2014).

9.3 Bone stress injuries in athletes

Bone stress injuries occur more commonly in those athletes exhibiting more than one component of the triad (Barrack et al. 2014). As recovery from a bone stress injury may take up to 6 weeks, this length of time either fully or partially immobilised means a reduction or cessation of training. In athletes with low BMD, early intervention can potentially prevent possible long-term consequences of the triad (De Souza et al. 2014).

Bone stress injuries may range from low-grade stress reaction to a stress fracture which is a partial or complete fracture of bone. This results from the bone's inability to withstand non-violent stress that is applied in a repeated, sub-threshold manner (McBryde 1985). It arises from accumulation of bone micro-damage which cannot be adequately repaired by the remodelling process. Any factor that increases the applied load, decreases bone strength or interferes with the repair process has the potential to increase the risk of stress fracture (Bennell et al. 1996). The diagnosis of stress fracture is based on the clinical findings of a history of exercise-related bone pain with local bony tenderness on examination.

Several studies have suggested that the relative risk for stress fracture is 2–4 times greater in athletes with existing or past menstrual disturbances than eumenorrhoeic controls (Grimston et al. 1991; Kadel et al. 1992; Bennell et al. 1995, 1996). However, most studies have been cross-sectional designs where women are specifically recruited according to certain criteria, such as stress fracture history or menstrual status. In these studies, cohorts are often small and categorisation of menstrual status is based on number of menses per year rather than on analysis of hormonal levels. Where hormonal assessment is included, most are single measurements, often non-standardised with respect to menstrual cycle phase.

A limited number of cross-sectional studies have compared regional bone density in small groups of female athletes with and without stress fractures, but results have been conflicting. This discrepancy may reflect differences in subject characteristics, type of sport, measurement techniques and bone sites tested. However, the findings of a prospective study indicate that low bone density is a risk factor for stress fractures in female track and field athletes (Bennell et al. 1995). In a prospective study of 259 young exercising females, 11% incurred a bone stress injury and the risk increased from 15% to 20% for single triad risk factors and 30% to

50% for multiple triad risk factors (Barrack et al. 2014). Key variables significantly related to bone stress injuries were: participating in >12 h/wk of exercise, low BMD, BMI <21, participating in a sport that requires a lean physique, menstrual dysfunction and elevated dietary restraint. This study highlights the association between bone stress injuries and the number of triad risk factors and provides support that bone density testing is indicated in those athletes with more than one triad risk factor (De Souza et al. 2014).

9.4 Bone health in athletes and healthy people

In an effort to provide an evidence-based approach to the prevention of osteoporosis, Osteoporosis Australia held a summit to develop clear, succinct, evidence-informed recommendations about calcium, vitamin D and exercise requirements for building healthy bones. The summit and its recommendations are detailed in the *Building healthy bones throughout life* supplement (see Ebeling et al. 2013a). The key message from the summit for building healthy bones at all stages of the lifecycle in healthy people was the benefits of adequate dietary calcium intake and optimal vitamin status, in combination with regular weight-bearing exercise and moderate sunlight exposure (Ebeling et al. 2013b). Underlying this generic message are specific recommendations for these variables for each stage of the life cycle. The next section addresses the evidence for the link between calcium, vitamin D and exercise and bone health in the general population and includes specific evidence, albeit limited, relevant to athletic populations.

9.4.1 Exercise effects on bone in healthy adults and athletes

Certain types of exercise are critical for maintaining both the architecture and mass of the skeleton. Some types of exercises may have little benefit on bone health, but they still have other health benefits such as decreasing risk of cardiovascular disease. As shown in Table 9.1, swimming and cycling are considered *non-osteogenic* (not bone-promoting) whereas exercises involving high impact forces and loading such as impact aerobics, basketball and gymnastics are *highly osteogenic* (Ebeling et al. 2013a, 2013b).

TABLE 9.1 The impact of selected exercises on bone health

Highly osteogenic	Moderately osteogenic	Low osteogenic	Non-osteogenic[a]
Basketball	Running	Leisure walking	Swimming
Netball	Jogging	Lawn bowls	Cycling
Impact aerobics	Brisk or hill walking	Yoga	
Tennis	Resistance training	Pilates	
Jump rope	Stair climbing		

[a]While certain exercises may have low or no osteogenic benefits, this should not be construed to imply that these exercises do not offer a wide range of other health benefits.
Source: Reprinted with permission from Ebeling et al. 2013a

Early cross-sectional studies in athletes highlighted the benefits of exercise on bone mass, particularly at weight-bearing sites (Heinrich et al. 1990; Kannus et al. 1994; Heinonen et al. 1995). When comparing athletes with non-athlete controls or different athletic groups, differences in body size are difficult to account for in cross-sectional analyses (Kerr et al. 2007). However, randomised controlled trials (RCTs) that examine the effects of different activities in athletic populations are difficult and impractical as they require a design that controls for training, diet and exercise loading. Therefore, most of the evidence for the effects of exercise on bone come from RCTs undertaken in sedentary adult populations (Snow-Harter et al. 1992; Heinonen et al. 1996; Kerr et al. 1996; Kerr et al. 2001; Kukuljan et al. 2009). Recent meta-analyses provide support for the positive effects on bone with weight-bearing exercise and resistance training at certain sites in pre- and post-menopausal women (Wallace & Cumming 2000; Martyn-St James & Carroll 2006; Martyn-St James & Carroll 2010; Kelley et al. 2013).

A summary of the key recommendations for exercise is provided in the *Building healthy bones throughout life* supplement (Ebeling et al. 2013a). Based on population studies, these recommendations state that all adults should endeavour to participate in a variety of weight-bearing physical activities, as well as other types of exercise to improve muscle strength and balance. The position stand from ACSM recommends weight-bearing activity (3–5 times/wk) as well as resistance exercise (2–3 times/wk) as a target for maintaining BMD (Kohrt et al. 2004). For elderly men and women, specific activities to improve balance and prevent falls was also to be included. For those with a history of fractures, more careful attention to the exercise prescription and adequate supervision (e.g. exercise physiologist, physiotherapist) is required.

9.4.2 Exercise effects on bone in children

Puberty is a critical time for bone mineral accrual. Peak bone mass is acquired during late adolescence and early adulthood (Baxter-Jones et al. 2011). About 26% of adult bone is accumulated during the 2 years around the period of peak bone accrual (Bailey et al. 1999).

Exercise interventions have shown beneficial effects on bone mineral accrual in the growing skeleton (MacKelvie et al. 2002).

There appears to be a greater benefit in bone accrual if the exercise is commenced prior to puberty and when the type of activity involves high impact, such as jumping (Fuchs et al. 2001; Iuliano-Burns et al. 2003). The outcome of these studies highlights the importance of maintaining a range of high-impact weight-bearing activities, such as team and racquet sports and jumping activities, throughout childhood. Whether these gains in bone mineral accrual are maintained into adulthood or reduce fracture risk later in life is unknown. To optimise bone health and other health benefits from exercise, participation in physical activity and sport throughout childhood should be actively encouraged.

9.5 Nutrition and bone health

Sufficient intake of calcium and adequate vitamin D status are critical for bone health; calcium and vitamin D insufficiency results in bone loss. Less is known about the effects of other nutrients and potentially whole foods or food intake patterns (e.g. fruit and vegetables, dairy food) working either independently or in isolation, which may be protective of bone. Other nutrients that have a role in bone metabolism include vitamins A, C and K, protein, phosphorus, potassium, magnesium and zinc. Food combinations and other food components may differ in their effects on bone compared with single nutrients.

To date, most nutrition intervention studies investigating diet and bone health have used a dietary calcium supplement, either with or without supplemental vitamin D, rather than manipulating food intake. A food-based intervention study is difficult to control for other dietary factors that might influence bone metabolism. For example, a milk intervention study provides additional energy, protein, phosphorus and calcium. Milk studies are further complicated by varying nutrient compositional differences (both seasonal and associated with country-specific fortification regulations), which further confounds interpretation. For example, vitamin D is added to milk in the USA, which is not required in many other countries. To date, few studies have been designed to address the potential effect of total food intake on bone.

However, considerable evidence now links the positive effect of particular dietary patterns adequate in calcium, phosphorus and vitamin D on bone health (Heaney 2009). In the Framingham Osteoporosis Study, participants with high intakes of vegetables and fruit had significantly higher BMD than those with other dietary patterns (Tucker et al. 2002). Several mechanisms have been proposed to explain how fruits and vegetables may improve bone health. These include the provision of specific nutrients such as flavonoids, creating an alkaline environment or as a possible marker of a healthy lifestyle (Nieves 2013). To date however, most of the evidence comes from epidemiological or cross-sectional studies rather than clinical trials (Weaver et al. 2012; Hanley & Whiting 2013). The possible beneficial effects of an alkaline diet on bone has been touted although this theory is not supported by well-designed long-term studies (Bonjour 2013). In the absence of convincing evidence for particular dietary patterns, promoting a healthy diet that includes a wide variety of nutritious foods, including fruits and vegetables, seems a sensible approach and is consistent with current dietary recommendations.

The effect of high dietary protein intakes has also been proposed for potential adverse effects on bone health. High protein intakes can induce metabolic acidosis and increase urinary calcium excretion, which may lead to bone loss (Barzel & Massey 1998). However, a recent meta-analysis on the effects of dietary acid load on bone disease showed that a high-protein diet did not have adverse effects on bone metabolism (Fenton et al. 2011). Several short-term well-designed studies have confirmed similar effects (Roughead et al. 2003; Kerstetter et al. 2005; Hunt et al. 2009). However, in a two-year placebo-controlled trial in older women, daily supplements of whey protein did not benefit or adversely affect hip bone density (Zhu et al. 2011). Hence the proposed effects of high-protein diets do not appear to translate into adverse effects on bone health, at least in older women.

9.5.1 Milk, milk products and bone health in children and adults

Dairy foods provide the major sources of calcium in most western diets. Milk intervention studies in pre-pubescent girls have shown positive effects on total body bone mineral accretion (Cadogan et al. 1997; Merrilees et al. 2000; Du et al. 2004). The evidence for calcium supplementation in healthy children is less convincing (Winzenberg et al. 2006). In studies of pre- and early-pubertal boys and girls, milk combined with exercise appears to have additive effects on bone accrual (Iuliano-Burns et al. 2003; Volek et al. 2003). Thus encouraging a healthy lifestyle approach with increased physical activity and adequate calcium is recommended during the growing years.

In adults, several RCTs that involved increasing milk intakes have demonstrated positive effects on retarding the rate of bone loss associated with ageing (Baran et al. 1990; Prince et al. 1995; Lau et al. 2001). One trial compared the effects of milk intake and calcium supplements on bone health. In this 2-year trial, Prince et al. (1995) randomised 168 elderly women to one of four treatment groups: placebo, milk powder (1000 mg/d Ca), calcium supplement (1000 mg/d Ca) and calcium supplement (1000 mg/d Ca) with exercise. The results demonstrated the equal effectiveness of calcium taken as either as a supplement or from milk powder in slowing the rate of bone loss at the hip. For men (>62 yrs), a milk drink fortified with calcium and vitamin D slowed the rate of cortical bone loss (Daly et al. 2006). Although milk intervention studies have shown benefits to reducing bone loss, any positive effects of milk consumption on fracture prevention in either older populations or on stress fractures in athletes have not been established. A recent meta-analysis found no association with milk intake and the risk of hip fracture in older women but concluded that more data was needed in men before any recommendations about the effects could be made (Bischoff-Ferrari et al. 2011).

9.5.2 Calcium supplementation and bone health in children and adults

A large number of well-designed trials have examined the effect of calcium supplementation on bone density. A review of 32 trials in postmenopausal women found positive effects on slowing bone loss with around 1000 mg/d of calcium supplements (Nordin 2009). However, more recently, several meta-analyses have raised concerns about the safety of calcium supplements in relation to the possible negative effects on cardiovascular events (Bolland et al. 2008, 2011; Li et al. 2012). These meta-analyses have provoked a lot of debate from the scientific community because no clear biological mechanisms for adverse cardiovascular effects have been proposed (Metcalfe & Nordin 2011; Prince et al. 2012). This debate has led organisations such as the National Osteoporosis Foundation and Osteoporosis Australia to advise caution when using calcium supplements and to recommend obtaining calcium from dietary sources. Where dietary calcium sources are not possible, the recommended dosage for calcium supplements is 500– 600 mg/d.

9.5.3 Calcium intakes and requirements at various life stages

The human skeleton retains the rather primitive function of serving as both a depot for the storage of excess calcium and as a reservoir, available to replenish calcium during times of deprivation. This portable supply of calcium, however, is a double-edged sword. When the calcium reserve, our skeleton, is called upon to meet dietary insufficiencies, bone strength is compromised.

Calcium balance is determined by the balance between dietary calcium intake and the amount of calcium absorbed from the intestine and excreted in the urine. Plasma calcium levels are tightly regulated by hormonal control. Hence, when negative balance occurs in response to low oestrogen levels, demineralisation from the skeleton will follow. Dietary calcium recommendations are based on the amount of dietary calcium needed to maintain calcium balance and optimal bone accretion rates.

Calcium balance studies suggest that there may be a threshold effect for calcium intake, which means that calcium retention increases up to a threshold, beyond which any additional intake of calcium does not result in increased calcium retention. Calcium absorption from the small intestine occurs by active absorption at low intake levels and by passive absorption at higher calcium intakes. Traditionally calcium requirements and recommendations have been determined from balance studies in which calcium intake and loss are equal. Little is known about calcium requirements for physically active people (Weaver 2000). The revisions to the nutrient reference standards for calcium in the USA and Canada used three approaches as the basis of setting the calcium recommendations for different population groups. These were derived from calcium balance studies; from a factorial model using calcium accretion on bone mineral accretion data; and from clinical trials, which investigated the response of change in BMD or fracture rate to varying calcium intakes. In 2006, Australia adopted the USA/Canadian recommendations with a slight modification. The approach used to set recommendations for calcium in Australia was based on the approach used by the Food and Agriculture Organization (FAO) of the United Nations and World Health Organization (WHO) in their 2001 revisions (FAO/WHO 2001). The Recommended Dietary Intake (RDI), the Australian equivalent to the RDA (Recommended Daily Allowance) in the US for both boys and girls (12–18) is 1300 mg/d, which is the same as for elderly men and women (>70). For both adult men (19–70) and women (19–50), the RDI is 1000 mg/d (Commonwealth Department of Health and Ageing et al. 2006). Postmenopausal women have slightly higher recommendations. The EAR (Estimated Average Requirement) is less than the RDI for all values (Commonwealth Department of Health and Ageing et al. 2006). See Chapter 2 for guidelines on the application of these standards. There are currently no specific recommendations for calcium intake for athletes so, until further studies are undertaken, population reference standards can be used as a benchmark for assessing adequacy.

9.5.4 Calcium requirements for athletes

While there is evidence for the importance of calcium in maintaining bone density across the lifespan, it is not clear athletes have higher requirements than others. In athletes, a key interest has been around the role of calcium and vitamin D in the maintenance of bone density and the prevention of stress fractures. A clinical review by Tenforde et al. (2010) highlighted the lack of evidence in determining the optimal intake for both calcium and vitamin D for athletes, particularly in the prevention of stress fractures. To date the only randomised trial to examine the effect of calcium and vitamin D supplements on stress fracture incidence has been conducted in 5201 Navy recruits (Lappe et al. 2008). The calcium and vitamin D group had a 20% lower incidence of stress fractures compared with the control group. In a 2-year prospective cohort study in 125 female competitive runners, higher intakes of calcium, skim milk and dairy products (as assessed by a food frequency questionnaire) were associated with lower rates of stress fractures (Nieves et al. 2010). From the available evidence, 1500 mg/d of calcium appears to be beneficial in reducing the risk of stress fractures (Tenforde et al. 2010). However, increasing calcium and vitamin D is not a replacement for addressing the other risk factors associated with the triad or RED-S.

9.6 Vitamin D

9.6.1 Synthesis and metabolism of vitamin D

Although vitamin D is classified as a vitamin, human requirements can be met entirely through cutaneous synthesis by exposure to sunlight (Zittermann 2003; Holick 2007). In this process, 7-dehydrocholesterol—present in the plasma membrane of epidermal and dermal cells—is photolytically converted to pre-vitamin D3 by the ultraviolet-B photons (UVB, 290–315 nm) in sunlight and then thermally isomerised to vitamin D3 (cholecalciferol). Vitamin D3 is subsequently ejected out of the plasma membrane, and drawn into the dermal capillary bed and into circulation by vitamin D-binding protein (VDBP). Newly synthesised vitamin D3 (as well as the vitamin D consumed in the diet) undergoes obligate hydroxylation to 25(OH)D (calcidiol) in the liver by the cytochrome P-450 system (CYP2R1). Its further hydroxylation in kidney tubules to the biologically active form, 1,25(OH)2D (calcitriol) by the cytochrome enzyme CYP27B1 is closely regulated and controlled by parathyroid hormone (PTH), serum calcium and phosphate concentrations, and fibroblastic growth factor (Hossein-Nezhad & Holick 2013). Hydroxylation of 1,25(OH)2D is also known to occur in extra-renal tissues (Hossein-Nezhad & Holick 2013) including bone, cartilage, muscle and blood (Nowak et al. 2012).

9.6.2 Dietary sources of vitamin D

Vitamin D is also found in the diet from limited natural sources including fatty fish, egg yolk and sundried mushrooms. In some countries, vitamin D is added to some foods including milk, yoghurt, margarine, ready-to-eat cereal and fruit juices (see Table 9.2). Dietary vitamin D includes both D3 derived from animal sources and D2 (ergocalciferol, derived from UVB exposure of fungi and yeast ergosterols). Both forms are well absorbed (50% bioavailable) into intestinal mucosal cells from micelles, incorporated into chylomicrons, and enter the circulation via the lymphatic system. Vitamin D absorption is diminished in individuals with malabsorption syndromes (Basu & Donaldson 2003) and those individuals following extremely low-fat diets (Mulligan & Licata 2010; Ross et al. 2010). Diet-derived vitamin D2 also undergoes similar hydroxylations in the liver and kidney similar to diet-derived and endogenously synthesised vitamin D3.

TABLE 9.2 Calcium, magnesium and vitamin D content of selected foods

Food	Serve size	Calcium	Magnesium	Vitamin D
		Mg	mg	IU
Dairy, egg and soy milk				
Cow's milk, skim	250 mL	306	0	0
Cow's milk, 2% fat	250 mL	285	27	98
Cheddar cheese	30 g	204	8	0
Mozzarella cheese	30 g	207	7	0
Swiss cheese	30 g	224	11	0
1 egg yolk	60 g	1	22	20–40
Soy milk, calcium-fortified	250 mL	368[a]	39[a]	100[a]
Yoghurt, natural, low fat	200 g	235	21	na
Grains				
Bread, most enriched	1 slice (30 g)	43	7	0
Bread, whole wheat	1 slice (30 g)	20	24	0

Cereal, fortified	30 g	125–1000[a]	variable	40–100[a]
Vegetables and legumes				
Avocado	100 g	14	27	na
Broccoli, boiled	100 g	33	18	na
Mixed vegetables, frozen, boiled (carrots, peas, corn)	100 g	31	17	
Spinach, English, boiled	60 g	65	83	na
Potato, new, boiled	100 g	4	19	na
Chickpeas, canned, drained	100 g (1/2 cup)	45	27	na
Baked beans in tomato sauce	100 g (1/2 cup)	40	25	na
Peanut, with skin, raw	20 g	10	32	
Meat and fish				
Lamb chop, trimmed, grilled	100 g	10	27	na
Beef rump steak, lean, grilled	100 g	6	27	na
Chicken breast, no skin, grilled	100 g	5	28	na
White fish, cod, bream, raw	100 g	22	28	104[b]
Salmon, Atlantic, raw (wild)	100 g	7	25	988[b]
Fruits				
Grapefruit juice, fortified[c]	250 mL (1 cup)	350	30	100
Orange juice, fortified[c]	250 mL (1 cup)	350	27	100
Nuts/seeds				
Almonds, raw with skin	20 g	50	52	na
Walnuts, raw with skin	20 g	18	30	na
Fruit nut and seed bar	40 g	25	32	na

[a]Varies by brand

[b]Lu et al., 2007

[c]Values for calcium and vitamin D fortified foods obtained from the food labels in the USA
na = Not available
Source: The USDA National Nutrient Database for Standard Reference (www.ars.usda.gov/main/site_main.htm?modecode = 12354500) and Food Standards Australia New Zealand NUTTAB 2010 (www.foodstandards.gov.au/science/monitoringnutrients/nutrientables/nuttab/Pages/default.aspx)

9.7 Indices of vitamin D status

Circulating calcidiol (i.e. 25-hydroxyvitamin D (25(OH)D)) concentration is currently considered the best indicator of vitamin D status (Zittermann 2003; Hollis 2005; Holick 2007, 2009; Cannell et al. 2008; Hossein-Nezhad & Holick 2013) because the conversion of vitamin

D3 into 25(OH)D is poorly regulated (no feedback regulation) and serum 25(OH)D concentration increases in proportion to vitamin D intake and cutaneous exposure (Holick 1981; Brown et al. 1999). Furthermore, circulating concentration of biologically active calcitriol 1,25(OH)2D is affected by other factors (including serum PTH, calcium and phosphate concentrations) and is not reflective of vitamin stores. Definitive cut-offs for vitamin D status are not yet scientifically established (i.e. as are other clinical thresholds such as serum calcium or blood glucose) but are based on suppression of PTH concentrations (Holick 2009) and clinical and disease risk markers (Cannell & Hollis 2008; Cannell et al. 2008; Hossein-Nezhad & Holick 2013).

The thresholds for diagnosing vitamin D deficiency, insufficiency, sufficiency and toxicity agreed on by most researchers and the suggested optimal range are shown in Table 9.3. The optimal range is thought to be the concentration where the human genome evolved (Hollis 2005; Cannell et al. 2008). In contrast, the Recommended Daily Allowance for the USA and Canada was established using 50 nmol/L as *adequate* based exclusively on bone markers and on the justification that serum concentrations greater than 75 nmol/L have not consistently been associated with health benefits. Current evidence, however, suggests that skeletal muscle function (Bischoff-Ferrari et al. 2004) and risk for chronic and acute illness may be reduced when vitamin D is maintained in the presumed optimal range (100–250 nmol/L) (Cannell et al. 2008; Hossein-Nezhad & Holick 2013). However, Heaney (2011), a leading researcher in this area, has argued that the distinction between deficiency and insufficiency is not useful or necessary but rather that maintenance of optimal stores associated with preventable disease (which is represented by a cut-off of at least 120 nmol/L), is a more realistic target.

TABLE 9.3 Vitamin D status and suggested reference ranges for serum vitamin D 25(OH)D

	nmol/L[a] (SI)
Deficiency[b]	50
Insufficiency[b]	75–80
Sufficiency	75–80
Optimal	100–250
Toxic	>375 plus hypercalcaemia

[a]To convert to ng/mL divide nmol/L by 2.496

[b]The cut-off for deficiency is the approximate concentration at which PTH rises abruptly and the cut-off for insufficiency is the concentration where PTH plateaus and calcium absorption is maximised (Zittermann 2003; Hollis 2005; Holick 2009; Hossein-Nezhad & Holick 2013).

9.8 Functions of vitamin D

Vitamin D functions as a modulator of an estimated 2000 genes involved in a variety of biological functions including calcium and phosphorus regulation, bone metabolism, skeletal muscle function, immune modulation and insulin and blood pressure regulation (Cannell et al. 2009; Holick 2011; Hossein-Nezhad & Holick 2013). It is estimated that close to 5% of the human genome is modulated by vitamin D (Zella et al. 2008), which is required in sufficient quantities to work effectively in gene expression. This action on immune function was confirmed in a study that measured the gene expression of a large number of genes in white-blood cells, which found that increases in serum 25(OH)D concentration following 2 months of oral vitamin D supplementation altered expression of 291 genes (Hossein-Nezhad et al. 2013).

9.8.1 Role of vitamin D in bone health

Vitamin D plays a central role in calcium and phosphate homeostasis and is essential for development and maintenance of healthy bone. Its main function is to maintain serum calcium concentration within the physiological range, which in turn helps prevent bone resorption and demineralisation when calcium intake is suboptimal. In this role, serum 1,25(OH)2D up-regulates the expression of genes that enhance intestinal calcium absorption, renal tubular calcium resorption (in association with PTH) and osteoclastic activity (Holick 2004, 2006, 2007). The potent effect of vitamin D on enhancing calcium absorption is evident in the research of Heaney and colleagues (2003), who found that fractional calcium absorption is >30% when serum 25(OH)D concentration is at least 75 nmol/L but only 10–15% when <50 nmol/L (i.e. in the deficient range). This effect is largely attributed to vitamin D-enhanced expression of intestinal calcium binding and epithelial calcium channel proteins (including calbindin). When dietary calcium is inadequate, 1,25(OH)2D interacts with its receptor in osteoblasts and induces the expression of receptor activator of nuclear factor κB ligand (RANKL), which induces the maturation of osteoclasts (Holick 2004, 2006). The net effect of suboptimal calcium intake is mobilisation of calcium from bone to maintain serum calcium homeostasis.

In bone remodelling

Vitamin D may also be involved in the formation of the bone matrix, although several studies have suggested that it is not essential in bone tissue formation (Brown et al. 1999). With deficient (or suboptimal) vitamin D status, serum ionised calcium concentration begins to decline and is immediately recognised by calcium-sensing cells in the parathyroid gland. This results in an up-regulation of parathyroid hormone (PTH) expression, synthesis and excretion. PTH is typically negatively associated with circulating 25(OH)D, and functions to promote the conversion of 25(OH)D to 1,25(OH)2D and both calcium resorption in kidney tubules and bone resorption (along with 1,25(OH)2D). PTH also promotes phosphorus loss in the kidney and decreases phosphorus absorption thereby altering the calcium to phosphorus product in blood. Ultimately, vitamin D deficiency results in mineralisation defects that, if prolonged, will lead to rickets in children and osteomalacia in adults (Holick 2006). Unlike osteoporosis, which decreases bone mass (and density), osteomalacia is characterised by defective bone mineralisation and soft bones.

Vitamin D is nevertheless essential for bone growth, development and maintenance. The actual relation between vitamin D intake/status and BMD is complex due to the multiple hormonal, biological and nutritional factors that also impact peak bone mass and bone loss. Association between higher serum 25(OH)D concentrations and higher BMD has been reported in large-scale studies of adolescents (Lamberg-Allardt et al. 2001; Outila et al. 2001), middle-aged adults (Bischoff-Ferrari et al. 2004) and postmenopausal women (Collins et al. 1998; Mezquita-Raya et al. 2001). In one study of over 13 000 ambulatory adults living in the USA, Bischoff-Ferrari and colleagues (2004) found a positive association between serum 25(OH) D concentration and BMD and/or BMC in the hip and lumbar spine in both younger and older adults. The higher 25(OH)D concentration in association with greater BMD throughout the reference range suggested that it may be advantageous to maintain circulating vitamin D concentrations in the upper portion of the reference range (90–100 nmol/L). A relation between serum 25(OH)D and BMD, however, is not consistently reported by all studies (Ross et al. 2010, 2011).

Link with vitamin D and stress fractures

In athletic populations, risk of stress fractures may be an important marker of bone health. Stress fracture is one of the most frequently seen types of overuse injuries in athletes and military recruits that may serve as a red flag for compromised bone health. Several studies provide evidence that sufficient vitamin D along with dietary calcium are important for prevention of bone injury and bone health (Tenforde et al. 2010). A prospective study in 125 female cross-country runners (18–26 yrs) found that higher intakes of calcium and dairy products predicted lower stress fracture rates, whereas higher intakes of vitamin D, calcium and dairy were associated with significant gains in hip BMD during two years of follow-up (Nieves et al. 2010). In another prospective study of US adolescent girls (n = 6 712), vitamin D but not calcium intake was associated with lower incidence of stress fracture, particularly in those regularly engaging in high-intensity activity (of whom 90% of the stress fractures occurred) (Sonneville et al. 2012). Based on vitamin D status—which may be more important than intake—several cross-sectional studies of Finnish military recruits found that vitamin D influences both peak BMD and stress fracture development. In one study of 220 male recruits, higher 25(OH)D concentrations were associated with greater BMC in the lumbar spine, femoral neck, trochanter and total hip (Valimaki et al. 2004). In two subsequent studies of male recruits, lower vitamin D status and higher PTH concentration (which is influenced by vitamin D status) were associated with stress fracture development (Ruohola et al. 2006; Valimaki et al. 2005). In the former study in 756 men, stress fracture risk was 3.6 times higher in recruits with serum 25(OH)D concentration <75 nmol/L (Ruohola et al. 2006). A final randomised double-blind trial in 5201 US female naval recruits found that supplementation with 800 IU/d of vitamin D plus 2000 mg/d calcium during basic training reduced stress fracture incidence by

20% (Lappe et al. 2008).

9.8.2 Function of vitamin D in skeletal muscle

Skeletal muscle function is important to consider along with bone health because muscle strength stabilises bone and may help prevent falls and bone fracture. Clinically, musculoskeletal pain and weakness are established (but often forgotten) symptoms of vitamin D deficient osteomalacia (Irani 1976) and rickets (Holick 2006) that resolve with repletion. Muscle pain and weakness, however, may be widespread in both younger and older members of the general population presumably without biochemical signs of osteomalacic bone involvement.

Recent studies in the general population have identified a strong link between low vitamin D status and persistent non-specific back and musculoskeletal pain (often considered to be fibromyalgia). A study conducted at a northern inner-city clinic in the USA found that 93% of patients with persistent non-specific musculoskeletal pain had 25(OH)D concentrations in the deficient range (<50 nmol/L) and 28% had severely deficient status (<20 nmol/L) (Plotnikoff & Quigley 2003). Another study from the north-west and midlands areas of England found that low vitamin D status (25(OH)D <25 nmol/L) was 3.5 times more common in women with widespread musculoskeletal pain (Macfarlane et al. 2005) than those without pain. Furthermore, several open clinical trials (i.e. not randomised) found substantial reductions in muscle pain with vitamin D3 supplementation (Al Faraj & Al Mutairi 2003; de Torrente de la Jara et al. 2004). In one trial, treatment with vitamin D3 supplements (5000–10000 IU/d for 3 m) resulted in clinically improved reductions in back pain in 100% of the patients who initially had low vitamin D status (Al Faraj & Al Mutairi 2003).

Several other cross-sectional studies in the general population have also found associations between low serum 25(OH)D concentration and muscle strength (Gerdhem et al. 2005; Foo et al. 2009; Grimaldi et al. 2013); jump height, velocity and power (Ward et al. 2009); aerobic fitness (Mowry et al. 2009; Ardestani et al. 2011); and balance and coordination (Dhesi et al. 2004). These studies suggested that at least some improvement in muscle strength occurs with vitamin D supplementation (Glerup et al. 2000a, 2000b; Ward et al. 2010; Diamond et al. 2013). Not all studies, however, have supplemented at the level necessary to correct hyperparathyroidism, which suggests supplementation did not correct deficiency. Hyperparathyroidism may also affect the action of vitamin D on both muscle strength and bone parameters.

Whether such muscle pain or weakness is common in athletes to a level that would deter performance is not yet established. Suboptimal vitamin D status, which ranges from insufficiency to severe deficiency, is reported in athletes (as reviewed below), which may support the possibility of accompanying muscle effects. Theoretically, low vitamin D status would be expected to compromise skeletal muscle function and thereby athletic performance. It is also possible, however, that compromised athletic performance or muscle changes are more difficult to detect in athletes given the multitude of factors that influence performance. Currently only a few studies have addressed vitamin D status and performance in athletic populations (Close et al. 2013a; Dubnov-Raz et al. 2014; Wyon et al. 2014). Of these four studies, two found improvement in various performance markers with oral supplementation. In previously deficient elite English soccer players, Close and colleagues (2013) found that oral supplementation with vitamin D at high doses (5000 IU/d) for 8 weeks improved 10-metre sprint and vertical jump performances but not 30-metre sprint time or squat ability. In professional dancers Wyon and colleagues (2014) found that vitamin D supplements (2000 IU/d) for 4 weeks improved quadriceps strength and power and vertical jump performance.

The action of vitamin D on skeletal muscle function and performance is modulated through both genomic and non-genomic events (Hamilton 2010; Barker et al. 2011; Boland 2011; Girgis et al. 2013). The genomic events occur through modulation of gene transcription and protein synthesis, ultimately influencing muscle cell proliferation and differentiation (particularly in type 11 fibres) and muscle contraction and relaxation. The non-genomic responses involve rapid sarcolemma calcium uptake and activation of mitogen-activated protein kinase signalling pathways. Animal and tissue culture studies found that vitamin D deficiency induces atrophy of fast-twitch muscle fibres, impairs sarcoplasmic calcium uptake and prolongs time to peak contractile tension and relaxation (Rivier et al. 1994). Type II skeletal muscle fibre atrophy is also observed in human biopsy studies of deficient patients (Sato et al. 2005). Oral vitamin D supplementation increases size and number of type II fibres (Sato et al. 2005) and increases ATP content of isolated muscle cells (Birge & Haddad 1975).

9.8.3 Other functions of vitamin D

Vitamin D also functions as an important regulator of inflammation and immunity, which has skeletal implications. Vitamin D up-regulates gene expression of broad-spectrum anti-microbial peptides (AMP)—important regulators in innate immunity—(Wang et al. 2004; Liu et al. 2006) and exerts an immunomodulatory effect on T and B-lymphocytes in acquired immunity (Zittermann 2003). AMPs, including cathelicidin, are important proteins in the innate immune system (Gombart et al. 2005) and help defend against acute illness including tuberculosis, influenza and the common cold (Cannell et al. 2006, 2008; Urashima et al. 2010). Vitamin D also works through the immune system to control inflammation, which results from the accumulation of fluid and immune cells in injured tissue. Although a detailed discussion of the mechanism of action of vitamin D in immune function is beyond the scope of this chapter, low vitamin D status may increase risk of acute illness (Laaksi et al. 2007; Halliday et al. 2011) and whole-body inflammation (Willis 2008), although further research is warranted.

9.8.4 Role of vitamin D in injury development and rehabilitation

Injury development in athletes is a complex process that may encompass injury to bone, muscle and cartilage and often involves pain and inflammation. An early recognition of a link between ultraviolet (UV) light (specifically UVB rays) and injury was alluded to in a German report in the 1950s. In this report, athletes experienced a significant reduction in chronic pain from sports injuries following 6 weeks of UV treatment with a 'central sun lamp'. The source of their injuries (i.e. bone, muscle or connective tissue) was not reported and vitamin D status not measured. Several recent studies, nonetheless, support a role of vitamin D in injury development. A preliminary observational study in American professional football players found that those who sustained a muscle injury during the season had lower circulating 25(OH)D concentration than those without muscle injury (50 versus 63 nmol/L) (Shindle et al. 2011). A recent study of college swimmers and divers observed that 77% of muscle and connective tissue injuries occurred after the seasonal drop in 25(OH)D in winter (Lewis et al. 2013). Another recent study in professional dancers—who are at high risk for injuries—found that oral vitamin D supplementation (2000 IU/d for 4 m) reduced injury incidence (Wyon et al. 2014).

Musculoskeletal rehabilitation following injury or surgery is also an emerging area that is linked to vitamin D status. Vitamin D insufficiency can delay rehabilitation (Kiebzak et al. 2007; Barker et al. 2011). Supplementation in deficient patients improves both muscle strength and the relative size and number of fast twitch (type II) muscle fibres (Sato et al. 2005). A most interesting study conducted in recreationally active patients undergoing anterior cruciate ligament (ACL) repair found that vitamin D insufficiency—25(OH)D concentrations <75 nmol/L—delayed recovery and dampened the increases in peak isometric force compared with those patients in a vitamin D sufficient state (Barker et al. 2011). Certainly these results are of interest because muscle weakness challenges physical rehabilitation in the sports medicine setting including the recovery from bone-related injury.

9.9 Vitamin D status and intakes of athletes

Suboptimal vitamin D status is widespread among the general population (Russell 1994; Holick 2007; Lips 2007; Boyages & Bilinski 2012; Hossein-Nezhad & Holick 2013). In athletes, the prevalence of deficiency and sufficiency is highly variable and is influenced by training location, sport (Larson-Meyer & Willis 2010) and skin colour (Hamilton et al. 2010; Shindle et al. 2011; Pollock et al. 2012) (see Figure 9.1). 25(OH)D concentrations are generally lower in the winter and among athletes who train predominantly indoors. Interestingly, athletes with the lowest vitamin D status reported were those training in Finland (Lehtonen-Veromaa et al. 1999), the UK (Close et al. 2013) and the Middle East (Hamilton et al. 2010). The highest status reported to date are in US college athletes during the autumn (Storlie et al. 2011; Halliday et al. 2011; Lewis et al. 2013).

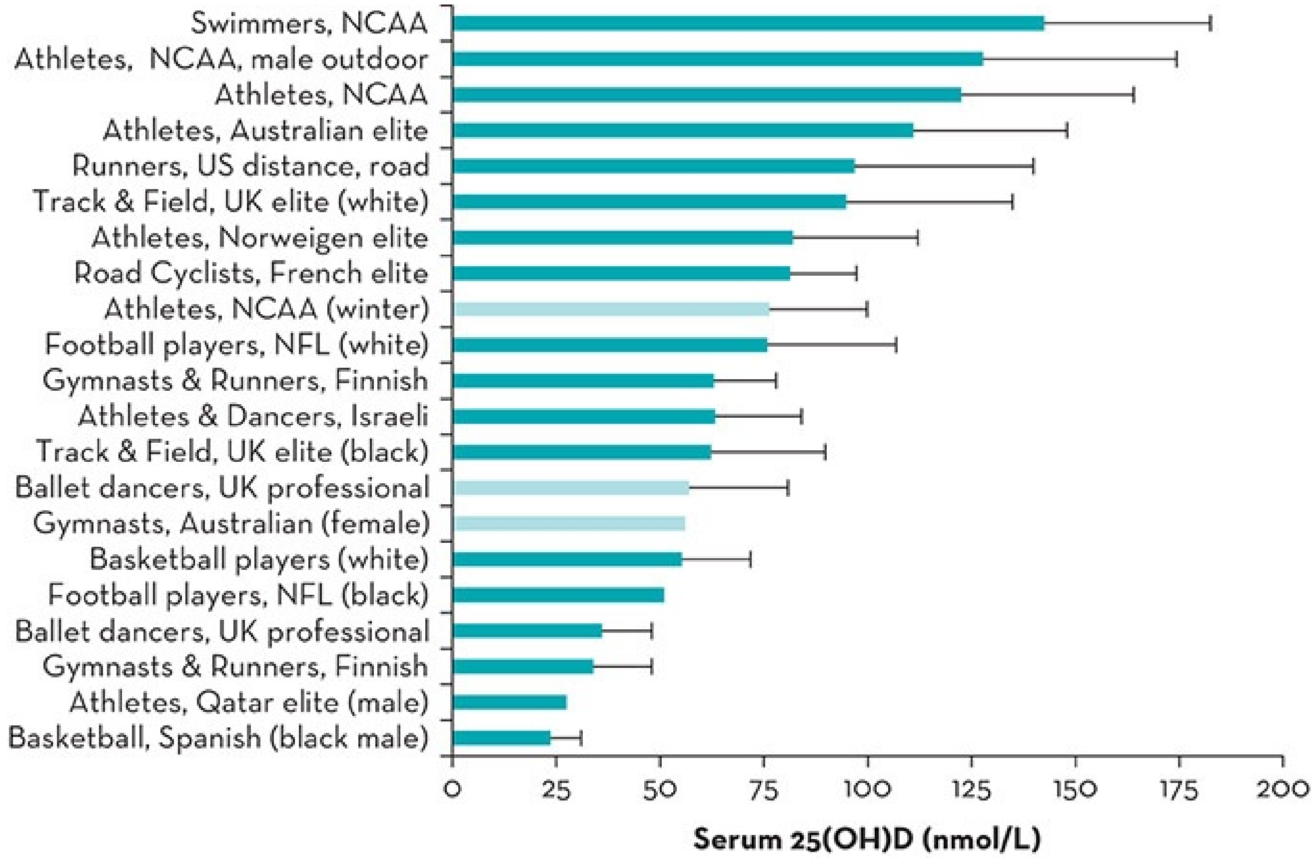

Figure 9.1 Vitamin D status in athletes from a variety of sports

Note: Data were collected in both male and female athletes (unless otherwise specified) and during sunny months (spring, summer or autumn: dark bars) and winter months (light bars). Error bars represent reported standard deviation.

Sources: Developed using references of Bergen-Cico & Short 1992; Lehtonen-Veromaa et al. 1999; Maimoun et al. 2006; Hamilton et al. 2010; Bescos Garcia & Rodriguez Guisado 2011; Helle & Bjerkan 2011; Storlie et al. 2011; Halliday et al. 2011; Pollock et al. 2012; Willis et al. 2012; Close et al. 2013b; Peeling et al. 2013; Wyon et al. 2014

Although the most probable reason for suboptimal vitamin D status is insufficient exposure to UVB light, poor vitamin D intake may contribute to poor vitamin D status. Intake studies over the last 15 years have found that many athletes do not come close to meeting the dietary recommendations for vitamin D of most countries (i.e. 200–400 IU) (Larson-Meyer & Willis 2010), particularly the newly revised US RDA of 600 IU (Ross et al. 2010). Two recent studies in US college athletes (Halliday et al. 2011) and adolescent females (Sonneville et al. 2012) found that few individuals met the US RDA for vitamin D from food sources. Furthermore,

studies in non-athletes have found that vitamin D intake is low in some individuals following vegetarian (Davey et al. 2003) and vegan (Outila et al. 2000) diets or consuming limited dairy products or fortified foods. In contrast, intake is higher in individuals who consume fish at least four times per week (Nakamura et al. 2000). However, there are limitations to interpreting the accuracy of vitamin D intake data because of variability of vitamin D in natural sources (including fish, egg yolks), differences in food fortification regulations in different countries and incomplete information on commercial foods that may be fortified with vitamin D in nutrient databases (Ross et al. 2010).

9.9.1 Factors influencing vitamin D status

Because adequate sunlight is necessary for vitamin D synthesis, anything that limits the amount or quality of sun exposure can compromise vitamin D status, as can factors that affect vitamin D absorption. These include amount and time of day of exposure, skin area exposed, sunscreen use, skin pigmentation, cloud cover, season, latitude, altitude and pollution (Holick 2007; Cannell et al. 2008). An increase in the zenith angle of the sun during early morning and late afternoon and during winter at latitudes >35 degrees north or south does not allow for synthesis because of the longer path for solar UVB photons, which are instead absorbed by the ozone layer (Hossein-Nezhad & Holick 2013).

Also, excess adiposity may compromise vitamin D status, which consistently correlates inversely with body fat percentage (Arunabh et al. 2003; Looker 2005; Sato et al. 2005; Snijder et al. 2005; Gallagher et al. 2013). Several studies had reported lower vitamin D levels in obese than non-obese individuals (Liel et al. 1988; Wortsman et al. 2000; Vimaleswaran et al. 2013). Similar observations have been reported among multi-sport athletes with higher adiposity although an association between vitamin D status and body adiposity is not apparent in the winter when 25(OH)D concentration drops substantially (Halliday et al. 2011). In athletes, it is not yet established whether lower serum 25(OH)D concentration is due to fat sequestration by excess adiposity (Liel et al. 1988; Wortsman et al. 2000) or volume dilution (from the larger body size of larger/overweight individuals) (Drincic et al. 2012). Preliminary studies, however, are in support of the mechanism of fat sequestration (Heller et al. 2014). Also it is not yet clear whether vitamin D stored in adipose tissue is released during winter or other periods of reduced UVB exposure (Holick 2007) or whether athletes with low body fat stores would be at risk for low status in less sunny seasons.

9.10 Vitamin D requirements for athletes

The RDA for vitamin D in the USA and Canada is 600 IU for children and adults (<70 yrs) and 800 IU for older adults (>70 yrs) (Ross et al. 2010). Although the RDA for adults is higher than recommended by most other countries—which ranges from 200 IU in Australia and New Zealand to the recently revised 400 IU in the UK (Table 9.4)—many vitamin D experts believe that the RDA, which was established exclusively on bone health (Ross et al. 2010), is not high enough to support non-skeletal health benefits (Heaney & Holick 2011; Holick et al. 2011) and optimise athletic performance (Cannell 2007). In contrast, the Endocrine Society of the USA recommends 1500–2000 IU/d for individuals not getting adequate sun exposure to keep concentrations in the sufficient range (Holick et al. 2011). There is no evidence to suggest that the vitamin D needs of athletes differ from those of the general population.

TABLE 9.4 Recommended intakes for calcium and vitamin D

Group	Australia & New	Nordic	United		World Health

	Zealand[a]	countries[b]	Kingdom[c]	USA & Canada[d]	Organization[c]
Calcium mg/day					
Children	500–1000	900	350–550	700–1000	500–700
Adolescents	1300	900	800 girls 1000 boys	1300	1300
Adults	1000	800	700	1000	1000
Older adults	1300 (women pm) 1300 (men >70 yrs)	800	700	1200 (women pm) 1200 (men >70 yrs)	1200 (women pm) 1200 (men >65 yrs)
Vitamin D IU/day (μg/d in square brackets)					
Children	200 [5]	300 [7.5]	280 [7] (<4 yrs)	600 [15]	200 [5]
Adolescents	200 [5]	300 [7.5]	400 [10]	600 [15]	200 [5]
Adults	200 [5]	300 [7.5]	400 [10]	600 [15]	200 [5]
Older adults	400 [10] (50–70 yrs) 600 [15] (>70 yrs)	400 [10] (>60 yrs)	400 [10]	800 [20] (>70 yrs)	400 [10] (51–65 yrs) 600 [15] (>65 yrs)

Notes: Recommended daily intake for children >1 y and adults defined as >18 or 19 y. Nutrient requirements are often higher after menopause (pm = postmenopause)

[a]Recommended Dietary Intake (calcium); Adequate intake (vitamin D)

[b]Nutrition Recommendation

[c]Recommended Nutrient Intake

[d]Dietary Reference Intake

Sources: Compiled using the following references: Department of Health, Committee on the Medical Aspects of Food Policy (COMA) 1991; Nordic Nutrition Recommendations 2004; World Health Organization & Food and Agricultural Organisation of the United Nations 2004; Commonwealth Department Health and Ageing et al. 2006; Ross et al. 2010

Because there are limited foods that contain vitamin D, most athletes will need sensible sun exposure, regular supplementation, or a combination of dietary intake, sun exposure and supplementation. The recommendation—to obtain 5 (in very fair-skinned people) to 30 (in darker-skinned people) minutes of sunlight exposure to arms, legs and torso 2–3 times a week at close to solar noon without sunscreen—usually leads to sufficient vitamin D synthesis (Holick 2007; Cannell et al. 2008). Fair-skinned individuals sunbathing in a bathing suit produce 10 000 to 20 000 IU of vitamin D in less than 30 minutes (Hollis 2005). Sunscreen or coverage with protective clothing should be used with additional sun exposure. Most recently, the time dose for sensible exposure has been altered and suggests sun exposure for approximately 25% to 50% of the time it would take to achieve a mild sunburn (defined as minimal erythemal dose). For example, if a Caucasian experiences sunburn in 20 minutes, 7–10 minutes of exposure, followed by sunscreen or full body coverage, should be sufficient to produce adequate vitamin D. A recent app, D-minder, can be used to provide athletes with information about sensible sun exposure and skin protection based on specific location (see Useful websites and resources).

9.11 Vitamin D intoxication

An understanding about the risk of vitamin D toxicity is important when working with athletes because some athletes, coaches and trainers believe that 'if a little is good, more is better'. Vitamin D intoxication from excess intake or supplementation, however, is extremely rare. Typical reported cases involve unintentional consumption of extremely high doses (Koutkia et al. 2001; Araki et al. 2011; Jacobsen et al. 2011). One report described a case of vitamin D intoxication that was caused by manufacturer error in which 188 640 IU of vitamin D3 had been added to a vitamin supplement instead of the intended 400 IU (Klontz & Acheson 2007). In this case, the patient was hospitalised with hypercalcaemia (>3.75 mmol/L) and elevated serum 25(OH)D (1171 nmol/L) and presented with classic toxicity signs and symptoms including fatigue, nausea, vomiting, constipation, back pain and forgetfulness. The dangers of hypercalcaemia include soft tissue calcification and resultant hypertension, renal and cardiovascular damage and cardiac rhythm abnormalities. On the other hand, doses of 10 000 IU per day for up to 5 months do not result in toxicity (Vieth 2004). Vitamin D intoxication from sunlight or artificial UVB exposure is not possible because feedback loops direct production of biologically inert photoproducts with prolonged exposure (Ross et al. 2010).

9.12 Clinical/dietary assessment, evaluation and treatment of vitamin D deficiency

Routine screening of vitamin D status and dietary intake of calcium and other nutrients required for bone health may be useful in athletes. If routine screening is not feasible, athletes with a history of bone and joint injury, stress fracture, skeletal weakness or pain, frequent illness or signs of overtraining should be assessed. Careful attention should also be given to high-risk groups, for example, athletes with restrained eating patterns (low energy availability) and those who spend the majority of time indoors (gymnasts, dancers, figure skaters, wrestlers). Vegan athletes with limited sunlight exposure may also warrant screening considering the potential for reduced bioavailability of vitamin D2 (vegan vitamin D) and limited intake of natural and fortified calcium- and vitamin D-rich foods.

Steps for assessment are outlined in Table 9.5. Although serum 25(OH)D concentration using a reliable assay is the usual biomarker, PTH may provide additional information when bone density is low, stress fracture is evident and/or vitamin D and calcium intakes are severely restricted. PTH concentration increases substantially when 25(OH)D concentration falls below 25–50 nmol/L (Holick 2009) and is independently related to bone density (Halliday et al. 2011) and stress fracture risk in active populations (Ruohola et al. 2006). Additional markers of bone health, including alkaline phosphatase, serum calcium or phosphorus or bone turnover, may be useful for clinical diagnosis or research purposes (Rajakumar et al. 2014). Bone-specific alkaline phosphatase, for example, is generally a good marker of osteomalacic bone involvement (Russell 1994; Holick 2006) as are serum phosphorus and calcium, which are typically low or low normal with osteomalacia but unremarkable with compromised BMD or osteoporosis. The history and physical examination should address all risk factors and assess symptoms of deficiency including idiopathic muscle pain and weakness, overtraining injury and frequency of infectious illness. Some medications interfere with vitamin D absorption or metabolism (see Cannell et al. 2008; Robien et al. 2013) (See Table 9.5).

TABLE 9.5 Clinical assessment

Anthropometrics and biological factors

- Age (vitamin D synthesis reduced by ~75% at age 70 y due to lower 7-dehydroxy cholesterol in cutaneous membrane)
- Body fat percentage (adiposity) and body mass index (excess body fat takes up and stores newly synthesised vitamin D; large body volume may dilute serum concentration)
- Weight history (recent weight change may impact BMD)

Biochemical (laboratory) data

- 25(OH)D concentration
- Parathyroid hormone (negatively associated with 25(OH)D concentration; detrimental to bone)
- Alkaline phosphatase (marker for hypovitaminosis D osteopathy; not typically found with osteoporotic BMD loss)
- Serum calcium and serum phosphorus (may be low or low-normal with osteomalacic osteopathy; reflective of compromised BMD)
- Serum magnesium
- Thyroid-stimulating hormone

Clinical

History

- Stress and other bone fractures
- Bone pain (may be indicative of osteomalacic osteopathy)
- Muscle pain, weakness, ‘heaviness in legs’ (common symptoms with severe hypovitaminosis D)
- Chronic injury
- Frequent viral and bacterial illness
- Photosensitivity
- Skin cancer/melanoma (sensible sun exposure does not increase risk but individuals should avoid further sun exposure)
- Family history skin cancer (may be prudent to avoid sun exposure)

Medications

- Anticonvulsants, corticosteroids, histamine H_2 receptor antagonists, theophylline, lipase inhibitors, anti-tuberculosis agents (may decrease vitamin D status)
- Thiazide diuretics, atorvastatin, lovastatin and simvastatin (may increase circulating vitamin D; thiazide diuretics taken with vitamin D supplements may cause hypercalcaemia)
- Sulphonamides, phenothiazines, tetracyclines, psoralens increase photosensitivity; may signal sun avoidance

Physical examination

- Idiopathic musculoskeletal pain (may be indicative of osteomalacic osteopathy)
- Muscle weakness (proximal limbs) (may be indicative of osteomalacic osteopathy)
- Undue pain on sternal or anterior tibial pressure (may be indicative of osteomalacic osteopathy)
- Lower limb deformities (knock knees, bowed legs) (may be indicative of osteomalacic osteopathy)
- Bowel function (steatorrhoea) (may alter vitamin D, calcium absorption)
- Skin pigmentation (or type)/hair colour (melanin absorbs UVB photons and prolongs exposure time required for vitamin D synthesis; pale complexion increases photosensitivity)
- Contraindications to sunlight (albinism, porphyrias, xeroderma pigmentosum)

Dietary intake

- Calcium

- Vitamin D and vitamin D-containing supplements (including multivitamin)
- Phosphate (nutritional supplements and soda/soft drink consumption may elevate PTH) (Zittermann 2003)
- Vitamin K
- Magnesium (commonly low in western diet; important for muscle and bone health; deficiency may mask elevated PTH)
- Other nutrients: vitamin C, omega-3/omega-6 fatty acids

Lifestyle and environmental

- Training regimen
- Training environment in relation to sunlight exposure
- Training latitude, altitude, climate
- Sunscreen use (SPF of 8 or 15 ↓ vitamin D synthesis capacity by 92–98%)
- Uniform or athletic clothing worn and SPF of clothing (tight weave, light colour and UVB protection inhibits UVB photons from reaching skin)
- Leisure sun exposure (frequency, duration and time of day)
- Tanning bed use (most emit UVB along with UVA)
- Beliefs concerning sunlight exposure

BMD = bone mineral density, SPF = sun protection factor, UVB/UVA = ultraviolet wavelengths, PTH = parathyroid hormone

Dietary assessment should primarily focus on estimating intake of calcium and vitamin D and possibly vitamin K and magnesium: the key nutrients important to bone health and muscle function. Suboptimal magnesium intake is common in the western diet, unless the athlete consumes ample nuts, seeds, legumes, whole grains and green leafy vegetables, and may influence bone and muscle function (Nielsen & Lukaski 2006) (see Table 9.2 for selected food sources).

Following a detailed assessment, recommendations for achieving/maintaining optimal vitamin D status can be individualised to each athlete's current 25(OH)D concentration, clinical history and symptoms, dietary and lifestyle habits. While most athletes can meet calcium requirements through the selection of a healthy diet, many may need sensible sun exposure, vitamin D supplements or a combination to maintain vitamin D status. Between 1500 and 2000 IU/d of vitamin D supplements is needed to maintain 25(OH)D concentrations in the sufficient range (Holick et al. 2011). Higher doses may be required in athletes who have excess adiposity or malabsorption syndromes, or who take medications affecting vitamin D metabolism or who are deficient (to increase stores). Known genetic variants in vitamin D-binding protein may help explain why some athletes do not respond as well to oral supplementation (Fu et al. 2009). Athletes with deficient status may benefit from short-term, high-dose loading regimens under supervision of a physician to more rapidly replenish stores. Examples of such protocols include 50 000 IU of vitamin D3 for at least 8 weeks (Holick et al. 2011) or 10 000 IU of D3/d for 8–10 weeks or until stores normalise. High-dose treatment with cod liver oil is not recommended (Cannell et al. 2008). Vitamin D3 may also be more effective than vitamin D2 (Armas et al. 2004; Heaney et al. 2011) and does not cause muscle damage as was recently shown following oral vitamin D2 supplementation (Nieman et al. 2014). Tanning beds that emit UVB radiation may also be appropriate for some athletes. These have been shown to resolve musculoskeletal pain and weakness associated with vitamin D deficiency (Koutkia et al. 2001) and help maintain adequate vitamin D stores in healthy individuals (Tangpricha et al. 2004) and in athletes (Halliday et al. 2011).

Summary

The skeleton represents a complex interaction between the hormonal environment and the stresses and strains placed upon it. Weight-bearing exercise or resistance training has been shown to be effective in slowing or preventing bone loss in both premenopausal and postmenopausal women. Adequate calcium intakes are effective in decreasing bone loss, at least in women, and should be increased after menopause or in an athlete with amenorrhoea. The prevalence of stress fractures is also higher among athletes who have menstrual disturbances compared with eumenorrhoeic women. The consequences of untreated menstrual disturbances on bone health are an increased risk of osteopaenia and potentially osteoporosis in later life.

Sports nutritionists and physicians should routinely assess for calcium intakes and vitamin D status and intake of other nutrients related to bone health (e.g. magnesium). Evidence confirms that maintaining adequate vitamin D status reduces the risk for stress fracture, acute infection and potentially the impairment of muscle function associated with vitamin D deficiency. More research is needed to determine whether insufficient vitamin D status increases risk for injury and whether vitamin D supplementation to correct deficiency and insufficiency can affect overall bone health, training and performance in athletes.

USEFUL WEBSITES AND RESOURCES

- **www.asbmr.org/education/bonecurriculum.aspx**
  The American Society for Bone and Mineral Research Bone Curriculum
- **https://www.anzbms.org.au**
  Australian and New Zealand Bone and Mineral Society
- **www.mja.com.au/open**
  Building healthy bones throughout life
- **www.mja.com.au/public/issues/190_06_160309/san10083_fm.html**
  Calcium and bone health: position statement for the Australian and New Zealand Bone and Mineral Society, Osteoporosis Australia and the Endocrine Society of Australia, 2009
- **www.cdc.gov/nutrition/everyone/basics/vitamins/calcium.html**
  Calcium and bone health (Centers for Diseases Control and Prevention (USA)
- **www.osteoporosis.org.au**
  Osteoporosis Australia
- National Institutes of Health Osteoporosis and Related Bone Diseases National Resource Center (USA)
- **www.niams.nih.gov/Health_Info/Bone**
  National Institute of Arthritis and Musculoskeletal and Skin Diseases
- **www.nof.org**
  National Osteoporosis Foundation (USA)
- **www.vitamindcouncil.org**
  Vitamin D Council (USA)

Practice tips

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- Optimal bone health in athletes relies on adequate calcium and vitamin D throughout life.
- A key role of the sports dietitian is to raise awareness and educate coaches and athletes about the possible long-term consequences of the female athlete triad (or RED-S) on bone health. Assessing menstrual history is important in an initial consultation with an athlete or when screening groups. The absence of menses for more than 6 months should be referred to a sports physician for assessment.
- The sports dietitian needs to be alert to the presence of eating disorders, which can cause serious and irreversible bone loss. Other dietary practices known to increase urinary calcium losses (such as excessive intakes of sodium, protein, caffeine and alcohol) should also be assessed. These are of greater importance when dietary calcium intake is low. In this situation, increasing the calcium intake is the priority because at higher calcium intakes (at least >800 mg/d), excessive sodium, protein and caffeine are of less importance. Cigarette smoking also has deleterious effects on bone health.
- There is currently no DRI/Nutrient Reference Value (NRV) for calcium for amenorrhoeic athletes. However, 1500 mg calcium/d has been proposed, which is consistent with the US National Institutes of Health consensus statement for postmenopausal women not taking oestrogen. Amenorrhoeic athletes would fit into the same category. During adolescence, it is essential that athletes consume adequate calcium as this is the time when peak bone mass occurs.
- What can be done when counselling female and male athletes who are not consuming enough calcium? Expounding the long-term risk of osteoporosis usually has little impact. However, if an individual has seen the debilitating effects of osteoporosis in their family or friends, they are usually more receptive to preventive strategies. Start by suggesting an increase in calcium intake by dietary means, but if this strategy is not possible or feasible, then a calcium supplement may be required. Referral to a sports physician for a bone density scan can be considered. An athlete with a low calcium intake together with a low bone density may be more receptive to change, once they have seen the results of their bone density scan.
- Female and male athletes consuming low energy intakes are at high risk of low calcium intakes. There are also many misconceptions in the general population about dairy foods, which are the major source of calcium. Any misinformation about calcium and dairy foods should be addressed in a clinical situation as well as other barriers and facilitators that might affect the choice of calcium-rich foods. Providing practical options for snacks and meals that increase calcium density (e.g. using calcium-fortified foods) is useful.
- Where it is difficult to meet calcium requirements by diet alone, a calcium supplement is warranted. Although bioavailability of calcium from supplements is lower than from dairy sources, taking supplements at bedtime or between meals maximises absorption and prevents the interference of inhibitory factors found naturally occurring in foods (e.g. phytic acid in cereal grains). Absorption of calcium supplements is most effective in doses of 500 mg or less (Heaney et al. 1975, 1988). Commonly prescribed calcium supplement brands are Citracal™ (calcium citrate), Sandocal™ (calcium lactate-gluconate) and Caltrate™ (calcium carbonate).
- Maintaining vitamin D status is also important for bone health. For individuals living in Australia and New Zealand, sunlight exposure is the major source of vitamin D. Suboptimal levels of vitamin D occur in people who are institutionalised or housebound, who actively avoid sunlight exposure, and who have excess adiposity. For athletes who train indoors, such as young gymnasts, it is important that they follow the recommendations for safe sun exposure to maintain their vitamin D status.
- Pharmacological intervention and biochemical tests of hormone status may be necessary for female athletes at risk of osteopaenia. Situations where medical referral is indicated include:
  - being amenorrhoeic for longer than 6 months
  - a history of anorexia nervosa or other diagnosed eating disorder
  - occurrence of stress fractures
  - being postmenopausal
  - a strong family history of osteoporosis.
- Male athletes with chronic low energy availability are at risk of low BMD. Chronic energy restriction during

puberty is a high-risk behaviour for low BMD and not reaching peak bone mass in late adolescence.

- Routine BMD screening of athletes is not recommended, as there is a small radiation dose using DXA. In athletes with low energy availability, disordered eating behaviours or disordered eating or amenorrhoea for>6 months, BMD should be measured using DXA (Mountjoy et al. 2014).
- Regular weight-bearing activity or weight training has a positive effect on bone density. Weight-bearing activities such as jogging, tennis, aerobics and walking have the greatest effects compared to cycling and swimming. Although these non-weight-bearing activities are excellent for aerobic fitness, they are unlikely to have much effect on bone mass.

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CHAPTER TEN

Prevention, detection and treatment of iron depletion and deficiency in athletes

Vicki Deakin and Peter Peeling

10.1 Introduction

Iron is an essential trace mineral in the body that is associated with numerous processes that may influence athletic performance (e.g. oxygen transport, energy production). Despite its importance, athletes (and in particular females and adolescents) are at risk of depleting their iron stores. If untreated, iron depletion may develop into iron deficiency anaemia (IDA), which severely affects an individual's training capacity. Recent findings suggest that maintaining an optimum iron status is far more important for athletes than has previously been realised. Even a mild shortfall in tissue iron status appears to reduce maximum oxygen uptake, aerobic efficiency and endurance capacity. Any athlete, particularly during adolescence, involved in regular high-intensity physical activity has a higher iron turnover, which can quickly result in depleted iron stores.

The body has no inherent mechanism to generate its own iron supply, and as a result, the daily iron requirements of an athlete must be found from food sources. Although iron is widely distributed in many food types, inappropriate food combinations can compromise absorption. For instance, high-carbohydrate (CHO) diets recommended for athletes undertaking high levels of physical activity may be high in compounds that inhibit iron absorption. To this end, dietary strategies to help prevent iron depletion should be implemented early in the training program in high-risk individuals. Ultimately, early detection of depleted iron stores and dietary intervention are warranted, since the recovery from depleted or exhausted iron stores is slow and may take some months to be replenished.

10.2 Stages of iron depletion

Several common haematological markers are used to categorise iron status in the general population. These cut-off values are also applied to an athletic population because there are no standardised reference values for athletes. Although iron depletion is a continuous process, traditionally three categories or stages of iron deficiency have been identified, based on haematological markers as shown in Table 10.1.

TABLE 10.1 Routine biomarkers used in clinical practice to evaluate iron status

Stage of iron deficiency	Iron measures in the blood	Cut-off value for adults
Stage 1		
Iron depletion: Iron stores in the bone marrow, liver and spleen are depleted	Serum ferritin	<35 µg/L
	Haemoglobin	>115 g/L
	Transferrin saturation	>16%
Stage 2		
Iron deficient erythropoiesis: Erythropoiesis diminishes as the iron supply to the erythroid marrow is reduced	Serum ferritin	<20 µg/L
	Haemoglobin	>115 g/L
	Transferrin saturation	<16%
Stage 3		
Iron deficient anaemia: Haemoglobin production falls, resulting in anaemia	Serum ferritin	<12 µg/L
	Haemoglobin	<115 g/L
	Transferrin saturation	<16%

Source: Peeling 2007

As iron stores deplete and serum ferritin—an indicator of iron stores—progressively decreases, the other iron-containing compounds (i.e. haemoglobin, transferrin) eventually become affected (as implied by the changing cut-off for each stage). In early stage iron depletion as iron stores decrease, transferrin saturation initially increases towards the upper end of the reference range and then decreases when erythropoiesis is compromised. The terminology and cut-off values for each stage of iron depletion vary in the literature, especially in studies on athletes. Terms including *marginal iron deficiency*, *early iron depletion*, *iron deficiency without anaemia*, *early functional iron deficiency* and *iron deficiency erythropoiesis* are often used interchangeably. Iron deficiency without anaemia is the most frequently used term that appears to encompass depleted iron stores and iron-deficient erythropoiesis. In the absence of standardised reference values for athletes, researchers have arbitrarily chosen varied cut-offs for serum ferritin (SF), ranging from <16 µg/L (Zhu & Haas 1998; Hinton et al. 2000) through to <35 µg/L (Peeling et al. 2007; Tan et al. 2012) to designate iron deficiency without anaemia. Clearly these different cut-offs confound interpretation of the true effects of different stages of iron status on performance measures and other health effects.

SF reflects the magnitude of the storage iron compartment, not the amount of iron available in the tissue or its functional pool. The emergence of a novel iron biomarker that measures tissue iron status—serum transferrin receptor (sTfR)—has allowed researchers to distinguish between depleted iron stores and iron-deficiency erythropoiesis and to investigate how inadequate iron in the tissue, compared to low SF, might affect the performance capacity of athletes. Calculating the ratio of sTfR to low SF is a way of combining sTfR and SF and is considered a more reliable determinant of iron status in individuals than either indicator used alone, especially in borderline cases (see section 10.8.3). These indicators have not yet been characterised in athletes, so there are no standardised cut-offs that represent the point at which low levels of tissue iron might limit performance capacity.

Interpreting iron biomarkers and the effect of iron depletion/deficiency on performance capacity in athletes is described in detail in sections 10.8 and 10.5 respectively.

10.2.1 Dilutional pseudo-anaemia or ‘sports anaemia’

An additional category unrelated to iron status but common in athletes is known as *dilutional pseudo-anaemia*, which has been used to describe a unique physiological response to training in an athletic population. The term *dilutional pseudo-anaemia* is the preferred terminology for the transient condition often referred to as *sports anaemia*, *athlete’s anaemia*, *runner’s anaemia*, *post-exercise anaemia* or *swimmer’s anaemia*. These alternative terms are misleading because dilutional pseudo-anaemia is not genuine iron deficiency but the result of a dilution effect on iron biomarkers. Periods of intense training can result in an increase in plasma volume—a ‘normal’ adaptive response to training—and a corresponding reduction in most haematological measures of iron status (Schumacher et al. 2002; Maczewska et al. 2004; Merkel et al. 2005). Training increases plasma volume at a greater rate than the increase in the red cell mass (Chatard et al. 1999). Previously, up to a 20% increase in plasma volume has been reported in the 48 hours after a race in marathon runners, and can persist for as long as a week (Brotherhood et al. 1975; Schmidt et al. 1989). Hence, the red cell mass appears diluted. However, in clinical practice, differentiating between dilutional pseudo-anaemia and true iron depletion using the usual iron biomarkers is often difficult.

10.3 How common is iron deficiency/iron depletion in athletes?

The three groups of athletes at the highest risk of iron depletion and deficiency are female athletes, distance runners and vegetarians (and those who eat little red meat) (Fogelholm 1995). However, athletes of both genders and across a range of sport types (e.g. endurance, team sports) have been shown to present with compromised iron stores (Milic et al. 2010; Tan et al. 2012). Adolescent athletes during a rapid growth spurt are at high risk if they undertake sports with endurance training, because of the additive effect of high iron requirements needed to support the increased red cell mass accompanying growth. However, the true prevalence of iron depletion or deficiency is unknown because reported prevalence data is usually based on SF measures alone, which are confounded by the lack of a universal cut-off for ferritin and the inconsistent terminology used for iron depletion. Furthermore, low SF levels in isolation do not necessarily confirm a diagnosis of iron deficiency or depletion.

10.3.1 Is the prevalence of low iron stores in athletes different from the general population?

Comparison of SF between athletes and control groups shows mixed results. Some authors have reported lower SF levels in athletes (Pate et al. 1993; Constantini et al. 2000; Dubnov et al. 2004), whereas others have reported higher values than untrained controls, particularly in males (Balaban 1992; Fogelholm et al. 1992; Schumacher et al. 2002).

In an analytical study of pooled iron status measures of athletes published from 1980 to 1994, lower SF levels were more frequently found in female athletes and in endurance athletes of both sexes (e.g. runners, rowers, swimmers) compared to untrained controls (Fogelholm 1995). However, in a more recent study of 126 female endurance athletes, the average SF values of athletes were higher than those of the controls (Malczewska et al. 2000). In this study, average SF levels were higher in females training an average of 2.5 h/d than in untrained controls. Nevertheless, 26% of all female athletes in this study were still considered iron deficient (based on SF<20 μg/L), compared to a 50% incidence of iron deficiency in the untrained controls.

10.3.2 Prevalence of depleted iron stores and iron deficiency anaemia in athletes

In athletes, the prevalence of depleted iron stores based on SF levels below reference cut-offs varies according to type of sport, age of the athlete and sex. The highest prevalence occurs in female and adolescent athletes, irrespective of the type of sport and intensity of training (Malczewska et al. 2000; Constantini et al. 2000; Dubnov & Constantini 2004; Merkel et al. 2005; Fallon 2004, 2007; Fallon & Gerrard 2007; Sandström et al. 2012; Koehler et al. 2012).

Male athletes, except for runners, usually have a lower prevalence than females because of their lower requirements and higher capacity to store iron (Fogelholm 1995; Schumacher et al. 2002; Koehler et al. 2012). However, in two studies, 45% of male gymnasts (Constantini et al. 2000) and 15% of male basketball players (Dubnov & Constantini 2004) had SF values below 20 μg/L.

The prevalence of IDA in athletes is usually quite low (<3%), and is similar between adult athletes and untrained controls (Weight et al. 1992; Fogelholm 1995; Sandström et al. 2012), with some exceptions (Dubnov & Constantini 2004; Merkel et al. 2005; Israeli et al. 2008).

10.4 Why is iron important to athletes?

Iron is the functional component of haemoglobin and myoglobin, and is therefore an essential nutrient for optimal oxygen delivery to the body's tissues and for facilitating oxygen diffusion to the cellular site of energy production, the mitochondria. In fact, the largest component of iron in the body is found in circulating blood's haemoglobin (60–70% of total iron) and in the muscle tissue myoglobin (10% of total iron) (Nielsen & Natchtigall 1998). As a result, it is also evident that iron is a key component in the production of new red blood cells (erythropoiesis): a process by which the total haemoglobin mass (tHb_{mass}) of the body increases. To highlight the importance of iron in this process, Garvican et al. (2011) treated an iron-deficient anaemic athlete with an intramuscular iron injection (Ferrum H, Fe polymaltose, 100 mg elemental Fe), followed by oral iron supplementation (FerroGradumet, 325 mg ferrous sulfate, 100 mg elemental Fe) for 15 weeks. The serum ferritin levels of the athlete increased from 9.9 to 23.4 ng/mL by 2 weeks post-injection, reaching a maximum level of 35.7 ng/mL after 7 weeks. In lieu of this, the authors also reported that the athlete's tHb_{mass} also increased by 49% within 2 weeks of the iron injection, increasing to 77% above baseline by the end of the 15-week period. Such results highlight the significance of iron to the erythropoietic pathway.

In addition to oxygen transport, iron also plays a key role in energy production through the electron transport chain, since mitochondrial enzymes and cytochromes are haem-containing proteins that promote oxidative phosphorylation. In combination with oxygen transport, it is clear that iron has a strong functional role in maintaining the key processes related to aerobic metabolism, which are highly associated with the endurance capacity of an athlete. Importantly however, it should also be considered that iron is involved in neural (Beard & Connor 2003), immune (Beard 2001) and cognitive function (Zimmerman & Hurrell 2007)—processes that may impact an athlete's ability to train or compete optimally, thereby further highlighting the importance of this trace mineral to the athlete.

10.5 Effects of iron status on performance and other health outcomes

The effects of iron depletion on performance capacity have been the focus of much research. Other effects of iron depletion, independent of its role in oxidative metabolism—such as fatigue, altered resistance to infection, and muscle and hormone dysfunction—could also limit training capacity. These effects further compound the negative influence of exhausted iron stores on the ability to undertake physical activity.

10.5.1 Effects of iron deficiency anaemia on performance

The debilitating effects of IDA on aerobic and endurance capacity, work capacity and energy efficiency in animal and human studies on both untrained and trained subjects are well presented by Haas and Brownlie (2001). These authors summarised that IDA impairs erythropoiesis, negatively impacting oxygen transport, resulting in a reduction to maximal oxygen consumption ($VO_{2\ max}$) and task duration prior to the onset of fatigue. It is also evident that even small decreases in haemoglobin of 1–2 g/100 mL can decrease performance by around 20% (Gardner et al. 1977). Additionally, disturbances in brain metabolism, muscle metabolism, immunity and temperature control can also occur in IDA, depending on severity of the depletion (Bothwell et al. 1979).

10.5.2 Effects of iron deficiency without anaemia on performance

Although it is well established that IDA impairs training capacity, the effects on performance in the early stages of iron depletion are less clear. Theoretically, with depleted iron stores, haemoglobin levels are not compromised, so oxygen delivery to the tissue should not be affected. However, given the role of iron at the cellular level, inadequate tissue iron may reduce oxidative capacity and also impair endurance capacity.

In order to assess this, a number of investigations have corrected depleted iron stores in athletes through supplementation, while collecting both pre- and post-supplementation test data for comparison of differences in athletic performance. Such methodology allows for comparison between iron-deficient and iron-replete populations, who act as their own controls. Using such protocols, the literature to date supporting a decreased aerobic or endurance capacity in athletes with depleted iron stores (low SF values) is equivocal, with a host of investigations showing either negative or no changes to athletic performance.

For instance, no significant improvement in aerobic capacity (using $VO_{2\ max}$ tests) in response to iron supplementation of athletes compared to controls was found in several well-designed randomised, double-blind, placebo-controlled trials (Rowland et al. 1988; Newhouse et al. 1989; Klingshirn et al. 1992; Zhu & Haas 1998; Peeling et al. 2007). One such study showed no significant improvement in endurance capacity (time to exhaustion) or energy efficiency in eight trained iron-depleted female athletes (sTfR = 5.9 mg/L, SF = 19 µg/L) after iron injection (5 × 2 mL of 100 mg elemental iron over 10 d) using a $VO_{2\ max}$ test and sub-max (70% $VO_{2\ max}$) test, despite substantial increases in iron stores in the injected group (Peeling et al. 2007). The authors suggested, however, that the 10-minute sub-max test at 70% $VO_{2\ max}$ might not have been specific enough to detect performance economy changes. Interestingly, in this study no improvements in mean sTfR in the iron injection group was seen, which implies that the functional role of iron was not yet affected, possibly explaining the lack of improvement in aerobic capacity seen here.

Conversely, Zhu and Haas (1998) demonstrated an increased energy efficiency during a 15-km running time-trial in active women with depleted iron stores after 8 weeks of oral iron supplementation (SF went from <15 to >35 µg/L) compared to controls. This increased energy efficiency was evident through a reduced utilisation of peak VO_2 by ~5%, suggesting that at the same exercise intensity, the iron-supplemented group were able to complete the 15 km time-trial at a lower level of physical exertion. Using a similar experimental design, Hinton et al.

(2000) found a significant improvement in endurance capacity (improved 15 km cycle time-trial by ~6.5%) and energy efficiency in untrained iron-deplete women, after 6 weeks of oral iron supplementation and 4 weeks of aerobic training (here, SF went from <11 to >14 μg/L and serum iron from <13 to >19 μmol/L). Subsequently, Hinton and Sinclair (2007) tested the effects of only 30 mg of elemental iron supplements for 6 weeks in iron-depleted recreational athletes (sTfR = 6.58 mg/L, SF = 11.67 μg/L), showing a significant improvement in ferritin, sTfR and aerobic function during a 60-minute sub-maximal exercise test at 60% $VO_{2\ max}$ compared to controls.

Finally, McClung et al. (2009) more recently used a different approach to assess the influence of depleted iron stores on physical performance, by tracking the iron status of 92 unsupplemented female military recruits over the course of a 9-week basic combat training course. These authors showed that iron status was progressively compromised over the duration of the training period (SF went from 26.9 to 21.5 μg/L; sTfR went from 1.18 to 1.55 mg/L), showing a significant negative correlation with 2-mile running performance. These authors suggested that such an outcome highlights the fact that a degraded iron status affects the ability to perform aerobically demanding tasks, adding to the proposition that iron depletion may negate the aerobic adaptation to a prolonged period of training.

10.5.3 Limitations in research design of performance studies

Failure of earlier studies to demonstrate a performance disadvantage of early iron depletion may have been attributed to several factors: (1) using varying cut-offs for SF; (2) small sample size and small effect size; (3) the influence of both psychological and training effects; (4) lack of a biomarker for measuring tissue iron depletion; and (5) low specificity of performance tests to measure endurance capacity, particularly in untrained subjects.

Furthermore, in these studies, endurance capacity was generally tested using exercise intensities >80% $VO_{2\ max}$, a level well above the anaerobic threshold of most subjects (Hinton et al. 2000). However, at this level of exertion, iron is not involved in energy production, so low tissue iron would not limit endurance capacity (Haas & Brownlie 2001). Regardless, in the study by Peeling and colleagues (2007), where a lower exercise intensity test was used (70% $VO_{2\ max}$), no improvement in endurance capacity was reported.

10.5.4 Summary of performance effects of iron deficiency

In summary, IDA impairs aerobic capacity, which can be corrected by iron supplementation that has a positive impact on haemoglobin production. Iron depletion without anaemia does not affect oxygen-carrying capacity ($VO_{2\ max}$) or limit the functional role of iron in the tissues to carry out oxidative metabolism, so aerobic capacity is not impaired. There is now limited although convincing evidence that tissue iron deficiency (which occurs at around Stage 2 iron deficiency erythropoiesis according to the cut-offs in Table 10.1)—represented by very low iron stores together with sTfR at levels greater than 8.0 mg/L—affects the functional role of iron in energy metabolism, can impair an athlete's capacity to undertake aerobic training, and can decrease work rate and energy efficiency. More studies are needed, however, to determine the level at which tissue iron deficiency might affect performance in highly trained athletes before definitive recommendations can be made.

10.5.5 Does iron depletion affect fatigue?

Symptoms of fatigue and lethargy (common complaints in IDA) are not usually reported in athletes without anaemia. In one study of 41 competitive athletes with persistent fatigue, iron depletion accounted for only 3% of unexplained fatigue (Reid et al. 2004). In this study, the most common conditions linked to persistent fatigue were immune deficiency (28%) and unresolved viral infections (27%). Overtraining is also implicated (Du Toit & Locke 2007). In another study of 50 elite athletes who presented with short-term fatigue and tiredness of around 1–3 weeks, the main contributing factors were training related (28%), or a combination of training effects and infection (30%) (Fallon 2007). In this study, iron depletion was a minor contributor, with only 1% of athletes presenting with low ferritin stores.

Nevertheless, some individuals with low SF values often complain of fatigue in the absence of any obvious clinical or physiological cause or anaemia. Such symptoms in an athlete can affect concentration and performance, as well as increasing the risk of injury. In two independent randomised, double-blind, placebo-controlled trials of untrained women with low SF values (mean SF ~30 μg/L), a significant improvement in unexplained fatigue was reported in the iron-supplemented group compared to controls (Verdon et al. 2003; Vaucher et al. 2012). The improvement was greatest in women with baseline SF ≤50 μg/L after iron supplementation in both studies. Similar improvements in fatigue were reported in female blood donors with iron depletion without anaemia after 4 weeks of iron supplements (8 mg elemental iron/d), compared with the placebo group (Waldvogel et al. 2012). One explanation for unexplained fatigue in relation to iron depletion is a depressed activity of iron-dependent enzymes, which can affect the metabolism of neurotransmitters (Beard & Connor 2003). In summary, the reasons for short-term and persistent fatigue in athletes may be related to iron depletion or deficiency, although the primary contributors are generally recognised as training or infection related.

10.5.6 Does iron depletion affect immune function?

Low iron status and inadequate intakes of other micronutrients including zinc and vitamins A, E, B6 and B12 are associated with immune dysfunction and possible exercise-induced immunosuppression (see Chapter 22). However, deficiencies of these micronutrients are rare in athletes and the extent by which low iron status, in isolation, may compromise the immune system is unknown.

10.6 Physiology of dietary iron absorption

Iron absorption and bioavailability (the amount of dietary iron available for metabolic functions) is influenced by the iron status of the individual and the type of iron consumed. Iron absorption and metabolism is regulated by the peptide hormone known as hepcidin, via its influence on the body's only known iron exporter, ferroportin (Ganz 2011). Hepcidin acts to internalise and degrade the ferroportin export channels in the gut and on the cell surface of macrophages, thereby reducing dietary iron absorption, and inhibiting the recycling of iron from senescent red blood cells in the macrophage (Nemeth et al. 2004a). Hepcidin levels within the body are tightly controlled in a homeostatic fashion. Elevations in the hormone activity result in response to inflammatory stimuli (namely the inflammatory cytokine interleukin-6: IL-6) or elevated iron levels; whereas attenuated hepcidin levels are usually invoked by stimuli such as hypoxia (Hintze & McClung 2011) or suppressed iron stores (Nemeth et al. 2004b; Kemna et al. 2005).

When iron stores are saturated, iron absorption is around 5–15% (Hallberg et al. 2000). In people with depleted iron stores or with high physiological requirements (e.g. to support growth, pregnancy or lactation), iron absorption increases to around 14–18% for mixed or meat based-diets (Hallberg & Hulthen 1996) but is still lower for vegetarian diets at 5–12% (Hurrell & Egli 2010). Iron status is more important than bioavailability in determining the amount of non-haem iron absorbed (Hunt et al. 2003; Hurrell & Egli 2010).

There are two types of iron in food: non-haem (inorganic) iron and haem iron, which are both absorbed in the small intestine but by different mechanisms. Non-haem iron is derived from plant foods, accounting for approximately 80% of the iron consumed from a meat-based diet (Craig 1994); however, its absorption process is very inefficient. The remaining iron intake is comprised of the more efficiently absorbed haem iron, derived primarily from the haemoglobin and myoglobin of meat sources. Meat contains 30–70% of its total iron as haem iron and the rest as non-haem iron (MacPhail et al. 1985; Rangan et al. 1997; Lombardi-Boccia et al. 2002). In Australian meat, 60–70% of the total iron in cooked lean rump steak, lamb, pork and chicken is haem iron, whereas beef sausage, liver and tuna contain around 20–35% haem iron and the rest as non-haem iron (Rangan et al. 1997). The haem iron content of lamb and beef is particularly high compared to that in chicken and fish, which explains why red meat is an excellent iron source. Although the contribution of haem iron in a meat-based diet is around 10–15% of total iron intake, it contributes around one-third of the total iron absorbed (Hallberg 1981), with the rest obtained from non-haem sources.

10.6.1 Dietary factors affecting dietary iron absorption

A large proportion of dietary iron is unavailable for absorption. The presence of vitamin C, vitamin A and muscle meat in a meal enhance iron absorption, whereas phytates (found in legumes and grains), calcium, and polyphenols and tannins (in tea and coffee) inhibit iron absorption (see Table 10.2). Because of the presence of these inhibitors and enhancers in food, iron bioavailability from the total diet can vary 10-fold from different meals with similar iron content (Hallberg et al. 2000).

TABLE 10.2 Components in food that affect bioavailability of iron, mainly from non-haem food sources

Iron enhancers	Iron inhibitors
Vitamin C-rich foods (ascorbic acid) (e.g. salad, lightly cooked green vegetables, some fruits, citrus fruit juices and vitamin C-fortified fruit juices)	Phytates (inositol hexaphosphate) (e.g. in cereal grains, wheat bran, legumes, nuts, peanut butter, seeds, bran, soy products, soy protein and spinach)
Some fermented foods with a low pH (e.g. sauerkraut, miso and some types of soy sauce)	Polyphenols (e.g. in strong tea and coffee, herb tea, cocoa, red wine and some cereals and legumes)
Peptides from partially digested muscle tissue, which enhance both haem and non-haem iron absorption (this is often called the ‘meat factor’ and is found in beef, lamb, chicken, pork and liver—and fish)	Calcium, which weakly inhibits both haem and non-haem iron absorption
Alcohol and some organic acids (e.g. in very low pH foods containing citric acid or tartaric acid, such as citrus fruit)	Peptides from partially digested plant and animal proteins (e.g. soy protein isolates and soy products, milk protein (casein and whey), egg protein)
Vitamin A and beta-carotene	

Sources: Hallberg 1981; Hallberg, Rossander-Hulthen et al. 1993; Cook et al. 1991; Hallberg & Rossander 1982a; Baynes et al. 1990; Gillooly et al. 1983; Lynch et al. 1994; Brune et al. 1992; Hurrell 1997; Layrisse et al. 1984; Gillooly et al. 1984; Hallberg & Rossander 1982b; Garcia-Casal et al. 1998, 2003

Haem iron is efficiently absorbed from single foods (15–35%), although absorption is mildly affected by the inhibitors and enhancers in foods (Hallberg et al. 1997; Hunt 2001). Iron-rich plant sources of non-haem iron, including cereals, nuts, seeds, legumes and spinach, contain substantial quantities of inhibiting components that reduce the solubility of iron and limit its absorption. This explains why the 2–15% non-haem iron absorption from single meals is much lower than haem iron absorption (Hallberg et al. 1997).

10.6.2 Inhibitors of dietary iron absorption

Phytates (salts of inositol hexaphosphate)—found in cereal grains, legumes, seeds, nuts, vegetables and fruit—are the main inhibitors of non-haem iron absorption (~50–80%) (Hallberg et al. 1989). Phytates concentrate in the bran and germ of cereal grains and legumes, and in soy protein isolates, which are added to some commercial foods (Hurrell 1997).

Although fibre itself does not inhibit iron absorption (Brune et al. 1992), foods rich in fibre are rich in phytates. High-bran cereals have more than 3000 mg phytates/100 g cereal compared to cornflakes, which have about 70 mg phytates/100 g cereal (Harland & Oberleas 1987). Vegetables and fruits are low phytate sources. Phytates in cereal grains are partially degraded in food processing and cooking, thereby increasing the bioavailability of the haem iron (Hurrell & Egli 2010).

Polyphenols found in plants, including phenolic acid and flavonoids (e.g. tannin), also inhibit non-haem iron absorption. Polyphenols are highest in black tea compared with other types of tea, such as herbal or green teas, and are also found in coffee, cocoa, red wine and many vegetables, including spinach and some cereals and legumes. In a western-style breakfast (toast and tea), the absorption of non-haem iron from bread was reduced by 60% because of concurrent consumption with black tea (Rossander et al. 1979). Coffee reduced absorption of non-haem iron by 35% from a hamburger meal, while tea reduced absorption by 62% (Hallberg & Rossander 1982a). One glass of red wine (high in polyphenols) reduced iron absorption from a bread-based meal by 75% (Cook et al. 1995). In one study of adult female runners, coffee/tea consumption was significantly associated with low iron stores (Pate et al. 1993).

Calcium inhibits both haem and non-haem iron absorption but the effect is weaker than for phytates and polyphenols (Hurrell & Egli 2010). This inhibitory effect does not justify avoidance of milk or dairy foods with iron-rich foods including breakfast cereal with milk (Reddy & Cook 1997; Grinder-Pedersen et al. 2004). Inadequate intakes of calcium in female and adolescent athletes are already a concern and removal of milk with breakfast cereal would further decrease calcium intakes.

Peptides derived from soy proteins, milk proteins (casein and whey) and egg protein (albumin) can inhibit non-haem iron absorption (Hurrell & Egli 2010) although the form of iron in soy is more bioavailable than the forms in other foods (Zhou et al. 2011).

10.6.3 Enhancers of dietary iron absorption

Dietary enhancers of iron absorption include vitamin C (ascorbic) acid, vitamin A, alcohol, some acidic foods and possibly peptides found in meat, seafood and poultry. Vitamin A, vitamin C and other organic acids and alcohol facilitate the conversion of ferric iron to the more soluble ferrous iron (the form of iron in non-haem food sources), which is the best form for absorption (Hurrell 1997).

Ascorbic acid (or vitamin C) is a potent iron enhancer. One glass of orange juice containing 100 mg of ascorbic acid consumed with a hamburger meal increased absorption of non-haem iron by 85% (Hallberg & Rossander 1982a). The effects of ascorbic acid are dose related. Amounts of up to 500 mg ascorbic acid supplement taken with food can negate the inhibitory effects of phytates, polyphenols, calcium and proteins in milk and milk products but levels higher than this have no additional effect (Hallberg et al. 1986, 1989). Based on this research, Hallberg et al. (1989) suggested that foods containing about 50 mg of ascorbic acid should be included with each main meal to enhance iron absorption and reduce the effect of inhibitors present (see Practice tips).

Peptides from partially digested muscle tissue (meat, seafood and poultry) also exert a weak promoting effect on the absorption of both haem and non-haem iron (Reddy et al. 2006). The haem iron in meat is efficiently absorbed, which explains why meat eaters have higher SF levels than vegetarians. Even small amounts of 'meat' (around 50 g) can increases non-haem iron absorption from a high phytate (200 mg), low-ascorbic acid meal, significantly reducing the inhibitory action of phytate (Baech et al. 2003). Other enhancers of non-haem iron absorption include alcohol and some organic acids (Gillooly et al. 1983), some fermented foods (Lynch et al. 1994) and foods rich in vitamin A and carotene (Garcia-Casal et al. 1998, 2003), although the enhancing effect on non-haem iron absorption is much weaker than muscle tissue and vitamin C.

10.6.4 Estimating iron bioavailability in meals or diets

Several algorithms have been developed and validated to calculate the bioavailability of iron and allow estimations of iron absorption from single food or whole diets, accounting for the effects of iron promoters and inhibitors (Hallberg & Hulthen 2000; Reddy et al. 2000; Armah et al. 2013). These can be used to predict iron absorption from various diets; to test the efficacy of iron intervention studies; and to evaluate the iron value/bioavailability of meals/diets when catering for any population, including athletes. Investigation of the bioavailability of iron from different diets designed for athletes or dietary intervention studies of athletes have not been conducted.

10.7 Causes of iron deficiency in athletes

In athletes, the higher iron requirements induced by adaptation to athletic training (i.e. erythropoiesis), haemolysis (red blood cell destruction), gastrointestinal bleeding, sweating and haematuria are all suggested as contributors for athletic-induced iron deficiency (Peeling et al. 2008). Additionally, inadequate iron intake and low iron bioavailability are further potential causes. More recently, the inflammatory response to exercise has been associated with an elevation in the iron regulatory hormone hepcidin, which has been proposed as a potential confound to post-exercise iron absorption from meals consumed within a 3–6 h window post-exercise (Roecker et al. 2005; Peeling et al. 2009a). Table 10.3 summarises physiological, medical and dietary factors linked with iron depletion that are specific to athletes.

TABLE 10.3 Physiological, medical and dietary risk factors associated with iron deficiency in athletes

Risk factors	Link with iron deficiency
Athletic/strenuous training	Increases iron requirements by stimulating an increase in vascularity, red cell mass and haemoglobin, thereby decreasing iron stores
	Potentially causes high iron loss from sweating, haematuria, GI bleeding, haemolysis, and/or elevated post-exercise hepcidin levels
Growth and pregnancy	Increases iron requirements by stimulating an increase in vascularity, red cell mass and haemoglobin
Infection, parasites	Increases iron requirements
Chronic inflammatory diseases (e.g. inflammatory bowel disease and other non-infective inflammatory diseases)	Increases hepcidin, which inhibits iron absorption, resulting in depleted iron stores
Hereditary defects	Thalassaemia, sickle cell anaemia as well as other nutritional deficiency, including folate and B12 deficiency
Other medical/physiological reasons for blood loss	Menstruation, ulcers, misuse of anti-inflammatory medications such as non-steroidal anti-inflammatories (NSAIDs) and aspirin, chronic use of antacids
Inadequate dietary iron intake/low dietary iron bioavailability	Poor food choices, unbalanced diets, low energy diets, high-CHO diets, avoidance of red meat/vegetarian or vegan diets, fad diets, low-haem iron intakes, avoidance of iron-fortified commercial foods

10.7.1 Physiological causes

Athletic training and strenuous exercise

Athletic training increases an individual's iron requirements due to an increased iron turnover in the tissue, an increase in iron loss from a number of exercise-associated mechanisms, and likely due to an increase in circulating hepcidin levels (subsequent to exercise-induced inflammation) that may reduce the absorption and recycling of iron. However, in the absence of a dietary reason, no single physiological factor can explain the magnitude of the decline in iron status commonly reported in athletes over extended periods of athletic training.

Previously, in a large study of 747 trained athletes divided into three groups (power, endurance and mixed), the lowest iron stores were confirmed in the endurance athletes of both sexes, especially in runners (Schumacher et al. 2002). However, attention has also been given to the prevalence of iron deficiency in team sport athletes, with Landahl et al. (2005) reporting 57% of their 28 elite female football players were iron deficient, and 29% considered as anaemic. Such an outcome was also supported by Milic et al. (2010), who showed that 55% of the female team sport athletes screened in their investigation had SF levels less than the commonly accepted cut-off value of 35 μg/L, with 23% of this population showing SF levels <22 μg/L. During their investigation, Milic and colleagues screened 359 athletes across a range of sporting disciplines (including sprint-type, endurance and team sport athletes), with the highest incidence of iron deficiency seen in the team sport cohort. As a result, it would appear that a large range of athlete types are susceptible to athletic-induced iron deficiency; it may be the intensity and duration of a training program that can lead to compromised iron stores, as shown by significant decreases in iron status reported after 12 weeks of resistance and weight training in young males and females (Deruisseau et al. 2004), and in trained cyclists after 6 weeks of high-intensity interval training (Wilkinson et al. 2002).

The higher iron requirements in athletes are a consequence of training adaptation associated with an increase in red cell mass, haemoglobin and metabolic iron use, all of which are dependent on dietary iron. Additionally, athletes involved in strenuous exercise may lose iron through a host of mechanisms such as sweating (Waller & Haymes 1996; Deruisseau et al. 2002), gastrointestinal bleeding (Lampe et al. 1987; Natchtigall et al. 1996), or from the haemolytic stress of the activity (Miller et al. 1988; Telford et al. 2003; Tan et al. 2012. Haemolysis has been related to iron loss due to the destruction of red blood cells as a result of the impact imparted through the body upon foot-strike during running-based activity, or from the circulatory stress of muscular contraction during exercise. A host of research has shown a haemolytic effect during running-based activity, with greater levels of haemolysis associated to greater levels of impact—for example, running versus cycling (Telford et al. 2003); uphill versus downhill running (Miller et al. 1988); hard surface versus soft surface running (Peeling et al. 2009b). Regardless, haemolysis is characterised by a rupturing of the red blood cell (RBC), which results in the release of its haemoglobin and associated iron into the surrounding plasma. Left unchallenged, oxidative tissue damage would occur (Reeder & Wilson 2005), compromising the RBCs' function to transport and supply oxygen around the body. However, in such instances the glycoprotein haptoglobin is employed to capture the free haemoglobin, which is then engulfed by macrophages in order to clear and salvage the lost content of the haemolysed RBC from the circulation (Kristiansen et al. 2001). Although this iron recycling mechanism appears to be efficient, the exercise-induced increases in circulating hepcidin levels may contradict such a process, since elevated hepcidin levels act to internalise and degrade the ferroportin export channels on the macrophage, thus likely rendering the salvaged iron trapped in the scavenger cell.

This action of elevated hepcidin levels on ferroportin export channels may also impact on the ability of the body to absorb iron in the gut. Recent work has suggested that the inflammatory process associated with exercise can result in the up-regulation of hepcidin activity in the 3–6 h post-exercise period (Peeling et al. 2009a). Clearly such timing is synonymous with that at which an athlete may be consuming meals that are high in iron-containing foods (e.g. lunch after a morning training session, or dinner after an afternoon training session). Currently, it has been established that the training session duration may impact the magnitude of hepcidin response, with longer sessions resulting in higher post-exercise hepcidin levels (Newlin et al. 2012). Despite this, the exercise intensity and/or mode do not seem to affect the magnitude of hepcidin response in the same manner (Sim et al. 2013). Currently, it is under investigation as to the influence of chronic elevations of hepcidin activity over the course of a training program on the iron status of an individual. It was recently shown that during a 9-week basic combat training course in military recruits, resting hepcidin levels were not impacted over time; however, it was noted that soldiers with lower iron stores did present with lower hepcidin levels (Karl et al. 2010). This is likely a result of the tight homeostatic regulation of this hormone in response to the body's iron stores. A number of investigations have also attempted to suppress post-exercise hepcidin elevations through the use of carbohydrate supplementation as a result of its commonly reported suppressive effects on the hepcidin regulatory cytokine, interleukin-6 (Robson-Ansley et al. 2011; Sim et al. 2012). However, in practically applicable dosages, it would appear that the carbohydrate ingestion of these investigations was not effective in attenuating the hepcidin response. Regardless, more research is required in this area to establish nutritional guidelines that may best complement post-exercise iron absorption and recycling via the attenuation of the well-documented post-exercise hepcidin response.

In summary, the physiological factors associated with athletic training can cause iron depletion. Although a number of mechanisms may explain the increased iron turnover in athletes, it has recently been established that the iron-regulatory hormone hepcidin may play a key role in this process, and future research should be encouraged into the role of this hormone in athletes.

Moderate physical activity

Short- to long-term moderate physical activity or sports participation has no significant effect on iron status in studies of untrained or trained adults (Schumacher et al. 2002) or in pubescent athletes and children. See Fogelholm (1999) for a review.

Growth, pregnancy and lactation, menstruation

The requirements for iron increase substantially in adolescents and children during a growth spurt, and for women in pregnancy and lactation, although a higher absorption of iron helps to compensate for this increase (Food & Nutrition Board 2000a). Menstruation is not usually a major causal factor of iron deficiency in athletes, except in those with heavy or frequent menstrual blood loss. However, in one randomised, double-blind study of endurance athletes who had sufficient iron intakes, the iron deficiency was principally attributed to menstrual blood loss (Malczewska et al. 2000). This is an unusual finding, since a reduction in severity and frequency of menses with increased training loads is not uncommon. Nevertheless, a pregnant or lactating athlete has very high iron requirements and is unlikely to meet these from dietary sources.

These physiological states cannot be overlooked as contributors to iron depletion, particularly in adolescent athletes, where reported iron intakes do not always meet increased requirements (Sandström et al. 2012).

Blood loss and other medical causes

Blood loss from other causes or sources—including injury, blood donation, ulcers, misuse of anti-inflammatory medication (such as NSAIDs and aspirin), and the bacterium *Helicobacter pylori*—cannot be discounted as possible causes of iron deficiency, nor can other pathological or hereditary factors.

Because non-haem iron absorption requires an acid environment, any condition or medication that decreases gastric acid secretion and is linked with gastrointestinal blood loss (e.g. *Helicobacter pylori* infection, chronic use of antacids) is associated with iron depletion and anaemia (Qu et al. 2010; Harris et al. 2013). Based on population data from the NHANES (1999–2000) study in the USA, *Helicobacter pylori* infection accounted for 35% and 51% of cases of iron deficiency and IDA (Cardenas et al. 2006), so athletes with chronically low SF levels may need to be checked for infection. The anaemia that occurs with chronic infections, inflammation or malignant disorders differs from IDA but cannot be discounted in an athletic population as a reason for low iron stores.

10.7.2 Dietary causes

In healthy athletes, habitual consumption of sub-optimal dietary iron intakes or diets that have a low iron bioavailability is a major cause of iron deficiency, especially for females. Meat-based diets with high iron bioavailability can match recommended iron requirements for most women, but as bioavailability decreases, meeting these requirements becomes untenable from diet alone (Hallberg et al. 1998). Based on estimates of bioavailability from different diets, these authors estimate that 20–40% of women on diets with medium to low iron bioavailability cannot meet their iron requirements. These figures relate to non-athlete women, and are likely to be even higher for both male and female athletes with increased iron requirements. Any athlete, irrespective of gender, will be at risk of iron depletion if they habitually follow the types of diets listed below.

Low-energy diets

Athletes on low-energy intakes are those most frequently reported as having diets that contain less-than-recommended intakes of iron (Haymes 1992; Martin et al. 2006). Based on a diet-modelling approach based on the Australian population food guide for adults, a 6000 kJ/d (~1400 kCal/d) meal plan (meat–based) met the RDI (RDA) for all micronutrients except for iron (O'Connor et al. 2011).

Vegetarian diets and very high carbohydrate diets

Vegetarian diets can provide as much or more total dietary iron than meat-based diets (Craig & Mangels 2009) because of the variety of iron-fortified foods now available (Gerrior & Bente 2001). Athletes following vegetarian-style diets or who eat little red meat have lower ferritin levels than meat eaters, despite similar or higher iron intakes (Snyder et al. 1989; Tetens et al. 2007). In the diet-modelling approach mentioned above, a 6000 kJ/d (~1400 kCal/d) vegan meal plan failed to meet the EAR for iron, which is higher for vegetarians than meat eaters (O'Connor et al. 2011).

High-CHO diets recommended for athletes, if high in phytates and low in haem sources, have low bioavailability and increase the risk of low iron stores. Several studies on athletes have confirmed an association between high CHO intakes and low iron stores (Telford et al.

1993; Pate et al. 1993), and low haem iron intake and low iron stores (Malczewska et al. 2001), although other physiological reasons may also be contributing.

Natural food eaters

Athletes who avoid commercial or packaged foods (such as breakfast cereals) may be missing out on iron-fortified foods. For example, one serve of porridge contains about half the iron of most commercial breakfast cereals, and is high in phytates, which strongly inhibit iron absorption. Contrary to popular belief, bread, breakfast cereals and other cereal-based products, not meat and meat products, provide the bulk of total iron in the Australian diet (McLennan & Podger 1998).

Fad diets and weight-control products

Athletes who follow unusual eating regimens, consistently miss meals and have erratic eating patterns can also consume sub-optimal intakes of iron. These types of dietary habits are not atypical of adolescents, a high-risk group for iron deficiency.

Are athletes eating inadequate dietary iron intakes?

Dietary iron intakes of athletes are usually higher than or no different from non-athlete controls for both males and females and in prepubescent athletes and school children participating in sports (Resina et al. 1991; Fogelholm 1995, 1999; Sureira et al. 2012), although athletes under-report dietary intakes in dietary surveys more than non-athlete controls (Haggarty et al. 1988).

Despite the potential bias from under-reporting in dietary surveys, female athletes are still likely to consume inadequate iron intakes. Male athletes tend to consume more dietary iron than females and easily meet or exceed iron intake recommendations (Fogelholm 1995).

10.8 Iron status assessment of an athlete: clinical perspectives

Confirmation of a diagnosis of true iron depletion in athletes requires the use of a number of diagnostic criteria, including a comprehensive assessment of the physiological, medical, biochemical and dietary factors implicated in the aetiology of iron deficiency. Blood tests alone do not always confirm a diagnosis of early iron depletion.

10.8.1 Laboratory measures of iron status and their interpretation

The diagnosis of iron depletion and deficiency has been complicated by the absence of a clear and reliable reference method for detecting early iron depletion, and the use of varying cut-offs for haematological markers at each stage. Several key haematological markers are used to assess iron status and define each stage in clinical practice: SF to measure depleted iron stores (stage 1), sTfR to measure functional iron deficiency (stage 2) and haemoglobin concentration to measure IDA (stage 3) (Skikne et al. 1990). The interpretation of these values and others commonly used in clinical practice are briefly described in sections 10.8.2–10.8.6.

Several haematological markers in the blood are used to fully assess iron status for research purposes, but only serum iron, SF, transferrin, transferrin saturation and haemoglobin are routinely measured in sports medicine practices, together with mean cell volume (MCV) and mean corpuscular haemoglobin concentration (MCHC). Early studies of athletes used either SF as a sole indicator of early iron depletion or a combination of these routine measures. More recent markers available include sTfR to measure the functional iron pool or sTfR/log SF index to determine the level of iron depletion.

Haematological markers of iron status collected on a single occasion do not always confirm iron deficiency. Repeated measures in combination with other diagnostic protocols, including clinical symptoms and dietary assessment, confirm the diagnosis (see Practice tips). Even when clinical symptoms are evident, iron depletion or deficiency using haematological markers can easily be misdiagnosed in athletes. Reference ranges and cut-offs for evaluating iron status in athletes are based on population reference standards and, although these may be inappropriate for some athletes, they still serve as the benchmark for interpretation. Iron status measures below reference ranges could represent adequate iron status, especially in adolescents, and are closely associated with maturational age, body mass, sex, genetic factors and physical activity (Rossander-Hulten & Hallberg 1996).

The extent of iron depletion is determined by examining several markers found in the three main iron-containing compartments of the body: storage iron, transport iron and red blood cells.

10.8.2 Storage iron

Serum ferritin

Iron is stored within the reticuloendothelial system as molecular complexes called ferritin and haemosiderin, a partially degraded form of ferritin. Normally, small amounts of SF circulate in the blood in concentrations between 15 and 300 μg/L in adults (Worwood 1991). As storage iron increases, there is a corresponding increase in SF, which is why it is the best marker of iron stores in healthy people and athletes (Borch-Iohnsen 1995; Handelman & Levin 2008).

In a survey of elite athletes aged 15–31 years training at the Australian Institute of Sport (AIS), the mean SF was 85 μg/L (range 16–288) in 105 males and 54 μg/L (range 8–205) in 166 females (Fallon 2004). The distribution of SF is skewed towards the lower end of the range in athletes and population surveys (Cook et al. 1974; Maes et al. 1997).

In athletes and non-athletes, SF varies with age, sex and physical activity, and is low in young children and in adolescents during the growth phase (Baynes 1996). Some male adolescent athletes have low SF, despite having high dietary iron intakes and using iron supplements (Telford et al. 1993), which is probably physiologically ‘normal’ at this age. In population studies, a slow increase in SF and body iron occurs through adolescence followed by a substantial rise in males in late adolescence (Bothwell 1995; Bergstrom et al. 1996; Cook et al. 2003). Females do not show a late adolescent increase. Further increases in body iron stores occur in men with ageing and in postmenopausal females, but not in younger women (Cook et al. 2003). SF has some day-to-day variability (Stupnicki et al. 2003) and can be distorted under different physiological conditions (see section 10.8.5).

In elite athletes, SF declines during the training season in both males and females. In 46 matched pairs of female athletes from the AIS, SF decreased by around 25% during training seasons, with greater declines seen in weight-bearing sports (netball and basketball) than non-weight-bearing sports (swimming and rowing) (Ashenden et al. 1998). This effect is also highlighted in section 10.5.2—McClung and colleagues (2009) showed the SF levels of army recruits declined over the course of a 9-week basic combat training course. SF is the best indicator of iron stores in the absence of inflammation or infection (Zimmerman 2008).

Diagnostic cut-offs for serum ferritin to detect stage of iron depletion

There is no internationally accepted cut-off for diagnosing each stage of iron depletion. In untrained adults, SF below 30 µg/L suggests depleted iron stores (Crosby & O'Neill 1984) and 12 µg/L denotes exhaustion of iron stores (Cook et al. 1974). In male and female adolescents and women (not athletes), functional iron deficiency (stage 2) based on the absence of stainable iron in the bone marrow and other low-iron status markers was detected at levels <15 µg/L (Hallberg, Hulthen et al. 1993; Hallberg, Bengtsson et al. 1993). In the absence of standard reference values for athletes, researchers have set cut-offs for SF ranging from <16 µg/L (Zhu & Haas 1998; Hinton et al. 2000) through <35 µg/L (Peeling et al. 2007; Tan et al. 2012) to designate iron deficiency without anaemia, so interpretation of low SF values needs to be cautious.

Diagnostic cut-offs for iron status indicators for adults cannot be applied reliably to adolescents because of the large variation reported during maturation (Bergstrom et al. 1996). Population reference ranges for adults, children and adolescents are available in most biochemistry or haematology textbooks.

10.8.3 Transport iron

Haematological markers involved in transferring iron in the circulation are transferrin, serum iron and haptoglobin. Serum iron and serum transferrin in isolation are of limited value in determining iron status. Instead, total serum iron-binding capacity is usually estimated at the same time so that the percentage saturation of transferrin can be calculated. The combination of transferrin saturation and sTfR and possibly the sTfR/log SF index may be more useful to determine early-stage iron depletion than serum iron and serum transferrin (Cook et al. 2003).

Serum iron

Serum iron is generally lower in females than males, and appears unaffected by the menstrual cycle, but shows a large diurnal variation of around 20% (Beaton et al. 1989). The highest values occur in the morning and the lowest in the later part of the day (Jacobs 1974). Population reference ranges for serum iron are 14–32 µmol/L for males and 10–29 µmol/L for females. There is a true sex difference in values for serum iron similar to serum haemoglobin, irrespective of physical activity. However, the mean serum iron concentration is reported to be higher in untrained people than in athletes (Brotherhood et al. 1975). Isolated reductions in serum iron are of dubious significance because of the wide diurnal variation (RCPA 2013).

Serum transferrin and transferrin saturation

Serum transferrin is the major plasma transport protein, which binds iron (Fe^{3+} only) and carries it from the storage sites to the bone marrow (Handelman & Levin 2008). Transferrin saturation is the ratio of serum iron to iron-binding capacity. In a non-athlete population, transferrin saturation is usually 20–40% saturated. Transferrin saturation, similar to serum iron, shows a wide diurnal variation and low specificity. In early stage iron depletion, as iron stores decreases, transferrin saturation initially increases towards the upper end of the reference range, and then decreases when erythropoiesis is compromised. A critical level of <16% transferrin saturation has been established for an adult as the level at which erythropoiesis is compromised (Bothwell et al. 1979; Finch & Cook 1984). A raised percentage transferrin saturation in isolation may be the earliest indicator of haemochromatosis (a chronic iron-overload disorder) (RCPA 2013).

Serum (soluble) transferrin receptor

Transferrin receptors (TfR), located on the surface of erythroblasts, control the uptake of iron delivered to the cell by transferrin. The remnant product of degradation of the receptor is released into the circulation in a soluble form and is called either serum or soluble transferrin receptor (sTfR). sTfR, which reflects the recruitment of receptors, provides a quantitative evaluation of mild tissue iron deficiency or functional iron depletion (Skikne et al. 1990; Ervasti et al. 2004), although the sensitivity of sTfR in isolation in detecting early and immediate stages of iron deficiency is low (Choi 2005).

In iron depletion, the sTfR concentration is increased because more TfRs are expressed on the surfaces of cells that require additional iron. As SF goes down, sTfR goes up in a linear manner (Skikne et al. 1990) and can be three to four times the 'normal' reference range in IDA (Huebers et al. 1990). sTfR is not as sensitive as SF to changes in plasma volume induced by strenuous activity (Schumacher et al. 2002) or growth spurts and is unaffected by an acute phase response (e.g. inflammation or infection) compared to SF (Baynes 1996; Hulthen et al. 1998). However, it is unreliable in people with anaemia, people with disorders in erythropoiesis (Beguin et al. 1993) and when measured in athletes in the days following prolonged high-intensity exercise. Large variability in sTfR levels was confirmed in 12 highly trained cyclists in the days following six days of simulated stage racing compared to baseline measures (Voss et al. 2013).

sTfR levels vary with age. In a large cross-sectional population-based study (990 females, 1060 males), the highest mean sTfR values were found in the group aged 13–17 years, compared to all older groups (von Schmiesing et al. 2009). These high levels correlated with the low ferritin values. This was the first study to determine sTfR reference values across a wide age range. The distribution and kinetics of sTfR in elite athletes have not been investigated.

sTfR has been mainly used in research, in combination with other iron status markers, to recruit subjects with functional iron deficiency, and as a benchmark to assess the effects of iron supplements or iron injections on performance measures (Brutsaert et al. 2003; Nikolaidis et al. 2003a; Peeling et al. 2007; Hinton & Sinclair 2007, Banfi et al. 2009; Sureira et al. 2012). Using the Orion Diagnostic Test kit, cut-offs for sTfR for each stage of iron depletion in healthy, non-athlete adults are Stage 1 (≤2.75), stage 2 >2.75 and stage 3 >2.8 (Suominen et al. 1998). Other diagnostic test kits have different cut-offs and give different values, further confounding interpretation and comparison between studies (Banfi et al. 2009).

sTfR is currently not used for screening for iron depletion in athletes or in general clinical practice. Based on the early work, sTfR in combination with the conventional iron status indicators did not provide additional information compared to SF tests based on values from hospital outpatients and medical students (Mast et al. 1998). Similar conclusions were reported in a study on elite athletes (Pitsis et al. 2002). At the AIS, sTfR is used only as an adjunct to other routine measures of iron status in cases of inflammation or persistent low ferritin levels after iron supplementation.

Transferrin receptor-ferritin ratio (sTfR-F index)

The ratio of sTfR to log SF, called the sTfR/log SF index, is considered a more reliable determinant of iron status than either indicator used alone, especially in borderline cases (Cook et al. 2003; Punnonen et al. 1997; Suominen et al. 1998). Using the Orion Diagnostic Test kit, a ratio of ≥1.8 for stage 1 (depleted iron stores) and ≥2.2 for stage 2 (functional iron deficiency or iron-deficient erythropoiesis) in 65 Finnish adults was the cut-off for differentiating between the stages (Suominen et al. 1998). These cut-offs using this testing kit have been used as benchmarks in two studies of athletes (Malczewska et al. 2001; Pitsis et al. 2002).

The advantages of these measures are that capillary rather than venous blood samples can be used (Cook et al. 2003), which could be useful for monitoring iron status in field studies. Conversion of the index into body iron stores (mg/kg of body mass) allows a quantitative estimation of early iron depletion for an individual and avoids the problem of using arbitrary cut-offs for SF that are population based (see Cook et al. 2003 for the conversion equation).

The sTfR-F index has demonstrated excellent validity in estimating body iron stores and measuring stage of iron depletion in non-athletes (Cook et al. 2003). However, because of the availability of different sTfR assays testing kits on the market, the cut-offs for estimating deficiency also vary and are not standardised, which limits interpretation between studies (Brugnara 2003; Banfi et al. 2009). Moreover, Stupnicki and colleagues (2003) challenged the validity of the sTfR-F index in athletes and found that the within-subject day-to-day error of the ratio was 50% higher in female athletes than controls. This error was attributed mainly to fluctuations in SF in response to exercise-induced changes in plasma volume. Despite the significant daily fluctuations in SF reported in this study and others (Malczewska et al. 2001), the index remained stable over 10 days of training. These authors suggest that if the index is used to monitor iron status in elite athletes, reference values based on untrained people will need to be adjusted. At present this index is used predominantly in research.

10.8.4 Red blood cell and other measures (full blood count, morphology and reticulocytes, hepcidin)

Full red blood cell count

Red blood cells undergo changes in number, size, haemoglobin concentration and composition in individuals developing iron deficiency. As iron depletes, red blood cell numbers eventually decrease if iron is limiting erythropoiesis, and low haemoglobin and abnormal red cell morphology occurs. Changes in red blood cell parameters of individual athletes who are developing iron deficiency are listed in Table 10.4. The technology needed to perform all tests listed in this table may not be readily available to sports medicine practices.

TABLE 10.4 Changes in red blood cell characteristics in individuals developing iron deficiency

The cellular haemoglobin content of reticulocytes is reduced.
% hypochromic and % microcytic cells are increased.
The haemoglobin content of RBC is reduced (MCH).
RBC MCV is reduced.
MCHC is reduced.
Microcytic, hypochromic RBC may start to appear as the severity of iron depletion progresses.
RBC = red blood count, MCV = mean cell volume, MCHC = mean corpuscular haemoglobin concentration

Source: Adapted from Pyne et al. 1997

Haemoglobin and haematocrit

Haemoglobin and haematocrit values decrease only when severe iron depletion is present (Bothwell et al. 1979). These measures are subject to wide diurnal variations and individual variability in physiologically ‘normal’ levels (Beatonet al. 1989).

Haemoglobin and haematocrit values are similar for boys and girls until puberty, when they are higher in boys than girls and similar to adult values. However, in the absence of anaemia, haemoglobin and haematocrit values are usually higher in adolescent athletes of both sexes

(Nikolaidis et al. 2003a), higher in adult athletes than in non-athletes (Brotherhood et al. 1975), higher in men than women (Sanborn & Jankowski 1994) and positively associated with a high body mass index (BMI) in athletes (Telford & Cunningham 1991).

Haemoglobin mass (Hb_{mass})

Haemoglobin mass (Hb_{mass}), which is a relatively new haematological measurement and used to detect autologous blood doping since 2007, also provides an additional and accurate marker for iron depletion and anaemia (Garvican et al. 2011). Hb_{mass} represents the absolute mass of circulating haemoglobin in the body. Compared to most iron biomarkers and haemoglobin, it is unaffected by shifts in plasma volume and diurnal variation (Prommer et al. 2008). Only small variations in Hb_{mass} (2–3%) were reported in individual cyclists over a competitive season (Schmidt et al. 2008; Garvican et al. 2012), although it is confounded by altitude training, maturational timing, blood donation, illness and injury. The variation associated with these variable is still relatively low (Gough et al. 2013). Although useful in combination with other iron markers to detect iron depletion and monitor response to intervention, its potential application is for the Athlete Biological Passport (ABP) to identify suspicious blood doping techniques (WADA 2010). It is not used in clinical practice.

Red cell morphology

With increasing severity of iron depletion, the number of red blood cells progressively decreases and the cells become microcytic and hypochromic. Occasional rod-shaped cells and target cells are also observed. Blood films may be examined to discriminate between different types of anaemia and disorders of iron metabolism that affect red blood cells (e.g. IDA and thalassaemia, which is an inherited disorder of haemoglobin synthesis).

Reticulocytes and haemoglobin content in the reticulocytes

Reticulocytes are red blood cells recently released from the bone marrow. Determination of the reticulocyte haemoglobin content (CHr) provides an early and reliable measure of functional iron deficiency and bone marrow iron stores, but is unreliable in subjects with elevated MCV and thalassaemia (Mast et al. 2002). Low haemoglobin in reticulocytes, which circulate for only 1–2 days, results in low haemoglobin in mature red blood cells. Iron supplements may be beneficial at this point to boost iron stores and to prevent further iron depletion (Ashenden et al. 1998); however, this measure is not used in clinical practice. Together with the sTfR-F index, CHr may allow for a more precise classification of iron depletion than other indicators (Brugnara 2003).

Hepcidin

The peptide hormone, hepcidin, synthesised in the liver, regulates systemic iron metabolism. Measured in the urine and plasma, it may be useful to differentiate between the anaemia of inflammation and other stages of iron deficiency and iron overload (Ganz 2005, 2011). High levels of circulating hepcidin inhibit intestinal iron absorption, activate release of iron from iron stores, and divert iron away from haemoglobin and red blood cell synthesis, resulting in decreased iron stores (Ganz 2005, 2011; Atanasiu et al. 2006). The mechanism of action is outside the scope of this text. For reviews, see Ganz (2005, 2011) and Hintze and McClung (2011). Hepcidin production and high circulating levels are induced by physiological stressors, including chronic infection, inflammation and—interestingly—chronic and acute high-intensity or endurance exercise. As explained in 10.7.1, chronically elevated hepcidin in combination with the haemolytic effect of exercise and inadequate/poor timing of dietary iron intake may explain the high prevalence of iron depletion reported in athletes (Peeling et al. 2008, 2009a).

In contrast, hepcidin production decreases when body iron requirements are high and erythropoiesis is active, as expected during growth and with low iron stores. Low pre- and post-exercise hepcidin levels have been previously reported in athlete cohorts with compromised iron stores (Peeling et al. 2009a), and during chronic hypoxic exposure (i.e. training at altitude), where an increased erythropoietic drive is present (Hintze & McClung 2011; Talbot et al. 2012). Currently, hepcidin levels in urine or plasma are not routinely measured in clinical practice or in athletes for the diagnosis of iron deficiency, making reference ranges difficult to present.

10.8.5 Errors in interpreting laboratory measures of iron status

The use of haematological markers to diagnose iron status from a single or one-off blood test can be unreliable or misleading, since these blood tests are susceptible to fluctuations from physiological and pathological conditions that confound their interpretation. Dehydration at the time of testing can cause a haemoconcentration effect, resulting in an apparent increase in the iron markers.

In the absence of dehydration, all haematological markers except SF, hepcidin and, to a lesser extent, sTfR decrease after acute short-term moderate to strenuous exercise; after prolonged aerobic exercise (such as cycling every day for three or more days) (Schumacher et al. 2002; Voss et al. 2013); and after a marathon race (Lampe et al. 1986; Fallon et al. 1999; Roecker et al. 2005). This decrease is associated with exercise-induced plasma volume expansion resulting in haemodilution.

Furthermore, since SF and hepcidin are acute phase reactants, infection, inflammation, liver disease, high alcohol consumption, reduced calorie intake and the physiological stress of strenuous exercise may also falsely elevate these measures (Fallon et al. 1999; Hallberg & Hulthen 2003). Minor infections can also induce an acute phase response. In one study of 1670 Swedish adolescents aged 15–16 years, 24.4% of those who had reported a mild common cold with fever during the preceding month had significant increases in SF, evidence of an acute phase response (Hulthen et al. 1998). In contrast, other indicators, such as haemoglobin, haematocrit, serum iron and transferrin, decrease with infection and inflammation (Finch & Huebers 1982). To avoid errors in iron status measures, blood should be taken prior to strenuous exercise and in the absence of active inflammation and infection.

10.8.6 Summary of laboratory measures of iron status

Blood tests alone do not always confirm a diagnosis of iron depletion, since iron status indicators that fall outside the diagnostic range may be ‘normal’ for some individuals. SF has been used routinely as a diagnostic measure in sports medicine practice and in research for the initial assessment and prospective monitoring of stage 1 and stage 2 iron depletion. Such protocol remains, in combination with haemoglobin, the iron test of choice for early detection of iron depletion, despite its limitations. Routine screening of iron status may be warranted in athletes in an elite training program. More recently, sTfR, or the sTfR/log SF index, has been added to discriminate tissue or functional iron deficiency (stage 2), although more studies are needed in athletes to assess the utility of this index as a diagnostic tool. Haemoglobin and haematocrit without SF have limited use in early detection of iron depletion, since significant decreases in these markers are observed only in IDA. Serum iron and transferrin saturation show a wide diurnal variation, although high levels of transferrin saturation in association with low SF are an indication of early iron depletion. Measuring and monitoring changes in immature red blood cells (reticulocytes) is useful for evaluating relative changes in iron status in individual athletes, but is not routinely used in clinical practice.

10.9 The presence of clinical symptoms

IDA is associated with symptoms of weakness, breathlessness and impaired aerobic and endurance capacity. Even with depleted iron stores, some athletes look pale, may have a slightly elevated resting pulse rate, feel ‘run-down’ or ‘washed out’, and exhibit changes in mood state or have a diminished appetite. These types of symptoms are non-specific and may be indicative of overtraining, immune deficiency, psychosocial stress, unresolved viral infections, non-fasting hypoglycaemia or sleep disorders; they may even be considered ‘normal’ in an athlete or adolescent. Conversely, athletes with low SF levels often have no symptoms. However, despite an absence of clinical symptoms, one study reported that 100 intercollegiate female athletes considered their performance to be worse when iron depleted (Risser et al. 1988). Experienced elite athletes are often overly concerned about fatigue or SF levels, routinely request blood tests and self-supplement with iron when tired, despite normal SF levels (Fallon & Gerrard 2007). In anxious elite athletes with a history of iron depletion, any sign of fatigue or lethargy is often perceived as iron depletion.

10.10 Dietary assessment

Because iron is found in so many foods and its intake is highly variable, determining usual intakes with reliability requires many days of data collection, which is not possible in clinical practice.

In clinical practice, dietary assessment of usual iron intake and bioavailability of iron of an individual athlete usually involves a dietary history, which includes an assessment of other physiological, training and medical factors that increase iron requirements, particularly blood loss, growth stage and training intensity. Techniques for assessing iron status are found in the Practice tips.

A habitual low intake and low bioavailability of iron, in combination with iron status measures below population reference values, can usually confirm a diagnosis of iron depletion or deficiency. Symptoms may or may not be present depending on the severity of iron depletion. However, any assessment of the usual diet and nutrient adequacy of an individual is imprecise, especially for iron, and must be interpreted cautiously because of the inherent problems in accurately assessing dietary intakes.

10.11 Dietary intervention for iron depletion and iron deficiency

Dietary intervention for the treatment of diagnosed iron depletion or deficiency usually requires iron supplements (or iron injection) in combination with advice about increasing dietary iron intake and improving iron bioavailability through different food combinations (see Practice tips).

10.11.1 Recommended dietary iron intakes for athletes

Average daily iron requirements for athletes in different sports have not been established and are likely to be highly variable. Daily iron losses in endurance runners have been estimated at 1.5–1.7 mg iron/d in men and 2.2–2.3 mg iron/d in women (Haymes & Lamanca 1989), although there is some debate about whether endurance athletes have truly low iron stores (Ashenden et al. 1998). Basal or obligatory iron losses in untrained adults are only 0.9–1.0 mg/d in men and 0.7–0.8 mg/d in women (excluding menstrual losses) (Bothwell 1996). To meet iron losses in distance runners, Haymes and Lamanca (1989) recommended iron intakes of 17.5 mg/d for men and around 23 mg/d in normally menstruating women, assuming iron absorption to be ~10% of dietary iron from the total diet.

Nutrient Reference Value (NRV)/Dietary Reference Intake (DRI) for iron are based on an average iron absorption of 18% from a mixed western diet (that includes animal foods) and 10% absorption from a vegetarian diet (Food and Nutrition Board 2000). Multiple levels of nutrient reference values have been set: the Recommended Dietary Intake (RDI), Estimated Average Requirements (EAR), Adequate Intake (AI) and Upper Level of Intake (UL) or Upper Intake Level (UIL) in the USA and Canada. These levels have different applications when used to assess and plan diets for individuals and groups, compared to previous usage (see Chapter

2). For assessment of an individual, the EAR is considered the best estimate of an individual's requirement (Food & Nutrition Board 2000).

Although the EAR may not be applicable for athletes without some adjustment, it can serve as a benchmark, in combination with the RDI, to examine the probability that usual intake is 'adequate' or 'inadequate' for an individual athlete (Food & Nutrition Board 2000b; Murphy et al. 2002). The Food and Nutrition Board (2000) suggests that the EAR for iron should be 1.3–1.7 times higher than normal EAR values for athletes and 1.8 times higher for vegetarians (non-athletes) to account for a low iron bioavailability diet of vegetarians. The EAR for female vegetarian athletes may be even higher than this because of increased requirements. Such high levels are unlikely to be met by dietary iron intake in this high-risk group without appropriate dietary planning.

Strategies to increase the total consumption of iron-rich food and enhance iron bioavailability

Iron-rich diets that have a high iron bioavailability are needed to prevent the development of iron depletion and to treat it. The Practice tips provide practical dietary strategies to achieve these goals using different food combinations.

10.12 Medical intervention: iron supplements

Many athletes self-administer or are provided with iron supplements daily or intermittently as potential ergogenic aids or as a preventive measure without being diagnosed with iron depletion. Indiscriminate use of high doses of iron supplements without iron deficiency is of concern for long-term safety, given the high prevalence of the genetic disorder haemochromatosis—1 in 250 people of Northern European background (NHMRC 2007). There is no indication or justification for taking iron supplements in the absence of iron deficiency (Goodman et al. 2011).

10.12.1 Iron supplements for clinical treatment of iron depletion

There is sufficient evidence from well-designed studies that support the use of iron supplements in therapeutic doses in athletes with low SF levels, although the level of SF at which supplementation should commence (from <12 to 35 µg/L) is still controversial (Nielsen & Natchtigall 1998) (see section 10.5.3).

Decisions about using iron supplements in athletes are usually made on a case-by-case basis and are not necessarily based only on low SF levels, but in combination with several other iron-status markers. Full recovery from depleted or exhausted iron stores takes around 3 months using high doses of oral iron supplements (Nielsen & Natchtigall 1998) and is not possible using dietary intervention alone in this timeframe. Hallberg (1998) calculates that it takes about 2 to 3 years to attain around 80% of iron stores in men and women by dietary means alone, based on predictive equations. Therefore, iron supplements are important, at least for rapid recovery in cases of depleted iron status.

In most cases, supplements are discontinued when the measurement of SF or serum transferrin receptor returns to 'normal' for the individual athlete, and diet therapy is maintained. The importance of habitually consuming foods with a high bioavailability of iron is crucial to continued recovery.

10.12.2 Dosage and duration of oral iron supplementation

Iron supplements are available in two main forms: ferrous and ferric. Ferrous salts—ferrous fumarate, ferrous sulphate and ferrous gluconate—are more readily absorbed than the ferric form (Hoffman et al. 2000). To replete exhausted iron stores, a dosage of around 100 mg elemental iron/d is necessary for at least three months (Nielsen & Natchtigall 1998), although doses of up to 300 mg/d have been used in severe cases (Balaban 1992). These doses are well above normal physiological levels.

Studies on non-athletes have found that iron supplements taken two to three times a week are just as effective in increasing SF and haemoglobin as supplements taken daily (Solomons 1997; Tee et al. 1999). In one study of 624 adolescent females in Malaysia with mild, moderate and borderline anaemia, 120 mg of elemental iron taken once a week was found to be as effective in increasing haemoglobin and SF levels as 60 mg of elemental iron taken once a day (Tee et al. 1999). Research is needed to determine whether similar oral intakes are as effective in athletes. Hallberg (1998), however, is more in favour of daily iron supplementation, which increases depleted iron stores more rapidly than weekly doses. Current practice also favours daily supplement use in athletes to promote rapid recovery.

10.12.3 Side effects of iron supplementation

Some people do not tolerate iron supplements, especially in high doses; they complain of constipation, black stools, abdominal cramping and, to a lesser extent, diarrhoea, nausea and vomiting. These adverse effects usually subside when iron is taken with food. Iron supplements inhibit zinc and copper absorption, so there may be an increased risk of deficiency of these minerals in athletes who continually take supplements, although evidence for inducing an actual zinc or copper deficiency has not been substantiated.

10.12.4 Safety of long-term excess iron: risk of iron overload and haemochromatosis

Although iron is metabolically essential in small doses, it is also highly toxic if unbound and in excess in the blood, as is seen in iron overload and in the genetic disorder haemochromatosis. Free iron is toxic and acts as a pro-oxidant, accelerating the production of free oxygen radicals, which react with cell membrane lipids, resulting in organ damage and cell death (Ganz 2005). Excess iron in the blood and elevated ferritin levels, even below the levels found in genetic haemochromatosis, are associated with diabetes, metabolic syndrome, hypertension, dyslipidaemia, elevated fasting insulin and glucose, and abdominal adiposity (Reddy & Clarke 2004; Basuli et al. 2014).

There is no evidence of a link between excessive dietary iron intake and iron overload in healthy subjects, including those consuming large intakes of red meat (Hallberg & Hulthen 2003; Heath & Fairweather-Tait 2003). However, there is some evidence of iron overload in people who regularly have iron injections and take iron supplements. In a survey of 1000 French male professional cyclists, 45% had SF levels >300 μg/L (Zotter et al. 2004), which is well above the diagnostic cut-off for iron overload. These high levels were largely attributed to repeated iron injections. High levels of SF from habitual iron supplement use were also reported in another study of professional cyclists (Deugnier et al. 2002). The long-term consequences of taking excessively high doses of iron supplements or iron injections in a healthy person are not known, but may mimic the effects of the genetic disorder haemochromatosis. Excess unabsorbed iron in the colon has been implicated in mucosal damage and possibly increases the risk of colorectal cancer (Reddy & Clarke 2004).

Hereditary haemochromatosis involves a group of genetic disorders associated with a deficiency of hepcidin, the hormone that regulates iron absorption and metabolism (Ganz 2005). As a consequence, iron is indiscriminately absorbed and slowly deposits in organs, resulting in irreversible damage. Only those homozygous for the gene express the condition, which occurs in around 0.3% of Caucasians (Bothwell 1995), although there is some evidence that carriers of the gene or its mutations—about 10–15% of Caucasians of northern European origin—are also at increased risk of health problems, particularly cardiovascular disease (Heath & Fairweather-Tait 2003). Clinical symptoms in the early stages of the condition are similar to those of iron deficiency and have been reported in individuals as young as 20 years (Worwood 1998). Over the last few years, we have picked up this condition in several young elite athletes from routine haematological screening or investigation of persistent fatigue. High levels of SF (>150–200 μg/L) or transferrin saturation greater than 45%, confirmed on two separate occasions, are alerts for genetic testing for haemochromatosis (RCPA 2013). In one male athlete with a confirmed case of homozygous (C282Y) haemochromatosis, SF was in the normal range, although transferrin saturation and serum iron were substantially elevated (Fallon & Gerrard 2007).

The overall high prevalence of haemochromatosis in northern European populations (around 1 in 250) and associated long-term health risks highlight the need for correct diagnosis of iron status in athletes and the importance of discouraging indiscriminate use of iron supplements, which are contraindicated in people carrying the gene.

10.12.5 Intramuscular iron therapy and intravenous iron infusion

One perceived issue with oral iron supplementation may be the prolonged nature of the treatment needed to re-establish adequate iron stores (i.e. ~2–3 months). Previous research has shown that 5 × 2 mL intramuscular iron injections administered over 10 days increased SF more rapidly than in the control group taking oral iron supplements for 30 days (Dawson et al. 2006). After about 15 days of treatment, SF values had significantly increased in the iron-supplemented group compared to baseline values but the magnitude of the response in the injected group was more than three times higher than in the iron supplemented group.

More recently, Garvican et al. (2013) assessed the efficacy of intravenous iron injections (2–4 injections of ferric carboxymaltose solution) and oral iron supplements over 6 weeks in distance runners. Here, it was again evident that the SF levels of both groups improved during the intervention and the magnitude of this increase was greater in the IV trial from the end of the first week. Such results are clearly promising should an athlete be in a compromised iron state close to competition, since this data shows improvements to iron status in as few as seven days. However, despite the apparent faster nature of these alternative supplement protocols, such methods should only be carried out by a trained sports physician. Iron injection may carry a minor risk of infection at the injection site, may cause discomfort during administration, and in severe cases may result in anaphylactic shock, which can ultimately result in severe and possibly fatal outcomes. To this end, if not time pushed, the well-established option of oral iron supplementation is clearly an efficacious method of addressing depleted iron stores; however, iron injection may be an option for the sports physician for rapid recovery.

Summary

Although IDA is uncommon in athletes, iron depletion or deficiency remains a problem, particularly in female athletes, in athletes who follow vegetarian-style diets, and in runners and endurance athletes. Recent evidence using transferrin receptor suggests that iron depletion without anaemia (i.e. iron deficiency) can reduce oxidative capacity and also impair endurance capacity in untrained people. When iron stores in both tissue and bone marrow become depleted, the capacity of muscle and other tissue to use oxygen and other iron-containing proteins for the oxidative production of energy can be compromised. However, more studies are needed on athletes to confirm these effects.

Treatment for iron depletion and deficiency (and IDA) requires a combination of iron supplements and dietary intervention and, in some cases, an iron injection may be administered for rapid recovery. Dietary intervention with a high iron intake alone is unlikely to provide rapid recovery so supplements are essential. However, routine use of iron supplements as a preventive measure is not recommended because of the risk of iron overload and the high prevalence of the genetic disorder haemochromatosis. For those athletes diagnosed with haemochromatosis, iron supplements are contraindicated and can cause irreversible damage to tissues and organs. For athletes without this disorder, an iron-rich diet with high bioavailability is the cornerstone to prevention and treatment of iron depletion and deficiency.

USEFUL WEBSITES

- **www.sportsdietitians.com.au/resources/upload/Iron_depletion_in_athletes.pdf**
  Sports Dietitians Australia, Iron depletion in athletes (fact sheet)
- **www.ausport.gov.au/ais/nutrition/factsheets/basics/iron_-_are_you_getting_enough**
  Australian Institute of Sport, Iron: are you getting enough? (fact sheet)
- **www.hematology.org/Patients/Anemia/Iron-Deficiency.aspx**
  American Society of Hematology, Information about iron status results for patient use

Practice tips

VICKI DEAKIN AND FIONA PELLY

OVERVIEW

Athletes involved in regular intensive training programs are at risk of depleting their iron stores, which can, if not detected and treated early, develop into the advanced condition of anaemia. Athletes have higher iron requirements and potentially higher iron losses than non-athletes. Sub-optimal intakes of iron are evident in athletes who follow very high-CHO, low-energy, fad and vegetarian diets, or who are natural food eaters. As athletes are encouraged to consume diets high in CHO, there is a risk that inhibitors of iron absorption (found in cereal grains, nuts and legumes) will reduce iron bioavailability. Therefore, those athletes at risk of iron depletion need practical strategies for maintaining high CHO intakes without compromising iron status. Food combinations that enhance iron absorption are important to achieve this outcome.

BIOCHEMICAL DETECTION OF IRON DEFICIENCY

Interpretation of biochemical indicators used to detect early iron depletion should be interpreted with care, as they can be affected by strenuous exercise just prior to testing, by chronic inflammation or infection, by hypohydration and even by mild infections. All low iron status measures in athletes should be treated as potential iron depletion.

MEDICAL OR PHYSIOLOGICAL CAUSES OF IRON DEFICIENCY

Assessing the contribution of any medical or physiological factors implicated in the aetiology of iron deficiency is important to target appropriate treatment or assign causation. These include:

- increased iron requirement (such as recent pregnancy, growth spurt or sudden increase in the intensity or duration of training)
- the habitual use of medications (including antacids) that decrease the acidity of the stomach.
- potential blood loss (e.g. from frequent nose bleeds, menorrhagia, ulcers, chronic use of anti-inflammatory medications, blood donation)—signs of blood loss after competition or heavy training include discoloured urine or blood in stools
- recent weight loss or illness
- malabsorption of iron (e.g. inflammatory bowel disease, or bacterial infection such as *Helicobacter pylori*).

DIETARY ASSESSMENT OF IRON DEFICIENCY

Diet-related risk factors that are linked to iron depletion include:

- infrequent consumption of red meat, poultry or seafood
- vegetarianism or very high CHO diets from mainly wholegrain cereals
- irregular or erratic eating patterns
- prolonged loss of appetite after physical activity
- low intake of bread, breakfast cereal and iron-fortified foods
- weight-reduction or fad diets, inappropriate food combinations or limited variety of food choices
- low intake of vitamin C- or vitamin A-rich foods with meals
- regular consumption of strong tea or coffee with most meals
- poor food knowledge, limited cooking skills, reliance on takeaway foods.

According to population reference standards, the requirements for iron intake are 1.3–1.7 times higher than EAR values for athletes and 1.8 times higher for vegetarians (i.e. non-athletes). Iron requirements for female vegetarian athletes may be even higher than this and will need to be modified when used to assess inadequacy of dietary iron intakes derived from food records or to plan diets.

DIETARY TREATMENT AND PREVENTION OF IRON DEFICIENCY

Although dietary iron absorption increases in people with iron deficiency, the amount absorbed is not sufficient to allow quick recovery, especially in athletes with high requirements and losses. If depleted iron stores have progressed in anaemia, it could take up to 2 or 3 years for the full recovery of iron stores on diet alone (Hallberg et al. 1998). Nevertheless, an iron-rich diet, in combination with iron supplements and strategies to enhance iron absorption, is still the cornerstone of treatment of iron deficiency.

SOURCES OF DIETARY IRON

The first strategy in treatment is to encourage iron-depleted athletes to increase total dietary iron intake and maintain high levels, especially after using supplements for treatment. Iron is found in a wide range of foods (see Table 10.5). The colour of meat is determined largely by its iron content; the 'redder' the meat, the higher the myoglobin (iron-containing pigment) content and hence iron content. Iron-enriched breakfast cereals and breads are important iron sources, although iron from meat and other animal foods is better absorbed. Many foods are fortified with iron. When eaten regularly, iron-fortified foods and, in particular, breakfast cereals provide a substantial amount of iron in the diet. One bowl of iron-enriched breakfast cereal has more than four times the iron content of a bowl of porridge.

TABLE 10.5 Iron content of Australian food (mg/100 g)

Food type	Serving size	Amount of total iron per serve (mg/serve)
Animal sources (good sources of haem and non-haem iron)		
Lean, cooked trimmed beef rump steak	1 small serve (100 g)	2.7
Lean, cooked trim lamb steak	1 small serve (100 g)	2.6
Egg	1 boiled egg (60 g)	1.0
Lean pork fillet, cooked	1 small serve, 1/2 cup (100 g)	0.6
Tuna, canned, in water	1/2 cup (100g)	1.3
Lean grilled chicken, no skin	1 small breast (100 g)	0.4
Fish, white flesh, cooked	1 small fillet (100 g)	0.3

Plant sources (good sources of non-haem iron)		
Commercial breakfast cereal (iron-fortified, such as Corn Flakes™)	Average serve (60 g)	4.2–10.2
Muesli (untoasted, not iron-fortified)	1 cup (100 g)	7.8
Fortified beverage base, chocolate (e.g. Milo™)	3 heaped teaspoons	4.3
Baked beans in tomato sauce	1 cup (275 g)	2.6
Bread (with added iron)	2 sandwich slices (60 g)	4.3
Bread (wholemeal or mixed grain, no added iron)	2 sandwich slices (60 g)	1.3
Plain muesli bar	1 bar (37 g)	0.7
Nuts (cashews, almonds)	50 g	1.9–2.5
Sweet corn, canned	1/2 cup (120 g)	0.6
Porridge (cooked oats)	1 cup (260 g)	2.3
Green vegetables (e.g. broccoli, spinach, silverbeet, cabbage, Chinese green vegetables)	1/2 cup (120 g)	0.8–4.7
Pasta/noodles, cooked	150 g	0.9
Rice, white, cooked	150 g	0.92
Dried fruit (prunes, apricots)	5–6 (50 g)	0.5–1.5
Fruit (fresh)	1 average piece	0.3–0.5

Source: Food Standards Australia New Zealand: NUTTAB 2010 – © Food Standards Australia New Zealand

Wholemeal bread has nearly twice the iron content of white bread, although the high phytate content reduces iron bioavailability. Legumes (such as lentils, baked beans, peanuts, soy beans) are good sources of iron, but high in inhibitory components (phytate and soy peptides). Dried fruit, sweet corn, green leafy vegetables (including broccoli, silverbeet, spinach) are also excellent sources with a low phytate content and hence high bioavailability.

The second strategy in treatment (and prevention) of iron depletion is to optimise iron absorption by improving food combinations in a meal.

LIMIT CONSUMPTION OF INHIBITORS OF IRON ABSORPTION WITH MEALS

Foods high in phytate (e.g. wheat-based cereals, soy flour and products, lentils, seeds, some nuts) and containing polyphenols (e.g. tea, coffee, cocoa, red wine) can strongly inhibit iron absorption from non-haem food sources, when consumed concurrently. Black tea and coffee contain almost twice as much polyphenol (expressed as tannin equivalents) as herb teas and green teas, which have similar amounts of tannin as red wine. Iron absorption is affected only when these beverages are consumed around the same time as iron-rich non-haem sources. White wine and other alcoholic beverages have only trace amounts of tannin and so have little effect on non-haem iron absorption. Athletes at risk of or diagnosed with iron depletion should avoid concurrent consumption of foods with high levels of inhibitors.

INCLUDE WITH MEALS FOODS THAT ENHANCE IRON ABSORPTION OR NEGATE THE EFFECT OF INHIBITORS

Inclusion of meat, poultry, seafood, vitamin C- or vitamin A-rich foods with meals (e.g. meat and salad on a sandwich, orange juice or fruit with breakfast cereal) enhances iron absorption substantially from non-haem foods. Vitamin C is a potent iron-enhancer and can override the inhibitory effects of phytates and tannins in a dose-dependent manner. Citrus fruit and juices are the richest natural sources of vitamin C. Many foods (including fruit juices) have added vitamin C, which enriches the food and also acts as an antioxidant to prevent vitamin C loss from oxidation. The presence of other organic acids in citrus fruits (e.g. grapefruit, orange, lemon, lime) also enhances iron absorption from non-haem foods.

Table 10.6 provides practical strategies to increase iron density and improve iron bioavailability.

TABLE 10.6 Summary of dietary strategies for enhancing iron intake and bioavailability

Eat lean red meat (beef, veal, lamb) in main meals at least three to four times a week. (This adds readily available iron to a meal and enhances absorption of iron from plant sources.) Add meat to high-CHO meals to enhance non-haem absorption (e.g. in a pasta sauce, in a stir-fry with rice or noodles or in meat and vegetable kebabs with rice).
Use lean red meat, poultry or seafood on sandwiches, preferably daily.
If vegetarian, ensure daily food choices are iron rich (e.g. choose iron-fortified cereal, baked beans, legumes, bread, fruits, green leafy vegetables). These are also excellent CHO sources.
Combine plant foods with a high phytate content (such as cereal grains, breads, breakfast cereal, soy products) with vitamin C-rich foods. Good sources of vitamin C are citrus fruits, fruit juice, strawberries, broccoli, cabbage, cauliflower and capsicum.
Eat iron-fortified breakfast cereal on most days and look for other foods that have been iron enriched. Porridge and some types of muesli, although excellent sources of CHO, are not iron fortified.
If diagnosed with depleted iron stores or iron deficiency, avoid or limit adding bran and wheat germ (rich in phytate) to meals and include a vitamin C-rich food to help negate the inhibitory effect of phytate.
If diagnosed with depleted iron stores or iron deficiency, avoid drinking strong tea, coffee and red wine (all rich in polyphenols) with meals and substitute with a vitamin C-rich beverage to help negate the inhibitory effects of these beverages.

TREATMENT WITH IRON SUPPLEMENTS

Where iron stores are exhausted, an iron-rich diet alone is insufficient to restore iron levels quickly. Iron supplements taken for at least three months in dosages of 80–100 mg elemental iron/day are necessary for full recovery of iron stores (see section 10.12.2). As most iron supplements have low iron bioavailability, consumption of vitamin C-rich food or a vitamin C supplement together with the iron supplement can enhance iron absorption. The ferrous form of an iron supplement has a higher bioavailability than the ferric form and is often well tolerated. When iron supplements are taken on an empty stomach, further elimination of interference from inhibitors is possible, although the incidence of gastrointestinal side effects may increase. In our experience, adverse side effects are infrequently reported in athletes on short-term, prescribed iron dosages.

FOLLOW-UP

Athletes should be discouraged from self-administering iron supplements, which may induce deficiencies of copper and zinc and are contraindicated in haemochromatosis. Biochemical measures of iron status in elite athletes are usually reviewed at 12 weeks, and supplements are discontinued when iron status indicators are within the individual’s usual levels or reference ranges.

FAILURE TO RESPOND TO INTERVENTION

Reasons for failure to respond to dietary iron intervention and iron supplements may relate to:

- poor compliance with the intervention
- inadequate dosage of iron supplement: <50 mg elemental iron is associated with little improvement of haemoglobin and iron status
- inadequate iron density in the diet to meet individual requirements (iron intakes may need to be set at higher

than previous levels)

- poor food combinations that inhibit iron absorption from food or iron supplements
- an underlying chronic medical condition (i.e. the anaemia associated with inflammation or chronic disease) that does not respond to iron supplements or parasite infection or *Helicobacter pylori* infection that was previously undetected
- incorrect original diagnosis (consistently low biochemical measures of iron status), and the usual or normal levels for an individual athlete or haematological intakes were unreliable at the time of testing.

To provide a valid measure of usual or baseline levels for individual athletes, biochemical indicators need to be monitored over several months. Large fluctuations in these measures are usually indicative of dietary, physiological and training influences.

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Introduction

Vegetarian diets are now part of mainstream eating in western countries. With the renewed interest in 'clean eating', vegetarian diets or near-vegetarian diets have regained popularity among recreationally active people and competitive athletes. To date, few studies have investigated the nutritional intakes of vegetarian athletes and the effects of vegetarian diets on performance. This commentary describes different types of vegetarian diets and their potential advantages and disadvantages in meeting the nutrient requirements of athletes and dietary goals for optimal sports performance. The potential effects of vegetarian diets on sports performance in athletes involved in high-intensity training programs are described.

Types of vegetarian diets

A vegetarian is a person who does not eat meat (including chicken) or seafood, or products containing these foods (Craig & Mangels 2009). Table C5.1 defines the different styles of vegetarian diets. Of these, the fruitarian and macrobiotic diets are the most restrictive and nutritionally inadequate for athletes. A lacto–ovo vegetarian diet includes plant foods and eggs, milk and dairy products. Some people, including athletes, describe themselves as vegetarians simply because they restrict or avoid eating red meat but may occasionally eat chicken and/or fish. In the general population, a substantial proportion of females restrict or avoid meat (especially red meat) (Baghurst et al. 2000) and convert to a vegetarian diet as a means of restricting energy intake to attain a lean body composition (Rodriguez et al. 2009). This behaviour may be a sign of disordered eating and reduced energy availability (Nattiv et al. 2007).

TABLE C5.1 Main categories of vegetarian-style diets

Type	Comments
Fruitarian	Consists of raw or dried fruits, nuts, seeds, honey and vegetable oil
Macrobiotic	Excludes all animal foods, dairy products and eggs Uses only unprocessed, unrefined, 'natural' and 'organic' cereals, grains and condiments such as miso and seaweed
Vegan	Excludes all animal foods, dairy products and eggs In its purest form, excludes all animal products including honey, gelatine, silk, wool, leather and animal-derived food additives
Lacto vegetarian	Excludes all animal foods and eggs Includes milk and milk products
Lacto–ovo vegetarian	Excludes all animal foods Includes milk, milk products and eggs

The reasons for being vegetarian vary and include health, religious and environmental considerations, economics, animal welfare, taste, smell, appearance or beliefs about the nutritional quality of animal foods. For an athlete, these reasons might extend to a perceived performance benefit, enhanced recovery or to assist in weight management. In a study of 351 athletes competing at the Delhi 2010 Commonwealth Games, 7% (n = 28) followed a vegetarian diet and 13% (n = 44) avoided red meat (Pelly & Burkhart 2014). In this study, athletes in weight-category sports were more likely to follow a vegetarian/vegan diet than athletes from other sports. Similar results were reported in a national survey of over 9000 US runners, where 8% of females and 3% of males were vegetarians (Williams 1997).

Effect of vegetarian diets on health outcomes

In adults, vegetarian diets are associated with a decrease in blood pressure in those with hypertension, lower cholesterol, lower body mass and lower rates of coronary heart disease, type 2 diabetes and metabolic syndrome compared with meat eaters (for a review see Craig & Mangels 2009). In Australian adolescents (age 14–<16 years), lower BMI and waist circumference and lower LDL were reported in those who consumed a vegetable-rich, near-vegetarian diet (i.e. red meat, chicken and fish consumed <1/wk) compared with non-vegetarian adolescents (Grant et al. 2008). However, other lifestyle factors not related to diet (e.g. physical activity) also partially explain the health differences reported between vegetarians and non-vegetarians (Le & Sabate 2014).

Effect of vegetarian diets on exercise performance

Limited data from well-controlled studies have been conducted that examine the effect of vegetarian diets on sports performance in athletes who are long-standing vegetarians (Hanne et al. 1986; Snyder et al. 1989; Nagel et al. 1989), mainly because of the difficulty in access to an adequate sample size of vegetarian athletes. Hanne et al. (1986) found no differences in aerobic or anaerobic capacities of 49 (29 male, 20 female) vegetarian athletes, compared with non-vegetarian controls. Similarly, no differences in maximal oxygen uptake were observed between nine female athletes following a modified vegetarian diet (<100 g of red meat/wk), compared with controls on a mixed diet (Snyder et al. 1989). No differences in finishing time or finishing order were reported in 50 lacto–ovo vegetarian runners compared with controls consuming a conventional western diet competing in a 1000-km stage foot race (Nagel et al. 1989). Both groups in this study consumed diets with the same macronutrient ratio, thereby controlling for any inherent macronutrient differences that may have existed between diets.

No differences in endurance performance were evident in an intervention trial over 12 weeks in eight well-trained male endurance athletes who consumed a lacto–ovo vegetarian diet for six weeks and then a mixed diet for another six weeks (Richter et al. 1991; Raben et al. 1992). Diets were isocaloric and formulated to contain similar macronutrient ratios and energy intakes. The results also indicated no differences in other parameters measured (e.g. immune function, strength, muscle glycogen, hormone level), except testosterone, which decreased significantly in those following the lacto–ovo vegetarian diet. The authors suggested that either the sudden switch in diet or the high soluble fibre content of the vegetarian diet, which has the capacity to bind cholesterol (a precursor to testosterone synthesis), might influence testosterone reduction (Raben et al. 1992).

In these studies, the results have been equivocal but may be affected by small sample size, low power and small effect size. Hence, any potential effects of vegetarian diets on exercise performance and exercise capacity are largely extrapolated from the known nutrient limitations of vegetarian diets, based on other studies of the general population. Further research is needed to directly compare the effect of vegetarian diets on performance capacity in well-trained athletes.

Vegetarian diets tend to be higher in nutrient-rich carbohydrates, antioxidants and phytochemicals, which are conducive to enhancing muscle refuelling and maintaining health and wellbeing in athletes engaged in strenuous daily training. They are generally higher in fibre and lower in energy, protein, fat, vitamin B12, riboflavin, vitamin D, calcium and bioavailable iron and zinc than omnivorous diets. These features increase the risk of nutrient inadequacy and potential effects of suboptimal nutrient status on performance for vegetarian athletes.

Diet-related considerations for vegetarian athletes

The position statement of the American Dietetic Association states that structured vegetarian diets, including vegan diets, are healthful and nutritionally adequate (Craig & Mangels 2009). Appropriately planned vegetarian diets can provide sufficient energy and an appropriate level of macronutrient intakes to support performance and health in athletes (Venderley & Campbell 2006). Female vegetarians have a higher frequency of suboptimal micronutrient intakes, particularly iron, zinc, calcium and energy, than male vegetarians and omnivorous controls. Hence, female vegetarian athletes are at likely to be at higher risk of suboptimal micronutrient status compared with male vegetarian athletes.

Revisions to population nutrient standards in Australia, UK, USA and Canada have indicated that people consuming vegetarian diets have higher requirements for zinc, iron and possibly calcium to adjust for the low bioavailability of these nutrients in such diets (Institute of Medicine 2000). For vegetarian athletes, requirements for these nutrients may be even higher than population standards. See Chapters 9, 10 and 11.

Energy

Planned vegetarian diets can provide sufficient energy to meet the added demands of daily training and competition (Venderley & Campbell 2006). However, vegetarian athletes (particularly children and adolescents with high energy requirements) engaged in heavy endurance training and wanting to increase muscle mass may have difficulty meeting daily energy requirements. Legumes, wholegrain cereals and grains, soybeans and many vegetables and fruits are high-fibre, relatively low-fat foods and very filling. For the vegan, incorporating energy-dense foods such as nuts, dried fruit, tofu, tempeh, textured vegetable protein and commercially prepared meat analogues can help increase energy density. The inclusion of milk, dairy products and eggs in a lacto–ovo vegetarian diet provides more accessible food choice options to increase energy density in this type of vegetarian diet compared to vegan diets.

Protein

Protein needs are higher for athletes than for sedentary people. This increased requirement is mainly attributed to the use of protein as an energy source during exercise and the need for essential amino acids to support muscle protein synthesis following exercise and for muscle hypertrophy (see Chapter 4). Although vegetarians have reported lower daily protein intakes than meat eaters in studies of the general population, average protein intakes still meet the recommendation for total daily protein intake of two times the RDA (see Chapter 4). However, the amount and 'quality' of protein intake at each meal and adequate protein ingestion after exercise provides an important stimulus to maximise muscle protein synthesis.

Of some concern for vegan athletes is the risk of inadequate intake of all essential amino acids from lower protein quality-plant-based vegetarian diets and potentially compromised muscle protein synthesis. Animal food sources and their products (e.g. dairy foods) are categorised as complete proteins, which implies that they contain all essential amino acids. In contrast, the main plant sources of protein—grains (cereals), nuts, seeds and legumes—are categorised as incomplete proteins and are limiting in one or more of the essential nutrients. For example, nuts, seeds and grains have inadequate amounts of the essential amino acids, lysine and isoleucine, which are found in adequate amounts in legumes. While it was once thought that these food sources of plant proteins had to be consumed together in the same meal to provide all essential amino acids, this type of meal planning is no longer considered necessary to meet essential amino acid requirements (Venderley & Campbell 2006).Nevertheless, for vegans, a variety of plant protein sources should be eaten on the same day and post exercise to ensure the availability of all essential amino acids (Mahan & Escott-Stump 2012).

In summary, while research on vegetarian diets in athletes is limited, well-planned and nutrient-adequate vegetarian diets can in principle support the macronutrient and energy demands of daily training and competition (Tipton & Witard 2007). Even in elderly men following a vegetarian diet with adequate macronutrient and energy intake, increases in muscle strength and muscle cross-sectional area were evident after 12 weeks of resistance exercise, suggesting that a plant-based diet did not influence the physiological response to training (Haub et al. 2002). However, many adolescent and young adult athletes who have recently embraced a vegetarian-style diet do not recognise that some meal planning and food combination is required and that nutrient intake may be compromised by removing meat from the diet. Some athletes on restrictive vegetarian diets struggle to meet daily protein intake goals and include adequate protein-rich alternatives to animal foods at strategic times throughout the day (i.e. during the post-exercise period) to optimise the response from daily training.

Micronutrients

Vegetarian diets can provide as much or more total dietary iron than meat-based diets (Craig & Mangels 2009), although the iron stores of vegetarians are lower than omnivores because of the low bioavailability of iron from a vegetarian diet. The recommended iron intake for vegetarians is 1.8 times those of non-vegetarians to compensate for this reduced bioavailability (Institute of Medicine 2000; Department of Health and Ageing et al. 2006). For female vegetarian athletes, the Estimated Average Requirement (EAR) may be more than three times higher than EAR values (Institute of Medicine 2000). Such high levels are unlikely to be met by dietary iron intake in this high-risk group without appropriate dietary planning.

The iron in plant foods is mainly non-haem iron. Unlike iron from haem sources, absorption of non-haem iron is sensitive to both naturally occurring inhibitors in foods (e.g. phytates, calcium, polyphenols) and enhancers (e.g. vitamin C) of iron absorption (see Chapter 10). Vegetarian athletes, especially women, are at higher risk of developing iron depletion and possible iron deficiency anaemia, compared with male athletes (Rodriguez et al. 2009). This risk is associated with either low iron and energy intakes or both or low iron bioavailability. Routine monitoring of iron status is advised for vegetarian athletes, particularly during periods of growth, when exposed to altitude and/or when engaged in strenuous endurance training. Chapter 10 provides strategies for improving total dietary iron and optimising iron bioavailability. Iron supplements are warranted only where iron depletion or iron deficiency anaemia has been diagnosed.

Zinc intake may also be suboptimal in athletes who consume vegetarian diets. Lower zinc intakes have been reported in female vegetarians compared to non-vegetarians (Janelle & Barr 1995; Haddad et al. 1999). In males, no differences in zinc intakes were found in lacto–ovo vegetarians compared to omnivores (Wilson & Ball 1999), although this one study does not confirm this outcome. Similar to iron bioavailability, zinc bioavailability is low in a vegetarian diet and may contribute to zinc deficiency (Venderley & Campbell 2006). For vegetarians, cereal grains and their products are the main zinc suppliers, followed by legumes, nuts, soy products, eggs and finally dairy foods.

An omnivorous and lacto–ovo vegetarian diet can easily meet vitamin B12 recommendations, because active vitamin B12 is found exclusively in animal foods and their products (including dairy and eggs). No active vitamin B12 is naturally found in any plant foods, including meat analogues or fermented soy products (such as tempeh) or mushrooms, contrary to popular belief (Herbert 1988). Strict vegetarians following vegan, fruitarian or macrobiotic diets have lower serum vitamin B12 levels than lacto–

ovo vegetarians or those who occasionally eat meat (Obeid et al. 2002; Grant et al. 2008) and can slowly develop vitamin B12 deficiency (Herbert 1994). For this group of vegetarians, consuming foods fortified with vitamin B12 (e.g. soy and rice milks, some breakfast cereals and meat analogues) or a daily vitamin B12 supplement is needed (Craig & Mangels 2009).

The major source of riboflavin in the diet is milk and milk products. For strict vegetarian athletes (fruitarians and vegan) who exclude soy milk and soy-milk products, consuming adequate riboflavin may be difficult as soy is also a good source of riboflavin. Janelle & Barr (1995) reported lower intakes of riboflavin in eight female vegans compared with 15 lacto–ovo vegetarians and 22 non-vegetarian controls, suggesting that riboflavin may be a limiting nutrient in vegetarians who avoid dairy and soy products.

Vegan females have been reported to have significantly lower calcium intakes compared to lacto–ovo vegetarians and omnivores (578, 875 and 950 mg calcium/d respectively) (Janelle & Barr 1995). However, this finding is not consistent (Haddad et al. 1999). Lacto–ovo vegetarians usually report similar or higher calcium intakes than omnivores (Craig & Mangels 2009). Calcium bioavailability from vegetable sources, including soy milk and its products, is lower than that from dairy sources, because of the inhibitors naturally present in many plant foods (e.g. phytates found in wholegrain cereals, some vegetables and legumes, including soy). Other inhibitors (such as oxalates found in spinach and rhubarb) are found in calcium-rich foods and provide negligible absorbable calcium from these foods. Low-oxalate vegetables including kale, broccoli and bok choy are alternative sources of calcium and more readily absorbed. High-salt (sodium) diets and high-protein diets can also influence calcium balance by increasing calcium excretion and may need attention in vegetarian athletes who have difficulty meeting recommended calcium intakes (Weaver et al. 1999). Vegetarians who avoid eating dairy products should increase intake of calcium-fortified foods to help meet daily calcium recommendations or consider the use of a calcium supplement.

Performance supplements of benefit to vegetarian athletes

Creatine supplements (creatine monohydrate) can enhance training capacity and recovery from repeated high-intensity workouts involving isolated muscular efforts with short recovery times (see Chapter 16). The response to creatine supplementation and subsequent uptake into muscle is elevated in vegetarians who have low levels of muscle creatine as a consequence of meat avoidance (Burke et al. 2003). Meat is a good natural source of creatine. Lukaszuk and colleagues (2002) showed that switching to a lacto–ovo vegetarian diet from an omnivorous diet substantially reduced muscle creatine levels within three weeks. In theory, the potential performance benefits of creatine supplementation in vegetarians with low-creatine muscle stores is likely to be greater than in those athletes who eat meat. However, in the few performance studies undertaken on vegetarians who are non-athletes, the performance effects have been equivocal

(Shomrat et al. 2000; Burke et al. 2003). Whether trained athletes who are vegetarian show an enhanced uptake of creatine or are more responsive to its performance-enhancing effects than non-athletes is currently unknown.

Similar to muscle creatine stores, vegetarians have lower muscle carnosine concentration than omnivores (Everaert et al. 2011). Because carnitine is mainly found in foods of animal origin (or animal flesh), vegetarians (lacto–ovo vegetarians and vegans) obtain negligible amounts of carnosine from the diet (Stephens et al. 2011). Carnosine is a dipeptide of β-alanine and L-histidine. Supplementation of β-alanine can compensate for low meat and fish intake, significantly raising the muscle carnosine concentration (Harris & Stellingwerff 2013). The positive cause and effect dose–response relationship between increased muscle carnosine and performance would suggest that vegetarians will benefit from β-alanine supplementation to a similar or greater extent as omnivores (Harris & Stellingwerff 2013). Further research, however, is needed to confirm this hypothesis.

Summary

Given the paucity of research undertaken on vegetarian diets in athletes, definitive conclusions about the beneficial or unfavourable effects of a vegetarian diet on athletic performance cannot be made. A well-planned lacto–ovo vegetarian diet and vegan diet appear to be adequate in meeting nutrient requirements of athletes. A fruitarian-style or macrobiotic-style vegetarian diet, even if well planned, will not meet nutrient requirements. Vegetarian diets are inherently high in carbohydrate, which is conducive to restoring and maintaining adequate glycogen stores in athletes involved in high-intensity training programs. When total daily energy requirements are met, plant sources of protein can also meet the higher protein requirements of athletes participating in high-intensity training programs. Including protein-rich vegetarian options at each meal and during post-exercise recovery periods is needed to optimise the response to training. Energy-dense plant foods high in protein should be encouraged for athletes with high daily energy and protein requirements.

Although animal foods are good sources of protein, iron, zinc and vitamin B12, and dairy products are rich sources of calcium, alternative plant sources that are either naturally rich or fortified with these nutrients are readily available. Food sources fortified with vitamin B12 should be included in a vegan diet to help meet requirements. Vegetarians, including vegetarian athletes, are at higher risk of non-anaemic iron deficiency. Routine assessment of iron status is warranted in vegetarian athletes.

Vegetarians have lower stores of muscle creatine and carnitine than non-vegetarians, which may reduce their performance capacity for maximal and sustained high-intensity exercise bouts, respectively. As the benefits of creatine and β-alanine supplementation (as a precursor to carnitine) are inversely related to muscle content, vegetarians are likely to experience greater gains in maximal exercise and sustained high-intensity exercise following supplementation than omnivores, although these effects are speculative.

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CHAPTER ELEVEN

Micronutrients: vitamins, minerals and antioxidants

Mikael Fogelholm

11.1 Introduction

Vitamins are organic compounds required in very small amounts (from a few micrograms to a few milligrams daily) to prevent clinical deficiency and deterioration in health, growth and reproduction. A distinct feature of vitamins is that the human body is not able to synthesise them. Classification of vitamins is based on their relative solubility: fat-soluble vitamins (A, D, E and K) are more soluble in organic solvents and water-soluble vitamins (B-complex, beta-carotene and C) in water.

Most vitamins participate in processes related to muscle contractions and energy expenditure (see Table 11.1). Vitamins of the B-complex group (e.g. thiamin, riboflavin, vitamin B6, niacin, biotin and pantothenic acid) act as cofactors for enzymes regulating glycolysis, citric acid cycle, oxidative phosphorylation, b-oxidation (breakdown of fatty acids) and amino acid degradation. Folic acid and vitamin B12 are needed for haem synthesis. Ascorbic acid (vitamin C) activates an enzyme regulating biosynthesis of carnitine, which is necessary for fatty acid transportation from cell cytosol into mitochondria. Finally, antioxidant vitamins (mainly vitamins C and E) and some minerals participate in the buffer system against free radicals, which are produced by increased energy turnover.

TABLE 11.1 Summary of the most important effects of vitamins on body functions related to athletic training and performance

	Cofactors and activators for energy metabolism	Nervous function, muscle contraction	Haemoglobin synthesis	Immune function	Antioxidant function	Bone metabolism
Water-soluble vitamins						
B group Thiamin	X	X				
Riboflavin	X	X			X[a]	
Vitamin B6	X	X	X	X		
Folic acid	X	X				
Vitamin B12	X	X				
Niacin	X	X				

05:34:55.

Pantothenic acid	X					
Biotin	X					
Vitamin C			X	X		
Vitamin A (beta-carotene)				X		
Fat-soluble vitamins						
Vitamin A				X	X	
Vitamin D						X
Vitamin E				X	X	

[a]Present in coenzymes for endogenous antioxidant actions

Minerals are inorganic substances found naturally on the earth. Based on their daily requirements, minerals are usually classified as macrominerals (e.g. sodium, potassium, calcium, phosphorus, magnesium), or trace elements (e.g. iron, zinc, copper, chromium, selenium). The daily dietary requirement for macrominerals is more than 100 mg/d, whereas trace elements are needed in much smaller quantities (less than 20 mg/d).

Several minerals and trace elements, such as magnesium, iron, zinc and copper, act as enzyme activators in glycolysis, oxidative phosphorylation and in the system responsible for maintenance of acid-base equilibrium (Table 11.2). Iron is needed for haem synthesis. Minerals (electrolytes) also affect muscle contraction.

TABLE 11.2 Summary of the most important effects of minerals on body functions related to athletic training and performance

	Cofactors and activators for energy metabolism	Nervous function, muscle contraction	Haemoglobin synthesis	Immune function	Antioxidant function	Bone metabolism
Sodium		X				
Potassium		X				
Calcium		X				X
Magnesium	X	X		X		X
Trace elements						
Iron	X		X	X	X[a]	
Zinc	X			X	X[a]	
Copper	X			X	X[a]	
Chromium	X					
Selenium				X	X[a]	

[a]Present in coenzymes for endogenous antioxidant actions

As evident from the introduction, micronutrients (vitamins and minerals) are essential for life. A well-balanced diet covers the needs for all micronutrients in healthy humans (NHMRC et al. 2006). Nevertheless, vitamin and mineral supplements—including vitamin C (especially), the B-complex vitamins, vitamin E and iron—are frequently used by athletes (Tsitsimpikou et al. 2009; Lun et al. 2012; Sousa et al. 2013). The most common motivations for vitamin supplementation are to enhance recovery and improve sports performance (Sousa et al. 2013).

The use of vitamin and mineral supplements at intakes below the Upper Intake Level (UIL) is only justified and necessary if a normal diet is unable to maintain an athlete's micronutrient status; if micronutrient supplements improve nutritional status and physiological functions of an athlete; and—most importantly—if supplements enhance athletic performance directly or indirectly (e.g. by enhancing recovery or by preventing infectious diseases).

This chapter aims to answer the following questions:

- Are micronutrient requirements increased in athletes?
- Are there any significant differences between indices of micronutrient status between athletes and untrained controls?
- If the supply of one or several micronutrients is marginal, would an athlete's functional capacity be less than optimal?
- If micronutrients are given in excess of population nutrient dietary recommendations, would this improve indices of micronutrient status, body functions and athletic performance?

Reviews on micronutrients and sports are addressed in chapters on measuring nutritional status of athletes (Chapter 2), calcium and vitamin D (Chapter 9) and iron (Chapter 10). Therefore, the above issues are not covered in this chapter. Chapter 16 provides additional information on the supplementation practices of athletes. Commentary 6 addresses the known effects of taking supplements of antioxidant vitamin and minerals, as a means to reduce the potential for oxidative damage associated with the production of free oxygen radicals termed reactive oxygen species (ROS) induced by high-intensity exercise.

11.2 Measuring vitamin and mineral status in athletes

11.2.1 What is nutritional status?

An organism has an adequate nutritional status if its cells, tissue, organs and anatomical systems, working together, can undertake all nutrient-dependent functions. The relationship between micronutrient supply and functional capacity is 'bell-shaped' (see Figure 11.1) (Brubacher 1989). The core of the above relationship is that the output (functional capacity) is not improved after the ‘minimal requirement for maximal output’ is reached. In contrast, too high an intake may in some cases reduce the output below the maximal level.

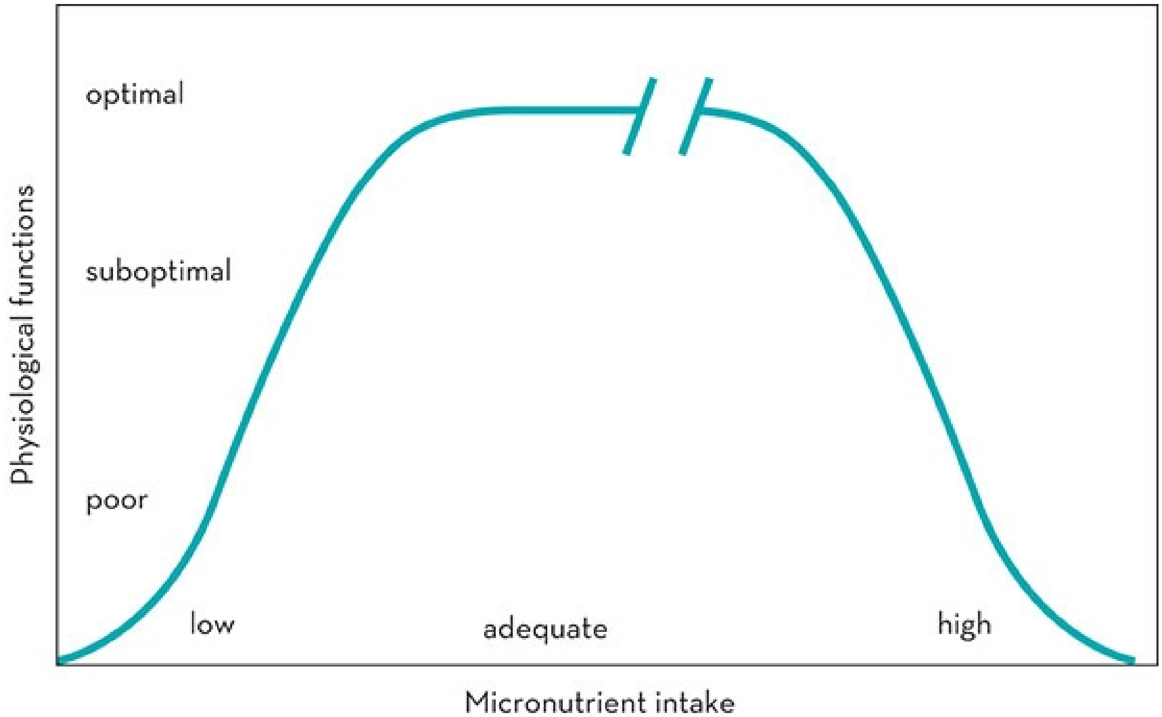

Figure 11.1 The relation between micronutrient supply and functional capacity

Source: Adapted from Brubacher 1989

Different body functions (single biochemical reactions, metabolic pathways, the function of anatomical systems and the function of the hosts themselves) reach their maximal output at different levels of supply. In other words, the supply (intake) needed for optimal function of an anatomical system (such as the muscle) may be quite different from the supply needed to maximise the activity of a single enzyme (Solomons & Allen 1983).

Short-term inadequacy of vitamin and mineral intake is characterised by the lowering of nutrient concentrations in different tissues and the lowering of certain enzyme activities (see Figure 11.2). However, functional disturbances (such as decreased physical performance capacity) may appear later (Solomons & Allen 1983; Fogelholm 1995). In the opposite case, very large intakes increase the body pool and activity of some enzymes, but do not necessarily improve functional capacity (Fogelholm 1995).

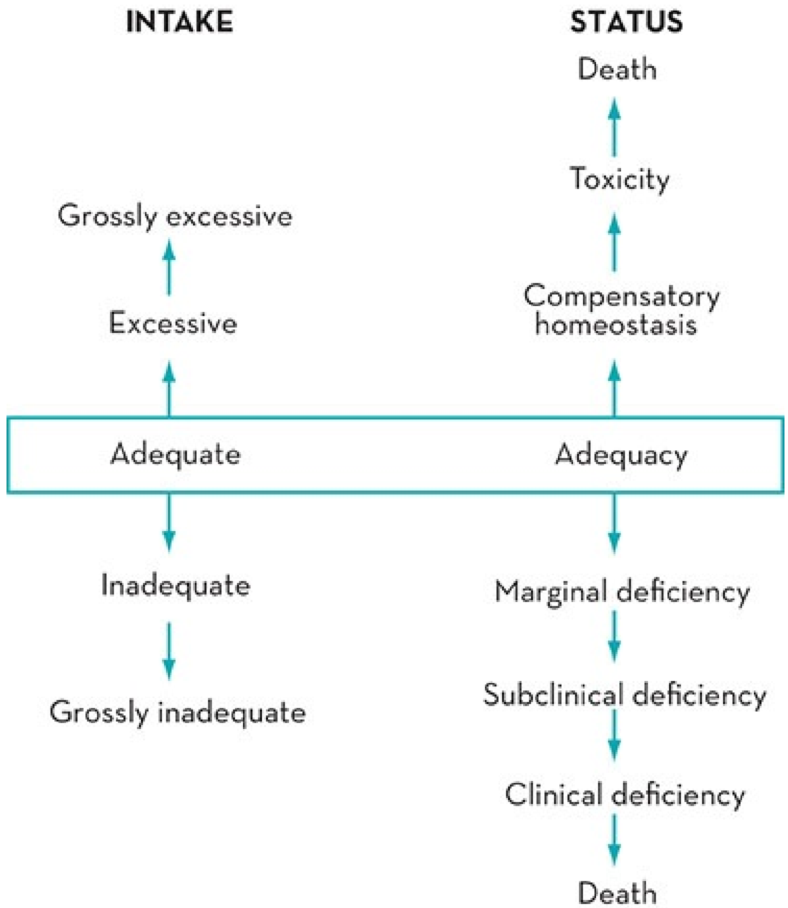

Figure 11.2 Dietary micronutrient intake and stages of micronutrient status

11.2.2 Assessment of nutritional status

Ideally, nutritional status should be assessed by combining the information obtained from different (clinical, anthropometric, dietary, biochemical) markers (see Chapter 2). Gross clinical deficiency of nutrients causes detectable changes in, for instance, the skin or eyes. Anthropometric and body composition data are indicators of energy balance. Chronic negative energy balance is also likely to be associated with inadequate micronutrient intake. Assessment of dietary intakes of individuals and groups provide only estimates of ‘usual’ intakes of foods or nutrients and are not accurate measures (see Chapter 2). Uncertainty about an individual’s nutritional requirements and inaccuracies related to dietary assessment, therefore, precludes the use of dietary intakes as a sole indicator of nutritional status (Aggett 1990).

Serum and plasma specimens are easy to collect and analyse, and therefore suitable for use with large numbers of people. Some water-soluble vitamins respond quickly to dietary intake (Singh et al. 1992b), and positive correlations between dietary intake and plasma concentration of ascorbic acid have been reported (Tur-Mari et al. 2006). The concentrations of minerals or trace elements in blood, serum or plasma are under strong homeostatic control (Solomons & Allen 1983), which makes them quite insensitive to marginal nutrient deficiency.

Also, factors unrelated to nutritional status, such as haemoconcentration, haemodilution or diurnal variation, can also affect blood chemistry. Therefore a single measurement of serum concentrations of micronutrients should be interpreted with caution (Tur-Mari et al. 2006).

Because serum and plasma concentrations are rather insensitive to marginal micronutrient deficiency, substantial research has been undertaken to identify other body compartments that would better represent the body's micronutrient status. Mononuclear leucocytes appear to be more sensitive indicators of vitamin C (Tur-Mari et al. 2006), magnesium (Weller et al. 1998) and zinc (Dolev et al. 1995) status, compared to the respective indicators in serum or plasma. Unfortunately, these interesting methods are still not used routinely. Data on micronutrient concentrations in the leucocytes of athletes are too scarce to lead to any definitive conclusions (Peake 2003).

Erythrocyte enzyme activation co-efficients (E-AC) are used as indicators for the status of thiamin (enzyme transketolase: TK), riboflavin (glutathione reductase: GR) and vitamin B6 (aspartate aminotransferase: AST, also called glutamate oxaloacetate transaminase: GOT). In these methods, the activity of an enzyme is measured with and without in vitro cofactor (vitamin B) saturation. The activation co-efficient (AC) is calculated by dividing saturated activity by basal activity (Bayomi & Rosalki 1976). High AC indicates inadequate nutritional status. These techniques are extremely interesting, because they are truly functional rather than static, as is the case with nutrient concentrations in plasma and serum (Solomons & Allen 1983).

Both physiological and analytical factors affect the sensitivity and specificity of blood tests as indicators of vitamin status. These factors lead to a notable analytical variation between laboratories. Ideally, each laboratory should make its own reference interval, or ascertain that the chosen reference interval is made by precisely comparable methods. Otherwise, the reported number of athletes with 'subclinical deficiency' could be wrong (Fogelholm et al. 1993b).

The enzymes that are used for assessment of thiamin, riboflavin and vitamin B6 status represent one metabolic pathway in the erythrocytes. Therefore, they do not necessarily indicate the function of other metabolic pathways in other tissues, for example glycolysis or the Kreb's cycle in the muscles. Moreover, when people follow a typical western mixed diet, erythrocyte concentrations of TK, GR and AST are not saturated in vivo by their cofactor. Consequently, after supplementation, their basal activity increases and activation co-efficient decreases (Guilland et al. 1989; Fogelholm et al. 1993b). However, these changes are not necessarily associated with improved physical performance (see section 11.6).

The relationships between exercise, free radicals and lipid peroxidation and oxidative damage have raised considerable interest. Breath pentane, thiobarbituric acid reactive substances (TBARS) and malondialdehyde (MDA) are indicators of lipid peroxidation caused by free radicals in humans. Unfortunately, these markers indicate neither the source nor the timing of lipid peroxidation (Alessio 1993), nor do they permit quantitative correlation of peroxidation (Weber 1990). MDA and TBARS, for instance, may arise from other pathways besides lipid peroxidation (Weber 1990). Moreover, during sleep and light daily activities, products of lipid peroxidation may be produced and remain in regions where blood-flow is limited (Weber 1990). When blood-flow during exercise is increased and blood is redistributed to new regions, these products may then be ‘washed out’.

11.3 Effects of exercise on vitamin and mineral requirements of athletes

11.3.1 Requirements of water-soluble vitamins and antioxidants

Physical activity increases energy expenditure. Van der Beek, in his 1991 landmark review, proposed that high metabolic activity might increase the turnover of several vitamins of the B-complex group. Indeed, some older data supported this concept. For example, Sauberlich and colleagues (1979) found that thiamin requirements in male subjects were 30% higher when daily energy intake was 15.1 MJ (3600 kcal), compared with 11.7 MJ (2600 kcal). However, an energy-related requirement for thiamin has also been questioned (Caster & Mickelsen 1991). Vitamin losses through sweat are minimal, even in physically active people (Brotherhood 1984).

Energy production involves reduction of molecular oxygen (O_2) in active skeletal muscles. The reduction is not always complete, and about 2–5% of the molecular oxygen turns into a free radical (superoxide radical, O_2^-) (Sjödin et al. 1990). The free radicals, called *reactive oxygen species* (ROS), are unstable, reactive and potentially harmful chemical substances with unpaired electrons in their outer orbitals (Sen 1995). An excessive production of free radicals, or an insufficient protection against them, has been linked to cell and mitochondria membrane damage and dysfunction, deterioration of the immune defence system, ageing, cancer and atherosclerosis (Jacob & Burri 1996). This radical-induced damage is termed oxidative stress and, in extreme cases, can lead to cell death. Excessive ROS may also induce oxidative damage to muscle proteins, diminish muscle force production and contribute to muscle fatigue during prolonged submaximal exercise for >30 minutes (Reid 2008; Powers et al. 2011).

In humans, indirect evidence of lipid peroxidation of the cell membrane (and hence the potential for oxidative stress) during physical exertion includes increased pentane exhalation in the breath (Simon-Schnass & Pabst 1988; Leaf et al. 1997), elevated MDA in serum and erythrocytes (McBride et al. 1998; Nakhostin-Roohi et al. 2008), increased TBARS (Machefer et al. 2007) and increased levels of oxidised glutathione (Goldfarb et al. 2007). Not all studies show an exercise-induced increase in indices of oxidative stress (Laaksonen et al. 1995; Margaritis et al. 1997). It is, however, apparent that lipid peroxidation is increased if the preceding exercise is intense and strenuous enough (Leaf et al. 1997).

All cells produce free radicals and have a sophisticated endogenous antioxidant system composed of enzymatic and non-enzymatic antioxidants which scavenge free radicals. These endogenous antioxidants work interactively with dietary antioxidants (mainly vitamins C, E and beta-carotene) to provide a protection system against free radical damage. Mena and colleagues (1991) showed that highly trained athletes have higher activities of several antioxidant enzymes, but the effects of moderate training are apparently much smaller (Tiidus et al. 1996). An increased need for endogenous defence against free radicals may, nevertheless, increase the requirement for antioxidant vitamins (Sen 1995; Jacob & Burri 1996). On the other hand, well-trained athletes may have a more developed endogenous antioxidant system than a sedentary individual, and hence the additional need for exogenous antioxidants (such as vitamins) may not be very high (Kanter 1998).

11.3.2 Requirements for minerals

The losses of magnesium and zinc through sweat may be important. The exercise-induced concentration of magnesium in sweat has varied between 3 and 36 mg/L in different studies (Beller et al. 1975; Costill et al. 1976, 1982; Verde et al. 1982; Wenk et al. 1989; Montain et al. 2007). Strenuous exercise increases magnesium requirements by 10–20% (Nielsen & Lukaski 2006). Similarly, the zinc concentration in sweat has varied between 0.6 and 1.5 mg/L (Beller et al. 1975; Aruoma et al. 1988; DeRuisseau et al. 2002; Montain et al. 2007), which would mean a 20–40% increase in the daily needs for each additional litre of sweat. Unfortunately, the interpretation of these data is difficult, because of several problems in sweat analyses. For instance, the composition of sweat varies with the collection site (Gutteridge et al. 1985; Aruoma et al. 1988). Moreover, it is uncertain whether a short sampling period represents long-term conditions. For instance, iron and zinc concentrations in sweat may decrease considerably after the first hour of exercise (DeRuisseau et al. 2002; Montain et al. 2007). The variation between different studies reflects the above-mentioned methodological problems.

In addition to sweating, micronutrients may be lost in urine or faeces. On one day of physical exercise, Deuster and colleagues (1987) found elevated (~22%) magnesium excretion compared with controls. Similarly, higher excretion of magnesium (21–76%) (Singh et al. 1990a; Nuviala et al. 1999) and zinc (Lichton et al. 1988; Deuster et al. 1989) has been found in trained athletes compared with controls. Results are not without contradiction: Nuviala and colleagues (1999) reported similar urinary zinc excretion in female athletes and untrained controls, and Zorbas and colleagues (1999) found increased copper excretion in trained subjects only when their activity was markedly restricted. Moreover, Dressendorfer and colleagues (2002) did not find increased urinary excretion of magnesium, iron, zinc and copper during very high-intensity training in male cyclists. Finally, because of an increase in free radical formation during strenuous exercise, the requirements for zinc and copper (in Zn_2Cu_2-superoxide dismutase enzyme) and selenium (in glutathione peroxidase) may also be greater than normal in physically active people (Koury et al. 2004).

In summary, it is biologically plausible to assume that athletic training, especially if it leads to very high energy expenditure, increases the requirements for micronutrients (at least through losses in sweat and urine, and perhaps in faeces) and through increased free radical production. Nevertheless, the data to date are insufficient to allow any quantification of micronutrient requirements in athletes. Because of a wide safety margin, the relevant cut-offs from the Dietary Reference Intakes (DRI)/Nutrient Reference Values (NRV) for vitamins and minerals, with adjustments for iron, can still be used for assessing the probability of nutrient inadequacy of dietary intakes in athletes, although with caution. Guidelines for using the appropriate cut-off as benchmarks for assessing nutrient requirements of individuals and groups are found in Chapter 2, section 2.9.

11.4 Biochemical indicators of vitamin and mineral status in athletes

Most biochemical measures that are indices of nutrient status in individuals are markers for risk and do not usually confirm a true diagnosis of nutrient overload or deficiency without other clinical and dietary assessment measures (see Chapter 2). One-off biochemical measures in individuals who are at the lower end or just below population reference ranges may be normal for some individual athletes. Biochemical measures of vitamins and minerals are used to confirm a diagnosis in individuals only in moderate to severe nutrient depletion states, which is unusual in an athlete population, with some exceptions (e.g. iron and vitamin D). They are used in research to compare relative differences between athletes and non-athlete controls and to allow an evaluation of nutrient requirements of groups of athletes involved in varying types of physical activity.

11.4.1 Biochemical indicators of water-soluble vitamin status

Vitamin B group

Cross-sectional studies comparing vitamin B status in athletes and untrained controls give inconclusive results. Guilland and colleagues (1989) found higher erythrocyte transketolase activation co-efficients (E-TKAC) in athletes than in untrained controls, whereas Fogelholm and colleagues (1992a, 1992b) found both lower and comparable E-TKAC in athletes. Keith and Alt (1991) found higher erythrocyte glutathione reductase activation co-efficients (E-GRAC) in female athletes, but Guilland and colleagues (1989) reported levels comparable to the controls. Only Guilland and colleagues (1989) have reported erythrocyte aspartate amino transferase activation co-efficient (E-ASTAC) values for both athletes and untrained controls; no difference was observed. All differences reported in the above studies were small, however, and most probably without any functional significance.

The effects of a 24-week fitness-type exercise program on micronutrient status in previously untrained female students have been studied (Fogelholm 1992). Indicators for thiamin and riboflavin status were unchanged throughout the study. E-ASTAC increased in exercise and decreased in control groups, which may indicate marginally impaired vitamin B6 status in the exercise group. However, the magnitude of the change (from 2.02 to 2.11) is not likely to affect functional capacity (Fogelholm et al. 1993a, 1993b).

The effects of varying the training volume on E-TKAC have also been studied in Finnish cross-country skiers (Fogelholm et al. 1992b). Despite very strenuous training in August and November, and clearly lighter training in February and in May, E-TKAC showed no seasonal variation.

Plasma total vitamin B6 concentration and/or pyridoxal-5′-phosphate concentration increase during short-term exercise, but decrease during very prolonged physical exertion (Crozier et al. 1994; Leonard & Leklem 2000). The physiological significance of these changes is not known.

Vitamin C

Reported data on vitamin C status in physically active people and untrained individuals are scarce. Serum ascorbic acid concentrations were the same (Guilland et al. 1989; Fogelholm et al. 1992a; Rokitzki et al. 1994a) or higher (Fishbaine & Butterfield 1984) in athletes than in controls. The pooled mean concentrations for serum or plasma ascorbic acid in the above studies were 59 and 56 µmol/L for athletes (n = 533) and controls (n = 193), respectively. Moreover, studies have not shown any differences between athletes' and controls' urinary excretion of ascorbic acid (Peake 2003), which further suggests that athletic training does not have any special negative effects on vitamin C status that a slightly increased dietary intake could not counterbalance.

11.4.2 Biochemical indicators of fat-soluble vitamin status

Vitamin A

Serum concentration for beta-carotene, the provitamin for retinol, is more responsive to marginal changes in status than retinol. Takatsuka and colleagues (1995) reported that hard physical activity was associated with lower beta-carotene concentrations in Japanese men and women. The results in this cross-sectional study were adjusted for age, body mass index (BMI), diet, smoking, serum cholesterol and serum triglycerides. Guilland and colleagues (1989) found comparable beta-carotene concentrations in athletes and controls, whereas Watson and colleagues (2005) reported higher concentration in athletes.

Vitamin D

Intake studies over the last 15 years have found that many athletes do not come close to meeting the population recommendations for vitamin D intake (i.e. 200–400 IU/d) (Larson-Meyer & Willis 2010), particularly the newly revised US RDA of 600 IU/d (Ross et al. 2010). Although the most probable reason for suboptimal vitamin D status is insufficient exposure to UVB light, poor vitamin D intake may still contribute to low vitamin D status (see Chapter 9).

Biochemical indices of vitamin D status in athletes have only been reported in a small number of studies. When serum 25(OH) vitamin D concentration was used as the indicator for vitamin D status, between 40% and 90% of the athletes studied showed deficient status (Lovell 2008; Willis et al. 2008; Constantini et al. 2010). These figures are similar to or slightly higher than the status in the general population and clearly a concern. Low vitamin D status is expected to be highest in those athletes who train inside and have little exposure to the sun in general, and is certainly of concern to bone health in young athletes (Peeling et al. 2013). See Chapter 9 for a review of the effects of low vitamin D status on performance and bone health.

Vitamin E

Moderate physical activity has not been shown to have any negative effects on vitamin E (alpha-tocopherol) concentration in plasma (Kitamura et al. 1997; Thomas et al. 1998) or in the muscle (Tiidus et al. 1996). Only a few studies have compared plasma alpha-tocopherol levels between competitive athletes and controls, and the results are contradictory. Watson and colleagues (2005) reported higher vitamin E levels in athletes than controls, Guilland and colleagues (1989) reported similar results, whereas Karlsson and colleagues (1992) reported lower levels in athletes than controls. Nevertheless, in this last study, the ratio between the alpha-tocopherol concentration between athletes and controls was the same as the ratio between free cholesterol concentration in plasma. Because alpha-tocopherol is transported in lipoproteins together with cholesterol, dissimilar blood lipid profiles in athletes and controls might explain the difference found by Karlsson and colleagues (1992).

11.4.3 Biochemical indicators of mineral status

Iron

Biochemical indicators of iron status in athletes have been studied extensively; see Chapter 10, Practice tips.

Magnesium

The data on mineral status in athletes and controls, published between 1980 and 1994, have been pooled in an analytical review (Fogelholm 1995). The mean serum magnesium concentration in athletes (n = 516 in seven studies) was 0.84 mmol/L, and in controls (n = 251) was 0.85 mmol/L. Later studies reported that athletes had similar (Crespo et al. 1995; Nuviala et al. 1999) or even higher (Nuviala et al. 1999) serum magnesium concentration, compared to sedentary controls. Lower (Casoni et al. 1990) or similar (Fogelholm et al. 1991) erythrocyte magnesium concentrations were found in athletes, compared with untrained controls. In a controlled intervention, a 24-week fitness-type exercise program did not affect serum or erythrocyte magnesium content in previously untrained young women (Fogelholm 1992).

Zinc

The mean serum zinc concentration, pooled from studies published between 1980 and 1994, was 13.5 μmol/L in athletes (*n* = 587 in 11 studies) and 13.7 μmol/L in controls (*n* = 244) (Fogelholm 1995). More recent studies have also found similar serum zinc levels in athletes and sedentary controls (Crespo et al. 1995; Nuviala et al. 1999). Following increased physical activity, three longitudinal studies found a negative time-trend for serum or plasma zinc (Miyamura et al. 1987; Lichton et al. 1988; Couzy et al. 1990). In contrast, serum zinc increased in competitive sailors during a three-week transatlantic race (Fogelholm & Lahtinen 1991). Other studies have not found any associations between training volume and serum zinc concentration (Lukaski et al. 1990; Ohno et al. 1990; Fogelholm 1992; Fogelholm et al. 1992a; Dolev et al. 1995).

Data on zinc levels from other biomarkers are scarce, but the few findings are interesting. Several studies have reported a positive association between high or increased physical activity and erythrocyte zinc concentration (Ohno et al. 1990; Singh et al. 1990b; Fogelholm et al. 1991; Fogelholm 1992). Dolev and colleagues (1995) reported maintenance of zinc concentration in erythrocytes, but an increased concentration in mononuclear leucocytes, in male military recruits during an 11-week training program. The increase in erythrocyte and leukocyte zinc level might be related to increased amounts of intracellular zinc-dependent enzymes, such as superoxide dismutase.

Copper, chromium and selenium

In studies published between 1980 and 1994, the pooled mean results of plasma copper concentration tended to be higher for females and for athletes: 16.3 and 15.1 μmol/L for female athletes and controls, and 14.7 and 14.1 μmol/L for male athletes and controls, respectively (Fogelholm 1995). Similar serum copper and ceruloplasmin results for female collegiate athletes, compared with controls, were also reported more recently (Gropper et al. 2003). Data on other trace elements are even scarcer. Anderson and colleagues (1988) reported similar plasma chromium concentrations in athletes and untrained controls. A competitive sailing crew had lower serum selenium concentration than controls (Fogelholm & Lahtinen 1991). Nevertheless, even the sailing crew's results were clearly within the reference range for selenium.

In summary, the available indices of magnesium, zinc, copper, chromium and selenium do not suggest compromised status in physically active people. Because of the lack of sensitivity of these indices, this interpretation must be taken with caution.

11.5 Does marginal deficiency of vitamins and minerals affect physical performance?

Studies involving marginal deficiency of vitamins and minerals test the hypothesis that physical performance is impaired when the status of one or several micronutrients is indicative of marginal deficiency.

11.5.1 Vitamins

Thiamin

Earlier studies have shown that subclinical thiamin deficiency is associated with increased exercise-induced blood lactate concentrations, especially after a pre-exercise glucose load (Sauberlich 1967). The deterioration of physical capacity in marginal deficiency, however, is less evident. Despite a five-week thiamin-depleted diet, Wood and colleagues (1980) did not find decreased working capacity in male students. E-TKAC decreased significantly, showing that the activity of this enzyme was affected faster than the activity of the enzymes of glycolysis and the citric acid cycle in working muscles.

Riboflavin

Significant changes in riboflavin supply affect both muscle metabolism and neuromuscular function. Data on the effects of marginal riboflavin supply are, however, scarce. In three studies (Belko et al. 1984, 1985; Trebler Winters et al. 1992), a four- to five-week marginal riboflavin intake lowered E-GRAC, but no relation between vitamin status and aerobic capacity was found. Similarly, Soares and colleagues (1993) did not find changes in muscular efficiency during moderate-intensity exercise after a seven-week period of riboflavin-restricted diet. Decreased urinary riboflavin excretion might be one mechanism to conserve riboflavin and prevent changes in riboflavin-dependent body functions during marginal depletion, which may explain the outcome of these studies.

Vitamin B6

In male wrestlers and judo athletes, a decrease in E-ASTAC indicated deteriorated vitamin B6 supply during a three-week weight-loss regimen (Fogelholm et al. 1993a). Maximal anaerobic capacity, speed or strength was, however, not affected. Also, Coburn and colleagues (1991) showed that the muscle tissue was quite resistant to a six-week vitamin B6 depletion.

Vitamin C

In one study (Van der Beek et al. 1990), a vitamin C-restricted diet reduced whole blood ascorbic acid concentration. Nevertheless, the marginal vitamin supply in the blood did not produce any significant effects on maximal aerobic capacity or lactate threshold in healthy volunteers. In contrast to the above study, Johnston and colleagues (1999) reported that a three-week experimental vitamin C depletion was associated with reduced work efficiency during submaximal exercise and that work performance increased after repletion with vitamin C supplements (two weeks, 500 mg/d).

Vitamin D

Although some studies have indicated marginal vitamin D deficiencies and intakes in athletes (Willis et al. 2008), there are no studies published that investigate the potential effects of marginal deficiency on athletic performance. Given the multiple roles of vitamin D in human metabolism, a prolonged inadequate intake and low biochemical status can increase the susceptibility for bone fracture (Tenforde et al. 2010), which has been confirmed in population studies (see Chapter 9).

Another link between inadequate vitamin D status and performance could be through impaired immune function and increased susceptibility to upper respiratory tract infections (Laaksi et al. 2007). Recently, He et al. (2013) studied the relationship between vitamin D status (assessed from serum vitamin D concentration) and respiratory infections in 225 endurance athletes during winter-season training. They found that athletes with low vitamin D status (serum 25-hydroxyvitamin D (25(OH) concentration <30 nmol/L) had significantly increased incidence of respiratory infections. Unfortunately, no indications of any dietary intakes were given. However, the results indicate that a randomised supplementation study with vitamin D in athletes is needed (Angeline et al. 2013).

Low vitamin D status may also compromise muscle function and has also been associated with muscle pain and a decrease in muscle strength. See Chapter 9, sections 9.8.1 and 9.8.2 for more detail.

Multivitamins

Finally, a combined depletion of thiamin, riboflavin, vitamin B6 and ascorbic acid has been found to affect both E-TKAC and aerobic working capacity (Van der Beek et al. 1988). Because of multiple depletion, the independent role of the single vitamins could not be demonstrated.

11.5.2 Minerals

Iron deficiency without anaemia may have a negative impact on endurance performance (see Chapter 10). The effects of marginal status of other minerals on physical performance in healthy volunteers have been studied only rarely. Lukaski and Nielsen (2002) studied the effects of dietary magnesium restriction in postmenopausal women. In magnesium-depleted status, the energy cost during submaximal exercise was increased, compared with adequate magnesium intake. Unfortunately, there are no comparable data with athletes or physically fit young adults.

Van Loan and colleagues (1999) investigated the effects of induced zinc depletion on muscle function in eight male subjects who were fed a formula diet (<0.5 mg Zn/d for 33 or 41 d). The mean serum zinc concentration decreased from an initial 11.0 µmol/L to 3.4 µmol/L at the end of depletion. Muscle endurance (total work capacity in isokinetic exercise tests) declined, but peak force was not affected. The authors speculated that the effects observed could have been related to decreased activity of lactate dehydrogenase, a zinc-dependent enzyme, and a concomitant increase in blood lactate concentration. In a more recent study, marginal zinc deficiency, induced by a low zinc diet, lowered peak oxygen uptake and respiratory exchange ratio during a maximal exercise test in young adult men (Lukaski 2005).

11.6 Effects of supplementation on biochemical indices of micronutrient status and physical performance

It is assumed that the current recommended dietary allowances and dietary reference intakes are appropriate even for athletes (American College of Sports Medicine 2010). Supplementation is warranted only if clear medical, nutritional and public health reasons are present (e.g. iron to treat iron deficiency anaemia or folic acid to prevent birth defects). Although the available data suggest that micronutrient deficiencies in athletes are rare and no more common than in untrained subjects, depletion studies show that even marginal micronutrient deficiency conditions may impair physical performance. Even a small potential risk of marginal micronutrient deficiency may be of concern for athletes and their coaches. Before making recommendations for or against micronutrient supplementation in athletes, two fundamental questions need to be reviewed:

- Are the observed indicators of micronutrient status compatible with optimal athletic performance?
- Would a nutrient intake beyond the daily dietary recommendations (supplementation) improve physical capacity?

11.6.1 Water-soluble vitamin supplementation

Thiamin

Thiamin supplementation (>7.5 mg/d) improves the activity of erythrocyte TK (Guilland et al. 1989; Fogelholm et al. 1993b). Nevertheless, despite improved indicators of thiamin status, several studies have shown that thiamin supplementation did not improve functional capacity in athletes who were not deficient (Telford et al. 1992a, 1992b; Webster 1998) or in young, moderately trained adults (Singh et al. 1992a, 1992b; Fogelholm et al. 1993b).

Riboflavin

Riboflavin supplementation improves the activity of erythrocyte GR (Weight et al. 1988b; Guilland et al. 1989; Fogelholm et al. 1993b) in athletes or trained students, even without indications of impaired vitamin status (Weight et al. 1988b). One early study suggested that supplementation and improved riboflavin status were related to improved neuromuscular function (Haralambie 1976). In contrast, later studies did not find any beneficial effects of riboflavin supplementation on maximal oxygen uptake (Weight et al. 1988a; Singh et al. 1992a, 1992b) or exercise-induced lactate appearance in the blood (Weight et al. 1988a; Fogelholm et al. 1993b).

Vitamin B6

Chronic supplementation of vitamin B6 increases the erythrocyte ASAT activity (Guilland et al. 1989; Fogelholm et al. 1993b) and plasma pyridoxal-5-phosphate (PLP) concentration (Coburn et al. 1991) even in healthy subjects. Supplementation of vitamin B6, in combination with other B-complex vitamins, has improved shooting target performance (Bonke & Nickel 1989) in male athletes. In contrast, a number of other studies did not find any association between improved indicators of vitamin B6 status and maximal oxygen uptake (Weight et al. 1988a, 1988b), exercise-induced lactate appearance in blood (Manore & Leklem 1988; Weight et al. 1988a, 1988b; Fogelholm et al. 1993b) or other tests of physical performance (Telford et al. 1992a, 1992b; Virk et al. 1999). In fact, an increase in the biochemical indicators of vitamin B6 status is not necessarily associated with a change in intramuscular vitamin B6 content (Coburn et al. 1991).

It appears that vitamin B6, either as an infusion (Moretti et al. 1982) or given orally as a 20 mg/d supplement (Dunton et al. 1993), stimulates growth hormone production during exercise. The hypothetical mechanism behind this effect is that PLP acts as the coenzyme for dopa decarboxylase, and high concentrations might promote the conversion of L-dopa to dopamine (Manore 1994). The physiological significance of the above effect on muscle growth and strength is not known.

Folate and vitamin B12

There are only a few studies linking folic acid or vitamin B12 supply to sports-related functional capacity. Folate supplementation and increased serum folate concentration did not affect maximal oxygen uptake (Matter et al. 1987), anaerobic threshold (Matter et al. 1987) or other measures of physical performance (Telford et al. 1992a, 1992b). Together with thiamin and vitamin B6 supplementation, elevated intake of vitamin B12 was, however, associated with improved target shooting performance (Bonke & Nickel 1989).

Other B vitamins (such as thiamin with pantothenic acid)

Data on supplementation of other B vitamins and performance are even scarcer. Webster (1998) reported no effect of a combined thiamin and pantothenic acid supplementation on a 2000 m time trial by cycle ergometer.

Vitamin C

In a review of supplementation studies, vitamin C supplementation ranging from 85 to 1500 mg/d for a period from one day to eight months have been reported (Peake 2003). About half of these studies showed no significant changes in plasma ascorbic acid concentration, whereas the remaining studies demonstrated significant increases. The non-responses may be attributed to higher dietary vitamin C intake that is close to saturation of plasma concentration; however most studies rarely report retrospective intakes.

In an early review, Gerster (1989) concluded that the majority of studies have not shown any measurable effects of vitamin C supplementation on maximal oxygen uptake, lactate threshold or exercise-induced heart rate in well-nourished subjects. In 2007, one study showed reduced levels of blood lactate concentration after a maximal exercise test after 90-day supplementation with vitamins C (1000 mg/d) and E (500 mg/d) and beta-carotene (30 mg/d) (Aguiló et al. 2007). However, a single study cannot be used for any recommendations about performance effects.

An interesting aspect of interaction between vitamin C and physical activity is related to the proposed connection with upper respiratory tract infections (URTI). Strenuous physical activity seems to increase the risk for URTI during the first week or two after exercise (Nieman & Pedersen 1999) (see Chapter 22). In a systematic analysis, Hemilä and Chalker (2013) concluded that vitamin C supplementation did not reduce the incidence of common colds in the general population. However, the results suggest that vitamin C supplementation (200–1000 mg/d) could be justified as a preventive measure in those athletes performing brief periods of strenuous exercise, such as marathon or long-distance cross-country skiing. A 2007 Cochrane review did not support a reduced incidence of colds during moderate training loads with vitamin C supplementation (Douglas et al. 2007). However, pooled results of the 2013 review have demonstrated a modest but consistent effect on reducing the severity and duration of symptoms of the common cold with regular vitamin C supplements. The optimal dose and protocol for use is not clear but likely to be between 500–1000 mg/d for vitamin C (Hemilä & Chalker 2013). However, chronic use of vitamin C supplements at high doses may have adverse effects on the oxidative defence system (see Commentary 6).

A potential mechanism for a reduced incidence of the common cold for some athletes involved in strenuous exercise could be maintenance of neutrophil function during heavy exercise (Robson et al. 2003). However, studies have not confirmed that supplementation with vitamin C alone (Bryer & Goldfarb 2006; Davidson & Gleeson 2006; Nakhostin-Roohi et al. 2008) or in combination with other antioxidants (Mastaloudis et al. 2004; Traber 2006) has an effect on inflammatory markers, including neutrophil function.

Several studies have also shown that vitamin C supplementation for more than two weeks (Bryer & Goldfarb 2006; Nakhostin-Roohi et al. 2008) or a mixture of antioxidants, including vitamin C (Mastaloudis et al. 2004; Traber 2006; Goldfarb et al. 2007; Machefer et al. 2007) reduces indicators of exercise-induced lipid peroxidation. High doses of vitamin C (500–1500 mg/d for at least 1 week) also reduces exercise-induced increases in the levels of the stress hormone cortisol (Peake 2003). The practical significance of this reduction, however, is not certain. Bryer and Goldfarb (2006) reported that vitamin C supplementation reduced post-exercise muscle soreness, but other studies have not confirmed this effect (Connolly et al. 2006; Mastaloudis et al. 2006; Bloomer et al. 2007). Theoretically the antioxidant properties of vitamin C might protect against reactive oxygen species during heavy exercise, although recent evidence from supplementation studies of antioxidant vitamins has shown mixed and possible adverse results (see Commentary 6). It is not known whether vitamin C supplementation has any effect on improving muscle soreness or recovery in regularly training athletes.

11.6.2 Fat-soluble vitamin supplementation

There are no data on the effects of retinol, vitamin D or vitamin K supplementation on athletes' performances or other parameters in those with sufficient status. Theoretically, low vitamin D status would be expected to compromise skeletal muscle function and thereby athletic performance. To date only three studies have been done on the effects of vitamin D supplements in athletes with vitamin D depletion which have shown mixed results (see Chapter 9). Based on several studies showing inadequate vitamin D status in female athletes, supplementation with around 10 or 20 µg (400 or 800 IU) vitamin D daily could be warranted (Willis et al. 2008). However, further research and particularly randomised interventions are needed to show if this kind of a supplementation has an effect on immune function or injuries in athletes (Tenforde et al. 2010).

Many studies have shown that alpha-tocopherol (vitamin E) supplementation (typical dose >100 mg/d) (>150 IU) reduces breath pentane excretion (Simon-Schnass & Pabst 1988) and serum MDA concentration (determined as TBARs) (Simon-Schnass & Pabst 1988; Meydani et al. 1993; Rokitzki et al. 1994b) during exercise. Both breath pentane and serum MDA were used as indicators of lipid peroxidation. Unfortunately, many of these studies were poorly designed. In an early review, Viitala and Newhouse (2004) concluded that studies with good research design did not support the hypothesis that vitamin E alone attenuates lipid peroxidation during exercise. Moreover, other studies have not supported the hypothesis that vitamin E attenuates exercise-induced muscle damage (McGinley et al. 2009).

In one controlled trial, vitamin E supplementation maintained aerobic working capacity at very high altitude (>5000 m) (Simon-Schnass & Pabst 1988). Later studies have not demonstrated that vitamin E intake exceeding daily recommendations would have any beneficial effects on athletic performance (Rokitzki et al. 1994b; Kanter & Williams 1998; Tiidus & Houston 1995; Buchman et al. 1999, Gaeini et al. 2006; Ristow et al. 2009).

11.6.3 Mineral supplementation

Iron supplementation does not necessarily improve performance in athletes with low iron stores (i.e. low ferritin) (see Chapter 10, section 10.11). However, the benefit of iron supplementation on performance in athletes with diagnosed iron deficiency anaemia is well established. Data on supplementation of minerals other than iron are scarce. Magnesium supplementation may improve performance in individuals with diagnosed magnesium deficiency (Nielsen & Lukaski 2006), but oral magnesium does not appear to have any beneficial effects in athletes with balanced magnesium status (Terblanche et al. 1992; Weller et al. 1998; Mooren et al. 2003). Similarly, in other studies (Singh et al. 1992b; Telford et al. 1992a), a supplement containing several minerals and trace elements failed to affect indicators of nutritional status and athletic performance.

Chromium supplementation, mainly as chromium picolinate, taken during training has been proposed to stimulate insulin function and to promote muscle growth and glycogen synthesis. In an early controlled trial, chromium (200 μg/d) or placebo was given to healthy, previously untrained students, during an 11-week weightlifting program (Hasten et al. 1992). Supplementation was associated with greater bodyweight gain in women, but not in men. Strength was not affected. Despite this preliminary finding, more recent studies with athletes have not found any significant effects of chromium supplementation on weight, body composition, strength or glycogen synthesis (Clancy et al. 1994; Trent & Thieding-Cancel 1995; Lukaski et al. 1996; Walker et al. 1998; Volek et al. 2006). Hence, it seems that short-term (4–12 wk) chromium supplementation with doses varying from 200 to 800 μg/d does not have any ergogenic effects in trained individuals.

A study by Kara et al. (2010) suggested that zinc supplementation may have antioxidant effects in wrestlers by reducing the formation of free radicals. The implications for health or athletic performance are, however, not known. In a systematic review, Hemilä (2011) studied the effects of zinc lozenges (tablets intended to be dissolved slowly in the mouth) on the duration of the common cold. According to his results, high intakes of zinc (>75 mg/d) may reduce the duration. The implications of this finding on athletes' training and health have not been studied.

11.7 Potential risks of taking vitamin and mineral supplements

11.7.1 Risks related to high intake of water-soluble vitamins

Although the scientific data do not support the hypothesis that high vitamin and mineral intake enhances performance in well-nourished athletes, the use of supplements is common (Tsitsimpikou et al. 2009; Lun et al. 2012; Sousa et al. 2013). Many athletes use supplements as a precaution, 'just in case', to ensure rapid recovery and performance. Nevertheless, very high, chronic intake of one or several micronutrients may involve risks. Therefore any precautionary use of supplements should remain within the safe recommended intake.

Thiamin, riboflavin and B6

Adverse reactions to chronic, elevated oral administration of thiamin are virtually unknown (Marks 1989; Meltzer et al. 2003). For chronic oral use, it is still safe to use at least 50–100 times the recommended daily intake: that is, above 100 mg daily. Riboflavin in large doses may cause a yellow discoloration of the urine, which might cause concern in people not aware of the source of the colour (Alhadeff et al. 1984). However, there is no evidence of any harmful effects even with oral riboflavin doses exceeding 100 times the recommended daily intake (Marks 1989; Meltzer et al. 2003). In contrast to thiamin and riboflavin, megadoses of vitamin B6 may have toxic and possibly permanent adverse effects. The most common disorder is sensory neuropathy (Bässler 1989). Long-term supplementation should not exceed 200 mg/d: that is, 100 times the recommended daily intake (Marks 1989; Meltzer et al. 2003).

Folate and B12

The effects of high doses of folate have not been widely studied, but some results indicate a possible interference with zinc metabolism (Marks 1989; Reynolds 1994). The current estimate of the safety dose is between 50 and 100 times the daily recommended intake (Marks 1989). The safety margin for vitamin B12 appears to be much larger, because even doses as high as 30 mg/d (10 000 times the recommended daily intake) have been used without noticeable toxic effects (Marks 1989). However, a Nordic group recommended limiting vitamin B12 intake to levels corresponding to no more than 100 times the recommended level (Meltzer et al. 2003).

Other B vitamins (e.g. nicotinic acid, biotin and pantothenic acid)

Acute oral intake of at least 100 mg of nicotinic acid per day (at least five times the recommended daily allowance) causes vasodilatation and flushing, which is a rather harmless effect (Marks 1989). The safe chronic dose appears to be at least 50 times the recommended allowance or 1 g/d. There are no reported toxic effects of biotin and pantothenic acid intake up to 100 times the recommended allowance (Alhadeff et al. 1984; Marks 1989).

Vitamin C

Very high (>1 g/d), chronic doses of vitamin C may in theory lead to the formation of oxalate stones, increased uric acid excretion, diarrhoea, vitamin B12 destruction and iron overload (Alhadeff et al. 1984). However, excluding diarrhoea, the risk for the above adverse effects is likely to be very low in healthy individuals, even with intake of several grams daily (Rivers 1989; Meltzer et al. 2003).

A study by Gomez-Cabrera and colleagues (2008) presented interesting results; it was the first to warn against unnecessary high vitamin C supplementation. The authors reported that vitamin C supplementation reduced endurance capacity by preventing some cellular adaptations to exercise. See Commentary 6 for more detail about this study. In a more recent review, Braakhuis (2012) suggested that the unwanted effects are seen if vitamin C supplementation exceeds 1 g/d, while a more modest supplementation (200 mg/d) has no harmful effects.

Evidence of the potential adverse effects of vitamin C and other antioxidant supplements on the antioxidant defence system, exercise performance and training adaptations is considered in Commentary 6.

11.7.2 Risks related to high intake of fat-soluble vitamins

Vitamin A

Chronic toxic intake of pre-formed retinol (vitamin A) will cause joint or bone pain, hair loss, anorexia and liver damage. The safety level for chronic use is estimated to be 10 times the recommended daily allowance: that is, 10 g retinol daily (Marks 1989). Because of an increased risk of spontaneous abortion and birth defects, the safe level during pregnancy may be only four to five times the daily recommendation. Beta-carotene, a water-soluble form of vitamin A, in contrast to pre-formed retinol, is not toxic. This provitamin is stored under the skin and is converted to active retinol only when needed.

Vitamin D and vitamin E

Vitamin D is potentially toxic, causing hypercalcaemia, hypercalciuria, soft-tissue calcification, anorexia and constipation, and eventually irreversible renal and cardiovascular damage. See Chapter 9 section 9.11 for more evidence of vitamin D intoxication from misuse of vitamin D supplements. Intakes of 10 times the recommended daily intake—75 µg/d (3000 IU/d)—are safe, but the tolerable upper intake level may be at least 250 µg/d (10 000 IU/d) (Vieth 2007). High calcium intakes may enhance the toxicity of vitamin D. Vitamin E, in contrast to vitamins A and D, is apparently not toxic for healthy individuals (Machlin 1989), although it may in high doses interfere with the normal function of the endogenous antioxidant system (see Commentary 6). The safety factor for long-term administration is at least 100 times the recommended daily intake: that is, at least 1 g/d in oral use.

11.7.3 Risks related to high intake of minerals

The body pool of macrominerals, and especially of trace elements, is under strong homeostatic control. Therefore, toxicity by dietary means or supplements is rare. If, however, toxic symptoms appear, they are severe and can even be fatal. Intakes needed to induce toxic effects are, luckily, extremely high. For instance, the first toxic symptoms (vomiting, diarrhoea) of zinc overload are seen after a huge intake of at least 4 g (normal dietary intake is 10–15 mg/d). Extremely high intakes of copper can lead to liver cirrhosis or hepatic necrosis, and too much selenium causes changes in skin and hair, and neurological abnormalities. The threshold between the daily allowance and toxicity level of selenium is relatively small.

Although severe toxicity is very rare, high mineral intake may interfere with nutrient absorption at the intestinal mucosa (O'Dell 1985). Chronic zinc intake of 50 mg/d decreases both iron and copper bioavailability. Analogously, high intake of either iron or copper interferes with zinc absorption. Therefore, a surplus of one trace element may cause marginal deficiency of another trace element, especially when the intake of the latter is marginal.

Because intestinal interactions are observed with rather low dosages, precautionary use of mineral supplements (not specifically to treat a diagnosed deficiency) that does not exceed five times the Recommended Dietary Intake (RDI)/Recommended Daily Allowance (RDA) is warranted (Meltzer et al. 2003).

Summary

The daily requirement of at least some vitamins and minerals is increased beyond normal levels in highly physically active people. The potential reasons for this increased requirement are excretion through sweating and urine, and perhaps faeces, and through increased free radical production. Unfortunately, it is not possible to quantify the micronutrient requirements of athletes. Measures on micronutrient status and supplementation trials are needed to evaluate whether or not normal dietary intakes in athletes are sufficient to cover their increased requirements.

Most studies have not revealed any significant differences in indices of micronutrient status between athletes and controls. The results suggest that athletic training, by itself, does not lead to micronutrient deficiency. These data should, however, be interpreted carefully. Because of the insensitivity of most indices to marginal deficiency, the results do not exclude the possibility of minor differences between athletes and controls. Also, the results of research studies do not exclude the possibility that some individual athletes (and controls) have inadequate status.

Physical performance can be affected even when micronutrient deficiency is only marginal. In depletion experiments, changes in one or more biochemical indices of micronutrient status were seen before or together with impairments in physical work capacity. Despite the associations between indices of micronutrient status and performance in depletion studies, a justification for undertaking routine laboratory assessment or screening of water-soluble vitamin, magnesium or trace element (excluding iron) status in athletes seems doubtful. Whether vitamin D concentration in serum should be routinely screened in athletes needs further research. Because the prevalence of even marginal deficiency is low, routine screening would lead to numerous false-positive diagnoses.

Supplementation of water-soluble vitamins is associated with improved corresponding indicators in the blood. Moreover, supplementation of vitamin C may decrease the severity and duration of the common cold and potentially attenuate stress response induced after very strenuous physical exercise. However, studies do not conclusively show that micronutrient supplementation would increase physical performance, and even harmful effects have been reported. Consequently, the levels of indicators of vitamin and mineral status seen in athletes are apparently compatible with optimal physical functions. The most likely exceptions in some individuals are iron and vitamin D.

The evidence for benefits of vitamin and mineral supplements on athletic performance is extremely limited and has not been an interest in sports nutrition research for some time. Also, high intake of single micronutrients may lead to physiological disturbances, especially if the diet is inadequate. Hence, an athlete should try to ensure adequate vitamin and mineral status mainly by dietary means rather than supplementation. If, for any reason, an athlete wants to use micronutrient supplements as a precaution, a multivitamin–mineral supplement with amounts not exceeding two times the RDA is likely to be both safe and adequate for optimal sports performance.

Practice tips

ALICIA NORRIS

ASSESSING DIETARY ADEQUACY OF MICRONUTRIENTS AND ANTIOXIDANTS

- Estimated Average Requirements (EAR) for micronutrients are appropriate to use with athletes to guide assessment of meeting adequate intakes for most micronutrients (Commonwealth Department of Health and Ageing et al. 2006). However, nutrient requirements (not recommendations) for antioxidant vitamins (C, E and beta-carotene) and minerals (zinc, iron and selenium) may need to be slightly higher in athletes than non-athletes, but are unlikely to exceed population reference requirements. (See Chapters 9 and 10.)
- Athletes who meet energy requirements and regularly consume a variety of nutrient-rich foods are likely to meet or exceed the population-based reference levels for micronutrients (Commonwealth Department of Health and Ageing et al. 2006). However, athletes who habitually restrict energy intake or limit food variety are at risk of suboptimal micronutrient intakes and should be monitored for adequacy and potential risk of low micronutrient status.
- Assessing usual intake of micronutrients in any individual or group requires a longer time period than needed when assessing intake of macronutrients because of the higher variability of day-to-day micronutrient intake. A combination of dietary assessment, clinical history and biochemical markers is needed to estimate risk of inadequate intake.
- Food frequency questionnaires (FFQs) are an acceptable method of assessing specific micronutrient intake in population groups and have been used or modified for use in groups of athletes (see Chapter 3). Braakhuis and colleagues (2011) have developed and validated an FFQ to estimate dietary antioxidant intake in groups of athletes, which could be adapted as a screening tool for assessment of individual athletes.
- Despite the known limitations of estimating micronutrient intakes over a short time and the inherent bias in measuring dietary intake data, the use of innovative technologies for measuring micronutrients and total dietary intake such as mobile devices can be useful for assessing and monitoring specific micronutrient intake (see Chapter 3 for limitations).
- Several diagnostic criteria in addition to risk of suboptimal intake should be investigated to confirm the micronutrient status of an individual athlete. A one-off low nutrient biomarker test may not necessarily be diagnostic of a clinical or even subclinical condition, especially for those nutrients under homeostatic control (i.e. most minerals or trace elements). Use of serial tests and several biomarkers, if available, gives more accurate information about individual status and reveals trends (Allen 2008).

STRATEGIES TO IMPROVE MICRONUTRIENT INTAKE IN ATHLETES

- The Suggested Dietary Targets (SDT) are used as recommended intakes from antioxidant vitamins and minerals (vitamins C, A, E, selenium and carotenoids) from food sources rather than supplements and are based on evidence of the amount required to reduce chronic disease risk (particularly CHD and cancer) for the general population. In the absence of specific recommendations for antioxidant vitamins and minerals for healthy athletes involved in high-intensity training programs, who have a high turnover of antioxidant nutrients, these targets are also useful benchmarks for the sports dietitian when designing specific nutrient-rich menu plans for athletes.
- An elite athlete is dependent on consistent high-level performance. A discussion of determinants of food choice (barriers and facilitators) and the impact of changes to food habits from a performance perspective is a strong motivation to promote an improvement in food choices in individual athletes, especially those who are fussy eaters and resistant to change. Highlighting the potential role of micronutrients on other health parameters such immune function, wound healing, reduced injury risk, enhanced recovery may also facilitate change.
- Consuming a varied intake of food from the five food groups and particularly promoting an increase in antioxidant-rich choices (cereals, grains, fruits and vegetables) on a daily basis is an effective strategy to increase micronutrient intake. Compared to most fruits, vegetables (particularly the dark and brightly coloured vegetables) are rich sources of antioxidants.
- Suggestions for improving micronutrient density in the diet include:
  - Increase vegetable intake: add to pizza, casseroles and stir-fries; grate vegetables into pasta sauces, muffins or slices; blend into soups or juices.

– Increase fruit intake: incorporate fresh, canned or dried fruits into cereals, smoothies or juices (particularly citrus fruits which are rich in vitamin C); fruit-based desserts such as crumble; tropical and berry fruits are high in antioxidants.
– Increase calcium intake: add extra milk, cheese and yoghurt to the daily menu through snack options. Use calcium-fortified lactose-free options if lactose intolerant (see Chapters 9 and 20).
– Increase iron intake: use iron-fortified breakfast cereal; add meat to sandwiches and wraps; incorporate lean red meat into a several meals during the week (see Chapter 10).
– Provide nutrient-dense alternatives when an athlete is following a special diet (such as FODMAPs, gluten-free) (see Practice tips, Chapter 20).

MICRONUTRIENT SUPPLEMENTS: ADVANTAGES AND DISADVANTAGES

Advantages

- A multivitamin/multimineral supplement (<2–3 times the RDI/RDA) may be indicated where habitual dietary intake of micronutrients is suboptimal and cannot be adequately addressed by dietary intervention. A low dosage is recommended.
- A therapeutic dose (~10 × RDI/RDA) may be required for short-term use when a deficiency of a micronutrient has been diagnosed (e.g. vitamin D, iron, folate) to promote rapid recovery. However, once nutrient status has recovered, dietary intervention is the cornerstone of long-term prevention and management to maintain improved status and avoid relapse.
- An adequate intake of selected vitamins and minerals is important for immune function (see Tables 11.1 and 11.2). Some researchers have suggested that supplementation of antioxidant vitamins and minerals, such as vitamins E, C, zinc and iron may be beneficial to prevent the potential oxidative stress associated with high intensity exercise. However, results are mixed, especially at high doses (see Commentary 6).
- Vitamin C and zinc supplements may have a beneficial effect on reducing the duration and severity of the common cold. The optimal dose and protocol for short-term use only is not clear but likely to be between 500 and 1000 mg/d for vitamin C and 75 and 100 mg/d for zinc, which could be achieved from a combination of diet and supplements (see Chapter 22).

Disadvantages

- Chronic supplementation with high doses of trace minerals and some vitamins (particularly vitamin C, vitamin E, vitamin B6, niacin, vitamin D and fat-soluble vitamin A) without a diagnosis of deficiency may have adverse and potentially harmful side effects.
- Recent research suggests that chronic supplementation with high doses of the antioxidant vitamins—vitamin C (~1000 mg/d) and vitamin E (~400 IU/d)—may alter normal redox (oxidation reduction) reactions and inhibit the positive effects of training adaptations (see Commentary 6).
- Other micronutrient supplements and naturally occurring compounds in food purported to have anti-inflammatory or antioxidant activity or performance benefits should be used with caution because of either potential inadvertent risk of contamination with illegal substances or unsupported claims of efficacy. Go to the Australian Institute of Sport website for evidence relating to the use and efficacy of micronutrient and other supplements with antioxidant activity (see www.ausport.gov.au/ais/nutrition/supplements/resources_new/supplements_and_sports_foods).

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Commentary 6

Antioxidant supplementation and exercise

JEFF COOMBES

Introduction

Naturally occurring (endogenous) and dietary antioxidants work together to inhibit the potential damage caused by free oxygen radicals produced by the reduction of molecular oxygen during energy metabolism. Excess production of free radicals has been linked to cell and mitochondria membrane damage and dysfunction, deterioration of the immune defence system, ageing, cancer, cardiovascular disease and diabetes mellitus. Exercise increases oxidation and the production of free radicals or reactive oxygen species (ROS) and hence increases the potential risk of oxidative damage. Well-trained athletes have a more developed endogenous antioxidant system than sedentary counterparts, as evident from the increased activity of endogenous antioxidant enzymes. This increased activity is an adaptation to the increased ROS production, thus providing a protective mechanism system against oxidative damage (see Chapter 11, Section 11.6).

Recreational and elite athletes may use antioxidant supplements to improve performance, promote recovery, provide health benefits or reduce the potential for oxidative damage. Specific antioxidant vitamin supplements (e.g. vitamins C and E) taken in usual recommended dosages may blunt the beneficial increases in the endogenous antioxidant system and interfere with the normal redox (oxidation reduction) reactions, which has adverse effects on cell function (e.g. insulin sensitivity) and muscle contraction. This adaptive response could be impaired by taking single or even mixed supplements of dietary antioxidants. However, the evidence of potential adverse effects from these and other antioxidant supplements is not consistent and is confounded by the complexity of the physiological interactions involved.

This commentary addresses the known effects of taking supplements of antioxidant vitamins (predominantly vitamins E and C) as a means to reduce the potential for oxidative damage associated with the production of ROS induced by high-intensity exercise. Only a few intervention studies investigating the antioxidant vitamins have been conducted on athletes or individuals who participate in regular exercise. Most intervention trials have been conducted on older population groups, who may be untrained or have an existing chronic disease. Hence, the recommendations for antioxidant supplement use by athletes are largely extrapolated from population studies.

Free radicals, reactive oxygen species and oxidative stress

ROS are defined as reactive intermediates derived from the incomplete reduction of oxygen, and are generated as a normal part of our metabolism (McCord 2000). Most ROS are free radicals and are very reactive. Superoxide (O_2^-, hydrogen peroxide (H_2O_2) and the hydroxyl radical (HO^-) are commonly known ROS (McCord 2000). The production of ROS, which are byproducts of oxidation of glucose, protein, lipid and DNA, occurs mainly in the mitochondria of contracting muscle and organs.

Transient increases in ROS reflect a health-promoting and normative physiological process. The presence of ROS provides the stimulus for normal signalling functions required for optimal cellular and physiological functioning, which includes cell growth and mitochondrial biogenesis. Also, ROS play a role in vasodilation and optimal muscular contraction as well as muscle adaptation to exercise. This process of redox signalling in muscle, induced by the presence of ROS, is a fundamental element in exercise physiology. See Powers et al. (2011) for a review of this process.

If the ROS generated are not mopped up by the body's antioxidant defences, they can react with surrounding cellular components including membrane lipids, cellular proteins and DNA. During exercise, whole body O_2 consumption can increase by 10- to 15-fold and the O_2 flux in active muscle has shown up to a 200-fold increase (Sen 1995). Thus, if superoxide production increases in proportion to oxygen consumption during exercise, the potential for oxidative stress is substantially increased.

A commonly used definition of oxidative stress was the point at which the build-up of ROS exceeds the capacity of the antioxidant defence mechanisms, leaving ROS to react and damage cellular components (Jenkins 2000). Although not yet accepted, several new definitions of oxidative stress have appeared in the literature that implicate the disruption in redox signalling as a proposed mechanism to explain the process of oxidative damage to cells. Powers and colleagues (2010) define oxidative stress as a disturbance in the redox balance in cells in favour of oxidants, with this imbalance resulting in oxidative damage to cellular components.

Oxidative stress is a predisposing factor linked to the pathogenesis of many chronic diseases (e.g. cardiovascular disease, some cancers, kidney disease, diabetes and complications of diabetes) (Small et al. 2012; Feng et al. 2013; Wu et al. 2014), fatigue and muscle damage (Powers & Hamilton 1999) and reduced immune function (Bishop et al. 1999).

Exercise and the link with oxidative stress

Acute exercise of high intensity and duration involving muscle contractile activity increases the production of ROS in both muscle and other organs, which can result in oxidative stress in both animals and humans. ROS activity has been detected by altered muscle and blood glutathione concentration and increases in protein, DNA and lipid peroxidation measures. Despite the potential for oxidative damage that occurs as a result of this increase in ROS, regular physical activity has well-known beneficial effects for health and performance outcomes. This apparently conflicting situation is referred to as the exercise-oxidative stress (EXOS) paradox (Powers et al. 2011).

Nevertheless, ROS have historically been viewed as harmful. Whether excess ROS cause an impairment in exercise performance and muscle damage is yet to be demonstrated.

What is an antioxidant?

In chemical terms, the definition of an antioxidant is any molecule or compound that inhibits oxidation. Humans have a complex antioxidant system that involves an interaction between endogenous antioxidants—for example, uric acid, bilirubin, plasma proteins and the enzymes superoxide dismutase, glutathione peroxidase catalase and coenzyme Q10—and dietary antioxidants ingested from food—for example, vitamin C, vitamin E, carotenoids (mainly β-carotene), polyphenols (e.g. flavonoids), glutathione and coenzyme Q10. Dietary sources are found mainly in plant foods including dark-coloured vegetables, citrus fruits, legumes, nuts, grains, seeds and oils. Antioxidants are also added to many commercial foods to help prevent chemical deterioration (oxidation reactions from oxygen either naturally present in food or from contact with the air). Some antioxidants (e.g. glutathione and coenzyme Q10) can be produced both endogenously or consumed in the diet (Halliwell & Gutteridge 1989).

Although vitamin antioxidants supplements are the most widely used and studied, there is a plethora of other supplements available that are marketed for their antioxidant actions. The claims made by manufacturers may be unsupported by scientific research. Dietary supplements are not subjected to the same regulatory or quality control standards as food products, so claims made by manufacturers may be misleading and unfounded (see Chapter 16).

The interaction between endogenous and dietary antioxidants is complex. Understanding the mechanism of antioxidant action is complicated when two antioxidant vitamins (e.g. vitamins E and C) are taken together. Mixed use of antioxidant supplements can induce varied cellular and systemic effects. For example, vitamin C recycles vitamin E, which can interfere with vitamin E activity. In high doses, antioxidants become pro-oxidants and oxidise themselves and other antioxidants.

Adaptation of endogenous antioxidants to repeated regular exercise

In response to athletic training, endogenous antioxidant defences eventually increase as a means to adapt and defend against increased production of ROS induced during repeated exercise. Three adaptive mechanisms have been proposed to help attenuate the increased production and possible accumulation of ROS:

1. the up-regulation of endogenously produced antioxidant enzymes
2. the increased de novo production of endogenous antioxidant molecules including glutathione, urate and coenzyme Q10
3. the mobilisation of antioxidant vitamins from tissue stores and their transfer through the plasma to sites undergoing oxidative stress.

A disturbance in antioxidant balance alters the redox environment, which has a profound effect on cellular function and muscle contraction. The physiological role of

ROS on cell signalling pathways involved in muscle adaptation to exercise are outside the scope of this commentary. See Powers et al. (2011) for a review.

Antioxidant supplements and their effect on exercise training adaptations

Theoretically, the antioxidant properties of vitamin C and vitamin E might protect against reactive oxygen species during heavy exercise, although evidence from supplementation studies of antioxidant vitamins has shown mixed and possibly adverse results. Evidence that antioxidant supplements could interfere with the adaptive oxidative system response associated with regular exercise was first reported by Sharman and colleagues in 1971 in a study of the effect of vitamin E supplementation (400 IU/d for 6 wk) on performance measures in 30 male adolescent swimmers. Although performance up to 10 minutes duration was not improved with vitamin E supplements in the swimmers compared to matched controls, there were significant decreases in the work recovery index in the supplemented group. However, this outcome may be biased because of inadequate matching of subjects and controls, the small sample and effect size, and the low sensitivity of the performance measures used at the time.

Since then, based on a review by Peternelj & Coombes (2011), 23 well-designed studies have reported an inhibition of exercise training-induced adaptations using antioxidant supplements either taken as single supplement (predominantly vitamins C and E, allopurinol, coenzyme Q10) or mixed in both animal and human studies. See Peternelj & Coombes (2011) for a summary of the research design, antioxidant dose and outcomes for each study. Of these 23 studies, three well-designed studies (outlined below) have frequently been cited as evidence that antioxidants interfere with cellular adaptations to exercise.

Gomez-Cabrera and colleagues (2008) were the first researchers to warn against the negative effects of high intakes of vitamin C supplements on performance capacity based on two intervention trials using rats and untrained men. The authors reported that after eight weeks of vitamin C supplementation (1000 mg/d) and regular exercise on a static bicycle for 3 days/wk, endurance capacity was significantly impaired in 14 previously untrained healthy men (aged 19–36 years). A simultaneous study on trained and untrained rats supplemented with vitamin C was conducted by these authors to determine any combined effect of the supplement and exercise on changes to muscle mitochondria activity. Analysis of rat muscle tissue after supplementation showed a reduction in antioxidant enzymes, which is indicative of an inhibition of the expected training-induced up-regulation. In the human subjects, a 50% inhibition of the training-induced increase in VO_{2max} was reported, compared with a 22% inhibition in the placebo group, although this difference was not significant.

The authors of the second study, Ristow and colleagues (2009), found that a combination of vitamins C and E supplements (1000 mg/d and 400 IU/d, respectively), taken daily for four weeks by 20 healthy young men, actually blocked the expected beneficial effects of four weeks of physical exercise on enhancing insulin sensitivity in both pre-trained men and untrained controls. As expected, exercise training in both trained men and untrained men (who did not take the antioxidant supplements) increased insulin sensitivity and ROS, which was paralleled by an increase in endogenous antioxidants. Those subjects that were taking antioxidant supplements did not demonstrate this expected adaptive response to exercise. Hence, in both trained and untrained men in this study, antioxidant supplementation blunted the health-promoting effects of exercise and up-regulation of endogenous defence responses.

Wray et al. (2009), in the third most-cited study, investigated interactions between exercise training and antioxidant supplements and effects on blood pressure. The authors found that after six weeks of single leg knee-extensor exercise, blood pressure in nine mildly hypertensive men was significantly lower at rest and during a subsequent aerobic exercise training program. However, supplementation with a *cocktail* containing vitamin C, vitamin E and alpha lipoic acid after exercise training blunted the blood pressure-lowering effect of exercise, returning subjects to a hypertensive state. Furthermore, vasodilation improved significantly in response to aerobic exercise training as expected, although it was inhibited when subjects took antioxidants.

A recent study involving 11 weeks of training and supplementation with 1000 mg/d vitamin C (ascorbic acid) and 235 mg/d (261 IU) vitamin E (DL-α-tocopherol acetate) in a group (n = 54) with a mixed exercise background highlights the difficulty in undertaking and interpreting the results of chronic training study (Paulsen et al. 2014). The authors found that antioxidant supplementation did not affect the endurance training-induced increases in VO_{2max} and running performance (20 m shuttle test). However, at the muscle level, the supplemented group showed a blunting of the training-induced changes in cellular proteins compared with the cohort that took a placebo. These cellular results were mirrored by a blunting of some of the physiological adaptations (e.g. changes in fat oxidation during submaximal exercise) in the supplemented group. The disconnect between cellular adaptations and performance outcomes was considered a reflection of study limitations (e.g. duration and sample size of the study), but also variability in the measurements and real changes in some parameters.

Other studies have failed to find any adverse effect of high dose vitamins E and C supplement on exercise training-induced adaptations in rats (Higashida et al. 2011) and moderately trained young men (Yfanti et al. 2010). No significant effects on physical adaptations to strenuous endurance training (i.e. VO_{2max}, maximum power and lactate) were reported in 12 young men after supplementation with vitamin E (400 IU/d) and vitamin C (500 mg/d) compared with controls. Endogenous antioxidant enzymes or other biomarkers of oxidative stress were not measured in this study. The type and lower dose of vitamin C supplements used, the small sample and small effect size (subjects were already moderately trained) may account for this lack of effect of the supplements on inhibiting training adaptations seen in other studies (Gomez-Cabrera et al. 2012). Given the complexity of the molecular, cellular and physiological mechanisms and the known individual variability in responses to exercise and differences in bioavailability of antioxidant action of supplements, it is not surprising that there will be mixed results. There appears to be a complex dose–response relationship (termed hormesis) in which both the low and high ends of the dose spectrum are associated with negative outcomes. However, the threshold at which any disturbance of redox potential induced by antioxidants supplements is unknown in exercising individuals. Such a relationship would be difficult to detect in trained athletes because of the adaptive response observed in athletes to higher levels of ROS induced by repeated bouts of acute exercise.

Details of the metabolic responses to antioxidant supplements are outside the scope of this text and can be found elsewhere (see Powers et al. 2011). An alteration of transcription factors associated with mitochondrial biogenesis is implicated (Gomez-Cabrera et al. 2008).

In summary, most studies suggest that supplementation with antioxidant vitamins (at relatively high doses) may preclude the health-promoting effects of exercise on insulin sensitivity and blood pressure in both young and older adults, inhibit the induction of the endogenous antioxidant defence system by interfering with ROS-mediated physiological processes and inhibit other markers of training adaptation. Vitamin C and vitamin E at doses of 1000 mg/d and 400 IU mg/d respectively may blunt the positive adaptive effects of exercise training on the oxidative system. Braakhuis and colleagues (2012) suggested that the unwanted effects are seen if vitamin C supplementation exceeds 1 g/d, while a more modest supplementation (200 mg/d) has no harmful effects.

Antioxidant supplements and their effect on health outcomes

Recent epidemiological evidence from randomised clinical trials and from systematic reviews have confirmed that antioxidant supplements do not prevent cancer, cardiovascular diseases or death (Bjelakovic et al. 2012). The evidence suggests that β-carotene, vitamin A and vitamin E supplements may actually increase mortality (Bjelakovic et al. 2014).

Epidemiological observations also suggest that populations consuming diets rich in fruits and vegetables have a lower incidence of some cancers and longer life expectancy; fruits and vegetables contain natural antioxidants (Soory 2012; Dolara et al. 2012). Dietary antioxidants have specific applications in ameliorating oxidative stress-induced inflammatory diseases such as diabetes mellitus and cardiovascular disease. Diets rich in antioxidant vitamins and minerals (vitamin A and carotenoids, vitamin C, vitamin E and possibly selenium) and polyphenols are cardio-protective (Wang et al. 2013). Polyphenolic compounds such as flavonoids, isoflavones, phenolic acids and lignan contribute to increased plasma antioxidant capacity, decreased oxidative stress markers and reduced total and LDL cholesterol (Soory 2012) via modulation of genes associated with metabolism, stress defence, detoxification and transporter proteins.

In one large intervention trial, the MRC/BHF Heart Protection study, over 20 000 UK adults with coronary disease, other occlusive arterial disease and diabetes were given a multiple antioxidant supplement (600 mg (670 IU) synthetic vitamin E (DL-α-tocopherol acetate), 250 mg vitamin C and 20 mg β-carotene daily) or matching placebo for a five-year treatment period (MRC/BHF Heart Protection Study Investigators 2002). No effect on major health outcomes was evident in the supplemented group. However, in the most recent Cochrane meta-analysis, pooled data from 78 randomised controlled trials of around 300 000 participants confirmed a higher risk of mortality in those taking antioxidant supplements (vitamins E, A and β-carotene) compared to placebo (Bjelakovic et al. 2012) but not vitamin C and selenium. This outcome is consistent with an earlier meta-analyses by Bjelakovic and colleagues in 2007.

Prior to the results of these studies, high-dose antioxidant supplements were perceived at worst as innocuous. According to recent dietary guidelines and meta-analyses, the evidence does not support the use of antioxidant supplements in the primary prevention of chronic diseases or in people with existing oxidative stress-induced inflammatory diseases such as cardiovascular disease and diabetes mellitus.

The effect of non-vitamin antioxidant supplements on exercise performance

The majority of studies have found that the most commonly consumed antioxidant supplements (i.e. vitamin C and vitamin E) at intakes exceeding daily recommendations have no beneficial effect on athletic performance (Peternelj & Coombes 2011) or reduced muscle soreness in trained athletes (see Chapter 11, Section 11.6.1). Other supplements that may have potential antioxidant activity and have shown some positive performance-enhancing effects include:

- coenzyme Q10 (Ylikoski et al. 1997; Bonetti et al. 2000; Cooke et al. 2008; Mizuno et al. 2008; Gokbel et al. 2010)
- N-acetylcysteine (Medved et al. 2004; Slattery et al. 2014)
- polyphenols, including quercetin (Davis et al. 2009, 2010; MacRae & Mefferd 2006; Nieman et al. 2007)
- resveratrol (Lagouge et al. 2006)
- polyphenolic compounds from grape extract (Lafay et al. 2009)
- beetroot juice (Bailey et al. 2009, 2010; Vanhatalo et al. 2010; Lansley et al. 2011) and blackcurrant juice (Braakhuis et al. 2013). (These foods (and extracted juices) are excellent sources of anthocyanin.)
- *Rhodiola rosea* plant (Skarpanska-Stejnborn et al. 2009)
- *Ecklonia cava* algae (Oh et al. 2010).

However, despite publication (often in low-ranked journals), many of these studies are of low quality and small sample size (small effect, low power) and may not conform to the academic rigour expected for publication in a high-ranked journal. Details of method of recruitment of subjects, inclusion/exclusion criteria and randomisation, allocation and concealment methods are often omitted. Hence, interpretation of potential performance benefits or disadvantages of antioxidant supplements based on intervention trials may be questionable and biased. Emerging evidence suggests that the antioxidant potential of phenolic compounds, for example, is unlikely to be the sole mechanism responsible for their protective benefits. It is likely that positive performance-enhancing effects or benefits are also be mediated by their interaction with various key proteins in the cell-signalling cascades (Vauzour et al. 2010). Chapter 16 provides summaries of the studies of several compounds with antioxidant activity that meet the criteria for acceptable level of evidence relating to performance benefits and for acceptable research design nominated by the Australian Institute of Sport sports supplement criteria (see www.ausport.gov.au/ais/nutrition/supplements).

Guidelines for authors, reviewers and journal editors when publishing or undertaking intervention trials, called Consolidated Standards of Reporting Trials (CONSORT), were developed in 2010 (see www.consort-statement.org/consort-2010). These guidelines are useful for students and new researchers to critically evaluate the quality of any intervention trial, including those of supplement effects.

The future: biomarkers of oxidation and redox reactions

As discussed earlier, disturbances in antioxidant balance from taking antioxidant supplements may alter the redox environment, decreasing insulin sensitivity and muscle contraction, and potentially increasing oxidative stress. Currently, there are no definitive tests that measure the true effect of redox control ability. Many of the currently used oxidative stress biomarkers (using blood or urine) are collected from a person at rest and are therefore not valid measures of redox ability, which is a dynamic system (Strobel et al. 2011). An exercise test is proposed to evaluate redox control (Nikolaidis et al. 2012), which is similar to the principle used for an oral glucose tolerance test to measure glycaemic control. This approach uses the change in oxidative stress biomarkers to quantify redox control ability rather than absolute measures. Indeed, the large between-individual variability in the oxidative stress response to exercise in a homogenous population implies that the test is worthy of further investigation (Mullins et al. 2013). Nutrigenomic assessments may also assist in understanding whether the responses to a particular antioxidant supplement is associated with specific genes.

Powers et al. (2010) have published guidelines for researchers investigating the impact of antioxidant supplements on exercise performance in both animal and human studies. These guidelines include an evaluation of oxidative stress biomarkers and a recommendation that all antioxidant experiments should include two or more acceptable biomarkers of oxidative damage in tissues. Future intervention trials of antioxidant supplements may need to consider both redox control ability during exercise and gene expression to determine intra- and inter-individual responses.

Summary

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Athletes often use antioxidant supplements to help prevent the potential adverse effects of exercise-induced oxidative stress, hasten recovery of muscle function and improve performance. Maintenance of redox control is critical for normal cell and muscle function. High doses of antioxidant vitamin supplements (e.g. vitamins E and C) have been shown to disrupt redox homeostasis and impair exercise-training adaptations in most studies.

Given the plethora of compounds in food and other supplements (often herbal) that also have antioxidant properties, it is recommended that when evaluating supplements in which manufacturers make claims of antioxidant efficacy, the focus should be on the activity of the specific ingredient present. The bioavailability of antioxidants in such supplements is likely to be highly variable and dose-dependent with little known about optimal dosages and efficacy, especially in athletes. There is a need for more rigour in research design when evaluating antioxidant activity and redox effects of all antioxidant supplements. Pharmacokinetics of antioxidant supplements should be measured to establish the dosage, timing and duration of use. In intervention trials, dietary intake of antioxidants should also be controlled to avoid confounding bias.

The most prudent recommendation for antioxidant vitamin therapy (i.e. vitamins C and E) to optimise the body's capacity to defend and adapt to increased ROS production during exercise is to consume a diet high enough in antioxidant-rich foods, rather than use these supplements. Dietary sources of vitamin C that contribute around 200 mg/d may be sufficient to reduce oxidative stress and provide other health benefits without impairing training adaptations. There is now increasing and consistent evidence from well-designed studies that high dosages of antioxidant vitamin supplements may cause more harm than good.

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CHAPTER TWELVE

Preparation for competition

Louise Burke

12.1 Introduction

The athlete's goals during competition are to perform to his or her optimum level. A range of factors can impair performance, including issues related to nutrition. 'Competition eating' is based on the principle of implementing nutrition strategies that can reduce or delay the onset of factors that cause fatigue or performance impairment. Of course, practical issues must also be considered, particularly in light of the frequency with which athletes are required to travel interstate or overseas to compete in their target events. Competition nutrition includes special eating strategies undertaken before, during and in the recovery from the event. In this chapter, we will review the strategies undertaken in the hours or days prior to competition, to prepare athletes to perform at their best.

12.2 Nutrition factors causing fatigue during performance

A variety of nutrition factors can reduce an athlete's ability to perform at their best during exercise (see Table 12.1). The risk and severity of an encounter during competition depends on issues including the:

- duration and intensity of the exercise involved
- environmental conditions, for example temperature and humidity
- training status of the athlete
- individual characteristics of the athlete
- success of nutrition strategies before and during the event.

TABLE 12.1 Nutrition factors associated with fatigue or a decline in performance

• Depletion of glycogen stores in the active muscle
• Hypoglycaemia
• Other mechanisms of 'central fatigue' involving neurotransmitters
• Dehydration
• Hyponatraemia
• Gastrointestinal discomfort and upset

It is relatively easy to investigate the physiological factors limiting the performance of simple exercise tasks (such as running or cycling) undertaken in an exercise physiology laboratory. Furthermore, factors identified in laboratory simulations of simple competitive events such as these may well apply to the real-life performances of athletes. However, it is more complicated to measure or predict factors limiting the performance of complex sporting events, particularly ball games and racquet sports, in which competition demands have a high degree of inter- and intra-athlete variability. Sports scientists try to pinpoint the likely risk that various factors will cause fatigue in a given sport or event, based on the available applied sports research, as well as accounts of the past competition experiences of the athletes involved. With this knowledge, athletes can then be guided to undertake specific competition nutrition strategies that will minimise or delay the onset of these problems.

12.3 Setting a competition nutrition plan to combat factors causing fatigue

Pre-competition nutrition strategies include dietary interventions that are implemented during the week prior to an event, as well as special tactics that are undertaken in the minutes or hours before the event begins. These nutrition strategies should target the specific physiological challenges that affect the performance of the athlete's sport. According to the characteristics of the event, strategies might aim to minimise fluid deficits, ensure fuel availability or prevent gastrointestinal discomfort. Ideally, an athlete should combat these challenges by undertaking a combination of nutrition strategies before, during and even in the recovery after an event. Although further systematic studies are needed to investigate the benefits of combining two or more strategies, it is logical that a multi-tasked approach would be superior to a single strategy. For example, it appears that the combination of carbohydrate (CHO) intake before, as well as during, an endurance event is superior to the performance benefits gained from undertaking each of these strategies in isolation (Wright et al. 1991; Chryssanthopoulous & Williams 1997; Burke et al. 1998). However, in practice, the athlete may not always be able to exploit each opportunity for a nutrition intervention. When one opportunity is missed or under-utilised, greater emphasis on tactics at another opportunity may help to compensate. For example, an athlete who has not been able to refuel muscle CHO stores before a prolonged event should place greater emphasis on consuming CHO during the event. Conversely, in events where opportunities to drink during exercise are limited and large fluid deficits occur, the athlete should pay extra attention to hydration before the event starts, and perhaps even experiment with hyperhydration (fluid overload). In any case, the athlete should set a complete plan for competition eating, using nutrition strategies that complement and enhance each other.

12.4 Pre-event fuelling

The depletion of body CHO stores is a major cause of fatigue during exercise (see Chapter 1 for review). Optimising CHO status in the muscle and liver is a primary goal of competition preparation. The key ingredients for glycogen storage are dietary CHO intake, and in the case of muscle stores, tapered exercise or rest (for review see Ivy 1991; Robergs 1991; Jentjens et al. 2003). The duration of pre-event fuelling will depend on the balance between the anticipated fuel needs of the event and the preparation time that can be devoted to the event.

12.5 Muscle glycogen storage

Since the application of the biopsy technique to exercise metabolism in the 1960s, sports scientists have been able to directly measure the glycogen content in isolated muscle samples, and thus determine the factors that enhance or impair storage. More recently, indirect techniques of muscle glycogen measurement involving nuclear magnetic resonance spectroscopy (Roden & Shulman 1999) have increased the practical opportunities to study such factors.

Studies of glycogen synthesis describe a biphasic response, consisting of a rapid early phase for the first 24 hours (non-insulin-dependent) followed by a slow phase (insulin-dependent) lasting several days (Ivy & Kuo 1998). Early studies proposed that activity of the glycogen synthase (GS) enzyme plays a key role in determining the rates of glycogen synthesis in the muscle (Danforth 1965), although this relationship does not totally explain all observations. For example, differences in post-exercise glycogen storage have been found in the absence of differences in GS activity (Ivy et al. 1988), and the increased GS activity observed immediately after glycogen-depleting exercise returns to normal in advance of the glycogen supercompensation that can be observed in trained muscle exposed to several days of high CHO intake (see Ivy & Kuo 1998). Insulin- or exercise-stimulated translocation of GLUT-4 protein transporter to the muscle membrane increases muscle glucose uptake and is also a determinant of glycogen resynthesis (McCoy et al. 1996). Factors affecting post-exercise glycogen storage are discussed in greater detail in Chapter 14.

Investigators have identified two separate glycogen pools within muscle, adding a new dimension to our understanding of glycogen synthesis and utilisation. The primer for glycogen synthesis is the protein glycogenin, which acts as both the core of the glycogen molecule and the enzyme stimulating self-glycosylation (Alonso et al. 1995). The initial accumulation of glucose units to glycogenin forms a glycogen type now known as proglycogen, which is of relatively smaller size. The storage of proglycogen is most prominent during the first phase of recovery and is sensitive to the provision of dietary CHO (Adamo et al. 1998). During the second phase of glycogen recovery, glycogen storage occurs mainly in the pool of macroglycogen, a glycogen molecule with greater amounts of glucose for each glycogenin core. It is an increase in the macroglycogen pool that appears to account for glycogen supercompensation in the muscle following 2–3 days of high CHO intake (Adamo et al. 1998).

Research indicates that the pools of proglycogen and macroglycogen are metabolically distinct. It is speculated that proglycogen is the small and dynamic intermediate form of glycogen, whereas macroglycogen is the larger storage form that increases on a relative basis as total glycogen increases (Adamo & Graham 1998). Adamo and Graham suggest that when the proglycogen pool has reached a critical limit in a favourable CHO environment, a portion is then synthesised into macroglycogen. Conversely, a study of a marathon race has shown that the pools of glycogen are utilised at separate rates (Asp et al. 1999). Future studies may allow us to exploit this information, and determine new factors and strategies that enhance the metabolic availability of glycogen pools or increase storage.

12.6 Fuelling up for non-endurance events

In the absence of muscle damage, muscle glycogen stores can be normalised by 24 hours of rest and an adequate CHO intake: up to 7–10 g/kg body mass (BM) per day (Costill et al. 1981; Burke et al. 1995). Such stores appear adequate for the muscle fuel needs of events of less than 60–90 minutes in duration; at least, CHO loading to achieve supercompensated glycogen levels does not enhance the performance of these events (for review, see Hawley et al. 1997b).

Typically, the resting value for muscle glycogen in trained muscle is 100–120 mmol/kg wet weight (ww). Allowing for a typical rate of GS at ~5 mmol/kg ww/h, the athlete should set aside 24–36 hours following their last training session to normalise fuel stores prior to non-endurance events. For many athletes this might be as simple as scheduling a day of rest or light training before the event, while continuing to follow high-CHO eating patterns. However, not all athletes eat sufficient CHO in their typical or everyday diets to maximise glycogen storage, particularly females who restrict their total energy intake to control body fat levels (Burke 1995). These athletes may need education or encouragement to temporarily remove their energy restriction and prioritise refuelling as the dietary goal on the day before competition. Similarly, some athletes may need to reorganise their training programs to allow a lighter training day or rest on the day prior to their event. At minimum, the athlete should avoid training sessions that cause significant muscle damage on the day before competition, since such damage interferes with glycogen storage (O'Reilly et al. 1987; Costill et al. 1990).

12.7 Carbohydrate loading for endurance events

CHO loading refers to practices that aim to maximise or supercompensate muscle glycogen stores prior to a competitive event that would otherwise deplete these fuel reserves. Such protocols may elevate muscle glycogen stores to ~150–250 mmol/kg ww. This will be an important strategy for events lasting more than 90 minutes, which would otherwise be limited by the depletion of muscle glycogen stores (for review, see Hawley et al. 1997b). The first CHO-loading protocols were an outcome of studies undertaken in the late 1960s by Scandinavian sports scientists to examine muscle properties using biopsy techniques. In a series of studies (Bergstrom & Hultman 1966; Ahlborg et al. 1967; Bergstrom et al. 1967; Hermansen et al. 1967), these researchers found that endurance, or capacity for prolonged moderate-intensity exercise, was determined by pre-exercise muscle glycogen stores. Several days of a low-CHO diet depleted the muscle glycogen stores and reduced cycling endurance compared with a normal-CHO diet. However, the subsequent high CHO intake over several days caused a supercompensation of glycogen stores and prolonged the cycling time to exhaustion. A clever research design involving one-legged cycling showed that glycogen supercompensation was localised to the muscle that had been previously depleted, and studies identified the activity of glycogen synthase enzyme as an important factor in glycogen synthesis. These pioneering studies produced the ‘classical’ 7-day model of CHO loading, involving a 3–4 day ‘depletion’ phase of hard training and low CHO intake and finishing with a 3–4 day ‘loading’ phase of high CHO eating and exercise taper. Early field studies of prolonged running events showed that CHO loading might enhance sports performance, not by allowing the athlete to run faster, but by prolonging the time that race pace could be maintained (Karlsson & Saltin 1971).

Studies extended to trained subjects produced a ‘modified’ CHO loading strategy. Sherman and colleagues showed that well-trained athletes were able to supercompensate muscle glycogen stores without a depletion or ‘glycogen stripping’ phase (Sherman et al. 1981). The runners in this study elevated their muscle glycogen stores with 3 days of taper and high CHO intake, regardless of whether this was preceded by a depletion phase or a more typical diet and training preparation. For well-trained athletes at least, CHO loading may be seen as an extension of ‘fuelling up’ (rest and high CHO intake) over 3–4 days. The modified CHO-loading protocol offers a more practical strategy for competition preparation, by avoiding the fatigue and complexity of extreme diet and training requirements associated with the previous depletion phase.

The time course of glycogen storage was more recently examined, investigating muscle glycogen concentrations after 1 and 3 days of rest and a high CHO intake (10 g/kg BM) in well-trained male athletes (Bussau et al. 2002). After 1 day, muscle glycogen content increased significantly from pre-loading levels of ~90 mmol/kg ww to values of ~180 mmol/kg ww, and thereafter remained stable despite another 2 days of rest and CHO intake. The authors concluded that this was an ‘improved 1-day CHO-loading protocol’ and that a 24-hour period of physical inactivity and a high CHO intake was sufficient for trained athletes to maximise muscle glycogen levels. However, the rates of glycogen storage achieved in this study were not abnormally high compared to other literature values (90 mmol/kg ww in the 24-hour period of observation), and subjects started the ‘loading’ phase with high glycogen values, since their last training session was undertaken on the day before the loading protocol officially started. In other words, the true loading phase was ~36 h. In essence, the study provides a midpoint to the glycogen storage observations of Sherman and colleagues (1981) and shows that the supercompensation is not linear and may not require a full 72-hour period in rested subjects. Therefore, rather than promoting a unique strategy for CHO loading, this study suggests that optimal refuelling is probably achieved within 36–48 hours of the last exercise session, at least when the athlete rests and consumes adequate CHO intake. Of course, it is not always desirable for athletes to achieve total inactivity in the days prior to competition, since even in a taper some stimulus is required to maintain previously acquired training adaptations (Mujika

& Padilla 2000).

An athlete's ability to repeat glycogen supercompensation protocols has also been examined. Well-trained cyclists who undertook two consecutive periods of exercise depletion, followed by 48 hours of high CHO intake (12 g/kg/d) and rest, were found to elevate their glycogen stores above resting levels on the first occasion but not the next (McInerny et al. 2005). Further studies are needed to confirm this finding and determine why glycogen storage is attenuated with repeated CHO loading.

Although CHO loading is so well known that the term has entered everyday language, it seems difficult for athletes to master, even in its simplified form. At least one study has shown that in real life, athletes may not have the knowledge to plan a suitable exercise taper; furthermore they fail to reach the daily CHO intake targets of 7–10 g/kg BM needed to maximise glycogen storage (Burke & Read 1987).

12.7.1 Carbohydrate loading and performance of endurance events

Theoretically, CHO loading could enhance the performance of exercise or sporting events that would otherwise be limited by glycogen depletion. An increase in pre-event glycogen stores can prolong the duration for which moderate-intensity exercise can be undertaken before fatiguing. It may also enhance the performance of a set amount of work (a set distance) by preventing the decline in pace or work output that would otherwise occur as glycogen stores decline towards the end of the task. In a review of CHO loading studies, Hawley et al. (1997b) summarised the findings that supercompensation of glycogen stores is beneficial for the performance of exercise of greater than 90 minutes' duration, with the majority of studies investigating exercise protocols involving cycling or running. Typically, CHO loading will postpone fatigue and extend the duration of steady state exercise by ~20%, and improve performance over a set distance or workload by 2–3% (Hawley et al. 1997b). Such an intervention would provide a substantial improvement in most simple endurance events such as marathons, prolonged cycling and triathlon races and cross-country skiing events. Shorter events of 45–90 minutes' duration do not show significant performance benefits from CHO loading (Sherman et al. 1981; Hawley et al. 1997a).

Some team and racquet sports extend for at least 60–90 minutes of playing time, although the work requirements of these activities are of a less predictable nature. The interaction of prolonged duration and intermittent high-intensity exercise, which is associated with an increased rate of glycogen utilisation, might be expected to result in muscle glycogen depletion during the event. Theoretically, performance in such sports would be enhanced by supercompensation of muscle glycogen stores. In real life, it is extremely difficult to undertake studies that measure the performance of complex and variable sports such as these. Sports that have been shown to benefit from pre-event enhancement of muscle glycogen stores include a soccer simulation (Bangsbo et al. 1992), an indoor soccer game (Balsom et al. 1999) and a real-life ice hockey game (Akermark et al. 1996). However, Abt and colleagues failed to show significant improvement in the performance of skill-based tasks in a simulation of a soccer match (Abt et al. 1998). Decisions about the benefits of CHO loading may be specific not only to the sport, but to the individual athlete, depending on the requirements of their position or style of play. Of course, the logistics of competition in many of these sports, where games may be played daily to twice a week, would prevent a full CHO-loading preparation before each event. Nevertheless, athletes in these sports should fuel-up prior to each competition as much as is practical, and perhaps experiment with an extended preparation before the most important games, such as the final of the tournament.

It is of note that the majority of studies of CHO loading have failed to employ a placebo in their design: most studies have used a cross-over design in which subjects are fed a high-CHO preparation on one occasion, and a low–moderate-CHO diet as a control. Since the benefits of CHO loading are well known, it might be argued that the responses to the high-CHO preparation will be psychologically as well as physiologically driven. Indeed, it is interesting to note that the two studies that have employed a placebo design, masking the true diets, failed to find benefits from CHO loading (Hawley et al. 1997a; Burke et al. 2000). Although this probably reflects that the exercise and dietary protocols used in the studies (e.g. intake of substantial amounts of CHO during the trial) are such that moderately enhanced glycogen stores do not provide additional benefits, it may also be a result of the removal of the placebo effect.

Finally, the interaction of various dietary strategies with CHO loading deserves systematic research. There is evidence that the benefits from supercompensation of glycogen stores persist, or add to the effects of consuming CHO during a sufficiently prolonged exercise task (Widrick et al. 1993; Rauch et al. 1995). Such studies need to be conducted over a greater range of exercise events.

12.7.2 Gender and carbohydrate loading

Most studies of glycogen storage have been conducted with male subjects, with the assumption that the results also apply to female athletes. However, one study reported that female athletes were less responsive to a CHO-loading protocol (they failed to supercompensate muscle glycogen stores compared to male subjects) and failed to show a performance benefit following this dietary strategy (Tarnopolsky et al. 1995). It is difficult to undertake gender comparison studies due to problems in matching males and females for important parameters such as aerobic capacity (relative versus absolute values). Furthermore, CHO loading studies such as that undertaken by Tarnopolsky and colleagues have been criticised for the failure to standardise or select a particular phase of the menstrual cycle, and particularly for the intake of smaller amounts of dietary CHO and restricted energy intake by females. Although females may increase CHO intake to achieve a similar relative energy contribution, total CHO intake will be restricted if total energy intake is low. Without a substantial increase in the intake of substrate, a female athlete could not expect to increase muscle fuel stores or gain the potential performance enhancement arising from increased fuel availability. Adequate energy intake is also important in promoting glycogen synthesis from a given CHO intake (Tarnopolsky et al. 2001).

Gender differences in substrate utilisation and storage are still being investigated and explained (for review, see Tarnopolsky 1999). There is some evidence that the menstrual status of female athletes affects glycogen storage, with greater storage occurring during the luteal phase rather than the follicular phase (Nicklas et al. 1989; Hackney 1990). Well-trained female athletes who were in the luteal phase of their menstrual cycle undertook a modified loading protocol, with the focus on an exercise taper and increased CHO intake over the previous 4 days (Walker et al. 2000). This program achieved a 13% increase in muscle glycogen stores compared with storage on a moderate CHO intake, and increased time to exhaustion during submaximal cycling. This study showed that when sufficient CHO is consumed in a favourable hormonal environment, female athletes can increase their glycogen stores and improve exercise endurance. However, although male subjects were not included in this study to allow a direct comparison, the researchers noted that the supercompensation achieved by the females was of a lower magnitude than commonly reported in studies of male athletes.

By contrast, other studies have shown that males and female subjects show similar increases in muscle glycogen following protocols involving 3–4 days of high CHO intake (9–12 g/kg BM/d) and rest (James et al. 2001; Tarnopolsky et al. 2001). However, such increases in glycogen storage did not lead to a significant performance enhancement in trained female runners (Paul et al. 2001; Andrews et al. 2003) or cyclists (Paul et al. 2001). Of course, these outcomes may have arisen because the performance protocol was insufficiently long to benefit from additional muscle glycogen stores, or not sensitive enough to allow the detection of small but worthwhile improvements in performance. In summary, the existence and practical significance of physiological limitations to glycogen storage in female athletes remains to be further investigated. In the meantime, it is likely that the main challenge related to the CHO-loading practices of females is the practical issue of consuming adequate CHO and energy intake.

12.8 The pre-event meal (1–4 hours pre-event)

Foods and drinks consumed in the 4 hours prior to exercise have a role in fine-tuning competition preparation. The pre-event menu can be eaten largely for comfort or confidence, or it may play an active role in preparation by contributing to refuelling and rehydration goals.

The goals of the pre-event meal are to (Hawley & Burke 1998):

- continue to fuel muscle glycogen stores if they have not fully restored or loaded since the last exercise session
- restore liver glycogen content, especially for events undertaken in the morning where liver stores are depleted from an overnight fast
- ensure that the athlete is well hydrated

- prevent hunger, yet avoid the gastrointestinal discomfort and upset often experienced during exercise
- include foods and practices that are important to the athlete's psychology or superstitions.

The pre-event meal menu should include CHO-rich foods and drinks, especially in the case where body CHO stores are suboptimal, or where the event is of the duration and intensity to challenge these stores. A CHO-rich meal consumed 4 hours before exercise significantly increases the glycogen content of muscle and liver depleted by previous exercise or an overnight fast (Coyle et al. 1985). Compared to results achieved after an overnight fast, the intake of a substantial amount of CHO (~200–300 g) in the 2–4 hours before exercise has been shown to prolong cycling endurance (Wright et al. 1991) and enhance performance of an exercise test undertaken at the end of a standardised cycling task (Neufer et al. 1987; Sherman et al. 1989). CHO availability is enhanced by increasing muscle and liver glycogen stores, as well as by the storage of glucose in the gastrointestinal space for later release. Since liver glycogen stores are labile and may be substantially depleted by an overnight fast, pre-exercise CHO intake on the morning of an event may be important for maintaining blood glucose levels via hepatic glucose output during the latter stages of prolonged exercise.

Despite these benefits, the issue of CHO intake prior to exercise is not straightforward. In fact, the elevation of plasma insulin concentrations following pre-exercise CHO feedings could be a potential *disadvantage* to exercise metabolism and performance. A rise in insulin suppresses lipolysis and fat utilisation, concomitantly accelerating CHO oxidation and causing a decline in plasma glucose concentrations at the onset of exercise. These metabolic alterations have been observed even when CHO was consumed 4 hours before exercise, and persisted despite the normalisation of plasma glucose and insulin concentrations at the onset of exercise (Coyle et al. 1985). However, such metabolic perturbations do not appear detrimental to performance. One safeguard is to ensure that the amount of CHO in the pre-event meal is substantial rather than minor; thus, any increase in CHO utilisation during exercise will be more than offset by the large increase in CHO availability.

In the field, it is not always practical to consume a substantial CHO-rich meal or snack in the 4 hours before a sporting event. For example, it is unlikely that an athlete will want to sacrifice sleep to eat heartily before an early morning race start. Most will settle for a lighter meal or snack before the event, and consume CHO throughout the event to balance missed fuelling opportunities. A smaller pre-event meal may also make sense for events or athletes predisposed to gastrointestinal discomfort. Foods with a low-fat, low-fibre and low–moderate protein content are the preferred choice for the pre-event menu since they are less prone to cause gastrointestinal upsets (Rehrer et al. 1992). Liquid meal supplements or CHO-containing drinks and bars are useful for athletes who suffer from pre-event nerves or an uncertain pre-event timetable. Above all, the individual athlete should choose a strategy that suits their situation and their past experiences, and can be fine-tuned with further experimentation.

12.9 Carbohydrate consumed in the hour before exercise

Not all studies of CHO intake before exercise have shown favourable outcomes. Foster and colleagues (1979) found that feeding 75 g of glucose 30 minutes prior to exercise impaired cycle time to exhaustion at 80% of $VO_{2\ max}$ compared to exercise in the fasted state. The pre-event feeding did not alter the length of time subjects were able to ride during more intense (100% of $VO_{2\ max}$) exercise. A rapid drop in blood glucose concentration was noted during the first 10 minutes of exercise after subjects had been fed CHO, but this response was transient and was not associated with fatigue. Although muscle glycogen content was not determined, the reduction in endurance following CHO feeding was attributed to an accelerated muscle glycogenolysis (Foster et al. 1979).

The results of this study have been so widely reported and publicised that warnings to avoid CHO intake during the hour prior to endurance exercise have become part of sports nutrition dogma (Inge & Brukner 1986; Wilmore & Costill 1994). However, reviews of the literature reveal that this is the *only* study to find a reduction in performance capacity after the ingestion of CHO in the hour before exercise (Coyle 1991; Hawley & Burke 1997). These reviews summarise that the findings of other investigations of pre-exercise CHO feeding range from no detrimental effect to improvements in performance in the order of 7–20% (Coyle 1991; Hawley & Burke 1997). In most cases, the decline in blood glucose observed during the first 20 minutes of exercise is self-corrected with no apparent effects on the athlete.

Nevertheless, there seems to be a small percentage of athletes who respond negatively to CHO feedings in the hour before exercise. These athletes experience an exaggerated CHO oxidation and decrease in blood glucose concentrations at the start of exercise, suffering a rapid onset of fatigue and symptoms of hypoglycaemia. Why some athletes experience such an extreme reaction is not known. One study has suggested that risk factors include the intake of small amounts of CHO (<50 g), increased sensitivity to insulin, a lower sympathetic-induced counter-regulation, and exercise intensity of low–moderate workload (Kuipers et al. 1999). However, a series of related studies failed to find any differences in the mild decline in blood glucose during exercise or the prevalence of blood glucose concentrations defined as hypoglycaemic (<3.5 mmol/L) with the systematic manipulation of the amount of CHO consumed 45 minutes before exercise (20–200 g) (Jentjens et al. 2003) or the intensity of exercise (55–90% $VO_{2\ max}$) (Achten & Jeukendrup 2003). Changing the timing of intake of 75 g CHO before exercise (15–75 minutes) caused differences in blood glucose concentrations at the beginning of exercise, with the later feeding times (45 and 75 minutes) being associated with a greater prevalence of hypoglycaemic levels of blood glucose. Nevertheless, differences self-corrected within 10 minutes of cycling (Moseley et al. 2003). The intake of sugars with a lower glycaemic index (GI) (such as trehalose or galactose) was associated with a reduced prevalence of hypoglycaemic levels of blood glucose compared with glucose feedings (Jentjens & Jeukendrup 2003). Finally, cyclists who developed hypoglycaemic blood glucose levels did not differ in insulin sensitivity compared with subjects whose lowest blood glucose level remained above 3.5 mmol/L (Jentjens & Jeukendrup 2002).

Not all athletes who experience a major decline in blood glucose concentrations experience hypoglycaemic symptoms; there is some evidence that sensitisation to low glucose levels may adapt the athlete to an increased threshold before symptoms are reported (Kuipers et al. 1999). Nevertheless, these effects are so clear-cut that at-risk athletes will be identified easily. Preventive action for this group includes a number of options:

- Experiment to find the critical time before exercise that CHO intake should be avoided.
- Consume a substantial amount of CHO in the pre-event snack/meal (>1g/kg).
- Include low GI, CHO-rich choices in the pre-event menu; these have an attenuated and sustained blood glucose and insulin response.
- Include some high-intensity sprints during the warm-up to the event to stimulate hepatic glucose output.
- Consume CHO during the event.

12.10 Pre-exercise carbohydrate and the glycaemic index

CHO-rich foods and drinks do not produce identical blood glucose and insulin responses; neither do athletes respond according to the dogma that 'simple' CHO types produce rapid and short-lived rises in blood glucose concentrations whereas 'complex' CHO-rich foods produce a flatter and sustained blood glucose rise. The glycaemic index (GI) offers a means to measure and utilise the individual blood glucose profiles achieved by consuming various CHO-rich foods and drinks. The GI provides a ranking of CHO foods based on the measured post-prandial blood glucose response compared to that of a reference food (either glucose or white bread). It has been shown to provide a reliable and consistent measure of relative blood glucose response to CHO-rich foods and meals. Furthermore, GI values have been used to manipulate meals and diets to produce a desired metabolic or clinical outcome, which is useful for the treatment for diabetes, hyperlipidaemia and, potentially, obesity (for reviews, see Wolever 1990; Brand Miller 1994).

Early studies of pre-exercise CHO feedings showed different outcomes according to the type of sugar ingested. Fructose, a low-GI saccharide, could be consumed before exercise without producing the metabolic impairments seen with glucose feedings (Hargreaves et al. 1985). Thomas and colleagues were the first to apply the GI of real foods to the arena of sports nutrition, by undertaking a manipulation of the glycaemic response to pre-exercise CHO-rich meals (Thomas et al. 1991). A 1 g/kg CHO meal in the form of a low-GI food (lentils), eaten 1 hour prior to cycling at 67% of $VO_{2\ max}$, was found to prolong time to exhaustion compared with the ingestion of a high-GI food (potatoes). These results were attributed to lower glycaemic and insulinaemic responses to the low-GI trial compared with the high-GI meal, promoting more stable blood glucose levels during exercise, reduced rates of CHO oxidation and increased free fatty acid (FFA) concentrations. Although muscle glycogen was not measured, the authors suggested that glycogen sparing may have occurred with the low-GI

CHO trial (Thomas et al. 1991). The results of the study have been publicised widely and are largely responsible for the general advice that athletes should choose pre-exercise meals based on low-GI, CHO-rich foods and drinks (Brand Miller et al. 1998). However, other studies have failed to find performance benefits following the intake of a low-GI pre-exercise meal. The literature pertaining to the GI of pre-exercise CHO feedings has been summarised in Table 12.2.

TABLE 12.2 Studies of GI and pre-exercise CHO intake

Study	Subjects	CHO feedings	Exercise protocol	Enhanced performance	Findings
Studies showing benefit of low-GI meals over high-GI, CHO-rich meals					
Thomas et al. 1991	Trained cyclists (*n* = 8 male)	Control (water) or 1 g/kg BM CHO: high-GI (potatoes) low-GI (lentils) glucose 60 min pre-exercise	Cycling to exhaustion @ 67% VO_2 max	Yes (low-GI versus high-GI foods) BUT No difference between low-GI and glucose	Compared with high-GI food, low-GI food reduced blood glucose changes (lower rise pre-exercise and smaller decline during exercise). Decrease in suppression of FFA also. CHO sparing/sustained CHO availability suggested as explanation for increased endurance (~20%) with low GI compared with high GI. No difference in performance time between low-GI and glucose trials.
DeMarco et al. 1999	Trained cyclists (*n* = 10 male + female)	Control (water) or 1.5 g/kg CHO: low-GI meal high-GI meal 30 min pre-exercise	Cycling for 120 min @ 70% VO_2 max, followed by time to exhaustion @ 100% VO_2 max	Yes	Low-GI meal maintained glucose during exercise and reduced CHO oxidation compared with high-GI meal. Time to exhaustion at maximal effort prolonged by 59%.
Wu and Williams 2006	Recreational runners (*n* = 8 male)	2 g/kg CHO: low-GI meal high-GI meal 3 h pre-exercise	Treadmill run at 70% VO_2 max until exhaustion	Yes	Time to exhaustion was longer (p <0.05) in low-GI trial (108.8 min) than high-GI trial (101.4 min). Fat oxidation was higher in low-GI trial.
Wong et al. 2008	Trained runners (*n* = 8 male)	1.5 g/kg CHO: low-GI meal high-GI meal 2 h pre-exercise	21 km run on treadmill: 5 km at @ 70% VO_{2max} + 16 km TT	Yes	21 km run time was faster with low-GI than high-GI trial (98.7 versus 101.5 min, p <0.01). CHO increased and fat oxidation decreased in high-GI trial compared with low-GI trial.
Moore et al. 2010	Well-trained cyclists (*n* = 10 male)	1 g/kg CHO: high-GI meal (cornflakes) low-GI meal (bran flakes) 45 min pre-exercise	40 km cycling TT	Yes	TT was completed in faster time with low-GI meal than high-GI meal (93 ± 8 min versus 96 ± 7, p = 0.009). Low-GI meal was associated with higher sustained CHO oxidation rates.
Studies showing no benefit of low-GI meals over high-GI, CHO-rich meals					
Thomas et al. 1994	Trained cyclists (*n* = 6 male)	1 g/kg CHO: low-GI powdered food high-GI powdered food low-GI cereal high-GI cereal 60 min pre-exercise	Cycling to exhaustion @ 67% VO_2 max	No	Inverse correlation between GI and decline in blood glucose, or suppression of FFA during exercise. No differences in performance times. No correlation between GI and endurance.
Febbraio and	Well-trained cyclists (*n* =	Placebo (low joule jelly) or 1 g/kg CHO:	120 min cycling @	No	Low-GI food reduced pre-exercise rise in blood glucose compared with high-GI food.

Stewart 1996	6 male)	high-GI (potatoes) low-GI (lentils) 45 min pre-exercise	70% VO_2 max + 15 min TT		No differences during steady state exercise in blood glucose or insulin. No differences in total CHO oxidation between the two CHO treatments, and similar muscle glycogen utilisation over 120 min. No differences in TT performance between three trials.
Sparks et al. 1998	Well-trained cyclists (*n* = 8 male)	Placebo (low-joule soft drink) or 1 g/kg CHO: high-GI (potatoes) low-GI (lentils) 45 min pre-exercise	50 min cycling @ 67% VO_2 max + 15 min TT	No	Low-GI food reduced pre-exercise rise in blood glucose and decline during exercise compared with high-GI food. Higher CHO oxidation during exercise with high-GI trial. No differences in work completed in 15 min TT between trials.
Burke et al. 1998	Well-trained cyclists (*n* = 6 male)	Placebo (low-joule jelly) or 2 g/kg CHO: low-GI (pasta) high-GI (potatoes) 120 min pre-exercise 10% CHO solution consumed during exercise to provide ~1g/kg CHO	120 min cycling @ 70% VO_2 max + 15 min TT	No	Low-GI food reduced rise in pre-exercise blood glucose compared with high-GI food. CHO fed during exercise minimised all differences in glucose, insulin, FFA during exercise. No difference in substrate oxidation or performance of TT between trials.
Wee et al. 1999	Active subjects (*n* = 8 male + female)	2 g/kg CHO: low-GI meal high-GI meal 3 h pre-exercise	Running to exhaustion @ 70% VO_2 max	No	High-GI meal caused a decline in blood glucose at the onset of exercise. Low-GI meal attenuated this response and reduced CHO oxidation during first 80 min of exercise. No differences in endurance.
Febbraio et al. 2000	Endurance-trained cyclists (*n* = 8 male)	Placebo (low-joule jelly) or 1 g/kg CHO: low-GI meal (muesli) high-GI meal (potatoes) 30 min pre-exercise	120 min cycling @ 70% VO_2 max + 30 min TT	No	High-GI meal caused self-correcting decline in blood glucose concentrations at onset of exercise compared with other trials, and suppression of FFA throughout steady state cycling. Increased CHO oxidation during high-GI trial due to increased glucose uptake and trend to increased muscle glycogenolysis. No differences in performance of TT between trials.
Wong et al. 2009	Trained runners (*n* = 9 male)	Placebo (low-joule jelly) or 1.5 g/kg CHO: high-GI meal low-GI meal 120 min pre-exercise 6.6% CHO solution consumed during exercise to provide ~48 g/h	21 km run on treadmill: 5 km at @ 70% VO_{2max} + 16 km TT	No	No difference in running times between trials (91.1, 91.8 and 92.9 for high-GI, low-GI and placebo, respectively, NS). Blood glucose concentrations, and rates of oxidation of fat and CHO were similar between trials.
Little et al. 2009	Trained athletes (runners, team sport) (*n* = 7 male)	Control (fast) or ~ 1.3 g/kg CHO: high GI (mashed potatoes) how GI (lentils) 3 h pre-exercise (+~ 0.2 g/kg of corresponding CHO at 'half-time' of protocol)	Intermittent running on treadmill to mimic soccer match (2 × 45 min halves + 15 min break with 5 × 1 min repeated sprint test in last 15 min)	No	Subjects were unable to tolerate original goals of 2 g/kg CHO pre-match and 0.25 g/kg at half time. Higher CHO oxidation with high-GI meal than control. Magnitude-based inference statistical analysis showed distance covered in repeated sprints measure at end of trial was likely to be further with both CHO trials compared to control (+ 247 ± 352 m with low GI and + 223 ± 385 m with high GI, which was 81% and 76% probability of being further than control).
Little et al. 2010	Trained athletes (runners, team sport)	Control (fast) or 1.5 g/kg CHO: high GI (mashed potatoes) low GI (lentils) 3 h	Intermittent running on treadmill to mimic soccer	No	Distance covered in in repeated sprints was greater in CHO trials (*p* <0.04) than control trial (~ 1500 m versus 1460 m), but no difference between low-GI and high-GI.

	(*n* = 16 male)	pre-exercise	match (2 × 45 min halves + 15 min break with 5 × 1 min repeated sprint test in last 15 min)		Generally, no difference in substrate use between low GI and high GI, perhaps due to high-intensity nature of exercise. Subset of 5 subjects underwent muscle biopsy at end of 75 min to measure glycogen availability prior to performance measure: muscle glycogen higher in both CHO trials but no difference due to GI of pre-exercise CHO. Lower RPE during performance test with low GI than control.
Hulton et al. 2012	Recreational team athletes (*n* = 9 male)	2 g/kg CHO: high-GI meal (high GI rice + chicken sauce + sports drink) low-GI meal (low GI rice + chicken sauce + apple juice) 3.5 h pre-exercise	Intermittent running on treadmill to mimic soccer match (2 × 45 min halves + 15 min break) + 1 km sprint at end	No	No difference in time to complete 1 km sprint between trials (210 ± 19 s for low-GI versus 216 ± 23 s for high-GI, NS). No difference in CHO oxidation between trials.
Bennet et al. 2012	Recreational team athletes (*n* = 10 male + 4 female)	1.5 g/kg CHO: high GI (mashed potatoes + bread) low GI (lentils + honey + berries) 2 h pre-exercise and repeated (2 g/kg CHO) of same meal within 1 h post-exercise in preparation for repeated exercise protocol 2 h later (e.g. 3 h between protocols)	2 × protocols with 3 h recovery: Intermittent running on treadmill to mimic soccer match (2 × 45 min halves + 15 min break with 5 × 1 min repeated sprint test in last 15 min)	No	Small period of lower CHO oxidation at beginning of low-GI trial. Main effect for performance of sprint measures at end of protocol: decreased on second protocol. No differences in distance covered in sprints between CHO trials (1328 ± 141 and 1283 ± 128 m with low-GI and 1333 ± 100 and 1303 ± 210 m with high-GI, NS).

BM = body mass, CHO = carbohydrate, FFA = free fatty acids, GI = glycaemic index, NS = not significant, RPE = ratings of perceived exertion, TT = time trial,

This literature shows that pre-exercise low-GI, CHO-rich meals generally achieve a lower post-prandial blood glucose response and a more sustained metabolic response throughout exercise compared to high-GI, CHO-rich foods (Sparks et al. 1998; DeMarco et al. 1999; Wee et al. 1999; Stevenson et al. 2006). These findings are not universal, however, with some studies finding that that glycaemic differences prior to the onset of exercise are shortlived and of transient consequence to metabolism (Febbraio & Stewart 1996). While some studies show that the GI of the pre-exercise meal(s) has no effect on glycogen utilisation during exercise (Febbraio & Stewart 1996; Stevenson et al. 2009), others show that glycogen utilisation is reduced (Wee et al. 2005) or possibly lower (Febbraio et al. 2000) following low-GI meals or diets. Reduced availability of FFA during exercise in response to high-GI CHO feedings has also been associated with increased utilisation of intramuscular triglyceride (Trennell et al. 2008; Stevenson et al. 2009). It has been suggested that these metabolic differences be further examined for outcomes related to loss of body fat and increased satiety in athletes (Stevenson et al. 2006).

Perhaps the greatest conflicts in the pre-event menu debate lie with the effect on exercise performance. As summarised in Table 12.2, some studies have reported that a low-GI meal before exercise enhances exercise capacity or performance compared with high-GI CHOs (Thomas et al. 1991; DeMarco et al. 1999; Wu & Williams 2006; Wong et al. 2008). However, other studies have failed to find benefits from the consumption of a low-GI pre-event meal (Thomas et al. 1994; Febbraio & Stewart 1996; Sparks et al. 1998; Burke et al. 1998; Wee et al. 1999; Febbraio et al. 2000; Wong et al. 2009) even when metabolism has been altered throughout the exercise. An important factor in the interpretation of these studies, as with many areas of sports nutrition research, lies in the issue of defining and measuring ‘performance’. It should be noted that three of the studies that report a beneficial exercise outcome following the lowering of the GI of the pre-event meal have measured time to exhaustion at a fixed work rate as their interpretation of performance (Thomas et al. 1991; DeMarco et al. 1999; Wu & Williams 2006). These findings are not readily applied to the world of competitive sport, where a successful performance is determined as being able to complete a set amount of work or set distance in the shortest possible time, and where the athlete is free to choose and vary their work rate. It is important if the results of studies are to be translated into practical advice for competitive athletes that the study design and variables should be chosen to mimic the situation of sport as closely as possible.

Finally, a central issue that is overlooked in the debate is the overall importance of pre-exercise feedings in determining CHO availability during prolonged exercise. In endurance exercise events, a typical and effective strategy used by athletes to promote CHO availability is to ingest CHO-rich drinks or foods during the event. Yet, in the typical pre-event meal study, athletes are expected to perform prolonged exercise while consuming water or with no intake at all. The final and practical message to the athlete can be considered only when the interaction of pre-exercise and during-exercise CHO intake has been studied. In one study, well-trained cyclists ate three different pre-event meals (high GI, low GI or placebo) on separate occasions before cycling for 120 minutes at 70% $VO_{2\ max}$, followed by a time trial to complete a set amount of work (Burke et al. 1998). During exercise, subjects ingested a 10% glucose solution, providing an intake of ~1 g CHO/kg/h. Despite pre-exercise differences in glucose, insulin and FFA concentrations between trials, there were no differences during exercise. Furthermore, there were no differences between any of the trials with regard to total CHO oxidation over the 120 minutes of steady state exercise, or the oxidation of the CHO consumed from the glucose drink. There were no differences in the time to complete the performance ride.

A similarly designed study involving a running protocol provides further support for the importance of CHO intake during exercise (Wong et al. 2009). In that study, runners consumed a CHO-electrolyte drink providing ~0.8 g CHO/kg/h during a 21-km treadmill time trial that was preceded either by a placebo meal or a meal providing 1.5 g/kg CHO from high- or low-GI sources. There were no differences in substrate utilisation, blood glucose profiles and running performances between trials.

In summary, it appears likely that when CHO is consumed during exercise according to sports nutrition guidelines, any effects of pre-exercise CHO intake on either metabolism or performance are negligible or at least greatly diminished. Each athlete must judge the benefits and the practical issues associated with pre-exercise feedings in their particular situation. In cases where an athlete may not be able to consume CHO during a prolonged event or workout, they may find it useful to choose a menu based on low-GI CHO foods to promote a more sustained release of CHO throughout exercise. However, there is no evidence of universal benefits from such menu choices, particularly where the athlete is able to refuel during their session, or where their favoured and familiar food choices happen to have a high GI. In the overall scheme, pre-event eating needs to balance a number of factors including the athlete's food likes, availability of choices and gastrointestinal comfort.

12.11 Pre-exercise hydration

Dehydration poses one of the most common nutrition problems occurring in sport. Chapter 13 details strategies to enhance fluid balance during training and competition sessions. Since, on most occasions, fluid intake during exercise will not match the rate of sweat loss, it is critical for the athlete to start the session well hydrated. Special attention is needed to ensure full restoration of fluid balance after previous exercise bouts, particularly if unusually large fluid losses have occurred. This can happen when athletes undertake deliberate dehydration to 'make weight' in weight-category sports (Chapter 7), or when an athlete moves to a hot environment and fails to adapt their fluid intake practices to cope with their new sweat losses. In the training scenario, at least, several studies employing measurements of urine specific gravity or osmolality show that some individuals begin the session in a hypohydrated state (Maughan et al. 2004; Godek et al. 2005; Shirreffs et al. 2005). Therefore, there is evidence that an education message promoting deliberate hydration strategies in the hours or days before exercise is warranted. However, the awareness of the dangers of hyponatraemia during prolonged exercise resulting from excessive fluid intake (see Chapter 13) provides a reminder that messages regarding fluid intake can be misinterpreted. Athletes need to be aware of fluid intake strategies that are commensurate with their fluid needs, and to undertake any overhydration strategies (section 12.14) under supervision. Chapter 14 deals with strategies to promote rapid restoration of fluid deficits.

12.11.1 Pre-exercise hydration and hyperhydration

Many athletes undertake events in which significant dehydration is inevitable, which poses a challenge to both their health and their performance. Such dehydration can occur when an athlete's sweat rate is extremely high, when there is little opportunity to drink during the event, or when these factors are combined. Some athletes have experimented with hyperhydration or 'fluid overloading' in the hours prior to the event to attempt to reduce the total fluid deficit incurred. This practice has been shown to increase total body water, expand plasma volume and ultimately enhance performance in a subsequent exercise trial (Moroff & Bass 1980).

However, there are some shortcomings and possible disadvantages to simple fluid overloading techniques. First, much of the fluid is excreted via urination, since the body has a well-developed system to regulate the volume and concentration of its fluid content both at rest and during exercise. Fluid overloading may have a detrimental effect on performance if it causes the significant interruption of having to urinate immediately before or in the early stages of the event. The discomfort of excess fluid in the gut has also been shown to impair the performance of moderate–high-intensity exercise (Robinson et al. 1995). As previously mentioned, if taken to extreme levels and in susceptible individuals, excessive fluid intake may lead to hyponatraemia. Clearly, fluid overloading just before an event is a strategy that needs to be researched before any more firm recommendations can be made to athletes.

In one study, heat-acclimatised subjects were required to double their usual fluid intake for a week, from a mean intake of 1980 mL/d to 4085 mL/d (Kristal-Boneh et al. 1995). This was found to reset normal fluid balance to retain an extra 600 mL of fluid. Several experimental trials in the heat found that superior hydration status increased heat tolerance and enhanced duration of work in the heat, allowed achievement of maximal aerobic workload at a lower heart rate and improved performance in a time trial. More work is needed to confirm these results, but they support the advice often given to athletes competing in a hot climate to increase their fluid intake over the days leading up to the event. Whether this advice merely ensures fluid balance, rather than promoting fluid overload, has not been tested adequately prior to this study.

Of course, the impact of any gain in BM as a result of increased fluid retention might need to be taken into account in sports that are weight-restricted (such as lightweight rowing) or weight-sensitive (such as running and uphill riding). In many sports, an increase in BM may increase the energy cost of the activity, impede the speed of acceleration and change of direction, and decrease the 'power to weight' (BM) ratio. Whether this occurs and whether the small impairments in performance are more than offset by the improvement in fluid balance need to be individually studied.

12.11.2 Glycerol hyperhydration

A method of hyperhydration under current study involves the consumption of a small amount of glycerol (1–1.2 g/kg BM) along with a large fluid bolus (25–35 mL/kg) in the hours prior to exercise. Glycerol, a three-carbon alcohol, provides the backbone to triglyceride molecules and is released during lipolysis. Within the body, it is evenly distributed throughout fluid compartments and exerts an osmotic pressure. When consumed orally, it is rapidly absorbed and distributed among body fluid compartments before being slowly metabolised via the liver and kidneys. When consumed in combination with a substantial fluid intake, the osmotic pressure will enhance the retention of this fluid and expansion of the various body fluid spaces. Typically, this allows a fluid expansion or retention of ~600 mL above a fluid bolus alone, by reducing urinary volume. Further information on glycerol as a hyperhydrating agent is found in reviews by Nelson and Robergs (2007) and van Rosendal and Coombes (2012).

The effect of glycerol hyperhydration strategies on thermoregulation and exercise performance is at present unclear. Studies investigating the effects on sports performance have shown both positive and neutral outcomes, with at least some of the inconsistency in the literature due to differences in study methodologies. However, the most promising scenario involves the use of glycerol to maximise the retention of fluid bolus just prior to an event in which a substantial fluid deficit cannot be prevented. In some, but not all, studies of this type, glycerol hyperhydration has been associated with performance benefits. Of particular interest are two studies undertaken with competitive athletes. In both studies, competitive cyclists were able to do more work in a time trial undertaken in the heat following hyperhydration with glycerol, compared with a trial using a fluid overload with a placebo drink (Hitchins et al. 1999; Anderson et al. 2001). Even if future studies confirm the beneficial effects of glycerol hyperhydration strategies, it may require careful research to determine the mechanisms behind the effect. At present the theoretical advantages of increased sweat losses and greater capacity for heat dissipation, and attenuation of cardiac and thermoregulatory challenges, are not consistently seen.

There are some concerns associated with glycerol hyperhydration strategies. As previously discussed, the cost of a gain in BM might need to be considered in some sports. Other side effects from the use of glycerol include nausea, gastrointestinal distress and headaches resulting from increased intracranial pressure. These problems have been reported among some, but not all, subjects in the current studies. Fine-tuning of protocols for individualised situations should always be undertaken, and may reduce the risk of these problems, but some individuals may remain at a greater risk than others.

The most important consideration with glycerol hyperhydration lies with its status within anti-doping codes. Since 2010, S5 (Diuretics and other masking agents) of the WADA Prohibited List has included glycerol within its examples of banned plasma expanders (www.wada-ama.org). This ban exists despite the conclusion of a meta-analysis of glycerol hyperhydration studies, that in comparison to the use of other plasma-volume expanding agents, it has very limited potential in increasing plasma volume and altering doping-relevant blood parameters (Koehler et al. 2013) and the findings of an intervention study that chronic (1 week) use of a glycerol hyperhydration protocol does not significantly alter doping-relevant blood parameters (Polyviou et al. 2012). Nevertheless, to distinguish normal dietary intake of glycerol and glycerol release from triglyceride breakdown from specific intake of glycerol to aid fluid balance, WADA has set a threshold for urinary glycerol at 1.3 mg/mL and advises that the small quantities of glycerol consumed in everyday foodstuffs and toiletries will not cause an athlete to test positive (www.wada-ama.org). In summary, the use of glycerol hyperhydration is not available to athletes who are competing under a WADA-linked anti-doping code. However, this ban may not be relevant to other people undertaking exercise in thermally stressful conditions, such as non-competitive/recreational sporting activities or occupational activities such as military exercises or fire-fighting.

12.12 Salt loading

Another protocol that has been shown to enhance fluid status prior to exercise is the consumption of a high sodium beverage. One strategy involves the intake of a moderate volume (10 mL/kg) of a high sodium beverage (~160 mmol/L) over a 60-minute period in the 2 hours prior to exercise. Studies involving trained males (Sims et al. 2007b) and females (Sims et al. 2007a) have shown that such a protocol increases plasma volume before exercise and is associated with less thermoregulatory and perceived strain during exercise and increased exercise capacity in warm conditions. Further studies are warranted.

12.13 Priming the stomach with a fluid bolus

Even if the athlete is not aiming to 'fluid overload', there can be good reasons to have a drink just before exercise. Effective rehydration during exercise depends on maximising the rate of fluid delivery from the stomach to the intestine for absorption. One of the factors affecting gastric emptying is gastric distension due to the volume of stomach contents. According to Noakes and colleagues, optimal delivery of fluid from the stomach can be achieved by beginning exercise with a comfortable volume of fluid in the stomach, and adopting a pattern of periodic fluid intake during the exercise designed to top up gastric contents as it is partially emptied (Noakes et al. 1991). Obviously, the athlete will need to experiment to determine what is a comfortable volume with which to prime gastric volume, and in particular how comfortable this feels once exercise has commenced. However, as a general rule of thumb, most athletes can tolerate a bolus of about 5 mL/kg BM (300–400 mL) of fluid immediately before the event starts. This may provide a useful start to fluid intake tactics during exercise (see Chapter 13).

Summary

The outcomes of pre-event nutrition strategies range from psychological wellbeing and confidence to optimal fluid and fuel status. The importance of these strategies will depend on the range and severity of physiological challenges that are likely to limit performance in the athlete's individual event. This may be determined by characteristics of the event itself, as well as the degree to which the athlete has been able to recover since their last workout or competition event. Nutrition strategies may include increased CHO intake during the day(s) prior to the event, as well as the extended fuelling-up known as CHO loading, which has been shown to enhance endurance and the performance of prolonged exercise events. The pre-event meal also provides an opportunity to refuel muscle and liver glycogen stores. There is some concern that pre-exercise CHO feedings may increase CHO utilisation during exercise, but the intake of substantial amounts of CHO can offset the increased rate of substrate use. The choice of low-GI, CHO-rich foods in the pre-event menu may also sustain the delivery of CHO during exercise, but this does not provide a guaranteed performance advantage, especially when additional CHO is consumed during the event. Pre-event preparation should also consider fluid balance, with strategies to rehydrate from previous dehydration associated with exercise or weight-making activities, as well as the potential for hyperhydration in preparation for events in which a large fluid deficit is unavoidable. A variety of eating practices can be chosen by the athlete to meet their competition preparation goals. These need to consider the practical aspects of nutrition such as gastrointestinal comfort, the athlete's likes and dislikes, and food availability. Above all, the athlete should experiment with their pre-event nutrition practices to find and fine-tune strategies that are successful. These may be individual to the athlete and their specific event. Pre-event nutrition practices should be undertaken as part of an integrated competition nutrition plan. Ideally, an athlete should combat these challenges of competition by undertaking a systematic plan of nutrition strategies before, during and even in the recovery after an event.

Practice tips

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PREPARING FOR COMPETITION

- The sports dietitian needs to carefully evaluate the physiological demand of competitive events to determine the best strategy for competition preparation. In general, athletes in endurance events lasting longer than 90 minutes should consider increasing CHO intake in the days before competition. This may also apply to athletes in team sports who are at risk of glycogen depletion due to the high intensity and prolonged nature of their sport and the likelihood of fatigue occurring late in the game.
- While endurance athletes may choose to CHO load, achieving maximal glycogen stores is not a main goal for team sport athletes who are undertaking a tournament-format competition. These athletes may simply increase their CHO intake on the day prior to the start of the tournament. Of course, these athletes may also need to maintain a high CHO intake throughout the course of the tournament, as multiple matches are typically played in succession, with few days of rest to fully recover.
- To pinpoint individual needs in preparation for competition, sports dietitians should undertake an assessment of the athlete's previous experiences, which includes the strategies used, their success or failure, specific complaints during exercise such as gastrointestinal issues or early onset of fatigue, and food preferences. Depending on the relationship between the athlete and sports dietitian, a specific plan can be developed, allowing fine-tuning across multiple competitions that includes feedback from the athlete. In the case of international competitions, a flexible approach is encouraged to accommodate the limited access to familiar foods when travelling.
- An assessment of this type can help the athlete to decide how many days are required for CHO-loading strategies. Endurance-trained individuals, who already practise high-CHO eating strategies, may only need an additional day of CHO loading at 10–13 g/kg/d to achieve their goals, while a novice runner may best consider a 3–day CHO load. Sports dietitians should be sensitive to individual preferences, and this is best assessed in a one-on-one counselling session, with athletes reporting from previous race experiences.
- CHO loading leads to water gain due to glycogen storage, and this typically leads to weight gain, occasional bloating and subjective feelings of heaviness. While these are common comments from athletes, there is no evidence of impaired performance due to these side effects. Thus, sports dietitians should fully disclose the chance of weight gain and may want to discourage athletes from measuring body mass during the loading phase to reduce stress. A positive experience with CHO loading in a race will be sufficient feedback to strengthen confidence for further optimisation of race preparation using CHO-loading regimens.
- CHO loading is not indicated if athletes have clinical issues such as diabetes type I or II, endocrine disorders, or where athletes are extremely fearful of weight gain. In these cases, a moderately high CHO intake (7–8 g/kg/d) would suffice for race preparation. However, in these cases, the sports dietitian may want to pay more attention to fuelling within the race or event.
- To succeed with CHO loading, practice is needed. It is recommended that strategies are first trialled for long

workouts before the regimen is tested during minor competitions and, ultimately, in preparation for key races.

- To assist with gastrointestinal comfort during CHO loading, athletes are encouraged to 'take care' of CHO targets as the first priority at each meal before eating other foods. Care must be taken so that athletes do not fill up on salads, raw vegetables or whole grain and leguminous options (e.g. beans) options, then find themselves unable to comfortably consume the foods needed to meet CHO targets. Adopting the culture of an Italian eating pattern may work best: first course (Primo) of risotto or pasta, second course (Secondo) of a small helping of meat/fish with steamed vegetables, followed by dessert.
- The fibre content of a CHO-loading diet may need to be reduced if athletes complain about discomfort due to 'overfilling', or gastrointestinal issues such as flatulence and diarrhoea. However, many athletes also find it is important to keep a certain level of fibre intake in order to maintain normal bowel function.
- Some athletes like to undertake a low-residue diet during their CHO-loading strategies to enable a small body mass 'loss' due to the temporary removal of fibre/residue from the gut. This may counteract some of the weight gain associated with the additional muscle water/glycogen retention, as well as reduce the need for a bowel movement on competition day (pre-event or during the event). Principles used in bowel preparation for surgery can be adapted into the CHO-loading plan: avoidance of vegetables and fruit that are raw, eaten with skins or include seeds; choice of 'white' versions of cereals, bread, rice and pasta; and avoidance of nuts and legumes. The addition of pureed or mashed fruits and vegetables and pulp-free juices provide some variety to meals based on refined grain foods, dairy and meats. Of course, this is not considered a 'healthy' long-term eating style but does not pose a problem for short-term use prior to major competitions.
- Simplicity and monotony should also be considered (see note, Table 12.3). There is no specific need for variety during a 1–3 day CHO-loading regimen. Athletes should simply do what feels best to them, and if they have found via experience that a certain breakfast or lunch 'sits well', they may like to repeat this over the loading days or use it also as a pre-event meal.
- When the competition involves travel, athletes are advised to take some of their most important foods with them, especially if the event is held in a location with limited availability of familiar food and difficulty in sourcing well-suited foods. Teaching dietary flexibility and literacy about the CHO content in foods is important for travel preparation and assists in achieving a successful implementation of CHO loading abroad.

TABLE 12.3 Three-day carbohydrate loading for a 60-kg female endurance athlete aiming for 10 g of carbohydrate per kilogram of body weight

Day 1	
Nutritional profile • 3100 kcal • 100 g protein • 600 g CHO (10 g/kg for 60 kg athlete) • 45 g fat • 1000 mg calcium • 25 mg iron • 25 g fibre	**Breakfast** Cereal, 1.5 cups Low fat milk, 1 cup Orange juice, 1 cup 1 bagel or 2 pieces of toast with little butter Local jam or honey, 1 tablespoon
	Morning snack Black or herbal tea, 2 cups Local honey, 2 tablespoons 1 granola bar (45 g CHO) 1 medium piece of seasonal fruit
	Lunch 2 sandwiches, 4 slices of bread + meat/cheese/hummus and seasonal salad *or* White pasta meal, 2 cups cooked + 1 cup tomato sauce + 100 g meat or wild fish Fruit juice (100%), 1 cup
	Afternoon snack Sports drink/lemonade, 2 cups 1 medium piece of seasonal fruit Bread, 2 slices with little butter Local jam or honey, 2 tablespoons
	Dinner

	Chicken breast, baked 110 g raw weight Seasonal steamed vegetables, 2 cups Olive oil, 1 tablespoon Soy sauce, 2 tablespoons White rice, cooked, 2 cups Low fat yoghurt with fruit, 1 cup Fruit juice, 1 cup
Day 2	
Nutritional profile • 3400 kcal • 115 g protein • 620 g CHO (10 g/kg for 60 kg athlete) • 60 g fat • 1500 mg calcium • 25 mg iron • 40 g fibre	**Breakfast** Low fat granola with dried fruit, 1 cup Low fat yoghurt with fruit, 1 cup 1 bagel or 2 pieces of toast with little butter Local jam or honey, 1 tablespoon
	Morning snack Black or herbal tea, 2 cups Local honey, 2 tablespoons Nut butter (1 tablespoon) and jam (1 tablespoon) Sandwich (1 slice bread) Piece of seasonal fruit
	Lunch White flour tortilla, 1 White rice, 1 cup Organic GMO-free corn, 1 cup Salsa 1/2 cup Avocado, 1/4 Coriander, handful Baked turkey breast, chopped, 1/4 cup Cheese, sprinkle Fruit juice, 1 cup
	Afternoon snack Sports drink/lemonade, 2 cups Piece of seasonal fruit Sports bar, 1
	Dinner Couscous, 2 cups cooked Olive oil, 1 tablespoon Wild fish, 110 g raw Steamed seasonal vegetables, 1 cup Fruit sorbet, 1 cup
Day 3	
Nutritional profile • 3270 kcal • 125 g protein • 620 g CHO (10 g/kg for 60 kg athlete) • 60 g fat • 1400 mg calcium • 26 mg iron • 40 g fibre	**Breakfast** Omelette with 2–3 free range eggs Seasonal vegetables, 1 cup Goat cheese, 1/4 cup 1 bagel or 2 pieces of toast with little butter Local jam or honey, 2 tablespoon Orange juice, 1 cup
	Morning snack Black or herbal tea, 2 cups Local honey, 2 tablespoons Granola bar, 1 (45 g CHO) Piece of seasonal fruit
	Lunch Grilled organic, GMO-free tofu, 120 grams raw Olive oil, 1 tablespoon Seasonal vegetables, 2 cups Pesto pasta, 2.5 cups

	Fruit juice (100%), 1 cup
	Afternoon snack Sports drink/lemonade, 2 cups Oatmeal, 1/2 cup dry Organic apple sauce, 1 cup
	Dinner Turkey breast (100 g raw) Pasta, 2 cups Tomato sauce, 1 cup Fresh local spinach, 1 cup Fruit sorbet, 1 cup

Note: Various options are presented to provide ideas for application but athletes may choose more monotony in order to manage the strategy with success.

Pre-event meal

- The pre-event meal is critical for all athletes and should be ingested at least 2–4 hours before the event. This can coincide with breakfast, lunch or late lunch, if competition is later in the day. CHO recommendations for the pre-event meal include a large range (1–4 g/kg) to account for differences in fuel needs for the event as well as the opportunity to tailor meal size to individual comfort or the timing of intake prior to the event. Pre-event meals ingested 4 hours before the start of competition can contain the upper end of this range (3–4 g/kg), with smaller meals (1–2 g/kg) being targeted for occasions when less time is available prior to the start of competition. Endurance athletes who want to benefit from additional CHO intake to top off their liver and muscle glycogen stores should plan for larger meals eaten well before competition.
- The type of CHO included in the pre-event meal can be important in dealing with the gastrointestinal issues many athletes experience before big events. Nervousness and unfamiliar food can sometimes lead to stomach cramps, nausea and diarrhoea. Keeping meal choices simple and familiar may be the most important concept for successful pre-event fuelling. Reducing the fibre content of the pre-event meal, along with moderating the amounts of fat and protein consumed, should assist with maintaining gastrointestinal functioning.
- Some athletes may also benefit from including CHO-rich foods with low GI in the pre-event meal, due to their better ability to sustain blood glucose concentrations during exercise. This effect is, however, unclear when CHOs are also ingested during exercise. Low-GI foods are often, but not always, high in fibre. Thus, athletes should trial these options in training first before applying them to events.
- Athletes who struggle to consume a pre-event meal should consider liquid meal supplements or start with easily digestible foods such as white rice or well-cooked oatmeal.
- As in the case of CHO-loading practices, athletes should experiment with the pre-event meal in training in order to become familiarised with successful strategies. Although the pre-event meal can be performance enhancing, particularly for endurance sports, ingesting foods in the wrong combination/amounts/timing can be debilitating and performance impairing.
- Athletes in weight-category sports should work with a sports dietitian to develop a vigorous nutrition plan for post-weigh-in recovery (see Chapter 7). Athletes in weight-sensitive sports, such as gymnastics or figure skating, may benefit from liquid supplements or a pre-event drink rather than solid food, since this may provide a feeling of ‘lightness’ rather than bloating.
- The small pre-event snack (~1 g/kg CHO) eaten 60 minutes before competition is a common preference for many athletes. In the case of endurance sports, this may need to be supplemented by additional fuel intake during the event. Examples for such pre-event snacks include a sports bar, two slices of bread with nut butter and jam, banana or apple sauce, or a cup of oatmeal.
- Many athletes also consume gels, bars, and sport drinks shortly (15–20 minutes) before the start of the event as a final source of CHO that may contribute to during-event fuel needs. All these fuelling strategies may come with advantages; however, athletes need to trial them first in training before applying them to competitive events.

Hydration before the event

- Athletes should begin their events in a well-hydrated state, requiring a good hydration strategy in the days and hours leading up to the event. Checking urine specific gravity in the mornings leading up to the event can be a helpful strategy to check that current drinking practices support optimal hydration.
- For athletes competing in multiple events per day, adequate and timely rehydration between events is critical to maintain performance.
- It is generally recommended that athletes consume 5–7 mL of fluid per kg of body mass about 4 hours before exercise and, if urine colour is dark, consume an additional 3–5 mL of fluid per kg of body mass in the final 2 hours. If athletes are thirsty, more fluid is advised. Sport drinks, which contain CHO and sodium, make the fluid not only tastier, but also more effective in achieving absorption and retention. In general, most athletes can tolerate 300–400 mL of fluid without stomach issues immediately before the event.
- Hyperhydration before the event should only be conducted under supervision. Excessive fluid intake can lead to hyponatraemia and may be fatal.

TABLE 12.4 Examples of pre-event meals (2–4 hours before event start)

Oatmeal with apple	• Oatmeal, 1 bowl • Apple sauce, 1/2 cup • Honey, 2 tablespoons • Almond, soy or regular milk, 1 cup • Juice or sports drink, 2 cups
Cereal with fruit and yoghurt	• Cereal, 1 bowl • Fresh seasonal fruit • Plain yoghurt, 1 cup • Honey, 1 tablespoon • 2 slices toast with butter and jam or honey • Sports drink, 2 cups
	• Eggs, fried, 2–3 • White rice, 1 cup • Spinach, 1/2 cup • Pancakes, 2 medium • Real maple syrup, 2 tablespoons • Juice or sports drink, 2 cups
Pasta	• White Italian pasta, 2 cups • Zucchini and carrots in olive oil with herbs • A few tomatoes • Grilled chicken breast, 100 grams raw • Granola bar • Juice or sports drink, 2 cups
Soup and sandwich	• Turkey sandwich with mustard, spinach • Wholewheat bread • Vegetable soup (minestrone or pureed carrot or pumpkin) • Juice or sports drink, 2 cups
Teriyaki rice bowl	• White rice, 2 cups • Soy sauce, 1 tablespoon • Teriyaki chicken, 100 grams • Cooked spring onion or spinach • Juice or sports drink, 2 cups

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CHAPTER THIRTEEN

Competition fluid and fuel

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13.1 Introduction

Although research has shown that ingestion of carbohydrate (CHO) and fluids can improve performance, this is not necessarily true for all individuals in all situations. The amount of fluid and the amount of CHO that should be ingested are dependent on the goals of the individual, the nature and duration of the event, the climatic conditions, the pre-event nutritional status, and the physiological and biochemical characteristics of the individual. The circumstances of each athlete, each sport and each competition must therefore be considered when making choices of what or whether to eat and drink and it is not possible to give 'one size fits all' advice. In a few situations, athletes can get it wrong, and performance can suffer if the type or amount of food and fluid ingested are inappropriate. In recent years, a small number of recreational participants in endurance events have died due to excessive consumption of fluids (and more recently there have been two reported cases in American football), so drinking too much may be even more harmful than drinking too little (Hew-Butler et al. 2008). Inappropriate intake of nutrients can also contribute to gastrointestinal problems during exercise, and the symptoms can be so severe that they will have an adverse effect on performance. Personalisation of sports nutrition for performance, as well as to support training and promote recovery and adaptation, is therefore required.

Food and fluid consumed during competition are part of a specific short-term nutritional strategy aimed at maximising performance at that particular time. When choosing foods and fluids to be consumed during competition, there is no need to take account of long-term nutritional goals, except perhaps (and even then to a limited extent) in extreme endurance events such as the Tour de France or in multi-day running events. In the Tour de France, prolonged exercise is performed on a daily basis over a period of about 21 days and the food consumed during each day's competition may account for about half of the total daily intake (Saris et al. 1989), but even this extreme event lasts for only 3 weeks. In the majority of competitive situations, a balanced diet is not the primary objective of sport nutrition and intake is targeted at minimising the impact of those factors that are responsible for fatigue and impaired performance. It is also recognised that sports nutrition practices and strategies during training may differ from those of competition, depending on the goals of the athlete. For example, training in a state of compromised muscle glycogen ('train low') is attracting interest in the research and applied sporting world (this and other similar ‘periodisation’ topics are discussed in Chapters 14 and 15).

New information continues to emerge, and this sometimes changes our understanding of the needs of athletes and the advice that is given to them. These new insights, however, have not resulted in fundamental changes in our understanding, and the challenge is more to provide athletes with the available information in a useful format than to generate further confirmations of what we already know. The most recent fluid replacement guidelines from the American College of Sports Medicine (Sawka et al., 2007) and the International Olympic Committee (Shirreffs & Sawka 2011), for example, are different in many ways from earlier guidelines from the same or similar organisations. In particular, there is an increased awareness of the need to individualise recommendations and for any guidelines to be sufficiently flexible to meet the needs of athletes with very different goals and different physical characteristics exercising in a range of environmental conditions. In other recent guidelines it has been recognised that the recommended amount of CHO depends on the duration of the event (with longer events requiring higher intakes) as well as the absolute intensity (slower athletes may need less than faster athletes) (Jeukendrup 2014) (see Figure 1).

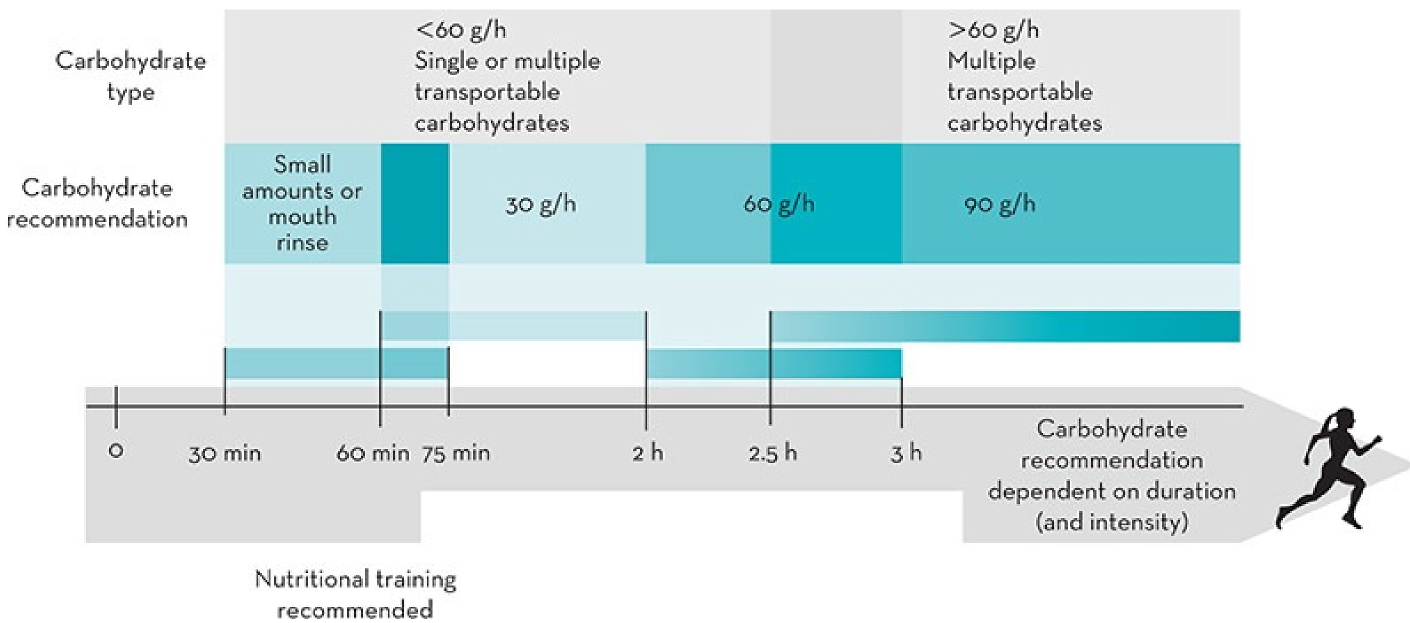

Figure 13.1 Carbohydrate intake recommendations during exercise

Source: Adapted from Jeukendrup 2014

Note: These recommendations may need to be reduced if the absolute intensity is low (e.g. slow running or walking). Especially with higher carbohydrate intakes, it is recommended to train with this regime at least once a week.

13.2 Fatigue during exercise

As outlined in Chapter 12, fatigue is a multifaceted phenomenon with every tissue and organ, particularly muscle and brain, playing important roles. In reality, a loss of performance can be due to a failure at one or more of the large number of links in the chain stretching from the central nervous system (CNS) to the intramuscular contractile machinery (Gandevia 2001). Intake of fuel and fluid during exercise integrates with pre-exercise nutrition strategies (see Chapter 12) to maximise performance potential and delay fatigue. At one end of the performance-fatigue continuum, muscle function is dependent on the force development and/or the sustained repeated contraction of myofibres fuelled by adenosine triphosphate (ATP) with the breakdown of phosphocreatine, the breakdown of glycogen or glucose to pyruvate or lactate, and the oxidation of CHO and fats serving to replenish the small but highly dynamic ATP pool. Other biochemical events in contraction are also influenced by muscle fuel stores: for example, glycogen availability also influences sarcoplasmic reticulum calcium release and excitation–contraction coupling (Nielsen et al. 2014).

However, and at the other end of the performance-fatigue continuum, in many situations the brain is the command centre that will make the athlete either slow down or maintain the pace for a little longer despite increasing muscle fatigue (and numerous other physiological disturbances out of the scope of this chapter). The idea that fatigue is fundamentally a phenomenon of the CNS rather than the peripheral tissues dates back to the observations of physiologists during the latter part of the nineteenth century. These observations highlighted that 'there are two types of fatigue, one arising entirely within the CNS, the other in which fatigue of the muscles themselves is superadded to that of the nervous system' (Bainbridge 1919). More recently, the role of the brain has been demonstrated by studying exercise in extreme environmental conditions and contrasting fatigue in those scenarios with more temperate conditions. The extensive literature that is available makes it clear that the availability of an adequate supply of CHO in the working muscles and in the bloodstream is central to the athlete's ability to sustain exercise performance in such conditions (for a recent review see (Stellingwerff & Cox 2014). In hot environments, however, fatigue occurs while substantial CHO stores remain (Parkin et al. 1999), and performance is limited more by factors associated with thermoregulatory function and hydration status. Nevertheless, subjects perform better in the heat after a few days on a high CHO diet than they do after a low CHO diet, even though CHO availability does not seem to be the limiting factor in the heat (Pitsiladis & Maughan 1999). The mechanisms by which performance is affected by these factors are not entirely clear, but there are well-recognised effects on the brain (Meeusen et al. 2006; Maughan, Shirreffs et al. 2007). As will be discussed in this chapter, CHO (and fluid ingestion) can help to delay fatigue at the level of the muscle and brain. First, we will evaluate the evidence that CHO improves performance during several types of exercise; we will then look into the mechanisms and discuss how CHO elicits these effects, and follow with a discussion of the practical implications and recommendations. Fluids will be discussed in a similar manner.

13.3 Carbohydrate supplementation and exercise performance

The ingestion of CHO during exercise has a number of effects on metabolism and can provide a number of benefits for performance. These effects are well, and moderately well, described in relation to prolonged bouts of moderate-intensity and intermittent-intensity exercise, respectively, but recent studies suggest that CHO ingestion may also be useful for the performance of high-intensity exercise of about 1 hour's duration. In general, during exercise longer than 2 hours, CHO feeding will prevent or delay hypoglycaemia, maintain high rates of CHO oxidation and increase endurance capacity compared with placebo ingestion. During shorter duration, higher intensity exercise (e.g. 1 h around 75% VO_{2max}), CHO ingestion during exercise can also improve exercise performance but the mechanism behind these performance improvements is completely different. The sections below will discuss the effects of CHO intake on performance during prolonged exercise (>2 h), high-intensity exercise (about 1 h duration) and intermittent exercise.

It is important to recognise that the contribution of CHO to energy metabolism increases with increasing exercise intensity but falls with increasing duration at a constant intensity. There is good evidence that depletion of glycogen in the active muscles is a likely cause of fatigue in cycling exercise lasting about 1–3 hours, though it is less clear that this is the case in exercise lasting shorter or longer than this or in running exercise of comparable duration. In exercise that results in fatigue within a few minutes, muscle glycogen concentration falls rapidly, but there is no opportunity or need for ingestion of CHO at such high intensities.

13.3.1 Carbohydrate ingestion and prolonged exercise performance

When exercise is prolonged (2 h or more), CHO ingestion is essential to prevent a decrease in performance. As will be discussed later in this chapter, larger amounts of CHO may be required for more prolonged exercise, but even small amounts (as little as 20 g/h) have been shown to result in performance improvements.

A detailed discussion of the large number of studies is outside the scope of this chapter and for such a discussion the reader is referred to a 2011 review (Vandenbogaerde & Hopkins 2011). This meta-analysis summarised the results of 88 randomised crossover studies in which CHO supplements were consumed before and/or during exercise and this provided 155 estimates for performance effects in time-to-exhaustion tests or in time trials with or without a preload. The authors concluded that CHO supplements with an appropriate composition and administration regimen can have large benefits on endurance performance. Beneficial effects of CHO ingestion are seen during constant effort prolonged cycling (Coggan & Coyle 1991) as well as during running (Tsintzas et al. 1993). The meta-analysed performance effects of CHO supplements ranged from clear large improvements of approximately 6% to clear moderate impairments of approximately 2% (Vandenbogaerde & Hopkins 2011). Later in this chapter we will discuss practical applications and guidelines for such prolonged endurance exercise.

13.3.2 Carbohydrate ingestion and high-intensity exercise performance

During shorter duration higher intensity exercise (e.g. 1 h around 75–90% VO_{2max}), CHO ingestion can also improve exercise performance. Interestingly, it was demonstrated that when glucose was infused into the systemic circulation, this glucose was taken up (and oxidised) at high rates without any performance effect (Carter et al. 2004). However, when subjects rinsed their mouth with a CHO solution, this resulted in performance improvements (Carter et al. 2004) similar to those seen with CHO ingestion (Jeukendrup 1997). Little, if any, CHO was ingested during this study, with each bolus being expectorated after a 5-second swill around the mouth. Therefore, the traditional metabolic effects of CHO were considered highly unlikely. Rather, the authors speculated that the CHO rinsed during the high-intensity exercise was detected by receptors in the oral cavity, which sent signals to the brain, altering efferent output so that a higher motor output could be sustained for longer. Since 2004, a growing number of studies have reported similar effects in high-intensity exercise typically lasting 45–70 minutes (for review, see Rollo & Williams 2011; Jeukendrup 2013). These support the findings of some, but not all, earlier studies reporting an ergogenic effect of ingesting CHO on high-intensity exercise performance. However, shorter duration 'endurance' exercise may not be influenced. Jeukendrup (2008) showed that ingestion of a 6% CHO-electrolyte drink did not improve performance in a cycling time trial that could be completed in about 25 minutes.

13.3.3 Carbohydrate ingestion and intermittent exercise

There is also accumulating evidence that CHO can improve intermittent exercise performance as well as other aspects of performance in team sports (e.g. skills, cognitive functioning). Studies in simulations of soccer, rugby, field hockey and basketball, where an intermittent running protocol was used, have consistently demonstrated that time to exhaustion or performance is enhanced with CHO ingestion compared with a water placebo (Phillips et al. 2010; Phillips et al. 2011). It has been argued that running capacity is only a minor component of team sports performance, and more recent studies have focused on the effects of CHO ingestion on aspects of cognitive performance, skills performance, jumping, sprinting and other movements characteristic of 'stop and go' sports. These studies have generally found mixed or no evidence supporting benefits of CHO ingestion during exercise for sprinting and jumping. Interestingly, however, several studies have demonstrated positive effects on agility, coordination and motor skills. Such components are notoriously difficult to measure, because of their inherent variability, so it is perhaps not surprising that studies have provided mixed results, with some studies finding positive effects and some studies finding no effects. Nevertheless, it is promising that several studies have demonstrated positive effects on these important aspects of team sports performance, which, in addition to the more consistent effects on running performance, should ultimately result in better performance for team sports players

(Cermak & van Loon 2013; Baker et al. 2014; Jeukendrup 2014).

13.4 The mechanisms: how carbohydrates work

It is clear from the section above that there is a vast body of literature to show that CHO feeding improves performance during activities lasting 45 minutes or longer. We will now explore the mechanisms responsible for these improvements, which, as was mentioned before, are different for different types of activities.

13.4.1 Prevention of hypoglycaemia

The blood glucose concentration is normally maintained within a narrow range by regulation of the addition of glucose to the circulation and its removal by peripheral tissues. Glucose can be added from the gastrointestinal tract after food intake or from the liver, which stores about 80–100 g of glycogen in the fully fed state and can also synthesise glucose from non-CHO sources. The primary hormones regulating the blood glucose concentration are insulin and glucagon, but it is increasingly recognised that a large number of other peptide hormones also play key roles in this process, either directly or by influencing the circulating insulin and glucagon levels. Important hormones in this respect are growth hormone, cortisol, somatostatin and the catecholamines. Because of the obvious difficulties in making the relevant measurements, there are limited data on the changes in liver glycogen content during prolonged exercise, but it is clear that a progressive fall occurs, with low levels being reached when subjects are exhausted (Hultman & Nilsson 1971).

It is important to maintain the circulating blood glucose concentration above about 2.5 mmol/L to provide a concentration gradient for transport into glucose-requiring cells. The cells of the central nervous system have an absolute requirement for glucose as a fuel, and when the blood glucose concentration falls below this level, the rate of uptake by the brain may not be sufficient to meet its metabolic needs. Hypoglycaemia leads to a variety of symptoms, including dizziness, nausea and disorientation and was one of the earliest medical problems identified, and subsequently published, in marathon runners suffering from fatigue and collapse at the end of a race. Levine and colleagues (1924) obtained blood samples from runners at the end of the 1923 Boston marathon race and observed that three of the 12 runners studied finished the race in a very poor condition; these individuals had a blood glucose concentration of less than 2.8 mmol/L. These same authors recognised that CHO feeding during the race could prevent the onset of hypoglycaemic symptoms; this was shown to be the case in the following year's race, and an improvement in performance was also reported when CHO was consumed (Gordon et al. 1925). CHO ingested during exercise will enter the blood glucose pool at a rate dictated by the rates of gastric emptying and absorption from the intestine; if this exogenous CHO can substitute for the body's limited endogenous glycogen stores, then exercise capacity should be increased in situations where liver or muscle glycogen availability limits endurance. Several studies have shown that the ingestion of even modest amounts of glucose during prolonged exercise will maintain or raise the circulating glucose concentration (Costill et al. 1973; Pirnay et al. 1982 Erickson et al. 1987) and delay fatigue.

13.4.2 Additional fuel to the exercising muscle during prolonged exercise

The liver is a relatively small organ, with a limited capacity to store CHO. Although the glycogen concentration in muscle tissue is much less, the total muscle glycogen store is relatively large, amounting to about 300–400 g in the average 70 kg athlete consuming a moderate CHO intake. The addition of those qualifications are important because it indicates the influence of body size, especially muscle mass, nutritional status and training status on muscle glycogen storage. In trained marathoners running at racing pace, the rate of CHO oxidation can be about 3–4 g per minute, but if this were sustained, the available CHO stores would be depleted long before the finish line was reached. Certainly in cycling (Hermansen et al. 1967) and perhaps also in running (Williams 1998), the point of fatigue in prolonged exercise coincides closely with the depletion of glycogen in the exercising muscles. Increasing muscle glycogen stores prior to exercise can also improve performance in both cycling (Ahlborg et al. 1967) and running (Karlsson & Saltin 1967). The picture has not changed significantly in the 50 years or so since the first studies showing this. In fact, one could argue that we have known this for almost 100 years since the first observation that dietary CHO intake can enhance exercise capacity (Krogh & Lindhard 1920).

Where performance is limited by depletion of the body's endogenous liver or muscle glycogen stores, exercise capacity should be improved when CHO is consumed. This assumes, of course, that the ingestion of CHO does not stimulate an increase in the rate of utilisation of endogenous CHO reserves, and the evidence indicates that this is indeed so. Ergogenic effects of CHO ingestion during exercise were initially attributed to a sparing of the body's limited muscle glycogen stores by the oxidation of the ingested CHO (Coyle et al. 1983; Hargreaves et al. 1984; Erickson et al. 1987), but other studies have failed to show a glycogen-sparing effect of CHO ingested during prolonged exercise (Coyle et al. 1986; Jeukendrup et al. 1999a). The current consensus view seems to be that there is probably little or no sparing of muscle glycogen utilisation, although liver glucose release is slowed (Erickson et al. 1987; Bosch et al. 1994; McConell et al. 1994; Howlett et al. 1998; Jeukendrup et al. 1999). Moreover, the primary benefit of ingested CHO is probably the maintenance of high CHO oxidation rates, allowing the exercise intensity to be maintained. When CHO oxidation rates drop below certain rates because endogenous glycogen stores are becoming depleted, fat oxidation will increase but may not be able to fully compensate for the reduction in ATP production. Maximal ATP production rates from fat are significantly lower than those achieved by oxidation of CHO and, therefore, the intensity will have to be reduced so that ATP production can match ATP use. For this reason, studies have attempted to maximise CHO delivery during exercise in order to maintain or increase CHO oxidation. These studies will be described in more detail later. Similar mechanisms may be responsible for the improved performance during prolonged intermittent exercise. During this type of exercise, muscle glycogen is broken down rapidly during the high intensity sections and depletion of muscle glycogen stores can occur before the end of a game.

We should also recognise that despite the current interest in strategies to promote fat utilisation during sports and exercise activities (Chapter 15), there are advantages to using CHO rather than fat as a fuel. Cardiovascular function may be stressed by the need to maintain a high skin blood flow to promote heat loss as well as maintaining blood flow to the muscle. Oxidation of CHO gives about 10% more energy per litre of oxygen consumed than does oxidation of fat. Using more CHO therefore results in a lower oxygen demand at any given running speed or power output. ‘Sparing’ CHO by increasing fat oxidation means that subjects must work at a higher fraction of their maximum aerobic capacity.

13.4.3 Effects of the central nervous system

In 2004, Carter and fellow scientists at the University of Birmingham asked cyclists to ride a 40 km time trial while rinsing their mouth with a CHO (non-sweet, tasteless maltodextrin) or placebo (water) solution without swallowing it (Carter et al. 2004). The rinsing protocol was standardised: subjects periodically throughout the exercise challenge rinsed their mouth for 5 seconds with the drink and then spat the bolus out into a bowl. The results were remarkable: it was unlikely that much, if any, CHO had been absorbed from the mouth rinse, yet performance was improved by about 3% (Carter et al. 2004), very similar to the 2–3% improvements observed with CHO ingestion during earlier studies employing similar high-intensity exercise protocols (Jeukendrup et al. 1997) (see Table 13.1).

TABLE 13.1 Summary of studies that investigated the effects of a carbohydrate mouth rinse on performance (in chronological order)

Authors	Number	Performance test	CHO type	Fed/fasted	Effect (+ indicates improvement)	Performance effect (statistical sign)
Carter et al. (2004)	9	Cycling time trial (1 h)	MD	4 h	+ 2.9%	Improved
Whitham & McKinney (2007)	7	1 h running	MD	4 h	–0.3%	NS
Rollo et al. (2008)	10	30 min running	Glu + MD mix	+ 10 h	+ 2.0%	Improved
Chambers et al. (2009)	8	1 h cycling	Glucose MD	+ 10 h + 10 h	+ 1.9% + 3.1%	Improved Improved
Beelen et al. (2009)	14	1 h cycling	MD	2 h	+ 0.5%	NS
Rollo et al. (2010a)	10	1 h running	Glu + MD mix	13 h	+ 2.0%	Improved

Pottier et al. (2008)	12	1 h cycling	sucrose	3 h	+ 3.7%	Improved
Rollo et al. (2010b)	10	1 h running	Glu + MD	3 h	+ 0.7%	NS
Fares and Keyser (2011)	13	TTE 60%W_{max}	MD MD	+ 10 h 3 h	+ 11.6% + 3.5%	Improved Improved
Lane et al. (2013)	12	Cycling TT (1h)	10%MD	10 h 2 h	+ 3.4% + 1.8%	Improved Improved

Glu = glucose, MD = maltodextrin, NS = not significant, TTE= time to exhaustion

The explanation remains uncertain, but it has been suggested that the alterations in power output commonly observed during a self-paced exercise task are under the influence of a ‘central governor’ that controls the recruitment of motor units during exercise to ensure that homeostasis is maintained (Noakes 2000; Kayser 2003). The ‘central governor’ is postulated to alter power output using afferent signals from peripheral physiological systems and receptors that detect changes in the external and internal environment (Lambert et al. 2005). It is therefore plausible that during exercise the positive central responses to an oral CHO stimulus could counteract the negative physical, metabolic and thermal afferent signals arising from muscles, joints and core temperature receptors that are sent to the brain and consciously or unconsciously contribute to central fatigue and an inhibition of motor drive to the exercising muscles (St Clair Gibson et al. 2001). For example, the dopaminergic system of the ventral striatum has been implicated in arousal, motivation and the control of motor behaviour (Berridge & Robinson 1998) and increased activity of this pathway during exercise has been postulated to attenuate the development of central fatigue (Gerald 1978; Chandler & Blair 1980; Davis et al. 2000). This would suggest that the beneficial effects of CHO feeding during exercise are not confined to its conventional metabolic advantage and may serve not as an energy substrate but as a positive afferent signal capable of modifying motor output.

In support of this theory, Chambers and colleagues (2009) used functional magnetic resonance imaging (fMRI) to investigate the responses of the human brain to a CHO and placebo mouth rinse (Chambers et al. 2009). The study revealed that tasting both a sweet (glucose) and non-sweet (maltodextrin) CHO solution activated areas of the brain, such as the anterior cingulate cortex and ventral striatum, which were unresponsive to artificial sweetener (saccharin). Other neuroimaging investigations have also reported that an oral CHO solution activates additional brain regions to those responding to an artificial sweetener (Frank et al. 2008; Haase et al. 2009), suggesting there may be taste transduction pathways that respond to CHO independently of those for sweetness. This is in line with the observation that performance was enhanced compared with the control condition with both a sweet and a non-sweet CHO (Carter et al. 2005).

13.5 From theory to practice

Although the knowledge that CHO feeding could improve the exercise performance and practices of endurance athletes using CHO supplementation for their races and training was established by the 1980s, many questions were still unanswered: What is the optimal amount of CHO? What is the best type of CHO? What is the optimal ingestion rate? What form should CHO be taken in? Do men and women respond in the same way? Is CHO as effective in less trained as in well-trained individuals? This section will describe the journey that researchers have taken to address these practical questions.

13.5.1 Type of carbohydrate

Although glucose was the form of CHO ingested during exercise in most of the early studies, glucose, sucrose and oligosaccharides have all been shown to be effective in maintaining blood glucose concentration and in improving endurance capacity when ingested during prolonged exercise (Maughan 1994). It appears that glucose can be replaced by a variety of other sugars, including sucrose, glucose polymers and mixtures of sugars, without markedly affecting the responses. These CHO are oxidised relatively rapidly, whereas others, such as fructose, are absorbed relatively slowly in the intestine. Furthermore, fructose must also be converted by the liver to glucose or lactate before it can be oxidised by muscle. Tracer studies show that the maximum rate of oxidation of orally ingested fructose is less than that for glucose, sucrose or oligosaccharides (Wagenmakers et al. 1993; Jeukendrup 2008; Jeukendrup 2014). Perhaps for this reason, the ingestion of solutions containing only fructose is not generally effective in improving performance of prolonged exercise (Maughan et al. 1989; Murray et al. 1989). Therefore, CHO can be divided into two categories: CHOs that are oxidised rapidly and those that are slower (Table 13.2). Even the fastest CHOs, however, cannot be oxidised at rates greater than about 60 g/h. This limits the energy provision from ingested CHOs even when consumed at high rates to ~240 kcal per hour, while energy expenditures can increase up to 1000 kcal per hour or more. The systematic study of why CHO could not be oxidised at higher rates identified eventually concluded that intestinal absorption was the main limiting factor (Jeukendrup 2004; Jeukendrup 2008).

TABLE 13.2 Slow and fast carbohydrates (based on their maximal exogenous carbohydrate oxidation rates during exercise)

Slow carbohydrates	Fast carbohydrates
Oxidised up to 0.6 g/min	Oxidised up to 1 g/min
Fructose	Glucose
Galactose	Sucrose
Isomaltulose	Maltose
Amylose (starch)	Maltodextrins (glucose polymers)
Trehalose	Amylopectin (starch)

There are theoretical advantages to the use of sugars other than glucose. Substitution of glucose polymers for glucose will allow an increased CHO content without an increased osmolality. If the osmolality of ingested drinks is too high, there will be a net efflux of water into the intestinal lumen, leading to a reduction in plasma volume and increasing the risk of gastrointestinal distress (Evans et al. 2009). The use of glucose polymers may also have taste advantages as these are less sweet than glucose or sucrose, but the available evidence suggests that the use of glucose polymers rather than free glucose does not alter the blood glucose response or the effect on exercise performance (Ivy et al. 1979; Coggan & Coyle 1983; Coyle et al. 1986; Maughan et al. 1987; Hargreaves & Briggs 1988). Similar effects are seen with the feeding of sucrose (Sasaki et al. 1987) or mixtures of sugars (Murray et al. 1987; Mitchell et al. 1988).

13.5.2 Multiple transportable carbohydrates

Limitations of the oxidation of exogenous CHO to around 60 g/h are related to intestinal absorption. Glucose needs the sodium-dependent transporter SGLT1 for absorption across the basolateral membrane of the intestinal lumen. When the transporter becomes saturated, delivery to the muscle and subsequent oxidation of ingested glucose cannot be further increased. Glucose, maltose, maltodextrins, sucrose and even amylopectin (a branched starch) can all be oxidised at a rate of approximately 1 g/min but no higher. As explained in the previous section, other CHO such as fructose, galactose, isomaltulose, trehalose and amylose (a straight chain starch), are oxidised at lower rates because their digestion and absorption occur at lower rates.

In search of a way to deliver more CHO to the muscle, Jentjens and colleagues (2004) undertook a study where they set out to saturate the SGLT1transporter with glucose (with an intake of 72 g/h) while adding fructose to the mix (at 36 g/h). Since fructose uses a completely separate transport mechanism through a different intestinal transporter (GLUT5), it was hypothesised that the combination of CHO sources could result in additional CHO oxidation. Indeed, this study demonstrated for the first time that oxidation rates well over 60 g/h could be achieved (1.26 g/min or 76 g/h). A series of studies followed in which different CHO blends were used and ingestion rates were increased even more. These studies confirmed that ingestion of multiple transportable CHOs resulted in (up to 75%) higher oxidation rates than CHO that use the SGLT1 transporter alone (for reviews see Jeukendrup 2010; Jeukendrup 2014; Jeukendrup & McLaughlin 2011). Interestingly, such high oxidation rates could not only be achieved with CHO ingested in a beverage but also as a gel (Pfeiffer et al. 2010a) or a low-fat, low-protein, low-fibre energy bar (Pfeiffer et al. 2010b) (both ingested with water).

Although these studies clearly demonstrated that more CHO could be delivered and utilised by the muscle at high rates, the outcome of key interest to athletes is whether it affects performance. Several investigations have now been conducted that link the increased exogenous CHO oxidation rates observed with multiple transportable CHO to delayed fatigue and improved exercise performance. In one study, in which subjects ingested 1.5 g/min of glucose:fructose or glucose only during 5 hours of moderate-intensity cycling, ratings of perceived exertion (RPE) were found to be lower with the CHO mixture than with glucose alone, while pedal cadence was also better maintained (Jeukendrup et al. 2006). Follow-up studies have reported reductions in fatigue and/or improvements in lab-based and field-based exercise performance with multiple transportable CHO (Currell & Jeukendrup 2008; Rowlands et al. 2012). It must be noted, however, that the benefits of multiple transportable CHO are only seen when the exercise duration is 2.5 hours or longer and CHO is ingested at high rates (90 g/h is usually recommended). Although glucose–fructose drinks will be just as effective as glucose when ingested at lower rates or when the exercise duration is shorter, there is no reason to believe that they would be more effective than a single CHO source.

13.5.3 Amount of carbohydrate

A very important question from a practical point of view is how much CHO should be ingested. Given the importance of that question, it is surprising that there are very few well-controlled dose–response studies on CHO ingestion during exercise and exercise performance. Serious methodological flaws associated with most of the older studies made it difficult to establish a true dose–response relationship; until a few years ago, it was assumed that a minimum amount of CHO was needed to provide a benefit (probably about 20 g/h based on one study) but that a dose–response relationship was absent (Rodriguez et al. 2009). More recently, however, evidence has been accumulating for a dose–response relationship between CHO ingestion rates, exogenous CHO oxidation rates and performance. In one recent carefully conducted study, endurance performance and fuel selection were measured during prolonged exercise while ingesting glucose at 15, 30, and 60 g/h (Smith et al. 2013). Twelve subjects cycled for 2 h at 77% VO_{2peak} followed by a 20 km time trial. The results suggest a relationship between the dose of glucose ingested and improvements in endurance performance. Furthermore, the exogenous glucose oxidation increased with ingestion rate and it is possible that an increase in exogenous CHO oxidation is directly linked with, or responsible for, exercise performance.

A large scale multicentre study by Smith and colleagues (2010) also investigated the relationship between CHO ingestion rate and cycling time trial performance to identify a range of ingestion rates that would enhance performance. In their study across four research sites, 51 cyclists and triathletes completed four exercise sessions consisting of a 2-hour constant load ride at a moderate to high intensity, followed by a 20 km time trial. Twelve different beverages (consisting of glucose:fructose in a 2:1 ratio) were compared, providing participants with 12 different doses of glucose:fructose CHO mixtures ranging from 10 to 120 g /h during the constant load ride. All sites used a common artificially sweetened, coloured and flavoured placebo (no CHO) and three different CHO doses, with a randomised application of the beverage treatments. The ingestion of CHO significantly improved performance in a dose-dependent manner and the authors concluded that the greatest performance enhancement was seen at a CHO ingestion rate between 60 and 80 g/h. Interestingly, these results are in line with an optimal CHO intake proposed by a recent meta-analysis (Vandenbogaerde & Hopkins 2010).

Based on the dose–response studies mentioned above, together with the multiple transportable studies, CHO intake recommendation for more prolonged exercise have been formulated and are listed in recently proposed guidelines in Figure 13.1. In summary, it appears that lower intakes are sufficient for shorter duration exercise but more prolonged exercise (>2.5 h) could benefit from large intakes of up to around 90 g/h. Some individuals may benefit from more than this, and individual tolerances should be established in training or minor races.

13.5.4 Training status

Another question that often arises is whether the results of these studies, often conducted in trained or even very well-trained individuals, translate to less trained or untrained individuals. A small number of studies have compared CHO ingestion in groups of trained and untrained people. For example, Jeukendrup and colleagues (1997) compared substrate use in trained and untrained men, who cycled at 60% of their maximal oxygen uptake, with the trained men exercising at significantly higher absolute exercise intensity. Glucose was ingested at regular intervals, with an average intake of ~1.1 g/min. Total CHO oxidation was similar in both groups but fat oxidation and energy expenditure were higher in the trained. Interestingly, even though the trained men were exercising at a 40% higher absolute power output, exogenous glucose oxidation was not different between the two groups (0.95 g/min in trained and 0.96 g/min in untrained) (Jeukendrup et al. 1997). In a follow-up investigation, van Loon and colleagues (1999) studied trained and untrained subjects at the same relative, but also absolute, exercise intensities. Again, no differences were found in exogenous CHO oxidation rates between trained and untrained; it must be noted, however, that the untrained subjects in both these studies (Jeukendrup et al. 1997; van Loon et al. 1999) had VO_{2max} values higher than the sedentary population, so guidelines may be extrapolated to athletes of different levels, but not necessarily to people who are sedentary. Although the absolute exercise intensity did not make a difference to glucose oxidation in the study by van Loon and colleagues (1999), it is possible that there is a threshold below which exogenous oxidation rates are lower, and all subjects in those studies always exercised above that absolute intensity threshold. It should also be noted that trained individuals exercise at a higher relative intensity (as a % of VO_{2max}) as well as a higher absolute intensity when racing over a fixed distance: this reflects in part the shorter time taken by the fitter participants.

Finally, it should be considered that it is the absolute exercise intensity and the absolute rates of CHO oxidation that determines exogenous CHO oxidation rates rather than the training status of the athlete. It is unlikely that the runner who completes a marathon in 5 hours would need an intake of 90 g CHO per hour as this would be close to, or could even exceed, the total CHO use, at that absolute exercise intensity.

13.5.5 The effect of exercise intensity

When the exercise intensity is low and total CHO oxidation rates are low, CHO intake recommendations might have to be adjusted downwards. There are actually surprisingly few studies on which to base firm recommendations. With increasing exercise intensity, the active muscle mass becomes more and more dependent on CHO as a source of energy. Both an increased muscle glycogenolysis and increased plasma glucose oxidation will contribute to the increased energy demands (van Loon et al. 2001). It is therefore reasonable to expect that exogenous CHO oxidation will increase with increasing exercise intensities. Indeed, an early study by Pirnay and colleagues (1982), reported lower rates of exogenous CHO oxidation at low exercise intensities compared with moderate intensities, with exogenous CHO oxidation tending to reach a plateau between 51 and 64% VO_{2max}. When the exercise intensity was increased from 60 to 75% VO_{2max}, no increase in exogenous CHO oxidation was observed (Pirnay et al. 1982).

It is therefore possible that lower rates of exogenous CHO oxidation are only observed at very low exercise intensities when the reliance on CHO as an energy source is minimal. In this situation, part of the ingested CHO may be directed towards non-oxidative glucose disposal (storage in the liver or muscle) rather than towards oxidation.

13.5.6 Do smaller persons need less?

The guidelines for CHO intake during exercise, presented here, are expressed in grams per hour of exercise without any correction for body mass. By contrast, some expert guidelines such as the 2009 position stand on nutrition for athletic performance by the American Dietetic Association (ADA) and American College of Sports Medicine (Rodriguez et al. 2009) have provided recommendations for CHO intake during exercise expressed in the format of g/kg/h. The rationale for this was unclear as there appears to be no correlation between body mass and exogenous CHO oxidation (Jeukendrup & Moseley 2010). The probable reason for this lack of correlation between body mass and exogenous CHO oxidation is that the limiting factor is CHO absorption, which is largely independent of body mass. It is likely, however, that the absorptive capacity of the intestine is modified by CHO content of the diet as it has been shown in animal studies that intestinal transporters can be up-regulated with increased CHO intake. There is emerging evidence that the same effects are seen in humans in response to changes in diet composition (Yau et al. 2014).

In summary, individual differences in exogenous CHO oxidation exist, although they are generally small. These differences are not related to body mass but, more likely, to a capacity to absorb CHO. This in turn could be diet-related because the gut is highly adaptable, as will be discussed in the next section. Since the contribution of exogenous CHO to energy supply is independent of body mass or muscle mass, but dependent on absorption and to some degree the absolute exercise intensity (at very low absolute intensities, low CHO rates may also restrict exogenous CHO oxidation), the advice given to athletes should be in absolute amounts. There is no rationale for expressing CHO recommendations during exercise for athletes per kilogram body mass.

13.5.7 Training the gut

Since the absorption of CHO limits exogenous CHO oxidation, and exogenous CHO oxidation seems to be linked with exercise performance, an obvious potential strategy would be to increase the absorptive capacity of the gut. Anecdotal evidence in athletes would suggest that the gut is trainable and that individuals who regularly consume CHO or have a high daily CHO intake may also have an increased capacity to absorb it. Intestinal CHO transporters can indeed be up-regulated by exposing an animal to a high-CHO diet (Ferraris 2001). To date there is limited evidence in humans. A study by Cox and colleagues (2010) investigated whether altering daily CHO intake affects substrate oxidation and in particular exogenous CHO oxidation. It was demonstrated that exogenous CHO oxidation rates were higher after the high CHO diet (6.5 g/kg bodyweight/day; 1.5 g/kg BW provided mainly as a CHO supplement during training) for 28 days compared with a control diet (5 g/kg bodyweight/day). This study provided evidence that the gut is indeed adaptable and this can be used as a practical method to increase exogenous CHO oxidation. We recently suggested that this may be highly relevant to the endurance athlete and may be a prerequisite for the first person to break the 2-hour marathon barrier (Stellingwerff & Jeukendrup 2011). Although more research is needed, it is recommended to practise the CHO intake strategy in training and dedicate at least some training to training with a relatively high CHO intake. This also allows opportunities to practise the physical actions of drinking while running or cycling at speed and an opportunity to become accustomed to exercise with fluid in the GI tract.

13.5.8 Timing of carbohydrate intake

In most sports, practical considerations dictate the timing and frequency of CHO (and fluid) intake during the event. During many endurance events (e.g. running and cycling), energy replacement occurs while the athlete is literally ‘on the run’. Saris and colleagues (1989) reported that about half of the daily energy intake of Tour de France cyclists was ingested during each day's cycling stage. Intake is generally much less in running events of comparable duration, as few runners are able to tolerate solid food, even when the exercise intensity is low. Intake during competition may be limited by consideration of the time lost in stopping or slowing down to consume food or fluid, or the impact of such ingestion on gastrointestinal discomfort. In other events, such as team sports, there are formal and informal pauses in play, and these may provide an opportunity to consume CHO/fluid. Athletes should be encouraged to make use of the opportunities provided in their sport to consume fluid and additional CHO. Experimentation and practice in training and in minor competitions will help to determine the best strategies for each situation.

Although pre-competition fuelling strategies have been covered in the preceding chapter, it is common to classify CHO feeding in the hour before exercise as 'during exercise fuelling' due to its close proximity (and that it often occurs during the active warm-up phase). Concerns have been raised that this CHO ingestion can have negative implications for performance due to the concomitant rise in plasma insulin concentrations, lowering of plasma glucose and suppression of fatty acid availability and oxidation. Indeed, a study by Foster and colleagues (1979) did report a reduction in cycling capacity in athletes who had consumed glucose in the hour before exercise compared to when they consumed water or a mixed liquid meal. On the basis of this one study, for some time it was believed by athletes and researchers alike that CHO feeding in the hour before competition would both increase the reliance on CHO oxidation at the onset of exercise, potentially leading to accelerated depletion of muscle glycogen, and also cause a sudden lowering of blood glucose concentrations, termed *rebound hypoglycaemia* (Koivisto et al. 1981). However, a body of research published since this time has shown that any metabolic disturbances following pre-exercise CHO feedings are transient and/or unimportant (Hargreaves et al. 2004). These studies show that CHO intake in the hour before exercise is not associated with a performance decrement and should be encouraged to help ensure optimal fuelling for the subsequent competition (Jentjens et al. 2003; Moseley et al. 2003).

13.6 Other effects of carbohydrate ingestion

Athletes in hard training are anxious to avoid any illness or injury that might interrupt their training program. These athletes may, however, be more susceptible than the sedentary individual to minor opportunistic infections, particularly those affecting the upper respiratory tract (Nieman & Pedersen 1999). While not serious in themselves, the disruption to training can have negative physical and psychological effects. Several reviews of the literature (Gleeson et al. 2004) suggest that exercise-induced increases in the release of catecholamines and glucocorticoids may be responsible for the reduced effectiveness of the immune system. Ingestion of CHO during exercise is effective in attenuating the rise in circulating catecholamine and cortisol concentrations that is normally observed during prolonged strenuous exercise, and has also been reported to reduce some of the immunosuppressive effects of exercise (Nieman 1997). In contrast to this finding, Bishop and colleagues (1999) reported that ingestion of a CHO drink before and during 90 minutes of an exercise session designed to simulate soccer match play had no effect on circulating cortisol concentration or on a number of markers of immune function. Notwithstanding the lack of an effect observed in this last study, it does seem that benefits may accrue to the athlete in hard training from the ingestion of CHO-containing drinks during each prolonged training session. The potential role of CHO ingestion during prolonged exercise as a strategy for staying healthy is covered in more detail in Chapter 22.

Another piece of evidence suggests that CHO ingestion during exercise may promote recovery of muscle glycogen stores in the post-exercise period (Kuo et al. 1999). In this study, rats performed two 3-hour swimming bouts, separated by 45 minutes of rest, to deplete muscle glycogen stores. A 50% glucose solution was administered by stomach tube at the end of each of the exercise bouts. CHO feeding resulted in glycogen supercompensation at 16 hours after exercise, an effect attributed to a stimulation of GLUT-4 protein expression in response to CHO. This suggests another reason for ingestion of CHO during exercise that is likely to result in substantial depletion of the muscle glycogen stores; this effect will be of particular significance when a second exercise bout—whether training or another competition—must follow after a short interval.

13.7 Carbohydrate intake recommendations

The American College of Sports Medicine recommends that athletes take between 30 and 60 g of CHO during endurance exercise (>1 h) (Sawka et al. 2007) or 0.7 g/kg/h (Rodriguez et al. 2009). The upper limit is determined by the observation that taking in more CHO does not result in greater use by the muscle; the lower rate is a reflection of studies that demonstrated performance benefits with those lower rates of CHO intake. This advice is very general and does not specify the sport or the level of the athlete that the recommendation is for, while the exercise duration guidelines are limited. More recently, specific guidelines have been formulated based on the research findings discussed previously in this chapter (see Figure 13.1). Although other factors do play a role, the main determinant of CHO needs during exercise is the duration of the activity. It is not necessary to ingest large amounts of CHO during exercise lasting approximately 30 minutes to 1 hour and a mouth rinse with CHO may be sufficient to get a performance benefit (see Table 13.1). In such activities, the performance effects with the mouth rinse were similar to those seen when ingesting the CHO drink, so there does not seem to be a disadvantage of consuming the drink, although occasionally athletes may complain of gastrointestinal distress when consuming larger amounts. With increasing duration, the recommended CHO intake goes up to 30–60 g/h and during exercise longer than 2.5 hours, a high intake of 90 g/h should be considered. With such intakes, however, it is essential to make sure that the CHO source is a mixture of glucose and fructose (or similar multiple transportable CHO). For athletes who compete at lower absolute intensities (speeds), the recommendations can be adjusted downwards. In addition, it is important to note that CHO intakes can be achieved with solids, semi-solids (gels) as well as liquids and can therefore be mixed and matched. The CHO intake needs to be balanced with an appropriate fluid intake and this will be discussed in the following sections.

13.8 Effects of hyperthermia and dehydration on performance

It is a matter of common experience that the perception of effort is increased, and exercise capacity reduced, in hot climates. This was recognised by the early pedestrians: in a challenge race held in Curacao in August, 1808, the local man chose to start the race at the hottest time of day to gain an advantage over his European opponent, Lieutenant Fairman. Notwithstanding his disadvantage, Fairman won, but he declared the event to be much more stressful than any other event he participated in (Thom 1813). More recently, and under more controlled conditions, the effects of increasing ambient temperature were quantified when Galloway and Maughan (1997) showed that exercise capacity at a fixed power output was greatly reduced at 31°C (55 minutes) compared to the same exercise performed at 11°C (93 minutes). They also observed that the exercise time was already reduced (to 81 minutes) at the comparatively modest temperature of 21°C. Parkin and colleagues (1999) have shown similar effects and also showed that there remained a substantial amount of muscle glycogen at the point of fatigue when the ambient temperature was high (40°C).

When the ambient temperature is higher than skin temperature, the only mechanism by which heat can be lost from the body is evaporation of water from the skin and respiratory tract. Complete evaporation of 1 litre of water from the skin will remove 2.4 MJ (580 kcal) of heat from the body, and sweat losses are determined primarily by the intensity and duration of exercise and by the ambient temperature and humidity. Data for typical sweat losses in a range of sports activities have been compiled by Rehrer and Burke (1996). However, sweat rates vary greatly between individuals, even when the metabolic rate is apparently similar, and high sweat rates are sometimes necessary even at low ambient temperatures if an excessive rise in body temperature is to be prevented (Maughan 1985). It is clear, too, that some individuals sweat at rates that are far higher than the maximum evaporative capacity, which is determined by the skin:environment temperature gradient, the local humidity and the rate of air flow over the skin. In this situation, sweat drips from the skin without contributing to heat loss.

Water losses are derived in varying proportions from plasma, extracellular water and intracellular water. Any decrease in plasma volume is likely to adversely affect thermal regulation and exercise capacity. When the metabolic rate is high, blood flow to the muscles must be maintained at a high level to supply oxygen and substrates, but a high blood flow to the skin is also necessary to convect heat to the body surface where it can be dissipated (Nadel et al. 1990). When the ambient temperature is high and blood volume has been decreased by sweat loss during prolonged exercise, there may be difficulty in meeting the requirement for a high blood flow to both these tissues. In this situation, skin blood flow is likely to be compromised, allowing body temperature to rise but preventing a catastrophic fall in central venous pressure (Rowell 1986). Muscle blood flow is also reduced, but oxygen extraction is increased to maintain oxidative energy metabolism (Gonzales-Alonso et al. 1999).

Coyle and colleagues have also investigated these factors and found that increases in core temperature and heart rate during prolonged exercise are graded according to the level of hypohydration achieved (Montain & Coyle 1992a). They also showed that the ingestion of fluid during exercise increases skin blood flow, and therefore thermoregulatory capacity, independent of increases in the circulating blood volume (Montain & Coyle 1992b). Plasma volume expansion using dextran/saline infusion was less effective in preventing a rise in core temperature than was the ingestion of sufficient volumes of a CHO-electrolyte drink to maintain plasma volume at a similar level. This suggests that oral intake achieves beneficial effects other than the maintenance of blood volume.

Hypohydration does not impair submaximal-intensity aerobic performance in cold–cool environments and sometimes impairs aerobic performance in temperate environments; however, it consistently impairs aerobic performance in warm–hot environments. Hypohydration begins to impair aerobic performance when skin temperatures exceed 27°C, and with each additional 1°C elevation in skin temperature, there is a further 1.5% impairment (Sawka et al. 2012). Furthermore, hypohydration exacerbates the negative effect of warm–hot environments on aerobic performance. Aside from endurance exercise, hypohydration can also compromise performance in high-intensity exercise (Nielsen et al. 1982; Armstrong et al. 1985). Although sweat losses during such brief exercise are small, prior dehydration (by as much as 10% of body mass) is common in weight-category sports where participants are often hypohydrated during competition (see Chapter 7). Nielsen and colleagues (1982) showed that prolonged exercise, which resulted in a loss of fluid corresponding to 2.5% of body weight, resulted in a 45% fall in the capacity to perform high-intensity exercise. It may be that even very small fluid deficits impair performance, but the methods used are not sufficiently sensitive to detect small changes.

Nybo and colleagues (2014) recently provided an extensive review of physiological mechanisms responsible for impaired performance in hot environments and proposed an integrated model. There are a number of physiological mechanisms impairing aerobic performance during heat stress and these factors may act independently or may influence each other (see Table 13.3). Hypohydration exacerbates all of the proposed physiological mechanisms thought to limit aerobic performance and these are listed in Table 13.3 compared with heat stress alone. For example, studies have shown repeatedly that hypohydration will increase cardiovascular strain more than hyperthermia alone and will make it more difficult to sustain the required cardiac output during aerobic exercise. It has also been demonstrated that hypohydration can impair skeletal muscle blood flow (Gonzalez-Alonso et al. 1998). Heat stress alone does not seem to have this effect. Others have shown that hypohydration elevates core temperature (Sawka et al. 1985), skeletal muscle glycogen usage (Logan-Sprenger et al. 2012), fatigue/discomfort (Montain et al. 1998), skeletal muscle motor-unit recruitment (Bigard et al. 2001) and other effects. Clearly, a multitude of physiological mechanism(s) contributing to impaired exercise-heat performance are further aggravated by hypohydration, thus providing mechanisms for the hypohydration-impaired aerobic performance during heat stress.

TABLE 13.3 Physiological mechanisms potentially contributing to impaired exercise performance in warm–hot environments (hypohydration may exacerbate these effects)

Mechanism	Effect	Influenced by dehydration?
Cardiovascular	Blood pressure Blood flow Oxygen delivery Metabolite removal	Yes Yes Yes Yes
Central nervous system	Cerebral metabolism Neurotransmitter levels Temperature	Yes Yes Yes
Peripheral muscular factors	Temperature Metabolic Afferent feedback	Yes Yes Yes
Psychological	Thermal comfort RPE Motivation Expectations	Yes Yes Yes Yes
Respiration	Hypocapnia Alkalosis Breathing sensations	Yes Yes Yes

Source: Modified from Nybo et al. 2014
Note: It should perhaps be noted that, if hypohydration is sufficiently severe, all physiological functions are likely to be compromised, though the effects are more selective at the levels of hypohydration normally encountered.

13.8.1 Hypernatraemia and hyponatraemia

The sweat loss that accompanies prolonged exercise leads to a loss of electrolytes and water from the body. Although the volume loss is easily estimated from changes in body mass after correction for substrate oxidation and respiratory water loss, electrolyte loss is rather more difficult to quantify and the extent of these losses has been the subject of much debate. The values for sweat electrolyte content in Table 13.4 show the great inter-individual variability in the concentration of the major electrolytes.

TABLE 13.4 Concentration, in mmol/L, of the major electrolytes present in sweat, plasma and in intracellular (muscle) water in man

	Plasma	Sweat	Intracellular
Sodium	137–144	40–80	10
Potassium	3.5–4.9	4–8	148
Calcium	4.4–5.2	3–4	0–2
Magnesium	1.5–2.1	1–4	30–40

Chloride	100–108	30–70	2

Note: The values are collated from a variety of sources (see Maughan 1994 for further details).

Sodium, the most abundant cation of the extracellular space, is the major electrolyte lost in plasma; chloride, which is also mainly located in the extracellular space, is the major anion. This ensures that the greatest fraction of fluid loss is derived from the extracellular space, including the plasma. Although the composition of sweat is highly variable, sweat is always hypotonic with regard to body fluids, and the net effect of sweat loss is an increase in plasma osmolality. The plasma concentration of sodium and potassium also generally increases during prolonged exercise, suggesting that replacement of these electrolytes during exercise may not be necessary. There may, however, be a need to replace some of the sodium lost in sweat when these losses are high. Some idea of the extent of the individual variability in sweat electrolyte losses that can occur comes from reports of salt losses in football (soccer) players during training (Maughan et al. 2004; Shirreffs et al. 2005). Some players lost less than 2 g of salt (sodium chloride) in a 90-minute training session, while others incurred losses of close to 10 g. Substantial salt losses were also seen in some, but not all players training in cold (5°C) conditions (Maughan et al. 2005). This may reflect, at least in part, the clothing worn in this environment.

Most participants in endurance events such as a marathon race or triathlon finish the event having lost more fluid than they consumed and are therefore relatively hypohydrated (Whiting et al. 1984). There have, however, been numerous publications in the scientific and medical literature over the last 2 decades drawing attention to the fact that some participants in endurance events consume more fluid than they lose and therefore complete the event in a hyperhydrated state (Noakes & Immda 2003). The main danger of excessive water intake is the development of hyponatraemia; while often asymptomatic, this condition, if severe, can result in nausea, collapse, loss of consciousness and even death. Early reports related almost exclusively to participants in ultra-endurance events where exercise intensity, and therefore sweat rate, was low and where opportunities for fluid intake were plentiful (Noakes et al. 1985; Hiller 1989; Noakes et al. 1990). Noakes and colleagues (1985) reported four cases of exercise-induced hyponatraemia; race times were between 7 and 10 hours, and post-race serum sodium concentrations were between 115 and 125 mmol/L. Estimated fluid intakes were between 6 and 12 litres, and consisted of water or drinks containing low levels of electrolytes; estimated total sodium chloride intake during the race was 20–40 mmol. Frizell and colleagues (1986) reported even more astonishing fluid intakes of 20–24 L of fluids (an intake of almost 2.5 L/hour sustained for a period of many hours, which is in excess of the maximum gastric emptying rate that has been reported) with a mean sodium content of only 5–10 mmol/L in two runners who collapsed after an ultramarathon run and who were found to be hyponatraemic (serum sodium concentration 118–123 mmol/L).

A study of 488 participants in the 2002 Boston Marathon revealed that 13% had a serum sodium concentration equal to or less than 135 mmol/L, and were therefore diagnosed as being hyponatraemic (Almond et al. 2005). Analysis of the results suggested that a substantial increase in body mass while running, a slow finishing time, and body-mass-index extremes were associated with hyponatraemia; although female runners with low body mass had earlier been suggested to be at particular risk of this condition, the results did not support this. These results suggest that medical staff should be alert to the possibility of hyponatraemia occurring in this situation, but this should not divert attention from the fact that most competitors will be both hypohydrated and hypernatraemic. It should also be recognised that, in the absence of any pre-race measurements, there is no way of knowing whether the hyponatraemia developed during the race or was present before the start. What is apparent is that participants in endurance events should not drink so much that they gain weight during the event (Sawka et al. 2007; Hew-Butler et al. 2008). Risk factors for the development of hyponatraemia identified by Hew-Butler et al. (2008) are shown in Table 13.5.

TABLE 13.5 Risk factors for the development of hyponatraemia

Athlete-related
Excessive drinking behaviour
Weight gain during exercise
Low body weight
Female sex
Slow running or performance pace
Event inexperience
Non-steroidal anti-inflammatory agents
Event-related
High availability of drinking fluids
>4 hours exercise duration
Unusually hot conditions
Extreme cold temperature

Source: Adapted from Hew-Butler et al. 2008

13.8.2 Fluid replacement and exercise performance

Most of the early studies carried out to investigate the effects of dehydration and rehydration on exercise in a military setting used very prolonged walking exercise as an experimental model and water as the fluid replacement. More recent studies have used a variety of exercise models more relevant to competitive sports situations, and most have investigated the effects of CHO-electrolyte drinks rather than of plain water. There have, however, been a few studies where the effects of plain water or of CHO-free electrolyte solutions have been investigated. In prolonged exercise at low intensity, water may be as effective as dilute saline solutions

(Barr et al. 1991) or nutrient-electrolyte solutions (Levine et al. 1991) in maintaining cardiovascular and thermoregulatory function. Maughan and colleagues (1996) had 12 male subjects exercise to fatigue at about 70% VO_{2max} on four occasions after appropriate familiarisation tests. When subjects ingested plain water (100 mL every 10 min) median exercise time was longer (93 min) than when no drink was given (81 min). Subjects also completed trials where dilute CHO-electrolyte drinks were given and these also resulted in extended exercise time compared to the no-drink trial. In a prolonged (90 min) intermittent high-intensity shuttle running test designed to simulate the demands of competitive soccer, McGregor and colleagues (1999) found that ingestion of flavoured water (5 mL/kg before the test and 2 mL/kg at 15-minute intervals) was effective in preventing a decline in performance of a soccer-specific skilled task. When no fluid was given, performance deteriorated.

It is clear that the addition of CHO has a number of potential benefits that may be important for performance (see sections 13.3–13.7). The separate effects of providing fluid and CHO were investigated by Below and colleagues (1995), who used an experimental model where subjects performed 50 minutes of exercise at about 80% VO_{2max} followed by a time trial where a set amount of work had to be completed as fast as possible. During the initial 50 minutes of exercise, subjects were given either a small volume (200 mL) of water, a small volume of water with added CHO (40% solution, 79 g of maltodextrin), a large volume (1330 mL) of flavoured water, or a large volume of water with the same amount of CHO as in the other CHO trial (as a 6% solution). They found water ingestion to be effective in improving performance; exercise time was 11.34 minutes on the placebo trial and 10.51 minutes on the water trial. Exercise time on the CHO trial was 10.55 minutes, indicating that CHO provision during exercise acted independently to improve performance, and the effects were found to be additive, with the shortest time (9.93 min) when the 6% CHO drink was given.

The results of these and other studies (see Maughan & Shirreffs 1998 for a review) suggest that fluid replacement is effective in improving exercise performance in a variety of different situations, and that an additional benefit is gained by the addition of CHO, and possibly also of electrolytes, to fluids ingested during exercise. The optimum formulation of drinks for use in different exercise situations has not, however, been clearly established.

13.9 Guidelines for replacing fluid during exercise

The major components of the sports drink that can be manipulated to alter its functional properties are shown in Table 13.6. To some extent, these factors can be manipulated independently, although addition of increasing amounts of CHO or electrolyte will generally be accompanied by an increase in osmolality, and alterations in the solute content will have an impact on taste characteristics, mouth feel and palatability.

TABLE 13.6 Variables that can be manipulated to alter the functional characteristics of a sports drink

CHO content: concentration and type
Osmolality
Electrolyte composition and concentration
Flavouring components
Other active ingredients

As well as providing an energy substrate for the working muscles, the addition of CHO to ingested drinks will promote water absorption in the small intestine, provided the concentration is not too high. Because of the role of sugars and sodium in promoting water uptake in the small intestine, it is sometimes difficult to separate the effects of water replacement from those of substrate and electrolyte replacement when CHO-electrolyte solutions are ingested. Below and colleagues (1995) have shown that ingestion of CHO and water had separate and additive effects on exercise performance, and concluded that ingestion of dilute CHO solutions would optimise performance. Most reviews of the available literature have come to the same conclusion (Murray 1987; Coyle & Hamilton 1990; Maughan & Shirreffs 1998).

13.9.1 Carbohydrate content and its influence on drink composition

The amount and types of CHO present in a drink will influence its efficacy when consumed during exercise. The optimum concentration of CHO to be added to a sports drink will depend on individual circumstances. High CHO concentrations will delay gastric emptying, thus reducing the amount of fluid that is available for absorption, but will increase the rate of CHO delivery. If the concentration is high enough to result in a markedly hypertonic solution, net secretion of water into the intestine will result, and this will actually increase the danger of dehydration (Evans et al. 2009). High concentrations of sugars (>10%) may also result in gastrointestinal disturbances (Davis et al. 1988). Where the primary need is to supply an energy source during exercise, increasing the sugar content of drinks will increase the delivery of CHO to the site of absorption in the small intestine. Beyond a certain limit, however, simply increasing CHO intake will not continue to increase the rate of oxidation of exogenous CHO (Wagenmakers et al. 1993; see section 13.5.3). Dilute glucose-electrolyte solutions may also be as effective, or even more effective, in improving performance in some exercise scenarios as more concentrated solutions (Davis et al. 1988; Watson et al. 2012), and adding as little as 90 mmol/L (about 16 g/L or 1.6%) glucose may improve endurance performance (Maughan et al. 1996).

The consequences of severe dehydration and hyperthermia are potentially fatal, but the symptoms of CHO depletion are usually nothing more than severe fatigue. It seems sensible, therefore, to favour more dilute solutions, especially when training or competing in warm weather. If greater quantities of CHO are required (as summarised in section 13.7) during prolonged endurance events of sufficient intensity, multiple transportable CHO-containing solutions are worthy of consideration (section 13.5.2).

13.9.2 Osmolality

It has become common to refer to CHO-electrolyte sports drinks as isotonic drinks, as though the tonicity was their most important characteristic. The osmolality of ingested fluids is important as this can influence the rates of both gastric emptying and intestinal water flux; both of these processes together will determine the effectiveness of rehydration fluids at delivering water for rehydration and substrate for oxidation (Schedl et al. 1994). Ingestion of strongly hypertonic drinks will promote net secretion of water into the intestine and, although this effect is transient, it will result in a temporary exacerbation of the extent of dehydration. The composition of the drinks and the nature of the solutes is, however, of greater importance than the osmolality itself (Maughan 1994).

Osmolality is identified as an important factor influencing the rate of gastric emptying of liquid meals, but there seems to be rather little effect of variations in the concentration of electrolytes on the emptying rate, even when this substantially changes the test meal osmolality (Rehrer 1990). The effect of increasing osmolality seems to be important only when nutrient-containing solutions are examined, and energy density is undoubtedly the most significant factor influencing the rate of gastric emptying of ingested liquids (Brener et al. 1983; Vist & Maughan 1994). There is some evidence that substitution of glucose polymers for free glucose, which will result in a decreased osmolality for the same CHO content, may be effective in increasing the volume of fluid and the amount of substrate delivered to the intestine. This is one reason for the inclusion of glucose polymers of varying chain length in the formulation of sports drinks. Vist and Maughan (1995) have shown that there is an acceleration of emptying when glucose polymer solutions are substituted for free glucose solutions with the same energy density. At low (about 40 g/L) concentrations, this effect is small, but it becomes appreciable at higher (180 g/L) concentrations; where the osmolality is the same (as in the 40 g/L glucose solution and 180 g/L polymer solution), the energy density is of far greater significance in determining the rate of gastric emptying. This effect may therefore be important when large amounts of energy must be replaced after exercise, but is unlikely to be a major factor during exercise where more dilute drinks are taken.

Water absorption occurs largely in the proximal segment of the small intestine, and, although water movement is itself a passive process driven by local osmotic gradients, it is closely linked to the active transport of solute (Schedl et al. 1994). Net flux is determined largely by the osmotic gradient between the luminal contents and intracellular fluid of the cells lining the intestine. Absorption of glucose is an active, energy-consuming process linked to the transport of sodium. The rate of glucose uptake is dependent on the luminal concentrations of glucose and sodium, and dilute glucose-electrolyte solutions with an osmolality that is slightly hypotonic with respect to plasma will maximise the rate of water uptake (Wapnir & Lifshitz

1985). Solutions with a very high glucose concentration will not necessarily promote an increased glucose uptake relative to more dilute solutions, but, because of their high osmolality, will cause a net movement of fluid into the intestinal lumen (Gisolfi & Summers 1990). This results in an effective loss of body water and will exacerbate any pre-existing dehydration. Other sugars, such as sucrose (Spiller et al. 1982) or glucose polymers (Jones et al. 1983; Jones et al. 1987), can be substituted for glucose without impairing glucose or water uptake, and may help by increasing the total transportable substrate without increasing osmolality. In contrast, iso-energetic solutions of fructose and glucose are isosmotic, and the absorption of fructose is not an active process in man: it is absorbed less rapidly than glucose and promotes less water uptake (Fordtran 1975). The use of different sugars that are absorbed by different mechanisms and that might thus promote increased water uptake is supported by evidence from an intestinal perfusion study (Shi et al. 1995), while the performance benefits of such CHO blends have been discussed previously in section 13.5.2. Although most of the popular sports drinks are formulated to have an osmolality close to that of body fluids (Maughan 1994), and are promoted as isotonic drinks, there is good evidence that hypotonic solutions are more effective when rapid rehydration is desired (Wapnir & Lifshitz 1985).

13.9.3 Electrolyte composition and concentration

The available evidence indicates that the only electrolyte that should be added to drinks consumed during exercise is sodium, which is usually added in the form of sodium chloride (Maughan 1994). Sodium will stimulate sugar and water uptake in the small intestine and will help to maintain extracellular fluid volume. There is much debate as to the optimum sodium concentration, and it has been argued that equilibration occurs so rapidly in the upper part of the small intestine that addition of high concentrations of sodium is not necessary (Schedl et al. 1994). Although most soft drinks of the cola or lemonade variety contain virtually no sodium (1–2 mmol/L), sports drinks commonly contain about 10–30 mmol/L sodium and oral rehydration solutions intended for use in the treatment of diarrhoea-induced dehydration, which may be fatal, have higher sodium concentrations, in the range 30–90 mmol/L. A high sodium content, although it may stimulate jejunal absorption of glucose and water, tends to make drinks unpalatable, and it is important that drinks intended for ingestion during or after exercise should have a pleasant taste in order to stimulate consumption. Specialist sports drinks are generally formulated to strike a balance between the twin aims of efficacy and palatability, although it must be admitted that not all achieve either of these aims.

13.9.4 Taste

Taste is an important factor influencing the consumption of fluids, and the choice of anion to accompany sodium may be important in this regard. The thirst mechanism is rather insensitive and will not stimulate drinking behaviour until some degree of dehydration has been incurred (Hubbard et al. 1984). This absence of a drive to drink is reflected in the rather small volumes of fluid that are typically consumed during exercise. In endurance running events, voluntary intake seldom exceeds about 0.5 L/h (Noakes 1993), and seems to be largely unrelated to the sweating rate. Because the sweat losses normally exceed this, even in cool conditions, a fluid deficit is almost inevitable whenever prolonged exercise is performed. The advent of mass participation marathons and other endurance events, however, has resulted in large numbers of very slow participants, many of whom will barely sweat but who will be exposed to essentially unlimited amounts of free drinks at frequent intervals. It is perhaps not surprising that intake exceeds loss in many of these individuals.

Anything that stimulates drinking behaviour is likely to be advantageous for those at risk of dehydration, and palatability is clearly important. Several factors will influence palatability, and the addition of a variety of flavours has been shown to increase fluid intake relative to that ingested when only plain water is available. Hubbard and colleagues (1984) and Szlyk and colleagues (1989) found that the addition of flavourings resulted in an increased consumption (by about 50%) of fluid during prolonged exercise. In a study of adolescent participants, Bar-Or and Wilk (1996) showed that the fluid intake during exercise of children presented with a variety of flavoured drinks is very much influenced by taste preference; under the conditions of this study, sufficient fluid to offset sweat losses was ingested when a grape-flavoured beverage was available but not when others were provided. In many of these studies, though, the addition of CHOs and/or electrolytes accompanied the flavouring agent, and the results must be interpreted with some degree of caution. The addition of CHO has a major impact on taste and mouth-feel, and a variety of different sugars with different taste characteristics can be added.

Effective post-exercise rehydration requires replacement of electrolyte losses as well as the ingestion of a volume of fluid in excess of the volume of sweat loss (Shirreffs et al. 1996). Drinks with a high sodium content can result in an unpalatable product. This can be alleviated to a large degree by substituting other anions for the chloride that is normally added, or by consuming foods and fluids that are naturally high in sodium along with adequate amounts of water.

13.9.5 Temperature of ingested drinks

As well as affecting the taste and perceived pleasantness of drinks, the temperature at which they are ingested may have implications for exercise performance. When cold drinks are ingested, heat must be added to raise them to body temperature: if the volume of fluid ingested is large and the temperature differential is also large, a measurable fall in body temperature will occur. It is well recognised that performance of prolonged exercise in warm environments can be improved by prior immersion in cool water to lower body temperature (Gonzales-Alonso et al. 1999), and there is emerging evidence that ingestion of cold drinks may have similar benefits. Wimer and colleagues (1997) found that, compared with the ingestion of approximately 1350 mL of water at 38°C, ingestion of the same volume of drinks at 0.5°C attenuated the rise in rectal temperature (Tre) during 2 hours of recumbent cycling at 51% VO_{2peak} in a temperate environment (26°C, relative humidity 40%). This observation was confirmed by Lee and Shirreffs (2007), who found that acute ingestion of 1 litre of drinks at 10°C during 90 minutes of cycling at 53% VO_{2peak} in a moderate environment (25°C, relative humidity 61%) was more effective in attenuating the rise in Tre than was ingestion of the same drink at 50°C. When drinks at 10 and 50°C were consumed in four smaller aliquots of 400 mL each at intervals during 90 minutes of cycling at 50% VO_{2peak} in a similar moderate environment (25°C, relative humidity 60%), the absolute rise in Tre at the end of exercise was similar. This can be explained by the initiation of appropriate thermoregulatory reflexes associated with ingestion of the cool and hot drinks (Lee et al. 2008). Lee, Shirreffs and Maughan (2008) subsequently reported that time to fatigue in a cycling test at 66% VO_{2peak} in a hot (35°C), humid (60%) environment was improved when a drink ingested at intervals before and during exercise was given at a temperature of at 4°C rather than at 37°C.

13.10 Monitoring individual fluid needs

It is a matter of everyday observation that some people sweat more than others, even when the exercise and environmental conditions are the same. The crusted salt deposits that can be seen on the exercise clothing worn by some athletes also show them to be salty sweaters. From the information presented in the preceding section, it is also apparent that some athletes drink much more than others and that the match between what athletes choose to drink and their fluid needs is far from perfect. In addition, it must be remembered that most laboratory studies are conducted on subjects who are rested, fed and well hydrated prior to exercise; in the real world, many athletes may begin exercise while still recovering from an earlier exercise session and may be hypohydrated at the start of exercise.

Pre-exercise hydration status can be assessed in several different ways, but urine markers, especially colour (Armstrong 2000) and osmolality (Shirreffs & Maughan 1998), are perhaps most reliable. Recent data suggest that many football players begin training with elevated urine osmolality, suggestive of significant hypohydration (Maughan et al. 2004; Shirreffs et al. 2005). A similar situation applies in the competitive environment, with as many as one-third of players providing urine samples with an osmolality in excess of 900 mosmol/kg upon reporting for a competitive game (Maughan et al. 2007).

It is often recommended that athletes should drink sufficient fluid during exercise to prevent any fall in body mass, as if this indicated a match between fluid intake and sweat losses. However, there is some loss of body mass during exercise due to substrate oxidation, and also some gain in body water due to the water of oxidation formed by oxidation of these fuels. It is also not apparent that there is an absolute need to replace all fluid losses. Two factors must be considered: the level of fluid deficit—or body mass loss—at which a decrement in performance occurs, and the potential benefit of the reduced body mass that follows from sweat loss in those sports where body mass must be supported. Body mass losses of up to 3% may be tolerable in cool environments without performance impairments, though smaller losses are tolerable in the heat (Coyle 2004). It might be more appropriate, therefore, to advise athletes to monitor their body mass losses during training or competition, and to drink sufficient to restrict body mass loss to not more than 1–2% of the initial value. This advice takes account of individual variations in sweating rate and in drinking behaviours, and also allows athletes to adjust drinking strategies to take account of the duration and intensity of an exercise session and of the climatic conditions. It does mean, though, that athletes must be willing to make these simple measures on themselves during training to allow anticipation of likely sweat losses in differing situations. The alternative strategy that has become popular is to advise athletes that they should simply drink according to the dictates of thirst. While this approach may work for some athletes in some situations, there are also drawbacks. It is difficult to plan for thirst, thirst is not always a good predictor of hydration status and, especially in longer events, gut function may be compromised during the later stages of the event. Therefore it is important to use the early stages to take up CHO and fluid and drinking ahead of thirst would be preferred.

13.11 Fluid intake: practical implications

It seems sensible to recommend that runners should drink enough to limit weight loss to not more than about 2% of body mass (Sawka et al. 2007) and perhaps even less than this on hot days. This recommendation puts some responsibility on the individual to experiment with different drinking strategies during training, but it cannot cater for the individual who enters a marathon to raise money for charity or for other reasons without having undertaken any preparation.

As outlined above, there are obvious benefits from the ingestion of fluids during exercise, and participants in endurance events should be encouraged to drink on a regular basis. There is, however, a need to apply common sense: slow runners on a cold day will lose little or no sweat and the primary need is for CHO intake. Drinking small amounts of concentrated CHO drinks will be an effective strategy. Fast runners in the heat may need to drink at a faster rate and are more likely to benefit from more dilute CHO-electrolyte drinks (Coyle 2004).

Appropriate CHO and fluid intake depends on many factors including the intensity and duration of exercise, the environmental conditions as well as individual sweat rates. In order to develop a sounds nutritional strategy some prior knowledge of sweat rates is required. This together with the CHO needs that can be predicted from Figure 13.1, should give the absolute quantities of fluid and CHO that should be ingested. The form in which the CHO and fluid is ingested depends on factors like preference, tolerance and practicality.

Summary

The intake of fluid and CHO offers benefits to the performance of a number of sports events and exercise activities. The effects of dehydration on performance are now well known, with the penalties ranging from subtle, but often important, decrements in performance at low levels of fluid deficit to the severe health risks associated with substantial fluid losses during exercise in the heat. Although evidence of the beneficial effects of CHO intake during exercise has existed for over 70 years, sports scientists are still to discover all the situations in which benefits occur and to explain the mechanisms involved. What is clear, though, is that a substantial body of evidence exists, and continues to grow, supporting performance benefits from CHO ingestion for a range of endurance activities, as well as intermittent, team sport events. It is also increasingly apparent that the ergogenic effects of CHO are not limited to the prevention of muscular fatigue and maintenance of power output; rather, aspects of motor skill, cognition and motor output are also evident in some situations. Optimal strategies for CHO and fluid intake during exercise will be determined by practical issues such as the opportunity to eat or drink during an event and gastrointestinal comfort. Variations in individual physiology and biochemistry will influence substrate use and sweat losses, so athletes must take responsibility for developing their personal plan based on individual circumstances.

Practice tips

MICHELLE MINEHAN

Plans for optimal replacement of fluid and fuel during sporting activities should be based on individual needs and fine-tuned with practice. However, the following strategies are suggested as a starting point for different types of sporting activities.

NON-ENDURANCE SPORTS: EVENTS OF LESS THAN 30 MINUTES IN DURATION

- The primary concern is minimal interference to competition.
- Recommendations:
  – Begin exercise in a well-hydrated condition.
  – Replace fluid losses between competition sessions.
- Athletes commonly approach competition in a hypohydrated condition as a result of failing to replace daily body fluid losses or as a result of deliberate dehydration strategies that are undertaken to ‘make weight’ in weight-limited sports (see Chapter 7). Exercising in a hypohydrated condition increases the risk of thermal injury and may reduce performance.
- Fluid ingested during exercise of less than 30 minutes’ duration will not benefit performance, as it will not become available to the body within the timeframe of the competition. However, the ingestion of fluid may offer some advantages such as to alleviate dry mouth and improve perceived exertion. Athletes must weigh up any perceived benefits of drinking during exercise against potential disadvantages such as increased body mass and having to slow down to drink.
- Athletes competing in tournament situations or multiple events should aim to rehydrate between sessions to avoid a progressive dehydration over the competition.

EVENTS OF 30–60 MINUTES IN DURATION

- The primary concerns are fluid intake with some opportunity for CHO provision.

- Recommendations:
  – Begin exercise well hydrated.
  – Using a fluid intake plan that has been practised in training, drink at a rate that is comfortable and practical to replace enough of the fluid lost via sweat to avoid problematic levels of dehydration.
  – Use a beverage that is cool (10–20°C), palatable and provides CHO.
  – Ingest beverage regularly to maintain gastric volume and increase fluid availability.
  – Make the most of opportunities to drink within the confines and environment of the sport.
  – Replace fluid losses between competition sessions.
- Theoretically, in hot environments, athletes should aim to drink enough to offset most of their fluid losses. In practice, athletes should drink according to what is comfortable and practical to achieve, preventing problematic levels of dehydration, but without exceeding the rate of their sweat losses so that they gain weight over the course of the event. Individuals vary enormously in their rates of gastric emptying, sweat loss and tolerance of fluid volume. Therefore, each athlete must devise an individualised drinking schedule that balances considerations of gastrointestinal discomfort and the loss of the time taken to drink against the benefits of reducing the level of dehydration. A guide to fluid requirements can be provided by weighing athletes before and after exercise sessions to estimate fluid losses (Figure 13.2). A fluid replacement plan can then be developed, and practised and refined during training. By experimenting and practising, it is possible for athletes to train themselves to tolerate greater volumes and learn to drink at a rate that matches as much of their sweat losses as considered practical and necessary. Hydration regimens should always be practised in training before trying them in competitive situations.
- The volume of fluid ingested is more important than the timing, but drinking regularly will help to maintain a high rate of gastric emptying, as fluids leave the stomach faster when gastric volume is high. It also makes sense to begin drinking early in competition to minimise dehydration rather than trying to reverse a severe deficit later in competition.
- CHO consumption is not necessary in short (30–45 minute) exercise sessions. However, mouth-rinsing with a CHO solution may provide performance benefits. Swilling a CHO solution for approximately 5 seconds at regular intervals should be considered.
- A supplementary source of CHO during exercise has been shown to improve performance of events of about 1 hour in duration, where fatigue would otherwise occur. A general recommendation of 30–60 g CHO per hour is suggested. Most sports drinks contain 60–80 g/L, making these rates of CHO ingestion easy to achieve. Gastric emptying slows as the energy density and osmolality of the fluid increase but solutions of up to 8% CHO can generally be tolerated, especially if a high gastric volume is maintained. Beverages that contain more than 8% CHO are more likely to cause gastrointestinal distress but individual tolerance varies. Combinations of sucrose, glucose, fructose and maltodextrins are all acceptable forms of CHO for ingestion, provided that fructose does not predominate.
- Sodium chloride replacement is not necessary in short exercise periods, but the inclusion of sodium chloride in a sports drink may promote fluid retention in the extracellular compartment, help maintain the osmotic drive to drink and improve the palatability of the drink. Fluids will be consumed in greater amounts when they taste palatable during exercise, are kept cool (10–20°C), are served in a user-friendly container and are readily accessible.
- The rules and conditions of some sports place restrictions on the opportunity to drink. Each athlete needs to identify opportunities to drink and practise strategies to utilise each opportunity. Sports such as netball, basketball and soccer restrict fluid intake to breaks in play, and drinks can only be taken from the sidelines. Players need to practise getting to the sidelines and distributing water bottles quickly. Other codes of football allow additional fluid to be provided by trainers on the field. Trainers need to monitor players and ensure fluids are distributed to all players. Players must make the effort to look for trainers and communicate their fluid needs. Opportunities for fluid intake in team sports are reviewed by Burke 2008. Athletes competing in individual sports need to practise skills such as drinking on the run, grabbing drinks from drink stations, and so on.

1. Weigh athletes before and after training in minimal clothing and after towel-drying.
2. Monitor volume of fluid consumed during training.
3. Determine change in body mass (BM) before and after any toilet stops.

sweat loss (mL) = change in BM (g) + fluid intake (mL) − urine losses (g)

Note that these calculations do not take into account the changes in BM that occur during prolonged exercise as a result of factors other than sweat loss. This includes weight changes due to oxidation of metabolic fuels, and the formation of water during such reactions. In prolonged bouts of intense exercise, these factors become substantial (e.g. 1–2% BM) and monitoring changes in BM without correcting for these factors will cause an overestimation of the true fluid deficit.

Figure 13.2 Quick method for estimating sweat loss

ENDURANCE SPORTS: EVENTS OF 1–2.5 HOURS IN DURATION

- The primary concerns are fluid replacement plus CHO provision.
- Recommendations:
  – Begin exercise well hydrated.
  – Use a fluid intake plan that has been practised in training, to drink at a rate that is comfortable and practical to replace enough of the fluid lost by sweating to avoid problematic levels of dehydration.
  – Take care not to drink excessive volumes of fluid (in excess of sweat rates) such that there is a gain in body mass over the session.
  – Use a beverage that is cool (10–20°C), palatable and provides CHO.
  – Begin ingesting fluid early in the exercise and continue to ingest the beverage regularly to maintain gastric volume and increase fluid availability.
  – Plan to consume 30–60 g CHO per hour of exercise.
- Sports drinks are intended to cater for the masses and suit the average sports event. For some individuals in some situations it may be desirable to vary the standard sports drink formula. On occasions when CHO needs take priority over fluid needs (e.g. in prolonged events carried out in cold conditions), a more concentrated solution might be useful. Alternatively, a more dilute preparation (e.g. 4%) might be appropriate for exercise in extremely hot conditions when fluid needs are of greatest priority. The intake of larger volumes of the sports drink in warm conditions will automatically increase the total amount of CHO consumed.
- Athletes use a variety of foods, fluids and gels during competition. Some provide a more concentrated source of CHO and will slow gastric emptying. However, solids can be desirable during prolonged competition as they increase the flavour options available, provide different textures and help to relieve hunger. Solids and gels also have the advantage of being a compact form of CHO, reducing the amount of sports drink an athlete must carry to enable refuelling. This is particularly useful for training sessions and for events conducted without the support of handlers or an intricate network of aid stations. Table 13.7 describes various food and fluids that may be used in competition.

TABLE 13.7 Food and fluid choices for endurance events

Description	Amount to provide 50 g CHO	Comments
Water		Does not assist with fuel needs, but may be drunk in addition to sports drinks or solid food to make up total fluid needs
Sports drinks 4–8% CHO + electrolytes	600–1000 mL	Best option for meeting fluid and CHO requirements simultaneously Has a good taste profile to encourage voluntary intake Provides small amounts of electrolytes
Soft drink 11% CHO	500 mL	May be more slowly absorbed due to CHO content Negligible source of electrolytes Provides alternative flavour during long events Cola drinks provide a small amount of caffeine
Fruit juices 8–12% CHO	500 mL	May be more slowly absorbed due to CHO content Negligible source of electrolytes Possible risk of gastrointestinal upset if juice is high in fructose
Sports gel 60–70% CHO	1½–2 gels	Concentrated CHO source Suitable for large fuel boost Experiment to avoid gastrointestinal discomfort Fluid requirements will need separate attention
Banana	2–3 medium	Solid foods may cause gastrointestinal concerns in some individuals, but may help to relieve hunger during long events Several portions are needed to provide substantial amounts of CHO Fluid requirements will need separate attention
Jelly beans	50 g	Compact CHO source Large amounts may cause diarrhoea Fluid requirements will need separate attention
Jam sandwich	2 thick slices + 4 teaspoons jam	Avoid adding fat sources (peanut butter, margarine) See comments for bananas
Chocolate bar	1½ bars	High in fat, so may be more slowly absorbed May help relieve hunger Fluid requirements will need separate attention
Muesli bar/cereal bar	1½–2 bars	Fat content varies from low to moderate See comments for bananas
Sports bars	1–1½ bars	Compact source of CHO Varying levels of fat May have various herbal additives of unknown function
Sports confectionery	5–10 pieces	Compact source of CHO Easy to carry Require chewing Fluid requirements will need separate attention

EVENTS OF GREATER THAN 2.5 HOURS IN DURATION

- The primary concerns are fluid plus CHO plus sodium provision.
- Recommendations:
  – Begin exercise well hydrated.
  – Use a fluid intake plan that has been practised in training, to drink at a rate that is comfortable and practical to replace enough of the fluid lost by sweating to avoid problematic levels of dehydration.
  – Take care not to overhydrate (to drink in excess of sweat losses) such that there is a gain in body mass over the session.
  – Use a beverage that is cool (10–20°C), palatable and provides CHO and sodium.
  – Begin ingesting fluid early in the exercise and continue to ingest beverage regularly to maintain gastric

volume and increase fluid availability.
– Plan to consume 30–90 g CHO per hour of exercise.
– Plan to replace sodium via sports drinks and foods.

- Hyponatraemia is a possibility in ultra-endurance events. A beverage (or foods) containing sodium chloride should be consumed, and will help to replace some of the sodium lost in sweat. However, the chief cause of hyponatraemia is excessive fluid consumption—drinking at a rate that exceeds the rate of sweat loss. Athletes should not drink in volumes that cause them to gain weight during an event. In fact, a loss of 1–2% body mass during prolonged events is likely to occur from factors unrelated to sweat losses and is acceptable.
- When competing at speed, consider higher rates of CHO consumption (90 g/h) using a mixture of glucose and fructose or other multiple transportable CHOs. Practise CHO consumption prior to competition so that the gut is trained to tolerate higher delivery rates. CHO intake can be achieved with a combination of solid food, gels and liquids.

SKILL-BASED SPORTS

- The primary concerns are fluid plus CHO provision.
- Recommendations:
  – Begin exercise well hydrated.
  – Use a fluid intake plan that has been practised in training, to drink at a rate that is comfortable and practical to replace enough of the fluid lost by sweating to avoid problematic levels of dehydration.
  – Use a beverage that is cool (10–20 °C) and palatable.
  – Plan to consume CHO in amounts similar to usual daily intake.
- Sports such as archery, shooting and bowling can involve long periods of competition, but the aerobic requirements of the sport are quite low. Drinking a fluid that is cool and palatable will encourage fluid intake. CHO from a variety of forms should be well tolerated as the aerobic demand is low. Athletes need to plan fluid and fuel replacement strategies that suit competition schedules.

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Commentary 7

Fluid guidelines

ALISON GARTH AND LOUISE BURKE

In recent years, there has been lively debate among scientific and sporting communities about the real-world application of guidelines for fluid intake during exercise. Although fluid guidelines developed by expert groups have progressed from highly prescriptive blanket recommendations to individualised plans, the practicality of implementing these recommendations during competitive events is complex. A recent review by Garth and Burke (2013) revealed that the patterns and drivers of fluid intake by athletes during competitive sporting events are multifaceted, leading to wide variability in fluid intake and sweat losses across and within different events.

Across all individuals and sporting activities, an array of factors influences an athlete's capacity (and desire) to replace fluid losses during exercise, many of which are beyond the athlete's control. These can include:

- thirst
- genetic predisposition to be an avid or reluctant drinker
- knowledge (or perception) of rates of fluid loss
- beliefs (or education) regarding dehydration or overhydration
- availability of fluids
- opportunity to drink
- palatability of drink (e.g. taste, temperature)
- gastrointestinal comfort
- (fear of) need to urinate
- using fluid as a vehicle for other ingredients (e.g. carbohydrate)
- desire for particular characteristics of fluid (e.g. temperature)
- desire/requirement to reduce body mass over session.

Adding further complexity to implementation of fluid guidelines is that each sport has its own unique set of opportunities and challenges for achieving fluid intake goals (see Table C7.1). In some situations, athletes can predict the influence of these factors and plan their fluid intake accordingly (e.g. distance between aid stations). However, other influences (e.g. race tactics, informal breaks in play) cannot be predicted and typically change between events in the same sport, making it challenging for athletes to plan their fluid intake.

TABLE C7.1 Factors influencing fluid intake during competitive sporting events

Sport category	Comments
Single-day endurance and ultra-endurance events (e.g. distance running, road cycling, triathlon)	• Fluid must be consumed while athlete is 'on the move' • **Access** to fluids is enhanced when drinks are provided by aid stations or handlers • Consideration of the time lost in obtaining and consuming fluid affects **opportunity** to drink • Technical skill requirements may limit **opportunity** to drink (e.g. downhill mountain bike riding, maintaining streamlined position in cycling time trial) • Practising drinking-on-the-move skills during training can increase **opportunity** to consume fluid during race • Devices such as fluid back-packs and spill-proof bottles may enhance **access** to fluid • Athletes may deliberately allow a fluid deficit to occur during exercise to enhance power-to-weight ratio
Multi-day ultra-endurance events (e.g. cycle tours, adventure racing)	• As above, with additional notes: – Potential to incur fluid deficit in 1 day that is carried to the next – Many adventure races require competitors to carry own supplies

Team and racquet sports 1. Outdoor: football codes, tennis, field hockey, cricket 2. Indoor: basketball, squash, badminton, squash	• **Opportunities** to drink include breaks in game or pauses in play that are specific to the sport (e.g. timeouts, change of ends, half-time, substitutions, free player rotations) • **Access** to fluid is increased when there is a drink supply close to the field of play and, where rules permit, when trainers can take drinks to players • In some team sports there is a culture of intravenous rehydration between periods of play within a game (or between games in a tournament) • Individual drink bottles may enhance hydration practices by increasing awareness of total fluid intake as well as **access** to fluid • In some team sports there is a culture that dehydration during training can 'toughen' players or assist with weight-loss goals; this may influence competition practices
Sprint, strength and power events (e.g. 100–1500 m sprint, jumps, throws, track cycling)	• There is no need or opportunity to hydrate during actual events • There are **opportunities** to drink fluid between rounds • **Access** to fluid may be limited to own supplies
Weight-making sports (e.g. wrestling, boxing, martial arts, lightweight rowing, weightlifting)	• Since most weight division sports involve competitive events of brief nature, fluid intake *during* competition will be limited to rehydration between weigh-in and competition (including the warm-up) or between individual bouts or races on the same day (e.g. martial arts tournament or horse-racing meet) • Although professional boxing provides opportunity for fluid intake in the break between rounds, this may be limited by gut comfort • Education is needed to minimise reliance on dehydration for weight-making
Skill sports (e.g. golf, shooting, archery)	• **Access** to fluid is reduced when sessions are undertaken in remote environment • **Access** to fluid is increased by provision of fluids at drink stations or by caddies • There may be **opportunity** to drink between rounds
Aquatic sports (e.g. swimming, water polo)	• There is no need or **opportunity** to hydrate during pool-based swimming races • **Access** to fluid in open-water swimming races is limited to feeding stations or feed boats • There may be **opportunity** to drink in water-polo limited to substitutions and between games • **Access to fluids** is enhanced by drink bottles on pool deck • There is potential for accidental ingestion of water from pool
On-water sports (e.g. rowing, kayak, sailing)	• There is no need or **opportunity** to hydrate during brief races (e.g. traditional rowing and kayaking) • **Access** to fluid is limited by boat space • **Opportunities** to drink are limited to breaks in activity since rowing or sailing activities typically involve both hands • In endurance rowing events, competitors often use fluid back packs
Winter sports (e.g. Alpine and Nordic skiing, skating, ice hockey)	• **Access** to fluid is reduced when session is undertaken in remote environment, or sub-zero temperatures (drinks freeze) • Voluntary fluid intake may be enhanced by provision of warm fluids • Intake of fluids may be consciously reduced to avoid the need to urinate in environments where there are no facilities
Motor sports	• **Access** to fluid and **opportunity** to drink are aided by use of fluid backpacks • **Opportunity** to drink is reduced during cornering due to abdominal pressure

This commentary will provide a brief overview of the key issues that influence fluid balance during competitive events for a selection of sporting events as well as provide insights into what the literature says athletes typically do (for a comprehensive review of this topic, we refer readers to the review by Garth & Burke 2013).

Single-day endurance and ultra-endurance events

In addition to the intensity and duration of the event, outdoor environmental conditions (e.g. heat, humidity, wind speed, altitude) and terrain have major influences on individual sweat losses and, therefore, the need to consume fluid during an event. In the majority of endurance and ultra-endurance events, fluid consumed can be a major source of ingredients such as carbohydrates, electrolytes and caffeine. Fluid intake can be challenging for these athletes, as they are required to drink 'on the go', which can lead to gastrointestinal upset and spillage of fluids. In elite races, opportunities to drink may be heavily influenced by race tactics whereby 'attacks' or 'surges' by competitors may occur if an athlete slows to drink. Access to fluids is typically dictated by placement of aid stations, although in many events, athletes may (or are required to) carry their own supplies in addition to fluids provided by event organisers.

The mode of exercise also influences sweat rates and fluid intake. During cycling events, wind speed can alter thermoregulatory regulation by improving cooling efficiency thereby reducing sweat rate; drafting during cycling reduces the power output required by an athlete, again reducing sweat rate. In single day endurance events, body mass (BM) changes of −1 to −3% are typically observed with fluid intakes of ~400–800 mL/h reported. BM changes appear higher in ultra-endurance events (−3 to −5%) with fluid intakes of ~300–1000 mL/h reported (typically lower intakes during running than cycling, which is likely due to differences in gastrointestinal comfort and access to fluids). It should be noted, however, that there is large individual variation between athletes in both endurance and ultra-endurance events with overhydration being a concern for some individuals.

Multi-day ultra-endurance endurance events

The factors influencing sweat losses and fluid intake during multi-day ultra-endurance events are similar to those noted for single-day events, with the additional challenge that intakes during 1 day of competition may contribute substantially to energy, carbohydrate and fluid requirements for the whole event. Additionally, deficits from 1 day can carry over to the subsequent days. Athletes competing in multi-day ultra-endurance events are typically observed to keep BM losses below 3% (unless conditions are very hot).

Fluid intakes are highly individualised but are reported to range between ~300 and 1000 mL/h.

Team sports

Variable and unpredictable high-intensity work patterns between and within sports underpin sweat losses during team sport matches. Even between matches of the same sport, individuals can have different game demands, which may be dictated by opposition game plans, playing times due to substitutions and overall match durations due to variations in overtime or informal breaks in play. In all sports, environmental conditions can have large influences on sweat rates; however, this is of more importance for team sports played outdoors than sports played inside climate-controlled arenas. Many sports also require athletes to wear heavy and impermeable protective clothing, which can increase sweat losses.

Depending on the competition format, team sport athletes may also commence competition with a fluid deficit carried over from the previous match. Opportunities to consume fluids during team sports are dictated by the rules of the game: in some sports, athletes have near ad libitum access to fluids (e.g. those with high rotations to and from the bench and officials who run fluid onto the field), while in others, they have only limited opportunity to drink (e.g. half time break in soccer or hockey). Body mass losses are typically maintained below 2% for athletes participating in indoor team sports outdoor team sports in cool to warm conditions (body mass losses up to 4% have been observed in outdoor sports played in hot conditions).

Fluid intakes typically range from 300–800 mL/h (although they have been observed to be >1000 mL/h in indoor sports with high fluid access).

Racquet sports

Similar to team sports, racquet sports are characterised by intermittent, high-intensity work patterns with variations in match duration and intensity between and within sports. Environmental conditions vary between sports, with some sports played on indoor climate-controlled courts, outdoor courts or even a combination of conditions (e.g. tennis stadium with a closing roof). Competition formats typically involve multiple matches over several days increasing the risk of carrying a fluid deficit from one match to the next.

Little is known about sweat rates of indoor racquet sports such as squash, table tennis or badminton as all of the current literature on racquet sports relates to tennis. Observations of tennis matches indicate much higher sweat rates in matches played in hot conditions (800–2000 mL/h) compared to cool to warm conditions (600–1000 mL/h).

Fluid intakes typically range between 800 and 1000 mL/h, keeping body mass losses for the majority of players below 2%.

Aquatic sports

Aquatic sports occur in lakes, rivers, oceans or temperature-controlled pools. There are both individual events (e.g. swimming) as well as team sports (e.g. water polo). Water provides high convective heat losses keeping sweat rates low. In many aquatic events, there is no need to consume fluid during exercise because of the short duration of the event. In long distance open water swimming events, athletes can access fluids from designated feed boats or pontoons. Access to fluids for athletes competing in water polo is limited to formal breaks in play and time outs.

Measuring fluid losses and intake is inherently difficult due to the wet aquatic environment and there is therefore very little published literature in the area. However, fluid intakes of water polo players and open water swimmers have been estimated to be ~400–500 mL/h with body mass losses well below 1%.

On-water sports

While there is no requirement to consume fluids during brief on-water events (e.g. Olympic rowing or kayaking races), opportunities during longer events (e.g. sailing regattas or endurance paddles) can be challenging as many activities require the use of both hands (e.g. winching, paddling, stroking). Environmental conditions such as heat, wind speed and sea-spray, as well as physical demands of different tasks or positions, all influence thermoregulation and therefore sweat rates. Race tactics can again influence opportunities to consume fluid during competition. Access to fluid is also limited by the capacity to carry fluids, as supplies must be considered in the context of weight and space restrictions. Research on fluid balance in on-water sports is sparse. Observations from sailing events suggest body mass losses are kept at <2% for the majority of athletes (although positional differences can lead to higher deficits if there is limited opportunity to consume fluids).

Fluid intakes are reported to range between 100 and 600 mL/h.

Winter sports

The huge breadth of winter sports presents a wide array of physiological and practical challenges and opportunities for fluid intake. Although competition occurs in cold environments, waterproof and insulating clothing or protective equipment can restrict heat loss resulting in increased sweat losses. The logistics of carrying fluids around remote environments, preventing drinks from freezing and the desire to avoid urinating (because of a lack of facilities or difficulty changing out of clothing), as well as a reduced thirst drive in the cold, can intentionally or unintentionally reduce fluid intake. Optimising power-to-weight ratio in some sports (e.g. aerial skiing or luge) as well as rules in team sports (e.g. ice hockey) can also influence opportunities (and desire) to drink. Research regarding fluid balance in winter sports is rare and the majority of the research focuses on ice-hockey competition, where fluid intakes typically reflect high sweat losses and overall BM losses are not high.

Summary

As has been highlighted in this commentary, when implementing fluid guidelines, there is a range of drinking strategies that athletes may use during competition to offset fluid losses. When working with athletes, it should be emphasised that strategies need to be individualised and flexible. Consider fluid as a source of other nutrients or characteristics within the plan.

Reference

Garth AK, Burke LM. What do athletes drink during competitive sporting activities? Sports Med 2013; 43(7):539–64.

James Carter is an employee of GSSI, a division of PepsiCo, Inc. The views expressed in this chapter are those of the author and do not necessarily reflect the position or policy of PepsiCo Inc.

CHAPTER FOURTEEN

Nutrition for recovery after training and competition

Louise Burke

14.1 Introduction

In many areas of sports nutrition, there is a familiar trajectory to the development of a new area of interest. The story often starts with a lack of interest and awareness of the area. However, a few key studies or anecdotal practices appear, pricking the interest of sections of the sports nutrition community. More targeted work follows, momentum is generated, and the resulting outputs allow the development of practical guidelines to take advantage of the new knowledge. By this stage, a market has been created to support the creation of a range of gadgets and resources that could facilitate the implementation of practical guidelines. Suddenly, the sports world is filled with experts, special facilities and must-have products. The overflow to the mass market and to recreational athletes usually leads to a simplification of the message into a few key practices that athletes come to feel are mandatory and universally applicable to all individuals and all situations. Then, just when mass saturation has occurred, new evidence emerges that the picture is far more complex than previously thought. We learn that we have over-diagnosed the importance, prevalence or symptoms of our issue. Worse still, we find that some of our newly recommended practices may not just be benign if they are unnecessary but may actually be harmful to our goals. This problem occurs because of a failure to understand the bigger context or the individuality of needs, and our drive to oversimplify what is a nuanced issue.

Many sports dietitians who observe real-life sports practice would attest that this scenario has occurred in the area of recovery nutrition whereby many modern athletes have taken a ‘one size fits all’ approach to aggressively consuming foods, fluids and supplements in the period after workouts or races. This chapter will provide a more nuanced consideration of several aspects of ‘recovery nutrition’ and some newer perspectives on post-exercise eating and drinking practices that can be truly beneficial. It will also articulate how these considerations integrate into the most recent guidelines for carbohydrate (CHO) intake in the training diet, including the possibility that withholding CHO may be beneficial on some occasions.

14.2 Differentiation of the goals of recovery eating

Athletes commonly undertake strenuous training programs involving one or more prolonged high-intensity exercise sessions each day, typically allowing 6–24 hours for recovery between workouts. In some sports, competition is conducted as a series of events or stages. In sports such as swimming or track and field, athletes are scheduled to compete in a number of brief races, or in a series involving heats, semi-finals and finals, often performing more than once each day. In tennis and team-sport tournaments, or cycle stage races, competitors may be required to undertake one or more lengthy events each day, with the competition extending for 1–3 weeks. Even where athletes compete in a weekly fixture, optimal recovery is desired to allow the athlete to train between matches or races to maintain conditioning and skills.

It is understandable, given the cluttered training and competition timetables of the modern athlete, that attention would be focused on strategies to make effective use of the period between exercise sessions. Indeed, sports nutrition has been a beneficiary of the flourishing industry that has spawned recovery centres, specialists and tailor-made foods and supplements. However, a sceptical view is that recovery eating is misunderstood and poorly practised by many athletes who see it as a requirement or excuse to eat or drink aggressively after every workout or event—often consuming nutrient-poor food and beverage choices or expensive products with a cocktail of exotic ingredients. At best, such misguided recovery eating can cause an unnecessary drain on the wallet. At worst, however, it can cause nutrition problems such as weight gain due to unnecessary or excessive intake of kilojoules or even a failure to promote optimal recovery and adaptation to a training program. Recent research has provided evidence that we require more knowledge and a more individualised and sophisticated approach to recovery eating.

An improved approach to post-exercise nutrition includes the recognition that it should be individualised and periodised within an athlete’s program. A new concept in recovery eating that underpins the need for different strategies for different exercise sessions is the identification of two separate goals for the post-exercise period:

- restoration of body losses/changes caused by the first session to restore performance levels for the next
- promotion of adaptive responses to the stress/stimulus provided by the session to gradually make the body become better at the features of exercise that are important for performance.

Obviously, the chief focus between events in a competition schedule is to restore homeostasis or depleted nutrient stores as quickly as possible to restore performance to optimal levels—or at least a level that is better than that of opponents or fellow competitors. This may also be the case for key training sessions where the athlete needs to achieve high performance outcomes in terms of intensity, speed, technique or skill. In contrast, some light workouts or easy competition scenarios may not create any major demands at all and it may not be necessary to undertake any special post-exercise eating practices. However, there may be additional scenarios in which the athlete may deliberately refrain from undertaking aggressive recovery eating strategies and may even avoid the intake of nutrients needed to restore homeostasis. These may arise because the athlete's priority is to promote the second recovery principle of adaptation to the training stimulus.

While the recovery goals of restoration and adaptation often overlap, or at least the strategies used to achieve one goal often also promote the other, several new themes have emerged in sports nutrition in which we recognise that better adaptation to an exercise stimulus might be achieved by contradicting the practices used to promote restoration. The first scenario addresses a concept popularly known as 'train low', which explores the hypothesis that greater adaptation to the same training stimulus can be achieved when the physiological/biochemical environment is not optimal. This concept has been most popularly applied to the issue of CHO availability, or more specifically, training with low muscle glycogen stores, which will be explored in more detail in section 14.5. However, there is some evidence that the 'train low' concept might have application in other areas, such as deliberately training with a fluid deficit to accelerate the processes underpinning acclimation to exercise in hot weather (Garrett et al. 2014).

The second scenario is explained by the concept that some of the 'damaging' processes that occur during exercise might be important in creating cellular signals for the processes that promote adaptive remodelling; this is particularly directed to the oxidative damage or inflammatory responses to exercise. Whereas a strategy that acutely addresses this damage may assist the body to *restore* its original function more quickly, it may also switch off processes that promote longer term adaptations to gradually *enhance* original function. This scenario is less well understood at the present time, but is best demonstrated by issues around the intake of antioxidant and anti-inflammatory nutrients/chemicals. The simplistic presentation of this theme is that these chemicals may be of benefit when they are used acutely to achieve a short-term recovery need (e.g. reducing the damaging effects of one competition bout to allow optimal performance in a successive event); however, their chronic application may blunt the adaptive processes to each exercise session, leading to a reduced adaptation to a longer period of training. Indeed, several studies have reported that chronic supplementation with antioxidants such as vitamins C and E (Gomez-Cabrera et al. 2008; Ristow et al. 2009) or coenzyme (Malm et al. 1997) interferes with the optimal response to a training program, leading to an impairment of performance or function compared with the intake of a placebo supplement. This area is covered in more detail in Chapter 11 and Commentary 6.

In addition to understanding key goals of recovery, a further differentiation in recovery eating arises from recognising the specific physiological stresses encountered in each workout or event. Each session differs in how much it causes the athlete to sweat, deplete muscle fuel stores, stimulate protein synthesis, or cause damage and disruption to the body. Recovery eating strategies need to address the degree to which restoration of homeostasis or promotion of adaptation from the specific session relies on:

- restoration of muscle and liver glycogen stores ('refuelling')
- replacement of the fluid and electrolytes lost in sweat ('rehydration')
- protein synthesis for repair and adaptation ('rebuilding')
- responses of other systems such as the immune, inflammatory and antioxidant systems.

Therefore, different types and amounts of nutrients will be needed to restore normal status. How important it is to deliver those nutrients to the body as soon as possible will depend on whether these nutrients are handled differently in the post-exercise phase and how long it will take to achieve restoration. The bottom line is that each session deserves its own recovery eating plan, and this may differ from athlete to athlete.

Finally, each recovery nutrition plan needs to be organised to integrate the athlete's overall nutritional goals, and to address any practical challenges that make it difficult to achieve. If recovery nutrition strategies make a frequent contribution to a long-term nutrition plan, they must be complementary to other goals related to physique management, needs for special nutrients, achievement of goals for healthy eating and the athlete's food preferences and social eating opportunities. On the other hand, if there is a special recovery plan for a single event, it may not need to take many of these longer-term issues into account. In many situations, optimal recovery after training or competition will occur only with a specially organised nutrition plan that recognises some of the practical factors that interfere with an athlete's post-exercise fluid and food intake plans (see Table 14.1). This is particularly important for the travelling athlete, who may be challenged by an inaccessible and foreign food supply. Special recognition of the needs of the travelling athlete is found in greater detail in Chapter 23.

TABLE 14.1 Practical factors interfering with post-exercise fluid and food intake

• Fatigue—interfering with ability/interest to obtain or eat food
• Loss of appetite following high-intensity exercise
• Limited access to (suitable) foods at exercise venue
• Other post-exercise commitments and priorities (such as coach meetings, drug tests, equipment maintenance and warm-down activities)
• Traditional post-competition activities (such as excessive alcohol intake)

This chapter provides detail about the recovery issues regarding refuelling and rehydration, since other sections of this book deal with protein/rebuilding (Chapter 4) and issues around antioxidant systems (Chapter 11 and Commentary 6) and the immune system (Chapter 22).

14.3 Factors in post-exercise glycogen storage

The depletion of muscle glycogen provides a strong drive for its own resynthesis (Zachwieja et al. 1991). Muscle glycogen resynthesis takes precedence over restoration of liver glycogen, and even in the absence of a dietary supply of CHO after exercise it occurs at a low rate—1–2 mmol/kg wet weight (ww) of muscle per hour—with some of the substrate being provided through gluconeogenesis (Maehlum & Hermansen 1978). High-intensity exercise that results in high post-exercise levels of lactate appears to be associated with rapid recovery of glycogen stores in the absence of additional CHO feeding (Hermansen & Vaage 1977). After moderate-intensity exercise, muscle glycogen synthesis is dependent on the provision of exogenous CHO. The rate of glycogen storage is affected by factors regulating glucose transport into the cell, such as the insulin- or exercise-stimulated translocation of GLUT 4 protein transporter to the muscle membrane (McCoy et al. 1996). It is also determined by factors regulating glucose disposal, such as the activity of glycogen synthase enzyme (Danforth 1965; McCoy et al. 1996). Changes in these factors are responsible for a bi-phasic muscle glycogen storage pattern, or a decline in glycogen storage rate over time (Ivy & Kuo 1998). Glycogen storage is impaired by damage to the muscle fibre, such as that caused by eccentric exercise or direct contact injury (Costill et al. 1988; Costill et al. 1991).

Liver glycogen stores are more labile than muscle glycogen stores, and may be depleted by an overnight fast as well as by a prolonged bout of exercise. Strategies to enhance the restoration of liver glycogen stores have been less well-studied due to practical problems in obtaining liver biopsy samples. Nevertheless it is considered that liver glycogen is restored by a single CHO-rich meal, and that fructose ingestion may cause a greater rate of liver glycogen synthesis than glucose intake (Blom et al. 1987).

Maximal rates of post-exercise muscle glycogen storage reported during the first 12 hours of recovery are within the range of 5–10 mmol/kg ww/h (Blom et al. 1987; Ivy et al. 1988; Reed et al. 1989). Coyle (1991) has commented that with a mean glycogen storage rate of 5–6 mmol/kg ww/h, 20–24 hours of recovery are required for normalisation of muscle glycogen levels following exercise depletion. In real life, the training and competition schedules of many athletes often provide considerably less time than this. If performance in subsequent exercise sessions is limited by muscle CHO stores, strategies that enhance rates of refuelling can be of value; several of the dietary factors that enhance or impair the rate of muscle glycogen storage are now discussed.

14.3.1 Amount of carbohydrate intake

The most important dietary factor affecting muscle glycogen storage is the amount of CHO consumed. According to a summary of studies that have monitored muscle glycogen storage following 24 hours of recovery from glycogen-depleting exercise, there is a direct and positive relationship between the quantity of dietary CHO and post-exercise glycogen storage, at least until the muscle storage capacity or threshold has been reached (Burke et al. 2004). Synthesising the data from a range of studies that have monitored glycogen storage followed exercise-induced depletion, including two studies which directly investigated 24-hour glycogen storage with different amounts of dietary CHO intake (Costill et al. 1981; Burke et al. 1995), it appears that the daily glycogen storage threshold occurs at a CHO intake of ~7–10 g/kg body mass (BM).

Although these figures have been integrated into the recommended CHO intakes for optimal muscle glycogen recovery, it is worth noting that they are derived from studies of glycogen storage during a passive recovery period where the exercise demand depleted muscle CHO stores (Burke et al. 2004). As a result, requirements for total daily CHO intake are lower for athletes whose training programs do not fully deplete glycogen stores and may be higher when the fuel requirements of continued heavy training are added to glycogen restoration needs. For example, well-trained cyclists undertaking 2 hours of training each day were found to have higher muscle glycogen stores after a week of a daily CHO intake of 12 g/kg BM than when consuming CHO intakes of 10 g/kg/d (Coyle et al. 2001). Furthermore, Tour de France cyclists riding at least 6 hours each day voluntarily consume CHO intakes of 12–13 g/kg/d (Saris et al. 1989) and the ingestion of substantial amounts of CHO during low- to moderate-intensity exercise has been reported to increase net glycogen storage during the session, particularly within non-active muscle fibres that have been previously depleted (Kuipers et al. 1987). Increased CHO intake may also be useful in the case of muscle damage (such as that following eccentric exercise), which typically impairs the rate of post-exercise glycogen resynthesis. Costill and colleagues (1991) reported that low rates of glycogen restoration in damaged muscles might be partially overcome by increased amounts of CHO intake during the first 24 hours of recovery.

Separate guidelines have been proposed to cover the CHO needs of the early phase (0–4 hours) of recovery. The 1991 consensus statement on nutrition for athletes by the International Olympic Committee (IOC) recommended that athletes should consume 50 g (~1 g/kg BM) of CHO every 2 hours until meal patterns are resumed (Coyle 1991). These guidelines were based on studies that failed to find differences in post-exercise glycogen storage following CHO intakes of 0.7 and 1.4 g/kg BM (Blom et al. 1987), or between 1.5 g and 3.0 g/kg BM (Ivy et al. 1988), fed at 2-hourly intervals. However, newer studies of early post-exercise recovery (Doyle et al. 1993; Piehl Aulin et al. 2000; Van Hall et al. 2000) achieved glycogen synthesis rates of up to 10–11 mmol kg ww/kg/h, or about 30% higher than previous literature values. Features of these more recent studies include larger CHO intakes (e.g. 1–1.8 g/kg/h) and repeated small feedings (e.g. an intake every 15–60 mins) rather than single or several large meals. Unfortunately, because these studies have not directly compared glycogen storage with different amounts of CHO and different feeding schedules, it is difficult to draw final conclusions about optimal CHO intake in the early recovery phase. Nevertheless, other studies (Van Loon et al. 2000; Jentjens et al. 2001) suggest that the threshold for early glycogen recovery is reached by a CHO feeding schedule providing 1.2 kg/h based on the failure to increase muscle glycogen storage when extra energy (protein) was consumed. These new recommendations were incorporated into the 2003 and 2010 IOC consensus guidelines (Burke et al. 2004, 2011).

14.3.2 Timing of carbohydrate intake

The highest rates of muscle glycogen storage occur during the first hour after exercise (Ivy et al. 1988), due to activation of glycogen synthase by glycogen depletion (Prats et al. 2009), exercise-induced increases in insulin sensitivity (Richter et al. 1989) and permeability of the muscle cell membrane to glucose. CHO feeding during these early stages appears to accentuate these effects by increasing blood glucose and insulin concentrations. Ivy and colleagues (1988) reported that the immediate intake of CHO after prolonged exercise resulted in higher rates of glycogen storage (7.7 mmol/kg ww/h) during the first 2 hours of recovery, slowing thereafter to the more typical rates of storage (4.3 mmol/kg ww/h). Although this study has been interpreted to highlight the significantly higher rates of glycogen synthesis in early recovery, it is unlikely that these differences (7.7 compared to 4.3 mmol/kg ww/h) are of physiological importance. The most important finding of this study is that failure to consume CHO in the immediate phase of post-exercise recovery leads to very low rates of glycogen restoration until feeding occurs. Thus the importance of early intake of CHO following strenuous exercise is to avoid delaying the provision of substrate to the muscle cell, more than to take advantage of a period of moderately enhanced glycogen synthesis. This strategy is most important when there are only 4–8 hours of recovery between exercise sessions, but may be of less impact when there is a longer recovery time (12 or more hours). For example, Parkin et al. (1997) investigated glycogen storage over 8 hours and 24 hours of recovery when high glycaemic index (GI) CHO meals were begun immediately after exercise, or delayed for 2 hours. They found no difference in glycogen restoration at either of these time points as a result of delaying the first CHO meal (Parkin et al. 1997).

Whether CHO is best consumed in large meals or as a series of snacks has also been studied. Studies investigating 24-hour recovery have found that restoration of muscle glycogen is the same whether a given amount of CHO is fed as two or seven meals (Costill et al. 1981) or as four large meals or sixteen hourly snacks (Burke et al. 1996). In the latter study, similar muscle glycogen storage was achieved despite marked differences in blood glucose and insulin profiles over 24 hours (Burke et al. 1996). In contrast, very high rates of glycogen synthesis during the first 4–6 hours of recovery have been reported when large amounts of CHO were fed at 15- to 30-minute intervals (Doyle et al. 1993; Van Loon et al. 2000; Van Hall et al. 2000; Jentjens et al. 2001) and attributed to the higher sustained insulin and glucose profiles achieved by such a feeding protocol. However, as previously noted, these outcomes were compared with other literature values of post-exercise glycogen restoration rather than directly tested against a control amount of CHO taken in less-frequent meals. One way to reconcile these apparently conflicting data is to propose that the effects of enhanced insulin and glucose concentrations on glycogen storage are most important during the first hours of recovery or when total CHO intake is below the threshold of maximum glycogen storage. However, during longer periods of recovery or when total CHO intake is above this threshold, manipulations of plasma substrates and hormones within physiological ranges do not add further benefit.

The practical implications of these studies are that meeting total CHO requirements is more important than the pattern of intake and, at least for long-term recovery, the athlete is advised to choose a food schedule that is practical and comfortable. A more frequent intake of smaller snacks may be useful in overcoming the gastric discomfort often associated with eating large amounts of bulky high-CHO foods, but may also provide direct benefits to glycogen storage during the early recovery phase. When the interval between exercise sessions is short, the athlete should maximise the effective recovery time by beginning CHO intake as soon as possible. However, when longer recovery periods are available, the athlete can choose their preferred meal schedule as long as total CHO intake goals are achieved. It is not always practical or enjoyable to consume substantial meals or snacks immediately after the finish of a strenuous workout.

14.3.3 Type of carbohydrate intake

Since glycogen storage is influenced by both insulin and a rapid supply of glucose substrate, it appears logical that CHO sources with a moderate to high GI would enhance post-exercise refuelling. This hypothesis has been confirmed in the case of single nutrient feedings of mono- and disaccharides; intake of glucose and sucrose after prolonged exercise both produce higher rates of muscle glycogen recovery than the low-GI sugar, fructose (Blom et al. 1987). The results of early investigations of real foods (Costill et al. 1981; Roberts et al. 1988) were confusing, since they used the structural classification of ‘simple’ and ‘complex or starchy’ CHOs to construct recovery diets; the conflicting findings are probably due to the failure to achieve a real or consistent difference in the GI of the diets.

The first comparison of foods based on published GI values reported greater glycogen storage during 24 hours of post-exercise recovery with a CHO-rich diet based on high-GI foods compared with an identical amount of CHO eaten in the form of low-GI foods (Burke et al. 1993). However, the magnitude of increase in glycogen storage (~30%) was substantially greater than the difference in 24-hour blood glucose and insulin profiles. We found that the meal consumed immediately after exercise produced a large glycaemic and insulinaemic response, independent of the GI of the CHO consumed, and overshadowed the differences in response to the rest of the diet. Other studies have confirmed an exaggerated glycaemic response to CHO consumed immediately after exercise compared with the same feeding consumed at rest (Rose et al. 2001). This has been explained as a result of greater gut glucose output and greater hepatic glucose escape, favouring an increase in muscle glucose uptake and glycogen storage. Therefore, while it appears that high-GI CHO foods achieve superior post-exercise glycogen storage, it cannot be totally explained in terms of an enhanced glucose and insulin response.

An additional or alternative mechanism to explain less efficient glycogen storage with low-GI CHO-rich foods is that a considerable amount of the CHO in these foods may be malabsorbed (Wolever et al. 1986; Jenkins et al. 1987). Indeed, the poor digestibility of a high amylose starch mixture (low GI) was proposed as an explanation for lower muscle glycogen storage observed during 13 hours of post-exercise recovery compared with intake of high-GI CHOs, glucose, maltodextrins and a high amylopectin starch (Joszi et al. 1996). This study concluded that indigestible forms of CHO provide a poor substrate for muscle glycogen resynthesis and overestimate the available CHO consumed by subjects. This issue needs to be further studied in relation to real foods. Nevertheless, a study of chronic exposure to a lower GI diet in recreationally active people found that muscle glycogen storage declined over 30 days compared with pre-trial values and was lower in comparison with the values on a high-GI trial (Kiens & Richter 1996).

Finally, there have been reports of enhanced glycogen synthesis associated with the use of a special glucose polymer with high molecular weight and a lower osmolarity when consumed in drink form (Piehl Aulin et al. 2000). This polymer, now available in commercial sports products, was found to promote higher muscle glycogen storage in the first two hours of recovery after glycogen-depleting exercise than the same amount of CHO from a glucose-based drink; these effects were no longer evident, however, in a subsequent 2-hour period (Piehl Aulin et al. 2000). The mechanism for the advantage was suggested to be a faster gastric emptying rate (Leiper et al. 2000). There may be targeted situations where such a product could be valuable. For example, after completing a submaximal cycling protocol to exhaustion, cyclists were able to produce more work during a 15-min time trial undertaken 2 hours later when they consumed 100 g CHO from the polymer than an equivalent amount of glucose (Stephens et al. 2008). Of course, the expense, specificity of use and lack of additional nutritional value of such a product would mean that it has only a small role to play in the everyday diet of most athletes.

In summary, when speedy restoration of glycogen is the goal, it seems prudent for low-GI foods to play a minor role in post-exercise meals. Typically, this does not need specific attention, since western eating patterns are generally based on high-GI CHO choices. However, the possible effect of the GI of recovery meals on metabolism during subsequent exercise requires further investigation (see Chapter 12) and may need to be taken into account.

14.3.4 Form of carbohydrate feeding

Both solid and liquid forms of CHO appear to be equally efficient in providing substrate for muscle glycogen synthesis (Keizer et al. 1986; Reed et al. 1989). Practical issues such as compactness and appetite appeal may be important in choosing CHO foods and fluids to meet the athlete’s total CHO intake goals. Liquid forms of CHO or CHO foods with a high fluid content may be particularly appealing when athletes are fatigued and appetite-suppressed.

14.3.5 Co-ingestion of other nutrients and total dietary energy intake

There is evidence that the relationship between CHO intake and glycogen storage is underpinned by consideration of total energy intake. In a study of CHO loading, female subjects showed a substantial enhancement of muscle glycogen storage associated with increased dietary CHO intake only after total energy intake was also increased (Tarnopolsky et al. 2001). The simplest way to consider this relationship is that dietary intake must provide for the body's immediate fuel requirements as well as storage opportunities. Greater proportions of available CHO substrates (such as dietary CHO) are likely to be oxidised to meet immediate energy needs during energy restriction, whereas CHO consumed during a period of energy balance or surplus may be available for storage within the muscle and liver.

It is possible that the co-ingestion of other macronutrients, either present in CHO-rich foods or consumed at the same meal, may directly influence muscle glycogen restoration as well as adding to energy intake. Although this hypothesis has not been systematically tested, factors that might directly or indirectly affect glycogen storage include the provision of gluconeogenic substrates, as well as effects on digestion, insulin secretion or the satiety of meals. The effects of protein have received most attention, but provided a source of debate for many years, with some studies reporting an increase in post-exercise glycogen storage when protein is added to a CHO feeding (Zawadzki et al. 1992; Van Loon et al. 2000; Ivy et al. 2002) and others finding no effect (Tarnopolsky et al. 1995; Roy & Tarnopolsky 1998; Carrithers et al. 2000; Van Hall et al. 2000). The overall effect was eventually explained via a meta-analytical approach to this literature. This work by Betts and Williams (2010) showed clear evidence that when CHO intake is below the targets for optimum glycogen storage during the first 4 hours of recovery (i.e. <1 g/kg/h), the co-ingestion of ~20–25 g protein can enhance glycogen storage; however, when CHO intake is adequate, the co-ingestion of protein has no further effect (see Figure 14.1). Of course, the intake of protein after exercise is important for other aspects of recovery via its stimulatory effects on protein synthesis (see Chapter 4).

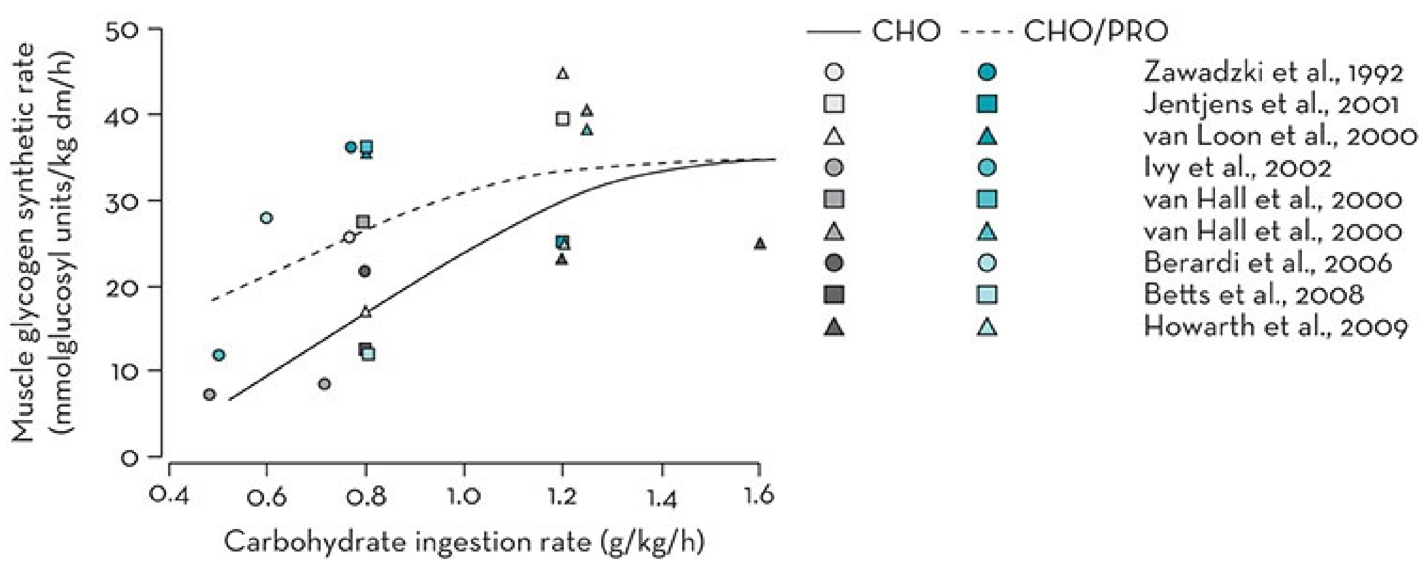

Figure 14.1 Reported rates of muscle glycogen resynthesis across nine studies that have compared muscle glycogen storage over >2–6 hours post-exercise with varied rates of carbohydrate (CHO) intake, with or without co-ingestion with protein (PRO). All studies have matched either for energy intake or CHO intake.

Source: Taken from Betts and Williams 2010, with permission

07:37:00.

14.3.6 Excessive intake of alcohol

Since there is evidence that some athletes, particularly in team sports, consume alcohol in large amounts in the post-exercise period (see section 14.6), it is important to consider the effect on recovery. With regard to muscle glycogen storage, rat studies have shown that intragastric administration of alcohol interferes with glycogen storage during 30 minutes of recovery from high-intensity exercise in oxidative but not non-oxidative fibres (Peters et al. 1996). Separate studies were undertaken in well-trained cyclists to monitor the effects of drinking a large amount of alcohol (~120 g or twelve standard drinks) after glycogen-depleting exercise on 8 hours and 24 hours of recovery (Burke et al. 2003). Muscle glycogen storage was impaired during both recovery periods when alcohol displaced an energy-matched amount of CHO from a standard recovery diet. This is a realistic comparison, since athletes typically forget about recovery programs and forgo well-planned meals during an alcohol binge and the pursuant hangover. Evidence for a direct effect of elevated blood alcohol concentrations on muscle glycogen synthesis was unclear, but it appeared that if an immediate impairment of glycogen synthesis existed, it might be compensated by adequate CHO intake and longer recovery time. It is likely that the most important effects of alcohol intake on glycogen resynthesis are indirect—by interfering with the athlete's ability, or interest, to achieve the recommended amounts of CHO required for optimal glycogen restoration. Athletes are therefore encouraged to follow the guidelines for sensible use of alcohol in sport (Burke & Maughan 2000) in conjunction with the well-supported recommendations for recovery eating.

14.3.7 Other promoters of glycogen storage

A common message from the literature on post-exercise refuelling/pre-exercise CHO loading is that the key dietary factor in glycogen storage is the amount of CHO consumed. Therefore, the logical approach to setting dietary guidelines around refuelling is to set a target for the CHO intake needed to store sufficient muscle glycogen for a given event or workout. While the theoretical limitations of this approach will be discussed later, most sports dietitians will recognise that there is a practical challenge to this approach: many athletes are unable to consume enough CHO to meet these targets. Sometimes this is due to their inability to eat sufficient food: for example when there is inadequate access to appropriate foods/drinks or poor appetite. However, in many situations of restricted energy budgets, athletes cannot meet the energy cost of these CHO targets while also meeting their needs for other macronutrients or micronutrient-rich foods. This is most often seen in female athletes as well as males in weight-dependent sports. Therefore, there is an interest in finding strategies that can promote greater glycogen storage from a given amount of CHO.

Both the 2003 and 2010 IOC consensus meetings on sports nutrition identified a need for studies to investigate strategies to increase glycogen storage from a suboptimal intake of CHO and/or energy (Burke et al. 2004, 2011). It was suggested under those circumstances that manipulations of the frequency and timing of eating or the type of CHO food and its accompaniments might increase the effectiveness of fuelling via alterations in blood glucose and insulin concentrations or other factors that affect glycogen storage. Very little research of this nature is available, apart from the now well-covered area of protein co-ingestion (Betts & Williams 2010). Some other opportunities exist, however, to enhance glycogen storage from a given amount of CHO; these include the use of high molecular weight glucose polymers (see 14.3.3), co-ingestion of large (9 mg/kg) amounts of caffeine (Pedersen et al. 2008), prior creatine loading (Robinson et al. 1999; van Loon et al. 2004), and the addition of fenugreek (Ruby et al. 2005). However, evidence for the benefits of some of these is equivocal, and the practical implications of these strategies may limit their use. For example, the side effects associated with supplementation of large doses of caffeine (e.g. interference with sleep) or creatine (e.g. weight gain) may prevent these combinations from being routinely used. In the case of the extract isolated from fenugreek (*Trigonella foenum-graecum*), an early study reported enhanced glycogen synthesis during 4 hours of recovery when it was added to a high dose of dextrose (Ruby et al. 2005). However, a follow-up investigation from the same group failed to find any refuelling advantages at 4 hours or 15 hours of post-exercise recovery when this product was consumed in combination with CHO intake (Slivka et al. 2008).

14.3.8 Effects of other recovery strategies on glycogen storage

Since athletes frequently undertake specialised activities after competition or key training sessions to promote various aspects of recovery, it is of interest to consider how these might interact with glycogen storage goals. For example, therapies that alter local muscle temperature to alleviate symptoms of exercise-induced muscle damage appear to have some effect on factors that are important in muscle glycogen synthesis, although the net effect is unclear. In one study, intermittent application of ice reduced net glycogen storage over 4 hours of recovery compared to a control leg (Tucker et al. 2012). In a companion study by the same researchers, the application of heat directly to the leg muscle was associated with greater refuelling (Slivka et al. 2012). Alterations in blood flow to the muscle secondary to temperature changes were presumed to play a role in these findings, although a reduction in muscle enzyme activities was also suspected to be a factor in explaining the outcomes of ice therapy. However, another study of cold water immersion following exercise failed to find evidence of impaired glycogen storage during the recovery period (Gregson et al. 2013). Potential interference or potentiation of glycogen storage by other activities undertaken in the post-exercise period should be considered: this may lead to a prioritisation of recovery goals so that better choices or combinations of activities can be made.

14.4 Guidelines for carbohydrate intake for training and recovery

The role of sports nutrition guidelines is to provide athletes with some general principles and ‘ball park’ targets for nutrient intakes or dietary practices that will support their health and performance goals. As new concepts and the evidence base to support them develop, these recommendations should be expected to evolve. Guidelines for CHO intake in the athlete’s everyday training diet, using the outputs of expert panels assembled by the IOC, provide an example of such evolution. The first IOC Consensus on Nutrition for Athletes in 1991 focused on the role of dietary CHO in restoring muscle glycogen levels, identifying targets to maximise daily glycogen storage on the basis that endurance athletes might have such goals. Targets were suggested for the early phase of recovery after exercise and for the total day—i.e. ‘Carbohydrate intake after exhaustive exercise should average 50 g per 2 hours of mostly moderate and high glycaemic CHO foods. The aim should be to ingest a total of about 600 g in 24 hours’ (Coyle 1991). Furthermore, it was suggested that ‘in the optimum diet for most sports, CHO is likely to contribute about 60–70% of total energy intake’. Hindsight and further research provided an opportunity to refine these concepts at the 2003 IOC consensus meeting. The following issues were identified (Burke et al. 2004):

- The communication of CHO intake targets in terms of an ideal percentage of energy intake is difficult for athletes and coaches to translate into food choices. Furthermore, it is conceptually flawed since the correlation between a targeted amount of dietary CHO and the percentage of total energy intake needed to achieve it is distorted by total energy intake (Burke et al. 2001); an athlete with a high energy intake might only need to devote 50% of energy intake to achieve a certain CHO target while an athlete with energy restrictions may not be able to achieve this target even if 80% of their reduced energy budget is consumed as CHO. Therefore, a specific recommendation was made to avoid the use of the percent energy terminology in setting dietary guidelines.
- Evidence that a chronic approach to undertaking each training session with well-fuelled glycogen stores leads to superior performance outcomes should be examined before we can confidently promote a focus on CHO intake in the athlete’s everyday diet. The available literature was examined to reveal only eight training studies in which diets providing high versus moderate CHO intakes were compared, with the duration of the investigation ranging from 5 days to 5 weeks (Burke et al. 2004). It was noted that only half of these studies found evidence of better performance or performance gains at the end of the training period, and that other researchers had claimed that this weakened the claim for benefits of high-CHO diets. However, the review noted the limitations of the methodologies of this literature including inadequate duration, difficulty in detecting small but worthwhile changes in performance, and the possibility that even the ‘moderate’ CHO intakes in some studies already achieved the fuel requirements of the training program, thus removing the need/benefit of a higher CHO intake. In responding to challenges that higher CHO intakes are not beneficial for sports performance (Noakes 1997), the

consensus group noted that none of the studies had found a benefit of moderate CHO intakes over the higher CHO diets. Furthermore, the group pointed out the conditions under which higher CHO intakes appear to be beneficial; in supporting training programs involving high volume and high-intensity training (Simonsen et al. 1991; Achten et al. 2004). Nevertheless, further research was encouraged and the potential for adaptations to periods of training with lower CHO intakes was acknowledged (Burke et al. 2004).

- Although the goal of promoting optimal training via daily restoration of glycogen stores was supported, it was recognised that many athletes do not undertake training sessions that completely deplete muscle fuel stores each day. Therefore, recommendations for CHO intake need to be scaled up and down according to the expected substrate needs of their training. The new guidelines offered a range of exercise bands with a sliding scale of suggested intakes of CHO, noting that this represented a starting point from which the athlete might fine-tune their eating practices based on what is achievable and beneficial according to trial and error.
- It was further recognised that the CHO intake needed to promote glycogen storage depends on the size of the muscle mass, and that the athletic population ranges from each end of the body size spectrum (e.g. from light/lean gymnasts and distance runners through to muscular rowers and football players). Therefore, guidelines for CHO intake should be scaled to the athlete's BM as the most commonly available proxy for muscle mass.

Full details of changes in the concepts and communication of guidelines for CHO intake to promote recovery and everyday training support can be found in the review by Burke and colleagues (2004).

The aggregation of activities of research and sports nutrition practice over the intervening period promoted further evolution of the guidelines at the third IOC consensus meeting in 2010. Changes in both the content and the communication of the guidelines were underpinned by the following issues:

- There is a need to integrate community health guidelines and popular dietary philosophies which recognise that unnecessarily high intakes of CHO, particularly from added sugars, can contribute to excessive energy intakes, obesity and other health concerns within all populations. This integration is needed both to continue to provide evidence-based guidelines to athletes and to reduce the perception that there is a 'disconnect' between sports nutrition and community health issues.
- In acknowledging a more moderate approach to CHO intake in general and promoting evidence-based scenarios in which higher CHO intakes are of benefit to sports performance, the guidelines should be re-written to highlight that dietary strategies to promote adequate muscle glycogen stores should be targeted to training sessions and competitive events where the goal is to perform optimally and the exercise demand includes high-intensity efforts, or skill and technique (see Table 14.2). The guidelines should also acknowledge that on other occasions, it may be less important to train with high glycogen stores, and that in some scenarios the athlete may deliberately train with

low glycogen stores to promote specific muscle adaptation (see section 14.5)

- Although the concept introduced in the 2004 guidelines of a sliding scale of CHO intake according to the anticipated glycogen requirements of a training/competition session was supported, the suggested CHO targets needed to be re-examined. The lack of actual measurements of the substrate requirements of the real-life workouts and sporting events undertaken by competitive athletes was acknowledged, with encouragement to sports scientists to collect or publish such data. It was noted that a pragmatic approach needed to be taken in light of observations of practitioners that many athletes were unable to consume CHO intakes within the previously set target ranges, particularly in cases of the restricted energy intakes needed to reduce body fat or maintain physique goals. Therefore, a widening of the range was suggested, with the acknowledgement that athletes with lower energy budgets may find their CHO intakes below these targets (Table 14.2).
- Several further changes to the terminology and application of the guidelines were made. It was suggested that just as 'energy availability' was coined to define an athlete's energy intake in relation to the energy costs of their specific exercise program, 'CHO availability' is a preferable way to discuss CHO intake. An athlete's CHO status should be considered in terms of whether their total daily intake and the timing of its consumption in relation to exercise maintains an adequate supply of CHO substrate for the muscle and central nervous system ('high CHO availability') or whether CHO fuel sources are depleted or limiting for the daily exercise program ('low CHO availability'). An outcome of this change in terminology would be that any intake of CHO could not be considered high or low on its own, and that the same CHO intake could provide high CHO availability for one athlete or one training day but represent low CHO availability in a different athlete or situation.
- Finally, it was recognised that an athlete's daily CHO targets should be dynamic rather than static. Each training session or competitive event has different substrate requirements (different costs on muscle glycogen and blood glucose utilisation) and different goals (determining whether it is important or indifferent to undertake it with high CHO availability, or even whether there may be merit in deliberate exposure to low CHO availability). Therefore, an athlete's CHO intake should be periodised over the weekly microcycle of training, the longer training mesocycle and annual macrocycle. A valuable tactic that could be used by athletes to manipulate a variable CHO intake is to arrange CHO intake around training sessions (pre-training snack, intake during longer sessions and post-training snack). As well as increasing CHO intake according to an increase in energy expenditure, this strategy can also specifically promote CHO availability around sessions where it is desired.

TABLE 14.2 2010 IOC guidelines for carbohydrate intake in the training diet

Situation	Carbohydrate (CHO) targets	Comments on type and timing of CHO intake
Total daily needs for fuel and recovery		

• When it is important to train hard or with high intensity, daily CHO intakes should match the fuel needs of training and glycogen restoration • These general recommendations should be fine-tuned with individual consideration of total energy needs, specific training needs and feedback from training performance • Since training commitments change due to periodisation in sport, an athlete's daily CHO intake should also be varied			
Light	• Low-intensity or skill-based activities	3–5 g/kg of athlete's body mass (BM)/d	• Intake may be timed to provide fuel intake around training sessions in the day (see below). Otherwise, as long as total fuel needs are provided, the pattern of meals/snacks may simply be guided by convenience and individual choice • CHO foods or meal combinations that are protein- and nutrient-rich will allow the athlete to meet other acute or chronic sports nutrition goals
Moderate	• Moderate exercise program (i.e. ~1 h per day)	5–7 g/kg/d	
High	• Endurance program (e.g. 1–3 h/d moderate–high-intensity exercise)	6–10 g/kg/d	
Very high	• Extreme commitment (i.e. >4–5 h/d moderate–high-intensity exercise)	8–12 g/kg/d	
Special timing of intake to support key training sessions • Consuming CHO around key training sessions to include pre-workout intake, and refuelling during and after the session, is likely to enhance performance and recovery • A focus on eating around training sessions helps the athlete to automatically match their energy and CHO intake with changing needs			
Pre-exercise fuelling	1–4 g/kg consumed 1–4 h pre-session		• Timing, amount and type of CHO foods and drinks should be chosen to suit individual preferences/experiences. • It may be useful to experiment with intended competition practices in some training sessions to fine-tune plans
Refuelling during training	45–75 min duration	Small amounts throughout training (including 'mouth rinsing')	• Even when supplementary muscle fuel is not needed, the brain responds to mouth contact with CHO; the athlete may feel better and train harder. This may be useful if the athlete is 'training low' with low muscle CHO stores
	1–2.5 h	30–60 g/h	• Practising with intended competition intake strategies will allow the plan to be fine-tuned and for the athlete to adapt to it
	2.5–3 h (when simulating optimal	Up to 90 g/h	• As above • Products providing multiple transportable CHOs

	competition practices and/or to support high daily energy needs)		(glucose:fructose mixtures) will achieve high rates of oxidation of CHO consumed during exercise
Post-exercise (especially when there is <8 h recovery between 2 fuel-demanding sessions)	1–1.2 g/kg/h in first hour		• Nutrient-dense forms of CHO (i.e. CHO-rich foods and food combinations that also provide protein and micronutrients) can promote other goals of recovery as well as nutritional goals for overall health and wellbeing

Source: Burke et al. 2011

These current guidelines for CHO intake in the training diet will continue to evolve, particularly around the new area of interest: deliberate training with low CHO availability.

14.5 Training with low CHO availability

New techniques that can examine molecular signalling within the muscle cell have allowed scientists to identify a range of pathways that underpin the various adaptations to training; this knowledge has led to new ideas about strategies to promote these adaptations. It is now recognised that muscle glycogen not only provides a fuel for exercise, but is also a point of regulation of the cellular signalling events that occur in response to endurance exercise (Philp et al. 2012).

Specifically, when endurance exercise is undertaken with low glycogen stores, there is an up-regulation of both the nuclear and mitochondrial responses to this exercise stimulus. It is believed that this occurs via enhanced activation of key cell signalling kinases (e.g. AMPK, p38MAPK), transcription factors (e.g. p53, PPARδ) and transcriptional co-activators (e.g. PGC-1α), leading to greater increases in enzymes and other proteins involved in functional changes that underpin the capacity for submaximal exercise (e.g. increased skeletal muscle fat oxidation). The amount or localisation of glycogen particles in the muscle can directly or indirectly result in a differential response to exercise, with mechanisms including changes in the activity of signalling proteins with a glycogen binding domain, changes in catecholamines and plasma free fatty acids, and changes in osmotic stress. Readers are directed to the excellent reviews by Philp et al. (2012) and Bartlett et al. (2015) for greater understanding of this new theme in exercise science. Although our understanding of the role of glycogen in response to training has expanded exponentially over the past decade, there are still many questions remaining as to how best to make use of this information.

14.5.1 ‘Train low’: exercising with low CHO availability

Exercising with low muscle glycogen stores amplifies the activation of signalling proteins that promote mitochondrial biogenesis and other training adaptations (Bartlett et al. 2015).

Exercising in a fasted state also promotes different cellular signalling responses than those associated with the consumption of CHO prior to and/or during the session (Civitarese et al. 2005), although this effect appears to be less robust than the differences seen by manipulating muscle glycogen content. These findings have led to the new concept of ‘train low, compete high’: that endurance athletes should train with low glycogen/low CHO availability to enhance the training response, but exercise with high CHO availability when performance is important, such as during competitive events. It is important to recognise that there are a number of potential ways to reduce CHO availability around training sessions, and these do not always require a low-CHO diet per se nor restrict CHO availability for all training sessions (see Table 14.3).

TABLE 14.3 Strategies to reduce carbohydrate availability to alter the molecular responses to endurance-based training sessions

Exercise–diet strategy	Main outcomes
Chronically low-CHO diet (CHO intake less than fuel requirements for training)	Reduction in muscle carbohydrate availability (endogenous and potentially exogenous sources) for all training sessions, depending on degree of fuel mismatch Chronic low CHO availability: whole-body effects including immune system and central nervous system
Twice-a-day training (low endogenous CHO availability for the second session in a day achieved by limiting the duration and CHO intake in recovery period after the first session)	Reduction in endogenous and exogenous CHO availability for the muscle during the second training session Acute reduction in CHO availability for immune and central nervous systems depending on duration of CHO restriction and muscle fuel requirements of second session
Training after an overnight fast	Reduction in exogenous CHO availability for the muscle for the specific session Potential reduction in endogenous CHO availability if there is inadequate glycogen restoration from previous day’s training Acute reduction in CHO availability for immune and central nervous systems depending on duration of CHO restriction and fuel requirements of the session
Prolonged training with or without an overnight fast and/or withholding CHO intake during the session	Reduction in exogenous CHO sources for the muscle for the specific session Acute reduction in CHO availability for immune and central nervous systems depending on duration of CHO restriction and fuel requirements of the session
Withholding CHO during the first hours of recovery	Could prolong the up-regulation of signalling activities to enable an enhanced period of adaptation to a session. Note that it will not interfere with training capacity of the first exercise session but will restrict fuel availability for the next session

Source: Adapted from Burke 2010

Note: Permutations and combinations of these strategies could alter exogenous and endogenous CHO supplies independently or interactively.

Our current interest in the potential benefits of the ‘train low’ approach emanates from a clever study by Danish researchers (Hansen et al. 2005) in which untrained males undertook a 10-week program of ‘kicking’ exercise while consuming a CHO-rich diet (~8 g/kg/d). Each leg did the same training program (5 × 1-hour sessions per week), but followed a different timetable. One leg was trained daily (refuelling between sessions) while the other leg did two sessions back-to-back every second day, with a rest day in-between (the second session on a training day was commenced with low glycogen stores). Compared to the leg that trained with normal glycogen reserves, the leg that commenced half of its training sessions with low muscle glycogen levels had a more pronounced increase in resting glycogen content and citrate synthase activity. Although the increase in maximal power was similar in each leg, the ‘train low’ leg had an almost twofold greater increase in exercise capacity (time to fatigue while kicking at 80% of peak power) compared to other leg.

As has been pointed out by several researchers (Burke 2010), there are difficulties in applying the results of this study to the real-life practices of athletes, especially elite competitors. Problems in translation include potential differences between untrained and well-trained individuals, and the application of one-legged kicking exercise to whole-body sporting activities. Importantly, the training sessions in the study were ‘clamped’ at a fixed submaximal intensity for the duration of the training program. This contrasts with the principles of modern training, which include the periodisation of different types of exercise into the program, underpinned by a philosophy of progressive overload.

Our group (Yeo et al. 2008) tried to address these issues in a more recent study. Two groups of well-trained cyclists consumed a CHO-rich diet (~8 g/kg/d) over 3 weeks of training: with each week consisting of three steady-state sessions (100 minutes at 70% $VO_{2\ peak}$) and three sessions of high-intensity intervals (8 × 5 minutes at maximal sustained power, with 1 minute’s recovery). One group alternated between these sessions (high group), while another seven subjects trained every second day, with the steady state session followed an hour later by the interval session (low group). Training intensity was measured as the self-selected power outputs achieved in the interval session, while performance was measured before and after the training block via a 1-hour time trial completed after an hour of steady-state cycling. We found that resting muscle glycogen concentrations, rates of whole-body fat oxidation during steady-state cycling and muscle activities of enzymes citrate synthase and β-hydroxyacyl-CoA-dehydrogenase (HAD) were increased only in the group that undertook the interval sessions with low glycogen. However, the intensity of the self-selected interval sessions and the total work completed in these workouts was reduced in the low group compared with the high training group. Overall, the 3 weeks of training produced a similar improvement in 1-hour cycling performance (~10%) in both groups (Yeo et al. 2008). A companion study conducted in another laboratory with a similar protocol produced almost identical results (Hulston et al. 2010).

Morton and colleagues (2009) studied three groups of recreationally active men who undertook four sessions of fixed-intensity ‘interval’ running over a 6-week period with either ‘high’ CHO availability (single day training), ‘train low’ (two training sessions, twice a week) or ‘train low plus glucose’ (as before, but with glucose intake before and during the second session). All groups recorded a similar improvement in VO_{2max} (~10%) and distance run during an intermittent running test (~18%), although the group who trained with low availability of exogenous and endogenous CHO sources showed greater metabolic advantages, such as increased activity of the enzyme succinate dehydrogenase. Observations of enhanced metabolic adaptations in the muscle, but similar gains in exercise capacity, have also been noted in studies comparing training in a fasted state versus training with a CHO drink in untrained or moderately trained groups (De Bock et al. 2008; Akerstrom et al. 2009).

The hypothesised cellular benefits of ‘train low’ strategies—enhanced metabolic adaptations to a given training stimulus—have been clearly demonstrated and lead to an increased ability to utilise fat as an exercise fuel and a reduced reliance on CHO. However, the available literature in well-trained athletes has failed to find evidence that this translates into a performance advantage. There are several explanations for this apparent disconnect between the enhanced cellular adaptation and exercise capacity; these include the brevity of the study period, the possibility that performance is not reliant or quantitatively linked to the markers that have been measured, our failure to measure other counterproductive outcomes and our focus on the muscular contribution to performance while ignoring the brain and central nervous system. And, of course, we are ultimately limited by our ability to measure performance in a reliable and valid way. Each of these factors probably contributes to the lack of evidence of performance advantages.

The most important issue to consider, however, is that the philosophy of the currently available studies (and the popular culture among athletes) is to expose 50–100% of training sessions to the ‘train low’ practice, including sessions that include high-intensity pieces or high-quality performances. As a result, there is the need to consider the ‘downside’ of training with low CHO, either chronically or for key training sessions (Burke 2009). There is plentiful evidence that ‘training low’ reduces the ability to train—increasing the perception of effort and reducing power outputs (Yeo et al. 2009; Hulston et al. 2010). This impairment of training outputs can be partially but not completely reversed by strategies such as the use of caffeine (Lane et al. 2013).

Nevertheless, most athletes and coaches fiercely guard the ability to generate high power outputs and work rates in training as an important component of the preparation for competition. Other potential downsides include the observation from our extensive work on exposure to high-fat diets (see Chapter 15) which induce metabolic adaptations that up-regulate fat utilisation and down-regulate CHO utilisation, leading to a reduction in ability to perform high-intensity exercise (Havemann et al. 2006). Finally, the effect of repeated training with low CHO status on the potential risk of illness, injury and overtraining need to be considered (see Chapter 22). In summary, there is evidence that ‘train low’ strategies need to be implemented judiciously if benefits are to be derived.

14.5.2 ‘Sleep low’: delaying the post-exercise restoration of muscle glycogen

Another manipulation of CHO availability in relation to exercise involves the deliberate withholding of CHO intake during the immediate recovery period in order to delay the restoration of muscle glycogen and prolong the period of elevated plasma free fatty acid concentrations. It has been proposed that maintaining this nutritional environment for some hours during the initial recovery period might delay the post-exercise return of cellular signalling pathways to baseline activity levels and thus achieve longer and greater transcription and translation of new proteins in response to the session (Hawley 2013). An advantage of this approach is that the exercise session might be undertaken with high CHO availability, thus limiting the interference with the quality of this session. Furthermore, it achieves an environment of low CHO availability for the subsequent training session. This strategy (dubbed 'sleep low' or 'recover low') is currently under investigation by our group as a potential enhancer of training outcomes when periodised into the training framework.

14.5.3 The periodised approach to CHO availability

Practitioners who work with elite athletes observe that just as they periodise their training programs, they probably already periodise the fuel availability for different sessions. Many athletes may undertake some of their workouts in a fuel-restricted manner, either by accident (doing morning sessions in a fasted state because it is impractical to refuel before or during the session) or design (restricting CHO in an energy-restricted diet). Equally, these athletes will ensure that other sessions are done with better fuel support. The view of many sports dietitians and researchers is that the optimal training outcome is achieved by cleverly mixing and matching nutritional strategies with each workout or session sequence to accentuate the specific goals of the session or training block. Thus, 'train low' activities could be periodised within the athlete's program to promote the outcome of sessions which target baseline conditioning effects (e.g. capacity to work at submaximal workloads). Other sessions that support training at higher-intensities/power outputs with the overlay of technical precision could be undertaken with strategies that promote high CHO availability. This approach could support the neural and biochemical adaptations needed to support good technique and fast movement patterns as well as help to develop tolerance to alterations in acid-base balance. Furthermore, training with the intake of CHO during the session can achieve gut adaptations to enhance the ability to absorb CHO and deliver it to the muscle as a substrate (Cox et al. 2010); this is good preparation for sports where the additional supply of CHO during the event provides a performance advantage. While we await research to validate the benefits of this more individualised and complex periodisation of training nutrition, there are documented case histories to support the theory and practice of this approach (Stellingwerff 2012, 2013).

14.6 Issues in post-exercise rehydration

As discussed in Chapter 13, most athletes finish a session of exercise at least mildly dehydrated because their fluid intake is less than their rate of sweat loss. High fluid deficits will occur in hot environments, during high-intensity exercise of prolonged nature, and/or in sporting activities in which there are limitations in terms of access to fluid or opportunity to drink (Garth & Burke 2013). Furthermore, in weight-division sports, such as boxing, wrestling and lightweight rowing, the athlete may have reduced their BM acutely before the competition weigh-in by purposely undertaking dehydrating techniques (see Chapter 7). Ideally, the athlete should replace body fluid losses after one exercise session so that the next workout can be commenced in fluid balance. In normal healthy people, the daily replacement of fluid losses and maintenance of fluid balance are well regulated by thirst and urine losses. However, under conditions of stress (e.g. exercise, environmental heat and cold, and altitude) thirst may not be a sufficient stimulus for maintaining euhydration (Greenleaf 1992). Furthermore, there may be a considerable lag of 4–24 hours before body fluid levels are restored following moderate to severe hypohydration. Adolph and colleagues first described the failure to fully replace fluid losses as 'voluntary dehydration' and noted that it was exacerbated by factors that reduced the availability or palatability of fluids (Rothstein et al. 1947). However, this phenomenon has been more recently renamed 'involuntary dehydration' to recognise that the dehydrated individual has no volition to rehydrate even when fluids and the opportunity are available (Nadel et al. 1990). The factors affecting self-chosen drinking patterns are multifaceted, and include behavioural issues such as social customs of drinking, as well as a genetic predisposition to be a reluctant or heavy drinker (Greenleaf 1992).

An additional challenge to post-exercise rehydration is that the athlete may continue to lose fluid during this phase, partly due to continued sweat losses, but principally due to urination. The success of post-exercise rehydration ultimately depends on the balance between fluid intake and urine losses. Ideally, an athlete should aim to fully restore fluid losses between exercise sessions so that the new event or workout can be commenced in a euhydrated state. This is difficult in situations where moderate to high levels of hypohydration have been incurred (deficits of 2–5% BM or greater) and the interval between sessions is less than 6–8 hours. Optimal rehydration requires a scheduled plan of fluid intake, to overcome physiological challenges such as inadequate thirst as well as practical problems such as poor access to fluids. A number of factors affecting post-exercise rehydration have been identified and will now be discussed.

14.6.1 Palatability of fluids

Numerous studies have reported that the palatability of fluids affects ad libitum intake, with quality, flavour and temperature being identified as important variables (for review, see Hubbard et al. 1990). Since many of these studies have investigated rehydration during exercise, it is uncertain whether the findings apply directly to post-exercise recovery (when the athlete is at rest). Perceptions may change with environmental conditions and with the degree of dehydration, and interestingly there is some evidence that perception of palatability or pleasure may not always correlate with total intake of a rehydration fluid. Hubbard and colleagues (1990) reviewed studies and concluded that while very cold water (0°C) may be regarded as the most pleasurable, cool water (15°C) may be consumed in larger quantities.

Flavouring of drinks has also been considered to contribute to voluntary fluid intake, with studies reporting greater fluid intake during post-exercise recovery with sweetened drinks than with plain water. For example, Carter and Gisolfi (1989) investigated fluid intake during the recovery after subjects had undertaken prolonged cycling to produce a fluid deficit of 2% BM. They found that subjects consumed significantly greater amounts of fluid when presented with a glucose-electrolyte drink than when plain water was provided. Water intake resulted in replacement of 63% of sweat losses, while the sweetened drink resulted in replacement of 79% ($p < 0.05$) of fluid losses. In both cases, ad libitum intake of fluid failed to meet total fluid losses, and the rate of intake decreased with time despite the continued fluid deficit. Whether subjects are responding to a sweet flavour or to energy replacement has not been systematically studied. There is some evidence that extreme sweetness and high CHO concentrations reduce voluntary intake, and that initial preferences for CHO-containing beverages may decrease after several hours (see Hubbard et al. 1990).

The addition of sodium to a beverage may also increase voluntary fluid intake, although there may be a concentration above which fluid intake is discouraged. Wemple and colleagues (1997) compared the volume of fluid consumed after dehydrating exercise when a flavoured water beverage was provided with 0, 25 or 50 mmol/L concentrations of sodium. Subjects replaced a mean level of 123%, 163% and 133% of the volume of their sweat losses respectively. Sodium concentration is not only important in influencing the volume of a fluid that is consumed voluntarily, but also the amount that is then retained. This is discussed in the section below.

14.6.2 Volume and pattern of drinking

As sweating and obligatory urine losses continue during the rehydration phase, fluids must be consumed in volumes greater than the post-exercise fluid deficit (exercise sweat loss) to restore fluid balance. A general finding of the previously reviewed studies is that replacement of fluid in volumes equal to sweat losses results in 50–70% rehydration over 2–4 hours of recovery (based on body weight restoration). Several studies have reported on the effect of the volume of rehydration fluids on restoration of body water deficit. Mitchell and colleagues (1994) dehydrated subjects by 2.5% BM by exercising them in a hot environment, then rehydrated them using a dilute electrolyte solution (15 mmol/L sodium) in volumes equivalent to 100% or 150% of their body weight deficit. Thirty per cent of the total volume was consumed as a priming dose, with the remainder of the fluid being consumed in five equal volumes at 30-minute intervals. Gastric emptying was measured, and fluid restoration was determined as body weight gain corrected for fluid remaining in the stomach. During the 3-hour rehydration period, the 150% rehydration protocol resulted in greater rates and amounts of fluid emptied from the stomach, and a greater net fluid restoration (68% restoration of BM losses versus 48% restoration). The rate of gastric emptying and regain of body weight decreased over time. Interestingly, there was no further restoration of fluid balance achieved through hours 2–3 of recovery with either protocol, due to an increase in urine production. Even the 150% rehydration protocol achieved restoration of only 68% of loss of BM, although this figure may be lower than results calculated from other studies due to the correction for gastric contents. Thus these authors conclude that even forced rehydration with a low-electrolyte beverage does not achieve restoration of fluid balance, and while ingestion of large volumes of fluid is more effective than an equal replacement of the lost fluid, the possible gastrointestinal fullness and discomfort that follow must be taken into account. This may be a practical consideration if a subsequent bout of exercise is scheduled within 2–4 hours post-recovery.

Similar conclusions were reported by Shirreffs and colleagues (1996) regarding the volume of fluid required to restore hydration after exercise. They studied subjects who exercised to dehydrate by 2% BM, then consumed volumes, equivalent to 50%, 100%, 150% and 200% of BM loss, of solutions containing sodium concentrations of 23 mmol/L or 61 mmol/L. Fluids were consumed in four equal volumes at 15-minute intervals, and fluid balance was monitored over 6 hours of recovery. They found that urine production was related to the volume of fluid consumed. In the case of the high-sodium fluids, consumption of 150% and 200% of fluid losses resulted in euhydration and hyperhydration, respectively, after 6 hours of recovery. All other trials resulted in residual hypohydration. These results show that a drink volume greater than the sweat loss during exercise must be ingested to restore fluid balance, but unless the sodium content of the beverage is sufficiently high this will result merely in an increased urinary output.

Whether the pattern of fluid intake influences rehydration has been investigated, comparing intakes of larger amounts of fluid in the immediate post-exercise period with the same total volume of fluid being spread out over 4–6 hours of recovery (Archer & Shirreffs 2001; Kovacs et al. 2002; Jones et al. 2010). In these studies, early replacement of large volumes of fluid was associated with better restoration of fluid balance during the first hours of recovery despite an increase in urinary output; however, differences in fluid restoration between hydration patterns disappeared over the course of the recovery period and the spaced fluid intake was eventually more effective in restoring fluid balance, via a reduction in urine losses. Of course, factors such as gastric comfort need to be considered when undertaking forced post-exercise rehydration practices, especially if the athlete needs to perform another exercise session within the next hours.

Finally, the consumption of a meal may be a useful adjunct to rehydration. Hubbard and colleagues (1990) note that food consumption may provide a social or psychological stimulus to increase voluntary fluid intake. Furthermore, the sodium content of the meal will enhance fluid retention (Maughan et al. 1996; Ray et al. 1998).

14.6.3 Replacement of electrolytes

Sodium is the principal electrolyte lost in sweat, and during prolonged bouts of heavy sweating, particularly in individuals with high sodium sweat content, a substantial loss of body sodium can occur in each session of exercise. For example, in several studies of team sport players undertaking training sessions in hot weather, individuals recorded losses in excess of 7 g of salt (sodium chloride) in a single session (Maughan et al. 2004; Shirreffs et al. 2005). Although western dietary patterns are generally considered to be excessive in salt intake, athletes who lose such large amounts of sodium during exercise may need to undertake special strategies to actively replace sodium losses during and after exercise (Bergeron 2003). Even if the loss of sodium during an exercise session is not substantial and will be replaced eventually by dietary means, there are good reasons for including sodium in recovery fluids and snacks.

When water is ingested following exercise-induced dehydration, there is a dilution of plasma osmolality and sodium content, which results in an increased diuresis and reduced thirst. Nose and colleagues (1988) compared rehydration with water plus sodium capsules with water plus a placebo capsule in subjects who had been dehydrated by approximately 2.5% BM. They found that intake of sodium (equivalent to a solution of approximately 80 mmol/L) achieved more rapid restoration of plasma volume than the water trial, due to a greater voluntary intake of fluid and lower urine output.

Maughan and Leiper (1995) dehydrated subjects by 2% BM via exercise in a hot environment, then observed them for 6 hours of recovery after they had consumed 150% of their fluid losses with test drinks providing varying levels of sodium. Fluid was consumed over a 30-minute period, beginning 30 minutes after the end of the exercise bout. After 90 minutes of recovery there was a significant treatment effect, with greater urine losses being observed with 2 mmol/L (no sodium) and 26 mmol/L sodium (low sodium) drinks than with the 52 and 100 mmol/L sodium solutions (see Figure 14.2). After 6 hours, the difference in mean urine output between the no-sodium and 100 mmol/L sodium drinks was in the order of 800 mL. Subjects were in fluid balance by the end of the recovery period when they consumed the two higher-sodium beverages, but were still in net negative fluid balance on the no- and low-sodium trials, despite the intake of a volume of fluid that was 1.5 times their estimated sweat losses. Retention of the ingested fluid was related to the sodium content, but there was no difference in net fluid balance between the 52 and 100 mmol/L sodium trials.

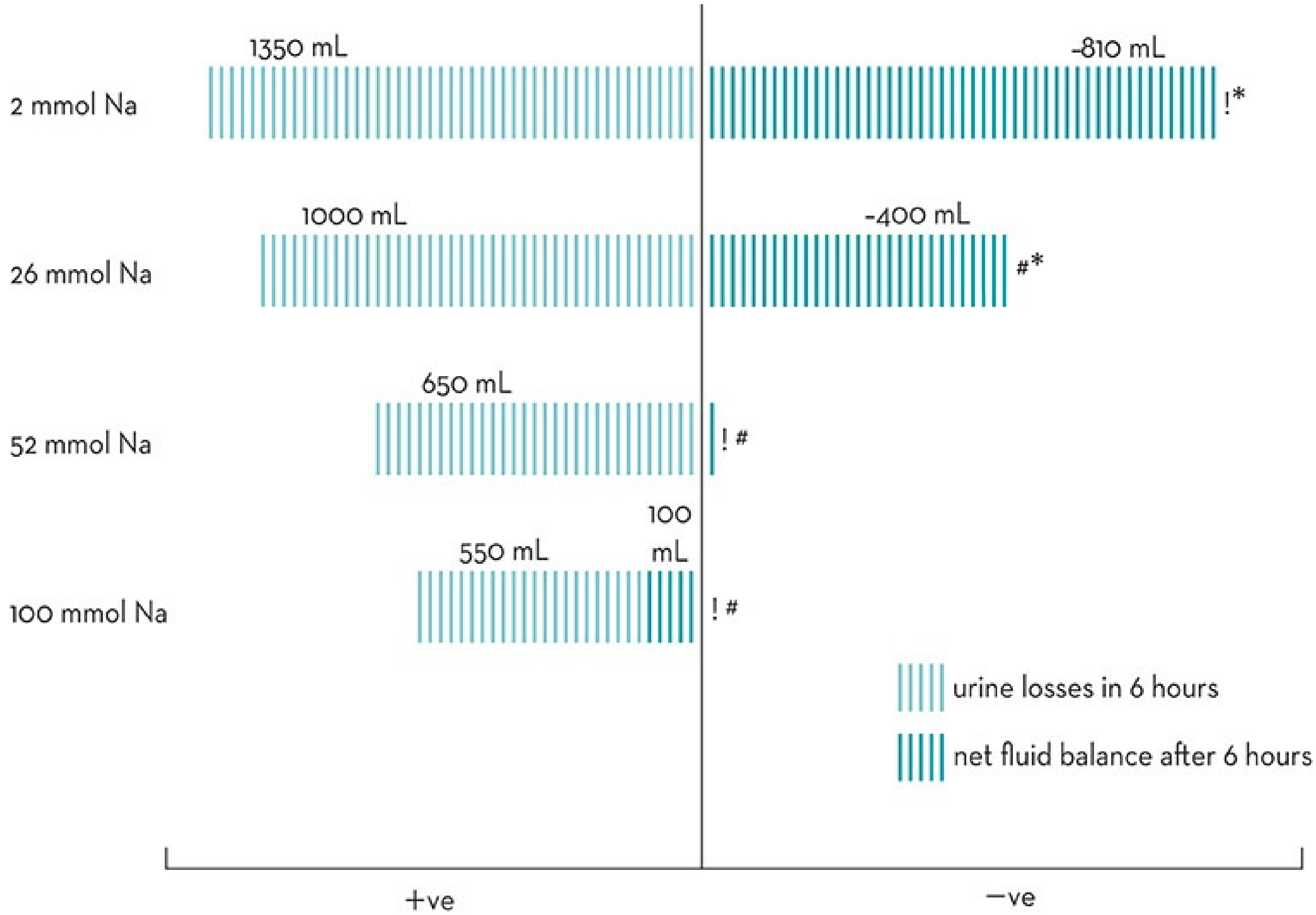

Figure 14.2 Effect of sodium content of fluid on urine losses and restoration of fluid balance following intake of volumes of fluid representing 150% of fluid deficit incurred by exercising in the heat. Fluids contain sodium content of 2 mmol/L, 26 mmol/L, 52 mmol/L and 100 mmol/L. * significantly different to pre-exercise, # significantly different to 2 mmol/L, ! significantly different to 26 mmol/L.

Source: Data redrawn from Maughan & Leiper 1995

There is some argument about the optimal sodium level for a post-exercise rehydration fluid. The World Health Organization recommends a sodium level of 90 mmol/L for the oral rehydration solutions used in the treatment of diarrhoea-induced dehydration (Walker-Smith 1992). However, this is based on the need to replace the sodium lost through diarrhoea as well as optimise intestinal absorption of fluid and retention of ingested fluid. Sodium losses in sweat vary markedly, with typical sweat sodium levels believed to be in the range of 20–80 mmol/L (Verde et al. 1982; Armstrong et al. 1987). Therefore, a post-exercise recovery drink with sodium levels of ~50 mmol/L may well be justified. Nevertheless, to be palatable and to have commercial appeal across a wide market, sports drinks have gravitated to a more moderate sodium content (10–25 mmol/L). When fluids are freely chosen, the interaction between the palatability of a drink (voluntary intake) and its sodium content (retention of fluid) are important. In one study, subjects chose to drink greater volumes (~2.5 L) when an orange juice/lemonade or sports drink was offered than when water or an oral rehydration solution were the recovery beverages (~1.7 L). However, urine losses were lowest with the oral rehydration solution (Maughan & Leiper 1993).

The effectiveness of such moderate sodium levels in restoring hydration appears to be slight. Gonzalez-Alonso and colleagues (1992) reported that a commercial sports drink (6% CHO, 20 mmol/L sodium) was more effective than plain water in promoting restoration of fluid levels after exercise-induced dehydration. Subjects were dehydrated by approximately 2.5% BM and were studied during 2 hours of rehydration while consuming a volume of fluid equal to this deficit, as two equal boluses consumed at 0 and 45 minutes of recovery. The sports drink trial achieved greater restoration of body weight (73% of the volume was retained) than the water trial (65% volume retained), principally due to decreased urine losses. Thus it appears from this and other studies that commercial sports drinks may confer some rehydration advantages over plain water, in terms of palatability as well as fluid retention. Nevertheless, where maximum fluid retention is desired, there may be benefits in increasing the sodium levels of rehydration fluids to levels above those provided in typical sports drinks (Maughan & Leiper 1995).

Alternatively, additional sodium may be ingested via sodium-containing foods or salt added to meals. Studies by Maughan and colleagues (1996) and Ray and colleagues (1998) have both shown that the intake of salt via everyday food choices enhances the retention of fluid consumed to rehydrate after exercise-induced dehydration. In the case of the study by Maughan and colleagues (1996), greater fluid retention (less urine production) was seen following the consumption of food plus fluid, than a drink only, potentially because of the greater electrolyte content. In this study, subjects were dehydrated by 2% BM by exercising in the heat. Over 60 minutes (beginning 30 minutes into the recovery period), they consumed fluid equal to 150% of sweat losses, either in the form of a sports drink (20 mmol/L sodium) or as a meal plus a low sodium drink (the water content of the meal was included to match the total fluid intake of the other trial). At the end of 6 hours of recovery, total urine production was lower following the meal plus drink and subjects were in a euhydrated state. The sports drink trial resulted in a net negative fluid balance of approximately 350 mL over the same period.

The addition of potassium (25 mmol/L) to a rehydration beverage is also effective in retaining fluids ingested during recovery from exercise-induced dehydration (Maughan et al. 1994). However, there is no additional benefit of a potassium-rich drink over a conventional sodium-rich sports drink (Pérez-Idárraga & Aragón-Vargas 2014). Furthermore, the replacement of sodium would appear to be a priority, since sweat losses of sodium can be significant. According to the review of Shirreffs and colleagues (2004) for the 2003 IOC Consensus on Nutrition for Athletes, there was no convincing justification for the addition of other electrolytes such as magnesium in recovery fluids. Although some magnesium is lost in sweat, the reduction in plasma magnesium concentration that accompanies exercise is likely to be due to a redistribution of compartmental fluids rather than large magnesium losses.

14.6.4 Characteristics of other rehydration fluids

In recent years, various common or commercially available beverages have been investigated for their efficacy in restoring rehydration in the hours after exercise. Motives for this interest include providing a new market/purpose for these drinks, helping athletes to take advantage of accessible fluids, or checking the potential of fluids consumed for other recovery benefits to also address rehydration goals. Milk has enjoyed such attention, probably for all three of these motives, and has been found to be better than water or conventional sports drinks (Shirreffs et al. 2007; Watson et al. 2008; Desbrow et al. 2014) in restoring fluid balance following exercise-induced dehydration. Soy milk also produces superior rehydration characteristics (Desbrow et al. 2014). The typical protocol for such studies has again involved the completion of an exercise task to achieve a standardised fluid deficit (~ 2% BM) followed by the intake of a prescribed volume of fluid (usually 100–150% of the fluid deficit) and the ongoing observation of urine characteristics, plasma osmolarity and BM changes over the next 3–6 hours. Although such studies have been able to identify net effects on fluid restoration, they are often unable to identify the mechanisms to explain the results. After all, although seemingly simple, rehydration may be affected by factors that alter rates of gastric emptying, intestinal uptake of fluid and fluid retention/urine production; these include a variety of features of the beverages including volume, energy density, osmolarity, sodium content, potassium content, CHO content, and the content and type of protein. In many of the comparative studies of rehydration beverages, several of these factors are altered at the same time.

Additional studies have been undertaken to manipulate several of these factors with the goal of either explaining the mechanism of effect on rehydration and/or developing a beverage with superior rehydration characteristics. In this regard, it seems that although the sodium content of milk contributes to its rehydration characteristics, the protein content is also important (Seifert et al. 2006; Shirreffs et al. 2007). Indeed, the addition of milk protein or whey protein to CHO-electrolyte drinks has been shown to enhance their rehydration characteristics (Seifert et al. 2006; James et al. 2011), independent of the energy density of the fluid or its sodium or potassium content (James et al. 2011). Both the CHO and protein content of fluids may increase the intestinal absorption of water via their sodium co-transported uptake; these effects are additive although of greater magnitude in the case of protein (Hobson & James 2015). Additional benefits to rehydration may be found via a slowing of gastric empting and a contribution to plasma osmolarity, both of which protect against the excessive production of urine that accompanies a sudden increase in plasma volume. There seems to be a ceiling for the beneficial effects of the protein content of a rehydration beverage within the same energy density: in one study, 40 g/L was not found to be better than 20 g/L (James et al. 2013). However, a liquid meal supplement with additional energy and protein was found to be superior to normal milk and soy milk in overall hydration characteristics due to the greater energy, protein and sodium content.

However, the effect of drink volume may also be important in determining the importance of protein content. Indeed when the rehydration beverage is consumed in volumes equivalent to 150% of the post-exercise fluid deficit (James et al. 2012, 2014, 2015), the addition of whey protein does not appear to enhance drink retention as seen when the volume of the rehydration beverage is equal to the fluid deficit (Seifert et al. 2006). Interestingly, the addition of milk protein seems to have a greater effect on the rehydration (i.e. fluid retention) characteristics of the beverage than does whey protein (James et al. 2011, 2013). Here it has been proposed that the inclusion of the casein sub-fraction reduces gastric emptying, thus slowing the rise in plasma volume and off-setting the decline in plasma osmolarity that accompanies the intake of large volumes of lower sodium fluids (James et al. 2012).

Coconut water, naturally rich in potassium, has also received attention as a rehydration beverage, and has been shown to be better than water (Pérez-Idárraga & Aragón-Vargas 2014) but equal to a sodium-based sports drink in terms of fluid retention (Saat et al. 2002; Kalman et al. 2012; Pérez-Idárraga & Aragón-Vargas 2014). Unfortunately, what is missing from many of the studies included in this section is consideration of the voluntary rather than planned intake of such beverages. Issues such as palatability, effect on thirst and gastric comfort need to be considered before we can be sure that athletes will consume beverages in sufficient volumes and thus achieve real-world efficacy on fluid restoration. Of course, the ability of the beverage to provide nutrients involved in other recovery processes is another practical consideration.

14.6.5 Caffeine and alcohol: potential diuretics

Diuresis may be stimulated by several factors commonly found in beverages consumed by dehydrated athletes. Gonzalez-Alonso and colleagues (1992) reported that consumption of a diet cola drink containing caffeine resulted in less effective regain of body fluid losses than their water or sports drink trials. Ingestion of diet cola in a volume equal to sweat losses resulted in restoration of 54% of BM losses; urine losses were significantly increased compared to the other trials and there was a small but significantly greater loss of fluid through continued sweating and respiratory losses (Gonzalez-Alonso et al. 1992). This finding fits with the common advice that caffeine-containing fluids are not ideal rehydration beverages and should be avoided in relation to exercise or other situations of dehydration (such as air travel). However, a large review of caffeine and hydration status found that there is a lack of rigorously collected data to show that caffeine intake impairs fluid status (Armstrong 2002). It concluded that the effect of caffeine on diuresis is overstated, and may be minimal in people who are habitual caffeine users. Indeed, a more recent study observed hydration status and urine losses in subjects who were first habituated to a daily caffeine intake of 3 mg/kg, then changed to 5 days of caffeine doses of 0, 3 or 6 mg/kg/d (Armstrong et al. 2005). There were no differences in BM, urine losses or serum osmolality as a result of the different intakes of caffeine, challenging the theory that caffeine consumption acts chronically as a diuretic. In addition, any small increase in fluid losses from caffeine-containing drinks may be more than offset by the increased voluntary intake of fluids that are enjoyed by the athlete, or part of social rituals and eating behaviours. If the athlete is suddenly asked to remove such beverages from their diet or post-exercise meals, they may not compensate by drinking an equal volume of other less familiar or well-liked fluids.

Alcohol has a diuretic effect, via mechanisms including the inhibition of anti-diuretic hormone; this is of particular consideration in the case of beverages with high alcohol content (i.e. large serve of alcohol in small volume of fluid) where the net effect is a body fluid deficit. Although other issues related to the consumption of alcohol will be further discussed below, it is noted that the capacity of people to enjoy beer in large volumes has been identified as a potentially useful factor in achieving rehydration goals. Of course, around the world, beers vary in their alcohol content with the typical alcohol content being from 4–6% (g/100 mL) and ranges varying from 0.5% (non-alcoholic beer) to 20%.

An early study reported that subjects consuming drinks containing 4% alcohol as a rehydration fluid reported greater urinary losses than when drinks containing 0, 1 or 2% alcohol were consumed (Shirreffs & Maughan 1995). Subjects exercised in the heat to dehydrate by 2% BM and rehydrated over a 60-minute period by consuming drinks equivalent to 150% of their fluid deficit of varying alcohol content. The total volume of urine produced over the 6-hour recovery period was related to the alcohol content of the fluid; however, only the 4% alcohol drink trial approached significance with a net retention of 40% of ingested fluid compared to 59% in the no-alcohol trial.

More recently, Flores-Salamanca and Aragón-Vargas (2014) conducted a similar study, but provided subjects with a volume equal to their post-exercise fluid deficit of either water, high-alcohol beer (4.6%) or non-alcoholic (0.5%) beer to drink during the first hour of recovery. Of the ~1.6 L of fluid consumed for rehydration, urine losses accounted for 1.2 L with the high-alcohol beer, compared with ~0.75 L with the other beverages. Although restoration of the fluid deficit did not occur with any of the beverages, subjects remained substantially dehydrated with the choice of the high-alcohol beer as a rehydration beverage (Flores-Salamanca & Aragón-Vargas 2014). It appears, however, that the diuretic effect of alcohol is dependent on the existing hydration status. Hobson and Maughan (2010) had subjects exercise to create a 2% BM fluid deficit then either rehydrate or remain dehydrated overnight before drinking 1 L of alcohol-free or 4% alcohol beer the following morning. They found that urine production over the next 4 hours was greater in the well-hydrated trials than the dehydration trials (~1.1L versus 200 ml), and that while urine volumes were similar with the alcoholic beer than non-alcoholic beer in the hypohydrated trial, an extra ~200 mL of urine was produced with the alcoholic beer when well-hydrated. They concluded that the diuretic effect of alcohol was blunted in cases where the subject was already in fluid deficit (Hobson & Maughan 2010).

Desbrow and colleagues (2013, 2014) have undertaken studies to systematically investigate whether the manipulation of the sodium and alcohol content of beer could produce a beverage with a compromise between high social acceptance and palatability and the negative effects on fluid retention. The same protocol was used to monitor post-exercise fluid balance following intake of different beer products in volumes equivalent to 150% of the fluid deficit over the first hour of recovery. In their first study, they compared low-alcohol beer (2.3%) with or without added sodium (25 mmol/L) and high-alcohol beer (4.8%) with or without sodium (Desbrow et al. 2013). After 4 hours of recovery, net fluid balance was significantly greater following the sodium-added light beer (−1.02 ± 0.35 kg) compared with high-alcohol beer, regardless of sodium content (~ −1.6 L), due to a significantly lower urine output (~ 1.5 L versus~2.2 L). In a second study, Desbrow and colleagues (2014) compared the effects of low-alcohol beer with 25 mmol/L or 50 mol/L sodium, and a mid-strength beer (3.5%) with or without 25 mmol/L sodium. Total urine output was significantly lower in the higher sodium low-alcohol beer (~1.4 L) compared to lower sodium light beer (~1.8 L), mid-alcohol + sodium (1.8 L) and mid-alcohol beer (2L). The net effect on restoration of BM was significant (+300–600 mL), although as in the previous study, BM remained lower than pre-exercise levels by ~1 kg or more. They concluded that electrolyte concentration of low-alcohol beer appears to have a more significant impact on post-exercise fluid retention than small changes in the alcohol content of beer. However, since subjects in all studies and trials were in negative fluid balance after 4 hours of recovery despite drinking large volumes of fluid, they concluded that beer is not a desirable rehydration beverage despite manipulations that can create small improvements in fluid balance (Desbrow et al. 2013).

14.6.6 Intravenous rehydration

In situations of severe dehydration, especially when fluid intake is compromised by gastrointestinal dysfunction and/or the collapse of the subject, rapid rehydration may be attempted via intravenous (IV) delivery of saline solutions. Before a recent change in anti-doping rules, however, it became fashionable for athletes to receive IV rehydration at the end of their event to aid recovery, or even during a game (i.e. at half time of a football match). This method is understandable for events such as the Tour de France and tennis tournaments, where moderate to severe fluid deficits are usual, the period between games or stages is brief, and the athlete has to juggle time to allow adequate sleep and recovery nutrition. IV saline given at half time has been associated with a moderate fluid deficit and the absence of gastrointestinal impediments to oral rehydration. Although intravenous rehydration is now banned in sports that are signatories to the World Anti-Doping Agency Code without the provision of a medical clearance, it is still permitted in other sports (e.g. American Football) that are covered by different rules. Many athletes and coaches believe that IV rehydration has advantages in enhancing recovery and the performance of subsequent exercise. This idea needs to be considered, and balanced against the expense and slight medical risk involved with IV procedures (especially when carried out in the field).

A series of studies compared the effects of oral and IV rehydration on restoration of fluid balance, and thermoregulation, metabolism and performance in subsequent exercise trials (Blom 1989; Castellani et al. 1997; Riebe et al. 1997; Casa et al. 2000a, 2000b; Maresh et al. 2001). Protocols involved exercise-induced dehydration, followed by no rehydration, or matched replacement of fluid via oral or IV delivery. After a period of recovery, subjects undertook a second bout of exercise in hot conditions. These studies showed that fluid replacement protocols achieved equal improvement in plasma volume and thermoregulation during subsequent exercise, compared with the no-rehydration trial (Castellani et al. 1997). However, oral rehydration was superior to IV rehydration in relation to reducing thirst and lowering the perception of effort in the second exercise trial (Riebe et al. 1997). In this study, exercise tolerance was better following oral rehydration compared with IV fluid replacement, apparently because of the reduced perception of workload. In the other series, oral and IV rehydration were equally effective in improving exercise performance compared with no rehydration, although it seemed that oral rehydration was associated with better return of physiological parameters (Casa et al. 2000a, 2000b; Maresh et al. 2001). Therefore, there is evidence that oral rehydration is at least as effective as IV therapy in treating moderate and uncomplicated situations of dehydration. The psychological sensation of drinking appears to provide an important component of recovery, enabling the athlete to feel better when tackling the next event or workout. On the other hand, IV hydration attenuates the loss of thirst that accompanies oral intake of fluid, even before fluid restoration is achieved.

14.7 Alcohol and recovery

Alcohol is strongly linked with modern sport. Although it is difficult to gain reliable data on the alcohol intake practices of people, there is reasonable evidence, including many testimonials and media evidence of binge-drinking behaviour by some athletes, particularly during the post-competition period (see Burke & Maughan 2000; O'Brien & Lyons 2000; Burke et al. 2003; O'Brien et al. 2005; Barnes 2014). This appears to be most prevalent in team sports where the culture may promote, or at least fail to discourage, post-game alcohol binges.

Heavy intake of alcohol may interfere with post-exercise recovery in a number of ways. The most important effects of alcohol are the impairment of judgement and reduced inhibition; heavy drinking has a major impact on the behaviour of athletes during the post-exercise recovery period, increasing high-risk behaviours that may lead to a poor image, accidents and injury and, sometimes, death. Alcohol consumption is highly correlated with accidents of drowning, spinal injury and other problems in recreational water activities (see O'Brien 1993; O'Brien & Lyons 2000) and is a major factor in road accidents. The intoxicated athlete is likely to be distracted from sound recovery strategies related to nutrition, injury treatment and sleep.

Previous sections discussed the evidence that alcohol may directly affect physiological processes such as rehydration (section 14.6.5) and glycogen storage (14.3.6). These sections concluded that alcohol intake is likely to interfere with optimal restoration of a fluid deficit, and although it may have a direct effect on glycogen storage, the larger effect on refuelling is likely to occur via the indirect outcome of inadequate CHO intake. Many sporting activities are associated with muscle damage and soft tissue injuries, as a direct consequence of the exercise, as a result of accidents, or due to the tackling and collisions involved in contact sports. Standard medical practice is to treat soft tissue injuries with vasoconstrictive techniques (such as rest, ice, compression and elevation). Since alcohol is a potent vasodilator of cutaneous blood vessels it has been suggested that the intake of large amounts of alcohol might cause or increase undesirable swelling around damaged sites, and might impede repair processes. Although this effect has not been systematically studied, there are case histories that report these findings.

As reviewed by Barnes (2014), the intake of large amounts of alcohol after exercise is likely to have a range of negative effects on anti-inflammatory responses, immune parameters, endocrine systems, vascular blood flow and the quantity and quality of sleep, all of which may impair recovery. In addition, we have undertaken a study of its effects on muscle protein synthesis: well-trained subjects undertook a laboratory protocol simulating the exercise requirements of a team sport before consuming 25 g whey protein, alcohol (1.5 g/kg or ~12 standard drinks) or alcohol and protein immediately after the workout and a CHO-rich meal 2 hours later (Parr et al. 2014). Rates of muscle protein synthesis increased above rest for all conditions, but compared to the trial in which only protein was consumed after exercise, there was a hierarchical and significant reduction with the alcohol + protein trial (24% reduction) and alcohol (37%). We concluded that alcohol consumption reduces the anabolic response and post-exercise muscle protein synthesis, even when co-ingested with protein and may therefore impair recovery and adaptation to training and/or subsequent performance.

The difficulties in undertaking research on alcohol intake by athletes that is translatable to real-life practices deserve comment. First, the ethical considerations are challenging and research boards usually place limits on the amount of alcohol that can be consumed and/or other activities and investigative procedures that can be undertaken. Of course, the amounts of alcohol consumed by subjects in these studies is substantial; indeed, in most studies of post-exercise alcohol consumption (Burke et al. 2003; Desbrow et al. 2013; Parr et al. 2014), blood alcohol concentrations reach levels that well exceed the limits for driving a motor vehicle. However, these intakes are usually well below the self-reported intakes of the binge practices of many athletes (Burke 1988; O'Brien et al. 1993; Prentice et al. 2014, 2015). In addition, the large variability in the responses to alcohol with many factors reduces the likelihood of detecting statistically significant effects in studies with small subject numbers, and the challenge of disguising the taste and effects of alcohol make it difficult to achieve a double-blinded application of treatments. Finally, most studies fail to include the usual real-life accompaniments to binge drinking such as the loss of sleep and failure to undertake recommended dietary and recovery practices. In this regard, some studies of alcohol use by athletes have introduced protocols in which they divide their participants into two groups: one that refrains from alcohol intake and the other that follows their normal post-game behaviours around alcohol use (O'Brien et al. 1994; Prentice 2014, 2015). Even then, these studies have failed to find consistent evidence of reductions in performance the day after a binge-drinking episode, with outcomes variously including a reduction in jump ability (Prentice et al. 2015) and aerobic performance (O'Brien et al. 1993) and no changes in jump ability, lower body strength or repeated sprint ability (O'Brien et al. 1993; Prentice 2014, 2015). Nevertheless, for many reasons, it is likely that acute and chronic episodes of binge drinking by athletes interfere with health and performance.

The alcohol intakes and drinking patterns of athletes merit further study and a well-considered plan of education, particularly to target binge-drinking practices that are often associated with post-competition socialising. In addition to being targeted for education about sensible drinking practices, athletes might be used as spokespeople for community education messages. Athletes are admired in the community and may be effective educators in this area.

Summary

Recovery after exercise poses various challenges to the modern athlete, with the type and severity of the recovery goals varying according to the specific session that was undertaken, the athlete's goals and other priorities in their nutrition plan. Therefore, unlike the scenario that has developed where some athletes aggressively consume foods and fluids after each training session or event, there are benefits in undertaking a more specific assessment of the real benefits or requirements for recovery nutrition practices, and tailoring a more specific plan. The unnecessary intake of food/drinks in the name of recovery can lead to unnecessary expense and/or unnecessary intake of energy in the face of a restricted energy budget.

Recovery after strenuous exercise involves an array of processes that are dependent on the provision of nutrients. The quantity of nutrients is important, but the timing of these is also important. In some cases, the stimulus for recovery is strongest in the period immediately after exercise and a lack of nutritional support at this time will reduce the total response to exercise. In other cases, there is no effective recovery until the nutrients are supplied. When the rate at which the recovery process is limited (e.g. glycogen synthesis) and there is a finite number of hours exercise between sessions, the delay in supplying nutrients may cause a substantial reduction in the return to optimal function for the subsequent session. Guidelines for post-exercise practices that optimise refuelling, rehydration and protein synthesis can be given, but must be customised to each session and to the goals of the individual athlete. Future research may identify similar guidelines for other processes involved in recovery, such as antioxidant, inflammatory and immune responses. Since athletes often compete in a foreign environment, the practical issues of food availability and food preparation facilities must be considered when making recommendations for post-exercise nutrition. The fascinating idea that adaptation to an exercise stimulus and restoration of optimal function are overlapping but different process may also lead to a different set of nutrition practices to promote a 'smarter' way to train. The concept that recovery processes might be deliberately withheld to create a scenario in which the next session involves 'train low' practices will continue to develop and may provide an opportunity to enhance adaptations to training.

Practice tips

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- Different types of exercise upset body homeostasis and/or provide a stimulus to which the body can adapt. The intake of nutrients may be needed to restore homeostasis and performance, or to optimise the adaptation processes. Recovery may need to integrate the processes of restoration of muscle and liver fuel stores, restoration of a fluid deficit, and the synthesis of new proteins in the muscle. Tables 14.4–Tables 14.6 provide an overview of the practical strategies that will promote each of these processes, and provide examples of situations in which proactive recovery might be valuable as well as others in which it might actually incur some disadvantages.

TABLE 14.4 Summary of guidelines for post-exercise refuelling

Strategies to maximise refuelling	• Start consuming CHO soon after the session finishes. Aim for a recovery snack/meal providing CHO equal to ~1 g per kg body weight (e.g. ~50 g for 50 kg female, 80 g for 80 kg male) • Continue with more snacks, drinks or meals to achieve a CHO target of 1 g/kg per hour for the first 4 hours of recovery, then resume an eating pattern that meets overall fuel and energy goals. Total CHO requirements can range from 3–12 g/kg BM per day (see Table 14.2)
Suitable choices for pro-active refuelling	Carbohydrate-rich selections include: • breads (including rolls, wraps, bagels, muffins and pizza bases) • cereals (porridge, packaged cereals and bircher muesli) • grains (rice, pasta, noodles, quinoa and couscous) • sweetened dairy (flavoured milk, flavoured yoghurt, custard) • fruits, starchy vegetables (potatoes) and legumes (baked beans, kidney beans and lentils) Mix and match these items into meals such as: • cereal with fruit • baked beans on toast • sandwiches/rolls/focaccia/wraps/pizza with thick portions of the bread • fruit smoothies (blended fruit, milk and yoghurt or ice-cream) • meals served with grains (curry and rice, pasta with sauce, stir fry with noodles) or where the grain is a major ingredient (pasta bake, risotto, paella) • desserts (cake/pudding with custard, fruit with yoghurt, crumbles with ice-cream Compact forms of CHO (useful when appetites are low, the gut is full/uncomfortable, or it is impractical to prepare/eat real foods): • sports drinks, liquid meals, gels and bars • confectionary • jam, honey and other sugary toppings • juices and soft drinks
Benefits	• Maximises muscle fuel for the next demanding workout or event
When should pro-active refuelling be practised?	• After races or fuel-depleting training sessions when the athlete is backing up for the next session in 8 hours or less • When total fuel needs are high: high-volume training, demanding competition schedule (e.g. cycling tour, tennis tournament)
Downsides	• May encourage the athlete to eat more kilojoules than needed (leading to weight gain) or a pattern of eating that is more risky for dental health • May encourage the athlete to choose nutrient-poor foods since these are more accessible or easy to eat immediately after exercise • May reduce the period of enhanced adaptation after exercise: there is a new theory that delaying the replacement of glycogen may allow the muscle to stay in an active state of adaptation for longer
When is it expendable?	• When sessions are light or low in intensity and muscle glycogen isn't likely to become depleted or limit performance • When the available recovery eating choices are low in nutritional value and it makes more sense to wait a little until you can have a more nutritious meal/snack • When the athlete has periodised some 'train low' sessions into the training program which may require a delay in refuelling in the attempt to prolong the adaptation to the session just done, or commence the next session with depleted glycogen stores

Source: Burke in press

TABLE 14.5 Summary of guidelines for post-exercise protein synthesis (repair and adaptation)

Strategies to maximise protein synthesis	• Consume a high-quality protein-rich food providing ~20–25 g of protein soon after the exercise session has finished. These targets might need to be expanded (e.g. 15–40 g) to account for the extreme range in athlete body size/muscle mass • Plan a pattern of snacks and meals to suit energy needs and other nutritional goals and lifestyle needs that incorporates this optimal protein serve every 3–5 hours • Include a protein-rich snack or meal before bed to allow protein synthesis to remain optimised overnight
Suitable choices for pro-active protein synthesis	Food sources of high-quality protein (providing ~10 g protein) include: • 2 small eggs • 25 g skim milk powder • 300 mL milk (full fat, reduced fat, skim) • 30 g (1.5 slices) of reduced fat cheese • 70 g cottage cheese • 200 g carton low-fat fruit yoghurt • 250 mL low-fat custard • 35 g lean beef, lamb or pork (cooked weight) • 40 g lean chicken (cooked weight) • 50 g grilled fish or canned tuna/salmon • 120 g tofu or soy meat • 400 mL soy milk Sports foods/supplements providing 10 g protein include: • 15–20 g high-protein powder or protein hydrolysate • 120–150 mL liquid meal supplement • 20–30 g high-protein sports bar Rapidly digested protein sources (liquid based, high in whey) that are particularly suitable for a post-exercise protein boost include: • milk, flavoured milk, milk shakes, dairy/fruit smoothies (esp. skim-milk boosted) • yoghurt and custard • whey-based protein drinks or liquid meal supplements
Benefits	• Maximises muscle protein synthetic response to exercise to promote adaptations to the training stimulus. Note that the elevated response lasts at least 24 hours
When should pro-active protein nutrition be practised?	• After competitive events or key training sessions (resistance sessions, high-intensity sessions) in which there is a major exercise stimulus or muscle damage • When gains in muscle mass and size are a priority
Downsides	• May encourage the athlete to think that expensive protein supplements are necessary • May require a reorganisation of eating practices since most western diets tend to be heavily loaded with protein at the evening meal rather than equally spread over the day
When is it expendable?	• When sessions are light or low in intensity and unlikely to promote great adaptation, and it doesn't suit the athlete's practical opportunities or energy restraints to include another food opportunity. In such a case, the athlete may not need to schedule a specific protein snack immediately after the session, but may still continue with an even spread of protein over the day to suit larger goals • If maximised protein-based recovery is desired but restricted energy requirements do not allow extra intake of food in the day, the athlete should consider changing the timing of training so that an existing

	meal can be consumed just after the workout to promote the enhanced protein synthetic response

Source: Burke in press

TABLE 14.6 Summary of guidelines for post-exercise rehydration	

Strategies to maximise rehydration	• Have a supply of fluids on hand that are palatable, suited to the conditions and suited to other recovery nutrition needs of the athlete
	• When the fluid deficit is moderate–large (e.g. >2 L) and the rehydration period is <6–8 h, have a planned fluid intake based on the deficit that needs to be replaced.
	  ◦ Typically, the difference between pre-exercise and post-exercise weight provides a quick guide to the fluid deficit (~1 kg = 1 L), although there are some variations to this guideline if the athlete commenced the exercise session over- or under-hydrated, or if the session was sufficiently lengthy to involve loss of body mass from fuel substrates (glycogen, fat)
	  ◦ It may require the intake of a volume of fluid that is ~125% of the estimated deficit to allow for ongoing fluid losses (urine losses, continued sweat losses)
	• Start to consume fluids soon after the session finishes and aim to consume the target volume over the next 2–4 hours. Where practical, it is best to spread fluids over this period rather than drink large volumes quickly. Such a pattern is better for gastrointestinal comfort as well as to maximise retention via smaller urine losses
	• Replace the electrolytes lost in sweat at the same time as consuming fluids, since this will maintain thirst and also maximise fluid retention via smaller urine losses. This can be achieved either by choosing fluids with added electrolytes (principally sodium) or by consuming salt-rich foods at the same time
	• Avoid excessive intake of alcohol since this is counter-productive to recovery goals and the diuretic effect of alcohol is likely to reduce the effectiveness of rehydration

Suitable choices for pro-active rehydration	Fluid preferences should be personalised to the individual athlete; useful ideas include: • in the case of recovery from prolonged exercise, choose flavour variants that are different to those offered during the session (offers an alternative to flavour fatigue) • in the case of extreme conditions, serve fluids at a palatable temperature (e.g. cool fluids in hot weather; warm fluids in cold conditions) Electrolyte (salt) replacement can be achieved in concert with fluid intake by: • choosing oral rehydration solutions or specialised high-electrolyte sports drinks • consuming fluids with meals or snacks that include salt-rich foods (e.g. breads, breakfast cereals, crackers, cheese, preserved meats) or have been seasoned with added salt (note that it is unnecessary to consume excessive amounts of salt, but rather to add a small amount scaled to the salt loss from sweat) When the athlete's total energy needs are low, low-energy fluids include: • water, low-energy soft drinks and mineral waters • oral rehydration solutions (these are lower in CHO content than sports drinks) Other recovery nutrients may be provided via fluid choices: • protein (milk/sweetened milks, protein shakes, fruit smoothies, soups containing milk or meats) • carbohydrate (sweetened milks, fruit smoothies, liquid meal supplements, soups containing pasta or legumes, fruit juices, sports drinks Since cola drinks or tea/coffee can provide a valuable source of fluid, and low levels of caffeine have minimal impact on hydration, it is unnecessary to totally avoid these drinks
Benefits	• Quickly restores hydration status for next demanding workout or event
When should it be undertaken?	• Following sessions that cause large sweat losses when the athlete is backing up for another session in 8 hours or less and will be exercising in hot conditions

Downsides	• May encourage the athlete to consume more kilojoules than needed (leading to weight gain) if the energy content of fluids is not accounted for • May cause gastrointestinal discomfort or the need for urination if large volumes of fluid are consumed quickly
When is it expendable?	• Just before bed, otherwise the athlete risks the interruption to sleep due to overnight toilet visits. It may be preferable to drink a little before bed, then rehydrate in the morning • When fluid losses are mild and further sessions are undertaken in cool conditions

Source: Burke in press

DECIDING ON RECOVERY GOALS

- The athlete should undertake an assessment of the exercise sessions within their training or competition programs to decide the extent to which various recovery processes need to be addressed after each session or as a whole, and whether their normal dietary practices will meet these needs. In the case where additional or immediate intake of nutrients will help to address an important recovery goal, the athlete should plan their intake of foods and drinks to achieve these goals, noting that many foods or combinations within a meal or snack can achieve a range of goals simultaneously (see Tables 14.4–Tables 14.6).
- When there is a need to undertake recovery nutrition strategies outside the normal eating plan, the athlete should give particular consideration to the practical challenges of storing, preparing or consuming items in the various locations in their sporting environment or lifestyle. The creativity and practical knowledge of a sports dietitian may be particularly useful in developing a plan to suit these situations. In addition, the practicalities offered by many packaged foods including specialised sports foods (e.g. liquid meal supplements, protein powders, sports bars) become useful in these circumstances where their expense may be better justified.
- Many athletes who live in environments in which there is little control over their meal-times/choices (e.g. boarding school or college dormitories, while travelling to foreign environments) may find it difficult to gain access to recovery-eating opportunities. This may be compounded if workouts or competitions are held at inhospitable hours. In some cases, special attention may be needed to arrange more flexible food availability. In other cases, it may be possible to change the time of the workout to better coordinate with catering options (i.e. choose a training time that allows the athlete to go straight to the next meal).

SPECIAL COMMENTS ABOUT ATHLETES WITH HIGH ENERGY REQUIREMENTS

- When energy needs are high, the athlete may undertake ‘recovery eating’ immediately after each training session as an additional opportunity for intake in the day as much as for any benefits of speedy intake of nutrients after exercise. A pattern of small, frequent meals over the day may assist the athlete to achieve high energy intakes without the discomfort of overeating.
- When energy/CHO needs are high, and appetite is suppressed or gastric discomfort is a problem, the athlete should focus on compact and low-fibre food forms. Fluids are also low in fibre and may be appealing to athletes who are fatigued and dehydrated. These include sports drinks, soft drinks and juices, commercial liquid meal supplements, milk shakes and fruit smoothies.

SPECIAL COMMENTS FOR ATHLETES WITH A RESTRICTED ENERGY BUDGET

- Recovery snacks should not contribute additional energy to a restricted energy budget. Rather, when rapid recovery is desirable, the energy-restricted athlete should change the timing of their existing meal structure to allow for immediate intake after exercise sessions. One option is to reschedule training sessions or meals so that the athlete is able to eat their normal meal as soon as possible after the workout. Where this is not practical, the athlete may be able to take a small snack from within their usual meal plan to consume immediately after training or as a pre-resistance-training snack (e.g. fruit and flavoured yoghurt usually consumed as a dessert with dinner), then consume the remainder of their meal at the usual time.

- Since the athlete may have increased requirements for protein and micronutrients as a result of their exercise program, it is important that foods consumed as recovery snacks contribute to overall nutrient intake goals as well as immediate recovery needs. Nutrient-rich choices (e.g. fruit, flavoured milk drinks and dairy foods, sandwiches with meat and salad fillings) are more valuable than lower nutrient choices (e.g. lollies, soft drink, bread with jam or honey).
- The energy-restricted athlete should also make use of foods with a high fibre content (e.g. fresh fruit rather than juice), high volume/low energy density (salad fillings added to sandwiches) or low GI (rolled oat cereals rather than cornflakes) to maximise the satiety value of meals and snacks. The addition of protein to meals and snacks (e.g. yoghurt with fruit, meat or cheese in sandwich) also improves satiety. Guidelines for low-fat eating are also important.
- The energy-restricted athlete is unlikely to have a sufficient energy budget to cover the guidelines for optimal intakes of some macronutrients (e.g. CHO for optimal daily glycogen synthesis). Specialised dietary advice from a sports dietitian is valuable in ensuring that the athlete has reasonable goals related to their energy requirements and physique goals, and is able to organise meal plans to optimise their nutrient intake within this energy budget. It may be valuable to cycle between nutritional goals—restrict energy during periods suitable for loss of body fat, while liberalising energy and CHO intake to promote better fuelling and recovery for key sessions or competition.

SPECIAL NOTES ABOUT REHYDRATION

- Athletes are often educated that the production of ‘copious amounts of clear urine’ is a desirable state and a sign of good hydration status. Measurements of urinary specific gravity or osmolality are sometimes undertaken to provide an indicator of euhydration and good hydration practices. Although this may be true in the long-term situation, the athlete is to be reminded that during the acute period of fluid replacement immediately following dehydration, mismatch of fluid and electrolyte replacement can lead to production of large amounts of dilute urine despite the continuing existence of substantial fluid deficits. Thus, in the case of significant fluid loss, the athlete should be aware of the need for electrolyte replacement, and should know that ‘urine checks’ over the first hours of fluid intake often provide false readings. Dietary strategies that minimise urine losses during the rehydration period not only enhance the speed of regaining fluid balance, but help the athlete to achieve better quality rest or sleep without frequent disturbances related to the need to urinate.

SPECIAL NOTES ABOUT ANTIOXIDANT AND ANTI-INFLAMMATORY NUTRIENTS AND FOODS

- The guidelines for recovery in terms of antioxidant, anti-inflammatory and immune system support are not yet as sophisticated as recommendations for other aspects of recovery. Although there is evidence that exercise causes oxidative damage and inflammation, and that the intake of large amounts of specific nutrients or food chemicals can reduce this, it is not yet clear how to deal with the balance between restoration of performance and promotion of adaptation to training. It may be prudent to limit the use of such supplementation to acute situations, such as competitive events, where the strategy is practised for brief periods with the priority on performance. In the case of everyday training, a diet based on nutrient sources of these protective chemicals is likely to be the best choice.

ALCOHOL INTAKE AND SPORT

- Alcohol is not an essential component of a diet. It is a personal choice of the athlete whether to consume alcohol at all. However, there is no evidence of impairments to health and performance when alcohol is used sensibly.
- The athlete should be guided by community guidelines, which suggest general intakes of alcohol that are ‘safe and healthy’. This varies from country to country, but in general, it is suggested that mean daily alcohol intake should be less than 40–50 g (perhaps 20–30 g per day for females), and that ‘binge’ drinking is discouraged.

Since individual tolerance to alcohol is variable, it is difficult to set a precise definition of ‘heavy’ intake or an alcohol ‘binge’. However, intakes of about 80–100 g at a single sitting are likely to constitute a heavy intake for most people.

- Alcohol is a high-energy (and nutrient-poor) food and should be restricted when the athlete is attempting to reduce body fat.
- Heavy alcohol intake is likely to have a major impact on post-exercise recovery. It may have direct physiological effects on rehydration, glycogen recovery and repair of soft-tissue damage. More importantly, the athlete is unlikely to remember or undertake strategies for optimal recovery when they are intoxicated. Therefore, the athlete should attend to these strategies first before any alcohol is consumed. No alcohol should be consumed for 24 hours in the case of an athlete who has suffered a major soft-tissue injury.
- Before consuming any alcohol after exercise, the athlete should consume a snack or meal providing CHO and protein goals. As well as addressing recovery needs, food intake will also help to reduce the rate of alcohol absorption and thus reduce the rate of intoxication.
- Once post-exercise recovery priorities have been addressed, the athlete who chooses to drink is encouraged to do so ‘in moderation’. Drink-driving education messages in various countries may provide a guide to sensible and well-paced drinking.
- Athletes who drink heavily after competition, or at other times, should take care to avoid driving and other hazardous activities.
- It appears likely that it will be difficult to change the attitudes and behaviours of athletes with regard to alcohol. However, coaches, managers and sports medicine staff can encourage guidelines such as these, and specifically target the ‘old wives’ tales’ and rationalisations that support binge drinking practices. Importantly, they should reinforce these guidelines with an infrastructure that promotes sensible drinking practices. For example, alcohol might be banned from locker rooms, and fluids and foods appropriate to post-exercise recovery provided instead. In many cases, athletes drink in a peer-group situation and it may be easier to change the environment in which this occurs than the immediate attitudes of the athletes.

SPECIAL COMMENTS ABOUT PERIODISING TRAINING WITH LOW CHO AVAILABILITY

- There is no single recipe for the implementation of a periodised approach to CHO availability in any sport, but the available evidence suggests that it should be approached on an individualised and careful basis to integrate the potential benefits of enhanced adaptation to an endurance training stimulus with the other goals of training. The fact that most studies that have incorporated ‘train low’ practices into 50–100% of workouts in a training block fail to achieve a performance benefit shows that it needs to be used selectively and occasionally rather than applied chronically.
- Many athletes already undertake some of their training sessions with low CHO availability (e.g. early morning training in fasted state, long training sessions undertaken with intake of water only, periods of high-volume training with 2–3 sessions/day where there is inadequate opportunity for restoration of glycogen between workouts). This may be because of practicalities, the culture of the sport or the trial-and-error activities that have allowed successful strategies to emerge. An understanding of the scientific underpinning of these sessions may help to consolidate/continue these sessions, to reorganise them for a more appropriate time or session, or to surround them with other sessions in which high CHO availability is targeted, thus optimising all the elements of such periodisation.
- The implementation of an individualised and periodised approach to CHO availability needs to be undertaken with good communication between the athlete, the coach and the sports science team. It should be remembered that training with low CHO availability interferes with training intensity, increases the perception of effort of exercise, and increases the stress on the immune system. Therefore it is best implemented around training sessions that involve lower intensity exercise, particularly during phases of the program in which the priority is to improve general conditioning. It is also likely that athletes respond differently to the same program. Therefore, it is also important to collect information around these sessions and the periods in which they are implemented to ensure that any concerns are identified before serious problems occur.

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CHAPTER FIFTEEN

Nutritional strategies to enhance fat oxidation during aerobic exercise

Louise Burke and John Hawley

15.1 Introduction

Compared with the finite stores of carbohydrate (CHO), endogenous fat depots in humans are large, and represent a potentially unlimited source of fuel for skeletal muscle metabolism during aerobic exercise. However, fatty acid (FA) oxidation by muscle is limited, especially during the high power outputs and intensities sustained by athletes in training and competition. It has been proposed that strategies that promote FA oxidation during exercise could attenuate the rate of muscle glycogen utilisation and improve sports performance; this is one of the key adaptations achieved by aerobic training. This chapter will review a variety of approaches have been attempted to achieve this goal, with particular focus on the low-carbohydrate high-fat (LCHF) diet, which has been enthusiastically revived in recent times. As background to this interest, it is important to understand the role of endogenous fat as an energy substrate for skeletal muscle during exercise, the effect of exercise intensity on the regulation of fat metabolism, and the processes that could limit FA during exercise.

15.2 Sources of fat as a muscle fuel

Lipids provide the largest nutrient store of chemical energy that can be used to power biological work (see Figure 15.1). As an energy source, triacylglycerol (TG) has several advantages over CHO: the energy density of lipid is higher (37.5 kJ/g for stearic acid versus 16.9 kJ/g for glucose), while the relative weight as stored energy is lower. TG also provides more adenosine triphosphate (ATP) per molecule than glucose (147 versus 38 ATP). However, it needs to be appreciated that the complete oxidation of FA requires more oxygen than the oxidation of CHO (6 versus 26 mol of oxygen per mole of substrate for glucose and stearic acid oxidation, respectively), making CHO a more economical fuel source when oxygen delivery to the muscle becomes limited. Indeed, the oxidation of CHO provides about 10% more ATP per L of oxygen than fat; and higher CHO stores have been associated with greater gross efficiency of exercise (power production per L of oxygen consumption or work done per kJ of energy expenditure) (Cole et al. 2014).

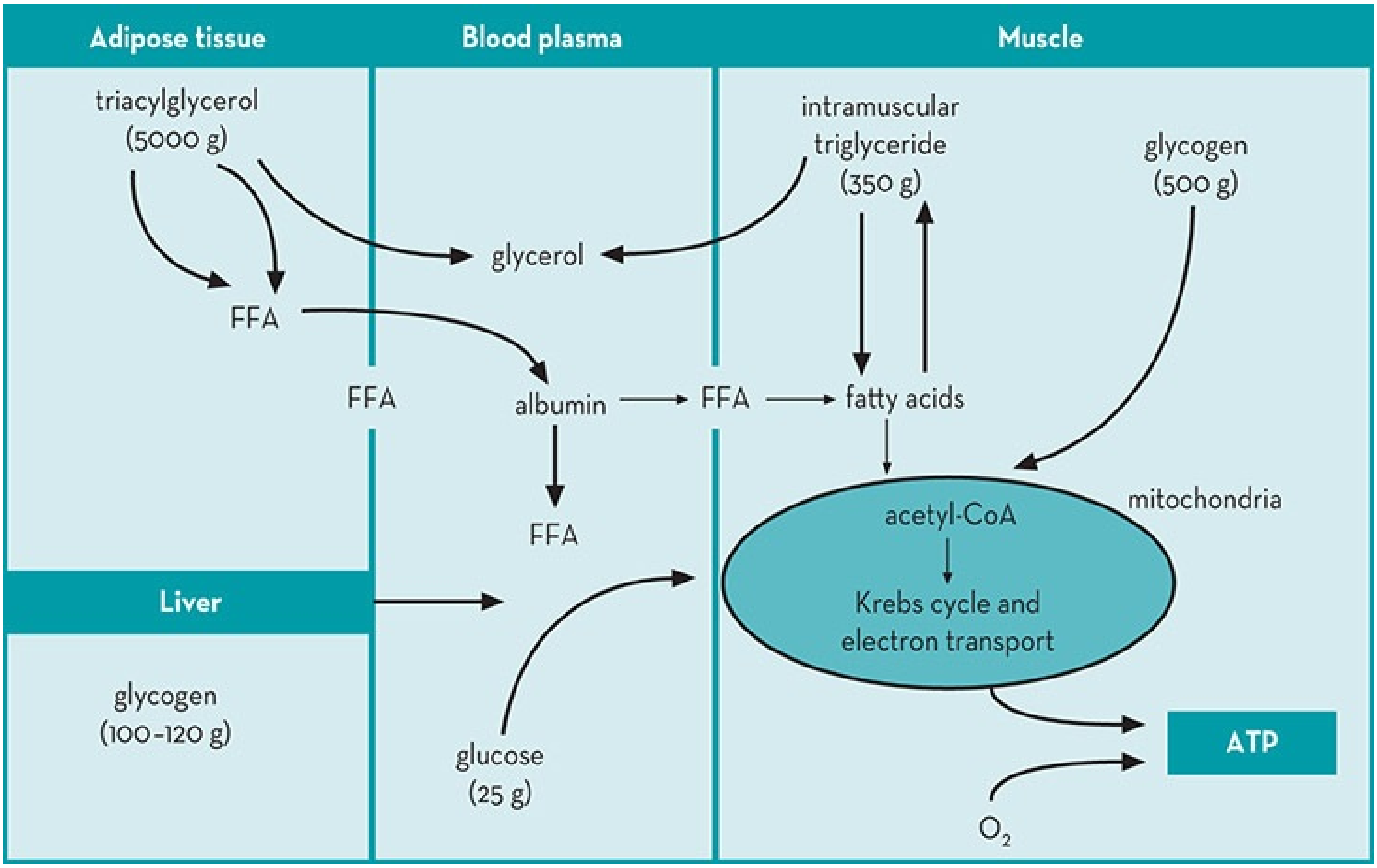

Figure 15.1 A schema of the major endogenous storage sites of carbohydrate and fat.

Source: Reprinted with permission from Coyle 1997

15.2.1 Adipose triacylglycerol

The size of the adipose tissue TG pool is difficult to estimate and obviously depends on the fat mass of each individual, but is likely to range from 50 000 to 100 000 kcal (200–400 MJ) in men and women with 10–30% body fat (see Figure 15.1). In order for this TG to be used as a substrate for oxidative metabolism, it has to be exported from adipose tissue and transported by the blood to the active tissues where it will be utilised.

15.2.2 Intramuscular triacylglycerol (IMTG)

Another important physiological store of TG can be found within the skeletal muscle (IMTG), mostly adjacent to the mitochondria: the total active muscle mass may contain up to 300 g of TG within the myocyte as small lipid droplets, although this amount can vary substantially due to individual differences in fibre type (type I fibres contain a greater concentration of IMTG than type II fibres), endurance training status (Kiens et al. 1993; Martin et al. 1993) and diet (Starling et al. 1997).

Intramuscular triacylglycerol and insulin resistance: the metabolic paradox

Elevated IMTG stores have been associated with reduced muscle insulin sensitivity in sedentary, obese and/or insulin-resistant individuals in some studies (Ebeling et al. 1998; Goodpaster et al. 2001) but not all studies (Bruce et al. 2003). In contrast, regular endurance training elicits an increase in IMTG concentration in healthy individuals (Kiens et al. 1993; Pruchnic et al. 2004) and is associated with increased insulin sensitivity (Staudacher et al. 2001). This greater storage of IMTG in the athlete represents an adaptive response to endurance training, allowing a greater contribution of the IMTG pool as a substrate for oxidative metabolism during exercise. In contrast, the elevated IMTG stores in the obese and/or type 2 diabetic patient seems to be secondary to a structural imbalance between FA availability, storage and oxidation (Kelley 2002). As such, the reported correlations between IMTG content and insulin resistance do not necessarily represent a functional relationship, as this association is strongly modulated by training status, habitual physical activity and muscle fibre composition. As 'fitness status' is better reflected through measures of skeletal muscle oxidative capacity rather than more traditional measures of whole body maximal aerobic power (Bruce et al. 2003), it has been suggested that IMTG content should be expressed relative to this muscle marker, rather than in isolation (Van Loon & Goodpaster 2006).

15.2.3 Circulating lipids

Finally, FA can also be derived from circulating TG (chylomicrons) and very low density lipoproteins (VLDL) formed from dietary fat in the post-absorptive state. Evidence suggests that if all the circulating VLDL-TG were taken up and oxidised, VLDL-TG degradation could contribute up to 50% of the lipid oxidised during submaximal exercise (Kiens 1998).

15.3 Processes that could limit fatty acid oxidation during exercise

Despite the vast stores of endogenous TG, the capacity for FA oxidation during exercise is limited. Unlike CHO oxidation, which is closely geared to the energy requirements of the working muscle, there are no mechanisms for matching the availability and utilisation of FA to the rate of energy expenditure (Holloszy et al. 1998). There are many potential sites at which the ultimate control of FA oxidation may reside (see Figure 15.2), with the relative importance of each site depending on many external factors, such as the aerobic training status of the individual, habitual dietary intake, ingestion of substrates (CHO and fat) before and during exercise, gender, and both the relative and absolute exercise intensity. A comprehensive analysis of the processes that potentially limit FA oxidation during exercise is beyond the scope of this chapter, and the reader is referred to the excellent review of Spriet (2014).

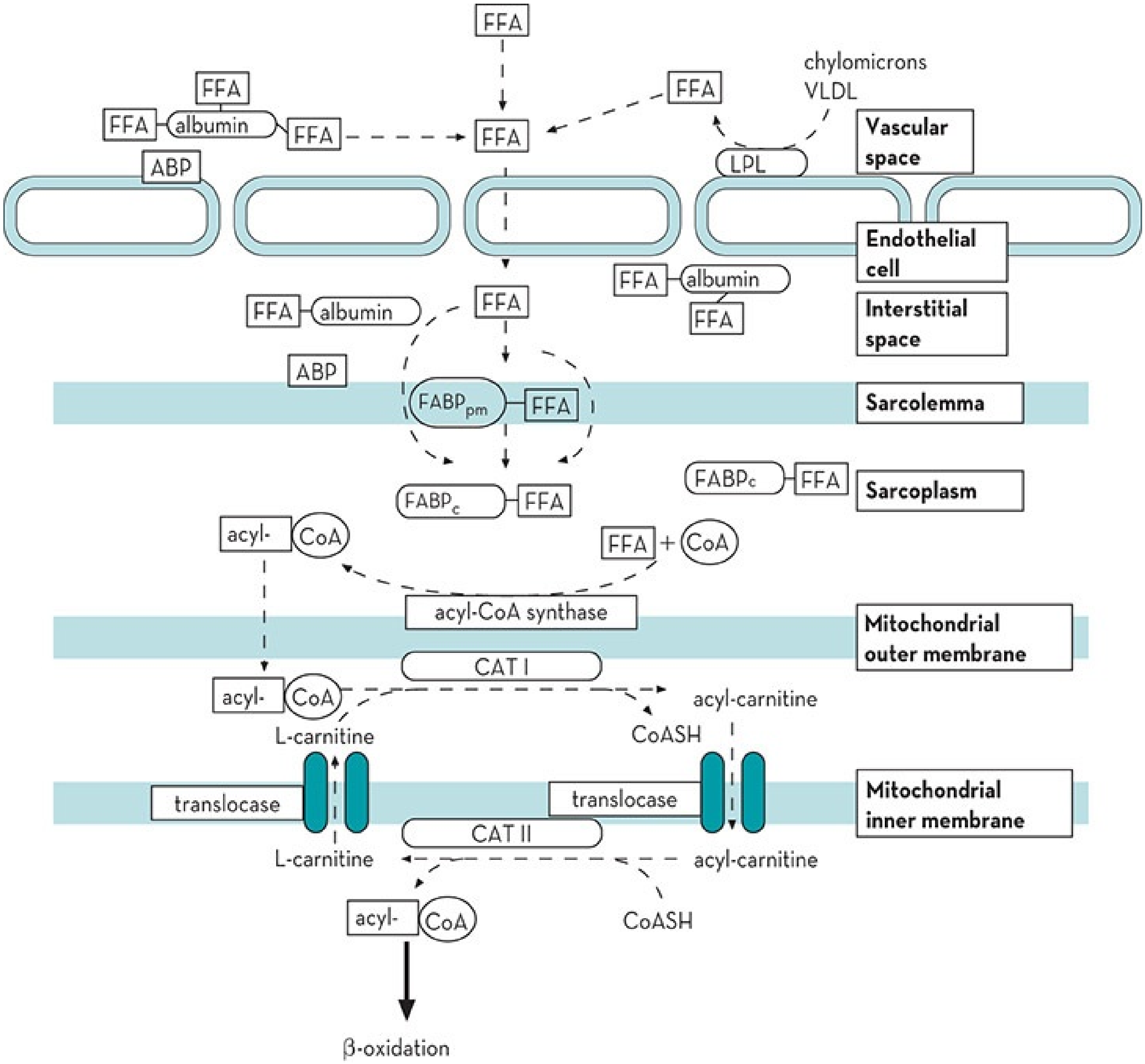

Figure 15.2 A schema of the transport of fatty acids from the vascular space to the inner mitochondria of the skeletal muscle where β-oxidation occurs. CAT I = carnitine acyl transferase I (also known as carnitine palmitoyltransferase 1: CPT1); CAT II = carnitine acyl transferase II; $FABP_C$ = fatty acid binding protein; $FABP_{pm}$ = plasma-membrane-bound fatty acid binding protein; FFA = free fatty acid; VLDL = very low density lipoprotein. The various processes are described in detail in the text.

Source: Reproduced with permission from Jeukendrup 1997

15.3.1 Mobilisation of fatty acids from adipose triacylglycerol: lipolysis

Triacylglycerols cannot be oxidised by skeletal muscle directly; first they must be hydrolysed into their components, non-esterified FAs (NEFA) and glycerol. This process, called lipolysis, is largely dependent on the activation of the enzyme hormone-sensitive TG lipase (HSL) in adipose tissue. Binding of hormone to plasma membrane receptors on adipocytes activates adenyl cyclase and initiates the lipolytic cascade (see Figure 15.2). Epinephrine and glucagon activate HSL, while high levels of plasma glucose and insulin inhibit the activity of the lipase and reduce lipolysis. FA and glycerol derived from lipolysis in adipose tissue are released into the circulation: FA are bound by serum albumin and transported to tissues for oxidation and production of ATP (discussed subsequently), while glycerol returns to the liver and can be either phosphorylated to glycerol 3-phosphate and used to form TG, or converted to dihydroxyacetone and enter the glycolytic or gluconeogenic pathways. An isoform of the enzyme HSL is also present in skeletal muscle, where it acts to break down IMTG stores.

15.3.2 Transport of fatty acids across the sarcolemmal membrane into skeletal muscle

During transport of FA from blood to muscle, several potential processes limit eventual FA uptake. These are the membranes of the vascular endothelial cells, the interstitial space between endothelium and muscle cell, and finally the muscle cell membrane. Although FA transport across the sarcolemmal membrane into the muscle fibre was originally thought to occur exclusively by simple passive diffusion along a concentration gradient, there is now good evidence of a long-chain FA (LCFA) transport system involving FA binding proteins (FABP), FA translocases (FAT) and FA transport proteins (FATP) (for review, see Glatz et al. 2001; Glatz et al. 2002). Of interest is the finding that FABP content is higher in type I (slow twitch) than type II (fast twitch) muscle fibres, and is also increased with endurance training. This suggests a functional relationship between the FA binding capacity and the degree of oxidative metabolism in the muscle (Kiens 1998). Once FA enter the cytoplasm of a muscle cell, they can either be esterified and stored as IMTG, or the FA can be bound to FABP for transport to the site of oxidation and activated to a fatty acyl-coenzyme A by the enzyme acyl-CoA synthase.

15.3.3 Oxidation of fatty acids

Whereas most fatty acyl-CoA is formed outside the mitochondria, the oxidative machinery is inside the inner membrane, which is impermeable to CoA. To overcome this problem, there exists a specific carnitine-dependent shuttle to carry acyl groups across the membrane (see Figure 15.3).

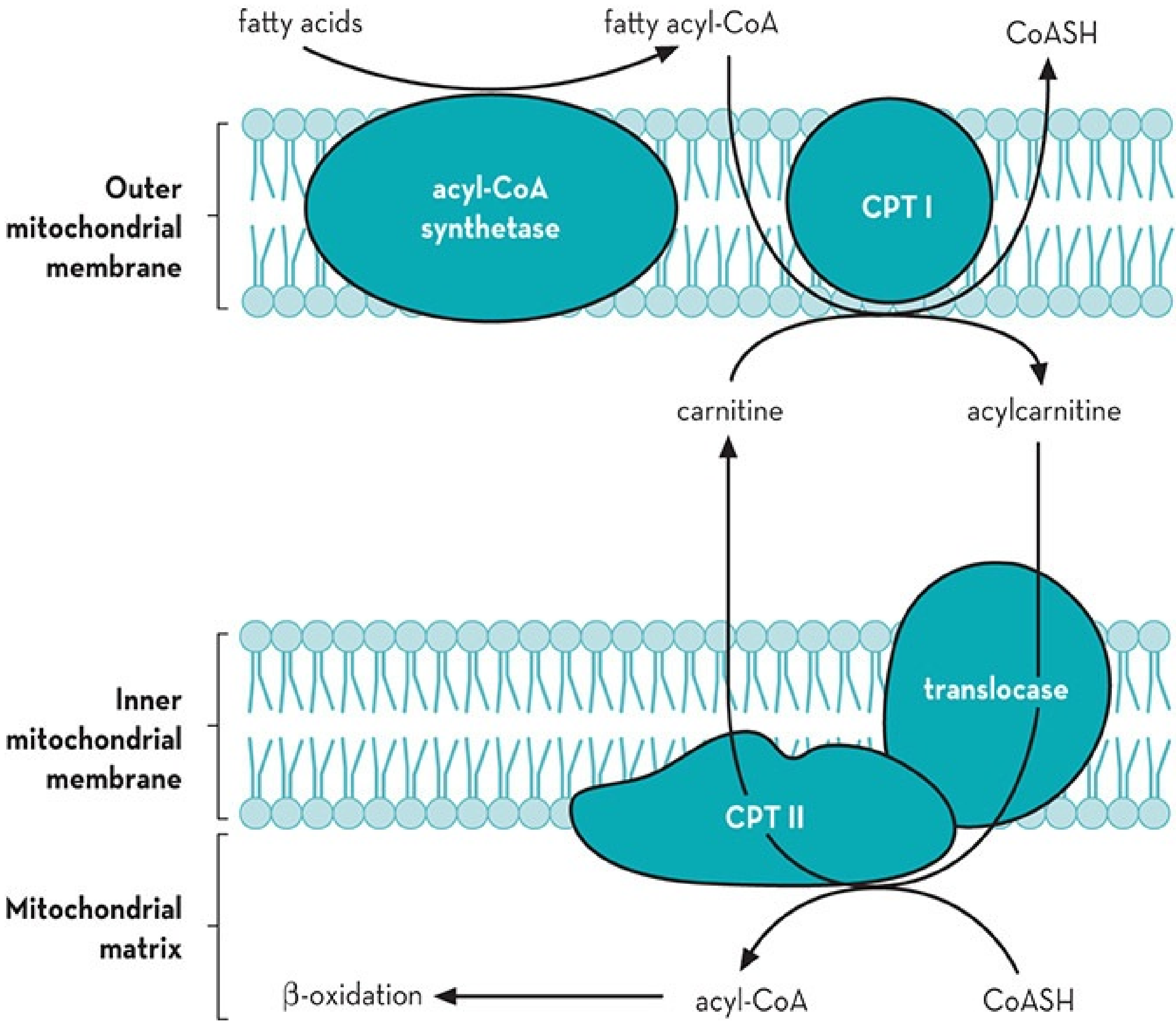

Figure 15.3 The transport of long-chain fatty acids from the cytosol through the inner mitochondrial membrane for oxidation is dependent on the carnitine palmitoyltransferase complex (see text for further details).

Source: Reproduced from TM Devlin, *Textbook of biochemistry* (fourth edition), Wiley-Liss, New York, 1997

Enzymes on both sides of the membrane transfer acyl groups between CoA and carnitine. On the outer mitochondrial membrane, the acyl group is transferred to carnitine catalysed by carnitine palmitoyltransferase I (CPTI). Acylcarnitine then exchanges across the inner mitochondrial membrane with free carnitine by a carnitine-acylcarnitine antiporter translocase. Finally, the fatty acyl group is transferred back to CoA by carnitine palmitoyltransferase II (CPTII) located on the matrix side of the inner membrane. This mitochondrial transport of fatty acyl-CoA functions primarily with chain lengths of C12–C18. Medium- and short-chain FA (MCFA and SCFA) can freely diffuse into the mitochondrial matrix and do not require a carnitine-dependent shuttle mechanism to allow transport across the mitochondrial inner membrane. There is some evidence to suggest that carnitine-dependent transport of LCFA into the mitochondria might be a rate-limiting step for FA oxidation (see below).

The process of β-oxidation, which occurs in the mitochondria, comprises four separate reactions in which the fatty acyl-CoA is sequentially degraded to acetyl-CoA and an acyl-CoA residue that has had 2C sequestered. The acetyl-CoA units enter the tricarboxylic acid (TCA) cycle and follow the same pathway as acetyl-CoA units from pyruvate. The rate at which FA are oxidised depends on the chain length and the degree of saturation: MCFA are oxidised more rapidly and more completely than LCFA.

15.4 Methods to quantify lipid metabolism during exercise

A background knowledge of some of the methods used to measure substrate metabolism during laboratory-based investigations is essential for the sports dietitian striving to comprehend and interpret the findings of such studies in a meaningful and practical manner to athletes and coaches. Our understanding of the regulation of fat metabolism has been advanced considerably by modern-day investigations that have used a combination of stable isotope techniques in association with conventional indirect calorimetry (Romijn et al. 1992, 1993). As the three most abundant FA are oxidised in proportion to their relative concentration in the plasma FA pool, total plasma FA kinetics can be estimated from stable isotope infusions of either oleate or palmitate. The rate of appearance (Ra) of an FA (palmitate) in the bloodstream gives an index of the release of FA into the plasma and represents the net balance between the rate of adipose tissue lipolysis and the rate of FA uptake and re-esterification. Glycerol, on the other hand, cannot be produced by the body other than from lipolysis. Furthermore, all glycerol released during lipolysis, whether from adipose tissue or skeletal muscle, appears in the plasma. Accordingly, the Ra of glycerol provides a useful indicator of the rate of whole body lipolysis. An estimation of total fuel utilisation (fat and CHO) during steady state exercise can be obtained from the respiratory exchange ratio, the volume of carbon dioxide produced (VCO_2) divided by the oxygen consumed (VO_2).

$$\text{CHO oxidation (g/min)} = 4.585\ VCO_2 - 3.226\ VO_2$$

$$\text{fat oxidation (g/min)} = 1.695\ VO_2 - 1.701\ VCO_2$$

Rates of substrate oxidation are usually expressed relative to an individual's body mass (BM) (or sometimes their lean muscle mass or fat-free mass). Accordingly, the rate of CHO oxidation (µmol/kg/min) is determined by converting the g/min rate of CHO oxidation to its molar equivalent, assuming 6 mol of O_2 are consumed and 6 mol of CO_2 produced for each mole (180 g) oxidised. Rates of FA oxidation (µmol/kg/min) are determined by converting the g/min rate of TG oxidation to its molar equivalent, assuming the average molecular weight of TG to be 855.26 g/mole and multiplying the molar rate of TG oxidation by three, because each molecule contains 3 mmol of FA.

Given the tracer-derived rates of total lipolysis and total FA released into the plasma, it is possible to distinguish peripheral lipolysis from adipose TG and intramuscular lipolysis.

$$\underset{(\mu\text{mol/kg/min})}{\text{IMTG FA oxidation}} = \underset{(\mu\text{mol/kg/min})}{\text{total FA oxidation}} - \underset{(\mu\text{mol/kg/min})}{\text{FA uptake (FARd)}}$$

For every three FA released from the IMTG pool, one glycerol molecule will be released into the plasma. Consequently, the minimum rate of release of glycerol from the IMTG pool gives an estimation of IMTG lipolysis and can be estimated from the following equation:

$$\text{intramuscular FA oxidation } (\mu\text{mol/kg/min}) \div (3\ \mu\text{mol FFA} \div \mu\text{mol glycerol})$$

The rate of total glycerol release (Ra glycerol) equals the glycerol released from adipocyte TG and glycerol released from the IMTG pool. Accordingly, it is possible to calculate the rate of adipose (peripheral) TG lipolysis from the following equation:

$$\underset{(\mu\text{mol/kg/min})}{\text{adipose lipolysis}} = \underset{(\mu\text{mol/kg/min})}{\text{total Ra glycerol}} - \underset{(\mu\text{mol/kg/min})}{\text{IMTG lipolysis}}$$

Using a combination of these techniques, it has been possible to estimate the effect of exercise intensity and duration on fat metabolism (Romijn et al. 1993).

15.5 The effects of exercise intensity on lipid metabolism

In the post-absorptive state, FA oxidation provides a major portion of the energy requirements for skeletal muscle: at rest, the rate of total FA oxidation is ~4 µmol/kg/min, which represents about 50% of oxygen consumption. The rate of lipolysis at rest is usually in excess of that required to provide resting energy requirements such that, at the onset of low- to moderate-intensity exercise, a significant increase in FA oxidation could occur even if there were no instant increase in lipolysis. During low-intensity exercise (25% of $VO_{2\ max}$), an intensity comparable to walking, most of the energy requirements can be met from plasma FA oxidation, with a small contribution from the oxidation of plasma glucose. At exercise of low intensity, the Ra of FA in plasma matches closely the rate of FA oxidation. Even when low-intensity exercise is sustained for 1–2 hours, the pattern of fuel utilisation does not change considerably. Presumably this is because the muscle energy requirements can be met almost exclusively from the oxidation of the FA mobilised from the large adipose TG stores, and lipolysis is not limited by blood flow.

With an increase in exercise intensity from 25% to 65% of $VO_{2\ max}$ (the pace that could be sustained by a trained person for up to 8 hours), total fat oxidation reaches its peak, despite a slight decline in the Ra of plasma FA. The higher rate of total FA oxidation at 65% compared to 25% of $VO_{2\ max}$ reflects a substantial increase in the oxidation of IMTG. Of interest is that even when the absolute rate of FA oxidation is at a peak, fat only contributes about 50% to the total fuel requirements of exercise, with the remainder of the energy coming from CHO (see Figure 15.4).

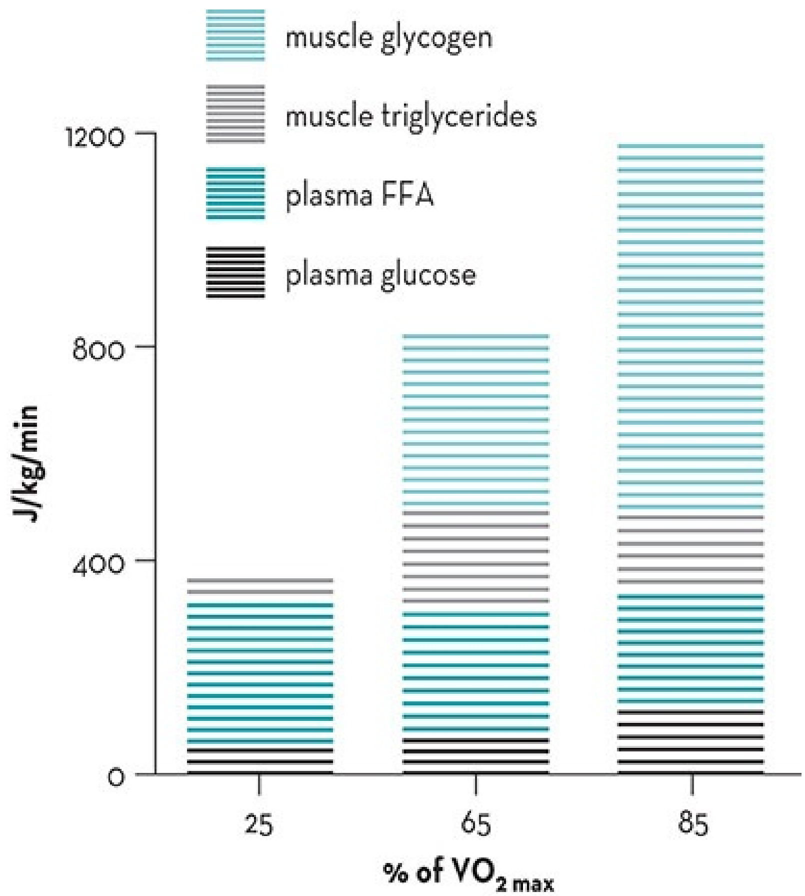

Figure 15.4 The effect of exercise intensity on the contribution from the four major substrates to energy expenditure.

Source: Redrawn from Romijn et al. 1993. Reproduced with permission from the American Physiological Society

During high-intensity exercise at 85% of $VO_{2\ max}$ (race pace for endurance events lasting ~90 minutes) there is a decline in total FA oxidation compared to moderate-intensity exercise (see Figure 15.4). This is largely due to a marked reduction in the Ra of plasma FA. It is likely the Ra for plasma FA decreases with increasing exercise intensity because of an insufficient blood flow and albumin delivery to transport FA from adipose tissue into the bloodstream. On the other hand, glycerol is water soluble and so its appearance in the plasma is not blood-flow dependent; consequently the Ra for glycerol is not affected. In addition, continuous high-intensity exercise is associated with high rates of glycogenolysis (see Figure 15.4) and the concomitant production of lactic acid that accumulates in muscle and blood. This increased glycolytic flux also acts to inhibit FA oxidation by skeletal muscle (see below).

15.5.1 Why can't fatty acid oxidation sustain intense exercise?

At rest, the Ra of plasma FA (lipolysis) normally exceeds the energy requirements of skeletal muscle. During low-intensity exercise, when lipolysis increases further, there is still a sufficient supply of FA to meet the muscles' energy demand. However, there is little further increase in lipolysis (the Ra FA) when exercise intensity increases to 65% of $VO_{2\ max}$; at such work rates Ra FA closely matches FA oxidation. During high-intensity exercise, lipolysis is markedly suppressed and the contribution of FA oxidation to the total energy requirement of exercise is diminished. These observations would support the notion that the reduced availability of FA (a reduction in lipolysis) may contribute to a part of the decline in FA oxidation during intense exercise.

To evaluate the extent to which decreased FA availability contributes to the lower rates of FA oxidation during intense exercise, Romijn and colleagues (1995) studied well-trained endurance subjects during 30 minutes of intense (85% of $VO_{2\ max}$) cycling once during a ‘control’ trial when plasma FFA concentration was normal (0.3 mM) and again when plasma FFA concentration was elevated to ~2 mM by an infusion of lipid (Intralipid™) and heparin. Total FA oxidation was increased 27% (from 26.7 to 34.0 µmol/kg/min) with the lipid infusion compared to control. However, the elevation of plasma FFA concentration (increased availability) during intense exercise only resulted in a partial restoration of FA oxidation, as the rates of total fat oxidation at 85% of $VO_{2\ max}$ were still lower than those observed in normal conditions at 65% of $VO_{2\ max}$. These findings indicate that FA oxidation is impaired during intense exercise because of a failure of lipolysis to meet the energy demands of the muscle. Therefore, in theory, TG lipolysis establishes the upper limit to FA oxidation during high-intensity exercise.

However, even when lipid is infused well in excess of the muscle requirements during high-intensity exercise, less than half of the total energy requirement is met by FA oxidation. This is because the muscle is also a major site of control of the rate of FA oxidation during such exercise. Specifically, the increased rate of glycogenolysis during intense exercise appears to inhibit the entry of LCFA into the mitochondria. Sidossis and colleagues (1997) reported that during cycling at 80% of $VO_{2\ max}$, the accelerated glycolytic flux associated with the high work rates resulted in high rates of pyruvate and acetyl-CoA formation, which inhibited CPT-I activity and, in turn, FA entry into the mitochondria. Coyle and colleagues (1997) also showed that CHO metabolism (glycolytic flux) regulates FA oxidation during exercise. Their subjects ingested CHO before exercise (in order to produce high concentrations of plasma glucose and insulin) and the rates of oxidation of a LCFA (palmitate) and a MCFA (octanoate) were subsequently determined. Unlike palmitate, which requires CPT-I for transport into skeletal muscle mitochondria, octanoate is not limited by mitochondrial transport. The increased glycolytic flux from pre-exercise glucose ingestion significantly reduced palmitate oxidation, but had no effect on octanoate oxidation. Even when FA availability is maintained by an infusion of lipid, CHO ingestion still inhibits LCFA oxidation (Sidossis et al. 1996), presumably because of the anti-lipolytic effects of elevated insulin concentrations. Taken collectively, these findings suggest that although the rate of lipolysis is important, the primary site of control of FA oxidation during moderate to intense exercise resides at the muscle tissue level (Wolfe 1998). Furthermore, increased glycolytic flux resulting from either CHO ingestion (Coyle et al. 1997; Horowitz et al. 1997, 1999) and the concomitant rise in plasma insulin, or an increase in exercise intensity (Romijn et al. 1995; Sidossis et al. 1997) directly inhibits

LCFA oxidation.

15.6 Nutritional supplements and acute strategies to enhance fat oxidation during exercise

There have been cyclical surges of interest among athletes, coaches and sports dietitians in various nutritional practices which, in theory at least, could promote FA oxidation, attenuate the normal rate of CHO utilisation and improve exercise capacity (for review, see Hawley et al. 2000b; Spriet 2014). This section will cover supplements and acute feeding strategies purported to achieve this effect. Although intravenous infusion of lipid (Intralipid^{TM}) accompanied by heparin is a potent lipolytic stimulant that increases FA oxidation and spares muscle glycogen stores during both moderate (Odland et al. 1996, 1998) and intense (Dyck et al. 1993, 1996) exercise, such a procedure is impractical in most sporting environments. Furthermore, intravenous infusions contravene the World Anti-Doping Agency Code. As such, a critique of this technique has not been included in this chapter. The reader is referred to the excellent reviews of Spriet and Dyck (1996) and Spriet (1999) for further information on this topic.

15.6.1 Caffeine ingestion before exercise

Caffeine is used by many athletes as an ergogenic aid to improve both short-term high-intensity and prolonged moderate-intensity exercise performance (see Chapter 16). Although the contemporary view is that caffeine exerts its major effects via direct effects on the central nervous system, reducing an individual's perception of effort, the early studies investigating the effects of caffeine ingestion on exercise capacity focused on changes in FFA availability and subsequent FA oxidation, the so-called 'metabolic theory' (Spriet 1997).

In a series of investigations conducted 30 years ago, Costill and colleagues were the first to report that the ingestion of moderate doses of caffeine (~5 mg/kg) ~1 hour before exercise stimulated lipolysis-enhanced rates of total FA oxidation (as estimated from RER measurements) and decreased the utilisation of muscle glycogen (Costill et al. 1978; Essig et al. 1980; Ivy et al. 1979). Caffeine was proposed to mobilise FA from adipose tissue and/or IMTG stores by increasing plasma epinephrine concentrations and/or directly antagonising adipocyte tissue adenosine receptors. The increased circulating FFA would then increase FA uptake and oxidation by muscle and spare endogenous CHO reserves. More recent evidence against the metabolic theory is provided by Graham and colleagues (2000), who quantified muscle metabolism by a combination of direct arteriovenous balance methods and muscle biopsies after ingestion of 6 mg/kg caffeine during 1 hour of submaximal exercise. They found that, although caffeine ingestion stimulated the sympathetic nervous system, it did not alter leg FA uptake, net muscle glycogenolysis or rates of CHO and fat metabolism in the monitored leg.

Another study from this group confirmed that intake of caffeine can prolong endurance during cycling at 80% $VO_{2\ max}$ without affecting muscle glycogen utilisation (Greer et al. 2000). However, others have found individual variability in the metabolic response to caffeine: half of a group of subjects was shown to ‘spare’ glycogen during the first 15 minutes of exercise following caffeine intake (9 mg/kg) compared with a placebo treatment, while glycogen utilisation was unaffected in the other half of the group following caffeine treatment (Chesley et al. 1998). Taken collectively, the results of these studies indicate that glycogen sparing following caffeine ingestion is a variable and short-lived response and that the concept of taking caffeine to enhance fat metabolism during exercise is now outdated.

15.6.2 Fat feeding before exercise

Several studies have investigated the effects of fat feeding before exercise on the subsequent rates of substrate oxidation and exercise performance. Costill and colleagues (1977) first reported that fat feeding in combination with intravenous administration of heparin stimulated lipolysis, elevated plasma FFA concentrations and decreased the rate of muscle glycogen utilisation by 40% compared to a control condition during 30 minutes of running at 70% of $VO_{2\ max}$. A more recent study from the same laboratory also reported muscle glycogen sparing with fat feeding and IV heparin compared to control during 60 minutes of cycling at 70% of $VO_{2\ max}$ (Vukovich et al. 1993).

In studies that have investigated performance as well as substrate use, Okano and colleagues (1996, 1998) reported only small differences in the rates of fat and CHO oxidation in response to high-fat or high-CHO meals ingested 4 hours before prolonged submaximal cycling (2 hours at 67% of $VO_{2\ max}$ followed by a ride to exhaustion at 78% of $VO_{2\ max}$). Furthermore, most of the differences in metabolism (a lower RER) after the fat feeding were evident only in the early stages of exercise, and did not result in an improved performance time. Whitley and colleagues (1998) also found that high-fat or high-CHO meals ingested 4 hours before exercise failed to substantially alter the pattern of fuel utilisation during 90 minutes of moderate-intensity cycling, or affect a subsequent 10-km cycle time trial. Wee and colleagues (1999) fed six endurance-trained runners a random order of either a high-fat, a high-CHO or a high-fat–high-CHO meal 3 hours before a run to exhaustion at 71% of $VO_{2\ max}$. Despite the rate of fat oxidation being elevated after the high-fat compared to the high-CHO and high-fat–high-CHO meals (19% and 14%, respectively), endurance time was 14% less after the high-fat meal. These workers concluded that CHO, rather than fat availability before exercise, exerts a predominant control over substrate selection during subsequent exercise.

Two studies have compared the effect of fat feeding versus CHO feeding on metabolism and performance during prolonged or intermittent exercise at higher intensities. Hawley and colleagues (2000a) reported that a high-fat feeding increased fat availability and elevated rates of FA oxidation during 20 minutes of cycling at 80% of $VO_{2\ max}$, but that the small reduction in

CHO oxidation after such a regimen did not enhance performance of a subsequent time trial lasting ~30 minutes. Similarly, consuming a high-fat or a high-CHO meal 3.5 hours prior to a soccer-specific protocol created differences in FA availability and fuel utilisation, but failed to change performance in a 1-km time trial at the end of the protocol (Hulton et al. 2013).

Finally, two studies have reported apparent benefits to exercise *capacity* following the intake of a high-fat meal. Pitsaladis and colleagues (1999) found that cycling time to exhaustion was prolonged (from 118 to 128 minutes) when their trained subjects ingested a high-fat (90% of energy) versus a high-CHO (70% of energy) meal 4 hours prior to exercise. As no differences in total CHO oxidation were reported between trials (383 versus 362 g for the CHO and fat meals respectively), it is difficult to explain the prolonged exercise time in that study. Similarly, a study with a complicated design (CHO loading + meal 4 hours prior to exercise + jelly 1 hour prior to exercise) reported that running to exhaustion at marathon pace was enhanced when a higher fat meal was paired with a CHO-containing jelly (~100 min) than when a higher CHO or a high-fat meal was paired with a placebo jelly (~90 and 92 min respectively) (Murakami et al. 2012). However, there was no trial in which both pre-event intakes were high in CHO, the 'high-fat' meal and 'high-CHO' meals weren't substantially differentiated (75 g CHO and 60 g fat versus 138 g CHO and 23 g fat), and subjects consumed water only during the endurance run. Therefore, the scientific and practical implications of these results could be challenged.

15.6.3 Medium-chain triglyceride ingestion during exercise

Medium-chain triglycerides (MCTs) are fats composed of medium-chain FAs (MCFA) with a chain length of six to ten carbon molecules. They are digested and metabolised differently from the LCFA that make up most of our dietary fat intake. Specifically, MCTs can be digested within the intestinal lumen with less need for bile and pancreatic juices than long-chain triglycerides, with MCFAs being absorbed via the portal circulation. MCFAs can be taken up into the mitochondria without the need for carnitine-assisted transport. As such, interest has focused on the potential ergogenic effect of ingesting MCFA solutions during exercise as a fuel source for endurance and ultra-endurance events. Jeukendrup and colleagues (1995) reported that the co-ingestion of MCT with CHO during prolonged exercise increased the rate of MCT oxidation, possibly by increasing its rate of absorption; the maximum rate of MCT oxidation was achieved at around 120–180 minutes of exercise, with values of 0.12 g/min.

Table 15.1 summarises the findings of studies that have examined the effect of the co-ingestion of MCT and CHO on ultra-endurance performance; the results show poor support for the benefits of MCT feedings and appear to depend on the amount of MCT that can be ingested and the prevailing hormonal conditions. Only one study, which involved the intake of large amounts of MCT, raising plasma FFA concentrations and allowing glycogen sparing, has reported a performance benefit with prolonged exercise (Van Zyl et al. 1996). However, these metabolic benefits appear to be compromised when exercise is commenced with higher insulin levels, as is the case when athletes consume a CHO-rich pre-exercise meal according to sports nutrition guidelines (Goedecke et al. 1999b; Angus et al. 2000). Indeed, the majority of studies have shown that the ability of MCT feedings to increase fat oxidation during exercise is minor or shortlived at best (Angus et al. 2000; Goedecke et al. 1999b; 2005; Jeukendrup et al. 1998; Vistisen et al. 2003).

TABLE 15.1 Studies of medium-chain triglycerides supplementation and ultra-endurance performance (for updates see www.ausport.gov.au/ais/nutrition/supplements)

Study	Subjects	Treatments	Exercise protocol	Enhanced performance	Comments
Van Zyl et al. 1996	6 endurance-trained cyclists Crossover design	2 L of • 4.3% MCT • 10% CHO • 10% CHO + 4.3% MCT Total MTC intake = 86 g	Cycling • 2 h @ 60% VO_2 max + 40 km TT (~70 min)	Yes	MCT + CHO enhanced TT performance times (65.1 min) compared with CHO (66.8 min) and MCT alone (72.1 min). MCT + CHO increased plasma FA and fat oxidation, while reducing glycogen utilisation.
Jeukendrup et al. 1998	9 endurance-trained male cyclists/triathletes Crossover design	20 mL/kg of • 10% CHO • 10% CHO + 5% MCT • 5% MCT • Placebo Total MTC intake = 86 g	Cycling • 2 h @ 60% VO_2 max + TT (~15 min)	No	No difference between CHO, CHO + MCT or placebo (~14 min) but MCT alone impaired performance (17.3 min). MCT + CHO showed slightly higher fat oxidation than CHO alone. No glycogen sparing.
Goedecke et al. 1998b	9 endurance-trained male cyclists Crossover design	1.6 L of • 10% CHO • 10% CHO + 1.7% MCT • 10% CHO + 3.4% MCT Total MTC intake = 26 or 52 g	Cycling • 2 h @ 63% VO_2 max + 40 km TT (~70 min)	No	No differences in TT performance. Two subjects experienced gastrointestinal distress with higher MCT intake. Higher FFA with MCT but no change in CHO oxidation.
Angus et al. 2000	8 endurance-trained male cyclists/triathletes Crossover design	1 L per hour of • 6% CHO + 4% MCT • 6% CHO • Placebo Total MTC intake = 42 g/h or ~120 g	Cycling • 100 km TT (~3 hr)	No	CHO enhanced performance over placebo, but addition of MCT did not provide further benefits. Four subjects experienced gastrointestinal problems with MCT. No differences in fat oxidation or plasma FFA between MCT and CHO + MCT. Suppression of fat oxidation may be due to high exercise intensity or pre-trial CHO meal causing high insulin concentrations.

Vistisen et al. 2003	7 well-trained male cyclists Crossover design	• 2.4 g/kg CHO • 2.4 g/kg CHO + 1.5 g/kg MCT/LCFA Total MTC intake = 93–128 g over 4 h	Cycling • 3 h @ 55% VO_2 max + 800 kJ TT (~50 min)	No	No difference in TT performance between CHO and CHO + MCT mixture (50.8 ± 3.6 versus 50.0 ± 1.8 min, NS). Significantly greater fat oxidation during first hour of ride, but not significantly different thereafter, indicating only minor differences in substrate utilisation as a result of the treatment. No major gastrointestinal side effects during trial, but with CHO + MCT trial problems experienced next day.
Goedecke et al. 2005	8 male endurance-trained cyclists Crossover design	1 hour pre-exercise: • 75 g CHO • 32 g MCT During exercise: 600 mL/h of • 10% CHO • 10% CHO + 4.2% MCT Total MTC intake = 148 g over 6 h	Cycling • 4.5 h @ 50% PPO + 200 kJ TT (~15 min)	No—in fact impaired	No difference in substrate utilisation during submaximal exercise. Half the subjects experienced gastrointestinal side effects with MCT. Overall, TT performance compromised in MCT trial (12.36 versus 14.30 minutes)

Critical to the whole issue is the ability of subjects to tolerate the substantial amount of MCT oils required to have a metabolic impact. Jeukendrup and colleagues found that the gastrointestinal tolerance of MCT is limited to a total intake of about 30 g, which would limit its fuel contribution to 3–7% of the total energy expenditure during typical ultra-endurance events (Jeukendrup et al. 1995). When intakes of MCT are increased beyond this level, subjects report gastrointestinal reactions that range in severity from minor (Van Zyl et al. 1996) to performance impairing (Jeukendrup et al. 1998; Goedecke et al. 2005). Differences in gastrointestinal tolerance between studies or within studies may reflect differences in the mean chain length of MCTs found in the supplements, or increased tolerance in some athletes due to constant exposure to MCTs. The intensity and mode of exercise may also affect gastrointestinal symptoms. Overall, it is unlikely that the provision of MCTs during exercise can promote a sufficiently large increase in rates of fat oxidation to support glycogen sparing and performance benefits, and there is a substantial risk that gastrointestinal disturbances may actually impair performance.

15.6.4 L-carnitine supplementation

The carnitine pool in a healthy individual is about ~100 mmol, of which ~95% is found in skeletal and cardiac muscle. Over half of the daily requirements for carnitine is provided by dietary intake of meat, poultry, fish and some dairy products, with the remainder being synthesised from methionine and lysine. Vegetarians have a lower or absent intake of dietary carnitine and have lower muscle carnitine stores (Stephens et al. 2011). Muscle carnitine plays a role in the oxidation of LCFA by assisting their transport into the mitochondria (see section 15.3.3) but also in the metabolism of CHO at high exercise intensities via its stabilising role on flux through the tricarboxylic acid cycle (for review, see Stephens et al. 2007a). Hereditary and acquired conditions associated with carnitine deficiency result in TG accumulation in the skeletal muscles, insulin resistance, an impaired utilisation of FA and reduced exercise capacity. These pathological changes can normally be reversed by carnitine supplementation. It has been hypothesised that increased availability of L-carnitine by supplementary ingestion would be of significant benefit both to endurance athletes and to individuals wishing to decrease their levels of adipose tissue.

There have been a number of well-controlled studies examining the effects of carnitine supplementation on metabolism and athletic performance in both moderately trained individuals and well-trained athletes (for review, see Heinonen 1996; Brass 2000). The doses administered in carnitine supplementation studies have varied between 2 and 6 g/d, with the length of administration from 5 days up to 4 months. The consensus from this literature is that carnitine supplementation per se has no effect on patterns of fuel utilisation either at rest or during exercise, does not reduce body fat content and does not enhance sports performance. The lack of efficacy of carnitine supplementation has been explained by its failure to increase muscle carnitine stores; an outcome that would be needed for it to influence muscle metabolism (Stephens et al. 2007a).

Recent work has created a new interest in carnitine supplementation. Muscle carnitine content was shown to increase when carnitine was infused intravenously in conjunction with insulin, with the stimulatory effect on muscle carnitine uptake occurring within physiological insulin range (Stephens et al. 2007b). Further study showed that this effect could be achieved by feeding carnitine (3 g/d) with a CHO drink (90 g CHO); however, due to the poor availability of dietary carnitine (< 15% for doses of this amount), it was calculated that it would take ~ 100 days to increase the muscle carnitine pool by ~ 10% via this method (Stephens et al. 2007c). Indeed, a study involving 168 d of twice-daily supplementation with 1.4 g carnitine combined with 80 g CHO was found to increase muscle total carnitine by 21% (Wall et al. 2011). This was associated with detectable changes in fuel utilisation during exercise to match the hypothesised effects of enhanced carnitine activity: a sparing of muscle glycogen at low exercise intensities (50% VO_{2max}) and evidence of greater efficiency of the TCA cycle at higher (80% VO_{2max}) intensities in the form of increased activation of the pyruvate dehydrogenase (PDH) complex and reduced lactate production (Wall et al. 2011). Work completed during a subsequent 30-minute cycling task was 11% higher than at baseline testing in the supplemented group, while no change was observed in a control group who consumed CHO only. A more recent study from the same group reported that an increase in muscle carnitine achieved via a similar supplementation protocol of carnitine and CHO increased fat utilisation during low-intensity exercise and prevented the gain in body fat that was seen in the control group who ingested CHO alone, in excess of energy requirements (Stephens et al. 2013). Therefore, there is growing evidence that it may be possible to increase fat utilisation and other metabolic advantages via carnitine use, in line with its theoretical benefits.

In summary, further research on carnitine supplementation and its potential benefits for body composition management and sports performance is indicated. However, it seems that benefits are seen only under specific conditions that require long-term compliance with a supplementation protocol that may not be suitable for some individuals (twice-daily intake of a reasonably substantial amount of CHO). Therefore, the practical implementation of the use of this supplement requires careful thought.

15.7 Adaptation to low-carbohydrate, high-fat (LCHF) diets

In the period since the publication of the last edition of this text, there has been an enthusiastic renewal of interest in the LCHF diet with promotion via peer-reviewed literature (Phinney 2004; Brukner 2013; Noakes et al. 2014; Volek et al. 2015), lay publications (Volek & Phinney 2012) and the highly developed information network of social media (Brukner 2013; Olsen 2014). We have noted that this interest comes from several scientists who are both researchers and devotees of this diet, and that the commentary in the scientific literature by these authors is more measured than their claims circulated via less formal sources (Burke 2015, in press). Claims related to health benefits from the LCHF diet are outside the focus of this chapter and will not be reviewed. However, the current promotion of the LCHF diet to athletes features the following claims:

1. The current sports nutrition guidelines and practices of sports dietitians promote a high-CHO intake at all times for all athletes, regardless of their event or exercise program.
2. Even the leanest athlete has plentiful stores of body fat which, following adaptation to the LCHF diet, can fuel sporting activities better than the body's limited CHO stores, leading to a range of better health and functional outcomes including enhanced performance.
3. The optimal diet for the athlete involves ketogenic adaptation, or chronic adaptation to a CHO-restricted diet (<50 g/d CHO) with high fat intakes (>80% of energy). Additionally recommended characteristics include maintenance of moderate protein intake at ~15% of energy or ~1.5 g/kg/d, with the note that intake of protein should not exceed 25% of energy intake or ketosis will be suppressed, and the need to ensure adequate intake of sodium and potassium at 3–5 g/d and 2–3 g/d respectively.
4. The benefits of the LCHF diet come from the adaptation to high circulating levels of ketone bodies, which provide an additional fuel source for the brain and muscle as well as achieve other health and functional benefits. Although this diet is associated with a brief period of discomfort and poor exercise tolerance, after 2–3 weeks of adaptation these symptoms disappear.
5. With such chronic keto-adaptation, it is considered unnecessary to consume CHO during

exercise, or perhaps to consume it in small amounts.

6. Many elite athletes and recreational sportspeople have switched to the LCHF diet and have achieved sporting success and/or performance improvements.

We have also noted that the current promotion of the LCHF diet for athletic performance has occurred in the absence of any new, or indeed existing, research, to show clear benefits to the performance of mainstream sports or high-trained competitors (Burke 2015, in press). To address the current claims, this chapter will finish with a summary of the research that has investigated strategies for adaptation to a high-fat diet.

15.7.1 Chronic administration of LCHF diets

The consumption of a high-fat (>60% of energy intake), low-CHO (less than 20% of energy) diet for 1–3 days increases FA oxidation during submaximal exercise, but is associated with an impairment of exercise capacity and performance since the increase in fat utilisation is insufficient to counteract the loss of contribution from the diminished muscle glycogen stores (see Hawley et al. 1998). However, longer periods of adaptation to high-fat diets may induce adaptive responses to reverse some of the mitochondrial adaptations that favour CHO oxidation and 'retool' the working muscle to increase its capacity for FA oxidation (Lambert et al. 1997). Studies that have investigated the effect of prolonged periods of adaptation to an LCHF diet in endurance-trained individuals are summarised in Table 15.2.

TABLE 15.2 Effect of up to 28 days of adaptation to low-carbohydrate high-fat (LCHF) diet on performance of endurance-trained individuals

Athletes and study design	LCHF adaptation protocol	Performance protocol	Nutritional status/strategies for performance	Performance advantage with LCHF
Ketogenic diets: 80–85% fat, <50 g/d CHO, 15–20% protein				
Well-trained cyclists (n = 5 M) Crossover design with order effect: control diet first (Phinney et al. 1983)	7 d HC (57% CHO) then 28 d LCHF (fat = 85% E, CHO = <20 g/d)	Cycling TTE at 60% VO_{2max}	Overnight-fasted + no CHO intake during exercise	**No**—NS difference in TTE between trials (151 versus 147 min for LCHF and HC) Group data skewed by one participant who increased time to fatigue by 156% on LCHF trial (see Figure 15.5)
Moderately trained off-road cyclists (n = 8 M) Crossover design (Zajak et al. 2014)	28 d HC (CHO = 50% E) LCHF (fat = 70% E, CHO = 15%) *Note: diet described by authors as ketogenic, but may not have truly achieved this state*	Cycling VO_{2max} test	Not stated	**No**—mixed results with small increase in VO_{2max} (56 versus 59.2 mL/kg/min for HC and LCHF, p < 0.01) but reduction in maximum workload (350 versus 362 W, p = 0.037). Small favourable change in body composition with LCHF (loss of ~1.8 kg with body fat loss from 14.9% to 11.0% BM, p < 0.01)
Non-ketogenic diets:~60–65% fat, 15–20% CHO, 15–20% protein				

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Moderately trained cyclists (*n* = 7 F) Crossover design (O'Keefe et al. 1989)	7 d LCHF (fat = 59% E, CHO =1.2 g/kg BM) HC (CHO = 6.4 g/kg BM)	Cycling TTE at 80% VO_{2max}	3–4 h after meal, no CHO intake during exercise	**No**—in fact, performance deteriorated with LCHF. Time to exhaustion reduced by 47% on LCHF trial
Well-trained cyclists (*n* = 5 M) Crossover design (Lambert et al. 1994)	14 d LCHF (fat = 67% E, CHO = 17% Ea) HC (CHO = 74% Ea)	Cycling 30-s Wingate test + TTE at 90% VO_{2max} + TTE at 60% VO_{2max}	Overnight-fasted + no CHO intake during exercise	**No**—two higher intensity tests **Yes**—submaximal cycling. Time to exhaustion increased by 87% on LCHF trial commenced with lower glycogen stores due to preceding exercise
Well-trained cyclists (*n* = 16 M) Parallel group design (Goedecke et al. 1999a)	15 d LCHF (fat = 69% E, CHO = 2.2 g/kg BM) HC (CHO = 5.5 g/kg BM)	Cycling 150 min at 70% VO_{2max} + 40-km TT Performance measured at *t* = 0, 5, 10 and 15 d	MCT intake 1.5 h before event (~14 g); MCT (0.3 g/kg/h) and CHO (0.8 g/kg/h) during exercise	**No**—TT performance increased over time in both groups as a result of training protocol. Significant improvements seen in both groups by day 10, but no difference in mean improvement between groups. Important finding of study: adaptations achieved after only 5 days of high-fat diet
Well-trained cyclists (*n* = 7 M) Crossover design (Rowlands et al. 2002)	14 d LCHF (fat = 66% E, CHO = ~2.4 g/kg) HC (CHO = ~8.6 g/kg, 70% CHO)	Cycling 5 h including 15 min TT + 100 km TT	LCHF = high-fat pre-event meal HC = high-CHO pre-event meal Both: 0.8 g/kg/h CHO during ride	**Yes**—submaximal intensity exercise **No**—higher-intensity exercise. Relative to baseline: HC showed small NS decreases in performance of both 15-minute TT and 100-km TT. LCHF showed larger but NS decrease in performance of 15-minute TT but small NS improvement in 100-km TT
Well-trained duathletes (*n* = 11 M) Crossover design (Vogt et al. 2003)	5 w LCHF (fat = 53% E, CHO = ~3.6 g/kg) HC (CHO = ~6.9 g/kg, 68% CHO)	Cycling 40-min incremental protocol + 20 min TT @ ~89% VO_{2max} Running (separate day) Outdoor 21 km TT	LCHF = high-fat pre-event meal HC = high-CHO pre-event meal Intake before and during half marathon not stated	**No**—Self-selected work output similar for cycling TT in both dietary treatments (298 ± 6 versus 297 ± 7 W, NS) for LCHF and HC respectively. Half marathon time not different between trials (80 min 12 s ± 86 s versus 80 min 24 s ± 82 s, NS)

Source: Adapted from Burke 2015

Note: [a]g/kg intakes unavailable; BM = body mass, CHO = carbohydrate, E = energy, F = female, HC = high carbohydrate diet, LCHF = low-carbohydrate high-fat diet, M = male, NS = not significant, TT = time-trial, TTE = time to exhaustion, VO_{2max} = maximal oxygen uptake

Since the study most frequently cited in support of the LCHF and athletic performance is that of Phinney and colleagues (1983), a careful review of the design, results and interpretation of the outcomes of this work is warranted. Here, under careful monitoring in a metabolic ward, five well-trained cyclists were tested once after following a week of a CHO-rich diet (~57% of energy) and again following 28 days of a severe CHO-restricted (<20 g/d) diet with energy contributions of 85% fat and 15% protein. Ketosis was demonstrated with this protocol: blood concentrations of beta-hydroxybutyrate increase from <0.05 to >1 mmol after a week and were maintained thereafter. Despite negligible intake, resting muscle glycogen stores were not depleted but rather reduced to ~45% of values seen with the high-CHO diet (76 versus 140 mmol/kg wet weight muscle). The outcome measure of 'performance' was a cycling test to exhaustion at ~63% VO_{2max}, and the study claimed that the LCHF diet was able to maintain exercise capacity (see Figure 15.5). Assessment of substrate utilisation during this cycling test showed a four-fold reduction in muscle glycogen utilisation, a three-fold decrease in the contribution of blood glucose and an increase in lipid oxidation to make up substrate needs. Gluconeogenic contributions from glycerol released from triglyceride use as well as lactate, pyruvate and certain amino acids were assumed to prevent hypoglycaemia during exercise as well as allow glycogen storage between training sessions.

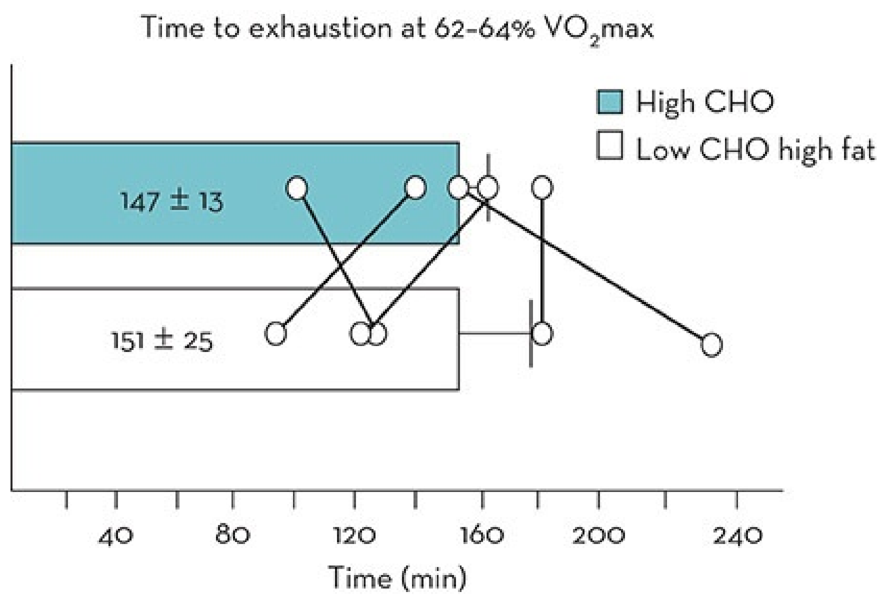

Figure 15.5 Exercise capacity (time to exhaustion at 62–64% VO_{2max}, equivalent to ~185 W) after 7 day of high-carbohydrate (CHO) diet followed by 28 days low-CHO high-fat diet. Data represent mean ± SEM from five well-trained cyclists (not significantly different) with individual data points represented by O

Source: Redrawn from Phinney et al. 1983

To avoid misinterpreting the results of the study, some direct quotes from the paper are provided. The researchers offered the following explanation of their findings: 'Metabolic adaptation to limit CHO oxidation can facilitate moderate submaximal exercise during ketosis to the point that it becomes comparable to that observed after a high CHO diet'. Furthermore, they noted that 'because muscle glycogen stores require many days for repletion, whereas even very lean individuals maintain appreciable calories stores as fat, there is potential benefit in this keto-adapted state for athletes participating in prolonged endurance exercise over two or more days'. However, they also commented on the results of VO_{2max} tests undertaken during each dietary phase with respect to the ketogenic diet: 'the price paid for the conservation of CHO during exercise appears to be a limitation of the intensity of exercise that can be performed … there was a marked attenuation of RQ value at VO_{2max} suggesting a severe restriction on the ability of subjects to do anaerobic work' (Phinney et al. 1983).

In a recent review of this study (Burke 2015, in press), we identified a number of limitations of the study itself and the way that it is currently being promoted. First, although the researchers were clear that their ketogenic diet did not enhance exercise capacity, noting at best that endurance at submaximal intensities was preserved at the expense of ability to undertake high-intensity exercise, the popular belief is that athletic performance was improved by the LCHF. Second, although excellent dietary control was achieved, little information was provided about the training program undertaken by the cyclists. It should be noted that the order effect in which all subjects undertook the LCHF trial after the high-CHO trial suggests that there was no mean benefit to performance associated with the additional 4 weeks of training. Finally, important information about this study is lost by the simple focus on the mean change in exercise capacity. The published findings of this study are largely skewed by the experience of a single subject who showed a large increase in endurance after the ketogenic diet and additional training (see Figure 15.5). Indeed, our review (Burke 2015) provided an alternative interpretation of the data using a magnitude-based inferences approach: the LCHF produced an unclear outcome, with the chances of a substantially positive, trivial and substantially negative outcome being 32%, 32% and 36%, respectively (Stellingwerff T, unpublished observations).

The other available studies of chronic (≥7 days) adaptation to an LCHF diet have involved a non-ketogenic diet in which CHO was restricted to less extreme levels (Table 15.2). In these studies, the rationale of the dietary treatment was to retool the muscle to increase fat utilisation, but to avoid ketosis and its confounding effect on the relationship between respiratory exchange ratio and substrate utilisation during exercise (Lambert et al. 1994). A range of adaptive responses to the LCHF diet was observed or confirmed in the trained individuals (summarised below), and their effects on exercise capacity/performance were tested under a range of different exercise scenarios and feeding strategies that would favour or reduce the benefit of enhanced fat oxidation (for review see Burke 2015). The variability in study design makes it difficult to make an all-encompassing assessment of the effect of LCHF on exercise, as is popularly desired. Although, theoretically, it could help to identify conditions under which adaptation to a high-fat diet may be of benefit or harm to sports performance, the small number of studies and small sample sizes prevents this opportunity from being fully exploited. Overall, however, no clear benefits from chronic adaptation to the LCHF diet can be found in the available literature.

In the meantime, attention is drawn to an important observation from one of the studies in which the time course of adaptation to the LCHF diet was investigated. Goedecke and colleagues (1999a) tracked changes in muscle fuel utilisation after 5, 10 and 15 days of exposure to such a diet and found that a substantial increase in fat oxidation and reduction in CHO utilisation was achieved by 5 days without further enhancement thereafter. This finding—that the trained muscle can be retooled to optimise muscle fat utilisation in a conveniently short period—led in part to the next phase of investigation in which attempts were made to separately optimise the muscle's capacity for lipid and CHO utilisation.

15.7.2 Fat adaptation and CHO restoration

It has been proposed that optimal performance of endurance/ultra-endurance sport might be achieved by undertaking a short-term adaptation to an LCHF diet followed by restoration of CHO for the competitive event. Such strategies are aimed at promoting simultaneous increases in fat and CHO availability and utilisation during exercise. Indeed, studies that have directly compared fuel utilisation during submaximal exercise after a fat adaptation protocol and then again after CHO restoration practices (e.g. CHO loading, a pre-event CHO-rich meal and CHO-intake during the event) have shown that the muscle re-tooling was robust enough to maintain an increase in fat utilisation during exercise despite CHO availability (see Table 15.3). Theoretically, high levels of CHO availability, coupled with an enhanced capacity to spare muscle glycogen use and increase the contribution of fat to muscle fuel use, could provide an advantage for the performance of endurance and ultra-endurance sports.

TABLE 15.3 Effect of adaptation (5–10 days) to high-fat low-carbohydrate diet (LCHF) followed by carbohydrate (CHO) restoration in trained individuals

Participant characteristics	LCHF adaptation protocol	CHO restoration	Performance protocol	Nutritional status/strategies for performance	Performance advantage with LCHF adaptation + CHO restoration
Well-trained cyclists/triathletes (*n* = 8 M) Crossover design (Burke et al. 2000)	5 d LCHF-adapt (fat = 68% E; CHO = 18% E, 2.5 g/kg BM) or HC (CHO = 74% E, 9.6 g/kg BM CHO)	1 d rest + high CHO (CHO = 75% E, 10 g/kg BM)	Cycling 120 min at 70% VO_{2max} + ~30-min TT (time to complete 7 J/kg BM)	Fasted + no CHO intake during exercise	**Perhaps** for individuals Two participants performed badly on HC trial, probably because of hypoglycaemia Plasma glucose better maintained on LCHF-adapt trial TT not significantly different between trials: 30.73 ± 1.12 versus 34.17 ± 2.62 minutes for LCHF and HC trial. However, mean difference in TT = 8% enhancement with LCHF trial (*p* = 0.21, NS; 95% CI = −6% to 21%)
Well-trained cyclists and triathletes (*n* = 8 M) Crossover design (Burke et al. 2002)	5 d LCHF-adapt (fat = 68% E; CHO = 18% E, 2.5 g/kg BM) or HC (CHO = 70% E, 9.3 g/kg BM CHO)	1 d rest + high CHO (CHO = 75% E, 10 g/kg BM)	Cycling 120 min at 70% VO_{2max} + ~30-min TT (time to complete 7 J/kg BM)	CHO intake 2 h before exercise (2 g/kg BM) and during exercise (0.8 g/kg/h)	**No**—Plasma glucose maintained in both trials due to CHO intake during exercise Difference in TT between trials was trivial: LCHF-adapt = 25.53 ± 0.67 minutes; HC = 25.45 ± 0.96 minutes (*p* = 0.86, NS). Mean difference in TT = 0.7% impairment with LCHF-adapt trial (95% CI = −1.7% to 0.4%)
Highly trained cyclists and triathletes (*n* = 7 M) Crossover design (Carey et al. 2001)	6 d LCHF-adapt (fat = 69% E; CHO = 16% E, 2.5 g/kg BM) or HC (CHO = 75% E, 11 g/kg BM)	1 d rest + high CHO (CHO = 75% E, 11 g/kg BM)	Cycling 240 min at 65% VO_{2max} + 60 min TT (distance in 1 hr)	CHO intake before exercise (3 g/kg BM) and during exercise (1.3 g/kg/h)	**No or perhaps for individuals** TT performance NS between trials: 44.25 ± 0.9 versus 42.1 ± 1.2 km for LCHF-adapt and HC trial. However, mean difference in TT performance = 4% enhancement with LCHF-adapt (*p* = 0.11, NS) (95% CI = –3% to 11%)
Highly trained cyclists and triathletes (*n* = 7 M) Crossover design (Noakes 2004)	5 d LCHF-adapt (fat = 69% E; CHO = 16% E, 2.5 g/kg BM) or HC (CHO = 75% E, 11 g/kg BM)	1 d rest + high CHO (CHO = 75% E, 11 g/kg BM)	Cycling 240 min at 65% VO_{2max} + 60-min TT (distance in 1 hr)	CHO intake before exercise (3 g/kg BM) and during exercise (1.3 g/kg/h)	**No**—Additional 6 subjects undertaken to test for Type 1 error in previous study (Carey et al. 2001) TT performance NS between trials: 42.92 ± 1.46 versus 42.94 ± 1.41 km for LCHF-adapt and HC trial (*p* = 0.98). Performance difference = 0.02 km or 0.1%
Trained cyclists	10 d LCHF-	3 d high	Cycling 150	MCT intake 1 h	**Yes**—Difference in TT performance =

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and triathletes (n = 5 M) Crossover design (Lambert et al. 2001)	adapt (fat = 65% E; CHO = 15% E, 1.6 g/kg BM) or HC (CHO = 53% E, 5.8 g/kg BM)	CHO (CHO = 65% E, 7 g/kg BM) + 1 day rest	min at 70% VO_{2max}+ 20-km (~30 min) TT	before event (~14 g); MCT (0.3 g/kg/h) and CHO (0.8 g/kg/h) during exercise	4% enhancement with LCHF-adapt: 29.35 ± 1.25 versus 30.68 ± 1.55 minutes for LCHF-adapt and HC ($p < 0.05$)
Well-trained cyclists (n = 7 M) Crossover design (Rowlands et al. 2002)	11.5 d LCHF-adapt (~2.4 g/kg, 15% CHO; 66% fat) or HC (CHO = ~8.6 g/kg, 70% E)	2.5 d high CHO (6.8 g/kg BM)	Cycling 5 h protocol including 15-min TT + 100-km TT	HCHO: High-CHO pre-event meal Both: 0.8 g/kg/h CHO during exercise	**Perhaps**—submaximal intensity exercise **No**—higher intensity exercise Relative to baseline testing: HC trial showed small NS decrease in performance of both 15-minute TT and 100-km TT. LCHF-adapt showed no change in 15-minute TT but small NS enhancement of 100-km TT
Well-trained cyclists (n = 8 M) Crossover design (Havemann et al. 2006)	6 d LCHF-adapt (fat = 68% E; CHO = 17% E, 1.8 g/kg BM) or HC (CHO = 68% E, 7.5 g/kg BM)	1 d rest + high CHO (8–10 g/kg)	Cycling 100 km TT, including 4 × 4 km sprints + 5 × 1 km sprints	CHO consumed during ride	**No**—in fact, performance enhancement of 1 km sprints Differences between 100-km TT performances NS (156 min 54 s versus 153 min 10 s for LCHF-adapt vs. HC). Difference between power output during 4-km sprints NS. However, power during 1-km sprints (undertaken at >90% PPO) was significantly reduced in LCHF-adapt trial

Source: Adapted from Burke 2015, in press

Note: All values are mean ± SEM; BM = body mass, CHO = carbohydrate, CI = confidence interval, E = energy, HC = high carbohydrate, M = male, MCT = medium-chain triglyceride, NS = not significantly different, PPO = peak power output, TT = time trial, VO_{2max} = maximal oxygen uptake

As discussed in the previous section, there is a range of permutations and combinations of dietary strategies and exercise protocols that can be investigated in combination with the fat adaptation and CHO restoration strategies to test the effect of such dietary periodisation on exercise capacity/performance. The available literature (Table 15.3) shows that although some of these scenarios might benefit more than others from enhanced capacity for fat utilisation during exercise, there is no clear evidence that fat adaptation strategies enhance exercise performance. Within this group of investigations, however, only one fully published study (Havemann et al. 2006) has employed an exercise test that bears any real resemblance to a sporting competition. The characteristics of this protocol include a sole focus on performance rather than a hybrid of metabolism and performance, and the need for participants to self-pace a protocol interspersing passages of high-intensity exercise against a background of moderate-intensity work; this reflects the stochastic profile of many real-life events. Again, this study merits special reflection before a general summary of the literature is provided.

In a crossover design, well-trained cyclists completed either a 6-day LCHF diet followed by a 1-day high-CHO diet, or 7 days of a high-CHO diet, before undertaking a laboratory-based cycling protocol designed to test some of the features of endurance sporting events (Table 15.3). Specifically, cyclists were required to undertake a 100-km self-paced cycling trial, during which there was a series of 4-km sprints undertaken at ~78–84% peak power output and 1-km sprints undertaken at >90% peak power output. Overall, differences in the performance of the 100-km time trial were not statistically significant, although the mean time for high CHO trial was 3 minutes 44 seconds or ~2.5% faster than the LCHF-adapted trial (153 min 10 s versus 156 min 54 s, $p = 0.23$). While there was no difference with regard to the 4-km sprint times, performance of the 1-km sprints was significantly impaired in the LCHF-adapted trial in all subjects, including the three subjects who were faster overall for the 100 km in this LCHF-adapted trial. The authors stated that although adaptation to the LCHF diet followed by CHO restoration increased fat oxidation during exercise, 'it reduced high-intensity sprint power performance, which was associated with increased muscle recruitment, effort perception and heart rate' (Havemann et al. 2006).

Although the mechanisms associated with the compromised performance in this study were unclear, speculations by the authors included 'increased sympathetic activation, or altered contractile function and/or the inability to oxidise the available CHO during the high intensity sprints'. Indeed, evidence for this latter suggestion was provided by our own laboratory in another study of the LCHF-adaptation + CHO restoration protocol (Stellingwerff et al. 2006). In examining changes within the muscle in response to this dietary strategy, we found an explanation for the reduction in glycogenolysis during exercise in the form of a reduction in the active form of pyruvate dehydrogenase (PDHa)—the enzyme complex responsible for linking the glycolytic pathway with the citric acid cycle. This study provided evidence of glycogen 'impairing' rather than 'sparing' in response to adaptation to an LCHF diet and a robust explanation for the impairment of key aspects of exercise performance as a result of this dietary treatment. As a result of these two studies, this author contributed to an editorial to 'wrap up' the interest in adaptation to LCHF diets for sports performance, with the conclusion that this strategy had little to offer competitive athletes in mainstream sports where performance required the ability to undertake high-intensity exercise throughout or at crucial parts of the event (Burke & Kiens 2006). Indeed, this author and other researchers moved on to other strategies of dietary periodisation and sports nutrition practice that might promote useful metabolic adaptations without the penalties associated with the LCHF diet.

15.7.3 Addressing the current promotion of the LCHF diet for athletes

In view of the enthusiastic promotion of LCHF diets in the present sports nutrition environment, this author has prepared a review of the existing research literature on fat adaptation protocols seen through the prism of knowledge that has evolved in other areas of sports nutrition as well as to address the claims made by key proponents of the LCHF 'revolution' (Phinney 2004; Volek and Phinney 2012; Noakes et al. 2014; Volek et al. 2015). A summary of this review (Burke 2015) is provided below:

1. Exposure to an LCHF diet causes key adaptations in the muscle in as little as 5 days to retool its ability to oxidise fat as an exercise substrate. Adaptations include, but are not limited to, an increase in IMTG stores, increased activity of enzymes which mobilise

triglycerides in muscle and adipose tissue and increases in key fat transport molecules (for extended review, see Spriet 2014). Together, these adaptations further increase the already enhanced capacity of the aerobically trained muscle to utilise endogenous and exogenous fat stores to support the fuel cost of exercise of moderate intensity. Rates of fat oxidation during exercise may be doubled by fat adaptation strategies.

2. The adaptive responses in the muscle to the LCHF diet are sufficiently robust that they persist in the face of at least 36 hours of aggressive dietary strategies to increase CHO availability during exercise (e.g. glycogen supercompensation, pre-exercise CHO intake, high rates of CHO during exercise). The time course of the 'washout' of retooling is unknown.
3. In addition to up-regulating fat oxidation at rest and during exercise, exposure to an LCHF diet down-regulates CHO oxidation during exercise, with the major effect being seen in a decrease in muscle glycogen use. The reduction in glycogen use is robust and independent of substrate availability; it is explained at least in part by a down-regulation of PDH activity, which represents a decrease in metabolic flexibility.
4. Despite the enhanced capacity for utilisation of a relatively limitless fuel source as an exercise substrate, fat adaptation strategies with or without restoration of CHO availability do not appear to enhance exercise capacity or performance per se. It is possible that the failure to observe benefits is due to type II statistical errors: failure to detect small but important changes in performance due to small sample sizes, individual responses and poor reliability of the performance protocol. While this explanation often looks attractive, in some cases, further exploration and enhanced sample size increases confidence in the true absence of a performance enhancement (Noakes 2004).
5. In the interest of objectivity, it is important to consider scenarios under which fat adaptation strategies may be beneficial to sports performance or other reasons to explain the testimonials of athletes who claim to have benefited from an LCHF diet (see Table 15.4). With specific regard to anecdotal reports of performance benefits, it is equally important to recognise that in many situations, closer scrutiny of the dietary practices of these individuals shows that they do not actually adhere to the principles of the LCHF diet (Edmond 2014).
6. The experience of athletes, at least in the short-term exposure to LCHF diets, is of a reduction in training capacity and increase in perceived effort, heart rate and other monitoring characteristics, particularly in relation to high-intensity/quality training which plays a core role in a periodised training program.
7. Fat adaptation strategies may actually impair exercise performance, particularly involving shorter high-intensity events or high-intensity phases during a longer event that require power outputs or intensities of 85–90% maximum level or above. This is likely to be due to the impairment of the muscle glycogen utilisation needed to support high work rates, even in scenarios where strategies to achieve high CHO availability are employed. As such, these strategies are not recommended to competitive athletes undertaking mainstream sport where high-intensity exercise makes up a large part or a crucial element

of successful performance.

8. The current claims of key proponents of the LCHF diet, particularly in less formal environments than the peer-reviewed literature, include items that are both untrue and unfair. This includes the misrepresentation of current sports nutrition guidelines as promoting high-CHO diets for all athletes at all times, as well as a misunderstanding of the metabolic demands of sporting events, at least for high-level competitors. This is not conducive to collaborative efforts to move forward sports nutrition knowledge and practice.
9. There is a need to undertake further research into the LCHF diet or other fat adaptation strategies to resolve the current debate. In particular, there has been little investigation of the currently promoted version of the LCHF: the ketogenic (<50 g/d CHO), high fat (80–85% of energy) diet. Furthermore, this author has argued that the current literature has failed to include the optimal ‘control’ diet (or additional intervention) in the form of the diet with periodised CHO availability (see Chapter 14). Future studies should investigate various LCHF strategies in comparison to this evolving model, to determine the best approaches for different individuals, different goals and preparation for different sporting events.
10. This author and others continue to undertake research to evolve and refine the understanding of conditions in which low CHO availability can be tolerated or actually beneficial (see Chapter 14. However, we also recognise that the benefits of CHO as a substrate for exercise across a larger range of exercise intensities, the better economy of CHO oxidation versus fat oxidation (ATP produced per L of oxygen combusted), and the potential CNS benefits of mouth sensing of CHOs (Chapter 13) can contribute to optimal sporting performance and should not be shunned.

TABLE 15.4 Scenarios or explanations for testimonials/observations of enhanced performance following change to a low-carbohydrate, high-fat (LCHF) diet

Scenarios favouring adaptation to LCHF diet	Other explanations for anecdotal reports of performance benefits as a result of switching to LCHF diet
• Individuals or events involving prolonged low-moderate exercise where there is no benefit or requirement for higher intensity efforts • Scenarios in which an event must be commenced with low muscle glycogen content • Individuals or events in which it is difficult to consume adequate CHO to meet goals for optimal CHO availability (e.g. GI upsets, logistical difficulties with accessing supplies during the event) • Individuals who are CHO-sensitive and likely to be exposed to low CHO availability	• Switch to LCHF has been associated with loss of body fat and increase in power:mass ratio • Previous diet and training were sub-optimal and switch has been associated with greater training and diet discipline • Order effect: natural progress in training and maturation in age and sporting experience • Previous program did not include accurate measurement of performance: awareness of performance metrics just commenced • Placebo effect/excitement about being part of new idea/culture • Misunderstanding of the LCHF diet or inability to practise its extreme requirements such that it is actually not being followed

Source: Burke 2015, in press

Summary

Many nutritional strategies have been employed in an attempt to promote FA oxidation, attenuate the rate of utilisation of endogenous CHO stores, and thereby enhance athletic performance. While there are some new and emerging options that still require further exploration (e.g. carnitine and CHO supplementation) others have been well investigated and show no clear evidence of benefit to sports performance (e.g. pre-exercise fat intake). A key area of interest in terms of chronic preparation lies in better interaction of exercise and nutrition in the form of the periodised training diet in which different training sessions are deliberately exposed to low or high CHO availability according to the goals of the session (see Chapter 14).

There is currently insufficient scientific evidence to recommend that athletes undertake the LCHF diet or other fat-loading strategies, despite the resurgence of interest in these strategies. Indeed, there is evidence that chronic restriction of CHO can impair competitive performance by interfering with the athlete's ability to undertake high-quality training and reducing the muscle's ability to utilise its CHO stores to support high-intensity events or pieces within competitive events. It is unfortunate that proponents of the LCHF diet promote the choice of one fuel source over the other, and a 'black versus white' vision of sports nutrition (Burke 2015, in press). The role of the sports dietitian is to integrate and individualise the range of dietary factors that can contribute to optimal sports performance.

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CHAPTER SIXTEEN

Supplements and sports foods

Louise Burke and Louise Cato

16.1 Introduction

Supplement use is a widespread and accepted practice among athletes, with a high prevalence of use of a large range of different types and brands of products. Such observations are illustrated by the results of a study of 77 elite Australian swimmers (Baylis et al. 2001), which found that 94% of the group reported the use of supplements in pill and powder form. When the use of specialised sports foods such as sports drinks was also taken into account, 99% of swimmers reported supplement use and a total of 207 different products was identified. According to other studies, supplement use is also widespread among athletes at high school and collegiate levels (Massad et al. 1995; Krumbach et al. 1999; Froiland et al. 2004).

16.2 Overview of supplements and sports foods

'Dietary supplements', 'nutritional ergogenic aids', 'sports supplements', 'sports foods' and 'therapeutic nutritional supplements'—these are some of the terms used to describe the range of products that collectively form the sports supplement industry. Just as there is a variety of names for these products, there is a variety of definitions or classification systems. Characteristics that can be used to categorise supplements include:

- function (e.g. muscle-building, immune-boosting, fuel-providing, to address medical issues)
- form (e.g. pills, powders, foods or drinks)
- availability (e.g. over-the-counter, mail order, Internet, multi-level marketing)
- scientific merit for claims (e.g. well-supported, unsupported, undecided).

For the purposes of this chapter, we will discuss supplements and sports foods that meet one or more of the following definitions:

- They provide a convenient and practical means of meeting a known nutrient requirement to optimise daily training or competition performance (e.g. a liquid meal supplement, sports drink, carbohydrate gel, sports bar).
- They contain nutrients in large quantities to treat a known nutritional deficiency (e.g. an

iron supplement).

- They contain nutrients or other components in amounts that claim to directly enhance sports performance or maintain/restore health and immune function—scientifically supported or otherwise (e.g. caffeine, creatine, glycerol, ginseng).

There is an ever-increasing range of supplements and sports foods readily accessible to athletes and coaches. It is of primary importance for sports dietitians to have a thorough working knowledge of the various sports foods and supplements in order to provide sound advice about appropriate situations of use, possible benefits, potential side effects and risks associated with use.

16.3 Regulation of supplements and sports foods

The regulation of supplements and sports foods is a contentious area and encompasses issues of manufacture, labelling and marketing. In addition to the concerns about efficacy and safety for the general consumer, athletes face the problem of contamination with prohibited substances, leading to a positive doping offence. There is no universal system of regulation of sports foods and supplements. Countries differ in their regulatory approach, with some involving a single government body (such as the Food and Drug Administration—FDA—in the USA), while others fall under several government agencies (e.g. Food Standards Australia New Zealand for food-based products and the Therapeutic Goods Administration for pill-based products in Australia).

Athletes need to have a global understanding of the regulation of dietary supplements, since regular travel and modern conveniences such as mail order and the Internet provide easy access to products that fall outside the scrutiny of their own country's system. Although it is outside the scope of this chapter to review the various regulatory issues in different countries, the important changes that can be attributed to the *Dietary Supplement Health and Education Act 1994* in the USA are worthy of special mention. This Act reduced the regulation of supplements and broadened the category to include new ingredients, such as herbal and botanical products, and constituents or metabolites of other dietary supplements. As a result, new groups of products have flooded the USA and international market: these include a variety of stimulants as well as 'pro-hormones' (compounds including androstenedione, DHEA, 19-norandrostenedione and other metabolites found in the steroid pathways that can be converted in the body to testosterone or the anabolic steroid nandrolone—see Blue & Lombardo 1999). These products will be discussed later, in the context of doping and inadvertent doping outcomes. The other important outcome of the *Dietary Supplement Health and Education Act 1994* was to shift responsibility from the supplement manufacturer to the FDA to enforce safety and claim guidelines. Since the passing of the Act, good manufacturing practice has not been enforced within the supplement industry, leaving non-compliant products or manufacturers to flourish unless there is specific intervention by the FDA.

Athletes and coaches often fail to understand that, in the absence or minimisation of rigorous government evaluation, the quality control of supplement manufacture is trusted to supplement companies. Large companies that produce conventional supplements such as vitamins and minerals, particularly to manufacturing standards used in the preparation of pharmaceutical products, are likely to achieve good quality control. This includes precision with ingredient levels and labelling, and avoidance of undeclared ingredients or contaminants. However, this does not appear to be true for all supplement types or manufacturers, with many examples of poor compliance with labelling laws (Gurley et al. 1998; Parasrampuria et al. 1998; Hahm et al. 1999) and the presence of contaminants and undeclared ingredients (see sections 16.4.4 and 16.4.5).

Although manufacturers are not meant to make unsupported claims about health or performance benefits elicited by supplements, product advertisements and testimonials show ample evidence that this aspect of supplement marketing is unregulated and exploited. For example, a survey of five issues of bodybuilding magazines found 800 individual performance claims for 624 different products within advertisements (Grunewald & Bailey 1993). It is easy to see how enthusiastic and emotive claims provide a false sense of confidence about the products. Most consumers are unaware that the regulation of such advertising is generally not enforced. Therefore, athletes are likely to believe that claims about supplements are medically and scientifically supported, simply because they believe that untrue claims would not be allowed to exist.

16.4 The pros and cons of using supplements and sports foods

The decision by an athlete to use a supplement or sports food should be made after careful consideration of several issues. Figure 16.1 provides an overview of important questions that should be answered regarding the safety, efficacy and legality of any product. It also characterises the balance between the arguments for and against the use of the product. The potential for both positive and negative outcomes will now be discussed in greater detail.

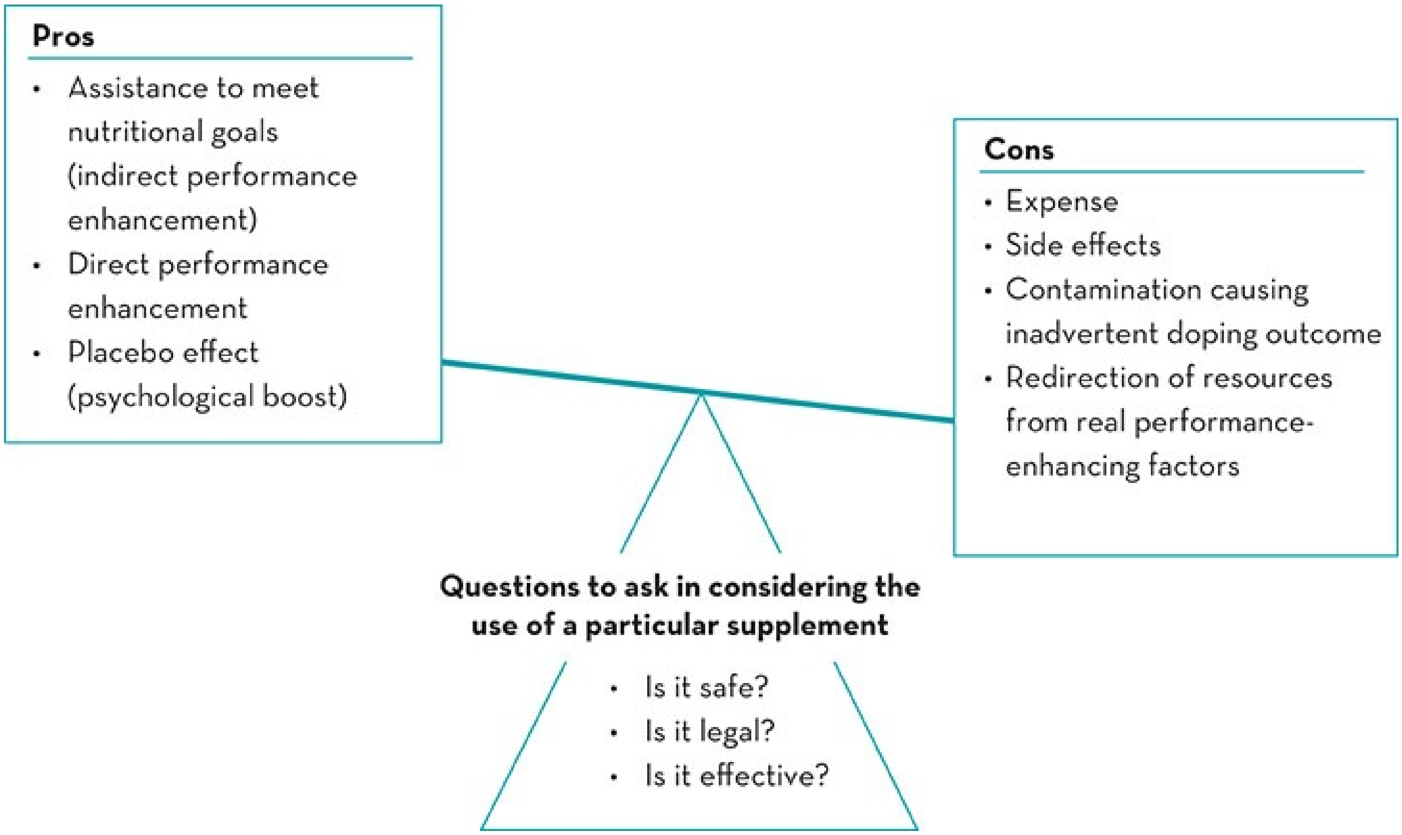

Figure 16.1 Issues to consider in the decision to take a supplement

16.4.1 Pros: true performance benefits

Some supplements and sports foods offer real advantages to athletic performance. Some products 'work' by producing a direct performance-enhancing (ergogenic) effect. Other products can be used by athletes to meet their nutrition goals and, as an indirect outcome, allow the athlete to achieve optimal health, recovery and performance. In some cases, these effects are so well known and easily demonstrated that beneficial uses of sports foods or supplements are clear-cut. For example, many studies support the benefits of consuming sports drinks to supply carbohydrate (CHO) and fluid during exercise (see Coombes & Hamilton 2000; Stellingwerff & Cox 2014). But even when indirect nutritional benefits or true ergogenic outcomes from supplement use are small, they are often worthwhile in the competitive world of sport (Hopkins et al. 1999). Of course, athletes need to be aware that it is the use of the product as much as the product itself that leads to the beneficial outcome. Therefore education about specific situations and strategies for the use of supplements and sports foods is just as important as the formulation of the product.

16.4.2 Pros: the placebo effect

Even in scenarios where a sports food does not produce a true physiological or ergogenic benefit, an athlete might attain some performance benefit because of a psychological boost or ‘placebo’ effect. The placebo effect describes a favourable outcome arising simply from an individual’s belief that they have received a beneficial treatment. In a clinical environment, a placebo is often given in the form of a harmless but inactive substance or treatment that satisfies the patient’s symbolic need to receive a ‘therapy’. In a sports setting, an athlete who receives enthusiastic marketing material about a new supplement or hears glowing testimonials from other athletes who have used it is more likely to report a positive experience.

Despite our belief that the placebo effect is real and potentially substantial, surprisingly few studies have tried to document this effect in relation to sport. In one investigation, weightlifters who received saline injections that they believed to be anabolic steroids increased their gains in lean body mass (BM) (Ariel & Saville 1972). Another investigation in which athletes were given either a sports drink or a sweetened placebo during a 1-hour cycling time trial found that performance was affected by the information provided to the subjects (Clark et al. 2000). The placebo effect caused by thinking they were receiving a CHO drink allowed the subjects to achieve a small but worthwhile increase in performance of 4%. Being unsure of which treatment was being received increased the variability of performance, illustrating that the greatest benefits from supplement use occur when athletes are confident they are receiving a useful product. Another fascinating study found that performance of a 10-km cycling time trial was altered in a dose–response relationship according to subjects’ beliefs about the amount of caffeine that they had consumed prior to their performance (Beedie et al. 2006). Although no caffeine was consumed in any of three experimental trials, mean power outputs were reduced by 1.3% compared to baseline testing in a trial in which subjects believed they had received a placebo treatment, while when they believed they had consumed 4.5 g/kg and 9.0 g/kg doses of caffeine, mean power increased by 1.3 and 3.1% respectively.

Additional well-controlled studies are needed to better describe the potential size and duration of the placebo effect and whether it applies equally to all athletes and across all types of performance. In the meantime, we can accept that the placebo effect exists and may explain, at least partially, why athletes report performance benefits after trying a new supplement or dietary treatment. Further reading on the available literature on the placebo effect and sports performance is provided in reviews by Beedie and Foad (2009) and Halson and Martin (2013).

16.4.3 Cons: expense

An obvious issue with supplement use is the expense, which in extreme cases can equal or exceed the athlete’s weekly food budget. Such extremes include the small number of athletes identified in many surveys (Baylis et al. 2001) who report a ‘polypharmacy’ approach to supplements, identifying long lists of products that often overlap in ingredients and claimed functions. For example, Australian 400-m runner Cathy Freeman lodged documents during a

Supreme Court case against her ex-manager, showing she spent A$3480 on supplements from health food shops in the 4 months leading up to her 2000 Sydney Olympic Games Gold medal (Magnay 2002). However, even a targeted interest in a small number of supplements can be expensive: the cost of some individual products can exceed A$50 per week to achieve the manufacturer's recommended dose or the amounts found to have a true ergogenic outcome in scientific studies. The issue of expense is compounded for teams and sports programs that have to supply the needs of a group of athletes.

Expense must be carefully considered when there is little scientific evidence to support a product's claims of direct or indirect benefits to athletic performance. But even where benefits do exist, cost is an issue that athletes must acknowledge and prioritise appropriately within their total budget. Supplements or sports foods generally provide nutrients or food constituents at a price that is considerably higher than that of everyday foods. At times, the expense of a supplement or sports food may be deemed money well spent, particularly when the product provides the most practical and palatable way to achieve a nutrition goal, or when the ergogenic benefits have been well documented. On other occasions, the athlete may choose to limit the use of expensive products to the most important events or training periods. There are often lower-cost alternatives to some supplements and sports foods that the budget-conscious athlete can use on less critical occasions; for example, a fruit smoothie fortified with milk powder or a commercial liquid meal replacement product is a less expensive choice to supplement energy and protein intake than most 'bodybuilder' products.

16.4.4 Cons: side effects

Since most supplements are considered by regulatory bodies to be relatively safe, in some countries there are no official or mandatory accounting processes to document adverse side effects arising from the use of these products. Nevertheless, information from medical registers (Perharic et al. 1994; Kozyrskyj 1997; Shaw et al. 1997) shows that while the overall risk to public health from the use of supplements and herbal and traditional remedies is low, there are cases of side effects including allergic reactions to some products, overexposure as a result of self-medication and toxicity due to contaminants. During the 1980s, deaths and medical problems resulted from the use of tryptophan supplements (Roufs 1992); products containing ephedra were a later source of medical concerns, sometimes causing deaths in susceptible individuals, leading to a ban by the US FDA in 2004 (Phillips 2004). Most recently, serious health incidents including deaths have been associated with the use of supplements containing the stimulant dimethylhexanamine (also known as DMAA) (Forrester 2013), while popular weight-loss supplements, hydroxycut and oxy-elite pro, have been implicated in cases of severe liver toxicity (CDC 2013). Many reports call for better regulation and surveillance of supplements and herbal products, and increased awareness of potential hazards (Perharic et al. 1994; Kozyrskyj 1997; Shaw et al. 1997).

16.4.5 Cons: doping outcomes

A number of ingredients found in supplements are prohibited by the anti-doping codes of the World Anti-Doping Agency (WADA) and other sports bodies. These include pro-hormones (steroid-related compounds such as androstenedione, DHEA and 19-norandrostenedione), stimulants (e.g. ephedrine, DMAA, sibutramine) and 'peptide hormones' (e.g. growth hormone releasing peptide 2 – GHRP2). Although some of these substances may be prohibited from sale in countries such as Australia, they can be bought as over-the-counter products in other locations such as the USA. Drug education programs highlight the need for athletes to read the labels of supplements and sports foods carefully to ensure that they do not contain such banned substances. This is a responsibility that athletes must master to reduce the risk of an 'inadvertent' anti-doping rule violation (ADRV). However, it is challenged in many cases by a general lack of urgency attributed to the perception that supplements are 'natural' and by the inability of athletes to recognise the different chemical and street names that can be used to describe various prohibited substances.

However, even when athletes take such precautions, inadvertent intake of banned substances from supplement products can still occur. This is because some supplements contain banned products without declaring them as ingredients; this is typically a result of contamination or poor labelling within lax manufacturing processes, although there have been cases where therapeutic doses of steroid compounds have been detected in supplements suggesting a deliberate strategy by manufacturers to gain 'street popularity' when testimonials of the favourable results of use are circulated (Geyer et al. 2008; Judkins & Prock 2012).

Credible evidence of product contamination was first uncovered by a study carried out by a laboratory accredited by the International Olympic Committee (Geyer et al. 2004). This study analysed 634 supplements from 215 suppliers in thirteen countries, with products being sourced from retail outlets (91%), the Internet (8%) and telephone sales. None of these supplements declared pro-hormones as ingredients, and they came from manufacturers that produced other supplements containing pro-hormones as well as companies that did not sell these products. Ninety-four of the supplements (15% of the sample) were found to contain hormones or pro-hormones that were not stated on the product label. A further 10% of samples provided technical difficulties in analysis such that the absence of hormones could not be guaranteed. Of the 'positive' supplements, 68% contained pro-hormones of testosterone, 7% contained pro-hormones of nandrolone, and 25% contained compounds related to both. Forty-nine of the supplements contained only one steroid, but forty-five contained more than one, with eight products containing five or more different steroid products. According to the labels on the products, the countries of *manufacture* of all supplements containing steroids were the USA, The Netherlands, the UK, Italy and Germany; however, these products were *purchased* in other countries. In fact, 10–20% of products purchased in Spain and Austria were found to be contaminated. Just over 20% of the products made by companies selling pro-hormones were positive for undeclared pro-hormones, but 10% of products from companies that did not sell steroid-containing supplements were also positive. The brand names of the 'positive' products were not provided in the study, but included were amino acid supplements, protein powders, and products containing creatine, carnitine, ribose, guarana, zinc, pyruvate, HMB, *Tribulus terrestris*, herbal extracts and vitamins/minerals. It was noted that a positive urinary test for nandrolone metabolites occurs in the hours following uptake of as little of 1 μg of nandrolone pro-hormones. The positive supplements contained steroid concentrations ranging from 0.01–190 μg per gram of product.

Since then, further reports of poor product labelling, deliberate adulteration and contamination have continued to emerge from independent studies (Geyer et al. 2008) and from reports by commercial companies that undertake specific audits of supplement products (Judkins et al. 2007). In some cases, the data suggest that the risks of product contamination have reduced, while in others there is evidence that there is merely a change in the prohibited ingredients involved. In the earliest instances, pro-hormone substances seemed to provide the greatest risk of the 'inadvertent' ADRV via supplement use, with a positive test for the steroid nandrolone being one of the possible outcomes. 'Body building' supplements promoting mass gain are the most likely products to contain these substances. More recently, stimulants have become the most prevalent supplement-related cause of ADRV, with many National Anti-Doping Agencies reporting an increase in cases of use of products such as DMAA (Outram 2015). Stimulant contamination or mislabelling is most likely to occur in products targeting weight loss or promoted as 'energy boosters', particularly in preparation for an exercise session ('pre-workout supplements').

Of course, there is no way to gather evidence to support an athlete's claim that the detection of banned substances via a urine test can be attributed to their unknowing ingestion of the product via a supplement. Even in cases where subsequent testing of a supplement has shown that it does contain prohibited substances, this does not prove that it was the only source of banned substances used by the athlete or that it was consumed inadvertently. Indeed, anti-doping codes involve strict liability clauses in which the athlete assumes responsibility for adverse analytical findings and can expect sanctions to apply. In various real-life cases, actual sanctions have varied from a suspended sentence to full imposition of a ban from participation in sport, differing according to the sporting authority/organisations involved, further referrals to courts of arbitration, and the detection of other mitigating circumstances. Regardless of the sanction, the athlete will carry an indelible reputation of having incurred an ADRV; even in cases where athletes have received large financial compensation from supplement manufacturers or reduced periods of suspension from their sporting activities, they have stated that the sullying of their reputation has caused overwhelming personal distress and loss of income. Therefore, the risk of an ADRV from supplement use is of major concern to sport.

It should also be pointed out that many people falsely believe that this problem only applies to elite or professional athletes, and in particular, that levels of sporting competition in which there is no/little monitoring of urine/blood are free from the risk of ADRV. In fact, anti-doping rules apply to all registered members/competitors of sporting organisations who are national signatories to an anti-doping code. While the risk of testing of biological samples to receive an ‘analytical positive’ may be negligible in lower-level competition, the recent WADA anti-doping codes have increased the number of instances in which other evidence of intended or actual use of prohibited substances can be used to make a case for an ADRV. This includes the purchase or importation of supplements containing banned substances by athletes, and there have been cases where the seizure of such goods by Customs authorities have led to an ADRV and doping sanctions applied to athletes in amateur and non-elite grades of sport (ASADA 2012).

Athletes should obtain specific information about the risks involved with supplement use in their environment. Many resources are provided by anti-doping agencies or companies who undertake third-party auditing of supplements and their manufacturers to educate or inform athletes about the risks of consumption of banned substances via supplement use (e.g. see www.usada.org/substances/supplement-411/, www.informed-sport.com, www.dopingautoriteit.nl/nzvt/nzvt-english). Risk reduction involves recognising the problem; identifying higher-risk products and manufacturers; and, in the case where a cost–benefit analysis justifies the use of a supplement or sports food, choosing a product that is inherently low risk or has been submitted to a batch testing program. While no supplement or sports food can be classified as entirely risk free, appropriate choices can help to minimise the risk to an acceptably low level. Further information can be found at the websites provided above.

16.4.6 Cons: displacement of real priorities

A more subtle outcome of reliance on supplements is the displacement of the athlete’s real priorities. Successful sports performance is the product of superior genetics, long-term training, optimal nutrition, adequate sleep and recovery, state-of-the-art equipment and a committed attitude. These factors cannot be replaced by the use of supplements, but often appear less exciting or more demanding than the enthusiastic and emotive claims made for many supplements and sports foods. Athletes can sometimes be side-tracked from the true elements of success in search of short-cuts from bottles and packets. Most sports dietitians are familiar with individual athletes who are reliant on supplements while failing to address some of the basic elements of good training and lifestyle.

16.4.7 Cons: special issues for young athletes

Success in sports involves obtaining an ‘edge’ over the competition, and children and adolescents may be uniquely vulnerable to the lure of supplements. The pressure to ‘win at all costs’, extensive coverage in lay publications and hype from manufacturers with exciting and emotive claims all play a role in the use of supplements by young athletes. The knowledge that famous or successful athletes and other role models use or promote supplements and sports foods adds to the allure.

An array of ethical issues arises in the consideration of supplement use by young athletes, including all the factors previously outlined in this section. Displaced priorities and the failure to build a foundation of sound training, diet and recovery strategies are particularly important since a long-term career in sport is underpinned by such an investment. The lack of information about the long-term safety of ingesting various compounds on a growing or developing body is a special concern.

Various expert groups have made strong statements against the use of supplements by young athletes. For example, a policy statement of the American Academy of Pediatrics (2005) on the use of performance-enhancing substances condemns the use of ergogenic aids, including various dietary supplements, by children and adolescents. Furthermore, the American College of Sports Medicine recommends that creatine not be used by people under 18 years of age (ACSM 2000). Finally, the recently released position statement of Sports Dietitians Australia (Desbrow et al. 2014) recommends a focus on core food groups and nutrition practices rather than the use of supplements, specifically noting that the use of supplements by developing athletes overemphasises their ability to manipulate performance in comparison with other training and dietary strategies. These policies are based on the unknown but potentially adverse health consequences of some supplements and the implications of supplement use on the development of the young athlete’s ethical framework. Some researchers consider supplements to be an ‘entry point’ to the decision to take more serious compounds, including prohibited drugs (Hildebrandt et al. 2012), although the current authors believe this hypothesis to be over-simplistic, with the complexity of the definition of ‘supplement’ and the varied reasons for the use of performance enhancing drugs making it difficult to establish robust evidence of a direct cause–effect relationship.

16.5 Finding proof of the efficacy of supplements and sports foods

The process of substantiating the performance benefits or outcomes from supplement use is difficult. To various audiences, ‘proof’ comes in different forms, including testimonials from ‘satisfied customers’ and scientific theories that predict the outcome from the use of a product. On evaluation, however, these methods are flawed in their ability to provide definite support for the actions of a supplement. The scientific trial remains the best option for measuring the potential benefits of the use of a product. Nevertheless, the limitations of scientific studies need to be understood before a full interpretation of the results can be applied to real-life sport.

16.5.1 Scientific theories

A common focus of the sports supplement industry is on compounds and nutrients that act as precursors, cofactors, intermediary metabolites or stimulants of key reactions in exercise metabolism. The rationale behind supplementation is that if the system is ‘supercharged’ with additional amounts of these compounds, metabolic processes will proceed faster or for longer time, thus enhancing sports performance. The marketing of many contemporary supplements is accompanied by sophisticated descriptions of metabolic pathways and biochemical reactions, with claims that enhancement of these will lead to athletic success.

To the scientist, a theory that links an increased level of a compound with performance enhancement may be a hypothesis that is worthy of testing, but it does not constitute proof for the idea. To the public, however, a hypothesis can be made to sound like a fait accompli, and athletes can be induced to buy products on the strength of a ‘scientific breakthrough’ that exists only on paper. In an era when sports scientists feel challenged by the apparent sophistication of the scientific theories presented by supplement companies, it is unlikely that athletes will possess sufficient scientific knowledge to be critical of these proposals.

While a ‘supercharging’ hypothesis may appear plausible at first glance, there are many reasons why it may not occur. Other issues to be considered include:

- Will oral ingestion of the compound increase concentrations at the sites that are critical?
- Does the present level of compound fall below the critical level for optimal metabolism?
- Is this reaction the rate-limiting step in metabolism or are other reactions setting the pace?

A scientific theory or hypothesis should be developed and fine-tuned before setting up a supplementation study. Since studies are expensive in time, money and resources, it is important that ideas that make it to trial are based on sound logic. But while a scientific theory should be developed in preparation for a study (or to explain the data collected in a study), it cannot be accepted as proof of the efficacy of a supplement until verified by actual research. See Table 16.1.

TABLE 16.1 Factors in conducting research on supplements and sports foods

Factors to consider in designing a research protocol to select independent and dependent variables of importance

- Subject variables—age, gender, level of training, nutritional status
- Measurement variables—validity and reproducibility of techniques, costs, availability of equipment, subjective versus objective measures, application to the hypothesis being tested
- Study design—acute versus chronic supplementation, lab versus field, ‘blinding’ of subjects and researchers, crossover versus parallel group design, placebo control
- Supplementation protocols—timing and quantity of doses, duration of the supplementation period

Strategies to eliminate or standardise the variables that might otherwise confound the results of a supplement study

- Recruit well-trained athletes, as the subject’s level of training may alter the effect of the supplement and will affect the precision of measurement of performance.
- Incorporate the use of a placebo treatment to overcome the psychological effect of supplementation.
- Use repeated measures or crossover design to increase statistical power; each subject acts as their own control by undertaking both treatment and placebo.
- Allow a suitable wash-out period between treatments.
- Randomly assign subjects to treatment and placebo groups and counterbalance the order of treatment.
- Employ a double-blind allocation of treatments to remove the subjective bias of both researcher and subjects.

- Standardise the pre-trial training and dietary status of subjects.
- Design the parallel conditions to mimic real-life practices of athletes, including the interaction of the supplement with other nutritional practices or supplements.
- Choose measurement variables that are sufficiently reliable to allow changes due to the supplement to be detected, and that are applicable to the hypothesis being tested.
- Choose a performance test that is highly reliable and applicable to the real-life performances of athletes.
- Choose a supplementation protocol that maximises the likelihood of a positive outcome.
- Interpret the results in light of what is important to sports performance.

16.5.2 Anecdotal support

Testimonials are a powerful force in the advertising and marketing of sports supplements, particularly products that target the bodybuilding or resistance-training industry. This is also true of supplements sold through multi-level marketing schemes, where individual distributors are encouraged to have a 'personal story' of how the product has enhanced their life. Testimonials for supplements and sports foods highlight the successful health or performance outcomes that people have achieved, allegedly as a result of their use of a supplement product. Often famous athletes or media stars supply these testimonials, but sometimes they also feature the exploits of 'everyday' people. Although people sometimes receive payment for their testimonials, in other cases the endorsement for a supplement is provided by hearsay, observation or direct recommendation from a 'satisfied customer'. Successful athletes and teams are perpetually being asked to nominate the secrets of their success by peers, fans or the media. In the following reviews of well-known ergogenic aids, there are many examples where public interest in a product can be traced back to the recommendation or testimonial of a winning sportsperson. Modern communication forums provided by social media and Internet chat rooms have added new opportunities for the sharing of anecdotal experiences with supplement use.

It is hard for athletes to understand that success in sport results from a complicated and multifactorial recipe, and that even the most successful athletes may not fully appreciate the factors behind their prowess. In many cases, it is likely that the athlete has succeeded without the effects of the supplements they are taking—and in some cases, perhaps, in spite of them! Unsupported beliefs and superstition are key reasons behind many decisions to use supplements. The idea that 'everyone is doing it' provides a powerful motivation to the athlete contemplating a new product. Sometimes, this manifests as a fear that 'others may have a winning edge that I don't have'. The ad hoc and undiscriminating patterns of supplement use reported by some athletes are testament to the power of 'word of mouth'.

Of course, the anecdotal experiences of athletes may be useful when considering the scientific investigation of a supplement. These experiences may support the case for expending resources on a study, or help in deciding on protocols for using a supplement or for measuring the outcomes. However, by itself, a self-reported experience provides very weak support for the benefits of a supplement. Many of the benefits perceived by athletes who try a new supplement result from the psychological boost or placebo effect that accompanies a new experience or special treatment.

16.5.3 The scientific trial

The scientific trial remains the 'gold standard' for investigating the effects of dietary supplements and nutritional ergogenic aids on sports performance. Scientists undertaking scientific trials should test the effects of the supplement in a context that simulates sports performance as closely as possible. Additional studies might be needed to elucidate the mechanisms by which these effects occur, but, overall, sports science research must be able to deliver answers to questions related to real-life sport.

It is beyond the scope of this chapter to fully explore the characteristics of good research design. However, there are many variables that interfere with the outcomes of research and that need to be considered. Table 16.1 summarises the issues that need to be addressed to control for these variables, along with other issues to consider in designing trials to test the effects of supplements on sports performance. Several factors that are important to consider in the interpretation of results will now be discussed.

Are we testing the athlete's definition of improvement?

In the world of sport, the difference between winning and losing can be measured in hundredths of seconds and in millimetres. To the athlete or coach, that hundredth of a second or millimetre seems a meaningful improvement in performance. This helps to explain why supplements that promise a performance boost are greeted with such enthusiasm—the chance of the tiniest improvement seems worth the investment. Unfortunately, the traditional framework of sports science research works on a different basis. The scientist aims to detect (i.e. declare statistically significant) an effect, with acceptably low rates for detection of non-existent effects (5%) and failed detection of a real effect (20%). Most scientific investigations of supplements are biased towards rejecting the hypothesis that the product enhances performance, due to small sample sizes and performance-testing protocols with low reliability. In effect, most intervention studies are able to detect only large differences in performance outcomes. Changes that are smaller than this large effect are declared to be 'not statistically significant' and are dismissed.

Hopkins and colleagues attempted to find some middle ground between what scientists and athletes consider significant (Hopkins et al. 1999). First, they established that an athlete's required improvement is *not* the tiny margin between the place-getters in a race (also known as between-athlete variation). Each athlete has their own day-to-day or event-to-event variability in performance, known as the within-athlete variation or coefficient of variation (CV) of performance. This variation would influence the outcome of an event if it were to be rerun without any intervention. By modelling the results of various sporting events in track and field, Hopkins initially suggested that 'worthwhile' changes to the outcome of most events require a performance change equal to ~0.4–0.7 times the CV of performance for that event; this was later reduced to 0.3. It should be noted that this ‘worthwhile’ change does not *guarantee* that an athlete would win an event, but would make a reasonable change to an athlete’s likelihood of winning—for example, improve the probability of winning, for an athlete who has a true probability of winning the race 20% of the time, to 30%.

Across a range of track and field events, Hopkins noted that the CV of performance of top athletes was within the range of 0.5–5%, thus making performance changes of up to 3% important to detect (Hopkins et al. 1999). Subsequent similar analysis across a large number of sports (Malcata & Hopkins 2014) has shown the range in the reliability of top athletes to vary according to the features of their sport and the level of the competitor. According to their review, skeleton and 1000-m speed-skating times were shown to have the lowest within-athlete variability (CV of 0.15% and 0.4%, respectively), apparently because of the effect of the initial phase of the race on race dynamics. Times in sprint and endurance sports also have relatively low variability (0.6–1.4%), reflecting the predominant contribution of mean power output to performance. Meanwhile there is higher variability in sports performed against water or wind resistance due to uneven effects on individual athletes. Sports requiring explosive power in a single effort, such as field events and weightlifting, have larger CVs for their performance measures (1.4–3.3%), likely reflecting substantial contributions of skill. Sports with the greatest within-athlete variability (~50%) were those with subjective scores (e.g. surfing). Overall, elite athletes are more reliable than their less-successful or less-well-trained counterparts. Sports scientists who are interested in the effects of a supplement on a particular event and population should try to determine the specific CV of their group to delineate the size of a worthwhile intervention. Overall, however, an intervention that can enhance performance by 0.5–1% is likely to have real-life significance in determining the outcome of competition, while improvements of 2–3% are robust in their effect (Malcata & Hopkins 2014).

Even though ‘worthwhile’ performance differences are larger than the tiny margins considered important by athletes, these changes are still outside the realms of detection for many of the studies commonly published in scientific journals. As discussed by Hopkins and colleagues (1999), a change of 0.7 of the CV in a parameter requires a sample size of about 32 for detection in a crossover study in which every athlete receives an experimental and placebo treatment. For a parallel group designed study, 128 subjects would be needed. Such sample sizes are beyond the patience and resources of most sport scientists!

To bridge the gap between science and the athlete on the issue of a significant performance change, Hopkins and colleagues have proposed a different approach to reporting and interpreting the results of intervention studies (Hopkins et al. 1999; Hopkins et al. 2009). Rather than using null hypothesis testing, they suggest using magnitude-based inferences in which the effect of an intervention, and the confidence intervals around this, are interpreted in light of the smallest worthwhile change. Rather than basing decisions around statistical significance, results should be interpreted in light of the likelihood that an effect is trivial, or has a small or large possibility of increasing or decreasing performance (Hopkins et al. 2009). While this approach is not without its critics (Welsh and Knight 2014), it has been embraced by many applied sports scientists.

Individual responses

Notwithstanding the general variability in performance, there is evidence that some treatments cause a range of different responses in individual athletes. In some cases, the same intervention can produce favourable responses in some individuals, neutral responses in others and, sometimes, detrimental outcomes to another group. For example, research has identified that some athletes are 'non-responders' to caffeine or creatine supplementation (Graham & Spriet 1991; Greenhaff et al. 1994). Real differences in responsiveness to a supplement can arise due to a number of factors. These include differences in background nutritional status or body stores of the supplement substance; for example, creatine loading protocols achieve greater increases in muscle creatine stores in people such as vegetarians who have lower starting stores (Watt et al. 2004). Genetic differences in metabolism may also account for a large fraction of the variability in responsiveness. For example, differences in the type, number and site of adenosine receptors around the body can explain differences in the responsiveness to caffeine, which acts as an adenosine antagonist, while differences in liver metabolism of caffeine explain differences in the half-life of this chemical (Yang et al. 2010).

Studies employing simple group analysis and small sample sizes are not appropriate for situations in which there is true variability in the size and direction of the response to an intervention. Such studies will fail to detect a difference in performance, even though this is a real outcome for some subjects in the group. Ideally, studies employing large sample sizes and co-variate analysis should be used; this approach will allow real changes to be detected and may also identify the characteristics of individuals that predict 'response' and 'non-response'. At present, such studies are rare.

It is useful to have metabolic or other mechanistic data to substantiate real differences in response, and to differentiate these from the general variability of performance. For example, it has been shown that subjects whose muscle creatine levels did not increase by at least 20% as a result of creatine supplementation did not show the functional changes and performance enhancements seen by the rest of the experimental group (Greenhaff et al. 1994). Undertaking repeated trials to confirm the robustness of a measured response to the supplement is also valuable, while probing for genetic polymorphisms in key proteins associated with the metabolism or response to a substance is becoming more common (see 16.7.2 Caffeine).

16.6 AIS Sports Supplement Framework

Some sporting organisations or institutions make policies or programs for supplement use on behalf of athletes within their care. This may range from a single sporting team to an entire sports program, such as that of the National Collegiate Athletics Association (Green 2001). The experiences and evolution of the Sports Supplement Program of the Australian Institute of Sport (AIS) are instructive in providing a case history and resources associated with managing the use of supplements and sports foods within a high-performance sport environment.

16.6.1 Evolution of the framework

The AIS Sports Supplement Program was initiated in 2000 to manage the supplement use of high-performance athletes within its scholarship program following an assessment of unsound practices within this population. The program was developed as a multidisciplinary project and conducted under the oversight of a Sports Supplement Panel that included experts in sports nutrition, physiology, medicine, strength and conditioning, policy and anti-doping from the AIS. Its activities were integrated into all aspects of the leadership roles of the AIS and the scholarship requirements of its athletes, with the following goals:

- to allow its athletes to focus on the sound use of supplements and specific sports foods as part of their specific nutrition plans
- to ensure that supplements and sports foods are used correctly and appropriately to deliver maximum benefits to the immune system, recovery and performance
- to give its athletes the confidence that they receive ‘cutting edge’ advice and achieve ‘state-of-the-art’ nutrition practices
- to ensure that supplement use does not lead to an inadvertent doping offence.

Elements of the AIS Sports Supplement Program included:

- an education arm, to provide information suited to different groups within its community (sports science/medicine professionals, athletes and coaches)
- a research arm, to continue to develop the evidence base on which decisions about supplement use would be made
- a provision arm, to allow a supply of low-risk sports foods and supplements to be made available to athletes within a supervised sports nutrition program
- a governance arm, to ensure that all components were managed with professionalism and transparency.

The program provided open-access information to its users and the general public about its activities and resources via the AIS sports nutrition section of the Australian Sports Commission website (www.ausport.gov.au/ais/nutrition/supplements). This action was taken partially to promote transparency as a government-funded organisation, but mostly to provide an enhanced level of high-quality evidence-based information on supplements and sports foods in the public domain. It was felt that this tactic would combat or at least provide a counterpoint to the marketing hype and anecdotal information found in many other websites. Furthermore, in addition to providing a community service, this information would inform discussions in high-performance sport and thus indirectly influence the knowledge and attitudes of AIS athletes towards the program and sports foods/supplements.

Over the decade 2000–2010, the program was further developed as a leadership activity of the AIS, with its principles and resources offered to national sporting organisations and the National (Sports) Institute Network in Australia via a non-exclusive, royalty-free licence. In

2011 a separate ‘members’ area’ version of the program was developed to provide upgraded resources to AIS staff and athletes, and to people nominated by the organisations within Australia’s High Performance Network who had become aligned with the program. The ‘members’ version’ was hosted at the password-protected site of the AIS website (Clearinghouse for Sport: secure.ausport.gov.au/clearinghouse/high_performance_sport/sport_science_and_medicine_dis These resources provided specific and additional details of best practice protocols for the use of supplements and sports foods, delivering a competitive advantage to the Australian high-performance sports system.

Informal evaluation of the impact of the AIS Sports Supplement Program suggests that it met its goal of developing a national and international reputation for providing high-quality education resources around the evidence-based use of sports foods and supplements by high-performance athletes. However, there is also evidence that it played a role in changing supplementation practices of the athletes with whom it directly worked. It should be noted that while the program educated athletes, including identifying supplements considered to be lacking an evidence-base via its ranking system (see section 16.6.2), it still relied on athletes to make the final decision about their use of these products. A survey of supplement use by elite Australian swimmers, of whom AIS swimmers made up one-third of the cohort, revealed significant differences in self-reported practices (Shaw et al. in press). Overall, 97% of swimmers reported taking supplements or sports foods over the preceding 12 months. AIS swimmers reported using a greater number of products overall, including a greater number of products considered to be ‘performance supplements’. However, they were less likely to use supplements considered to be lacking scientific support and more likely to report sports dietitians and sports physicians as advisors of their supplement use. The AIS swimmers sourced a greater percentage of their supplements from an organised program (94 ± 16%) than other swimmers (40 ± 32%) ($p < 0.001$), who sourced a greater percentage (30 ± 30%) of their dietary supplements from supermarkets. These findings suggested a positive impact of the program, with the authors suggesting that providing access to evidence-based supplements and sports foods was an important aspect of its implementation.

In November 2012, a new funding and governance model (Australia’s Winning Edge 2012–2022) was launched within the high-performance environment of Australian sport. This involved the closure of AIS sports programs at the end of 2013 and terminated the model under which the AIS controlled and delivered daily training and competition preparation programs to Australian athletes and teams. The operation of the AIS Sports Supplement Program also ceased, with the responsibility for the use of sports foods and supplements within Australia’s high-performance sports environment then falling under the oversight of national sporting organisations and other sporting organisations/agencies. During the transition period of 2013, the AIS identified areas and activities in which it could provide leadership to assist Australian sporting organisations and agencies achieve the governance requirements of the Australian Sports Commission and its Winning Edge performance targets. The spotlight on poor supplement practices among several professional Australian sporting codes or teams highlighted the risks involved with this area of sports nutrition/medicine/science and the value of the expertise gathered by the AIS during the implementation of the Sports Supplement Program from 2000 to 2013.

Following interaction with key stakeholders in the Australian sports community, the AIS Sports Supplement Program was remodelled into the AIS Sports Supplement Framework, with the aims of:

- providing information and tools to assist Australian sporting organisations/agencies develop and implement their own sports supplement programs and guidelines to direct the sports food and supplement use by high performance athletes under their governance
- facilitating the implementation of other activities related to the safe, effective and legal use of supplements and sports foods in the Australian high-performance sports environment.

Experts on sports foods and supplements within the AIS have formed two groups to assist with the present and ongoing role of the framework: the AIS Sports Supplement panel to provide oversight and the framework management team to drive its daily activities and further development. External experts have been invited to join these groups to provide transparency and ensure the framework is developed with a national perspective to support Australia's Winning Edge goals.

16.6.2 The AIS ABCD classification system

A key part of the AIS Sports Supplement Program/Framework is the ABCD classification system for supplements and sports foods; it is a ranking system with four tiers, each of which has a suggested level of use that would be justified within a specific sports supplement program. The system is underpinned by a risk:benefit assessment by a panel of experts in sports nutrition, medicine and sports science of the evidence base for the beneficial uses of each product. Although the hierarchy of categories was developed for long-term use, there is a regular assessment of supplements and sports foods to ensure that they are placed in the category that best fits the available scientific evidence. The hierarchical system allows the program to avoid the 'black and white' assessment that any particular product works or fails to live up to claims. Rather, the available science is reviewed to place supplements into categories ranked from what is most likely to provide a benefit for little risk, to what provides least benefit and a definite risk. Table 16.2 provides a summary of the AIS Sports Supplement Program at the time of publication.

TABLE 16.2 Summary of ABCD categories of AIS Sports Supplement Framework 2015

Group A supplements		
Overview of category	**Sub-categories**	**Examples**

Evidence level: Supported for use in specific situations in sport using evidence-based protocols	**Sports foods**—specialised products used to provide a practical source of nutrients when it is impractical to consume everyday foods.	Sports drink Sports gel Sports confectionery Liquid meal Whey protein Sports bar Electrolyte replacement
Use within supplement programs: Provided or permitted for use by some athletes according to best-practice protocols	**Medical supplements**—used to treat clinical issues, including diagnosed nutrient deficiencies. Require individual dispensing and supervision by appropriate sports medicine/science practitioner.	Iron supplement Calcium supplement Multivitamin/mineral Vitamin D Probiotics (gut/immune)
	Performance supplements—used to directly contribute to optimal performance. Should be used in individualised protocols under the direction of an appropriate sports medicine/science practitioner. While there may be a general evidence base for these products, additional research may often be required to fine-tune protocols for individualised and event-specific use.	Caffeine B-alanine Bicarbonate Beetroot juice Creatine
Group B supplements		
Overview of category	**Sub-categories**	**Examples**
Evidence level: Deserving of further research and could be considered for provision to athletes under a research protocol or case-managed monitoring situation	**Food polyphenols** Food chemicals which have purported bioactivity, including antioxidant and anti-inflammatory activity. May be consumed in food form or as isolated chemical.	Quercetin Tart cherry juice Exotic berries (acai, goji etc.) Curcumin
Use within supplement programs: Provided to athletes within research or clinical monitoring situations	**Other**	Antioxidants C and E Carnitine HMB Glutamine Fish oils Glucosamine
Group C supplements		
Overview of category	**Sub-categories**	**Examples**
Evidence level: Have little meaningful proof of beneficial effects **Use within supplement programs:** Not provided to athletes within supplement programs May be permitted for individualised use by	Category A and B products used outside approved protocols. The rest—if you can't find an ingredient or product in Groups A, B or D, it probably deserves to be here. Note that the framework will no longer name Group C supplements or supplement ingredients in this top line layer of information. This will avoid the perception that these supplements are special.	Fact sheets and research summaries on some supplements of interest that belong in Group C may be found via on the 'A–Z of Supplements' page in the AIS Sports Nutrition section of the ASC website (www.ausport.gov.au/ais/nutrition/supplements) The fact sheet will identify that such ingredients have been placed in the Group C category without drawing undue attention to them.

an athlete where there is specific approval from (or reporting to) a sports supplement panel		
Group D supplements		
Overview of category use within AIS system	**Sub-categories**	**Examples**
Evidence level: Banned or at high risk of contamination with substances that could lead to a positive drug test **Use within supplement programs:** Should not be used by athletes	**Stimulants** World Anti-Doping Agency (WADA) list: list.wada-ama.org	Ephedrine Strychnine Sibutramine Methylhexanamine (DMAA) Other herbal stimulants
	Prohormones and hormone boosters WADA list: list.wada-ama.org	DHEA Androstenedione 19-norandrostenione/ol Other prohormones *Tribulus terrestris* and other testosterone boosters Maca root powder
	GH releasers and 'peptides' WADA list: list.wada-ama.org Technically, while these are sometimes sold as supplements (or have been described as such) they are usually unapproved pharmaceutical products.	
	Other WADA list: list.wada-ama.org	Glycerol used for re/hyperhydration strategies—banned as a plasma expander
		Colostrum—not recommended by WADA due to the inclusion of growth factors in its composition

The remainder of this chapter provides a summary of the current scientific support for a range of products within the various supplement categories.

16.7 Supplements in Group A of the AIS Sports Supplement Program

According to the judgments of the expert panel of the AIS Sports Supplement Framework, products listed in Group A have scientific support to show that they can be used within an athlete's nutritional plan to provide direct or indirect benefits to performance. They can be further divided into sports foods, medical/health supplements and performance supplements.

16.7.1 Sports foods and medical/health supplements

Products that meet the definition of sports foods and medical/health supplements are summarised in Table 16.3, which also identifies other chapters in this book to which the reader is directed for more in-depth information about their evidence-based use by athletes. This summary will consider some general philosophies that cover their use within sport. Typically, sports foods that provide a practical way to meet goals of sports nutrition are among the most valuable special products available to athletes. Table 16.3 summarises the major classes of sports foods, their typical format and composition, and the scenarios in which this nutrient profile can address some sports nutrition goals. Substantiation for many of these nutrition goals is well accepted and often includes situations where a measurable enhancement of performance can be detected as a result of the correct use of the sports food. It should be noted in many cases 'real food' options provide similar nutrition components to the athlete and are a perfectly viable option. The advantage of specialised sports foods comes from situation-specific practicality or convenience.

TABLE 16.3 Sports foods and medical supplements used to meet nutritional goals

Supplement	Form	Typical composition	Sports-related use	Chapter
Sports drink	Powder or liquid	5–8% CHO 10–35 mmol/L sodium 3–5 mmol/L potassium	Optimum delivery of fluid + CHO during exercise	13
			Post-exercise rehydration	14
			Post-exercise refuelling	14
Sports gel or sports confectionery	Gel 30–40 g sachets or larger tubes or jelly–type confectionary (generally 5–6 g each in pouch of ~40–50 g)	60–70% CHO (~25 g CHO per sachet) or 90% CHO confectionery items (~5 g per piece) Some contain caffeine or electrolytes	Supplement high-CHO training diet	14
			CHO loading	12
			Post-exercise CHO recovery	14
			May be used during exercise when CHO needs exceed fluid requirements	13
Electrolyte replacement supplements	Powder sachets or tablets	< 2% CHO 50–60 mmol/L sodium 10–20 mmol/L potassium	Rapid and effective rehydration following dehydration undertaken for weight-making	7
			Replacement of large sodium losses during ultra-endurance activities	13
			Rapid and effective rehydration following moderate to large fluid and sodium deficits (e.g. post-exercise)	14
Protein supplement	Powder (mix with water or milk) or liquid protein-rich bar, usually low in CHO	Powder of 70–90% protein content from whey or other animal or vegetable sources of high biological value protein (e.g. milk, pea, soy) May have other ingredients: additional branched chain amino acids Powders or bars with other ingredients from group C are considered non-optimal due to additional cost and increased risk of contamination	Post-exercise recovery following key training sessions or events where adaptation requiring protein synthesis is desired	14
			Achievement of increase in lean mass such as during growth or response to resistance training	4
			Portable nutrition for travelling	23

			athlete	
Liquid meal supplement	Powder (mix with water or milk) or liquid	1–1.5 kcal/mL 15–20% protein 50–70% CHO low to moderate fat vitamins/minerals: 500–1000 mL supplies RDI/RDAs	Supplement high-energy/CHO/nutrient diet (especially during heavy training/competition or weight gain)	4, 14
			Low-bulk meal replacement (especially pre-event meal)	12
			Post-exercise recovery—provides CHO and protein to meet goals	14
			Portable nutrition for travelling athlete	23
Sports bar	Bar (50–60 g)	40–50 g CHO 5–10 g protein Usually low in fat and fibre Vitamins/minerals: 50–100% of RDA/RDIs May contain creatine, amino acids	CHO source during exercise	13
			Post-exercise recovery—provides CHO, protein and micronutrients	14
			Support for high-energy/CHO/nutrient diet	14
			Portable nutrition (travelling)	23
Vitamin/mineral supplement	Capsule/tablet	Broad range 1–4 × RDI/RDAs of vitamins and minerals	Micronutrient support for low-energy or weight-loss diet	6
			Micronutrient support for restricted variety diets (e.g. vegetarian)	20
			Micronutrient support for unreliable food supply or disrupted eating patterns (e.g. travelling athlete)	23
Iron supplement	Capsule/tablet	Ferrous sulfate/gluconate/fumarate	Supervised management of iron deficiency (including treatment and prevention)	10
Calcium supplement	Capsule/tablet	Calcium carbonate/phosphate/lactate	Calcium supplementation in low-energy or low dairy food diet under supervision	9
			Treatment/prevention of osteopenia	9
Vitamin D	Capsule/tablet	Vitamin D3	Supervised management of vitamin D deficiency (including treatment and prevention for athletes as risk of deficiency)	9
Probiotics	Capsule/tablet or liquid	*Bifidobacterium lactis Lactobacillus fermentum Lactobacillus casei* —Shirota strain	Treatment prevention of gastrointestinal upset for athletes in high-risk situations, including travel	22, 23
			Prevention of upper respiratory tract infection	22

CHO = carbohydrate, RDA = Recommended Dietary Allowance, RDI = Recommended Dietary Intake

Although most sports foods have specialised uses in sport, some products have crossed successfully into the general consumer market. The case of sports drinks has become topical and merits comment. On one hand, if the marketing of these products leaves the sports nutrition message intact, this outcome may not cause undue concern. However, there is growing discussion that the unnecessary intake of sports foods by certain populations (e.g. overweight people or children) or outside their intended use (as a lifestyle beverage perceived to be 'healthy') can contribute to health issues due to unnecessary intake of energy or nutrient-poor CHO, as well as represent a waste of money. Clearly, there is a potential truth to some of these concerns, which speaks to the importance of responsible marketing and specific education regarding evidence-based use of these products. However, it should also be appreciated that there is support for the use of products such as sports drinks by 'weekend warriors' and recreational exercisers when consumed in the correct scenario. Indeed, the benefits of fluid intake and CHO replacement during a workout are determined by the physiology of exercise rather than the calibre of the person who is exercising. In fact, most studies of sports drinks have been undertaken on moderately trained to well-trained performers rather than elite athletes; therefore, where evidence of a performance benefit does exist, it is directly relevant to these sub-elite populations.

Medical and health supplements are also a potential tool for supporting the performance goals of athletes, when they are used to prevent or treat a situation of poor nutritional status or health. Table 16.3 summarises the various scenarios for the evidence-based use of various micronutrient supplements in sports nutrition, and again it is noted that there is the potential for false marketing, or misapplication and misuse by athletes. This is most likely to occur when athletes diagnose their own health or nutrition problems rather than seek appropriate expertise for assessment and monitoring. The reader is directed to the related chapters in this book for further discussion.

16.7.2 Caffeine

Caffeine is a drug that enjoys social acceptance and widespread use around the world. This acceptance now includes its use in competitive sport following the removal of caffeine from the WADA list of prohibited substances in January 2004. Caffeine is the best-known member of the methylxanthines, a family of naturally occurring stimulants found in the leaves, nuts and seeds of a number of plants. Major dietary sources of caffeine—such as tea, coffee, chocolate, cola and energy drinks—typically provide 30–200 mg of caffeine per serve, whereas some non-prescriptive medications contain 100–200 mg of caffeine per tablet. The recent introduction of caffeine (or guarana) to 'energy drinks', confectionery and sports foods/supplements has increased the opportunities for athletes to consume caffeine, either as part of their everyday diet or for specific use as an ergogenic aid.

The complex range of actions of caffeine on the human body has been extensively researched and emanate primarily because of its role as an adenosine receptor antagonist

(Tarnopolsky 1994; Spriet 1997; Fredholm et al. 1999; Graham 2001a, 2001b). Briefly, caffeine has several effects on skeletal muscle, involving calcium handling, sodium–potassium pump activity, elevation of cyclic-AMP and direct action on enzymes such as glycogen phosphorylase. Increased catecholamine action, and the direct effect of caffeine on cyclic-AMP, may both act to increase lipolysis in adipose and muscle tissue, causing an increase in plasma free fatty acid concentrations and increased availability of intramuscular triglyceride. It has been proposed that an increased potential for fat oxidation during moderate-intensity exercise promotes glycogen sparing. However, studies have found this effect to be short-lived or confined to certain individuals, and are unable to explain the ergogenic effects seen with caffeine supplementation (for review, see section 15.6.1). Caffeine is likely to achieve its greatest effects on athletic performance via central nervous system effects, such as a reduced perception of effort or an enhanced recruitment of motor units. Breakdown products of caffeine such as paraxanthine and theophylline may also have actions within the body. Caffeine supplementation is a complex issue to investigate due to the difficulty in isolating individual effects of caffeine, and the potential for variability between subjects.

The effects of caffeine on exercise and sporting activities has received extensive scientific attention for a century (Rivers 1907), with excellent reviews by Spriet (1997), Graham (2001a, 2001b), Doherty and Smith (2004, 2005) and a complete text (Burke et al. 2013) covering a large range of studies. Burke (2008) and colleagues (2013) have concentrated on the literature that is most relevant to sports performance rather than exercise capacity per se, reviewing studies involving trained individuals and defined tasks rather than exercise-to-fatigue protocols. Due to the large volume and turnover of such research, we will present our summary of the available peer-reviewed literature as a continually updated online publication at the website: www.ausport.gov.au/ais/nutrition/supplements. Some of the key information from this resource is presented to summarise our current knowledge of the effects of the quantity, timing and source of caffeine on exercise performance.

Type of sport

One of the fascinating things about caffeine is that it appears to enhance the performance of a diverse array of sports. There is evidence of benefits to the outcomes of prolonged predominantly sustained-intensity events of ~30–60 min, or those most often typified as endurance (>90 min) and beyond; these are seen across cycling, running and cross-country skiing protocols both when simulated in the laboratory (Cox et al. 2002; Kovacs et al. 1998) or undertaken in the field (Berglund & Hemmingsson 1982; Bridge & Jones 2006). Interestingly, although endurance events are seen as the traditional recipients of the benefits of caffeine use, the available research for such protocols might now be seen as less well-explored than other types of sport, particularly in the case of field studies. This may simply reflect the notion that the benefits are well accepted here and that other sports merit more investigation.

Indeed, there is now a fertile area of research into the benefits of caffeine supplementation on team sports such as soccer, rugby codes, volleyball and ice hockey, with results from field studies (Del Coso et al. 2012, 2013, 2014; Duncan et al. 2012) supplanting findings from earlier studies based in the laboratory (Gant et al. 2010; Roberts et al. 2010). Caffeine can be seen to enhance the skill outputs in tennis (Hornery et al. 2007; Strecker et al. 2007) and golf (Stevenson et al. 2009) as well as reaction time in combat sports (Souissi et al. 2012). It has also been shown to improve performance in sporting events of 1–30 min of sustained maximal and supra-maximal intensity such as swimming (MacIntosh & Wright 1995; Vandenbogaerde & Hopkins 2010), rowing, (Anderson et al.2000; Bruce et al. 2000), track cycling (Wiles et al. 2006) and middle-distance running (Kilding et al. 2012; Wiles et al. 1992). In fact, the only sports in which caffeine supplementation does not appear to be beneficial are those requiring a single high-intensity effort such as jumps, sprints and throws, although there may be benefits to sessions which require multiple efforts of this type (Bellar et al. 2013). Although a range of mechanisms may contribute to these outcomes, most share the feature of involving some level of fatigue across the event, which may be masked by caffeine use.

Caffeine dose

An obvious interest of athletes is to find the dose of caffeine that elicits the greatest benefit to their specific performance for the minimum level of risk or side effect. Studies that have investigated a dose–response relationship with caffeine supplementation, by examining the performance outcomes following the intake of different doses of caffeine by the same well-trained subjects, deserve specific comment. In briefer, supramaximal sports, it has been found that 6 and 9 mg/kg of caffeine had similarly positive effects on a time trial performance (Anderson et al. 2000; Bruce et al. 2000). In events of longer duration (~1 hour or greater), similar benefits to endurance or performance were identified at caffeine doses of 5, 9 and 13 mg/kg (Pasman et al. 1995) and at 3 and 6 mg/kg (Graham & Spriet 1995). Kovacs and colleagues (1998) demonstrated a threshold of the performance benefits to a cycling time trial at a caffeine dose of 3.2 mg/kg dose (i.e. no further benefit with 4.5 mg/kg). Cox and colleagues (2002) found that intakes of as little as 1–2 mg/kg caffeine enhanced the performance of a time trial at the end of 2 hours of cycling to the same degree as an intake of 6 mg/kg. In some trials, large amounts of caffeine (9 mg/kg) reduced endurance compared to smaller doses (Graham & Spriet 1995).

It should also be noted that many recent studies have found benefits to the performance of real-world and simulated sports when caffeine intakes of 3 mg/kg are provided to athletes (Del Coso et al. 2012, 2013, 2014). Overall, it appears that, for the performance of events lasting 30 minutes or longer, benefits are seen at low doses of caffeine (1–3 mg/kg), and there do not seem to be further benefits at doses higher than this. Further research of this type is warranted so that athletes can identify the *smallest* dose of caffeine that produces a worthwhile benefit to their performance across the whole range of sporting applications.

Timing of intake of caffeine

Caffeine is rapidly absorbed, reaching peak concentrations in the blood within 1 hour after ingestion. It is slowly catabolised (half-life 4–6 hours) with peak concentrations being maintained for 3–4 hours (Fredholm et al. 1999). Studies have reported that the ergogenic benefits of caffeine may be sustained for up to 6 hours post-ingestion (Bell & McLellan 2002), even after a prior bout of exhaustive exercise (Bell & McLellan 2003). Therefore, the traditional approach to caffeine supplementation has been to consume the caffeine dose 1 hour prior to exercise. In the case of shorter high-intensity exercise (1–20 min), intermittent sports simulating team and racquet sports, and sustained activities of both non-endurance and endurance duration (<60 and >90 min respectively), this protocol of intake is often associated with a benefit to endurance or performance (see Burke et al. 2013 and tables on caffeine supplementation at www.ausport.gov.au/ais/nutrition/supplements). Of course, pre-exercise intake makes practical sense in the case of brief events where is not opportunity to consume a supplement or sports/food during exercise and/or insufficient time for peak caffeine concentrations to appear in the blood if this was considered to be important.

More recently, sports scientists and athletes have recognised that beneficial outcomes occur when caffeine is consumed in a variety of protocols around sporting events, perhaps mimicking the way that it is consumed in everyday or 'social' use. In fact, early studies (Ivy et al. 1979) demonstrated the potential benefits of consuming caffeine both before and throughout an exercise task, but the interest in investigating the effects of divided or progressive doses of caffeine during exercise is more contemporary (Kovacs et al. 1998; Cox et al. 2002; Hunter et al. 2002; Conway et al. 2003). Conway and colleagues (2003) investigated the performance of a time trial following 90 minutes of submaximal cycling with supplementation of 6 mg/kg caffeine, either 1 hour pre-trial or 3 mg/kg pre-trial and 3 mg/kg at 45 minutes of exercise (= 3 + 3 mg/kg). There was a large improvement in the mean time to complete the task with caffeine (28.3, 24.2 and 23.4 min for placebo, 6 mg/kg and 3 + 3 mg/kg, respectively); however, differences between trials were not found to be statistically significant.

We showed that similar performance benefits of a 3% improvement in time trial performance were achieved when six doses of caffeine of 1 mg/kg were spread throughout a 2-hour submaximal cycling bout prior to the time trial, when 6 mg/kg of caffeine was consumed 1 hour prior to the cycling bout, or when small amounts of caffeine (~1.5 mg/kg) were consumed over the last third of the protocol (Cox et al. 2002). It may be that subjects become more sensitive to small amounts of caffeine as they become fatigued. Further investigations are required to confirm this and the potential for strategic timing of intake of caffeine in various sporting and exercise activities.

Source of caffeine

Because of its use as a vehicle for caffeine intake in the general population, a number of studies have investigated the effects of coffee on exercise, providing subjects with decaffeinated coffee with or without added caffeine. While some of these studies have shown that coffee intake can enhance performance (Costill et al. 1978; Wiles et al. 1992; Trice & Haymes 1995; McLellan & Bell 2004), others have failed to find a detectable effect (Graham et al. 1998; Vanakoski et al. 1998). Only two of these studies (Graham et al. 1998; McLellan & Bell 2004) included trials in which the performance responses to caffeinated coffee were compared to responses to pure caffeine. Graham and colleagues found that runners increased their treadmill running time to exhaustion following the intake of pure caffeine compared to their endurance following the intake of the same amount of caffeine consumed in coffee. The differences in performance occurred despite similar appearance rates of caffeine and other caffeine metabolites; it has been suggested that other components within coffee might antagonise the responses to the caffeine or counteract its ergogenic effects by directly impairing performances. However, McLellan and Bell (2004) demonstrated that the consumption of one cup of coffee (either decaffeinated or caffeinated) prior to the consumption of pure caffeine failed to dampen the ergogenic effect gained from the larger caffeine doses. Further research is needed on this issue, since in the 'real world', athletes often consume coffee before competing or training, either as part of their normal social and dietary patterns, or as an intentional source of caffeine as an ergogenic aid.

The most important reason that coffee should not be considered the preferred source of caffeine for athletes is its large, and otherwise unknown, variation in caffeine dose. Although food tables generally provide a mean value for the typical caffeine content of coffee, in fact, there is a variable quantity of caffeine in the coffee bean and differences in the way that coffee drinks are prepared for consumption by individuals—even among the same 'types' of coffee drinks prepared under seemingly standard conditions. For example, Desbrow and co-workers found that the caffeine content of a single espresso purchased from a large number of retail outlets varied from 25 to 214 mg/serve (Desbrow et al. 2007). This makes it an unreliable source of a targeted caffeine intake.

Several studies have used energy drinks as a caffeine source for sports performance (Del Coso et al. 2012, 2013, 2014), with the results confirming the advertised benefits of these drinks in enhancing activity patterns. Only one study (Cox et al. 2002) has investigated the effect of the use of cola-containing beverages on the performance of road cycling, to mimic the patterns practiced in real life, where athletes consume a sports (CHO-electrolyte) drink for the first two-thirds of their event before switching to de-fizzed cola drinks. We found that the consumption of ~750 mL of Coca-cola towards the end of our cycling protocol enhanced the performance of a time trial to a similar extent as larger amounts of caffeine taken before and/or during the protocol. A separate study was then conducted to test whether this effect was achieved by the presence of the caffeine (~1–2 mg/kg BM) or the increase in CHO concentration between the cola and sports drink. The second study confirmed the finding of a 3% enhancement of performance of the time trial with the intake of Coca-cola, finding that the majority of this effect (2%, $p < 0.05$) could be explained by the caffeine content (Cox et al. 2002).

Finally, recent investigations have tested the efficacy of caffeinated gum, prepared commercially for military uses, on sports performance, noting both the practical ease of achieving a targeted caffeine dose (100 mg/pellet) and rapidity of increase in plasma caffeine concentrations due to the uptake of caffeine from buccal cells via this source (Bellar et al. 2012; Lane et al. 2014; Paton et al. 2015). Overall, it would seem that the most important factors in selecting a caffeine source for sports performance are the practicality and confidence with which a modest caffeine dose can be achieved. Sports products (gels and confectionary), commercial beverages (cola drinks and energy drinks) and specialised gums are readily available and provide the athlete with a number of alternatives.

Responsiveness to caffeine intake

As with many drugs, individuals respond differently to caffeine, with the outcomes ranging from positive to neutral (Graham and Spriet 1991; Wiles et al. 2006) and even negative outcomes, particularly when taken in high doses (>6 mg/kg). Side effects include increases in heart rate, tremors and headaches, and it may cause a direct impairment of performance in some scenarios or individuals. A range of polymorphisms in genes related to adenosine receptors or liver metabolism of caffeine have been identified, and account for some of the differences in habitual caffeine intake patterns (Cornelis et al. 2007) as well as the variability in performance benefits following caffeine supplementation (Womack et al. 2012). Further work in this area is needed.

Traditionally in caffeine research or in actual competition use, athletes withdraw from caffeine use for 24–48 hours prior to exercise to remove their habituation to repeated use. However, no consistent differences in the performance effects of caffeine between regular users and non-users of caffeine, or as a result of withdrawal from regular caffeine use, are apparent (Graham 2001a). Indeed, a study which directly compared the effects of caffeine on a 1-hour cycling time-trial, with or without a 4-day withdrawal period, found that a 3 mg/kg dose enhanced performance regardless of the preparation strategy (Irwin et al. 2011). Avoiding or withdrawing caffeine prior to a performance trial may be associated with side effects such as headaches and fatigue. In fact, there are suggestions that the benefits of caffeine seen in controlled studies may be overstated if they reflect the reversal of adverse withdrawal symptoms rather than an ergogenic effect of caffeine per se (James & Rogers 2005). Withdrawal can also increase the risk that subsequent caffeine intake attracts the negative effects often seen with large caffeine doses (irritability, tremor, heart rate increases).

Summary of caffeine and sports performance

- Since January 1 2004, caffeine has been absent from the WADA banned list, allowing athletes to consume it in their everyday lives or as a performance supplement without penalties. However it remains on the WADA monitoring list, with surveillance of urinary caffeine concentrations during in-competition testing, to check for evidence of abuse by athletes. Athlete education is needed to ensure that such abuse does not occur; indeed, the levels of caffeine intake associated with performance benefits are modest and well within the patterns of community use deemed to be ‘normal’ or ‘healthy’. Therefore, the consumption of large amounts of caffeine by athletes, particularly associated with the co-ingestion of other stimulants such as in commercial ‘pre-workout supplements’, is unnecessary and should be discouraged.
- There is sound evidence that caffeine enhances endurance and provides a small but worthwhile enhancement of performance over a range of sporting activities. The most likely mechanism is that central nervous system effects provide fatigue resistance or a

masking of the perception of effort. Most studies of caffeine and performance have been undertaken in laboratories; studies that investigate performance effects in elite athletes under field conditions or during real-life sports events are welcome. Caffeine is also likely to be a useful training aid, allowing the athlete to undertake better and more consistent training. Rather than promoting the intake of large amounts of caffeine, however, good sports nutrition practice may see athletes making use of strategic timing of intake of ‘everyday’ amounts of caffeine around their training or competitive events to support training goals.

- Recent studies confirm that beneficial effects from caffeine occur at very modest levels of intake (1–3 mg/kg BM or ~70–150 mg caffeine), when it is taken before and/or during exercise. Furthermore, there is little evidence of a dose–response relationship to caffeine—that is, performance benefits do *not* appear to increase with increases in the caffeine dose. This information is an advance on the traditional caffeine supplementation protocols, which provided intakes of 6–9 mg/kg BM (e.g. 400–600 mg) 1 hour prior to the exercise. Further research is needed to define the range of caffeine intake protocols that provide performance enhancements across various sports or exercise activities. The requirement for withdrawal from caffeine intake in the days leading into competition to potentiate the performance benefits is now considered unnecessary.
- The effects of caffeine supplementation differ between individuals. Some people are non-responders and others experience negative side effects such as tremors, increased heart rate, headaches and impaired sleep. Such side effects are more common at higher doses—for example, exceeding 6–9 mg/kg BM—and may cause a direct impairment of performance. They may also indirectly impair exercise outcomes—for example, disturbing the sleep patterns of athletes who compete in a multi-day sport and need to recover between days of competition.
- Coffee is not an ideal vehicle for caffeine supplementation by athletes, chiefly because of the variability of caffeine content. A range of other commonly available caffeine sources with known and moderate caffeine doses can be used to achieve caffeine intakes known to enhance performance; these include caffeinated sports foods, cola drinks and energy drinks. Athletes are encouraged to find a pattern of caffeine use in their everyday lives and sports performance that is consistent with good health, wellbeing and performance goals.

16.7.3 Creatine

When the first edition of this book was written in 1994, creatine was the latest ‘hot supplement’, with testimonials from gold medal winners at the 1992 Barcelona Olympic Games. Unlike many supplements that draw the attention of athletes, creatine enjoyed some scientific support, with the 1992 publication of a study that showed that muscle creatine stores could be increased by the intake of large doses of an oral source of creatine (Harris et al.

1992). Since then, creatine supplements have become a phenomenon in sports nutrition—a performance aid that records huge annual sales and has been the topic of over 500 investigations, book chapters and reviews. It is not often that scientists and athletes are excited by the same product.

Creatine (methylguanidine-acetic acid) is a compound derived from the amino acids methionine, glycine and arginine. It is stored primarily in skeletal muscle at typical concentrations of 100–150 mmol/kg/dry weight (dw) of muscle. About 60–65% of this creatine is phosphorylated and it is this phosphocreatine (PCr) which provides a rapid but brief source of phosphate for the resynthesis of adenosine triphosphate (ATP) during maximal exercise, and is therefore an important fuel source in maximal sprints of 5–10 seconds. Other functions of phosphocreatine metabolism are the buffering of hydrogen ions produced during anaerobic glycolysis and the shuttling of ATP, generated by aerobic metabolism, from the muscle cell mitochondria to the cytoplasm, where it can be utilised for muscle contraction. Creatine metabolism is covered in more detail in Chapter 1 and in reviews of creatine metabolism and supplementation (Greenhaff 2000, 2001; Hespel et al. 2001; Hespel & Derave 2007).

The daily turnover of creatine, eliminated as creatinine, is approximately 1–2 g/d. This can be partially replaced from dietary creatine intake, found in animal muscle products such as meat, fish and eggs, and typically consumed in amounts of ~1–2 g/d in an omnivorous diet. Additional creatine needs are endogenously synthesised from arginine, glycine and methionine, principally in the liver, and transported to the muscle for uptake. Creatine is transported into the muscle against a high concentration gradient, via saturatable transport processes that are stimulated by insulin (Green et al. 1996a, 1996b). High dietary creatine intakes temporarily suppress endogenous creatine production. Vegetarians who do not consume a dietary source of creatine are believed to have a reduced body creatine store, suggesting that they do not totally compensate for the lack of dietary intake (Green et al. 1997). The reason for the variability of muscle creatine concentrations between individuals is uncertain. There are some suggestions that females typically have higher muscle creatine concentrations (Forsberg et al. 1991) and it appears that creatine stores decline with ageing. The effect of training on creatine concentrations also requires further study.

Protocols for creatine supplementation

Creatine monohydrate is a highly bioavailable supplemental form of creatine. The watershed study by Harris and colleagues showed that muscle creatine levels were increased as a result of supplementation with repeated doses of creatine, large enough to sustain plasma creatine levels above the threshold for maximal creatine transport into the muscle cell (Harris et al. 1992). The protocol provided four to six doses of 5 g creatine (monohydrate) for 5 days to increase total muscle creatine concentrations by 20%, and reach an apparent muscle threshold of ~150–160 mmol/kg dw. About 20% of the increased muscle creatine content was stored as PCr and saturation occurred after 2–3 days. Increases in muscle creatine stores were greatest in those who had the lowest pre-supplementation concentrations and when coupled with intensive daily exercise.

Although this discovery appears to be recent, in fact, studies showing that oral creatine doses are largely retained in the body were available 90 years ago (Chanutin 1926). However, it is only now that muscle biopsy procedures and imaging techniques are available to enable scientists to monitor muscle stores of creatine and investigate the success of creatine loading protocols. Over the past decade, a number of studies have refined our knowledge of supplementation protocols. Rapid loading is achieved by consuming a daily creatine dose of 20–25 g (0.3 g/kg/d), in split doses (e.g. 4 × 5 g), for 5 days. Alternatively, a daily dose of 3 g/d will achieve a slow loading over 28 days (Hultman et al. 1996). Elevated muscle creatine stores are maintained by continued daily supplementation of 2–3 g (Hultman et al. 1996). Across studies, there is evidence that the creatine loading response varies between individuals, with ~30% of individuals being 'non-responders' or failing to significantly increase muscle creatine stores (Spriet 1997; Greenhaff 2000). Co-ingestion of substantial amounts of CHO (75–100 g) with creatine doses has been shown to enhance creatine accumulation, a result of insulin stimulation of muscle Cr transport (Green et al. 1996a, 1996b). This protocol assists individuals to reach the muscle creatine threshold of 160 mmol/kg dw. Creatine appears to be trapped in the muscle; in the absence of continued supplementation, it takes ~4–5 weeks to return to resting creatine concentrations (Hultman et al. 1996). Many studies have reported an acute gain in BM of ~1–2 kg during rapid creatine loading (Buford et al. 2007). This is likely to be primarily a gain in body water, and is mirrored by a reduction in urine output during the loading days (Hultman et al. 1996).

Effects of creatine supplementation on performance

Many studies have investigated the effect of creatine supplementation on muscle function exercise and performance. Studies vary according to the characteristics of subjects (e.g. gender, age, training status), the mode of exercise, and whether supplementation involved an acute loading intervention or a chronic effect on training adaptations. It is beyond the scope of this chapter to summarise the large and growing body of literature on creatine supplementation and performance effects—even those that are carried out in trained individuals with relevance to sports activities. We will provide this information via an updated online resource (www.ausport.gov.au/ais/nutrition/supplements). However, we offer the following summary of this literature and of the more recent reviews; Mujika & Padilla 1997; Kraemer & Volek 1999; Branch 2003; Rawson & Volek 2003; Bemben & Lamont 2005; Hespel and Derave 2007):

- The major benefit of creatine supplementation appears to be an increase in the rate of phosphocreatine resynthesis during the recovery between bouts of high-intensity exercise, producing higher phosphocreatine levels at the start of the subsequent exercise bout. Creatine supplementation can enhance the performance of repeated 6–30-second bouts of maximal exercise, interspersed with short recovery intervals (20 seconds to 5 minutes), where it can attenuate the normal decrease in force or power production that occurs over the course of the session.
- Oral creatine supplementation cannot be considered ergogenic for single-bout or first-

bout sprints because the likely benefit is too small to be consistently detected. The exercise situations that have been most consistently demonstrated to benefit from creatine supplementation are laboratory protocols of repeated high-intensity intervals, involving isolated muscular efforts or weight-supported activities such as cycling.

- In theory, acute creatine supplementation might be beneficial for a single competitive event in sports involving repeated high-intensity intervals with brief recovery periods. This description includes team games and racquet sports. Similarly, chronic creatine supplementation may allow the athlete to train harder at exercise programs based on repeated high-intensity exercise, and make greater performance gains. These benefits may apply to the across-season performance of athletes in team and racquet sports, as well as the preparation of athletes who undertake interval training and resistance training (e.g. swimmers and sprinters).
- For many specific sports, the benefits of creatine are theoretical, since few studies have been undertaken with elite athletes or as 'field studies'. Performance enhancements may not always occur in complex games and sports; even if changes in strength or speed are achieved by creatine-assisted training, these may not translate into improvements in game outcomes (e.g. goals scored).
- Acute creatine loading is associated with an increase in BM of ~0.6–1.0 kg. Performance enhancements will occur in weight-bearing and weight-sensitive sports (such as lightweight rowing and rock climbing) only if gains in muscular output compensate for increases in BM. This gain is believed to be acute retention of intracellular water and may not be as evident when a slower loading protocol (e.g. loading over 4 weeks) is used. Whether any attenuation of this effect is due to masking by other changes in body composition over this time, or whether there is an equilibration of total body water, is unknown. This acute change is likely to be interpreted as an increase in lean mass on DXA estimates of body composition although it does not represent a real increase in muscle protein content.
- Evidence that creatine supplementation is of benefit to endurance exercise is inconsistent, but the concept has not been well investigated, perhaps because of the initial reports that it was associated with a weight gain that might reduce performance in 'weight-bearing' sports (e.g. distance running, hill climbing). However, there are several concepts that require further investigation. First is the ability of enhanced muscle creatine stores to support the capacity for high-intensity bursts that regularly occur during many endurance sports and often dictate the outcome of events. Indeed, one study has reported that creatine supplementation enhanced the performance of sprints undertaken at the end of a simulated road cycling protocol (Vandebuerie et al. 1998). Second, there is evidence that creatine supplementation enhances the capacity of the muscle to store or supercompensate glycogen stores (Robinson et al. 1999; van Loon et al. 2004): the benefits of additional glycogen in endurance sports in which muscle glycogen is limiting (e.g. marathon or 50-km race walking event) merit investigation.
- Early theories about the chronic effects of creatine supplementation considered that sports

performance was enhanced by allowing the athlete to train harder as a result of improvements in cellular energetics. However, more recent investigations suggest that there are other mechanisms underpinning the value of creatine for training support. Creatine loading increases cellular water and osmolarity, with cell swelling being a potent stimulator of anabolism. One study of creatine supplementation in sedentary individuals found that it significantly upregulated genes and protein content of a large array of enzymes—these included factors involved in osmosensing and signal transduction, cytoskeleton remodelling, regulation of protein and glycogen synthesis, satellite cell proliferation and differentiation, DNA replication and repair and cell survival (Safdar et al. 2008). This finding has exciting implications if it further upregulates the signalling cascade associated with a bout of exercise. Indeed, creatine supplementation in combination with strength training has been shown to amplify the training-induced increase in satellite cell number and myonuclei concentration in human skeletal muscle fibres, thereby allowing an enhanced muscle fibre growth in response to strength training (Olsen et al. 2006). Further investigation is warranted, since it appears that the potential benefits of creatine supplementation on health and exercise are wider and more complex than previously thought.

- Creatine supplementation may offer a range of additional benefits to sports performance via an array of other direct and indirect mechanisms. These include the potential for enhanced cellular hydration to support thermoregulation and performance in hot conditions (Dalbo et al. 2008); attenuation of some of the negative effects of immobilisation and enhancement of the rate of rehabilitation for the injured athlete (Hespel & Derave 2007) and, perhaps, even neuroprotection in the case of brain injury (Sullivan et al. 2000).

Concerns with creatine use

Whether there are short- or long-term side effects and problems associated with creatine has been hotly debated since 1992. Although this should be a fundamental question asked of all supplements, it appears that there has been some less-than-objective reporting of this issue in relation to creatine. There are two separate issues: problems related to creatine supplementation per se, and problems related to the creatine supplements commonly used by athletes. These will be considered separately.

The proposed side effects associated with creatine loading include anecdotal reports of nausea, gastrointestinal upset and headaches. Although these are sometimes noted as a subject experience in studies of creatine supplementation and performance, they are not typically reported as major issues in larger retrospective or prospective studies comparing creatine users and non-users. Other reported effects associated with acute creatine supplementation include a weight gain of 1–2 kg (Buford et al. 2007), which can be of detriment to performance in weight-sensitive sports (Balsom et al. 1993), and a reduced range of movement around joints (Sculthorpe et al. 2010). These effects are plausible, particularly in light of increased water retention within skeletal muscle cells, but may not occur or be as obvious to detect during slower loading protocols. An increased risk of muscle strains/cramps and injuries is frequently purported to occur with creatine use; however, studies have failed to find evidence of an increased prevalence or risk of these problems when comparing creatine users and non-users within groups of athletes (Greenwood et al. 2003b; Kreider et al. 2003). Indeed, in one study of collegiate football, creatine users reported a lower incidence of some of these issues over a season of play. Similarly, despite early warnings that creatine loading would cause muscle cramps and impair performance in hot conditions by binding intracellular water and making it unavailable, it is now known that creatine supplementation is neutral, or in fact beneficial, for tolerance for hot weather exercise by maintaining haematocrit, aiding thermoregulation and reducing exercising heart rate and sweat rate (Dalbo et al. 2008; Lopez et al. 2009).

Although it is commonly suggested that creatine supplementation may cause renal impairments, these are limited to case reports in a few patients with pre-existing renal dysfunction. Longitudinal studies have reported that creatine intake had no detrimental effects on renal responses in various athletic populations (Poortmans et al. 1997; Mayhew et al. 2002). The safety of creatine use has been regularly considered and several reviews have concluded that the benefits to exercise capacity, and more importantly its use in medical and clinical scenarios, merit further attention and respect (Brosnan & Brosnan 2007; Buford et al. 2007; Kim et al. 2011; Gualano et al. 2012). Importantly, a suggestion from a French food safety agency that creatine supplementation is carcinogenic has been discredited (see Hespel et al. 2001). Nevertheless, expert bodies such as the American College of Sports Medicine have taken a cautious view on the pros and cons of creatine supplementation (ACSM 2000). It is reasonable that creatine supplementation should be targeted to well-developed athletes who use it in evidence-based scenarios. Adolescent athletes are able to make substantial gains in performance through maturation in age and training, without the need to expose themselves to the expense or small potential for long-term consequences of creatine use.

Most concern is directed to creatine users who undertake protocols outside the evidence-based regimens outlined in this chapter. This includes long-term creatine users who self-medicate with doses far in excess of recommended creatine loading and maintenance protocols. In addition, it is noted that despite the cost-effectiveness and efficacy of creatine monohydrate as a supplemental form of creatine, there are many alternative forms on sale including creatine ethyl ester, creatine nitrate, creatine phosphate, creatine serum and alkaline creatine. These are typically more expensive than creatine monohydrate and typically lack evidence of benefits above creatine monohydrate (Buford et al. 2007). In some cases, there may be significant problems with the product: for example, a creatine serum product was found to contain almost no creatine at all (Harris et al. 2004). Finally, it is noted that creatine is often included in many multi-ingredient supplements such as the so-called ‘pre-workout trainers’ where problems such as the failure to provide a therapeutic dose and the presence of large and/or unidentified amounts of stimulants are common issues.

16.7.4 Bicarbonate and citrate

Anaerobic glycolysis provides the primary fuel source for exercise of near-maximal intensity lasting longer than approximately 20–30 seconds. The total capacity of this system is limited by the progressive increase in the acidity of the intracellular environment, caused by the accumulation of lactate and hydrogen ions (see Chapter 1). When intracellular buffering capacity is exceeded, lactate and hydrogen ions diffuse into the extracellular space, perhaps aided by a positive pH gradient. Since the 1930s, it has been recognised that dietary strategies that decrease blood pH (e.g. intake of acid salts) impair high-intensity exercise, while alkalotic therapies improve such performance (Dennig et al. 1931; Dill et al. 1932). In theory, an increase in extracellular buffering capacity should delay the onset of muscular fatigue during prolonged anaerobic metabolism by increasing the muscle's ability to dispose of excess hydrogen ions. The two most popular buffering agents are sodium bicarbonate and sodium citrate; loading with these salts is permitted for human sports performance, although it is banned in dog and horse racing. It is difficult to detect the use of bicarbonate- or citrate-loading strategies by athletes, since urinary pH varies according to dietary practices such as vegetarianism and high CHO intake (Heigenhauser & Jones 1991).

Protocols for supplementation with bicarbonate and citrate

Athletes have practised bicarbonate loading for over 70 years (McNaughton 2000), with sodium bicarbonate being ingested in the form of the household product 'bicarb soda' or as pharmaceutical urinary alkalinisers such as Ural. The general protocol for bicarbonate loading is to ingest 0.3 g of sodium bicarbonate/kg BM 1–2 hours prior to exercise; this equates to 4–5 teaspoons of bicarbonate powder. Sodium citrate is also usually ingested in doses of 0.3–0.5 g/kg BM, but since it seems to have much lower efficacy as a buffering agent in terms of performance enhancement (Carr, Hopkins et al. 2011), it will receive less focus in this chapter.

The major side effect of bicarbonate supplementation is gastrointestinal (GI) distress including nausea, stomach pain, diarrhoea and vomiting (McNaughton et al. 2008). Indeed, sports scientists often observe that bicarbonate loading is not practised by athletes who could potentially benefit from an enhanced buffering capacity in a competitive setting due to the fear or personal experience of such GI upsets. More importantly, some studies have shown that the gut disturbances associated with bicarbonate use by some individuals can lead to an impairment of sports performance or counteraction of other ergogenic practices (Carr, Gore et al. 2011). Previous advice to athletes to overcome this issue was to consume the bicarbonate dose with plenty of fluids to reduce the risk of hyperosmotic diarrhoea. A recent study systematically investigated bicarbonate supplementation protocols, varying the time taken to consume the load (spreading it over 30–60 mins), the form of the bicarbonate (flavoured powder or capsules) and the co-ingestion of various amounts of fluid or food with the bicarbonate (Carr, Hopkins et al. 2011). The strategy that optimised blood alkalosis and reduced the occurrence of gastrointestinal symptoms was to consume bicarbonate capsules in a spread-out protocol, commencing 120–150 min before the start of exercise and, if practical, at the same time as consuming a meal composed of carbohydrate-rich food choices and some fluid. Athletes should practise with such strategies to fine-tune a successful protocol for their situation.

Some athletes need to compete in heats, semis and finals to decide the outcome of their event. Swimmers, rowers and track athletes may compete several times over a series of days, and sometimes more than once on the same day. Whether buffering protocols can be repeated on each occasion and how such protocols are best undertaken are questions that require investigation. Issues of interest include the determination of the lowest effective dose of bicarbonate or citrate, in order to minimise side effects such as gastrointestinal discomfort or disturbances to post-race recovery. After all, although an acute supplementation protocol may enhance the performance of the immediate race, side effects in the post-race period could jeopardise the outcomes of the following events. It is also important to investigate whether subsequent doses are still effective and without side effects at this level. It is possible that a lower dose may be effective in a repeated supplementation protocol, especially if the first dose has not been completely washed out. It is also possible that a subsequent dose may have a reduced or absent effect. In this case the athlete might need to decide if the priority is to enhance performance to make the final, or to trust that they will make the final without aid and save any supplementation protocols for the most important race.

In the case of bicarbonate, a ‘serial’ loading protocol has been investigated as an alternative to the repetition of acute protocols. McNaughton and colleagues studied the effect of 5–6 days of bicarbonate supplementation with a total of 500 mg/kg/d, spread into four doses over the day (McNaughton et al. 1999a). This protocol was found to achieve an increase in plasma base excess, which was sustained over the days of bicarbonate intake. Further, in recreational participants, it enhanced the performance of a prolonged sprint undertaken on the 1–2 days *after* the bicarbonate supplementation ceased compared to the pre-trial performance (McNaughton & Thompson 2001). The persistence of the ergogenic outcome may be a desirable feature for sports involving a series of competition events or when the athlete wants to finish their intake of bicarbonate (and the risk of gastrointestinal side effects) on the day prior to their competition, while maintaining the benefit to performance. Further study of such buffering strategies is needed to confirm these findings and investigate its application over a wider range of exercise protocols.

Effect of buffering protocols on performance

Bicarbonate or citrate loading strategies are ‘purpose-built’ for strategies that might otherwise be limited by disturbances to acid-base balance. The obvious candidates are events involving sustained high-intensity exercise lasting 1–7 minutes (‘sustained power’ sports), such as middle distance swimming, middle distance running and rowing events. However, bicarbonate loading may also benefit the performance of longer events (e.g. 30–60 mins) involving sustained exercise just below the so-called anaerobic threshold if it can support the athlete for periods in which the pace is increased (i.e. surges during the event, the final sprint to the end).

Similarly, the repeated-sprint performance typical of team and racquet sports, and even combative sports, may also be enhanced by improved buffering.

Over the past 40 years, sports scientists have investigated this potential in a large number of studies with various levels of application to real-life sport. Table 15.4 summarises the results of some relevant investigations of bicarbonate or citrate supplementation and sports performance published in the last decade which meet the criteria of applied sports research summarised in section 15.5. In particular, we have focused on studies employing well-trained subjects and a performance measure that simulates the requirements of sport. Different scenarios have been included (acute supplementation prior to a sports-related performance, serial supplementation prior to a performance, acute supplementation prior to exercise simulating a training session, and chronic supplementation for support of training sessions). Some studies involved the investigation of independent and additive effects of bicarbonate/citrate supplementation with another supplement. This summary shows that there is reasonable but not unanimous support for the benefits of bicarbonate/citrate loading for each of the sporting scenarios previously mentioned. Nevertheless, reviews of the larger body of literature have concluded in different ways that bicarbonate/citrate loading can be of benefit to some athletes:

- An early meta-analysis (Matson & Tran 1993) concluded that the ingestion of sodium bicarbonate has a moderate positive effect on exercise of 30 s to 7 min, with the mean performance of the bicarbonate trial being 0.44 standard deviations better than the placebo trial. Ergogenic effects were related to the level of metabolic acidosis achieved during the exercise, showing the importance of attaining a threshold pH gradient across the cell membrane from the combination of the accumulation of intracellular H^+ and the extracellular alkalosis.
- Requena and co-workers (2005) undertook a narrative review which concluded that athletes competing in high-intensity sports involving fast motor unit activity and large muscle mass recruitment (athletics events, cycling, rowing, swimming and many team sports) could benefit from bicarbonate loading.
- More recently, Peart and colleagues (2012) performed a meta-analysis of bicarbonate loading to take into account more recent literature and to differentiate between trained and untrained populations. A total of 58 effect sizes from 395 participants (348 male and 47 female) were identified from the literature. Studies included for analysis used an acute dosage protocol of 0.2–0.4 $g \cdot kg^{-1} \cdot$body weight^{-1} 60–120 minutes before exercise. Exercise was defined as either a single or repeated bout protocol (e.g. repeated sprints and sports specific simulations) with performance measures assessing included time to exhaustion, average and peak power, performance time, total work and distance completed and frequency of events. Compared to Matson and Tran's previous ES of 0.44, Peart et al. also showed the overall effect of bicarbonate supplementation to be moderate (weighted effect size of 0.36). The overall ES was significantly higher in untrained subjects ($p = 0.007$) and also higher in single bout exercise compared to repeated bout protocols ($p = 0.013$). Other variables that resulted in a higher effect size included the use

of a time to exhaustion or total performance measure. The authors suggest that training adaptations such as an improved muscle buffering capacity in trained subjects may be why a significantly lower ES in trained subjects was observed.

- Carr, Hopkins and Gore (2011) conducted a meta-analysis of bicarbonate- and citrate-loading studies, using a different data extraction process in which performance data (speed, distance and time) were converted to a common metric (percentage change in mean power). This mixed-model approach allows (1) the effect on performance of specific characteristics to be isolated and quantified and (2) estimation of random differences in magnitude of performance effect. The 38 studies and 137 estimates for sodium bicarbonate produced a possibly moderate performance enhancement of 1.7% (90% CL ± 2.0%) with a typical dose of ~0.3 g/kg/BM in a single 1-minute sprint, following blinded consumption by male athletes. In the 16 studies and 45 estimates for sodium citrate, a typical dose of 1.5 mmoL/kg BM (~0.5 g/kg/BM) had an unclear effect on performance of 0.0% (±1.3%). Study and subject characteristics had the following modifying small effects on the enhancement of performance with sodium bicarbonate: an increase of 0.5% (±0.6%) with a ~0.1 g/k/BM increase in dose; an increase of 0.6% (±0.4%) with five extra sprint bouts; a reduction of 0.6% (±0.9%) for each 10-fold increase in test duration (e.g. 1–10 minutes); reductions of 1.1% (±1.1%) with non-athletes and 0.7% (±1.4%) with females. The repeated bout exercise finding is in contrast to Peart's meta-analysis, which may be due to uneven distribution of trained and untrained subjects in the analysis. It was suggested that single-bout exercise is less likely to benefit from bicarbonate supplementation, as the performance enhancement arises from improved acid-base recovery between bouts.
- Popular theories about bicarbonate and citrate loading include the likelihood that anaerobically trained athletes should show less response to protocols because their intrinsic buffering capacity is already better, and the risk that performance of *prolonged* high-intensity exercise will be impaired if bicarbonate/citrate supplementation leads to increased rates of glycogen utilisation. However, studies that have examined bicarbonate or citrate loading using sports-specific protocols and well-trained subjects (see Table 16.4) fail to support these theories. Some, but not all, studies of well-trained athletes have found performance improvements following bicarbonate/lactate loading prior to brief (1–10 minutes) or prolonged (30–60 minutes) events involving high-intensity exercise (see Table 16.4).

TABLE 16.4 Studies of bicarbonate or citrate loading with relevance to sports performance

Study	Subjects and design*	Dose	Performance measure	Performance enhancement	Summary
Acute supplementation before sports-related performance protocol					
Stoggl et al. 2014	12 endurance-trained males (running and X-country skiing)	300 mg/kg sodium bicarbonate @ 90 min pre-	X-country sprint skiing (Simulation of competition final	No	Active recovery coupled with pre-exercise bicarbonate supplementation or carbohydrate supplementation did not lead to

	Crossover design	exercise (Study also tested effect of CHO supplementation and active rather than passive recovery)	with 3 × 2–5 min efforts with 10–25 min recovery) • 3 × TTE @ 19 km/h treadmill run @ 5% include with 25 min recovery between		detectable improvements in performance as assessed by time to fatigue at high-speed treadmill running, regardless of sustaining blood-buffering capacity. Although bicarbonate supplementation increased pH, there was no significant difference in running capacity in the first or subsequent bouts. Only the application of active recovery led to enhanced performance and attenuated fatigue when compared with the passive recovery intervention.
Afman et al. 2014	7 basketball players Crossover design	2 × 200 mg/kg sodium bicarbonate @ 90 and 20 min pre-game and/or 75 g carbohydrate consumed 45 min pre-game	Basketball 60 min simulated game • Sprint performance • Goals scored in lay up	Yes No	Ingestion of sodium bicarbonate reduced fatigue and maintained sprint performance into the final quarter relative to placebo; sprint performance was also improved with carbohydrate (carbohydrate: –0.07 ± 0.04 s; *p* <0.01 and bicarbonate: –0.08 ± 0.05 s; *p* =0.02). Neither condition resulted in improvement of skill-based shooting performance (lay up).
Hobson et al. 2014	20 male club level rowers Crossover design	300 mg/kg sodium bicarbonate: 200 mg/kg @ 4 h pre-exercise + 100 mg/kg BM @ 120 min pre-exercise	Rowing • 2000 m TT on ergometer	Perhaps	No statistically significant difference in TT performance between conditions, however magnitude based inferences suggest a *likely* positive outcome (84% likelihood) with bicarbonate supplementation. Additionally, 12 athletes improved their performance (3.4 ± 3.7 s), 1 athlete had no change, and 7 athletes had a worse performance (2.3 ± 1.7 s). Authors suggest a responder vs non-responder outcome.
Flueck et al. 2014	6 male and 3 female elite wheelchair-racing athletes Crossover design	500 mg/kg sodium citrate spread @ 90–120 min pre-exercise and/or 6 mg/kg caffeine @ 60 min pre-trial	Wheelchair ergometer • 1500 m TT	No	No benefit to TT performance noted with citrate, caffeine or combination of products. Five of 9 subjects suffered gut side effects with ingestion of sodium citrate alone or in combination with caffeine.
Russell et al. 2014	10 competitive adolescent male swimmers (at least regional level) Crossover design with 4 treatments (acute + chronic citrate + placebo for each application)	500 mg/kg sodium citrate @ 120 min pre-exercise	Swimming • 200 m in preferred stroke	Perhaps	Similar performance in both placebo trials confirmed the reliability of swimming performance. Although swimming performance did not improve across the group, 5 swimmers were determined to be responders, based on enhancement of >0.4% in swimming time. Mean race improvement was 1% across these swimmers, who were also found to be older, heavier and have achieved a greater post-race blood lactate concentration than the other swimmers. See also serial supplementation summary.

Christensen et al. 2014	11 male and 1 female international level rowers (6 lightweight and 6 heavyweight) Crossover (Latin square) design	300 mg/kg sodium bicarbonate @ 75 min pre-exercise (and/or 3 mg/kg caffeine @ 45 min pre-trial)	Rowing • 6 min ergometer test	No	Compared with placebo trial (1865 ± 104 m), a greater distance was covered with caffeine (1878 ± 97, p <0.05) and combined caffeine/bicarbonate (1877 ± 97, p <0.05) but not for bicarbonate alone (1860 ± 96, p = 0.1). Therefore bicarbonate was not associated with performance enhancement of 6-min ergometer test, but neither did it negate the effectiveness of caffeine, as seen in some other studies.
Ducker et al. 2013a	24 competitive male team sport athletes Parallel group design (n = 6 for bicarbonate, β-alanine, combined and placebo)	300 mg/kg sodium bicarbonate @ 60 min pre-exercise (and/or 28 d chronic supplementation β alanine @ 80 mg/kg BM/d)	Team sport • 3 sets @ 6 × 20 m sprints on 25 s with 45 s between sets (tests conducted pre and post 28 d supplementation)	Yes	Effect size comparison revealed that bicarbonate supplementation resulted in 'very likely' faster total sprint time (mean = 1.28-s improvement). Combined β-alanine and bicarbonate supplementation resulted in 'possible' improvement (mean = 0.58 s faster). Benefits for mean sprint time for each set separately were 'very likely' for bicarbonate, and there was a likely enhancement of first and best sprint times for sets 2 and 3 (~0.05-s improvement). Similar trends with 'likely' benefits with combined bicarbonate and β-alanine for each parameter. Small sample sizes noted. See also β-alanine summary.
De Salles et al. 2013	14 well-trained junior swimmers: 7 male, 7 female Crossover design (bicarbonate) embedded with parallel group design (β-alanine)	300 mg sodium bicarbonate @ 90 min pre-exercise (and/or 1 w @ 3.2 g/d + 3.5 w @ 6.4 g/d; Total = 201.6 g)	Swimming • 100 m TT • 200 m TT	Yes Yes	Magnitude-based inference analysis showed that the likelihood of bicarbonate having a positive effect on 100 m race time was 97%, while the addition of bicarbonate to β-alanine had a 72% likelihood of achieving extra benefit. Six of the 7 swimmers improved race time with bicarbonate alone (2.8% faster). 200-m swim time improved with bicarbonate (2.3%) and bicarbonate achieved a further improvement of 0.6% when added to β-alanine with all subjects showing faster race times. No improvements were seen with the placebo group
Mero et al. 2013	13 highly trained male swimmers (international and national level) Crossover (bicarbonate) before and after β-alanine	300 mg/kg sodium bicarbonate @ 60 min pre-exercise (first swim) (pre- and post-28 d @ 4.8 g/d β-alanine)	Swimming • 2 × 100 m, 12 min apart	No	Prior to β-alanine supplementation, there was no effect of bicarbonate supplementation on swim 1, but there was less attenuation of performance in swim 2 such that performance was ~1.5 s or 2.4% faster (p <0.05). Following β-alanine supplementation, bicarbonate failed to provide any change in performance of swim 1 or swim 2.
Hobson et al.	20 well-trained	300 mg/kg	Rowing	Perhaps	Magnitude-based inference

07:37:13.

2013	male rowers Crossover (bicarbonate) with simultaneous parallel (β-alanine) design	sodium bicarbonate: 200 mg/kg @ 4 h pre-exercise + 100 mg/kg BM @ 120 min pre-exercise (and/or 30 d @ 6.4 g/d β-alanine)	• 2000 m TT on ergometer		analysis showed that bicarbonate supplementation had a likely benefit on rowing performance (3.2 ± 8.8 s improvement over placebo). Whereas, β-alanine supplementation was very likely to be beneficial to 2000-m rowing performance (6.4 ± 8.1 s effect compared with placebo) and there was a small (1.1 ± 5.6 s) but possibly beneficial additional effect when combining chronic β-alanine supplementation with acute sodium bicarbonate supplementation compared with chronic β-alanine supplementation alone.
Bellinger et al. 2012	14 highly trained cyclists Crossover (bicarbonate) with simultaneous parallel (β alanine) design	300 mg/kg sodium bicarbonate @ 90 min pre-exercise (and/or 28 d @ 65 mg/kg/d β alanine	Cycling • 4 min TT on cycling ergometer	Yes	Average power output was increased compared to baseline for placebo + bicarbonate (+3.1% ± 1.8%) and β alanine + bicarbonate (3.3% ± 3.0%), but not for β alanine + placebo (1.6% ± 1.7%). Note that 2 subjects experienced mild paraesthesia (side effect of β alanine supplementation), which may have compromised blinding of the study.
Driller et al. 2012	8 male cyclists Crossover design	300 mg/kg BM spread @ 30-90 min pre-exercise See also serial bicarbonate supplementation protocol	Cycling • 4 min TT on cycling ergometer	Yes	Acute bicarbonate loading protocols resulted in higher power output (acute 431.5 ± 37.9 W vs placebo 418.3 ± 41.5 W, *p* <0.05), with equal improvements also seen with serial supplementation protocol (427.5 ± 43.6 W). Minor gastrointestinal symptoms were reported in 3 subjects for the acute loading but not serial supplementation.
Joyce et al. 2012	8 highly trained male swimmers Crossover design	300 mg/kg BM/d sodium bicarbonate spread @ 90–105 min pre-exercise See also serial bicarbonate supplementation protocol	Swimming • 2 × 200 m TT 24 h apart	No	Plasma bicarbonate concentration was increased by acute loading prior to first TT but was returned to baseline after 24 h. There was no enhancement of TT performance with acute loading compared with placebo trial (1:59.57 ± 0:06.21 vs 1:59.02 ± 0:05.82 and no difference in further TT after 24 h. Mild symptoms of gut discomfort were noted but were not reduced by a serial loading protocol. Suggested that highly trained athletes may already have enhanced muscle-buffering capacity and benefit less from bicarbonate loading.
Carr et al. 2012	4 male + 3 female club level rowers Crossover design	300 mg/kg @120 min pre-exercise See also serial bicarbonate supplementation protocol	Rowing • 2 × 2000 m TT on ergometer, 48 h apart to simulate regatta	No	Differences in performance (2000-m mean power, stroke rate or RPE, and characteristics of each 500 m segment) between acute, placebo and serial loading were trivial, despite the increase in blood bicarbonate concentrations with supplementation. Performance was

					a little less reliable with acute supplementation compared with placebo or serial supplementation.
Zabala et al. 2011	10 elite motocross (BMX) cyclists Crossover design	300 mg/kg sodium bicarbonate @ 90 min pre-exercise	BMX simulation • 3 × 30 s cycling Wingate tests with 15-min recovery • Counter movement jump	No No	Bicarbonate maintained enhanced buffering capacity across the 3 trials, but did not improve power or fatigue characteristics of Wingate sprints or jump height. Authors concluded that test protocol may not have produced sufficient H+ efflux to benefit from additional buffering. The authors noted that some aspects of the results might be explained by the use of lower calibre rowers (e.g. club rowers) compared to studies of national level rowers.
Carr, Gore et al. 2011	8 well-trained rowers Crossover design	300 mg/kg sodium bicarbonate and/or 6 mg/kg caffeine pre-exercise	Rowing • 2000 m ergometer TT	No	Performance was enhanced by ~2% with caffeine but GI symptoms associated with bicarbonate counteracted this, leading to unclear performance outcome.
Sale et al. 2011	20 male cyclists Crossover (bicarbonate) with simultaneous parallel (β alanine) design	300 mg/kg sodium bicarbonate pre-exercise and/or 4 w @ 6.4 g/d B-alanine	Cycling • TTE @ 110% power max on ergometer	Perhaps	β-alanine enhanced cycling capacity. Addition of bicarbonate increased TTE by 6 s (4% increase), which did not reach statistical significance but according to magnitude based inferences, has a 70% probability of being a meaningful improvement.
Cameron et al. 2010	25 male rugby players Crossover design	300 mg/kg sodium bicarbonate @ 65 min pre-warm-up	Rugby Union • Rugby-specific repeated sprint test + 9 min rugby-specific play	No	No difference in performance of rugby-specific test between trials. High risk of gut side effects noted: this may impair performance.
Ching-Lin et al. 2010	9 male collegiate tennis players Crossover design	300 mg/kg sodium bicarbonate @ pre-exercise and 100 mg/kg during exercise	Tennis • Simulated tennis match with Loughborough Tennis skill test performed pre and post	Yes	Consistency scores for a number of strokes declined significantly post match with placebo but were maintained in the bicarbonate trial. The match-induced declines in the consistency scores were significantly larger in the placebo trial than those in the bicarbonate trial.
Tan et al. 2010	12 elite female water polo players Crossover design	300 mg/kg sodium bicarbonate pre-exercise	Waterpolo • 59 min match simulation with 56 × 10 m sprint swims	No	Percentage difference in mean sprint times with bicarbonate vs placebo were not substantial (0.4 ± -1.0, p =.51).
Siegler & Hirscher 2010	10 male amateur boxers Crossover design	300 mg/kg sodium bicarbonate pre-exercise	Boxing • Sparring: 4 × 3 min rounds with 1 min recovery	Yes	Significant increase in punches successfully landed in bicarbonate trial.
Siegler & Gleadall-	6 male and 8 female competitive	300 mg/kg sodium	Swimming • 8 × 25 m with 5	Yes	Total swim time was 2% faster in the bicarbonate trial (159.4 ± 25.4

Siddall 2010	swimmers Crossover design	bicarbonate pre-exercise	s recovery (simulation of 200 m race controlled for variability in turns)		vs 163.2 ± 25.6 s; *p* <0.04).
Pruscino et al. 2008	6 elite male swimmers Crossover design	300 mg/kg sodium bicarbonate spread @ 30–120 min pre-exercise 6 mg/kg caffeine 45 min pre-exercise	Swimming • 2 × 200 m TT on 30 min recovery	No, for a one-off 200 m TT Yes, for repeat 200 m TT	Bicarbonate enhanced performance, with and without caffeine on repeat performance. Effect was less evident for a single effort. Majority of athletes recorded fastest TT for single and repeat performance from the combination of bicarbonate and caffeine.
Zabala et al. 2008	9 elite male BMX riders Crossover design	300 mg/kg sodium bicarbonate @ 90 min pre-exercise	BMX simulation • 3 × 30 s cycling Wingate tests with 30 min recovery	No	Bicarbonate maintained enhanced buffering capacity across the three trials, but did not improve power or fatigue characteristics of Wingate sprints or perceived effort. Riders reported enhancement of perceived readiness for next trial, however. Authors concluded that test protocol may have been too short and recovery period too long to sufficiently challenge buffering capacity.
Lindh et al. 2008	9 elite male swimmers Crossover design	300 mg/kg sodium bicarbonate spread @ 60–90 min pre-exercise	Swimming • 200 m TT	Yes	Swimming TT with bicarbonate trial was 1.6% faster than placebo trial in internationally competitive swimmers.
McClung et al. 2007	4 female and 12 male national level endurance runners Crossover design	sodium bicarbonate spread @ 90–120 min pre-exercise	Running • 1000 m track run	Perhaps – but due to belief effect?	Study design tested the independent effects of bicarbonate and belief by comparing performance in 4 trials: receiving bicarbonate when told they had received an active agent or inert agent, and inert agent when told they were receiving active ingredient or not. Fastest run achieved when told and received bicarbonate (184.7 ± 24.1 s) with next best time when told they had received active agent but didn't receive it (185.1 ± 22.1 s). Times when told no active agent were similar despite receiving bicarbonate (188.5 ± 24.4 s) or not (187.9 ± 22.4 s). Overall, the only statistically significant effect was a main effect of being told they had received an active agent.
Artioli et al. 2007	9 national level judo athletes Crossover design	300 mg/kg sodium bicarbonate @ 120 min pre-exercise	Judo • 3 × judo-specific throwing fitness tests on 5 min recovery	Yes	Bicarbonate supplementation increased total throws completed, primarily in bouts two and three.
Artioli et al. 2007	14 national level judo athletes	300 mg/kg sodium	Judo • 4 × Wingate	Yes	Greater mean power with bicarbonate supplementation in

	Crossover design	bicarbonate @ 120 min pre-exercise	anaerobic upper body tests on 3 min recovery		bouts three and four and greater PPO in bout four.
Bishop & Claudius 2005	7 female team sports players Crossover design	2 × 200 mg/kg bicarbonate @ 90 min and 20 min pre-exercise	Team sport simulation • Intermittent cycling protocol of 2 × 36 min 'halves' involving repeated 2 min blocks (4 s sprint, 100-s active recovery at 35% VO_2 peak, and 20 s rest)	Yes	Bicarbonate supplementation failed to produce any effect on performance in first half, but caused trend towards improved total work in the second half (p =0.08). In particular, subjects completed significantly more work in 7 of 18 4 s sprints in the second half of the bicarbonate trial.
Van Montfoort et al. 2004	15 competitive male distance runners Crossover design	300 mg/kg sodium bicarbonate or 525 mg/kg sodium citrate @ 90–180 min pre-race	Running • Treadmill run to exhaustion at speed designed to last 1–2 min	Yes for bicarbonate. Perhaps for citrate	Analysis estimated likelihood of treatments increasing endurance compared to placebo by at least 0.5% (considered to be the smallest worthwhile improvement). Bicarbonate produced 2.7% enhancement of endurance (96% chance of improvement); citrate enhanced endurance by 0.5% (50% chance). Overall, authors concluded that bicarbonate is most effective, and citrate is possibly not as effective. No difference in gastrointestinal symptoms.
Bishop et al. 2004	10 female recreational team sports players Parallel group design	300 mg/kg sodium bicarbonate @ 90 min pre-exercise	Cycling • 5 × 6 s maximal sprints, every 30 s	Yes	Compared to the control group there was a significant increase in total work for five sprints and peak power output in sprints 3–5.
Mero et al. 2004	8 male + 8 female national level swimmers Crossover design (30-d washout of creatine)	300 mg/kg bicarbonate @ 120 min pre-exercise + 6 d @ 20 g/d creatine	Swimming • 2 × 100 m swims with 10 min passive recovery	Yes (?)	Faster time for second swim with creatine/bicarbonate trial than with placebo: 1 s reduction in performance from first swim in placebo compared with 0.1 s drop-off in supplement trial (p <0.05). Study unable to indicate individual effect of bicarbonate.
Oopik et al. 2004	10 male collegiate runners Crossover design	500 mg/kg sodium citrate @ 120 min pre-exercise	Running • 5000 m track run •	No	Citrate consumed in 1.5 L water increased BM but did not enhance performance of a 5000 m run in field conditions (1100.0 ± 79.1 and 1082.7 ± 62.0 s in citrate and placebo trials, respectively, p =0.09).
Price et al. 2003	8 active male runners Crossover design	300 mg/kg sodium bicarbonate @ 60 min pre-exercise	Team sport simulation • Intermittent cycling of 30 min @ repeated 3 min blocks (90 s @ 40%, 60 s @ 60% VO_{2max} and 14 s @ maximal sprint)	Yes	Significant main effect with greater PPO achieved in 14 s sprints across protocol in bicarbonate trial, whereas placebo trial showed gradual decline in PPO across time. Blood lactate levels elevated to 10–12 mmol/L by 10 minutes and remained elevated across rest of protocol. Such values are higher than is generally reported in team

					sports; thus movement patterns may not reflect the true workloads or physiological limitations of team sports.
Oopik et al. 2003	17 male collegiate distance runners Crossover design	500 mg/kg sodium citrate @ 120 min pre-exercise	Running • 5000 m treadmill run	Yes	Performance significantly faster (p <0.05) for citrate trial (1153 s) compared with placebo trial (1183 s). High risk of gastrointestinal distress. Blood lactate concentration higher after race with citrate trial. No change in RPE.
Bowman et al. 2002	7 female and 3 male collegiate swimmers Crossover dose–response design	100, 200 or 300 mg/kg @ 60 min pre-race (45 min before warm-up)	Swimming • 1100–400 m TT depending on main event, short-course pool	Yes at all doses	Swimmers improved TT performances at all doses: compared with control, times were 97.0 ± 1.27% for 0.1 g/kg, 98.19 ± 1.75% for 0.2 g/kg and 99.10 ± 1.02% for 0.3 g/kg, all p <0.05. Side effects increased with increasing dose.
Stephens et al. 2002	6 well-trained male cyclists/triathletes and 1 cross-country skier Crossover design	300 mg/kg sodium bicarbonate @ 120 min pre-exercise	Cycling • 30 min @t 77% VO_{2max} + TT (~30 min) or ergometer	No	Increase in blood lactate but no difference in TT performance time, muscle glycogen utilisation or lactate.
Shave et al. 2001	7 elite male + 2 elite female athletes Crossover design	500 mg/kg sodium citrate @ 90 min pre-race	Running • 3000 m	Yes	Performance time significantly faster (p <0.05) for citrate trial (610.9 s) compared with placebo trial (621.6 s). High risk of GI distress.
Schabort et al. 2000	8 endurance-trained male cyclists and triathletes Crossover design	200 mg/kg, 400 mg/kg and 600 mg/kg sodium citrate @ 60 min pre-exercise	Cycling • 40 km TT including 500 m, 1 km and 2 km sprints	No	Increasing citrate dose increased blood pH but no effect on sprint performances or overall 40 km TT performance (58:46, 60:24, 61:47 and 60:02 minutes for citrate (200, 400 and 600 mg/kg doses) and placebo.
McNaughton, Backx et al. 1999	10 well-trained male cyclists Crossover design	300 mg/kg sodium bicarbonate @ 90 min pre-exercise	Cycling • 60 min TT	Yes	14% more work completed with bicarbonate
Potteiger, Nickel et al. 1996	8 trained male cyclists Crossover design	500 mg/kg sodium citrate @ 90 min pre-exercise	Cycling • 30 km TT	Yes	Reduction in TT time (57:36 minutes vs 59:22). Sodium citrate raised pH values from 10 km onwards and improved power output in the initial 25 min.
Potteiger, Webster et al. 1996	7 well-trained male runners Crossover design	300 mg/kg sodium bicarbonate and 500 mg/kg sodium citrate @ 120 min pre-exercise	Running • 30 min at LT + TTE at 110% LT	No	Both citrate and bicarbonate supplementation increased blood pH during steady-state run. No differences in run to exhaustion: 287 s, 172.8 s, 222.3 s for bicarbonate, citrate and placebo respectively.
Tiryaki & Atterbom 1995	11 collegiate female runners + 4 trained non-	300 mg/kg sodium citrate or sodium	Running • 600 m	No	No performance effect despite significant changes to acid-base status.

	athletes Crossover design	bicarbonate @ 150 min pre-exercise			
Bird et al. 1995	10 trained middle-distance runners Crossover design	300 mg/kg sodium bicarbonate (half at 120 and half at 90 min pre-exercise)	Running • 1500 m	Yes	Performance in bicarbonate trial improved compared with placebo trial (253.9 vs 256.8 s, p <0.05).
Pierce et al. 1992	7 male collegiate swimmers Crossover design	200 mg/kg bicarbonate, sodium chloride placebo or control @ 60 min pre-exercise	Swimming • 100 yards freestyle • 2 × 200 yards swims 20 min recovery between each race (simulation of competition program)	No No	No difference in swim times between trials.
McNaughton & Cedaro 1991	5 highly trained male rowers Crossover design	300 mg/kg sodium bicarbonate @ 95 min pre-exercise	Rowing • 6 min maximum effort on ergometer	Yes	Increased work and distance rowed in bicarbonate trial (1861 m vs 1813 m). Increased lactate levels.
Goldfinch et al. 1988	6 trained male runners Crossover design	400 mg/kg sodium bicarbonate @ 60 min pre-exercise	Running • 400 m	Yes	Improved running time (56.94 s vs 58.63 for placebo and 58.46 for control). Elevated post-exercise values for pH and base excess.
Wilkes et al. 1983	6 varsity track male athletes Crossover design	300 mg/kg sodium bicarbonate @ 150 min pre-exercise	Running • 800 m	Yes	Improved running time (2:02.9 min vs 2:05.1 for placebo and 2:05.8 for control). Elevated post-exercise values for pH, lactate and blood bicarbonate.
Serial supplementation (e.g. several day protocol) prior to sports-related performance protocol					
Russell et al. 2014	10 competitive adolescent male swimmers (at least regional level) Crossover design with 4 treatments (acute + chronic citrate, 2 placebo)	4 d @ 300 mg/kg sodium citrate split into 3 doses with last dose being 300 mg @ 120 min pre-exercise (See also acute bicarbonate supplementation)	Swimming • 200 m in preferred stroke	No	Similar performance in both placebo trials confirmed the reliability of swimming performance. Despite an increase in base excess with the serial supplementation, there was no enhancement in swimming performance.
Tobias et al. 2013	37 judo and jiu-jitsu athletes Parallel group design for both supplements	7 d @ 500 mg/kg/day sodium bicarbonate split into 4 doses (and/or 4 w @ 6.4 g/d β-alanine)	Combat sports simulation • 4 × 30 s upper body Wingate tests @ 3 min	Yes	Bicarbonate supplementation increased total work done compared to placebo (+8% p = 0.002). β-alanine supplementation also enhanced work (+7% p = 0.003), and co-ingestion resulted in the greatest gains (+14%, p < 0.0001). Similarly, mean power for the 2nd, 3rd and 4th bouts was improved with both supplements individually and in combination.
Carr et al.	4 male and 3	Serial: 3 d @	Rowing	No	Despite the increase in blood

2012	female club level rowers Crossover design	500 mg/kg/d sodium bicarbonate split into 5 doses (See also acute bicarbonate supplementation)	• 2 × 2000 m TT on ergometer @ 48 h apart to simulate regatta (testing on day 2 and day 4)		bicarbonate, serial bicarbonate supplementation failed to enhance performance in terms of 2000 m mean power, stroke rate or RPE, or characteristics over each 500 m segment. The authors suggest that the lower calibre of rower in this study may explain the small effect size of performance changes. The reliability of performance in response to chronic supplementation protocol was high; suggesting that individual variability in responsiveness can be quickly established from a trial.
Driller et al. 2012	8 male cyclists Crossover design	400 mg/kg BM spread over 4 doses, for 3 d pre-exercise (See also acute bicarbonate supplementation protocol)	Cycling • 4 min TT on ergometer	Yes	Both serial and acute bicarbonate loading protocols resulted in higher power output (acute 431.5 ± 37.9 W and serial 427.5 ±43.6 W versus placebo 418.3 ± 41.5 W, $p < 0.05$), with no difference between supplementation protocols. Minor gastrointestinal symptoms were reported in 3 subjects for the acute loading, but no symptoms reported for serial loading. Serial loading may offer a practical alternative to the acute loading approach, with potential for minimisation of gastrointestinal side effects.
Joyce et al. 2012	8 highly trained male swimmers Crossover design	300 mg/kg/d sodium bicarbonate divided in 3 daily doses for 3 d and final dose on day 4 (See also acute bicarbonate supplementation protocol)	Swimming • 20 m TT on day 4, followed by a further TT 24 h later	After 3 d: no Further 24 h later: no	Plasma bicarbonate concentration was increased after 3 d serial loading, although to a smaller extent than when same daily amount was taken as acute dose 90 min before the TT. However bicarbonate concentrations returned to baseline 24 h after last dose. There was no enhancement of TT performance immediately after serial loading compared with placebo trial (1:58.53 ± 0:05.64 vs, 1:59.02 ± 0:05.82) and no difference in between this TT and a further TT after 24 h. Mild symptoms of gut discomfort were noted but not different to those noted in acute loading protocol.
Duncan et al. 2014	8 resistance-trained males Crossover design	300 mg/kg sodium bicarbonate @ 60 min pre-exercise	Resistance exercise 3 sets @ 80% 1RM to failure • Bench press • Back squat	No Yes	Participants completed more repetitions to failure in the back squat after $NaHCO_3$ ingestion (p = 0.04) but not for bench press (p = 0.679). Mean ± SD of total repetitions was 31.3 ± 15.3 and 24.6 ± 16.2 for back squat and 28.7 ± 12.2 and 26.7 ± 10.2 for bench press in $NaHCO_3$ and placebo conditions, respectively.
Carr et al. 2013	12 resistance-trained males Crossover design	300 mg/kg sodium bicarbonate	Resistance exercise 4 sets @ 10–12	Perhaps	Protocol involved feeding strategies designed to reduce gut symptoms (pre-event snack and fluid intake).

		spread @ 50–80 min pre-exercise	reps to failure • Back squat, leg press and knee extension		Supplementation resulted in a greater number of repetitions for all exercises combined (bicarbonate 163.7 ± 15.1 vs placebo 156.7 ± 14.5 repetitions), however no significant difference between repetitions for individual exercises.
Siegler et al. 2010	8 males trained in intermittent sports Crossover design	300 mg/kg sodium bicarbonate @ 60 min pre-exercise	Running • 4 sets @ 5 × 24 s treadmill sprints on 1 min intervals with 1 min rest between sets	No	Increase in buffering capacity did not translate into clear performance benefits due to considerable inter-individual responses, including negative effects due to gut distress
Webster et al. 1993	6 resistance-trained males Crossover design	300 mg/kg sodium bicarbonate @ 105 min pre-exercise	Resistance exercise • 4 sets @ 12 reps leg press @ 70% 1RM + 5th set to failure	No	Repetitions performed in the final exercise set were not significantly different between trials (NaHCO3: 19.6 ± 1.6, placebo: 18.2 ± 1.1 repetitions)
Gao et al. 1988	10 male collegiate swimmers Crossover design	250 mg/kg sodium bicarbonate 60 min hour pre-exercise	Swimming • 5 × 100 yard swim with 2 min rest	Yes	Faster times in fourth and fifth swim ($p < 0.05$). Supplementation also associated with higher post-race blood lactate concentrations.
Chronic supplementation as support for training outcomes/adaptation					
Driller et al. 2013	12 elite male rowers Parallel group design	4 w @ 300 mg sodium bicarbonate @ 90 min pre-HIT session (training = 2/w HIT sessions)	Rowing • 2000 m TT on ergometer	No	Both bicarbonate and placebo groups improved performance after 4 w training but differences between groups for 2000 m performance time or power, peak power output or power at 4 mmol/L lactate threshold were trivial. Supplementation was taken with food to reduce risk of gastrointestinal upset and produced a change in blood biochemistry for HIT sessions. There was no difference in training volume or intensity between groups.
Edge et al. 2006	16 moderately trained females (team sports and rowing) Parallel group design	8 w @ 400 mg/kg sodium bicarbonate spread 30–90 min pre-HIT training (3/w HIT with 6–12 reps @ 2 min sprint + 1 min rest)	Team sport relevance Cycling • TTE @ 100% pre-training VO_{2peak}	Yes	Both groups increased VO_{2peak}, lactate threshold and TTE after training. However, bicarbonate group had significantly greater improvements in lactate threshold (26 versus 15%) and TTE (164 versus 123%) than the placebo group. Both groups undertook same training load; effect potentially explained as greater adaptation to same stimulus when disturbance to cell homeostasis was reduced by enhanced buffering.

*Randomised, double-blinded, placebo-controlled unless noted.

BM = body mass, HIT = high-intensity interval training, LT = lactate threshold, PO = power output, PPO = peak power output, RPE = rating of perceived exertion, TT = time trial, TTE = time to exhaustion, W/kg = watts per kilogram

Note: For updates see www.ausport.gov.au/ais/nutrition/supplements

Until further research can clarify the range of exercise activities that might benefit from bicarbonate or citrate enhancement, individual athletes are advised to experiment in training and minor competitions to judge their own case. It is important that experimentation is conducted in a competition-simulated environment, including the need to undertake multiple loading strategies for heats and finals of an event; the athlete needs to discover not only the potential for performance improvement but also the likelihood of unwanted side effects.

Finally, there is some preliminary evidence that bicarbonate loading may be used to enhance training outcomes. Edge and colleagues (2006) monitored moderately trained female team athletes over an 8-week program that included three interval training sessions per week: one group undertook an acute bicarbonate loading protocol prior to each of these sessions, while a matched group took a placebo. While the groups increased their maximal aerobic capacity equally, the bicarbonate-loading group recorded a larger increase in lactate threshold and the ability to exercise at the pre-training level of $VO_{2\ peak}$ for a longer duration than the placebo group. The mechanism of such improvements may include an enhanced ability to train harder (see Table 16.4 for summary of studies of acute supplementation and training capacity). However, Edge and colleagues (2006) also suggest that benefits could accrue if enhanced alkalosis during high-intensity training reduces damage to muscle protein. Further work is warranted.

16.7.5 β-alanine

Carnosine is a dipeptide, formed from the amino acids histidine and β-alanine (the only naturally occurring amino acid in 'β' form). It is found in large amounts in the brain and muscle, particularly type II fibres, where it has a number of known roles as well as the potential for other important but as yet under-appreciated activities. It is currently best known for its antioxidant properties as well as accounting for about 10% of the muscle's ability to buffer the acidity (H^+ ions) produced by high intensity exercise. However, it also plays a role in increasing the efficiency of calcium handling in the muscle to enhance the efficiency of excitation-contraction coupling, via the process known as the carnosine shuttle (Blancquaert et al. 2015). Some dietary carnosine is absorbed intact, but most is quickly broken down to β-alanine and histidine. Within the muscle, β-alanine availability is the limiting factor in carnosine production. Therefore, intake of β-alanine provides an efficient strategy to increase muscle carnosine content. Dietary sources of carnosine and β-alanine include meats, especially 'white' (fast twitch) meat such as the breast meat of poultry and of sea animals that are exposed to hypoxia, such as whale. Vegetarians have lower resting muscle carnosine concentrations than meat eaters (Harris et al. 2007).

Protocols for β-alanine supplementation

The first studies of β-alanine supplementation showed that consumption of 5–6 g/d (~65 mg/kg) β-alanine can increase muscle carnosine content by ~60% after 4 weeks and ~80% after 10 weeks of supplementation (Harris et al. 2006). Further dose-response studies (Stellingwerff et al. 2012a) and reviews of the available supplementation literature (Stellingwerff et al. 2012b) have identified further themes regarding the dynamics of supplementation and washout:

- There is a linear correlation between total grams of β-alanine consumed (of daily intake ranges of 1.6-6.4 g β-alanine/day) versus both the relative and absolute increases in muscle carnosine that is independent of body mass and initial muscle carnosine content (Stellingwerff et al. 2012b).
- The efficiency of β-alanine loading (i.e. the increase in muscle stores per given dose) is only about 2–3%, but appears to be increased in trained muscle (Bex et al. 2014) and when co-ingested with a CHO-rich meal (Bex et al. 2014). The fate of the majority of the dose is unknown, but since little appears in the urine, there are suggestions that it may simply be oxidised in the body.
- So far, a ceiling in the storage of carnosine has not been reported, but researchers suggest that an increase of 30–50% is a reasonable goal; this can be achieved by a total intake of 200–300 g of β-alanine over a period of ~4–10 weeks (e.g. ~6.4 g/d for 6 weeks).
- Many athletes experience paraesthesia (a 'prickling sensation') when doses of >800 mg of β-alanine powder are consumed. It is of interest to note that this is equivalent to the β-alanine content of a chicken breast and is thus presumed to be associated with the rapid increase in plasma β-alanine levels. This side effect appears to be increased by prior exercise but can be circumvented by the use of a slow-release preparation (e.g. CarnoSyn™) (Harris et al. 2008) and/or by taking the β-alanine supplement as split doses over the day, with meals. This strategy is generally recommended since it is likely to enhance uptake as well as reduce side effects.
- The washout of loaded carnosine stores is slow, taking ~15 weeks to return to baseline (Baguet et al. 2010). Once elevated, muscle carnosine stores can be maintained with an intake of ~1.2 g/d (Stegen 2014).

Effects of β-alanine supplementation on performance

Like creatine loading, supplementation with β-alanine may enhance sports performance via 'acute' and/or 'chronic' effects; elevated muscle carnosine may directly enhance performance of a competitive event and/or it may improve competition performance as a result of greater gains from the training program. It is more difficult, however, to distinguish between these effects since it may take at least 4 weeks to achieve worthwhile increases in muscle carnosine levels; thereby creating the opportunity for effects on training. In another similarity to creatine loading, the mechanism(s) underlying the benefits of elevated muscle carnosine on exercise capacity are not fully understood, making it difficult to identify the full range of sporting activities to which it could be applied.

If the major benefit of increasing muscle carnosine is to enhance intracellular buffering capacity, then β-alanine supplementation could be considered useful to athletes who undertake events which are limited by the alterations to intracellular pH as a result of high rates of ATP regeneration from anaerobic glycolysis (Derave et al. 2010). These include:

- sustained competitive events lasting 1–7 minutes (e.g. rowing, swimming, track cycling, middle distance running)
- repeated bouts of high-intensity work (sprints, lifts) which cause an exercise-limiting increase in H^+ ions over time (e.g. interval training and resistance training, racquet and team sports)
- high-intensity effort(s) undertaken within or at the end of prolonged exercise that is principally undertaken at intensities below those associated with pH-related limitations (e.g. road cycling, distance running).

These are the same scenarios to which bicarbonate/citrate buffering strategies are targeted (see section 16.7.3) with β-alanine supplementation representing an opportunity to provide a chronic manipulation of buffering capacity rather than undertake a series of acute applications. This can overcome some of the challenges previously identified with the use of bicarbonate loading for competition situations. In addition, it may provide a more suitable or sustained approach to training support either by increasing the capacity to train hard or by reducing the damage associated with high levels of muscle acidity. Athletes may also use a combination of bicarbonate and β-alanine supplementation strategies in competition scenarios to see if there is an additive effect of enhancing both intra- and extra-cellular buffering capacity. Finally, increased muscle carnosine concentrations may offer other advantages to training adaptations and competition performance arising from its other roles in the muscle (e.g. antioxidant activity or calcium handling in the muscle). The events to which these might be best applied are less evident.

The available literature of β-alanine supplementation and exercise performance is summarised in Table 16.5, with our criteria for inclusion being the involvement of exercise protocols that are relevant to sport and involve subjects that are specifically trained for a sport. We have ignored studies involving ‘physically active’ cohorts who undertook simplistic exercise protocols such as measurements of VO_{2max} or time to exhaustion at a fixed power output. Given the range in study methodologies in terms of subject populations, performance tests and dosing protocols, it is not surprising that there are diverse outcomes. There is evidence of enhanced training adaptations and/or performance of high-intensity exercise in some studies but these findings are not universal. Variability in study design such as the duration and dosing of β-alanine supplementation protocols; the number, type and calibre of subjects; and the mode and duration/intensity of exercise protocols make it difficult to determine the overall effects.

- Others have tried to summarise this literature via narrative reviews (Artioli et al. 2010; Bellinger et al. 2014; Blancquaert et al. 2015; Quesnele et al. 2014) or a meta-analytical approach (Hobson et al. 2012). Such reviews are supportive of the hypothesis and scientific data underpinning the potential performance benefits of β-alanine supplementation but note the difficulties of defining the applications of this product. Further confounding issues in conducting studies and interpreting the outcomes include the difficulties in controlling other variables in chronically applied treatments and the complexity of understanding benefits from products with a range of potential mechanisms of action. It has been suggested that significant ergogenic effects have been more easily detected in untrained individuals performing exercise bouts under laboratory conditions, but that highly trained athletes may also achieve modest but potentially worthwhile benefits to competition performance due to the direct effects of β-alanine supplementation or to superior adaptations to a training program (Bellinger et al. 2014; Blancquaert et al. 2015). The meta-analysis of the available β-alanine supplementation literature in 2010 concluded that it is clearly beneficial to the performance of high-intensity exercise protocols lasting 1–4 minutes ($p = 0.001$), whereas the effect was absent in shorter events ($p = 0.312$) and less obvious in longer events ($p = 0.046$). Of course, as shown in Table 16.5, the literature continues to accumulate and will gradually help to define the best uses of β-alanine supplementation.

- It is noticeable from our summary table that there has also been considerable interest in the separate and additive effects of β-alanine supplementation and bicarbonate loading on sports protocols that challenge the muscle's buffering capacity. The current literature provides examples where neither, either or both provide benefits to sports-related protocols (Table 16.5), but again a clear interpretation of the literature is challenged by differences in the statistical power of the study design, the calibre of subjects, the exercise test and the analytical approach. Further work on the individual and additive effects of supplements is welcomed, since in the real world, athletes often combine the use of products that 'work' for the same type of exercise scenario. Until there is clear delineation of the evidence-based uses of β-alanine, it is typically acknowledged that the use of pure forms of this amino acid is considered safe, with the only commonly reported side effect of paraesthesia being controlled by the use of sustained release products and/or split dose protocols. Of course, as with all commercial products, there is a risk of problems when β-alanine is included in multi-compound supplements with less rigour over the full disclosure of ingredients.

*

Gross et al. 2014	9 elite male alpine skiers Parallel group design	5 w @ 4.8 g/d	Alpine skiing elements • countermovement jumps (CMJ) • 90 s cycling bout at 110% VO_{2max} • maximal 90 s box jump test	Yes Perhaps Perhaps	β-alanine improved maximal (+7 ± 3%, $p < 0.04$) and mean power (+7 ± 2%, $p < 0.04$) in the CMJ and showed a trend to enhanced aerobic contribution to 90-s cycling test (reduced oxygen deficit (-3 ± 8 %, $p = 0.06$) and lactate accumulation (-12 ± 31%) with

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					increased aerobic energy contribution (+1.3 ± 2.9%, *p* = 0.07). Performance in the last third of the 90-s jumping test was improved (+7 ± 4%, *p* = 0.02). No changes in performance seen in placebo group. Note small subject numbers.
Chung et al. 2014	27 well-trained trained male cyclists/triathletes Parallel group design	6 w @ 6.4 g/d Total = 268.8 g	Cycling • 1 h TT on ergometer	No	Proton MRS measurement of muscle carnosine showed significant increases in muscle carnosine in β-alanine group (161.0% in soleus and 143.0% in gastrocnemius). No significant benefit of β-alanine on 1 h cycling TT performance. Small effect of β-alanine on lactate accumulation associated with systemic acidosis.
De Salles Paneilli et al. 2014	20 endurance-trained males and 20 untrained young males Parallel group design	4 w @ 6.4 g/d Total = 179.2 g	Cycling • 4 × 30 s Wingate	Yes	Significant benefit of β-alanine in both trained and untrained populations for repeated 30 s Wingate sprints in terms of total work done and peak power.
James et al. 2014	19 trained competitive (Cat 1) male cyclists	4 w @ 6.4 g/d Total = 179.2 g	Cycling • 20 km cycling TT	No	No significant benefit of β-alanine on 20 km cycling TT performance.
De Salles Painelli et al. 2013	16 well-trained junior swimmers: 12 male, 6 female Parallel group design	1 w @ 3.2 g/d and 4 w @ 6.4 g/d Total = 201.6 g	Swimming • 100 m TT • 200 m TT	Yes Yes	Compared with placebo group, improvement in 100 m performance in β-alanine group approached significance (*p* = 0.07): all 9 swimmers in this group swam faster (mean = 2.1% faster) while only 4 of 7 in placebo group improved (mean = 0.3% slower). For 200 m, improvement in β-alanine group significantly different to placebo group (*p* = 0.002: 8 of 9 swimmers swam faster (mean = 2.0% faster) while only 4 of 7 in placebo group improved (mean = 0.1% slower).
De Salles Painelli et al. 2013	14 well-trained junior swimmers: 7 male, 7 female Parallel group design (β-alanine) with crossover design (bicarbonate)	1 w @ 3.2 g/day + 3.5 w @ 6.4 g/d Total = 201.6 g (+ 300 mg sodium bicarbonate @ 90 min pre-exercise)	Swimming • 100 m TT • 200 m TT	Perhaps Yes	Magnitude-based inference analysis showed that the likelihood of β-alanine having a positive effect on 100 m race time was 65%, with the addition of bicarbonate having a 72% likelihood of achieving extra benefit. All 7 swimmers improved race time with β-alanine (1.4% faster) and the combination with sodium bicarbonate (2.8% faster). 200-m swim time improved with all supplementation protocols (1.5% for β-alanine and 2.1% for β-alanine and bicarbonate combined) but no changes were seen with placebo group. All swimmers showed faster times with β-alanine

					supplementation and a further improvement with the addition of sodium bicarbonate.
Ducker et al. 2013b	18 male middle distance runners (club level) Parallel group design, blinded to subjects	28 d @ 80 mg/kg/d (~6 g/day) Total = ~168 g	Running • 800 m running on track	Yes	Post-supplementation race times were significantly faster following β-alanine (*p* = .02), with post- versus pre-supplementation race times being faster after β-alanine (-3.64 ± 2.70 s, -2.46 ± 1.80%) but not placebo (-0.59 ± 2.54 s, -0.37 ± 1.62%). Split times at 400 m were significantly faster (*p* = .02) post-supplementation in the β-alanine group, compared with placebo. β-alanine supplementation had no effect on blood biochemistry.
Ducker et al. 2013a	24 competitive male team sport athletes Parallel group design with *n* = 6 for bicarbonate, β-alanine, combined and placebo	28 d @ 80 mg/kg/d or ~6 g/d Total = ~168 g (300 mg/kg sodium bicarbonate @ 60 min pre-exercise)	Team sport • 3 sets @ 6 × 20 m sprints on 25 s with 45 s between sets	Yes	Effect size comparison revealed that effect of β-alanine supplementation by itself produced trivial effects on total sprint time, mean sprint time, first sprint and fastest sprint. Whereas bicarbonate alone was associated with very likely improvements in these factors, combination of β-alanine with bicarbonate reduced the size of this improvement to 'likely benefit'. Small sample sizes noted. See also bicarbonate summary.
Ducker et al. 2013c	16 well-trained male rowers (international and national level) Parallel group design	28 d @ 80 mg/kg/d or ~6–7 g/d Total = ~168–196 g	Rowing • 2000 m TT on ergometer	Perhaps	Small likelihood of benefit with β-alanine on 2000-m rowing ergometer performance (49% likelihood) and significantly quicker split times at 750 and 1000 m. No significant effect of β-alanine on biochemistry.
Hobson et al. 2013	22 well-trained rowers (club level) Parallel group design for β-alanine with crossover design for sodium bicarbonate	30 d @ 6.4 g/d Total = 192.0 g (and/or 300 mg/kg sodium bicarbonate: 200 mg/kg @ 4 h pre-exercise + 100 mg/kg BM @ 120 min pre-exercise	Rowing • 2000 m TT on ergometer	Yes	Magnitude-based inference analysis showed that β-alanine supplementation was very likely (96% likelihood) to be beneficial for 2000-m rowing performance (6.4 ± 8.1-s improvement compared with placebo) and there was a small (1.1 ± 5.6 s, 67% likelihood) but possibly beneficial additional effect when it was combined with acute sodium bicarbonate supplementation.
Howe et al. 2013	16 highly trained cyclists (VO_{2peak} = 67 mL/kg/min) Parallel group design	4 w @ 65 mg/kg/d or ~4.5 g/d Total = 126.0 g 2 high intensity interval training	Cycling • 4 min TT on ergometer	Perhaps	β-alanine supplementation was 44% likely to increase average power output during the 4 min cycling TT when compared with the placebo, although this was not statistically significant (*p* = .25). No significant effect of β-alanine on blood biochemistry.

		sessions/w and habitual training			
Tobias et al. 2013	37 judo and jiu-jitsu athletes Parallel group design for both β-alanine and bicarbonate supplements	28 d @ 6.4 g/d Total = 179.2 g (and/or 7 d @ 500 mg/kg/d sodium bicarbonate split into 4 doses)	Combat sports simulation • 4 × 30 s upper body Wingate tests @ 3 min	Yes	β-alanine supplementation enhanced work (+7% *p* = 0.003) compared to placebo. Bicarbonate supplementation also increased total work done (+8% *p* = 0.002) and co-ingestion resulted in the greatest gains (+14%, *p* < 0.0001). Similarly, mean power for the 2nd, 3rd and 4th bouts was improved with both supplements individually and in combination.
Bellinger et al. 2012	14 highly trained male cyclists (VO_{2peak} = ~65 mL/kg/min) Parallel group design for β-alanine with crossover design for sodium bicarbonate	28 d @ 65 mg/kg/d or ~4.6 g/d Total = ~128.8 g 2 high intensity interval sessions per week and habitual training	Cycling • 4 min TT on ergometer	No	No significant benefit for β-alanine alone although magnitude based inference analysis suggests a 37% likelihood of improvement with 0% chance of negative effect. Significant benefit for β-alanine combined with bicarbonate (3.3%) but this was not significantly different from bicarbonate alone (3.1%).
Chung et al. 2012	23 highly trained male and 18 female swimmers (international and national level) Parallel group design	4 w @ 4.8 g/day and 6 w @ 3.2 g Total = 268.8 g	Swimming • Training sets according to race event (Testing at 0, 4, 10 w) • Race performance (100 - 400 m) at major meet (Testing at 0, 12 w)	Perhaps Unclear	Study involved real-world, uncontrolled environment. Small improvement of β-alanine supplementation on training test set performance at week 4 (1.3%) but unclear effects at week 10. Minimal effect of β-alanine on competition performance.
Saunders, Sale et al. 2012	16 elite and 20 non-elite males (team sport athletes) Parallel group design	4 w @ 6.4 g/d Total = 179.2 g	Team sport • Loughborough intermittent shuttle test	No	Lack of deterioration of performance in both groups indicate that the exercise protocol may not have been fatiguing enough to test for any effect from β-alanine.
Saunders, Sunderland et al. 2012	17 young males (amateur footballers) Parallel group design	12 w @ 3.2 g/d Total = 268.8 g	Team sport • YoYo Level 2 intermittent recovery test	Yes	Significant improvement in β-alanine group with distance covered (+34.3%, *p* < 0.001) compared to placebo (-7.3%, *p* = 0.24). Note range of improvement in distance covered for β-alanine group varied from 0.0%–72.7%.
Kern & Robinson 2011	37 collegiate males (22 wrestlers and 15 gridiron footballers) Parallel group design	8 w @ 4 g/d or placebo	Intermittent sport (team sport, combat sport) • 6 × 50-yard shuttle run	Perhaps	Both groups reported non-significant improvements in performance (300 m shuttle improved 1.1 s β-alanine versus 0.4s placebo compared to baseline). β-alanine group showed slightly greater improvement in body composition: gain in lean mass of 0.5 kg vs 0.4 kg loss in

					placebo.
Baguet et al. 2010	18 elite male rowers Parallel group design	7 w @ 5 g/d	Rowing • 2000 m ergometer	Yes	Strong positive correlations between muscle carnosine concentrations and speed in all-out rowing effort over 100 m, 500 m, 2000 m and 6000 m. Muscle carnosine concentrations increased by 45.3% (soleus) and 28.2% (gastrocnemius). Trend to enhanced rowing TT in the β-alanine group (p = 0.07).
Van Thienen et al. 2009	Moderate–well-trained male cyclists Parallel group design	8 w @ 2–4 g/d β-alanine	Cycling • 110 min submax cycling and 10 min TT • 30 s sprint	No Yes	Mean power output during the TT (~300 W) was similar between placebo and β-alanine group at baseline and 8 weeks testing. However, compared with placebo, during the final sprint after the TT, the β-alanine group increased peak power output by 11.4% (7.8–14.9%, p = 0.0001), whereas mean power output increased by 5.0% (2.0–8.1%, p = 0.005). Blood lactate and pH values were similar between groups at any time.
Hoffman et al. 2008a	8 resistance-trained men Crossover design with 4 week washout	30 d @ 4.8 g/d β-alanine (dose gradually built up from 2.4 g/d over first 8 d)	Resistance exercise • 1 RM squat	No	Authors acknowledged that 4 week washout may not be sufficient for return to baseline of muscle carnosine content. Although 4 weeks of β-alanine supplementation resulted in greater increase in volume of training and number of repetitions during training set than placebo treatment, there was no difference in gains in strength or BM at end of treatment.
Hoffman et al. 2008b	Collegiate football players Parallel group design	30 d @ 4.5 g/d β-alanine	Team sport • 60 s Wingate test • 3 line drills (200-yard runs with 2 min rest)	No No	Authors claim trend to lower fatigue rate in Wingate test in β-alanine group on day 1 of camp than placebo group. Higher training volume and reduced self-reported ratings of soreness during camp.
Derave et al. 2007	15 competitive sprint-trained track and field athletes (400 m time < 53 s) Parallel group design	4–5 w @ 4.8 g/d sustained release β-alanine (dose gradually built up from 2.4 g/d over first 8 days)	Running • 400 m run (indoor track)	No	Non-invasive measurement of muscle carnosine (proton magnetic resonance spectroscopy) found average increase of ~40% in calf muscles resulting from supplementation. Response was not related to initial carnosine concentration and did not appear to reach maximal threshold as a result of supplementation. B-alanine supplementation associated with attenuation of muscle fatigue (dynamic knee contractions test). Failure to see clear enhancement of 400 m performance perhaps related to under-powering of study design, failure of sufficient increase in muscle carnosine/muscle

					buffering or failure of H+ ion accumulation to limit 400 m performance. No relationship between change in 400 m performance, change in muscle carnosine or post-race lactate concentration.
Hoffman et al. 2006	33 resistance-trained male collegiate football players Parallel group design (creatine, creatine + β-alanine, placebo)	10 w @ 3.2 g/d of β-alanine added to 10.5 g/d creatine or 10.5 g/d creatine alone	Football (gridiron) • 1 RM squat • 1 RM bench press • 2 × 30 s cycling Wingate • 20 × vertical jump	No No No No	Addition of β-alanine to creatine supplementation appeared to enhance training volume and DXA-determined loss of body fat and gain of LBM. However, no differences in cycling and jump performance over the training period in any group. Improvement in strength with creatine supplementation but no further improvement with creatine + β-alanine.

*Randomised, double-blinded, placebo-controlled unless noted: habitual training maintained unless noted.

BM=body mass, DXA=dual energy X-ray absorptiometry, LBM=lean body mass, PWC=physical work capacity, RM=repetition maximum, TT=time trial, TTE=time to exhaustion

Note: For updates see www.ausport.gov.au/ais/nutrition/supplements

16.7.6 Beetroot juice/nitrates

The current interest in beetroot juice arises from research by Professor Andy Jones and his colleagues from the University of Exeter, UK, which used this juice as a rich and consistent source of nitrate (Burke 2013). Nitrate (NO^{3-}) is one of a family of compounds containing nitrogen and oxygen that is both produced by our bodies and found in the diet. Estimates of the average dietary intake of adults in the US and Europe is 1–2 mmol/d (~60–120 mg/d) with vegetables providing about 80% of this total and smaller contributions coming from processed meats and the water supply. Vegetarians and people who follow 'heart-friendly' eating plans high in vegetable content such as the DASH diet are likely to consume higher nitrate intakes. Further information on dietary nitrate intake can be found in the review by Hord and colleagues (2009).

As summarised by Jones (2014), dietary nitrate is rapidly absorbed in the stomach and small intestine, with plasma nitrate levels peaking after ~1–2 hours. However, a significant proportion (~25%) of plasma nitrate is extracted by the salivary glands and concentrated in saliva where commensal facultative anaerobic bacteria found in the tongue convert it to nitrite in an oxygen independent reaction. The swallowing of nitrite into the acidic stomach environment starts the process in which nitrite is further processed into reactive nitrogen species including nitric oxide (NO). Plasma nitrite concentrations peak at ~2.5 h following the intake of dietary nitrate and return to baseline within 24 hours. Factors that interfere with salivary nitrate handling processes, such as the use of antibacterial mouthwashes and gum to reduce mouth levels of bacteria or the prevention of the swallowing of saliva, will reduce this rise in plasma nitrite.

It was previously believed that the synthesis of NO, which regulates a large number of important processes in the body, was reliant on a pathway involving oxygen and the amino acid arginine and catalysed by various NO synthases (Moncada & Higgs 1993). However, the discovery of the alternative nitrate–nitrite–NO pathway, which operates under oxygen-independent and acidic conditions, is important in the sporting context, since relative hypoxia and acid-base disturbances are frequently found in the muscle during exercise. The discovery of this well-developed pathway of production of NO supports the theory that dietary nitrate is useful for health and function rather than being a toxic substance as was originally proposed in relation to epidemiological connections between high intake of cured meats (e.g. sausages, ham, etc. with added nitrite and nitrate salts) and cancer (Sindelar & Milkowski 2012).

The gas NO is a very important chemical in our bodies with activities including relaxation of the tone of blood vessels (thereby regulating blood pressure, the susceptibility of vessels to vascular disease and tissue oxygenation), regulation of platelet aggregation (thereby reducing the risk of atherosclerosis) and immune system activities (especially to reduce infection in the mouth, gut and skin). The recent review by Dyakova et al. (2015) summarises a number of roles of NO in skeletal muscle with implications for health and disease. Nitrate supplementation has been shown to enhance some of known functions of NO, even in healthy people. For example, supplementation with dietary nitrate sources or sodium nitrate has been shown to reduce blood pressure even in individuals with normal blood pressure (see Jones 2014).

The first study of nitrate supplementation and exercise capacity, undertaken by Larsen and colleagues in 2007, found that healthy subjects who ingested sodium nitrate for three days were able to undertake submaximal exercise at a lower oxygen cost than following a placebo treatment (Larsen et al. 2007). Exercise economy or efficiency, defined as power output/speed for a given volume of oxygen consumption, is both an important determinant of performance in many sports and a generally stable characteristic within individual athletes. Thus, these findings were noted with interest by Jones and colleagues, who initiated a series of studies of nitrate supplementation on exercise economy and performance in healthy and trained populations and precipitated a productive research and commercial interest in beetroot juice as a natural source of dietary nitrate (for review see Burke 2013; Jones 2014). As shown in Table 16.6, the original studies were forced to use blackcurrant juice as a 'placebo' for the highly distinguishable beetroot juice. However, the eventual (patented) preparation of a nitrate-depleted beetroot juice as well as a commercial concentrated 'shot' form of juice, have provided researchers with a practical form of nitrate supplementation, a blinded control for nitrate supplement studies, and confirmation that the nitrate content of beetroot juice is the active ingredient underpinning the observed physiological and exercise benefits.

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Thompson et al. 2015	16 male team sport players	Chronic 7 d supplementation • 2 × 70 mL/d BJ shot (~12.8 mmol) nitrate/d Placebo: nitrate-depleted BJ	Team sport simulation on cycle ergometer • 2 × 40 min 'halves' of repeated 2 min blocks (@ 6 s 'all-out' sprint, + 100 s active recovery + 20 s rest) • Cognitive tests simultaneously	Yes Yes	Total work done during the sprints of the intermittent sprint test was greater in BJ (123 ± 19 kJ) compared to placebo (119 ± 17 kJ; $p < 0.05$). Reaction time of response to the cognitive tasks in the second half of the IST was improved in BJ compared to placebo (BJ first half: 820 ± 96 vs second half: 817 ± 86 ms; placebo first half: 824 ± 114 vs second half: 847 ± 118 ms; $p < 0.05$). There was no difference in response accuracy.
Boorsma et al. 2014	8 male elite 1500 m runners Study involved two protocols of acute supplementation (day 1) and preload + acute (day 8) in crossover design	Acute supplementation @ 2.5 h pre-exercise • 3 × 70 mL BJ shot (19.5 mmol nitrate) • 8 d chronic pre-load + acute: 7 d (2 × 70 mL BJ/d or 13 mmol nitrate/d) + acute dose (3 × 70 mL BJ shot or 19.5 mmol nitrate) @ 2.5 h pre-exercise Placebo: nitrate-depleted BJ shot	Running • 1500 m on indoor 200 m track	No, but perhaps some responders	The 1500 m TT was unaffected by acute BJ supplementation (4:10.7 min ± 1.5 s and 4:10.4 min ± 2.5 s) for BJ and placebo respectively or the pre-load + acute (4:10.5 min ± 2.2s and 4:11.4 min ± 2.7 s). There was no difference in exercise economy (VO_2 cost of exercise @ different workloads during a VO_{2max} test). However, 2 subjects showed a consistent improvement in 1500 m time with both protocols suggesting an individual response (5.8 and 5.5 s for acute and 7 and 0.5 s for pre-loaded protocol). Note that large doses of nitrate were provided, including a pre-load ensuring that nitrate status was significantly increased.
Hoon et al. 2014	10 well-trained male rowers Crossover design	Acute supplementation @ 2 h pre-exercise • 70 mL BJ shot (4.2 mmol) • 2 × 70 mL BJ shot (8.4 mmol) Placebo = nitrate-depleted BJ	Rowing • 2000 m ergometer TT	Perhaps with higher dose	The single dose demonstrated a trivial effect on TT time compared to placebo (mean difference: 0.2 ± 2.5 s). A possibly beneficial effect was found with the double dose compared with the single dose (mean difference -1.8 ± 2.1 s) and with placebo (-1.6 ± 1.6 s).
Lane et al. 2014	12 male and 12 female well-trained cyclists Crossover design Latin Square with BJ and/or caffeine	2 d chronic pre-load + acute supplementation • 2 d (70 mL BJ shot or 8.4 mmol	Cycling • TT on ergometer simulating Olympic course	No No	There was no effect of BJ when used alone (-0.4%, $p = 0.6$ compared to baseline) or combined with caffeine (-0.9%, $p = 0.4$ compared to caffeine). Caffeine alone and caffeine + BJ resulted in improved power output,

	supplementation	nitrate) + acute dose @ 2 h pre-trial (2 × 70 mL BJ shot or 8.4 mmol nitrate) (and/or 3 mg/kg BM caffeinated gum 40 min pre-trial) Placebo: nitrate-depleted BJ	• 29.35 km females • 43.83 km males		however there was not a significant difference between these two conditions.
Martin et al. 2014	9 male and 7 female team sport athletes Crossover design	Acute supplementation @ 2 h pre-trial • 70 mL BJ shot (minimum 0.3 g nitrate) Placebo: nitrate-depleted BJ	Team sport simulation • High-intensity intermittent sprint test (HIIST): 8 s sprints with 30 s recovery to fatigue	No	Fewer sprints were performed with supplementation (BJ = 13 ± 5 sprints vs placebo = 15 ± 6, *p* = .005). Similarly, less total work was done with supplementation (BJ = 49.2 ± 24.2 kJ vs placebo = 57.8 ± 34.0, *p* = .027). The authors conclude the lack of an effect of nitrate at near-maximal oxygen uptake suggests that nitrate does not have an ergogenic effect at greater exercise intensities.
Muggeridge, Howe et al. 2014a	9 male trained cyclists Crossover design	Acute supplementation @ 3 h pre-trial: • 70 mL BJ shot (5 mmol nitrate) Placebo: nitrate-depleted BJ Trials occurred at simulated 2500 m altitude	Cycling • 15 min @ 60% max work rate + 16.1 km TT	Yes	Trials occurred at simulated 2500-m altitude. A athletes were not exposed to altitude apart from the trials. VO_2 during steady-state exercise was lower in the BJ trial (2542 ± 114 mL/min) than in the placebo trial (2727 ± 85 mL/min, *p* = 0.049). TT performance was significantly faster after BJ (1664 ± 14 s) than after placebo (1702 ± 15 s, *p* = 0.021).
Muggeridge, Sculthorpe et al. 2014b	9 male trained cyclists Crossover design	Acute supplementation @ 2.5 h pre-trial • 2 × 60 mL nitrate gel (~8.1 mmol nitrate) • And/or UVA exposure (20 J/cm^2 whole body irradiation) Placebo: nitrate-depleted BJ	Cycling • 15 min @ 60% max work rate + 16.1 km TT	Yes, when combined with UVA exposure	During steady-state exercise, VO_2 was reduced following BJ + UVA light (*p* = 0.034) but not for BJ + placebo light or placebo + UVA light compared to placebo + placebo light. Performance in the TT was significantly faster for BJ + UVA light (1447 ± 41 s *p* = 0.005), but not for placebo + UVA light (1450 ± 40 s) or BJ + placebo light (1455 ± 47 seconds) compared to placebo + placebo light (1469 ± 52 s).Exposure to UVA light alone did not alter the physiological or performance measures. However a combination of UVA and BJ did improve cycling TT performance, which may be due to a larger increase in NO availability.

Peeling et al. 2014	5 elite female kayakers Crossover design	Acute supplementation @ 2 h pre-trial • 2 × 70 mL BJ shots of 9.6 mmol nitrate Placebo: nitrate-depleted BJ	Kayak • 500 m on water (field study)	Yes	TT performance was improved by 1.7% (p = 0.01) with BJ supplementation. Crossover study allowed for slight differences in environmental conditions in field situation to be controlled for.
Christensen et al. 2013	10 male highly trained cyclists Crossover design Single blind	4 and 6 d chronic supplementation • 500 mL/d beetroot juice ~0.5 g nitrate/d spread over day Placebo: 500 mL blackcurrant juice	Cycling • 120 min submaximal + 400 kcal TT • 6 × 20 s sprints @ recovery 100 s	No No	Measurements of VO_2 kinetics and exercise economy were the same for BJ and placebo. Peak and mean power during repeated sprinting were similar in BJ and placebo. TT performance was similar with an average completion time of 18:20 min (BJ) and 18:37 min (placebo), and average power outputs of 290 ± 43 W (BJ) and 285 ± 44 W (placebo).
Muggeridge et al. 2013	8 male trained kayakers Crossover design Single blinded	Acute supplementation @ 3 h pre-trial • 70 mL BJ shot (5 mmol nitrate) Placebo: tomato juice	Kayak • 15 min steady state at 60% max work rate + 5 × 10 s all-out sprint on ergometer • 1 km TT ergometer	No No	VO_2 during steady-state exercise was lower in the BR trial than in the PLA trial (p = .010). However there was no difference in either peak power in the sprints (p = .590) or TT performance between conditions (PLA: 277 ± 5 s, BR: 276 ± 5 s, p = .539).
Wylie et al. 2013	14 male team sport players Crossover design	1 d pre-load + acute supplementation • 1 d (2 @ 2 × 70 mL BJ shot with total of 28.4 mmol nitrate) + 2 × 70 mL BJ shot with 8.2 mmol nitrate @ 2.5 h pre-exercise + 1 × 70 mL BJ shot with 4.1 mmol nitrate @ 1.5 h pre-exercise Placebo: nitrate-depleted BJ	Team sport • YoYo intermittent recovery level 1 test (IR1)	Yes	Distance covered in the YoYo test was 4.2% greater (p < 0.05) with BJ supplementation.
Bescos et al. 2012	13 trained cyclists and	3 d chronic supplementation	Cycling	No	No differences in either the mean distance (nitrate: 26.4 ± 1.1 km;

07:37:13.

	triathletes Crossover design	• Sodium nitrate (10 mg/kg BM) Placebo: sodium chloride	• 40 min ergometer TT		placebo: 26.3 ± 1.2 km; *p* = 0.61) or mean power output (nitrate: 258 ± 28 W; placebo: 257 ± 28 W; *p* = 0.89) between treatments, despite increases in plasma nitrite and nitrate with supplementation. 7 subjects showed minimal response to supplementation (low responders) and 6 showed a large increase in plasma nitrate (<50%) (high responders).
Cermak, Gibala et al. 2012	12 male cyclists Crossover design	6 d chronic supplementation • 2 × 70 mL/d BJ shot (~8 mmol) nitrate/d Placebo: blackcurrant cordial	Cycling • 30 min @ 45% Wmax + 30 min @ 65% Wmax + 10 km TT	Yes	Increase in mean power in TT with BJ compared with placebo (294 ± 12 vs 288 ± 12 W, *p* < 0.04) and faster time (953 ± 18 vs 965 ± 18 s, *p* < 0.005. VO_2 at 65% Wmax and 45% Wmax was reduced with BJ.
Cermak, Res et al. 2012	20 male trained cyclists Crossover design	Acute supplementation @ 2.5 h pre-exercise • 2 × 70 mL BJ shot (8.7 mmol nitrate) Placebo: nitrate-depleted BJ shot	Cycling • 1 h TT on ergometer	No	No difference between conditions for TT performance (65.5 ± 1.1 vs 65 ± 1.1 s), power output (275 ± 7 vs. 278 ± 7 W), and heart rate (170 ± 2 vs 170 ± 2 beats/min), despite elevated pre-trial plasma nitrate conditions for BJ supplementation.
Peacock et al. 2012	10 male junior cross country skiers Crossover design	Acute supplementation 2.5 h pre-trial • Potassium nitrate (gelatine capsule) total 614 mg nitrate Placebo: capsule with maltodextrin	Skiing • 5 km ski TT on treadmill	No	No significant difference in 5 km TT performance between nitrate (1005 ± 53 s) and placebo treatments (996 ± 49 s, *p* = 0.12). In an accompanying test, the oxygen cost of submaximal running was not significantly different between placebo and nitrate trials at 10 km/h (both 2.84 ± 0.34 L/min) and 14 km/h (3.89 ± 0.39 vs 3.77 ± 0.62 L/min).
Wilkerson et al. 2012	8 male well-trained cyclists Crossover design	Acute supplementation @ 2.5 h pre-exercise • 500 mL BJ (total 6.2 mmol/L)	Cycle • 50 mile (80.47 km) TT on ergometer	No	BJ resulted in a 0.8% improvement in TT time, however this was not significant. Power output was not different between conditions. Oxygen uptake trended lower with supplementation (*p* = 0.06). Authors note the high training status of the cyclists may have reduced the physiological and performance response to NO_3 compared to moderately trained athletes. Note responders and non-responders were separated

					for analysis. Responders had significantly greater PO and mean reduction of TT performance of 2%.
Lansley et al. 2011	9 male trained cyclists Crossover design	Acute supplementation 2.5 h pre-trial • 6.2 mmol nitrate as 500 mL beetroot juice (BJ) (Beet it) • Placebo: nitrate depleted BJ	Cycling • 4 km TT • 16 km TT	Yes Yes	2.8% improvement in 4 km TT (6.45 vs 6.27 min: 279 vs 292 W, *p* <0.05) for placebo and BJ trials. 2.7% improvement in 16.1 km TT (27.7 vs 26.9 min: 233 vs 247 W, *p* < 0.01) for placebo and BJ trials. Note that 7–11% improvement in power output was achieved with no increase in oxygen cost of exercise, that is increased economy.

*Randomised, double-blinded, placebo-controlled unless noted.

Note: For updates www.ausport.gov.au/ais/nutrition/supplements

BJ = beetroot juice, BP = blood pressure, PO = power output, PPO = peak power output, TTE = time to exhaustion, TT = time trial, UVA = ultraviolet light A

Initial studies of beetroot juice supplementation employed a short period of standardisation to a nitrate-depleted diet (minimal intake of vegetables) to ensure that the supplement achieved a substantial increase in nitrate levels. More recent studies have shown that beetroot juice supplementation still increases nitrate levels in people consuming varied diets although the degree of increase in nitrate levels is smaller and more variable (Jones 2014). While there has been a parallel interest in the use of beetroot juice/nitrate supplementation to increase exercise tolerance in clinical populations (see Clements et al. 2014), the immediate focus of this chapter is the literature on trained populations employing an exercise protocol of relevance to sports performance (see Table 16.6).

Most studies of beetroot juice supplementation in athletes have employed an acute dosing protocol in response to an early results which showed that physiological benefits in healthy populations achieved by chronic (up to 15 d) supplementation with beetroot juice could be replicated with a single dose consumed about 2–2.5 h prior to exercise (Vanhatalo et al. 2010). However, more recent recognition of the lack of consistent benefits of beetroot juice supplementation on performance in more highly trained populations has led to speculation that a larger dose may be needed to achieve ergogenic effects (Jones 2014). Indeed, dose-response studies (Wylie et al. 2013; Hoon et al. 2014) have reported that a critical pre-exercise dose of at least 8 mmol (equivalent to two beetroot juice 'shots') is needed to achieve a physiological or performance benefit. Furthermore, it has been suggested that in well-trained individuals a more chronic or 'pre-loaded' protocol in which supplementation is undertaken for several days prior to a competitive event might be superior to a single pre-exercise dose (Jones 2014).

The summary of performance outcomes from nitrated supplementation in trained populations (Table 16.6) shows that while several studies have found an enhancement of sports performance following nitrate supplementation in groups or 'responders' within groups, the benefits are not clear-cut or consistent. However, this may simply reflect the variety of confounding characteristics within the currently available studies such as differences in the calibre of subjects, supplementation doses and protocols, and the types of sports/exercise protocols. A meta-analysis of exercise-related studies published prior to August 2012 (Hoon et al. 2013), reported that in time-to-exhaustion protocols, the pooled effect of nitrate supplementation was a moderate (ES = 0.79, 95% CI: 0.23–1.35) but significant (p = 0.006) improvement. The effect on time trial performance was small (ES = 0.11, 95% CI: –0.16–0.37), non-significant by statistical assessment (p = 0.43), but suggested to be meaningful in the world of elite sport. A further qualitative analysis reported that the effects of nitrate supplementation appeared to be greater in recreationally active participants and when a chronic supplementation over several days is undertaken (Hoon et al. 2014). Of course, while this approach is useful in providing a 'bigger picture' of the current literature, it is still skewed by the limitations of the available literature.

A number of potential mechanisms of action have been reported to explain the benefits of nitrate supplementation on sports performance: these include an enhancement of muscle oxygenation, changes in calcium handling or ATP-ase activity underpinning muscle contraction and enhanced mitochondrial efficiency (see Jones 2014). An understanding of such mechanisms and the conditions under which the nitrate–nitrite–NO pathway might be more important could help to identify the likely protocols, athlete populations and sports scenarios in which nitrate supplementation is likely to be more valuable. For example, if the mechanism involves adaptation in the muscle mitochondria to increase exercise efficiency by reducing proton leakage (Larsen et al. 2011), then it is understandable that supplementation over a couple of days will be needed to achieve this effect. A worthwhile increase in plasma nitrite concentrations may also be needed to achieve a physiological change, explaining the benefits of larger doses, longer supplementation protocols and the existence of responders versus non-responders who are differentiated by this outcome. According to Jones (2014), observation in elite athletes of higher baseline concentrations of plasma nitrite than their lesser-trained counterparts could account for the lower response to a given nitrate dose.

The prominence of the nitrate–nitrite–NO pathway in relatively hypoxic and/or acidic environments might explain the trend to see performance benefits in exercise protocols involving higher intensity exercise where the local muscle environment achieves these conditions. Indeed, Jones (2014) suggests that the benefits of nitrate supplementation are best seen in sports protocols of 5–30 minutes duration, although recent observations of benefits to work outputs during a simulation of an 80-minute team sport competition suggest that scenarios involving repeated bouts of high-intensity exercise might also respond. However, highly trained athletes, who have increased NO production capacity, may have less need to rely on an enhanced nitrate–nitrite–NO pathway and their superior blood flow and microperfusion of muscle may reduce the relative hypoxia/acidity in muscles during higher-intensity exercise (Jones 2014). Since the benefits seen are more prominent in type II muscle fibres, endurance athletes who have a lower proportion of such fibres may be less responsive to nitrate supplementation.

As with many supplements, more research is needed to better understand the potential benefits of nitrate supplementation, protocols for use during training or competition, and the range of scenarios in which to use it. Until then, the potential disadvantages or side effects of the use of beetroot juice seem minor, and may actually represent an increase in the intake of fruits and vegetables by some athletes. In concentrated form and larger doses, beetroot juice intake sometimes causes mild gastrointestinal discomfort, and the harmless and temporary pink colouration of urine and stools. Use of sodium nitrate supplements may be associated with a greater risk of misjudging dosages. Some athletes may also mistakenly (or deliberately) use sodium nitrite as a supplement and expose themselves to other risks. Excessive intake of sodium nitrate, and especially sodium nitrite, can produce toxic effects such as methemoglobinaemia ('blue baby syndrome') (Clements et al. 2014). Although there is occasional reference to the epidemiological association between nitrosamine and some cancers when considering nitrate supplementation (Lundberg et al. 2011), this does not appear to be a concern in the case of the intake of vegetable sources of nitrate.

Summary

Sports dietitians frequently observe a chaotic pattern of use of supplements and sports foods by athletes and coaches, and the appearance on the market of an almost never-ending range of products that are claimed to achieve benefits needed to enhance sports performance. The poor regulation of supplements and sports foods in many countries allows athletes and coaches to be the target of marketing campaigns based on exaggerated claims and hype rather than documented benefits. However, scientific study has identified a number of products that offer true benefits to performance or the achievement of nutritional goals.

A systematic approach to educating athletes and coaches about supplements and sports foods, and managing their provision to athletes and teams, can allow sportspeople to include the successful use of these products with the activities that underpin optimal performance.

- The use of supplements is commonplace among athletes, from junior to elite ranks, in the competitive sporting environment. During an athlete's development, it is important that they understand that a supplement, by the very definition of the word, is 'an extra that is used in addition to something to complete or enhance'. In dietary terms, a supplement may be beneficial when it enhances a nutritious diet that has been designed to optimise performance in training and competition over a period of time.
- Under the World Anti-Doping Code's principle of strict liability, athletes are responsible for any substance found in their body. Any decisions around supplement use should therefore be the responsibility of the individual. The role of the sports dietitian is to help the athlete balance the potential benefits and risks of supplement use. Dietitians should have the knowledge to advise athletes on the effectiveness of a given

supplement in improving performance (in light of evidence-based research), correct dosage for performance gains, the safety of a supplement, the legality of its use in the relevant sport during training and competition, and the potential impact and side effects of use when the whole diet is taken into account.

- The increase in supplement use by elite athletes in recent years has resulted in the need for many sporting bodies and organisations to develop and implement strategies and policies around supplement provision and use. Sports dietitians looking for guidance on how to manage and monitor supplement use by their athletes would be well advised to contact their relevant national sporting organisation(s) to access their resources or protocols. Tools such as the Australian Institute of Sport's Supplement Framework provide guidelines and information on education, research and governance to aid in the development of policies and protocols.
- A number of programs exist to help athletes make the best possible choice in relation to the safety of their supplements and sports nutrition products. Third party auditing programs such as Informed Choice and the NSF Certified for Sport® have been designed to assist manufacturers to minimise the risk that their supplement or sports food contains a banned substance. Of course, no auditing program can provide a 100% guarantee that a supplement product is free of banned substances as ingredients or contaminants, and athletes must remain aware of their liability if an anti-doping rule violation occurs.
- Many pre-workout and recovery products contain several purportedly active ingredients in the one product; these are often referred to as 'polysupplements' or 'multi-ingredient' products. To ensure that athletes consume only appropriate doses of effective ingredients, and avoid the intake of unnecessary (or unspecified) ingredients, it is preferable to source supplements in their individual form.
- A recent trend in food technology is the inclusion of ingredients that are considered supplements in a sporting context, into standard supermarket products. Examples of this include the caffeine contained in energy drinks and the addition of protein powder to several products including muesli bars, cereals and smoothies. Education around these foods should be included when talking to athletes about supplement use.
- It is good practice to regularly ask athletes about their supplement use. A nutrition screen should include questions surrounding the use of supplements and sports foods (including brands, quantities used, timing of use) and an assessment of the athlete's knowledge regarding the proposed benefits of each product taken. This should be reviewed regularly to ensure athletes have not altered their usage or introduced new supplements since the last meeting, thereby placing themselves at risk.
- Any advice or products provided to athletes should be well documented by the sports dietitian. Athletes may act outside of advice provided or start using products of their own accord, so it is important that records are made of communications that occur around supplement use.

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CHAPTER SEVENTEEN

Nutritional issues for young athletes: children and adolescents

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17.1 Introduction

The prospect of sporting success is appealing to many young athletes and their families. Organised sport provides many benefits to young people including regular physical activity, social interaction and the development of confidence and self-esteem. Indeed, a commitment to regular physical activity is promoted in population dietary guidelines, regardless of age. While population dietary guidelines and nutrient reference values—that is, Estimated Average Requirements (EAR)—are appropriate for addressing the micronutrient and energy requirements of children and adolescents involved in most sports, those involved in high-intensity sports training programs have higher nutrient and energy requirements. The basic principles of sports nutrition recommended to adult athletes also apply to younger athletes.

Meeting higher nutrient and energy requirements during periods of rapid growth and coping with changing body composition, metabolic and hormonal fluctuations is often challenging for young athletes. Social, emotional and stage of development and maturational timing has a major influence on how individuals view themselves and their sporting achievements. Childhood and adolescence is a critical time in life to develop long-term food habits and the connections between food, exercise and body image. New practices, beliefs and meanings associated with food may be acquired during this time, such as the adoption of vegetarianism, fad diets and supplement use. To ensure that the young athlete fulfils his or her potential, eating patterns that integrate the unique requirements for sporting success and ensure healthy growth and development support need to be supported.

This chapter considers how the physiological response to exercise and the nutritional requirements of children and adolescents participating in sport may differ from adult athletes. Identifying the body composition changes during growth, development and maturation is crucial to advising young athletes about expected changes in body composition, nutrient requirements and response to exercise. In this text, children refers to those aged 5–12 years, and adolescents are those aged 13–18 years, according to population categories (NHMRC 2013).

17.2 Growth and developmental changes in the young athlete

Growth is divided into three additive components: infancy, childhood and puberty (Karlberg 1989a, 1989b, 1990). Identifying stage of growth and development changes to body composition during maturation of young athletes is fundamental for the sports dietitian to estimate nutrient requirements and advise on the expected physical response to a training program. Large individual variations in stage of growth and development are evident in young athletes of the same age even in the same team, although training requirements and coach expectations are often the same for all athletes. It is not uncommon to see adolescent athletes compete in team sports with adult athletes, particularly the early maturers with specialised skills and talent. These adolescent athletes may appear physically mature but may still have immature or poorly adapted physiological, metabolic and enzyme responses to training (see section 17.3). Risk of injury is likely to be higher in an adolescent compared with an adult athlete, particularly stress fracture and joint injury (see Chapter 22).

17.2.1 During infancy

After birth, most infants make growth adjustments (increased or decreased velocity) within their first two years to reach their genetic determined growth potential, according to population estimates (Karlberg 1989b; Rogol 1995). As a result of these growth adjustments, infants can 'shift centiles' during the first 12–18 months of life. Substantial growth in bone length occurs in the second year of life. Some environmental factors and/or diseases may delay the onset of the subsequent growth hormone (GH)-dependent growth spurt during this phase (Karlberg 1989b). Further, the position of an infant on the growth curve at two years tends to remain constant during infancy (Smith et al. 1976; Karlberg 1989b; Cooper et al. 1995). Growth is stable during the third year of life, when the infancy component of growth has virtually disappeared and growth is predominated by the GH-dependent childhood component.

17.2.2 During childhood

The onset of the GH-dependent childhood component is characterised by a growth spurt between the ages of 6 and 12 months. There is an obvious and abrupt increase in growth rate, which results in a slowing of the rapid deceleration in growth velocity. The peak velocity during this growth spurt is approximately 17 cm/yr (Karlberg 1989a, 1989b; Wollmann & Ranke 1996).

The childhood component, often referred to as prepubertal growth, is nearly identical in girls and boys, as shown in Figure 17.1. During this phase, there is an initial rapid decrease in velocity, and then a slow, consistent decline in velocity to reach a minimal prepubertal height velocity of about 5 cm/yr in both sexes (Karlberg 1989a, 1989b; Prader 1992). A mid-childhood growth spurt at 7 to 8 years has been observed in two-thirds of healthy children. This transient acceleration in height is thought to be due to adrenal androgens, involves growth in all dimensions, and is similar for both genders. The subsequent decline in growth immediately before puberty often highlights this mid-childhood spurt (Karlberg 1989a, 1989b; Prader 1992).

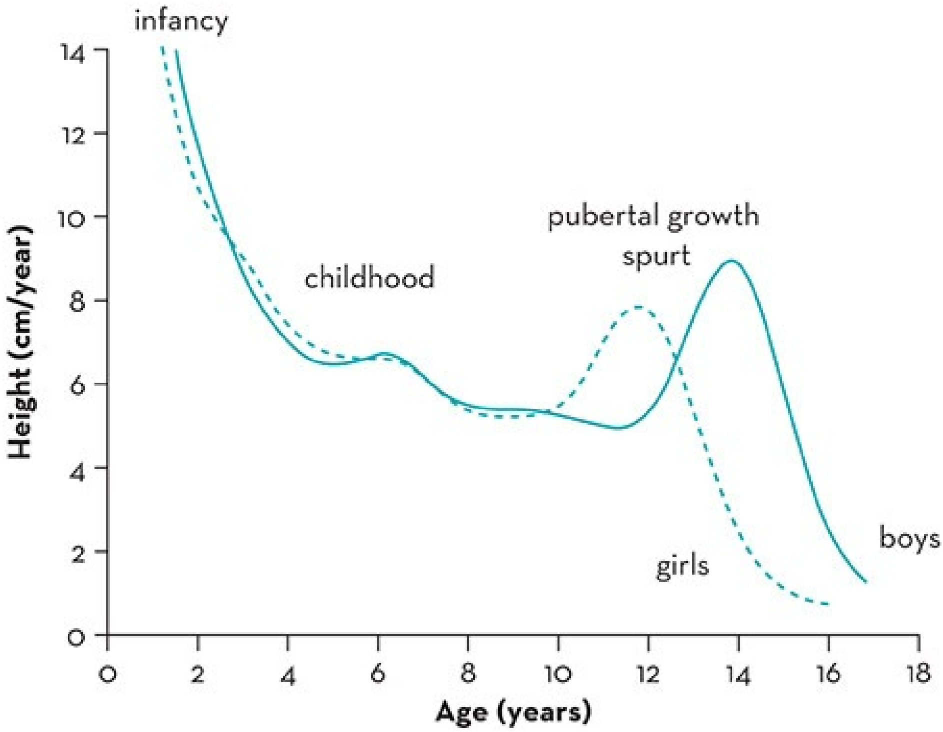

Figure 17.1 Skeletal growth velocity. The childhood component, often referred to as prepubertal growth, is nearly identical in girls and boys. During this phase, there is an initial rapid decrease in velocity, and then a slow consistent decline in velocity. The pubertal component is characterised by a growth spurt. Girls generally start their growth spurt and attain their peak height velocity (PHV) approximately two years earlier than boys

Source: Adapted from Preece et al. 1992

The childhood patterns of long-bone growth are very stable and orderly compared to those of infancy and puberty. Changes in percentile levels with increasing age are minimal (Maresh 1955; Karlberg 1989b). Growth in the childhood component is dominated by growth of the legs compared to trunk growth, which remains constant (Karlberg 1989b). At 12 years of age, the gain in height for both sexes due to childhood growth is about equal (Karlberg 1989b).

17.2.3 During puberty

Growth during puberty is more complex than in the childhood years (when growth rate is relatively constant). Similar to the infancy phase of growth, there is also a growth spurt during puberty. In contrast to the infancy component, however, the growth spurt during puberty is dominated by an acceleration in trunk length; growth of the legs proceeds with constant velocity. Approximately 15% of the final stature is achieved during the pubertal growth spurt and 97% of final stature has been attained by the time of menarche (Faulkner et al. 1993; Bass et al. 1999). Pubertal growth is often considered to be a growth-promoting event and the final height-limiting process. This is because although height velocity is markedly increased, the rate of skeletal maturation is also increased, which eventually leads to fusion of the epiphyseal cartilage (Bourguignon 1988). This pubertal growth spurt is directly related to hormonal changes that accompany sexual development, and is characterised by three phases (Cara 1993):

- minimal height velocity just before the spurt (prepubertal growth lag)
- maximal growth—PHV
- decreasing height velocity (epiphyses fuse and final height is achieved).

Girls generally start their growth spurt and attain their PHV two years earlier than boys (see Figure 17.1). Early in puberty, girls tend to be taller than boys because they begin their spurt earlier and grow faster in the early phases of the growth spurt (Buckler 1990; Preece et al. 1992).

17.3 Physiological characteristics of children and adolescents that influence response to exercise or physical training

The physiological, metabolic and hormonal systems of young athletes are also undergoing development and have not yet reached full functional capacity, even in late adolescence. Unfortunately, few longitudinal studies have been conducted on young athletes, particularly children and prepubescent adolescents, that assess the effects of high-level sports training and resistance exercise on growth, maturational timing and nutrient requirements. Table 17.1 identifies the main physiological factors specific to children and prepubertal adolescents that affect the capacity to undertake high-performance training programs, compared with adults. Some young athletes are involved in fairly rigorous training programs most days from a fairly young age (e.g. gymnastics, ballet, football, swimming, athletics) and have a wide range of individual responses to training.

TABLE 17.1 Physiology of the young athlete compared with adult physiology

Physiological component	Functional capability
Thermoregulatory response	Altered mechanisms to regulate body heat compared to adults
Fluid regulation	Smaller plasma volume than adults in which to lose water as sweat
Ability to undertake aerobic metabolism	It is unclear whether children and pre-pubescent adolescents can reach the same peak VO_2 max as adults (this is confounded by measurement error and bias)

	Higher heart rate and smaller cardiac output than adults, which suggests a lower capacity to train at high levels of intensity for a long duration Smaller increases in blood pressure with exercise than adults No or limited information on adaptive effects of substrates use or enzymes, except in some groups of adolescent athletes
Ability to undertake anaerobic metabolism	Can tolerate only low levels of lactic acid but have more muscle mitochondria than adults
Bone	Bone mineral deposition and skeletal maturation (occurs earlier in girls than boys)
Sweat mechanism	Lower sodium content in sweat than adults Lower sweat rate with exercise than adults
Muscle fibres	Smaller muscle mass than adults

Compared with adults, children are better suited to undertake aerobic types of activity than anaerobic because of their higher levels of muscle mitochondria, although they have a reduced capacity to clear lactic acid. A smaller body size and cardiac output suggests a lower capacity for high-intensity energy expenditure over long periods than adults. This reduced capacity is reflected in the reduction to event distances such as triathlon for junior participants (age-matched categories from 16–19 y) than comparative adult events. Those participants with above-average skills progress into youth and junior *elite* categories and qualify for events, which mimic Olympic sprint distance racing format (see www.usatriathlon.org/elite-international/junior-elite.aspx). Triathlons for youth have age categories starting as young as 8 years with age-matched categories up to the junior level. In triathlon and most endurance sports (e.g. cycling, long-distance running), the junior cut-off is less than 19 years.

17.3.1 Changes to muscle mass and effects of resistance exercise in young athletes

Strength work is also limited in children because of their smaller muscle mass relative to other tissue as well as potential instability of joints and ligaments. At birth muscle mass comprises around 25% of lean body mass. This increases to around 40% in a young adult and thickens with ageing and physical activity (or use). During adolescence, muscle growth is mostly in length, not breadth. Thickening of muscle fibres rather than lengthening predominates in post-adolescence. This *normal* physiological development of muscle accounts, to some extent, for the delay often experienced by adolescent athletes who want to increase muscle mass while their muscles are still lengthening.

The ability of the young athlete to undertake a resistance training program that is intended to increase muscle mass and strength is limited, to some extent, by their stage of maturation. Although there is evidence of an increase in strength and muscle hypertrophy, especially during the pre-adolescent phase in response to appropriate training stimulus (i.e. structured resistance exercise), the greatest gains occur once growth has ceased (see Chapter 4). Although research is limited in young athletes, enlarged muscle cross-sectional areas (CSA) following 10–12 weeks of isometric strength training for the upper (Fukunaga et al. 1992) or lower extremities (Mersch & Stoboy 1989) have been reported in healthy prepubertal children. These increases are mainly attributed to neural adaptation rather than hypertrophic factors (Granacher et al. 2011). Muscle hypertrophy is more common in pre-adolescent males following resistance exercise compared with females at the same growth stage because of increasing levels of circulating androgens (Faigenbaum et al. 2009). In contrast to adolescents, children have a more limited capacity to increase muscle mass (or muscle hypertrophy) in response to resistance training, although resistance training in children can increase muscle strength, improve motor skills and reduce risk of injury (Bloemers et al. 2012; Fukunaga et al. 2013). Similar benefits from resistance training are reported in obese and overweight children and adolescents with the added advantage of substantial improvement in body composition (Schranz et al. 2013).

Population guidelines for physical activity in children and adolescents focus predominantly on aerobic fitness and recommend an accumulation of up to 60 minutes/day of moderate to vigorous physical activity (WHO 2010). These guidelines do not adequately address resistance training so are suboptimal for children and adolescents who need to improve muscle strength and movement skills. This includes young athletes. Early concerns raised by Kraemer and colleagues in their landmark review in 1989 that maximal or near maximal resistance exercise (i.e. lifts) increases risk of structural injury in young athletes has not been substantiated to date. Nevertheless, even in 1989, these authors recommended that adult resistance programs should never be used for youth, which is still supported in the recent official position statement of the UK Strength and Conditioning Association on youth resistance training (see Lloyd et al. 2014).

17.3.2 Potential effects of exercise in a hot environment for young athletes

Historically, children and adolescents are considered less effective in regulating body temperature, have demonstrated a lower exercise tolerance in the heat and therefore been considered at a higher risk of hyperthermia and heat stress than adults (Drinkwater et al. 1977; Bar-or et al. 1980; Sports Medicine Australia 2008). Indeed, death from heat-related illness ranked second to head injury in secondary schools (Meyer & Bar-Or 1994).

Factors differentiating children and adolescents from adults in relation to thermoregulation include:

- a greater surface area-to-body mass ratio: causes a greater heat gain from the environment on a hot day (Drinkwater et al. 1977)
- more metabolic heat produced per unit mass during physical activities (Astrand 1952)
- a lower sweat rate: evaporation of sweat is the mechanism for keeping the body cool (endothermic reaction); in children the threshold when sweating commences is higher than

in an adult (Shibasaki et al. 1997)

- a lower cardiac output: cooling is also brought about by convection (heat transfer) in the blood; low cardiac output is associated with a slower rate of blood flow to the skin (and hence heat loss).

However, in some studies of children, where rigorous methods have been used to match exercise intensities, fitness levels, environmental conditions and hydration status of adults, children have demonstrated similar capacity to adults to tolerate heat during exercise in hot conditions (Shibasaki et al. 1997; Inbar et al. 2004; Rowland et al. 2008). Their mechanisms for dissipating heat loads during exercise differs from adults (Falk & Dotan 2008; Rowland 2008). Children and adolescents rely more on peripheral blood redistribution (radiative and conductive cooling) rather than sweating (evaporative cooling) to maintain thermal equilibrium. There is also evidence that adolescents who undertake regular training adapt by enhanced peripheral vasodilation (Roche et al. 2010), which is likely to improve non-evaporative cooling. While the timing of the transition from child-like to adult-like thermoregulatory mechanism is likely to be related to pubertal development, it appears that these changes do not become physiologically evident until puberty is completed (Falk et al. 1992).

The limited studies published that have investigated the impact of childhood adiposity on thermoregulation have observed higher core temperatures in lean children than those classified as overweight (Leites et al. 2013). These findings, however, may merely be a consequence of greater heat production at the same relative aerobic intensity in lean individuals when assessed at the same relative exercise intensity. To isolate the influence of adiposity on changes in core temperature during exercise from possible confounding effects, Morris and Jay (2013) recommend that future studies use heat production per unit body mass when undertaking exercise in lean and obese groups as a comparative measure.

In summary, most children can tolerate mild to moderate heat stress. However, when the environmental temperature is very hot or when the environment has a high vapour pressure (i.e. high relative humidity, thereby discouraging evaporative cooling mechanisms) and the children are not adapted to exercising in the heat, the risk of heat stress is high. Children may not recognise the signs or symptoms of heat stress, forget to drink unless reminded and continue to exercise to keep up with their peers.

17.3.3 Potential effects of exercise in a cold environment on young athletes

Children also have difficulty adapting to very low temperatures and are at risk of hypothermia. Young athletes who participate in aquatic activities and in land-based activities where there is a low temperature and high wind chill factor are at risk. Risk is dependent, to some extent, on the adiposity of the child and stage of maturation of the thermoregulatory response. Lean children are at higher risk of hypothermia than fatter children because of the low levels of subcutaneous or thermal-insulating fat. Children of small stature are at high risk because the smaller the individual, the larger the body surface area per unit body mass. The larger the surface area, the greater the heat loss.

In swimming pool or aquatic activities

The loss of heat from the body is 25–40 times faster in water than in air. In one early study, children aged 8–9 years were taken out of the water after 18–20 minutes because of a drop in core temperature (Sloan & Keatinge 1973). In the same study, adolescents of around 18 years could swim for 30 minutes longer than the children. An increase in water temperature of pools by 1–2° C above the usual temperature set for adults helps maintain core temperature in children undertaking swimming training or learning to swim.

17.4 Fluid recommendations for young athletes

Fluid intake remains an important priority for sports nutrition advice to young athletes. Poor hydration status is the main reason for the high prevalence of heat-illness associated with sport and physical activity in young athletes (CDC 2011). Heat illness is also influenced by unexpected physical exertion, insufficient cooling between exercise bouts and inappropriate choices of clothing, including protective equipment and poor-quality sports uniforms (Council on Sports Medicine and Fitness & Council on School Health 2011). Unfortunately, there is no evidence to determine the extent to which (if at all) fluid intake may modulate the risk of heat illness in young athletes. Fluid monitoring studies on children and adolescents are scarce and often fail to report participants who actually experience heat illnesses (Somboonwong et al. 2012). In contrast, field studies indicate that adolescent athletes can experience significant changes in fluid (≥4% body weight) during training and competition in the heat (Silva et al. 2011; Aragon-Vargas et al. 2012). Fluid shifts of this magnitude have the potential to impact on exercise performance (Walsh et al. 1994) providing further justification to monitor fluid consumption and balance in young athletes. It appears prudent to apply the recommended fluid intake guidelines for adults, which suggest that during exercise athletes should drink to avoid weight changes >2% of their pre-exercise body mass (Sawka et al. 2007).

17.4.1 Type of fluid

Sports drinks are readily available and marketed as an alternative to soft drinks to the general population, and particularly to children and adolescents (O'Dea 2003). However, for the active young athlete engaged in routine physical activity, the use of sports drinks in place of water on the sports field is unnecessary for several reasons:

- sweat sodium losses are generally lower in young athletes compared to adults (Meyer et al. 2012) so replenishment of electrolyte loss is not needed

- excess consumption of sports drinks may lead to excessive caloric consumption and increased risk of overweight and obesity (Committee on Nutrition & Council on Sports Medicine and Fitness 2011).

Sports drinks have been produced for specific sporting situations. Designed for rapid delivery of water and nutrients (CHO and electrolytes) during periods of prolonged, vigorous sports participation, they are also intended to minimise possible gastrointestinal complications associated with the consumption of solid food. Sports drinks are an ideal choice if gastrointestinal disturbances occur before or during exercise and/or in recovery between several events or races or in endurance events, such as triathlon, which has increasing popularity with young athletes. Of some concern is that many young athletes consume caffeinated energy drinks close to their participation in competition. O'Dea (2003) suggests that young athletes' understanding of the differences between sports drinks and caffeinated energy drinks is limited. These drinks contain higher concentrations of carbohydrate and other stimulants and are not advisable for young athletes to consume in association with sport or exercise. Following exercise, milk is also a suitable recovery beverage for young athletes because it has a similar osmolality to sports drinks (~300 mOsmoles)—the same osmolality as blood plasma—and contains similar amounts of carbohydrate and protein. More detail on fluid intake and choices is available in Commentary 7.

17.5 Energy recommendations for young athletes

During childhood and adolescence, adequate energy is required to meet both growth and development needs, as well as the substrate demands associated with everyday physical activity, training and competition (Petrie et al. 2004; Unnithan & Goulopoulou 2004; Aerenhouts et al. 2011). Estimating the energy needs of an young athlete with precision is difficult because of the wide range of metabolic variability within and between individuals (Petrie et al. 2004), and the methodological difficulties in estimating energy expenditure (Burke et al. 2001). Energy expenditure estimates are used to determine energy requirements. Furthermore, the energy expenditure associated with the exercise commitments of young athletes vary substantially because of individual differences in total training and competition load, in seasonal influences, and in participation in more than one competitive sport (Petrie et al. 2004).

Most population recommendations for energy requirements are based on predictive estimates of resting energy expenditure multiplied by an activity factor. In Australia and the USA, the equation used is based on the weight and height within the 10–17 years age category and does not allow for pubertal maturation (Schofield 1985). The energy expenditure values calculated are a crude estimate of energy requirements for individuals (Meyer et al. 2007). Indeed, adolescent athletes often report much lower energy intakes than those suggested using predictive equations (Aerenhouts et al. 2011; Gibson et al. 2011), although this may be attributed to under-reporting, which is inherent in all dietary intake data collection methods.

Energy requirements need to be adjusted for the specific growth stage of a young athlete, which varies considerably. Even in well-nourished children and adolescents, large differences in timing of growth spurts is evident between individuals of the same age, and between different cultures and genetic backgrounds. The energy needs for growth consist of two parts: the energy deposited in growing tissues and the energy expended to synthesise those tissues (Torun 2005). The energy deposited in growing tissues is small and has been commonly estimated as 8.6 kJ/g of daily weight gain—for example, for a male age 15 years gaining 6 kg/yr = ~140 kJ/d (WHO 1983). The energy expended to synthesise new tissue is incorporated in measures of total energy expenditure such as doubly labelled water. Measurements of total energy expenditure of adolescents indicate that the energy changes associated with physical activity and/or training are likely to have a much greater influence on energy demands than the increases associated with growth (Torun 2005). Suggested energy requirements (incorporating total energy expenditure plus energy deposited in growing tissues) for adolescent populations with different levels of habitual physical activity have been published elsewhere (Torun 2005).

Prolonged periods of low energy availability (see Chapter 5) in young athletes can have several adverse health consequences, including delayed puberty, menstrual irregularities, poor bone health, short stature, the development of disordered eating behaviours and increased risk of injury (Meyer et al. 2007; Nattiv et al. 2007; Bass & Inge 2010). Methods of estimating energy availability are found in Commentary 3.

Estimating energy balance (and energy availability) in maturing athletes is complex and confounded by changing metabolic requirements associated with stage of growth and maturational timing. Some evidence suggests that an acceleration in growth can occur immediately following retirement in former elite adolescent athletes such as gymnasts (Caine et al. 2001). While this apparent *catch up* may suggest that exercise delays growth, an alternative explanation is that this response is normal for gymnasts who are late maturers. For some other groups of children and adolescents, there is increased risk of overweight/obesity, metabolic disorders such as type 2 diabetes, hyperlipidaemia, atherosclerosis and hypertension, and an increased risk of injury (AIHW 2012; NHMRC 2012), which is strongly assessed with a positive energy balance. While weight loss may be desirable for an overweight child or pre-pubescent athlete, severe and prolonged energy restriction is not recommended. Weight maintenance rather than weight loss is more appropriate.

Energy balance is estimated by assessing energy intake and comparing it with concurrent serial estimates of exercise energy expenditure (e.g. Sensewear™) and other markers. A variety of methods are available for estimating energy expenditure and energy intake in a clinical and research setting (see Chapter 5). Comparing dietary intake data to population nutrient and energy reference values, matched for age and weight status, can be used a guideline for assessing the probability of inadequacy, although there are limitations when applied to athletes (Heaney et al. 2010). Energy availability values can be compared to population measures of growth, development and physiological function to determine energy adequacy. Other indicators of the adequacy of energy availability may include measures of self-reported fatigue, maturational timing, menstrual dysfunction and bone mineral density.

The dietary intakes of young athletes for selected sports are described in Table 17.2. Typically these studies involve the collection of prospective dietary intake data as a case study (i.e. without a control group of non-athletes for comparison). The dietary intake data indicate that younger athletes report large variability in dietary intakes (even for individuals competing in the same sport). Also, it appears that female athletes such as gymnasts and swimmers are particularly vulnerable to the consequences of low energy availability.

TABLE 17.2 Self-reported average energy intake of young athletes

Reference	Sport	Number of subjects	Age range (yrs)	Height (cm)	Weight (kg)	(MJ/d)	(kcal/d)
Females							
Aerenhouts et al. 2011	Sprinting (up to 400 m): 3-year follow-up	24 (baseline)	13–15	167.1	52.6	8.6	2.1
		18 (3 years)	16.5–18.5	169.1	56.4	8.8	2.1
Benardot et al. 1989	Gymnastics	29	7–10	134.9	30.6	6.9	1.7
Benson et al. 1985	Dance	92	12–17	160.2	46.8	8.1	1.9
Berning et al. 1991	Swimming	22	15–18	N/A	58.2	15.0	3.6
Heaney et al. 2010	Elite athletes (7 different sports)	72	19.2 ± 3.6 (mean ± SD)	N/A	68.7	10.6	2.5
Perron & Endres 1985	Volleyball	26	13–17	N/A	N/A	7.5	1.8
Males							
Aerenhouts et al. 2011	Sprinting (up to 400 m) –3 year follow-up	24	12–16	172.2	57.9	11.2	2.7
		22	15–19	177.4	66.4	13.2	3.2
Berning et al. 1991	Swimming	42	15–18	182.0	75.1	19.1	4.5
Hawley & Williams 1991	Swimming	9	11–14	N/A	56.4	12.9	3.1
Hickson et al. 1987	US Football	46 88	11–14 15–18	N/A	60.9 75.9	10.6 14.1	2.5 3.4
Rico-Sanz et al. 1998	Soccer	8	15–19	170	63.0	16.6	4.0
Schemmel et al. 1988	Running	4	7–10	N/A	N/A	8.2	2.0
		14	11–14			10.6	2.5
		4	15–18			11.4	2.7
	Wrestling	4	7–10			7.9	1.8

		50	11–14			10.3	2.5
		20	15–18			11.3	2.7

N/A = *value not available*

17.6 Body image and dieting behaviours of young athletes

Participation in sport develops and promotes healthy self-perceptions of body image and psychological wellbeing for most adolescents (Ekeland et al. 2005). Despite this positive effect, some young athletes involved in sports emphasising leanness or aesthetic requirements (e.g. ballet, gymnastics, ice skating) have unusual eating attitudes and are at high risk of disordered eating behaviours (Ferrand et al. 2005; Rouveix et al. 2007; Monthuy-Blanc et al. 2012). Their perception about an ideal body physique and their own physique for their sport may be distorted.

Similar to other age-matched adolescents, adolescent and prepubescent athletes may engage in inappropriate dietary and training behaviours to control body shape and weight, to increase lean body mass or decrease fat mass or even delay growth (Bonci et al. 2008). Clear differences exist between adolescent males and females. Females tend to focus on low body fat levels and low body weight whereas males favour increased muscularity and are not so concerned about weight (Botta 2003). Both male and female adolescent athletes find it difficult to dissociate the food intake required for supporting and enhancing sport performance and that required to achieve societal *ideals* of physique. Adolescent males, in particular, are influenced by testimonials from other athletes (e.g. body builders) and are susceptible to using supplements that claim increases in lean body mass, strength or power. While restrictive dieting is common in both male and female adolescent athletes (Huon 1994; Boutelle et al. 2002; Neumark-Sztainer et al. 2006; Martinsen et al. 2010), those with high energy requirements are vulnerable to the physical and psychological consequences of restrictive eating behaviour.

17.7 Protein recommendations for young athletes

In adult athletes, high-quality protein (and carbohydrate) consumed frequently during the training day and recovery period after strenuous exercise is recommended (Tarnopolsky 2004; American Dietetic Association et al. 2009; Phillips & Van Loon 2011). Such a pattern promotes activation of the protein synthesis pathway and provides substrate for lean tissue accretion. While these considerations focus on maximising the response to the stimulus of exercise training, children and adolescents have additional protein requirements needed to support growth and development (Meyer et al. 2007; Aerenhouts et al. 2011). Protein requirements are likely to be even higher in young athletes who consume restricted energy diets with low energy availability.

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With suboptimal energy intake (either from inadequate total daily energy intake or long gaps with no food between meals), endogenous protein (i.e. lean body mass) is mobilised as well as liver glycogen to maintain homeostasis of blood glucose, potentially reducing the availability of protein for its primary functions (Petrie et al. 2004; Campbell et al. 2007). Matching total daily energy intake with expenditure helps preserve endogenous protein and prevent use of protein as a substrate for glycolysis. This is the physiological rationale behind frequent consumption of food (and high-quality protein) throughout the day.

Population recommendations for protein requirements (i.e. Estimated Average Requirements (EAR)) for children and adolescents are set according to body mass and age and account for the increased requirements for growth and development (NHMRC & NZMH 2006). EAR values are based on a factorial method, which uses estimates of whole body protein turnover with an additional estimate provided for growth and maintenance on a fat-free mass basis (NHMRC & NZMH 2006). The EAR and extrapolated Recommended Dietary Intake (RDI) or recommended Dietary Allowance (RDA) for protein for adolescents is only slightly higher than adults, as seen in Table 17.3 (NHMRC & NZMH 2006). Notably, the low levels of evidence used to determine the EAR and hence RDI values for the adolescent age group have been acknowledged (NHMRC & NZMH 2006). Therefore, the population reference values serve as useful initial criteria to assess the probability of inadequacy of protein intake for most young athletes.

TABLE 17.3 Australia and New Zealand Nutrient Reference Values for protein[a]

	Age range (years)	EAR (g/kg/d)	RDI (g/kg/d)
Children (both sexes)	4–8	0.73	0.91
Boys	9–13	0.78	0.94
	14–18	0.76	0.99
Girls	9–13	0.61	0.87
	14–18	0.62	0.77

[a]Intake to support growth and maintenance of fat-free mass; should be used for active adolescents
EAR = Estimated Average Requirement, RDI= Recommended Dietary Intake

Source: National Health and Medical Research Council & New Zealand Ministry of Health, 2006

TABLE 17.4 Australia and New Zealand Nutrient Reference Values for iron

	Age range (years)	EAR (mg/d)	RDI/RDA (mg/d)
Children (both sexes)	4–8	4	10
Boys	9–13	6	8
	14–18	8	11
Girls	9–13	6	8
	14–18	8	15

Source: National Health and Medical Research Council & New Zealand Ministry of Health, 2006

Some adolescent athletes, however, may need slightly higher protein requirements than the upper end of EAR requirements recommended for the general population, particularly if they have well-developed musculature, are above the reference weight for age or are involved in high-intensity training. This has been confirmed by nitrogen balance studies on young athletes including sprinters and soccer players. In these studies, positive nitrogen balance was achieved with reported daily protein intakes between 1.35 and 1.6 g/kg/d (Boisseau et al. 2007). In a more recent longitudinal study of adolescent sprint athletes, a dietary protein intake within this range was needed to achieve a positive nitrogen balance, irrespective of the adolescent's growth rate or increase in fat-free mass (Aerenhouts et al. 2011). These reported values are higher than the RDI/RDA.

In reality, adolescent and younger athletes who meet their energy requirements and reside in a developed country either meet or exceed the range of protein intakes reported above (Gibson et al. 2004; Petrie et al. 2004; Heaney et al. 2011). This range (i.e. ~1.2–1.6 g/kg/d) is similar to the recommended protein intakes for adult athletes (see Chapter 4). Hence protein supplements are usually not needed (Gibson et al. 2004; Petrie et al. 2004; Aerenhouts et al. 2011; Heaney et al. 2010). A justifiable reason for recommending protein-containing supplements to adolescent athletes is that they provide a convenient source of protein within the athlete's lifestyle rather than to meet protein requirements.

Over the past two decades, innovative improvements in technology and research methodology have allowed new insights into the skeletal muscle response to food, exercise and their interaction (Phillips et al. 2012) (see also Chapter 4). This research has been conducted on adult and elderly subjects but not children or adolescents. Whether adolescents and children have similar cellular responses to food and exercise at the skeletal muscle site is largely unknown so any similarity is inferred based on comparable physiological function.

17.8 Carbohydrate (CHO) recommendations for young athletes

The benefit of a high-CHO diet on performance enhancement of adult athletes involved in high-intensity training programs and endurance sport is well documented (see Chapters 12, 13 and 14). However, potential differences in CHO storage and substrate utilisation during exercise may exist between children and adults. These differences may be attributed to maturational timing, training load and the patterns of sport participation in young athletes. Despite these potential differences, no definitive CHO recommendations are available for children or adolescents involved in high-level sport training.

During the early 1970s, several Scandinavian studies have used muscle biopsy techniques to compare adaptations to exercise training between young male athletes (11–16 y) and adult athletes (Eriksson et al. 1973, 1974; Eriksson & Saltin 1974). The initial findings indicated that young athletes adapted to training in a similar way to adults (i.e. increased muscle glycogen, glycogen utilisation during maximal tests, oxidative and anaerobic enzyme capacity).

However, the magnitude of the increase in anaerobic activity as measured by enzyme levels was lower in adolescents compared with adults, which suggests that capacity for adaptation is still developing (Eriksson et al. 1973, 1974; Eriksson & Saltin 1974). Similarly, other researchers have demonstrated that at high-intensity exercise loads, younger individuals (i.e. 6–12 y) rely more on oxidative or aerobic metabolism than anaerobic metabolism to meet energy demands (Taylor et al. 1997). In addition, when CHO metabolism is occurring during exercise, the relative oxidation of ingested (exogenous) CHO is higher in pre- and early-pubescent boys than in men (Timmons et al. 2003, 2007).

However, not all researchers have demonstrated changes in enzyme capacity or agree that reduced enzyme rates indicate that glycolytic metabolism in humans is maturity-dependent (Haralambie 1982; Petersen et al. 1999). In an investigation into the activity of 22 enzymes related to energy metabolism, glycolytic enzymes, including fructose-6-phosphate kinase, showed no significant differences in their activity in adults compared to adolescents (Haralambie 1982). Other authors have raised concerns that many early studies used absolute values of specific enzyme activity as markers of CHO metabolism. Adjusting these values relative to fat-free mass (FFM), which varies during adolescence, is a more accurate measure (Brandou et al. 2006). In a study comparing trained pre-pubertal and pubertal female swimmers, Petersen et al. (1999) demonstrated no differences in in-vivo glycolytic metabolism in response to a high-intensity exercise task, particularly when muscle cross-sectional area was considered (Petersen et al. 1999). Taken collectively, the existing evidence relating to the impact of maturation on energy metabolism suggests little to warrant deviation from the current adult recommendations for dietary CHO.

The duration and intensity of an exercise session determines CHO utilisation patterns and refuelling requirements. Therefore, dietary CHO needs should be considered in light of the training loads and competition characteristics that are typically undertaken by young athletes. These can differ from those undertaken by adults in a number of ways; some sports feature different rules, game durations or race lengths for younger competitors while others have different competition formats (e.g. sports carnivals, representative competitions, trials). While the training commitments involved in junior sport may be less compared with adult sport (e.g. fewer sessions in a week, shorter duration sessions), it is also possible that adolescent athletes may participate in several different sports. These different energy demands and subsequent CHO requirements must be either added or adjusted depending on whether the participation is concurrent or seasonal (e.g. cricket in summer and football in winter months).

Increasing muscle CHO availability in adults through CHO loading or increasing ingestion throughout exercise may improve performance in events of more than 90 minutes duration (Burke et al. 2001). Given the shorter duration of events undertaken by young athletes, the recommendations encouraging high CHO intakes should only be employed in relevant situations.

17.9 Dietary fat recommendations for young athletes

Adequate dietary fat intake is needed to meet requirements for fat-soluble vitamins and essential fatty acids, and helps to provide energy to support growth and maturation (Petrie et al. 2004). Adipose tissue (or stored body fat) and the triacylglycerol (breakdown product of fat) stored within muscle is the major endogenous energy store. Increasing physical activity is associated with an increased capacity for oxidation of fatty acids and reduced reliance on carbohydrate stores, which is a normal training adaptation (Shaw et al. 2010). Typically, the need for pro-active replenishment of fat stores after exercise has not been considered because of the relatively large (adipose) stores that can be found in even the leanest athletes (Burke et al. 2004). Recently, a renewed interest in the role of the intramuscular triacylglycerols on exercise performance has emerged as well as the effect of training in a carbohydrate-depleted state (Bergman et al. 2010; Shaw et al. 2010). This practice can further promote adaptations to enhance the capacity to use fat as an exercise fuel (Hawley 2011). However, the effect of strategies which promote *fat adaptation* on endurance exercise performance in adolescent athletes remains unstudied.

Since chronically high fat intakes are associated with a positive energy balance and consequent high risk of overweight and obesity (NHMRC 2012), the recommendation for type and total fat intake by adolescent athletes should be in accordance with public health guidelines. While no nutrient reference values exist for total fat intake, the Acceptable Macronutrient Distribution Range (AMDR) to reduce chronic disease risk of 20–35% of total energy, with saturated and trans fatty acids providing no more than 10% of total energy (NHMRC & NZMH 2006), can be used for young athletes. Dietary surveys of adolescent athletes suggest that dietary practices typically provide a fat intake of at least 30% of total energy intake (Croll et al. 2006; Juzwiak et al. 2008), which is close to population targets.

Some female adolescent athletes with very high training demands are at risk of impaired menstrual function (Nattiv et al. 2007). Given that fat is the most energy-dense macronutrient, an increase in fat intake from predominantly unsaturated fat sources by some individuals can help increase total energy intakes and energy availability. However, the causes of amenorrhoea and oligomenorrhoea in athletes are complex and multifactorial and a full dietary review is warranted in such cases.

17.10 Iron recommendations for young athletes

Depleted iron stores, without clinical symptoms, are observed frequently in studies conducted on adolescent and young adult athletes (particularly endurance athletes) (Gropper et al. 2006; Rodenberg & Gustafson 2007). The prevalence of iron depletion is no different to non-athletic counterparts (Sandstrom et al. 2012). Interpreting one-off measures of iron status markers (i.e. serum ferritin) in athletes, especially adolescent athletes, should be done with caution for several reasons: athletes generally have lower levels of ferritin than non-athletes, sex differences are evident between males and females during adolescence, cut-off values for ferritin are not standardised in studies of young athletes, and ferritin levels can be falsely positive even in mild infection or under physiological stress (see Chapter 10). In one recent study of 193 elite young athletes (96 males, 97 females; 16.2 ± 2.7 y) from 24 different sports, iron depletion was evident in 31% of male and 57% of female athletes, based on a serum ferritin cut-off of <35 μg/L (Koehler et al. 2012). However, when a cut-off of <12 μg/L for serum ferritin was applied to the same group of athletes, 4% of male and 7% of females were iron depleted and diagnosed as anaemic (Koehler et al. 2012). This is slightly higher than the reported prevalence rates for iron deficiency anaemia for adult athletes and untrained controls, which is around 3% (Weight et al. 1992; Fogelholm 1995; Sandström et al. 2012). Regardless of its aetiology, iron deficiency, with or without anaemia, is of concern. Even with a mild tissue shortfall in iron, at least in studies of adult athletes, endurance performance capacity and aerobic adaptation to a prolonged period of training is adversely affected (Rodenberg & Gustafson 2007). Detection and treatment of early iron depletion in children and adolescents is warranted (see Chapter 10). Because of higher iron requirements for growth in adolescents compared to older athletes, the progression of low iron stores into a state of iron deficiency is more rapid.

In adolescent female endurance athletes, suboptimal iron status is mainly attributed to low iron intake, low iron bioavailability in combination with high requirements associated with training and blood loss (e.g. increased red cell mass, menstruation, haematuria, haemolysis) (Gropper et al. 2006; Koehler et al. 2012). In contrast, suboptimal iron status in adolescent male athletes is associated more with high physiological requirements (i.e. training and growth) than with diet (see Chapter 10). The EAR for iron for developing girls accounts for iron lost from menstruation. When setting population reference values, the age cut-off for menarche is 14 years (NHMRC & NZMH 2006). The Food and Nutrition Board (2000) suggests that the requirements for iron for athletes should be 1.3–1.7 times higher than EAR values and 1.8 times higher than the EAR for vegetarians (non-athletes) to account for a low iron bioavailability diet of vegetarians. Clearly young athletes who are vegetarian are at high risk for iron deficiency (see Chapter 10).

Results from dietary intake studies of young athletes indicate males typically exceed RDI/RDA cut-offs for iron. In contrast, although the average dietary iron intake of females either meet or exceed RDI/RDA cut-offs, individual intakes vary considerably (Juzwiak et al. 2008; Heaney et al. 2010; Martínez et al. 2011). The use of iron supplements should be guided by an appropriately qualified health professional (e.g. general practitioner, sports physician) with regular follow-up.

17.11 Calcium and vitamin D recommendations for young athletes

Calcium and vitamin D are key nutrients for bone health. Adequate dietary calcium for growth is essential for optimising peak bone mass. It maintains the structural or mineral content of bone and, in the presence of adequate intake, reduces bone resorption. Activated vitamin D (calcitriol) is required for the regulation of calcium homeostasis. Vitamin D enhances calcium absorption in the gut and is essential for normal calcium metabolism. Adolescence is the period in which the level of bone remodelling is highest (MacKelvie et al. 2002). Bone mineral accrual occurs during adolescence and early adulthood where peak bone mass is reached. Chapter 9 describes this process and the risk factors for athletes in more detail.

Calcium requirements for children and adolescents are highest during a growth spurt. The rate of skeletal calcium accretion during adolescence is estimated to be around 300 mg/d (Matkovic 1991). After accounting for urinary and sweat losses and assuming a net calcium absorption from food of approximately 25–35%, the RDI/RDA for males and females (age 14–18 y) is 1300 mg/d (NHMRC & NZMH 2006). At this age range, there is no difference in calcium requirements between sexes, despite differences in maturational timing. Girls begin the growth spurt earlier than boys, whose growth spurt extends for a longer time. Weight-bearing exercise and, to a lesser extent, resistance exercise, increases bone mineral content in adolescents compared to untrained controls, although the effect is small (<6% difference) and confounded by the normal increase that accompanies growth (Nichols et al. 2001; Stear et al. 2003; Weeks et al. 2008). However, this relatively small additional bone mineral accretion is unlikely to substantially increase calcium requirements in exercising adolescents above the requirements of inactive adolescents. Nevertheless, even small increases in bone mineral content at this crucial time of optimising peak bone mass is likely to result in significantly greater bone density in active compared to inactive individuals by the end of adolescence (Bailey et al. 1999). There are currently no specific recommendations for calcium intake for athletes so, until further studies are undertaken, population reference standards can be used as a benchmark for assessing adequacy.

Similar to adult athletes, young athletes are at high risk of vitamin D deficiency if they have little exposure to the sun and live at latitudes >35 degrees, spend long periods training indoors, are dark-skinned, use sunscreen or wear protective clothing (see Chapter 9). In one study of adolescent female gymnasts training indoors, six out of 18 athletes had vitamin D deficiency (i.e. serum vitamin D levels below 50 nmol/·L) (Lovell 2008). Generally, dietary sources of vitamin D are a minor contributor to vitamin D status compared to exposure to sunlight but become a more important source during winter, when the sun is not strong enough to provide enough UV rays. Heaney et al. (2010) reported that over 90% of female athletes (including many adolescents) from a variety of sports had vitamin D intakes below current recommendations.

Although vitamin D is best known for its role in bone health, it has many functions in other physiological system (e.g. immune system, muscular system) (see Chapter 9, section 9.8). Vitamin D insufficiency is also linked to skeletal muscle function, muscle pain and weakness and may potentially increase susceptibility to injury and slow rate of rehabilitation from injury. In one study of girls aged 12–14 years in a secondary school in the UK, low serum vitamin D status was associated with reduced muscle power and force in standardised fitness measures (Ward et al. 2009).

In another large randomised double-blind, placebo-controlled study of over 5000 female navy recruits with a high risk of stress fractures, 8 weeks of vitamin D (800 IU/d) and calcium (2000 mg/d) supplements reduced the risk of stress fractures, compared with a placebo group (Lappe et al. 2008). A follow-up study of the same young navy recruits found double the risk of stress fractures of the tibia and fibula in women with serum vitamin D (i.e. 1,25-dihydroxyvitamin D3 [1,25(OH)2D3]) of <20 ng/mL compared with those with concentrations >40 ng/mL (Burgi et al. 2011). In another prospective study of US adolescent girls (*n* = ~6 700), vitamin D but not calcium intake was associated with lower incidence of stress fracture, particularly in those regularly engaging in high-intensity activity (among whom 90% of the stress fractures occurred) (Sonneville et al. 2012). The results of these studies has implications for those young athletes at high risk of stress fractures who avoid exposure to sunlight.

Adolescent female athletes with amenorrhoea (or delayed menarche) or low energy availability are also at risk of suppressed bone mineral accrual (Barrack et al. 2010). This can present as osteopaenia, which is evident in several studies of adolescent females (Carbon et al. 1990; Nattiv et al. 2007). Amenorrhoeic adolescent athletes have significantly impaired bone microarchitecture compared with eumenorrhoeic athletes and non-athletic controls (Ackerman et al. 2011). Restoring energy availability by increasing energy intake, reducing energy expenditure or both is the priority for treatment and recovery of amenorrhoea (Nattiv et al. 2007). In some female adolescents, hormone replacement therapy is warranted. Meeting calcium and vitamin D requirements in amounts that meet, rather than exceed, existing recommendations for the general population is recommended (Nattiv et al. 2007). This recommendation suggests that calcium and vitamin D requirements are either not likely to be elevated in athletes with low bone density or have a weaker effect on improving bone density, compared with an intervention that increases energy availability.

Despite increases in bone mineral accrual in active adolescents associated with weightbearing and resistance exercise (see Chapter 9, section 9.4.2), it is unlikely that adolescents require more calcium than population requirements. However, it is of concern that the reported calcium intakes of many adolescent male and female athletes are well below the RDA and possibly EAR (Juzwiak et al. 2008; Martínez et al. 2011; Gibson et al. 2011), despite the under-reporting inherent in estimating nutrient intakes from food records used in these studies. In population surveys, calcium intake of boys and girls is fairly similar during childhood until early adolescence when females substantially decrease intake of dairy foods and hence calcium intake compared with males (Magarey & Bannerman 2003; Savige et al. 2007). These food choice behaviours are likely to be no different in groups of young athletes.

17.12 Dietary supplements and nutritional ergogenic aids for young athletes*

The judicious use of specific dietary supplements and nutritional ergogenic aids may improve sporting performance in adults. However, their effectiveness and potential long-term effects have not been rigorously studied in younger populations, largely due to the ethical concept of beneficence (i.e. cost versus benefit). Despite this lack of scientific evidence, supplement use with the intent to improve sports performance among young athletes is common (McDowall 2007; Evans-Jr et al. 2012). The prevalence of supplement use among US children and adolescents (<18 y) as a performance enhancer was 1.6% (Evans-Jr et al. 2012). Muscle-building supplements, vitamin/mineral tablets, herbal supplements, guarana and creatine and high-protein milk supplements are frequently used by young male athletes (O'Dea 2003).

Adolescent athletes take 'performance-enhancing' dietary supplements for several reasons, such as pressure to achieve results, the pursuit of physical ideals, peer, social and marketing pressure. Also, while some young elite athletes do not believe that dietary supplements are required to be successful in their sport, they still consider them important for certain training adaptations such as strength gains (Bloodworth et al. 2012).

Philosophically it is inappropriate for young athletes to be encouraged to consume dietary supplements for performance enhancement. This view is consistent with those of leading sporting organisations and expert groups (Council on Sports Medicine and Fitness 2005; Meyer et al. 2007; International Olympic Committee 2010). This recommendation excludes the clinical use of dietary supplements (e.g. calcium, iron, vitamin D) when taken under appropriate guidance from suitably qualified health professionals (e.g. a medical practitioner, sports dietitian). Apart from issues related to safety, the use of supplements in developing athletes overemphasises their ability to manipulate performance. Essentially younger populations have the potential for greater performance enhancement through maturation and experience in their sport, along with adherence to proper training, nutrition and rest regimens. It can also be argued that discouraging the use of dietary supplements downplays the 'win at all costs' mentality and sets an important example for young athletes.

([*] *The definition of dietary supplements and ergogenic aids excludes sports foods and sports drinks.*)

Summary

Young athletes have unique nutritional requirements as a consequence of undertaking daily training and competition in addition to the demands of growth and development. Dietary education and recommendations for this group of athletes should reinforce eating for long-term health. More specifically, the developing athlete should be encouraged to moderate eating patterns to reflect daily exercise demands and provide a regular spread of high-quality carbohydrate and protein sources over the day, especially in the period immediately after training. Particular consideration should also be given to dietary calcium, vitamin D and iron intake of young athletes because of the elevated risk of deficiency and high requirements. The nutrient needs of young athletes should be met by food rather than supplements. The use of dietary supplements by developing athletes overemphasises their ability to manipulate performance in comparison to other training and dietary strategies.

Practice tips

RUTH LOGAN AND BEN DESBROW

DEVELOPING TRUST AND RAPPORT

- Many young athletes (and their parents) are unaware about sports nutrition and the strategies used to enhance athletic performance and recovery. Some athletes and parents have beliefs and attitudes about food choices that do not match sports nutrition recommendations. Coaches often refer young athletes who are consistently tired or underperforming. Young athletes are usually accompanied by a parent (or coach) so can be apprehensive or reluctant to reveal their true eating habits, especially in front of the coach. Parents accompanying children at a consultation may be defensive.
- Follow-up sessions for older adolescent athletes on a one-to-one basis without a parent or coach present helps build rapport and encourages self-reliance. A thorough assessment of current dietary practices and recovery habits highlight areas for improvement. Recommendations should be achievable and conform with the family's culture and usual eating environment.

DIETARY ASSESSMENT

- The capacity of the young athlete to undertake a high-intensity training program is limited to some extent by their stage of maturation, which is highly variable for the same age group training in the same team.
- Growth (height) should be assessed at regular intervals. Assessment of other anthropometric measures such as weight, circumferences and skinfolds of an individual or group of young athletes needs to be justified and cautious, especially in adolescents (both male and female) who are sensitive about body image. Measuring skinfolds of children is not justified. Absolute sum of skinfolds and relative changes are used for evaluation of skinfold data. Normative population data for height and weight, for the country of origin, is used for evaluating growth in children and young adolescents (see www.cdc.gov/growthcharts). Substantial deviation from an individual's growth curve should be investigated. Growth retardation or a delay in maturation can be a consequence of prolonged energy restriction and malnutrition.
- The principles of dietary assessment of an athlete are outlined in the practice tips in Chapter 2. Compared with adult athletes, young athletes are more likely to participate in several different sports including school, club, national or representative sport, so assessment of all levels of physical activity is important.
- In clinical practice, a thorough nutritional assessment for a young athlete usually involves a diet history (with a parent contributing, for younger children). In some situations where children are erratic eaters, a food record method or 24-hour recall may be used, which may be undertaken using electronic (i.e. digital food record or image-based food record) or manual aids. Any form of recording of dietary intake requires extensive training and can substantially distort usual eating behaviour (see Chapter 2). Using a combination of dietary intake assessment tools including food intake checklists and 24-hour recalls of food is useful for young athletes at least to provide a window into eating habits. Short-term recall techniques are more accurate than food records for young athletes.
- Enquiring about body composition goals and beliefs/attitudes about nutrition and sport (if any) at a consultation will assist you to better target and prioritise nutrition advice and determine the most effective dietary and behavioural intervention.
- Supplement use should always be checked to ensure that young athletes are not violating anti-doping regulations, although many forget the specific brand names of any supplements taken. Some supplements may contain banned substances, have adverse side effects if misused, or have questionable quality control checks. Many sports organisations now have their own supplement policies. One role of the sports dietitian is to provide advice on appropriate supplement use, be familiar with the supplement policy of sports organisations and to reinforce the World Anti-Doping Code (WADA 2009). Athlete support personnel should also be aware that harsher penalties apply for anti-doping rule violations when such individuals are found to be administering or trafficking to a minor (clause 10.3.2, WADA 2009).
- Any cessation of menses in adolescent female athletes for more than three months should be cause for concern. Referral to a sports physician for further assessment is warranted.
- Alcohol misuse by adolescents is of concern in the general population, and in adolescent athletes. Those adolescents involved in team sports with adult athletes are likely to be exposed to alcohol and to peer pressure to consume alcohol when underage.
- Changes in usual eating behaviour often occurs during competition away from home. Encourage parents and

young athletes to prepare a suitable supply of foods and fluids for competition. Venue canteens do not always provide suitable options.

COUNSELLING AND NUTRITION EDUCATION STRATEGIES FOR YOUNG ATHLETES

- Practical and interactive nutrition education sessions targeting types and timing of food intake for different situations (rather than focusing on nutrients) with accompanying scientific justification can modify food choice behaviours, especially in young athletes. Communal cooking nights are popular with junior athletes in teams and provide an opportunity for nutrition education in an informal setting (see Chapter 25 for guidelines). Determining the differences in eating habits between girls and boys at different ages, the influence of the sports culture and ethnic background on food choice is essential to effective nutrition education for any group of athletes. Dietary strategies and suggestions for change need to be realistic and cost effective and acceptable to athletes and parents.
- Cooking nights for parents of young athletes are also an ideal opportunity for a social event for the parent and a platform for the dietitian to address key diet-related concerns and issues relating to the young athlete.
- Many young athletes, especially adolescents, are focused on their weight and body image. A counselling approach that emphasises sporting performance, and also recognises and addresses the importance of growth, weight change and body image issues, even at very young ages, helps to allay this focus.
- Positive reinforcement is recommended when counselling children and adolescents. Offer praise for changes made, rather than constant criticism about existing poor practices and policing food choices.
- Children should accept some responsibility for their food choices and eating behaviour. Encourage food providers, parents and coaches to collaborate with young athletes to be self-reliant. Athletes benefit from being involved in assessing their own diets, identifying problems, setting goals and developing their own strategies.

MEETING NUTRITIONAL NEEDS

- Nutrient requirements during childhood and adolescence are higher than in adulthood, particularly during growth spurts and high training loads. The energy requirements of growth in combination with the added energy and nutrient turnover imposed by sport impose a high risk of nutrient depletion and potential nutrient deficiency for young athletes.
- Specific recommendations for energy intakes for young athletes do not exist. A variety of methods are available for estimating energy expenditure and hence energy requirements in a clinical and research setting (see Chapter 5). Population reference guidelines matched for age and sex or predictive equations can be used as a guideline (see Practice tips, Chapter 5). The increasing availability of other physical activity monitors that allow prediction of total energy expenditure (e.g. motion sense technology, heart rate monitors, accelerometers, GPS monitors) may be popular in young athletes. Estimates of energy expenditure from these devices are not representative of energy requirements for individuals but may be useful to show variation in energy expenditure at different times of the day.
- Chronic negative energy balance or low energy availability is contraindicated in young athletes and linked to long-term health risks. At-risk athletes, including gymnasts, dancers, swimmers, distance runners and those in weight-class sports, should be assessed regularly.
- To meet daily energy and nutrient requirements for training and recovery, young athletes need to eat at least three nutritious meals each day with between-meal snacks. Specific examples of suitable snacks that are convenient, portable and accessible, including appropriate fast food choices at take-away outlets and school canteens, can be provided. Suggesting that high-fat, sugary foods should be avoided is overly restrictive and may promote cravings for these foods. Ideally, nutrition goals should encourage the inclusion of nutrient-rich foods to meet energy requirements with inclusion of discretionary foods according to population dietary guidelines.
- Young athletes who need to increase body mass or with high daily energy expenditures often have difficulty meeting energy requirements or may lose their appetite after hard training sessions. These athletes require individual strategies on how to add low-bulk, high-energy and nutrient-dense foods to their diet without feeling too full. Some examples include dried, stewed or juiced fruits, high-energy fruit, nut and grain bars, milkshakes

and/or commercial liquid meal supplements, such as Sustagen Sport™. Full-cream milk products may be needed to meet high energy requirements. Adequate dairy food intake, with milk as the primary beverage, is associated with diets of high nutritional quality and higher intake of total daily energy (Fayet et al. 2013).

- Most young athletes easily meet total protein daily requirements. Of concern for potential suboptimal protein intakes are those athletes who are vegans or who regularly miss meals, particularly breakfast. There is a limited storage pool of protein. Including protein-rich foods at each meal is recommended to prevent catabolism of protein for glycolysis. Encourage young athletes to include both carbohydrate and protein at breakfast and in their recovery snacks after exercise.
- There are no specific recommendations for CHO intake for young athletes, although children use more CHO as an energy substrate than adults. Carbohydrate-rich foods need to be eaten at regular intervals each day to meet high the requirements for exercise and prevent protein catabolism. Be aware of influential adults who may have fixed views on restricting CHO intake due to societal pressure and common misconceptions.
- Sugary food, especially sticky food, confectionery and sports gels, increases acid production in the mouth, which causes dental caries. Similar to soft drink and cordial, sports drinks are also a high-risk beverage, particularly when consumed in bidons, because of the proximity of the front teeth. Rinsing the mouth with water does not necessarily protect teeth but reduces acid production and induces salivary flow, which offers some reduction in risk. Children should be discouraged from relying on sports drinks during competition (see section 17.4.1). Sports Dietitians Australia provides guidelines for minimising risk of dental caries (www.sportsdietitians.com.au/factsheets/dentalhealth).
- Depleted iron stores in young athletes can be present without clinical symptoms. Blood counts and iron status screening may be warranted in any young athlete with symptoms of fatigue and lethargy who habitually consumes a low iron intake with low bioavailability (see Chapter 10). These symptoms may also be attributed to poor recovery practices, insufficient rest days, poor sleep patterns, depression or anxiety or an underlying infection. Referral to a sports physician may be needed.
- Early detection and intervention of iron depletion is important to prevent iron deficiency. Some young athletes eliminate red meat from their diet. Strategies to increase iron intakes and bioavailability are outlined in Chapter 10.
- Calcium requirements are high during childhood and adolescence and difficult to meet in young athletes who limit or avoid dairy foods, without adequate substitution. Calcium intake of boys and girls is fairly similar during childhood until early adolescence when females substantially decrease intake of dairy foods and hence calcium intake, compared with males. Suboptimal calcium intakes are evident in adolescent female athletes (see Chapter 9).
- Vitamin and mineral supplements should be discouraged except in those young athletes who have difficulty meeting nutrient requirements from food sources, such as vegetarians and those on restricted-energy diets or with a diagnosed deficiency.
- Performance-enhancing supplements are not recommended for young athletes because limited studies have been conducted on the safety of intake in this age group.

FLUID AND IMPAIRED TEMPERATURE REGULATION

- Young athletes are at higher risk of hypothermia and hypothermia than adults. Poor hydration may potentiate heat-illness associated with sport and activity in younger athletes.
- Monitor weight changes (or fluid losses) before and after exercise to determine fluid loss. Coaches may need reminding to provide young athletes with frequent opportunities to drink, to cool down or keep warm when training in extreme environmental conditions.
- Fluid guidelines for young athletes involved in regular training programs should be individually determined. To help ensure adequate hydration, advise young athletes to drink periodically throughout the day at school until not thirsty, use water bottles at school, pass urine at recess and lunchtime, and check that urine is a pale, straw colour.
- Young athletes who train after school are often hypohydrated or hungry at the start of training. Being light headed or having a headache are symptoms of low blood sugar and inadequate fluid intake.
- Water is the fluid of choice during training and competition for young athletes. Sports drinks may be useful in those who have large carbohydrate requirements and are unable to eat before training. They can also be useful to replace fluid and electrolytes during extended sessions of activity, under very hot and/or humid conditions.

- The use of caffeine-containing energy drinks by young athletes should be discouraged.

COMPETITION EATING

- Guidelines for the pre-competition meal, eating between events and recovery should be provided (see Chapter 12). Young athletes are discouraged from experimenting with new foods at competition, but experiment with these at training. Providing a list of suggested food and beverage options and adhering to familiar foods and fluids during competition is well received.
- Eating (either foods and fluid or a combination of both) immediately or as soon as possible after training and competition (especially when competing on consecutive days or more than once in the one day) promotes rapid recovery of fuel reserves (see Chapter 14). A liquid meal supplement is useful for those young athletes who lose their appetite, don't feel like eating or cannot tolerate solid food in between events on the same day.

FOLLOW-UP

- Several consultations are usually needed with young athletes. One visit may be insufficient to develop a rapport and influence change in food choice. Behavioural change is a learned response that requires encouragement and support. Shorter, more frequent consultations provide the opportunity to do this.
- Children and adolescents respond well to take-home tasks or activities using current technology to assess their intake of fluid and food. This helps to improve self-reliance and practise suggested strategies, as well as providing encouragement and positive reinforcement.

MULTIDISCIPLINARY APPROACH

- A multidisciplinary, team approach has advantages in clinical counselling and education of young athletes. Other members of the medical team, including the sports physician and psychologist, may be involved, and regular communication is essential between service providers and the coach. Coaches are often present at the interview and can either inhibit or enhance communication. When counselling and educating young athletes, communication and involvement of the parent is essential for support, but should not interfere with your communication with the individual athlete.

SPECIFIC ISSUES

- Where there is evidence of an eating disorder, a multidisciplinary approach is essential.
- Rapid weight loss is discouraged. Weight maintenance or gradual weight loss (where required) is recommended. Both the child and parents need to understand the dangers of rapid weight loss. Regular monitoring is essential.
- Adolescent athletes wanting to bulk up should be encouraged to eat five to six energy-dense meals or snacks per day. High-energy milk-shakes and smoothies made with fortified milk (high-calcium milks), high-energy snacks like bars and liquid meal supplements are easy options.

USEFUL WEBSITES

- **www.sportsdietitians.com.au/factsheets**
  Sports Dietitians Australia: various fact sheets

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CHAPTER EIGHTEEN

Nutrition issues for the masters athlete

Peter Reaburn, Thomas Doering and Nattai Borges

18.1 Introduction

Masters athletes are typically older than 35 years and systematically train for and compete in organised forms of sport specifically designed for their age group (Reaburn & Dascombe 2008). Enjoyment, health, fitness and social benefits and competition appear the primary drivers for participation (Reaburn & Dascombe 2008, 2009).

A substantial increase in the number of older athletes competing at high performance levels in individual and multisport events designed for masters athletes has occurred in the last 20 years. This is evident from the increase in participants in the World Masters Games held every four years since its inception in Toronto, Canada, in 1985. In 2009, in Sydney, Australia, there was an almost four-fold increase in the number of participants compared with the original games. By 2013 in the games held in Torino, Italy, there were more participants, more countries participated and there was an increase in the number of events offered, compared with the previous event.

This increased participation may be linked to the strength of evidence confirming the effects of physical activity, particularly vigorous activity, on decreasing the risk of chronic lifestyle-related diseases, increasing longevity and improving health outcomes in older people (for reviews, see Lee et al. 2010; Grundy et al. 2012). Government policy in many countries is providing specific guidelines for older people to engage in sport and physical activity. Based on evidence from large cohort studies, participation in sport at any age is associated with a 20–40% reduction in all-cause mortality compared with non-participation (Khan et al. 2012).

Specific nutrient requirements for masters athletes, particularly in relation to micronutrients, have not been determined and are difficult to estimate because of large differences in the rate of ageing between individuals and the many different types of sports undertaken (from lawn bowls to ironman triathlon). Because of the limited research available on the dietary intakes and nutrient requirements of older athletes, nutrient recommendations for masters athletes are based on:

- the known physiological changes associated with ageing and their impact on nutrient requirements
- extrapolation from studies on younger athletes

- reported dietary intakes of older healthy populations
- population nutrient reference standards for older age groups
- the presence of any medical condition that requires specific dietary intervention
- the possible use of medications that may interact with nutrient absorption.

18.2 Physiological changes in masters athletes

The ageing process, at least in sedentary individuals, is accompanied by many physiological changes that affect nutrient and energy requirements as well as food preferences. Occurring with ageing are a substantial loss of lean body mass (muscle mass and bone mass), decreased absorption and utilisation of some nutrients, reduced immunity, gastric atrophy, decreased sensitivity to taste and smell, and a reduced thirst sensitivity. Whether these declines are a result of the ageing process itself, or the inactivity that often accompanies ageing, is unknown. In the general population, fat mass (FM) and percentage of body fat increases with ageing to around 80 years (Jackson et al. 2012), and decreases thereafter (Going et al. 1994). Similar observations are evident in ageing male and female endurance runners, swimmers and cyclists, although increases in FM are much less than that of the general population (Van Pelt et al. 2001; Peiffer et al. 2008; Wroblewski et al. 2011).

The progressive loss of muscle mass and decreases in muscle response, strength and function are inevitable with ageing. The reasons for these changes are varied, although a failure of satellite cell activation (Walker et al. 2012) and protein synthesis impairments in ageing skeletal muscle (Dickinson et al. 2013) are two of the many factors that are implicated (Walker et al. 2012). Satellite cells are mononuclear cells that are adjacent to muscle fibres. These are activated by injury and exercise and help to repair damaged muscle and maintain its function.

Physical activity appears to have little effect on sparing or preventing loss of fat-free mass (FFM) (Tanaka & Seals 2003, 2008; Brisswalter & Nosaka 2013). Despite the stimulatory effects of physical activity, particularly load- and impact-bearing activity, on satellite cell activity and building and maintaining muscle (and bone), there is still a decrease in muscle mass with ageing in endurance athletes (Proctor & Joyner 1997; Reaburn & Dascombe 2008; Brisswalter & Nosaka 2013) and in those involved in a lifetime of sprint or power training (Reaburn 1994; Faulkner 2008).

Furthermore, bone mineral density in postmenopausal athletic women is still lower than in younger athletic women; however, continuing athletic participation can maintain bone mineral density compared with age-matched non-athletes (Beshgetoor et al. 2000). In women, bone loss occurs earlier than in men (around 45 years) and at a faster rate (1%/yr). In men, bone loss commences around 50 years and continues at the rate of around 0.3% per year thereafter. Also, because men usually have a larger bone mass than women, they lose less bone and are therefore at a lower risk of degenerative osteoporosis than women of the same age. There is good evidence that exercise can retard the rate of bone loss, and that the sporting discipline and musculoskeletal loading patterns may influence the rate of decrease in bone mineral density. Studies on master sprint runners have reported significant elevations in whole body total, femoral neck and lumbar spine bone mineral density compared to middle and long distance runners (Gast et al. 2013). However, exercise does not prevent overall total loss of bone mass with ageing.

The ageing athlete is at higher risk for heat stress, cold intolerance and hypohydration than younger athletes. As the fine-tuning of the thermoregulatory responses deteriorates with ageing, masters athletes are at similar risk of heat loss and overheating as children. The difficulty in regulating temperature homeostasis is in part related to age-related changes occurring in the functioning of the cardiovascular system, which are independent of disease (e.g. atherosclerosis) or disease process (e.g. atherogenesis). These changes include a decline in maximal cardiovascular function (associated with a decreased maximal heart rate and decrease in maximal oxygen consumption) and a diminished contractile function of the heart (Shibata et al. 2008). These changes lower cardiac output and diminish blood flow to the skin, thus reducing the capacity to remove heat. Hence, older athletes are at risk of heat stress.

Table 18.1 summarises the major physiological changes that occur with ageing that are likely to affect nutrient requirements.

TABLE 18.1 Major age-related changes that may influence nutrient requirements of masters athletes

Age-related change	Nutritional implication
Decreased muscle mass	Decreased energy requirements
Decreased aerobic capacity	Decreased energy requirements
Decreased muscle glycogen stores	Decreased energy requirements
Decreased bone density	Increased need for calcium and vitamin D
Decreased gastric acid	Increased need for vitamin B12, folic acid, calcium, iron and zinc
Decreased skin capacity for synthesis	Increased need for vitamin D
Decreased calcium bioavailability	Increased need for calcium and vitamin D
Decreased hepatic uptake of retinol	Decreased need for vitamin A
Decreased efficiency in metabolic use of vitamin B6	Increased need for vitamin B6
Increased oxidative stress status	Increased need for carotenoids and vitamins C and E
Increased levels of homocysteine	Increased need for folate and vitamins B6 and B12
Decreased thirst perception	Increased fluid needs
Decreased kidney function	Increased fluid needs

18.3 Nutrient and energy recommendations for masters athletes

Specific nutrient and energy requirements for masters athletes have not been determined. In the absence of definitive requirements, population nutrient references, in combination with extrapolated data from studies of the nutrients requirements of younger athletes, are used for making nutrient recommendations and planning diets for older athletes. Population nutrient standards (NRV in Australia and New Zealand and DRI in the USA and Canada) for macro- and micronutrients for different age groups are appropriate for older athletes for most nutrients (Institute of Medicine 2001, 2002, 2010; Commonwealth Department of Health and Ageing et al. 2006). Recommendations for protein, vitamin D and calcium are slightly higher in men and women over 70 years than between 51 and 70 years (Institute of Medicine 2002, 2010). Recommendations for other nutrients are no different for younger adults, except for energy. However, these age ranges are wide and do not necessarily account for the additional metabolic and physiological changes imposed by athletic training of high intensity or duration. Reference standards may need some adjustment for master athletes participating in high-intensity training. For reviews of nutrient recommendations for the ageing athlete, see Rosenbloom and Dunaway (2007) and Gille (2010).

Only a small number of studies have been published on the dietary intakes of masters athletes (see Table 18.2). Based on these studies, masters athletes who are aerobically fit consume a more nutrient-dense diet than their less-fit peers.

TABLE 18.2 Summary table of studies examining dietary intakes of ageing athletes

Study	Age (years)	Sample	Comparative reference	Compared to reference (RDA or control group)
Butterworth 1993	72.5 ± 1.8	Female endurance (*n* = 12)	RDA	↓ Calcium (Ca) ↓ Energy intake
Butterworth 1993	72.5 ± 1.8	Female endurance (*n* = 12)	Age-matched, healthy controls	↓ Energy intake ↑ CHO, fat, protein ↑ Fibre ↑ Vitamins B6, E, folate ↑ Riboflavin, thiamin, niacin ↑ Ca, Ph, Mg, Fe, Zn, Na, K, Cu
Hallfrisch 1994	66.6 ± 1.3	Male endurance (*n* = 16)	Age-matched, healthy controls	↑ Energy intake ↑ CHO, protein
Reaburn & Le Bon 1995	50.6 ± 4.2	Male endurance (*n* = 14) Female runners (*n* = 15)	RDA	↑ Protein (M & F), fat (M & F) ↑ Riboflavin, niacin, thiamin (M & F) ↑ Vitamins A, C (M & F) ↑ Na, K (M & F) ↓ Ca (F), Fe (F), Zn (M & F), Mg (M & F)
Chatard 1998	63 ± 4.5	Male endurance (*n* = 18)	RDA	↓ Mg, vitamin D, Ca ↑ Energy intake ↑ CHO, fat, protein ↑ Fe, vitamins A, B1, B12, C, E
Beshgetoor 2000	49.6 ± 7.9	Female endurance	RDA	↓ Calcium (Ca)
Beshgetoor & Nichols 2003	48.4 ± 2.4	Female endurance (non-supplement users)	RDA	↓ Vitamin E ↓ Calcium (Ca)

M = male, F = female

18.4 Energy recommendations for masters athletes

Estimates of energy expenditure are used to determine energy requirements in any population group. Daily energy requirements decrease with ageing because of:

- decreases in energy expenditure required for physical activity and resting metabolic rate (RMR) (Roberts & Rosenberg 2006; Manini 2010)
- decreases in FFM (Going et al. 1994; Jackson et al. 2012)
- decreases in physical activity levels and training volume levels (Starling et al. 1999; Van Pelt et al. 2001; Weir et al. 2002).

Physical training (aerobic and/or resistance training) in previously untrained people increases energy requirements because it helps maintain and reduce the rate of loss of metabolically active muscle that occurs with ageing (Fiatarone Singh 2002; Lucas & Heiss 2005). Older masters cyclists (>55 yrs) have, in some instances, reported higher training volumes than younger masters cyclists (35–44 yrs) (Peiffer et al .2008). As such, energy requirements should be based upon reported activity levels, rather than average population estimates.

Masters athletes undertaking regular physical training have reported higher energy intakes than sedentary controls (Butterworth et al. 1993; Hallfrisch et al. 1994; Beshgetoor & Nichols 2003), especially when matched for energy intake per kilogram of body mass (BM) (Van Pelt et al. 2001). Reported energy intakes of masters athletes (aged 55–75 yrs) were between 10 300 kJ/d (Hallfrisch et al. 1994) and 11 500 kJ/d (Chatard et al. 1998), which are still lower than those reported in younger athletes (Van Erp-Baart et al. 1989; Burke et al. 1991; Hawley et al. 1995). Despite the potential for under-reporting of energy intake in dietary surveys, some masters athletes may be at risk of suboptimal energy intakes or poor recovery practices similar to younger athletes, despite stable weight.

18.5 Macronutrients

In the general population, total macronutrient intakes—carbohydrate (CHO), fat and protein—tend to decrease with ageing (Wakimoto & Block 2001). Dietary surveys of masters athletes (Van Pelt et al. 2001; Beshgetoor & Nichols 2003; Sallinen et al. 2008) and middle-aged highly aerobically fit athletes (Brodney et al. 2001) have found macronutrient intakes close to the population goals. Although using relative percentages of energy from macronutrients to the total energy of the diet is no longer recommended for use in athletes (see Chapter 2), these population targets or Acceptable Macronutrient Distribution Ranges (AMDR) are used as a basis for evaluating dietary balance in epidemiological surveys. In one epidemiological study of over 10 000 men and women (aged 20–87 yrs) divided into in low, moderately and highly aerobically fit cohorts, Brodney and colleagues (2001) reported significantly higher percent CHO and lower percent fat intakes in both men and women in the highest fit group, compared with the low and moderately fit groups. However, in this study, the relative contribution from CHO was still below the CHO target recommended for athletes. More recently, Sallinen and colleagues (2008) reported relatively high fat intakes (36% or total energy) and low CHO intakes (43% of total energy) in male Finnish national level masters strength and power athletes (mean age 52 ± 5 yrs) but not in the older athletes (72 ± 4 yrs). These studies suggest that such imbalances may not be conducive to enhancing sports performance and not optimal for promoting better health outcomes.

18.5.1 Carbohydrate

The rationale for recommending high-CHO diets for training and recovery is described in detail in Chapters 12 and 14. In ageing endurance runners, glycogen storage per unit of muscle is lower than in similarly trained younger runners, while glycogen utilisation per unit of energy expenditure is higher during submaximal exercise (Meredith et al. 1989). However, following regular endurance training, older individuals are able to increase muscle glycogen storage and recover glycogen stores at similar rates as younger athletes (Tarnopolsky et al. 1997). Healthy, previously sedentary older individuals who undertake aerobic exercise training can improve glucose tolerance and the rate of insulin-mediated glucose disposal, and increase skeletal muscle glucose transporter (GLUT-4) activity (Hughes et al. 1993; Cox et al. 1999), similar to younger athletes.

Recommendations for the amount of CHO needed for training and recovery in masters athletes are similar to those recommended for younger adult athletes (see Chapter 14), since CHO absorption and utilisation is unaffected by ageing, in the absence of any disease (Saltzman & Russell 1998; Elahi & Muller 2000). Diets rich in CHO, preferably from a mixture of low to moderate glycaemic index foods, are recommended. Diets high in fibre have also been associated with less weight gain in population-based studies (Fogelholm et al. 2012). However, for some susceptible older individuals, very high fibre intakes often exacerbate abdominal discomfort and flatulence. Nevertheless, the health and performance benefits of consuming high CHO intakes for training and competition usually outweigh any discomfort, which can be treated on a case-by-case basis by a sports dietitian.

18.5.2 Fat

Population dietary guidelines that recommend a reduction of total and saturated fatty acid intakes are also appropriate for the masters athlete. Ageing people retain the ability to digest, absorb and utilise fat (Saltzman & Russell 1998; Toth & Tchernof 2000). Some older athletes report consuming relatively high-fat diets compared with population targets and controls (see Table 18.2). Relatively low fat intakes are still important for masters athletes, particularly those involved in endurance training, so that more energy can be derived from CHO and protein for the same energy budget. Very low daily total fat intakes (i.e. less than 20% of energy from fat) may compromise the intake of fat-soluble vitamins (A, D, E and K) and decrease satiety between meals.

18.5.3 Protein

According to Australian nutrient references, estimated protein requirements for older men (>70 yrs) are slightly higher than younger men (19–70 yrs) as indicated by the EAR of 0.86 g protein/kg BM, compared to 0.68 g protein/kg BM. Females have slightly lower cut-offs than this (Commonwealth Department of Health & Ageing et al. 2006).

In adult athletes, protein turnover and hence requirements are higher than these population reference cut-offs (see Chapter 4). Although there is minimal consensus, masters athletes (mainly endurance and power athletes) may have slightly lower daily protein requirements than younger adult athletes because of:

- a decline in muscle mass with ageing (Reaburn 1994; Proctor & Joyner 1997)
- a decrease in whole-body protein turnover and decreased protein synthesis post-exercise and feeding (Rennie et al. 2010).
- a decrease in use of protein as a fuel because of reduced training intensity and/or volume reported (Trappe et al. 1995; Weir et al. 2002)
- possible age-related reduction in the absorptive capacity of the gut for amino acids and peptides at low doses (Saltzman & Russell 1998).

Based on a comprehensive review of daily protein needs in older athletes, master athletes undertaking resistance training maintain nitrogen balance with protein intakes between 1.0–1.3 g/kg BM/d (Lucas & Heiss 2005), by consuming around 50% of their protein from foods with high biological value (i.e. meat, fish, egg, milk) (Tarnopolsky 2008). These values are well above the EARs for older males but lower than the protein recommendations for younger adult athletes engaged in similar training programs (see Chapter 4).

Immediately following training or competition (within 30 minutes), the early provision of CHO and protein maximises glycogen and muscular recovery as well as strength development. Nutritional recommendations for post-exercise for masters athletes are based upon guidelines for younger athletes (Tarnopolsky 2008) (see Chapter 12). Although the daily protein recommendation/d is similar between young adult and masters athletes, the timing and amount of this protein ingested post-exercise for older individuals is probably more important. Recent evidence suggests that older untrained men exhibit resistance to anabolic stimuli—termed anabolic resistance—in response to food intake after exercise, compared with younger athletic controls (Dickinson et al. 2013; Burd et al. 2013). This resistance results in an impairment of muscle protein synthesis, which may be attributed to deficits in anabolic cellular signalling (Guillet et al. 2004; Cuthbertson et al. 2005), decreased protein absorption and decreased metabolic efficiency (Durham et al. 2010). Sarcopaenia—the degeneration of muscle mass, function and strength that occurs with ageing—has been attributed to many factors including this diminished muscle protein response to food intake (Pennings et al. 2011).

Feeding food containing CHO with or without protein in younger adult athletes stimulates substantial increases in muscle protein synthesis during the acute post-exercise recovery phase (see Chapters 13 and 14) as well as during continuous endurance-type exercise (Beelen et al. 2011). Several dietary interventions, based on the amount and type of protein and amino acid consumed and the absorption characteristics of the protein source, have been investigated in older untrained men to measure the specific effect on anabolic resistance (Pennings et al. 2011, 2012; Yang et al. 2012). Yang and colleagues (2012) demonstrated that 40 g whey protein resulted in an enhanced muscle protein synthesis compared with 10–20 g whey protein in 37 elderly untrained men (71 ± 4 yrs) undertaking resistance exercise. Furthermore, whey protein showed a better response than casein or casein hydrolysates using similar experimental protocols, which is probably linked to its faster rate of absorption (Penning et al. 2011). These results have implications for older men and women involved in masters sports. These studies and others (Drummond et al. 2008) suggest that, in contrast to younger adults, in whom post-exercise rates of muscle protein synthesis are saturated with 20 g of protein, exercised muscles of older adults need slightly higher protein doses from high-quality, rapidly absorbed protein (and carbohydrate) to address anabolic resistance. Hence a whey protein supplement (together with CHO) may be the best choice post-exercise for rapid recovery during competitions with repetitive events (or training periods of high intensity and duration) for masters athletes.

Dietary surveys of masters athletes indicate that daily protein recommendations are easily met when adequate energy intake and a variety of high-quality protein sources (dairy, meats, eggs, fish) are consumed (Lemon 2000; Van Pelt et al. 2001; Tarnopolsky 2008). Protein intakes of 1.25–1.45 g/kg BM/d reported in dietary surveys of masters athletes from a variety of training backgrounds confirm adequate intakes within current recommendations (see Table 18.2). However, older athletes in heavy training, particularly those involved with strength and power sports, may require slightly higher protein intakes than these reported intakes because of muscle damage (Lemon 2000; Tarnopolsky 2008).

Studies on the effects of very high protein daily diets as a means of stimulating protein synthesis and preventing losses of FFM that occurs with ageing have been equivocal (Volpi et al. 1998; Welle & Thornton 1998). The consensus is that very high protein diets (above nutrient reference standards) do not provide a significant increase in protein synthesis or conserve muscle mass (Starling et al. 1999; Campbell et al. 2001). Moreover, without sufficient physical activity, increasing protein consumption may increase energy and fat intake beyond energy expenditure and is associated with an increase in body fat (Vinknes et al. 2011).

Potential adverse effects of very high protein intakes include impaired kidney function, increased calcium loss, atherogenic effects and increased urinary calcium excretion. These outcomes may be a risk for people with a genetic predisposition. Protein supplements and high salt diets increase excretion of urinary calcium, which can also increase risk of kidney stones in susceptible people. For most people, there is no convincing evidence that very high protein diets and the additional nitrogen and possibly calcium excretion that accompanies a high protein diet cannot be handled by a normally functioning kidney (Lemon 2000). Higher intakes of protein of around 1.8 g/kg BM/d may have adverse effects in healthy non-exercising older adults (Walrand et al. 2008).

In summary, the beneficial effects of high protein intakes appear to outweigh the possible adverse effects. Protein intakes in older persons of 1.0–1.3 g/kg BM/d, in combination with adequate calcium intake, could be beneficial for bone health (Lucas & Heiss 2005). Lemon (2000) has confirmed that protein intake of 1.2–1.7 g/kg BM/d will not adversely affect cardiovascular health.

18.6 Micronutrients

The requirements for and effects of micronutrients (i.e. vitamins, trace minerals and anti-oxidants) on performance in adult athletes are considered in Chapter 11. Because there is limited research on micronutrient (vitamins, minerals and antioxidants) requirements for masters or older athletes, nutrient recommendations are largely based on extrapolated data from studies of ageing sedentary population. For reviews of micronutrients and ageing, see Park et al. (2008) and Marian and Sacks (2009).

Irrespective of age, athletes have slightly increased requirements for several vitamins and minerals compared to non-athletes, especially endurance athletes (see Chapter 11). However, depending on their age and intensity or duration of their training programs, masters athletes may be at higher risk of micronutrient deficiencies than younger athletes for several reasons:

- an age-related decrease in nutrient absorption reported for several micronutrients (vitamins B6, B12, D and calcium) (Institute of Medicine 2001, 2010; Saltzman & Russell 1998)
- the wide variation in nutrient requirements between individuals
- the potential use of medications that interfere with nutrient absorption or utilisation
- the presence of chronic disease.

From our experience working with older athletes, the middle-aged and elderly athlete is likely to be nutritionally aware and has already modified food intake to address any existing health issues. The determinants of food choice change with ageing and changing lifestyle. However, awareness does not necessarily translate into behavioural change and often some foods or food groups are restricted and micro-nutrient intake is suboptimal. Suboptimal intakes—less than the Recommended Daily Allowance (RDA)—of calcium, iron, zinc, magnesium and vitamins D and E have been reported in older endurance athletes, despite adequate energy intakes (see Table 18.2). Intakes below the RDA but greater than the EAR suggest a possibility of inadequate intake but do not confirm a deficiency. Most studies used the RDA as a benchmark for adequacy, which is no longer appropriate. Nevertheless, these nutrients may still be of concern.

Usually when energy intakes are met, micronutrient intakes are satisfactory. With prolonged low energy intakes or suboptimal nutrient consumption, those micronutrients with low storage pools in the body, particularly those involved in energy metabolism—for example, thiamin (vitamin B1) and iron—diminish rapidly with high-intensity exercise resulting in a depleted or deficiency state that can negatively affect performance (see Chapter 11). Furthermore, in any individual athlete, risks of micronutrient deficiencies may be compounded by chronic medical conditions, use of medications or changes in gut function that can impair nutrient absorption (McCabe 2004).

Section 18.6 considers the role and potential changes in requirements of key micronutrients associated with ageing, the influence of exercise on nutrient requirements relevant for an older athlete and suggested recommendations for at risk groups.

18.6.1 Vitamins

Vitamin A

Vitamin A (retinol) is derived from two sources (retinol and carotenoids) and involved in tissue growth, vision, blood cellular components, bone production, antioxidant defence and immune function. In a healthy ageing population, the clearance of vitamin A is decreased by about 50% compared to younger adults. The RDA cut-offs (700 µg/d for females and 900 µg/d for males) for vitamin A (or retinol equivalents) for >51 years are well in excess of requirements and will easily meet the needs of masters athletes. However, masters athletes may experiment with supplements of vitamin A (or other antioxidants) because of their potential role in reducing oxidative stress induced by exercise, and its link with reducing risk of chronic degenerative disease including atherosclerosis, diabetes and some cancers (Herrera 2010; McCormick 2010). Misuse of vitamin A supplements, particularly pre-formed vitamin A —the fat-soluble form—is associated with vitamin A toxicity (Fortes et al. 1998). Increasing vitamin A-rich foods from dietary sources (liver, fish, eggs, carrots, leafy greens, tomatoes) is more appropriate than from supplements and avoids the potential pro-oxidative effects associated with taking a supplement that can potentially damage cells and tissues (Braakhuis et al. 2012; see Chapter 11).

Vitamin B2

Vitamin B2 (riboflavin) assists other micronutrients in oxidative metabolism and the electron transport system within mitochondria to maintain skin health. Masters athletes who are following low-energy diets or have a sudden increase in training volume or intensity may have higher riboflavin requirements than suggested by population reference standards (Manore 2000). However, the few dietary surveys undertaken suggest adequate riboflavin intakes using RDA (not EAR) cut-offs (Beshgetoor & Nichols 2003), with one exception. In one study of 14 women (50–67 yrs) undertaking an exercise program, riboflavin intakes met the RDA, although urinary riboflavin excretion was significantly lower than controls, which may be indicative of riboflavin depletion (Winters et al. 1992). Low riboflavin intakes have been reported in older people who limit dairy products or are vegans who eat none of the major sources of riboflavin such as meat, fish and poultry (Russell & Suter 1993). Thus, masters athletes who also avoid meat and limit dairy foods may be at risk of suboptimal riboflavin intakes, although deficiency is unlikely and would usually occur alongside other deficiencies associated with gross malnutrition.

Recommended requirements (i.e. the EAR) for riboflavin from population references for people aged 31–70 years are the same as for younger adults, although these values are based on the lower overall energy intakes observed in normal ageing populations. Thus, in masters athletes with a relatively high energy intake, these values may not apply.

Vitamin B6

Vitamin B6 (pyridoxine and related compounds) acts as a coenzyme in energy metabolism including amino acid metabolism, glyocogenesis, glycogenolysis as well as red blood cell formation. Requirements for vitamin B6 increase as the intake of protein increases, which corresponds to its amino acid metabolism. The EAR for vitamin B6 is slightly higher for older people (>51 yrs) than younger adults (19–50 yrs). However, dietary surveys of the general population indicate that vitamin B6 intakes in many elderly people do not always meet the RDA cut-offs, which suggests a possibility of inadequacy for some people (Russell & Suter 1993), although clinical deficiency is rare.

Older athletes, similar to younger athletes, lose vitamin B6 through urinary 4-PA excretion (Manore 2000), thus suggesting increased turnover. Furthermore, serum concentrations of vitamin B6 decrease with ageing and high intakes may be needed to support immune system functioning, especially in masters athletes undertaking high-intensity training (Meydani et al. 1990, 1991). The main dietary suppliers in the western style diet include fortified ready-to-eat breakfast cereals and whole grains, potato and other starchy vegetables, meat, fish or poultry and non-citrus fruits.

Vitamin B12

Vitamin B12 (cobalamin) is essential for normal blood function and neurological function and involved in synthesis of fatty acids and DNA synthesis. Dietary surveys of masters athletes consuming a mixed diet have reported that vitamin B12 intakes easily met or exceeded RDA cut-offs, which were routinely used for assessing nutrient adequacy in these older studies (Chatard et al. 1998; Beshgetoor & Nichols 2003). Moreover, vegans are at risk of vitamin B12 deficiency so ageing vegan athletes, or athletes who only occasionally eat meat or dairy products, are likely to be at even higher risk from possibly years of suboptimal intake. Thus, for masters athletes who are vegans, vitamin B12 supplementation and use of fortified foods is recommended.

Recommended requirements (i.e. EAR) for vitamin B12 also increase with ageing in both males and females. This increase is linked to the age-related atrophic gastritis seen in approximately 30% of those over 60 years (Pfeiffer et al. 2007). Gastritis decreases gastric and intrinsic factor secretion needed for B12 absorption. The incidence of pernicious anaemia as a result of malabsorption and deficiency of B12 increases in ageing individuals. Estimated requirements for vitamin B12 for masters athletes who are vegetarians or have atrophic gastritis may be slightly higher than non-athletes (Sachek & Roubenoff 1999; Pawlak et al. 2013). From an athlete perspective, pernicious anaemia is a risk for older vegan athletes and if present, substantially impairs capacity for training and endurance performance, similar to every type of anaemia (American College of Sports Medicine et al. 2009a).

Vitamin C

Vitamin C (ascorbic acid) is involved with collagen and connective tissue formation, enhances iron absorption, improves immune function, reduces cellular damage, is an antioxidant and is involved with steroid and catecholamine synthesis. There is no evidence that vitamin C absorption or utilisation is impaired with ageing (Chernoff 2014). Dietary surveys of an ageing sedentary population and masters athletes have indicated that vitamin C intakes easily meet or exceed population reference intakes (Butterworth et al. 1993; Hallfrisch et al. 1994; Reaburn & Le Bon 1995; Chatard et al. 1998; Beshgetoor & Nichols 2003).

For those athletes engaged in strenuous and prolonged exercise, the American College of Sports Medicine (2009a) suggests an intake of 10–1000 mg/d, which is much higher than EAR cut-offs of 60–75 mg/d. Many athletes take vitamin C supplements either intermittently or regularly (see Chapter 11). However, for susceptible people, high doses of vitamin C supplements (>1000 mg/d) can increase the risk of kidney stones and gout and potentially impair copper absorption (Alhadeff et al. 1984). These doses have also been associated with 'runner's diarrhoea' (Hoyt 1980).

A recent review of the effect of vitamin C supplements on physical performance found that of the 12 studies evaluated, four found impaired performance and four found no improvements in athletes supplementing with >1000 mg/d (Braakhuis 2012). This review concluded that five or more servings of fruit and vegetables/d such as oranges, strawberries, leafy greens, broccoli, kale and sweet potatoes can deliver around 200 mg/d of vitamin C and is a better option than supplements. This amount meets vitamin C requirements for most people.

Vitamin D

Vitamin D has multiple roles and is involved in enhancing calcium absorption and reabsorption (and bone formation) as well as the regulation of cellular growth, immune function, inflammation and protein synthesis. Apart from obtaining small amounts of dietary sources of vitamin D (from fortified cereals/dairy, margarines and eggs), most vitamin D is manufactured in the liver and, to a lesser extent, the kidney, by the action of ultraviolet light on a vitamin D precursor in the skin. Ultimately, the skin supplies 80–90% of the body's vitamin D needs. Ageing reduces the capacity of the skin to synthesise vitamin D precursors and decreases renal production of the activated form (vitamin D3) (Aloia et al. 2011) so masters athletes training at low latitudes or indoors are at high risk of low vitamin D status.

Musculo-skeletal pain and impaired muscle strength and weakness are known but often overlooked symptoms of vitamin D deficiency (Larson-Meyer 2010). These symptoms, similar to fatigue, are not uncommon in an older athlete for other reasons but can mask vitamin D deficiency. Vitamin D supplementation in older vitamin D-deficient females over 70 years has been shown to lead to significant improvements in both muscle strength and walking distance (Verhaar et al. 2000). However, performance enhancement in the absence of deficiency is unlikely. While the evidence on performance enhancement in younger athletes is equivocal, the evidence between low vitamin D status and stress fractures and muscle injury is supported (see Chapter 9).

The evidence suggesting that vitamin D may provide protection against bowel cancer, cardiovascular disease, type 2 diabetes and infectious diseases has recently been challenged in a systematic review of nearly 500 studies (Autier et al. 2014). The authors found the benefits of vitamin D were unconfirmed in 172 randomised controlled trials and concluded that low vitamin D status is likely to be a consequence of the inflammatory effects involved in the occurrence and progression of disease, which might explain why low vitamin D status is reported in people with a wide range of disorders.

Nevertheless, ageing populations report consuming less vitamin D than younger people and have a high risk of low vitamin D status (Muir & Montero-Odasso 2011). In winter, dietary sources of vitamin D become more important, when exposure to sunlight is insufficient to produce adequate vitamin D. The limited number of studies examining dietary intakes in masters athletes (see Table 18.2) suggest they have adequate intakes compared with population benchmarks but may still have suboptimal vitamin D status, particularly in the older age groups or in those with a longstanding chronic inflammatory illness.

Slightly higher vitamin D recommendations for both males and females over 70 years are reflected in population nutrient reference values (Institute of Medicine 2010) compared with younger adults. Until further research is available in masters athletes, these can be used for athletes in all age groups to assess the probability of inadequacy of intake.

In summary, the high prevalence of suboptimal vitamin D status in older populations suggests that older masters athletes, likely to be at risk of low vitamin D status, have a blood test to check vitamin D status. For reviews of vitamin D and sports performance, see Powers et al. (2011) and Moran et al. (2013).

Vitamin E

Vitamin E is another antioxidant like vitamins C and A. Antioxidants have a protective role against the damaging effects of free oxygen radicals or reactive oxygen species (ROS) induced by prolonged or eccentric exercise such as running, cycling and weight training (see Chapter 11).

Whether vitamin E supplements are worthwhile for antioxidant protection and/or its other effects or physiological functions in masters athletes is questionable. Reviews of nutrient requirements for elderly exercisers (Sachek & Roubenoff 1999) and normal older individuals (Marian & Sacks 2009) have suggested that those people with cardiovascular disease undertaking endurance training may consider vitamin E supplements of 100–200 mg tocopherol equivalents/d. Older studies have reported benefits from recommended doses of vitamin E supplementation for improving immune function (Meydani et al. 1990a) and reducing incidence of cataracts (Robertson et al. 1989) and cardiovascular disease (Rimm et al. 1993; Marian & Sacks 2009) for elderly people.

In contrast to these protective effects of early studies using vitamin E supplements, several meta-analyses have revealed unfavourable health outcomes (Vivekananthan et al. 2003; Miller et al. 2005). Miller and colleagues (2005) reported an increase in all-cause mortality when vitamin E was supplemented at high doses (>400 IU/d for at least 1 yr). In dose−response analyses, all-cause mortality progressively increased when vitamin E dosage exceeded 150 IU/d, but at dosages less than this mortality was decreased (Miller et al. 2005), so there appears to be a threshold level.

In another well-designed controlled trial in 40 healthy young men with no existing disease, a significant decrease in glucose infusion rates into muscle cells and a decrease in insulin sensitivity with high dosage of both vitamin C and E supplements was reported (Ristow et al. 2009). In this study, a combination of vitamin C and vitamin E supplements (1000 mg/d and 400 IU/d, respectively) taken over four weeks actually blocked the beneficial effects of exercise on enhancing insulin sensitivity in both pre-trained and untrained controls. It is well known that physical exercise, even short-term exercise, increases ROS and insulin sensitivity and can potentially reduce the progression of metabolic syndrome into type 2 diabetes (Pan et al. 1997). However, when these high-dose supplements were taken concurrently in this study, the formation of ROS was inhibited and hence the capacity of ROS to counteract insulin resistance was suppressed. According to the authors, the outcome of supplement use with exercise may actually increase the risk of type 2 diabetes rather than decrease it, which has implications for an older population with existing risk factors. There is also some evidence that single antioxidant use in people with type 2 diabetes has been linked to increased prevalence of hypertension (Ward et al. 2007) and overall mortality (Bjelakovic et al. 2007). Based on this and other evidence (Ristow et al. 2009; Lavie & Milani 2011), use of antioxidants as supplements in people with risk factors for type 2 diabetes or with diagnosed diabetes may counteract the beneficial effects of exercise on increasing insulin sensitivity (see Chapter 11 and Commentary 6). Further research on at risk and older age groups is required to test the long-term consequences of this effect.

In summary, the effects of short-term, high-dose vitamin E and other antioxidant supplements on health outcomes or improving performance have not supported their widespread use (see Chapter 11). Habitual supplementation with high-dose vitamin E and other antioxidants in athletes or in any population may therefore be unnecessary except in older people with malabsorptive conditions such as Crohn disease.

Folate

Like other B vitamins, folate acts as a coenzyme in nucleic acid and amino acid metabolism and supports the formation of red and white blood cells. In masters athletes with gastric atrophy, which is common in elderly persons (Hershko et al. 2007), the associated decrease in stomach acid production may lead to decreased folate absorption. Otherwise, both healthy masters athletes and the normal population are likely to meet the RDA. Sachek and Roubenoff

(1999) suggested that masters athletes should consume at least 200 μg/d of folate, as long as vitamin B12 intakes were adequate, since high levels of folate can mask vitamin B12 deficiency. Major sources of folic acid in the diet include liver, meat, milk, green leafy vegetables, fortified cereals, mushrooms, asparagus, peas and beetroot.

Folate supplementation may be linked to increased risk of heart disease and cancer (see Marian & Sacks, 2009 for review) although the evidence is conflicting. Importantly, some medications for arthritis (NSAID), for diabetes (metformin) and for lowering cholesterol are associated with anti-folate activity, suggesting that folate deficiency may be an issue for masters athletes on these medications.

18.6.2 Minerals

Calcium

Calcium is needed for bone health, nerve conduction, muscle excitation and contraction; it is a cofactor in glycogenolysis and works with vitamin K in blood coagulation and wound healing.

Although weight-bearing exercise promotes bone density (Marques et al. 2011, 2013), masters athletes who already have low bone density, oestrogen insufficiency and possibly longstanding suboptimal calcium intakes are likely to be at high risk of stress fractures when undertaking repetitive impact activities (McDonnell et al. 2011; Wentz et al. 2012). Repetitive impact activities such as running—particularly in hot, humid conditions—are also associated with substantial calcium loss from sweat. Thus, adequate calcium intakes are important in masters athletes, particularly peri-menopausal and postmenopausal women, to help maintain bone health (see Chapter 9).

Calcium absorption decreases with ageing in both genders, largely because of decreased vitamin D absorption and production (Aloia et al. 2011). Calcium balance deteriorates in women at menopause because of the decline in calcium absorption and increased calcium excretion. Calcium bioavailability is further affected in people with atrophic gastritis, a common problem in older persons resulting in a decrease in gastric acid production (Hershko et al. 2007).

The EAR/RDA reflect the increased requirement for calcium in postmenopausal women and the degenerative bone loss and reduced absorptive capacity that accompanies ageing, irrespective of gender. In masters athletes, dietary intake studies suggest that many athletes are either not meeting the RDA or are eating less calcium compared with controls (see Table 18.2).

Calcium supplements may be needed in those athletes who have difficulty meeting high calcium requirements, including those who are lactose intolerant, dislike milk and dairy products, or are allergic to milk. The American College of Sports Medicine and colleagues' (2009a) guidelines for younger athletes recommend that supplementation with both calcium and vitamin D is determined after nutrition assessment. They further suggest that athletes with disordered eating or amenorrhoea, who are at risk of osteopaenia, take a combined supplement of calcium and vitamin D at dosages of 1500 mg/d of elemental calcium and 400–800 IU/d of vitamin D.

Supplements of calcium citrate or malate are better absorbed than calcium carbonate, at least in studies of postmenopausal women with low dietary intakes of calcium, and appear more effective than calcium carbonate in reducing bone demineralisation and lowering bone fracture rates in this age group (see Chapter 9). Furthermore, calcium citrate is the preferred form used in elderly people with intestinal absorption problems (Straub 2007). Indeed, Lewis and Modlesky (1998) have recommended that additional calcium supplements (~1000 mg/d) for premenopausal and late postmenopausal women who are already consuming a relatively high calcium intake (i.e. 70–1000 mg/d) can reduce bone loss.

Iron

In athletes of any age, particularly those involved with endurance exercise, iron is an integral component of the oxygen-carrying capacity of haemoglobin in the blood and myoglobin within muscle. Moreover, iron is also found within the mitochondrial cytochrome complex and thus is involved in aerobic metabolism.

Iron deficiency anaemia substantially reduces performance capacity and maximal aerobic power in athletes and in animal studies (see Chapter 10). Even a mild shortfall in tissue iron status (i.e. stage 2 iron depletion) appears to reduce maximum oxygen uptake, aerobic efficiency and endurance capacity, at least in young adult athletes (see Chapter 10). Iron absorption from non-haem food sources (mainly plants) is also affected by atrophic gastritis (Garry et al. 2000), which impairs vitamin B12, calcium and zinc absorption. Also, the presence of inhibitors naturally occurring in foods, especially tea, when consumed with iron-rich meals, binds non-haem iron, and further reduces its bioavailability (see Chapter 10).

Iron stores usually increase with ageing in both males and females until they reach a steady state level (Garry et al. 2000; Liu et al. 2003). There is no evidence to suggest that iron absorption reduces with ageing (O’Connor et al. 2011). Healthy older people, therefore, require less dietary iron than younger people, which is reflected in population reference values. Dietary iron deficiency is rare in healthy older populations of men (but not women) and may co-exist with the anaemia of chronic illness or inflammation in both men and women (Pieracci & Barie 2006).

Nevertheless, our unpublished observations found that the dietary iron intakes in many ageing Australian male and female endurance athletes were below Recommended Dietary Intakes (RDI) (Reaburn & Le Bon 1995), although a study of 25 middle-aged female endurance cyclists and runners in the USA showed iron intakes met or exceeded the RDA (Beshgetoor & Nichols 2003). Decreased dietary iron intake with ageing appears a consistent finding in different cultures, particularly in women (Wakimoto & Block 2001; Martins et al. 2002; Marriott & Buttress 2003) so iron depletion may still be a problem in older female athletes.

The population reference values for iron are 1.3–1.7 times higher for athletes, and 1.8 times higher for vegetarians (non-athletes) to account for the low bioavailability of iron from vegetarian diets (Institute of Medicine 2001). Hence, iron depletion may still be of concern in both male and female masters athletes, particularly those involved in endurance training programs.

Zinc

Dietary zinc is involved in tissue repair, antioxidant and immune function, which is particularly applicable to the ageing athlete, who is susceptible to injury. Zinc is also a cofactor in the synthesis and catabolism of CHO, fats, proteins and nucleic acids. Most zinc losses occur through urine, faeces and sweating so requirements may be high in an athlete of any age training in hot, humid climates. Interestingly, the turnover of zinc is increased in anaerobic exercise, which is likely to be linked to its metabolic role in glycolysis (Lukaski 1995). These functions might suggest a high zinc requirement, particularly for masters athletes undertaking high-intensity interval training. However, there is no evidence to suggest that zinc balance is different between younger and older adult populations on a similar type of diet. While zinc absorption decreases with ageing, its excretion diminishes, so zinc balance appears better maintained in older people compared with younger adults (Turnland et al. 1986).

Many older people including older athletes may not be eating enough zinc and are at risk of zinc deficiency if they limit meat intake. Suboptimal or borderline intakes of zinc have been reported in masters athletes (Table 18.2). Similar to non-haem iron absorption, zinc absorption is reduced by the presence of naturally occurring inhibitors in food such as phytates (found in whole grain cereals, soy, peanut) and also by calcium and iron supplements taken with a meal. The EAR for zinc for vegetarians can be as much as 50% higher than for meat eaters, particularly strict vegetarians whose main staples are grains and legumes, because of the inhibitory effects of phytates on zinc absorption (Institute of Medicine 2001).

In summary, masters athletes may be at risk of zinc deficiency if they sweat profusely, consume high-CHO or vegetarian-style diets with a high phytate content or routinely take calcium or iron supplements with meals, which inhibit zinc absorption.

18.7 Water

Modest (<2% BM) weight loss over a short time has been used as a benchmark in athletes to define the point at which athletic performance is impaired (Roberts 2012). A recent meta-analysis suggests that exercise-induced dehydration of up to 4% weight loss does not hinder cycling time in younger adult cyclists in short-term time trials (Goulet 2011). However, this amount of weight loss may be too high for older athletes who are at higher risk of hypohydration and heat stress for several reasons:

- a decrease in total body water with ageing and subsequent decrease in plasma volume thereby increasing the risk of heat stress (Popkin et al. 2010)
- reduced efficiency of renal anti-diuretic hormone receptors, which increases urinary excretion rate leading to increased loss of fluid (Rolls & Phillips 1990), thereby

increasing the risk of hypohydration

- reduced thirst sensation, caused by a decrease in osmoreceptors that are sensitive to blood concentrations of fluid-regulating hormones and electrolytes (Hansen et al. 2011)—hence the potential for a decrease in fluid intake
- decreased ability to regulate temperature homeostasis associated with a diminished contractile function of the heart; the resultant lowering of cardiac output diminishes blood flow to the skin thus reducing the removal of heat and increasing the risk of heat stress (Inoue et al. 1998)
- decrease in sweat production with ageing together with a delay in the onset of the sweat response (Dufour & Candas 2007; Viveiros et al. 2012), which decreases the cooling effect of vaporisation of sweat from skin.

Buono (1991), however, reported that lifelong aerobic exercise retarded the decrease in peripheral sweat production normally associated with ageing. Similarly, Viveiros and colleagues (2012) found that well-trained middle-aged runners (54 ± 2 yrs) compensated for a decreased sweat rate by an increased number of heat-activated sweat glands.

Other physiological factors exacerbate risk of hypohydration and heat stress in masters athletes, because thirst sensors are unreliable. In one study, older people (56 ± 3 yrs) undertaking strenuous hill-walking of between 10 and 35 km/d over 10 consecutive days became progressively hypohydrated compared with younger walkers (24 ± 3 yrs). This was evident from a two-fold increase in urine osmolarity, which is the method used to measure fluid balance (Ainslie et al. 2002). The reasons for inadequate fluid intake were not assessed in this study so linking this outcome to reduced thirst was not confirmed. In another study, reduced thirst was confirmed in older 'average fit' males (~60 yrs) who were progressively hypohydrated and rated themselves less thirsty than younger controls (~20 yrs) (Meischer & Fortney 1989).

Guidelines recommended for fluid intake before, during and after competition for younger adult athletes can also be applied to masters athletes (see Commentary 7) but perhaps need to be applied more aggressively than in younger athletes in those undertaking prolonged physical activity in the heat. For older athletes, a pre-exercise hydration plan is recommended to help minimise the risk of dehydration, especially for those training or competing in the heat (Goulet 2012). This plan involves prior measurement of weight loss (or sweat loss) during training and ensuring a matching fluid volume is replenished regardless of thirst. Older athletes are also recommended to drink small volumes regularly when training and competing to replenish losses (Ferry 2005).

Similar to younger athletes including children, older athletes consume more fluid when offered sports drinks compared with water. This response was confirmed during a contrived cycling protocol under laboratory conditions where 27 older male and female recreational exercisers (aged 54–70 yrs) consumed more fluid and restored plasma volume faster when offered a sports drink than those offered only water (Baker et al. 2005). In situations where hypohydration and heat stress are a risk, beverages containing 6–8% CHO (e.g. sports drinks) are the best choice during prolonged exercise (>1 hour), especially for older athletes (Sawka et al. 2007).

18.8 Medications: nutrient interactions

Between 14% and 25% of participants in masters sport reported a pre-existing medical condition (primarily hypertension, asthma and coronary heart disease), with up to 34% taking medication for cardiovascular, respiratory and inflammatory conditions (Farquharson 1990; Reaburn et al. 1995). For these athletes, there may be interference with nutrient absorption or drug-nutrient interactions that influence nutrient requirements. Some prescription and over-the-counter medications interfere with nutrient absorption, metabolism and excretion and mask or alter the usual physiological response to exercise. For example, beta-blockers prescribed for hypertension blunt the elevation in heart rate that occurs with exercise. Conversely, certain foods or nutrients in food interfere with medication effect. For review of drug–nutrient interactions, see Bellows & Moore (2013) and Hermann (2013).

Drug–nutrient interactions and adverse effects increase exponentially with ageing and with multiple medication use or polypharmacy (Salazar et al. 2007). Given the complexity of drug–nutrient interactions, it is important that a sports dietitian works closely with a sports physician to minimise any interference.

18.9 Supplements

The use of dietary supplements has increased substantially in most industrialised countries and in older people, despite no or a weak association with all-cause mortality, cancer and cardiovascular disease mortality (Sebastian et al. 2007; Messerer 2008). Interestingly, older people more likely to take supplements, particularly vitamins, were health conscious and physically active (Sebastian et al. 2007).

Dietary supplement use is widespread in athletes and the general population, particularly older adults (see Chapter 16). In one study, 75% of ultramarathon runners (mean age 44.5 ± 9.8 yrs) used vitamins and/or other supplements (Hoffman & Fogard 2012). In another study of 598 male and female masters athletes, close to 61% used supplements. Of these, 36% took vitamins and 30% took minerals (Streigel et al. 2006). The main reasons reported for supplement use were for injuries (25.5%), health reasons (19.9%), success in sport (18.3%), increased endurance and performance (17.3%) and increased strength (10.3%). In this study, supplement users trained more than non-users, which suggests that the main reasons for use were sports related. In an earlier study of 25 female endurance athletes (mean age 50.4 yrs), neither supplement users nor non-supplement users met the RDA for calcium and vitamin E, despite meeting energy recommendations (Beshgetoor & Nichols 2003). The authors concluded that at least for the subjects in this study, calcium supplements or a multivitamin/mineral supplements were indicated.

For ageing exercisers involved in regular physical training who are considering using micronutrient supplements, a daily multivitamin–mineral supplement that provides no more than 100% of the RDA is suggested, rather than single supplements, unless diagnosed with a deficiency (Sachek & Roubenoff 1999). Despite recent suggestions that multivitamin−mineral supplements, especially antioxidants, may reduce oxidative stress and risk of neurodegenerative vascular disease, cancer, diabetes and potential mortality, the current consensus is that such supplementation may not benefit mortality in older populations (Herrera et al. 2010; McCormick 2010; Macpherson et al. 2013). Moreover, they recommend a well-balanced diet rich in fruit and vegetables.

Antioxidant supplements have been widely suggested for use by ageing populations (Chernoff 2014; Park et al. 2008; Marian & Sacks 2009). In masters endurance cyclists and runners (66.1 ± 5.8 yrs, n = 8), the effect of 21 days of multivitamin and multimineral supplement on muscle activity and cycling efficiency during a high-intensity 10-minute cycling test was improved cycling efficiency and reduction in fatigue (Louis et al. 2010). The reasons for this might be attributed to the antioxidant content of the supplements, although the concentration of antioxidants in the supplements was fairly low. However, antioxidant supplements (i.e. vitamin E and C) at usual suggested doses for people with metabolic syndrome or type 2 diabetes may counteract the beneficial effects of exercise on enhancing insulin sensitivity (see Chapter 19).

In summary, and as recommended by the American College of Sports Medicine, American Dietetic Association and Dieticians of Canada Joint Statement (2009a), athletes of any age at greatest risk of poor micronutrient status are those who restrict energy intake or engage in severe weight loss practices, eliminate one or more food groups, or consume unbalanced or low micronutrient-dense diets. These athletes, regardless of age, may benefit from a daily multivitamin–mineral supplement.

Summary

There are no specific recommendations for nutrient intakes for masters athletes. The EAR is considered the best estimate of nutrient requirements for individuals and groups and the recommended targets for CHO and protein for younger athletes are applicable to healthy masters athletes, providing energy requirements are met. The EAR cut-offs account for the physiological changes to nutrient requirements that occur with ageing. However, some modifications may be needed to adjust for the presence of chronic disease and/or use of medications that potentially interfere with nutrient bioavailability in those masters athletes who participate in physical activity of high intensity and/or high volume.

The limited number of dietary intake studies on masters athletes suggests that intakes of several micronutrients (vitamins B6, B12, D and E, folate, calcium, iron and zinc) may be suboptimal or at least below RDA/RDI cut-offs in some groups. The RDA in isolation is no longer recommended as a cut-off to assess the ‘adequacy’ or ‘inadequacy’ of nutrient intakes and may over-estimate true requirements. Nevertheless, for those healthy older (and younger) master athletes who habitually consume low micronutrient intakes, improving intake is best achieved through dietary means rather than supplements. If an ageing athlete wants to use micronutrient supplements for any reason, a multivitamin–mineral supplement that provides no more than 100% of the RDA is likely to be both safe and adequate for optimal sports performance.

Dietary guidelines for older people are available in most western countries and provide a basis for advising masters athletes about food choice. With the increased participation in both recreational and competitive sport in the ageing population, more studies are needed on masters athletes at different levels of activity to further examine the interactive effects of diet, ageing, physical activity, medication use and presence of chronic disease before definitive nutrient recommendations can be made. Epidemiological studies on these interactions in masters athletes are required.

Practice tips

VICKI DEAKIN AND ALISON CAIAFA

INTRODUCTION

- Most masters athletes are involved in sport for fun, camaraderie, socialising, fitness and the pure pleasure of being active. Masters athletes usually have a competitive spirit and may seek specific nutrition advice to enhance their sports performance. Most are as healthy as younger athletes, just a little slower and less flexible. Nutrition advice for the older masters athletes needs to consider the physiological and diet-related risk factors associated with ageing and exercise (e.g. high risk of heat stress and hypohydration, changing nutrient requirements with age) as well as the influence of any medical condition or medications that affect training capacity or nutrient absorption. Otherwise, general sports nutrition advice for performance enhancement of master athletes is similar to younger adult athletes.

THIRST, FLUID INTAKE AND RISK OF DEHYDRATION

- Thirst sensation, sweat rates and hence thermoregulation decreases with ageing. Risks of heat stress and hypohydration are higher than in younger athletes. An older masters athlete who has been training or competing for many years may be out-of-date in their knowledge about current practice and have entrenched behaviours about food and fluid intakes for training or competition. For some athletes, fluid may be purposely restricted to minimise night-time urination (although this is probably not effective, as concentrated urine irritates the bladder and encourages more frequent urination). These beliefs increase the risk of chronic low fluid status at any time. Conversely, older athletes participating in endurance events may over-consume fluid and are often at higher risk of hyponatraemia than younger athletes if they take a long time to complete an event.
- There are no fluid guidelines specific for the ageing athlete. Assessment of daily fluid intake practices and training practices is useful to estimate fluid balance. Fluid intakes should be individualised because of varying fitness levels and environmental conditions encountered, similar to younger athletes (see Commentary 7). During competition, estimated fluid requirements need to be applied more aggressively than in younger athletes because of compromised thermoregulation in the older athlete.
- A daily training and competition fluid plan is recommended with an emphasis on increasing frequency of fluid intake, rather than large bolus intakes before, during or after exercise. For recovery after exercise, a low fat, calcium-fortified milk-based drink (e.g. milk shake) consumed soon after exercise or competition is an ideal option for improving fluid retention.

BONE HEALTH

- Although weight-bearing activity, and to a lesser extent resistance exercise, helps to minimise bone loss that occurs with ageing, an adequate calcium and vitamin D intake is still important because of increased requirements in older people. Vitamin D status is mainly determined by exposure to sunlight. Adults more than

65 years produce four times less vitamin D in the skin compared with adults aged 20–30 years (Institute of Medicine 2010). For moderately fair-skinned people, a walk with arms exposed for 6–7 minutes mid-morning or mid-afternoon in summer, and with as much bare skin exposed as feasible for 7–40 minutes (depending on latitude) at noon in winter, on most days, will help maintain adequate vitamin D levels in the body in a healthy adult (Nowson et al. 2012). Masters athletes who are dark skinned, train indoors or train at low latitudes are at risk. It is prudent to assess vitamin D status of masters athletes who either cover up in the sun or have limited sun exposure during the summer months.

- Good food sources of vitamin D include oily fish (e.g. tuna, sardines, mackerel), cod liver oil, liver, eggs, cheese and margarine. Several countries fortify foods with vitamin D (e.g. margarine in Australia and the UK, and milk in the USA), although vitamin D intake from dietary sources and fortified foods alone is likely to be insufficient to meet requirements. When sun exposure is minimal, there is more reliance on vitamin D intake from dietary sources; supplementation of at least 600 IU/d for people aged <70 yrs and 800 IU/d for people aged >70 yrs is recommended (Nowson et al. 2012).

SARCOPAENIA

- Loss of muscle strength, power and function are inevitable with ageing; the decline starts around 40 years, then accelerates after age 65–70 years, even in people who exercise regularly. Older adults can substantially increase muscle strength and power after resistance exercise training, at least in the short term. Moderate- to high-intensity resistance and impact training is also effective in improving muscle endurance although eventually, as people get older, muscle losses increase and the response to exercise stimulus reduces.
- Diet intervention has minimal to no effect on muscle status in healthy masters athletes who are matching daily energy and protein intake with expenditure.

MASTERS ATHLETES WITH OTHER CHRONIC DISEASES OR RISK FACTORS: ARE SUPPLEMENTS INDICATED?

- A masters athlete may have an existing chronic medical condition such as diabetes, hyperlipidaemia, hypertension or an arthritic condition. He/she may have outdated beliefs about diet and disease, or still be following dietary advice provided many years ago that is no longer valid.
- Masters athletes with other age-related conditions (e.g. arthritis, failing eyesight and short-term memory loss) may seek your advice about the effects of specific foods or supplements to alleviate symptoms. The role of the dietitian in this situation is to assess the safety of the product or supplement, inform the athlete of the current evidence, and then allow the individual to decide (see Chapter 16).
- Some supplement use in masters athletes, including sports supplements, may be based solely on a perceived improvement in sports performance or recovery and unrelated to health issues. In athletes, particularly endurance athletes, the use of antioxidants supplement is common (see Chapter 11). However, antioxidants vitamin C and E taken as supplements in the usual recommended amounts may counteract the beneficial effects of exercise on enhancing insulin sensitivity (Ristow et al. 2009; Powers et al. 2011). This effect has implications for masters athletes, who are overweight or obese, with metabolic syndrome or type 2 diabetes.
- In summary, most of the research on sports supplements has been conducted on younger athletes or untrained older groups and not on older masters athletes with or without existing health problems. Nevertheless, short-term, high-dose antioxidant supplements in combination with physical activity may do more harm than good, although further studies are required to confirm these effects in the older athlete.

LOSS OF APPETITE IN OLDER MASTERS ATHLETES

- Loss of appetite is a common physiological response to intense exercise and can last for a long time after cessation. Taste and smell, which also diminish with ageing, affect the enjoyment of food and often reduces energy intake. For those masters athletes with poor appetites, disinterest in food or suboptimal energy intakes, advice on nutrient and energy-dense foods, frequent snacking and appropriate food supplements is needed. Eating almost immediately after exercise, particularly a hard training session, is just as important to promote rapid recovery for the older athlete as for the younger athlete. See Chapter 14 for practical food ideas.

NUTRIENT AND ENERGY REQUIREMENTS IN MASTERS ATHLETES

- With ageing, most micronutrient requirements remain the same as a younger adult, except for vitamin B6, B12, D and calcium, which marginally increase. Conversely, energy requirements usually decrease, although they are highly variable in masters athletes, depending on age and activity levels. Determining energy requirements needs to be individualised. Estimates of the energy cost of different activities and total energy expenditures using multi-system technology (e.g. Sensewear™, heart rate monitors) may be distorted in older people but useful to measure variability in energy expenditure with different types of exercise (see Practice tips, Chapter 5).
- For masters athletes who are overweight (or obese) or predisposed to weight gain, there is less room for 'indulgence' foods, particularly when incidental activity is low, despite regular exercise. Hence, nutrient-dense foods in combination with a lower energy intake are an important part of healthy eating. See Chapter 6 for tips for meeting nutrient needs within a limited energy budget.
- Recommendations for CHO intakes needed for training and recovery in masters athletes are the same as younger adult athletes since CHO absorption and utilisation is unaffected by ageing in a healthy person.
- Recommendations for total daily protein intake for masters athletes, particularly those undertaking resistance exercise training, are higher than EAR cut-offs at 1.0–1.3 g/kg BM/d but still slightly lower than the recommendations for younger athletes undertaking the same type of activity. Consuming protein together with CHO-rich foods immediately following resistance training maximises muscle and strength development, similar to younger athletes.

GUIDELINES FOR COMMENCING EXERCISE OR PHYSICAL ACTIVITY IN OLDER PEOPLE

- A dietitian is often asked for exercise advice. If an older individual with an existing disease is starting an exercise program after previously being sedentary, consultation and clearance with a physician is usually mandatory for membership of gyms and fitness centres. Guidelines from the American College of Sports Medicine and the American Heart Association suggest that moderate-intensity aerobic activity, muscle-strengthening activity, reducing sedentary behaviour and risk management should be the focus for physical activity and health promotion in older adults (Nelson et al. 2007). These guidelines suggest that:
  - aerobic activity should be undertaken at an intensity that accounts for the older adult's aerobic fitness
  - activities are provided that maintain or increase flexibility
  - balance exercises are included for older adults at risk of falls.

These guidelines further recommend an activity plan for older adults that will eventually achieve the recommended physical activity levels, which integrate both preventive and therapeutic recommendations.

HOW MUCH PHYSICAL ACTIVITY IS REQUIRED TO REDUCE RISK FACTORS?

- In 2009, the American College of Sports Medicine (2009b, 2009c) revised its recommendations for the amount and intensity of physical activity required for people of all ages, including older adults, to improve health outcomes, prevent transition from overweight to obesity and prevent weight regain after weight loss. They are not intended for high-level fitness or the type of sports training undertaken by a masters athlete.
- Given the increased prevalence of chronic diet/lifestyle-related diseases in the population and obesity in older postmenopausal women, many older adults are either commencing exercise for the first time or increasing the intensity and volume of exercise in the hope of improving health outcomes. Exercise guidelines that address these risk factors are summarised in Table 18.3. For example, to achieve clinically significant weight loss with no restriction in energy intake, moderate to vigorous exercise for >250 min/wk is needed (see Table 18.3).
- Clearly, the recommendations in Table 18.3 for weight loss and risk reduction for chronic disease may be

unachievable for many older people in terms of time and physical capacity; hence the importance of encouraging greater levels of incidental activity in an individual's daily life, in addition to some structured exercise.

- Nevertheless individuals with a high BMI (<35) and good aerobic fitness have lower all-cause and cardiovascular mortality than those with normal BMI and poor fitness according to a recent systematic review (Fogelholm 2010).
- For those older adults who are overweight or obese, modest energy restriction and higher total energy expenditure in combination with behavioural therapy is the most effective strategy to improve weight control, reduce other risk factors and potentially improve health outcomes and quality of life (American College of Sports Medicine et al. 2009b, 2009c).
- In summary, the older a person becomes, the more dissimilar they are from others in their age group, so expect a large variety of fitness levels, personal goals and nutrition knowledge. For the masters athlete, providing a list of contacts with local and or national sporting organisations, other allied health professionals who are knowledgeable in masters sport and the specific issues experienced by older athletes with existing medical conditions is helpful.

TABLE 18.3 Preventive and therapeutic amounts and intensity of exercise required to reduce risk factors and control weight

Objective	Frequency and intensity	Duration
To increase cardiovascular fitness and decrease risk factors for cardiovascular disease, hypertension, osteoporosis and type 2 diabetes and falls in older adults	Aerobic: a minimum of 5 d/wk for moderate aerobic intensity or a minimum 3 d/wk of vigorous intensity Muscle strengthening: at least 2 d/wk Flexibility and balance: at least 2 d/wk for those at risk of falls, include exercises that maintain and improve balance	At least 30 min/d (up to 60 mins for greater benefit) – moderate intensity. This can be performed in bouts of at least 10 min each. At least 20 min/d – vigorous activity 8–10 exercises involving major muscle groups with 8–12 repetitions
To lose weight in adults	Most days of the week: moderate to vigorous	>250 min/wk to achieve clinically significant weight loss
To prevent the transition from overweight to obesity in adults	Most days of the week: moderate to vigorous	150–250 min/wk
To prevent weight regain after weight loss in obese adults	Most days of the week: moderate to vigorous	>250 min/wk but insufficient evidence to judge the effectiveness of physical activity alone

Note: Moderate intensity = physical activity at a level that causes the heart to beat faster and some shortness of breath, but that you can still talk comfortably while doing; Vigorous intensity = physical activity at a level that causes your heart to beat a lot faster and makes talking more difficult between breaths
Sources: Adapted from American College of Sports Medicine et al. 2009b, 2009c

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CHAPTER NINETEEN

Special needs: the athlete with diabetes

Pat Phillips and Vicki Deakin

19.1 Introduction

Diabetes mellitus is a group of metabolic disorders resulting from defects in insulin secretion, insulin action (sensitivity) or both, which cause abnormalities in carbohydrate, protein and fat metabolism that are life threatening, if untreated. If poorly controlled or improperly treated, blood glucose rises to high levels in the acute phase and can cause hyperglycaemia (high blood glucose), dehydration, ketosis, weight loss and blurred vision. Long-term complications in poorly controlled diabetes include cardiovascular disease; nephropathy leading to renal failure; retinopathy with potential loss of vision; peripheral neuropathy with risks of foot ulcer and amputation and autonomic neuropathy causing gastrointestinal, genitourinary and cardiovascular symptoms and sexual dysfunction (American Diabetes Association 2014a, 2014b). Hence, the long-term treatment goals, irrespective of the type of diabetes, are to maintain glycaemic control, glycaemia and to delay or prevent the onset and progression of these long-term complications, and ultimately improve quality of life (American Diabetes Association 2014b).

A person with diabetes can become an elite athlete and train and compete at a high level of exercise intensity and endurance. Several renowned athletes with diabetes, including Sir Steven Redgrave, who won five successive gold medals in rowing from 1985 to 2000, and Wasim Akram, the Pakistani fast bowler, provide testament to overcoming any perceived barriers imposed by this condition. As hypoglycaemia (low blood glucose) is the most common problem faced by an athlete with diabetes on insulin, certain sports that are solo in nature, take place in mid-air or water or limit an individual's ability to recognise the symptoms of hypoglycaemia—which can be masked by exercise—may be contraindicated or undertaken with caution with a training partner. For athletes participating in these types of sports, coaches and training partners need to recognise the symptoms of hypoglycaemia and know how to treat it.

This chapter focuses on clinical practice guidelines and medical nutrition recommendations for athletes with diabetes, particularly those on insulin, undertaking different levels of physical activity. Although the carbohydrate (CHO) and protein recommendations before, during and after exercise are similar for athletes with or without diabetes, the metabolic response to exercise in an athlete on insulin is affected by the levels of blood glucose and circulating insulin at the start of exercise. Regular self-monitoring of blood glucose in response to exercise, as well as on-going consultation with a team of qualified clinicians—including a dietitian with experience in diabetes management, a diabetes educator and a physician—is critical for an athlete with diabetes on insulin—to assist the athlete to fine-tune insulin dosage and nutrition to match energy requirements for exercise.

19.2 Definition and description of diabetes mellitus

Diabetes mellitus is a group of metabolic disorders resulting in defects in insulin secretion, insulin action (sensitivity) or both (Selvin et al. 2014). Most cases of diabetes fall into two broad categories: type 1 and type 2. Type 1 diabetes or immune-mediated diabetes is less common than type 2 diabetes, affecting around 5–10% of people with diabetes. Type 1 diabetes (previously named insulin-dependent diabetes or IDDM, type I diabetes or juvenile-onset diabetes) mainly affects a younger age group and has an abrupt and often life-threatening onset. It typically involves the autoimmune destruction of pancreatic beta cells, which is the site of insulin production. Management involves lifelong treatment with exogenous insulin, diet modification and self-monitoring of blood glucose levels. There are several other types of diabetes involving genetic defects of the pancreatic beta cells, genetic defects in insulin action, endocrinopathies and drug- or chemical-induced types (American Diabetes Association 2014a).

The most common form is type 2 diabetes (previously named non-insulin-dependent diabetes or NIDDM, type II diabetes or adult-onset diabetes), which affects around 90% of people with diabetes (AIHW 2011). With this type, individuals have insulin resistance and usually have relative (rather than absolute) insulin deficiency. Type 2 diabetes is a progressive condition, which may take 5–10 years of deteriorating glucose tolerance before being diagnosed (Goldberg 1998). In Australia, the prevalence of type 2 diabetes is highest in older adults, in the Indigenous population, and in several other ethnic groups (South Pacific Islanders, Southern Europeans, and people from Middle Eastern, North African and Southern Asian countries) and, in particular, in people who are overweight or obese (Dunstan et al. 2001; Thow & Waters 2005; AIHW 2011). An increased incidence was reported in adolescents and children in Australia in 2004 (McMahon et al. 2004). Similar increases around the same time were reported in children and adolescents in the UK (Ehtisham & Hatt Molnár 2004; Ehtisham et al. 2004), in the US (Alberti et al. 2004) and to a lesser extent in European children (Molnár 2004; Malecka-Tendera et al. 2005).

This increased prevalence of diabetes at all ages parallels the epidemic of obesity, which is a significant contributor (Garber 2012). Obesity is strongly linked to diabetes because it causes some degree of insulin resistance. In the early stages of type 2 diabetes, there are often no symptoms and insulin is not required because autoimmune destruction of the beta cells in the pancreas does not occur. However, lethargy, polyuria, nocturia and polydipsia, the typical symptoms of hyperglycaemia, can occur and usually worsen with progression of the condition. Lifestyle management for type 2 diabetes is similar to that for type 1 and involves diet modification and weight loss, if overweight or obese. In most cases, medication (oral hypoglycaemic agents and possibly insulin) is warranted.

19.3 Physiological effects of exercise

For either form of diabetes, regular physical activity including both resistance and at least moderate levels of aerobic activity improves insulin action and glycaemic control, increases insulin sensitivity and reduces associated cardiovascular risk factors (Colberg et al. 2010; American Diabetes Association 2014a). This effect is independent of exercise-related changes in body mass (BM) (Duncan et al. 2003). However, the acute physiological effects of moderate to strenuous levels of physical activity can induce either hypoglycaemia (low blood glucose) or hyperglycaemia (high blood glucose), especially in an athlete on insulin.

In terms of glucose metabolism, acute exercise is a potent stimulus for glucose uptake and utilisation by muscle tissue (Ebeling et al. 1993; Colberg et al. 2010). These mechanisms and the requirements for utilisation of fuel for muscular activity are no different from those of an athlete without diabetes. In athletes without diabetes who exercise, levels decrease and glucagon and other counter-regulatory hormones increase, promoting an increase in hepatic glucose production (gluconeogenesis). The amount of glucose produced by gluconeogenesis matches the amount required by the increased metabolic requirements of other physiological systems: the respiratory, neural and cardiovascular systems. As a result, blood glucose levels can remain relatively stable for up to 1–2 hours of continuous exercise without food intake. With no food consumed during exercise and during prolonged moderate- to high-intensity exercise of more than 60–90 minutes duration, blood glucose levels will fall when hepatic glucose production lags behind glucose utilisation and liver glycogen becomes depleted. For athletes with diabetes (and non-diabetic athletes), there is a substantial benefit from consuming CHO foods during exercise to prevent declines in blood sugar levels and promote recovery for the next training session (Perrone et al. 2005). For reviews about the effects of acute (and chronic) exercise on glycaemic control, see Colberg et al. (2010) and the American Diabetes Association (2014a).

19.3.1 Effects of long-term physical activity or exercise training on insulin sensitivity

Of relevance to a person with diabetes is that repeated aerobic and resistance exercise increases insulin sensitivity (Wasserman et al. 2002; Colberg et al. 2010). Insulin sensitivity can be sustained for 12–15 hours after a single bout of moderate to vigorous exercise and even longer by regular exercise training (Devlin 1992). In a group of obese adults with type 2 diabetes, 1 week of aerobic training improved whole body and peripheral insulin sensitivity (Winnick et al. 2008).

The main functions of insulin are to:

- stimulate glucose uptake into most cells of the body
- inhibit glucose release from the liver
- inhibit the release of fatty acids from storage deposits
- facilitate protein synthesis in the body's cells
- stimulate resynthesis of muscle glycogen after exercise.

An increase in insulin sensitivity improves these actions, and is associated with a reduction in insulin dosage or in other oral hypoglycaemic medications (see section 19.5.2).

19.3.2 Acute effects of exercise in athletes with type I diabetes

During exercise in athletes with type 1 diabetes, insulin absorption from injection sites continues and other counter-regulatory hormones are secreted (catecholamines, glucagon, growth hormone and cortisol). These changes regulate glucose metabolism and affect cardiovascular response, body temperature, fluid and electrolyte homeostasis.

The magnitude of these metabolic and hormonal effects induced by exercise is determined by the:

- intensity and duration of exercise
- degree of metabolic control before exercise
- type and dose of insulin injected before exercise
- site of insulin injection
- timing of the previous insulin injection
- timing of the previous meal.

19.3.3 Exercise in the presence of hyperinsulinaemia

In athletes with diabetes on insulin, plasma insulin levels may not decrease during exercise. This induces hypoglycaemia, as insulin prevents the appropriate rise in hepatic glucose production and accelerates the exercise-induced stimulation of glucose uptake (see Figure 19.1).

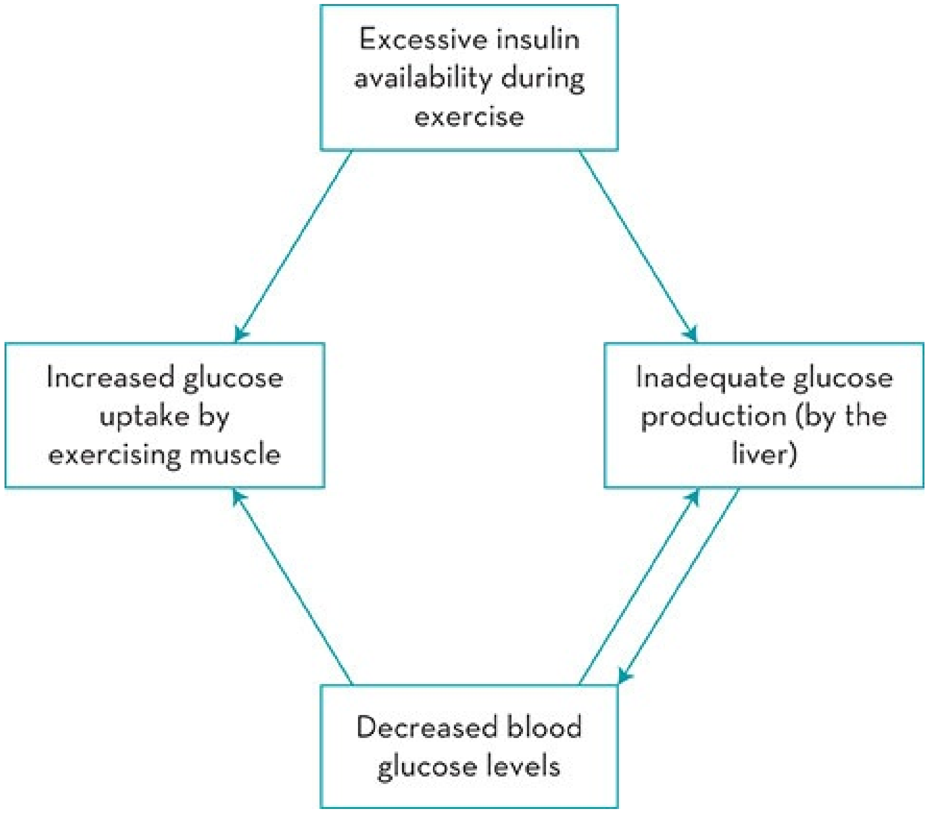

Figure 19.1 Metabolic response to exercise in diabetes

Elevated insulin levels also prevent the increase in mobilisation of lipids during exercise, leading to a reduced availability of non-esterified fatty acids as a fuel source (Wasserman et al. 1995). Hyperinsulinaemia may occur when the usual dose of short-acting insulin (if injected shortly before exercise) reaches its peak action during exercise. This effect can potentially be exaggerated if the previously injected limb is exercised, which may promote an increase in insulin absorption from the injection site, especially when using an intramuscular site (Koivisto & Felig 1978). Subcutaneous injection, particularly in the abdomen, usually addresses this problem (see section 19.5.1). If an athlete adjusts both dietary intake and insulin and monitors glycaemic response under different training conditions, hyperinsulinaemia can be avoided.

19.3.4 Exercise in the presence of hypoinsulinaemia

In this situation, the inhibitory effect of insulin on hepatic glucose production and its stimulating effect on glucose uptake are both reduced. In addition, the counter-regulatory hormone response to exercise is higher than normal during insulin deficiency. These changes substantially increase hepatic glucose production and diminish glucose utilisation by the exercising muscle, and result in marked hyperglycaemia. During extremely strenuous acute exercise, hyperglycaemia can be induced by excessive production of these counter-regulatory hormones, which stimulate a surplus of production of hepatic glucose beyond the limits of peripheral utilisation. This can occur in type 1 diabetes even in the presence of insulin.

In a diabetic athlete without adequate insulin, lipid mobilisation and ketogenesis induces ketonaemia. When both dietary intake and insulin doses are adjusted appropriately, this situation can be avoided.

19.4 Medical nutrition therapy (MNT) for athletes with type 1 diabetes

In 2014, the American Diabetes Association released updated evidence on nutrition therapy in the management of adults with existing type 1 and type 2 diabetes. In this position statement, Evert and colleagues (2014) reinforced the importance of an individualised MNT undertaken by a registered dietitian familiar with the components of diabetes. Nutrition goals for training and competition recommended for athletes with diabetes are no different from those recommended for athletes without diabetes (Colberg et al. 2010). An overall training diet for athletes with diabetes, based on CHO-rich foods (see Chapter 12) and low-fat foods, is compatible with diabetes management and athletic performance. Low-CHO diets are counterproductive to maintaining capacity for athletic performance and recovery. However, the amount of carbohydrate and available insulin may be the most important factor influencing glycaemic response after eating and should be considered when developing an eating plan for an athlete with type 1 diabetes (Evert et al. 2014).

Athletes with diabetes should be encouraged to adjust insulin dosage according to their lifestyle and training program rather than distorting their eating patterns to suit the insulin dosage. Although it is not necessary to follow a rigid pattern of food intake, a reasonably consistent eating routine facilitates diabetic management. Maintaining consistent training and eating routines day to day also assists in the establishment of insulin dosage and fine-tuning of food intake. A well-trained athlete who regularly exercises at the same time each day will usually need less adjustment to food and insulin than a person who exercises only occasionally (Franz et al. 2002). For athletes with recently diagnosed type 1 diabetes, the stabilisation and adjustment phase is usually the first priority and an interruption to training and competition routines may be expected in the short term.

Any individual with type 1 diabetes who participates in regular physical exercise of moderate to high intensity needs to consider:

- the macronutrient content and timing of meals and snacks
- the insulin dose and its predicted peak period of activity in relation to their exercise
- routine monitoring of blood glucose levels to assist with the adjustment of food and insulin to prevent and manage hyperglycaemia and hypoglycaemia.

Evidence of the efficacy of different types of eating plans (e.g. the Mediterranean style diet, DASH diet, low-carbohydrate diet, vegetarian-style diet) for adults with diabetes are described in the 2014 position statement from the American Diabetes Association on nutrition management of diabetes (see Evert et al. 2014). With the exception of the low-CHO diet, these eating plans are appropriate for athletes with diabetes.

19.4.1 Type of carbohydrate recommended for athletes with diabetes

The type of CHO—low or high glycaemic index foods—consumed in the total diet has a minimal effect on achieving glycaemic control (Evert et al. 2014) because of the wide variability in individual responses to CHO-containing foods. Contrary to popular belief, dietary sucrose does not increase blood glucose levels more than isocaloric amounts of starch (Franz & Wylie-Rosett 2007). However, sucrose or high-sugar-containing foods should be minimised to avoid displacing nutrient-dense foods (Evert et al. 2014). Glucose polymers—which are the main ingredient in sports drinks—rather than free glucose, do not elevate the blood glucose response (Maughan et al. 1987). Although fructose does not elevate blood glucose levels at the same rate as glucose or sucrose, high concentrations can cause gastrointestinal upset. Sports drinks, which contain a combination of glucose polymer and other sugars, are an ideal fluid and CHO choice during and after prolonged endurance exercise for athletes with diabetes. Nevertheless, for the athlete with type 1 diabetes, high intakes of sports drinks or other sugar-containing foods should be minimised to avoid displacing nutrient-dense foods (Evert et al. 2014) and adequately covered with insulin.

19.4.2 Recommended carbohydrate intakes for athletes with diabetes

The CHO content in food and the amount consumed, rather than the fat and protein content, remains the cornerstone of glycaemic control in an athlete with diabetes. The amount of CHO consumed is still the main determinant of post-prandial blood glucose levels. In contrast, proteins in food do not increase blood glucose levels post-prandially. The key strategy for achieving glycaemic control is to monitor CHO consumption, by CHO counting, CHO exchanges or experience-based estimations (Dietitians Association of Australia 2006; Evert et al. 2014).

Table 19.1 provides guidelines for the amounts of CHO (in grams) recommended for athletes undertaking exercise of varying intensity and duration. Examples of foods containing 50 g of CHO are found in Chapter 14, Table 14.4. In principle, the recommended CHO intake before, during and after exercise for an athlete with diabetes is no different from that recommended for an athlete without diabetes (see Chapters 12, 13 and 14). These CHO recommendations are derived from laboratory studies of groups of people who are unlikely to be elite athletes and are being tested under contrived experimental conditions. They can be applied to those diabetic athletes who have good metabolic control with usual blood glucose levels between 4 mmol/L and 8 mmol/L (Franz et al. 1994). If blood glucose levels are chronically very high or often low, food choices and/or insulin dosage should be modified.

TABLE 19.1 Recommended dietary carbohydrate (CHO) intakes (g) before exercise

Exercise intensity and duration	Blood glucose (mmol/l)	Dietary CHO (g)
Brief high-intensity (<30 min) (e.g. weightlifting, sprints)	6–10	No food required
Light (e.g. walking 30 min, easy-pace aerobics 60 min)	<6 >6	15 No food required
Moderate (<45 min) (e.g. swimming, jogging, tennis, basketball)	<6 6–10 10–14 14+	30–45 15 No food required Exercise not advised
Moderate (>60 min) (e.g. football, cycling)	10–14 plus reduced insulin dosage >13–14 and ketones 17 (no ketones)	10–15 g/h Exercise contraindicated Exercise not advised
Strenuous (<60 min) (e.g. triathlon, marathon, canoeing, kayaking, cross-country skiing, cycling)	<6 6–10 10–14 14+	45 30–45 15–30 Exercise not advised
Strenuous (>60 min) (e.g. triathlon, marathon, canoeing, kayaking, cross-country skiing, cycling)	<6 6–10 10–14	50 g/h 25–30 g/h 10–15 g/h

Note: Exercise at any level is contraindicated if ketones are present
Source: Adapted from Armstrong 1992

Adapting these CHO recommendations based on individual responses to exercise of differing intensities and duration is done by trial and error together with blood glucose monitoring. CHO (and protein) recommendations need to be adapted to the individual and to the energy demands of each sporting activity.

19.4.3 Recommendations for carbohydrate intakes for brief, high-intensity sports and light training

Short-term, high-intensity exercise such as sprinting or running less than 1500 m, swimming less than 400 m and other sports involving sudden, intense activity (such as weightlifting), usually require minimal or no adjustment to food intake provided the pre-event blood glucose level is between 5.6 and 10.0 mmol/L (Armstrong 1992; Franz et al. 1994). This is also the case for low-level activity, such as a 30-minute leisurely walk or bike ride.

For a blood glucose level 10–14 mmol/L, there is probably no need for increased food (or CHO) if the exercise is of less than 1-hour duration. If the blood glucose level is above 14 mmol/L, the urine should be tested for ketones. Exercise is contraindicated in the presence of ketones. Blood glucose levels should be below 14 mmol/L before the activity is commenced (Armstrong 1992).

19.4.4 Recommendations for carbohydrate intakes for moderate-intensity exercise of short duration (15–30 minutes)

A meal 1 hour (for users of mealtime analogue bolus insulin) or 3 hours (for users of neutral bolus insulin) before starting exercise is recommended according to the principles of preparation for exercise (see Chapter 12). If insulin action is likely to peak at the time of the event, an additional 10–15 g CHO taken 20–30 minutes beforehand is usually sufficient to prevent hypoglycaemia, provided the blood glucose level is between 5.6–10.0 mmol/L at the commencement of exercise (Armstrong 1992; Franz et al. 1994). If the blood glucose level is more than 14 mmol/L, postponement of exercise is recommended.

After exercise, 15–30 g CHO food may be required to maintain adequate blood glucose levels, prevent post-exercise hypoglycaemia and promote recovery (see Chapter 14).

19.4.5 Recommendations for carbohydrate intakes for moderate- to strenuous-intensity exercise of medium to long duration (<30 minutes)

A meal before exercise of an additional 15 g CHO from foods with a high to moderate glycaemic index for each 20 minutes of planned activity should be tried initially. A meal 1 hour (for users of mealtime analogue bolus insulin) or 3 hours (for users of neutral bolus insulin) before exercise, with the support of an additional 15 g CHO from high glycaemic index foods taken just before competition is recommended.

Consuming some type of CHO at around 60–90 minutes or even earlier into a prolonged bout of physical activity helps prevent hypoglycaemia during exercise and maintains exercise capacity (Kahn & Vinik 1988; Armstrong 1992). In practice, the form of CHO (solid or liquid) consumed makes no difference to performance outcomes and depends on individual preferences and opportunities for consumption during the activity. The amount of CHO recommended during moderate to intense exercise for an athlete with type 1 diabetes is no different to that recommended for an athlete without diabetes (see Chapter 13). For those athletes with type 1 diabetes who do not tolerate solid food, a sports drink is useful during exercise to deliver CHO (Andrade et al. 2005; Perrone et al. 2005). In one study of 16 adolescent athletes with type 1 diabetes, an average intake of 92 mL sports drink containing 8% CHO maintained blood sugar levels and fluid status during a 60-minute cycle ergometer test at 55–60% % VO_{2max} with no change in insulin dosage (Perrone et al. 2005). For four athletes in this study, a sports drink containing 10% CHO during exercise was required to prevent exercise-induced hypoglycaemia. The 8% sports drink was not sufficient to prevent hypoglycaemia. No gastrointestinal symptoms were reported in any athletes in this study using both concentrations.

19.4.6 Pre-event carbohydrate loading

Adequate diabetic control is required before an athlete with diabetes should attempt any form of CHO loading.

Dietary and training techniques for CHO loading of glycogen stores for competition are considered in detail in Chapter 13. As CHO loading is dependent on the availability of insulin to store muscle glycogen, this technique should be used with caution in an athlete requiring insulin. The insulin dose should be adjusted to match the changes in diet and the tapering effects of exercise before competition. For the athlete on insulin who has poor diabetic control, stabilising blood glucose is already difficult and CHO loading can lead to further deterioration in glycaemic control.

19.4.7 Post-exercise re-feeding

CHO and small amounts of protein should be eaten soon after the completion of exercise (where possible) to promote muscle glycogen and protein resynthesis (see Chapter 14). Around 1.0–1.5 g CHO/kg BM (and 10 g protein) shortly after exercise and again at 60 minutes is recommended to maximise recovery of fuel reserves for the next training session (Armstrong 1992).

19.4.8 Fluids

Athletes with type 1 diabetes tend to become preoccupied with replacing CHO and forget their requirements for fluid. Thirst is not necessarily a reliable indicator of fluid status in athletes with well-controlled diabetes or even without diabetes. Because the thirst mechanism is less sensitive during exercise, it is important to drink water before becoming thirsty. Conversely, excessive thirst could be a sign of hyperglycaemia in an athlete with diabetes and not necessarily indicative of fluid status. Guidelines for the composition and volume of suitable fluids for everyday training and specific competition for an athlete with diabetes are the same as those for non-diabetic athletes (see Chapter 13 & Commentary 7).

19.4.9 Alcohol

Alcohol is a potent inhibitor of hepatic glucose production and may precipitate late and severe hypoglycaemia in a person with diabetes on insulin or oral hypoglycaemic agents. Excess alcohol also impairs an individual's ability to recognise symptoms of hypoglycaemia. For athletes with diabetes, when consuming alcohol, the recommended intake is the same as that for the general population: 1 drink/d or less for women and 2 drinks/d or less for men as an occasional addition to the regular meal plan (Evert et al. 2014). When soft drinks are used as mixers with alcohol, or when sweet wines or liqueurs are consumed, the opposite effect—hyperglycaemia—can be induced.

19.5 Insulin adjustments for athletes with type 1 diabetes

The main determinant of the glycaemic response to exercise is insulin availability. The following schedules are typical of current management of type 1 diabetes.

- *Intensive schedules (basal-bolus)* use bolus injections of short- or ultra-short-acting insulin before each meal, and intermediate- or long-acting insulin once or twice daily (before bedtime and breakfast).
- *Continuous subcutaneous insulin infusion (SCII)* uses a pump that delivers a continuous infusion of short- or ultra-short-acting insulin via a subcutaneous needle with bolus doses activated by patient before meals.
- *Split-mixed schedules of basal and bolus insulin* before breakfast and the evening meal are less used now.

19.6 Available insulins

Available insulin preparations are either purified bovine insulins or human insulins obtained by recombinant DNA technology.

- *Ultra-short and short-acting insulins* are soluble insulins (clear solution); short-acting insulins are the only type that can be given intravenously, for example in diabetic ketoacidosis.
- *Insulin lispro*, *insulin aspart* and *insulin glulisine* are insulin analogues obtained by recombinant DNA technology; their ultra-short onset of action allows them to be given immediately before meals.
- *Insulin glargine* and *insulin detemir* are insulin analogues, which provide a basal insulin level that is constantly delivered over 24 hours and often allows once-daily dosing.
- *Intermediate insulins* have a prolonged duration of action resulting from complexing insulin with a protein (protamine, isophane insulin).
- *Mixed insulins* (also called *biphasic insulins*) combine a short- or ultra-short-acting insulin in varying proportions (75–50%) with an intermediate-acting insulin.

Fewer people use neutral bolus or isophane basal insulins since insulin analogues have become available. The short-acting analogues have an onset of action within minutes after subcutaneous injection, peak at 1–2 hours and have a duration of approximately 4 hours. The common bolus neutral (clear) insulins have an onset of action of about 30 minutes after subcutaneous injection, peak at 2–3 hours and have a duration of approximately 7 hours. Both kinds of bolus insulin are given before each meal, using a syringe or a pen injector. The basal insulins (Isophane™ or analogue) are usually given before bed. The intermediate-acting insulins (Isophane™ or Lente MC™) have an onset of action after 1–2 hours, peak at 6–10 hours and have a duration of 16–20 hours. The profile of the long-acting analogue basal insulins is essentially flat with the duration of a single injection up to or more than 24 hours. Table 19.2 summarises the onset and duration of action, and time to peak activity, in these different types. Pre-mixed preparations of neutral and isophane insulins are also available. The amount of neutral insulin in these preparations varies from 20–50%, with the remainder being isophane insulin.

19.6.1 Site of insulin injection

The site of injection (subcutaneous or intramuscular) may affect insulin absorption. In one study, intramuscular injection by adults followed by cycle exercise induced a marked increase in insulin absorption and a substantial fall in plasma glucose compared with subcutaneous thigh injection (Frid et al. 1990). No effects of 30 minutes of intense exercise on insulin absorption results were found in a study of 13 adults using subcutaneous thigh injection of basal-bolus insulin—glargine—on the evening before the study (Peter et al. 2005). Abdominal bolus subcutaneous injection is the probably best site to minimise potential effects.

TABLE 19.2 Types, onset, peak and duration of action of currently available insulins

Insulin type	Onset of action (hours)	Time to peak activity (hours)	Duration of action (hours)
Ultra-short-acting			
Insulin lispro, insulin aspart, insulin glulisine	0.25	1	35–45
Short-acting			
Neutral	0.5	2–5	6–8
Intermediate-acting			
Non-mixed	1–2.5	4–12	12–24
Mixed with short-acting insulin	0.5–1	2–12	12–24
Mixed with ultra-short-acting insulin	0.25	1	12–24
Long-acting			
Insulin glargine	*	*	24
Non-mixed	*	*	24–36

* Generally the profiles of long-acting basal insulins are essentially flat
Source: Adapted from Diabetes Australia 2012/13

19.6.2 Insulin adjustment

The challenge for the management of an individual athlete using bolus injection or a CSII insulin pump is monitoring how varying levels of exercise affect blood glucose levels and how to adjust insulin to match (Heiemann et al. 2001).

Smaller total daily insulin dosages are required in athletes with type 1 diabetes who undertake regular physical activity, largely because of an increase in insulin sensitivity associated with exercise. When physical activity is less than 30 minutes duration, it is usually not necessary to reduce insulin dosage. For more prolonged physical activity, the insulin is adjusted to reduce the insulin operating at the time of the exercise, usually by 15–40%. The exact reduction depends on the intensity and varies considerably between individuals.

19.6.3 Insulin adjustment using multiple dose insulin schedules

For physical activity undertaken in the morning, the short-acting insulin dose should be reduced before breakfast by 15–40%. When physical activity is undertaken in the afternoon, basal insulin (if used before breakfast) should be reduced. If an athlete is on a combination of short-acting insulin before each meal and basal insulin at night, the pre-lunch short-acting dose is reduced for afternoon exercise.

When physical activity is continuous over several hours (e.g. a 3- to 4-hour cycle or a 2-hour run), then a more substantial reduction in insulin dose is required. In this situation, reducing both morning short-acting (clear) and basal doses by up to 50% could be indicated. For some events, such as a marathon or triathlon, insulin has been omitted completely, without adverse effects (Meinders et al. 1988). However, although it is important to avoid hypoglycaemia during exercise, an excessive reduction in insulin doses can lead to hypoinsulinaemia resulting in hyperglycaemia with consequent increases in fluid loss via kidneys, possible ketosis and a high risk of dehydration and heat stress.

19.6.4 Insulin adjustment using a continuous subcutaneous insulin infusion (CSII): insulin pumps

Insulin pumps contain a reservoir or cartridge filled with short-acting insulin analogue. The dose is delivered subcutaneously in the abdomen, buttocks, legs or upper arm through a needle or infusion catheter, which is repositioned every 3 days. This mode of delivery provides adjustable 24-hour insulin as ‘basal’ delivery with variable bolus insulin at meal times or when required. The pump can be programmed to accommodate the decreased insulin requirement in response to exercise, so is associated with a lower risk of hypoglycaemia, compared to using multiple daily injections (Colberg 2002). Although expensive, it is estimated that around 20% of people with diabetes in the US are using insulin pumps. In Australia, CSII use is increasing, especially in children and adolescents undertaking sport, because of its advantages in offering more flexibility in dosage and in eating habits. For athletes undertaking non-contact sports, the pump can usually remain connected. Disconnection for up to 1–2 hours is possible in those undertaking water or contact sports (Smart & Morrison 2008).

19.7 Monitoring blood glucose levels

Individuals vary considerably in their metabolic response to physical activity and their glycaemic control. It is important to consider the type, intensity and duration of the sports activity (e.g. training or competition) when making decisions about appropriate insulin dosage and how frequently blood glucose is monitored. A management plan based on food requirements after physical activity and corresponding insulin requirement can then be formulated. Once a response pattern is established for a particular individual to a particular type of exercise, it is likely to be similar on future occasions and, therefore, appropriate adjustments can be predicted.

If physical activity is unplanned and insulin has already been administered, then adjustments to CHO intake during or prior to exercise will be needed. Adjustments to insulin doses and food intake after exercise may also be needed depending on the intensity and duration of the exercise.

19.8 Special problems for the athlete with type 1 diabetes

Very low and high blood glucose levels occur from time to time in any individual with diabetes on insulin. Although habitual physical activity improves glycaemic control, the acute metabolic effects of exercise, particularly high-intensity exercise, are often disturbed in an athlete on insulin compared to a non-diabetic athlete. During exercise, hypoglycaemia or hyperglycaemia can occur and late-onset hypoglycaemia after exercise is not uncommon, especially in children and adolescents during growth spurts. In individuals with poor metabolic control, blood glucose levels can be extreme. On competition day, further high blood glucose levels may unexpectedly occur, which is attributed to the physiological and psychological stress of competition with increases in circulating catecholamines and other catabolic hormones. For instance, it is not uncommon for an athlete on insulin who is more prone to hypoglycaemia on a regular training day to develop hyperglycaemia at competition, even though undertaking the same nutritional practices as a training day.

19.8.1 Hypoglycaemia

The most common problem for the athlete with diabetes on insulin is hypoglycaemia. Children and adolescents, particularly during growth spurts, are more susceptible to hypoglycaemia than adults. Most children and adolescents on insulin who exercise for prolonged period (e.g. >30 minutes) experience a substantial drop in blood glucose levels (Riddell et al. 1999). The associated symptoms—increased sweating, signs of anxiety, nausea and disorientation—are easily confused or masked by the usual physiological effects of exercise. If symptoms of hypoglycaemia are not recognised, then severe hypoglycaemia can develop. Untreated hypoglycaemia, associated with too much insulin or too little food, may lead to hypoglycaemic coma.

19.8.2 Late hypoglycaemia after exercise

A further problem affecting an athlete on insulin is the occurrence of late hypoglycaemia after exercise. This can occur up to 36 hours after exercise and is particularly prevalent in very active children (Shehadeh et al. 1998; Tupola et al. 1998). The onset of delayed hypoglycaemia is associated with increased insulin sensitivity for several hours after cessation of exercise (see section 19.2), and possibly incorrect insulin adjustment or inappropriate nutrition intervention (Riddell & Iscoe 2006). If appropriate snacks are used prior to exercise, then reducing the next insulin dose usually avoids this effect. Too much alcohol also precipitates late hypoglycaemia and can mask the early symptoms of hypoglycaemia and therefore increase the risk of hypoglycaemic coma.

19.8.3 Dietary treatment of hypoglycaemia

While foods containing 15 g of quickly absorbed CHO (with a high glycaemic index), such as glucose, may relieve the early symptoms of hypoglycaemia under normal circumstances, it may be necessary to have two or three times this amount to treat hypoglycaemia induced after strenuous exercise (Franz 1991). Dietary treatment must be continued until stable blood glucose levels are achieved.

19.8.4 Hypoglycaemic coma

A 50% dextrose solution is injected intravenously to treat hypoglycaemic coma. Fluid or food must not be administered by mouth. If intravenous glucose cannot be given, an intramuscular injection of 1 mg glucagon is used, which induces glucose release from liver glycogen. However, glucagon injections may not work after prolonged exercise if hepatic glycogen stores are depleted, after a marathon or triathlon for example. Rapid medical assistance is crucial for anyone in hypoglycaemic coma.

19.8.5 Impaired temperature regulation

A further risk associated with hypoglycaemia is impaired temperature regulation. If there is risk of hypothermia (in sports such as cross-country skiing) or hyperthermia and potentially hypohydration (in marathon events), then particular care should be taken to maintain fluid balance and glycaemic control.

19.8.6 Hyperglycaemia

The usual causes of hyperglycaemia are infection, overconsumption of food, inadequate insulin and, in some cases, alcohol consumption (when sweet wines, liqueurs or soft drinks are consumed inappropriately). Exercise is not recommended in individuals with severe hyperglycaemia (see section 19.7) as it increases glycogenolysis and lipolysis, leading to even higher blood glucose levels and possibly ketosis. An athlete who exercises with very high blood sugar levels can become confused and disoriented and is at high risk of dehydration from the effects of polyuria. Exercise is contraindicated if ketones are present at the start of exercise.

Hyperglycaemia after exercise is likely to be caused by an overconsumption of food (particularly CHO). Often athletes with diabetes who fear the onset of hypoglycaemia during exercise consume excess food before or during exercise as a preventive measure. Frequent monitoring of blood glucose levels in this situation is again important (Monk et al. 1995; Franz et al. 2002).

19.8.7 Risk of metabolic problems with exercise in athletes with type 1 diabetes

Athletes with appropriate glycaemic control, who participate in ongoing training program of cardiovascular fitness, have no long-term diabetes complications and are aware of hypoglycaemia are at low risk of metabolic problems, especially if exercise is fairly consistent in timing intensity and duration. Such athletes have adopted appropriate schedules for modifying carbohydrate intake and insulin doses when undertaking exercise of more than 30 minutes duration. After prolonged exercise (> 60minutes duration), they are aware of the potential for hypoglycaemia during the following night and arrange appropriate precautions (e.g. setting the alarm to waken them during the night to check blood glucose and take extra carbohydrate if necessary). Some may have a companion or parent who can wake them and treat if necessary.

Conversely, athletes with poor glycaemic control who have irregular, inconsistent exercise schedules (particularly prolonged, intensive or anaerobic) are at high risk of hypo- or hyperglycaemic problems. These problems are made much more likely and more serious if there is misuse of alcohol or the use of mood-altering drugs. If such athletes do not have a reliable companion to monitor them overnight, hypo- or hyperglycaemic coma may not be detected and may become life-threatening (APEG and ADS Guidelines 2011). During a growth spurt, young athletes need extra food to match increased energy requirements and possibly insulin adjustment to avoid large fluctuations in blood glucose levels.

19.8.8 Long-term complications with poorly controlled diabetes

Poorly controlled diabetes is associated with several long-term complications, which include cardiovascular disease; nephropathy leading to renal failure; retinopathy, with potential loss of vision; nephropathy, leading to renal failure; peripheral neuropathy, with risks of foot ulcer and amputation; and autonomic neuropathy, causing gastrointestinal, genitourinary and cardiovascular symptoms and sexual dysfunction (American Diabetes Association 2014b). These complications increase with ageing and with duration of diabetes but can be reduced substantially by improving glycaemic control.

19.8.9 Cardiovascular disease

Poorly controlled diabetes is associated with damage to the large blood vessels (macrovascular disease), which accelerates atherosclerosis and can also damage smaller vessels leading to microvascular disease associated with nephropathy, retinopathy and neuropathy. Any athlete with established type 1 diabetes contemplating a vigorous training program should have a detailed cardiovascular assessment beforehand. With the increased participation in sport by older people, there is an even stronger case for formal cardiac assessment before commencing an exercise program. One of the key findings of the AusDiab study in Australia in 1999–2000 was that about half of the people diagnosed with diabetes did not know they had it (Dunstan et al. 2001). For most people, however, the cardiovascular and psychological benefits of regular physical activity far exceed the risks (NHMRC 2009; Colberg et al. 2010; APEG and ADS 2011).

19.8.10 Peripheral neuropathy

Many people with type 1 diabetes also develop peripheral neuropathy (damage to nerves in the extremities) over time. This results in loss of sensation in the feet and, in older individuals, may be associated with peripheral vascular disease. This increases the risk of damage to the feet, which is not recognised by the usual symptoms of soreness and pain. Foot ulcers associated with repetitive exercise (pounding on hard surfaces or rubbing of footwear) are a high risk for such an athlete with diabetes. Foot ulcers are particularly difficult to heal and often require long hospital admission. The appropriate choice of footwear and advice from a podiatrist are important for all athletes with diabetes involved in weight-bearing exercise (Farrell 2003).

19.8.11 Autonomic neuropathy

Autonomic neuropathy is an abnormality of the autonomic nervous system and may be induced by longstanding, poorly controlled diabetes; it may result in loss of ability to detect hypoglycaemia or in abnormal cardiovascular and thermoregulatory responses to exercise. This complication of diabetes may limit exercise capacity and increase the risk of an adverse cardiovascular event during exercise. Disturbances in blood pressure control that are linked to autonomic neuropathy are more common in people at the start of an exercise program than in those who exercise regularly. A medical assessment to determine the extent of damage from autonomic neuropathy is strongly recommended prior to embarking on an exercise program that involves moderate to strenuous activity.

19.9 Physical activity for people with type 2 diabetes

Physical activity (both aerobic and resistance training, such as weightlifting) is effective in improving glycaemic control, reducing HbA_{1c} (glycated haemoglobin), increasing insulin sensitivity and reducing cardiovascular mortality risk in people with type 2 diabetes (Colberg et al. 2010). There is now unequivocal evidence that type 2 diabetes can also be prevented or at least delayed by physical activity (Gill & Cooper 2008; Orozco et al. 2008) and by weight loss in the majority of at-risk people (Nield et al. 2008b). The National Health and Medical Research Council (NHMRC) (2009) recommends at least 2.5 hours of moderate- to vigorous-intensity physical activity a week (50–70% of maximum heart rate), distributed over 3 days and with no more than 2 consecutive days without physical activity to achieve a preventive and treatment effect. This level of physical activity is required to improve glucose control, independent of weight loss (Orozco et al. 2008). The risk reduction associated with increased physical activity is greatest in those at increased baseline risk of the disease, such as the obese, those with a positive family history and those with impaired glucose tolerance and/or fasting glucose (Gill & Cooper 2008). The prescription for resistance exercise for a risk reduction effect is at least twice weekly on non-consecutive days, but more ideally three times a week, along with regular aerobic exercise (Colberg et al. 2010) (see Colberg et al. 2010 for details of a training program). Nevertheless, some people do not respond to these physical activity interventions. Pooled data from six large-scale diabetes prevention intervention trials involving both physical activity at these levels and weight loss found 2–13% of participants with impaired glucose tolerance still developed the disease (Gill & Cooper 2008).

A combination of dietary intervention with structured exercise and behaviour modification is more effective in improving glycaemic control (Boule et al. 2001; Nield et al. 2008a, 2008b), promoting greater weight loss (Wing 2002) and maintaining long-term weight control (Boule et al. 2001; Saris et al. 2003), than dietary intervention or physical activity alone. While short-term benefits of diet interventions with no exercise are effective, long-term adherence to diet alone (and maintenance of weight loss) in people with type 2 diabetes is poor (Nield et al. 2008a). Moreover, to maintain weight loss after intervention with exercise alone, a large amount of exercise (up to 7 h/wk) at moderate to high intensity is required in previously obese people to sustain the weight lost (Saris et al. 2003). Clearly, these targets are unrealistic for most people.

Type 2 diabetes occurs more frequently in older people who have a variety of age-related disabilities that limit the feasibility of exercise. Nevertheless, increasing numbers of the ‘young elderly’ (biological age <65 yrs) are participating in regular exercise, and often at competitive levels (see Chapter 18). For both elderly and young elderly people with type 2 diabetes, a cardiac assessment prior to commencing an exercise program is warranted. This age group is also likely to have some degree of peripheral vascular disease, so care of the feet in impact sports should be addressed. Hypoglycaemia can occur in people with type 2 diabetes who use insulin or one of the sulfonylurea medications (which enhance insulin secretion). The dosage of these drugs is reduced in people who undertake regular physical activity. Hypoglycaemia is unlikely to occur in individuals using only metformin; insulin-sensitising agents (e.g. acarbose, glitazones or glucagon-like peptide agents); or gliflozins. Because many individuals with type 2 diabetes are overweight or obese, it is usually appropriate to encourage weight reduction and emphasise medication reduction during exercise rather than increasing CHO intake.

19.10 High-risk sports

Individuals with diabetes on treatment with insulin or sulfonylureas are at risk of hypoglycaemia and disorientation. Sporting activities that would pose a risk to themselves or to those around them are best avoided. Hypoglycaemia may occur in any sporting situation where the individual is alone and does not recognise the symptoms or in a situation where glucose cannot be readily administered as treatment. These situations include hang-gliding, scuba diving, solo yachting or motor car racing. For these reasons, Australian authorities do not recommend scuba diving for people with type 1 diabetes (APEG & ADS 2011). Other activities where there is a lesser element of risk—such as cross-country skiing, surfboard riding or long-distance running—should be undertaken only if the individual is accompanied by someone who is able to recognise and treat hypoglycaemia.

19.11 Insulin abuse and sport

Insulin is an anabolic agent that promotes protein synthesis and inhibits protein breakdown in muscle. It also transports electrolytes and fluid into muscle cells, which makes them swell and gives the muscle a better definition. The misuse and abuse of insulin was reported over 20 years ago in bodybuilders and is still happening today. Use of insulin as an anabolic agent has been claimed in other sports, including cross-country skiing. Deliberately using insulin to produce hypoglycaemia to initiate the physiological release of growth hormone has also been claimed. These practices are not without risk, as evidenced in a bodybuilder who developed severe and permanent brain damage after inducing hypoglycaemia by using intravenous insulin (Elkin et al. 1997). There are also risks associated with sharing of needles to give insulin. The forms of insulin used for injection can be detected from blood tests. Athletes with diabetes who are prescribed insulin or other diabetic drugs need medical clearance in case of drug testing. Insulin, its analogues and other anabolic agents are banned substances.

Summary

Athletes with diabetes are able to participate in virtually all sports, apart from a few exceptions that pose a risk because of hypoglycaemia. The dietary recommendations for training and competition for athletes with diabetes are similar to those for athletes without diabetes, provided the blood glucose range is within acceptable levels at the commencement of exercise. Older athletes, and those with longstanding diabetes, need to be screened for complications of the condition, such as cardiovascular disease, retinopathy and neuropathy. All athletes need to be properly instructed on strategies to avoid and treat exercise-induced hypoglycaemia. Frequent monitoring of blood glucose levels should be encouraged. Exercise should not be undertaken in the presence of hyperglycaemia and/or ketosis. Sports coaches of athletes with diabetes should be conversant with the effect of diabetes on athletic performance and, in particular, should be able to recognise and adequately treat hypoglycaemia.

USEFUL WEBSITES

- **www.diabetessociety.com.au**
  Guidelines developed by the Australian Diabetes Society
- **www.apeg.org.au**
  Australasian Paediatric Endocrine Group (APEG): represents those involved in management and/or research of children with diabetes mellitus

- **www.diabetesaustralia.com.au**
  Diabetes Australia: The latest national evidence-based guidelines for management of diabetes
- **http://daa.asn.au**
  Dietitians Association of Australia: A range of publications for health professionals and the general public
- **www.nhmrc.gov.au**
  National Health and Medical Research Council: Evidence-based guidelines
- **www.glycemicindex.com**
  Australian site: Information about the glycaemic index of food and the GI symbol used on Australian food labels
- **www.diabetes.org**
  American Diabetes Association: Position statements for all aspects of management of diabetes
- **www.ausport.gov.au**
  Ausport: Useful information about exercise, sport and athletics in Australia
- **www.diabetes-exercise.org/index.asp**
  Diabetes, Exercise and Sports Association: UK support site for people with diabetes and health professionals
- **www.runsweet.com**
  Runsweet: UK website specific to diabetes and sport

Practice tips

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- Individuals with diabetes present in many ages, sizes and shapes, with exercise habits varying from light and infrequent to multiple sessions per day of moderate to high intensity. An individualised plan formulated by a team consisting of a sports dietitian, the athlete's doctor, diabetes educator and coach is essential to achieving the best overall health and performance outcomes for the athlete.

FORMULATING A PLAN

- Liaise closely with the care team to ensure a clear understanding of the goals of training and the nutrition and insulin regimens that will best support those goals. Establish a plan for how fine-tuning will take place and who you will communicate with regarding such changes.
- Combining the training diary, food and blood glucose record is an effective way to gather useful information, particularly during times of adjustment. Review the athlete's knowledge of carbohydrate foods and carbohydrate counting as required. Care needs to be taken when athletes use carbohydrate-counting databases, to ensure that the information is accurate and based on food data from their country of origin. The same care is needed when athletes use 'common food' lists on their insulin pumps.

IMPACT OF THE INSULIN REGIMEN

- Be confident about an athlete's insulin regimen and the onset, peak and duration of action of different insulins used by the individual; this is critical to developing a successful management plan.
- Multiple daily injections:

- Short or ultra-short acting insulin—the type of short-acting insulin affects timing of the pre-training meal (see section 19.5).
- Is the long-acting insulin taken once or twice daily? (Twice-daily insulin allows for greater manipulation to reduce late onset hypoglycaemia post-exercise).
- If insulin, taken with the meal prior to training, peaks during the training session, additional carbohydrate may be required depending on the individual and length of the training session.

- Insulin pump:
  - Temporary basal rates may be selected on the pump and can be used to reduce the risk of post-training and late-onset hypoglycaemia.
  - Insulin sensitivity settings in the pump can be changed as required.
  - A pump provides greater flexibility with carbohydrate intake.
  - Insulin bolus delivery can be altered, with features allowing a bolus to be split into an initial dose with the remainder delivered over a set time period. This may help with post-training recovery meals, which can be staggered over a few hours.
  - Some insulin pumps will allow storage of CHO content data, which may be useful for common pre/post-training meals.

DIETARY FOUNDATIONS

- The recommended diet for an athlete with diabetes is no different from that recommended for a non-diabetic athlete.
- The primary nutrition goal is for nutrient and energy requirements to match training needs and body composition goals, with insulin and medications adjusted accordingly, rather than distorting food intakes to suit insulin or medication doses.
- Individuals show substantial variation in blood glucose response to exercise, so recommendations need to be made on a case-by-case basis.
- Over-consumption of CHO foods in response to fear of hypoglycaemia during or after exercise can often lead to hyperglycaemia. Athlete re-education and review of the insulin regimen may be required.
- Ensure the athlete has appropriate CHO available for treating hypoglycaemia at all training sessions. Supplies need to be quickly accessible. Symptoms and treatment of hypoglycaemia should be well understood by the coach or a training partner. Aim for an initial ingestion of 15 g of high glycaemic index carbohydrate. This may need to be followed with a further 15 g (or more) of carbohydrate depending on the individual and the circumstances (e.g. timing of next meal or snack and severity of the hypoglycaemic event (see section 19.7.1).

Eating before-exercise

- A CHO-containing meal should be consumed 1–3 hours before exercise, with the amount of CHO dependent upon the type and expected duration of exercise as well as the starting blood glucose levels (see Chapter 12 and Table 19.1).
- If insulin action is likely to peak during the exercise session, an additional 10–15 g CHO taken 20–30 minutes beforehand is usually sufficient to prevent hypoglycaemia, provided the blood glucose level is between 5.6–10.0 mmol/L at the commencement of exercise (see section 19.4.4).

Eating during exercise

- For continuous aerobic exercise lasting longer than 60 minutes, 30–60 g CHO per hour is recommended for optimal performance; to prevent hypoglycaemia, additional insulin is usually not required. These recommendations are the same as those for athletes without diabetes (see Chapter 13). Small amounts every 20–30 minutes may be better than one single feed.

Eating after exercise

- Dietary recommendations after exercise are the same as for athletes without diabetes (see Chapter 14). Including high glycaemic index CHO may help to prevent post-exercise hypoglycaemia.

Carbohydrate loading

- CHO loading, particularly involving a depletion phase, should not be attempted unless appropriate glycaemic control has been regularly maintained. Insulin and medication doses require careful adjusting, depending on the method of loading used and the style of exercise tapering practised before a competition. Even for an athlete with diabetes with well-controlled glycaemia, this adjustment is difficult and the glucose response is often erratic. CHO loading is not recommended for children with type 1 diabetes as they are unlikely to be competing in the sort of events that benefit from CHO loading (e.g. marathon or even half-marathon events).

SPECIFIC CONSIDERATIONS

- *Blood glucose level (BGL) testing* Recording blood glucose levels during training may be difficult and unreliable. Consider the circumstances of each individual athlete: are they in the water, on a bike or rowing? What strategies can help the athlete get the most accurate recordings with minimal disruptions to training? Residual amounts of CHO on the athlete's fingers from bars, gels or sports drinks can contribute to false high blood glucose readings during exercise. Use of continuous glucose monitoring systems (CGMS) may be beneficial at various points in the training cycle.
- *Hypoglycaemia* The increase in insulin sensitivity associated with repeated physical activity means that an athlete with type 1 diabetes is likely to need less total daily insulin once the training pattern is established. Regular checks of blood glucose levels before, during and after training will assist with fine-tuning insulin doses as well as monitoring for hypoglycaemia. Hypoglycaemic reactions may occur up to 36 hours after training. Athletes with type 2 diabetes who are on medication may also need to reduce their dosage. Exercise should be delayed if blood glucose is <3.5 mmol/L (APEG & ADS 2011).
- *Strength training* can induce hyperglycaemia because of the hormonal response to training which stimulates glycogenolysis. This is usually transient.
- *Alcohol intake* Delayed hypoglycaemia is a normal physiological response to alcohol. However in a person with insulin-requiring diabetes, the 'morning-after' effects of alcohol, in combination with insulin, can accelerate a hypoglycaemic state into a potentially life-threatening coma. There is also a risk of hyperglycaemia in the short term if alcohol is consumed with high sugar mixers such as soft drink or juices.
- *Competition* If the exercise is anaerobic, undertaken in the heat or in the presence of competition stress, increased insulin may be required to reduce risk of hyperglycaemia.

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CHAPTER TWENTY

Athletes with gastrointestinal disorders, food allergies and food intolerance

Jeni Pearce and Stephanie Gaskell

20.1 Introduction

Gastrointestinal disturbances occur in some athletes before, during and after competition and training, and are more common in those athletes involved in high-intensity exercise. Most studies on the effects of exercise on gastrointestinal function and physiology have been conducted on subjects undertaking recreational exercise with existing symptoms or gastrointestinal conditions rather than asymptomatic subjects. The results of these studies indicate that exercise can alter blood flow and functional capacity of the gastrointestinal (GI) tract, leading to uncomfortable and debilitating side effects (see Commentary 8).

Typical symptoms of gastrointestinal disturbance (e.g. excessive flatulence, abdominal distension, intermittent diarrhoea, constipation), however, may not necessarily be associated with exercise but linked to an underlying undiagnosed condition such as irritable bowel syndrome (IBS), coeliac disease and inflammatory bowel disease (IBD) or food intolerance. Of concern to sports service staff and coaches, especially when travelling with athletes away from home, is the occurrence of an unexpected adverse reaction to food or a natural ingredient in food. Adverse food reactions range from a mild to moderate reaction (which is localised in the GI tract) to an immune-mediated allergic reaction. Extreme hypersensitivity can induce respiratory distress, skin irritations and anaphylactic shock. Adverse reactions to food that induce mild side effects at home can worsen at competition. Failure to detect, treat and manage chronic GI symptoms and food-related adverse reactions (FRAR) can result in a reduced capacity to train and compete, in withdrawal from competition, and in nutrient deficiencies and adverse long-term health outcomes.

The number of athletes requiring special diets for food allergies and intolerances has increased at each Olympic Games since the 2000 Games in Sydney (see Chapter 25). Whether this increase is a true reflection of an increased prevalence is unclear because many athletes either self-diagnose or choose to avoid or limit specific foods. Testimonials from athletes and the media about excluding specific foods, food ingredients or other naturally occurring compounds in foods (e.g. gluten) to improve wellbeing and enhance sports performance may also contribute to this increase.

This chapter considers the detection, diagnosis and principles of medical nutrition management of GI conditions relevant to athletes: IBS; IBD (Crohn disease and ulcerative colitis); FRAR, including coeliac disease, food allergy, food intolerance (e.g. the low FODMAP diet); and food aversions. Detailed medical nutrition management is outside the scope of this chapter and can be found in any current textbook of dietetic practice. Exclusion of many specific foods and several food groups can occur in athletes with chronic GI conditions and referral to a sports dietitian is essential to avoid chronic deficiency, stress from following food plans limited in variety, unnecessary restriction negatively impacting performance (e.g. avoidance of CHO-based food or key sources of recovery protein) and poor compliance. The risk of nutrient deficiencies and even malnutrition is possibly higher in athletes with undiagnosed and untreated GI tract conditions than the general population because of increased nutrient requirements.

20.2 Causes of gastrointestinal disturbances associated with acute exercise

Avoidance of food before, during or after exercise to reduce adverse GI symptoms is common in many athletes. Such practice is counterproductive to performance and recovery. In the absence of any GI condition, infection or inflammation that affects gut function, symptoms associated with high-intensity exercise are associated with:

- the nutrient composition, osmolality and quantity of food or beverage consumed
- a delay in gastric emptying
- changes to gastric motility
- reduced blood flow to the gut
- very high lactic acid levels (or acidosis).

Nevertheless, the gut can be trained and adapted over time to accommodate increasing volumes of food and/or carbohydrate load with no or occasional adverse effects. Training also improves clearance of lactic acid and reduces the risk of acidosis. The effect of different exercise intensities on gastric emptying and gastric motility is considered in Commentary 8 and varies considerably between individuals.

20.3 Physiological and dietary factors affecting gastric emptying and gut comfort

Gastric emptying is controlled by numerous stimuli mediated by nerves and hormones, which influence the release of digestive secretions from the GI tract. Receptor sites in the mucosal cells in the duodenum and jejunum have sensors which are activated by the presence of acidic chyme, osmotic pressure of food contents (food is hyperosmolar), fat, peptides and amino acids to slow down the rate of gastric emptying. A bolus of food which has been hydrolysed by gastric juice and enzymes empties from the stomach in a fixed amount at a fixed rate (around 3 mL every 20 s). Water empties at a slightly faster rate than the food bolus at around 1 L/h (or 5 mL every 20 s). The presence of food mixed with water delays this rate of gastric emptying. Fat is slow to empty because it forms an oily layer on the top of the hydrolysed food contents. Usually a low-fat (high-CHO) meal is emptied between 1–2 hours after eating while a high fat meal can take up to 6 or more hours. Specific actions of hormones (both intestinal and gastric in origin) which are stimulated by the presence of food and specific macronutrients and nerves in the GI tract can be found in any physiology textbook.

Table 20.1 summarises the main factors that affect the rate of gastric emptying that are relevant to athletes. The outcome of a delay or increase in gastric emptying may cause uncomfortable side effects.

TABLE 20.1 Factors that change the rate of gastric emptying

Increase rate of gastric emptying	Decrease rate of gastric emptying
Large meal or large fluid volume (linked to stomach distension)	Dehydration High energy content of a meal (particularly high-fat meal) High osmolality (high solute load of food or beverage) Intense exercise (>70% VO_{2max}) Stress and anxiety

During exercise at >70% VO_{2max}, the rate of gastric emptying can be delayed because of decreased peristalsis in the mid-oesophagus and upper part of the small intestine (see Commentary 8). For some athletes, residual food in the stomach causes uncomfortable or irritating side effects during exercise. In one survey of 55 male triathletes, more GI symptoms (vomiting and reflux) were reported in those consuming foods high in dietary fibre, fat or protein before competition (Rehrer et al. 1992a). Those triathletes who ate a meal shortly before the event (30 m prior to the swim) were more likely to vomit.

Foods in liquid form empty faster than solid form thereby minimising any adverse effects when consumed before or during exercise. Foods and beverages with a high osmolality (hyperosmolar) (the concentration of molecules or ionic particles in a solution) delay gastric emptying. All foods and most beverages are hyperosmolar. As a benchmark, the osmolality of blood serum is around 300 mOsmoles. Full-cream milk, ice-cream and grape juice is around 275, 1150 and 11700 mOsmoles respectively. Hyperosmolar solutions are thought to cause GI upset because of water retention in the GI tract. In contrast, beverages that are iso-osmolar or isotonic (at the same concentration as blood serum), such as sports drinks (6–8% mixed sugars plus glucose polymers), empty quickly and induce no or minimal adverse effects (see Chapter 13). Hyperosmolar beverages such as fruit juice, cordial and soft drink (between 10–15% sugars, mostly sucrose) delay gastric emptying, which can induce adverse effects such as osmotic diarrhoea when consumed in large amounts during or even just before exercise.

Adverse side effects have also been reported from consuming lower sugar content solutions. Complaints of stomach upset and side ache were higher in athletes consuming an 8% beverage carbohydrate−electrolyte solution compared with a 6% solution consumed during intermittent, high-intensity exercise (76% VO_{2max}) (Shi et al. 2004). Triathletes participating in a half-triathlon who had vomited or experienced the urge to vomit had consumed the hyperosmolar beverage (Rehrer et al. 1992b). Clearly some athletes are more sensitive than others. These studies suggest that a CHO concentration of >6% from simple sugars in a liquid solution is the threshold for no or minimal delay in gastric emptying and low incidence of adverse effects. Since Rehrer's study in 1992, the formulation of sports drinks has evolved to increase the CHO concentration to 6–8% without increasing the osmolality.

Large quantities of food with a high-fat or high-energy content empty slowly from the stomach even at rest and can cause stomach distension. However, during exercise, higher intakes of CHO consumed as gels (glucose and fructose combinations) at a concentration of 1.4 g/min were tolerated by the majority of athletes undertaking simulated races of 16 km (Pfeiffer et al. 2009). These results suggests that a high concentration of CHO presented in a dense bolus rather than in large volumes can reduce adverse side effects. Alternatively, the athletes who participated in this study were accustomed to eating during exercise and had adapted to the high solute load. Nevertheless a higher score of nausea was reported by athletes who consumed the higher overall concentration of CHO (90 g CHO/h) compared with those consuming 60 g CHO/h (Pfeiffer et al. 2009).

20.3.1 Gut trainability

Anecdotal evidence suggests that the rate of gastric emptying in athletes can be slightly increased over time and athletes will gradually tolerate increasing volumes of food and drink during exercise without adverse symptoms (Gisolfi 2000). This adaptive response may be attributed to the repeated experience of ingesting food and drink during exercise. Two studies of endurance athletes confirm an increase in gastric emptying rate and oro-caecal transit time, which provides some evidence for this adaptive effect (Carrio et al. 1989; Harris et al. 1991). Gut adaptations may take up to 20 days (Jeukendrup, pers. comm. June 2014). Adaptation relies on athletes routinely eating and drinking during hard training sessions and not changing these habits at competition, when stress or anxiety levels can further increase the risk of adverse GI reactions.

20.4 Lower gastrointestinal tract conditions

The aetiology of lower GI tract conditions varies and is complex. Stress and anxiety, specific food or food components, and GI infection may trigger and exacerbate adverse symptoms. Genetic susceptibility is evident in people diagnosed with IBS, coeliac disease and IBD (Crohn disease and ulcerative colitis). Changes to gut function associated with exercise described earlier are unlikely to increase the risk of developing these conditions. Diet is the cornerstone of management for these GI tract conditions. Accurate diagnosis of these conditions is also essential to exclude the adverse GI effects consequent to other GI infections (i.e. parasitic, bacterial or viral) which may persist for some time, when untreated.

20.4.1 Irritable bowel syndrome (IBS)

IBS, also termed irritable colon syndrome, substantially impacts quality of life. It is characterised by chronic and relapsing symptoms including abdominal pain, bloating, excessive flatulence and altered bowel movement (constipation, diarrhoea, either alone or alternating) with no abnormal pathology. The diagnostic criteria are listed in Table 20.2.

TABLE 20.2 Rome III criteria for diagnosis of irritable bowel syndrome

Pattern and duration of symptoms	>6 m history of symptoms with presence of symptoms on at least 3 d/m for the preceding 3 m
Symptoms	Recurrent abdominal pain or discomfort with two or more of the following: • improvement with defecation • onset associated with change in frequency of stool elimination • onset associated with change in stool form
Symptoms that suggest IBS but are not included in criteria	Abnormal stool frequency Abnormal stool form Straining at defecation Urgency or incomplete emptying of bowel Passing mucus Bloating

Source: Adapted from Gibson 2012

Individuals with IBS may have co-existing GI disorders such as symptomatic reflux disease and functional dyspepsia as well as extra-intestinal symptoms (e.g. irritable bladder, fibromyalgia, chronic fatigue syndrome, chronic pelvic pain, temporomandibular joint problems) (Gibson 2012). The diagnosis of IBS is by a medical specialist.

The pathogenesis of symptoms in IBS is unclear but may be attributed to visceral hypersensitivity, psychological stress and anxiety, abnormal brain activation, altered colonic motility, abnormal gas propulsion and expulsion and abnormal colonic flora. Several dietary factors may trigger or aggravate the condition including citrus fruits, insufficient or excessive dietary fibre, excess caffeine/coffee, fat, alcohol and malabsorption, although the response varies between individuals. Heavily spiced foods (curry, chilli, cinnamon, horseradish) or fat-rich foods also induce adverse abdominal symptoms in some individuals with IBS. High fat intake increases sensitivity of mucosal cells in the colon. This effect appears exaggerated in IBS sufferers compared to healthy controls (Simren et al. 2007). Salicylates, amines, glutamate and protein may also trigger IBS episodes, which may be indicative of potential food allergy or intolerance (Gibson 2012).

Around two-thirds of IBS patients report reduced thresholds to pain or discomfort relating to rectal distension compared with controls (Prior et al. 1993; Dong et al. 2004; Kanazawa et al. 2011). Substantial individual variation exists in what is considered a normal stool frequency and consistency for people diagnosed with IBS. *Normal* elimination ranges from up to three formed and non-urgent stools/d to one stool in 3 days without abdominal pain.

20.4.2 Medical and dietary intervention for IBS

Treatment options for the prevention and management of symptoms of IBS include medical and psychological intervention (e.g. stool-bulking agents, anti-diarrhoeals, anti-spasmodics, low dose anti-depressants, cognitive behavioural therapy) and dietary intervention (e.g. avoidance of known dietary irritants and high dietary fibre intake, a low FODMAP diet—see below). Individuals with severe debilitating symptoms benefit from a multidisciplinary clinical approach involving expertise from a general practitioner, gastroenterologist, nurse, dietitian and psychologist (Gibson 2012).

The benefit of increasing fibre is questionable for overall IBS symptom improvement and IBS-related constipation. High fibre intake may contribute to bloating, abdominal distension and flatulence, particularly when increased suddenly. Ispaghula (psyllium), a type of soluble fibre, may be tolerated in those with constipation-predominant IBS, when introduced slowly; however it can also increase bloating and flatulence.

The low FODMAP™ diet for IBS

The low FODMAP diet is now recognised as one of the most successful evidence-based dietary therapies for the treatment and management of IBS. The effectiveness of the low FODMAP diet supports the theory of selected and individual dietary triggers as irritants for IBS and symptom relief (Shepherd & Gibson 2006; Shepherd et al. 2008; Gibson & Shepherd 2010; Staudacher et al. 2012; Barrett 2013; Halmos et al. 2014). A low FODMAP diet is recommended as the preferred dietary therapy for IBS management in Australia and is increasing in use and acceptance worldwide (New Zealand, the USA, the UK and Canada). FODMAP is an acronym for:

- Fermentable
- Oligosaccharides (fructans, galacto-oligosaccharides [GOS])
- Disaccharides (lactose)
- Monosaccharides (fructose in excess of glucose)
- Polyols (sorbitol, mannitol, maltitol, xylitol and isomalt)

These naturally occurring compounds in food and added substances to food are found in everyday foods such as milk and yoghurt, wheat, apples, pears, onions, garlic, some stone fruits, some vegetables (asparagus) and, importantly for athletes, in some sports foods and beverages (sports drinks, gels and powders). FODMAPs share three common features:

- partial to no absorption in the small intestine
- osmotically active compounds
- rapid fermentation by GI bacteria.

These undigested or partially digested compounds induce undesirable side effects, such as osmotic diarrhoea when consumed in large quantities. The human gut does not absorb oligosaccharides. Dietary restriction of fructose, fructans (e.g. in wheat, onions) and galacto-oligosaccharides is likely to reduce symptoms in the majority of IBS sufferers (Shepherd & Gibson 2006). These naturally occurring or added compounds in commercial foods in the FODMAP list do not cause IBS; however restricting the intake provides an opportunity for reducing symptoms.

One of the first studies of the FODMAPs concept was undertaken in Australia, where 75% of patients with IBS who were prescribed a diet restricted in fructose and fructans reported sustained relief from GI symptoms (Shepherd & Gibson 2006). Other studies since then have confirmed an alleviation of symptoms at similar frequencies (Shephard et al. 2008; Staudacher et al. 2011). A low FODMAP diet may also be effective in managing GI symptoms in 70% of those with quiescent Crohn disease and ulcerative colitis (Barrett 2013).

Food sources of FODMAP components

Detailed information about food sources of FODMAPS is found elsewhere (e.g. www.med.monash.edu/cecs/gastro/fodmap). Athletes diagnosed with IBS do not necessarily need to exclude all foods in the FODMAP list—hence the importance of isolating the known irritants in individuals. Hydrogen and methane breath testing can be used to detect fructose, lactose and polyol malabsorption but there are no definitive laboratory tests available for detecting the other compounds. Detection may be based on self-report of known food irritants, which is not reliable. Ideally, a detailed food and symptom record helps to isolate dietary triggers. This method is not necessarily valid or reliable as a diagnostic technique because collecting accurate dietary intake data for the few weeks requires a motivated and literate client and is time consuming for both client and clinician.

Only a small number of monosaccharides found in nature (i.e. glucose, fructose and galactose) are absorbed by the human gut. The capacity of the human gut to digest and absorb dietary carbohydrate and other compounds in the FODMAP list is variable between individuals and can be affected in several ways as described below.

Oligosaccharides (specifically fructans, galacto-oligosaccharides) malabsorption

Oligosaccharides contain 2 to 20 sugar molecules and are usually soluble in water. Unlike ruminants, humans have lost the enzymes needed to digest oligosaccharides, fibre and resistant starch so there is no absorption. Undigested starches (resistant starch), different types of fibre and oligosaccharides become a substrate for gut bacteria, which leads to production of volatiles resulting in flatulence and discomfort. As discussed earlier, some individuals are more sensitive than others to undigested remnants of these molecules and can suffer debilitating symptoms.

Disaccharide malabsorption

The three most important disaccharides in human nutrition are sucrose, lactose and maltose. Sucrose (common table sugar extracted from sugar beets or sugar cane) is seldom malabsorbed (Kneepkens et al. 1984) or associated with adverse effects unless there is inflammation or substantial bowel resection. Specific enzymes from the brush border of the upper small intestine *digest* these disaccharides into monosaccharides ready for absorption. Maltose is seldom found naturally in food, is formed from starch hydrolysis during digestion and may be added to processed foods. It is unlikely to be eaten in large quantities in the human diet. In contrast, lactose found in milk and dairy products is consumed in large amounts by children and adults in many cultures. Declining lactase activity is a normal physiological response after weaning in humans and other mammals. Lactose malabsorption (or lactose intolerance) is higher in populations of Asian, African, South American and Mediterranean origin compared to others (e.g. Caucasians). Section 20.7.3 considers lactose intolerance in more depth.

Fructose malabsorption

Fructose malabsorption is a normal phenomenon affecting approximately 34% of healthy individuals without IBS, compared to 45% of individuals with IBS and ~70% of people with ileal Crohn disease (Barrett 2013). Large individual variation exists in the absorptive capacity of fructose with some individuals failing to absorb as little as 5 g (equivalent to one medium apple or pear). Fructose malabsorption occurs when fructose is consumed in excess of glucose or there is a high fructose load. The addition of glucose present in the food consumed and/or galactose facilitates fructose absorption, possibly by facilitated diffusion or active transport. Glucose stimulates fructose absorption in a dose-dependent mechanism. Equimolar amounts of glucose and fructose can override malabsorption of fructose and diminish GI symptoms (Rumessen et al. 1986; Truswell et al. 1988). Athletes with fructose sensitivity need to check the composition and ratio of sugars when selecting sports drinks or gels, which contain a combination of monosaccharides, disaccharides and glucose polymers. High-fructose corn syrup, used as a sweetener in processed foods, popular in North America, should be avoided.

Exercise may also decrease the absorptive capacity of fructose as reported in a study of 10 healthy male volunteers (18–38 y) with no history of GI problems (Fujisawa et al. 1993). Unfortunately, this study did not report any effects on GI symptoms.

Polyols malabsorption

Unlike oligosaccharides, polyols (sugar alcohols) are slowly absorbed by around 30% of the population by passive diffusion in the small intestine with no or minimal adverse effects reported by malabsorbers (Barrett 2013; Yao et al. 2013). Polyols occur naturally in foods in small amounts and are used as sugar substitutes and humectants in processed foods. They contain around 10 kJ/g compared to 20 kJ/g of other sugars (Thomas & Bishop 2007). The two most commonly occurring polyols in the diet are sorbitol and mannitol. Interestingly, individuals with IBS often exhibit adverse symptoms to these polyols, which are attributed to either active fermentation or an osmotic effect resulting in fluid retention in the small intestine (Yao et al. 2013). The restriction of polyols is included in the introduction of a low FODMAP diet irrespective of an individual's absorptive capacity. Other polyols are erythritol, maltitol, lactitol, xylitol and isomalt.

When polyols are consumed in large amounts of around 30 g/d in adults (e.g. from chewing gum, cereal bars, sports bars) they can induce a laxative effect because of the high solute load (i.e. osmotic effect) (Thomas & Bishop 2007). Manufactured foods containing polyols may be labelled *excess use/consumption may have a laxative effect*.

20.4.3 Coeliac disease

Coeliac disease (CD), also referred to as gluten-sensitive enteropathy, is an inflammatory autoimmune response to the specific peptide fractions of proteins (glutenins and prolamins), commonly called gluten, found in wheat, rye and barley. Although an allergy to oats is rare, some coeliac sufferers react to the specific protein in oats (avenins), although Coeliac Associations from several countries (e.g. the UK, Australia, the USA, Finland, the Netherlands and Canada) suggest that uncontaminated oats should not be excluded in a person with coeliac disease. Oats can be contaminated if grown in close proximity to a wheat crop or by cross-contamination in the food processing.

Onset of CD may be acute and induced by infection, inflammation or stress in an individual with a genetic predisposition or it may be present for years with minimal or nonspecific symptoms. Inflammation causes damage to the mucosal cells and villi of the small intestine, impairing digestion and absorption of many nutrients. Gluten sensitivity, another related condition, presents with non-specific symptoms without the local and systemic immune response or the intestinal damage evident in people with CD. Gluten intolerance is the term used to describe individuals with either condition who experience GI symptoms following gluten ingestion with or without CD. A variant of coeliac disease, dermatitis herpetiformis, involves the skin, and is accompanied by muscle and joint pain. Other autoimmune conditions may co-exist with coeliac disease (e.g. type 1 diabetes and collagenous colitis, which affects the ascending caecum).

If CD is untreated, chronic inflammation and immune system involvement eventually results in enough damage to compromise the normal secretory, digestive and absorptive function of mucosal cells. Atrophy and flattening of the villi, an outcome of extensive damage, further reduces absorptive capacity of nutrients and results in reduced secretions from the gall bladder and pancreas.

Symptoms are variable to non-existent and can include abdominal distension, pain, diarrhoea, constipation, fatigue and nausea. Symptoms usually occur soon after consumption of gluten. A deficiency of vitamin B12 and iron, despite adequate intake, is an alert for the presence of undiagnosed CD, particularly in those people with non-specific symptoms. Other signs include unexplained weight loss, lethargy and chronic fatigue.

Diagnosis of coeliac disease

A combination of symptoms, specific tests for the presence of antibodies, histological assessment (i.e. lymphocyte counts in the superficial epithelial cells of the distal duodenum) and family history confirm the diagnosis. An antibody test and a biopsy of the intestinal site does not always confirm CD, unless the disease is active. Many people may have avoided gluten for a long time before testing and therefore testing for coeliac serology or taking biopsies is unreliable. To stimulate an inflammatory response prior to testing, an adequate intake of gluten (usually four slices of gluten-containing bread or equivalent in adults consumed most days for six weeks prior to testing) is the usual challenge. Alternatively, another option—which does not require a gluten challenge—to exclude a diagnosis of CD is a gene test for markers of CD. An absence of HLA DQ2 and DQ8 gene markers helps to exclude CD, although the presence of these genes does not provide a definitive diagnosis in isolation. Only 1 in 40 people with a positive gene test develops CD (Anderson et al. 2013). A gluten-free diet should not be started until CD diagnosis is confirmed.

Treatment for coeliac disease

Lifelong adherence to a gluten-free or low gluten diet is the only treatment for CD because the autoimmune response to gluten is permanent. Improvement in symptoms is usually experienced within 2–8 weeks of commencing the diet. Once gluten is removed, the autoimmune process and the intestinal mucosa progressively return to normal or near-normal function. Inadvertent gluten consumption is the most common cause for a non-response to diet therapy; however, other coexisting disorders may contribute. Untreated CD is associated with an increase in morbidity and mortality with long-term complications including osteoporosis and lymphoma. Development of the several forms of anaemia (iron, vitamin K, folate, vitamin B12) due to malabsorption and increased bone damage (osteomalacia, osteopaenia, fractures) are avoidable in well-controlled coeliac disease. Some athletes are already at risk of iron deficiency and stress fractures (see Chapter 9 and 10), which may increase with untreated coeliac disease.

Is a wheat- or gluten-free diet nutritionally adequate for an athlete?

Exclusion of wheat-based products (e.g. flour, bread, wheat-based breakfast cereals), a staple grain for athletes in western countries, increases risk of suboptimal intake of several macro- and micronutrients, particularly carbohydrate, fibre, B vitamins, iron and zinc, unless adequate substitutes are consumed. There are no known dietary intake studies of athletes who avoid or limit gluten or wheat products. In an Australian study of 55 diet-experienced adults with coeliac disease (80% females), compliant with a gluten-free diet for 12 months, and 50 newly diagnosed coeliacs (18–71 y, 24% males), males and females in both groups ate a nutrient-dense diet and achieved most of the EAR/AI cut-offs, which suggests a 50% probability of nutrient adequacy (Shepherd & Gibson 2013). Of concern in this study was the high incidence of intakes <RDI/AI for fibre, folate, thiamin, calcium, iron (in women) and zinc (in men). However, retrospective dietary analysis of the newly diagnosed group found that consumption of these same nutrients were inadequate before commencing the diet, with the exception of thiamin.

The results of this and other dietary studies of people following gluten-free diets (Thompson et al. 2005; Wild et al. 2010) have implications for athletes, who are likely to have higher iron, folate and B vitamin requirements than sedentary people. Thiamin, folate and several B vitamins act as cofactors in energy metabolism, have a high turnover and potential increased requirement for athletes involved in endurance training programs (see Chapters 10 and 11). In Australia, bread flour is fortified with thiamin (and iodine). Most wheat-based commercial breakfast cereals are fortified with other B vitamins (including thiamin), iron and folate. Gluten-free foods have less fibre, less folate and less thiamin than the wheat-based alternatives. Hence athletes following gluten- or wheat-free diets are at high risk of suboptimal intake of these nutrients.

20.5 Inflammatory bowel disease

Inflammatory bowel disease (IBD) is a chronic immune condition of unknown aetiology involving inflammation of the small and/or large intestine. The two main types of IBD are Crohn disease and ulcerative colitis; a less common type is microscopic colitis. Although there is a low prevalence of these diseases in the population, the peak incidence for both diseases occurs in late adolescence and early adulthood (Loftus 2004; Thomas & Bishop 2007).

Periods of remission are followed by periods of relapse. Currently there is no cure for IBD; however, ongoing medical management helps reduce disease activity and improves quality of life and participation in sport. Anaemia, fatigue, hypoproteinaemia and electrolyte abnormalities can occur depending on the severity and duration of IBD attacks. Diagnosis is confirmed by endoscopy and histology.

20.5.1 Crohn disease

Crohn disease involves inflammation of the full thickness (all layers of the mucosa) of different or multiple segments of the small and large intestine but can involve any part of the GI tract from mouth to anus. The terminal ileum (referred to as ileitis) and the colon (Crohn colitis) are the predominant sites affected. Healthy segments may be separated by inflamed sections. Crohn disease may be accompanied by ulceration, fistulas, thickening of the sub-mucosal intestinal wall leading to narrowing and partial or complete obstruction in the lumen. Typical symptoms of active disease include abdominal pain, discomfort, diarrhoea, fever, nausea, vomiting, loss of appetite, weight loss and growth impairment in children and adolescents.

20.5.2 Ulcerative colitis

Ulcerative colitis involves inflammation of the lining of the large intestine and rectum. If only the rectum is involved, which is the most common clinical manifestation of the disease, it is referred to as ulcerative proctitis. Typical symptoms include fatigue, abdominal pain, discomfort, diarrhoea, rectal bleeding and the presence of mucous in the stool. Ulcerative colitis increases the risk for colorectal cancers, especially if the disease is untreated and present for a long time and affects the majority or all of the colon (Thomas & Bishop 2007). Iron deficiency associated with chronic or acute blood loss or the anaemia of inflammation is a common occurrence. Severe attacks require hospital admission and support and can be fatal.

20.5.3 Nutrition-related concerns for athletes with untreated IBD

Malnutrition and weight loss are evident in people with untreated IBD, especially Crohn disease, and are risk factors for postoperative complications. Enteral or parenteral nutrition may be needed when malnutrition is evident. It is unlikely that an athlete with active Crohn disease will participate in exercise because of the presence of adverse symptoms (nausea, diarrhoea, lack of appetite); however, physical activity is not considered harmful when the inflammation resolves or abates (Loudon et al. 1999). The degree and severity of IBD symptoms dictate participation in sport.

People (and possibly athletes) with IBD may adopt restrictive-energy diets or eliminate known food aggravates because of fear of evoking pain or to control a flare up. This practice contributes to suboptimal nutrient intake, weight loss and loss of lean body mass, even when the condition is inactive. Maldigestion and malabsorption is expected when mucosal inflammation is active, as well as after gut resection or bypass, which occurs in the majority of Crohn disease sufferers at some stage. People with untreated IBD are at risk of developing osteomalacia, osteopaenia, lactose intolerance and deficiencies of fat-soluble vitamins as well as magnesium, iron and zinc (Garrow et al. 2000). The incidence of anaemia is high. Although chronic malabsorption is the main reason for malnutrition and nutritional risks, psychological and social factors and medication side effects also contribute, as outlined in Table 20.3.

TABLE 20.3 Nutrients at high risk of deficiency in patients with untreated IBD

Nutritional risk	Symptoms or markers of risk	Reasons
Protein energy malnutrition	Nausea, poor appetite, weight loss, increased urea (nitrogen loss)	Low total energy intake, malabsorption, side effects from medication, elevated protein needs for tissue repair and regeneration
Vitamin B12	Fatigue, anaemia (pernicious)	Malabsorption because of resection or chronic inflammation of the

		terminal ileum
Vitamin B6	Fatigue, anaemia	Increased requirement for physical activity Malabsorption
Folate	Fatigue, anaemia	Malabsorption, medication use (e.g. methroxate, sulfasalazine), low residue diet (prolonged use), suboptimal intake
Zinc	Poor wound healing, taste changes	Malabsorption, potential loss into fistula Resection of jejunum (or part thereof)
Iron	Fatigue, anaemia	Malabsorption, blood loss from ulceration or epithelial cell damage, poor intake, chronic blood loss
Vitamin D and calcium	Osteopaenia, osteoporosis, osteomalacia	Malabsorption, steroid therapy for suppression of inflammation, avoidance of dairy
Sodium and potassium	Blood markers, nausea, electrolyte disturbances	Vomiting and diarrhoea
Fat soluble vitamins	Steatorrhoea	Malabsorption

20.5.4 Medical nutrition therapy for athletes with IBD

The main objective for medical intervention for IBD is to control inflammation and prevent complications associated with malabsorption. Maintaining nutritional status during active and remission states helps prevent malnutrition and promote recovery. Medical nutrition therapy for active IBD is outside the scope of this textbook and can be found elsewhere (Mahan et al. 2012). For an athlete with active IBD, improving health and wellbeing and achieving and maintaining appropriate body mass for their sport is the first step to address capacity to train. Specific nutrition intervention for performance enhancement is introduced slowly and according to individual response and adaptation. Individuals should be encouraged to follow a normal healthy training diet; however, the diet may need to be modified during relapse. While there is no specific diet therapy for IBD, small, frequent, high-energy, nutrient-dense meals and snacks are beneficial and possibly better tolerated than large meals (Thomas & Bishop 2007). Fortified meals, liquid meal replacements, nutritional supplements and sports drinks are beneficial when appetite is poor and fluid requirements are high (see Practice tips). If there is chronic diarrhoea, recurrent vomiting or a fistula in place, measuring hydration status helps ensure fluid needs are met (see Commentary 7). During times of remission, a return to a normal diet is recommended.

Some individuals with IBD have co-existing IBS. A low FODMAP diet is effective in managing the symptoms of IBS so may be indicated for IBD sufferers during the active phase of the condition.

20.6 Other factors influencing GI comfort

20.6.1 Belching, bloating and flatulence

Abdominal distension, belching, bloating and flatulence are normal physiological responses. An average of 700 mL of flatus (gas) is passed each day and approximately 200 mL is found in a healthy GI tract, with amounts varying daily and between individuals (Mahan et al. 2012). Excessive gas can become uncomfortable in those who eat high-carbohydrate and high-fibre diets or suddenly switch to a higher fibre intake. The main cause of belching and flatulence is swallowing air (aerophagia) which occurs when food is eaten too quickly; from drinking carbonated drinks; or gulping air when talking, laughing or exercising. Some air is removed by belching. The rest is swallowed and most of the oxygen is reabsorbed in the stomach. The remaining nitrogen moves into the anaerobic intestine where it can cause distension, bloating, rumbling and pain, followed eventually by rectal flatulence. Another source of flatus is carbon dioxide, which partly comes from sodium bicarbonate, produced from a chemical reaction between hydrochloric acid and fatty acids in the stomach. More volatile gases (e.g. carbon dioxide, methane, nitrogen indoles, skatoles, hydrogen sulphide, methyl sulphides) are produced by fermentation of undigested dietary carbohydrate and fibre in the colon. These also contribute to rectal flatulence.

Undigested starch, oligosaccharides and foods high in fibre, particularly legumes and cruciferous vegetables (cauliflower, cabbage, broccoli) are the main flatus producers. Malabsorption of disaccharides secondary to gut inflammation or lactose intolerance may also contribute. Often ignored are the sugar substitutes (e.g. sorbitol, mannitol) consumed in substantial amounts in carbohydrate-modified foods (e.g. as chewing gum) and added to processed food for their humectant properties (see section 20.4.2). These polyols have been shown to induce GI symptoms—bloating, abdominal pain and flatulence (Hyams 1983). Sorbitol is found naturally occurring in apples, pears, stone fruit as well as some medicines and cough mixtures which may help explain the anecdotal reports of excess flatus after ingestion of these foods and pharmaceutical products. However, individual response to food varies. Usually over time and repeated use of known offending foods, the gut adapts and abdominal side effects diminish. It may take around 2–3 weeks for the GI tract to adjust when switching from low to high intakes of fibre-rich foods. Swallowing air can be minimised by eating slowly, chewing with the mouth closed and avoiding drinking through straws.

Another often-overlooked reason for GI discomfort in women is stage of the menstrual cycle. GI symptoms are reported to be highest in the pre-menstrual period and may increase or abate when menses commences (Bernstein et al. 2014). Symptoms may exacerbate in IBD and IBS sufferers (Moore et al. 1998; Kane et al. 1998; Bernstein et al. 2012). In one study, around 30% of women reported an increase in GI symptoms at menstruation (Moore et al. 1998). Females with IBD reported similar increases at menstruation while those with Crohn disease had increased incidence of diarrhoea at this time, compared with those with ulcerative colitis or IBS (Kane et al. 1998). Symptoms may be linked or confused with the typical symptoms of pre-menstrual syndrome (PMS), for example, fluid retention, bloating, abdominal pain.

Medical referral is recommended if GI and/or PMS symptoms are severe and substantially impair training capacity.

20.7 Food-related adverse reactions (FRAR)

The term *food-related adverse reactions* (FRAR) includes food allergy (immune-mediated) and food intolerance (non-immune mediated). Figure 20.1 shows the pathophysiology of adverse food reactions. Idiosyncratic and psychological reactions to food also induce adverse GI tract reactions and other generalised systemic symptoms. Generalised symptoms include skin rash, urticaria, asthma or breathing difficulties, vasoconstriction, and in those people who are extremely hypersensitive, life-threatening anaphylaxis. The degree of sensitivity varies between individuals for both immune-mediated and nonimmune-mediated reactions. Individuals may react differently to the same allergen (i.e. have different symptoms on repeated exposure), which may be dose dependent or related to other psychological or physiological stressors.

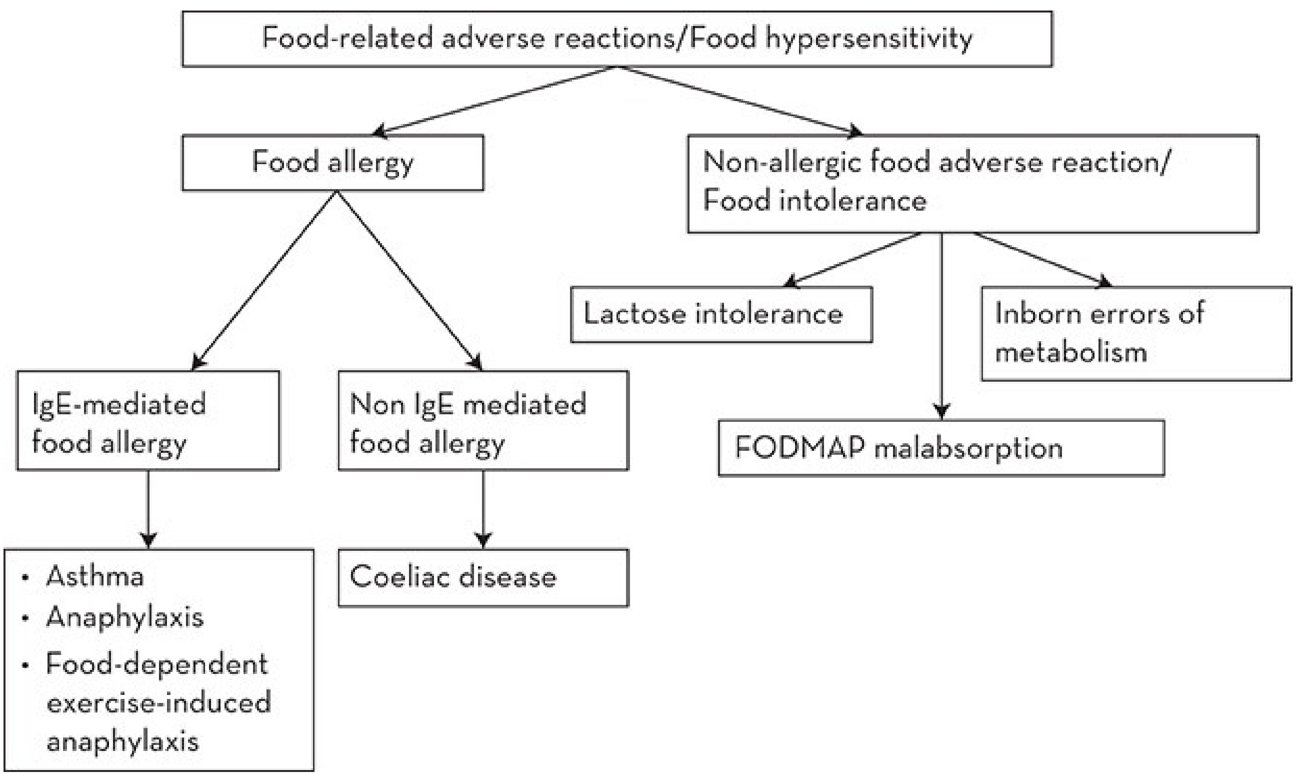

Figure 20.1 Pathophysiology of food-related adverse reactions

Source: Adapted from Mahan et al. 2012

Differentiating between a physiological allergic response or intolerance to food and a psychological aversion is often difficult. An individual's claims of adverse reactions to specific foods and avoidance of those foods may be indicative of an underlying eating disorder, fear or panic attack, a lifelong aversion or a misdiagnosis in infancy or childhood. A clinician with a good understanding of food composition can readily detect inconsistencies in reported responses to different foods and suspect an incorrect diagnosis. The prolonged omission of specific foods or food groups can compromise nutritional status with an athlete likely to be at increased risk because of their high energy and nutrient requirements. Appropriate methods of detection, isolation and diagnosis of food-related allergens and irritants is essential and should be confirmed in those athletes claiming an adverse reaction to food.

20.7.1 Food allergy

Food allergy involves an immunological or allergenic response to a food (or food component) that the body identifies as harmful or foreign. Food that the immune system recognises as a harmful substance (i.e. an allergen) is usually a protein or peptide fraction in food that induces adverse symptoms in susceptible individuals. Peanuts are a common allergen with serious and often immediate symptoms in those who are hypersensitive. A typical food allergy response involves an IgE antibody or cell-mediated immune reaction which occurs consistently after ingestion, inhalation or touch of a particular food, causing functional changes to target organs. The onset of the immunological effects can be within minutes to one hour or more (usually IgE-mediated response) or be delayed (cell-mediated response), which occurs many hours (~48–72 h) after ingestion. A range of adverse GI tract or generalised systemic effects of varying severity can occur. Of concern to coaches and support staff are those athletes who are extremely hypersensitive and exhibit symptoms of respiratory distress, skin irritations and anaphylactic shock. Food-induced anaphylaxis is an acute, often severe and potentially fatal response that usually occurs within minutes of exposure to the food. Examples of common food allergens and typical symptoms are listed in Table 20.4. Further information on the mechanism of the immunological response to food can be found elsewhere (Thomas & Bishop 2007; Mahan et al. 2012).

TABLE 20.4 Examples of food-related-adverse reactions (FRAR) associated with an allergenic immune-mediated reaction

Allergen	Comments
Milk	Adverse reactions usually occur with cow's milk Other mammalian milk may not be allergenic The response varies and length of exclusion required is very individual Soy milk is substituted
Egg yolk	Tolerance usually occurs by 5 years of age in 80% of cases with 20% of sufferers continuing to react as adults During infancy, all eggs, egg products and food made with eggs are avoided (e.g. egg lecithin, egg pasta, cakes) Egg white can be used as a binding agent
Peanut and other tree nuts	Peanut (a legume) allergy induces a severe and potentially anaphylactic reaction, compared with other food allergens Peanut oil is avoided An allergy to tree nuts may or may not accompany a peanut allergy and includes walnut, almond, hazelnut, cashew, pistachio, pine nuts and brazil nuts

Soy	Soy allergy is often outgrown and affects less than 0.5% of the population It may be present with other food allergies
Wheat	Wheat allergy may be an intolerance and needs to be separated from coeliac disease and IBS
Fish and shellfish	The incidence of allergy to fish is unknown Allergic reactions can be severe from shellfish and fish or when combined and may be related to specific fish species or type (e.g. white fish, not tuna) Shellfish allergy appears more commonly in adults than children
Kiwifruit	Adverse reactions include rash and loose bowel movements in children, especially under the age of two years after consuming more than one kiwifruit/d, and in some adults with intakes above two kiwifruit In some individuals, the reaction causes anaphylaxis
Sesame seed	An emerging allergen Very little is known
Celery	Reaction ranges from mild to severe (anaphylaxis) Four specific allergens have been identified: three are inactivated by cooking The other allergen, Api g 2, is not denatured by cooking, by gastric acid or by digestive enzymes (Gadermaier et al. 2011), which may be the main trigger for hypersensitivity, although the amount present is very low

Coeliac disease and other autoimmune inflammatory bowel diseases (such as ulcerative colitis, other forms of colitis, gastroenteritis) also evoke cell-mediated immunoglobulins or non IgE-mediated reactions. Some people with coeliac disease experience severe food-related hypersensitivity reactions when they ingest even small amounts of gluten-containing foods.

Some individuals have allergic reactions to several food allergens (e.g. eggs and nuts, gluten and milk, peanut and other tree nuts, eggs and soy).

20.7.2 Food intolerance

Other adverse reactions to food that are not immune mediated fall into the categories of food intolerance and psychological aversion. Food intolerance affects those people with inborn errors of metabolism (e.g. galactosaemia, phenylketonuria) and with maldigestion where there is mucosal inflammation in the epithelial cells of the GI tract, which impairs digestive enzyme activity (e.g. IBS, Crohn disease). Lactase deficiency is the most well-recognised digestive enzyme deficiency worldwide (Mahan et al. 2012). Some people with food intolerance are very hypersensitive and exhibit similar symptoms to those with food allergy (see Table 20.5), although most experience the typical symptoms of GI discomfort. Food intolerance can be caused by toxic, pharmacologic, metabolic or idiosyncratic reactions to the naturally occurring chemical compounds in food (e.g. lactose, FODMAPS, salicylates, amines) or to those substances added to food during processing (e.g. monosodium glutamate, food colours), preservatives (e.g. sulphites). Allergy-like intoxications can be mistaken or confused with food allergies because the symptoms are similar.

TABLE 20.5 Examples of specific food compounds associated with adverse reactions

Cause	Food sources	Symptoms
Monosodium	Found naturally in mushrooms, tomatoes	Flushing (key characteristic),

07:37:45.

glutamate (MSG)	Used as a flavour enhancer in processed food Used extensively in Asian cuisine as a tenderiser	headache, abdominal pain, chest pain, nausea several hours after ingestion
Sulphites	Found naturally in dried fruit Used as a preservative to prevent browning in wine, vinegars, beer, jam, baked goods (apple pie), some fresh and dried fruits and vegetables, added to some meat to retain redness	Asthma (usually in asthmatics), potential anaphylaxis
Salicylates	Found naturally in many fruits and vegetables especially dried fruit (currants), oranges, herbs and spices (especially curry powder, mustard, paprika) Found in medications such as aspirin	Rhinitis, urticaria, asthma, GI inflammation, rash, itching, headaches, can cause anaphylaxis
Caffeine	Chocolate, coffee, tea (to a lesser extent), energy drinks	Sweating, palpitations, rapid pulse, increased respiration
Amines (e.g. tyramine, serotonin, tryptamine)	Chocolate, cheese (mature and blue), fermented food, (sauerkraut), meat and yeast extracts, red wine and fruit (citrus, banana, avocado and pears)	Powerful vasoconstrictors: migraine, headache, nausea and giddiness
Food colourings[a] (tartrazine) and food additives/preservatives)	Soft drink, processed foods, includes sports drink powders, ready to drink, gels	Hives, rash, asthma (in susceptible individuals)
Lactose, fructose	Lactose (milks and dairy products) Fructose (some fruit such as apples, pears, mangoes, fruit juices and vegetables including asparagus and artichokes, sports drinks, high-fructose corn syrup)	Mainly GI symptoms

[a] Food colourings that induce adverse effects are no longer approved for use in processed foods. Only a limited number of colours are approved for use in food processing based on Codex Alimentarius guidelines. See www.codexalimentarius.org

The food and food compounds listed in Table 20.5 are mainly associated with food intolerance and cause non-immunological reactions to food, which range from gut irritation to systemic reactions. Some compounds induce a histamine reaction, which causes pharmacological reactions including vascular constriction and respiratory problems.

20.7.3 Lactose intolerance

Undigested lactose, the carbohydrate present in the milk of all mammals and in most products made from milk, is a common cause of adverse GI symptoms in many adults. Lactase, the specific enzyme secreted by the brush border intestinal cells, converts lactose into the monosaccharides, glucose and galactose, ready for absorption. Its activity is highest in the neonatal and suckling period, then declines rapidly during childhood. Most adults have some degree of lactose intolerance, called primary lactase deficiency, because they are homozygous for a recessive autosomal allele that regulates the physiological decline in lactase activity. However, adults whose ancestry derives from central or northern Europe or from other milk-drinking populations in the distant past (e.g. North Africa, Arabia) retain the ability to digest lactose throughout life. These populations are heterozygous or homozygous for the dominant allele that encodes for lactase activity.

Even in these population groups, lactose intolerance can be acquired as a result of GI inflammation (e.g. from infectious diarrhoea, active coeliac disease or IBD). This intolerance is a temporary condition, termed secondary lactose intolerance, in those who normally tolerate lactose. It usually resolves on recovery—unless there is gross ileal damage or partial resection of the small intestine. Another cause of lactose intolerance is congenital lactase deficiency—a rare condition. Lactose intolerance is unlikely in ulcerative colitis. However, people with IBD and Crohn disease often avoid milk and dairy foods in the belief that these foods exacerbate GI symptoms. Although they may be lactose tolerant, these food choice behaviours may be entrenched and difficult to change. Severe infection or intestinal damage can also result in the malabsorption of other disaccharides, particularly sucrose and maltase, which can exist alongside lactose intolerance.

With declining or absent lactase, lactose is maldigested and becomes a substrate for fermentation by gut bacteria, resulting in adverse symptoms including abdominal distension, flatulence and explosive watery diarrhoea (osmotic diarrhoea). To relieve these symptoms, high lactose-containing foods are limited but rarely need to be avoided. Small amounts of lactose are usually well tolerated in individuals with decreased lactase activity. Regular intakes of lactose improve tolerance because of colonic adaptation.

Food sources of lactose and alternative lactose-free or low-lactose substitutes for dairy products are found in the Practice tips (see Table 20.8). Other sources of lactose are found in pharmaceutical preparations (e.g. as fillers in tablets as bulking agents), in sports bars, cereal bars and a range of recovery sports drinks and formulations that contain casein or milk solids. Calcium supplements may be indicated for those athletes who are unable to meet high calcium intakes from food sources alone because of avoidance of dairy foods, a rich source of calcium (e.g. female athletes with amenorrhoea, male and female adolescents)(see Chapter 9). Regular bone scans may be necessary for long-term sufferers of lactose intolerance (and dairy food avoiders), especially endurance or weight class athletes who restrict energy intake.

20.8 Considerations for athletes with food-related adverse reactions (FRAR)

In athletes, the adverse symptoms of a food reaction may be masked or exacerbated by high-intensity exercise. A correct diagnosis of the food allergen or irritant can be successfully managed by dietary intervention. The general population alter their diet based on the belief of an adverse reaction to food or food ingredient and athletes are likely to share similar beliefs. Athletes can be easily influenced by others to avoid or include specific foods if performance enhancement is a potential outcome. For some athletes, these practices may be harmless and improve the quality of the diet: for example, avoiding wheat products may reduce intake of high-fat fried battered foods. However, when many foods or several food groups are modified without appropriate substitution (e.g. avoiding grains, fortified breads and cereals), the quality and nutrient density of the diet is compromised. Other concerns include the potential effects of the inadvertent ingestion of a food allergen leading to delayed life-threatening reactions causing illness, especially during training or competition.

20.8.1 Life-threatening FRAR

Anaphylaxis is the most serious and life-threatening adverse reaction to food. Symptoms can include swelling of the mouth, tongue and throat, loss of consciousness, collapse and death, and may be dose-related. Strenuous vigorous exercise undertaken by an athlete with multiple allergies, with poorly controlled asthma, or with an upper respiratory tract infection can often increase the risk of onset and severity of an adverse reaction to food.

20.8.2 Food-dependent, exercise-induced anaphylaxis (FDEIA)

Some athletes are susceptible to exercise-induced anaphylaxis following submaximal exercise of short duration with no food allergy involvement (Robson-Ansley & Du Toit 2010). Others may have an exercise-associated food allergy, which is a rare type of food allergy where a food triggers an anaphylactic response or reaction when an individual participates in exercise within 2–4 hours after ingestion or even within an hour after exercise (Briner 1992; Du Toit 2007; Robson-Ansley & Du Toit 2010). The higher the intensity of the exercise the sooner and more severe the reaction (Robson-Ansley & Du Toit 2010). In the absence of any exercise, the specific trigger food may not be associated with any symptoms. The cause of this combined reaction is linked to increased GI permeability, redistributed blood flow from the GI tract to the muscle and increased osmolality (Robson-Ansley & Du Toit 2010).

Once the allergen is diagnosed, the food source of the allergen is avoided both before and for at least an hour after exercise. Avoidance of the allergenic food is also recommended 4–6 hours prior to exercise in case of a delayed responses with exercise (Robson-Ansley & Du Toit 2010). Briner (1992) suggested physical allergy and exercise-induced anaphylaxis may be more common than expected.

20.8.3 Asthma and food allergy

Although links between food allergy and asthma have been touted, there is limited evidence that there are specific food triggers for asthma although existing asthmatic symptoms may be aggravated by some allergens (e.g. sulphites) (Mahan et al. 2012). Avoidance of known allergenic or trigger foods compounds is likely to be beneficial in severe asthma management and brittle asthma (the acute and severe end of the asthma spectrum with repeated attacks, without an obvious trigger) (Ayres et al. 1998). Food triggers such as milk, eggs and fish are likely to be less important as potential irritants for asthmatics than household dust and mites, pets and pollens.

20.9 Diagnosis of FRAR

Where a food allergy is suspected, specialised tests are available by medical referral to identify the IgE-specific antibodies. Correct diagnosis is essential to provide the appropriate nutrition care plan. A false positive result results in unnecessary food restrictions and lifestyle changes.

20.9.1 Dietary assessment and testing for a food allergy or intolerance

A clinical and dietary history followed by a detailed food and symptom record—including timing of food ingestion and onset of symptoms—helps to isolate specific foods that cause problems. Usually a minimum of 2–3 weeks of dietary intake data collection including a scale of the severity of symptoms is needed. Because of the chemical complexity of food, the variety of meals and food combinations consumed, and the often delayed and inconsistent occurrence and severity of symptoms, isolation of the specific food allergen or irritant is difficult. An in-depth overview of specific methods of data collection for food allergy sufferers is found in dietetic reference texts such as Thomas and Bishop (2007) and Mahan et al. (2012).

Once the trigger food or food groups are identified, the next step is medical referral for appropriate testing (e.g. skin pick or patch testing, sublingual test, blood test, biopsies, food challenges). Other routine pathology tests might include a full blood count, a stool test, hydrogen breath analysis, tests for intestinal parasites and genetic tests for allergies or the presence of other GI conditions (CD, gluten sensitivities, PKU and other amino-acid abnormalities).

20.10 Medical nutrition therapy for athletes with FRAR

Medical nutrition therapy may involve an exclusion diet involving avoidance of a single food or group (e.g. milk and dairy foods) or multiple food exclusions (several foods at the same time where the suspected food has yet to be identified). All food variants of the suspected food (raw, cooked, processed, protein derivatives) are avoided. Although useful for diagnosis, the use of multiple exclusion diets as a diagnostic approach for an athlete in training is time consuming, unsociable, expensive and should only be continued if clear evidence of improvement justifies the continuance. There is no agreement internationally on the period of elimination for recovery of normal gut function where there is extensive inflammation; 2–12 weeks has been reported (Mahan et al. 2012; Thomas & Bishop 2007). In some cases, exclusion diets result in rapid symptom improvement while in other cases, there is no obvious effect.

When or if symptoms improve with an elimination diet, excluded foods are then reintroduced, one at a time, to identify which food has provoked the adverse reaction and symptoms are closely monitored. Re-exposure to the food(s) believed to have provoked the symptoms occurs through a food challenge or planned reintroduction in a supervised medical setting. One of three food challenges is usually used: 1) consuming the suspected food with the patient aware of its potential allergenic effect; 2) giving the suspected food as single-blind-placebo-controlled challenge; and 3) as a double-blind placebo-controlled challenge. This challenge approach for an individual with a history of hypersensitivity poses a high risk of adverse reactions and should be undertaken with experienced medical support. In contrast, for people with a suspected food intolerance rather than a food allergy (such as FODMAP), ingestion of small amounts of the suspected food may be tolerated at home.

The primary focus for medical nutrition therapy is the relief of symptoms and restoration of a healthy nutrient-dense diet. For an athlete, this therapy needs to be adjusted for training and competition and needs to meet high energy requirements. Ideally, food exclusion and challenges should be undertaken during off-season or during non-competitive phases.

20.10.1 Diet-related risk factors for athletes on exclusion diets

Athletes who commence exclusion diets for the initial treatment of FRAR can be categorised into three levels of risk for suboptimal nutrient or energy intake that may affect capacity to train and compete.

- *Low risk* One food allergen is identified, such as a fruit or vegetable. Exclusion does not influence energy, macro and micronutrient intake (e.g. an allergy to kiwifruit).
- *Moderate risk* The food allergen is ubiquitous in the food supply. Avoidance or restriction does not limit food choice substantially and appropriate substitute foods are readily available (e.g. fish, seafood, tree nuts, egg white).
- *High risk* The food allergen is ubiquitous in the food supply, is a staple or major contributor to macro- and micronutrient intake. Elimination results in major lifestyle and dietary changes (e.g. wheat, gluten, soy, egg, milk, peanuts).

20.10.2 Are rotation diets an effective treatment option?

Testimonials from people with food reactions suggesting beneficial effects from either spacing individual allergenic foods or groups of foods in a rotational pattern (over a number of days or weeks) or avoiding the offending food or compound for any number of days for IgE-mediated food allergies sound convincing. Currently no scientific evidence supports either practice. These practices are different from elimination or exclusion diets. For people with non-mediated allergy that is dose-related, some relief or reduction in symptoms may be provided by limiting the quantity of a food known to be reactive.

Summary

Gastric emptying of food varies depending on the composition and volume of food and fluids ingested, the experience and everyday food habits of the athlete and the genetics and physiology of the athlete. Some athletes experience adverse GI symptoms associated with exercise or may have underlying conditions involving food allergies or intolerances. To diagnose and treat adverse food-related intolerances and allergies, the sports dietitian benefits from specific training and experience in this complex area. Athletes often self-diagnose and are convinced that they have a food allergy or adverse GI reaction. Identification of true sensitivity and allergy by a qualified medical professional/specialist and/or with the assistance of a dietitian is encouraged for athletes experiencing such symptoms. Other conditions such as an eating disorder, psychological stress or depression may be underlying. Identification of the food allergen or irritant is difficult and treatment often complex requiring a multidisciplinary team. Dietary and medical team support for the athlete may not differ from that required to support a non-athlete.

The role of the sports dietitian is to assist the athlete and coach to understand the diagnosis, provide practical management of food choice around training, competition and performance as well as provide expert nutritional support to avoid nutrient deficiencies, energy deficiency, and unnecessary restriction and food challenges.

One of the greatest risks, even for athletes experienced in managing their food allergy, is an inadvertent exposure to the allergen. Areas where the sports dietitian can provide valuable input to service organisations to athletes and assist in reducing the risk of an FRAR include are with local restaurants and caterers, catering at training camps, international travel arrangements and through support at large-scale events such as Commonwealth Games, World Championship and Olympic Games.

USEFUL WEBSITES

IBS and FODMAPS

- **www.med.monash.edu/cecs/gastro/fodmap**
  Monash University, School of Medicine, Nursing and Health Sciences: Information about the low FODMAP diet including useful resources (i.e. FODMAP smartphone app)
- **http://shepherdworks.com.au/disease-information/low-fodmap-diet**
  Shepherd Works: Provides an overview of the low FODMAP diet with further useful resources (e.g. cookbooks which can be ordered online)

Coeliac disease

- **www.coeliacsociety.com.au**
  Coeliac Society: Australian resource for patients that describes coeliac disease diagnoses, treatment, management options
- **www.celiactravel.com/coeliac-societies.html**
  Celiac Travel: Provides information for travellers (e.g. tips, recipes, restaurant cards/translation cards)

Food allergy and intolerance

- **www.foodallergy.org**

Food Allergy Research and Education: US site for people with food allergies

- **www.allergyaction.co.uk**
  Allergy Action: Information to help allergic people choose food and travel safely
- **www.allergyuk.org**
  Allergy UK: Information for people with different allergies—UK based
- **www.allergyfacts.org.au**
  Allergy & Anaphylaxis Australia: Shares information, educates, advocates, supports research and provides guidance and support for people with allergies

Sports specific resources for athletes with GI problems

- **www.sportsdietitians.com**
  Sports Dietitians Australia: Food allergy and adverse reactions fact sheets available specific for use with athletes
- **www.nutritionstrategies.com.au/shop**
  Nutrition Strategies: FODMAP and Gluten Status Sports Supplement Guide: a guide to suitable sports nutrition supplement products for athletes experiencing symptoms of irritable bowel syndrome
- **www.sportsdietitians.com.au/content/2535/TheLowFODMAPsDiet**
  Sports Dietitians Australia: Fact sheet about low FODMAP diet for athletes
- **www.sportsdietitians.com.au/content/2598/AthletesWithCoeliacDisease**
  Sports Dietitians Australia: Fact sheet for athletes with coeliac disease
- **www.sportsdietitians.com.au/content/2553/RunnersGut**
  Sports Dietitians Australia: Fact sheet on runner's gut

Practice tips

JENI PEARCE AND STEPHANIE GASKELL

BACKGROUND AND DATA COLLECTION

- A detailed clinical and dietary history of an athlete with a suspected food reaction, followed by a food and symptom record that includes timing of ingestion and onset/type of GI symptoms, is the cornerstone to helping identify the link between exercise, food and potential food allergens or irritants in any individual. A family history of GI conditions or food allergies/intolerances should also be investigated. A sports dietitian may be the first point of contact for an athlete with GI problems associated with exercise or food, rather than a medical practitioner. GI symptoms may be an alert for other underlying psychological concerns (e.g. anxiety, depression, disordered eating behaviour or an overt eating disorder).

COMMON GASTROINTESTINAL COMPLAINTS

- Symptoms of bloating, burping, flatulence and abdominal distension are normal GI reactions and indicative of healthy gut function.
- If these symptoms are excessive, dietary therapy can assist in management, although medical investigation may be indicated to exclude other conditions or diseases.

GASTROINTESTINAL SYMPTOMS ASSOCIATED WITH EXERCISE

- A trial and error approach is used for those athletes who experience GI symptoms associated with eating before or during exercise. For example, foods high in fibre, protein, fat and fructose can increase risk of GI upset in susceptible athletes. Selecting foods with a different ratio of glucose and fructose may benefit those with fructose malabsorption.
- Athletes need to repeatedly trial food and drinks choices intended for consumption in competition during training sessions that closely match the intensity and duration of the match or competition event to adapt the gut to eating. Routinely eating during hard training sessions is necessary for gut adaptation, which can take up to 20 days.
- To adapt the gut to food, start with liquid foods (e.g. meal replacement formula) or beverages that are isotonic (e.g. sports drinks) before training or competition, then progress to low-fat, high-CHO options (see Chapter 13). Sports drinks are optional during training or competition, but other solid food options need to be slowly introduced for longer events.
- If there is no or a limited response from dietary modification or no apparent gut adaptation, medical referral is indicated to check for the presence of any underlying condition. Pharmaceutical intervention, particularly if vomiting is recurrent, may be needed because of the risk of oesophageal erosion.

IRRITABLE BOWEL SYNDROME

- Diagnostic criteria for IBS are described in Table 20.1. Medical management may include a combination of counselling or other techniques for relaxation and stress reduction; medication to normalise gut motility and decrease visceral hypersensitivity; and dietary intervention.
- Although the aetiology in unknown, infection and psychological factors can trigger an IBS episode.
- Food intolerance associated with malabsorption may be implicated—hence the success of the low FODMAP diet as a treatment option (see section 20.4.2).
  - If athletes have constipation-predominant IBS, which is accompanied by low fibre intake, increase fibre (and fluid) as tolerated. Use of a fibre concentrate such as ispaghula (psyllium), rice or oat bran, may be of benefit; however symptoms of bloating and wind may increase with these options. Dosage for psyllium varies (typically between 10–30 g/d). To reduce side effects, it is best consumed in small amounts spread over the day (e.g. 2–3 time/d), rather than in a bolus dose.
  - If diarrhoea is an ongoing symptom, probiotics may be of benefit (see Chapter 22).
- Limit caffeine and alcohol consumption on an empty stomach, if reflux is a symptom.

GENERAL GUIDELINES FOR RECOMMENDING A LOW FODMAP DIET TO ATHLETES

- The low FODMAP diet is now recognised as one of the most successful evidence-based dietary therapies for the treatment and management of IBS. It may also assist in the management of abdominal bloating, pain, wind, reflux and elimination problems. For a review of the low FODMAP diet and foods to avoid or limit, see Monash University's online information (www.med.monash.edu/cecs/gastro/fodmap). A FODMAP app for android and iPhones, released in 2012, is available at this URL.
- Dietitians benefit from undertaking specific training prior to initiating a low FODMAP diet for athletes. Prolonged restriction of most foods on the FODMAPs list may be required for 1–8 weeks to achieve a more complete picture of improvement and reduction in symptoms (Gibson 2012; Barrett 2013). Once symptoms resolve, the usual process is to reintroduce specific food challenges to help identify the main dietary trigger(s) and an individual's tolerance levels.
- Avoiding all foods on the FODMAPS list may not be necessary. Lactose intolerance can be diagnosed by a diet and symptom history and/or a hydrogen/methane breath test. Dietary restriction of foods containing an excess amount of fructose compared to glucose, high levels of fructans, galacto-oligosaccharides and polyols is likely to reduce symptoms in the majority of IBS sufferers. See section 20.4.2.
- Individuals with diagnosed fructose malabsorption do not need to avoid all foods containing fructose. Symptoms usually occur after consuming foods containing fructose in excess of glucose such as apples, pears, mangoes, artichokes and asparagus. Table 20.6 provides practical strategies for those athletes with symptomatic fructose malabsorption. Oligosaccharides such as fructans (long chains of fructose units) and

galacto-oligosaccharides are not absorbed effectively by the human gut. They are found in wheat, rye and barley. When consumed in large amounts, they can induce GI symptoms in susceptible people.

- Many alternative food options are available for athletes on a low FODMAPS diet. Encourage athletes to experiment with new and different foods and flavours rather than practice avoidance behaviour.

TABLE 20.6 Guidelines for limiting fructose, fructans, galacto-oligosaccharides and polyols from the diet

• Avoid/limit foods containing fructose in excess of glucose (e.g. mangoes, apples, pears, watermelon, honey, asparagus, artichokes), foods with a high fructose load such as fruit juice, dried fruit, some jams, confectionery (jellies and soft jubes), wine, some stocks and sauces as they can contain excess fructose and oligosaccharides (onion, garlic).
• Avoid consuming large amounts of fruit at one time (2–3 hours between serves may be well tolerated).
• Onion, leeks and garlic are well-known triggers.
• Avoid processed food that contains high fructose corn syrup or corn syrup solids.
• Avoid commercial food products that have added inulin or oligosaccharide as a thickener (e.g. some yoghurts). Check the label for these additives.
• Limit sorbitol rich-foods (e.g. prunes, many dried fruits, muesli bars with dried fruit) or added sorbitol used as a carbohydrate-modified sweetener in commercial food (food additive code 420, E420). Low-calorie chewing gum may contain sorbitol. Mushrooms and cauliflower may be problematic because of high levels of naturally occurring mannitol (food additive code 421, E421).
• Wheat, rye and barley consumed in large amounts contain small amounts of these compounds. Limiting small amounts (such as breadcrumbs or when used as ingredients in recipes) is not required. Wheat glucose, caramel colour, wheat dextrose, wheat starch and wheat maltodextrin do not need restriction.

COELIAC DISEASE, GLUTEN AND WHEAT SENSITIVITY

- Coeliac disease may be present and undiagnosed for a long time in people with low-grade non-specific symptoms. Often it is detected in athletes when they are referred for assessment of fatigue or iron depletion. Unlike wheat and gluten sensitivity, coeliac disease invokes an inflammatory cell-mediated immune response to the protein fraction in wheat, barley rye and oats and requires removal of these foods and their by-products. A low-gluten or gluten-free diet is recommended depending on individual sensitivity.
- Improvement of symptoms generally occurs within 2–8 weeks of avoidance of gluten.
- Many processed foods contain hidden sources of the by-products of these grains, which are used as thickeners or bulking agents in soups, sauces, desserts and canned vegetables (e.g. wheat starch, flour thickener, wheat dextrin and maltodextrin, wheaten cornflour). Processed meats (sausages and packaged slices of meat) may also contain gluten as flour or breadcrumb toppings.
- A range and variety of gluten-free sports food and food products (cereal bars and gels) are now available. See FODMAP and gluten status sports supplement guide (www.nutritionstrategies.com.au/shop) for more information.
- Athletes on a gluten-free diet are at risk of suboptimal intakes of iron, thiamin and folic acid.
- Travel overseas for competition poses a high risk for potential inadvertent gluten intake. Country-specific websites provide lists of gluten-free foods (see Useful websites).
- The amount of gluten present in a food labelled gluten-free varies in different countries. In Australia and New Zealand, gluten-free on a food label is defined as the absence of ‘detectable gluten’ (i.e. ‘zero’ gluten) at

amounts of <3 ppm. Recent improvements in technology have resulted in more sensitive detection levels for gluten in food with the potential to progress into the parts per billion ranges, which reduces the number of foods that can be labelled as gluten-free. Codex Alimentarius (the International Standard) supports <20 ppm of gluten in a food as the cut-off for gluten-free on food labels.
- Regular reviews (every 6–12 months) with a sports dietitian are recommended to assess nutritional status, compliance, tolerance under stress or competition and during and after periods of illness and injury, when the condition may exacerbate.

IRRITABLE BOWEL DISEASE (CROHN DISEASE AND ULCERATIVE COLITIS)

- Athletes with IBD may have difficulty maintaining weight and lean mass and are at risk of malnutrition. Avoidance of many foods and loss of appetite is a common behaviour in people with these conditions. Table 20.7 provides strategies to help improve appetite and promote weight gain.
- A combination of biochemical (urine and blood) markers (i.e. transferrin, ferritin, MCV, CK and elevated urea) and the Subjective Global Assessment (Massironi et al. 2013) may be used for athletes suspected of underlying malnutrition. Rapid loss of lean body mass is likely to be more indicative of early malnutrition in an athlete than these biochemical markers, which can be affected by hard exercise. Iron deficiency is a common problem in athletes and in IBS and IBD sufferers—hence low iron status markers may confound the diagnosis when used as a sign of early malnutrition.
- Regular assessment of body composition and nutritional status is warranted in athletes with IBD.
- Referral for specialised medical and dietetic management is recommended for athletes with IBD.
- The sports dietitian may provide this specialised medical nutrition management service as part of this team or assist the athlete to meet their energy requirements to prevent further weight loss or promote weight gain.

TABLE 20.7 Practical nutrition strategies for athletes with IBD and poor appetite or tolerance food intake

Use high-energy and high-protein supplements, rather than liquid meal replacements recommended by athletes Hospital formulations are indicated (e.g. Sustagen Hospital Formula™, Ensure™, Resource™, Proform™, Modulen™)	Encourage athlete with IBD to eat more at the time of day when they feel well
Consider a multivitamin or multimineral supplement	Ensure food is well chewed
Use specified vitamin or mineral supplements only when a deficiency is diagnosed	Use high-energy beverages (often liquids are better tolerated than solid food, add Polycose™ to beverages)
Avoid known aggravates (e.g. caffeine, alcohol, high fat, spicy foods, foods high in FODMAPs if disease is active and IBS is associated with IBD, or there is diagnosed intolerance)	Small, frequent, energy-dense meals (more appealing)
A low residue or low fibre diet may be of benefit (when disease is active or there is narrowing/strictures present)	Fibre recommendations based on individual symptoms and history

FOOD INTOLERANCE AND FOOD ALLERGY

- Correct identification of the allergen or irritant is essential to manage the FRAR and prevent adverse reactions.
- The type of reaction (Figure 20.1 and Table 20.4) is highly individual and ranges from a mild rash, nausea and vomiting to life-threatening anaphylaxis. Major organs (digestive tract, respiratory system and skin) may be involved. Maintaining quality of life is a key consideration.
- The services of a clinical dietitian and food allergy specialist and/or gastroenterologist may be needed.
- Self-management of symptoms and avoidance of an inadvertent reaction is important, especially in those athletes who travel regularly overseas.
- Alerting coach and support staff of expected reactions of an athlete with a food allergy and treatment protocols

is obligatory.

- Regular reassessment is recommended to reevaluate reaction response and potential alterations in tolerance.

LACTOSE INTOLERANCE

- Consuming lactose-containing foods with food (e.g. milk on breakfast cereal) is better tolerated than drinking milk by itself. Lacteeze™ drops and chewable Lacteeze™ tablets, which contain the lactase enzyme, are an alternative to lactose-free foods. These are useful when competing away from home, particularly international venues, and for use at catered venues where lactose-free foods may not be available.
- Whey protein beverages, where the CHO is removed (whey protein isolate), fermented and low- or reduced-lactose milks are suitable. In western countries, most lactose-free milks such soy, rice and almond milk and their products are fortified with calcium and match the calcium content of cow's milk products.
- Food containing lactose (see Table 20.8) can be tolerated in small amounts.

TABLE 20.8 Lactose content of food

Food	Serving	Lactose content (g)
Milk (any fat content)	1 cup (250 mL)	10–12
Milk evaporated	1/2 cup (125 mL)	13
Milk (sweetened condensed)	1/2 cup (125g)	20
Butter	1 tsp (5g)	Trace
Margarine (table)	1 tsp (5g)	Trace
Cheese (cheddar)	30g	0–2
Cheese (parmesan)	30g	Trace
Cheese (cottage)	1/2 cup	2–4
Cheese (cream)	30g	1
Ice cream	1/2 cup (125g)	3–9
Sour cream	1/2 cup (125g)	4–5
Cream (whipped)	1/2 cup	3–4
Cream (pouring)	1 tbsp (15–20 mL)	1
Yoghurt (whole milk)	1 cup (200g)	10–12
Yoghurt (low-fat)	1 cup (200g)	5–19
Custard	1/2 cup (125g)	6
Sustagen Kid Essentials™	2 tbsp (25g)	Trace

INADVERTENT CROSS-CONTAMINATION WITH FOOD ALLERGENS

- Cross-contamination of food allergens is a risk factor for athletes when:
  – serving utensils are shared or repeatedly used to prepare foods without appropriate cleaning
  – cooking equipment is not adequately cleaned
  – food is prepared on the same chopping board.
- Buffet-style eating is a high-risk eating environment for an athlete who is highly sensitive to an allergen. Pre-

plated meals are a better option.

- High standards of food hygiene are essential for caterers. Education of catering staff about the potential severity of food allergy and inadvertent contamination through poor food safety practices may help prevent an incident.
- Travelling overseas and eating out either overseas and at home in local restaurants is a high risk factor for an athlete with FRAR because of varying food safety practices. Advise restaurants in advance if possible.
- Question food-service staff and chefs thoroughly on ingredients used and preparation methods.
- Information on food labels in other countries vary. Although Codex Alimentarius is an international standard for food labels, different countries vary in labelling laws and quality control of food production. For example, the ingredient list may not identify the original source (e.g. lecithin from egg or soy, egg-containing mayonnaise added to salads or coleslaw).

TREATMENT OF ATHLETES WITH SEVERE FRAR REACTIONS

- Athletes who have known FRAR reactions should advise coaches and support staff of emergency procedures and location of emergency equipment (e.g. injectable adrenaline syringes—EpiPen), the use of medications such as antihistamines (used to dampen responses) and emergency plans.
- During competition away from home, an athlete who has rapid or even delayed hypersensitivity reactions to food needs to alert a team mate to inform support staff, in the event of a delayed response at night. Exercise can mask some of the generalised allergic reactions (see section 20.8.2).
- Coaches, nutrition or sports science support staff should know the location of EpiPens and be able to administer the injection in athletes who have a known food allergy. Medical staff may not be on site at meal times or other eating occasions, especially when travelling to competitions.
- At competition venues, where athletes are likely to have severe reactions and anaphylactic reactions, injectable EpiPen (injections of epinephrine) should be available.

THE USE OF SPORTS FOODS AND SUPPLEMENTS IN ATHLETES WITH GUT PROBLEMS AND FRAR

- There are many sports foods and alternative beverages (i.e. sports drinks and liquid meal supplements) that are gluten-free, lactose-free or have glucose in greater amounts than fructose for the fructose malabsorbers. Athletes need to understand labels and seek assistance if unsure.
- Sports drinks are iso-osmolar and the best choice for athletes with GI problems as well as providing rapid uptake of electrolytes, carbohydrate and fluid during exercise.
- GI symptoms can be induced by caffeine, buffers (sodium bicarbonate), glycerol and alcohol as well as some medication (e.g. iron supplements, non-steroidal anti-inflammatory drugs [NSAIDs])
- Athletes with GI problems or diagnosed malabsorption should trial sports nutrition products and determine individual tolerance under medical or dietetic supervision.

RECOMMENDATIONS FOR EFFECTIVE NUTRITION INTERVENTION FOR ATHLETES WITH CHRONIC FOOD-RELATED ADVERSE REACTIONS

- Referral and collaboration with an immunologist and/or gastroenterologist is recommended to confirm a diagnosis and treatment plan.
- Dietary management needs to be individualised.
- Referral to a psychologist is beneficial if there is an underlying issue that may be contributing to, or a consequence of, recurrent GI symptoms.

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Commentary 8

Gastrointestinal issues and disorders associated with exercise

KIERAN FALLON

Effects of exercise on the gastrointestinal (GI) system

Gastrointestinal motility and blood flow

Exercise is associated with a decrease in lower-oesophageal sphincter pressure (Peters et al. 1988). A reduction in duration, frequency and amplitude of oesophageal contractions has been observed with exercise intensities above 60% $VO_{2\ max}$ (Soffer et al. 1993). More recent research suggests an increase in peristaltic velocity during cycling at 70% of maximum power output accompanied by reduced peristaltic contraction pressure at the mid-oesophagus and reduced duration of peristaltic contractions at the mid- and distal oesophagus (Van Nieuenhoven et al. 1999).

The effect of exercise on gastric emptying has been addressed in Chapter 20. An early study indicated that at exercise intensities up to 70% VO_{2max}, the gastric emptying rate was not disturbed (Fordtran & Saltin 1967). Since then, the majority of studies have demonstrated that physical activity at similar intensities can delay transit time of food in the small intestine and have attributed that effect to decreased propulsion or peristalsis (Brouns & Beckers 1993). In contrast, the transit time of gut contents speeds up in the colon, probably because of increased mucosal secretion, which dilutes colonic contents. It appears, however, that for some athletes, these opposing effects may compensate for each other because oro–caecal transit time does not appear to be affected by exercise in cyclists who have no exercise-related gastrointestinal symptoms (Van Nieuenhoven et al. 1999). Also, in a study involving one hour of high-intensity cross-country running, exercise had no effect on small bowel and colonic transit time, which were similar for both symptomatic and asymptomatic groups (Rao et al. 2004). In practice, some individuals appear more susceptible to gastric disturbances than others. The reasons for this variability are linked to changes in blood flow to the GI tract.

Symptoms of disturbances to GI function, including abdominal pain, diarrhoea and bleeding, are associated with a reduction in local blood flow. During exercise, splanchnic vasoconstriction occurs, allowing more blood to be shunted to muscle and skin for oxygen and substrate transport, and thermoregulation, respectively. Blood flow to the GI tract from splanchnic and coeliac circulations can decrease during exercise, depending on exercise intensity. A 60–70% decrease in splanchnic blood flow was measured in athletes working at 70% $VO_{2\ max}$ (Rowell et al. 1964). At $VO_{2\ max}$ splanchnic flow can be reduced by up to 80%. The reductions in blood flow are worsened by concomitant dehydration.

This effect (i.e. splanchnic hypoperfusion) can occur within the first 10 minutes of strenuous exercise and recovers within the first 10 minutes after exercise (van Wijck et al. 2011). Its severity is proportional to exercise intensity, confirming the observations in Rowell's study, but is not critical during submaximal exercise in healthy individuals. However, the majority of symptomatic athletes who appear to develop splanchnic hypoperfusion during submaximal exercise, suggest that they have a degree of hyper-responsiveness of the splanchnic bed to exercise (Ter Steege et al. 2012). However, blood flow tends to improve in the trained athlete and when exercise intensity is reduced (Clausen 1977). Hence, adaptation may be operating in highly trained athletes.

Gastrointestinal bleeding

Splanchnic hypoperfusion is also associated with loss of integrity between intestinal epithelial cells, which can lead to intestinal blood loss (van Wijck et al. 2012). Reperfusion following ischaemia may also contribute to reduction of epithelial cell integrity cells (Derikx et al. 2008).

Overt GI bleeding (macroscopic bleeding), although fairly uncommon, is a dramatic GI symptom that may occur in response to hard physical activity. Direct evidence of bleeding comes from the presence of mucosal erosions at endoscopy after strenuous running (Choi et al. 2001) and melaena. It is most frequently reported in long-distance runners (Sullivan 1986). In one study, up to 22% of runners had evidence of GI bleeding based on a positive test for faecal occult blood following a marathon (Schwartz et al. 1990). In another study, almost 85% showed a positive test following an ultra-marathon (Baska et al. 1990). Bleeding, again based on the same test, has also been reported in 12 competitive cyclists, with 8.4% of 310 stool samples collected intermittently over one year showing positive results (Dobbs et al. 1988). Yet other studies have reported much lower incidences of blood in the stools of athletes. In one report, none of the 63 runners in a marathon reported blood in the stool (Halvorsen et al. 1990). In another marathon only 6% of all runners surveyed (n = 125) reported blood in the stool after the event (McCabe et al. 1986). However, these reports were based on self-observation rather than laboratory testing and are likely to underestimate the presence of blood in the stool because of the difficulty in visual detection.

The site of bleeding has been assessed by endoscopy in several studies. Gaudin et al. (1990) assessed seven runners by upper-GI endoscopy before and after training runs of between 12–30 miles. All runners had gastric mucosal lesions after the run, consistent with ischaemia; only four had no lesions prior to exercise. All runners had reported previous GI symptoms prior to the study, which make conclusions from this study uncertain. Schwartz and colleagues (1990) examined seven runners who had occult blood loss following a marathon using oesophago–gastro duodenoscopy and colonoscopy. Three runners were examined within 48 hours after the race; two had gastric antral erosions, and another had erosion of the mucosa at the splenic flexure. The other four runners, examined between 4 and 30 days following the race, had normal results. Bleeding is associated with the intensity of exercise, but not with use of aspirin or NSAIDs (McMahon et al. 1984). Occasional cases of severe intestinal ischaemia have led to bowel resection (Moses 2005). This uncommon complication should be considered in cases of persistent pain and diarrhoea following exercise.

The cause of macroscopic bleeding should be determined by investigation. The extent of investigation required for microscopic bleeding (i.e. the presence of red blood cells in faeces) associated with exercise is age dependent, with those aged over 40 years requiring full investigation.

Preventive strategies include gradual increases in training, careful hydration and, in cases of recurrent upper-GI bleeding, medication. Use of H_2 receptor antagonists (Baska et al. 1990) and proton pump inhibitors (Thalmann et al. 2006) can significantly lower the rate of bleeding.

Gut ischaemia and changes to intestinal permeability

Changes in intestinal permeability have been related to gut ischaemia consequent to exercise (van Nieuwenhoven et al. 2004). The clinical effects of short-term changes appear minimal; however, a state of increased permeability of intestinal epithelial cells (or cell junctions) can occur in response to hard training sessions and inadequate recovery between sessions. In this situation, there is evidence of elevations in acute phase reactants and an increase in endotoxin absorption (Bosenberg et al. 1988). This consequence of high-intensity exercise is further exacerbated by frequent use of non-steroidal anti-inflammatory drugs (NSAIDs), which can potentially induce enteropathy. Administration of NSAIDs increases intestinal permeability within 12 hours and inflammation in the small intestine within 10 days (Park et al. 2011). NSAID-induced enteropathy from prolonged administration may lead to iron deficiency anaemia, gastrointestinal bleeding, low serum albumin, diarrhoea and abdominal pain, although most cases fortunately have no symptoms and minimal consequences. More severe consequences such as intestinal perforation are rare but can be life threatening (Adebayo & Bjarnason 2006).

Effects on changes to GI function

The long-term clinical significance of repeated episodic alterations in gastrointestinal function induced by exercise has not been well studied and is likely to be minimal. The exceptions are the contribution of repeated blood loss between iron depletion and consequent anaemia, and the positive association between physical activity and reducing risk of colorectal cancer.

Several meta-analyses have demonstrated a positive relationship between physical activity and reducing risk of colon cancer (Samad et al. 2005, Wolin et al. 2009). No such relationship has been demonstrated between physical activity and rectal cancer. In the later meta-analysis, 52 studies of males and females (30–50 y) were analysed using a summary measure of total physical activity. An inverse association between physical activity and colon cancer was found with an overall relative risk (RR) of 0.76 and little gender difference. Although early qualitative evidence suggests that vigorous-intensity physical activity is associated with the greatest risk reduction of colon cancer, walking alone at moderate intensity may be sufficient to reduce risk (Wolin et al. 2007). Further studies are required to determine the type, intensity and duration of physical activity that may afford the greatest risk reduction.

Gastrointestinal disturbances during competition

GI symptoms are common in athletes and have been extensively studied in marathon runners and triathletes (Sullivan 1981, 1986; Keeffe et al. 1984; Worobetz & Gerrard 1985; Halvorsen et al. 1990). Riddoch and Trinick (1988) summarised the frequency of GI disturbances during or immediately after running from four surveys of recreational and competitive runners, marathon runners and triathletes; they included loss of appetite (28–50%), heartburn (9.5–24%), nausea (6–20%), vomiting (4–6%), abdominal cramps (~19–39%), urge for a bowel movement (30–42%) and diarrhoea (14–27%). In these studies, upper- and lower-GI symptoms were more likely to occur during a hard run than an easy run. Lower-GI symptoms were more common than upper-GI symptoms, which is consistent with other surveys. The most common strategies used by athletes to minimise symptoms were to run in a fasted state and to have a bowel movement prior to running.

Rehrer and colleagues (1992) investigated dietary intake in relation to GI complaints in 55 male triathletes in a half-Ironman distance triathlon. In these athletes, hyperosmotic beverage consumption during the event was associated with increased incidence of severe GI symptoms. Dietary fibre ingestion before competition increased the likelihood of GI symptoms. The ingestion of a large food volume or a high-fat and high-protein pre-race meal may also increase the risk of upper-GI distress in some athletes, because of a slower rate of gastric emptying of fat and protein, compared with carbohydrate.

Table C8.1 provides guidelines for food choices to minimise exercise-related GI symptoms in those athletes who have existing problems. These guidelines may also be useful for many asymptomatic athletes at important competitions, where GI problems often arise for the first time, particularly in novice athletes.

TABLE C8.1 Recommendations for minimising exercise-related GI symptoms during training or competition

- Avoid solid food during the last 2–3 hours prior to exercise.
- Avoid high-fibre foods prior to exercise.
- Use liquid foods (i.e. liquid meal supplement) as a meal prior to competition or a hard training session.
- Whenever fluid is of first priority, drinks should be low in CHO.
- When training in hot conditions and rapid delivery of fluid and CHO is required, the optimal concentration of a beverage should be in isotonic (e.g. sports) drinks.
- Athletes suffering frequently from diarrhoea or the urge to defecate during exercise may benefit from liquid meal supplements (low in fibre content) during the last meal and perhaps the evening meal on the day before competition.
- Drinking and eating during exercise should be practised in training.

Source: Adapted from Brouns et al. 1987

Upper gastrointestinal tract

Major upper-GI problems include gastro-oesophageal reflux disease, gastritis, peptic ulcer and functional dyspepsia.

Gastro-oesophageal reflux disease

Gastro-oesophageal reflux disease (GERD) is a common disorder, with about one-third of adults experiencing heartburn once a month and 5–10% of adults experiencing heartburn each day.

The major factor preventing GERD is maintenance of contraction of the lower oesophageal sphincter. The most common mechanism for reflux is transient lower oesophageal sphincter relaxations, which are normally triggered by gastric distension (Bredenoord et al. 2013). Several forms of exercise can induce gastro-oesophageal reflux (GER), which may be asymptomatic. GER increases with intensity of exercise and is more common in endurance athletes. It is more likely to occur with exercise in those who have GER at rest. Yazaki and colleagues (1996) compared the incidence of GER in normally asymptomatic athletes involved in rowing and running, by measuring intra-oesophageal pH before, during and after rowing, fasted running and post-prandial running. While GER was reported in two out of 17 athletes prior to exercise, it occurred after or during exercise in 70% of rowers, in 45% of fasted runners and in 90% of runners who had eaten just prior to exercise. This study suggests that athletes with a history of GER should avoid even small meals close to the time of exercise. Fortunately, GER and GERD are easily treated in the majority of cases. Medical management generally involves:

- avoidance of aggravating factors such as eating large meals, lying down soon after eating and ingestion of alcohol, some of which may be specific to the individual
- weight loss, if overweight
- antacids or medications to reduce acid secretion (e.g. H_2 receptor antagonists, proton pump inhibitors)
- prokinetic agents, which increase the rate of gastric emptying.

Athletes with established reflux oesophagitis should avoid drugs that are directly damaging to the oesophageal mucosa, including non-steroidal anti-inflammatory drugs (NSAIDs) and iron supplements.

Functional dyspepsia

Functional dyspepsia is defined as chronic or recurrent pain or discomfort in the upper abdomen, with no clinical or endoscopic evidence of organic disease. It is a common but poorly understood condition. The symptoms are varied, but four subgroups have been identified, as indicated in Table C8.2.

TABLE C8.2 Subgroups of functional dyspepsia, including symptoms

Subgroup	Symptoms
Ulcer-like dyspepsia	Nausea, vomiting, epigastric pain More severe cases may be accompanied by severe pain and haematemesis
Reflux-like dyspepsia	Heartburn, reflux of stomach contents into the mouth
Dysmotility-like dyspepsia	Early satiety, post-prandial fullness, nausea, retching and/or vomiting that is recurrent, bloating in the upper abdomen (not accompanied by visible distension), upper abdominal discomfort, often aggravated by food
Unspecified dyspepsia	Symptoms do not fill the criteria of other categories

Source: Talley et al. 1991

The diagnosis of functional dyspepsia should only be made following clinical evaluation and, ideally, an upper-GI endoscopy. This excludes other disorders, including peptic ulceration, reflux oesophagitis and malignancy.

Despite the high incidence of this condition in the general population, its incidence in the athlete population has not been studied, but the condition is often seen in sports medicine practice.

Gastritis and peptic ulcer

Gastritis refers to inflammation of the mucosa of the stomach, which may be acute or chronic. Acute gastritis is associated with severe illness, where it is termed ‘acute erosive gastritis’ or ‘stress-induced gastritis’. In the athlete population or those suffering from rheumatological disorders, direct irritants such as aspirin, NSAIDs and alcohol can induce gastritis. Acute gastritis is also associated with infection by the bacterium *Helicobacter pylori*. In milder cases, which occur more frequently, symptoms include nausea, vomiting and epigastric pain. Severe pain and haematemesis occur in more advanced cases. Chronic gastritis is often associated with *H. pylori* infection; active inflammation is apparent in about 30–50% of *H. pylori* infections.

Management of most acute cases involves avoidance of further irritation and the prescription of antacids or medications that inhibit acid secretion. Those cases associated with *H. pylori* infection are treated by the above-mentioned measures, with the addition of antibiotics.

Peptic ulcer is a generic term for a common condition, which includes both duodenal and gastric ulcers. The aetiology is multifactorial, but involves the action of acid and digestive enzymes, reduction of mucosal defences and infection by *H. pylori*. *H. pylori* infection is found in 80% of cases of duodenal ulcers and 70% of gastric ulcers. NSAIDs are responsible for large numbers of gastric ulcers. Diagnosis is based on clinical history and examination and is usually confirmed by endoscopy. There is some evidence from a cross-sectional study that the prevalence of duodenal ulcers is significantly lower in males who walk or run more than 10 miles (22 km) per week (Cheng et al. 2000).

Lower gastrointestinal tract

Diarrhoea

The urge for a bowel movement, reported in 30–42% and diarrhoea in 14–27%, are the most frequent gastrointestinal symptoms reported by runners. These symptoms appear to be related to more prolonged bouts of exercise, which tend to be associated with hyperthermia and dehydration. While initially thought to be related to shortened colonic transit time, this hypothesis is now controversial. These symptoms are now thought to be directly related to intestinal hypoperfusion and inappropriate food choices. Suggestions for prevention are outlined in Table C8.1. Should these prove ineffective, athletes may use anti-diarrhoeal medication prior to the onset of exercise as a preventive measure. The efficacy of such medication in this situation has not been documented.

Inflammatory bowel disease

The term ‘inflammatory bowel disease’ includes ulcerative colitis and Crohn disease, both of which are chronic relapsing diseases of the intestine. The main relevance to the sporting population lies in the age distribution of both diseases. While the prevalence is low in the general population (5–8 per 100 000 people for ulcerative colitis and 2–4 per 100 000 people for Crohn disease), the peak incidence for both diseases occurs in late adolescence or early adulthood (Loftus 2004). The prevalence in the athletic population is not known.

The cause of inflammatory bowel disease is unknown, but both genetic and environmental factors are implicated. Tobacco smokers are twice as likely to develop Crohn disease as non-smokers, but only half as likely to develop ulcerative colitis. Users of the oral contraceptive pill have about double the risk of developing Crohn disease (Timmer et al. 1998).

While any preventive effect of physical activity is inconclusive (Peters et al. 2001) it appears that physical activity is not harmful for patients with inflammatory bowel disease (Loudon et al. 1999).

Irritable bowel syndrome

Irritable bowel syndrome (IBS) is a condition characterised by abdominal pain and altered bowel habit in the absence of any abnormality on radiological, endoscopic or laboratory investigations. Approximately one-third of patients have constipation, one-third diarrhoea and one-third alternate between these symptoms. All have abdominal pain. IBS is the most common chronic gastroenterological disorder and has a prevalence of between 10–30% (Daley et al. 2008). It is more common in the young and therefore may be frequently found in the young active population, although the prevalence in this age group has not been studied.

The cause is unknown, but involves abnormalities in intestinal motor function, intestinal sensory function and central nervous system–enteric nervous system regulation. Infection and psychological factors have been implicated in triggering IBS and in determining which patients present for treatment. True allergic or immunologically mediated responses to foods have little role in the aetiology of ICS. Food intolerance, however, may play a role, with dairy products and grains implicated as common precipitants of symptoms (Nanda et al. 1989). Exercise can improve symptoms of constipation in irritable bowel syndrome without an improvement in quality of life scores (Daley 2008).

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CHAPTER TWENTY-ONE

Special needs: the Paralympic athlete

Elizabeth Broad and Siobhan Crawshay

21.1 Introduction

Sport and exercise have long been recognised as important components in the therapy and rehabilitation of people with physical or intellectual impairments. However, since the inception of the Summer Paralympic Games in 1960 and the Winter Paralympic Games in 1992, the number of individuals with impairments training and competing at high levels has increased dramatically. Local, national and international competitions are held in summer and winter disciplines throughout the world. Performance standards continue to improve and Paralympic athletes therefore need to train to the same intensity and with the same structure and planning as able-bodied athletes do. Even in non-Paralympic sports, athletes with impairments are accomplishing challenging feats, such as individuals with paraplegia swimming the English Channel and completing the Hawaii Ironman Triathlon.

Despite the growth of Paralympic sports internationally, scientific research about these athletes has failed to match that of their able-bodied counterparts. Applying sports nutrition principles to some of these athletes is straightforward as their events and nature of their impairment mean that there is little to no physiological difference to able-bodied athletes. Others require the sports dietitian to adapt these principles and/or their practical application on an individual basis. This chapter outlines some common areas where such adaptations may be required and provides simple strategies to help optimise training capacity, health and ultimately the performance of Paralympic athletes.

21.2 Paralympic sports

The large range of sports on offer to Paralympic athletes presents opportunities for so many types of impairment. At the Summer Paralympic Games, athletes compete in 22 disciplines, and at the Winter Paralympic Games six disciplines, as shown in Table 21.1. Within each sport, there are certain criteria, or classification rules, that govern the type of impairment that an athlete must have to participate. For example, wheelchair rugby is limited to those athletes who mobilise in a wheelchair and have an impairment involving all four limbs (usually quadriplegia). Where a sport may be restricted to, or is primarily undertaken by, specific classes of impairment, this has been highlighted in Table 21.1. In many cases, there are other impairments that may also be classified for that particular sport, but these are not as common.

TABLE 21.1 Current Paralympic sports

Summer sports	Classification of disability
Archery	Physical impairment
Athletics	All impairment types (wide range)
Boccia	Cerebral palsy or other physical impairment, who require a wheelchair for daily mobility
Para-canoe	Physical impairment
Cycling	All impairment types (wide range)
Equestrian	Physical impairment or visual impairment
Football five-a-side	Visually impaired
Football seven-a-side	Cerebral palsy
Goalball	Visually impaired
Judo	Visually impaired
Powerlifting	Lower body impairments (e.g. paraplegia, amputee)
Rowing	Physical impairment
Sailing	Physical impairment
Shooting	Physical impairment
Sitting volleyball	Physical impairment
Table tennis	Physical and intellectual impairments
Para-triathlon	All impairment types (wide range)
Wheelchair basketball	Lower body impairments (e.g. paraplegia, amputee)
Wheelchair fencing	Amputees, spinal cord injuries and cerebral palsy
Wheelchair rugby	Quadriplegia, or limb amputee/deficiency affecting all four limbs
Wheelchair tennis	Spinal cord injuries, amputees
Winter sports	**Classification of disability**
Alpine skiing	Physical impairment or visual impairment
Biathlon	Physical impairment or visual impairment
Cross-country skiing	Physical impairment or visual impairment
Ice sledge hockey	Physical impairment in lower part of the body
Para-snowboard	Physical impairment in lower part of the body
Wheelchair curling	Physical impairment in lower part of the body, who require a wheelchair for daily mobility

The rules governing each sport are outlined on the International Paralympic Committee

(IPC) website (www.paralympic.org). At times, rules may differ from those of able-bodied sports in order to ensure the safety of competitors.

21.2.1 Classification of disabilities

For Paralympic classifications of athlete disability, see Table 21.1.

21.3 Classes of Paralympic athletes

Paralympic athletes comprise six primary classes (amputees, cerebral palsy, spinal injured, visual impairment, intellectual impairment and Les Autres), which are generally grouped together under a functional system for the purpose of competition. Each class, and associated specific sports nutrition considerations, is outlined below. In addition to the Paralympic Games, there are several large-scale International competitions for individuals with impairments, such as the Special Olympics (intellectual impairments), Deaflympics (hearing impairments) and Transplant Games (organ transplants); the scope of this chapter does not allow for a full discussion on all forms of impairment.

Each Paralympic sport has its own system or approach to classification, which categorises athletes according to their physical capability (Buckley 2009; see also www.paralympic.org). This system helps to ensure that the vast array of athletes can compete at the highest level of which they are capable, with elite competition in mind. All athletes with impairments must hold an internationally authorised classification to participate in competition. Those athletes whose impairment is progressive are reclassified when required. Hence, the running of a competition for athletes with impairments is more complex than that for able-bodied athletes as athletes may only compete against similarly classified individuals. The IPC website (www.paralympic.org) contains useful information on classification.

Exercise science/nutrition research findings in Paralympic populations are few and far between, and can be confusing because of the limited numbers of individuals tested, the large individual variation in functional capacity (e.g. a group of 10 athletes with spinal cord injuries can exhibit a wide range of energy expenditure and exercise capacity), the different levels of training undertaken by individual athletes in the past compared to current day, and a weighting towards clinical or rehabilitation settings as opposed to elite athletes. Consequently, in the majority of cases, sports dietitians working with Paralympic athletes must apply their sports nutrition knowledge from able-bodied athletes in a manner which requires an in-depth understanding of the specific sport/activity and the manifestations of the athlete's impairment, which becomes a very individualised approach.

21.3.1 Spinal cord injuries

Spinal-injured athletes include athletes with traumatic spinal cord (SC) lesions, poliomyelitis and spina bifida. Figure 21.1 explains the spinal cord segments and likely effects of a disruption along the cord. Traumatic SC lesions generally result from an accident (e.g. sporting or motor vehicle). Movement and sensory function is disrupted below the level of the SC lesion, with the resulting impairment depending upon the location and completeness of the lesion. Spina bifida is a congenital developmental defect of the spinal column in which vertebral arches have failed to fuse, resulting in an abnormal gap in the spinal column (Winnick 1995; Goodman et al. 1996). Since nerve tissue can protrude through this gap, it can become damaged, resulting in varying degrees of paralysis and loss of sensation in the legs and lower trunk. As with traumatic spinal cord injuries, the manifestation depends on the location of the gap (or lesion) and the amount of damage to the spinal cord (Winnick 1995). Hence, the physiological response of individuals with spina bifida and traumatic spinal lesions are very similar, and for the purposes of this chapter will be grouped as spinal cord injuries (SCI).

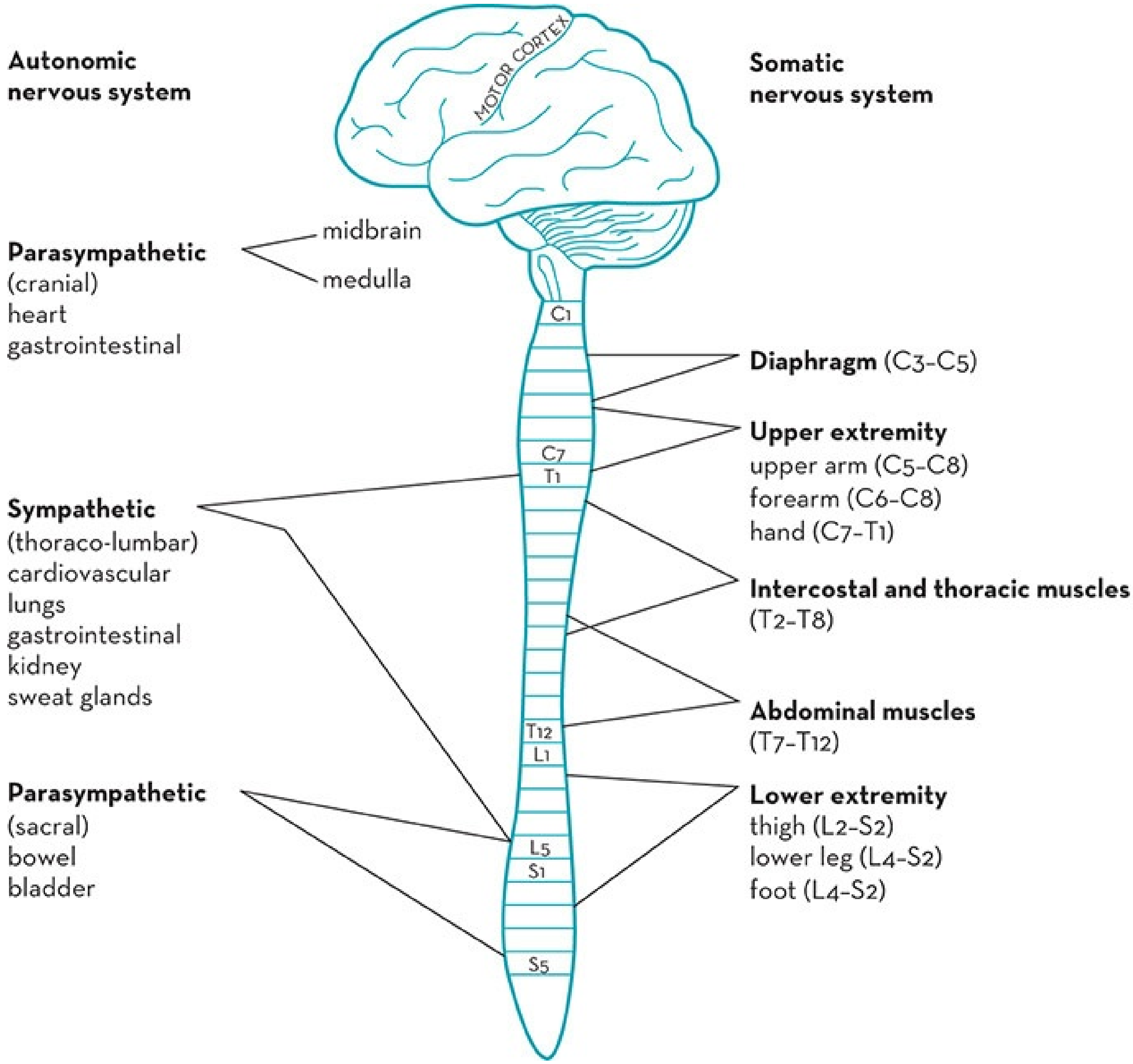

Figure 21.1 Spinal cord injury innervation

Source: Adapted from Glaser 1985, p. 269, reprinted with permission

The *normal* physiological response to exercise may be disrupted in people with SCI (especially T6 and above) through a number of mechanisms: ineffective vasoregulation below the SC lesion, increased total peripheral resistance, reduced ability to increase stroke volume because of blood pooling through the absence of a musculo-skeletal pump from lower limbs, and less total active muscle mass. Hence, individuals with paraplegia, poliomyelitis or spina bifida tend to have dampened peak oxygen uptakes (Shephard 1988) and anaerobic capacity (Van der Woude et al. 1997), the degree of change is proportional to the level and completeness of their SC lesion. As Figure 21.2 indicates, under normal conditions quadriplegics are unable to raise their maximal heart rates (HR) above 120–130 bpm due to the disruption of their sympathetic nervous system (Wicks et al. 1983; Hjeltnes 1984; Coutts & Stogryn 1987; Bhambhani et al. 1994, 1995). Similarly, their $VO_{2\ peak}$ and lactate levels fall well below those of other athletes undertaking similar work. Even athletes with SC lesions between T1 and T6 tend to have a more restricted peak HR and $VO_{2\ peak}$ relative to able-bodied people and those with Cerebral Palsy (Wicks et al. 1983; Hopman et al. 1993; Barfield et al. 2005). Although this may limit their absolute exercise capacity, their ability to maintain power output and speed at any given percentage of their maximal capacity remains similar to able-bodied individuals. Ratings of perceived exertion are useful in these populations to determine exercise intensity relative to maximum.

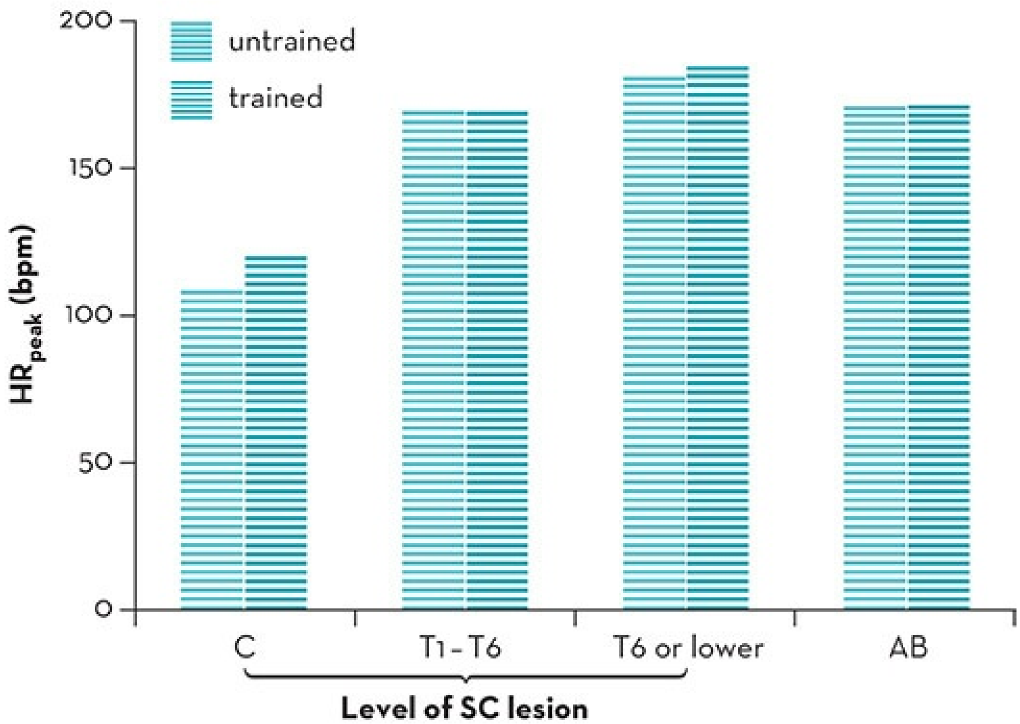

Figure 21.2 Comparative peak heart rates in untrained and trained spinal cord injury and able-bodied athletes

Source: Adapted from data by Wicks et al. 1983; Hjeltnes 1984; Coutts & Stogryn 1987; Bhambhani et al. 1994, 1995; Price & Campbell 2003

Resting metabolic rates (RMR) of SCI individuals have been reported to be up to 30% lower than reference standards (Abel & Platen 2003), primarily because of the reduced active muscle mass (Noreau & Shephard 1995). Consequently, metabolism is relative to the level of SC lesion. Energy expenditure of exercise is also lower than in able-bodied individuals (e.g. 17–27 kJ/min at heart rates ranging 102–145 bpm) (Abel & Platen 2003), due to the smaller absolute muscle mass used. Price (2010) presents a useful review of energy expenditure of SCI, including reference estimates of energy expenditure during exercise. Energy expenditure during exercise is reported to be 26–85% less than those of able-bodied individuals, and to vary with the type of exercise and the level of SC injury as well as experience/competitive level of the athlete (e.g. elite wheelchair tennis and basketball players exhibit around 10% greater energy expenditure during exercise than recreational players).

Whether SCI athletes use a different fuel mix to able-bodied athletes during exercise remains uncertain. Only one study has investigated muscle glycogen status in people with SCI.

Skrinar and colleagues (1982) took muscle glycogen samples from the deltoid muscle before and after one hour of submaximal exercise in four SCI subjects. Although the authors did not control for the effects of pre-exercise diet and used the deltoid muscle (which is not the primary muscle used in wheelchair ambulation), it was interesting to note that the mean pre-exercise glycogen level (92 mmol/kg wet weight) was lower than reported resting glycogen values in able-bodied athletes (120–140 mmol/kg wet weight). The mean change over the one hour of exercise was 30 mmol/kg wet weight (Skrinar et al. 1982), which is comparable to leg muscle glycogen losses in able-bodied athletes at the same relative exercise intensity (Karlsson & Saltin 1971). The researchers suggested that the lower pre-exercise glycogen levels in the wheelchair athletes could be attributed to a greater dependence on glycogen stores for simple daily ambulation. An alternative explanation could be that upper-body musculature may have a lower capacity to store muscle glycogen compared with that in able-bodied athletes. Goosey-Tolfrey and colleagues (2006) studied quadriplegics (both wheelchair rugby and tennis) and found these athletes have relatively high anaerobic profiles when compared with paraplegic athletes. The well-established understanding that carbohydrate (CHO) is the primary fuel source at high intensities gives reason for CHO being an important consideration in SCI athletes and the same guidelines for CHO intake should be followed for able-bodied athletes.

Knechtle and colleagues (2003) reported that peak fat oxidation rates occurred at an intensity of 75% of $VO_{2\ peak}$ in both trained SCI athletes and trained cyclists undertaking cycle ergometry. However, the same group reported lower peak fat oxidation rates at a lower intensity (55% $VO_{2\ peak}$) in well-trained SCI athletes undertaking wheelchair ergometry (Knechtle et al. 2004a) and handbike cycling (Knechtle et al. 2004b). Hence, it is apparent that more studies are required to determine fat and CHO oxidation during exercise in SCI athletes.

The neurological interruption resulting from an SC lesion means that evaporative cooling from sweating is reduced in these athletes (Hopman et al. 1993; Price 2006). The potential surface area available for cooling through sweating is related to the level and completeness of the SC lesion—the higher the lesion, the smaller the surface area (Normell 1974). Although lower sweat rates may appear to be advantageous for minimising dehydration and slowing the decline in stroke volume in the SCI, such a decrease actually presents a disadvantage for controlling body temperature. In addition, some medications, such as anti-cholinergics (used to control muscle spasms) can also impair thermoregulatory abilities (Kennedy 1995). Finally, the unique body positioning and movements in a wheelchair may further limit heat dissipation (Sawka et al. 1984).

In hot conditions, SCI individuals will experience a net heat gain, especially those with high-level SC lesions (Petrofsky 1992; Broad 1997). Consequently, the body temperature of an individual with an SCI is more difficult to regulate effectively in the heat (Price & Campbell 2003), even when they are not actively exercising (such as in archery and shooting events) (Broad 1997). The metabolic heat production of exercise exacerbates this heat load, and the ability to manage this depends on the individual's sweating capacity (Price 2006). Active cooling, both before and during exercise, reduces elevations in body temperature and ratings of perceived exertion (Webborn et al. 2005) and results in improved exercise performance (Goosey-Tolfrey et al. 2008a; Webborn et al. 2008). Methods include keeping athletes in the shade as much as possible, spraying them with water and applying cooling devices (e.g. head pieces, neck wraps, cooling jackets) and using ice drinks. Cooling needs to be applied regularly, especially when exercising or competing in the heat, as the metabolic heat produced may exceed the cooling potential of the mechanism used (Armstrong et al. 1995; Broad 1997). Whether this challenge to exercise in the heat exacerbates CHO usage during exercise has not yet been determined. Fluid intake and total sweat losses should be monitored when active cooling is utilised as it can substantially reduce fluid consumption (Goosey-Tolfrey et al. 2008b) and hence may increase the risk of dehydration.

In cold environments, active heating and heat loss prevention are necessary since the ability to vasoconstrict peripheral blood vessels is usually impaired (Petrofsky 1992), resulting in rapid heat loss in a higher-level SCI athlete. In these circumstances it is important for athletes to be well covered and support personnel to be observant to prevent hypothermia. In ambient conditions (20–22°C), there is little difference in the thermoregulatory response to exercise between those with low-level SC lesions and able-bodied individuals, although those with high SC lesions continue to exhibit obvious difficulties controlling body temperature during exercise (Price & Campbell 1999; Price 2006).

Bone density of lower limbs in SCI individuals is around 25% lower than that of able-bodied people (Thompson 2000; Miyahara et al. 2008; Sutton et al. 2009), and is negatively correlated with number of years since injury (Miyahara et al. 2008). However, bone density is higher in the arms in wheelchair athletes compared to able-bodied controls (Miyahara et al. 2008; Sutton et al. 2009).

Gastrointestinal transit times can also be more prolonged in athletes with an SCI (Krogh et al. 2000). This can impact on decisions regarding timing and type of food intake before and after exercise, interaction of food with medications, timing of medication ingestion, and protocols surrounding ingestion of sports nutrition supplements.

21.3.2 Cerebral palsy and acquired brain injuries

Cerebral palsy (CP) is a group of permanent disabling symptoms primarily affecting voluntary control of musculature, resulting from non-progressive damage to the motor control areas of the brain (Winnick 1995). The most common symptoms include impaired muscle coordination, reduced capacity to maintain posture and balance, and reduced ability to perform normal movements and skills. These neuromuscular symptoms are often accompanied by disturbances in sensation, cognition, communication, perception and/or behaviour, and/or by a seizure disorder such as epilepsy (Rosenbaum et al. 2007). A number of neurological conditions which result in similar functional disturbances are grouped within this classification, including stroke and acute head trauma. The symptoms resulting from CP vary widely, from barely noticeable alterations in gait and neuromuscular control through to greatly restricted ability to walk and undertake activities of daily living such that an electric wheelchair is required for mobilisation. For some individuals, the CP only affects one limb (monoplegia). More commonly, it will affect one side of the body (hemiplegia) or the lower limbs more than the upper limbs (diplegia). The symptoms exhibited by those with CP can be classed in three ways:

- *Spastic CP*, the most common, involves shortened muscles and a decreased range of motion around the joints.
- *Athetoid* (or *dyskinetic*) *CP* involves uncontrolled involuntary muscle contractions which often occur all over the body.
- *Ataxic CP* results in an inability to activate the correct muscle patterns, whereby the ability to walk and spatial awareness are impacted.

With this wide variation in neuromuscular function comes a wide range of energy expenditure: at rest, during activities of daily living (including ambulation) and during exercise. Very little research has been undertaken in athletes with CP. In a study of non-athletes using doubly labelled water (used to determine energy expenditure), those individuals who were wheelchair dependent had lower total energy expenditures (8430 kJ/d) than those who were ambulating (10 360 kJ/d) (Johnson et al. 1997). Regardless of ambulatory status, having athetosis increased estimated RMR (Johnson et al. 1996); however, athetosis also reduced the amount of day-to-day activity undertaken. As a result, total energy expenditure did not differ compared to non-athetotic CP individuals, and the biggest predictor of total energy expenditure was ambulation status (Johnson et al. 1997). For the sports dietitian, estimates of energy expenditure will need to account for ambulation status and involvement of athetosis along with the sport the athlete is undertaking.

Limited research in CP athletes means that macronutrient recommendations should be estimated in relationship with total energy expenditure. Individuals with spastic CP exhibited an elevated heart rate response to exercise compared to able-bodied controls, even when gross motor function was considered excellent (Suzuki et al. 2001). How this impacts on exercise capacity remains uncertain, and most likely will vary substantially according to the limbs impacted by CP and the sport undertaken.

Fluid requirements and sweat rates can vary substantially in individuals with CP. Sweat gland numbers and patterning may vary between the affected sides of the body compared to the unaffected side (Unnithan & Wilk 2005; Korpelainen et al. 1993). Consequently, determination of individual fluid requirements and sweat rates is equally important in athletes with CP as it is in able-bodied athletes.

Lower bone density is likely in those with CP if their mobility is more confined due to diminished load-bearing activity. Careful attention should be paid to optimising bone density through sufficient calcium and vitamin D status.

From a practical perspective, the sports dietitian needs to be aware of any potential limitations in feeding capacity, especially with more debilitating forms of CP. Modified food textures may be necessary for those with feeding difficulties, and some may even require enteral feeds. Fine motor control in preparing and cooking food, and fatigue associated with standing and working around the kitchen, may also benefit from modifications in the kitchen and in recipes.

21.3.3 Amputees

Athletes classified as amputees may have had a physical amputation of a limb (or more than one limb), primarily due to cancer or traumatic injury, or have a congenital limb malformation. Amputees are involved in most sports. Lower limb amputees may use prosthetic limbs, or may ambulate predominantly in a wheelchair.

The ability to accurately estimate energy expenditure in amputees includes difficulty ascertaining height (bilateral lower limb amputation), ambulation via wheelchair, and loss of a major proportion of limbs. Wherever possible, an assessment of lean body mass (e.g. via dual energy X-ray absorptiometry—DXA) will enable more accurate prediction of energy expenditure via the Cunningham equation (Cunningham 1980). Alternatively, Osterkamp (1995) reviewed the proportional energy expenditure contribution of different body proportions/limbs, which may be useful in adjusting estimates of resting metabolic rate. Athletes with an amputation may have a higher energy expenditure in ambulation due to the balancing required of prosthetic limbs (Genin et al. 2008) or the lack of limb (for arm amputees) (Yizhar et al. 2008). For example, Schmalz and colleagues (2002) showed that the VO_2 of treadmill or free walking (an indicator of energy cost of exercise) is 25% higher in transtibial amputees using a prosthesis than able-bodied individuals, and that of transfemoral amputees using prostheses was 55–65% greater. To counterbalance this, the lesser muscle mass will reduce RMR so overall energy expenditure may not differ substantially from that of able-bodied counterparts.

Assessing body composition in athletes with an amputation is most accurate using surface anthropometry (raw data without conversion to percent body fat) and/or DXA. The innate assumptions of bioelectrical impedance analysis and any form of body density assessment (such as BodPod) are violated with the loss of part or all of a limb. Modifications may be required to standard surface anthropometry procedures in regards to the sites measured, the side from which measurements are taken and assessment of stretch stature, depending on the exact nature and location of the limb deficit. Sitting height is an appropriate alternative to stretch stature in bilateral lower limb amputees, especially when the athlete is still growing. Changes in body mass/composition can have a substantial impact on the fitting of prosthetic limbs and must be considered if body composition change is required.

Medically, athletes with an amputation who wear prosthetic limbs are at higher risk of pressure wounds at the site of contact with the prosthesis. Bone density is often lower in the affected limb compared to an intact limb.

From a practical perspective, athletes with an upper limb amputation may require adaptive tools to assist with cooking. Those with lower limb amputations fatigue more rapidly when required to walk long distances or stand for prolonged periods of time, such as doing grocery shopping or cooking, so this should be considered if undertaking practical nutrition education sessions with them.

21.3.4 Visual impairment

Athletes with visual impairment (also known as VI athletes) are generally classified into one of three categories, ranging from least to most visually impaired. Some athletes require the support of a guide in order to participate in their sport, and some sports are only for those with visual impairments (such as goalball and five-a-side football). There are many potential causes of visual impairment, including traumatic injuries, retinitis pigmentosa, hereditary retinal disorders (such as albinism and congenital nystagmus) and optic nerve hypoplasia. For some athletes, vision can continue to deteriorate over time leading to a change in classification.

Physiologically, there are no known differences between visually impaired athletes and able-bodied athletes during exercise, hence sports nutrition principles can be readily applied. Visual impairments can lead to disrupted sleep patterns associated with the lack of light sensory mechanisms and a natural circadian rhythm that exceeds 24 hours. These potential effects can negatively impact on recovery and contribute to variable day-to-day performance times (Squarcini et al. 2013). Individuals with albinism may have reduced vitamin D status due to their increased skin sensitivity to sunlight and avoidance of exposure. The intake of omega-3 fatty acids may be beneficial for a number of progressive eye disorders, as will nutritional strategies which optimise nervous system function due to increased reliance on all other forms of sensory input (such as hearing, smell and touch).

There remain many practical considerations, both at home and during travel, for athletes with a visual impairment. These athletes are usually unable to drive, thereby increasing their reliance on public transport and/or other support to get to and from training and supermarkets. This can prolong travel times substantially, resulting in the need to be organised with portable, practical recovery strategies. An inability to use visual cues to monitor hydration status may result in greater attention to sensory cues and education of the athlete through monitoring of urine specific gravity. Limitations in reading food labels results in greater specificity in education surrounding food choices. Increased vigilance is required around choice of supplements when athletes rely on the advice of salespeople. Visually impaired athletes may also be more reluctant to prepare their own food because of safety considerations, requiring a range of strategies to be considered to best support their needs.

21.3.5 Intellectual impairment

Intellectual impairment is classified as individuals with an IQ less than 75, with age of onset under 18 years combined with limitations in adaptive behaviours, such as self-care, communication, social interaction and learning. There are a wide range of diagnoses within this category, including Down syndrome, Asperger syndrome and autism. Following 10 years of non-inclusion at the Paralympic Games, individuals with an intellectual impairment competed in swimming, athletics and table tennis at the London 2012 Paralympics. In contrast, the Special Olympics attracts over 7000 participants from over 170 countries.

Individuals with an intellectual impairment can exhibit lower maximal oxygen consumption (VO_{2max}) than able-bodied individuals, which is particularly evident in those with Down syndrome due to autonomic dysfunction. Regardless, physiological adaptation to training is similar to that of any other athlete (Tsimaras et al. 2001). Estimates of energy expenditure during exercise and ambulation may require adaption if there is an observable biomechanical inefficiency, such as uncoordinated or exaggerated gross motor movement (Lante et al. 2010), as energy expenditure may be higher than that of able-bodied athletes.

Undertaking body composition assessment in athletes with an intellectual disability can be difficult. They tend to be sensitive to touch or uncomfortable with the close presence of another individual. Many have difficulty staying still for an assessment such as a DXA scan. Assessments should be carefully explained and possibly demonstrated to these athletes beforehand and appropriately justified from a professional perspective.

The mode of sports nutrition education may require modification in this group of athletes. Attention spans are short and ability to understand complex explanations is limited, requiring a flexible approach to education with inclusion of carers wherever possible to reinforce messages.

21.3.6 Les Autres

The class Les Autres includes a varied range of impairments that don’t fit into other categories, such as dwarfism, multiple sclerosis, Freidrich ataxia, osteoporosis, severe burns, muscular dystrophies, musculo-skeletal or neural pathway damage and lower-limb deformities. The variation in impairments within this category makes it impossible to generalise regarding physiological, practical or medical issues. The dietitian is encouraged to take time researching and understanding the specific impairment and to modify practice according to the presentation of the athlete.

21.4 Specific considerations

21.4.1 Body composition assessment

In general, the measurement and manipulation of body composition are as relevant to performance for athletes with impairments as they are for any able-bodied athlete. Traditional approaches to measuring body composition may need to be modified for some athletes. For example, the measurement of body mass (BM) needs to allow for muscle atrophy and bone resorption in individuals with SCI, spina bifida and severe cases of CP. Furthermore, where muscle atrophy is present, it can be difficult to differentiate the body fat component of skinfold thicknesses (Gass 1988), and other injuries (such as burns) can influence skinfold thickness. Hence, regression equations used in able-bodied athletes to estimate percentage body fat from underwater weighing, bioelectrical impedance, BodPod and skinfold thicknesses are inappropriate for many athletes with impairments because they have not been validated in this population (see Chapter 3).

Measurements of BM, girths and sum of skinfolds at the same time are useful to assess and track body composition changes over time for most Paralympic athletes. BM may be difficult to measure on a set of portable scales, therefore access to seated or wheelchair scales for wheelchair-dependent individuals is needed. Skinfold sites may need to be altered from conventional sites that are unused or atrophied. Landmarking may be inaccurate such as supraspinale site in an SC individual. Muscle atrophy results in poor differentiation in skinfolds, particularly thigh and calf sites. A trained International Society for the Advancement of Kinanthropometry practitioner can make the necessary modifications to surface anthropometry protocols. This may involve reducing the total number of sites measured (e.g. sum of four skinfolds for SCI individuals—bicep, tricep, subscapular and abdominal) or measuring the left side in hemiplegic CP (which affects the right side of the body).

If available, DXA is the gold standard to assess body composition and is invaluable in tracking changes over time when a standardised procedure is used (see Chapter 3). Waist girth correlated strongly with body fat in 19 female wheelchair athletes using DXA as the validation measurement (Sutton et al. 2009). In this study, BMI, although a crude measure, was also considered a reasonable technique for measuring changes in body fat. Anthropometric equations for converting skinfold measures to percent body fat in both male and female athletes with an impairment are not applicable, except for a modification of the Wither's equation in females (Sutton et al. 2009). Because DXA allows measurement of lean body mass, estimates of resting energy expenditure can also be determined with reasonable accuracy. Compartmentalising body composition is useful for asymmetric individuals and for designing and adjusting prosthetic limbs to ensure even balance in amputees. However, it is not suitable for individuals with frequent muscle spasm, such as athetotic CP, because of the need to remain still during measurement.

No normative anthropometric data are available for Paralympic athletes, presumably because of the large within-sport variability in body shape and size among athletes with impairments.

21.4.2 Medical

As a group, athletes with impairments have higher incidences of medical problems than other athletes. For example, individuals confined to wheelchairs have a high risk of pressure sores, urinary tract infections and infections as a result of wounds, and may have other medical complications depending on the cause of their impairment. Amputees are at high risk of developing wounds around the attachment site of the prosthesis. An individual who has acquired an impairment as a result of an accident may also have internal injuries and problems such as a colostomy, ileostomy or a portion of intestine or bowel removed. Individuals with CP have a higher incidence of epileptic seizures, especially under stress and fatigue, which are attributed to the neurological effects of the disorder. This group often has feeding difficulties, and therefore requires specific dietary intervention.

A common result of SCI is a lack of bladder and/or bowel control (Goodman et al. 1996). Limited mobility as a result of CP, lower-limb amputations or other disorders also disturb normal bowel function. The management of toileting is usually carefully controlled by the athlete to coincide with their sports activity and training time and may be acutely affected by dietary changes, especially changes in fluid intake. SCI athletes have low bone density in their lower limbs, a high prevalence of low vitamin D status and consequent high risk of fracture. In these athletes with bone fractures, healing is slow, which is attributed to the reduced blood flow and nutrient delivery to lower limbs. Given these risk factors, vitamin D status and calcium intake should be routinely checked.

The adverse side effects of some prescribed medications, particularly on appetite and gut function (see Table 21.2), although necessary for maintaining quality of life and functionality, impose further medical and diet-related problems. Essential prescribed medications may contravene the World Anti-Doping Authority (WADA) list of banned substances and require a therapeutic use exemption (TUE) (see below). Some athletes are not compliant with prescribed medication. Others may seek alternative therapies (including herbal and nutritional supplements) to achieve a similar outcome. The sports dietitian needs to be aware of the reasons why an athlete may use a supplement or seeks alternative therapies and assist them in finding effective solutions that don’t contravene WADA regulations.

TABLE 21.2 Frequently used medications among Paralympic athletes

Primary classification	Common medications	Possible side effects
Bowel management/stool softeners and laxatives	Docusate sodium (Coloxyl™) Sennoside (Senokot™) Enemas (Bisalax™)	Stomach cramps, nausea, diarrhoea
Bladder control/antiseptics	Oxybutynin hydrochloride (Ditropan™) Macrodantin™ Nitrofurantoin Cranberry juice or tablets	Dry mouth, constipation (not related to cranberry juice/tablets)
Antispasmodics/muscle relaxants	Diazepam (Valium™) Dantrolene sodium (Dantrium™) Gabapentin/Pregabalin Phenobarbitone Baclofen (Clofen™, Lioresal™) Botox injections	Reduced alertness, lethargy, dry mouth, constipation, upset stomach

Pain relief/antidepressants	Amitriptyline hydrochloride (Tryptanol™) Carbamazepine (Tegretol™) Gabapentin/Pregabalin Medications containing codeine	Gastrointestinal ulceration and bleeding, constipation, urinary retention
Other	Medications for sleep support (e.g. Melatonin, Zolpidem, Temazepam): often used by visually impaired athletes trying to normalise body clock, and by amputees to help manage phantom limb pain Botulinum toxin (or Botox): may be used in the management of spasticity (often in cerebral palsy athletes)	Weak, shallow breathing, confusion, slurred speech, nausea, agitation and potential aggression, headache, nausea Botox side effects (e.g. headache, malaise, slight nausea) are rarely reported and usually transient

Source: Adapted from MIMS Australia 2005

Many pain medications are prohibited for use by athletes (see https://www.wada-ama.org). Where used for therapeutic reasons, a therapeutic use exemption (TUE) must be granted prior to use.

Therapeutic use exemption (TUE)

There are situations when an athlete is prescribed medications or administers medications by a method that is prohibited by WADA. In this case, the athlete must apply for and be granted a TUE. A TUE will only be granted if there is no unfair advantage for the athlete by taking the substance or using the administration method. Although athletes are ultimately responsible for ensuring that any substance used is permitted, dietitians and other health professionals need to be aware of medications taken and the TUE procedure.

21.4.3 Use of vitamin, mineral or sports supplements in Paralympic athletes

Similar to able-bodied athletes, the need for nutritional supplements in athletes with impairments is determined on a case-by-case basis. Vitamin and mineral supplements, for example, may be needed when athletes have difficulty consuming a variety of foods or sufficient energy to meet adequate nutrient intakes or when gastrointestinal tract function is impaired or nutrient absorption is decreased. Also, some medications compromise the metabolism of specific vitamins and minerals (e.g. phenytoin and vitamin D metabolism). Specific medical conditions may require specialised nutrition supplements and therapeutic intervention (such as arginine for pressure wounds).

Sports nutrition supplements such as sports drinks and sports foods have the same potential to provide a practical source of energy, protein and other nutrients to support performance for Paralympic athletes as able-bodied athletes. However, limited research has been conducted on Paralympic athletes to provide any definitive guidelines. Spendiff and Campbell (2003) were the first researchers to demonstrate that consuming an 8% CHO solution prior to exercise improved time trial performance following one hour of steady state exercise in wheelchair athletes. A follow-up study of eight male paraplegic athletes who consumed a 4% then 11%

CHO solution 20 minutes prior to endurance exercise of 1.3 hour reported similar physiological and metabolic responses to those seen in able-bodied individuals. No significant difference (p = 0.08) in performance and power output was reported between the two concentrations of drinks (Spendiff & Campbell 2005). In this study, a tendency towards greater power output with the 11% CHO solution led the authors to recommend a higher concentration solution, which is a logical option when fluid volume may need to be small. Acceptability of the 11% solution was not assessed in this study; however there was no reported hypoglycaemic response, which was attributed to the timing of ingestion prior to exercise.

Only two studies have been published to date on the use of ergogenic aids in Paralympic athletes. The first study (double-blind, placebo-controlled, cross-over design) found no improvement in race time or decrease in lactic acid in four male and two female 800 m wheelchair track athletes consuming 20 g of creatine monohydrate over 6 days in two treatment protocols prior to testing (Perret et al. 2006). However, performance times also increased in the placebo group and a small effect size may account for the absence of an improvement. More recently, no improvement in performance was evident in wheelchair track and field athletes over a 1500 m distance taking either caffeine or sodium citrate, or the two supplements combined using similar protocols (Flueck et al. 2014). However, two studies do not necessarily exclude a potential benefit, which has been consistently reported in well-designed studies using these supplements in able-bodies athletes (see Chapter 15). Dietitians may consider the potential efficacy in experienced Paralympic athletes of ergogenic aids that are appropriate for the physiological demands of their sport. Caution is advised in the initiation of usual dosing protocols with regards to potential side effects and the body size of the athlete.

21.5 Dietary intakes and potential issues for athletes with impairments

There is limited research on sports nutrition requirements and recommendations specific to Paralympic athletes, however in the majority of cases, sports nutrition principles can be readily transferred. Athletes with impairments may have lower daily energy requirements; require modification to CHO and protein recommendations per kilogram of BM; have specific hydration issues; and be more sensitive to dietary changes that affect bladder and bowel function compared with able-bodied athletes. These modifications depend on the nature of the impairment and the mobility restrictions imposed by the impairment.

21.5.1 Reported dietary intakes

Only a few studies have reported the dietary intakes of athletes with impairments. Daily energy intakes of SCI athletes were reported to be between ~6300 and 12550 kJ (Goosey-Tolfrey & Crosland 2010; Krempien & Barr 2011). CHO intakes were 3.4–4.4 g/kg BM/d, and protein

1.0–1.5 g/kg BM/d. In 32 elite Canadian athletes with SCI, 25% of men and women reported intakes of calcium, thiamin, riboflavin and vitamin D less than the Estimated Average Requirement (EAR) based on average nutrient intake derived from a three-day food record (Krempien & Barr 2011). Further, the men's food choices in this training camp indicated a higher incidence of suboptimal intake of several B vitamins, folate magnesium and zinc than the women. Hence one in four of these athletes may be at high risk of nutrient inadequacy if this intake was similar at home. It is not surprising that lower energy requirements make it more difficult for athletes to achieve nutrient requirements. However, it is equally possible that athletes with impairments unnecessarily restrict energy intake to better manage potential gains in fat mass, particularly in those with mobility restrictions. These studies highlight the need to educate these athletes about optimising the nutrient density in their diet to achieve nutrient requirements within a smaller energy budget.

In contrast, Innocencio da Silva Gomes and colleagues (2006) assessed the dietary intake of the 15-member Brazilian amputee men's soccer team and reported an average energy intake of 16000 kJ/d, CHO intake of 7.3 g/kg BM/d; protein intake of 3.1 g/kg BM/d based on a six-day diet record. Some athletes did not meet the RDA for calcium. These average values match or exceed recommended dietary targets for endurance athletes.

21.5.2 Fibre, timing of food intake and bowel control

Large bowel (and sometimes small bowel) function can be impaired when mobility is limited and gastrointestinal tract damage has occurred. As a result, frequency of bowel movement, stool formation and consistency can be adversely affected. Some athletes may have a colostomy or ileostomy. Constipation is common and the use of bowel softeners is frequent among wheelchair athletes. Timing and frequency of bowel motions need to be fairly well controlled to occur at a convenient time; therefore the type and amount of dietary fibre may require adjustment. A consistent dietary and fluid intake in combination with routine training of bowel habits facilitates control.

From a practical perspective, coaches need to be aware of athletes' bowel management routines when considering the timing of training and the time available between a meal and the next training session.

21.5.3 Fluid intake

Similar to able-bodied athletes, providing recommendations on fluid intake before, during and after exercise based on assessment of individual sweat rates (and weight loss) is one method but not always practical. Appropriate scales (such as seated scales) are used for those unable to stand. Other ways to assess hydration status, such as colour of urine may be an option for detecting suboptimal fluid intake. However, urine colour is influenced by some medications so may give a false negative. Refractometers for urine specific gravity assessment are more accurate and are considered best practice to assess daily hydration status.

While maintaining hydration levels is important for all athletes, the practical consequences of consuming adequate fluid require more attention for many athletes with impairments. Proximity to appropriate toilet and water facilities and use of urine collection bags require consideration, especially in wheelchair athletes. An athlete with an ileostomy may not tolerate large fluid volumes, which can dump and fill the ileostomy bag too rapidly. As discussed earlier, some Paralympic athletes have minimal sweat rates, so the focus may be more on effective cooling strategies as opposed to ingesting fluid. Heat gain from exercise and/or the environment increases the drive to drink. It is not uncommon for these athletes (especially quadriplegics) to gain weight during a training session as fluid intake exceeds sweat rates. This should not necessarily be considered dangerous in terms of the risk of hyponatraemia provided the volume of fluid ingested remains relatively low. Dietitians need to consider the type of fluid and the relative concentrations of fuel and electrolytes that optimise performance without over-hydrating the athlete.

21.5.4 Body composition management

Extremes of underweight and over-fatness can be evident in athletes with impairments. While some body composition alterations are related to the impairment itself, the genetic background of the athlete is also a contributing factor similar to able-bodied athletes. Each athlete will need to be considered on an individual basis according to the nature of the impairment, the potential impact of the impairment on metabolic rate and the impact body composition has on exercise performance in their sport.

Athletes with intellectual impairments often have BM control problems for a wide variety of reasons, including poor food choices from lack of insight, food used as a comforter, either limited or excess voluntary activity and genetic disorders (such as Down syndrome). When working with intellectually impaired athletes, it is imperative to work together with their primary carers to assist in managing the foods provided to the athlete.

Many non-wheelchair-dependent lower-limb amputees or athletes with CP appear to have higher energy demands, possibly related to the greater biomechanical demand of walking. In contrast, wheelchair-dependent athletes (including SCI and CP) can have reduced metabolic rates and energy costs of exercise, yet have similar micronutrient requirements as other athletes (or higher, especially when wound healing is active such as pressure wounds).

The ability to accurately estimate energy expenditure in some Paralympic athletes is limited by the lack of published data on energy expenditure during exercise, ambulation/activities of daily living, and resting metabolic rates. To estimate energy expenditure in athletes with impairments, a combination of current reported dietary intake and standard methods is used (as outlined in Chapter 5). Estimating energy requirements using indirect methods and designing menu plans with a specified kilojoule intake is mostly guesswork, which highlights the importance of regular monitoring of diet, body composition changes, fatigue levels, training capacity and recovery. Menu plans may require shifting food timing to optimise preparation for and recovery from training sessions in practical, nutrient-dense ways, and departing from more traditional eating practices where a lot of nutrients are delivered in proportionally larger meals.

21.6 Eating difficulties and behaviours observed in some athletes with impairments

Regardless of whether you are working with an individual or a group of athletes with impairments, you should be aware of various aspects of feeding or eating that differ from able-bodied athletes. Examples include but are not limited to the following:

- Individuals with impaired hand function (e.g. quadriplegia, arm amputees/arm deformities) may have difficulty handling standard serving and eating utensils—for example, reaching into a buffet and serving food onto the plate, especially if the buffet height and depth is at the wrong level. Similarly, plastic utensils are difficult to control even for someone who has good manual dexterity. At buffets, chopped meat is easier to handle, as is penne pasta compared with spaghetti. Consideration of this is necessary when liaising with hotels, restaurants or any catering situation involving these athletes.
- People with some forms of CP and classes of 'Les Autres' experience feeding difficulties. Some of this group may require enteral feeding or texture-modified food to supplement their limited oral intake.
- Athletes with visual impairments generally work on a 'clock' system to locate foods on their plates (e.g. meat at 12 o'clock, peas at 6 o'clock) and may use their hands to assist with eating. High standards of personal hygiene when eating and access to hand sanitation is an important for food safety; this may need to be reinforced and encouraged, especially in young athletes. They may also require assistance to assist with serving food (especially in buffets and unfamiliar environments).
- Athletes with intellectual impairments can display some 'unusual' responses to food and eating. Examples include unfounded fears of a specific food because of the way it looks, inability to use eating utensils in the customary manner and eating excessive quantities of foods, but not comprehending why such behaviour is inappropriate. Resolving these issues takes time, patience and understanding and can challenge a dietitian's patience.
- When undertaking practical sessions with athletes with impairments, such as amputees and CP, be conscious of fatigue associated with ambulation in these groups. Chairs should be made available during cooking sessions to enable rest periods, and supermarket tour durations shortened.

Summary

Working as a sports nutrition professional with athletes with an impairment is both challenging and rewarding. Sports nutrition principles can be readily transferred to the Paralympic athlete population provided the dietitian has a good understanding of the needs of the athlete and the physiological basis of the sport they are competing in. The lack of sport- and athlete-specific research should not be considered an impediment, but rather a reason to explore the knowledge that we have and apply it very specifically to the individual.

Practice tips

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- The sports dietitian gains an insight into the physiological and functional capacity of each Paralympic athlete under his/her care from direct observation while training and competing or by sharing meals. This experience helps the dietitian to better target nutrition intervention for performance enhancement and to consider the feasibility and practicality of dietary advice. For most athletes, the general principles of sports nutrition apply. However, in some athletes, novel approaches are needed. Paralympic athletes tend to be adaptable and knowledgeable about their personal needs, and can help the dietitian devise solutions. Therefore, being open-minded about the athlete's food choices and eating behaviours, and modifying rather than imposing your own solutions, is an effective strategy.
- Be aware of the terminology to use with the athlete and specifically understand the effects that inappropriate language can have on individuals with impairments. It is also important to understand their specific sport and its rules/requirements, and their classification within the sport.
- Communication boards may be useful with some athletes in a consult. Ensure resources are practical and easy to read and understand (i.e. enlarge documents or provide in electronic format for use on touchscreen for visually impaired athletes). Online shopping is convenient if accompanied by some guidance and a weekly menu plan.
- While the technique used to assess nutritional status can be applied to an athlete with impairment (see Chapter 2), additional information—including duration, type, cause and severity of impairment, mobility, medications, feeding issues, practical support tools used at home and medical history (i.e. comorbidities)—is required. Therefore, screening or consultation forms can be adapted to reflect these additional issues. If the athlete is within the first 12 months of acquiring their impairment (e.g. post-trauma), they will still be undergoing many physical and psychological adjustments.
- Sum of skinfolds and girths provide more useful information about body composition than BM alone or BMI (note that skinfold measurements may be undertaken on fewer sites than in able-bodied athletes if this is more practical for the specific athlete).
- Some athletes are accompanied by a support person or carer, who assists in many ways (e.g. such as riding the front of a tandem bike for a visually impaired cyclist or assisting an athlete with CP get into position for a throwing event). It is important to respect this relationship and the intimate knowledge carers have about the athlete, and to involve them in discussions.
- Energy requirements of some athletes with impairments may be low and this is associated with a high risk of suboptimal nutrient intakes and deficiency. Consuming nutrient-dense foods is a high priority in athletes following a low energy diet. Additional intervention is needed in this group who travel or eat out frequently and are susceptible to weight gain. In contrast, others with higher energy requirements may struggle to match intake with expenditure, especially when they are not interested in shopping, preparing or eating food or have other medical issues that affect food intake. In these cases, energy supplements, energy-dense food and fluids are indicated.
- Some athletes with impairment who have bladder emptying problems, or use urinary catheters, intentionally

limit fluid intake at times where practical access to a bathroom is restrictive (e.g. long-haul flights, training sessions outdoors, in the afternoon/evening before bed) and hence may chronically dehydrate. When advising about fluid intake, investigate how the athlete manages bladder control and the likely effects of increasing fluid intake. Be willing to trial strategies in conjunction with the athlete to come up with the best solution to prevent potential kidney damage, decrease risk of dehydration and improve performance capacity.

- Electrolyte drinks are palatable, help retain fluid balance, reduce the risk of dehydration and urinary tract infections and improve recovery time after long-haul flights and hard training sessions. Urinary specific gravity is the best way of regularly monitoring hydration status.
- Overheating is a risk for SCI athletes because of an impaired sweat response associated with the SC lesion. Using cooling methods during training and competition, particularly in the heat, is an effective strategy for those athletes to manage their core temperature as well as prevent excessive fluid intake. Fluid requirements may actually be low, so encouraging them to drink more is not the optimal solution.
- Bowel management may be an ongoing problem in some athletes, especially those with SCI and those in wheelchairs. Athletes have their own routine, which should be considered when suggesting changes to diet or timing of supplements (e.g. stool softeners). Dietary fibre and fluid intake, as well as consistency with meal patterns and food choices, assist bowel control. Bowel management is a challenge during travel to competition, and at competition venues overseas (see below).
- When working as one of the support staff with a team, or in an unfamiliar environment for the athletes, you may be required to provide practical assistance. For example, visually impaired athletes may require assistance ordering food and locating food and utensils in a new eating environment. The ‘clock system’ is an easy way to describe location of food on a plate. If operating from self-contained units, it is best to room them with sighted athletes. Depending on the impairment, assistance with carrying cutting up food and other lifestyle-management issues may be required.
- Vitamin D status of Paralympic athletes may be suboptimal, particularly in SCI athletes. (see section 21.4.2). Routine screening and monitoring is indicated in other athletes with impairments with limited sun exposure.
- Recommendations for the use of sports nutrition supplements should involve consideration of potential interactions with medications and their energy content with respect to energy requirements of the athlete.
- The use of ergogenic aids requires careful consideration in Paralympic athletes. Aside from assessing actual need, it is important to consider the nuances of the individual event, which may differ from their able-bodied counterparts. Potential interactions with medications and side effects may require a more graduated approach when trialling a new supplement.

TRAVELLING WITH PARALYMPIC ATHLETES

When travelling with Paralympic athletes, extra time is needed for the transportation of athletes to venues. With flights, wheelchair-dependent individuals usually board first and are the last to deplane, which means longer periods of time in transit. The athletes also face the challenges of eating different foods, being in an unfamiliar environment and not having access to their usual home setup or toileting facilities. Strategies for the sports dietitian and other support staff to streamline the travel experience for athletes with impairments are listed below.

- With air travel, split the team into smaller groups, rather than travelling together. This reduces the logistics for airline staff and helps to distribute the workload for the support staff.
- Provide additional support for athletes with intellectual impairments around meal times—for example, assist with meal choices at buffets, personal hygiene and food safety practices.
- Some athletes may require foods to be processed or fluids to be thickened. Ensure that appropriate equipment is available (e.g. mini-blender for pureeing when self-catering), and liaise with kitchen staff so they can provide suitable meals.
- Ensure the timing of training sessions and meals allows for the athlete’s personal care routine (e.g. bowel management routines).
- Organise for recovery snacks and fluids to be available at venues to ensure availability of familiar post-exercise food choices.
- Ensure the setup of the dining hall/buffet is suitable for the athlete group. Consider the location of accommodation relative to dining areas to minimise walking/pushing distance, the height of the buffet tables (may need to be lower for wheelchair athletes) and the serving utensils used (lightweight and not requiring too much hand rotation). Where relevant to the athlete group, check that menu options do not require cutting (i.e.

no large pieces of meat), can be easily handled with fork or spoon only, and that metal cutlery is available (rather than plastic cutlery, which can be difficult to control).

USEFUL WEBSITE

- **www.paralympic.org**
  International Paralympic Committee: provides information on events, class of athletes, history of Paralympic sports

USEFUL BOOKS

- Vanlandewijck YC, Thompson WR, eds. The Paralympic athlete. Chichester, UK: Wiley-Blackwell, 2011.
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CHAPTER TWENTY-TWO

Immunity, infective illness and injury

Gary Slater, David Pyne and Kevin Tipton

22.1 Introduction

Consistency in training is a common trait of successful athletes. An athlete who is injured or ill may be required to reduce training load or withdraw from training and competition. With insufficient training stimulus, a detraining effect occurs, which involves a partial to complete loss of the metabolic and physiological adaptations to training. The rate of loss depends on the athlete's stage of training and the duration of time of cessation or reduced training load.

Training-induced immunosuppression has been reported in some athletes. The immune system may be compromised during periods of high-intensity training or when there is insufficient food or fluid intake, thereby increasing the susceptibility for opportunistic infection and slowing the rate of healing and recovery from injury. All macronutrients (CHO, protein and essential fats) and several micronutrients are involved in immune function (i.e. iron, zinc, selenium, magnesium, most fat-soluble vitamins, vitamin B6, B12 and folate). Athletes likely to be at risk of opportunistic infections are those with poor nutritional status, low energy availability and a chronic unmanaged illness that decreases appetite or impairs nutrient absorption.

The immune system also plays an important role in the repair of tissue after injury as part of a coordinated inflammatory response. Knowing how the immune system promotes recovery from injury, and the effect of various nutritional strategies on these processes, forms part of an overall treatment and management plan for athletes. Nutritional strategies alongside therapeutic interventions such as non-steroidal anti-inflammatory drugs and corticosteroids, and various physical modalities such as ultrasound and cryotherapy, are key considerations in injury treatment and management.

This chapter focuses predominantly on nutritional considerations that help to minimise the adverse implications of a potentially suppressed immune system associated with repeated high-intensity exercise; help to prevent short-term infective illnesses common in athletes (e.g. upper respiratory tract infection (URTI), gastrointestinal tract infection (GIT), skin infection); and help athletes recover from injury. Chapters 20 and 23 consider gastrointestinal (GI) tract conditions and GI diseases in more detail, including the treatment of GI symptoms caused by pathogens.

22.2 Physiological and metabolic effects of cessation of training

A consequence of illness and injury is detraining culminating in the loss of training-induced adaptations that underpin higher levels of exercise performance. Illness or injury at the time of competition can have a direct effect on competitive performance, but in the context of training, the influence of detraining could extend to weeks or even months. In physiological terms, studies have shown that athletes who detrain even for one month can experience adverse effects on body composition, fitness and metabolism (Ormsbee & Arciero 2012). Moreover, cessation of resistance training can decrease all components of muscular strength (Bosquet et al. 2013). The magnitude of detraining appears to be larger in older individuals (>65 years) and in active people for maximal force and maximal power compared with recreational athletes. Training status, age and duration of the training cessation are all factors that will determine the individual degree of impairment in physiological and performance measures (Bosquet et al. 2013). Strategies to counteract the negative effects of detraining include cross-training, taking short off-season breaks and employing active recovery rather than complete rest or a break from training. Nutritional strategies during this period are also important to balance caloric and nutrient intakes with reduced energy expenditure. Strategies for modifying training during recovery from injury that limit the risk of training-induced impairments are outside the scope of this text and are found in the most recent position statement on maintaining immune health in athletes (Walsh et al. 2011).

22.3 The immune system: role in infection and injury

The immune system involves a combination of physical, cellular and soluble components that work together to defend against pathogens and repair tissue damaged from injury or illness. The physical components of the immune system include the skin, epithelial cells, cilia (hair-like strands that line the respiratory tract) and mucosal fluids. The cellular components include white blood cells (leucocytes) that are subdivided into lymphocytes, monocytes and granulocytes (neutrophils). A number of proteins including immunoglobulins, complement, lactoferrin and lysozyme form the soluble components of the immune system. Pathogenic agents include viruses, bacteria, fungi, allergens and toxins. Body tissues and cells can be damaged by trauma, cuts, burns and surgery, and in the context of exercise and sport, from the effects of an acute or chronic injury. An immune and inflammatory response typically involves a coordinated process of cellular and soluble components that work synergistically to destroy and remove infected or damaged cells. In some individuals the control or regulation of the immune response is compromised leading to autoimmune disorders such as diabetes or rheumatoid arthritis, where the immune system attacks host tissues and cells, resulting in inflammation and damage. The normal immune response can be compromised during periods of high-intensity training, environmental stress (e.g. extremes of ambient temperature and/or altitude), heightened exposure to pathogens (via air-borne transmission, blood-borne transmission, direct physical contact and common source exposure such as contamination food or fluids) or when there is insufficient intake of food and/or fluid.

Infection by a pathogen or an injury stimulates the production of cytokines (regulators of the immune system), which are modulated by nutrients and trigger several inflammatory mechanisms involved in the recovery and healing process (Moreira et al. 2007). In an infection, these cytokines are responsible for initiating an acute-phase reaction, which is evident from the presence of typical symptoms such as fever, loss of appetite, decreased food intake, cellular hypermetabolism and multiple endocrine and enzyme reactions. This reaction to infection is often termed cytokine-induced malnutrition. Some athletes with a high frequency of opportunistic infection may have an underlying undiagnosed condition that increases their susceptibility to infection (Spence et al. 2007). Deficiency of the immunoglobulin M (IgM) is evident in some individuals with recurrent infections, atopy, coeliac disease and systemic lupus erythematosus, but can also occur in asymptomatic individuals. Individuals with a deficiency in another immune protein, complement C3, are also susceptible to bacterial infection.

Following an acute injury, a similar inflammatory process to that observed with an infective illness can occur. Inflammation, muscle damage and oxidative stress have been implicated in the recovery after injury (Bessa et al. 2013). The term *inflammation* is widely used in sports medicine and exercise sciences although its meaning and implications are not always clear. Essentially, *inflammation* is an umbrella term referring to the clinical, physiological, cellular and molecular processes that underpin the repair of damaged tissues (Scott et al. 2004). Physiological processes involved with inflammation include coagulation, vasodilation, local metabolic changes, angiogenesis and occasionally muscle spasm. A positive or negative outcome of injury is influenced by interactions of various cell types including platelets, mast cells, endothelial cells and white blood cells (leucocytes). At a clinical level, inflammation typically manifests as localised heat, swelling, redness and/or pain at the site of injury. The clinician and dietitian need an understanding of these inflammatory processes, and how they inform choices of medical and/or nutritional intervention.

22.4 The immunosuppressive effect of high-intensity exercise

The physiological stress of high-intensity training programs and competition can induce transient impairment in immune function leading to an increased risk of illness. In some athletes, the immune system can become compromised leading to immunosuppression, which increases the vulnerability to infective illness (such as URTI and GIT) and potentially compromises the inflammatory response to injury. Anecdotal reports and some laboratory-based studies have given rise to the so-called ‘J-curve relationship’ between exercise intensity and underlying immune function (Nieman 1994). In this model, moderate-intensity exercise (undertaken by both recreational and elite athletes) can enhance immune function, lowering the risk of illness; in contrast, repeated high-intensity exercise typical of elite athlete training programs can transiently lower immune function, increasing the risk of illness becoming established in the hours after exercise.

Explanations for the impairment of immune function centre on changes in the concentrations and functional activities of immune cells and soluble components, or alterations in other immunoregulatory control mechanisms. Lower salivary immunoglobulin A (IgA) secretion and a higher anti-inflammatory cytokine response to antigen exposure are considered key determinants of infection risk in athletes (Gleeson & Bishop 2013). Treg (regulatory) cells in the respiratory tract play an essential role in regulating mucosal immunology (Liu et al. 2009). These form the so-called ‘first line of defence’ against infection in the respiratory and GI tracts. The common mucosal immune system—in which cells from inductive sites, such as Peyer’s patches in the intestines, traffic to mucosal surfaces following interaction with antigen-presenting cells (Chen et al. 2014)—is important although this sequence of events is not easily studied in athletes. An abnormal immune response could reflect an increased pro-inflammatory response to a bacterial component, a decreased immune regulatory response, or a combination of the two. Exercise-induced changes in plasma concentrations of immune messengers (cytokines and C-reactive protein) and stress-related hormones (catecholamines and cortisol) can also influence the immune response (Menicucci et al. 2013).

Impairments in the immune system and host defence can be identified in several different ways. At a clinical level, immunosuppression can manifest as fatigue, lassitude, disturbed sleep, alterations in mood, and/or presentation of one or more of the classical signs and symptoms of illness. For example, a common cold often presents with a sore throat, headache, runny then congested nose, watery eyes, cough, chest congestion and occasionally a fever and elevated heart rate. At a sporting level, an athlete will report impaired training or competitive performance (possibly physical, cognitive and/or technical) and often prolonged recovery after exercise. A clinician will often make a diagnosis based on a routine consultation, patient history and physical examination, although pathology testing can be undertaken in selected cases.

The key time points for preventive action are the extended periods of prolonged and/or intensive training in the months prior to major competitions for individual athletes, and seasonal games or tournament-style competitions in team sports. In individual sports, pre-competition periods are usually characterised by high-volume and high-intensity training prior to the taper in the weeks immediately prior to the competition. There can also be a seasonal effect with an increased frequency of illnesses in the winter months. Not all athletes show decrements in immune function and an adaptive response is likely in healthy athletes responding positively to training and competitive loads.

The apparent susceptibility of athletes to infection is also linked to other non-nutritional risk factors including high training volume and intensity, exposure to pathogens, poor health status, possibly inadequate sleep, poor recovery practices, and anxiety (Pyne et al. 2000). These risk factors work synergistically and need to be addressed when providing advice to individual athletes. Referral of high-risk athletes to other service providers (e.g. a psychologist) may be needed if anxiety and poor sleep are evident.

22.5 Effect of infective illness on sports performance

Illness of limited duration with mild symptoms, such as the common cold, is unlikely to have major consequences for training capacity and competition performance. When athletes are unwell for several days, with moderate to severe symptoms, sporting performance will be impaired. However, few studies have directly quantified the effect of common infections on athletic performance. In a retrospective study over three years on 69 swimmers from the Australian swim team, Pyne et al. (2005) reported that symptoms of mild infections (predominantly respiratory and, to a lesser extent, GI and viral skin conditions) that occurred for two or more consecutive days in the final six weeks before international competition were likely to have a small adverse effect on competitive performance in male (but not female) swimmers. It was unclear why this response was observed in the males but not the females, although the analysis was based on mean responses and predicting the likelihood of benefits or harm, rather than individual outcomes. Nevertheless, in this study, the chance of harm for individuals, even for mild symptoms of illness prior to competition, was substantial.

Severe symptoms of infective illness in the weeks prior to competition that require medical intervention and complete cessation of training are likely to have a more adverse effect on competitive performance than mild to moderate symptoms. The taper period, which may involve team travel with a corresponding increased risk for opportunistic infection, is a critical time to implement preventive strategies.

22.5.1 Nutrition and the immune system

Micronutrient deficiencies and low energy intake can negatively influence immune function and potentially increase the susceptibility to pathogenic illnesses. Practically all immune systems are adversely affected by protein-energy malnutrition, although non-specific defences and cell-mediated immunity are the most severely affected (Calder 2013). Although an athlete is unlikely to be diagnosed with malnutrition per se, it is not uncommon for athletes to have a deficiency or depletion of single or several micronutrients involved in immune function (e.g. iron and vitamin D). Deficiency of single nutrients results in an altered immune response that is evident when the deficiency state is relatively mild (Chandra 2002). Although there is evidence of compromised immune function with micronutrient deficiencies in the general population (Taylor & Camargo 2011), there is limited evidence in athletes. Clinical trials are difficult to conduct with injured or ill athletes because of limited access and typically small sample sizes so the link between dietary practices, nutritional supplements and immune function is inconclusive (Romeo et al. 2010). Most recommendations are extrapolated from evidence obtained in studies of other population groups.

22.5.2 The effect of suboptimal macronutrients and low energy availability on the immune system

A well-balanced diet rich in macro- and micronutrients is needed to maintain a robust and resilient immune system with sufficient intake of energy and recovery nutrition practices to tolerate the demands of training and competition (Nieman 2008; Gleeson 2013).

Low energy intake/availability

A low energy diet (and subsequent diagnosis of low energy availability) is a risk factor for compromised immune function and increases the potential susceptibility to infection (Montero et al. 2002). Other consequences and symptoms of prolonged inadequate energy intake include poor nutritional status, fatigue, dehydration, delayed growth, impaired tissue repair (Montero et al. 2002), menstrual dysfunction (Thein-Nissenbaum et al. 2011) and psychological stress (Hagmar et al. 2013).

A systematic review of 103 exercise and nutrition studies showed that increasing exercise or physical activity did not necessarily alter ad libitum daily energy intake or macronutrient composition in healthy adults (Donnelly et al. 2013). Despite potential under-reporting evident in most dietary intake studies, some athletes either habitually or intermittently consume suboptimal energy intakes and are diagnosed with low energy availability (see Chapter 5). Commentary 3 provides guidelines on techniques to measure low energy availability in athletes.

Personal circumstances, lifestyle, cultural, socioeconomic and religious factors are also determinants of food choice, which may alter energy intake. For example, during Ramadan, abstinence from all food and fluids from dawn to sunset for one month is practised. During this time, experienced Muslim athletes can maintain their usual training load with few adverse physiological or immunological effects reported and without decrements in fitness measures (Mujuka et al. 2010). In one study of 15 elite male Judo athletes, small but significant changes in inflammatory, hormonal and immunological markers were evident at the end of the fasting period of Ramadan, although most markers were still in the normal range (Chaouachi et al. 2009). The lower biomarker levels reported are of little clinical relevance and, in the absence of a control group in this study, may be indicative of a normal physiological response to training load and/or eating more food in the evening to compensate. Fatigue levels of these athletes were significantly higher during the middle of Ramadan compared to the end, providing further support for adaptation to intermittent fasting.

However, individuals with prolonged energy restriction (i.e. chronic low energy availability) might be at increased risk of an impaired immune response to infectious agents (Gardner et al. 2011). Purported mechanisms associated with influenza during periods of energy restriction include impaired natural killer cell function and altered inflammation. Athletes in weight class or aesthetic sports, who chronically or intermittently restrict energy intake and lose substantial body mass, have shown reductions in immune cell activation, even with short-term weight loss (Umeda et al. 2004). Decreases in salivary levels of the immune proteins, IgG, IgM and complement C3, were still evident seven days after competition in 49 college judoists who had a mean weight loss of ~3 kg just prior to competition (Umeda et al. 2004). Although acute changes in these immunoglobulins were small, the potential cumulative effects of repeated weight cycling might have more substantial effects on the immune system than observed in this cross-sectional study. Further studies are needed to confirm these findings in other athletes and to investigate any increase in the frequency of infection or acute or chronic energy restriction.

Probiotics and immune function

Probiotics are live bacteria that are present in some fermented dairy foods and nutritional supplements, which may prevent or limit the effects of URTI and GIT infections. Probiotics complement the normal GI flora by enhancing gut immunity against GIT infection (West et al. 2009). Several studies of active people have reported modest reductions in frequency, severity and/or duration of both respiratory and GI illnesses from clinical trials using probiotics (Pyne et al. 2012). A systematic review of 37 studies concluded that specific probiotics can also provide benefits for individuals with symptoms of irritable bowel syndrome and antibiotic-associated diarrhoea (Hungin et al. 2013). Probiotics may also be beneficial for athletes who experience acute symptoms of GI discomfort (i.e. nausea, distension, diarrhoea, abdominal pain) associated with eating before or during high-intensity exercise (see Chapter 20).

22.6 Injury

Injuries are an unfortunate but unavoidable aspect of participation in sport. Around 10% of athletes competing at major, multi-sport games incur an injury (Engebretsen et al. 2010, 2013). Reports on injury rates of elite athletes indicate that around half of the injuries received are severe, resulting in cessation of training for around three weeks (on average) to 74 days (Malliaropoulos et al. 2010). Total or partial immobilisation (involving a limb injury) induces changes to site-specific musculotendon remodelling resulting in skeletal muscle atrophy and subsequent reduction in strength and function. These physiological changes may also increase the risk of re-injury (Silder et al. 2008).

Several treatment options for injury include a combination of rest, ice, massage, heat, electrical stimulation, acupuncture and manual therapy. Nutrition intervention is often overlooked, or at least underappreciated, during the acute post-injury period and subsequent rehabilitation phase. Although specific studies of nutrition intervention for promoting recovery from sports-related injury are scarce in groups of athletes, there is adequate evidence from other populations and from physiological models of recovery from injury that can be applied to an athletic population. For example, evidence associated with wound healing, trauma, surgery and muscle-unweighting models—such as bed rest, limb casting and space flight—is applicable to exercise-induced injuries (Adams et al. 2003). Also, the strategies used in the eccentric muscle damage model, including nutrition intervention, are applicable to the muscle changes that occur during exercised-induced muscle injuries (Sousa et al. 2014). However, efficacy of nutrition intervention in athletes with acute exercise-induced muscle damage with sustained immobilisation has not been validated using this model.

Skeletal muscle atrophy occurs within the first 2–3 weeks after total immobilisation (Adams et al. 2003) and can start as early as a few days after injury (Edgerton et al. 1995; Wall et al. 2014). Skeletal muscle loss of ~150 g/d (~1 kg/wk) has been reported (Wall & van Loon 2013). Type II muscle fibres are more susceptible to atrophy than type I muscle fibres (Edgerton et al. 1995; Hvid et al. 2010). Hence, countermeasures (i.e. nutrition intervention and cross-training) to inhibit further muscle loss associated with inactivity should be applied early post-injury (de Boer et al. 2007).

The following section considers the physiology of recovery and rehabilitation after severe injury and the efficacy of nutrition interventions (i.e. energy availability, protein intake and supplements of specific amino acids, fish oil and creatine) involved in optimising the recovery process.

22.6.1 Stages of injury: overview

Most exercise-induced injuries that involve cessation of training or full to partial immobility of a limb go through two main, although not clearly defined, physiological stages in the recovery process. The first stage is characterised by inactivity and muscle atrophy. During this time, metabolic changes in the tissues as a consequence of disuse result in declines in strength and function (Jones et al. 2004). Loss of muscle, specific to the injured site, accounts for these functional decrements, although other tissues, including tendons and ligaments, may also weaken (de Boer et al. 2007).

The second stage is defined by the return of mobility and active rehabilitation. This stage is characterised by increasing activity of the injured limb and recovery of atrophied skeletal muscle and ancillary tissues, with associated return of functionality (Jones et al. 2004). Unfortunately, complete recovery of strength and function following injury-induced immobilisation takes much longer than the time it takes to atrophy (Jones et al. 2004). For example, morphological deficits are still evident two years following hamstring strain injury (Silder et al. 2008) and at ten years after surgery for anterior cruciate ligament reconstruction (Snow et al. 2012).

22.6.2 Stage 1: Tissue repair, inactivity and atrophy

Stage 1 involves the healing process associated with the repair of damaged tissue (e.g. wounds; fractured bones; torn or damaged tendons, ligaments and muscles) and minimising the effects of atrophy associated with immobilisation or reduction in activity.

Tissue repair

Wound healing is a complex process involving several physiological pathways including inflammation, cell proliferation and remodelling. These processes are not discrete. A detailed description of this process is beyond the scope of this chapter and can be found elsewhere (see reviews by Lin et al. 1998; Barnes et al. 1999; Arnold & Barbul 2006; Lorenz & Longaker 2008; Demling 2009).

Repair processes begin almost immediately after injury and continue for days to weeks depending on its severity (Lorenz & Longaker 2008). An inflammatory response is the initial physiological response to injury, which is an essential component of the healing process (Lin et al. 1998). An elevation in acute phase inflammatory biomarkers (e.g. serum ferritin, C-reactive protein, cytokines) may last for a few hours to several days (Lin et al. 1998).

Given that inflammation is a critical component of the healing process, chronic intake of very strong anti-inflammatory medications that suppress or eliminate the inflammatory response during the recovery phase may not be ideal for optimal recovery. In very severe injuries such as a major burn injury, excessive inflammation occurs. In contrast, most exercise-induced injuries, in healthy people and athletes, would not be severe enough for uncontrolled inflammation to be a problem requiring immunosuppression (Lin et al. 1998). Excessive use of nutrient supplements (e.g. fish oil) that have anti-inflammatory properties may be contraindicated, at least in the early phase of tissue repair (Galland 2010; Lopez 2012a, 2012b). The nutrient implications for promoting tissue repair are considered in section 22.6.3.

Tissue atrophy associated with inactivity

Many exercise-induced injuries involve limbs that require immobilisation and a reduction in overall physical activity. During immobility, the balance between the rate of muscle protein synthesis (MPS) and muscle protein breakdown changes, which contributes to a reduction in muscle mass, strength and functional capacity (Phillips et al. 2009), particularly in weight-bearing limbs (de Boer et al. 2007; Glover et al. 2008). The outcome is a negative net muscle protein balance. The consequent muscle loss is attributed primarily to a decrease in basal MPS, although there is some evidence of a transient increase in muscle protein breakdown in the first few days of immobility (Urso et al. 2006; Tesch et al. 2008; Wall et al. 2014). Thus to be effective in minimising muscle loss, interventions focus on inhibiting the decrease in MPS. There is also a reduced response of MPS to nutritional anabolic stimuli termed *anabolic resistance*. Immobility decreases the ability of myofibrillar protein synthesis to respond to amino acids (Glover et al. 2008). Anabolic resistance to insulin during immobility has also been reported (Rasmussen et al. 2006; Wilkes et al. 2009).

As well as muscle loss, bone, tendons and ligaments are negatively affected by immobilisation and reduced activity (de Boer et al. 2007). Decreased collagen synthesis in tendons from immobilisation results in impairment in tendon mechanical properties (de Boer et al. 2007). However, other aspects of tendon, bone and ligament healing in relation to collagen synthesis remain to be determined. Declines in tendon function are perhaps an underappreciated detrimental aspect of limb immobilisation associated with injury. These processes are illustrated in Figure 22.1.

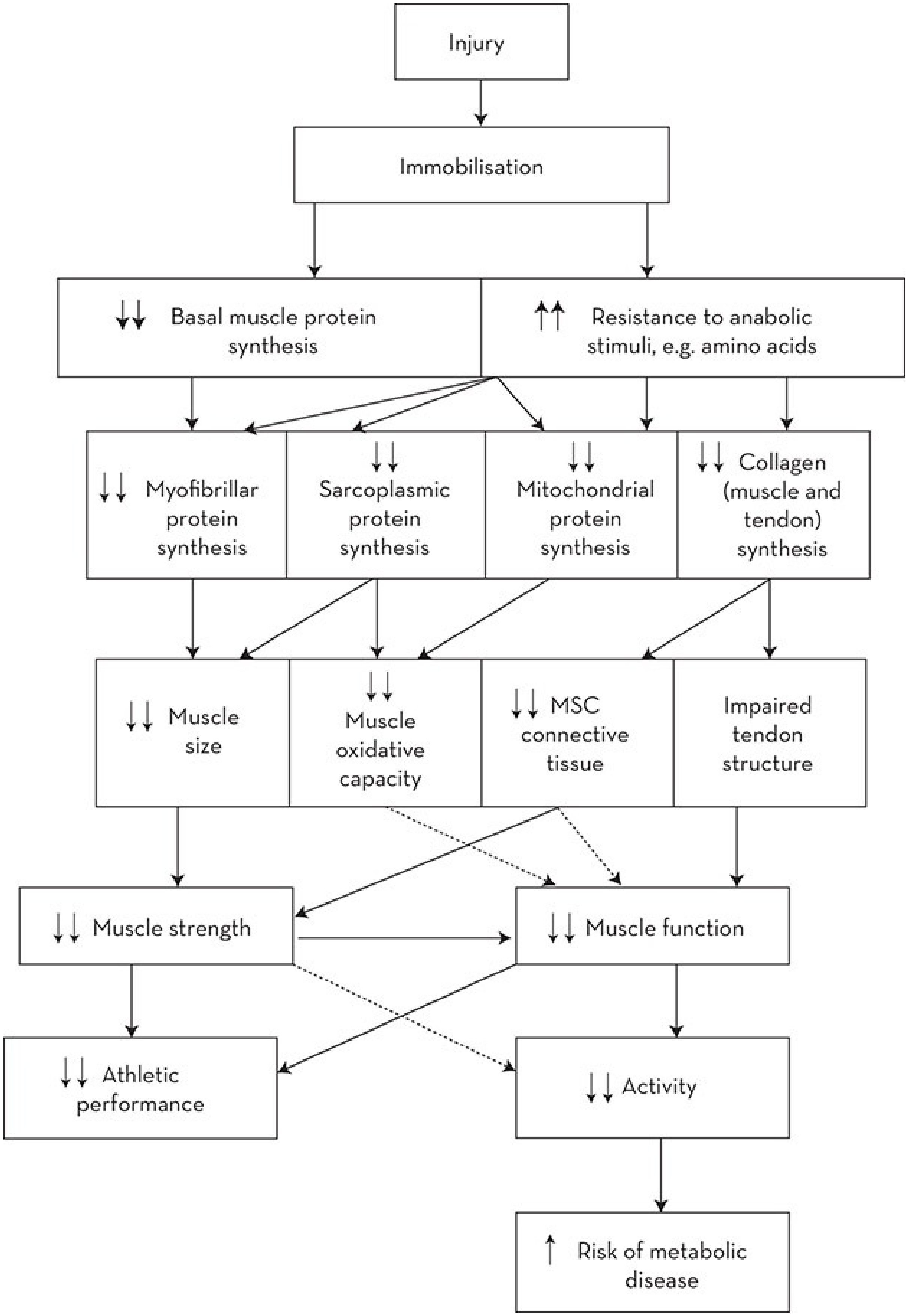

Figure 22.1 Flow diagram of the metabolic and functional changes during immobilisation associated with exercise-

07:40:09.

induced injury. Decreased basal synthesis of muscle and tendon proteins, as well as decreased stimulation from amino acids, leads to a quick and dramatic decrease in muscle size and strength, tendon structure and function

Source: Adapted from Tipton (2010)

In addition to loss of muscle strength, nearly all aspects of mitochondrial metabolism and function are impaired in muscle that is immobilised and inactive. Oxidative capacity, enzyme and signalling pathways involved in mitochondrial biogenesis are depressed or reduced (Abadi et al. 2009). Insulin sensitivity decreases (Stuart et al. 1988; Richter et al. 1989), which is linked to a decrease in GLUT4 content in immobilised muscle (Op't Eijnde et al. 2001). Ultimately, muscle wastage and loss of muscle strength occurs.

22.6.3 Nutrition support for stage 1: Tissue repair, inactivity and atrophy

Generally, the principles of nutrition intervention are similar for both stages with minor modification depending on the location and severity of the injury or surgical procedures required.

Effect of total protein intake on muscle protein synthesis (MPS)

Insufficient total dietary protein intake impedes wound healing (Arnold & Barbul 2006; Demling 2009) and is required for synthesis of collagen and other proteins in the repair process (Lorenz & Longaker 2008). Protein intake stimulates MPS, both at rest and following exercise, resulting in positive muscle protein balance (Biolo et al. 1997; Tipton et al. 1999a; Wilkinson et al. 2007; Tang et al. 2009). However, because of the anabolic resistance that occurs in immobilised muscle, increasing total protein intake does not have the same stimulatory effect on MPS in the injured muscle that is observed in non-injured muscles of athletes (Lorenz & Longaker 2008). In non-injured athletes, the maximal rate of MPS is highest in response to intakes of high biological value protein at between 20–25 g (0.25–0.30 g/kg BM) consumed in any one meal (Witard et al. 2014). This rate of MPS is suppressed in injured immobilised athletes, because of reduced sensitivity of inactive muscle to hyperaminoacidaemia, which suggests that greater amounts of protein may be necessary to stimulate maximal rates of MPS after injury. In summary, it may be prudent during immobilisation to adjust protein and amino acid intake and distribute throughout the day to maximise uptake.

Effects of specific amino acids on MPS

The essential amino acids (EAA), particularly leucine (LEU), appear primarily responsible for activating the cascade that ultimately results in MPS (Tipton et al. 1999b), at least in the absence of anabolic resistance. In an animal model, LEU ingestion ameliorated muscle loss during immobilisation (Baptista et al. 2010), but the effect in humans remains contentious. High dose EAA ingestion (16.5 g EAA, 3.1 g LEU, 3 × /d (between meals), as part of a high-energy, carbohydrate-rich snack, reduced loss of lean body mass in weight-bearing limbs in healthy adult males during 28 days of bed rest (Paddon-Jones et al. 2004) and during joint immobilisation in healthy adults, adaptations to unloading were faster in tendons than muscle (Kubo et al. 2010). Damage compromises collagen synthesis, the primary protein in tendon, and impairs tendon function. Significant deterioration was observed within 2 weeks of injury (de Boer et al. 2007). While stretching stimulates collagen synthesis, it may be contraindicated when the tendon is severely damaged.

The rate of collagen synthesis in tendon and muscle does not appear to be nutritionally regulated (Babraj et al. 2005a), which suggests that consuming protein or specific amino acids would have little impact on healing. While the amino acid, proline, comprises one-third of collagen proteins, proline supplements also appear ineffective (Barbul 2008), although high doses (~20 g/d in divided doses) of the proline precursor, arginine, can enhance wound collagen deposition (Barbul et al. 1990).

Bone collagen synthesis, an important aspect of bone healing, in contrast, does respond to increased protein intakes (Babraj et al. 2005b). Protein supplementation enhanced recovery from hip fracture surgery (Delmi et al. 1990; Schurch et al. 1998; Eneroth et al. 2006). However, these results come from studies of elderly patients, many of whom exhibit protein-energy malnutrition (Eneroth et al. 2006). Efficacy of either high protein intakes (as total protein or amino acid supplements) for young, healthy athletes on bone healing is still to be determined. Nevertheless, these data suggest that protein feeding may enhance bone formation following injury, albeit at this time only in theory.

Timing of protein or amino acid intake on MPS

Timing of protein intake over the day–every three hours–may result in greater stimulation of muscle protein synthesis in athletes during periods of resistance training than ad libitum protein intake (Moore et al. 2012; Areta et al. 2013). This type of structured timing may exert similar stimulatory effects in athletes with muscle atrophy from disuse or injury (Paddon-Jones et al. 2004). Most athletes are reported to eat protein in an ad libitum pattern (Garcia-Roves et al. 2000; Burke et al. 2003). For those athletes able to undertake cross-education or unilateral strength training of the uninjured body part, ingestion of high biological value protein (such as whey protein supplements) at ~0.3 g/kg/BM acutely after exercise promotes maximal protein synthesis (Hendy et al. 2012). Similar amounts of protein need to be consumed (either from food or supplements) when electrical stimulation is used as an adjunct therapy. Electrical stimulation of muscle also stimulates MPS and, when applied in sufficient amounts, can prevent muscle disuse atrophy (Dirks et al. 2014).

The effect of protein status prior to injury on muscle loss and recovery

Habitual low-protein intake prior to injury negatively influences recovery time. Also, a sudden decrease in protein intake just prior to injury, which develops into a prolonged and large negative nitrogen balance, can promote even higher muscle loss than expected with a positive protein balance (Quevedo et al. 1994). Chronic sub-optimal protein intakes are unlikely in most healthy athletes who meet energy requirements (see Chapter 4). However, those athletes in weight category sports who cut substantial amounts of BM by restricting energy intakes and following unplanned diets are at risk of negative nitrogen balance and potential low recovery rates post-injury.

Energy requirements during recovery from injury

The maintenance of energy balance during recovery from injury is important to maximise recovery rates. Unfortunately, the first impulse of most injured athletes is to reduce energy intake to avoid weight gain and thereby risk a slow recovery rate. When mobility is limited, it is likely that total energy expenditure will decrease, particularly if a leg is immobilised because exercise is more difficult or less convenient (McBeath et al. 1974; Waters et al.1987). To compensate, other forms of exercise that do not involve the immobilised muscles are engaged to help prevent a detraining effect (see Section 22.2). Furthermore, a more subtle reduction in energy expenditure may be attributed to reduced protein turnover (Ferrando et al. 1996). Surprisingly, despite the common belief of excessive weight gain during injury, the evidence suggests that athletes regulate energy balance during periods of prolonged physical inactivity (Bergouignan et al. 2010).

Nevertheless, energy requirements vary at different stages of recovery and are determined by the severity of the injury. Adequate energy intake is required, especially in the early stages of injury when energy requirements may be high from trauma (such as in a fracture of the femur). When the injury is severe, energy requirements can be up to ~20% higher than usual requirements (Frankenfield 2006). Thus, depending on the severity of the injury and the extent of mobility, energy intake may not need to be substantially reduced.

A negative energy balance during recovery and immobilisation inhibits MPS. MPS rates and associated synthetic intracellular signalling proteins are down-regulated in response to an acute, moderate energy deficit (Pasiakos et al. 2010). Decreased MPS synthesis is the major contributor to muscle loss (Glover et al. 2008). If a negative energy balance is sustained, an accelerated loss of muscle mass occurs (Biolo et al. 2007). In contrast, in non-injured people, positive energy balance provides the anabolic stimuli for MPS and recovery (Forbes et al. 1986; 1989). Based on this premise, it is tempting to suggest a positive energy balance for an injured immobilised athlete, which accelerates muscle atrophy, most likely via activation of systemic inflammation (Biolo et al. 2008). A negative outcome of an ongoing positive energy balance may be an increase in fat mass, although excessive weight gain does not appear to be a problem as mentioned above. To address the common fear of excessive weight or fat gain during injury, routine assessment of energy intake, body mass and body composition during immobilisation and rehabilitation is warranted.

Omega-3 fatty acids (n-3 FA)

n-3 FA have anti-inflammatory and immune-modulatory properties and are used therapeutically for this purpose when given as supplements in high doses (e.g. fish oil) (Galli & Calder 2009). Fish oil supplements depress the inflammatory response, which has both positive and negative implications for recovery in injured athletes. A positive immunosuppressive effect of regular use of fish oil supplements is well documented for people with ongoing excessive inflammation (e.g. chronic arthritis) (Galli & Calder 2009), although there are non-responders.

However, long-term use of n-3 FA supplements at therapeutic doses for their inflammatory-limiting effects in injured athletes may be counterproductive to the recovery process, although there is no direct evidence to support this assumption, except in animal studies. Impaired wound healing with n-3 FA supplementation is reported in rat studies (Albina et al. 1993).

Another potentially positive effect of n-3 FA supplements may be an attenuation of muscle loss, although this effect needs confirmation in human studies (see Figure 22.2 for proposed mechanism of action). Evidence from animal studies has demonstrated that fish oil supplements may ameliorate muscle atrophy in rats during immobilisation (You et al. 2010). Preliminary evidence in human studies also suggests that high doses of n-3 FA supplements (4 g Lovaza™/d containing 1.86 g/d eicosapentaenoic acid) and 1.50 g/d docosahexaenoic acid enhanced the rate of MPS in elderly healthy adults (Smith et al. 2011a) and in healthy young and middle-aged adults (Smith et al. 2011b). The authors proposed that n-3 FA supplements enhanced the metabolic effects of insulin and amino acid infusion into the muscle. Figure 22.2 illustrates the proposed mechanism of action. No similar effects are available from human studies of injury associated with immobility or in injured athletes. Recommendations for dosage of the inflammatory-limiting effect of n-3 FA supplements vary from 1–2g/d to 4g/d (Rangel-Huerta et al. 2012).

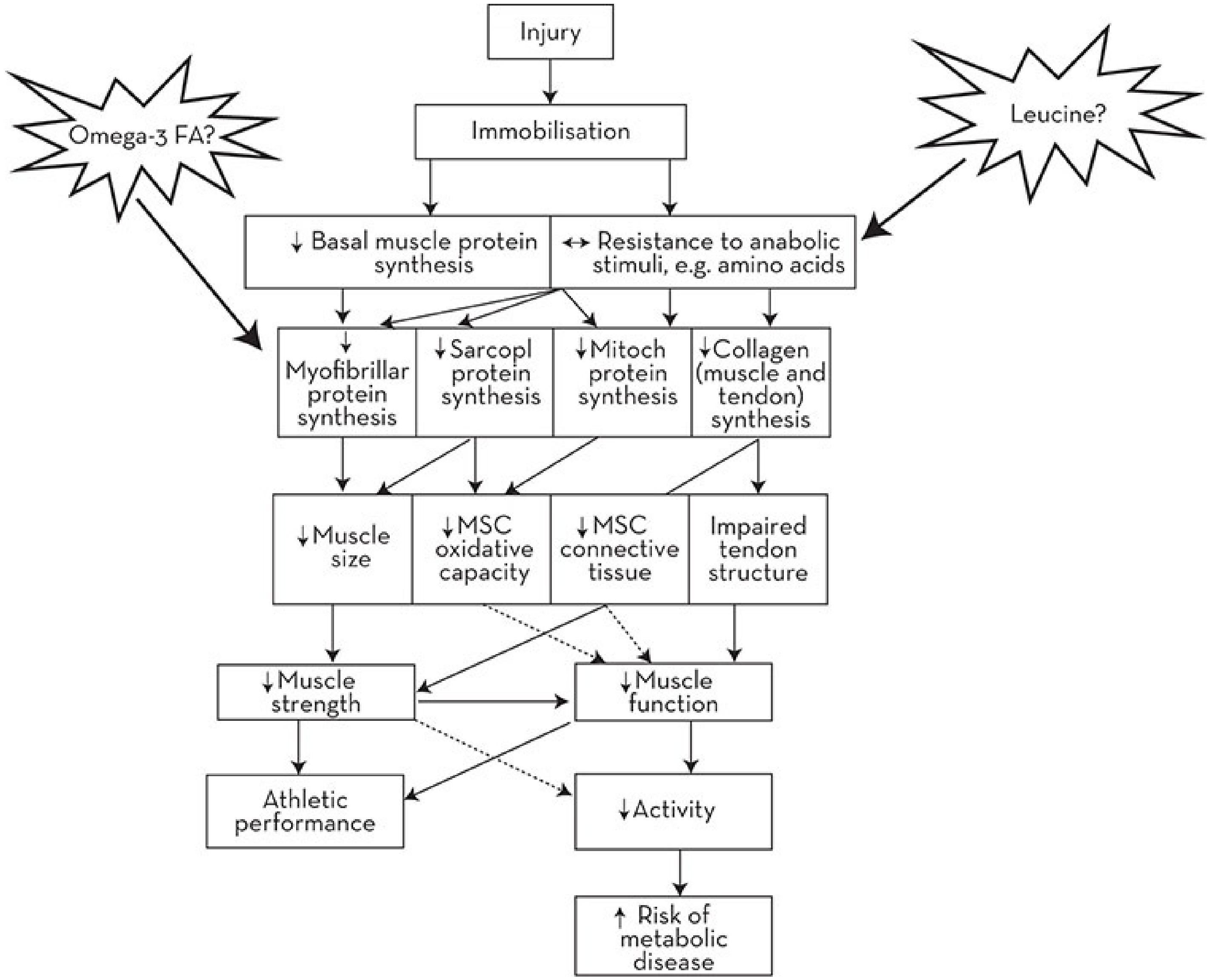

Figure 22.2 Flow diagram of the theoretical metabolic and functional changes during immobilisation

Source: Adapted from Tipton (2010)

n-6 FA also have immune modulatory effects and oppose the inflammatory-limiting effects of n-3 FA. However, avoidance of foods rich in n-6 FA is not practical so a low ratio of dietary n-6:n-3 FA is a more useful recommendation for enhancing anti-inflammatory effects of the eicosanoids (Simopoulos et al. 1999). In practical terms, this ratio implies consuming less n-6 FA (e.g. predominantly found in vegetable oils) and consuming more oily fish and other foods rich in n-3 FA (e.g. mackerel, tuna, salmon, walnuts, flaxseed oil). Some evidence suggests that n-6 FA may also have some anti-inflammatory properties, which are mediated through a different pathway to n-3 FA (Calder & Deckelbaum 2008; Calder & Yaqoob 2009). This concept of an anti-inflammatory effect of n-6 series FA challenges earlier research and recommendations of leading health authorities. Hence there is no current consensus about low ratio dietary n-6:n-3 FA as protective in inflammatory conditions (see www.heart.org). Nevertheless, adequate dietary intake of total fat, fat-soluble vitamins and particularly the essential FAs—linoleic acid (n-6 FA) and alpha-linolenic acid (n-3 FA)—should be encouraged in healthy as well as injured athletes.

In summary, unless the inflammatory process is excessive following injury—an unlikely event for most healthy exercisers—there is little reason to use n-3 FA supplements. Based on the current evidence available, there is some justification for caution in using n-3 FA supplements in athletes. Thus, a ubiquitous recommendation for using n-3 FA supplements does not seem warranted.

Specific micronutrients

Sufficient intake of calcium and vitamin D during healing from fractures is important to optimise bone formation. Wound healing is impaired in malnutrition and linked to sub-optimal micronutrient status of, for example, zinc, vitamin C and vitamin A. Two recent reviews (Stechmiller 2010; Wild et al. 2010) provide an evaluation of the evidence between micronutrients and wound healing. Vitamin C, for example, is associated with hydroxyproline synthesis, which is necessary for collagen formation and is an important component in healing of wounds and repair of damage to bone, ligaments and tendon injuries. There is no clear evidence for the use of micronutrients supplements or higher micronutrient requirements as a means of accelerating recovery from injury (Arnold & Barbul 2006).

Creatine (monohydrate) supplements

Creatine supplements may potentially decrease the rate of muscle loss associated with immobilisation. Creatine is used therapeutically in muscle-wasting conditions such as muscular dystrophy (Tarnopolsky 2007), although the evidence for its therapeutic benefits in attenuating muscle loss in athletes is limited. One of its uses as a sports supplement is to enhance resistance exercise-induced muscle hypertrophy (Hespel & Derave 2007). Thus, the use of creatine to counter muscle loss during immobilisation has been investigated for this potential effect. Unfortunately, the evidence in the limited number of studies to date is equivocal. Creatine supplementation during two weeks of lower-limb casting did not attenuate muscle loss in healthy volunteers (Hespel et al. 2001) or in patients following total knee arthroplasty (Roy et al. 2005). However, in a more recent study, loss of muscle in arms immobilised for seven days was ameliorated with creatine supplementation (Johnston et al. 2009). The varied results between studies were likely to be attributed to differential responses of arms and legs to immobilisation. More research is needed to determine the impact of creatine supplementation on muscle loss in various muscle groups before definitive recommendations can be made.

The effect of alcohol intake on recovery

Wound healing is impaired by ethanol ingestion (Radek et al. 2009), which is probably mediated by decreasing the inflammatory response (Jung et al. 2011). Ethanol excess impairs MPS in animal studies using rats (Lang et al. 2000) and increases muscle loss with immobilisation (Vargas & Lang 2008). A similar impairment to MPS with alcohol ingestion after resistance exercise was recently documented in humans (Parr et al. 2014). Given these effects, limiting or avoiding alcohol during immobilisation is encouraged.

22.6.4 Stage 2: Rehabilitation and hypertrophy

The second stage of recovery from injury is rehabilitation when mobility commences. The metabolic and functional requirements during rehabilitation are different from that during enforced inactivity of the limb. Even with active rehabilitation, the recovery of muscle size and strength usually takes longer than the time it takes to lose it (Jones et al. 2004).

Overall energy expenditure during rehabilitation is likely to increase, particularly when crutches are used. The energy cost of ambulation on crutches may be in the range of two to three times higher than regular walking (Waters et al. 1987). Higher levels of activity during rehabilitation provides further stimulus to restore basal levels of MPS (Phillips et al. 1999, 2002). The type of exercise undertaken determines the response of particular protein types at the injured site (Wilkinson et al. 2008). Resistance exercise leads to hypertrophy of atrophied muscles by stimulating synthesis of myofibrillar proteins (Wilkinson et al. 2008). Tendon collagen synthesis also increases (Christensen et al. 2008). High energy requirements are also needed to support the higher energy cost required for MPS (Waterlow et al. 1978; Giordano & Castellino 1997). During rehabilitation-induced muscle growth, muscle protein catabolism also increases, which is probably an adaptation to promote muscle remodelling (Jones et al. 2004), although this adaptation seems counterproductive to muscle synthesis. It is well documented that resistance exercise increases muscle protein breakdown (Biolo et al. 1995; Phillips et al. 1999), although overall muscle protein turnover is increased during this stage. These increases in muscle protein turnover also contribute to increased energy requirements during recovery. In summary, energy requirements are higher during rehabilitation from a combination of increased physical activity and metabolism. Energy intake should not be restricted during rehabilitation. These processes are illustrated in Figure 22.3.

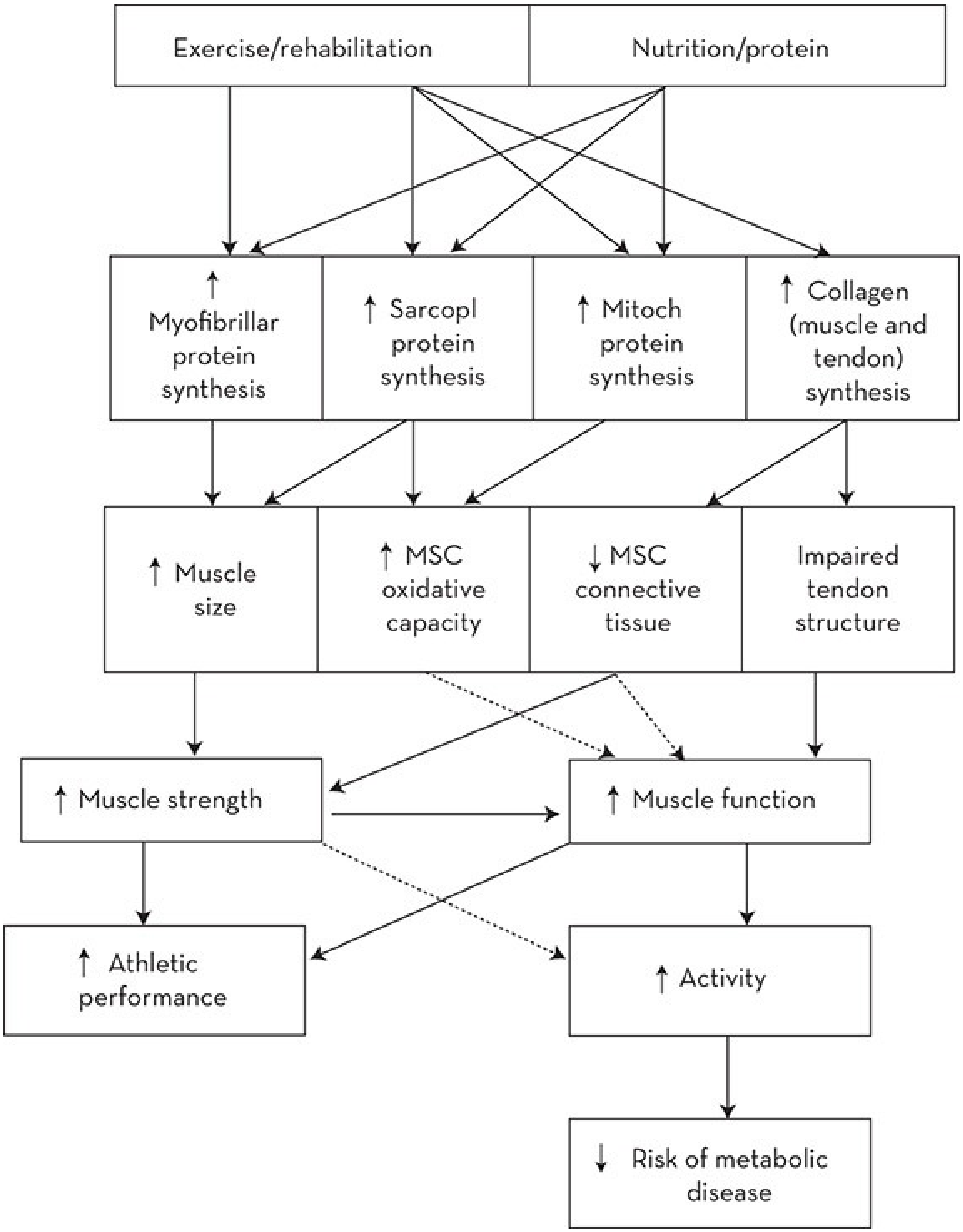

Figure 22.3 Flow diagram of the metabolic and functional changes in muscle and tendon when activity is restored following immobilisation associated with injury

Source: Adapted from Tipton (2010)

22.6.5 Nutrition support for stage 2: Rehabilitation

Following injury-induced immobility, the primary nutritional goal is to support muscle growth and increased strength with rehabilitation and retraining. Generally, nutritional strategies for rehabilitation are similar to those of any other exerciser desiring an increase in muscle mass.

Meeting energy and protein requirements is critical (Tipton & Wolfe 2004; Tipton & Witard 2007; Tipton & Ferrando 2008; Burd et al. 2009; Little & Phillips 2009; Phillips 2009; Rennie et al. 2010). Recommended protein intakes for stage 1 continue during stage 2; they include moderate serves of high biological value protein at each meal and snack, and rapidly digested proteins rich in leucine (e.g. whey protein) following resistance/rehabilitation training. See Chapter 4 for protein recommendations for athletes wanting to bulk up. Limiting carbohydrate intake during rehabilitation is not recommended because low intakes during and following exercise impair MPS (Churchley et al. 2007; Howarth et al. 2010) and net muscle protein balance (Howarth et al. 2010) when muscle glycogen levels are low. Further, there is some evidence that performance during resistance exercise can be enhanced with carbohydrate provision before and/or during training (Lambert et al. 1991; Haff et al. 1999). Thus, both total and timing of carbohydrate intake around rehabilitation exercise may be important, with increments on quantities to reflect training progression.

Creatine (monohydrate) is often touted as important for optimal recovery from disuse, although the limited studies of its efficacy in people with exercise-induced injury have shown equivocal results. In one study of 22 healthy young individuals, creatine supplementation resulted in enhanced recovery of muscle mass and function during the ten weeks following two weeks of immobilisation (Hespel et al. 2001). However, in another study of injured adults, creatine supplements for 12 weeks following anterior cruciate ligament surgery did not improve muscle mass or strength (Tyler et al. 2004). The therapeutic response to creatine in these studies may be confounded by the effects of the type and severity of the injury, the duration of immobilisation and the inflammatory response, and limited by small sample sizes. Unfortunately, there were no measurements of inflammatory biomarkers in both studies. Immobilisation was much longer in the Tyler and colleagues study in which creatine had no effect, which implies that length of immobilisation may make the muscle resistant to the anabolic effects of creatine. However, this metabolic mechanism is hypothetical and has certainly never been demonstrated. Hence, the efficacy of creatine for recovery of muscle following disuse atrophy is still undecided.

22.7 Maxillofacial fractures

Sport-related maxillofacial fractures make up a substantial proportion of all fractures (Antoun & Lee 2008). Football, cycling, ball sports and skiing are high-risk sports for these fractures (Tanaka et al. 1996). Surgery is the preferred treatment in the majority of cases. The average time spent in hospital is 3.5 days, although 40 or more days may be needed before resumption of sport (Roccia et al. 2008). During this recovery time, substantial loss of muscle mass has been reported, which is likely to be attributed to poor nutritional status as a consequence of the inability to eat solid food (Worrall 1994) and possibly reduced energy intake and activity. If nutritional status is compromised before injury and during recovery, the rate of recovery can also be retarded, as previously discussed. Dietetic referral is recommended to devise a modified texture meal plan that meets energy and nutrient requirements (see Table 22.1).

TABLE 22.1 Tips for creating liquid and soft diets

Liquid	Soft
• Include both sweet and savoury options and a variety of different combinations to minimise flavour fatigue. • Use commercial or hospital-based liquid meal supplements where needed to meet high protein energy requirements. • High-fat foods boost energy intake and can increase palatability (e.g. butter, cream, melted cheese). • Use energy or CHO-modified beverages (e.g fruit juice or Polyjoule™ (glucose polymer)). Breakfast and snack suggestions • A variety of smoothies made with milk, fruit, protein or Ensure™ powder, chocolate powder, peanut butter, Weetabix, blended oats, ice cream, honey • Baby food squeeze pack: vegetable or fruit Lunch/dinner suggestions • Soup with added cheese, cream, Polyjoule. This could also contain pureed meat, potato, pasta/rice and vegetables	• Blending or pureeing different wet dishes increases variety. Recipe books for introducing solids to babies are good source of ideas when meal planning. • While wet dishes work best, most regular family meals can be blended with the addition of liquid stock or milk. Breakfast • Scrambled egg • Porridge • Soft/soggy cereals (e.g. breakfast biscuit softened in milk, bircher muesli) Snacks/dessert suggestions • Ice cream, custard, jelly, smoothies • Rice pudding • Sweet pancakes/pikelets Lunch/dinner suggestions • Scrambled egg, avocado (mashed) • Mashed potato and other root vegetables • Well-cooked pasta and rice • Savoury pancakes

Summary

- The physiological stress of high-intensity training programs and competition can induce immunosuppression and increase the susceptibility of some athletes to infective illness (such as URTI and GIT infection) and an altered inflammatory response to injury.
- Some athletes may have subclinical rather than overt micronutrient deficiencies, especially if they have an undiagnosed medical condition that increases nutrient requirements or absorption.
- Athletes benefit from a meal plan with sufficient calories and intake of macronutrients and micronutrients (particularly iron, zinc, and vitamins A, D, E, B6, B12 and folate) to enhance immune system function, reduce the risk of illness and promote recovery after illness or injury.
- Benefits of ingesting foods and supplements rich in polyphenols (flavonoids) and probiotics to treat URT and GIT infection are emerging.
- During immobilisation, muscle mass is lost, which is attributed to a reduction in MPS and muscle response to anabolic stimuli. Low energy availability and low protein intake exacerbates this loss during recovery and rehabilitation.
- Moderate serves of high biological value proteins rich in leucine at most daily meals/snacks may ameliorate muscle loss and promote MPS.
- Energy requirements at the rehabilitation stage may be higher than expected, especially for an athlete on crutches.
- Caution is recommended for use of high-dose n-3 FA supplements to assist recovery and healing.
- Creatine (monohydrate) supplementation may help limit muscle atrophy and subsequent muscle hypertrophy during rehabilitation although any benefit may depend on the type of injury and the length of immobilisation.
- Close monitoring of body composition and energy intake is warranted during both stages of injury so

that nutritional intervention can be adjusted to optimise recovery rate.

Practice tips

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- While illness management sits in the domain of the medical practitioner, nutrition providers may be asked for accompanying nutrition intervention.
- For an athlete with a high frequency of infective illnesses (e.g. URTI and GIT infection), a nutritional consultation usually involves assessment of usual food choices (and nutrient density), energy availability and timing and amounts of food intake relative to training. The object of the consultation is to determine whether there is risk of long-term inadequate intake of energy, macronutrients and micronutrients (particularly iron, zinc, and vitamins A, D, E, B6, B12 and folate) that may predispose an athlete to infection during periods of high-intensity activity.
- In the absence of definitive recommendations for these nutrients in an athlete population, population reference standards can be used to assess the probability of nutrient inadequacy of retrospective diet (see Chapter 2).
- For athletes who are following restricted energy intakes (e.g. for weight loss in weight category sports), nutrient density of food choices to support immune function is important, although it may not fully compensate for very low energy intakes—an independent risk factor for vulnerability to infection. Early preparation to meet weight loss goals and a slow rate of weight loss minimises the need for extreme energy-restricted diets.
- Injured athletes with very high energy requirements should be reviewed for excessive fibre intakes, which may contribute to a low energy intake.
- Avoiding a very low fat diet and including foods rich in n-3 FA, such as oily fish, avocado and nuts, may help support immune function. Vitamin D deficiency can adversely affect immune function.
- Probiotics (in the form of a supplement, enriched food or beverage—e.g. yoghurt drink) may support immune function and help treat an infective illness (e.g. URTI and GIT infection). The delivery format chosen depends on practicalities such as cost, the need for refrigeration and the availability of bioactive strains.
- Vitamin C and zinc supplements may have a beneficial effect on reducing the duration and severity of URTI. The optimal dose and protocol for use is not clear but likely to be between 500 and 1000 mg/d for vitamin C (Hemilä & Chalker 2013) and 75–100 mg/d for zinc (Hemilä 2011), which could be achieved from a combination of diet and supplements. Chronic use of these supplements at these high doses can adversely affect the normal adaptation to training and the iron/zinc balance. See Chapters 10 and 11.
- Colostrum is not recommended for athletes under the World Anti-Doping Code (WADA 2013), despite claims of potential immune benefits, because of inclusion of other prohibited growth factors.
- Where possible, the mouth should be kept moist during training with sips of water to help maintain adequate saliva: the first line defence.
- Good personal hygiene and adequate sleep are the cornerstones to minimising risk of illness and are worthy of review.
- Because high-intensity exercise can suppress the immune system in some athletes, it may be prudent to alert the athlete’s coach about this effect where a modification in training intensity may be warranted.

INJURY MANAGEMENT

- Recovery from injury is a multi-stage process, characterised by varied nutrient and energy requirements. The energy cost of trauma, surgery, stage of the healing process and decrease in mobility should be considered in planning nutrition intervention.

- Unwanted weight gain is often the primary concern for athletes facing an extended rehabilitation period.

STAGE 1: HEALING, INACTIVITY AND MUSCLE ATROPHY

- Pain medication prescribed post-surgery and inactivity are associated with constipation.
- Although vitamin C, vitamin A and zinc are involved in wound healing, supplements do not speed it up. Similar to recovery from infection, adequate nutrient and energy intake provides the impetus for promoting wound healing.
- Social support, meal preparation and shopping may be difficult for the injured athlete, depending on home circumstances. Facilitating planning and support for these tasks along with ideas for easily prepared meals and snacks may improve recovery.
- Low energy availability can retard the rate of recovery of bone and soft tissue injury and the healing process. Low calcium intake and vitamin D deficiency negatively influence bone remodelling (see Chapter 9).
- Where extended bed rest or immobilisation is required, high-biological-value proteins and foods or snacks rich in essential amino acid (particularly leucine) consumed in small amounts over the day may theoretically reduce some muscle loss. Where budget makes this feasible, and particularly in those who need to limit energy intake, the use of branched chain amino acid supplements at regular intervals (e.g. at mid-meals) may provide additional stimulus for muscle protein synthesis. Where electrical stimulation or unilateral training is used, timing the protein intake around these sessions in the same way may be beneficial.
- n-3 FA supplements (e.g. fish oil) have anti-inflammatory effects but are potentially contra-indicated to promote wound healing.
- A fractured jaw presents practical challenges to meeting energy and nutrient requirements. Jaw wiring/banding, or a combination of both, prevents wide opening of the mouth and chewing. Trauma, surgery, restriction of choices and boredom all combine to exacerbate the risk of substantial weight loss. Diets typically start out with liquids only and progress over a period of weeks to a soft and then full diet. Tables 22.1 and 22.2 provide tips for modifying texture and an example of a liquid diet for an athlete with energy requirements of around 15 000 kJ/d.

TABLE 22.2 Example of a liquid diet

Meal	Food	Notes
Breakfast	Breakfast smoothie: • 300 mL full-cream milk • 1 tablespoon skim milk powder • 1 ripe banana • 200 g plain yoghurt • 2 breakfast biscuit 1 large glass of orange juice	• Use different fruit to add variety
Morning tea	220 g creamed rice blended with 1 serve of Ensure™* and a little extra milk to thin to appropriate consistency	• Use yoghurt or custard for variety • Stir 53 g of Ensure™ powder into

07:40:09.

		creamed rice
Lunch	Cream of chicken and mushroom soup Baby food: chicken pasta and veg casserole (store-bought) ‘Spider’ drink: 1 large glass of soft drink with 2–3 scoops of ice-cream 200 g thick custard (full-fat)	• Blend soup and baby food together then add 14 g (2 serves) of Beneprotein™* • Let the ice-cream melt a little before drinking
Afternoon tea	Smoothie: • 250 mL full-cream milk • 1 scoop ice cream • 1 ripe banana • 1 serve of protein powder • 1 tablespoon of peanut butter • 1 tablespoon of honey 1 tub of fruit puree	
Dinner	1–2 serves of instant mashed potato (made with extra milk to thin to appropriate consistency + 1 tablespoon of margarine). Stir through 2 eggs 1 large glass of cordial or fruit juice	• Lightly beat eggs before stirring into potato
Supper	Crème caramel Malt milkshake: • 250 mL full-cream milk • 250 mL full-cream milk • 1 tablespoon skim milk powder • 2 scoops of ice-cream • 1 teaspoon vanilla essence • 1 tablespoon of malt powder	• Choose other dairy desserts, like chocolate mousse, flavoured yoghurt, custard • Use other flavours to add variety to milkshakes: Milo™, strawberry topping, chocolate powder

Nutrient	**Average/d**
Energy (kJ)	15 000
Protein (g)	145
Carbohydrate (g)	480

*Available from pharmacists

STAGE 2: REHABILITATION AND HYPERTROPHY

- Some body composition changes are unavoidable but appropriate advice and monitoring can help set realistic expectations to avoid excessive loss of lean tissue and body fat gain. Anthropometric measures such as limb girths combined with skinfolds and regional assessment of dual energy X-ray absorptiometry (DXA) body composition scans are helpful monitoring tools for assessment of atrophy and hypertrophy. Whole-body assessment using DXA body composition scans and changes in skinfolds and body weight provide feedback on energy balance.
- After the acute phase of injury, energy requirements are likely to be reduced, unless crutches are used. Using low-energy-density foods and high-biological-value protein

intake throughout the day (rather than a bolus meal) may assist with satiety.

- Inflammation is a normal physiological process associated with healing and recovery. The physiotherapist and/or sports physician can confirm a chronic or excessive inflammatory response to an injury. In this situation, there may be a therapeutic indication for use of creatine supplements.
- During recovery from injury, indirect calorimetry can determine resting metabolic rate and predict daily energy requirements at both stages. Activity monitors (e.g. Sensewear™) are an alternative option. See Commentary 3.

THE ROLE OF THE NUTRITION PROVIDER IN MINIMISING THE RISK OF INJURY

Poor nutritional status (bordering malnutrition) and inappropriate body composition are risk factors for injury and infection. Regular nutrition and body composition assessments can be used to assess risk of injury in athletes. The two situations below provide an insight into the ways that a nutrition provider can contribute to risk management of an athlete.

Example 1: Tendinopathy

Patellar tendinopathy is a common injury in basketball, volleyball and badminton. As part of a screening process, anthropometric measures including high body mass index, the Q angle, tibia length relative to stature and a higher absolute waist girth or waist-to-hip ratio increase the risk of this injury and can be identified by the nutrition provider and modified by dietary intervention.

Example 2: Stress fractures

Bone stress injuries are common in sports where there is a high training load. The nutrition provider can:

- routinely investigate energy availability and adjust diet appropriately to training load
- alert medical staff to risk factors such as substantial weight or body composition changes or menstrual disturbances
- refer to medical provider for bone mineral density check if these physiological or dietary risk factors are evident.

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CHAPTER TWENTY-THREE

Medical and nutritional issues for the travelling athlete

Ryan Kohler

23.1 Introduction

Athletes and support staff involved in national and international sporting events may encounter various nutritional and medical challenges associated with travel to competition and being away from home. Nutrition and medical illness in this context are not mutually exclusive. Early preparation and attention to detail is paramount to the success of any travel plan. The input of the medical team, particularly the sports dietitian, in preparing and educating athletes prior to travel can help prevent the onset of potential travel-related problems. Travel increases the risk of common infectious illness such as cold and flu, food-and water-borne infections, environmental and psychological stress and jet lag. Medical practitioners with experience in international travel should be consulted to address preventive strategies. Web-based resources are useful in providing medical and nutritional information specific to various geographical locations.

23.2 Jet lag and travel fatigue

The body's circadian pacemaker located in the hypothalamic suprachiasmatic nuclei maintains the timing of the sleep–wake cycle according to a 24-hour rhythm (Rajaratnam et al. 2013). When athletes fly between continents across multiple time zones, misalignment between the circadian pacemaker and the sleep–wake cycle occurs. The phenomenon is called 'jet lag', which is characterised by symptoms of fatigue, sleep–wake disturbance, mood changes, bowel disturbance and impaired cognitive function (Rajaratnam et al. 2013). Some people are more susceptible to jet lag than others.

The effects of jet lag can be minimised by adapting to the destination sleep time before departure (if practical), avoiding late-night departures and planning flights to maximise sleep time. For athletes with a history of sleep difficulty when flying, a limited supply of a medically prescribed hypnotic medication (e.g. temazepam 10–20 mg, 30 m before intended sleep time) is effective during and immediately after travel. Other strategies on arrival at the destination include early morning exposure to bright light in combination with melatonin. Burke et al. (2013) found that a combination of taking 5 mg of melatonin six hours before bedtime followed by early morning bright light exposure (prior to wake up) produced the greatest adaptation to jet lag compared with sleeping in dim light. Imposing a new sleep–wake cycle to the new time zone as soon as possible at the destination helps prevents lag effects, so athletes need to avoid sleeping during the day, especially in the early afternoon (Samuels 2012).

Travel in a westward direction causes less jet lag than eastward because travelling west results in a longer day (Manfredini et al. 1998). The adage ‘west is best’ applies. Sufficient time is needed prior to any event for athletes to adjust or acclimate. Acclimatisation varies between individuals—usually one full day is required for every (one-hour) time zone crossed (French 1995). Up to three time zones may be crossed before an effect on performance is evident.

Jet stress refers to external problems in transit as a result of air travel and is independent of the number of time zones crossed. The air in the aircraft cabin is relatively dry, so an increased fluid intake is required to counteract increased respiratory fluid losses. Drinks that contain caffeine (such as coffee) were previously thought to contribute to hypohydration, but more recent work has shown that moderate intakes of caffeine by habitual caffeine users do not affect urine losses and hydration status (Armstrong 2002; Armstrong et al. 2005). The effects of alcohol are increased in the pressurised cabin environment so alcohol is best avoided, especially in excessive amounts. Airlines tend to provide a limited variety of foods during travel, but special meals can be arranged in advance. Usually a minimum of 24 hours’ notice is required by most airlines for individual catering requests, but a longer period is advised for teams. During long flights, jet stress can be minimised by making the flight as comfortable as possible (with bulkhead and aisle seats), applying relaxation techniques, increasing fluid intake and consuming small amounts of carbohydrate-rich foods and taking frequent walks around the aisles. Planning and timing of flights to avoid delays or lengthy periods in transit lounges is also beneficial.

Travel of long duration in a plane, car or bus can disrupt normal routines (such as sleep patterns, meal times, training times), leading to travel fatigue and increased risk of travel-related oedema and venous thrombosis. Travel-related venous thrombosis and oedema are linked to immobility and possibly hypohydration. However, sub-optimal fluid intake was excluded as a risk factor in a simulated study mimicking eight hours of air travel in 71 healthy volunteers (11 of whom had prior history of coagulation activation after travel) (Schreijer et al. 2008). Low-ankle-pressure graduated-compression tights (or *flight socks*) have been shown to reduce ankle oedema, symptoms of leg pain, discomfort and swelling in people flying for >5 hours (Hagan & Lambert 2008). In this study, participants also reported an improvement in energy levels, in post-flight sleep, in ability to concentrate and in overall alertness. Flight socks increase lower limb venous blood flow during prolonged seated immobility and may be of some value in prevention of venous thrombosis in the calves (Charles et al. 2011; Sugerman et al. 2012). Frequent movement of the legs (e.g. walking in the aisle in a bus or plane) or simply moving the feet up and down when seated promotes blood flow to the calves, which is also an effective risk reduction strategy for venous thrombosis (Sugerman et al. 2012).

23.3 Illness associated with travelling

Travel and competition away from home and overseas involves close contact between athletes and the public and an increased risk of acquiring infection from unusual organisms, especially those spread by the faecal–oral route and by insect vectors. Some elite athletes in the final preparation leading into competition and travel (and in intense training periods) have compromised immunity and are highly susceptible to infections when they travel (see Chapter 22). This susceptibility decreases over time as the immune system adapts. The risk can be minimised when travelling by adopting preventive strategies to limit cross-contamination (e.g. washing hands frequently, avoiding drinking from the water fountain on the plane, not sharing food or drink bottles). In one study, a 2–3-fold higher incidence of illness was reported in 259 elite rugby union players during a 16-week tournament overseas following travel to a foreign country with a >5-hour time difference from the home country (Schwellnus et al. 2012a). Infections were responsible for 54.5% of illness (Schwellnus et al. 2012b). Upper respiratory tract was most frequently reported area of infection followed by the gastrointestinal tract and skin (Schwellnus et al. 2012b). A higher incidence of infection rates among athletes at international football competitions compared with at home have been reported in other studies (Dvorak et al. 2011; Theron et al. 2013).

The occurrence of respiratory, skin and gastrointestinal conditions predominates in able-bodied athletes (Schwellnus et al. 2012a). In contrast, Paralympic athletes have a higher incidence of skin, gastrointestinal and urinary system problems (Derman et al. 2013). Common causative infective pathogens are *Escherichia coli*, *Salmonella*, *Shigella* and *Campylobacter* (Shah et al. 2009). Viruses account for approximately 17% of gastroenteritis cases, mostly due to norovirus, which is often associated with cruise ship travel (Koo et al. 2010). Antibiotic chemoprophylaxis is a debated practice with no real consensus. However, there is agreement for chemoprophylaxis in the following two situations: travel to a high-risk area (e.g. South Asia) and when a trip does not allow for any deviations in the planned schedule. If an antibiotic is prescribed as a preventive measure, rifampicin (150 mg once daily) taken 24 hours before departure or doxycycline (50 mg daily) for approximately 2–3 weeks' duration is recommended (Nair 2013). Doxycycline can induce photosensitivity so UV protection or avoidance of sun exposure is indicated (Drucker 2011). Another prophylactic antibiotic, flouroquinolone (ciprofloxacin), may be associated with tendinopathy and tendon rupture and is best avoided in athletes (Del Rosso 2010).

23.3.1 Strategies to minimise the risk of infective illness

Prevention of infective illness focuses on addressing personal hygiene and food safety practices, vaccinations and chemoprophylaxis (Nair 2013). Table 23.4 provides a list of strategies to help minimise the risk of food- and water-borne illness. In many developing countries, the drinking water is unsafe. Hence the food prepared using that water, such as salads and vegetables, is also unsafe. This risk extends to cleaning teeth from the same unsafe water, which is often ignored or forgotten because of ingrained habits. Tap water is ‘safe’ for consumption if boiled for at least five minutes; however, this is rarely practical. Water-purifying tablets purchased in pharmacies are another alternative, but are not always effective.

At competition time, athletes are renowned for ignoring any illness and not reporting symptoms to their coach or support staff for fear of exclusion from competing or letting down the team. This common behaviour should be addressed at the team brief prior to travel to encourage athletes to report any illness symptoms early so that they can be isolated from the rest of the group. Establishing contact with medical services and medical support staff in the destination country prior to departure is important in the event of a problem.

Some nutritional supplements are promoted commercially to prevent and treat infective illness in the general population. However, the evidence to support such use may be weak or controversial or have mixed results. The efficacy of probiotics as a means to reduce the incidence of viral infection, particularly upper respiratory tract infection (URTI) has shown positive results in endurance athletes (Cox et al. 2010) and healthy active men and women (West et al. 2013). The risk of an URTI was significantly lower (27% reduction) in a randomised double-blind placebo-controlled trial of 465 healthy active men and women taking a daily supplement of a specific probiotic (*Bifidobacterium animalis lactis* Bl-04) for 150 days, compared with placebo (West et al. 2014) (see Chapter 22). Zinc supplementation (75 mg taken within 24 h of developing cold symptoms) can reduce the duration of symptoms in healthy adults (Singh & Das Rashmi 2013). However, no firm recommendation can be made to support prophylactic zinc supplementation as a preventive measure (Singh & Das Rashmi 2013). There are limited data to support the preventive or therapeutic use of ginseng, echinacea or large doses of vitamin C (Walsh et al. 2011).

23.4 Medical and dietetic treatment of food- and water-borne illness

Table 23.5 provides dietary strategies for the management of the acute and common symptoms of food- and water-borne illness—diarrhoea, nausea and vomiting. If diarrhoea is incapacitating or fluid intake is unable to keep up with losses, anti-diarrhoeal medication (e.g. Imodium™ 2 mg with each loose stool) is indicated, provided no blood or mucus is present. If diarrhoea is associated with fever, blood or mucus, antibiotics may be required. A doctor with knowledge of antibiotic resistance patterns in the region should prescribe these. Treatment with azithromycin (500 mg for 3 d) is effective (Nair 2013). Flouroquinolones (ciprofloxacin) are best avoided by athletes because of the risk of tendinopathy and tendon rupture (Del Rosso 2010). If low-grade diarrhoea persists for more than three days, the parasite *Giardia lamblia* (or another gut parasite) may be the underlying organism. For *Giardia lamblia*, antibiotics such as metronidazole (400 mg, 3 times/d for 7 d) or tinidazole (2 g stat dose with a meal) are indicated. Some microorganisms cause delayed symptoms that may not occur until return to the country of origin. Athletes who experience changes to usual gut function on return from travel should seek medical advice.

23.5 Changes to usual food habits associated with travel

Travelling interstate or overseas for competition often changes usual eating behaviour and food choice and can compromise an athlete's nutrition goals and familiar practices when competing. For some elite athletes, any deviation from usual competition practices, including food intake, can adversely affect performance from a psychological as well as physiological perspective. Long bus trips to competition encourage a holiday mentality. Athletes frequently snack on inappropriate food choices when away from home. Convenience stops at roadside diners encourage further purchases of snack foods or convenience meals. Also, athletes may be self-catering when away from home, have limited money and cooking skills, and rely on eating out or takeaways. Changes to bowel habits and constipation are common complaints.

Travelling overseas causes further disruption to usual eating behaviour and induces other physiological disturbances because of jet lag, different climatic conditions and time differences as previously described (see section 23.2). On arrival in the foreign country, the athlete is faced with other challenges relating to food choice, such as unfamiliar foods, unavailability of usual foods and different eating customs. Unless the athlete is multilingual, an inability to understand the menu and ask questions about food and its content can be daunting. These are just a few of the possible complications to expect. Adequate food choices are usually provided at most large-scale competitions, where athletes stay in residential environments (see Chapter 25). However, at smaller competitions, athletes may be living in local hotels, eating local foods and responsible for their own meals and snacks.

In summary, when travelling to competition, athletes should aim to eat similar foods to home and, where possible, avoid large variations in their usual food intake and food choices. Prior to travel overseas, providing athletes (as well as coaches and support staff) with opportunities to become familiar with the food and culture of the destination, the risks and challenges of eating away from home, and reinforcing the message about the role of nutrition before, during and after competition on sports performance and recovery can avert potential problems. Carrying familiar foods to competition that are portable and do not breach customs regulations is recommended. Conducting cooking classes with the team just prior to departure helps instil confidence in athletes with poor cooking skills and provides an ideal opportunity to plan the menu. Recommendations for suitable snacks, meals and recipes is also well received. In some cases, working with team management to provide suitable options may be an effective way of ensuring good food choice. See Practice tips for detailed guidelines.

Summary

Many athletes suffer from jet lag, travel fatigue and illness when they travel. The increased risk of infective illness returns to baseline when arriving home, which suggests that the cause is related to the destination and not the travel itself. Illness affects mainly the respiratory and gastrointestinal systems in able-bodied athletes. In Paralympic athletes, non-respiratory illnesses are more common. Chemoprophylaxis is an acceptable practice in high-risk areas for traveller's diarrhoea. Probiotic supplementation may offer increased protection against viral infection in athletes. Education of athletes and support staff about the physiological effects and risk factors associated with travel and ways to reduce those risks can help avert potential travel-related problems. Education provides an opportunity to reinforce the importance of a high standard of personal hygiene, food hygiene and food safety practices to reduce risk of illness, cross-contamination and a potential team outbreak.

USEFUL WEBSITES

- **www.prokerala.com/travel/jetlag.php**
  Jet Lag and Jetlag Calculator
- **www.masta-travel-health.com**
  Masta Travel Well
- **www.traveldoctor.com.au**
  The Travel Doctor

Practice tips

CHRISTINE DZIEDZIC

PLANNING TRAVEL

- Determine all aspects of an athlete's journey (including travel arrangements, accommodation, catering, training and event schedules and venues) well in advance to ensure appropriate and specific practical advice and goal setting can be provided. Table 23.1 provides a checklist of issues to investigate prior to departure.
- Encourage athletes to use a diary to plan and document proposed meal and snack times around travel, training and competition.

TABLE 23.1 Pre-travel checklist

Vaccinations (where necessary) Influenza vaccination is recommended for all travellers
Existing medical conditions of athletes (e.g. coeliac disease) and related nutrition care plans
Existing allergies of athletes and associated treatment medication (e.g. Epipen, antihistamines)
Itinerary, including modes of transport, travelling times and likely breaks in the journey (e.g. meal stops, refuelling stops, overnight stopovers)

Type of accommodation and meal arrangements
Local travel, training and event schedule
Team/travel manager's details
Familiarity with place of destination (e.g. climate, time zones, altitude, food safety and availability)
Local customs relevant to the athlete (e.g. clothing, language, food culture)
Baggage limits, including equipment
Food, fluids, nutrition supplements and hand sanitiser to be taken or provided by team management (where applicable), noting customs regulations

LOCAL TRAVEL

- Where there is an option, staying in self-contained accommodation provides greater flexibility in food choices and preparation. Discuss portable foods to carry or purchase, as well as cooking utensils (such as a hand-held blender for smoothies, a rice cooker or an electric wok) to bring, where applicable. Chapter 25 provides additional practical suggestions for self-catering and how to choose foods that meet dietary goals when eating out.
- During road travel, carry a small insulated bag with portable snacks (see Table 23.2).
- When travelling by train or bus, check the availability of food en route and the frequency of food stops. Advise athletes to carry snacks to avoid reliance on take-away food and long gaps with nothing to eat.
- For teams travelling to training camps or competitions that last for several days, contact accommodation venues in advance and request 'in-house' menus and offer assistance in menu planning (see Chapter 25). Provide suggestions for food choices and menu plans from the existing menu to catering staff, managers, coaches and athletes prior to departure. Some (but not all) caterers welcome these suggestions and may be receptive to preparing special meals for athletes.

TABLE 23.2 Suitable snacks for travelling
Fruit (if travelling locally) or canned fruit
Dried fruit and nuts
Rice crackers, pretzels
Low-fat cheese sticks
Canned tuna or salmon
Sandwiches, rolls, mini bagels, fruit bread, pikelets
Popcorn (made with minimum fat/oil)
Muesli and cereal bars, breakfast biscuits, energy bars (low-fat)
Dry cereal (in single-serve boxes)
Low-fat fruit yoghurt or fromage frais
Low-fat rice pudding
Low-fat flavoured or plain milk
Water, fruit juice, sports drinks, liquid meal supplements (such as Sustagen Sport™)

OVERSEAS TRAVEL

- In countries where the risk of travellers' diarrhoea is high, probiotic supplementation in the weeks leading up to departure and continuing throughout travel, or consuming yoghurts containing probiotic cultures, may be useful in reducing the incidence of travellers' diarrhoea (Fedorak & Madsen 2004).
- Airlines provide special meal options (such as low-fat, vegetarian), which are ordered at the time of booking or at least 48 hours prior to departure.
- When travelling to unfamiliar countries where usual foods are unavailable, the athlete should pack or send a supply of food staples. Liquid meal supplements, dried milk powder, breakfast cereals, cereal and sports bars are portable and useful travel items (see Table 23.2). Check customs regulations before taking or sending any food overseas.
- Prior to departure, athletes can benefit by familiarising themselves with the food choices and dietary customs of their destination by eating in local restaurants offering similar cuisine or participating in destination-themed cooking classes.
- Encourage athletes to include a personal first aid and 'survival kit', especially when travelling overseas. Include the usual first aid items, supplements (e.g. multivitamins/minerals, probiotics), electrolyte-replacement sachets (in case of vomiting and diarrhoea), antibacterial wipes or sanitiser gel for hands (for hygiene) and a food thermometer. This kit is a good idea for domestic travel as well.

PREVENTING JET LAG

Table 23.3 provides dietary strategies to help prevent and minimise the effects of jet lag.

TABLE 23.3 Dietary strategies to help prevent and minimise the effects of jet lag
Adapt meal and snack times to destination time 24–48 hours before departure, and during the flight.
Bring plenty of activities (e.g. books, iPod, games) to prevent boredom eating.
Keep a food diary to remind you of when you last ate.
Adopt regular meal and snack patterns upon arrival and sustain a higher fluid intake until well hydrated.
To induce sleep
High-carbohydrate, low-protein meals during transit may help induce drowsiness.
Drink sufficient fluid in transit as respiratory fluid losses are increased in the dry pressurised cabin. Bring an empty drink bottle that can be refilled on the plane and sipped continuously.
Avoid, or limit intake of, alcoholic beverages because the adverse effects of alcohol (i.e. dehydration, hypoglycaemia, light-headedness) are exacerbated in the airplane environment.
Pre-arrange special meals that provide carbohydrate and are low in fat.
Pack portable, high-carbohydrate snacks (including fruit) in your hand luggage. Note that Customs regulations usually require fruit or other fresh food disposal at the destination.
To promote alertness
Consider the responsible use of caffeine. An intake of 80–200 mg is considered effective.

PREVENTING FOOD- OR WATER-BORNE ILLNESS

- The risk of food- and water-borne illness is minimised in any situation (either at home or away from home) by paying attention to personal hygiene (frequent hand washing), by avoiding foods that are high risk for contamination and by adopting hygienic food handling and storage practices.
- High-risk foods for contamination are uncooked fresh foods (meat, fish, vegetables, eggs), unpasteurised dairy products and reheated foods. Cross-contamination of foods can occur when foods are left uncovered in the refrigerator, and by re-using cutting boards, cooking and eating utensils, tea towels and dishcloths.

Table 23.4 lists practical tips to raise awareness of food hygiene and food safety issues that can help prevent food-borne illness for travelling athletes, especially when travelling to foreign countries.

TABLE 23.4 Preventing nutrition-related illnesses when travelling

Wash hands with soap frequently, and for at least 30 seconds, especially before eating. Ideally use an air dryer to dry hands, otherwise a clean towel.
If the local water supply is unsafe, drink bottled water. Soft drink is 'safe', but should not be used as a substitute for water.
Avoid ice in drinks unless you are sure the water is safe for drinking.
Avoid eating salad or raw vegetables unless the food has been washed in water that is safe for drinking.
Peel all fruit.
Avoid other raw foods such as oysters, shellfish and raw fish (as eaten in sushi).
Avoid buying foods from local stalls and markets, where hygiene may be poor.
If possible, check that raw food (such as meat) is kept refrigerated, and that cooked food is kept hot (>60°C) and has not been re-warmed or kept warm for more than two hours.
Avoid buffet food that is not served very hot or chilled, or which has been sitting for extended periods of time.
In countries where food hygiene and safety standards are questionable, select foods that have been 'cooked to order' rather than pre-cooked and re-heated, such as take-away or fast-food.

STRATEGIES FOR TREATING DIARRHOEA AND VOMITING

- Replacing fluid and electrolytes is the main priority if travellers' diarrhoea or vomiting occurs. Low-fat, low-fibre foods can be gradually introduced as tolerated. Table 23.5 provides detailed strategies for dietary treatment of nausea, vomiting and diarrhoea.

TABLE 23.5 Diet therapy for nausea, vomiting and diarrhoea

Nausea/vomiting
Withhold food in the short term, but maintain fluid intake.
Consume small, frequent meals (such as dry crackers, plain rice or noodles or dry toast at first, then progress to light, low-fat meals as tolerated).
Elevate the head, eat slowly.
Diarrhoea
Maintain a high fluid intake using bottled water and electrolyte replacement drink or sports drink. Avoid milk, caffeine drinks (such as coffee and cola), gaseous drinks (such as soft drinks) and fruit juice.
In the acute stage, avoid very high-fibre foods and spicy foods and other foods specific to the individual that cause flatulence.
Avoid fatty foods and high sugar foods.
Introduce small amounts of low-fibre foods initially (e.g. plain rice, noodles, dry white toast, dry low-fat crackers or canned pears) and, once tolerated, slowly increase dietary fibre intake back to normal.
Consider taking probiotics prior to and during travel, to reduce the incidence of diarrhoea and upper respiratory tract infections.

Source: Adapted from Deakin, 1998

MEETING DIETARY GOALS

- Monitoring body mass (with or without skinfold assessment) is useful to help prevent substantial body composition changes, a common occurrence observed when athletes are away from home. The reduced access to fresh produce, easy access to inappropriate food choices, buffet-style eating in restaurants or unfamiliar foods can make it difficult for athletes to maintain an ideal body mass or meet their usual training eating plan. Dietary strategies to address this changing environment are best given prior to departure and reinforced by coaches and managers while away (see Chapter 25).
- Monitoring of morning body mass and the characteristics of morning urine samples (urine specific gravity) can be used to gauge hydration status (see Chapter 14).
- Daily recording of food intake works well for some athletes (although not many) to control eating behaviour while away. It also provides an opportunity to more accurately assess individual food choices and eating behaviours on return, than using retrospective dietary assessment methods, which allows the sports dietitian to address any concerns for future travel.
- Regular communication with the athlete living or travelling overseas for extended periods via email or social media is invaluable for providing encouragement, support and preventing diet- and often health-related problems.
- On return, a follow-up appointment with an individual athlete or a team debrief is invaluable to assist with evaluating any food-related issues and to check the efficacy of the strategies provided. This session reinforces dietary goals and helps to better target the dietary intervention strategies for future travel.

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CHAPTER TWENTY-FOUR

Altitude, cold and heat

Megan L Ross and David T Martin

24.1 Introduction

Athletes may be required to train and compete in demanding environmental conditions. For endurance athletes, extreme temperatures and altitude can influence normal physiological responses to exercise and, in some cases, sporting performance is detrimentally affected. In the case of heat and altitude, acclimatisation over 1–3 weeks can improve endurance capacity; however, the benefits of cold exposure for sporting performance have not been clearly established. The advantage of altitude and/or heat exposure to promote adaptations that facilitate exceptional performances at sea level has yet to be established. It remains unclear whether athletes who participate in shorter anaerobic events can also enjoy performance advantages following acclimatisation to moderate-altitude training and heat exposure. In addition to the known acute physiological effects of heat, cold and altitude, there are other, less well-documented, psychological aspects of these environmental conditions that can influence how athletes eat, train and compete.

Specialised nutritional practices for athletes exposed to extreme conditions have long been a topic of interest. Winter sport athletes often find themselves at moderate altitude in cold conditions; open-water swimmers and team athletes who compete outside in the winter often experience cold; and many summer athletes are forced to cope with extremely hot conditions. This chapter considers the physiological effects associated with exercise in extreme temperatures and at moderate altitude and highlights nutritional strategies that may support performance capacity.

24.1.1 Background

The detrimental effects of altitude and heat on sporting performance have been the topic of many comprehensive scientific review papers (Saunders et al. 2009; Nybo et al. 2014). Recently, the Medical Commission of the International Olympic Committee (IOC) published a consensus statement addressing important sport-related risks associated with exercise in both hot and hypoxic conditions (Bergeron et al. 2012). Although the links between cold acclimatisation and sporting performance are not strong, cold temperatures can lead to physiological adaptations that are worth considering (Shephard 1985). In addition to these in-depth review papers focusing on physiological effects and health implications of extreme environmental conditions, there are also some excellent publications discussing nutritional implications of heat (Burke 2001), cold (Meyer et al. 2011) and altitude (Kechijan 2011) from a sport perspective.

24.1.2 General stress response

As discussed in many insightful scientific review papers (Shephard 1993; Boning 1997; Pandolf 1998; Milledge 2006; Saunders et al. 2009), living and exercising in extreme environments involves additional stressors that challenge homeostasis. It is therefore no surprise that the sympathetic branch of the autonomic nervous system is more active both at rest and during submaximal exercise at altitude (Mazzeo et al. 1995) and in hot (Febbraio 2001) and cold (Shephard 1993) conditions. Although extreme temperatures and hypoxia invoke many unique physiological responses and adaptations (Shephard 1993; Pandolf 1998; Saunders et al. 2009), some modifications to metabolism at rest and exercise are similar. For instance, resting metabolic rate (RMR) can increase following exposure to hypoxia, cold and heat. Similarly, in all three conditions, fluid loss can be greater both at rest and during exercise. In the case of heat, fluid requirements are greatly increased during submaximal exercise (Burke 2001; Meyer et al. 2011). Under the influence of a greater sympathetic tone, there tends to be an increased reliance upon carbohydrate (CHO), which is especially evident at altitude and in the heat (Febbraio et al. 1996; Katayama et al. 2010). In extreme environments, athletes sometimes find it difficult to obtain uninterrupted sleep, will likely experience a greater perceived exertion for a given workload and have a limited exercise capacity. Over time, the athlete becomes more familiar with the extreme conditions and with appropriate nutrition and recovery, physiological adaptations will occur that support increased exercise capacity.

24.2 Altitude physiology

Altitude is associated with a decrease in barometric pressure and a corresponding reduction in oxygen availability. The severity of altitude can be categorised as low (1000–2000 m), moderate (2000–3000 m), high (3000–5000 m) and extreme (>5000 m) (Levine & Stray-Gundersen 2002). In an attempt to provide oxygen to metabolically active tissue, the physiological response to a standard exercise task at altitude is exaggerated (Saunders et al. 2009). More specifically, even mild hypoxia causes ventilation, heart rate and perception of effort to be elevated from sea-level values. Also, the normal endocrine response (e.g. adrenaline, noradrenaline, cortisol) to exercise is typically amplified at altitude, which promotes an increased use of CHO as a fuel, compared to the response when exercising at sea level (Berglund 1992). Altitude also increases oxidative stress, which is independent of exercise (Heinicke et al. 2009). Within the first few days of altitude exposure there is a loss in plasma volume as a result of increased blood pressure and diuresis. Hypoxia stimulates the production of the glycoprotein erythropoietin (EPO) by the kidneys, promoting many adaptations beneficial for athletes including erythropoiesis (Gore et al. 2013). If athletes continue to train at moderate altitude for 2–3 weeks and are exposed to an ideal diet and training program, numerous physiological adaptations can occur that facilitate tissue oxygenation. The most widely documented and discussed adaptation to moderate altitude training is an increase in red blood cell mass, which can increase by approximately 1% for every 100 hours of exposure (Gore et al. 2013). Athletes may enjoy the benefits of elevated haemoglobin mass for 2–3 weeks following both live high–train high (natural) and live high–train low (artificial) altitude camps. Altitude-induced increases in muscle buffering capacity and accumulated oxygen deficit have also been documented (Gore et al. 2001; Garvican et al. 2011) that have implications for strength and power athletes. Over the past 50 years these acute and chronic physiological responses to altitude training have been thoroughly documented and reviewed (Rusko et al. 2004; Levine & Stray-Gundersen 2006; Wilber 2007; Lundby et al. 2012; Gore et al. 2013; Gore et al. 2014).

Although the physiological adaptations to altitude training are now widely accepted, the best way to use altitude exposure to facilitate performance gains remains a popular and sometimes controversial topic among coaches and sport scientists. From a nutritional perspective, understanding the unique stresses and adaptations associated with altitude exposure provide rationale for targeted modifications of the diet. In some cases, the full extent of potential adaptations may not be revealed without sufficient dietary support.

24.3 Altitude training

Although aerobic exercise capacity is impaired in hypoxic conditions, multi-week camps at altitude generally have a positive influence on both sprint and endurance performance (Girard & Chalabi 2013; Gore et al. 2013). Similar to other types of training, it has long been recognised that some athletes do not respond favourably to altitude (Saunders et al. 2009). It appears that adaptations to altitude training depend upon a large number of factors including nutritional status. It can be a challenge for the sport scientist and coach to ensure that the diet and training program provides the athlete with the best chance for a positive adaptation to altitude training. By adhering to contemporary recommendations, athletes are more likely to increase their chances of maximising positive adaptations.

24.3.1 Ideal conditions for adaptation

Australian Olympic athletes have been exposed to structured and supervised altitude training camps for more than 30 years. It has been proposed that genetics contribute to whether an athlete responds favourably to altitude (Chapman et al. et al. 1998). However, many Australian sport scientists designing and supporting altitude camps have observed that a 'non-responder' to one altitude camp can become a dramatic 'responder' to a subsequent training camp, highlighting the importance of non-genetic factors. Diet, sleep, overall health and other environmental factors may influence whether athletes experience beneficial adaptations to altitude training. In addition to focusing on the diet during an altitude training camp, it may be equally important for the athlete to pursue health, recovery and a healthy diet prior to departing to the mountains.

Adequate iron status

Low iron status prior to and during altitude training can compromise adaptive effects of altitude training. Athletes with low iron stores (i.e. ferritin <30 ng/mL) may not respond to adaptations to altitude training, many of which are iron-dependent. The general recommendation for athletes with low iron status is to increase iron intake and endogenous iron stores from an iron rich-diet and, if required, use oral iron supplementation for 1–2 months prior to altitude exposure (Friedmann et al. 1999). Whether iron-sufficient athletes can benefit from additional iron intake is not clear. The risks associated with consuming an iron-rich diet are low.

Adequate energy and carbohydrate intake

Athletes pursuing weight loss who consume prolonged energy-restricted diets are at risk of further weight loss at altitude, which compromises adaptive responses, particularly erythropoiesis. Energy requirements are higher at altitude (see section 24.4.2). In a clinical setting, EPO efficacy was improved when long-term haemodialysis patients were given energy supplements equivalent to 475 kcal (~2000 kJ) daily (Hung et al. 2005). Hence, for these patients, increasing energy intake acted as a stimulus for EPO activity. Similar inhibitory effects of EPO efficacy may occur in healthy athletes who follow energy-restricted diets.

The risk of infection increases at altitude (Mishra & Ganju 2010), which is exacerbated when training sessions are not supported with adequate CHO or energy intake (Mazzeo 2005). Low blood glucose also compromises immune system function (see Chapter 22). Hydration practices that involve CHO-electrolyte drinks are useful to prevent hypoglycaemia.

Injury and illness

Prolonged inflammatory responses associated with injury may interfere with altitude adaptations and slow the rate of wound healing. Athletes recovering from injury benefit from remaining at low altitudes until tissue healing is complete. Similarly, serious viral and bacterial infections can impair adaptation to altitude training. Athletes with persistent colds, flu or gastrointestinal problems are best to fully recover before going to altitude. Although direct evidence supporting this recommendation is limited, indirect evidence from a clinical setting has confirmed a link between elevated inflammatory cytokines and EPO efficacy

(Drueke 2001).

Training intensity

Athletes who gradually increase training volume and intensity during an altitude training camp tend to have fewer side effects and better training adaptations compared with those who train aggressively (Wilber 2007). High-intensity training within the first week of an altitude camp can promote excessive fatigue. Sleep disruptions usually occur during the first 3–5 days of acute low to moderate altitude exposure and are more prolonged at higher altitudes (Kinsman et al. 2005). A reduction in training load to submaximal in the early phase of training at altitude helps reduce these side effects. Once the sleep cycle has stabilised, athletes are more likely to respond to strenuous training in the altitude environment.

24.4 Dietary recommendations at altitude

Nutrition to support performance at altitude has long been topical for those working with soldiers, athletes and mountaineers (Simon-Schnass 1992; Kayser 1994; Askew 2004; Kechijan 2011). Higher energy and fluid requirements at altitude compared to sea level underpin the design of high altitude diets, which need to provide maximum support for exercise, recovery and ultimately adaptation. Research relating to dietary recommendations at altitude has focused mainly on high-altitude mountaineers, which may not be entirely relevant for athletes training at low–moderate altitude. Nevertheless the benefits of consistent fluid consumption and a well-balanced diet with an emphasis on adequate energy, nutrient-rich sources of CHO and protein, iron-rich foods (particularly from haem sources) and antioxidant-rich foods (e.g. fresh fruit and vegetables) are supported (Askew 2004; Kechijan 2011).

24.4.1 Fluid

Fluid requirements increase when training in a hypoxic environment because of dry and possibly cold air (low humidity), increased water lost through sweat or lungs from an increased breathing frequency associated with both living or exercising in hypoxia, and an increase in basal metabolic rate (BMR) and metabolic cost of exercising at altitude. Exposure to dry air at simulated altitude (10% relative humidity (RH) at ~2000 m) for 6 hours at rest resulted in a significantly greater loss of total body water than in control conditions (65% RH, 0 m) (Hashiguchi et al. 2013). Due to the dry air, any fluid loss associated with thermoregulation during exercise is augmented at altitude. At higher altitudes (>4000 m), there is an acute reduction in total body water and subsequent decrease in plasma volume (Pugh 1964). Interestingly, this fluid loss and resultant haemoconcentration increases the oxygen-carrying capacity of the blood, which may signify a beneficial adaptation preceding the eventual altitude-induced polycythaemia (Grover et al. 1986). Although acute altitude sickness is rare at moderate altitude, it is worth noting that hydration status and redistribution of total body water may contribute to symptoms. In a cohort of 192 soldiers exposed to moderate altitude (1500–2500 m), 15% developed at least mild symptoms associated with acute mountain sickness and 29% developed high altitude headaches. Using survey methodology, researchers established that both hydration status and sleep duration were associated with undesirable symptoms, indicating that behavioural factors can influence initial adaptations to moderate altitude (Norris et al. 2012).

For an athlete undertaking high-intensity training at altitude (live – high – train – high), sports drinks can be useful. Hydration status of male skiers was monitored during a training camp at 1800 metres and the results indicated that sports drinks were more effective than water in preventing loss of plasma volume and maintaining fluid balance (Yanagisawa et al. 2012). Other studies have found that sports drinks maintain the drive to drink, minimise urinary fluid loss during exercise and maintain the extracellular fluid volume space (Nose et al. 1988; Maughan & Leiper 1995). In some cases, the consumption of tea at altitude has been discouraged based on the assumption that the caffeine in tea will act as a diuretic. Fortunately for those who enjoy tea, research has not been able to provide any evidence of compromised hydration in tea drinkers at altitude. In contrast, drinking tea was associated with a positive influence on mood for climbers residing at the Mt Everest base camp (Scott et al. 2004).

Although attention to fluid intake during exercise at altitude has merit, some athletes concerned with dehydration tend to consume excessive amounts of fluids prior to going to sleep. This practice invariably results in a restless night of sleep, as the athlete is required to make frequent trips to the toilet. By focusing on hydration in the morning and afternoon, it is possible to adopt a more moderate approach to fluid consumption during evening meals.

24.4.2 Energy intake

During altitude training camps, a decrease in volitional food intake can occur because of hypoxia-related appetite suppression and/or a conscious desire to lose weight. A prolonged decrease in energy intake combined with a corresponding increase in RMR will ultimately result in a non-desirable energy deficit that may compromise adaptations to training and lead to weight loss.

In a widely cited paper, Butterfield and colleagues (1992) measured the BMR in seven men at sea level and after arriving at an altitude of 4300 metres. Within the first 2 days, BMR increased by ~25% compared to sea level measures, then gradually reduced to ~15% after 2 weeks of altitude exposure. After the first week of altitude exposure, subjects received a 335 kcal (~1400 kJ)/d (~10%) increase in energy intake, which corrected the negative nitrogen balance observed in the subjects in the first week. In another study, Butterfield (1999) found that when body mass (BM) was maintained by active nutrition intervention during high-altitude training, the increase in BMR was 7% compared with 28% when subjects ingested food ad libitum and lost weight. Nutritional strategies for preventing BM loss and loss of lean BM (or fat-free mass) at high altitude are published elsewhere and are focused mainly on studies of mountaineers (Wing-Gaia 2014). As mentioned above, extrapolating results from high altitude studies to moderate-altitude environments where athletes participate in training camps needs caution.

Although some athletes may intentionally use altitude training camps as an opportunity to reduce body weight and body fat levels, there is a risk that highly motivated and competitive individuals will adopt an excessively low energy intake. Hence, evaluating weight satisfaction attitudes in athletes prior to long periods of altitude training and monitoring weight or body composition of at-risk athletes may help avert inappropriate weight-reduction behaviours. For a variety of reasons, altitude training camps often utilise high-volume training programs with limited high-intensity exercise. Hence, high-energy (and high-CHO) intakes are required.

24.4.3 Macronutrients

Carbohydrate

CHO as a substrate for physical activity increases at altitude. Hence, CHO requirements at altitude appear to be higher than at sea level, although no definitive recommendations have been determined. Since appetite can be suppressed at altitude, at least in the early period of acclimatisation, it would be prudent for athletes (and support staff) to ingest CHO-dense foods at each meal and for snacks consumed between meals and during training. High-CHO diets may also improve physical and mental intolerance to hypoxia (Consolazio et al. 1969). These recommendations assume endurance athletes exercising at moderate altitude respond similarly to moderately trained individuals who have participated in high altitude research. CHO intake recommendations should take into consideration overall higher energy intake requirements.

Protein

In an animal model, hypoxia impairs muscle protein synthesis (MPS) regardless of maintenance of energy balance (Brugarolas et al. 2004). Prolonged suboptimal intake of energy and protein accompanied by weight loss promotes a decrease in lean BM. In normoxic conditions in trained athletes, MPS substantially increases during the recovery period after exercise. Although research in this area is limited, available evidence supports the hypothesis that hypoxia can compromise MPS post-exercise.

The type and timing of protein intake is important for stimulating MPS during recovery (see Chapter 4). At sea level, 20–25 grams of high-quality protein consumed after exercise is sufficient to maximally stimulate MPS (Witard et al. 2014). Athletes who develop a negative energy balance at altitude will require slightly higher protein intakes than normally recommended for athletes engaged in heavy training at sea level. Branched chain amino acids, particularly leucine (found naturally in high-quality protein-rich foods), are beneficial for regulating post-prandial MPS (Koopman et al. 2006). At moderate and high altitude, protein supplements containing leucine may improve retention of lean BM. However, only three studies have investigated the efficacy of protein supplements at altitude. More research is needed to substantiate the effect of specific amino acids before definitive recommendations can be made. For a review of these studies, see Wing-Gaia (2014).

Fat

The digestion of a relatively high proportion of energy from dietary fat appears to be well tolerated at high altitude. In one study of soldiers living and exercising at 3800 metres, 324 g/d dietary fat, which contributed 47% of the total energy of the diet, was not associated with any gastrointestinal problems or with constipation or diarrhoea (Rai et al. 1975). Even at higher altitudes (up to 4700 m), this high fat intake was well accepted by the soldiers in this study and not associated with any metabolic disturbances. Mountaineers are often dependent on high-fat foods (e.g. chocolate, nuts) because of their high energy density, palatability and portability. High fat intake may also be of benefit for those elite athletes training in altitude training camps who have difficulty maintaining weight and eating a higher volume of food to meet their higher energy requirements.

24.4.4 Micronutrients

Several micronutrients (e.g. iron, antioxidant nutrients—specifically vitamin C and E) taken as supplements have been touted to improve adaptations and performance at altitude. This section considers the evidence, although limited, for those micronutrients that have been investigated in the context of altitude training.

Iron

At altitude or simulated altitude environments, iron requirements increase to meet the induced haematopoietic response. Athletes with low iron stores (low ferritin) prior to altitude training can quickly develop iron deficiency within 1–2 weeks of altitude training (Garvican et al. 2012). Routine use of iron supplements for athletes without a diagnosis of iron depletion, prior to and during altitude training, is controversial (Nielsen & Nachtigall 1998). Although relatively safe, iron supplementation is not without side effects and health risks, such as iron overload and haemochromatosis (see Chapter 10). Prior to training at altitude, athletes are often advised to increase dietary iron intake, particularly from haem iron sources (see Chapter 10, Table 10.5). For athletes diagnosed with low iron stores, an oral iron supplement in conjunction with vitamin C (to maximise iron absorption) is needed for rapid recovery. In one study of distance runners with mild to severe iron depletion, 120 to 130 mg of elemental iron/d, divided into two doses, improved iron stores in 4–6 weeks (Garvican et al. 2014). This dosage is higher than the usual therapeutic dose (see section 10.12.2) but was well tolerated by this group and led to rapid improvement of iron stores. In a recent consensus statement, the IOC recommended that iron status of athletes should be checked 8–10 weeks prior to training or competition at moderate altitude or simulated hypoxic environment (Bergeron et al. 2012).

Antioxidants

Hypoxia increases oxidative stress, which is exacerbated by even single bouts of exercise even at low–moderate altitudes. Impaired oxidative status has been observed in elite endurance athletes following an 18-day live high–train low altitude camp at 2500–3500 metres (Pialoux et al. 2010). Biomarkers of oxidative stress remained impaired for 14 days after the athletes returned to normoxic (1200 m) conditions, suggesting that antioxidant supplementation may be beneficial. Additional intakes of antioxidant-rich foods, rather than antioxidant supplements, are recommended to attenuate the increased oxidative stress (Powers et al. 2011), although this recommendation has not been tested in athletes training at altitude. In summary, although the use of antioxidants for altitude training is intuitively appealing, antioxidant supplements may be contraindicated for other reasons (see Chapter 11 and Commentary 6). Until further research becomes available, it is prudent for the elite athlete to consume antioxidants from food sources.

Vitamin D

Athletes competing in winter sports who habitually live and train at altitude, particularly those residing at low latitudes, are at high risk of low vitamin D status (Farrokhyar et al. et al. 2014). Decreased skin exposure to UV light because of clothing protection and possibly low vitamin D intake contribute to low vitamin D. Low vitamin D status can compromise muscle and immune function (Todd et al. 2014) (see Chapter 9). Screening the vitamin D status of 'winter' athletes who primarily train and compete at altitude may be warranted.

24.5 Altitude: summary

In summary, there are many unique physiological demands associated with living and training at altitude that have the potential to be positively influenced by nutrition. Although it is tempting to focus only on food choices during altitude exposure, the diet 3–6 weeks prior to an altitude training camp may be equally important, especially from the perspective of maintaining or improving iron status and overall health. Nutrition for general health can be beneficial, as both illness and injury have been shown to compromise efficacy of EPO-induced haematological adaptations. The additional energy and hydration needs faced by the athlete upon arrival at altitude should be considered as well as an increase in antioxidant-rich foods. Athletes who are healthy, iron sufficient and supported by a well-balanced diet are likely to achieve the desired physiological adaptations expected at altitude training camps.

24.6 Cold environments

Cold conditions, alone or with altitude exposure, can cause adverse conditions such as breathing discomfort associated with reactive airways, freezing and non-freezing cold injury, and hypothermia (Shephard 1972, 1985; Anderson & Daviskas 2000; Castellani et al. 2006;

Cappaert et al. 2008). Factors known to predispose athletes to hypothermia include participation in water sports and rain, low body fat and low muscle mass, and limited muscle glycogen stores (Castellani et al. 2006). As expected, winter athletes tend to modify their behaviour (e.g. avoid wet windy conditions; use hat, gloves, warm boots, waterproof clothing) to minimise the consequences of cold weather. However, the appropriate selection of clothing is sometimes difficult throughout different training and competition scenarios. High endogenous heat production that induces thermoregulatory sweating may increase the risk of hypothermia if athletes are unable to remove wet clothing promptly. Increased water losses—not only through heavy sweating, but cold-induced diuresis and the evaporation of water vapour during respiration—add to the challenge of maintaining fluid balance (Marriott et al. 1996), although the negative impact of dehydration on athletic performance in cold conditions has been shown to be negligible (Cheuvront et al. 2005). Summer athletes can be particularly unprepared to cope with cold, wet, windy conditions and may be at risk for hypothermia, compromised muscle function and poor coordination due to cool body and muscle temperature. The heat loss in water (~4.2 kJ/kg) is approximately four times greater than air at the same temperature (~1.0 kJ/kg), but varies according to the duration of exposure, the temperature of the water (or, subsequently, relative wind-chill) and the size of the body surface exposed. Ironically, an athlete may attempt to warm up by exercising at a greater intensity only to find that they become considerably colder as an increase in wind speed (cycling, cross-country skiing) or water speed (swimming) accelerates heat loss due to convection.

Scientists examining the physiological impact of cold exposure and cold acclimatisation on exercise tend to focus on military and occupational scenarios (Shephard 1993). Rarely do elite athletes find themselves chronically exposed to cold conditions for sustained periods of time. Instead, it is common for athletes to experience a few hours of training or competing in the cold. In addition, athletes who travel frequently can experience cool temperatures due to excessive air conditioning or hotel rooms with inadequate heating and minimal blankets, which can lead to a restless night's sleep.

Researchers investigating the physiological consequences of cold conditions have focused on the endocrine system, energy expenditure at rest, shivering energetics, cognitive function and motor control (Shephard 1985, 1993). Altitude studies performed in the field rarely control for temperature, making it difficult to understand the impact of cold alone on resting energy expenditure and dehydration. Recently, the concept of cold acclimatisation has been recognised (Castellani et al. 2006; van der Lans et al. 2013). Upon cold exposure, there can be increased production of thyroid hormones and catecholamines. In addition, RMR can be elevated due to both shivering and non-shivering thermogenesis, promoting an increase in glucose and glycogen oxidation. Indeed, the oxygen cost of shivering has been reported to be as high as 2.2 L/min, which is ~6 times greater than RMR or ~50% VO_{2max} in one individual during whole body cold-water immersion (Eyolfson et al. 2001). The topic of brown adipose tissue recruitment during cold exposure has attracted renewed interest. Brown fat can become metabolically active in the cold independent of exercise and is responsible for the increased energy requirement associated with non-shivering thermogenesis (van der Lans et al. 2013).

24.6.1 Dietary recommendations in the cold

In cold conditions, energy requirements increase to meet the corresponding increase in RMR that occurs because of shivering and non-shivering thermogenesis. Indeed, underfeeding in cold conditions may impair heat production, but it is likely that shivering-induced thermogenesis can be rapidly restored with glucose feeding (Gale et al. 1981). Adequate feeding to maximise muscle glycogen stores before exercising in the cold is strongly advised. Fluid intake should also be increased because cool air tends to be dry, contributing to insensible fluid loss. From a practical perspective, consuming warm foods and liquids can be both pleasant and effective in warming the body. Indeed, in one study, warm drinks were shown to improve mood in a group of mountaineers (Scott et al. 2004). There is also a tendency for an increase in appetite, resulting in greater voluntary food intake during multiple days of cold exposure (Moellering & Smith 2012).

24.7 Hot environments

Environmental heat is a powerful stimulus causing an increase in the physiological and perceptual strain associated with exercise. Numerous studies have documented a decrease in exercise capacity in hot conditions or when thermoregulation is impaired (Nybo et al. 2014). However, athletes often train in natural (i.e. acclimatisation) or man-made (i.e. acclimation) heat to induce physiological adaptations that support performance (Lorenzo et al. 2010; Racinais et al. 2014). The acute and chronic effects of the heat on exercise have been a popular research topic for more than a century but the precise mechanisms responsible for impaired exercise capacity continue to be debated. Many scientific review papers comprehensively discuss the physiological responses to exercise in moderate and extreme temperatures (Mundel 2008).

The primary challenge of exercising in hot conditions is heat dissipation with little, if any, thermal gradient between the body and the environment. During exercise, there is a demand for oxygenated blood by skeletal muscle and other metabolically active tissues. However, blood flow to the skin is generally required to cool the body. The dual requirement of blood flow to muscle and skin during exercise in the heat cannot always be met during high-intensity exercise, resulting in premature fatigue and elevated core temperature (Nybo et al. 2014). Sweating with inadequate fluid intake leads to a decrease in plasma volume, which can further compromise thermoregulation (Sawka et al. 2001). Heat acclimatisation over 2–3 weeks exposure is associated with an increase in plasma volume, more effective skin blood flow, greater sweat rates and consequently a lower body core temperature for a given workload in hot conditions (Lorenzo et al. 2010; Garrett et al. 2012). There is also evidence that heat acclimatisation attenuates the increased energy expenditure and affords a lower heart rate at submaximal workloads (Pandolf 1998).

Sensory input from thermal receptors, predominately in the skin, is believed to cue behavioural responses necessary to defend against excessively high core temperatures (Schlader et al. 2009). Thermal sensation and thermal comfort are linked to changes in central drive, leading to motor unit recruitment and work output regulating the rate of heat storage in an attempt to prevent thermoregulatory catastrophe. The observation that self-selected cycling power output is reduced early during a prolonged time trial in hot conditions, well before the occurrence of acidosis, elevated core temperature or glycogen depletion, indicates that perceptions and anticipatory mechanisms are involved (Tucker et al. 2004, 2006).

The brain down-regulates skeletal muscle recruitment in hot conditions, resulting in a reduction in metabolic heat production (Nybo 2007, 2009). As a result, athletes often self-select a lower power output during exercise in the heat. Although protein and CHO oxidation and hence oxidative stress can be higher in the heat than in temperate conditions for a given workload, athletes often reduce exercise intensity to compensate. Hence, energy requirements may be lower in hot conditions than expected. Additional research is required to establish any definitive modifications to macronutrient and micronutrient recommendations when exercising in the heat.

24.8 Dietary recommendations in the heat

Living and exercising in a hot environment promotes fluid loss and increases fluid requirements. The effects of fluid imbalance, the type of beverage and its solute load consumed during and after exercise in hot conditions are considered in this section.

There appears to be minimal impact of exercising in the heat on RMR and total daily energy expenditure and hence energy requirements in trained people. Total daily energy expenditure is more likely to be influenced by changes in activity levels rather than an increase in RMR in response to high ambient temperature in trained athletes (Burstein et al. 1996; Burke 2001; Tharion et al. 2005). Some athletes reduce voluntary food intake in the heat or when training in a hot climate creating a negative daily energy balance (Burstein et al. 1996), which may persist for several days. Rapid weight loss can significantly decrease performance capacity (see Chapter 6).

Sport scientists examining the effects of hot environments on dietary intake of athletes have focused mainly on fluid and energy intake prior to, during and following exercise. Research on macronutrient and micronutrient requirements of highly trained athletes exercising in unusually hot conditions or in a hot climate (without prior acclimation) is scarce. Recommendations for nutrient intakes are largely extrapolated from the known substrate utilisation of fuels for exercise in the heat rather than dietary studies.

24.8.1 Fluid

The current guidelines on fluid replacement and exercise have shifted from the previous approach of generic recommendations for athletes involved in various types and intensity of exercise to an approach specific to the individual athlete (Sawka et al. 2007; Shirreffs & Sawka 2011). These guidelines promote personally tailored strategies for individual athletes across a range of sporting scenarios to optimise health and performance outcomes. When defining a fluid plan for an individual athlete, sweat rates and hence predicted fluid losses (which increase in proportion to ambient temperature), level of fitness, exercise intensity, heat acclimatisation and clothing worn are considered (for a comprehensive review, see Baker & Jeukendrup 2014). Further contextual information about the exercise task (mode and duration) and the opportunity to consume fluid during sports competition is needed to guide prescription and timing of appropriate fluids to minimise the fluid deficit (<2% hypohydration recommended).

Athletes who are not acclimatised to the heat may be tempted to ingest an increased volume of water, particularly during competition. The addition of sodium to sports beverages can replace sodium losses associated with sweating, prevent hyponatraemia, promote the maintenance of plasma volume and enhance intestinal absorption of glucose and fluid (for review, see Maughan et al. 1994). The CHO content of a beverage ingested during exercise in the heat appears to have little effect on fluid availability or exercise performance, provided the CHO concentration is not extreme. The change in plasma volume and exercise performance in the heat did not differ between ingesting beverages containing 0, 4.2 and 7% CHO respectively (Febbraio et al. 1996). However, a 14% CHO solution reduced plasma volume and elevated rectal temperature. Accordingly, exercise performance tends to fall (see Figure 24.1) (Febbraio et al. 1996). It is important, therefore, to keep the concentration of CHO (<~10%) in a fluid beverage during exercise in the heat, even though CHO utilisation is augmented in these circumstances. In terms of volume and frequency, a practical recommendation might be 400 mL every 15 minutes for those who sweat heavily. CHO beverages should also be included during recovery to replenish intramuscular glycogen stores and promote hydration.

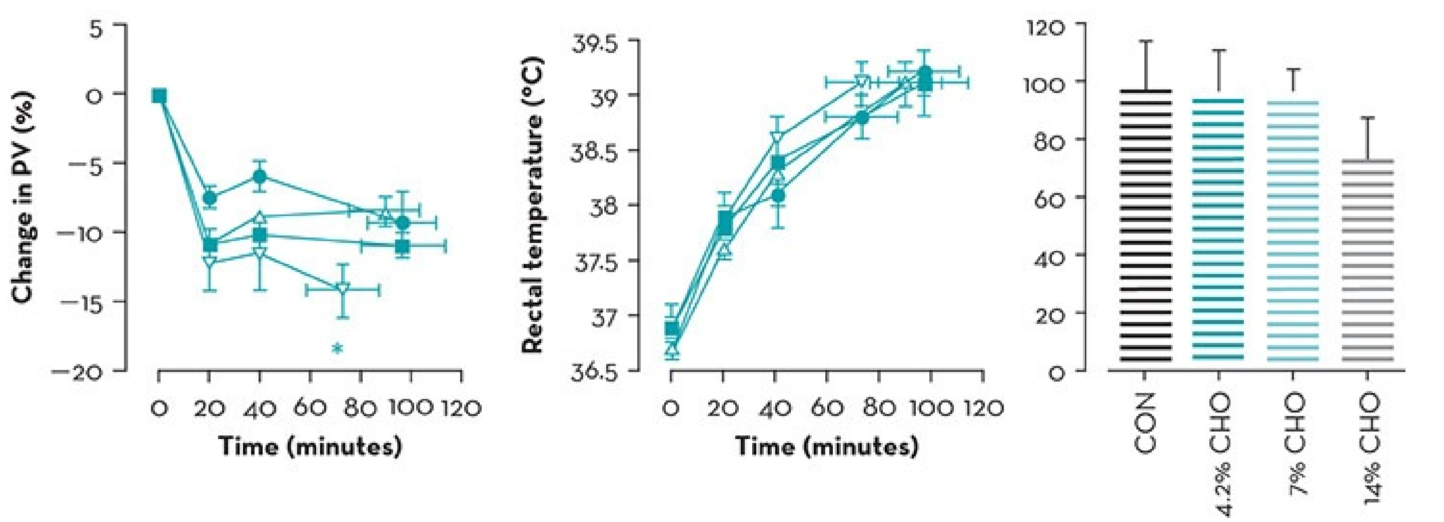

Figure 24.1 The change in plasma volume (left), rectal temperature (middle) and time to exhaustion (right), while consuming a placebo (CON, -■-), 4.2% CHO (LCHO, -●-), a 7% CHO (-△-) or 14% CHO (HCHO, -▽-) beverage during fatiguing exercise at 70% VO_{2max} in 33°C conditions. * denotes difference (p < 0.05) from other trials. Data

expressed as means ± SE. Adapted from Febbraio 1996

However, caution should be used when recommending specific amounts of fluid during prolonged events or training sessions (>3 h), when the cumulative effect of mismatches between fluid needs and replacement can compromise an athlete's health (hypohydration or hyponatraemia) (Sawka & Noakes 2007). Importantly, drinking a CHO-electrolyte beverage may increase ad libitum fluid consumption because of increased palatability.

A higher concentration of sodium chloride in sweat is reported during exercise in hot, compared with temperate, environments (Dziedzic et al. 2014). Heat acclimatisation improves sodium chloride reabsorption, thus resulting in lower sodium chloride concentration for a given sweat rate (Allan & Wilson 1971). In unacclimatised athletes, either the addition of 20–50 meq/L sodium to pre-exercise hydration beverages or consuming small amounts of salty food is central to the restoration of fluid balance following exercise-induced hypohydration to replace sodium lost. The additional salt helps stimulate thirst, enhance palatability and retain consumed fluids compared to plain water (Sawka & Noakes 2007; Baker & Jeukendrup 2014). However, the recommended concentration of electrolytes to help replace sweat electrolyte losses during exercise is slightly lower (~20–30 meq/L sodium and ~2–5 meq/L potassium). Individualising whole-body fluid balance may involve monitoring acute changes in BM (to estimate sweat rates) or the frequency and osmolality of urine. A practical method of estimating sweat electrolyte content, and importantly electrolyte losses, is the regional absorbent-patch method. Use of this method requires technical expertise and includes a standardised protocol to determine the amounts of dietary sodium needed to replace losses (Dziedzic et al. 2014).

Since sweat rate and respiratory rate are increased during exercise in the heat, the process of dehydration is accelerated. Fluid losses may be as high as 2–3 L/h in extreme conditions (Burke 2001). In circumstances where endogenous heat production and high environmental temperature result in fatigue prior to CHO stores being depleted, adequate fluid ingestion, irrespective of whether the fluid contains CHO or protein, can delay the rise in body-core temperature. Exercise-induced dehydration is associated with an increase in core temperature, reduced cardiovascular function and impaired exercise performance. These effects are reduced, or indeed prevented, by adequate fluid ingestion, which also improves exercise performance in the heat (Below et al. 1995). Fluid ingestion also reduces muscle glycogen use during prolonged exercise (see Figure 24.2, (Hargreaves et al. 1996; Gonzalez-Alonso et al. 1999).

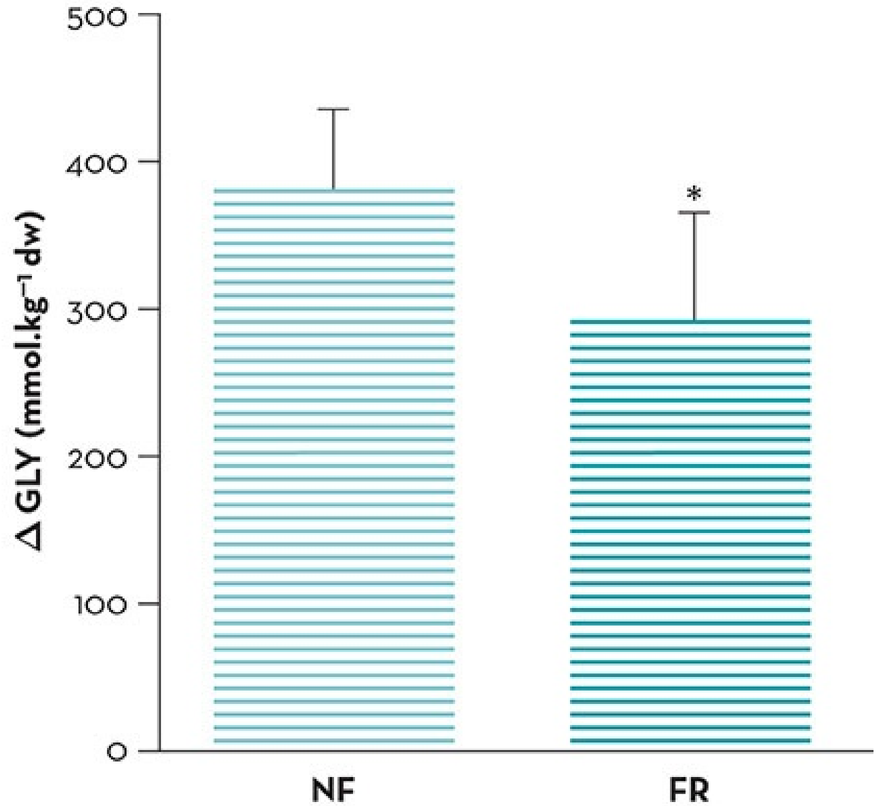

Figure 24.2 Net muscle glycogen utilisation (△ GLY; pre-post-exercise) during 120 minutes of exercise in the absence (NF) or presence (FR) of fluid ingestion. * indicates difference (p < 0.05) compared with NF. Data expressed as mean ± SE (n = 5). From Hargreaves 1996, used with permission

It is well established that the co-ingestion of food and beverages at mealtimes is beneficial for the retention of fluids, but when time is limited or the athlete experiences gastrointestinal distress, ideal food and fluid ingestion practices may not be practical (Sawka et al. 2007). Instead, one option, which is relatively high in energy, water and electrolytes, is milk (Baker & Jeukendrup 2014). Indeed, milk has been shown to be highly effective in the acute recovery from exercise-induced hypohydration (Shirreffs et al. 2007; Watson et al. 2008; James et al. 2011).

Rapid fluid replacement, according to the current guidelines, recommends that 150% of BM losses should be consumed post-exercise (Sawka et al. 2007). Shirreffs and colleagues (2007) observed urine outputs to be ~600 mL lower with milk over 4 hours of recovery than when an equal volume of water or CHO-electrolyte beverage was consumed. This meant that subjects returned to euhydration only when milk was consumed. Despite the rapid return to a hydrated state, there was no improvement in endurance capacity detected following the use of milk as a rehydrating agent (Watson et al. 2008). However, measurement of differences in performance associated with small increments in hypohydration is difficult to detect statistically, especially in events lasting less than 1 hour (Goulet 2012), so any real effect may have been missed. When addressing the compositional differences of rehydration beverages, James and co-workers (2011) attributed the augmented fluid retention to the protein component in milk , which is responsible for slowing the rate of gastric emptying. This consequently delays the rate of fluid absorption compared to that of water or CHO-electrolyte beverages. Although speculative, the inclusion of milk beverages during the acute-phase recovery from exercise in the heat is likely to contribute to greater muscle glycogen and muscle protein re-synthesis.

Functional dehydration

The rigorous consumption of sport drinks or water will reduce the acute weight loss (sweat loss) that normally accompanies prolonged exercise. For cyclists and runners who often want to be as light as possible, especially when they need to run or cycle uphill, the mass associated with excessive fluid consumption is undesirable. As a result, some athletes have adopted a minimalist approach to fluid consumption in races or stages that provide an uphill challenge. In attempts to explore whether it is best to be slightly heavier and hydrated, or lighter and dehydrated, Ebert and colleagues (2007) recruited eight competitive male cyclists, who were asked to perform maximal cycling hill climbing efforts on a treadmill. Cyclists rode at 55% of the power output that corresponded to $VO_{2\ max}$ for 2 hours and then performed a maximal duration hill climb at an exercise intensity that was ~105% of lactate threshold power. Cyclists either consumed 1.2 L or 0.2 L per hour during the 2-hour ride and power output was monitored using instrumented power cranks (SRM) during the hill climb. CHO intake during both trials was held constant. Performance time was significantly better during the high fluid intake trial despite the observation that cyclists were nearly 2 kg heavier during this trial. Results from this study highlight the importance of aggressive drinking strategies when performing endurance events where minimal BM is desired.

Temperature of drinks

Lowering the temperature of the ingested fluid to a cold (Lee et al. 2008; Burdon et al. 2010) or ice-slurry beverage (Dugas 2011; Siegel et al. 2012; Burdon et al. 2013) is a practical and effective way of optimising thermoregulation during exercise in the heat. Cold beverages increase palatability, which consequently increases fluid intake and promotes a positive influence on hydration status (Mundel et al. 2006; Burdon et al. 2012), whereas ice-slushy beverages offer an alternative strategy for enhancing heat storage capacity from the ingestion of relatively smaller fluid volume (see Figure 24.3). In addition to cooling, typical recommended beverages (e.g. sports drinks) deliver nutrients (CHO, electrolytes) that promote rapid delivery of water and sensory advantages, which can independently influence performance in other ways. Indeed, it has been observed that oral temperature-sensitive regions of the brain were activated in response to perceived pleasant sensations of a cold fluid when placed in the mouth (Guest et al. et al. 2007). Moreover, the merits of cold sensations induced by L(-)-menthol mouth-rinsing during exercise in the heat (Mundel & Jones 2010) has been shown to improve performance through the stimulation of orophalangeal cold receptors. These cold receptors give the perception that air inhaled or water consumed feels cool, which may enhance desirable central drive. It has since been shown that when rinsed in the mouth, an ice-slushy provided a sensory effect of improving perceptions of effort and comfort despite a lack of temperature or cardiovascular alterations or meaningful improvement in endurance performance (Burdon et al. 2013). Indeed, the first study to integrate physiological and perceptual cooling through ingesting a menthol-slushy before and during a 20-km cycling time trial showed positive performance effects, but more work is needed in this area to confirm any real advantage (Riera et al. et al. 2014). A practical benefit of these strategies might appeal to athletes who would otherwise be advantaged by increases in BM if large volumes of fluid were to be ingested, or when hydration or fuel status is not compromised by lack of beverage ingestion. Practical strategies to maintain fluid balance before, during and after exercise have also been addressed in Chapter 13 and Commentary 7.

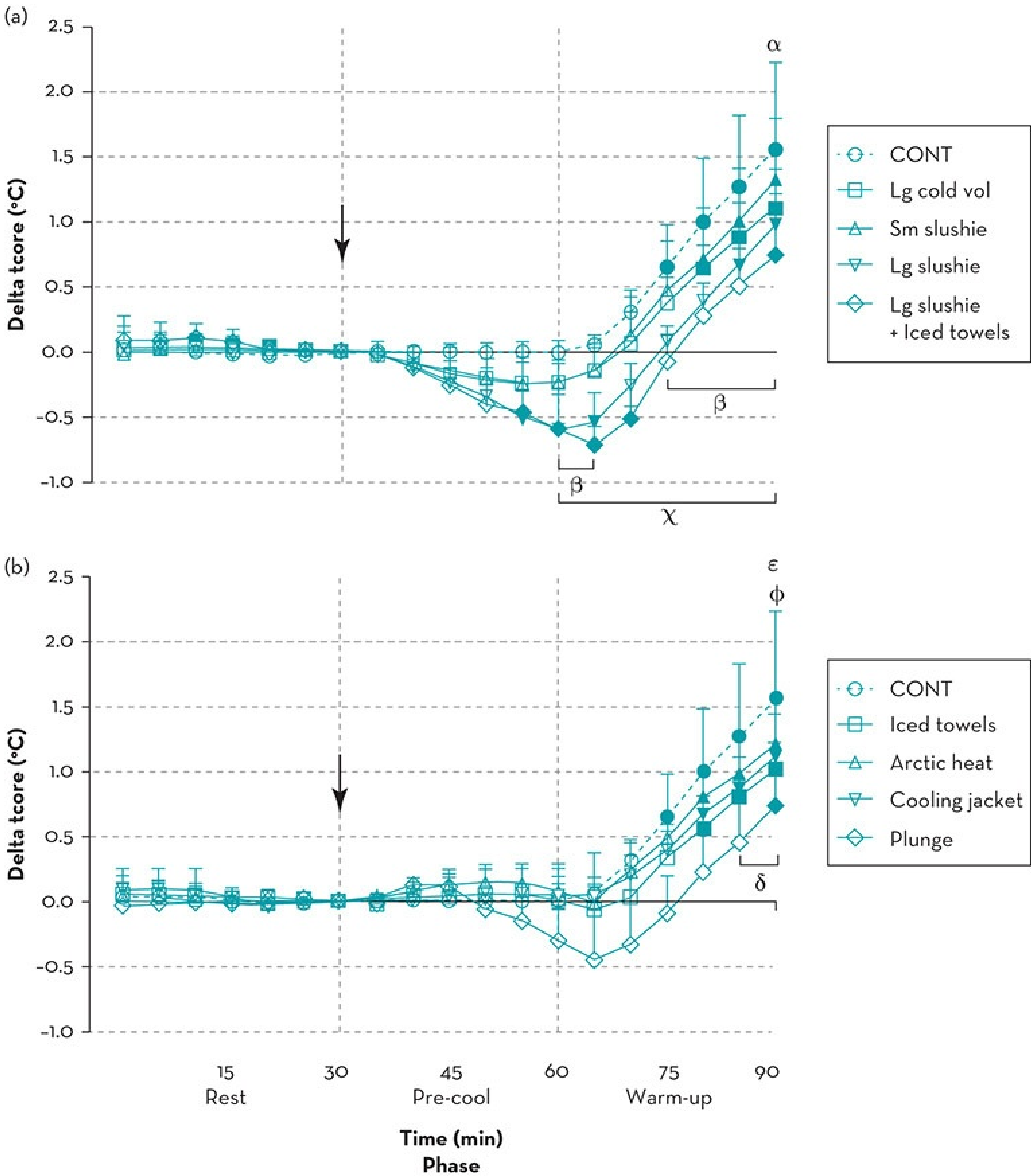

Figure 24.3 Relative change in core temperature in response to pre-exercise cooling. This figure demonstrates the effectiveness of ingesting cold and ice-slurry beverages to increase the heat storage capacity of the body during heat and exercise exposure. Internal and combination pre-cooling (a) and external pre-cooling (b) during 30-minute heat stabilisation phase, 30-minute pre-cooling and 30-minute cycling warm-up. Significant time effects (from *t* = 30 min) are denoted by dark symbols. Significant treatment effects between large cold volume are denoted by alpha (α); large slushy are denoted by beta (β); large slushy + iced towels are denoted by chi (χ); iced towels are denoted by delta (δ); arctic heat are denoted by epsilon (ε); and, RMIT/AIS Jacket are denoted by phi (φ) when compared with control (CONT). Commencement of pre-cooling phase indicated with an arrow (↓). Statistical significance is set

07:40:20.

at the level of $p < 0.05$. Data represented by mean ± SD. From Ross 2011, reproduced with permission

24.8.2 Macronutrients and energy intake

Carbohydrate

CHO utilisation from intramuscular glycogen use and glycolysis during exercise is augmented, whereas lipid utilisation generally decreases in an athlete with heat stress. Mediated by an increase in circulating plasma epinephrine (Febbraio 2001), endogenous glucose production is increased and whole-body CHO oxidation is enhanced. An elevated core temperature appears to be responsible for the shift to CHO oxidation during submaximal exercise. The increased utilisation of CHO as a substrate during exercise in the heat is ameliorated with heat acclimatisation (Febbraio et al. 1994). Unlike submaximal exercise in comfortable temperatures, exercise in extreme heat is not limited by CHO availability. However, during mild heat stress it is likely that CHO availability and rapid delivery of CHO for glucose oxidation and maintaining glucose homeostasis is important.

The ingestion of a CHO-electrolyte beverage can benefit exercise performance in warm/hot conditions. One unique mechanism that influences performance capacity involves the physiological effects from stimulation of taste receptors in the mouth. Interestingly, these receptors appear to produce an ergogenic effect regardless of whether the fluid is ingested or not (see Burke & Maughan 2015). However, CHO ingestion and subsequent absorption is also needed to maintain fuel status in the post-absorptive state (Burke et al. et al. 2011). Current guidelines for CHO ingestion recommended during exercise account for the duration and intensity of exercise, rather than the influence of environmental exposure. As such, the amount and type of CHO recommended during exercise may need adjustment for warm/hot conditions (see Chapter 13). In recent review of fluid balance studies, Baker and Jeukendrup (2014) reported that plasma volume restoration is enhanced by consuming a protein–CHO beverage after exercise in hot conditions, compared to water or CHO ingestion alone.

Protein

Similar to the early effects of altitude training, exercising in the heat can impair MPS and increase protein degradation above that of exercising in temperate conditions (Snow et al. 1993; Febbraio et al. 1994; Febbraio 2001). The enhancement of muscle protein synthesis is primarily determined by the essential amino acid content of the ingested protein source (see Chapter 4). Protein ingestion after all forms of exercise enhances muscle protein synthesis and net protein balance and should be a viewed as a core and critical component of ‘recovery’ nutrition for all athletes. As discussed above, adding CHO to a post-exercise snack or beverage has other advantages. See Chapter 4 for guidelines on food choices, timing and quantities of protein intake for different sports situations.

During prolonged high-intensity exercise in the heat (and in high-intensity exercise in temperate conditions), blood flow to the gut can be compromised (gut ischaemia), which increases gastrointestinal permeability and susceptibility to absorption of endotoxins (see Commentary 8). Gut ischaemia is one factor associated with some of the uncomfortable side effects (diarrhoea, gut pain) and gastrointestinal bleeding reported in athletes after a marathon or triathlon. The protein-dense supplement, bovine colostrum, has been shown to reduce gastrointestinal hyperpermeability associated with exercise in the heat in rats (Prosser et al. 2004). However, any similar relationship between this supplement and changes to gut physiology in humans is not confirmed because of measurement difficulties (Carrillo et al. 2013; Morrison et al. 2014). Nevertheless, the clinical effects of short-term changes in gastrointestinal permeability appear minimal. Disturbances to gut function and blood flow tends to improve in the trained athlete and when exercise intensity is reduced, suggesting that adaptation may be operating in highly trained athletes.

24.8.3 Micronutrients and other nutrition supplements

Antioxidants

High-intensity exercise increases oxidative stress and the production of free radicals (see Commentary 6). Only a few intervention studies have been conducted on athletes or individuals who participate in regular exercise investigating the effects of antioxidants supplements (see Commentary 11 for a review of key studies). These studies are mostly undertaken in temperate conditions. Although controlled trials on the effects of exercise in the heat in relation to oxidative responses are scarce because of ethical issues, one such trial found that exercise-induced dehydration both with and without environmental heat stress (33.9°C and 23.0°C) increased oxidative stress markers in seven healthy, trained, male cyclists (Hillman et al. 2011). Maintenance of hydration tended to attenuate some, but not all, oxidative stress markers measured, independent of temperature. Although the sample and effect size was small, this study highlights the importance of adequate hydration during prolonged exercise in the heat.

As discussed in Chapter 11 and Commentary 6, there may be disadvantages to chronic antioxidant supplementation to ameliorate the increase in reactive oxygen species (ROS) associated with exercise. Chronic supplementation with high doses of the antioxidant vitamins —vitamin C (~1000 mg/d) and vitamin E (~400 IU/d), either in combination or as taken singly —can disrupt the complex balance of antioxidant defences and reduce the effectiveness of oxidative signals that stimulate desired adaptations to training. Nevertheless, the evidence for potential adverse effects from antioxidant nutrients and other antioxidants taken as supplements is not consistent and is confounded by the complexity of the physiological interactions involved. Whether high-dose antioxidant supplements interfere with the adaptive effects of training in the heat requires further investigation. The current consensus is to increase antioxidant intake from food sources rather than from supplements (see Commentary 11).

Caffeine

Caffeine is found in many beverages (e.g. coffee, tea, energy drinks), foods, supplements and chewing gum. It acts as a cardiac and central nervous system stimulant that can enhance athletic performance when used appropriately (Lopez & Casa 2009). High intakes of caffeine either from strong coffee or supplements are used as an ergogenic aid by many athletes in different sports (see Chapter 16). Some athletes believe that caffeine acts as a diuretic and affects fluid-electrolyte balance, contributing to dehydration. However, one study on 59 active college-aged men demonstrated that caffeine, when ingested prior to an exercise heat-tolerance test (38°C, 56% RH), had no effect on fluid-electrolyte balance, on reducing exercise heat tolerance or increasing risk of heat illness (Roti et al. 2006). Contemporary research findings do not support the notion that athletes need to refrain from consuming moderate levels of caffeine from a hydration or performance perspective when exercising in the heat (Armstrong et al. 2007; Ganio et al. 2011a, 2011b) Caffeine beverages do not independently raise internal temperature or predispose an individual to heat stress (Ganio & Armstrong 2012).

Caffeine supplementation at moderate to low doses (~3 mg/kg BM) can improve performance in the heat in some (Pitchford et al. 2014) but not all studies (Cohen et al. 1996). At this dose, side effects are minimal. However, practitioners should be cautious about recommending caffeine supplements or high intakes of caffeine-containing beverages to athletes who are unaccustomed to its use or consumption, particularly in hot conditions, because of the potential risk of uncomfortable side effects.

Glycerol supplements

Pre-exercise hyperhydration involving the deliberate intake of 1.2 g/kg BM glycerol suspended in 26 mL/kg BM of water has been proposed as a technique that can prevent or attenuate the effects of exercise-induced dehydration (van Rosendal et al. 2010). This co-ingestion of glycerol with the large volume of water can increase fluid retention over that of water alone (Goulet et al. 2007). However, replacing the water with a CHO-electrolyte solution in this glycerol suspension may further enhance fluid absorption and have metabolic benefits, such as an improvement in thermoregulation (Goulet 2009). On one study, large, sustained reductions in body temperature were evident in 12 well-trained male cyclists who consumed a chilled (4°C) glycerol-CHO-electrolyte beverage prior to undertaking three 46.4-km laboratory-based cycling trials in hot humid conditions (~33.3°C, ~50% RH) (Ross et al. 2012). While glycerol has been shown to be an effective hyperhydrating agent in this and other studies (Goulet al. et 2007), the exact mechanism of fluid retention remains to be elucidated. A meta-analysis concluded that the use of glycerol hyperhydration in hot conditions provides a small but worthwhile enhancement effect (effect size 0.35) on prolonged exercise performance (3% increased power output) above hyperhydration with water (Goulet et al. 2007). However, not all studies have reported performance benefits. Finally, while the reported incidence of side effects associated with glycerol consumption is low, a prolonged (90-min) ingestion protocol is recommended to prevent potential adverse symptoms, which may include light-headedness, nausea and varying levels of gastrointestinal distress (from bloating to diarrhoea). Athletes intending to use glycerol at the dosages suggested above need to trial it in training before use at competition in case of adverse side effects.

The World Anti-Doping Code (2015) prohibits the use of glycerol supplements for its plasma expander effects, which mask the use of illicit drugs (https://www.wada-ama.org/ Section C5, 2015 Prohibited list – International Standard). Specifically, the code states that the ban is on glycerol supplements per se, although some everyday foods, beverages, personal care products, medicinal tablets and cough syrups contain small amounts. These sources will not cause an athlete to test positive. Glycerol hyperhydration is therefore discouraged for use by high performance athletes (because of the ban) but may have some benefit for individuals who are at high risk of dehydration/heat stress during prolonged exercise in the heat.

24.9 Heat: summary

Many sports require athletes to train and compete in hot conditions. Although a large amount of research has documented the profound effects of heat on exercise physiology and performance capacity, translating knowledge into practice remains a challenge. Fluid and CHO support for training and competition in hot climatic conditions remain key topics of interest and the focus of most nutrition intervention. Additionally, acclimatisation and familiarisation to training and performing in hot conditions is highly recommended. Substantial adaptations to heat exposure occur within 2–3 weeks in healthy athletes, assuming a well-thought-out training program, a well-balanced diet high in antioxidants, and, ideally, an individualised fluid plan. Ingesting fluids that are cool can induce a thermoregulatory advantage and potential performance advantage. Although there is no direct evidence from human studies, colostrum may be advantageous for preparing the gastrointestinal track for the repercussions of an exercise-induced thermal stress, and milk has repeatedly been demonstrated to be advantageous as a recovery drink, helping the athlete attend to both fluid and energy requirements and stimulating muscle protein synthesis. Although research examining the efficacy of caffeine supplementation for exercise in hot conditions is generally positive, experimenting in training under similar environmental conditions is recommended.

Practice tips

SUSIE PARKER-SIMMONS

The nutritional recommendations listed below need to be modified for individual athletes depending on their history and experience training or competing in conditions of heat, cold and altitude.

- *Increase energy intake* Altitude, cold and heat can increase energy requirements. This increase is substantial at high altitude but may be offset by a reduction in the intensity of exercise in hot conditions. For those athletes who have a suppressed appetite on exposure to moderate altitudes or in a hot climate, more frequent access to and ingestion of high-energy-dense snacks and beverages and larger meals than usual may be needed to meet higher energy requirements associated with the additive effects of training and hypoxia.

- *Increase carbohydrate intake* During early acclimatisation at altitude, additional CHO or even CHO-based supplements are required, even at rest. Examples include sport drinks or other CHO-rich fluids, dried fruit, grains, cereals, sports bars or gels. These foods may also be appropriate for situations where appetite is suppressed in the heat or at altitude since they are a compact source of fuel.
- *Increased fluid requirements* In hot environments and at the start of altitude training, athletes are at highest risk of disturbances in thermoregulation and dehydration. Fluid losses and hence requirements increase in hot and cold environments. The current position statement on recommending fluid intake to athletes is to provide an individualised fluid or drinking plan. For athletes training in extreme temperatures or altitude, this should be initiated during the early phase of acclimation when the risks are high. It is not uncommon for cyclists and other endurance athletes (e.g. long-distance runners/walkers) to routinely have suboptimal fluid intakes during long training sessions. These athletes are at high risk for hypohydration when they compete or train at altitude or in the heat unless they are prepared to change these entrenched behaviours. Additional hydration strategies may include aggressive rehydration after each exercise session, hyperhydration strategies in the heat such as glycerol loading (although banned by WADA for its potential to mask illicit drugs) and increasing electrolyte levels of the drinks.
- *Measure fluid balance* Hydration tests are useful in the early phase of acclimatisation and again when adaptation has occurred (to prevent overdrinking and the potential for hyperhydration). Suitable field tests include urine specific gravity (first void), fluid balance testing during training, urine colour and haemoglobin testing (see Commentary 7).
- *Type and timing of fluid intake* See Chapter 13 and Commentary 7. The CHO and protein content of milk beverages consumed during the acute-phase recovery from exercise-induced hypohydration in the heat or cold enhance fluid retention and may contribute to greater muscle glycogen and MPS synthesis. Exercising in the heat and at altitude or a simulated hypoxic environment impairs protein synthesis, increases protein catabolism and increases CHO utilisation at a greater rate than in temperate climates and at low altitudes. Combinations of foods/beverages that contain CHO and small amounts of protein consumed almost immediately after exercise is a critical component of 'recovery' nutrition for all athletes but especially those training in extreme environmental conditions.
- *Keeping cool in a hot environment* During prolonged exercise in extreme heat where there is a risk of hypohydration, cold beverages (cold sports drinks (~≥ 10% CHO), ice slushies), increase the drive to drink and hence increase fluid consumption and have potential performance benefits as well as a positive influence on hydration status. Using mouth washes with ice slushies/plain ice or menthol also give the perception of a cooling effect (see section 24.8.1). Cold water/ice is likely to be adequate for short duration training sessions in the heat.
- *Increase in oxidative stress* Although there is no consensus in studies of athletes, sufficient dietary intake of vitamin antioxidants (vitamin C, E and β-carotene) and other exogenous antioxidants may have benefits for blunting the potential damaging effects of oxidative stress, which is exacerbated by training at altitude and in hot climates. Recent evidence suggests that antioxidants taken as supplements may do more harm than good (see Commentary 6).
- *Assess risk for suboptimal vitamin D status at altitude* Screening and monitoring vitamin D status of high-risk athletes may be worthwhile in athletes who spend most of the year training at altitude. Serum 25 (OH) D concentrations is the best indicator for checking the variability of vitamin D levels. The vitamin D content in food is not high enough to meet daily requirements, although some countries are now fortifying more foods with vitamin D as a public health measure. Effects of low vitamin D status and intervention strategies are summarised in Chapter 9.
- *Assess risk of low iron status prior to altitude training or prolonged competition at altitude* Training at altitude stimulates the hormone erythropoietin (EPO), which increases the number of red blood cells, haemoglobin and plasma volume, resulting in substantial increases in iron requirements. Measuring iron status at 8–10 weeks prior to altitude training or prolonged competition at altitude may be warranted to allow time to restore low iron stores. Treatment for iron depletion (or deficiency) for athletes training at altitude is usually according to therapeutic guidelines (see Chapter 10). For some cases, higher-than-usual therapeutic doses of elemental iron (e.g. 120–130 mg/d) in divided doses may be needed for rapid recovery, depending on the individual athlete's risk and tolerance of potential side effects.

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CHAPTER TWENTY-FIVE

Catering for athletes

Fiona Pelly and Joanne Mirtschin

25.1 Introduction

Catering for athletes and providing advice to caterers is an exciting and challenging task. The main role for the dietitian is to design menus and recipes that meet the nutrient needs of athletic groups and satisfy a wide variety of individual tastes, expectations, dietary regimens and cultural preferences. Working directly with caterers provides the opportunity to select suitable foods for consumption and potentially influence the eating behaviour of athletes. Unfortunately, sporting venues and institutions that supply food for athletes do not usually employ a food-service dietitian or cater adequately for athletes' needs. Commonly, the available foods contradict sports and general nutrition principles.

This chapter focuses on the processes involved in menu planning, the impact these have on athletes' food choices and the role of the dietitian in this process. The dining hall of the athletes' village at the Olympic and Commonwealth Games and the Australian Institute of Sport (AIS) provide examples of working models of large-scale residential catering for elite athletes. Practical tips and strategies for the self-catering of athletes living away from home are also included.

25.2 Determinants of food choice of athletes

Determinants of food selection by individuals are complex and influenced by many factors. Eating habits are acquired over a lifetime and dietary change requires a conscious effort, practice and commitment to maintain this change (Nestle et al. 1998). The most commonly reported influences on food choice reported in the general population are taste, cost, convenience, health and nutrition beliefs, social relationships and food quality (Furst et al. 1996; Falk et al. 2001). In athletes, health beliefs, performance expectations, concerns over body composition and taste are important determinants of food choice and are also linked to consuming low-fat diets (Smart & Bisogni 2001). In a study of 418 athletes competing at the 2006 Commonwealth Games in Melbourne, Australia, the two most important determinants of food choice reported were familiarity with foods and nutrient composition (Pelly et al. 2006b). In another study of people eating in an army cafeteria, the main determinants of food choice were taste and quality, followed by size of portions (related to cost), length of the serving line, individual cravings, nutrient density of the food, amount of time available to eat and appetite (Sproul et al. 2003).

25.2.1 Taste and sensory factors

Taste is a key determinant of food choice, especially in males, and rates high for most people, irrespective of age and gender (Hess 1997; Buscher et al. 2001; Sproul et al. 2003). Respondents in surveys report high-fat, high-sugar and energy-dense foods to be associated with feelings of personal satisfaction, reward, free choice and indulgence (Nestle et al. 1998). These food preferences are often established in early childhood when high-fat foods are offered in positive contexts associated with celebrations or used as rewards (Nestle et al. 1998).

There are few studies on taste preferences of athletes. In one study, Guinard and colleagues (1995) found no difference in the preferences for fatty animal products (dairy and meat) between male athletes and sedentary controls. In contrast, in another study, female athletes reported a significantly lower preference for high-fat food, compared with sedentary controls (Crystal et al. 1995). Ratings of taste of dining hall food by athletes have been shown to decrease over the competition period (Burkhart & Pelly 2013a).

Caterers to athletic groups therefore should consider that taste and other sensory appeal of food strongly influences food selection, although all too often they are more concerned about cost and minimising the time and effort that catering staff spend on food preparation (Reichler & Dalton 1998).

25.2.2 The food-service setting

Self-serve, buffet-style food service is frequently used when feeding large groups of athletes, which can easily cater for individual needs and food preferences (Modulon & Burke 1997). With this style of food presentation, people tend to choose a high-carbohydrate diet with a favourable macronutrient balance (Hickson et al. 1987) but tend to overeat (Stroebele & De Castro 2004)—which was also confirmed in US college hockey players, particularly younger compared with older athletes (Smart & Bisogni 2001).

Although meals selected from a menu with a wide range of choices may be the first preference for most people, it is not feasible or economical when serving large groups of athletes. A buffet-style self-service is more suitable because it provides readily available food, is cost effective, allows for bulk cooking and low wastage, and offers flexibility in food choice and serve size. Feedback from athletes competing at the Olympic and Commonwealth Games suggests that a buffet-style food service is adequate to meet their needs with high ratings received for presentation, variety, temperature and ease in finding suitable items (Pelly et al. 2009; Burkhart & Pelly et al. 2013a). However, those with previous experience of this type of dining tend to be more critical of the food provided (Burkhart & Pelly 2013a). Athletes living in residence, where buffet-style food is always served, often complain about boredom and lack of menu variety.

Some athletes have reported negative attitudes towards eating in an institutionalised environment, which they equate with low quality, limited variety and an unpleasant physical environment (Meiselman et al. 2000; Stroebele & De Castro 2004). Providing a pleasant physical environment (colour and soft lighting) and attractive food presentation (including garnishes, attractive serving bowls or tablecloths) can change the perception of the dining experience and alter food selection.

Access to food and beverages also influences food choice. In one study, twice as much water was consumed when placed on the dining table rather than six metres away in the same room, or in a different room (Engell et al. 1996). In another study, Meiselman and colleagues (1994) found that intakes of socially desirable foods (e.g. confectionery, potato chips) decreased when located away from the main buffet. Providing easy access to specific foods with limited waiting time influences food selection.

25.2.3 The influence of others

Peer acquaintance also influences food choice and volume consumed. People who are less familiar with each another can be self-conscious about communal eating and are often hesitant to overeat (Clendenen et al. 1994). The opposite behaviour is observed when food is consumed with friends, where meal portions were reported to be around 44% larger than when eaten alone (Stroebele & De Castro 2004). People also tend to adapt to the eating behaviour of their companions, particularly if they are trying to make an impression. Young athletes are easily influenced by others, particularly their coach, and may modify their food intake to either please or impress. This behaviour was evident in the younger members of a US college hockey team who were more willing than the more experienced players to try new combinations of foods (Smart & Bisogni 2001). In a survey of athletes at the 2006 Commonwealth Games, the presence of teammates and the coach was reported to have a greater influence on food choice for athletes from Asian, African and Caribbean backgrounds than from other regions (Pelly et al 2006b). Social factors strongly influence food choice, especially in adolescents and young adults. For example, individuals are more likely to choose healthier food if it is considered the social norm (Robinson et al. 2013). Recognising and modifying the social determinants of food choice among a group of athletes is likely to have a strong influence on improving food choice in individuals.

25.2.4 Gender and age

Gender and age also influence food choice (Connor 1994). Differences in the influence of smell, visual appeal, familiarity and nutrient composition on food choice were found between male and female athletes in a study of 418 athletes competing at the 2006 Commonwealth Games and in residential accommodation (Pelly et al. 2006b). Male athletes prefer a cooked breakfast and hot meals at lunchtime, whereas female athletes prefer cereal and/or toast for breakfast and sandwiches for lunch. Adult female athletes appear to be more conscious of energy intake than male athletes and non-exercisers (Georgiou et al. 1996). The difference in food choices between sexes may be a reflection of the influence of different body perceptions between male and females, particularly adolescents (see Chapter 8).

Adolescents are conservative in their food choices and often avoid spicy foods and dishes with unusual flavours. Adolescent girls appear to be more adventurous in trying new foods than boys of the same age. Self-selected food choices by adolescents may also be nutritionally inadequate when living in residence away from parental influence (Garrido et al. 2007).

25.2.5 Cultural background and dietary regimen

The home and cultural environment has a major influence on food choice. Individuals with a strong sense of cultural identity are more likely to choose foods that they associate with their national cuisine (Robinson et al. 2013). Athletes who travel a lot for competition are faced with the challenge of eating unfamiliar foods and are often reticent to sample different foods when away from home. Those accustomed to western diets encounter difficulties when competing in Asian countries, where bread and breakfast cereals are often unavailable. Similar difficulties arise when Asians and Africans experience western food. This is reflected in the number and type of requests for culturally specific items (e.g. Vegemite™, kimchi, biltong, ugali) requested by athletes at the Delhi 2010 Commonwealth Games (Burkhart & Pelly 2013a). Religious beliefs (e.g. kosher, halal), medical problems (e.g. intolerances, allergies) and personal preferences (e.g. vegetarian diets) also influence food selection. In a study on the self-reported dietary regimens of an international selection of athletes, 33% followed a regimen based on religious beliefs (predominately halal), avoidance of allergens, vegetarianism and therapeutic reasons. Vegetarianism is more common in athletes from non-western countries; although the majority followed a western style of eating, 22% still ate their traditional style (Pelly & Burkhart 2014).

Although it is difficult, if not impossible, to present a menu plan that satisfies every individual in a large group setting, it is possible to offer enough choices to accommodate different types of diets and cultural differences (e.g. vegetarian, non-red-meat eaters). At elite-level competition, where there is a large range of athletes from different cultures, specialist chefs are often employed.

25.2.6 Sport and stage of competition

The type and culture of sport also influences food choice. In one study, athletes involved in power/skill sports placed less emphasis on the nutrient content of the food compared to those involved in team, weight-conscious and endurance sports where dietary intake and a lean body composition are perceived to be important to overall performance (Pelly et al. 2006b). However, those in power/weight sports may be more critical of the food provided and rate the availability of all items, and in particular sports foods, lower than other athletes (Burkhart & Pelly 2013a). Those in endurance sports report following a diet based on the nutrient composition of the food more than those involved in other sports. Power/sprint athletes and weight-conscious athletes report following high-protein and low-fat diets respectively (Pelly & Burkhart 2014).

Often oversized meals are served to athletes, irrespective of their body mass (BM) or sport because public perceptions about feeding athletes can be distorted. Despite the apparent intensity and duration of physical activity in some sports, energy requirements, and hence food intakes, are not that much greater in athletes than in untrained people (see Chapter 5). Such large meals are quite daunting for those athletes who are watching their BM or skinfolds. For these athletes, foods available need to be nutrient dense and potentially low fat so that adequate nutrient intakes can be met from smaller amounts of food. For athletes with large BM (e.g. heavyweight rowers), the serve sizes in restaurants are too small for their energy needs. Providing written guidelines to caterers about serve sizes and meal sizes for different athletes helps address these misconceptions.

Food choices of athletes also vary with stage of competition and season. Athletes at the 2006 Commonwealth Games rated nutrient composition and stage of competition much higher than sensory appeal of food (e.g. smell, visual appeal) as factors influencing sports performance at competition (Pelly et al. 2006b). In another study of US college hockey players, the key determinants of food choice during competition were health beliefs and effects of diet on performance, compared with taste during the off-season (Smart & Bisogni 2001). During tapering, athletes are usually concerned about weight gain, eating less food and often following extremely low-fat diets. In those athletes in weight-class sports, extreme dieting, fluid restriction and other weight-loss behaviours (e.g. excess time in the sauna, training in plastic clothes) may be practised (see Chapter 7).

25.3 Food provision in a catering environment

Developing a catering policy to standardise the food supply, menus and recipes for athletes is a valuable strategy for quality control and for providing consistent messages about food provision, safety and nutrition, and has been used successfully at the AIS, the Olympic Games (Pelly et al. 2011; Pelly et al. under review) and Commonwealth Games (Pelly et al. 2006a; Burkhart & Pelly 2013a), world championships and International Masters games. The approach by organisers of food catering for athletes at events, particularly international events and even national events, varies considerably. Currently there are no existing nutrition standards or guidelines.

The challenge for catering organisations at any large-scale competition is to provide food suitable for thousands of athletes, representing hundreds of nations, who are competing in a wide variety of sports events. For example, at the Sydney 2000 Olympic Games around 1.16 million meals were served over 28 days with around 49 000 meals/d. The history of food provision at the Olympic Games is provided in detail in *Evolution of food provision to athletes at the Summer Olympic Games* (Pelly et al. 2011).

On a smaller scale, the Australian Institute of Sport (AIS) provides an excellent and successful model for meal catering for athletes in short- and long-stay residence, which could be used as a benchmark for feeding elite athletes when self-catering or competing away from home in other environments. Meal service for athletes and staff is delivered by private contractor, which includes a proviso for employment of a food-service dietitian to advise and oversee the menu plan. The mission statement in the dining hall is 'Feeding athletes for today, educating them for tomorrow'. The next section builds on the AIS model and provides strategies for food provision and menu design and delivery when catering for athletes in both short- and long-stay residential environments.

25.3.1 The menu cycle

The length of the menu cycle is determined by the length of time an athlete spends in residence. A weekly menu cycle is satisfactory for those in residence for less than a month. At competition events such as the Olympic Games, residents live in the athletes' village for up to one month (Pelly et al. 2011). A five- to ten-day menu cycle is common practice to avoid 'menu fatigue' and repetition (Pelly 2013). For those living in permanent residence, the menu cycle may be much longer. For example, at the AIS, where the athletes are usually in one- to two-year residence, the menu rotation is four weeks.

25.3.2 Variety and nutritional balance

The menu for both short and long-stay residential facilities must cater for training and performance requirements, and cultural and special dietary needs of athletes. Offering a variety of dishes addresses the variable nutrient needs between individuals and allows a wide range of personal choice. At permanent residences such as the AIS—which caters for approximately 300–600 athletes, coaches, staff and visitors—the menu options include four choices of hot meals at dinner—two high-CHO dishes, plain meat, a wet dish (e.g. curry or casserole)—and a selection of salads, fresh fruit and assorted breads. The four meal choices also include options for special dietary requirements (e.g. vegetarian and gluten-free food selection as well as avoidance of some allergens). To increase food diversity and provide visually appealing meals, a variety of meats are offered, using different cooking methods each week. Also, legumes—a relatively inexpensive, high-satiety food—are added to vegetarian or meat dishes to add carbohydrate and protein. A variety of flavours from different regions of the world with various textures and 'mouth feel' are available to maintain interest. The menu and most recipes are low in fat and high in carbohydrate to meet the dietary goals of athletes.

At large-scale international competition events such as the Olympic Games, the main dining hall in the athletes' village provides a rotational menu consisting of traditional hot and cold items, pizza, pasta, Mediterranean, African and Asian cuisine, and all-day cereal, yoghurt, bread, pastries, fruit, salad, condiments and beverages. Hot menu items are grouped into cultural areas. A restaurant for 'casual dining' and street food carts are located elsewhere in the village. The main dining hall is open 24 hours to cater for late night and early morning events, and also provides an overnight menu ('supper'). Sufficient low-energy food choices are needed. While a wide range of individual foods are offered at large-scale competitions (769 items at the Sydney 2000 Olympic Games), there shoud be adequate provision of sports foods, high energy drinks/meal replacements, snack foods, culturally specific, gluten-free foods and hot desserts (Pelly et al. 2009; Burkhart & Pelly 2013a; Pelly et al. under review). Sponsorship rights at large events may exclude the provision of familiar brands of sports food and beverages used by athletes at home.

25.3.3 Cost of food

Cost is a primary concern in menu planning. Cost-cutting measures include buying food in bulk at wholesale prices, adapting the menu to incorporate cheaper items, bulking-up casseroles and minced meat dishes using cheaper ingredients such as legumes and grains, purchasing in-season produce, and cooking meals in bulk and 're-inventing' the dish at a later stage. Leftover foods are often re-used, provided food safety and hygiene have been addressed (see section 25.3.6). Self-catering for a group is a cheaper option than eating out.

25.3.4 Seasonal variations

At residential facilities, the menu should be adapted to suit the season. In winter, athletes prefer hot meals such as thick soups, casseroles and curries. In summer, lighter dishes including stir-fries, cold meats and salads are preferred. To boost carbohydrate intake from salad dishes, rice, couscous, burghul, pasta, potatoes and corn are usually added. At major events, there is usually sufficient variety to provide both hot and cold choices to suit personal preference and seasonal conditions.

25.3.5 Timing of meals

Offering meals to coincide with training and competition schedules creates problems for catering management and staffing. At permanent residential facilities, athletes may train late in the day and not be present for meals until late at night. Because of staffing costs, meal times are usually held within a relatively short timeframe. However, there should be enough flexibility to arrange pre- and post-match meals and cater for athletes who regularly eat late. There is often the need for food provision over 24 hours, including options of portable snacks and boxed lunches. At the Olympic Games, the competition venue may be a considerable distance from the village and food must be provided for travelling to and while at the venue.

25.3.6 Food safety

In a large-scale catering facility, all kitchen staff should be trained in food safety and hygiene practices. Concerns over food safety at past Olympic Games have led to an onsite Hazard Analysis and Critical Control Points (HACCP) laboratory with testing of food samples and GPS tracking of food (Pelly 2007; Wanik 2009). For this reason, food is generally not permitted to be removed from the dining hall. Training in food safety practices, including food handling skills, appropriate storage and re-use of food and personal hygiene, are crucial for athletes who self-cater. Food safety training is also relevant for athletes who travel to countries where the risk of food-borne illness is high.

25.3.7 Food sustainability

At the population level, awareness of the environmental and economic impact of food provision and supply (i.e. food sustainability) has increased and is now being addressed with provision of food to athletes at major events. Food sustainability practices include, for example, the use of local, sustainable and seasonal food, reduced packaging, recycling, and general waste reduction. Food wastage is high with buffet-style eating. At the London Olympic Games, sustainability standards were incorporated into the catering tender (Pelly et al. under review).

25.3.8 Self-catering

Many young athletes have a limited budget and cannot afford restaurant meals, so self-catering is the best option. Restaurant dining or take-away food provides a convenient meal for athletes who are tired and hungry. However, take-away food choices do not always provide a suitable pre-exercise or recovery meal. Restaurant meals also encourage overeating and are more suitable at the end of competition. For some groups, restaurants and take-away foods are the only available option. Sometimes accommodation includes pre-paid meals. In these situations, liaison with restaurant staff in menu planning prior to competition prevents unsuitable foods being offered. If the timing of the training or competition schedule is tight, notifying the restaurant ahead of time averts waiting time or meals and opportunities for athletes to consume unsuitable foods.

Athletes who are unadventurous or inflexible in their eating habits usually resort to poor food choices when away from home. If trips away are frequent (e.g. every weekend), performance capacity and possibly nutritional status can be affected. A supplementary food supply including additional snacks and favourite foods is recommended to help minimise reliance on take-away foods.

25.4 Nutrition support to catering

The aim of nutrition support is to guide caterers to provide the most suitable food and meals, which assists and educates athletes to make appropriate food choices. Strategies that influence food choice behaviours of athletes in a catering environment include modifying catering policy to provide appropriate meals and healthy food choices, providing point-of-choice nutrition information (i.e. with nutrition labels) and encouraging individuals in meal and portion size selection (Glanz & Hoelscher 2004; Seymour et al. 2004). In practice, effective nutrition support delivered by dietitians to caterers includes training catering staff in understanding the nutrient needs of athletes, upgrading staff skills in food safety practices, ongoing menu review and recipe modification, and access to dietitians or nutrition experts within the dining environment (Pelly et al. 2009; Pelly et al. 2006a; Burkhart & Pelly 2013b, 2013c; Pelly et al. under review).

The interaction with sports dietitians and catering staff has increased substantially since the initiation of a comprehensive nutrition program at the Sydney 2000 Olympic Games (Pelly et al. 2011); however, the amount of input varies between events. At the 2006 Commonwealth Games, catering and medical dietetic services were successfully integrated to provide a comprehensive nutrition support service that was valued highly by athletes (Pelly et al. 2006a). Skills required by dietitians to undertake this role include food-service and clinical expertise, cultural awareness and a good understanding of performance requirements (Burkhart & Pelly 2013c). Challenges to implementing food-service modifications include the absence of a clearly defined nutrition policy, changes to catering providers between events and food sponsorship issues.

At the AIS, the food-service dietitian reinforces dietetic advice given to athletes from the sports nutrition department in a dining environment by translating advice into appropriate food choices, develops new recipes, designs the menu plan, checks on adherence to recipes by chefs and trains food-service staff.

25.4.1 Education of food-service staff

Catering staff may not have experience feeding athletes. Providing written guidelines with menu suggestions and recommended serve sizes encourages appropriate food provision. Several recipe books are available for catering for large groups of people and specifically for athletes (see Practice tips). Providing nutrition education sessions to food-service staff helps them to feel involved with the sport service team and improves their receptiveness to menu changes and complaints. Food-service staff can then assist athletes in food and portion size selection. Nutrition education sessions have been successfully conducted for catering staff at the Commonwealth and Olympic Games (see Pelly et al. 2006a, 2009).

25.4.2 Menu review and analysis

Menu planning allows the translation of nutrition principles into real food choices for athletes. Helping athletes select food provides an ideal opportunity to raise awareness of food choice, influence existing beliefs and attitudes and ultimately improve food selection. Menu reviews may be conducted both quantitatively (using dietary analysis software) or qualitatively (variety, spread across food groups, provision of macronutrients). Menus are usually designed by caterers or chefs and reviewed by a designated nutrition expert who assesses the suitability for the target group. Prior to the Sydney 2000 Olympic Games, a custom database ('Foodweb') was designed to enable dietary analysis of food production data, classify dishes based on macronutrient content and special dietary needs, and create point-of-choice nutrition labels (Pelly et al. 2009). The Foodweb database was linked to the caterer's recipe database so that the analysis dynamically reflected changes to the preparation of foods. The development of a similar database would be of benefit to both residential and competition events in the future.

Since 2007, independent review of the Olympic food menu has been conducted by a group of international nutrition experts Professionals in Nutrition and Exercise Science (PINES), with feedback provided to the International Olympic Committee (IOC). This process initially focused on a suitability of the menu from a cultural and performance perspective, but has extended to encompass the total food provision at these events (including food for travelling to and at venues). While this review helps caterers plan suitable food provision for athletes, there are still issues with the implementation of these recommendations on site (Pelly et al. under review). Ideally, close and ongoing involvement in the early planning stage to the final implementation of food service in this environment is crucial to a successful outcome.

Based on our observations, the proportion of athletes requesting special dietary requirements at major events (i.e. Olympic/Commonwealth and other international events) has increased (Pelly & Burkhart 2014). An inadequate provision of modified foods (e.g. lower energy/low fat items, particularly dessert) was the key finding of a recent food menu review (Pelly et al. under review). Another role of the dietitian is to assist catering staff (who may not necessarily be chefs) to modify traditional recipes to meet special dietary requirements without compromising the flavour, texture or structure of the finished product.

25.4.3 Use of nutrition labels

Providing nutrition information about meals and snacks in a dining environment influences food selection of diners (Mayer et al. 1989; Seymour et al. 2004). Although the use of nutrition labels of meals in dining halls is a well-established method of providing nutrition information, their success in altering food choice has produced mixed results (Davis-Chervin et al. 1985; Sproul et al. 2003; Steenhuis et al. 2004a, 2004b).

Nutrition labels were first introduced at the 1992 Barcelona Olympic Games and are now mandatory for food provided in the main dining hall at both Olympic and Commonwealth Games. Studies conducted at major competition events have found that athletes rate the labels as useful and read them the majority of the time (Pelly et al. 2006a, 2009), particularly when in an unfamiliar environment (Burkhart & Pelly 2013b). However, label-reading behaviour does not necessarily alter food choice, especially in young males (Sproul et al. 2003) and may increase overall energy intake in restrained eaters (Aaron & Mela 1995).

Evidence from non-athlete populations suggests that label readers are predominantly female, have a high level of education (Cowburn & Stockley 2005; Belser et al. 2012), already eat a healthy diet and are health conscious (Kreuter & Brennan 1997). In the competition dining hall environment, athletes with less education, from specific countries (i.e. India/Sri Lanka, Africa) and from team and weight category sports tend to use food labels. In our study of 352 athletes, females rated the importance of providing nutrition information higher than males (Burkhart & Pelly 2013b).

Nutrition labels may include energy and nutrient composition, serve size, an ingredient list, suitability for a range of dietary regimens and identification of potential allergens. See Burkhart & Pelly (2013b) for a summary of information provided on labels at past sporting events. Label formats targeted at athletic populations vary considerably resulting in a lack of uniformity and standardisation (Burkhart & Pelly 2013b; Pelly et al. under review). It is well documented that consumers have difficulty understanding food labels and the nutrition information panel on foods (Cowburn & Stockley 2005). Athletes are likely to be no different. Training athletes on how to read food labels (prior to an event) as well as providing brochures and nutrition booths (or a nutrition information centre) at the dining hall is more likely to influence food choice than displaying labels (Seymour et al. 2004).

Figure 25.1 is an example of a nutrition label used for athletes.

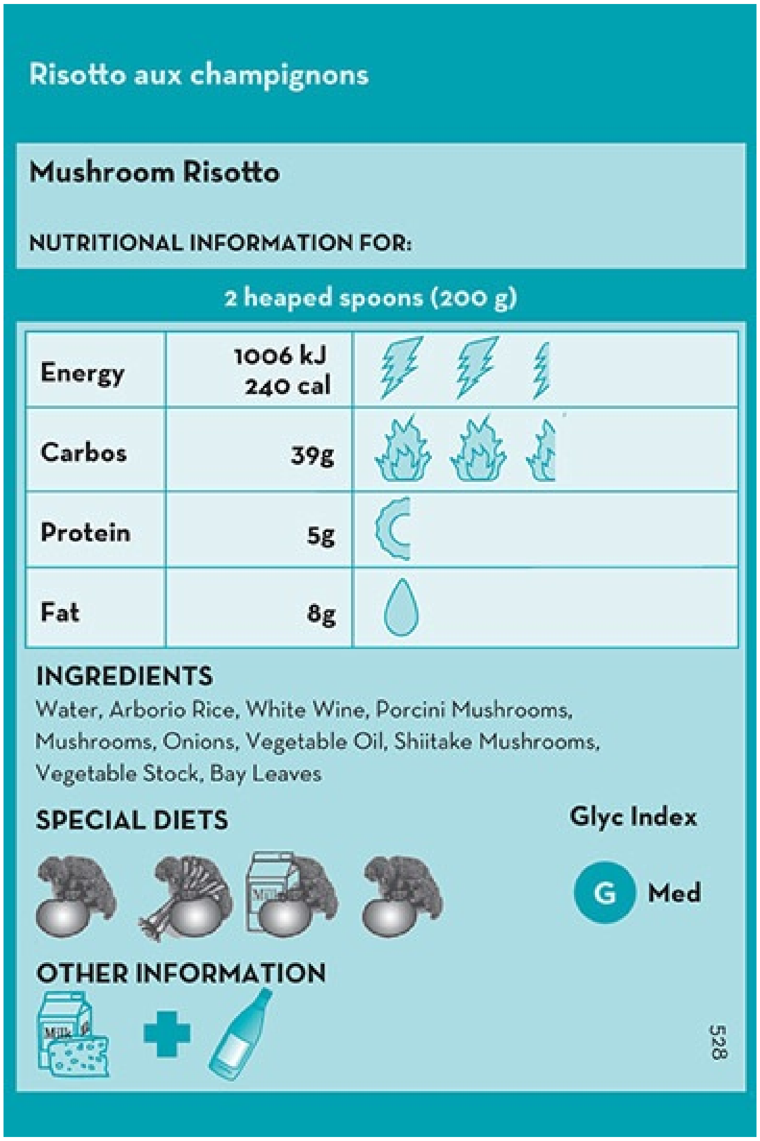

Figure 25.1 Example of a nutrition label from the Sydney 2000 Olympic Games

Source: Reproduced with permission from Pelly F & Denyer G, University of Sydney, Australia

25.4.4 Nutrition information centre

A nutrition information centre located within the dining hall at competition events is a valuable addition to enhance the nutrition service to athletes (Pelly et al. 2006a; Pelly et al. 2009; Burkhart & Pelly 2013c). While some athletes receive nutrition advice in their home countries, many have had no access to any nutrition services (Burkhart & Pelly 2013c). The dining hall is an ideal location to assist athletes with appropriate food choices and provide nutrition advice. An evaluation of the service provided by the nutrition centre at the 2006 and 2010 Commonwealth Games suggested that it was highly valued by athletes and officials (Pelly et al. 2006a; Burkhart & Pelly 2013c). Sports dietitians located at the nutrition desk provided athletes with information about the menu, conducted individual dietary assessment, provided meal plans, met special dietary requests and answered sports and clinical nutrition enquiries. Consultations at the Delhi 2010 Games were predominately for athletes trying to make or manage their weight prior to competition, while enquiries focused on provision of suitable foods for those athletes with food allergies and intolerance (Burkhart & Pelly 2013c). At the 2006 Commonwealth Games, the special diet requests were predominately for gluten- and nut-free items, soy and oat milk, garlic, unseasoned food and probiotics (Pelly et al. 2006a). In this centre, sports dietitians are in an ideal location to provide immediate feedback to athletes on their questions and complete quality assurance and compliance checks on the menu.

25.4.5 Menu websites and other technologies

Viewing the menu in advance of competition can assist athletes with planning their competition diet before arriving at the venue. For the Sydney 2000 Olympic Games, a website was developed for athletes to see the menu well in advance of the games (Pelly et al. 2009). For people with special requirements, a search could be done for all dishes excluding a certain ingredient (e.g. meat) or for specific dietary needs or religious requirements (e.g. halal). Other features included ‘Ask a sports dietitian’, sample meal plans and information sheets. Since 2000, there are been varying degrees of technological support for information about the menu. At the London 2012 Games, an internet site was available for athletes to search the menu; however, smart phone and tablet apps have not been used to date.

25.5 Education of athletes

Dietitians who travel with athletes need to be multi-skilled and offer services other than just menu planning, including shopping, cooking and team management. For example, cooking classes with the team just prior to departure helps instil confidence in those athletes with poor cooking skills and provides an ideal opportunity to plan the menu. Providing guidelines on suitable snacks, meals and recipes is also well received. In some cases, working with the team management to provide suitable options may be an effective way of ensuring good food choice.

Dietitians working at the AIS conduct regular education sessions with residential and visiting athletes. The food-service dietitian is responsible for promoting nutrition education via multiple media. Examples include converting nutrition information into user-friendly terms for the AIS television channel and digital menu screens in the dining hall. Planning frequent international theme nights encourages athlete participation in menu planning, alleviates the monotony of eating in the dining hall and allows athletes to get used to the cuisine of a particular country in advance of their visit.

Summary

07:38:06.

Catering for athletes provides an opportunity to directly influence the food supply and food choice. Nutrition support in the catering environment can be multifaceted and can assist both catering staff and athletes' confidence in food selection and preparation. The involvement of a dietitian in food-service provision provides an ideal opportunity to translate the nutrition goals of competition into practical food choices.

Practice tips

FIONA PELLY AND JOANNE MIRTSCHIN

THE ROLE OF THE FOOD-SERVICE DIETITIAN

The tasks of a food-service dietitian in catering for athletes may include the following:

- development, analysis or review of a menu or entire food provision service considering all cultural, therapeutic and performance needs
- analysis and modification of recipes
- education of athletes, coaches and food-service providers
- education and professional development for catering staff
- management of a quality assurance program, which includes evaluation by formal surveys, focus groups and verbal feedback from representative athletes
- production of education material to complement the food service
- design or input into nutrition labelling and supportive material
- input into the food safety program
- development of specialised meal plans incorporating menu items
- design portable lunch/dinner packs suitable for different dietary needs
- liaison with catering management, team dietitians, coaches, trainers and other health professionals
- in the dining hall of an athletes' village, the dietitian may conduct individual consultations with athletes, group sessions with teams and dining hall tours.

MENU PLANNING FOR ATHLETIC GROUPS

A menu must appeal to athletes' taste preferences, meet their nutrient requirements, and address any special dietary and sporting needs for a range of ages and cultural backgrounds. To be acceptable to an organisation, it must be cost effective, minimise wastage and satisfy budgetary constraints, food safety regulations and available cooking facilities.

When catering for large groups of athletes from differing sports, the preferred option is buffet-style self-service. Traditional western styles of eating, with the emphasis on meat as the main part of the meal, does not encourage high carbohydrate consumption. Strategies that enhance carbohydrate intake and improve nutrient balance in a buffet-style servery include offering carbohydrate-rich foods first in the line-up (pasta, rice, potatoes and vegetables), and meat dishes towards the end of the line. Placing bread close to soup, for example, and offering a variety of breads (e.g. focaccia, crusty and Italian-style loaves), further encourages high carbohydrate choices. Adding legumes to casseroles and soups, thick layers of pasta to lasagne, noodles to salads and soups, and using thick pizza bases further improves contribution of carbohydrate. Sandwiches, fruits, yoghurts, cereals and cereal bars can be provided for in-between meals. At least one low-fat/energy alternative for almost every type of traditional/authentic menu item should be offered at each meal period. Traditional 'popular' menu items such as lasagne, schnitzel and Thai-style curries can be produced without the usual high-fat ingredients.

For a smaller group of athletes, one evening meal choice is acceptable, provided that extra carbohydrate-rich foods are offered as an accompaniment. In a small group, individuals with special needs, such as vegetarians, can be catered for individually. The use of different-coloured ingredients improves aesthetic appeal and encourages consumption.

For large-scale catering, sufficient basic or plain menu options should be available to accommodate the taste preferences of younger athletes. When feeding a smaller group, meat dishes and their accompanying sauces can be served separately to cater for fussy eaters. To cater for differences in food choice between

males and females, hot items such as tinned spaghetti and baked beans can easily be incorporated into the breakfast menu by using pre-prepared foods.

It is important to consider the cultural mix of the target group when menu planning. Religious practices such as food avoidance and fasting, and timing of events such as Ramadan should be factored into food provision. Use books such as *Food and culture* (Kittler & Sucher 2011) as a guide to suitable menu items and aim to include a few culturally specific items. Consider the provision of items that will meet specific dietary regimens such as gluten free and lactose free.

RECIPE MODIFICATION FOR CATERERS

Australian cookbooks written specifically for sportspeople are readily available (Inge & Roberts 1996; O'Connor & Hay 1996, 1998; Modulon & Burke 1997; Burke et al. 1999, 2001, 2004, 2009). Any low-fat cookbook or recipe is suitable for athletes, although additional carbohydrate-rich foods may need to be added to the final meal. Modifying existing recipes with lower fat or alternative ingredients is possible with most recipes, without a large change in texture or flavour. It may be necessary to provide assurance that taste profiles can be maintained and, at times, enhanced with some changes to cooking methods. An efficient strategy to minimise modifying recipes is to ensure that one recipe suits multiple dietary requirements. For example, using mushroom oyster sauce instead of a regular variety in a stir-fry has the potential to satisfy a vegetarian, gluten-free and no seafood option without compromising the flavour of the original item. Some restaurants tend to remove fat and sugar out of their menu entirely, which is not always necessary or appropriate for some groups.

GUIDELINES FOR ATHLETES WHEN CHOOSING FOODS FROM A BUFFET

Athletes living and eating in a communal environment frequently overeat, eat sub-optimal amounts of one or more food groups (mostly vegetables) or habitually eat a poorly balanced diet when offered food from a buffet service. The disadvantages of a live-in environment are lack of involvement with food, loss of cooking skills and no knowledge of the ingredients in a recipe. These issues are addressed by providing education sessions when athletes arrive, using nutrition labelling and involving athletes in menu planning. Table 25.1 provides the information given to athletes at the AIS about developing eating strategies in the dining hall.

TABLE 25.1 Tips for athletes eating from a dining hall buffet

1.	Know your nutritional goals and how to choose foods to achieve these goals.
2.	Be focused and organised when planning your meal times and snacks. Don't leave it to chance. Stick to this plan.
3.	Treat the dining hall like a restaurant. Look at the menu, then make a decision as you wait in the queue.
4.	Don't concern yourself with the amount and type of food that other athletes are consuming.
5.	Don't pile a bit of everything on your plate. This type of eating is haphazard and unbalanced, and you may end up eating more than you need.
6.	Eat what you need from a balanced food selection. The more colourful, the better!
7.	Read any nutrition information provided to learn more about your meal choice and to help make food choices.
8.	Relax. There is plenty of food for everyone and menu items will be repeated.
9.	Plan for healthy snacks between meals—especially if you have high-energy needs.

10.	Don’t hang around the dining hall once you have finished your meal. You’ll end up eating things that you don’t need.

Source: Adapted from the AIS dining hall information sheet

EDUCATING CATERING STAFF

The content and format of education sessions for catering staff need to reflect the particular catering situation and background of the staff involved and can cover standard recipes, service standards, guidelines for healthy eating, special nutrition needs of athletes, menu planning, recipe modification, and food hygiene and safety practices. Aspects of these education programs are easily adapted into written guidelines to give to new caterers. This information is also useful for team managers responsible for organising meals.

MENU PLANNING FOR SELF-CATERING

Prior to departure, the dietitian needs to arrange a meeting with the team, coach and manager to determine the catering requirements, competition or training schedules, the food skills of the team and the allocated budget. For most local or national competitions where the athletes are not professional, self-catering is usually the preferred option because of financial constraints. Table 25.2 provides some suggestions for those working with travelling teams.

TABLE 25.2 Tips for dietitians to prepare athletes for self-catering

- Discuss catering requirements with the athletes/manager/coach.
- Determine the cooking skills of the group. Plan pre-camp cooking sessions if feasible. Alternatively coordinate during the camp if travelling with the team.
- Determine in advance the cooking equipment, food budget, shop access and cooking facilities available.
- Plan the menu around the length of stay, competition or training program.
- Designate responsibility for shopping and cooking of food among the athletes. Consider contacting the local supermarket to see if they will deliver.
- Develop a list of basic ingredients and cooking equipment that can be taken from home and shared among the team.

Planning for dining out

- Book a restaurant in close proximity to the accommodation.
- Confirm that the menu provides appropriate and suitable foods for the age, budget and food preferences of the team.
- If a large group of athletes is booked in advance, send suitable menu selections, or ask to view the menu to see if you need to suggest some slight variation for your group.
- For a large group of athletes, arrange a set à la carte menu with the option of one or two choices, or a buffet-style servery.
- Inform restaurants of the estimated time of arrival.
- Ensure there is plenty of extra bread served with the meal.
- Ensure water and juices are kept well stocked during the meal.

Source: Adapted from Modulon & Burke 1997

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Abbreviations

3-MH	3-methylhistidine
AA	amino acids
ABP	athlete biological passport
AC	activation co-efficient
ACL	anterior cruciate ligament
ACSM	American College of Sports Medicine
ADP	adenosine diphosphate
AI	adequate intake
AIS	Australian Institute of Sport
AMDR	acceptable macronutrient distribution range
AMP	adenosine monophosphate
AMPK	AMP-activated protein kinase
ASADA	Australian Sports Anti-Doping Authority
AST	aspartate aminotransferase

AT	adaptive thermogenesis
ATP	adenosine triphosphate
a-v	arteriovenous
BCAA	branched-chain amino acid
BCAAT	branched-chain aminotransferase
BCOAD	branched-chain oxo-acid dehydrogenase enzyme
BED	binge-eating disorder
BIA	bioelectrical impedance analysis
BIS	bioelectrical spectroscopy
BM	body mass
BMC	bone mineral content
BMD	bone mineral density
BMI	body mass index: weight (kg) ÷ height (m^2)
BMR	basal metabolic rate
CAT	carnitine acylcarnitine translocase
CHO	carbohydrate
CHr	reticulocyte hemoglobin content
CP	creatine phosphate, also cerebral palsy

CPT	carnitine palmityltransferase
CrP	creatine phosphate
CSA	cross-sectional areas (in muscle)
CT	computed tomography
CV	coefficient of variation
DHEA	dehydroepiandrosterone
DIT	diet-induced thermogenesis
DLW	doubly labelled water
DRI	dietary reference intake
DT	drive for thinness
DXA	dual energy X-ray absorptiometry
EA	energy availability
E-AC	erythrocyte enzyme activation co-efficient
EAR	estimated average requirement
E-ASTAC	erythrocyte aspartate amino transferase activation co-efficient
EC	excitation–contraction
EDNOS	eating disorders not otherwise specified

EE	energy expenditure
EEE	energy cost of exercise
EER	estimated energy requirement
E-GRAC	erythrocyte glutathione reductase activation co-efficients
EI	energy intakes
EI:BMR	the ratio of reported energy intakes (EI) to predicted basal metabolic rates (BMR)
eIF2Bepsilon	eukaryotic initiation factor 2B
E_{in}	energy in (energy intake)
EMS	eosinophilic myalgia syndrome
EPOC	excess post-exercise oxygen consumption
E-TKAC	erythrocyte transketolase activation co-efficients
EXOS	exercise-oxidative stress
E_{out}	energy out
FA	fatty acid/s
FBR	fractional breakdown rate
FDA	Food and Drug Administration
FED-NEC	feeding or eating disorders not elsewhere classified
FES	

	functional electrical stimulation
FFA	free fatty acids
FFM	fat-free mass
FFQ	food frequency questionnaire
FHA	functional hypothalamic amenorrhea
FM	fat mass
FRAR	food-related adverse reactions
FSR	fractional synthetic rate
FT	fast twitch
GER	gastro-oesophageal reflux
GERD	gastro-oesophageal reflux disease
GH	growth hormone
GI	gastrointestinal, also glycaemic index
GnRH	gonadotropin-releasing hormone
GOT	glutamate oxalacetate transaminase
GR	glutathione reductase
GS	glycogen synthase
GTF	glucose tolerance factor

HAD	3-hydroxyacyl coenzyme A dehydrogenase
HBV	high biological value
HMB	beta-hydroxy beta-methylbutyrate
HPLC	high pressure liquid chomatography
HSL	hormone-sensitive TG lipase
IBD	inflammatory bowel disease
IBS	irritable bowel syndrome
ICS	irritable colon syndrome
IDA	iron deficiency anaemia
IGF-I and IGF-II	insulin-like growth factors I and II
IMP	inosine 59-monophosphate
IOC	International Olympic Committee
IPC	International Paralympics Committee
ISAK	International Society for the Advancement of Kinanthropometry
ISCD	International Society for Clinical Densitometry
LBM	lean body mass
LCFA	long-chain fatty acid
LDL	

	low-density lipoprotein
LEU	leucine
LH	luteinising hormone
LMI	lean mass index
MAP	mean arterial pressure
MAPK	p38 mitogen-activated protein kinase
MCFA	medium-chain fatty acids
MCHC	mean corpuscular haemoglobin concentration
MCT	medium-chain triglyceride
MCV	mean cell volume
MDA	malondialdehyde
MES	minimum effective strain
MET	metabolic equivalent of task
MNT	medical nutrition therapy
MPS	muscle protein synthesis
MRI	magnetic resonance imaging
mRNA	messenger RNA
MSDS	materials safety data sheet

mTOR	mammalian target of rapamycin
MVC	maximum voluntary contraction
MW	minimal weight
NBAL	nitrogen balance
NCAA	National Collegiate Athletic Association
NEAT	non-exercise activity thermogenesis
NEFA	non-esterified fatty acid
NHANES	National Health and Nutrition Examination Survey
NHMRC	National Health and Medical Research Council
NO	nitric oxide
NRV	nutrient reference value
NSAID	non-steroidal anti-inflammatory drug
PAL	physical activity level
PCOS	polycystic ovarian syndrome
PDH	pyruvate dehydrogenase
PHV	peak height velocity
PINES	Professionals in Nutrition and Exercise Science
PLP	

	plasma pyridoxal-5-phosphate
POMC	pro-opiomelanocortin
PQCT	peripheral quantitative computed tomography
PRPP	phosphoribosyl pyrophosphate
PTH	parathyroid hormone
PYY	peptide YY
QUS	quantitative ultrasound
RBC	red blood cells
RCT	randomised controlled trials
RDA	recommended dietary allowance
RDI	recommended dietary intake
RED-S	relative energy deficiency in sports
RER	respiratory exchange ratio
RMR_{ratio}	ratio between measured and predicted resting metabolic rate
RMR	resting metabolic rate
RNI	recommended nutrient intake
ROS	reactive oxygen species
RPE	ratings of perceived exertion

RQ	respiratory quotient
SC	spinal cord
SCI	spinal cord injury
SD	standard deviation
SDT	suggested dietary targets
SF	serum ferritin
SMR	sleep metabolic rate
SNRI	selective noradrenaline reuptake inhibitor
SPA	spontaneous physical activity
SR	sarcoplasmic reticulum
SSRI	selective serotonin reuptake inhibitor
sTfR	serum transferrin receptor
TBARS	thiobarbituric acid reactive substances
TBK	total body potassium
TBW	total body water
TCA	tricarboxylic acid
TEA	thermic effect of activity
TEE	

	thermic effect of exercise
TEF	thermic effect of food
TEM	technical error of measurement
TEM	thermic effect of a meal
TfR	transferrin receptors
TG	triacylglycerol
TGA	Therapeutic Goods Administration (Australia)
TK	transketolase
tRNA	transfer RNA
TUE	therapeutic use exemption
UIL	upper intake level
UL	upper level of intake
URTI	upper respiratory tract infection
UV	ultraviolet
UWW	underwater weighing
VDBP	vitamin D-binding protein
VLCD	very low calorie diet
VLED	

	very low energy diet
VO2 max	maximal oxygen uptake
VO2 peak	peak oxygen uptake
W	watt
WADA	World Anti-Doping Agency
WHO	World Health Organization
ww	wet weight

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